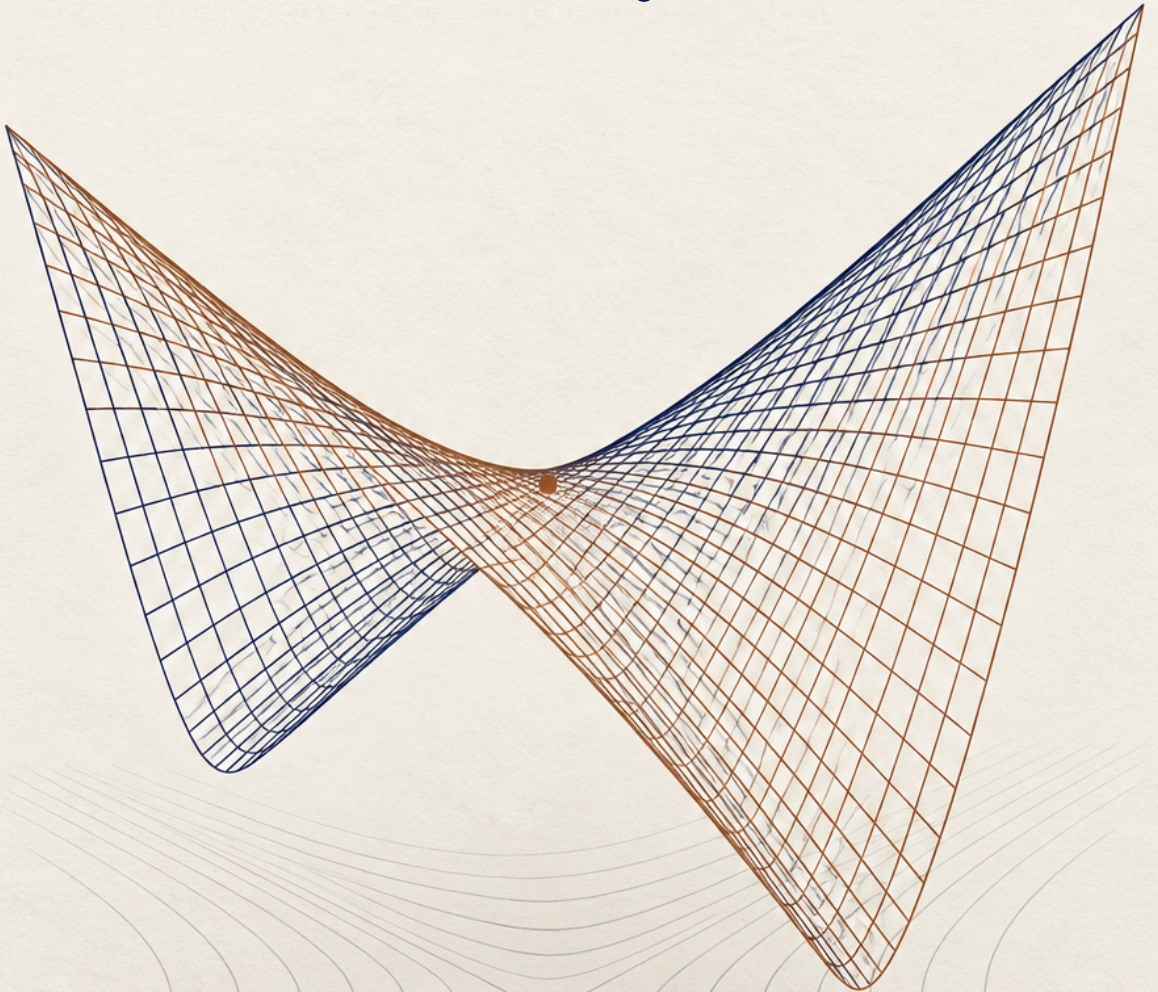


LECTURES ON

Optimization and Optimal Control Theory



Quanyan Zhu

New York University

Lectures on Optimization and Optimal Control Theory

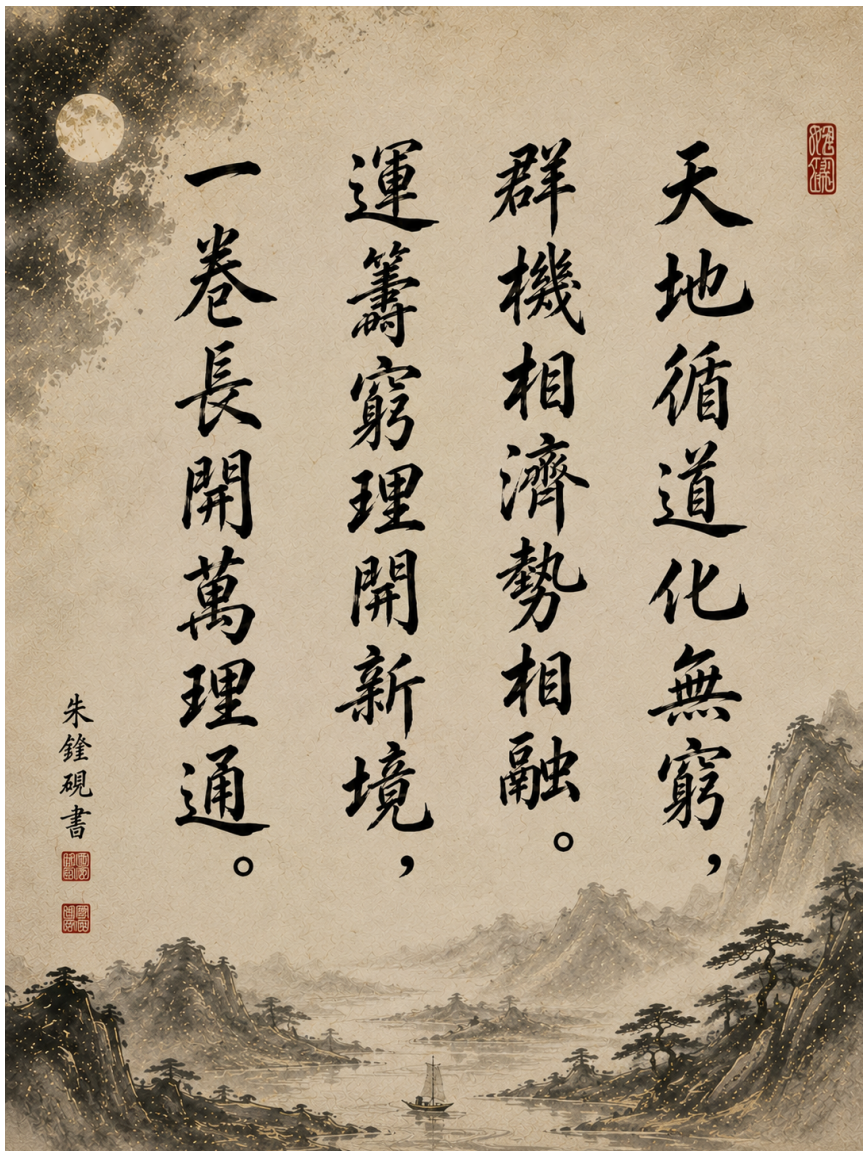
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群機相濟勢相融。
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朱銓硯書



Contents

1	Introduction	13
1.1	Introduction	13
1.2	Examples	19
1.3	A Historical Timeline of Optimization Theory	27
1.4	Organization of the Book	31
1.5	Distinctive Features of the Book	33
1.6	Further Reading	34
1.7	Exercises	36
I	Foundations and Optimization Theory	39
2	Mathematical Foundations of Optimization Theory	41
2.1	Mathematical Foundations: Vector Spaces, Norms, and Differentiation	41
2.2	Extreme Values of Functionals and Compactness	62
2.3	Gâteaux and Fréchet Differentials	72
2.4	Local Optimality Conditions in Finite-Dimensional Optimization	82
2.5	Variational Inequality Formulation	101
2.6	Duality for the General Constrained Problem	103
2.7	Further Reading	110
2.8	Exercises	111
3	Classical Optimality Conditions	119
3.1	Constrained Optimization and Optimality Concepts	119
3.2	Method of Lagrange	123
3.3	Fritz–John and Mangasarian–Fromovitz Conditions	139
3.4	Further Reading	160
3.5	Exercises	160
II	Convex Optimization	165
4	Basic Convex Optimization	167
4.1	Convex Optimization: Basic Concepts and Geometry	167
4.2	Introduction to Convex Optimization	187
4.3	Constraint Qualifications and Slater’s Condition	189
4.4	Basic Convex Program Characterization	191
4.5	Duality and the Dual Function	201
4.6	Transformation of Nonconvex Problems into Convex Problems	218

4.7	Hahn-Banach Theorems	231
4.8	Further Reading	238
4.9	Exercises	238
5	Generalized Convex Optimization	249
5.1	Generalized Saddle-Point Theorem	249
5.2	Saddle-Point Characterization without Slater's Condition	252
5.3	Introduction to Non-Smooth Convex Programs	283
5.4	Further Reading	291
5.5	Exercises	291
6	Vector Space Convex Optimization	297
6.1	Linear Functionals and Dual Spaces	297
6.2	Cones	300
6.3	General Convex Programming in Vector Spaces	305
6.4	Case Study: Optimal Control with Terminal Inequality Constraints	319
6.5	Case Study: Optimal Extraction of an Exhaustible Resource	321
6.6	Further Reading	324
6.7	Exercises	324
III	Algorithms	333
7	Algorithms for Unconstrained Problems	335
7.1	Numerical Algorithms for Unconstrained Optimization	335
7.2	Convergence Results	340
7.3	Nonconvex Gradient Descent and the Łojasiewicz Framework	352
7.4	Gradient Methods with Noise or Errors	360
7.5	Newton's Method	379
7.6	Further Reading	386
7.7	Exercises	387
8	Algorithms for Constrained Problems	391
8.1	Feasible Direction Methods	391
8.2	Projection Methods	396
8.3	Homotopy and Barrier Method	405
8.4	Primal–Dual Algorithms	413
8.5	Primal–Dual Algorithms for Network Utility Maximization	421
8.6	Further Reading	428
8.7	Exercises	428
IV	Deterministic Optimal Control	433
9	Calculus of Variations	435
9.1	From Classical Examples to the Calculus of Variations	435
9.2	The Calculus of Variations: Functional-Analytic Setting	438
9.3	First-Order Necessary Conditions	442
9.4	Extensions of the Euler–Lagrange Equation	452
9.5	Second-Order Conditions	465
9.6	Second-Order Sufficient Conditions	468

9.7	Extensions: First-Order Necessary Conditions	473
9.8	Further Reading	478
9.9	Exercises	479
10	Discrete-Time Optimal Control	483
10.1	Discrete-Time Maximum Principle	483
10.2	Neural Network Training as a Finite-Dimensional Constrained Optimization Problem	491
10.3	Convex Optimal Control with State Inequality Constraints	496
10.4	Convex Optimal Control with State Constraints: Normal-Cone Formulation	498
10.5	Infinite-Horizon Limit and Stationary Case	501
10.6	Dynamic Programming Principle: Intuition, Structure, and Formalization	502
10.7	Shortest-Path Problems: A Constructive Illustration	508
10.8	A Canonical Example: Resource Allocation over Time	509
10.9	Time Consistency and Representation	518
10.10	Further Reading	529
10.11	Exercises	529
11	Continuous-Time Optimal Control	533
11.1	Optimal Control Problems and the Minimum Principle	533
11.2	Examples of Optimal Control Problems	542
11.3	Saddle-Point Maximum Principle	547
11.4	Kalman Filtering via the Minimum Principle	551
11.5	Dynamic Programming and Hamilton–Jacobi Equations	556
11.6	Turnpike Properties of Optimal Control	566
11.7	Further Reading	576
11.8	Exercises	577
12	Optimal Control under Incomplete Information	581
12.1	Discrete-Time Optimal Control under Imperfect State Measurements	582
12.2	Dynamic Programming and Certainty Equivalence	590
12.3	Continuous-Time Optimal Control under Imperfect Measurements	604
12.4	Notes and Further Reading	613
12.5	Exercises	615
V	Stochastic Optimal Control	619
13	Estimation Theory	621
13.1	Pre-Hilbert and Hilbert Spaces	621
13.2	The Projection Theorem	624
13.3	The Gram–Schmidt Procedure	628
13.4	Estimation of Random Variables and Vectors	636
13.5	Sequential Linear Least Squares Estimation	643
13.6	Further Reading	647
13.7	Exercises	647
14	Kalman Filtering	651
14.1	Discrete-Time Kalman Filter	652
14.2	The Stochastic Linear Regulator Problem	667
14.3	Asymptotic Properties of the Linear Regulator	674

14.4	Stochastic Control of Linear Systems with Partial Observations	675
14.5	Stability of the Closed-Loop System	679
14.6	Further Reading	683
14.7	Exercises	683
15	Stochastic Control	685
15.1	Elements of Stochastic Control Systems	685
15.2	Open-Loop and Feedback Control	688
15.3	Transition Probabilities and the Markov Property	693
15.4	Linear Gaussian Stochastic Systems	695
15.5	Optimal Control under Complete Information	699
15.6	Case Studies of Optimal Control Under Complete Information	704
15.7	Information States and Dynamic Programming	714
15.8	Dual Control with an Unknown Parameter	720
15.9	Finite-Horizon LQG with Imperfect State Information	723
15.10	Infinite-Horizon Steady-State LQG Control	737
15.11	Further Reading	742
15.12	Exercises	742
	Appendices	749
A	Linear System Theory	749
A.1	State-Space Models	749
A.2	Solution of the State Equation	750
A.3	Stability	752
A.4	Controllability and Reachability	754
A.5	Observability and Duality	755
A.6	Canonical Decomposition and Realizations	757
A.7	Notes and Further Reading	758
	Exercises	758
B	Probability Theory	761
B.1	Probability Spaces and Events	761
B.2	Random Variables and Distributions	763
B.3	Expectation, Moments, and Covariance	764
B.4	Gaussian Random Vectors	766
B.5	Conditional Expectation and Conditional Distributions	767
B.6	The Hilbert Space of Random Variables and the Projection Theorem	769
B.7	Sequences of Random Variables and Limit Theorems	770
B.8	Stochastic Processes	771
	Exercises	773
	Bibliography	775
	Index	781

List of Figures

1.1	Anatomy of a minimization problem	14
1.2	Optimization as a systems loop	16
1.3	Resource allocation under capacity	21
1.4	The brachistochrone problem	23
1.5	Epidemic control with dynamics	24
1.6	Optimal stopping and threshold rules	26
1.7	Historical timeline of optimization and optimal control	27
1.8	Organization of the book	32
2.1	Foundational dependencies in optimization	42
2.2	Unit balls of common norms	44
2.3	Open, closed, and neither sets	47
2.4	Relative interior of a set	48
2.5	Continuity in normed spaces	52
2.6	Completeness and the limits of Cauchy sequences	55
2.7	Compactness via sequences	64
2.8	Nonattainment on the unit ball	68
2.9	Coercivity and existence of minimizers	70
2.10	Gâteaux versus Fréchet differentiability	75
2.11	Cones under nonnegative scaling	85
2.12	Tangent cone at a polyhedral corner	86
2.13	Tangent cone of a half-space	87
2.14	Tangent cone to the unit disk	88
2.15	Tangent cone of an equality constraint	89
2.16	Tangent cone of an inequality constraint	90
2.17	Normal cone to a half-space	90
2.18	Normal cone to the unit disk	91
2.19	Tangent and normal cones	91
2.20	The critical cone	99
2.21	Fenchel duality construction	103
3.1	Euler–Lagrange proof geometry	125
3.2	LICQ at a feasible point	125
3.3	Quadratic penalty method geometry	136
4.1	Convexity as a line-segment property	168
4.2	Convex function chords and epigraph	175
4.3	Slater’s condition and KKT geometry	190
4.4	Water-filling interpretation of KKT	202

4.5	Dual supporting hyperplanes, inactive case	206
4.6	Supporting hyperplanes, active case	207
4.7	Degenerate absolute-value problem	207
5.1	Feasible-set decomposition	250
5.2	Tangent and normal cones at an optimum	251
5.3	Separation proof for the saddle theorem	253
5.4	Sensitivity of KKT multipliers	281
5.5	Subgradients and the subdifferential of $ x $	284
5.6	Subdifferential of a maximum of affine functions	285
5.7	Soft thresholding from the subdifferential	289
6.1	The dual cone and its half-angle	302
6.2	Saddle inequality and strong duality	309
6.3	The differentiable KKT system	318
6.4	Duality reduces the control problem	320
6.5	Front-loaded optimal extraction path	323
7.1	Anatomy of a descent algorithm	336
7.2	Contraction factor versus step size	350
7.3	Constant versus diminishing step size	351
7.4	Relative error of steepest descent	352
7.5	Zig-zagging steepest-descent iterates	353
7.6	Gradient descent on a double-well objective	358
7.7	Convergence rates versus the Łojasiewicz exponent	360
7.8	Phase-space picture of the ODE method	366
7.9	Tracking argument in the proof	366
7.10	Mean-square error of stochastic gradient descent	378
7.11	Newton's method versus gradient descent	386
8.1	Descent proposal and feasibility projection	397
8.2	Projected gradient on a box-constrained quadratic	403
8.3	Logarithmic-barrier central path	409
8.4	Homotopy continuation for a scalar system	411
8.5	Saddle-point structure of the Lagrangian	415
8.6	Projected primal–dual method on a QP	419
8.7	A general network utility maximization instance	422
8.8	Three-source network utility instance	425
8.9	Distributed primal–dual congestion control	426
9.1	Admissible perturbations of a curve	436
9.2	Dido's isoperimetric problem	436
9.3	Fermat's principle: reflection and refraction	437
9.4	The catenary curve	437
9.5	The brachistochrone problem	438
9.6	Proof of the fundamental lemma	446
9.7	Geometry of the transversality condition	456
9.8	A forced corner in an extremal	466
10.1	The forward–backward sweep	486
10.2	Neural network as a discrete-time system	491

10.3	Common activation functions and their derivatives	493
10.4	Shortest path on a layered graph	509
10.5	Cost of process noise: open loop vs. feedback	516
10.6	American option exercise threshold	528
11.1	Where the pieces fit	534
11.2	Pontryagin's principle coupling	534
11.3	Two-point boundary-value optimality system	539
11.4	Double integrator switching curve	546
11.5	Kalman filtering as an optimal control problem	552
11.6	Structure of the Kalman–Bucy filter	556
11.7	Control–estimation duality	556
11.8	The turnpike phenomenon	566
11.9	Turnpike as a Hamiltonian saddle point	574
12.1	Past–future cost decomposition	592
12.2	Forward recursion for the cost-to-come	593
12.3	Selection step in the LQ case	602
13.1	Projection theorem decomposition	624
14.1	Predict–correct cycle of the Kalman filter	653
14.2	Innovations geometry of the Kalman filter	654
14.3	Prediction variance decay	665
14.4	Predict–update covariance cycle	666
14.5	LQG separation structure	679
15.1	Information feedback in stochastic control	686
15.2	Information filtration timeline	688
15.3	Open-loop versus feedback control	693
15.4	Backward dynamic programming	703
15.5	Optimal stopping threshold policy	713
15.6	Reduction to a belief-state MDP	717
15.7	Orthogonal decomposition for certainty equivalence	732
15.8	Cost-to-go meets cost-to-come	735
15.9	LQG separation principle	736

List of Algorithms

Algorithm 7.1	BFGS Quasi-Newton Method	338
Algorithm 7.2	Generic Iterative Descent Method	339
Algorithm 7.3	Gradient Descent with Backtracking (Armijo) Line Search	339
Algorithm 7.4	Stochastic Gradient / Robbins–Monro Method	375
Algorithm 7.5	Newton’s Method with Backtracking Line Search	380
Algorithm 8.1	Feasible Direction Method for Convex Programs with Mixed Constraints	395
Algorithm 8.2	Distributed Primal–Dual Congestion Control Protocol	427
Algorithm 15.1	Kalman filter	728

Chapter 1

Introduction

This chapter presents a systems-theoretic view of optimization and optimal control, treating them as fundamental methods for abstraction, modeling, analysis, and design of complex decision-making systems. Optimization is framed not merely as a computational tool, but as a unifying language for reasoning about input-output behavior, performance limits, and trade-offs under structural, informational, and resource constraints. From this perspective, system behavior is evaluated in terms of achievable outcomes rather than internal mechanisms, allowing principled comparison and ranking of feasible decisions.

A series of illustrative case studies, including resource allocation, trajectory optimization, epidemic control, and optimal stopping, demonstrates how optimization principles naturally arise across engineering and socio-technical applications, and how they serve both descriptive roles (explaining observed system behavior) and prescriptive roles (guiding design and intervention).

Building on this systems perspective, the chapter frames the existence question for finite-dimensional optimization problems. Classical results such as the Weierstrass extreme value theorem are introduced alongside more general existence guarantees based on lower semicontinuity and coercivity, clarifying how compactness, continuity, and their relaxations ensure well-posed optimization problems and provide a foundation for subsequent analysis, characterization, and computation.

1.1 • Introduction

Optimization and optimal control study decision-making problems in which one seeks to choose variables, functions, or control inputs so as to optimize a specified performance criterion subject to constraints. These problems arise throughout engineering, economics, operations research, and the natural sciences, where limited resources and competing objectives must be balanced in a principled manner.

At an abstract level, an optimization problem can be formulated as

$$\min_{x \in C} f(x), \tag{1.1}$$

where $f : C \rightarrow \mathbb{R}$ is the *objective function* and C is the *feasible (or constraint) set*. The objective function quantifies performance, cost, or utility, while the constraint set encodes admissibility requirements on the decision variable x . This formulation implicitly assumes that objective values can be compared via a total order, here induced by the usual ordering on $\mathbb{R}$.

The use of the notation $\min$ further assumes that an optimal solution exists and that the minimum is attained; when this is not guaranteed, the problem is more precisely described in

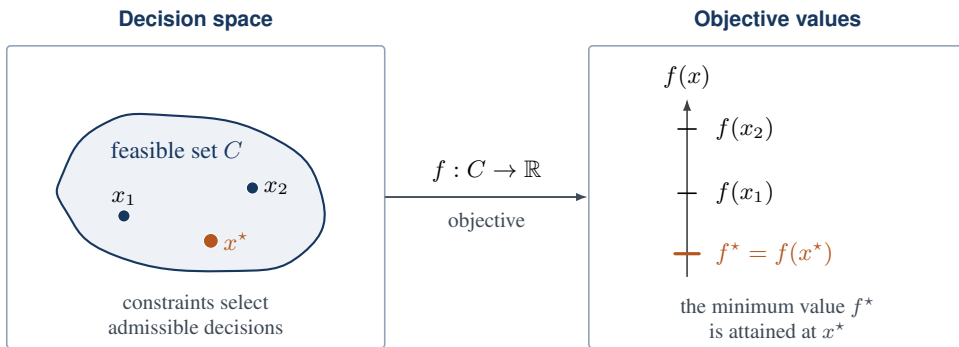


Figure 1.1: Anatomy of a finite-dimensional minimization problem. The feasible set C collects admissible decisions, the objective function places each feasible decision on the ordered real line, and a minimizer x^* exists when the lowest attainable value f^* is achieved by some feasible point.

terms of an infimum.

An element $x^* \in C$ is called an *optimal solution* (or *minimizer*) if

$$f(x^*) \leq f(x), \quad \forall x \in C. \quad (1.2)$$

That is, an optimal solution achieves the smallest objective value among all feasible decisions.

The set of all optimal solutions is denoted by

$$\mathcal{X}^* := \{x \in C \mid f(x) = \inf_{y \in C} f(y)\}, \quad (1.3)$$

and may be empty, contain a single element, or consist of multiple solutions, depending on the structure of f and C .

The *optimal value* of the problem is defined as

$$f^* := \inf_{x \in C} f(x), \quad (1.4)$$

which represents the best achievable performance according to the objective function. When there exists $x^* \in C$ such that $f(x^*) = f^*$, the infimum is said to be *attained*, and x^* is an optimal solution.

Figure 1.1 summarizes this abstraction: constraints determine admissible decisions, the objective function assigns ordered scalar values to those decisions, and optimality asks whether the best attainable value is achieved by some feasible point.

All definitions extend analogously to maximization problems of the form $\max_{x \in C} f(x)$, by replacing $\inf$ with $\sup$ and reversing the inequality in the definition of optimality.

These basic concepts provide the foundational language for optimization and optimal control. Subsequent sections study the conditions under which optimal solutions exist, whether they are unique, how they can be characterized, and how they can be computed or approximated in practice.

1.1.1 ■ Finite- and Infinite-Dimensional Formulations

In finite-dimensional optimization, the decision variable x belongs to $\mathbb{R}^n$, and the objective function is a mapping

$$f : \mathbb{R}^n \rightarrow \mathbb{R}, \quad (1.5)$$

with $C \subseteq \mathbb{R}^n$. Under suitable assumptions on f and C , one may characterize optimal solutions using tools such as first- and second-order optimality conditions, duality theory, and iterative algorithms.

In many applications, however, the decision variable is not a finite-dimensional vector but a function. In such cases, the objective is a *functional*

$$f : V \rightarrow \mathbb{R}, \quad (1.6)$$

defined on an infinite-dimensional vector space V . A common example is

$$V = C^k([a, b], \mathbb{R}^n), \quad k \geq 0, \quad (1.7)$$

the space of k -times continuously differentiable functions from the interval $[a, b]$ into $\mathbb{R}^n$.

In this infinite-dimensional setting, optimization problems are studied within the calculus of variations and optimal control theory. The decision variable represents a trajectory or control signal evolving over time, and optimality conditions are expressed in terms of differential equations and variational arguments rather than finite-dimensional gradients.

The feasible set C may impose a variety of constraints, including pointwise bounds, integral constraints, boundary conditions, or differential equations that govern admissible trajectories. In optimal control problems, these differential constraints typically represent the system dynamics.

A basic example of a functional optimization problem is

$$J(x) = \int_a^b g(x(t), t) dt, \quad (1.8)$$

where $g : \mathbb{R}^n \times [a, b] \rightarrow \mathbb{R}$ is a given integrand representing instantaneous cost or utility. The decision variable is a function $x(\cdot)$ defined on the interval $[a, b]$, and the objective evaluates the aggregate performance accumulated over time.

The admissible functions $x(\cdot)$ are typically restricted to a suitable function space, such as $C([a, b], \mathbb{R}^n)$ or $C^1([a, b], \mathbb{R}^n)$, and may be required to satisfy pointwise constraints of the form

$$|x(t)| \leq d, \quad \forall t \in [a, b], \quad (1.9)$$

as well as boundary conditions such as $x(a) = x_a$ and $x(b) = x_b$. Problems of this type arise naturally in the calculus of variations and provide a direct generalization of finite-dimensional constrained optimization to function spaces.

Another class of problems arises when the admissible functions $x(\cdot)$ are required to satisfy differential constraints of the form

$$\dot{x}(t) = \phi(x(t), t), \quad x(a) = x_a, \quad (1.10)$$

where ϕ specifies the evolution of the state over time. In this case, the optimization problem becomes

$$\min_{x(\cdot)} \int_a^b g(x(t), t) dt \quad \text{subject to} \quad \dot{x}(t) = \phi(x(t), t). \quad (1.11)$$

The differential equation restricts the feasible set to trajectories that are dynamically consistent with the prescribed evolution law.

Such problems form a bridge between the calculus of variations and optimal control theory. When the dynamics depend on an additional control input $u(\cdot)$, the problem naturally extends to a full optimal control formulation involving both state and control variables, with optimality characterized by conditions such as the Euler–Lagrange equations or Pontryagin’s Maximum Principle.

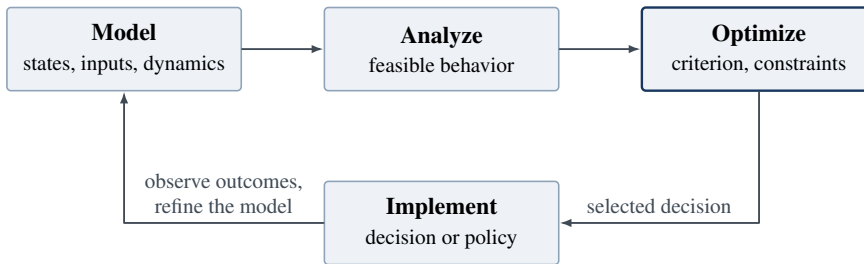


Figure 1.2: Optimization as a systems loop: modeling defines admissible behavior, analysis identifies structure, optimization selects a decision, and implementation feeds observed outcomes back into the model.

1.1.2 ■ Systems and Optimization

Optimization and optimal control are fundamentally *systems methods*. In this context, a *system* is an abstraction that characterizes how inputs are mapped to outputs, possibly through internal states, constraints, and interconnections. The defining feature of the systems viewpoint is the focus on *input–output behavior*: decisions are evaluated according to how they shape observable outcomes and aggregate performance, rather than by a complete description of internal mechanisms.

This systems viewpoint contrasts with a *non-system* or *mechanistic* approach, which is characteristic of many areas of physics and biology. In those disciplines, the primary objective is often to *understand what is inside the box*: to uncover the underlying structure, laws, or mechanisms that govern observed phenomena. Such approaches emphasize explanatory fidelity, causal decomposition, and internal consistency, even when the resulting models are too complex to support direct decision-making or control.

By contrast, systems methods deliberately abstract away internal detail in favor of externally relevant behavior. From a systems perspective, the internal structure of the box matters only insofar as it constrains or explains the input–output relationships of interest. This abstraction allows one to reason about performance, robustness, and trade-offs without requiring a complete or microscopically accurate model of the underlying processes.

Within this framework, optimization is not an end in itself, but a *tool for system analysis and design*. It enables the systematic study of which input–output behaviors are achievable under given constraints, the quantification of performance limits, and the synthesis of decision rules that shape system behavior toward desired objectives. Optimal control extends this framework by embedding optimization within feedback architectures, allowing inputs to be adapted based on observed outputs and evolving system behavior. Consequently, optimization and control problems are typically addressed through four closely related stages of a system: *modeling*, *analysis*, *design*, and *implementation*.

Figure 1.2 summarizes this systems loop.

System Modeling

The modeling stage concerns the abstraction of a real-world system into a mathematical formulation suitable for optimization. This abstraction requires identifying the essential components of the system, such as states, inputs, outputs, and disturbances, while deliberately omitting secondary details that are negligible, unobservable, or analytically intractable. The objective is not to replicate reality in full detail, but to capture the causal and structural relationships that govern system behavior and are most relevant to the control objective.

In optimization and optimal control, modeling typically involves three core elements. First,

one specifies the *decision variables*, representing the degrees of freedom available to the decision-maker. These may be static variables, time-varying trajectories, or control inputs acting on a dynamical system. Second, one defines a *performance criterion* that quantifies desired system behavior, such as cost, energy expenditure, risk, or utility, often accumulated over time. Third, one introduces *constraints* that encode physical laws, resource limitations, safety requirements, or informational restrictions.

A central modeling consideration is the choice of an appropriate level of fidelity. High-fidelity system models may incorporate nonlinear dynamics, uncertainty, and detailed interactions, but can be analytically intractable or computationally prohibitive. Low-fidelity models may sacrifice accuracy in favor of analytical clarity and insight. Effective systems modeling strikes a balance between these extremes, preserving qualitative structure while enabling rigorous analysis and tractable solution methods.

System Analysis

Once a system model has been specified, optimization provides a framework for *system analysis*: the study of fundamental performance limits and structural properties implied by the model. From a systems viewpoint, analysis seeks to determine what input–output behaviors are achievable under the given constraints, how performance depends on system parameters, and what trade-offs are inherent in the system architecture. Importantly, system analysis focuses on properties that are intrinsic to the model and therefore independent of any specific implementation or algorithm.

At a mathematical level, system analysis gives rise to a set of foundational questions that underlie any optimization problem.

Existence. The existence question concerns whether the optimization problem admits at least one optimal solution. In finite-dimensional settings, existence is often established using compactness and continuity arguments, such as the Weierstrass extreme value theorem. In infinite-dimensional or optimal control problems, existence typically requires additional structure, such as coercivity, convexity, or weak compactness of the feasible set. Establishing existence is essential, as it ensures that the modeled system admits a well-defined notion of optimal behavior.

Uniqueness. When an optimal solution exists, one may ask whether it is unique. Uniqueness is typically guaranteed under strict convexity of the objective function and convexity of the feasible set. In optimal control problems, uniqueness may also depend on the system dynamics and terminal conditions. Non-uniqueness is common in practice and may reflect symmetries in the system, redundancy in the control inputs, or indifference among multiple optimal input–output behaviors.

Characterization. Characterization refers to identifying conditions that optimal solutions must satisfy. In finite-dimensional optimization, these conditions include first-order stationarity, second-order optimality conditions, and complementary slackness relations. In the calculus of variations and optimal control, characterization takes the form of differential equations and variational inequalities, such as the Euler–Lagrange equations, Pontryagin’s Maximum Principle, and Hamilton–Jacobi–Bellman equations. These characterizations provide structural insight into optimal system behavior and form the basis for both theoretical understanding and algorithmic design.

Computation. The computation question concerns how optimal solutions can be obtained algorithmically once existence and characterization have been established. Only a limited class of

optimization and optimal control problems admit closed-form solutions, such as linear-quadratic regulators or unconstrained convex programs. In general, optimal solutions must be computed numerically.

Classical computational approaches exploit problem structure and include gradient-based and Newton-type methods, interior-point and active-set methods for constrained optimization, dynamic programming for sequential decision problems, and iterative schemes based on variational principles or fixed-point equations. In optimal control, computation often involves solving boundary-value problems, Riccati equations, or Hamilton–Jacobi–Bellman equations, either exactly or approximately.

In contemporary settings, computation increasingly incorporates *learning* and *data-driven methods*. When system models are incomplete, unknown, or too complex for direct analysis, data may be used to approximate value functions, policies, gradients, or system responses. Learning-based methods, including reinforcement learning, approximate dynamic programming, and policy optimization, can be viewed as computational tools for solving or approximating optimization and optimal control problems using observed data rather than explicit models.

The feasibility of computation therefore depends not only on problem dimension and structure, but also on data availability, sample efficiency, and the interaction between learning and optimization. Issues such as convergence, generalization, robustness, and computational complexity play a central role, particularly in large-scale, uncertain, or real-time environments. As a result, the computational question is no longer merely whether an optimal solution can be computed in principle, but whether it can be computed reliably, efficiently, and adaptively given limited information and data.

System Design and Implementation

System design and implementation translate the results of analysis and computation into concrete decision mechanisms that operate within a real system. While analysis characterizes what is achievable in principle and computation addresses how optimal solutions can be obtained algorithmically, design and implementation concern how these solutions are realized as operational system components.

System Design. From a systems perspective, design is the process of synthesizing *decision rules* that shape system behavior in accordance with specified performance objectives. In optimization and optimal control, this typically takes the form of policies, controllers, or decision mappings that assign inputs based on available information, such as system states, outputs, or measurements.

Design is inherently constrained by the system abstraction. Information availability, actuation limits, communication constraints, and uncertainty all restrict the class of admissible decision rules. As a result, system design often involves structural choices, such as selecting feedback versus open-loop architectures, centralized versus distributed control, or model-based versus data-driven decision mechanisms. These choices determine how optimization results are embedded into the system and how decisions interact with the system’s input–output behavior.

Importantly, design does not necessarily require exact optimality. In many applications, the objective is to achieve satisfactory or robust performance under uncertainty rather than strict optimality with respect to an idealized model. Approximate, hierarchical, or receding-horizon designs are therefore common and reflect practical trade-offs between performance, complexity, and robustness.

Implementation. Implementation concerns the realization of designed decision rules in real systems. This stage accounts for non-idealities that are not captured fully in the abstract model,

including finite computation time, numerical precision, delays, noise, incomplete information, and model mismatch. From a systems viewpoint, implementation is not a purely technical afterthought, but a critical component that influences achievable input–output behavior.

In optimal control and learning-based systems, implementation may involve real-time computation, online adaptation, or continual data acquisition. Constraints on memory, computation, and communication can significantly affect how optimization-based decisions are executed. Consequently, implementation considerations often feed back into modeling and design, motivating simplified models, reduced-order representations, or approximate solution methods.

Design–Implementation Interaction. System design and implementation are rarely sequential and are best viewed as part of an iterative loop. Limitations encountered during implementation may require revisions to the system model, performance criteria, or control architecture. Conversely, advances in computation, sensing, and learning can enable more sophisticated designs and richer system abstractions.

1.1.3 ■ Descriptive and Prescriptive Roles of Optimization

A useful distinction in systems and decision sciences is between *descriptive* and *prescriptive* approaches. Descriptive methods aim to model, explain, or predict how a system behaves, whereas prescriptive methods aim to determine how a system *should* behave in order to achieve specified objectives. Descriptive approaches are common in disciplines such as physics, biology, and statistics, where the primary goal is to understand the input–output behavior of a system as it is observed in nature. In this setting, models are evaluated based on their explanatory power, predictive accuracy, or consistency with empirical data. Learning and data-driven techniques are often used in a descriptive role to identify system dynamics, estimate parameters, or forecast future outputs given observed inputs.

While optimization and optimal control are often associated with *prescriptive* decision-making, optimization also plays an important *descriptive* role in the modeling and understanding of systems. In a descriptive role, optimization is used to explain or predict observed system behavior by positing that the system evolves so as to optimize an implicit objective. Classical examples include physical systems characterized by energy minimization principles, biological systems modeled through fitness maximization, and economic systems described by utility optimization. In these settings, the optimization problem is not a control prescription imposed by an external decision-maker, but a modeling hypothesis that captures the governing input–output behavior of the system. When valid, such models provide explanatory insight and predictive power.

In its prescriptive role, optimization is used to determine how inputs or decisions *should* be chosen to achieve desired outcomes under specified constraints. Here, the objective function reflects design goals rather than emergent system tendencies, and optimization serves as a normative tool for synthesizing decision rules or control policies.

From a systems perspective, the distinction lies not in the mathematical form of the optimization problem, but in its interpretation. In descriptive optimization, the objective function encodes hypothesized system principles that explain observed behavior. In prescriptive optimization, the objective function encodes desired performance criteria that guide intervention and design. Modern learning-based methods often blur this boundary: data may be used to infer implicit objectives or value functions descriptively, and those same constructs may then be used prescriptively to shape future system behavior.

1.2 ■ Examples

1.2.1 ■ Resource Allocation

A canonical class of optimization problems arises in *resource allocation*, where a limited resource must be distributed among multiple agents or activities. Such problems provide a simple yet powerful illustration of how objective functions, feasible sets, and ordering assumptions enter an optimization model.

Problem formulation. Consider N agents sharing a divisible resource of total capacity $C > 0$. Agent i receives an allocation $x_i \geq 0$ and derives utility $f_i(x_i)$ from that allocation. A central planner seeks to allocate the resource so as to maximize total utility:

$$\max_{x_1, \dots, x_N} \sum_{i=1}^N f_i(x_i) \quad (1.12)$$

$$\text{s.t.} \quad \sum_{i=1}^N x_i \leq C, \quad (1.13)$$

$$x_i \geq 0, \quad i = 1, \dots, N. \quad (1.14)$$

Decision variables and feasible set. The decision variable is the allocation vector $x = (x_1, \dots, x_N) \in \mathbb{R}^N$. The constraints define the feasible set

$$\mathcal{C} = \left\{ x \in \mathbb{R}^N \mid x_i \geq 0 \text{ for all } i = 1, \dots, N, \sum_{i=1}^N x_i \leq C \right\},$$

which captures both the nonnegativity requirements and the scarcity of the shared resource. In the abstract optimization problem $\min_{x \in \mathcal{C}} f(x)$, this set $\mathcal{C}$ plays the role of the constraint set C .

Objective function and ordering. The objective function is given by $f(x) := \sum_{i=1}^N f_i(x_i)$, which aggregates individual utilities into a scalar performance measure. Because $f : \mathcal{C} \rightarrow \mathbb{R}$ is real-valued, the usual total order on $\mathbb{R}$ induces a total preorder on feasible allocations. In particular, for any two feasible allocations $x, y \in \mathcal{C}$, either $f(x) \leq f(y)$ or $f(y) \leq f(x)$ holds. This comparability allows feasible allocations to be meaningfully ranked and enables the formulation of a well-defined optimization problem.

The abstract utility function f_i admits concrete interpretations depending on the application domain. In each case, f_i maps an individual resource allocation x_i to a scalar measure of performance or benefit, thereby allowing heterogeneous preferences to be aggregated into a single system-level objective.

Congestion control. Here x_i denotes the bandwidth allocated to user i in a communication network. A widely used choice is $f_i(x_i) = \log(x_i + 1)$, which reflects diminishing marginal utility of bandwidth and leads to *proportional fairness*. Under this objective, any feasible reallocation that increases one user's bandwidth by a given percentage must decrease the aggregate bandwidth of others by at least the same percentage. This choice of f_i yields tractable convex optimization problems and aligns well with observed behavior in distributed congestion-control protocols [111, 78].

Portfolio optimization. In financial settings, x_i represents the capital invested in asset i . A common quadratic utility model is $f_i(x_i) = \mu_i x_i - \frac{1}{2} \sigma_i^2 x_i^2$, where μ_i is the expected return and

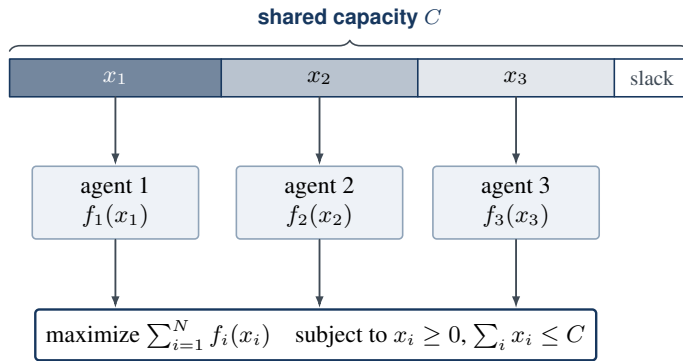


Figure 1.3: Resource allocation (shown for $N = 3$) splits a shared capacity C into individual allocations $x_i \geq 0$, possibly leaving some capacity unused, evaluates each allocation through a utility $f_i(x_i)$, and chooses a feasible split that maximizes aggregate system value.

σ_i^2 quantifies risk or variance [101]. The linear term rewards expected gain, while the quadratic penalty captures risk aversion. Summing such utilities across assets leads to classical mean-variance portfolio optimization, where the objective encodes a trade-off between return and risk in a single scalar criterion.

Urban planning. In urban planning problems, the allocation variable x_i typically represents a limited resource such as budget, land area, housing units, or infrastructure capacity assigned to project or district i . The function $f_i(x_i)$ models the *social benefit* or *economic value* generated by that allocation. The aggregate objective $f(x) = \sum_{i=1}^N f_i(x_i)$ represents total social welfare, economic output, or quality-of-life metrics across the city or region.

Scheduling. In scheduling problems, the allocation variable x_i typically represents processing time, machine capacity, or computational resources assigned to task i . The function $f_i(x_i)$ measures the *value* of allocating that amount of time or capacity to the task [94].

Typical formulations include:

- *Completion-value models*, such as $f_i(x_i) = v_i \min\{x_i, d_i\}$, where d_i denotes the processing time required for completion and v_i is the priority or reward associated with task i .
- *Delay-penalty models*, for example $f_i(x_i) = -c_i \max\{d_i - x_i, 0\}$, which penalize insufficient allocation through tardiness or deadline violations.
- *Concave utility models*, such as $f_i(x_i) = w_i \log(1 + x_i)$, capturing diminishing returns from allocating processing time beyond a core requirement.

In this setting, maximizing $\sum_{i=1}^N f_i(x_i)$ corresponds to prioritizing tasks, balancing throughput and fairness, or minimizing aggregate delay costs.

Although the interpretations differ across domains, the mathematical role of f_i is the same: it maps an individual allocation to a scalar measure of value. By aggregating these functions, the planner induces a total order over feasible allocations, enabling comparison, ranking, and optimization within a common framework.

From a *prescriptive* perspective, problem (1.12) specifies how a planner *should* allocate resources to achieve efficiency, fairness, or other design objectives. From a *descriptive* perspective,

similar optimization problems are used to explain observed allocation patterns by hypothesizing that systems evolve toward configurations that maximize aggregate utility.

This example illustrates how concrete modeling choices give operational meaning to the abstract components of optimization: the feasible set $\mathcal{C}$ encodes system constraints, the objective function f induces an ordering over allocations, and optimal solutions represent preferred system-level outcomes.

1.2.2 ■ Shortest-Time (Brachistochrone-Type) Problem

A classical example illustrating optimization over *functions* rather than finite-dimensional vectors is the *brachistochrone problem*, which originates in the calculus of variations and predates modern optimal control [6, 77].

Physical setting. Consider a particle sliding without friction under uniform gravity from a starting point a to an endpoint b in a vertical plane. The particle is constrained to move along a curve $y = y(x)$ connecting these two points, where the vertical coordinate y is measured downward from the initial height so that $y(x) > 0$ along the trajectory. Among all admissible curves satisfying the boundary conditions $y(a) = y_a, y(b) = y_b$, the objective is to determine the curve along which the particle reaches b in the shortest possible time.

Modeling and objective functional. Under conservation of energy, the speed of the particle at height y is proportional to $\sqrt{y}$. The infinitesimal time required to traverse an arc-length element ds is therefore proportional to $\frac{ds}{\sqrt{y}}$. Since the arc length satisfies $ds = \sqrt{1 + (y'(x))^2} dx$, the total travel time along a curve $y(\cdot)$ can be written as the functional

$$J[y] = \int_a^b \frac{\sqrt{1 + (y'(x))^2}}{\sqrt{y(x)}} dx = \int_a^b g(y(x), y'(x)) dx, \quad (1.15)$$

where $g(y, y') := \frac{\sqrt{1 + (y')^2}}{\sqrt{y}}$.

Here, the *decision variable* is the function $y(\cdot)$ itself, the *feasible set* consists of sufficiently smooth curves satisfying the boundary conditions and the positivity constraint $y(x) > 0$, and the objective is to minimize the scalar functional $J[y]$. The integral induces a total order on admissible trajectories via their travel times.

Analysis via Euler–Lagrange equations. The problem (1.15) is an infinite-dimensional optimization problem. Necessary conditions for optimality are obtained by applying the Euler–Lagrange equation

$$\frac{d}{dx} \left(\frac{\partial g}{\partial y'} \right) - \frac{\partial g}{\partial y} = 0.$$

Exploiting the fact that g does not depend explicitly on x , one may further derive a first integral of motion, leading to a closed-form characterization of the optimal curve.

Solving the resulting equations shows that the minimizing trajectory is a *cycloid*, that is, the curve traced by a point on the rim of a rolling circle. Remarkably, an arc of the same family of curves is optimal for every admissible pair of endpoints, whatever their horizontal and vertical separation, provided the particle moves under gravity alone.

The brachistochrone problem illustrates several ideas that are central to optimal control and systems theory. First, the optimization is performed over entire trajectories rather than static

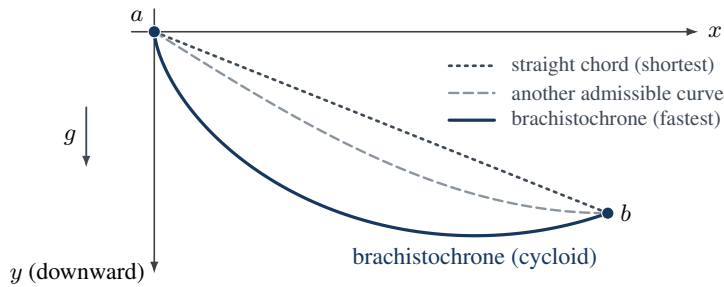


Figure 1.4: The brachistochrone problem optimizes over an entire trajectory: one chooses a curve $y(\cdot)$ joining a to b to minimize the travel-time functional $J[y]$ in (1.15). The straight chord is shortest in length but not fastest. The minimizer is a cycloid, drawn here exactly for these endpoints: its steep initial descent builds speed early, and for this aspect ratio it even dips below b before rising to meet it.

decision variables, highlighting the infinite-dimensional nature of many control problems. Second, the performance criterion is expressed as an objective functional defined by an integral over space or time, a structure that recurs throughout the calculus of variations and optimal control. Third, the form of the cost functional is dictated by physical modeling principles, in this case, conservation of energy, showing how system dynamics shape optimization objectives. Finally, the problem demonstrates how simple underlying principles can give rise to highly nontrivial optimal solutions, such as the cycloid.

Historically, this problem marks one of the earliest instances where system behavior is explained and predicted by an underlying optimization principle. Conceptually, it provides a bridge between static optimization, the calculus of variations, and modern optimal control, where similar integral cost functionals arise naturally in dynamical systems and trajectory planning.

1.2.3 ■ Epidemic Control Model

A canonical example of optimal control in dynamical systems arises in *epidemic control*, where intervention policies are designed to mitigate the spread of an infectious disease while accounting for limited resources and societal costs [87, 70].

System dynamics. Consider a controlled susceptible–infected–removed (SIR) model given by

$$\dot{S}(t) = -\beta S(t)I(t), \quad (1.16)$$

$$\dot{I}(t) = \beta S(t)I(t) - u(t), \quad (1.17)$$

$$\dot{R}(t) = u(t), \quad (1.18)$$

where $S(t)$, $I(t)$, and $R(t)$ denote the susceptible, infected, and removed (population recovered, isolated, or vaccinated) compartments at time t , respectively. The parameter $\beta > 0$ represents the transmission rate of the disease. The control input $u(t)$ models an intervention such as treatment, isolation, or vaccination that transfers individuals from the infected to the removed compartment.

Decision variables and feasible controls. The decision variable in this problem is the control trajectory $u(\cdot) : [t_0, t_1] \rightarrow \mathbb{R}_{\geq 0}$, which specifies the intervention intensity over time. Admissible controls are typically required to satisfy pointwise constraints $u(t) \in \mathcal{U} := [0, u_{\max}]$, reflecting limited treatment capacity, vaccine supply, or logistical constraints.

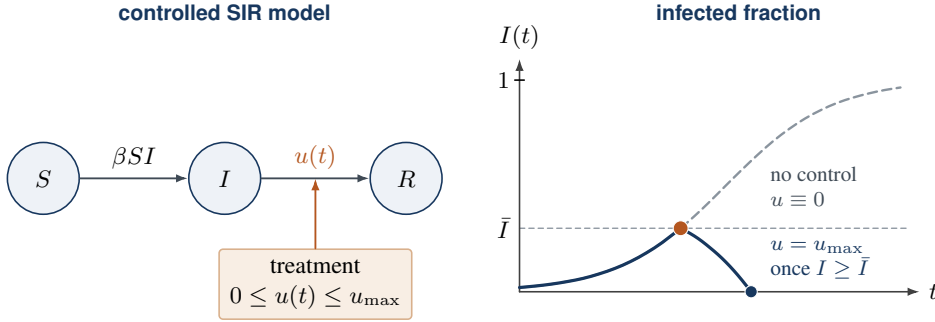


Figure 1.5: Epidemic control couples decisions to dynamics. Left: the intervention $u(t)$ moves individuals from the infected to the removed compartment at a bounded rate. Right: solutions of the controlled SIR model with $\beta = 0.6$, $S(0) = 0.98$, $I(0) = 0.02$. Without intervention ($u \equiv 0$) nothing leaves the infected compartment and $I(t) \rightarrow 1$. The causal rule that applies $u_{\max} = 0.2$ from the first time I reaches $\bar{I} = 0.3$ (marked) caps the outbreak and drives I to zero. An optimal control balances such reductions in infection against the effort required to intervene.

In addition, admissible controls are assumed to be *causal* (or nonanticipative): the control action at time t may depend only on information available up to time t , such as current or past infection levels, but not on future states of the system.

Objective functional. The planner seeks to minimize a cumulative cost functional of the form

$$J(u) = \int_{t_0}^{t_1} \ell(I(t), u(t), t) dt, \quad (1.19)$$

where the running cost ℓ encodes trade-offs between public health outcomes and intervention effort. Typical choices include $\ell(I, u, t) = c_I I(t) + c_u u(t)^2$, where the first term penalizes high infection levels and the second term penalizes aggressive or costly interventions. Time-dependent costs may also be used to reflect changing priorities or seasonal effects.

Optimization problem. The epidemic control problem can thus be written as

$$\min_{u(\cdot) \in \mathcal{U}} J(u) \quad \text{subject to the SIR dynamics and given initial conditions.}$$

This is an infinite-dimensional optimization problem, as the decision variable is a function of time rather than a finite-dimensional vector.

Necessary conditions for optimality are typically derived using Pontryagin's Maximum Principle, which introduces adjoint variables associated with the state dynamics and yields a system of coupled state–costate equations. The resulting optimal control balances the marginal benefit of reducing infections against the marginal cost of intervention.

This example illustrates several defining features of optimal control in socio-technical systems: the explicit coupling between dynamics and decision variables, the role of constraints and causality, and the use of integral cost functionals to encode long-term objectives. Epidemic models of this form have been widely used to inform public health policy, resource allocation, and real-time decision-making during outbreaks.

1.2.4 ■ Optimal Stopping

We next consider a discrete-time *optimal stopping problem*, in which a decision-maker observes the evolution of a stochastic process and must decide *when to stop* it so as to optimize an expected performance criterion [16, 22]. This formulation isolates the temporal tradeoff between waiting for additional information and committing to a terminal action, and it admits a clean dynamic programming characterization.

Let $\{X_k\}_{k \in \mathbb{Z}_{\geq 0}}$ be a discrete-time stochastic process defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and adapted to the filtration $\{\mathcal{F}_k\}_{k \geq 0}$, where $\mathcal{F}_k$ represents the information available to the decision-maker up to time k . The process is assumed to be Markovian, with transition kernel $\mathbb{P}(X_{k+1} \in A \mid X_k = x)$ for $A \in \mathcal{B}(\mathcal{X})$, on a state space $\mathcal{X}$ (typically $\mathbb{R}^n$ or a finite set).

At each time k , after observing X_k , the decision-maker chooses one of two actions:

- *continue*, in which case the process evolves to X_{k+1} according to the transition kernel; or
- *stop*, in which case the process is terminated permanently and a terminal reward is realized.

A stopping rule is represented by a stopping time $\tau : \Omega \rightarrow \mathbb{Z}_{\geq 0} \cup \{\infty\}$, such that $\{\tau = k\} \in \mathcal{F}_k$ for all k . Let $\mathcal{T}$ denote the set of all admissible stopping times with respect to $\{\mathcal{F}_k\}$.

Objective function. The objective is to maximize the expected reward accrued at the stopping time:

$$V(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x[G(X_\tau, \tau)], \quad (1.20)$$

where $G : \mathcal{X} \times \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ is a terminal reward function and $\mathbb{E}_x$ denotes expectation conditioned on the initial state $X_0 = x$.

A common and practically relevant specialization decomposes the reward into a terminal payoff and accumulated running costs:

$$G(X_\tau, \tau) = R(X_\tau) - \sum_{k=0}^{\tau-1} c(X_k), \quad (1.21)$$

where $c : \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$ is a per-stage continuation cost and $R : \mathcal{X} \rightarrow \mathbb{R}$ is the reward obtained upon stopping. This structure captures the tradeoff between waiting longer (incurring additional cost but potentially reaching a more favorable state) and stopping earlier.

The value function satisfies the Bellman optimality equation

$$V(x) = \max\{R(x), -c(x) + \mathbb{E}[V(X_{k+1}) \mid X_k = x]\}. \quad (1.22)$$

This recursion reflects the fundamental binary choice at each time step: stop immediately and receive reward $R(x)$, or continue and defer the stopping decision. Equation (1.22) induces a partition of the state space into:

- a *stopping region*, where $R(x) \geq -c(x) + \mathbb{E}[V(X_{k+1}) \mid X_k = x]$;
- a *continuation region*, where continuing yields higher expected value.

The optimal stopping time is the first time the process enters the stopping region.

Canonical example: timing of medical intervention. Let X_k represent a patient's health state evolving stochastically over discrete clinical visits $k = 0, 1, 2, \dots$. Continuing observation provides additional diagnostic information but carries the risk of further health deterioration,

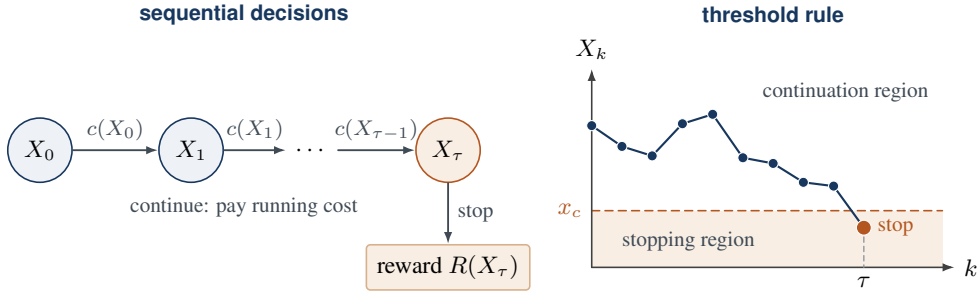


Figure 1.6: Optimal stopping compares the value of continuing with the value of stopping. Left: at each stage the decision-maker either continues, paying the running cost $c(X_k)$, or stops and collects the terminal reward $R(X_\tau)$. Right: in many applications this comparison induces a stopping region $\{x \leq x_c\}$ bounded by a critical level x_c , and the optimal rule stops at the first time τ the sample path of X_k enters that region.

while stopping corresponds to initiating an irreversible intervention such as surgery or aggressive treatment. The terminal reward $R(x)$ captures post-intervention health outcomes and quality of life, whereas the running cost $c(x)$ captures the physiological risk, discomfort, or mortality hazard associated with delaying action. The optimal stopping policy balances the value of information against the cost of waiting and typically induces a threshold structure: intervention is optimal once the health state falls below a critical level.

Canonical example: when to quit a project. Consider a project whose latent quality or viability evolves stochastically over discrete decision epochs $k = 0, 1, 2, \dots$, represented by a state process X_k . At each stage, the decision-maker may either continue the project, incurring an operating cost $c(X_k)$ while observing new performance signals, or stop the project permanently and receive a terminal payoff $R(X_k)$, which may represent salvage value, opportunity cost recovery, or reallocation of resources. Continuing allows the agent to learn about the project's future potential, but also exposes the organization to further losses if the project is fundamentally unviable. The optimal policy is characterized by a stopping region in the state space: the project should be terminated once expected future returns fall below the continuation value. In many models, this policy takes the form of a threshold rule, whereby projects are abandoned when their estimated quality or belief state drops below a critical level.

In both cases, optimal stopping theory formalizes the trade-off between *information acquisition* and *irreversibility*, providing a normative benchmark for deciding when waiting is beneficial and when decisive action is optimal.

Behavioral considerations. Optimal stopping decisions are often made by humans and organizations, often without explicit modeling and analysis. Examples include deciding when to sell an asset, abandon a project, intervene medically, or terminate a failing strategy. Unlike the idealized decision-maker assumed in (1.22), human agents are subject to systematic behavioral biases, most notably the *sunk-cost fallacy*, loss aversion, and present bias [118, 32]. These effects lead decision-makers to continue suboptimal processes because of past investments rather than future expected value. From this perspective, optimal stopping theory provides not only a normative benchmark for rational decision-making, but also a diagnostic tool for identifying and quantifying departures from rational behavior in real-world systems.



Figure 1.7: A conceptual timeline showing the parallel development of two strands unified in this book. The left strand traces optimization and convex analysis, from stationarity and multipliers to linear and convex programming, constraint qualifications, convex analysis, and efficient algorithms. The right strand traces the calculus of variations and optimal control, from the brachistochrone and the Euler–Lagrange equation through Hamilton–Jacobi theory to dynamic programming, the maximum principle, filtering, and stochastic and learning-based control. Each entry is dated individually; the list is selective, and many contributions are necessarily omitted.

1.3 ■ A Historical Timeline of Optimization Theory

To provide a high-level roadmap, Figure 1.7 summarizes selected major contributors to the development of optimization theory, highlighting the core conceptual shifts that shaped each stage of the field. The list is not exhaustive, and many important contributions are necessarily omitted for brevity.

1.3.1 ■ Foundations of Optimization Theory

The origins of optimization theory lie in the calculus of variations. The subject is often traced to the brachistochrone problem posed by Johann Bernoulli in 1696, which asks for the curve of fastest descent and thereby frames optimization over an entire trajectory rather than a single variable [49]. Building on such problems, Leonhard Euler (1707–1783) introduced the idea that optimal solutions to continuous problems can be characterized by infinitesimal variations, leading to the Euler–Lagrange equations. This variational viewpoint established optimality as a condition of stationarity with respect to admissible perturbations and provided the first systematic framework for reasoning about extrema in infinite-dimensional spaces; see, e.g., [77, 28, 49].

Building on this foundation, Joseph–Louis Lagrange (1736–1813) introduced the method of Lagrange multipliers as a systematic approach to constrained optimization. His central insight was that constraints need not be treated externally, but can be incorporated directly into the objective function through auxiliary variables. This reformulation reduces constrained optimization problems to stationary conditions of an augmented Lagrangian. In modern notation, optimality is characterized by $\nabla f(x) + \sum_i \lambda_i \nabla g_i(x) = 0$. Through this construction, Lagrange reframed optimization as a geometric problem, in which optimal solutions arise from the tangency between level sets of the objective and the constraint surfaces. This geometric perspective extends Euler’s variational ideas to settings with explicit constraints and remains foundational in both finite- and infinite-dimensional optimization; see [80, 26].

The variational tradition matured over the nineteenth and early twentieth centuries into the direct classical foundation of optimal control. William Rowan Hamilton and Carl Gustav Jacob Jacobi developed the theory of the Hamilton–Jacobi equation, while Karl Weierstrass and Adrien–Marie Legendre established rigorous necessary conditions for variational problems [49, 77]. In the early twentieth century the Chicago school of Oskar Bolza and Gilbert Bliss, and especially Constantin Carathéodory, brought the subject to maturity; Carathéodory’s approach anticipated both dynamic programming and the maximum principle [77]. This lineage leads directly to the Hamilton–Jacobi–Bellman equation and to modern optimal control.

A further conceptual transformation occurred in the early twentieth century with the emergence of convexity and duality as organizing principles. Leonid Kantorovich (1912–1986) founded linear programming and introduced dual prices, or shadow prices, providing the first rigorous convex optimization framework with clear economic and geometric interpretations [65, 58]. Around the same time, John von Neumann (1903–1957) developed the minimax theorem and fixed-point methods, revealing deep structural connections among optimization, convexity, and game theory [121, 120]. Collectively, these developments elevated optimization from a collection of problem-specific techniques to a coherent mathematical theory governed by duality, equilibrium, and geometric regularity; see also the modern convex-analytic perspective in [26].

1.3.2 ■ Optimization as a Science for Operations Research, Decision-Making, and Control

The modern theory of constrained optimization began to crystallize in the late 1940s, driven in part by emerging decision problems in operations research arising from logistics, resource allocation, and planning during and after World War II. These developments established optimization as a mathematical framework for rational decision-making under constraints, with strong connections to economics, engineering, and systems analysis; see, e.g., [34, 80, 58]. A landmark of this period was George B. Dantzig’s development of the simplex method in 1947, which provided the first practical algorithm for solving linear programs and established linear programming as a cornerstone of operations research [38, 80]. Within this context, Fritz John derived in 1948 what are now known as the Fritz–John optimality conditions [60], providing necessary conditions

for constrained optimization without assuming any constraint qualification. These conditions generalize Lagrange’s approach by allowing for the possibility that the multiplier associated with the objective function may vanish, thereby capturing degenerate cases in which standard multiplier rules fail. While mathematically general, the Fritz–John conditions lack normalization and therefore do not directly yield economically or geometrically interpretable multipliers; see classical nonlinear programming treatments in [10, 15, 8].

This limitation motivated the seminal work of Harold Kuhn and Albert Tucker in the early 1950s, also strongly influenced by developments in operations research and mathematical programming. In 1951, they introduced the Karush–Kuhn–Tucker (KKT) conditions [73], which refine the Fritz–John framework by imposing regularity assumptions that normalize the multipliers and restore a clear geometric and economic interpretation. Essentially the same optimality conditions had in fact been obtained over a decade earlier by William Karush in his 1939 master’s thesis on the minimization of functions subject to inequality constraints [67]; his priority went unrecognized until the 1970s and is now acknowledged in the compound name. The KKT conditions extend Lagrange’s method to inequality constraints and provide necessary first-order conditions for optimality:

$$\nabla f(x^*) + \sum_i \mu_i \nabla g_i(x^*) = 0, \quad \mu_i \geq 0, \quad \mu_i g_i(x^*) = 0.$$

Their validity, however, depends critically on the satisfaction of appropriate constraint qualifications, highlighting the central role played by the geometry of the feasible set; see [15, 10].

To formalize these regularity requirements, additional conditions were introduced. In 1950, Morton Slater proposed what is now known as Slater’s condition [110], which requires the existence of a strictly feasible point satisfying $g_i(x) < 0$ for all inequality constraints. For a convex problem this assumption is enough to force strong duality and to guarantee that KKT multipliers exist, providing a rigorous foundation for duality theory in convex optimization [26, 58]. Building on this geometric viewpoint, Mangasarian and Fromovitz introduced in 1967 the Mangasarian–Fromovitz Constraint Qualification (MFCQ) [82], a strictly weaker condition requiring that the equality-constraint gradients be linearly independent and that some direction annihilate them while strictly decreasing every active inequality constraint. These developments clarified that dual variables arise from the geometry of tangent and normal cones, rather than from algebraic constraint representations alone; see [81, 102].

During the 1960s and 1970s, convex analysis emerged as a unifying language for optimization theory. Building on the theory of convex conjugate functions introduced by Werner Fenchel [47] and the proximal and duality constructions of Jean–Jacques Moreau [86], this program was brought to maturity largely through the work of Ralph Tyrrell Rockafellar. His monograph *Convex Analysis* [119], together with later developments in variational analysis [102], introduced subgradients, normal cones, epigraphs, and Fenchel duality as central objects. Within this framework, the KKT conditions appear naturally as first-order optimality conditions in convex geometry, and stability and sensitivity properties of solutions can be analyzed systematically.

In parallel with these advances in static optimization, the 1950s also witnessed the extension of optimization theory to dynamical decision problems. In the mid-1950s, Richard Bellman introduced dynamic programming and the principle of optimality, showing that multistage decision problems admit recursive value-function representations through the Bellman equation and, in continuous time, the Hamilton–Jacobi–Bellman equation [12, 16]. Almost simultaneously, in 1956, Lev Pontryagin and his collaborators developed the Pontryagin Maximum Principle, which characterizes necessary conditions for optimal control via adjoint dynamics and Hamiltonian maximization; see [96, 77, 6]. These results may be interpreted as infinite-dimensional analogues of the KKT conditions posed on spaces of trajectories, linking static optimization to feedback control and system dynamics.

A closely related line of work established the optimization foundations of estimation and

filtering. Norbert Wiener’s theory of optimal filtering of stationary processes [124] recast signal estimation as a variational problem, and in 1960 Rudolf Kalman introduced the recursive state-space filter [64] together with the dual theory of the linear-quadratic regulator [63]. The resulting duality between optimal estimation and optimal control, mediated by Riccati equations, is a recurring theme in later chapters on filtering and stochastic control.

The following decades codified optimal control for engineering practice and extended it to uncertain and sequential settings. Bryson and Ho [30] and Athans and Falb [6] developed the adjoint, gradient, and linear-quadratic techniques that made optimal control a practical design methodology. In parallel, the theory of Markov decision processes, advanced by Howard and others, recast sequential decision-making under uncertainty as dynamic programming [99, 16], while Wonham’s separation principle and the rigorous continuous-time framework of Fleming and collaborators established the foundations of stochastic optimal control [48]. These developments underlie Parts IV and V of this book.

At roughly the same time, Rufus Isaacs developed the theory of differential games during the late 1950s and early 1960s [59], extending dynamic optimization to adversarial settings involving multiple decision-makers with opposing objectives. The resulting Hamilton–Jacobi–Isaacs equation generalizes Bellman’s dynamic programming framework to min–max problems and completes the conceptual progression from static constrained optimization to dynamic and strategic decision-making; see [45].

1.3.3 ■ Algorithmic and Computing Foundations of Optimization

Iterative methods for optimization have a long history: the gradient (steepest-descent) method dates back to Augustin–Louis Cauchy in 1847 [33], and Newton-type methods are older still. What changed beginning in the late 1970s and accelerating through the 1980s was a decisive convergence of mathematical structure and computational tractability. This period marked the transition from primarily analytic characterizations of optimality to algorithmic frameworks with provable performance guarantees. A pivotal result was Leonid Khachiyan’s ellipsoid method in 1979, which established for the first time that linear programs are solvable in polynomial time [69]. Though impractical, it was soon followed by Narendra Karmarkar’s interior-point method in 1984, which combined polynomial-time guarantees with practical efficiency and initiated the modern theory of interior-point algorithms [66]. Building on these advances, seminal work by Yurii Nesterov, Arkadi Nemirovski, and others extended interior-point methods to broad classes of convex optimization problems, establishing polynomial-time complexity bounds and oracle-based complexity theory; see, e.g., [89, 14, 13]. These developments clarified not only how convex optimization problems can be solved efficiently, but also which problem classes admit efficient algorithms and which do not. Optimization thus evolved into an algorithmic science grounded in complexity, scalability, and numerical stability.

In the 1990s and early 2000s, optimization theory became a foundational infrastructure across engineering and data-driven disciplines. Convex optimization emerged as a universal modeling language for control systems, signal processing, communications, and operations research. Problems ranging from state estimation and model predictive control to resource allocation and network design were unified under a common optimization framework, supported by increasingly mature numerical solvers and software ecosystems; see [26, 34, 80]. During this phase, emphasis shifted from deriving optimality conditions to designing scalable, reliable, and robust algorithms suitable for large-scale implementation.

During the 2010s, optimization assumed a central role in machine learning. Empirical risk minimization, regularization, support vector machines, sparse recovery, and large-scale regression all rely on optimization as their computational backbone. First-order methods, stochastic gradient algorithms, proximal methods, and accelerated schemes—the analysis of the stochastic schemes

tracing back to the stochastic approximation scheme of Robbins and Monro [100]—enabled learning from massive datasets, often in regimes where classical assumptions of smoothness or strong convexity no longer hold. In this period, optimization theory increasingly interacted with statistics, leading to refined analyses of generalization, robustness, and algorithmic stability; see [23, 90].

From 2020 onward, the scope of optimization expanded further with the rise of large-scale artificial intelligence systems. Training and deploying deep neural networks, reinforcement learning agents, and multi-agent systems involve optimization in nonconvex, stochastic, and high-dimensional settings. As a result, optimization theory has shifted its focus toward questions of scalability, adaptivity, and interaction. Topics such as bilevel optimization, constrained learning, distributionally robust optimization, and game-theoretic optimization have become central, reflecting the need to reason about learning dynamics, strategic behavior, and safety in complex AI-driven systems; see [109, 14, 13].

In this modern view, optimization is no longer merely a tool for solving isolated mathematical problems. It functions as the organizing principle for decision-making in large, adaptive, and interconnected systems, providing the theoretical and computational foundation for machine learning, control, signal processing, and artificial intelligence; see also the systems perspective in [55, 85].

1.4 - Organization of the Book

The book is organized into five parts, preceded by this introductory chapter and followed by two appendices. The overall structure follows a single conceptual arc: it begins with the existence, characterization, and computation of solutions to *static, finite-dimensional* optimization problems, and then lifts these ideas to *dynamic, infinite-dimensional* optimal control, first in a deterministic and then in a stochastic setting. Figure 1.8 summarizes this organization and the dependencies among the parts.

Chapter 1: Introduction. The present chapter establishes the systems viewpoint that runs throughout the book, introduces the abstract optimization problem and its finite- and infinite-dimensional instances, and raises the existence question (Weierstrass, lower semicontinuity, and coercivity) that decides whether optimization problems are well posed; the results themselves are proved in Chapter 2. It also situates the subject historically and previews the material to come.

Part I: Foundations and Optimization Theory. This part builds the mathematical language used everywhere else. Chapter 2 develops the analytical and geometric foundations of optimization—norms, inner products, convergence, differentiability, and the geometry of feasible sets—in a way that serves both finite- and infinite-dimensional problems. Chapter 3 derives classical optimality conditions, including first- and second-order stationarity, Lagrange multipliers, and the Karush–Kuhn–Tucker conditions, which characterize candidate solutions once existence is assured.

Part II: Convex Optimization. Convexity is the structural property that makes optimization tractable and turns necessary conditions into sufficient ones. Chapter 4 treats basic convex optimization, convex sets and functions, separation, and duality. Chapter 5 extends the theory to generalized convexity, broadening the class of problems for which the convex machinery applies. Chapter 6 recasts convex optimization in general *vector spaces*, so that the same duality and optimality principles apply verbatim to problems whose decision variables are functions

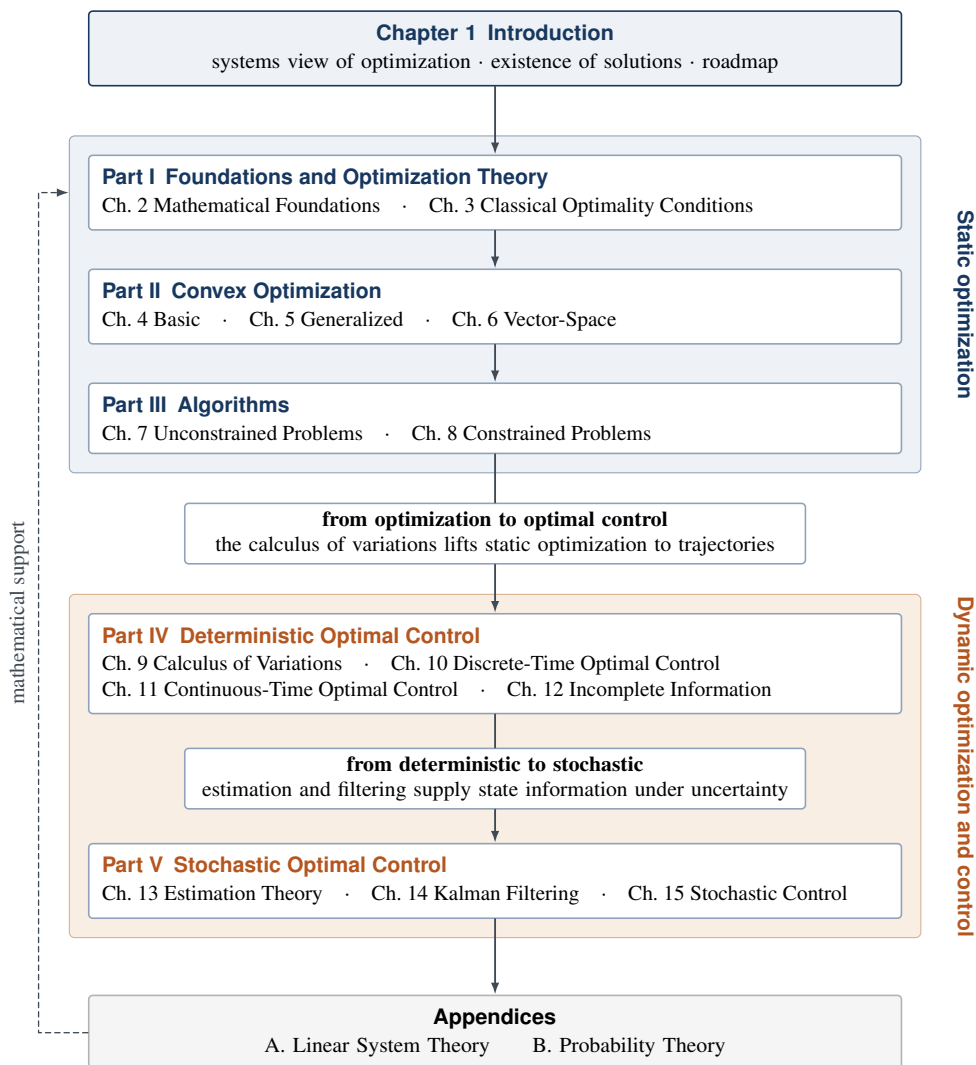


Figure 1.8: Organization of the book. Chapter 1 sets the systems perspective and existence theory. Parts I–III develop static optimization: foundations and optimality conditions, convex programming (culminating in the vector-space formulation), and algorithms. Parts IV–V extend the same variational principles to dynamic problems posed over trajectories, first deterministic and then stochastic optimal control. The appendices supply the linear-system and probability background used throughout.

rather than finite-dimensional vectors. This vector-space formulation is the bridge that carries the finite-dimensional theory into the infinite-dimensional control problems of Parts IV and V.

Part III: Algorithms. Having established when solutions exist and how they are characterized, this part addresses how they are computed. Chapter 7 develops iterative methods for unconstrained problems, including gradient, Newton, and quasi-Newton schemes and their convergence analysis. Chapter 8 treats constrained problems through penalty, barrier, interior-point, and Lagrangian-based methods, connecting the algorithms back to the optimality conditions of Part I.

Part IV: Deterministic Optimal Control. This part makes the transition from static optimization to dynamic decision-making over time. Chapter 9 introduces the calculus of variations, the historical and conceptual entry point to optimizing over functions, and derives the Euler–Lagrange equations. Chapters 10 and 11 develop discrete-time and continuous-time optimal control, presenting dynamic programming, the Bellman and Hamilton–Jacobi–Bellman equations, and Pontryagin’s maximum principle. The part closes with Chapter 12 on optimal control under incomplete information, which treats the deterministic, set-membership theory and motivates the stochastic and estimation-theoretic tools of Part V.

Part V: Stochastic Optimal Control. The final part incorporates uncertainty. Chapter 13 develops estimation theory, and Chapter 14 specializes it to the Kalman filter, the recursive estimator for linear–Gaussian systems. Chapter 15 combines estimation with control to treat stochastic optimal control, including the linear-quadratic–Gaussian problem and the separation principle, closing the loop between sensing, estimation, and decision-making.

Appendices. Appendix A reviews linear system theory and Appendix B reviews probability theory. These provide self-contained references for the background invoked in the main text, so that the development remains accessible without external prerequisites.

1.4.1 ■ How to Read the Book

Parts I–III are largely sequential and are recommended for all readers, since they establish the vocabulary, optimality conditions, and algorithms used throughout. Readers primarily interested in optimal control may proceed directly from Part I and the vector-space material of Chapter 6 to Parts IV–V. Parts IV and V build on the calculus of variations introduced in Chapter 9, and Part V in addition relies on the estimation background in Appendix B. The appendices may be consulted as needed rather than read linearly.

1.5 ■ Distinctive Features of the Book

Several features distinguish this book from standard treatments of optimization and of optimal control, which are usually presented as separate subjects in separate texts.

Shared foundations for optimization and control. Rather than treating optimization and optimal control as distinct disciplines, the book develops a common foundation that serves both. The existence theory, optimality conditions, convex-analytic tools, and duality principles introduced for finite-dimensional problems are precisely the ones later specialized to dynamic control problems. A reader thus learns one theory, not two, and sees optimal control as optimization carried out in a richer space.

An explicit transition from optimization to optimal control. The book is deliberately arranged so that optimal control emerges as a continuation of optimization rather than as a separate axiomatic theory. The calculus of variations (Chapter 9) is presented as the natural bridge: the Euler–Lagrange equations are the infinite-dimensional analogue of first-order stationarity, and Pontryagin’s maximum principle and the Hamilton–Jacobi–Bellman equation appear as the trajectory-space counterparts of the Karush–Kuhn–Tucker conditions and dynamic programming. This makes the conceptual continuity between the two subjects explicit.

Unification of finite- and infinite-dimensional problems. By formulating convex optimization in general vector spaces (Chapter 6), the book treats finite-dimensional programs and infinite-dimensional variational and control problems within a single framework. Norms, inner products, dual spaces, and separation arguments are developed abstractly, so that a minimizer in $\mathbb{R}^n$ and an optimal trajectory in a function space are understood as instances of the same underlying principles. This unification is difficult to achieve when the two settings are studied in isolation.

Beyond the convex optimization taught in a first course. While convexity is a central organizing theme, the book does not stop at the convex problems typically covered in an introductory course. It also addresses generalized convexity, nonconvex phenomena such as multiple local minima and saddle structure, existence under weakened regularity (semicontinuity and coercivity), and the genuinely nonconvex problems that arise in optimal control and estimation. The aim is to prepare the reader for the problems encountered in research and practice, not only for the idealized convex case.

Self-contained mathematical development. The book is written to be self-contained. The analytical tools it relies on—compactness and existence arguments, elements of convex and functional analysis, linear system theory, and probability—are developed in the text or collected in the appendices. Proofs are provided rather than merely cited, and worked examples and exercises accompany each chapter. A motivated reader with a background in calculus and linear algebra can therefore follow the development from first principles through to modern optimal control and estimation.

1.6 ■ Further Reading

This chapter provides an overview of system optimization and optimal control. Students are encouraged to pursue further reading in these areas using the references listed below, which offer complementary perspectives, deeper theoretical foundations, and pathways to advanced topics and applications.

Core optimization concepts and the optimal value function. The abstract formulation of optimization problems, the definition of the optimal value function, and the distinction between minima and infima form the conceptual core of this book. A modern and highly accessible introduction to these ideas, with strong emphasis on convexity, duality, and geometry, is given by Boyd and Vandenberghe [26]. Complementary perspectives that emphasize algorithmic and modeling viewpoints can be found in Chong and Zak [34] and Luenberger and Ye [80]. For students interested in the economic interpretation of optimal values, constraints, and dual variables, Intriligator [58] provides a clear bridge between optimization theory and economic reasoning.

Existence, compactness, and analytical foundations. The existence results of Section 2.2 rely on compactness, coercivity, and semicontinuity. Classical vector-space and topological approaches to these questions are treated in depth in Luenberger [79] and Rudin [104]. For a more advanced and general treatment, especially relevant to infinite-dimensional optimization, Brezis [28] provides essential background in functional analysis.

Rockafellar's *Convex Analysis* [119] and the monograph *Variational Analysis* by Rockafellar and Wets [102] are foundational references for lower semicontinuity, coercivity, and stability of optimal values under perturbations. These works underpin much of modern optimization theory, including the generalized Weierstrass theorem discussed in this book.

Nonlinear, nonsmooth, and parametric optimization. Beyond convex problems, many systems lead to nonlinear or nonsmooth optimization models. Classical treatments of nonlinear programming include Bertsekas [15], Bazaraa et al. [10], Avriel [8], and Mangasarian [81]. For problems involving nonsmooth objectives or constraints, Clarke [35] provides a definitive treatment based on generalized gradients.

Issues of parameter dependence and stability of solutions are addressed in Zlobec [130], which is particularly relevant when studying how optimal values change under model uncertainty or perturbations. Numerical aspects of optimization, including gradient and Newton-type methods, are treated systematically in Nocedal and Wright [90].

Calculus of variations and optimal control. Sections 2.2 and 2.3 introduce optimization over functions and dynamical systems. The calculus of variations perspective is developed concisely and rigorously in Liberzon [77], which serves as a natural bridge between static optimization and optimal control. A classical systems-oriented presentation is given by Athans and Falb [6].

For linear-quadratic control problems, Anderson and Moore [3] provide a definitive reference, clarifying the role of Riccati equations and state feedback in optimal control design.

Dynamic programming and sequential decision-making. Dynamic programming provides a unifying framework for optimal control, optimal stopping, and Markov decision processes. Bertsekas's two-volume treatment [16] is the standard reference for deterministic and stochastic dynamic programming. A comprehensive treatment of Markov decision processes is given by Puterman [99].

Approximate and learning-based approaches to dynamic optimization are developed in Bertsekas and Tsitsiklis [20] and Powell [98]. These references are particularly relevant for large-scale or data-driven systems where exact computation of value functions is infeasible.

Stochastic programming and uncertainty. When uncertainty enters explicitly into optimization models, stochastic programming provides a principled framework. Classical references include Birge and Louveaux [22] and the more recent and comprehensive treatment by Shapiro, Dentcheva, and Ruszczyński [109]. These works complement the stochastic decision problems discussed in Section 1.2.4.

Systems perspective and applications. Optimization and control should be viewed as system-level tools embedded within a broader modeling–analysis–design–implementation cycle. This perspective is developed explicitly in systems-oriented texts such as Haberfellner et al. [55] and Mobus and Kalton [85].

Application-driven studies illustrate how abstract optimization concepts are translated into concrete decision problems. Epidemic control models are treated classically by Morton and Wickwire [87]. Risk-aware optimization is developed by Rockafellar and Uryasev [101]. Variational inequalities and equilibrium problems, which arise naturally in constrained and multi-agent systems, are treated in Facchinei and Pang [45].

Robust and modern convex optimization. Finally, modern developments in convex and robust optimization are presented in Ben-Tal and Nemirovski [14] and Ben-Tal, El Ghaoui, and Nemirovski [13]. These references are particularly important for systems subject to uncertainty, modeling error, or adversarial disturbances, and provide a forward-looking complement to the classical theory developed in this book.

Additional mathematical background. For students seeking stronger mathematical foundations, Strang [113] provides an intuitive and applied introduction to linear algebra, while Royden and Fitzpatrick [103] offer a rigorous treatment of real analysis relevant to convergence and continuity arguments. Bishop [23] is recommended for readers interested in connections between optimization, probabilistic modeling, and learning.

1.7 ■ Exercises

Exercise 1.1 (Anatomy of an Optimization Model). For each situation, identify (a) the decision variable and the space it lives in, (b) the feasible set, (c) the objective function, and (d) whether the problem is finite- or infinite-dimensional.

- (i) A data center allocates a total power budget C among N server clusters; cluster i processes $f_i(x_i)$ requests per second when given power $x_i \geq 0$.
- (ii) A vehicle must travel between two fixed points in fixed time T while minimizing fuel, where the instantaneous fuel-burn rate depends on the acceleration profile $a(\cdot)$.
- (iii) A hospital chooses a treatment intensity over a horizon $[0, T]$ to keep a patient's tumor size below a threshold while minimizing cumulative toxicity.
- (iv) An investor selects fixed fractions of wealth to place in n assets to maximize expected return subject to a bound on portfolio variance.

Exercise 1.2 (Finite- versus Infinite-Dimensional Decisions). For each of the following, state the decision variable, decide whether the optimization is over a finite-dimensional vector or over a function, and name the natural space of admissible decisions.

- (i) Choosing how many units of each of 12 products to manufacture next month.
- (ii) Choosing the thrust profile of a rocket during ascent.
- (iii) Choosing the shape of a cable hanging between two poles so as to minimize potential energy.
- (iv) Choosing the reorder quantities for a warehouse at the start of each of 52 weeks.

Explain in one sentence what changes, conceptually, when a decision becomes a function of time rather than a finite list of numbers.

Exercise 1.3 (Descriptive versus Prescriptive Modeling). Recall the distinction between using optimization to *explain* observed behavior and using it to *prescribe* decisions. For each scenario, state which role optimization plays and identify the (possibly implicit) objective.

- (i) A soap film spanning a wire loop settles into a minimal-area surface.
- (ii) A logistics company routes its delivery trucks to minimize total distance.
- (iii) Light travels between two media along the path of least time.
- (iv) A central bank sets an interest-rate policy to balance inflation and unemployment.

Exercise 1.4 (Choice of Objective Shapes the Solution). A divisible resource of size C is shared among N users, with user i receiving $x_i \geq 0$ and $\sum_i x_i \leq C$.

- (i) Write the allocation problem when the planner maximizes total throughput $\sum_i x_i$. What allocations are optimal, and why is the outcome often considered unfair?
- (ii) Write the problem when the planner instead maximizes $\sum_i \log(1 + x_i)$. Argue informally why this promotes a more balanced (proportionally fair) allocation.
- (iii) Explain how the same feasible set can yield very different “optimal” system behavior depending only on the modeler’s choice of objective.

Exercise 1.5 (From Dynamics to an Optimal Control Problem). A tank’s water level $h(t)$ evolves as $\dot{h}(t) = u(t) - d(t)$, where $u(t) \geq 0$ is a controllable inflow and $d(t) \geq 0$ is a known demand. Over a horizon $[0, T]$ the operator wants to track a target level h^* while limiting pumping effort.

- (i) Formulate an optimal control problem: specify the decision variable, the dynamic constraint, any pointwise constraints, and a suitable objective functional.
- (ii) Identify which quantities are states, which are inputs, and which are exogenous.
- (iii) State one reason this problem is infinite-dimensional and one modeling choice that would make it finite-dimensional.

Exercise 1.6 (Formulating an Optimal Stopping Problem). An owner observes the fluctuating market price X_k of an asset at discrete times $k = 0, 1, 2, \dots$ and must decide when to sell. Selling at time k yields X_k (in present value), while continuing incurs a per-period holding cost $c > 0$.

- (i) Formulate the decision as the choice of a stopping rule, and write the objective as an expected payoff.
- (ii) Explain what information the decision at time k may depend on (causality), and why the owner cannot use future prices.
- (iii) Describe, in words, the qualitative form you would expect the optimal policy to take, and relate it to a real behavioral pitfall such as the sunk-cost fallacy.

Exercise 1.7 (Competing Objectives and Trade-offs). A city plans investments $x = (x_1, \dots, x_n)$ across n districts under a total budget B . Two goals compete: total economic output $g(x)$ and an equity measure $e(x)$ that rewards balanced investment.

- (i) Explain why “maximize both g and e ” is not, by itself, a well-defined optimization problem.
- (ii) Propose a scalarized objective that expresses a chosen trade-off, and describe how a single parameter sweeps out a family of compromise solutions.
- (iii) Contrast this formulation with imposing a hard constraint on $e(x)$ instead of including it in the objective.

Part I

Foundations and Optimization Theory

Chapter 2

Mathematical Foundations of Optimization Theory

This chapter develops the mathematical foundations underlying local optimality in finite- and infinite-dimensional optimization problems. Beginning with vector spaces, norms, and topological notions such as openness, closedness, and convergence, the chapter establishes the geometric and analytical framework required for optimization theory. It then introduces continuity and linear transformations in normed spaces, emphasizing the role of boundedness and operator norms. Special attention is given to differentiability concepts in infinite-dimensional spaces, contrasting Gâteaux and Fréchet differentials and clarifying why Fréchet differentiability provides the appropriate notion of a stable first-order approximation. The gradient and Hessian are derived as coordinate representations of the first and second Fréchet differentials in Euclidean spaces, leading naturally to first- and second-order Taylor expansions. These tools are then used to formulate local optimality conditions over general feasible sets through feasible directions, tangent cones, and normal cones, adopting an explicitly geometric viewpoint. The chapter highlights how the local geometry of the feasible set, particularly at its boundary, restricts admissible perturbations and naturally leads to cone-based optimality conditions. Throughout, the material serves as a rigorous bridge between classical finite-dimensional optimization and the variational and optimal control frameworks developed in subsequent chapters.

2.1 ■ Mathematical Foundations: Vector Spaces, Norms, and Differentiation

This section reviews the mathematical background required for the subsequent development of optimality conditions. We begin with vector spaces and subsets, which provide the geometric setting for optimization. We then introduce normed spaces, which allow us to quantify notions of size, distance, and convergence. Finally, we review differentiability in both general normed spaces and the finite-dimensional setting, where derivatives reduce to gradients. These ingredients build on one another in a definite order, sketched in Figure 2.1: algebraic structure acquires geometry through a norm, geometry supports limits and continuity, and continuity in turn makes differentiation and optimality conditions meaningful.

2.1.1 ■ Vector spaces and sets

Optimization problems are formulated over collections of objects, including numbers, vectors, functions, or trajectories, that can be added and scaled. This algebraic structure is captured by the notion of a vector space, while constraints are described through subsets of that space.

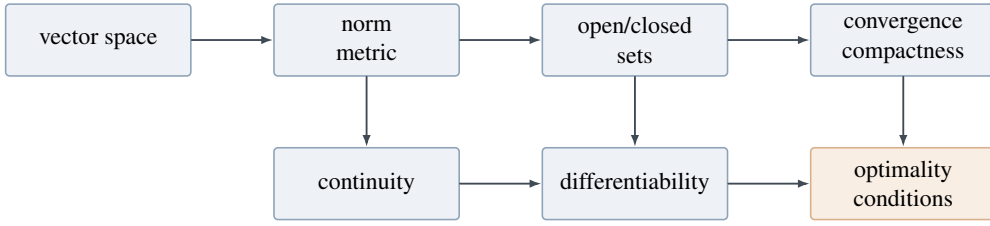


Figure 2.1: Foundational dependencies used throughout optimization: algebraic structure becomes geometric through norms, then supports limits, continuity, differentiation, and optimality conditions.

Let X be a nonempty set. We say that X is a *real vector space* if it is equipped with two operations:

- *vector addition* $(x, y) \mapsto x + y$ from $X \times X$ to X ,
- *scalar multiplication* $(\alpha, x) \mapsto \alpha x$ from $\mathbb{R} \times X$ to X ,

satisfying the following axioms for all $x, y, z \in X$ and all $\alpha, \beta \in \mathbb{R}$:

- (1) $x + y = y + x$ (commutativity),
- (2) $(x + y) + z = x + (y + z)$ (associativity),
- (3) there exists $0 \in X$ such that $x + 0 = x$ (zero vector),
- (4) for each $x \in X$ there exists $-x \in X$ such that $x + (-x) = 0$ (additive inverse),
- (5) $\alpha(x + y) = \alpha x + \alpha y$ (distributivity over vectors),
- (6) $(\alpha + \beta)x = \alpha x + \beta x$ (distributivity over scalars),
- (7) $(\alpha\beta)x = \alpha(\beta x)$,
- (8) $1x = x$.

Elements of X are called *vectors* and are denoted by $x, y, \dots$, while elements of $\mathbb{R}$ are called *scalars*. The zero vector is typically denoted by 0 .

Remark 2.1. The vector space axioms ensure that linear combinations of vectors are well defined. As a result, expressions of the form $\alpha_1 x_1 + \dots + \alpha_k x_k$ are meaningful for any finite collection of vectors $x_i \in X$ and scalars $\alpha_i \in \mathbb{R}$.

Typical examples include:

- $X = \mathbb{R}^n$, the space of finite-dimensional decision vectors;
- spaces of sequences, such as ℓ^2 ;
- spaces of functions, such as $C([a, b])$, the space of continuous functions on an interval.

The vector space X represents the *ambient decision space* in which candidate solutions live.

A subset $D \subset X$ specifies which vectors are admissible. In optimization, D is called the *feasible set* or *constraint set*: only vectors $x \in D$ are allowed as solutions. For example,

constraints such as $x_i \geq 0$ or $\|x\| \leq 1$ restrict the feasible domain without changing the ambient space X .

Definition 2.2 (Linear subspace). A subset $M \subset X$ is a linear subspace if $\alpha x + \beta y \in M$, $\forall x, y \in M$, $\forall \alpha, \beta \in \mathbb{R}$.

Equivalently, a linear subspace contains the zero vector, is closed under vector addition, and is closed under scalar multiplication. Geometrically, subspaces are the higher-dimensional analogues of lines or planes passing through the origin.

Example 2.3 Subspaces versus affine sets.

In $\mathbb{R}^3$, the set $M = \{(x_1, x_2, x_3) : x_1 + x_2 + x_3 = 0\}$ is a linear subspace (a plane through the origin). In contrast, the set $\{x : x_1 + x_2 + x_3 = 1\}$ is not a subspace, since it does not contain the origin. $\square$

Linear subspaces arise naturally when studying directions of motion, tangent spaces, and nullspaces of linear constraints.

2.1.2 ■ Normed spaces

To study limits, continuity, and derivatives, a vector space must be equipped with a notion of *size* or *magnitude*. This is provided by a norm, which allows us to quantify distances between vectors and to formalize the idea of “small perturbations.” Norms bridge algebraic structure (vector spaces) and analytical structure (limits and approximation). Let X be a real vector space.

Definition 2.4 (Normed space). A normed space is a pair $(X, \|\cdot\|)$, where $\|\cdot\| : X \rightarrow \mathbb{R}_{\geq 0}$ is a function satisfying, for all $x, y \in X$ and all $\alpha \in \mathbb{R}$,

$$(1) \quad \|x\| = 0 \text{ if and only if } x = 0 \quad (\text{positivity}),$$

$$(2) \quad \|\alpha x\| = |\alpha| \|x\| \quad (\text{homogeneity}),$$

$$(3) \quad \|x + y\| \leq \|x\| + \|y\| \quad (\text{triangle inequality}).$$

The quantity $\|x\|$ should be interpreted as the “length” or “size” of the vector x . The triangle inequality ensures that the direct displacement from 0 to $x + y$ is no larger than moving first to x and then to y .

Remark 2.5. The norm axioms are precisely those needed to control how vectors behave under addition and scaling. These properties are essential for defining derivatives, which rely on comparing the magnitude of changes to the magnitude of the perturbations.

The same vector space can carry many different norms, and each one leads to a different notion of distance, geometry, and convergence. The clearest way to picture this is through the *unit ball* $\{x : \|x\| \leq 1\}$, whose shape encodes what the norm regards as “small”: the ℓ^1 ball is a diamond, the ℓ^2 ball a disk, and the ℓ^∞ ball a square, so that a single vector may be judged small under one norm and large under another (Figure 2.2).

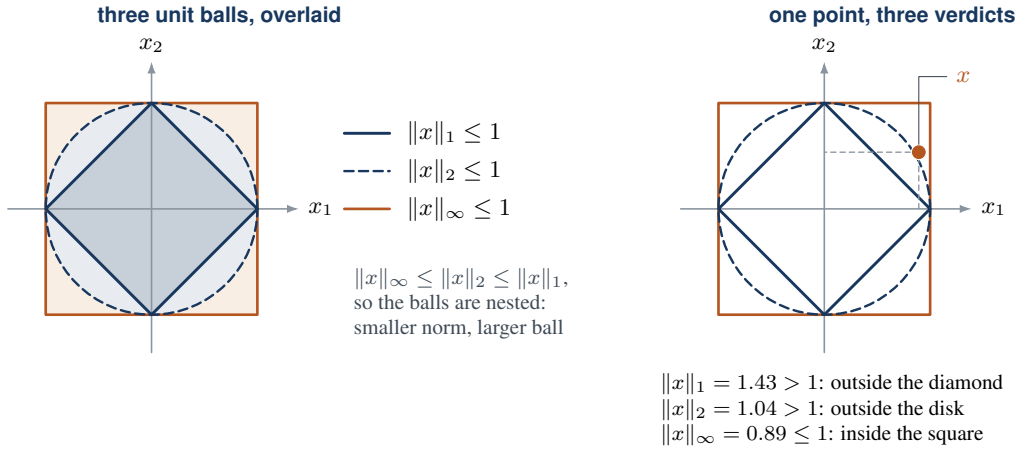


Figure 2.2: Different norms on $\mathbb{R}^2$ produce different unit balls and therefore different local geometries: the same point may count as “small” for one norm and not for another.

Example 2.6 Euclidean space.

The space $\mathbb{R}^n$ equipped with the Euclidean norm $\|x\|_2 = \left(\sum_{i=1}^n x_i^2\right)^{1/2}$ is a normed space. This is the standard setting of finite-dimensional calculus and optimization. Other norms, such as

$$\|x\|_1 = \sum_{i=1}^n |x_i|, \quad \|x\|_\infty = \max_i |x_i|,$$

are also valid norms on $\mathbb{R}^n$ and induce different geometries. □

Example 2.7 Continuous functions with the sup norm.

Let $C[a, b]$ denote the space of continuous real-valued functions on $[a, b]$. Define $\|x\|_\infty := \max_{a \leq t \leq b} |x(t)|$. Then $(C[a, b], \|\cdot\|_\infty)$ is a normed space. This norm measures the maximum deviation of a function over the interval and is fundamental in uniform approximation and optimal control. □

Example 2.8 Differentiable functions.

Let $D[a, b]$ denote the space of continuously differentiable functions on $[a, b]$. The norm $\|x\| := \max_{a \leq t \leq b} |x(t)| + \max_{a \leq t \leq b} |\dot{x}(t)|$ turns $D[a, b]$ into a normed space. This norm simultaneously controls the size of a function and the size of its derivative, which is essential in stability and sensitivity analysis. □

Example 2.9 Functions of bounded variation.

A function $x : [a, b] \rightarrow \mathbb{R}$ is said to have *bounded variation* if there exists $K > 0$ such that for every partition $a = t_0 < t_1 < \dots < t_n = b$, $\sum_{i=1}^n |x(t_i) - x(t_{i-1})| \leq K$. The total variation of x is defined as

$$\text{TV}(x) := \sup_{\{t_i\}} \sum_{i=1}^n |x(t_i) - x(t_{i-1})|.$$

The space $BV[a, b]$ of functions of bounded variation becomes a normed space with the norm $\|x\| := |x(a)| + \text{TV}(x)$. Such spaces arise naturally in optimal control and variational problems with discontinuous solutions. $\square$

Remark 2.10. In finite-dimensional spaces such as $\mathbb{R}^n$, all norms are equivalent in the sense that they induce the same open sets, convergence, and continuity. Thus, local optimality and differentiability are independent of the particular norm chosen.

In infinite-dimensional spaces, different norms can induce genuinely different topologies. As a result, continuity and differentiability properties may depend on the chosen norm, making it a modeling decision rather than a technical detail.

2.1.3 ■ Open and closed sets

Let $(X, \|\cdot\|)$ be a normed space. The norm induces a metric $d(x, y) = \|x - y\|$ and hence a topology on X . The notions of interior, openness, closure, and closedness formalize how sets behave under small perturbations and are fundamental to analysis and optimization.

For any $x \in X$ and $r > 0$, the *open ball* centered at x with radius r is defined by $B_r(x) := \{y \in X : \|y - x\| < r\}$. The collection of all open balls forms a basis for the topology induced by the norm.

Example 2.11 Open balls under different norms.

In $\mathbb{R}^n$ equipped with the Euclidean norm, $B_r(x)$ is the usual open Euclidean ball. Under the ℓ^∞ norm, $B_r(x)$ is an open hypercube centered at x . $\square$

Interior points and open sets.

Definition 2.12 (Interior point). Let $D \subset X$. A point $x \in D$ is called an *interior point* of D if there exists $r > 0$ such that $B_r(x) \subset D$. The set of all interior points of D is called the *interior* of D and is denoted by $\text{int}(D)$.

Definition 2.13 (Open set). A set $D \subset X$ is said to be *open* if $D = \text{int}(D)$, that is, if every point of D is an interior point.

Example 2.14 Open sets.

It is instructive to test the definition against a few familiar sets, including one that fails to be open.

- The empty set $\emptyset$ and the entire space X are open.
- Every open ball $B_r(x)$ is an open set.
- In $\mathbb{R}^2$, a line segment or a single point has empty interior and is therefore not open.

□

Openness of the domain is essential for defining derivatives, which rely on infinitesimal perturbations in all directions. When the domain is not open, perturbations may be restricted near the boundary, leading to constrained or one-sided notions of differentiation. This issue motivates the use of feasible directions, tangent cones, and normal cones in constrained optimization.

Closure points and closed sets.

Definition 2.15 (Closure point). Let $D \subset X$. A point $x \in X$ is called a closure point of D if for every $r > 0$, $B_r(x) \cap D \neq \emptyset$. The set of all closure points of D is called the closure of D and is denoted by $\overline{D}$.

Equivalently, $x \in \overline{D}$ if and only if there exists a sequence $\{x_k\} \subset D$ such that $x_k \rightarrow x$.

Definition 2.16 (Closed set). A set $D \subset X$ is said to be closed if $D = \overline{D}$, that is, if it contains all of its closure points.

Example 2.17 Closed sets.

The same familiar sets, now examined through their closure points, illustrate closedness and its failure.

- The empty set and the entire space X are closed.
- In $\mathbb{R}^n$, the closed ball $\{x : \|x\| \leq r\}$ is closed.
- Any finite set of points is closed.
- The open interval $(0, 1)$ is not closed in $\mathbb{R}$, since its closure is $[0, 1]$.

□

Openness and closedness are not opposites, and a set can be neither. Figure 2.3 contrasts the three cases: an open set, in which every point admits a surrounding ball that still lies inside the set; a closed set, which contains the limit of every convergent sequence drawn from it; and the half-open interval $[0, 1)$, which fails both tests at once.

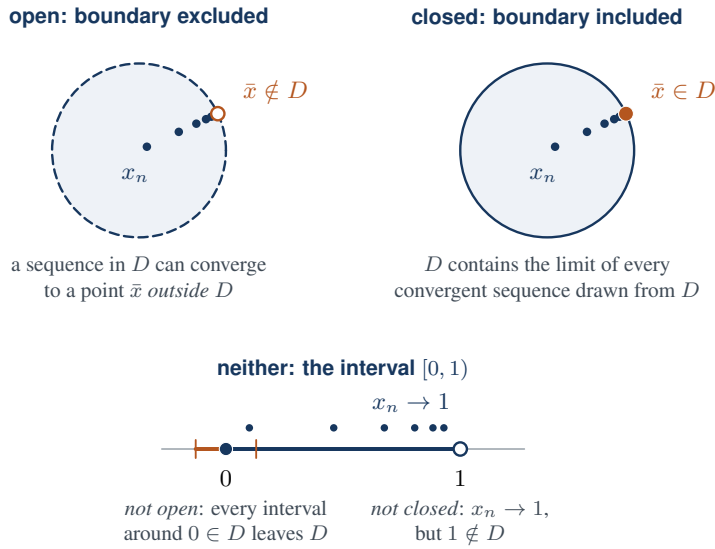


Figure 2.3: Open, closed, and neither-open-nor-closed sets, seen through convergent sequences. In each panel a sequence $\{x_n\} \subset D$ approaches a boundary point $\bar{x}$ (the dots crowd together as they converge). The set is *closed* exactly when it always contains such limits (filled dot, top right): the open disk omits its boundary limit (hollow dot, top left), while the half-open interval $[0, 1)$ includes 0 but not the limit 1, so it is neither open nor closed.

Stability properties of open and closed sets.

Proposition 2.18. *Let $(X, \|\cdot\|)$ be a normed space.*

- (1) *If $D_1, \dots, D_m \subset X$ are open sets for some finite $m \in \mathbb{N}$, then their intersection $\bigcap_{i=1}^m D_i$ is open.*
- (2) *If $\{D_\alpha\}_{\alpha \in A}$ is an arbitrary family of open subsets of X , then their union $\bigcup_{\alpha \in A} D_\alpha$ is open.*
- (3) *If $C_1, \dots, C_m \subset X$ are closed sets for some finite $m \in \mathbb{N}$, then their union $\bigcup_{i=1}^m C_i$ is closed.*
- (4) *If $\{C_\alpha\}_{\alpha \in A}$ is an arbitrary family of closed subsets of X , then their intersection $\bigcap_{\alpha \in A} C_\alpha$ is closed.*

In constrained optimization, feasible sets often lie in lower-dimensional affine subspaces, in which case they may have empty interior in the ambient space.

Definition 2.19 (Relative interior). *Let $D \subset V \subset X$, where V is a linear or affine subspace. A point $x \in D$ is called an interior point of D relative to V if there exists $r > 0$ such that $B_r(x) \cap V \subset D$. The set of all such points is called the relative interior of D .*

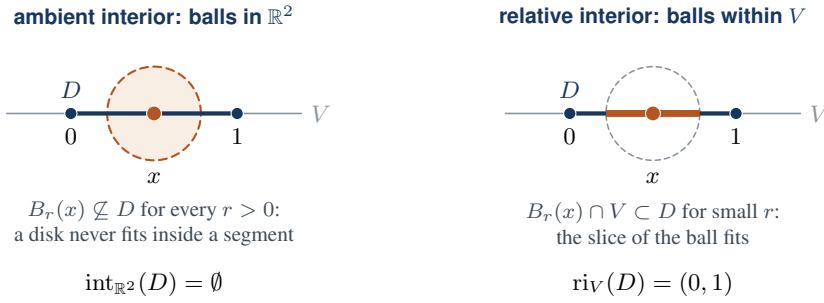


Figure 2.4: Relative interior is computed after restricting attention to the affine subspace that contains the feasible set.

Example 2.20 Relative interior of a segment.

In $\mathbb{R}^2$, the line segment $D = \{(x, 0) : 0 \leq x \leq 1\}$ has empty interior in $\mathbb{R}^2$, but its relative interior with respect to the line $V = \{(x, 0) : x \in \mathbb{R}\}$ is the open interval $(0, 1)$. □

The distinction is a matter of *where* one is allowed to place the test ball, as illustrated in Figure 2.4: a full two-dimensional ball never fits inside a one-dimensional segment, so the ambient interior is empty, yet the intersection of a small ball with the line V does fit, so the relative interior is nonempty.

Relative notions of interior and openness distinguish boundary effects imposed by constraints from those inherent to the ambient space. They play a central role in convex analysis and in the formulation of optimality conditions.

2.1.4 ■ Convergence

Many existence proofs in analysis and optimization proceed by constructing a sequence of vectors that converges to a limit and then showing that the limit inherits the desired properties. For this reason, the notion of convergence plays a central role in normed spaces.

Convergence of sequences. Let $(X, \|\cdot\|)$ be a normed space.

Definition 2.21 (Convergence). A sequence $\{x_n\} \subset X$ is said to converge to a vector $x \in X$ if $\lim_{n \rightarrow \infty} \|x_n - x\| = 0$. In this case, we write $x_n \rightarrow x$ and call x the limit of the sequence.

Convergence in a normed space thus reduces to convergence of a real-valued sequence $\{\|x_n - x\|\}$ to zero.

Remark 2.22. If $x_n \rightarrow x$, then $\|x_n\| \rightarrow \|x\|$. Indeed, by the triangle inequality, $|\|x_n\| - \|x\|| \leq \|x_n - x\| \rightarrow 0$.

Example 2.23 Componentwise convergence in $\mathbb{R}^n$.

In $X = \mathbb{R}^n$, a sequence $x_n = (x_{n,1}, \dots, x_{n,n})$ converges to $x = (x_1, \dots, x_n)$ if and only if each component converges: $x_{n,i} \rightarrow x_i$, for $i = 1, \dots, n$. This equivalence relies on the fact that all norms on $\mathbb{R}^n$ are equivalent. $\square$

In infinite-dimensional spaces, however, convergence may depend delicately on the chosen norm and need not coincide with componentwise convergence.

Example 2.24 Componentwise convergence does not imply norm convergence.

Let X be the space of finitely nonzero sequences equipped with the norm $\|x\| = \sum_{i=1}^{\infty} |x_i|$. Define $e_i = (0, \dots, 0, 1, 0, \dots)$, where the 1 appears in the i th position. Then $e_i \rightarrow 0$ componentwise, but $\|e_i\| = 1$ for all i , so $\{e_i\}$ does not converge to 0 in norm. $\square$

This illustrates that convergence in norm is stronger than componentwise convergence in general.

Proposition 2.25 (Uniqueness of limits). *If a sequence $\{x_n\}$ converges in a normed space, then its limit is unique.*

Proof. Suppose $x_n \rightarrow x$ and $x_n \rightarrow y$. Then $\|x - y\| = \|x - x_n + x_n - y\| \leq \|x - x_n\| + \|x_n - y\| \rightarrow 0$. Hence $\|x - y\| = 0$ and therefore $x = y$. $\square$

Convergence can equivalently be expressed in terms of open balls.

Proposition 2.26. *A sequence $\{x_n\} \subset X$ converges to $x \in X$ if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $x_n \in B_\varepsilon(x)$, for all $n \geq N$.*

This formulation makes explicit the topological nature of convergence and connects it directly to the notion of neighborhoods.

Closed sets and convergence. Convergence provides a useful characterization of closed sets.

Proposition 2.27 (Sequence characterization of closed sets). *A set $F \subset X$ is closed if and only if every convergent sequence $\{x_n\} \subset F$ has its limit in F .*

Proof. $(\Rightarrow)$ If F is closed and $x_n \rightarrow x$ with $x_n \in F$, then x is a closure point of F and hence belongs to F .

$(\Leftarrow)$ Suppose F is not closed. Then there exists $x \in \overline{F} \setminus F$. Since x is a closure point of F , for each $n \in \mathbb{N}$ we may choose $x_n \in F$ such that $\|x_n - x\| < \frac{1}{n}$. Then $x_n \rightarrow x$, but $x \notin F$, contradicting the assumed property. $\square$

2.1.5 ■ Transformations and continuity

The objects that make vector spaces useful in analysis, control, and optimization are *transformations*: rules that map elements of one space into another. Optimization problems, in particular, are formulated in terms of selecting vectors that optimize a prescribed transformation, typically a real-valued one. Understanding the structure and continuity of transformations is therefore essential.

Definition 2.28 (Transformation). *Let X and Y be vector spaces, and let $D \subset X$. A transformation from X into Y with domain D is a mapping $T : D \rightarrow Y$ that assigns to each $x \in D$ a unique element $T(x) \in Y$. If the domain is not specified explicitly, it is understood that $D = X$.*

Thus, a transformation is simply a function whose domain and codomain are vector spaces. The additional algebraic structure becomes relevant when studying how the mapping interacts with vector addition and scalar multiplication.

Definition 2.29 (Injective and surjective mappings). *A transformation $T : D \rightarrow Y$ is said to be*

- *one-to-one (injective) if $T(x_1) = T(x_2)$ implies $x_1 = x_2$,*
- *onto (surjective) if for every $y \in Y$ there exists $x \in D$ such that $T(x) = y$.*

These properties describe whether information is lost or fully covered by the transformation.

Functionals. A particularly important class of transformations arises when the codomain is the scalar field.

Definition 2.30 (Functional). *A transformation $f : X \rightarrow \mathbb{R}$ (or $\mathbb{C}$) is called a functional on X .*

Functionals are central in optimization, since an optimization problem consists of minimizing or maximizing a functional over a feasible set.

Example 2.31 Functionals.

Functionals arise from many everyday operations that assign a single number to each element of a space, on both finite- and infinite-dimensional spaces.

- On any normed space X , the mapping $f(x) = \|x\|$ is a functional.
- On $C[0, 1]$, the mappings

$$f_1(x) = x(1/2), \quad f_2(x) = \int_0^1 x(t) dt, \quad f_3(x) = \max_{0 \leq t \leq 1} x(t)$$

are functionals.

□

Definition 2.32 (Linear transformation). *A transformation $T : X \rightarrow Y$ is said to be linear if*

for all $x_1, x_2 \in X$ and all scalars α_1, α_2 ,

$$T(\alpha_1 x_1 + \alpha_2 x_2) = \alpha_1 T(x_1) + \alpha_2 T(x_2).$$

Linearity captures the idea that the transformation preserves the algebraic structure of the space.

Example 2.33 Linear transformations.

Linear transformations range from finite-dimensional matrices to differential and integral operators on function spaces.

- Any matrix $A \in \mathbb{R}^{m \times n}$ defines a linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $T(x) = Ax$. This is the canonical example of a linear transformation in finite-dimensional spaces.
- The differentiation operator $T(x) = \dot{x}$ is linear on suitable function spaces, such as the space $C^1[a, b]$ of continuously differentiable functions on $[a, b]$.
- Let $C[a, b]$ denote the space of continuous real-valued functions on $[a, b]$. Let $k : [a, b] \times [a, b] \rightarrow \mathbb{R}$ be a function that is continuous in both variables (called the *kernel*). Define the transformation $T : C[a, b] \rightarrow C[a, b]$ by $(Tx)(t) := \int_a^b k(t, \tau) x(\tau) d\tau$, $t \in [a, b]$. Then T is a linear transformation. Linearity follows from the linearity of the integral: for any $x_1, x_2 \in C[a, b]$ and scalars α_1, α_2 we have $T(\alpha_1 x_1 + \alpha_2 x_2) = \alpha_1 T(x_1) + \alpha_2 T(x_2)$. Such operators are called *integral operators* and arise frequently in optimal control, integral equations, and systems theory.

□

2.1.6 - Continuity of Transformations

When both the domain and codomain are normed spaces, the associated norms provide a natural way to measure the effect of small perturbations in the input on the output. Continuity formalizes the requirement that a transformation respond stably to such perturbations.

Definition 2.34 (Continuity at a point). Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces, and let $T : X \rightarrow Y$ be a mapping. The transformation T is said to be continuous at $x_0 \in X$ if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$\|x - x_0\|_X < \delta \implies \|T(x) - T(x_0)\|_Y < \varepsilon.$$

If T is continuous at every point $x_0 \in X$, then T is said to be continuous.

Continuity guarantees that arbitrarily small changes in the input result in arbitrarily small changes in the output. This property is essential in optimization and control, where stability, robustness, and sensitivity to perturbations play a central role. The definition is illustrated in Figure 2.5: the challenge is a tolerance ε on the output, and the response is a tolerance δ on the input that meets it.

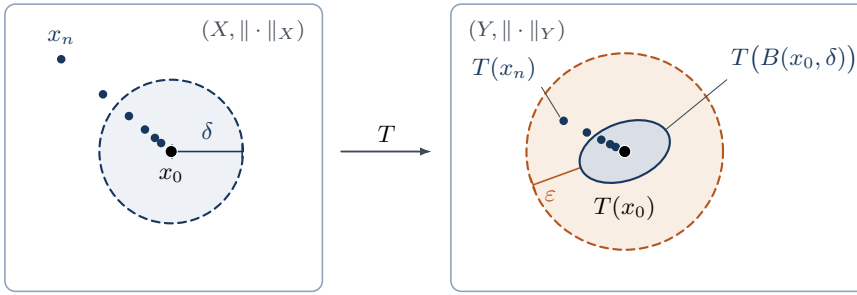


Figure 2.5: Continuity of T at x_0 . Given any tolerance $\varepsilon > 0$ on the output, continuity supplies an input tolerance $\delta > 0$ such that the whole ball $B(x_0, \delta)$ is carried inside $B(T(x_0), \varepsilon)$. Equivalently (Proposition 2.35), every sequence $x_n \rightarrow x_0$ has $T(x_n) \rightarrow T(x_0)$: limits may be passed through T .

In normed spaces, continuity admits an equivalent and often more convenient characterization in terms of convergent sequences.

Proposition 2.35 (Sequence characterization). *Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces. Let $T : X \rightarrow Y$ be a transformation between normed spaces. The mapping T is continuous at $x_0 \in X$ if and only if for every sequence $\{x_n\} \subset X$ satisfying $x_n \rightarrow x_0$, we have $T(x_n) \rightarrow T(x_0)$.*

Remark 2.36. This result states that continuity is equivalent to the preservation of limits: limits may be passed through a continuous transformation. The sequence characterization is particularly useful in analysis, where continuity is often verified via limiting arguments.

For linear transformations, continuity admits especially strong and convenient characterizations. In particular, continuity is completely determined by the behavior of the mapping near the origin and is equivalent to a global norm bound.

Theorem 2.37 (Continuity and boundedness of linear maps). *Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces, and let $T : X \rightarrow Y$ be a linear transformation. Then the following statements are equivalent:*

- (1) T is continuous at 0.
- (2) T is continuous at some point $x_0 \in X$.
- (3) T is continuous on X .
- (4) (Boundedness) There exists a constant $C > 0$ such that

$$\|T(x)\|_Y \leq C\|x\|_X \quad \forall x \in X. \quad (2.1)$$

Proof. We prove the implications $(3) \Rightarrow (2) \Rightarrow (1) \Rightarrow (4) \Rightarrow (3)$.

$(3) \Rightarrow (2)$. If T is continuous on X , then it is continuous at any point $x_0 \in X$.

$(2) \Rightarrow (1)$. Assume T is continuous at some $x_0 \in X$. Let $h \in X$ and set $x = x_0 + h$. By linearity, $T(x) - T(x_0) = T(x_0 + h) - T(x_0) = T(h)$. Given $\varepsilon > 0$, continuity at x_0 provides $\delta > 0$ such

that $\|h\|_X < \delta$ implies $\|T(h)\|_Y = \|T(x_0 + h) - T(x_0)\|_Y < \varepsilon$. This is precisely continuity at 0.

(1) $\Rightarrow$ (4). Assume T is continuous at 0. Apply the definition of continuity with $\varepsilon = 1$. Then there exists $\delta > 0$ such that $\|x\|_X < \delta \Rightarrow \|T(x)\|_Y < 1$. Let $x \in X$ be arbitrary. If $x = 0$, (2.1) holds trivially. If $x \neq 0$, define $u := \frac{\delta}{\|x\|_X} x$, so that $\|u\|_X = \delta$. For any $0 < \eta < 1$, we have $\|\eta u\|_X = \eta\delta < \delta$, hence $\|T(\eta u)\|_Y < 1$. Using linearity, $\|T(\eta u)\|_Y = \eta\|T(u)\|_Y < 1$, so $\|T(u)\|_Y < 1/\eta$. Letting $\eta \uparrow 1$ yields $\|T(u)\|_Y \leq 1$. Therefore,

$$\|T(x)\|_Y = \left\| T\left(\frac{\|x\|_X}{\delta} u\right) \right\|_Y = \frac{\|x\|_X}{\delta} \|T(u)\|_Y \leq \frac{1}{\delta} \|x\|_X.$$

Thus (2.1) holds with $C := 1/\delta$.

(4) $\Rightarrow$ (3). Assume (2.1) holds. Let $x_0 \in X$ and let $x \in X$. By linearity, $T(x) - T(x_0) = T(x - x_0)$. Using (2.1), $\|T(x) - T(x_0)\|_Y = \|T(x - x_0)\|_Y \leq C\|x - x_0\|_X$. Given $\varepsilon > 0$, choose $\delta := \varepsilon/C$. Then $\|x - x_0\|_X < \delta$ implies $\|T(x) - T(x_0)\|_Y < \varepsilon$, proving continuity at x_0 . Since x_0 was arbitrary, T is continuous on X . $\square$

Remark 2.38 (Operator norm and Lipschitz property). If T is linear and bounded, the smallest constant C in (2.1) is called the *operator norm* of T and is denoted by

$$\|T\|_{\text{op}} := \sup_{\|x\|_X \leq 1} \|T(x)\|_Y = \sup_{x \neq 0} \frac{\|T(x)\|_Y}{\|x\|_X}.$$

The inequality (2.1) can then be written as $\|T(x) - T(y)\|_Y \leq \|T\|_{\text{op}} \|x - y\|_X$, $\forall x, y \in X$, so every bounded linear map is (globally) Lipschitz continuous.

Example 2.39 Linear maps in finite dimensions.

Let $X = \mathbb{R}^n$ and $Y = \mathbb{R}^m$, each equipped with any norm. For a fixed matrix $A \in \mathbb{R}^{m \times n}$, define $T(x) = Ax$. Then T is a linear transformation and is continuous on $\mathbb{R}^n$. Indeed, there exists a constant $C > 0$ such that $\|Ax\| \leq C\|x\|$, $\forall x \in \mathbb{R}^n$. In fact, in finite-dimensional spaces, *every* linear transformation is continuous, regardless of the norm chosen. $\square$

Example 2.40 A nonlinear continuous mapping.

Let $X = Y = \mathbb{R}$ with the usual absolute value norm, and define $T(x) = x^2$. Then T is continuous at every point $x_0 \in \mathbb{R}$. Indeed, if $x_n \rightarrow x_0$, then $x_n^2 \rightarrow x_0^2$, showing continuity via the sequence characterization. This example illustrates that continuity does not require linearity. $\square$

Example 2.41 Integral operator on a function space.

Let $X = Y = C[a, b]$ with the supremum norm $\|\cdot\|_\infty$, and let $k : [a, b] \times [a, b] \rightarrow \mathbb{R}$

be a continuous kernel. Define $(Tx)(t) := \int_a^b k(t, \tau)x(\tau) d\tau$. Then T is a linear and continuous transformation. Moreover, there exists a constant $C > 0$ such that $\|Tx\|_\infty \leq C\|x\|_\infty$, $\forall x \in C[a, b]$, where $C = \max_{t \in [a, b]} \int_a^b |k(t, \tau)| d\tau$. Such operators arise naturally in integral equations and optimal control. $\square$

Example 2.42 A discontinuous linear transformation.

Let $X = C^1[0, 1]$, the space of continuously differentiable functions, regarded as a subspace of $C[0, 1]$ and equipped with the supremum norm $\|\cdot\|_\infty$ inherited from it. Define the linear functional $T(x) = x'(0)$. Then T is *not* continuous with respect to the supremum norm. Indeed, take $x_n(t) = \frac{1}{n} \sin(n^2 t)$. Then $\|x_n\|_\infty = \frac{1}{n} \rightarrow 0$, so $x_n \rightarrow 0$ uniformly; yet $x'_n(t) = n \cos(n^2 t)$, so $x'_n(0) = n$ and hence $T(x_n) = n$ does *not* converge to $T(0) = 0$. The obstruction is that the supremum norm controls the size of a function but not the size of its derivative. This example demonstrates that, in infinite-dimensional spaces, linearity alone does not guarantee continuity. $\square$

2.1.7 ■ Banach Spaces

Let $(X, \|\cdot\|)$ be a normed linear space. One of the central questions in analysis is whether limits of sequences generated by natural approximation or iteration procedures exist within the space. This motivates the study of Cauchy sequences.

Definition 2.43 (Cauchy sequence). A sequence $\{x_n\} \subset X$ is called a Cauchy sequence if $\|x_n - x_m\| \rightarrow 0$ as $n, m \rightarrow \infty$, that is, for every $\varepsilon > 0$ there exists an integer N such that $\|x_n - x_m\| < \varepsilon$ for all $n, m \geq N$.

The Cauchy condition is *intrinsic*: it depends only on distances between elements of the sequence, and not on the existence or identification of a candidate limit. This feature is particularly important in infinite-dimensional spaces, where explicit limits are often difficult or impossible to compute. It also exposes the gap that completeness is designed to close: being Cauchy says the terms bunch together, not that anything in the space is being approached. Figure 2.6 contrasts the two possibilities.

Lemma 2.44. Every Cauchy sequence in a normed linear space is bounded.

Proof. Since $\{x_n\}$ is Cauchy, there exists $N \in \mathbb{N}$ such that $\|x_n - x_N\| < 1$ for all $n \geq N$. Then, for all $n \geq N$, $\|x_n\| \leq \|x_N\| + \|x_n - x_N\| \leq \|x_N\| + 1$. Thus $\{x_n\}$ is bounded. $\square$

Proposition 2.45 (Convergent sequences are Cauchy). Let $(X, \|\cdot\|)$ be a normed linear space. If a sequence $\{x_n\} \subset X$ converges to some $x \in X$, then $\{x_n\}$ is a Cauchy sequence.

Proof. Assume that $x_n \rightarrow x$ in X . By definition of convergence, for every $\varepsilon > 0$ there exists an integer N such that $\|x_n - x\| < \frac{\varepsilon}{2}$ for all $n \geq N$. Then, for any $n, m \geq N$, the triangle inequality

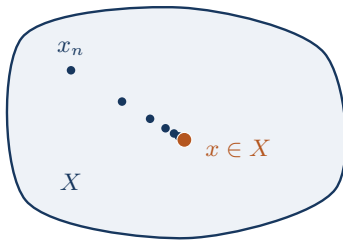
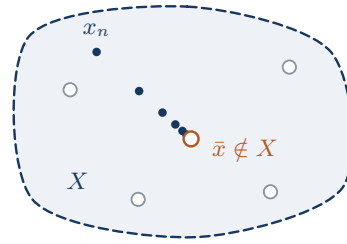
(a) Complete: the limit lies in X .(b) Incomplete: the limit escapes to $\overline{X}$.

Figure 2.6: Why completeness is the dividing line. A Cauchy sequence is defined intrinsically, by distances between its own terms, so both pictures show a legitimate Cauchy sequence in X . In a Banach space (a) the limit is guaranteed to be an element of X , and an iterative argument that produces the sequence has thereby produced a point of X . In an incomplete normed space (b) the sequence still bunches together, but its limit is a “hole” of X (hollow dots) that exists only in the completion $\overline{X}$: the argument stops one step short of existence.

yields

$$\|x_n - x_m\| = \|(x_n - x) + (x - x_m)\| \leq \|x_n - x\| + \|x_m - x\| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\{x_n\}$ is a Cauchy sequence. $\square$

Proposition 2.45 shows that convergence is a *stronger* notion than the Cauchy property: convergence requires not only that the elements of the sequence become arbitrarily close to one another, but also that they approach a specific limit belonging to the space.

The converse statement, namely, that every Cauchy sequence converges, is not true in general. Importantly, when this failure occurs, it is not due to a defect of the sequence, but rather to a structural deficiency of the ambient space.

Example 2.46 A Cauchy sequence that does not converge.

Let $X = (0, 1)$ be the open interval, viewed as a metric space with distance $d(x, y) := |x - y|$. (Note that $(0, 1)$ is not a linear space, so this illustrates the phenomenon in its simplest scalar form; genuinely normed instances follow below.) Define the sequence $\{x_n\} \subset X$ by $x_n = \frac{1}{n+1}$. $\square$

Proposition 2.47. *The sequence $\{x_n\}$ is Cauchy in X but does not converge in X .*

Proof. For any $n, m \in \mathbb{N}$,

$$|x_n - x_m| = \left| \frac{1}{n+1} - \frac{1}{m+1} \right| \rightarrow 0 \quad \text{as } n, m \rightarrow \infty,$$

so $\{x_n\}$ is a Cauchy sequence. However, $x_n \rightarrow 0$ as $n \rightarrow \infty$, and the limit 0 does not belong to $X = (0, 1)$. Hence, $\{x_n\}$ does not converge in X . $\square$

Completeness and Banach Spaces

The preceding discussion shows that, although convergence always implies the Cauchy property, the converse may fail in general normed spaces. This observation leads naturally to the notion of completeness, which characterizes spaces in which the Cauchy criterion for convergence is both necessary and sufficient.

Definition 2.48 (Complete normed space). Let $(X, \|\cdot\|)$ be a normed linear space. The space X is said to be complete if every Cauchy sequence $\{x_n\} \subset X$ converges to a limit $x \in X$.

Definition 2.49 (Banach space). A Banach space is a normed linear space that is complete.

Completeness is a structural property of the space, not of individual sequences. It asserts that whenever a sequence exhibits intrinsic convergence behavior, namely, that its elements become arbitrarily close to one another, there exists an element of the space that realizes this limiting behavior. In other words, the space is closed under limits of its Cauchy sequences.

Conceptual significance. The definition of completeness formalizes the idea that the norm topology on X fully captures convergence phenomena internal to the space. If X is not complete, then there exist sequences whose limiting behavior is well-defined in an ambient or extended space, but whose limits lie outside X . Completeness rules out such pathologies.

Incomplete Normed Spaces: Cautionary Examples

Example 2.50 Continuous functions with the integral norm.

Let X be the space of continuous functions on $[0, 1]$ equipped with the norm $\|x\| = \int_0^1 |x(t)| dt$. Define a sequence $\{x_n\} \subset X$ by

$$x_n(t) = \begin{cases} 0, & 0 \leq t \leq \frac{1}{2} - \frac{1}{n}, \\ nt - \frac{n}{2} + 1, & \frac{1}{2} - \frac{1}{n} \leq t \leq \frac{1}{2}, \\ 1, & t \geq \frac{1}{2}. \end{cases}$$

□

Proposition 2.51. The sequence $\{x_n\}$ is Cauchy in $(X, \|\cdot\|)$ but does not converge to any element of X .

Proof. A direct computation shows that $\|x_n - x_m\| = \frac{1}{2} \left| \frac{1}{n} - \frac{1}{m} \right| \rightarrow 0$, so the sequence is Cauchy. The pointwise limit is the discontinuous step function

$$x(t) = \begin{cases} 0, & t < \frac{1}{2}, \\ 1, & t \geq \frac{1}{2}, \end{cases}$$

which is not continuous and therefore does not belong to X . □

Completeness depends not only on the underlying set of functions, but critically on the choice of norm. The same collection of functions may be complete under one norm and incomplete under

another.

Example 2.52 Finitely supported sequences.

Let X be the space of sequences $x = (\xi_1, \xi_2, \dots, \xi_n, 0, 0, \dots)$ with only finitely many nonzero entries, equipped with the norm $\|x\| = \max_i |\xi_i|$. Define

$$x_n = \left(1, \frac{1}{2}, \frac{1}{3}, \dots, \frac{1}{n-1}, 0, 0, \dots\right).$$

□

Proposition 2.53. *The sequence $\{x_n\}$ is Cauchy in X but has no limit in X .*

The sequence converges pointwise to $(1, 1/2, 1/3, \dots)$, which lies outside X . Thus, X is not closed under limits of Cauchy sequences and is therefore incomplete.

Classical Examples of Banach Spaces

We now present several fundamental examples of Banach spaces that arise throughout analysis. In each case, completeness follows from a close alignment between the norm-induced topology and the underlying structure of the space.

Example 2.54 The space $C[0, 1]$.

Let $C[0, 1] := \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$. Define the supremum norm by $\|f\|_\infty := \sup_{t \in [0, 1]} |f(t)|$. Then $(C[0, 1], \|\cdot\|_\infty)$ is a Banach space. □

The norm $\|\cdot\|_\infty$ induces the topology of uniform convergence. If $\{f_n\}$ is Cauchy in this norm, then it converges uniformly to some function f . Since uniform limits of continuous functions are continuous, it follows that $f \in C[0, 1]$, establishing completeness.

Example 2.55 The sequence spaces ℓ^p .

Let $1 \leq p < \infty$ and define

$$\ell^p := \left\{ x = (x_1, x_2, \dots) \mid \sum_{k=1}^{\infty} |x_k|^p < \infty \right\},$$

with norm

$$\|x\|_p := \left(\sum_{k=1}^{\infty} |x_k|^p \right)^{1/p}.$$

For $p = \infty$, define

$$\ell^\infty := \left\{ x = (x_1, x_2, \dots) \mid \sup_{k \in \mathbb{N}} |x_k| < \infty \right\},$$

with norm $\|x\|_\infty := \sup_{k \in \mathbb{N}} |x_k|$. Then $(\ell^p, \|\cdot\|_p)$ is a Banach space for all $1 \leq p \leq \infty$. $\square$

If $\{x^{(n)}\}$ is Cauchy in ℓ^p , then each coordinate sequence $\{x_k^{(n)}\}$ is Cauchy in $\mathbb{R}$ and hence convergent. The Cauchy property provides uniform control over the ℓ^p norms, which guarantees that the coordinatewise limit belongs to ℓ^p and that convergence holds in norm.

Example 2.56 The function spaces L^p .

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $1 \leq p < \infty$. Define

$$L^p(\Omega) := \left\{ f : \Omega \rightarrow \mathbb{R} \text{ measurable} \mid \int_{\Omega} |f|^p d\mu < \infty \right\},$$

where functions that agree almost everywhere are identified. Equip $L^p(\Omega)$ with the norm

$$\|f\|_p := \left(\int_{\Omega} |f|^p d\mu \right)^{1/p}.$$

For $p = \infty$, define

$$L^\infty(\Omega) := \left\{ f : \Omega \rightarrow \mathbb{R} \text{ measurable} \mid \operatorname{ess\,sup}_{\omega \in \Omega} |f(\omega)| < \infty \right\},$$

with norm $\|f\|_\infty := \operatorname{ess\,sup}_{\omega \in \Omega} |f(\omega)|$. Then $(L^p(\Omega), \|\cdot\|_p)$ is a Banach space for all $1 \leq p \leq \infty$. $\square$

Completeness of $L^p(\Omega)$ follows from measure-theoretic convergence principles. If $\{f_n\}$ is Cauchy in L^p , then it admits a subsequence that converges almost everywhere to a measurable function f . Norm bounds and Fatou-type arguments ensure that $f \in L^p(\Omega)$ and that $f_n \rightarrow f$ in the L^p norm. The identification of functions that agree almost everywhere is essential for completeness.

The spaces $C[0, 1]$, ℓ^p , and $L^p(\Omega)$ illustrate three complementary mechanisms by which completeness arises: uniform convergence preserving structure, coordinatewise convergence under global norm control, and almost-everywhere convergence combined with integral bounds. Together, they form the foundational Banach spaces of analysis.

Closed Sets and Finite-Dimensional Subspaces

The relationship between completeness and topological closure plays a fundamental role in functional analysis. In Banach spaces, completeness is inherited precisely by closed subsets.

Theorem 2.57 (Closed subsets of Banach spaces). *Let $(X, \|\cdot\|)$ be a Banach space and let $Y \subset X$ be a linear subspace. Equip Y with the norm $\|y\|_Y := \|y\|$, $y \in Y$, obtained by restricting the norm of X to Y . Then $(Y, \|\cdot\|_Y)$ is complete if and only if Y is closed in X .*

Proof. Since $\|\cdot\|_Y$ is by definition the restriction of $\|\cdot\|$ to Y , a sequence in Y has exactly the same Cauchy and convergence behavior whether we regard it as a sequence in $(Y, \|\cdot\|_Y)$ or as

a sequence in $(X, \|\cdot\|)$. The only distinction between the two viewpoints is *where the limit is allowed to live*, and this is precisely what the equivalence turns on.

($\Leftarrow$) *If Y is closed, then Y is complete.* Let $\{y_n\} \subset Y$ be a Cauchy sequence in $(Y, \|\cdot\|_Y)$. By the remark above it is also Cauchy in X . Since X is a Banach space, it is complete, so there exists a limit $x \in X$ with $y_n \rightarrow x$. Each y_n lies in Y and Y is closed, so by the sequence characterization of closed sets (Proposition 2.27) the limit satisfies $x \in Y$. Thus the Cauchy sequence $\{y_n\}$ converges to a point of Y . As $\{y_n\}$ was arbitrary, $(Y, \|\cdot\|_Y)$ is complete.

($\Rightarrow$) *If Y is complete, then Y is closed.* By Proposition 2.27 it suffices to show that whenever a sequence $\{y_n\} \subset Y$ converges in X to some $x \in X$, the limit x belongs to Y . So suppose $y_n \rightarrow x$ in X . A convergent sequence is Cauchy (Proposition 2.45), hence $\{y_n\}$ is Cauchy in X and therefore Cauchy in $(Y, \|\cdot\|_Y)$. Because Y is complete, there is some $y \in Y$ with $y_n \rightarrow y$ in Y , and consequently $y_n \rightarrow y$ in X as well. The sequence $\{y_n\}$ now converges in X to both x and y ; by uniqueness of limits (Proposition 2.25) we conclude $x = y \in Y$. Hence Y contains the limit of every convergent sequence drawn from it, so Y is closed. $\square$

Theorem 2.57 shows that completeness is a *relative* property: a subspace of a Banach space is complete precisely when it contains the limits of all sequences that converge with respect to the ambient norm. Any failure of closedness corresponds to the existence of Cauchy sequences in the subspace whose limits lie outside the subspace.

Example 2.58 A closed and complete subspace.

Let $X = C[0, 1]$ equipped with the supremum norm $\|f\|_\infty = \sup_{t \in [0, 1]} |f(t)|$, which makes X a Banach space. Define the subspace $Y = \{f \in C[0, 1] \mid f(0) = 0\}$.

The norm on Y is the restriction of $\|\cdot\|_\infty$ to Y , so convergence and Cauchy properties in Y are understood with respect to uniform convergence. If $\{f_n\} \subset Y$ converges in $\|\cdot\|_\infty$ to some $f \in C[0, 1]$, then $f(0) = \lim_{n \rightarrow \infty} f_n(0) = 0$, which implies $f \in Y$. Hence Y is closed in X , and by Theorem 2.57, the space $(Y, \|\cdot\|_\infty)$ is complete. $\square$

We now turn to a fundamental structural result that distinguishes finite- and infinite-dimensional normed spaces.

Theorem 2.59 (Completeness of finite-dimensional subspaces). *Let X be a normed linear space and let $Y \subset X$ be a finite-dimensional subspace. Then Y is complete.*

Proof. Let $n = \dim Y$ and fix a basis $\{e_1, \dots, e_n\}$ of Y . Every $y \in Y$ has a *unique* coordinate representation

$$y = \sum_{i=1}^n a_i e_i, \quad a = (a_1, \dots, a_n) \in \mathbb{R}^n,$$

and we measure coordinate vectors with the maximum norm $|a|_\infty := \max_{1 \leq i \leq n} |a_i|$. The idea of the proof is to bound the given norm $\|\cdot\|$ on Y above and below by the coordinate norm $|\cdot|_\infty$, and then to transfer completeness from $\mathbb{R}$ (one coordinate at a time) back to Y .

Step 1: An upper bound (the coordinate map is continuous). For any $y = \sum_i a_i e_i$, the triangle

inequality and homogeneity of the norm give

$$\|y\| = \left\| \sum_{i=1}^n a_i e_i \right\| \leq \sum_{i=1}^n |a_i| \|e_i\| \leq \left(\sum_{i=1}^n \|e_i\| \right) \max_{1 \leq i \leq n} |a_i| = M |a|_\infty,$$

where the constant $M := \sum_{i=1}^n \|e_i\| > 0$ depends only on the basis. In particular $\left\| \sum_i a_i e_i \right\| \rightarrow 0$ whenever $|a|_\infty \rightarrow 0$.

Step 2: A lower bound. We claim there is a constant $m > 0$ with

$$\left\| \sum_{i=1}^n a_i e_i \right\| \geq m |a|_\infty \quad \text{for all } a \in \mathbb{R}^n. \quad (2.2)$$

Define $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}$ by $\varphi(a) := \left\| \sum_{i=1}^n a_i e_i \right\|$. Using Step 1 together with the reverse triangle inequality,

$$|\varphi(a) - \varphi(b)| \leq \left\| \sum_{i=1}^n (a_i - b_i) e_i \right\| \leq M |a - b|_\infty,$$

so φ is (Lipschitz) continuous on $\mathbb{R}^n$. Consider the unit sphere of the coordinate norm, $S := \{a \in \mathbb{R}^n : |a|_\infty = 1\}$, which is closed and bounded, hence compact by the classical (finite-dimensional) Heine–Borel theorem. The continuous function φ therefore attains a minimum on S ; set $m := \min_{a \in S} \varphi(a) = \varphi(a^*)$ for some $a^* \in S$. This minimum is *strictly positive*: were $m = 0$, we would have $\left\| \sum_i a_i^* e_i \right\| = 0$, hence $\sum_i a_i^* e_i = 0$, and linear independence of $\{e_i\}$ would force $a^* = 0$, contradicting $|a^*|_\infty = 1$. Thus $m > 0$. Finally, for arbitrary $a \neq 0$ apply the bound on S to the normalized vector $a/|a|_\infty \in S$ and use homogeneity:

$$\varphi(a) = |a|_\infty \varphi\left(\frac{a}{|a|_\infty}\right) \geq |a|_\infty m,$$

which is (2.2); the case $a = 0$ is trivial.

Step 3: Completeness via coordinates. Let $\{y^{(k)}\}_{k \geq 1} \subset Y$ be Cauchy in $(Y, \|\cdot\|)$, and write $y^{(k)} = \sum_{i=1}^n a_i^{(k)} e_i$ with coordinate vectors $a^{(k)} \in \mathbb{R}^n$. Applying the lower bound (2.2) to the difference $y^{(k)} - y^{(l)} = \sum_i (a_i^{(k)} - a_i^{(l)}) e_i$ gives, for every pair k, l ,

$$\max_{1 \leq i \leq n} |a_i^{(k)} - a_i^{(l)}| = |a^{(k)} - a^{(l)}|_\infty \leq \frac{1}{m} \|y^{(k)} - y^{(l)}\|.$$

The right-hand side tends to 0 as $k, l \rightarrow \infty$, so each of the n coordinate sequences $\{a_i^{(k)}\}_k$ is Cauchy in $\mathbb{R}$. Since $\mathbb{R}$ is complete, the limits $a_i^* := \lim_{k \rightarrow \infty} a_i^{(k)}$ exist for $i = 1, \dots, n$. Set

$$y^* := \sum_{i=1}^n a_i^* e_i \in Y.$$

Applying the upper bound of Step 1 to $y^{(k)} - y^*$ yields

$$\|y^{(k)} - y^*\| = \left\| \sum_{i=1}^n (a_i^{(k)} - a_i^*) e_i \right\| \leq M \max_{1 \leq i \leq n} |a_i^{(k)} - a_i^*| \xrightarrow{k \rightarrow \infty} 0.$$

Hence $y^{(k)} \rightarrow y^*$ with $y^* \in Y$. Every Cauchy sequence in $(Y, \|\cdot\|)$ thus converges to an element of Y , so Y is complete. $\square$

Remark 2.60. The two inequalities established in the proof, namely $m \|a\|_\infty \leq \|\sum_i a_i e_i\| \leq M \|a\|_\infty$, say precisely that the given norm $\|\cdot\|$ and the coordinate norm $\|\cdot\|_\infty$ are *equivalent* on Y . This is the finite-dimensional phenomenon alluded to below: on a finite-dimensional space all norms are equivalent, so they induce the same convergent sequences, the same Cauchy sequences, and hence the same topology. The compactness of S used in Step 2 is exactly the ingredient that fails in infinite dimensions, which is why completeness can be lost there.

Example 2.61 Finite-dimensional subspaces of function spaces.

Let $X = C[0, 1]$ with the supremum norm, and consider the subspace $Y = \text{span}\{1, t, t^2, \dots, t^n\}$. Then Y is finite-dimensional and hence complete by Theorem 2.59. In particular, every Cauchy sequence in Y converges to an element of Y .

□

Example 2.62 Finite-dimensional subspaces of sequence spaces.

Let $X = \ell^2$ and define $Y = \{(x_1, \dots, x_n, 0, 0, \dots) \mid x_i \in \mathbb{R}\}$. Then Y is a finite-dimensional subspace of ℓ^2 and is therefore complete. Moreover, Y is closed in ℓ^2 , illustrating the compatibility of Theorems 2.57 and 2.59.

□

Theorem 2.59 shows that incompleteness is an inherently infinite-dimensional phenomenon. Finite-dimensional normed spaces share the regularity properties of Euclidean spaces: all norms induce the same topology, and every Cauchy sequence converges. Pathological behaviors associated with nonconvergence arise only in infinite-dimensional settings.

Normed spaces vs. Banach spaces: local structure and global existence

A normed vector space is sufficient for many structural and geometric arguments, but it becomes inadequate the moment an argument relies on convergence. The distinction between a general normed space and a Banach space is therefore not cosmetic: it marks the boundary between *local reasoning* and *global existence guarantees*. In short, Banach spaces are needed precisely when limits matter.

At the level of geometry and algebra, completeness plays no role. Convexity, norm inequalities, separation of convex sets, Hahn–Banach extensions, epigraph constructions, and subgradient characterizations are all local statements. They compare finitely many points, hyperplanes, or inequalities at a time. As long as a norm is defined, these arguments remain valid. This is why large parts of convex analysis, finite-dimensional optimization, and formal duality theory can be developed entirely within normed spaces, without assuming completeness.

The situation changes fundamentally once a proof constructs a sequence and then appeals, explicitly or implicitly, to its limit. Iterative methods, approximation schemes, and infinite expansions naturally generate Cauchy sequences. In a merely normed space, such sequences may converge only in a completion, with the limit lying outside the space itself. Completeness is exactly the property that prevents this escape. A Banach space guarantees that every Cauchy sequence produced by an algorithm, iteration, or variational argument converges to an element of the space.

This necessity is most transparent in fixed-point theory. The Banach contraction principle, Picard iteration, Newton–Kantorovich methods, and value iteration in dynamic programming

all rely on showing that an iteratively defined sequence is Cauchy. Without completeness, the argument stops short of existence: the limit may exist in an abstract completion, but not in the original space. In this sense, Banach spaces are not chosen for elegance, but because they are the minimal setting in which iterative reasoning is logically sound.

The same issue arises in infinite series and operator theory. Neumann series, resolvent expansions, and power series of operators require completeness to ensure convergence in norm. Without it, formal algebraic manipulations may produce expressions whose sums do not belong to the space under consideration. This is why functional analysis insists on Banach spaces even when working only with bounded linear operators.

Existence results in infinite-dimensional optimization and variational problems also force completeness. Minimizing sequences often exist under mild assumptions, but turning a minimizing sequence into an actual minimizer requires convergence. Lower semicontinuity and coercivity alone are not sufficient if the ambient space is incomplete: the candidate minimizer may simply lie outside the space. Banach (and often reflexive or Hilbert) structure is what closes this gap.

Finally, the central theorems of functional analysis, including the closed graph theorem, the open mapping theorem, and the uniform boundedness principle, are all false in incomplete normed spaces. Each depends on passing to limits of sequences of operators or function values. Their failure without completeness underscores that Banach spaces are not a technical convenience, but the natural habitat for global operator-theoretic statements.

The calculus of variations highlights this distinction especially clearly. At a purely formal level, many variational computations do not require a Banach space. Gâteaux derivatives, Euler–Lagrange equations, first variations, and necessary optimality conditions rely only on local perturbations and algebraic structure. In this sense, the “calculus” part of the calculus of variations does not demand completeness.

Existence theory, however, is a different matter. The direct method of the calculus of variations relies on constructing a minimizing sequence, extracting a convergent subsequence, and passing to the limit in the functional. Each step implicitly requires a topology strong enough to guarantee convergence and a space complete enough to ensure that the limit remains admissible. Without completeness, the argument collapses at the second step.

This is why Sobolev spaces play a central role in variational problems. Spaces such as $W^{1,p}$, H^1 , and L^p are Banach (and often reflexive or Hilbert) spaces not by accident, but because variational limits naturally live there. Weak convergence, which replaces compactness in infinite dimensions, is meaningful only in complete spaces. The very notion of a weakly convergent minimizing sequence presupposes Banach structure.

More broadly, in calculus of variations the admissible space is not merely a container for functions; it encodes the regularity one expects the minimizer to possess. Banach spaces serve as a contract between the functional and its minimizers: they are precisely large enough to contain limits of minimizing sequences, yet structured enough to retain integrability, differentiability, and boundary conditions. Incomplete spaces break this contract.

Table 2.1 collects the dividing line in one place. The test is not the sophistication of the argument but its logical shape: an argument that compares finitely many objects needs only a norm, while an argument that manufactures a sequence and then speaks of its limit needs completeness.

2.2 ■ Extreme Values of Functionals and Compactness

Optimization theory is fundamentally concerned with the maximization or minimization of real-valued functionals over prescribed feasible sets. A central question, which precedes any algorithmic or analytical considerations, is the *existence* of an optimizer. In finite-dimensional spaces, existence is often guaranteed by classical results. In infinite-dimensional normed spaces,

Table 2.1: Normed spaces versus Banach spaces. Completeness is irrelevant to arguments that compare finitely many objects, and indispensable to arguments that construct a sequence and pass to its limit.

	NORMED SPACE SUFFICES	BANACH SPACE REQUIRED
<i>Shape of the argument</i>	compares finitely many points, hyperplanes, or inequalities	constructs a sequence, then invokes its limit
<i>Representative results</i>	convexity and norm inequalities; separation of convex sets and Hahn–Banach extension; epigraphs and subgradients; Gâteaux derivatives, first variations, and Euler–Lagrange equations; formal duality	Banach fixed point and Picard iteration; Newton–Kantorovich; value iteration; Neumann and resolvent series; the direct method of the calculus of variations; closed graph, open mapping, and uniform boundedness
<i>Without completeness</i>	nothing breaks: these statements are local	the limit exists only in the completion $\overline{X}$, so existence in X is never established

however, existence depends critically on topological properties of both the functional and the feasible set.

Existence, however, cannot be taken for granted, not even for smooth objectives on elementary feasible sets. Two examples make the point before we turn to the conditions that guarantee it.

Example 2.63 Unbounded objective.

Consider $\min_{x \in \mathbb{R}} x$. The objective is linear and continuous, but the feasible set $\mathbb{R}$ is unbounded below: for any candidate x there is a smaller $x' < x$ with $f(x') < f(x)$, so the objective decreases without bound. Formally $\inf_{x \in \mathbb{R}} x = -\infty$, and no minimizer exists. Boundedness of the feasible set is therefore essential for existence.

□

Example 2.64 Infimum not attained.

Consider $\min_{x \in \mathbb{R}} e^{-x}$. The objective is continuous and bounded below, since $e^{-x} > 0$ and $\lim_{x \rightarrow +\infty} e^{-x} = 0$, so $\inf_{x \in \mathbb{R}} e^{-x} = 0$. Yet no x satisfies $e^{-x} = 0$: the infimum is finite but not attained. Boundedness of the objective alone is thus insufficient for existence.

□

Remark 2.65 (Minimum versus infimum). These examples turn on the distinction between the *minimum*—a value $f(x^*)$ achieved at some $x^* \in C$ —and the *infimum*, the greatest lower bound of $\{f(x) : x \in C\}$, which need not be attained. Writing $\min_{x \in \mathbb{R}} e^{-x} = 0$ is therefore incorrect; the accurate statement is $\inf_{x \in \mathbb{R}} e^{-x} = 0$. Conflating the two is a common source of error in optimization analysis.

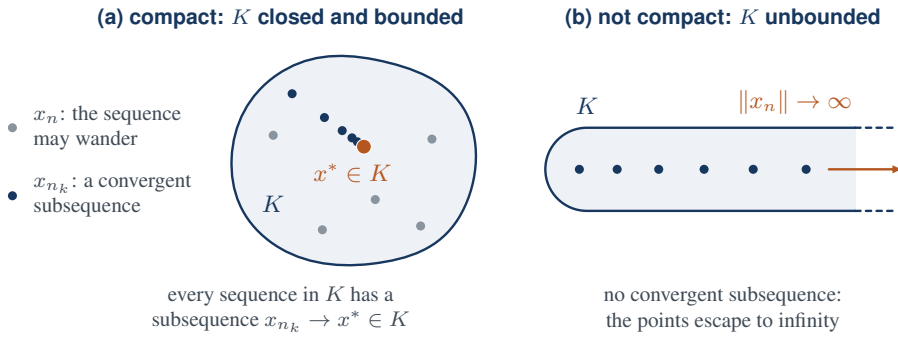


Figure 2.7: Compactness as a sequential property (Definition 2.67). (a) In a compact set K , a sequence $\{x_n\}$ may wander, yet it must contain a subsequence $\{x_{n_k}\}$ that converges to a limit x^* lying inside K . (b) An unbounded set is not compact: a sequence can run off to infinity, leaving no convergent subsequence. A set can also fail to be compact by not being closed, then the subsequential limit lands on an excluded boundary point, as in Figure 2.3. In finite dimensions, K is compact exactly when it is closed and bounded.

2.2.1 ■ Weierstrass Extreme Value Theorem

In $\mathbb{R}^n$, the classical Weierstrass Extreme Value Theorem states that a continuous function defined on a closed and bounded set attains both a maximum and a minimum. This result underpins much of finite-dimensional optimization theory, as feasibility and continuity alone suffice to ensure existence.

The goal of this section is to generalize this principle to normed spaces and to clarify why such generalizations are necessarily more restrictive in infinite-dimensional settings.

A key observation is that full continuity is not required to guarantee the existence of maximizers. Instead, a weaker condition, upper semicontinuity, is sufficient.

Definition 2.66 (Upper and lower semicontinuity). Let $(X, \|\cdot\|)$ be a normed space and let $f : X \rightarrow \mathbb{R}$.

- f is upper semicontinuous at $x_0 \in X$ if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|x - x_0\| < \delta \implies f(x) < f(x_0) + \varepsilon.$$

- f is lower semicontinuous at x_0 if $-f$ is upper semicontinuous at x_0 .

An equivalent characterization of upper semicontinuity is $\limsup_{x \rightarrow x_0} f(x) \leq f(x_0)$. If a functional is both upper and lower semicontinuous, then it is continuous.

Existence of extrema also requires structural properties of the feasible set.

Definition 2.67 (Compact set). A subset K of a normed space X is said to be compact if every sequence $\{x_n\} \subset K$ admits a subsequence that converges to an element of K .

Figure 2.7 illustrates this sequential notion: no matter how a sequence in K wanders, compactness guarantees a subsequence that settles onto a limit belonging to K . In finite-dimensional spaces, compactness is equivalent to being closed and bounded. In general normed spaces, however, closed and bounded sets need not be compact. Compact sets are necessarily closed and complete.

Example 2.68 Compactness on the real line.

The closed bounded interval $K = [0, 1]$ is compact. Any sequence $\{x_n\} \subset [0, 1]$ is bounded, so by the Bolzano–Weierstrass theorem it has a convergent subsequence $x_{n_k} \rightarrow x^*$; and because $[0, 1]$ is closed, the limit satisfies $x^* \in [0, 1]$. This is exactly the mechanism in Figure 2.7(a): the sequence may wander through the interval, but some subsequence must accumulate at a point of the set.

Dropping either hypothesis destroys compactness, in the two ways the figure distinguishes. The half-line $K = [0, \infty)$ is closed but *unbounded*: the sequence $x_n = n$ has $|x_n| \rightarrow \infty$, so no subsequence converges. This is the escape-to-infinity failure of Figure 2.7(b).

The half-open interval $K = (0, 1]$ is bounded but *not closed*: the sequence $x_n = 1/n$ lies in K and converges to 0, so every subsequence also converges to $0 \notin K$. The limit slips out through the missing endpoint, exactly the situation of Figure 2.3. In $\mathbb{R}^n$, keeping both closedness and boundedness is therefore necessary and sufficient: this is the Heine–Borel theorem. $\square$

Example 2.69 A closed and bounded set that is not compact.

In infinite-dimensional spaces the equivalence “compact = closed and bounded” breaks down. Let $X = \ell^2$ be the space of square-summable sequences with norm $\|x\|_2 = (\sum_k x_k^2)^{1/2}$, and let $B = \{x \in \ell^2 : \|x\|_2 \leq 1\}$ be its closed unit ball, which is both closed and bounded.

Consider the standard basis vectors $e_n = (0, \dots, 0, 1, 0, \dots)$, with the 1 in the n th position. Each e_n lies in B , yet $\|e_n - e_m\|_2 = \sqrt{2}$ whenever $n \neq m$. No subsequence of $\{e_n\}$ can be Cauchy, so none converges. Thus B contains a sequence with no convergent subsequence and is not compact, even though it is closed and bounded.

This is why existence results in infinite-dimensional optimization, such as those of the calculus of variations and optimal control, cannot rely on closedness and boundedness alone; they require compactness in a weaker topology or additional structure on the problem. $\square$

Theorem 2.70 (Weierstrass). *Let X be a normed space and let $K \subset X$ be compact. If $f : X \rightarrow \mathbb{R}$ is upper semicontinuous, then f attains a maximum on K .*

Proof. Let $M := \sup_{x \in K} f(x)$, allowing the possibility that $M = +\infty$. By definition of the supremum, there exists a sequence $\{x_n\} \subset K$ such that $f(x_n) \rightarrow M$. Since K is compact, there exists a subsequence $\{x_{n_k}\}$ converging to some $x \in K$.

Upper semicontinuity of f implies $f(x) \geq \limsup_{k \rightarrow \infty} f(x_{n_k}) = M$. Since $f(x) \leq M$ by definition of M , we conclude that $f(x) = M$ and that the maximum is attained at x . $\square$

In finite dimensions the hypotheses are mild, and existence can hold even where smoothness- or interior-based local analysis is inconclusive.

Example 2.71 Interior saddle, extrema on the boundary.

Let $f(x, y) = xy$ on $C = \{(x, y) : x^2 + y^2 \leq 1\}$. The function is smooth, and its only interior critical point is the origin, where the Hessian $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is indefinite; hence the origin is a saddle and there is no interior local minimizer or maximizer. Nonetheless f is continuous and C is compact, so Weierstrass guarantees global minimizers and maximizers, here $f = \pm \frac{1}{2}$, attained on the boundary at $\pm(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ and $\pm(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$: extrema may exist even when local analysis is inconclusive. $\square$

Example 2.72 Extreme values on a set with empty interior.

Let $f(x, y) = x^2 - y^2$ on the unit circle $C = \{(x, y) : x^2 + y^2 = 1\}$. The feasible set is compact but has empty interior. Despite the absence of admissible perturbations in arbitrary directions, f attains both a minimum and a maximum on C ; the theorem does not rely on interior-point arguments. $\square$

Example 2.73 A nowhere-differentiable objective.

Let f be the Takagi (*blancmange*) function on $C = [0, 1]$,

$$f(x) = \sum_{k=0}^{\infty} 2^{-k} \operatorname{dist}(2^k x, \mathbb{Z}),$$

where $\operatorname{dist}(s, \mathbb{Z})$ is the distance from s to the nearest integer. The series converges uniformly, so f is continuous; yet f is differentiable at *no* point of $[0, 1]$. Compactness of C nonetheless forces f to attain its extrema, so smoothness plays no role in the existence of extrema. $\square$

Example 2.74 Extrema from the topology of the constraint set.

Let $f(x, y) = x/\sqrt{x^2 + y^2}$ on the unit circle $C = \{(x, y) : x^2 + y^2 = 1\}$. The function cannot be extended continuously to the origin, but on C it reduces to $f = x$ and is continuous, so Weierstrass yields extrema that arise purely from the topology of the constraint set rather than from properties of the ambient space. $\square$

Dually, applying the theorem to $-f$ shows that a *lower* semicontinuous function attains its *minimum* on a compact set, the natural regularity condition for minimization problems.

Compactness is a strong topological property that plays a central role in existence results for optimization problems. In finite-dimensional spaces, closed and bounded sets are compact, which guarantees that continuous functions attain their extrema. In infinite-dimensional normed spaces, however, compactness is exceptional rather than typical: many natural feasible sets, including closed and bounded ones, fail to be compact. As a result, maximizing or minimizing a continuous

functional may lead only to a supremum or infimum that is not attained.

Example 2.75 Failure of attainment on the unit ball of $C[0, 1]$.

Let $C[0, 1]$ denote the space of continuous real-valued functions on $[0, 1]$, equipped with the supremum norm $\|x\|_\infty = \max_{t \in [0, 1]} |x(t)|$. Define a linear functional $f : C[0, 1] \rightarrow \mathbb{R}$ by

$$f(x) = \int_0^{1/2} x(t) dt - \int_{1/2}^1 x(t) dt.$$

The functional f is continuous, since

$$|f(x)| \leq \int_0^{1/2} |x(t)| dt + \int_{1/2}^1 |x(t)| dt \leq \|x\|_\infty,$$

which shows that f is bounded by the norm.

Consider the closed unit ball $B := \{x \in C[0, 1] : \|x\|_\infty \leq 1\}$. For any $x \in B$, we have $f(x) \leq 1$, so $\sup_{x \in B} f(x) = 1$.

However, this supremum is not attained by any function in B . Indeed, achieving $f(x) = 1$ would require

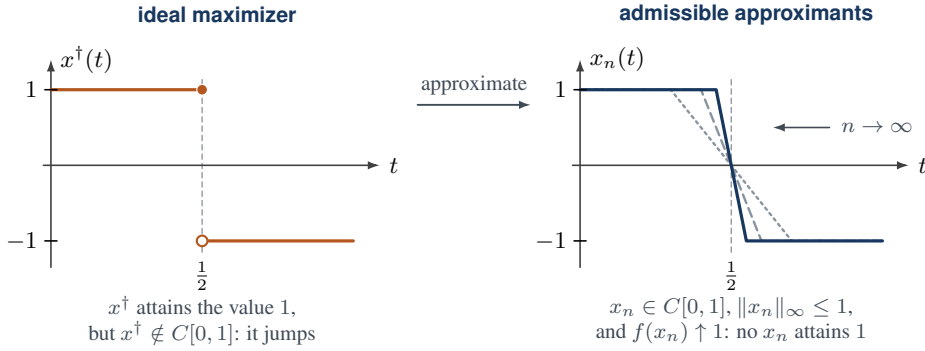
$$x(t) = 1 \quad \text{for } t \in [0, 1/2], \quad x(t) = -1 \quad \text{for } t \in (1/2, 1],$$

which corresponds to a discontinuous step function. No continuous function can exhibit this jump while maintaining $\|x\|_\infty \leq 1$. Although one can construct a sequence of continuous functions that approximate this step arbitrarily well, no admissible function attains the limiting value.

This example shows that the closed unit ball of $C[0, 1]$ is not compact. Even though the functional is continuous and the feasible set is closed and bounded, the absence of compactness prevents the existence of a maximizer. Figure 2.8 shows the mechanism: a sequence of admissible continuous functions steepens toward the ideal step maximizer, so the values $f(x_n)$ climb to the supremum 1, yet the limiting step function jumps and therefore escapes the space. $\square$

One way to recover existence of optimal solutions in infinite-dimensional settings is to restrict attention to a finite-dimensional subspace. In finite dimensions, closed and bounded sets are compact, and continuous functionals necessarily attain their extrema. This observation underlies many approximation and discretization methods in optimization and control.

Example 2.76 Polynomial subspace restriction.



The supremum is approached by continuous functions, but the limiting step function is not feasible.

Figure 2.8: Nonattainment on the closed unit ball of $C[0, 1]$: compactness fails because the natural maximizing limit leaves the space.

Consider the functional $f : C[0, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \int_0^{1/2} x(t) dt - \int_{1/2}^1 x(t) dt.$$

This functional is linear, since it is defined by integration against a fixed bounded measurable function, and it is continuous with respect to the supremum norm. Indeed, for any $x \in C[0, 1]$,

$$|f(x)| \leq \int_0^{1/2} |x(t)| dt + \int_{1/2}^1 |x(t)| dt \leq \|x\|_\infty,$$

so f is a bounded linear functional on $C[0, 1]$.

Let $B := \{x \in C[0, 1] : \|x\|_\infty \leq 1\}$ denote the closed unit ball of $C[0, 1]$ equipped with the supremum norm. We claim that f does not attain its maximum on B .

Formally, maximizing $f(x)$ under the pointwise constraint $|x(t)| \leq 1$ suggests choosing $x(t)$ to align with the sign of the integrand. This leads to the step function

$$x^\dagger(t) = \begin{cases} 1, & 0 \leq t < \frac{1}{2}, \\ -1, & \frac{1}{2} < t \leq 1, \end{cases}$$

with the value at $t = \frac{1}{2}$ arbitrary. For this function,

$$f(x^\dagger) = \int_0^{1/2} 1 dt - \int_{1/2}^1 (-1) dt = 1,$$

which is the supremum of f over B .

However, $x^\dagger$ has a jump discontinuity at $t = \frac{1}{2}$ and therefore does not belong to $C[0, 1]$. Consequently, $x^\dagger \notin B$, and the supremum value $\sup_{x \in B} f(x) = 1$ is not

attained by any function in B . Thus, although f is continuous, the unit ball of $C[0, 1]$ is not compact, and the extreme value theorem does not apply in this setting.

Now restrict the admissible set to the subspace $\mathcal{P}_n \subset C[0, 1]$ of polynomials of degree at most n . Every element of $\mathcal{P}_n$ can be written uniquely as $x(t) = a_0 + a_1t + \cdots + a_nt^n$, with coefficients $(a_0, \dots, a_n) \in \mathbb{R}^{n+1}$. The mapping from the coefficient vector to the corresponding polynomial is linear and bijective, so $\mathcal{P}_n$ is linearly isomorphic to $\mathbb{R}^{n+1}$. In particular, $\dim(\mathcal{P}_n) = n + 1 < \infty$.

Define the restricted feasible set $B_n := \{x \in \mathcal{P}_n : \|x\|_\infty \leq 1\}$. Since $\mathcal{P}_n$ is finite-dimensional, all norms on $\mathcal{P}_n$ are equivalent, and the set B_n is closed and bounded in $\mathcal{P}_n$. By the Heine–Borel theorem, B_n is compact.

Because f is continuous, the Weierstrass extreme value theorem guarantees the existence of a maximizer $x_n^* \in B_n$ such that $f(x_n^*) = \max_{x \in B_n} f(x)$.

As the degree n increases, the maximizing polynomials x_n^* can approximate the discontinuous step function that maximizes f over the full unit ball of $C[0, 1]$. However, for each fixed n , the restriction to $\mathcal{P}_n$ restores compactness and ensures existence of an optimal solution. $\square$

2.2.2 ■ Existence via Coercivity

The Weierstrass Extreme Value Theorem does not require the ambient space to be a Banach space. A normed vector space is sufficient. The decisive assumption is compactness of the feasible set, which guarantees the existence of convergent subsequences and ensures that limits remain feasible. Completeness of the ambient space plays no role, since every compact set is automatically complete.

When compactness is unavailable, existence must be recovered indirectly. This requires additional structure: coercivity to prevent minimizing sequences from escaping to infinity, lower semicontinuity to pass to limits, and completeness of the ambient space (typically Banach) to ensure that limiting points exist within the space. In this sense, Banach space assumptions become essential precisely when compactness is no longer assumed.

Definition 2.77 (Coercive functional). Let $(X, \|\cdot\|)$ be a normed space. A function $f : X \rightarrow \mathbb{R}$ is said to be coercive if

$$\|x\| \rightarrow \infty \implies f(x) \rightarrow +\infty.$$

Equivalently, all sublevel sets $\{x \in X : f(x) \leq c\}$, $c \in \mathbb{R}$, are bounded.

Coercivity ensures that any minimizing sequence must remain in a bounded region of the space. In finite dimensions, boundedness is sufficient to extract convergent subsequences. In infinite-dimensional spaces, however, additional structure is required to recover compactness.

In finite dimensions this structure is automatic: coercivity makes the sublevel sets compact, and Weierstrass then delivers a minimizer. Figure 2.9 contrasts a coercive objective, whose sublevel sets are bounded and whose minimizer is attained, with a non-coercive one whose infimum escapes to infinity.

Lemma 2.78 (Compactness of sublevel sets). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous and coercive. Then for every $c \in \mathbb{R}$ the sublevel set $L_c := \{x \in \mathbb{R}^n : f(x) \leq c\}$ is compact.

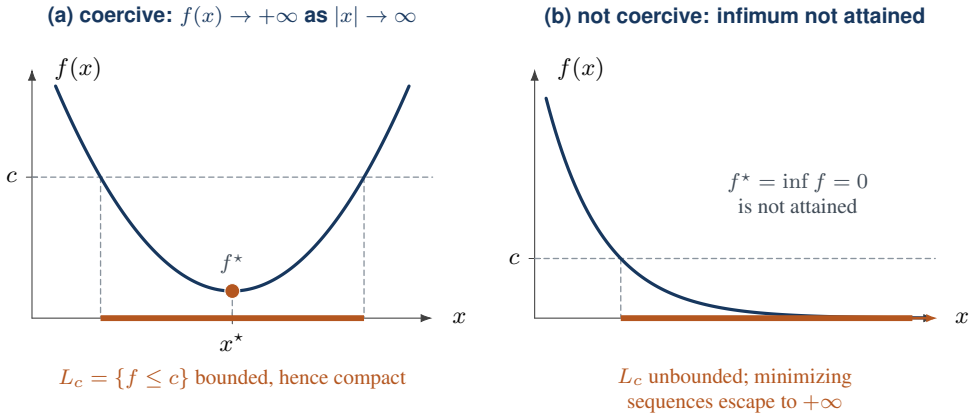


Figure 2.9: Coercivity as a growth condition guaranteeing existence. **(a)** A coercive objective grows to $+\infty$ in every direction, so each sublevel set $L_c = \{x : f(x) \leq c\}$ is bounded, hence compact, and a global minimizer x^* is attained. **(b)** $f(x) = e^{-x}$ fails to be coercive; its sublevel sets are unbounded rays, the minimizing sequence escapes to infinity, and the infimum $f^* = 0$ is never attained.

Proof. By coercivity there is $R > 0$ with $f(x) > c$ for all $|x| > R$, so $L_c \subseteq \overline{B}_R(0)$ is bounded. Continuity makes L_c closed, being the preimage of $(-\infty, c]$. In $\mathbb{R}^n$ closed and bounded sets are compact, so L_c is compact. $\square$

Theorem 2.79 (Existence for coercive functions). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous and coercive. Then $\min_{x \in \mathbb{R}^n} f(x)$ admits at least one minimizer.

Proof. Let $m := \inf_{x \in \mathbb{R}^n} f(x)$ and fix any $c > m$. The sublevel set L_c is nonempty and compact by Lemma 2.78, so by Weierstrass there is $x^* \in L_c$ with $f(x^*) = \min_{x \in L_c} f(x)$. Any $x \notin L_c$ satisfies $f(x) > c > m$ and cannot minimize f ; hence x^* is a global minimizer. $\square$

Theorem 2.80 (Existence via compact level sets). Let $D \subseteq \mathbb{R}^n$ and $f : D \rightarrow \mathbb{R}$ be continuous. If some sublevel set $L_c := \{x \in D : f(x) \leq c\}$ is nonempty and compact, then $\min_{x \in D} f(x)$ admits at least one minimizer.

Proof. Since L_c is compact and f continuous, Weierstrass gives a minimizer of f on L_c ; any minimizer over D lies in L_c , so the minimum is attained on D . $\square$

Example 2.81 Quadratic functions.

For $f(x) = x^\top Qx + b^\top x + c$ with Q symmetric positive definite, $x^\top Qx \geq \lambda_{\min}(Q)|x|^2$ with $\lambda_{\min}(Q) > 0$, so $f(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$ and f is coercive. By Theorem 2.79, f has a global minimizer on $\mathbb{R}^n$. $\square$

Example 2.82 Norm-based growth.

For $p > 1$, $f(x) = |x|^p$ is coercive. More generally, any f with $f(x) \geq \alpha|x|^p - \beta$ for some $\alpha > 0$ and $\beta \in \mathbb{R}$ is coercive. $\square$

Example 2.83 Coercivity without convexity.

Let $f(x) = |x|^2 + 10 \sin(|x|)$. Its radial profile $g(r) = r^2 + 10 \sin r$ has $g''(r) = 2 - 10 \sin r$, which changes sign, so f is nonconvex and has several local minima; nevertheless the quadratic term dominates as $|x| \rightarrow \infty$ (indeed $f(x) \geq |x|^2 - 10$), so f is coercive and admits a global minimizer. Coercivity is independent of convexity. $\square$

Example 2.84 Coercivity suppressed by a compact constraint.

Let $f(x, y) = e^x + y^4 - xy$ on $C = \{(x, y) : x^2 + y^2 \leq 1\}$. On $\mathbb{R}^2$ the function is neither coercive nor bounded below, but the compact constraint set removes every escape direction, so Weierstrass still yields extrema on C even though global coercivity fails. $\square$

Remark 2.85 (Compactness vs. completeness). The Weierstrass theorem does *not* require the ambient space X to be complete. The essential assumption is the compactness of the feasible set. Indeed, compactness guarantees that every sequence admits a convergent subsequence with limit in the feasible set, which is all that is needed in the proof.

In contrast, coercivity-based existence results do not assume compactness a priori. Instead, compactness must be recovered indirectly from boundedness, lower semicontinuity, and structural properties of the space. This is where Banach space assumptions become essential.

Theorem 2.86 (Existence via coercivity in Banach spaces). *Let X be a Banach space and let $f : X \rightarrow \mathbb{R}$ be coercive and weakly sequentially lower semicontinuous, that is, $f(x) \leq \liminf_{k \rightarrow \infty} f(x_k)$ whenever $x_k \rightharpoonup x$. Suppose that every closed and bounded subset of X is weakly sequentially compact. Then f attains a minimum on X .*

Remark 2.87. Weak sequential lower semicontinuity is genuinely stronger than lower semicontinuity in norm: a norm-lower-semicontinuous functional need not be lower semicontinuous along weakly convergent sequences, because weak limits are tested only against continuous linear functionals. The gap closes for *convex* functionals, whose norm-closed sublevel sets are convex and hence weakly closed; for them lower semicontinuity in norm already implies the weak form assumed here. This is why the direct method of the calculus of variations is usually stated for convex integrands.

Proof. Let $m := \inf_{x \in X} f(x)$. By definition of the infimum, there exists a minimizing sequence $\{x_n\} \subset X$ such that $f(x_n) \rightarrow m$.

We first show that the sequence $\{x_n\}$ is bounded. If this were not the case, then $\|x_n\| \rightarrow \infty$ along a subsequence. Since f is coercive, this would imply $f(x_n) \rightarrow +\infty$, contradicting the fact

that $f(x_n) \rightarrow m < \infty$. Hence, $\{x_n\}$ is bounded in X .

By assumption, every bounded sequence in X admits a weakly convergent subsequence. Therefore, there exists a subsequence $\{x_{n_k}\}$ and an element $x^* \in X$ such that $x_{n_k} \rightharpoonup x^*$. By definition of weak convergence, this means that $\varphi(x_{n_k}) \rightarrow \varphi(x^*)$ for every continuous linear functional $\varphi \in X^*$. In other words, although the subsequence $\{x_{n_k}\}$ need not converge in norm, it converges in the sense of all linear observations of the space.

We now invoke weak sequential lower semicontinuity of f , which gives exactly $f(x^*) \leq \liminf_{k \rightarrow \infty} f(x_{n_k})$ for the weakly convergent subsequence just extracted. Since $f(x_{n_k}) \rightarrow m$, we obtain $f(x^*) \leq m$.

On the other hand, by definition of m as the infimum of f over X , we always have $f(x^*) \geq m$. Combining the two inequalities yields $f(x^*) = m$. Thus, x^* is a minimizer of f , and the minimum of f over X is attained. $\square$

Theorem 2.86 illustrates a fundamental distinction between finite- and infinite-dimensional optimization. In finite dimensions, compactness follows automatically from closedness and boundedness. In infinite dimensions, compactness is typically absent and must be replaced by weaker forms of compactness, such as weak sequential compactness, together with coercivity and lower semicontinuity.

This result forms the foundation of the *direct method of the calculus of variations* and underlies existence results for a wide range of optimization, control, and partial differential equation problems. In practice, the assumption of weak sequential compactness is often ensured by working in reflexive Banach spaces, such as $L^p(\Omega)$ or Sobolev spaces $W^{1,p}(\Omega)$ with $1 < p < \infty$, or in Hilbert spaces.

2.3 ■ Gâteaux and Fréchet Differentials

Differentiability in infinite-dimensional spaces admits several distinct notions. The two most important for optimization and control are the *Gâteaux differential* and the *Fréchet differential*. They generalize directional derivatives and total derivatives, respectively, from finite-dimensional calculus.

Throughout this subsection, let X be a vector space, Y a normed space, and let $T : D \subset X \rightarrow Y$ be a (possibly nonlinear) transformation.

2.3.1 ■ Gâteaux differential.

The Gâteaux differential captures the rate of change of a transformation along a fixed direction.

Definition 2.88 (Gâteaux differential). Let $x \in D$ and $h \in X$ be such that $x + \alpha h \in D$ for all sufficiently small α . If the limit

$$\delta T(x; h) := \lim_{\alpha \rightarrow 0} \frac{T(x + \alpha h) - T(x)}{\alpha}$$

exists in Y , it is called the Gâteaux differential of T at x in the direction h . If this limit exists for every $h \in X$, then T is said to be Gâteaux differentiable at x .

For fixed x , the mapping $h \mapsto \delta T(x; h)$ defines a transformation from X to Y . If T is linear, then $\delta T(x; h) = T(h)$ for all x and h .

Gâteaux differentials of functionals. When $Y = \mathbb{R}$, the transformation T is a real-valued functional.

Definition 2.89. Let $f : X \rightarrow \mathbb{R}$. If the limit

$$\delta f(x; h) = \left. \frac{d}{d\alpha} f(x + \alpha h) \right|_{\alpha=0}$$

exists, it is called the Gâteaux differential of f at x in direction h .

Example 2.90 Finite-dimensional case.

Let $X = \mathbb{R}^n$ and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ have continuous partial derivatives. Then

$$\delta f(x; h) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x) h_i,$$

which coincides with the classical directional derivative. □

Example 2.91 Integral functionals.

Let $X = C[0, 1]$ and define $f(x) = \int_0^1 g(x(t), t) dt$, where g_x exists and is continuous in both arguments. Then

$$\delta f(x; h) = \int_0^1 g_x(x(t), t) h(t) dt.$$

□

The limitations of Gâteaux differentiability are best understood through concrete examples. These examples illustrate that differentiability along every direction does not guarantee stability, continuity, or a valid linear approximation in a neighborhood.

Example 2.92 Directional derivatives without continuity.

Let $X = \mathbb{R}^2$ and define

$$T(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

For a direction $h = (h_1, h_2)$, a direct computation gives

$$\frac{T(\alpha h) - T(0)}{\alpha} = \frac{h_1^2 h_2}{\alpha^2 h_1^4 + h_2^2} \xrightarrow{\alpha \rightarrow 0} \delta T(0; h) = \begin{cases} h_1^2/h_2, & h_2 \neq 0, \\ 0, & h_2 = 0. \end{cases}$$

Thus the directional derivative exists for *every* direction h , yet the map $h \mapsto \delta T(0; h)$ is *not linear* in h ; in particular T is not Gâteaux differentiable in the linear sense.

Moreover, T is *not continuous* at $(0, 0)$: along the parabola $y = x^2$, $T(x, x^2) = \frac{x^4}{2x^4} = \frac{1}{2}$, which does not converge to $T(0, 0) = 0$ as $(x, y) \rightarrow (0, 0)$. Thus, despite

having directional derivatives in every direction, T fails to behave continuously near the point.

Lesson: directional derivatives may exist in every direction without being linear in the direction, and without implying continuity. $\square$

Example 2.93 Direction-wise linearity without uniform approximation.

Let $X = \ell^2$, the space of square-summable sequences, and define $f(x) = \sum_{n=1}^{\infty} n x_n^2$. This series is finite only on the dense domain $D = \{x \in \ell^2 : \sum_n n x_n^2 < \infty\}$, on which we work.

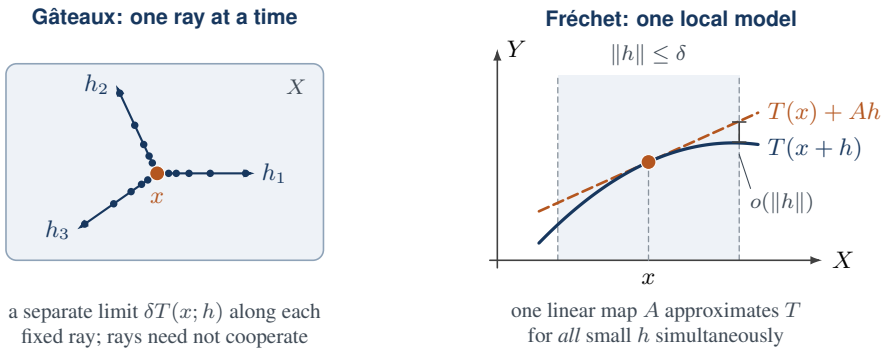
For fixed $x \in D$ and direction $h \in D$, the limit $\delta f(x; h) = \lim_{\alpha \rightarrow 0} \frac{f(x+\alpha h) - f(x)}{\alpha}$ exists and is given by $\delta f(x; h) = 2 \sum_{n=1}^{\infty} n x_n h_n$, so f has a directional derivative in every such direction. However, the map $h \mapsto \delta f(x; h)$ need not be a *continuous* linear functional: at a point $x \in D$ with $\sum_n n^2 x_n^2 = \infty$ (for instance $x_n = n^{-3/2}$), its coefficient sequence $(2n x_n)$ does not belong to ℓ^2 , so the functional is unbounded. Consequently, the first-order approximation $f(x+h) \approx f(x) + \delta f(x; h)$ is not uniformly valid for small $\|h\|$.

Lesson: Directional derivatives exist, but the linear approximation is unstable. $\square$

The fundamental limitation of Gâteaux differentiability is that it probes the behavior of a transformation only along individual directions. Although the directional limit $\lim_{\alpha \rightarrow 0} \frac{T(x+\alpha h) - T(x)}{\alpha}$ may exist for every fixed direction h , the resulting linear approximations need not be compatible across different directions. In particular, the quality of the approximation may depend on the direction in which h approaches zero, and no control is imposed on how these directional behaviors interact.

From the standpoint of local approximation, this means that Gâteaux differentiability does not ensure the existence of a single linear operator that provides a reliable first-order model of T in a neighborhood of x . The transformation may admit directional derivatives in all directions while still failing to behave approximately linearly when perturbations approach zero along more general paths. Consequently, Gâteaux differentiability alone does not imply continuity of T , nor does it guarantee stability of the linearization.

The Fréchet differential addresses this deficiency by enforcing a uniform first-order approximation with respect to all sufficiently small perturbations. Rather than fixing a direction and shrinking the step size, Fréchet differentiability requires that a single bounded linear operator approximate the increment $T(x+h) - T(x)$ for all perturbations h with small norm. This uniform control ensures that the linear approximation is stable, direction-independent, and valid throughout an entire neighborhood of the base point. Figure 2.10 contrasts the two notions: the Gâteaux differential inspects the function one ray at a time and lets the answer depend on the ray, whereas the Fréchet differential commits to a single linear model that must fit every small perturbation at once.



Gâteaux differentiability checks rays one at a time; Fréchet differentiability demands a single first-order model that is accurate throughout a whole neighborhood, and therefore yields a genuine local linear approximation.

Figure 2.10: Gâteaux differentiability probes one direction at a time, while Fréchet differentiability supplies a stable first-order model for all small perturbations.

2.3.2 ■ Fréchet Differential

Fréchet differentiability captures the appropriate notion of total differentiability in normed spaces and provides the correct foundation for continuity, sensitivity analysis, and local linear modeling in infinite-dimensional settings.

Definition 2.94 (Fréchet differential). Let X and Y be normed spaces and let $D \subset X$ be open. The transformation $T : D \rightarrow Y$ is said to be Fréchet differentiable at $x \in D$ if there exists a bounded linear operator $A : X \rightarrow Y$ such that

$$\lim_{\|h\| \rightarrow 0} \frac{\|T(x+h) - T(x) - Ah\|_Y}{\|h\|_X} = 0.$$

The operator A is called the Fréchet differential of T at x and is denoted by $\delta T(x)$.

The defining limit requires that the error of the linear approximation $T(x+h) \approx T(x) + Ah$ be negligible compared to $\|h\|$ as $h \rightarrow 0$, uniformly over all directions. The boundedness of A ensures that the linear approximation itself depends continuously on the perturbation.

Remark 2.95 (Interpretation). Fréchet differentiability asserts that near x , the transformation T behaves like an affine mapping whose linear part is A . In contrast to directional derivatives, the approximation must hold simultaneously for all small perturbations, not merely along fixed rays.

Example 2.96 Finite-dimensional case.

Let $X = \mathbb{R}^n$ and $Y = \mathbb{R}^m$, and let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be continuously differentiable in the classical sense. Then T is Fréchet differentiable at every x , and $\delta T(x)h = J_T(x)h$, where $J_T(x)$ is the Jacobian matrix of T at x . Thus, Fréchet differentiability reduces to ordinary total differentiability in finite-dimensional spaces. $\square$

Example 2.97 Integral operator.

Let $X = Y = C[a, b]$ equipped with the supremum norm, and let $k : [a, b] \times [a, b] \rightarrow \mathbb{R}$ be continuous. Define $(Tx)(t) = \int_a^b k(t, \tau)x(\tau) d\tau$. Then T is Fréchet differentiable everywhere with $\delta T(x)h = \int_a^b k(t, \tau)h(\tau) d\tau$. In this case, T is linear, and the Fréchet differential coincides with T itself. $\square$

Proposition 2.98 (Uniqueness). *If the Fréchet differential of T at x exists, it is unique.*

Proof. Suppose A and B both satisfy the definition. Then for all h ,

$$\|Ah - Bh\| \leq \|T(x+h) - T(x) - Ah\| + \|T(x+h) - T(x) - Bh\| = o(\|h\|).$$

Dividing by $\|h\|$ and letting $\|h\| \rightarrow 0$ yields $\frac{\|(A-B)h\|}{\|h\|} \rightarrow 0$. Since $A - B$ is linear and bounded, this implies $A = B$. $\square$

Proposition 2.99. *If T is Fréchet differentiable at x , then T is Gâteaux differentiable at x , and $\delta T(x; h) = \delta T(x)h$, $\forall h \in X$.*

Proof. Let $A = \delta T(x)$. For fixed h ,

$$\frac{T(x + \alpha h) - T(x)}{\alpha} = Ah + \frac{o(\alpha\|h\|)}{\alpha}.$$

Letting $\alpha \rightarrow 0$ yields the existence of the directional limit and the stated expression. $\square$

This result shows that Fréchet differentiability is a strictly stronger notion: it implies directional differentiability in every direction, with a derivative that depends linearly on h .

Proposition 2.100. *If T is Fréchet differentiable at x , then T is continuous at x .*

Proof. From the definition,

$$\|T(x+h) - T(x)\| \leq \|\delta T(x)h\| + o(\|h\|) \leq C\|h\| + o(\|h\|),$$

where C is the operator norm of $\delta T(x)$. The right-hand side tends to zero as $\|h\| \rightarrow 0$, proving continuity at x . $\square$

Remark 2.101 (Why this matters for optimization). Fréchet differentiability guarantees the existence of a stable linear model that controls first-order variations of T under all small perturbations. This property is essential for sensitivity analysis, Newton-type methods, and the derivation of first-order optimality conditions in infinite-dimensional optimization problems.

2.3.3 ■ Gradient and Hessian

Fréchet differentiability provides the appropriate notion of a first-order linear approximation in general normed spaces. For real-valued functionals, this framework unifies the familiar gradient in $\mathbb{R}^n$ with more abstract notions of sensitivity in infinite-dimensional spaces, and it naturally leads to second-order objects such as the Hessian when higher-order differentiability holds.

Throughout this section, let $(X, \|\cdot\|)$ be a real normed space.

Definition 2.102 (Bounded linear functional). A mapping $\ell : X \rightarrow \mathbb{R}$ is called a bounded linear functional if it is linear and there exists a constant $C > 0$ such that $|\ell(x)| \leq C\|x\|$, $\forall x \in X$.

Boundedness is equivalent to continuity for linear mappings between normed spaces. Intuitively, a bounded linear functional produces outputs that vary continuously and proportionally with small changes in the input.

Definition 2.103 (Dual space). The set of all bounded linear functionals on X is called the dual space of X and is denoted by X^* .

Elements of the dual space act on vectors in X by producing real numbers. They encode linear measurements, sensitivities, and directional responses, and play a central role in optimization, variational analysis, and optimal control.

Example 2.104 Euclidean spaces.

Let $X = \mathbb{R}^n$ equipped with any norm. For any fixed vector $a \in \mathbb{R}^n$, the mapping $\ell(x) = a^\top x$ is a bounded linear functional. By the Cauchy–Schwarz inequality, $|\ell(x)| = |a^\top x| \leq \|a\|_2 \|x\|_2$, so $\ell \in (\mathbb{R}^n)^*$. Conversely, every bounded linear functional on $\mathbb{R}^n$ can be written uniquely in this form. Hence, $(\mathbb{R}^n)^* \cong \mathbb{R}^n$, and the dual space can be identified with the space itself. $\square$

This identification underlies the classical notion of the gradient: the gradient vector represents the unique linear functional that best approximates a scalar function at first order.

Example 2.105 Coordinate projections.

On $X = \mathbb{R}^n$, the coordinate projection $\ell_i(x) = x_i$ is a bounded linear functional. The collection $\{\ell_1, \dots, \ell_n\}$ forms a basis of $(\mathbb{R}^n)^*$ dual to the standard basis of $\mathbb{R}^n$. Gradients and Jacobians are naturally expressed in this dual basis. $\square$

Example 2.106 Integral functionals on continuous functions.

Let $X = C[a, b]$ be the space of real-valued continuous functions on $[a, b]$, equipped with the supremum norm $\|x\|_\infty = \max_{t \in [a, b]} |x(t)|$. For any fixed continuous function $w : [a, b] \rightarrow \mathbb{R}$, define the mapping $\ell(x) = \int_a^b w(t) x(t) dt$. Then ℓ is linear. Moreover, it is bounded, since

$$|\ell(x)| \leq \int_a^b |w(t)| |x(t)| dt \leq (b - a) \|w\|_\infty \|x\|_\infty.$$

Hence ℓ is a bounded linear functional on X , and therefore $\ell \in X^*$.

This example shows that every continuous weighting function w induces an element

of the dual space via integration. Such functionals naturally arise as first variations in calculus of variations, optimal control, and sensitivity analysis, where w plays the role of a costate or adjoint variable. $\square$

The preceding example might suggest that the dual space X^* consists precisely of integral functionals with continuous weights. While these form an important subclass, they do *not* exhaust the entire dual space.

Example 2.107 Evaluation functionals.

Again let $X = C[a, b]$. For any fixed $t_0 \in [a, b]$, the evaluation mapping $\ell(x) = x(t_0)$ is linear and bounded, since $|\ell(x)| = |x(t_0)| \leq \|x\|_\infty$. Thus $\ell \in X^*$.

This example demonstrates that the dual space contains functionals that extract local, pointwise information about functions. In particular, the evaluation functional cannot be represented as an integral against a continuous weight function, highlighting that X^* is strictly larger than the class of integral functionals defined above. $\square$

The full dual space $X^* = C[a, b]^*$ admits a complete characterization via the *Riesz–Markov representation theorem*. In its classical form, the theorem states the following.

Theorem 2.108 (Riesz–Markov representation theorem). *Let K be a compact Hausdorff space, and let $C(K)$ denote the space of real-valued continuous functions on K equipped with the supremum norm. For every bounded linear functional $\ell \in C(K)^*$, there exists a unique finite signed regular Borel measure μ on K such that $\ell(x) = \int_K x \, d\mu$, $\forall x \in C(K)$. Moreover, this correspondence is isometric, in the sense that $\|\ell\| = |\mu|(K)$, where $|\mu|$ denotes the total variation measure of μ .*

Specializing to $K = [a, b]$, every bounded linear functional $\ell \in C[a, b]^*$ admits a unique representation of the form $\ell(x) = \int_a^b x(t) \, d\mu(t)$, $\forall x \in C[a, b]$, for some finite signed Borel measure μ on $[a, b]$. Under this representation, integral functionals with continuous weights correspond to measures that are absolutely continuous with respect to Lebesgue measure, while evaluation functionals correspond to Dirac measures concentrated at individual points. More generally, arbitrary elements of X^* may involve both continuous and singular components.

Conceptually, the Riesz–Markov theorem reveals that bounded linear functionals on $C[a, b]$ act not merely by pointwise evaluation or smooth averaging, but by aggregating function values against generalized weights in the form of measures. Thus, the dual space X^* can be identified isometrically with the space of finite signed Borel measures on $[a, b]$, providing a precise geometric and measure-theoretic description of all continuous linear sensitivities on $C[a, b]$.

Example 2.109 Mixed interior–boundary functional.

Let $X = C[a, b]$ with the supremum norm. Fix a continuous weight $w \in C[a, b]$ and

constants $\alpha, \beta \in \mathbb{R}$. Define

$$\ell(x) = \int_a^b w(t) x(t) dt + \alpha x(a) + \beta x(b).$$

This mapping is linear. Moreover,

$$|\ell(x)| \leq ((b-a)\|w\|_\infty + |\alpha| + |\beta|) \|x\|_\infty,$$

so ℓ is bounded and hence $\ell \in X^*$.

By the Riesz–Markov representation theorem, there exists a unique finite signed Borel measure μ on $[a, b]$ such that $\ell(x) = \int_a^b x(t) d\mu(t)$, $\forall x \in C[a, b]$. In this case, the representing measure is given explicitly by $\mu = w(t) dt + \alpha \delta_a + \beta \delta_b$, where $w(t) dt$ denotes the absolutely continuous part with respect to Lebesgue measure, and δ_a, δ_b are Dirac measures concentrated at the endpoints a and b .

Thus, μ decomposes into an interior component capturing distributed sensitivity across the interval and atomic components capturing boundary sensitivity. Such mixed measures arise naturally as first-order sensitivity objects in variational problems with endpoint penalties, optimal control problems with terminal costs, and constrained optimization with boundary terms. $\square$

Remark 2.110. In infinite-dimensional spaces, not every linear functional is bounded. Boundedness is therefore essential: it guarantees continuity and ensures that elements of the dual space behave well under limits. This distinction does not arise in finite-dimensional spaces, where all linear functionals are automatically bounded and the dual space has the same dimension as the primal space.

The first Fréchet differential as a dual element

Let $f : D \subset X \rightarrow \mathbb{R}$ be a real-valued functional defined on an open set D . Recall that the dual space X^* is defined as the set of all *bounded linear functionals* from X to $\mathbb{R}$.

If f is Fréchet differentiable at a point $x \in D$, then by definition there exists a linear mapping $A : X \rightarrow \mathbb{R}$ such that

$$\lim_{\|h\| \rightarrow 0} \frac{|f(x+h) - f(x) - A(h)|}{\|h\|} = 0. \quad (2.3)$$

The limit condition (2.3) implies that the remainder $f(x+h) - f(x) - A(h)$ is of smaller order than $\|h\|$ as $h \rightarrow 0$.

In particular, there exists $\varepsilon > 0$ and a constant $C > 0$ such that $|A(h)| \leq C\|h\|$, for all $\|h\| < \varepsilon$. Since A is linear, this local bound extends to all of X , which shows that A is bounded. Therefore, by definition of the dual space, $A \in X^*$.

The functional A is unique and is called the *Fréchet differential* of f at x , denoted by $\delta f(x)$. Equivalently, $\delta f(x) \in X^*$ and $f(x+h) = f(x) + \delta f(x)(h) + o(\|h\|)$ as $h \rightarrow 0$.

Thus, the first Fréchet differential is itself an element of the dual space: it assigns to each direction $h \in X$ the first-order sensitivity of f at x along h through a bounded linear functional.

In finite-dimensional spaces, identifying X^* with X allows one to represent $\delta f(x)$ as a gradient vector. In infinite-dimensional spaces, $\delta f(x)$ should be interpreted directly as a bounded linear functional, without necessarily admitting a vector representation.

Gradient in $\mathbb{R}^n$ as the Fréchet differential.

Now specialize to $X = \mathbb{R}^n$ equipped with the standard inner product $\langle u, v \rangle = u^\top v$ and the Euclidean norm $\|\cdot\|_2$. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be Fréchet differentiable at x . By definition, there exists a *bounded linear functional* $Df(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f(x+h) = f(x) + Df(x)(h) + o(\|h\|_2), \quad h \rightarrow 0. \quad (2.4)$$

Thus $Df(x)$ denotes the Fréchet differential at x , while $Df(x)(h)$ denotes its action on the increment h .

To obtain the gradient, we show that $Df(x)$ admits a unique coordinate representation.

Lemma 2.111 (Representation of linear functionals on $\mathbb{R}^n$). *Let $\ell : \mathbb{R}^n \rightarrow \mathbb{R}$ be a linear functional. Define a vector $g \in \mathbb{R}^n$ by $g_i := \ell(e_i)$, $i = 1, \dots, n$, where $\{e_1, \dots, e_n\}$ is the standard basis of $\mathbb{R}^n$. Then $\ell(h) = g^\top h$, $\forall h \in \mathbb{R}^n$. The vector g is unique.*

Proof. Any $h \in \mathbb{R}^n$ can be written as $h = \sum_{i=1}^n h_i e_i$. By linearity,

$$\ell(h) = \ell\left(\sum_{i=1}^n h_i e_i\right) = \sum_{i=1}^n h_i \ell(e_i) = \sum_{i=1}^n h_i g_i = g^\top h.$$

Uniqueness follows by evaluating at $h = e_i$. $\square$

Theorem 2.112 (Gradient as the coordinate representation of the Fréchet differential). *If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is Fréchet differentiable at x , then there exists a unique vector $\nabla f(x) \in \mathbb{R}^n$ such that*

$$Df(x)(h) = \nabla f(x)^\top h, \quad \forall h \in \mathbb{R}^n. \quad (2.5)$$

Consequently,

$$f(x+h) = f(x) + \nabla f(x)^\top h + o(\|h\|_2), \quad h \rightarrow 0.$$

Proof. Since f is Fréchet differentiable at x , the operator $Df(x)$ exists and is a bounded linear functional on $\mathbb{R}^n$. Applying Lemma 2.111 to $\ell = Df(x)$ yields a unique vector $g \in \mathbb{R}^n$ such that $Df(x)(h) = g^\top h \forall h$. Define $\nabla f(x) := g$. Substituting this representation into (2.4) gives the claimed expansion. $\square$

Remark 2.113 (Connection to partial derivatives). Fix $i \in \{1, \dots, n\}$ and consider increments of the form $h = \alpha e_i$. Dividing (2.4) by α and letting $\alpha \rightarrow 0$ yields $\frac{\partial f}{\partial x_i}(x) = Df(x)(e_i)$, and using (2.5) this equals $Df(x)(e_i) = \nabla f(x)^\top e_i = (\nabla f(x))_i$. Hence

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix}.$$

Moreover, for any direction h ,

$$\delta f(x; h) = \lim_{\alpha \rightarrow 0} \frac{f(x + \alpha h) - f(x)}{\alpha} = Df(x)(h) = \nabla f(x)^\top h,$$

recovering the classical directional derivative formula.

Second Fréchet differential and the Hessian in $\mathbb{R}^n$.

Assume now that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is Fréchet differentiable in a neighborhood of x and that the mapping $x \mapsto Df(x) \in (\mathbb{R}^n)^*$ is itself Fréchet differentiable at x . By definition, there exists a bounded linear operator $D^2f(x) : \mathbb{R}^n \rightarrow (\mathbb{R}^n)^*$ such that

$$Df(x+h) - Df(x) = D^2f(x)(h) + o(\|h\|_2), \quad h \rightarrow 0. \quad (2.6)$$

The operator $D^2f(x)$ is called the *second Fréchet differential* of f at x .

Second differential as a bilinear form. For any $u, v \in \mathbb{R}^n$, define

$$\delta^2 f(x)(u, v) := (D^2f(x)(u))(v). \quad (2.7)$$

Since $D^2f(x)$ is linear in u and $D^2f(x)(u)$ is a linear functional acting on v , the mapping $\delta^2 f(x) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is bilinear and bounded.

Thus, the second Fréchet differential defines a bounded bilinear form that captures the second-order variation of f .

In finite-dimensional Euclidean space, every bounded bilinear form admits a matrix representation. Therefore, there exists a unique matrix $\nabla^2 f(x) \in \mathbb{R}^{n \times n}$ such that

$$\delta^2 f(x)(u, v) = u^\top \nabla^2 f(x) v, \quad \forall u, v \in \mathbb{R}^n. \quad (2.8)$$

The matrix $\nabla^2 f(x)$ is called the *Hessian* of f at x .

Remark 2.114 (Connection to second partial derivatives). If f is twice continuously differentiable in a neighborhood of x , then the entries of the Hessian are given by

$$(\nabla^2 f(x))_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(x), \quad i, j = 1, \dots, n.$$

In this case, $\nabla^2 f(x)$ is symmetric.

If f is twice Fréchet differentiable at x , then for small $h \in \mathbb{R}^n$ we have the second-order expansion

$$f(x+h) = f(x) + Df(x)(h) + \frac{1}{2} \delta^2 f(x)(h, h) + o(\|h\|_2^2), \quad h \rightarrow 0. \quad (2.9)$$

Substituting the gradient and Hessian representations yields the familiar form

$$f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top \nabla^2 f(x) h + o(\|h\|_2^2).$$

The gradient $\nabla f(x)$ describes the first-order sensitivity of the objective and determines descent directions, while the Hessian $\nabla^2 f(x)$ captures local curvature. Second-order optimality conditions and Newton-type methods are direct consequences of this Fréchet-based expansion.

Fréchet expansions and their relation to Taylor series

The notion of Fréchet differentiability leads naturally to a general, vector-space formulation of Taylor expansion in normed spaces.

Definition 2.115 (Fréchet–Taylor expansion of order k). Let $(X, \|\cdot\|)$ be a normed space, let $D \subset X$ be open, and let $f : D \rightarrow \mathbb{R}$. Suppose that f is k times Fréchet differentiable at a point $x \in D$. Then there exist mappings $\delta^j f(x) : X^j \rightarrow \mathbb{R}$, $j = 1, \dots, k$, with the following properties:

- (1) Each $\delta^j f(x)$ is a bounded, symmetric, j -linear form.
- (2) $\delta f(x) = \delta^1 f(x)$ coincides with the first Fréchet differential and belongs to the dual space X^* .
- (3) For $j \geq 2$, $\delta^j f(x)$ is induced by the j th Fréchet differential via repeated evaluation.

Moreover, for all $h \in X$ sufficiently small,

$$f(x+h) = f(x) + \sum_{j=1}^k \frac{1}{j!} \delta^j f(x)(h, \dots, h) + o(\|h\|^k), \quad h \rightarrow 0. \quad (2.10)$$

The expression (2.10) is a polynomial approximation of degree k in the increment h . Each coefficient is an intrinsic object determined by the local behavior of f at x and depends only on the normed space structure of X . The remainder term $o(\|h\|^k)$ means that

$$\lim_{\|h\| \rightarrow 0} \frac{\left| f(x+h) - f(x) - \sum_{j=1}^k \frac{1}{j!} \delta^j f(x)(h, \dots, h) \right|}{\|h\|^k} = 0.$$

The first-order term $\delta f(x) = \delta^1 f(x)$ is a bounded linear functional on X and therefore belongs to the dual space X^* . This object captures the leading-order sensitivity of f to perturbations in X . Higher-order terms $\delta^j f(x)$, $j \geq 2$, are bounded multilinear forms on X , representing successive derivatives of f in a vector space.

The Fréchet–Taylor expansion is intrinsic and applies in finite- and infinite-dimensional settings. When $X = \mathbb{R}^n$, every bounded multilinear form admits a coordinate representation. Under the canonical identification $(\mathbb{R}^n)^* \cong \mathbb{R}^n$, the first Fréchet differential corresponds to the gradient $\nabla f(x)$, and the second Fréchet differential corresponds to the Hessian $\nabla^2 f(x)$. In this case, (2.10) reduces to the classical Taylor expansion

$$f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top \nabla^2 f(x) h + \dots + o(\|h\|^k).$$

2.4 ■ Local Optimality Conditions in Finite-Dimensional Optimization

This section reviews core concepts from finite-dimensional optimization that serve as a template for the calculus of variations and optimal control. In those settings, the decision variable is a *function* rather than a vector, but the underlying logic is the same: one studies local optimality by examining the effect of admissible perturbations, first to first order and then to second order. These ideas reappear in the derivation of Euler–Lagrange equations, Pontryagin’s Maximum Principle, and Hamilton–Jacobi–Bellman theory.

Let $D \subseteq \mathbb{R}^n$ be nonempty and let $f : D \rightarrow \mathbb{R}$. We study local optimality when the feasible set D is an arbitrary subset of $\mathbb{R}^n$. Unlike the unconstrained case, perturbations of a candidate solution must remain inside D , motivating feasible directions and their cone closures.

For $x \in \mathbb{R}^n$ and $r > 0$, define the open ball $B_r(x) := \{y \in \mathbb{R}^n : \|y - x\| < r\}$. The *interior* of D is $\text{int}(D) := \{x \in D : \exists r > 0 \text{ such that } B_r(x) \subseteq D\}$.

We first fix terminology for optimality over the feasible set.

Definition 2.116 (Optimal value). The scalar $f^* := \inf_{x \in D} f(x)$ is the optimal value of the problem, the smallest value the objective can achieve over D , whether or not it is attained.

Definition 2.117 (Global minimizer). A point $x^* \in D$ is a global minimizer (or optimal point) if $f(x^*) = f^*$, in which case the optimal value is attained. The set of all global minimizers, $X_{\text{opt}} := \{x \in D : f(x) = f^*\}$, may be a single point, several points, or empty when the infimum is not attained.

Definition 2.118 (Local minimizer over a set). A point $x^* \in D$ is a local minimizer of f over D if there exists $r > 0$ such that

$$f(x^*) \leq f(x), \quad \forall x \in D \cap B_r(x^*). \quad (2.11)$$

Definition 2.119 (Isolated local minimizer). A local minimizer $x^* \in D$ is isolated if there exists $r > 0$ such that x^* is the only local minimizer of f in $D \cap B_r(x^*)$.

A local minimizer is a point that cannot be improved by any sufficiently small feasible perturbation, even though lower values of f may exist elsewhere in D . Every global minimizer is a local minimizer, but the converse fails in general: nonconvex objectives may exhibit many valleys of varying depth separated by ridges or flat regions, and this multiplicity of locally optimal solutions is a primary source of difficulty in nonconvex optimization. Isolated minimizers, by contrast, are separated from competing minima by a neighborhood and arise when f has strict curvature near x^* ; they play a central role in stability analysis and in the convergence of numerical algorithms. When both D and f are convex, every local minimizer is global, a fundamental property underlying the tractability of convex optimization [130, 26, 106].

2.4.1 ■ Feasible directions and first-order geometry

To develop intuition for constrained optimization, it is useful to begin with the idea of *feasible directions*. Informally, a direction at a point $x \in D$ is feasible if one can move a small distance from x along that direction without immediately leaving the feasible set D . Such directions represent admissible infinitesimal motions that respect the constraints.

A simple way to formalize this intuition is the following. A vector $d \in \mathbb{R}^n$ is called a feasible direction for D at x if there exists $\varepsilon_0 > 0$ such that $x + \varepsilon d \in D$, for all $\varepsilon \in (0, \varepsilon_0)$. We denote the set of such directions by $\mathcal{F}_D(x)$.

Feasibility is a local notion: only arbitrarily small displacements matter. The set $\mathcal{F}_D(x)$ is always a cone, since $d \in \mathcal{F}_D(x)$ implies $\lambda d \in \mathcal{F}_D(x)$ for all $\lambda > 0$. If $x \in \text{int}(D)$, then every direction is feasible and $\mathcal{F}_D(x) = \mathbb{R}^n$. At boundary points, however, $\mathcal{F}_D(x)$ is typically a strict subset of $\mathbb{R}^n$.

While this pointwise notion captures the basic idea of feasibility, it can be too restrictive at nonsmooth boundary points. At such points, the set of directions that are feasible in the strict sense above may be smaller than the set of geometrically admissible first-order directions. In particular, there may exist directions that cannot be followed exactly while remaining in D , but that arise as limits of feasible displacements along nearby directions. Conversely, nonsmooth boundary points may admit many admissible first-order directions, forming a tangent cone with a richer geometric structure than suggested by pointwise feasibility alone. Such directions are relevant for first-order optimality conditions and must be taken into account.

To capture all admissible first-order motions, one is therefore led to the concept of the *tangent cone*, which consists of all directions that can be approximated arbitrarily well by feasible displacements. We will introduce the tangent cone formally later and show that it provides the correct geometric object for first-order analysis, especially in the presence of nonsmooth constraints.

With this intuition in mind, we now state the first-order necessary condition. Let $f : D \rightarrow \mathbb{R}$ be differentiable at a point $x^* \in D$. If x^* is a local minimizer of f over D , then the directional derivative of f at x^* must be nonnegative along every direction that remains feasible to first order. In particular, for every feasible direction $d \in \mathcal{F}_D(x^*)$, one has $\nabla f(x^*)^\top d \geq 0$.

Geometrically, this condition states that the gradient $\nabla f(x^*)$ forms a *non-obtuse* angle with all feasible directions. At interior points, where all directions are feasible, this forces $\nabla f(x^*) = 0$. At boundary points, the gradient need not vanish, but it cannot point into the feasible set in a way that would locally decrease the objective.

In its most general and precise form, this condition is stated using the tangent cone rather than the pointwise feasible-direction set. The tangent-cone formulation subsumes the above condition and remains valid even at nonsmooth boundaries. We will return to this refined geometric viewpoint and present a fully rigorous treatment in a later subsection.

2.4.2 ■ Tangent and normal cones

When the feasible set D is not open, not every infinitesimal perturbation of a point $x \in D$ is admissible. In constrained optimization, the key question is therefore not *how f behaves in all directions*, but rather how it behaves along directions that remain feasible to first order. Tangent and normal cones provide a precise geometric language for this idea.

In first-order analysis, only the *direction* of a perturbation matters, not its magnitude. This motivates the following definition.

Definition 2.120 (Cone). A set $C \subset \mathbb{R}^n$ is called a cone if it is closed under nonnegative scalar multiplication, that is, $d \in C \implies \lambda d \in C$, $\forall \lambda \geq 0$.

Equivalently, if a vector d belongs to C , then every vector pointing in the same direction as d also belongs to C , regardless of scale.

Cones encode *admissible directions* rather than feasible displacements. Indeed, in first-order optimality analysis we are concerned with whether it is possible to move away from a point in a given direction while remaining feasible, not with how far we move. If a direction is admissible for an infinitesimal step, then the same direction remains admissible for any sufficiently small step size. This invariance under scaling is precisely the defining property of a cone.

Remark 2.121. A cone need not be convex, closed, or symmetric. In particular, if $d \in C$, it does not necessarily follow that $-d \in C$. This asymmetry reflects the fact that constraints may restrict motion in one direction but not in the opposite one.

Example 2.122 Cones.

Several familiar sets are cones. The whole space $\mathbb{R}^n$ is a cone, and so is any linear subspace. The half-space $\{d \in \mathbb{R}^n : d_1 \geq 0\}$ is a cone but not a subspace, since it is not closed under negation. By contrast, the unit ball $\{d : \|d\| \leq 1\}$ is *not* a cone, because scaling a nonzero boundary vector by $\lambda > 1$ pushes it outside the ball. $\square$

Cones play a central role in optimization because first-order feasible perturbations naturally form cones. This observation leads directly to the definition of the tangent cone, which captures all directions along which the constraint set can be locally approached.

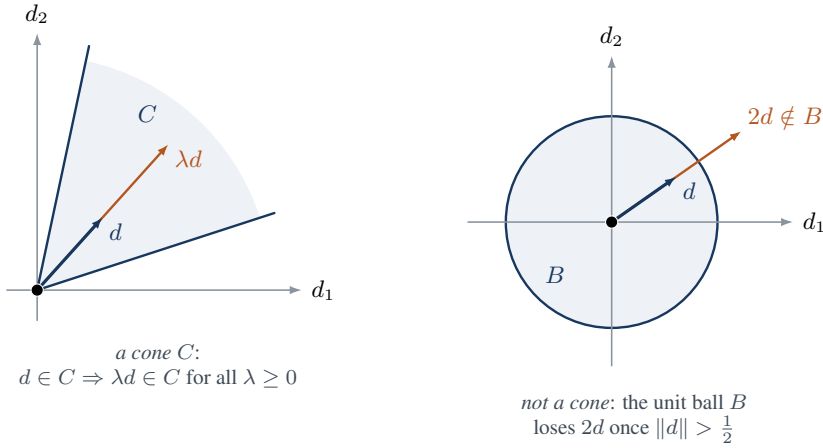


Figure 2.11: A cone is closed under nonnegative scaling: any admissible direction d drags the entire ray $\{\lambda d : \lambda \geq 0\}$ along with it (left). The unit ball fails this test, since scaling any nonzero direction by a large enough factor pushes it outside the set (right).

Let $D \subset \mathbb{R}^n$ be a feasible set and let $x \in D$. Intuitively, the tangent cone consists of all directions in which one can *begin to move* from x while staying in D , up to first order.

Definition 2.123 (Tangent cone (Bouligand/contingent cone)). *The tangent (or contingent) cone to D at x is*

$$T_D(x) := \left\{ d \in \mathbb{R}^n : \exists \varepsilon_k \downarrow 0, \exists d_k \rightarrow d \text{ such that } x + \varepsilon_k d_k \in D \forall k \right\}.$$

The use of sequences allows for small deviations and captures limiting directions that may not be feasible in a strict pointwise sense but are feasible as limits of admissible motions. The tangent cone is always a closed cone, as the remark below records, but it need not be convex.

Remark 2.124 (Interior and boundary cases). If $x \in \text{int}(D)$, then $T_D(x) = \mathbb{R}^n$, since every direction is feasible. If instead x lies on the boundary of D , then $T_D(x)$ is typically a strict subset of $\mathbb{R}^n$ that reflects the local geometry of the constraints.

Example 2.125 Corner point and tangent cone for a polyhedral set.

Let $D := \mathbb{R}_+^2 = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0\}$, and consider the boundary point $x := (0, 0) \in D$.

Let $d = (d_1, d_2) \in \mathbb{R}^2$ be an arbitrary direction. For $\varepsilon > 0$ sufficiently small, $x + \varepsilon d \in D$ if and only if $d_1 \geq 0$ and $d_2 \geq 0$. Hence, the set of feasible directions at x in the pointwise first-order sense is exactly $\mathbb{R}_+^2$.

Now consider the tangent (Bouligand) cone $T_D(x)$, defined by

$$d \in T_D(x) \iff \exists \varepsilon_k > 0, \varepsilon_k \downarrow 0, \exists d_k \rightarrow d \text{ such that } x + \varepsilon_k d_k \in D \forall k.$$

Since $\varepsilon_k > 0$, the condition $x + \varepsilon_k d_k \in D$ implies $(d_k)_1 \geq 0$ and $(d_k)_2 \geq 0$ for all

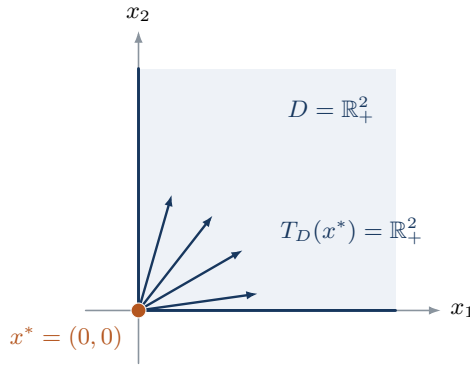


Figure 2.12: At the polyhedral corner $x^* = (0, 0)$ of $D = \mathbb{R}_+^2$, the tangent cone coincides with the feasible set itself: a direction is tangent exactly when both components are nonnegative, so $T_D(x^*) = \mathbb{R}_+^2$. The sequence-based definition adds no directions beyond the pointwise-feasible ones.

sufficiently large k . Passing to the limit and using $d_k \rightarrow d$, we obtain $d_1 \geq 0$ and $d_2 \geq 0$. Thus, $T_D(x) \subseteq \mathbb{R}_+^2$.

Conversely, if $d_1 \geq 0$ and $d_2 \geq 0$, then for any sequence $\varepsilon_k \downarrow 0$ with $\varepsilon_k > 0$, $x + \varepsilon_k d = \varepsilon_k d \in D \forall k$, and hence $d \in T_D(x)$. Therefore, $T_D(x) = \mathbb{R}_+^2$.

This example shows that at a polyhedral corner, such as the origin of $\mathbb{R}_+^2$, the sequence-based definition of the tangent cone does not introduce any additional directions beyond those that are feasible in a pointwise first-order sense. $\square$

Remark 2.126 (Closedness of the Bouligand tangent cone). The Bouligand (contingent) tangent cone $T_D(x)$ is always a closed subset of $\mathbb{R}^n$. This property follows directly from its definition, which is formulated in terms of limits of sequences of admissible directions and step sizes. As a result, no additional closure operation is required: every limit of tangent directions is itself a tangent direction.

Example 2.127 Half-space.

Let $D = \{x \in \mathbb{R}^2 : x_1 \geq 0\}$.

Interior point. Let $x = (x_1, x_2)$ with $x_1 > 0$. Consider an arbitrary direction $d = (d_1, d_2) \in \mathbb{R}^2$. For any sufficiently small $\varepsilon > 0$, $(x + \varepsilon d)_1 = x_1 + \varepsilon d_1 > 0$, since $x_1 > 0$ and εd_1 can be made arbitrarily small. Thus $x + \varepsilon d \in D$ for all small $\varepsilon > 0$, regardless of d , and hence $T_D(x) = \mathbb{R}^2$.

Boundary point. Now let $x = (0, x_2)$ lie on the boundary of D . A direction $d = (d_1, d_2)$ belongs to $T_D(x)$ if and only if there exist sequences $\varepsilon_k \downarrow 0$ and $d_k \rightarrow d$ such that $(x + \varepsilon_k d_k)_1 = \varepsilon_k (d_k)_1 \geq 0$ for all k . Dividing by $\varepsilon_k > 0$ and letting $k \rightarrow \infty$ yields $d_1 \geq 0$. Conversely, if $d_1 \geq 0$, then $x + \varepsilon d \in D$ for all

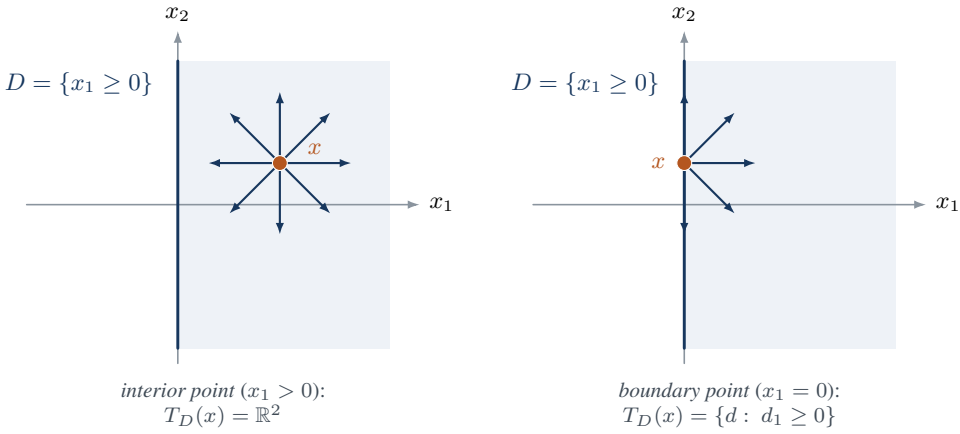


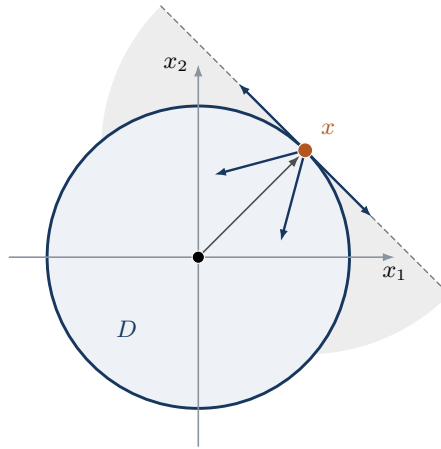
Figure 2.13: Tangent cone of the half-space $D = \{x_1 \geq 0\}$. At an interior point every direction is feasible, so $T_D(x) = \mathbb{R}^2$ (left). At a boundary point only the directions pointing into the half-space survive the limit, giving the closed half-plane $T_D(x) = \{d : d_1 \geq 0\}$ (right).

sufficiently small $\varepsilon > 0$. Therefore, $T_D(x) = \{d \in \mathbb{R}^2 : d_1 \geq 0\}$. □

Example 2.128 Unit disk.

Let $D = \{x \in \mathbb{R}^2 : \|x\|_2 \leq 1\}$ and let x satisfy $\|x\|_2 = 1$. A direction $d \in \mathbb{R}^2$ is tangent if there exist $\varepsilon_k \downarrow 0$ and $d_k \rightarrow d$ such that $\|x + \varepsilon_k d_k\|_2 \leq 1$ for all k . Squaring both sides gives $\|x\|_2^2 + 2\varepsilon_k x^\top d_k + \varepsilon_k^2 \|d_k\|_2^2 \leq 1$. Since $\|x\|_2 = 1$, this simplifies to $2\varepsilon_k x^\top d_k + \varepsilon_k^2 \|d_k\|_2^2 \leq 0$. Dividing by $\varepsilon_k > 0$ and letting $k \rightarrow \infty$ yields $x^\top d \leq 0$. Conversely, if $x^\top d \leq 0$, then the inequality holds for all sufficiently small $\varepsilon > 0$. Hence, $T_D(x) = \{d \in \mathbb{R}^2 : x^\top d \leq 0\}$. This condition expresses that feasible directions must not increase the radius to first order. □

Remark 2.129 (Unit norm constraint and the role of the origin). Although the boundary point x in the unit disk satisfies $\|x\|_2 = 1$ and is not the origin, the tangent cone $T_D(x)$ is nevertheless defined as a subset of $\mathbb{R}^2$ with apex at the origin. This is because the tangent cone encodes *directions of motion* rather than feasible points. A direction d is feasible if small displacements of the form $x + \varepsilon d$ remain in D for $\varepsilon > 0$ sufficiently small. The base point x enters only through this translation, while the cone itself lives in the vector space of directions. Equivalently, after translating the feasible set by $-x$, the tangent cone at x coincides with the tangent cone at the origin of the translated set. Thus, the fact that $\|x\|_2 = 1$ affects which directions are admissible, but not the location of the cone.



shaded: the half-plane $x + T_D(x)$, with $T_D(x) = \{d : x^\top d \leq 0\}$

Figure 2.14: Tangent cone to the unit disk at a boundary point x with $\|x\|_2 = 1$. Feasible directions are exactly those that do not increase the radius to first order, i.e. that make a nonacute angle with the outward radius x : $T_D(x) = \{d : x^\top d \leq 0\}$, drawn translated to x as the shaded half-plane on the disk side of the tangent line.

Example 2.130 Equality constraint.

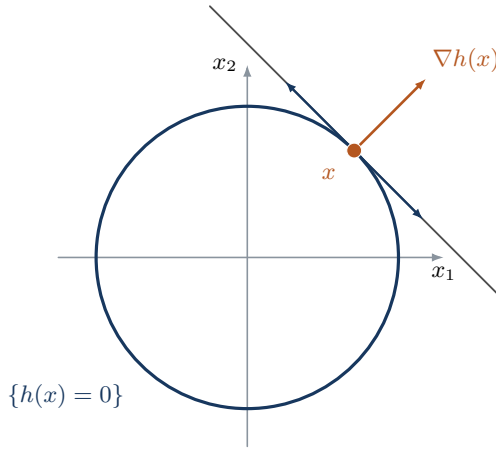
Let $D = \{x \in \mathbb{R}^n : h(x) = 0\}$, where $h : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable and $\nabla h(x) \neq 0$. A direction d is tangent to D at x if there exist sequences $\varepsilon_k \downarrow 0$ and $d_k \rightarrow d$ such that $h(x + \varepsilon_k d_k) = 0$ for all k . Using the first-order Taylor expansion of h at x , $h(x + \varepsilon_k d_k) = h(x) + \varepsilon_k \nabla h(x)^\top d_k + o(\varepsilon_k)$. Since $h(x) = 0$, dividing by ε_k and letting $k \rightarrow \infty$ yields $\nabla h(x)^\top d = 0$. Conversely, suppose $\nabla h(x)^\top d = 0$. Since $\nabla h(x) \neq 0$, the implicit function theorem yields a smooth curve $t \mapsto x(t) \in D$ with $x(0) = x$ and $\dot{x}(0) = d$ (the perturbation $h(x + \varepsilon d) = o(\varepsilon)$ can be corrected back onto the manifold $h = 0$ at order $o(\varepsilon)$). Hence $d \in T_D(x)$, and therefore $T_D(x) = \{d \in \mathbb{R}^n : \nabla h(x)^\top d = 0\}$. This set coincides with the classical tangent space of the smooth constraint manifold defined by $h(x) = 0$. $\square$

Example 2.131 Inequality constraint.

Let $D = \{x \in \mathbb{R}^n : g(x) \leq 0\}$, where $g : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable.

Interior point. If $g(x) < 0$, then x lies in the interior of D . Any sufficiently small perturbation remains feasible, and hence $T_D(x) = \mathbb{R}^n$.

Boundary point. If $g(x) = 0$ and $\nabla g(x) \neq 0$, a direction d is tangent at x if there exist sequences $\varepsilon_k \downarrow 0$ and $d_k \rightarrow d$ such that $g(x + \varepsilon_k d_k) \leq 0$ for all k . Using the first-order Taylor expansion, $g(x + \varepsilon_k d_k) = g(x) + \varepsilon_k \nabla g(x)^\top d_k + o(\varepsilon_k)$. Since $g(x) = 0$, dividing by ε_k and letting $k \rightarrow \infty$ yields $\nabla g(x)^\top d \leq 0$. Conversely, if $\nabla g(x)^\top d <$



line through x : $x + T_D(x)$, with $T_D(x) = \{d : \nabla h(x)^\top d = 0\}$

Figure 2.15: For a smooth equality constraint $D = \{h(x) = 0\}$ with $\nabla h(x) \neq 0$, the tangent cone is the full *tangent line* (in general, the tangent space) $\{d : \nabla h(x)^\top d = 0\}$, orthogonal to $\nabla h(x)$. Because both d and $-d$ are admissible, the cone is a subspace, recovering the classical tangent space of the constraint manifold.

0, then $g(x + \varepsilon d) < 0$ for all sufficiently small $\varepsilon > 0$, so $d \in T_D(x)$; the boundary case $\nabla g(x)^\top d = 0$ follows by approximating d with such strict-descent directions and using that $T_D(x)$ is closed. Therefore, $T_D(x) = \{d \in \mathbb{R}^n : \nabla g(x)^\top d \leq 0\}$. This set is a closed half-space whose boundary coincides with the tangent space of the active constraint manifold $g(x) = 0$. □

While the tangent cone describes feasible directions of motion, the normal cone collects vectors that form nonacute angles with all feasible directions.

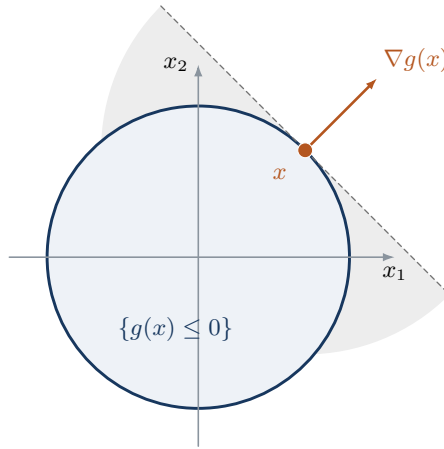
Definition 2.132 (Normal cone). *The normal cone to D at x is defined as the polar cone of $T_D(x)$:*

$$N_D(x) := \{v \in \mathbb{R}^n : v^\top d \leq 0 \quad \forall d \in T_D(x)\}.$$

Geometrically, vectors in $N_D(x)$ point *outward* from the feasible set and are orthogonal (or obtuse) to all admissible directions. Unlike tangent cones, normal cones are always closed and convex.

Example 2.133 Half-space revisited.

For $D = \{x \in \mathbb{R}^2 : x_1 \geq 0\}$ and $x^* = (0, 1)$, $N_D(x^*) = \{\lambda(-1, 0) : \lambda \geq 0\}$. The normal cone consists of outward-pointing normals to the constraint boundary. □



shaded: the half-plane $x + T_D(x)$, with $T_D(x) = \{d : \nabla g(x)^\top d \leq 0\}$

Figure 2.16: For an active inequality constraint $D = \{g(x) \leq 0\}$ at a boundary point with $g(x) = 0$ and $\nabla g(x) \neq 0$, the tangent cone is the closed half-space $\{d : \nabla g(x)^\top d \leq 0\}$ pointing into the feasible region. Its boundary is the tangent line to the active constraint, and $\nabla g(x)$ is the outward normal.

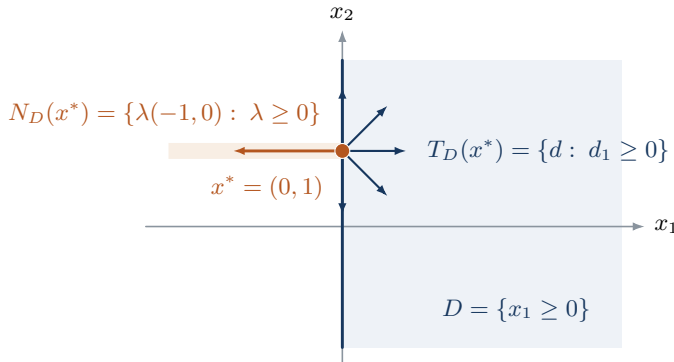
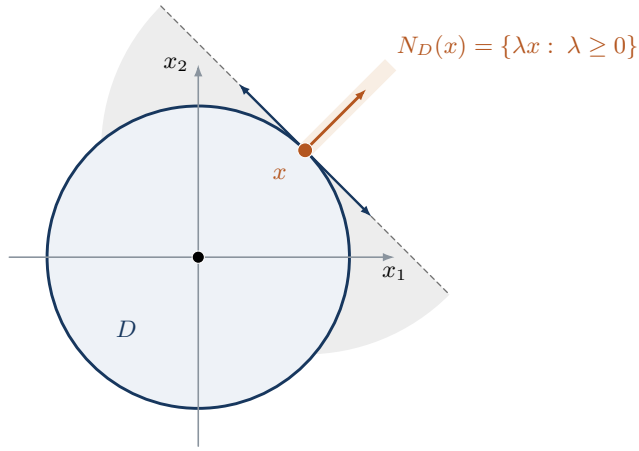


Figure 2.17: Normal cone to the half-space $D = \{x_1 \geq 0\}$ at the boundary point $x^* = (0, 1)$. It is the polar of the tangent cone $\{d_1 \geq 0\}$: the single outward ray $N_D(x^*) = \{\lambda(-1, 0) : \lambda \geq 0\}$, which makes a nonacute angle with every feasible direction.

Example 2.134 Unit disk.

For $D = \{x : \|x\|_2 \leq 1\}$ and $\|x\|_2 = 1$, $N_D(x) = \{\lambda x : \lambda \geq 0\}$. The normal cone is generated by the outward radial direction. $\square$

Tangent and normal cones allow first-order optimality conditions to be expressed in a coordinate-free geometric form that highlights the duality between feasible directions and supporting normals. At a smooth boundary point, as shown in Figure 2.19, the tangent cone is the half-plane beneath the tangent line and the normal cone is the outward normal ray; optimality requires the negative gradient to lie in that ray, so that no feasible direction decreases the objective.



shaded: the tangent half-plane $x + T_D(x)$; the normal ray is its polar

Figure 2.18: Normal cone to the unit disk at a boundary point x with $\|x\|_2 = 1$. It is generated by the outward radial direction: $N_D(x) = \{\lambda x : \lambda \geq 0\}$, the polar of the tangent half-plane $T_D(x) = \{d : x^\top d \leq 0\}$.

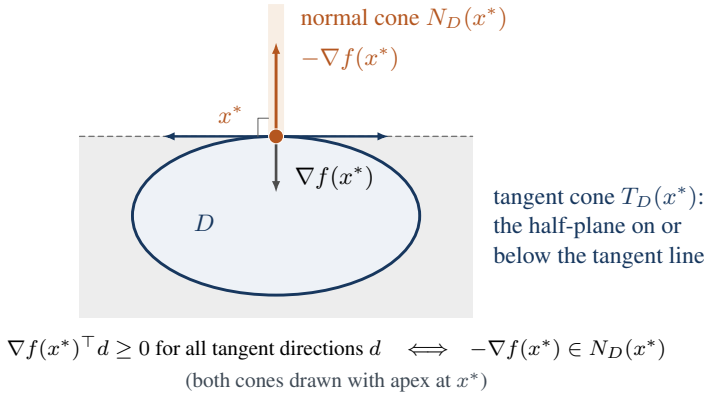


Figure 2.19: At a boundary point, first-order optimality is governed by the tangent cone of feasible directions and the polar normal cone: at a smooth boundary point the tangent cone is the half-plane under the tangent line, and the normal cone is the outward normal ray.

Theorem 2.135 (First-order optimality condition: primal and dual forms). *Let $D \subset \mathbb{R}^n$ and let $f : D \rightarrow \mathbb{R}$ be differentiable at a point $x^* \in D$.*

(i) (Necessary condition) *If x^* is a local minimizer of f over D , then*

$$\nabla f(x^*)^\top d \geq 0, \quad \forall d \in T_D(x^*). \quad (2.12)$$

(ii) (Equivalent dual form) *The condition (2.12) holds if and only if*

$$-\nabla f(x^*) \in N_D(x^*), \quad (2.13)$$

where $N_D(x^*)$ denotes the normal cone to D at x^* .

Proof. (i) *Necessary condition.* Assume that x^* is a local minimizer of f over D . By definition, there exists a neighborhood U of x^* such that $f(x) \geq f(x^*)$, $\forall x \in D \cap U$.

Let $d \in T_D(x^*)$ be arbitrary. By definition of the tangent (Bouligand) cone, there exist sequences $\{\varepsilon_k\}_{k \geq 1}$ and $\{d_k\}_{k \geq 1}$ such that $\varepsilon_k \downarrow 0$, $d_k \rightarrow d$, and $x^* + \varepsilon_k d_k \in D$ for all k .

Since $\varepsilon_k \rightarrow 0$ and $d_k \rightarrow d$, we have $x^* + \varepsilon_k d_k \rightarrow x^*$. For k sufficiently large, $x^* + \varepsilon_k d_k \in U$. By local optimality of x^* , this implies $f(x^* + \varepsilon_k d_k) \geq f(x^*)$, $\forall k$ sufficiently large.

Dividing both sides by $\varepsilon_k > 0$ yields $\frac{f(x^* + \varepsilon_k d_k) - f(x^*)}{\varepsilon_k} \geq 0$.

Because f is differentiable at x^* , its first-order expansion gives $f(x^* + h) = f(x^*) + \nabla f(x^*)^\top h + o(\|h\|)$ as $h \rightarrow 0$. Applying this with $h = \varepsilon_k d_k$ yields

$$\frac{f(x^* + \varepsilon_k d_k) - f(x^*)}{\varepsilon_k} = \nabla f(x^*)^\top d_k + \frac{o(\varepsilon_k \|d_k\|)}{\varepsilon_k}.$$

Since $d_k \rightarrow d$, the sequence $\{d_k\}$ is bounded, and hence $o(\varepsilon_k \|d_k\|)/\varepsilon_k \rightarrow 0$. Taking the limit as $k \rightarrow \infty$ therefore gives

$$\lim_{k \rightarrow \infty} \frac{f(x^* + \varepsilon_k d_k) - f(x^*)}{\varepsilon_k} = \nabla f(x^*)^\top d.$$

Because each finite-difference quotient on the left-hand side is nonnegative, its limit must also be nonnegative. Thus, $\nabla f(x^*)^\top d \geq 0$.

Since $d \in T_D(x^*)$ was arbitrary, the inequality holds for all $d \in T_D(x^*)$, which proves (2.12).

(ii) *Equivalence with the dual condition.* Recall that the normal cone to D at x^* is defined as $N_D(x^*) := \{v \in \mathbb{R}^n : v^\top d \leq 0 \forall d \in T_D(x^*)\}$.

($\Rightarrow$) If (2.12) holds, then for all $d \in T_D(x^*)$ we have $\nabla f(x^*)^\top d \geq 0$, equivalently $(-\nabla f(x^*))^\top d \leq 0$. By definition of the normal cone, this implies $-\nabla f(x^*) \in N_D(x^*)$.

($\Leftarrow$) Conversely, if $-\nabla f(x^*) \in N_D(x^*)$, then $(-\nabla f(x^*))^\top d \leq 0$, $\forall d \in T_D(x^*)$. Multiplying by -1 yields $\nabla f(x^*)^\top d \geq 0$, $\forall d \in T_D(x^*)$, which is exactly (2.12).

Thus, the primal and dual conditions are equivalent. $\square$

Remark 2.136 (Geometric interpretation of first-order optimality). The primal condition (2.12) states that no tangent direction to the feasible set at x^* can locally decrease the objective. Equivalently, the dual condition (2.13) expresses the same requirement in geometric terms: the negative gradient must lie in the normal cone to D at x^* , so the gradient cannot point into the feasible region. This primal–dual equivalence forms the geometric basis of first-order constrained optimality and underlies the Karush–Kuhn–Tucker conditions.

Remark 2.137 (Regularity required for first-order optimality). The first-order optimality condition of Theorem 2.135 requires only that f be differentiable at the point x^* . Continuity of the gradient is not needed. Indeed, the proof relies solely on the first-order expansion $f(x^* + h) = f(x^*) + \nabla f(x^*)^\top h + o(\|h\|)$, which is the defining property of differentiability at a point. In contrast, second-order sufficient conditions require additional regularity to control the behavior of the Hessian in a neighborhood of x^* .

Corollary 2.138 (Strict first-order condition and strict local optimality). *Under the assumptions of Theorem 2.135, suppose that*

$$\nabla f(x^*)^\top d > 0 \quad \forall d \in T_D(x^*) \setminus \{0\}. \quad (2.14)$$

Then x^ is a strict local minimizer of f over D .*

Equivalently, the strict condition (2.14) holds if and only if

$$-\nabla f(x^*) \in \text{int } N_D(x^*), \quad (2.15)$$

where $N_D(x^*)$ denotes the normal cone to D at x^* .

Proof. We first show that (2.14) implies that x^* is a strict local minimizer, arguing by contradiction.

Suppose x^* is not a strict local minimizer. Then there is a sequence $\{x_k\} \subset D$ with $x_k \neq x^*$, $x_k \rightarrow x^*$, and $f(x_k) \leq f(x^*)$ for every k . Put $h_k := x_k - x^*$, $\varepsilon_k := \|h_k\| > 0$ and $d_k := h_k/\varepsilon_k$, so that $\varepsilon_k \downarrow 0$, $\|d_k\| = 1$ and $x^* + \varepsilon_k d_k = x_k \in D$. The unit vectors d_k lie in the compact unit sphere, so after passing to a subsequence we may assume $d_k \rightarrow d$ with $\|d\| = 1$. Definition 2.123 applied to the sequences $\{\varepsilon_k\}$ and $\{d_k\}$ then gives $d \in T_D(x^*) \setminus \{0\}$.

Differentiability of f at x^* yields

$$0 \geq \frac{f(x_k) - f(x^*)}{\varepsilon_k} = \nabla f(x^*)^\top d_k + \frac{o(\varepsilon_k)}{\varepsilon_k},$$

and letting $k \rightarrow \infty$ gives $\nabla f(x^*)^\top d \leq 0$. This contradicts (2.14), since d is a nonzero tangent direction. Hence x^* is a strict local minimizer.

It is worth recording the uniform bound that underlies this argument. Because $T_D(x^*)$ is closed, the set $K := T_D(x^*) \cap \{d \in \mathbb{R}^n : \|d\| = 1\}$ is closed and bounded, hence compact. If $K \neq \emptyset$, the continuous function $d \mapsto \nabla f(x^*)^\top d$ is strictly positive on K by (2.14), so it attains a minimum

$$\alpha := \min_{d \in K} \nabla f(x^*)^\top d > 0. \quad (2.16)$$

The growth of f away from x^* along feasible sequences is therefore at least linear with rate α , and not merely strict.

We now prove the equivalence with the dual condition. By definition of the normal cone, $N_D(x^*) = \{v \in \mathbb{R}^n : v^\top d \leq 0 \ \forall d \in T_D(x^*)\}$. Condition (2.14) is equivalent to $(-\nabla f(x^*))^\top d < 0$ for all $d \in T_D(x^*) \setminus \{0\}$. If $v := -\nabla f(x^*)$ satisfies this, then with α as in (2.16) every u with $\|u\| < \alpha$ obeys $(v + u)^\top d \leq -\alpha + \|u\| < 0$ on K and hence $(v + u)^\top d \leq 0$ on all of $T_D(x^*)$, so $v + u \in N_D(x^*)$ and $v \in \text{int } N_D(x^*)$. Conversely, if $v \in \text{int } N_D(x^*)$ and $d \in T_D(x^*) \setminus \{0\}$, then $v + \varepsilon d/\|d\| \in N_D(x^*)$ for small $\varepsilon > 0$, whence $v^\top d \leq -\varepsilon\|d\| < 0$. Both directions use only closedness of $T_D(x^*)$. This proves (2.15). $\square$

The strict inequality (2.14) states that every nonzero tangent direction at x^* leads to a first-order increase in the objective. Thus, there are no flat directions and no descent directions compatible with the constraints. Geometrically, the gradient $\nabla f(x^*)$ points strictly outside the feasible region, and the negative gradient lies in the interior of the normal cone. As a result, any nearby feasible point must have strictly larger objective value, making x^* a strict local minimizer.

This situation can occur only at boundary points of the feasible set. At interior points, the tangent cone equals $\mathbb{R}^n$, and the strict inequality would be violated by considering both d and $-d$. Hence strict first-order optimality reflects the presence of active constraints and represents a purely first-order mechanism for strict local optimality.

Example 2.139 Interior point: unconstrained case.

Let $D = \mathbb{R}^2$ and let $f(x) = x_1^2 + x_2^2$. The unique minimizer is $x^* = (0, 0)$. Since x^* is an interior point, $T_D(x^*) = \mathbb{R}^2$ and $N_D(x^*) = \{0\}$. The gradient satisfies $\nabla f(x^*) = (0, 0)$. The cone condition $\nabla f(x^*)^\top d = 0 \geq 0$ for all $d \in \mathbb{R}^2$ holds

trivially, and equivalently $-\nabla f(x^*) = 0 \in N_D(x^*)$. This recovers the classical unconstrained optimality condition $\nabla f(x^*) = 0$. $\square$

Example 2.140 Boundary point of a half-space.

Let $D = \{x \in \mathbb{R}^2 : x_1 \geq 0\}$, let $f(x) = x_1 + x_2^2$, and consider the point $x^* = (0, 0)$. The tangent and normal cones are $T_D(x^*) = \{d : d_1 \geq 0\}$ and $N_D(x^*) = \{\lambda(-1, 0) : \lambda \geq 0\}$. The gradient is $\nabla f(x^*) = (1, 0)$. For any feasible direction $d \in T_D(x^*)$, we have $\nabla f(x^*)^\top d = d_1 \geq 0$, so the primal cone condition holds. Equivalently, $-\nabla f(x^*) = (-1, 0) \in N_D(x^*)$. Geometrically, the negative gradient points directly outward, normal to the constraint boundary, preventing descent into the feasible region. $\square$

Example 2.141 Corner point: multiple active constraints.

Let $D = \mathbb{R}_+^2 := \{x : x_1 \geq 0, x_2 \geq 0\}$, let $f(x) = x_1 + x_2$, and let $x^* = (0, 0)$. The tangent and normal cones are $T_D(x^*) = \mathbb{R}_+^2$ and $N_D(x^*) = \{v : v_1 \leq 0, v_2 \leq 0\}$. The gradient is $\nabla f(x^*) = (1, 1)$. For any $d \in T_D(x^*)$, $\nabla f(x^*)^\top d = d_1 + d_2 \geq 0$, so the primal condition holds. Equivalently, $-\nabla f(x^*) = (-1, -1) \in N_D(x^*)$. At a corner, the normal cone is higher-dimensional, reflecting the presence of multiple blocking constraints. $\square$

Example 2.142 Smooth equality constraint.

Let $D = \{x \in \mathbb{R}^2 : x_1 + x_2 = 1\}$, let $f(x) = x_1^2 + x_2^2$, and consider $x^* = (\frac{1}{2}, \frac{1}{2})$. The tangent and normal cones are $T_D(x^*) = \{d : d_1 + d_2 = 0\}$ and $N_D(x^*) = \{\lambda(1, 1) : \lambda \in \mathbb{R}\}$. The gradient is $\nabla f(x^*) = (1, 1)$. For any $d \in T_D(x^*)$, $\nabla f(x^*)^\top d = d_1 + d_2 = 0$, so the primal cone condition holds. Equivalently, $-\nabla f(x^*) = (-1, -1) \in N_D(x^*)$. This example illustrates how Lagrange multipliers emerge naturally from normal-cone geometry. $\square$

Example 2.143 Nonconvex feasible set.

Let $D = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 = x_1^2\}$, let $f(x) = x_2$, and consider $x^* = (0, 0)$. The tangent and normal cones are $T_D(x^*) = \{(d_1, 0) : d_1 \in \mathbb{R}\}$ and $N_D(x^*) = \{(0, \lambda) : \lambda \in \mathbb{R}\}$. The gradient is $\nabla f(x^*) = (0, 1)$. For all $d \in T_D(x^*)$, $\nabla f(x^*)^\top d = 0$, and equivalently $-\nabla f(x^*) = (0, -1) \in N_D(x^*)$. Despite the nonconvexity of D , the cone-based first-order condition remains valid and correctly captures the local geometry. $\square$

Example 2.144 From cone optimality to a primal–dual saddle point.

Consider the constrained minimization problem

$$\min_{x \in D} f(x), \quad D := \{x \in \mathbb{R}^n : g_i(x) \leq 0, i = 1, \dots, m\},$$

where f and g_i are differentiable. Let $x^* \in D$ be a local minimizer. By Theorem 2.135, first-order optimality is equivalently characterized by the normal-cone condition

$$-\nabla f(x^*) \in N_D(x^*). \quad (2.17)$$

Assume that a constraint qualification holds at x^* (e.g., the Mangasarian–Fromovitz constraint qualification). Then the normal cone to D at x^* admits the explicit representation

$$N_D(x^*) = \left\{ \sum_{i \in \mathcal{A}(x^*)} \lambda_i \nabla g_i(x^*) : \lambda_i \geq 0 \right\}, \quad (2.18)$$

where $\mathcal{A}(x^*) := \{i : g_i(x^*) = 0\}$ is the active constraint set at x^* . Combining (2.17) and (2.18), there exists a vector $\lambda^* \in \mathbb{R}^m$ with $\lambda_i^* \geq 0$ and $\lambda_i^* g_i(x^*) = 0$ such that

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) = 0. \quad (2.19)$$

Define the Lagrangian

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x).$$

Equation (2.19) is precisely the stationarity condition $\nabla_x \mathcal{L}(x^*, \lambda^*) = 0$, while the properties $\lambda_i^* \geq 0$ and $\lambda_i^* g_i(x^*) = 0$ correspond to dual feasibility and complementary slackness, respectively.

Thus, the dual variables λ^* arise as the coefficients of the normal-cone decomposition of $-\nabla f(x^*)$, and the primal–dual optimality conditions are a coordinate representation of the geometric normal-cone condition (2.17). $\square$

2.4.3 ■ Second-order optimality conditions (unconstrained/interior case)

Second-order conditions quantify curvature. In unconstrained problems (or at interior feasible points), they are expressed in terms of the Hessian. Throughout this subsection, let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable on an open set $\mathcal{U} \subset \mathbb{R}^n$.

Definition 2.145 (Interior local minimizer). A point $x^* \in \mathcal{U}$ is a local minimizer of f if there exists $r > 0$ such that $f(x^*) \leq f(x)$, for all $x \in B_r(x^*) \subset \mathcal{U}$.

Theorem 2.146 (Second-order necessary conditions). Assume that $x^* \in \mathcal{U}$ is a local minimizer of f . Then

$$(i) \nabla f(x^*) = 0,$$

$$(ii) \nabla^2 f(x^*) \succeq 0, \text{ i.e., } d^\top \nabla^2 f(x^*) d \geq 0, \forall d \in \mathbb{R}^n.$$

Proof. Fix $d \in \mathbb{R}^n$ and define $\phi(\varepsilon) = f(x^* + \varepsilon d)$ for ε small enough that $x^* + \varepsilon d \in \mathcal{U}$. Since x^* is a local minimizer, $\varepsilon = 0$ is a local minimizer of the univariate function ϕ . By the chain rule, $\phi'(\varepsilon) = \nabla f(x^* + \varepsilon d)^\top d$ and $\phi''(\varepsilon) = d^\top \nabla^2 f(x^* + \varepsilon d) d$. The first-order necessary condition for ϕ at $\varepsilon = 0$ gives $\phi'(0) = \nabla f(x^*)^\top d = 0$ for all d , hence $\nabla f(x^*) = 0$. The second-order necessary condition gives $\phi''(0) \geq 0$, i.e., $d^\top \nabla^2 f(x^*) d \geq 0$ for all d , which means $\nabla^2 f(x^*) \succeq 0$. $\square$

Theorem 2.147 (Second-order sufficient conditions). Assume $f \in C^2(\mathcal{U})$ and that $x^* \in \mathcal{U}$ satisfies

$$(i) \nabla f(x^*) = 0,$$

$$(ii) \nabla^2 f(x^*) \succ 0.$$

Then x^* is a strict local minimizer of f ; that is, there exists $r > 0$ such that $f(x) > f(x^*)$, for all $x \in B_r(x^*) \setminus \{x^*\}$.

Proof. Let $H = \nabla^2 f(x^*) \succ 0$. Then there exists $m > 0$ such that $d^\top H d \geq m\|d\|^2$, $\forall d \in \mathbb{R}^n$. By Taylor's theorem,

$$f(x^* + h) = f(x^*) + \nabla f(x^*)^\top h + \frac{1}{2}h^\top H h + o(\|h\|^2) \quad (h \rightarrow 0).$$

Using $\nabla f(x^*) = 0$ and the bound above yields

$$f(x^* + h) \geq f(x^*) + \frac{m}{2}\|h\|^2 + o(\|h\|^2).$$

For $\|h\|$ small enough, $|o(\|h\|^2)| \leq \frac{m}{4}\|h\|^2$, hence $f(x^* + h) \geq f(x^*) + \frac{m}{4}\|h\|^2 > f(x^*)$ whenever $h \neq 0$ is sufficiently small. This proves strict local minimality. $\square$

Example 2.148 Saddle point: second-order necessary condition violated.

Let

$$f(x) = e^{x_1} - x_1 - (e^{x_2} - x_2), \quad x \in \mathbb{R}^2.$$

Then $\nabla f(x) = (e^{x_1} - 1, -(e^{x_2} - 1))$ and $\nabla^2 f(x) = \text{diag}(e^{x_1}, -e^{x_2})$.

At $x^* = (0, 0)$, $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) = \text{diag}(1, -1)$, which is indefinite. Hence the second-order necessary condition of Theorem 2.146 fails. Geometrically, f curves upward in the x_1 direction and downward in the x_2 direction. Indeed, since $e^\varepsilon - \varepsilon > 1$ for every $\varepsilon \neq 0$,

$$f(0, \varepsilon) = (e^0 - 0) - (e^\varepsilon - \varepsilon) = 1 - (e^\varepsilon - \varepsilon) < 0 = f(0, 0),$$

while $f(\varepsilon, 0) = (e^\varepsilon - \varepsilon) - 1 > 0 = f(0, 0)$; the value both increases and decreases arbitrarily close to x^* , so x^* is a saddle point. $\square$

Example 2.149 Degenerate Hessian: necessary but not sufficient.

Let $f(x) = x_1^4 + x_2^2$, with $x \in \mathbb{R}^2$. Then $\nabla f(x) = (4x_1^3, 2x_2)$ and

$$\nabla^2 f(x) = \begin{pmatrix} 12x_1^2 & 0 \\ 0 & 2 \end{pmatrix}.$$

At $x^* = (0, 0)$, $\nabla f(x^*) = 0$ and

$$\nabla^2 f(x^*) = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} \succeq 0,$$

but the Hessian is singular and therefore does not satisfy the second-order sufficient condition.

Nevertheless, x^* is a strict local minimizer since $f(x^* + h) = h_1^4 + h_2^2 > 0$ for all $h \neq 0$. Strict minimality is enforced by fourth-order curvature in the x_1 direction, which is invisible to the quadratic approximation.

This example shows that positive semidefiniteness of the Hessian is necessary but not sufficient for strict local minimality when higher-order terms dominate the local geometry. $\square$

Example 2.150 Failure of second-order sufficiency without C^2 smoothness.

Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} x^2 \left(1 + 2 \sin\left(\frac{1}{x}\right) \right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

First derivative at the origin. Since $|f(x)| \leq 3x^2$, the difference quotient satisfies

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} x \left(1 + 2 \sin(1/x) \right) = 0,$$

because $1 + 2 \sin(1/x)$ stays bounded. Thus $x^* = 0$ is a stationary point.

No second derivative at the origin. For $x \neq 0$,

$$f'(x) = 2x + 4x \sin\left(\frac{1}{x}\right) - 2 \cos\left(\frac{1}{x}\right).$$

The second-order difference quotient is

$$\frac{f'(x) - f'(0)}{x} = 2 + 4 \sin\left(\frac{1}{x}\right) - \frac{2}{x} \cos\left(\frac{1}{x}\right),$$

whose term $-\frac{2}{x} \cos(1/x)$ oscillates with unbounded magnitude as $x \rightarrow 0$. Hence $f''(0)$ does not exist: the quadratic Taylor coefficient on which the second-order test relies is not even defined, which is precisely the loss of C^2 smoothness.

The origin is not a local minimizer. Although $f(x) = x^2 + 2x^2 \sin(1/x)$ looks like a positive quadratic perturbed by a higher-order term, the perturbation is itself of order x^2 and can overwhelm the leading term. Choose $x_k \rightarrow 0$ with $\sin(1/x_k) = -1$ (for instance $1/x_k = -\frac{\pi}{2} + 2\pi k$). Then

$$f(x_k) = x_k^2(1 + 2(-1)) = -x_k^2 < 0 = f(0).$$

Thus arbitrarily close to 0 there exist points with strictly smaller value than $f(0)$, so 0 is not a local minimizer.

This example shows that when the Hessian fails to be continuous—indeed, when $f''(0)$ does not exist—stationarity together with a nominally positive quadratic term does not guarantee a local minimum. A genuine second-order lower bound requires curvature to be controlled throughout a neighborhood, that is, C^2 smoothness. $\square$

2.4.4 ■ Second-order conditions with constraints

In constrained optimization, second-order optimality conditions must reflect the fact that not all perturbation directions are admissible. Unlike the unconstrained case, where every direction in $\mathbb{R}^n$ can be explored locally, constraints restrict feasible motion to directions contained in the tangent cone. Consequently, curvature information provided by the Hessian is relevant only along directions that are feasible to first order.

A further refinement is required because even among feasible directions, some directions already increase the objective to first order and therefore cannot lead to a local decrease. Second-order conditions become meaningful only along directions for which the first-order variation vanishes.

Definition 2.151 (Critical cone). Let $x^* \in D$ satisfy the first-order necessary condition $-\nabla f(x^*) \in N_D(x^*)$. The critical cone at x^* is defined as

$$\mathcal{C}(x^*) := \{d \in T_D(x^*) : \nabla f(x^*)^\top d = 0\}.$$

The critical cone consists of directions that are feasible to first order and along which the objective function exhibits no first-order change. These are precisely the directions along which second-order effects determine whether the point x^* is locally minimizing. Figure 2.20 makes the roles explicit: any direction with $\nabla f(x^*)^\top d > 0$ is already ruled out by the linear term, so curvature has the final word only on the critical cone, where the linear term is silent.

Definition 2.152 (Locally polyhedral feasible set). The set $D \subset \mathbb{R}^n$ is locally polyhedral at $x^* \in D$ if there exists $r > 0$ such that

$$D \cap B_r(x^*) = (x^* + T_D(x^*)) \cap B_r(x^*),$$

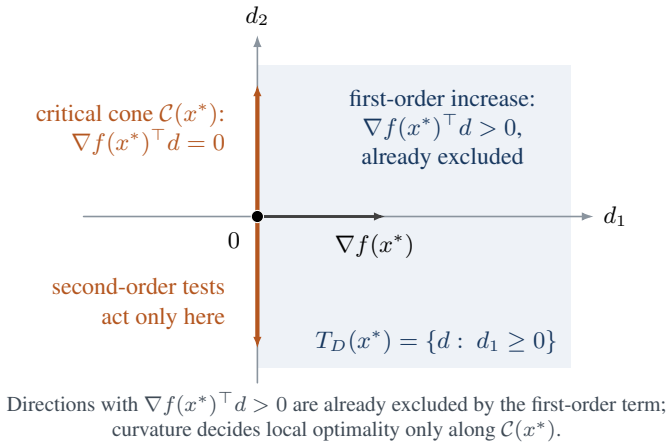


Figure 2.20: The critical cone refines the tangent cone to the directions where curvature, rather than the linear term, decides local behavior.

that is, near x^* the feasible set coincides with the translate of its own tangent cone. Polyhedra, half-spaces, orthants, affine subspaces and $\mathbb{R}^n$ itself are locally polyhedral at each of their points.

Theorem 2.153 (Second-order necessary condition along the critical cone). *Let $f \in C^2$ in a neighborhood of x^* , let D be locally polyhedral at x^* , and let x^* be a constrained local minimizer of f over D . If $-\nabla f(x^*) \in N_D(x^*)$, then $d^\top \nabla^2 f(x^*) d \geq 0$, $\forall d \in \mathcal{C}(x^*)$.*

Proof. Let $d \in \mathcal{C}(x^*)$, so that $d \in T_D(x^*)$ and $\nabla f(x^*)^\top d = 0$; we may assume $d \neq 0$. Because D is locally polyhedral at x^* , the whole segment is feasible: there is $r > 0$ with $x^* + \varepsilon d \in D$ for every $\varepsilon \in (0, r/\|d\|)$. This is the point at which local polyhedrality is used, and it lets us move along d exactly rather than along an approximating sequence $d_k \rightarrow d$.

Fix such a sequence $\varepsilon_k \downarrow 0$ and put $x_k := x^* + \varepsilon_k d \in D$. Since x^* is a constrained local minimizer, $f(x_k) \geq f(x^*)$ for all large k , while Taylor's theorem gives

$$f(x_k) = f(x^*) + \varepsilon_k \nabla f(x^*)^\top d + \frac{1}{2} \varepsilon_k^2 d^\top \nabla^2 f(x^*) d + o(\varepsilon_k^2).$$

The linear term vanishes because d is critical. Subtracting $f(x^*)$, dividing by $\varepsilon_k^2 > 0$ and letting $k \rightarrow \infty$ therefore yields $0 \leq \frac{1}{2} d^\top \nabla^2 f(x^*) d$, which is the claim. Since $d \in \mathcal{C}(x^*)$ was arbitrary, the result follows. $\square$

Remark 2.154 (Why local polyhedrality cannot be dropped). For a curved feasible set the conclusion genuinely fails, because the second-order expansion then also feels the curvature of D itself. Take $D = \{(x_1, x_2) : x_2 = -x_1^2\}$, $f(x) = x_2 + \frac{1}{2}x_1^2$ and $x^* = (0, 0)$. On D we have $f = -\frac{1}{2}x_1^2$, so x^* is not even a local minimizer; yet $T_D(x^*) = \{(d_1, 0)\}$, $N_D(x^*) = \{v : v_1 = 0\}$ and $\nabla f(x^*) = (0, 1)$, so the first-order condition $-\nabla f(x^*) \in N_D(x^*)$ holds, the critical cone is $\mathcal{C}(x^*) = \{(d_1, 0)\}$, and $d^\top \nabla^2 f(x^*) d = d_1^2 > 0$ on $\mathcal{C}(x^*) \setminus \{0\}$. Both the necessary and the sufficient condition below would be contradicted.

The repair used in smooth nonlinear programming is to replace $\nabla^2 f(x^*)$ by the Hessian of the *Lagrangian*, whose multiplier terms carry exactly the curvature of the constraint surfaces; this is the form derived in Chapter 3. The alternative, pursued in variational analysis, is to add a second-order tangent-set correction to the expansion.

Remark 2.155. This condition states that the Hessian of f must be positive semidefinite when restricted to directions that are both feasible to first order and neutral with respect to the first-order change in the objective. In smooth nonlinear programming with equality and inequality constraints, this result appears in the more familiar form that the Hessian of the Lagrangian is positive semidefinite on the critical subspace, under suitable constraint qualifications.

The first-order condition eliminates directions along which the objective can be decreased linearly. The second-order condition then rules out quadratic descent along directions where the linear term vanishes. Together, these conditions exclude both linear and quadratic escape directions from the candidate minimizer.

Example 2.156 Critical cone and second-order test for a constrained minimum.

Let $D = \{x \in \mathbb{R}^2 : x_1 \geq 0\}$ and $f(x) = x_2^2$, and consider the point $x^* = (0, 0)$.

The gradient and Hessian are

$$\nabla f(x) = (0, 2x_2), \quad \nabla^2 f(x) = \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}.$$

At x^* , $\nabla f(x^*) = 0$, so the first-order condition $-\nabla f(x^*) \in N_D(x^*)$ is satisfied.

The tangent cone at x^* is $T_D(x^*) = \{d \in \mathbb{R}^2 : d_1 \geq 0\}$. Since $\nabla f(x^*) = 0$, the critical cone coincides with the tangent cone: $\mathcal{C}(x^*) = T_D(x^*)$.

For any $d = (d_1, d_2) \in \mathcal{C}(x^*)$, we have $d^\top \nabla^2 f(x^*) d = 2d_2^2 \geq 0$. Thus the second-order *necessary* condition holds along all critical directions. This is consistent with, but does not by itself establish, local minimality: the *sufficient* condition of Theorem 2.157 fails here, since $d^\top \nabla^2 f(x^*) d = 0$ for the critical direction $d = (1, 0)$. Minimality can nonetheless be seen directly, since $f(x) = x_2^2 \geq 0 = f(x^*)$ shows that x^* is a (non-strict) global minimizer of f over D . $\square$

Theorem 2.157 (Second-order sufficient condition on the critical cone). *Let $D \subset \mathbb{R}^n$ be locally polyhedral at $x^* \in D$ (Definition 2.152), and let $f \in C^2$ in a neighborhood of x^* . Assume that*

- (1) x^* satisfies the first-order necessary condition $-\nabla f(x^*) \in N_D(x^*)$,
- (2) the Hessian is positive definite on the critical cone, i.e., $d^\top \nabla^2 f(x^*) d > 0$, $\forall d \in \mathcal{C}(x^*) \setminus \{0\}$.

Then x^ is a strict local minimizer of f over D . Remark 2.154 shows that the polyhedrality hypothesis cannot be omitted.*

Proof. Suppose, for contradiction, that x^* is not a strict local minimizer. Then there is a sequence $x_k \in D$ with $x_k \rightarrow x^*$, $x_k \neq x^*$ and $f(x_k) \leq f(x^*)$ for every k . Define $\varepsilon_k := \|x_k - x^*\| \downarrow 0$ and $d_k := (x_k - x^*)/\varepsilon_k$. By compactness of the unit sphere, we may extract a subsequence (not relabeled) such that $d_k \rightarrow d$ with $\|d\| = 1$; since $x^* + \varepsilon_k d_k = x_k \in D$, the limit satisfies $d \in T_D(x^*)$.

Local polyhedrality supplies the extra information the argument needs: for all k large enough that $\varepsilon_k < r$ we have $x_k \in D \cap B_r(x^*)$, hence $\varepsilon_k d_k \in T_D(x^*)$ and therefore $d_k \in T_D(x^*)$, the

tangent cone being closed under positive scaling. The first-order condition $-\nabla f(x^*) \in N_D(x^*)$ then gives

$$\nabla f(x^*)^\top d_k \geq 0 \quad \text{for all large } k, \quad (2.20)$$

and in the limit $\nabla f(x^*)^\top d \geq 0$.

Case 1: $\nabla f(x^*)^\top d > 0$. Taylor expansion at x^* gives

$$f(x_k) - f(x^*) = \varepsilon_k \nabla f(x^*)^\top d_k + o(\varepsilon_k) = \varepsilon_k (\nabla f(x^*)^\top d + o(1)),$$

which is strictly positive for all large k , contradicting $f(x_k) \leq f(x^*)$.

Case 2: $\nabla f(x^*)^\top d = 0$. Then $d \in \mathcal{C}(x^*)$ with $\|d\| = 1$, so hypothesis (2) gives $d^\top \nabla^2 f(x^*) d =: 2\beta > 0$. The second-order expansion, together with the sign information (2.20), yields

$$f(x_k) - f(x^*) = \varepsilon_k \nabla f(x^*)^\top d_k + \frac{1}{2} \varepsilon_k^2 d_k^\top \nabla^2 f(x^*) d_k + o(\varepsilon_k^2) \geq \frac{1}{2} \varepsilon_k^2 d_k^\top \nabla^2 f(x^*) d_k + o(\varepsilon_k^2),$$

where the linear term was discarded because it is nonnegative. Since $d_k \rightarrow d$, we have $d_k^\top \nabla^2 f(x^*) d_k \rightarrow 2\beta$, so the right-hand side is at least $\frac{\beta}{2} \varepsilon_k^2 > 0$ for all large k . This again contradicts $f(x_k) \leq f(x^*)$.

Both cases are impossible, so x^* is a strict local minimizer. $\square$

Example 2.158 Second-order sufficiency via the critical cone.

Consider the constrained optimization problem $\min_{x \in D} f(x)$ with $f(x) = x_1^2 + x_2^2$ and $D = \{x \in \mathbb{R}^2 : x_1 \geq 0\}$. Let $x^* = (0, 0)$. The gradient and Hessian of f are $\nabla f(x) = (2x_1, 2x_2)$ and $\nabla^2 f(x) = 2I$, where I denotes the 2×2 identity matrix, so that $\nabla f(x^*) = (0, 0)$.

The tangent cone to D at x^* consists of all directions that do not point into the infeasible half-space: $T_D(x^*) = \{d \in \mathbb{R}^2 : d_1 \geq 0\}$. Since $\nabla f(x^*) = 0$, the first-order necessary condition $\nabla f(x^*)^\top d \geq 0$, $\forall d \in T_D(x^*)$ is satisfied trivially. The corresponding normal cone is $N_D(x^*) = \{\lambda(-1, 0) : \lambda \geq 0\}$.

Because the gradient vanishes, the critical cone coincides with the tangent cone: $\mathcal{C}(x^*) = \{d \in T_D(x^*) : \nabla f(x^*)^\top d = 0\} = T_D(x^*)$. For any nonzero $d \in \mathcal{C}(x^*)$, we have $d^\top \nabla^2 f(x^*) d = 2(d_1^2 + d_2^2) > 0$. Hence, the Hessian is positive definite when restricted to the critical cone.

By the second-order sufficient condition for constrained optimization, x^* is a strict local minimizer of f over D . Indeed, for any feasible perturbation $x^* + h$ with $h \neq 0$ and $h_1 \geq 0$, $f(x^* + h) = h_1^2 + h_2^2 > 0 = f(x^*)$. $\square$

2.5 ■ Variational Inequality Formulation

Many first-order optimality conditions in constrained optimization can be expressed compactly as a *variational inequality* (VI). This formulation provides a unifying language that connects optimization, equilibrium problems, and nonlinear analysis, and it plays a central role in modern optimization theory.

Definition 2.159 (Variational inequality). Let $D \subset \mathbb{R}^n$ be a closed set and let $F : D \rightarrow \mathbb{R}^n$ be a vector-valued mapping. The variational inequality problem $\text{VI}(D, F)$ consists of finding a point

$x^* \in D$ such that $F(x^*)^\top(x - x^*) \geq 0, \forall x \in D$.

Geometrically, the condition states that the vector $F(x^*)$ forms a nonacute angle with every feasible displacement $x - x^*$ within the set D .

A key observation is that first-order necessary conditions for constrained minimization problems are special cases of variational inequalities.

Theorem 2.160 (Local minimizers satisfy a variational inequality). *Let $D \subset \mathbb{R}^n$ be convex and let $f : D \rightarrow \mathbb{R}$ be differentiable at $x^* \in D$. If x^* is a local minimizer of f over D , then x^* solves $\text{VI}(D, \nabla f)$, that is,*

$$\nabla f(x^*)^\top(x - x^*) \geq 0 \quad \forall x \in D.$$

Proof. Let $x^* \in D$ be a local minimizer of f over D . By the first-order necessary optimality condition in tangent-cone form (Theorem 2.135), we have $\nabla f(x^*)^\top d \geq 0, \forall d \in T_D(x^*)$.

Because D is convex, for any point $x \in D$, the entire line segment joining x^* and x is contained in D , that is, $x^* + t(x - x^*) \in D, \forall t \in [0, 1]$. In particular, for all sufficiently small $t > 0$, the point $x^* + t(x - x^*) \in D$, which shows that the displacement $d := x - x^*$ is a feasible first-order direction at x^* . Equivalently, d belongs to the tangent cone $T_D(x^*)$.

Substituting $d = x - x^*$ into the tangent-cone optimality condition yields $\nabla f(x^*)^\top(x - x^*) \geq 0, \forall x \in D$. This is precisely the variational inequality condition stated in the theorem. $\square$

Remark 2.161 (Failure for nonconvex feasible sets). The convexity assumption on D cannot be dropped. If D is not convex, there may exist points $x \in D$ such that the displacement $x - x^*$ does not belong to the tangent cone $T_D(x^*)$.

For example, let $D = \{(-1, 0), (1, 0)\} \subset \mathbb{R}^2$ and let $x^* = (-1, 0)$. Then x^* is an isolated point of D , and hence $T_D(x^*) = \{0\}$. However, the displacement $x - x^* = (2, 0)$ corresponding to the other feasible point $x = (1, 0)$ does not belong to $T_D(x^*)$.

This example shows that when D is nonconvex, local first-order optimality conditions apply only to directions in the tangent cone and cannot, in general, be expressed as a variational inequality over all $x \in D$. Convexity is precisely the property that allows local first-order geometry to be written as a global variational inequality.

For convex feasible sets, local optimality implies a variational inequality that holds globally over the entire feasible set. The variational inequality formulation is equivalent to the normal-cone condition derived earlier.

Theorem 2.162 (VI and normal cone equivalence). *Let $D \subset \mathbb{R}^n$ be closed and convex, and let $F : D \rightarrow \mathbb{R}^n$. A point $x^* \in D$ solves $\text{VI}(D, F)$ if and only if $-F(x^*) \in N_D(x^*)$.*

Proof. By definition of the normal cone for convex sets, $N_D(x^*) = \{v \in \mathbb{R}^n : v^\top(x - x^*) \leq 0 \forall x \in D\}$. Thus,

$$-F(x^*) \in N_D(x^*) \iff F(x^*)^\top(x - x^*) \geq 0 \forall x \in D,$$

which is precisely the variational inequality condition. $\square$

The variational inequality formulation highlights the geometric meaning of optimality. At a solution point, the gradient of the objective cannot point into the feasible set in any direction that

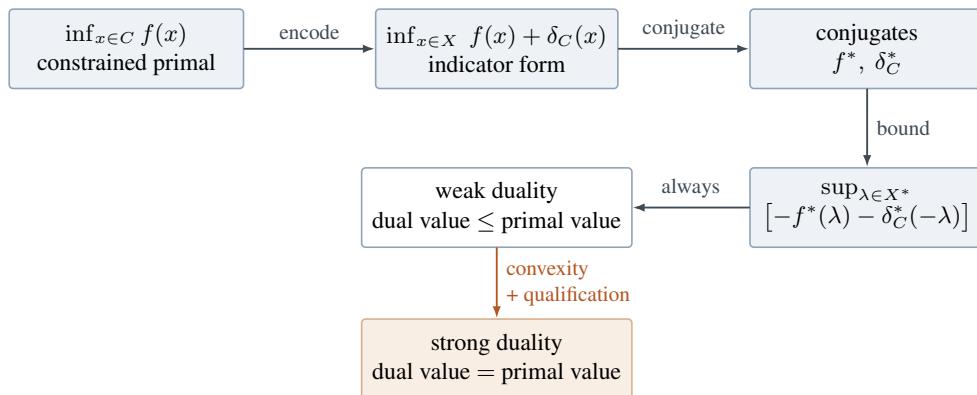


Figure 2.21: Fenchel duality rewrites constraints through an indicator function, passes to conjugates, and produces dual lower bounds; convexity and a qualification condition remove the gap.

would locally reduce the objective value. Equivalently, every admissible first-order displacement forms a nonacute angle with the gradient, so that no feasible infinitesimal motion leads to descent. This geometric viewpoint unifies gradient-based optimality conditions with the supporting hyperplane interpretation of convex sets.

Beyond optimization, variational inequalities arise naturally as equilibrium conditions in a wide range of applications. In economics and game theory, they describe market equilibria and Nash equilibria through balance conditions between competing agents. In transportation and network flow models, they characterize user-equilibrium traffic patterns in which no traveler can reduce cost by unilaterally changing routes. In mechanics, variational inequalities govern contact and obstacle problems, where constraints prevent penetration or overlap. In control and dynamical systems, they capture constrained evolution laws and steady-state equilibria under inequality constraints.

In subsequent chapters, the variational inequality formulation will serve as a conceptual bridge to Euler–Lagrange equations, complementarity systems, and Hamiltonian optimality conditions in continuous-time problems. This framework provides a unifying language that connects finite-dimensional optimization, variational analysis, and optimal control.

2.6 - Duality for the General Constrained Problem

Optimization problems with constraints admit two complementary descriptions: a *primal* formulation expressed in the original decision space, and a *dual* formulation expressed in a space of linear functionals acting on that space. The dual problem encodes supporting hyperplanes, separation, and variational structure, and often provides structural insight, optimality conditions, and computational advantages.

To develop this viewpoint in full generality, we begin without imposing convexity assumptions. Convexity will be introduced later when strong duality and optimality characterizations are discussed. At this stage, the goal is purely structural: to understand how a constrained problem can be rewritten so that linear functionals naturally appear as dual variables. Figure 2.21 previews the route taken in this section: the constraint is absorbed into an indicator function, the problem is passed to conjugates, and the conjugates produce a lower bound (weak duality) that becomes tight (strong duality) once convexity and a qualification condition hold.

2.6.1 ■ Problem formulation

Let X be a real locally convex topological vector space. Its continuous dual, denoted by X^* , is the vector space of all real-valued linear functionals on X that are continuous with respect to the given topology. For $\lambda \in X^*$ and $x \in X$, the duality pairing is written $\langle \lambda, x \rangle := \lambda(x)$.

Let $f : X \rightarrow (-\infty, +\infty]$ be an arbitrary extended-real-valued function, and let $C \subset X$ be nonempty. We consider the constrained optimization problem

$$\inf_{x \in C} f(x). \quad (\text{P})$$

Introducing the indicator function

$$\delta_C(x) = \begin{cases} 0, & x \in C, \\ +\infty, & x \notin C, \end{cases}$$

problem (P) can be written equivalently as

$$\inf_{x \in X} [f(x) + \delta_C(x)]. \quad (\text{P}')$$

2.6.2 ■ Conjugate Functions

Definition 2.163 (Fenchel conjugate). Let $g : X \rightarrow (-\infty, +\infty]$ be an extended-real-valued function. The Fenchel conjugate (or Legendre–Fenchel transform) of g is the function $g^* : X^* \rightarrow (-\infty, +\infty]$ defined by

$$g^*(\lambda) = \sup_{x \in X} (\langle \lambda, x \rangle - g(x)), \quad \lambda \in X^*. \quad (2.21)$$

The value $g^*(\lambda)$ measures how well the affine functional $x \mapsto \langle \lambda, x \rangle$ can dominate g . Geometrically, $g^*(\lambda)$ is the support function of the epigraph of g evaluated in the direction $(\lambda, -1)$.

Definition 2.164 (Proper function). Let $g : X \rightarrow (-\infty, +\infty]$ be an extended-real-valued function. We say that g is proper if

- (1) $g(x) > -\infty$ for all $x \in X$, and
- (2) there exists at least one point $x_0 \in X$ such that $g(x_0) < +\infty$.

Equivalently, g is proper if it is not identically $+\infty$ and never takes the value $-\infty$.

Proposition 2.165 (Basic properties). For any function $g : X \rightarrow (-\infty, +\infty]$, the conjugate g^* satisfies:

- (1) g^* is convex and lower semicontinuous.
- (2) (Fenchel–Young inequality)

$$g(x) + g^*(\lambda) \geq \langle \lambda, x \rangle \quad \forall x \in X, \lambda \in X^*. \quad (2.22)$$

- (3) If g is proper, convex, and lower semicontinuous, then $g^{**} = g$.

Proof. We prove the statements in order.

(1) Convexity of g^* .

For each fixed $x \in X$, define $\phi_x(\lambda) := \langle \lambda, x \rangle - g(x)$ for $\lambda \in X^*$. For fixed x , the mapping $\lambda \mapsto \phi_x(\lambda)$ is affine and continuous on X^* , since it is the sum of a continuous linear functional and a constant.

By definition, $g^*(\lambda) = \sup_{x \in X} \phi_x(\lambda)$, so g^* is the pointwise supremum of affine functions. Since the supremum of affine functions is convex, it follows that g^* is convex.

(2) Lower semicontinuity of g^* .

Each function ϕ_x is continuous in λ . A pointwise supremum of continuous functions is always lower semicontinuous. Hence g^* is lower semicontinuous on X^* .

(3) Fenchel–Young inequality.

Fix $x \in X$ and $\lambda \in X^*$. By definition of g^* as the supremum $g^*(\lambda) = \sup_{y \in X} (\langle \lambda, y \rangle - g(y))$, retaining only the term $y = x$ gives $g^*(\lambda) \geq \langle \lambda, x \rangle - g(x)$. Rearranging yields $g(x) + g^*(\lambda) \geq \langle \lambda, x \rangle$, which proves the Fenchel–Young inequality.

(4) Biconjugation inequality.

Define the biconjugate

$$g^{**}(x) = \sup_{\lambda \in X^*} (\langle \lambda, x \rangle - g^*(\lambda)).$$

Applying the Fenchel–Young inequality $g(x) + g^*(\lambda) \geq \langle \lambda, x \rangle$ for all $\lambda \in X^*$ implies $\langle \lambda, x \rangle - g^*(\lambda) \leq g(x)$. Taking the supremum over λ yields $g^{**}(x) \leq g(x) \forall x \in X$.

(5) Equality under convexity.

If g is proper, convex, and lower semicontinuous, then the Fenchel–Moreau theorem asserts that $g^{**} = g$.

In general, without convexity or lower semicontinuity, g^{**} coincides with the greatest lower semicontinuous convex function that is everywhere less than or equal to g . Equivalently, g^{**} is the lower semicontinuous convex envelope of g . $\square$

In particular, for the objective function f we have $f^*(\lambda) = \sup_{x \in X} (\langle \lambda, x \rangle - f(x))$, and for the indicator function δ_C , $\delta_C^*(\lambda) = \sup_{x \in C} \langle \lambda, x \rangle$, which is the support function of C .

Example 2.166 Quadratic function in $\mathbb{R}^n$.

Let $X = \mathbb{R}^n$ and $g(x) = \frac{1}{2} \|x\|^2$. Then

$$g^*(\lambda) = \sup_{x \in \mathbb{R}^n} \left(\langle \lambda, x \rangle - \frac{1}{2} \|x\|^2 \right).$$

The supremum is attained at $x = \lambda$, yielding $g^*(\lambda) = \frac{1}{2} \|\lambda\|^2$. Thus the quadratic function is self-conjugate. $\square$

Example 2.167 Indicator function.

Let $C \subset X$ be nonempty. The indicator function

$$\delta_C(x) = \begin{cases} 0, & x \in C, \\ +\infty, & x \notin C, \end{cases}$$

has conjugate

$$\delta_C^*(\lambda) = \sup_{x \in C} \langle \lambda, x \rangle,$$

which is the support function of C . If C is bounded, this function is finite everywhere. If C is unbounded in the direction λ , then $\delta_C^*(\lambda) = +\infty$. $\square$

Example 2.168 Norm and dual norm.

Let X be a normed space and define $g(x) = \|x\|$. Then

$$g^*(\lambda) = \begin{cases} 0, & \|\lambda\|_* \leq 1, \\ +\infty, & \text{otherwise,} \end{cases}$$

where $\|\cdot\|_*$ denotes the dual norm. Thus the conjugate of a norm is the indicator function of the dual unit ball. $\square$

Example 2.169 Linear function.

Let $g(x) = \langle \lambda_0, x \rangle$. Then

$$g^*(\lambda) = \begin{cases} 0, & \lambda = \lambda_0, \\ +\infty, & \lambda \neq \lambda_0. \end{cases}$$

Thus the conjugate of a continuous linear functional is the indicator function of the single point λ_0 in the dual space. $\square$

2.6.3 ■ The Fenchel Dual Problem

We return to the unconstrained reformulation

$$\inf_{x \in X} [f(x) + \delta_C(x)]. \quad (P')$$

This problem is of the general Fenchel type $\inf_{x \in X} [g(x) + h(x)]$, with $g = f$ and $h = \delta_C$.

Theorem 2.170 (Fenchel duality – weak form). *Let $g, h : X \rightarrow (-\infty, +\infty]$ be extended-real-valued functions. Then*

$$\sup_{\lambda \in X^*} [-g^*(\lambda) - h^*(-\lambda)] \leq \inf_{x \in X} [g(x) + h(x)]. \quad (2.23)$$

Proof. For any $x \in X$ and $\lambda \in X^*$, the Fenchel–Young inequality gives $g(x) + g^*(\lambda) \geq \langle \lambda, x \rangle$ and $h(x) + h^*(-\lambda) \geq \langle -\lambda, x \rangle$. Adding the two inequalities yields $g(x) + h(x) + g^*(\lambda) + h^*(-\lambda) \geq 0$, and rearranging gives $-g^*(\lambda) - h^*(-\lambda) \leq g(x) + h(x)$. Taking the infimum over x and the supremum over λ yields (2.23). $\square$

Applying this result to (P') with $g = f$ and $h = \delta_C$, we obtain the associated dual problem:

Definition 2.171 (Fenchel dual problem). *The Fenchel dual of*

$$\inf_{x \in X} [f(x) + \delta_C(x)]$$

is

$$\sup_{\lambda \in X^*} [-f^*(\lambda) - \delta_C^*(-\lambda)]. \quad (\text{D})$$

Since $\delta_C^*(\lambda) = \sup_{x \in C} \langle \lambda, x \rangle$, we obtain the equivalent representation

$$\sup_{\lambda \in X^*} \left[-f^*(\lambda) - \sup_{x \in C} \langle -\lambda, x \rangle \right]. \quad (2.24)$$

Using the identity $\sup_{x \in C} \langle -\lambda, x \rangle = -\inf_{x \in C} \langle \lambda, x \rangle$, the dual problem may also be written as

$$\sup_{\lambda \in X^*} \left[-f^*(\lambda) + \inf_{x \in C} \langle \lambda, x \rangle \right]. \quad (2.25)$$

The next result expresses the fundamental inequality underlying Fenchel duality. It asserts that every dual feasible point produces a global lower bound for the primal objective. This inequality is entirely algebraic and does not require convexity. Convexity enters only when one seeks equality of primal and dual values.

Example 2.172 Linear equality constraints.

Let $X = \mathbb{R}^n$ and $C = \{x \in \mathbb{R}^n : Ax = b\}$, where $A \in \mathbb{R}^{m \times n}$.

The support function of C satisfies

$$\delta_C^*(\lambda) = \begin{cases} \langle \lambda, x_0 \rangle, & \text{if } \lambda \in \text{Range}(A^\top), \\ +\infty, & \text{otherwise,} \end{cases}$$

where x_0 is any point satisfying $Ax_0 = b$.

Equivalently, introducing a multiplier $\mu \in \mathbb{R}^m$ with $\lambda = A^\top \mu$, we have, by the dual form (D), $\delta_C^*(-\lambda) = \sup_{Ax=b} \langle -\mu, Ax \rangle = -\langle \mu, b \rangle$, so the dual becomes

$$\sup_{\mu \in \mathbb{R}^m} [-f^*(A^\top \mu) + \langle \mu, b \rangle],$$

which coincides with the classical Lagrangian dual (after the relabeling $\mu \mapsto -\mu$, this is $\sup_{\mu} [-f^*(-A^\top \mu) - \langle \mu, b \rangle]$). $\square$

Example 2.173 Norm ball constraint.

Let X be a normed space and $C = \{x \in X : \|x\| \leq 1\}$. Then

$$\delta_C^*(\lambda) = \sup_{\|x\| \leq 1} \langle \lambda, x \rangle = \|\lambda\|_*,$$

where $\|\cdot\|_*$ denotes the dual norm.

The dual problem becomes

$$\sup_{\lambda \in X^*} [-f^*(\lambda) - \|\lambda\|_*].$$

Thus norm constraints in the primal correspond to norm penalties in the dual. $\square$

2.6.4 ■ Weak and Strong Duality

Proposition 2.174 (Weak duality). *Let $f : X \rightarrow (-\infty, +\infty]$ be an arbitrary extended-real-valued function and let $C \subset X$ be nonempty. Then for every $\lambda \in X^*$,*

$$-f^*(\lambda) - \delta_C^*(-\lambda) \leq \inf_{x \in C} f(x). \quad (2.26)$$

Consequently,

$$\sup_{\lambda \in X^*} [-f^*(\lambda) - \delta_C^*(-\lambda)] \leq \inf_{x \in C} f(x). \quad (2.27)$$

Proof. Fix $\lambda \in X^*$. By definition of the Fenchel conjugate, $f^*(\lambda) = \sup_{y \in X} (\langle \lambda, y \rangle - f(y))$, so for every $x \in X$ we have $f^*(\lambda) \geq \langle \lambda, x \rangle - f(x)$. Rearranging yields

$$f(x) \geq \langle \lambda, x \rangle - f^*(\lambda), \quad \forall x \in X. \quad (2.28)$$

Restricting (2.28) to $x \in C$ and taking the infimum over $x \in C$ gives $\inf_{x \in C} f(x) \geq \inf_{x \in C} (\langle \lambda, x \rangle - f^*(\lambda))$. Since $f^*(\lambda)$ does not depend on x , the right-hand side equals $\inf_{x \in C} \langle \lambda, x \rangle - f^*(\lambda)$. By definition of the conjugate of the indicator function, $\delta_C^*(-\lambda) = \sup_{x \in C} \langle -\lambda, x \rangle = -\inf_{x \in C} \langle \lambda, x \rangle$, so $\inf_{x \in C} \langle \lambda, x \rangle = -\delta_C^*(-\lambda)$. Combining these yields $\inf_{x \in C} f(x) \geq -f^*(\lambda) - \delta_C^*(-\lambda)$, which proves (2.26). Taking the supremum over $\lambda \in X^*$ preserves the inequality and yields (2.27). $\square$

Weak duality always holds, but equality of the primal and dual values requires additional structure. In particular, convexity and a suitable qualification condition ensure that no duality gap occurs.

Example 2.175 Weak duality with nonzero duality gap.

Let $X = \mathbb{R}$ and take $f(x) = -x^2$ with $C = [-1, 1]$.

The primal problem is $\inf_{x \in [-1, 1]} (-x^2)$. Since $-x^2$ is minimized at $x = \pm 1$, we obtain $\inf_{x \in C} f(x) = -1$.

We now compute the dual. The conjugate of f is

$$f^*(\lambda) = \sup_{x \in \mathbb{R}} (\lambda x + x^2).$$

Since the quadratic is unbounded above, $f^*(\lambda) = +\infty$ for all $\lambda \in \mathbb{R}$. Hence $-f^*(\lambda) = -\infty$, and the dual problem becomes $\sup_{\lambda \in \mathbb{R}} (-\infty) = -\infty$.

Thus the dual value equals $-\infty$, whereas the primal value is -1 . The duality gap is therefore infinite.

This example shows that without convexity, weak duality may be strict. $\square$

Define the *duality gap* by

$$\inf_{x \in C} f(x) - \sup_{\lambda \in X^*} [-f^*(\lambda) - \delta_C^*(-\lambda)].$$

Strong duality holds when this gap is zero.

Theorem 2.176 (Convex strong duality). *Let $f : X \rightarrow (-\infty, +\infty]$ be proper, convex, and lower semicontinuous, and let $C \subset X$ be nonempty, closed, and convex. Assume the qualification condition*

$$\text{ri}(\text{dom } f) \cap \text{ri}(C) \neq \emptyset, \quad (2.29)$$

where ri denotes relative interior. Then

$$\inf_{x \in C} f(x) = \sup_{\lambda \in X^*} [-f^*(\lambda) - \delta_C^*(-\lambda)].$$

Remark 2.177 (Reading the qualification condition). Condition (2.29) is stated in the form that is standard when X is finite-dimensional, where every nonempty convex set has nonempty relative interior. In a general locally convex space the relative interior may be empty, and (2.29) must be replaced by a genuinely topological qualification: it suffices, for instance, that f be continuous at some point of C , or that $\text{int}(\text{dom } f) \cap C \neq \emptyset$. All the applications in this book are covered by one of these two forms.

Proof. The statement is a direct consequence of the Fenchel–Rockafellar duality theorem applied to the pair f and δ_C . Under convexity and the qualification condition (2.29), the infimum of the primal problem equals the supremum of the dual problem. $\square$

Example 2.178 Strong duality in the convex case.

Let $X = \mathbb{R}$ and take $f(x) = \frac{1}{2}x^2$ with $C = [1, +\infty)$.

The primal problem is $\inf_{x \geq 1} \frac{1}{2}x^2$. Since f is increasing on $[0, \infty)$, the minimizer is $x^* = 1$ and $\inf_{x \in C} f(x) = \frac{1}{2}$.

We compute the dual. The conjugate of f is $f^*(\lambda) = \frac{1}{2}\lambda^2$.

The support function of C is $\delta_C^*(-\lambda) = \sup_{x \geq 1}(-\lambda x)$. If $\lambda \geq 0$, then $\sup_{x \geq 1}(-\lambda x) = -\lambda$; if $\lambda < 0$, the supremum is $+\infty$. Thus the dual reduces to

$$\sup_{\lambda \geq 0} \left[-\frac{1}{2}\lambda^2 + \lambda \right].$$

Maximizing the concave quadratic gives $\lambda^* = 1$ with dual value $\frac{1}{2}$.

Hence the dual value equals the primal value, and the duality gap is zero.

□

2.7 ■ Further Reading

This chapter draws on foundational ideas from real analysis, linear algebra, and nonlinear optimization. Concepts such as continuity, Fréchet differentiability, gradients, Hessians, and cone-based optimality conditions are most naturally understood within the broader mathematical frameworks provided by these subjects. The following references are recommended for readers who wish to deepen their understanding.

A rigorous treatment of limits, continuity, and differentiation underlies all first- and second-order optimality conditions. Classic references include Rudin [104] and Royden and Fitzpatrick [103], which develop the ε – δ foundations of analysis in $\mathbb{R}^n$. For a more modern perspective emphasizing normed spaces, linear operators, and Fréchet differentiability, Brezis [28] provides an authoritative introduction that is particularly relevant for infinite-dimensional extensions.

Gradients, Hessians, and second-order conditions rely fundamentally on linear algebraic structures such as inner products, symmetric matrices, eigenvalues, and quadratic forms. Strang [113] offers an intuitive and geometric introduction, while Horn and Johnson [57] provide a comprehensive and rigorous reference on matrix analysis. For an optimization-oriented treatment of linear algebra, Boyd and Vandenberghe [26] give an especially useful discussion of positive (semi)definite matrices and their role in curvature, convexity, and stability.

The systematic use of derivatives, feasible directions, tangent cones, and normal cones in optimization is developed in detail in Luenberger [79] and Ruszczyński [106]. These texts extend the local optimality conditions presented here to general nonlinear programming problems and introduce constraint qualifications, Lagrange multipliers, and second-order theory in a unified manner.

A particularly important unifying framework is provided by variational inequalities, which reformulate first-order optimality conditions as geometric projection or equilibrium problems involving monotone operators and normal cones. This perspective is developed systematically in Facchinei and Pang [45], where variational inequalities are shown to encompass constrained optimization, complementarity problems, and a wide class of equilibrium models arising in economics, mechanics, and network systems. Variational inequalities also provide a natural bridge to optimal control, differential inclusions, and constrained dynamical systems.

As a general guide, students seeking strong analytical foundations should begin with Rudin [104]. For linear algebraic intuition behind gradients and Hessians, Strang [113] is recommended. Luenberger [79] serves as an effective bridge between analysis and optimization, while Ruszczyński [106] and Facchinei and Pang [45] provide deeper and more advanced treatments, particularly for readers interested in variational inequalities and equilibrium formulations.

2.8 ■ Exercises

Exercise 2.1 (Infimum and Supremum).

- (i) Find the infimum and supremum of the real-valued function $f(x_1, x_2) = x_1 + x_2$ over the set $D = \{x : x_1^2 + x_2^2 \leq 1, x_1 > 0, x_2 > 0\}$.
- (ii) Does there exist a minimizer and a maximizer to this problem?

Exercise 2.2 (Limsup and Liminf). Let $\{x_n\}_{n=1}^{\infty}$ be a sequence of real numbers. Define $y_m = \sup_{n \geq m} x_n$ with the convention that $y_m = \infty$ if $\{x_n\}_{n=1}^{\infty}$ is unbounded above. Note that the sequence $\{y_m\}_{m=1}^{\infty}$ is a decreasing sequence, and it allows us to define the following:

$$\limsup_{n \rightarrow \infty} x_n = \begin{cases} \infty & \text{if } \{x_n\}_{n=1}^{\infty} \text{ is unbounded above} \\ -\infty & \text{if } \{y_m\}_{m=1}^{\infty} \text{ is unbounded below} \\ \lim_{m \rightarrow \infty} y_m & \text{otherwise.} \end{cases}$$

In the same way, we can define $z_m = \inf_{n \geq m} x_n$, and

$$\liminf_{n \rightarrow \infty} x_n = \begin{cases} -\infty & \text{if } \{x_n\}_{n=1}^{\infty} \text{ is unbounded below} \\ +\infty & \text{if } \{z_m\}_{m=1}^{\infty} \text{ is unbounded above} \\ \lim_{m \rightarrow \infty} z_m & \text{otherwise.} \end{cases}$$

With the definitions above, calculate $\limsup_{n \rightarrow \infty} a_n$ and $\liminf_{n \rightarrow \infty} a_n$ for $a_n = (-1)^n(n + 5)/n$.

Exercise 2.3 (Limits of nonnegative and positive functions). Let $q : (0, \varepsilon_0) \rightarrow \mathbb{R}$ be a function such that $q(\varepsilon) \geq 0 \forall \varepsilon \in (0, \varepsilon_0)$, and assume that the (one-sided) limit $\lim_{\varepsilon \downarrow 0} q(\varepsilon)$ exists and is finite.

- (i) Show that $\lim_{\varepsilon \downarrow 0} q(\varepsilon) \geq 0$.
- (ii) Is the conclusion in part (i) still true if the assumption $q(\varepsilon) \geq 0$ is replaced by $q(\varepsilon) > 0$ for all $\varepsilon > 0$? Justify your answer by either a proof or a counterexample.

Exercise 2.4 (Review of Linear Algebra). Consider the following matrix:

$$\mathbf{A} = \begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

- (i) Find the eigenvalues and eigenvectors of matrix $\mathbf{A}$.
- (ii) Find the determinant and trace of $\mathbf{A}$.
- (iii) Is matrix $\mathbf{A}$ positive semi-definite?
- (iv) Find the range space and the null space of matrix $\mathbf{A}$.

Exercise 2.5 (Matrix Decomposition). Consider a symmetric matrix,

$$\mathbf{B} = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

Find $\mathbf{P}$ and $\mathbf{\Lambda}$ such that $\mathbf{B} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{-1}$, where the columns of $\mathbf{P}$ are orthonormal and $\mathbf{\Lambda}$ is a diagonal matrix.

Exercise 2.6 (Fréchet and Gâteaux Differentials). Let $X = \mathbb{R}^n$ and let $f(x) = f(x_1, x_2, \dots, x_n)$ be a functional on $\mathbb{R}^n$ having continuous partial derivatives with respect to each variable x_i .

- (i) Show that the Gâteaux differential of f is $\delta f(x; h) = \sum_{i=1}^n \frac{\partial f}{\partial x_i} h_i$.
- (ii) Show that the Fréchet differential of f is the same as the Gâteaux differential.

Exercise 2.7 (Unconstrained Optimization). Consider the following unconstrained minimization problem $\min_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) := x_1^2 - x_1 x_2 + x_2^2 - 3x_2$. Find a local minimum. Is the local minimum also a global minimum?

Exercise 2.8 (Weierstrass' Theorem). Does there exist a minimizer and a maximizer for the function $f(\mathbf{x}) = x_1^3 + 2x_2^2 + 3x_3 + 4x_4^5$ over the set $D = \{\mathbf{x} \in \mathbb{R}^4 : x_1^2 + 2x_2^2 + 3x_3^2 + 4x_4^2 \leq 1\}$.

Exercise 2.9. Let $f(\mathbf{x}) = -2x_1^2 - 2x_1 x_2 - 2x_2^2 + x_1 + x_2 + 3$.

- (i) Find the minimum and the maximum of f over $\mathbb{R}^2$ if they exist.
- (ii) Show that $f(\mathbf{x})$ can be made to approach $-\infty$ by choosing $\mathbf{x}$ appropriately.

Exercise 2.10 (Open Sets). Identify which of the following sets are open, which are closed, and which are neither.

- (i) $E = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$
- (ii) $E = \{(x, y) \in \mathbb{R}^2 : x^2 - 2x + y^2 = 0\} \cup \{(x, 0) \in \mathbb{R}^2 : x \in [2, 3]\}$
- (iii) $E = \{(x, y) \in \mathbb{R}^2 : y \leq x^2, 0 \leq y < 1\}$
- (iv) $E = \{(x, y) \in \mathbb{R}^2 : x^2 - y^2 < 1, -1 < y < 1\}$

Exercise 2.11 (Open Sets). Let $n \in \mathbb{N}$, $\mathbf{a} \in \mathbb{R}^n$. Let $s, r \in \mathbb{R}$ with $s < r$, and set $V = \{\mathbf{x} \in \mathbb{R}^n : s < \|\mathbf{x} - \mathbf{a}\| < r\}$. Use definition to prove that V is open.

Exercise 2.12 (Continuity). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2 - 3x + 5$. Use the definition of continuity to prove that f is continuous at $x = 2$.

Exercise 2.13 (Continuous Functions). Let $K > 0$, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the condition

$$|f(x) - f(y)| \leq K|x - y|, \quad (2.30)$$

for all $x, y \in \mathbb{R}$.

- (i) Show that f is continuous on $\mathbb{R}$.
- (ii) A function that satisfies condition (2.30) is said to be Lipschitz continuous. Show that (1) $f(x) = \frac{1}{1+x^2}$ and (2) $f(x) = \sqrt{1+x^2}$ are Lipschitz continuous.

Exercise 2.14 (Separation Theorems). ([BV] Problem 2.20) Suppose $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, with $\mathbf{b} \in \text{Range}(\mathbf{A})$. Show that there exists an $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{x} \geq 0$, $\mathbf{A}\mathbf{x} = \mathbf{b}$ if and only if there exists no λ with $\mathbf{A}^T \lambda \geq 0$, $\mathbf{A}^T \lambda \neq 0$, $\mathbf{b}^T \lambda \leq 0$.

Exercise 2.15. Method of Lagrange may not produce candidates for optimality if the gradients of constraints are linearly dependent at a local optimum. Verify this claim for the program $\max_t t$, s.t. $t^2 = 0$.

Exercise 2.16. Consider optimizing x_1/x_3 subject to $x_1 + x_2 - 1 \leq 0$ and $-x_1 + x_2x_3 + 1 \leq 0$, $x_1 \geq 0, x_2 \geq 0, x_3 \geq \frac{1}{2}$. Show that the point $x_1^* = 1, x_2^* = 0, x_3^* = 1$ satisfies various necessary conditions for local optimality, but it is not locally optimal. Sketch the feasible set of this program in $\mathbb{R}^3$. Find a better feasible point than x^* .

Exercise 2.17 (Gordan's Lemma). Let A be an $m \times n$ matrix. Then exactly one of the following two conditions hold:

- (i) There is an $x \in \mathbb{R}^n$ such that $Ax < 0$ (all components of Ax are strictly negative.)
- (ii) There is a $\mu \in \mathbb{R}^m$ such that $A^T\mu = 0, \mu \neq 0$, and $\mu \geq 0$.

Hint: You can assume that there exist x and u satisfying simultaneously (i) and (ii), and reach a contradiction. Use Prop. 3.3.10 in [15] for a suitable minimax problem to show that if (i) fails to hold, then (ii) must hold.

Exercise 2.18. Consider $f(x) = 4x_1^2 - \sin(x_1 + x_2) + 2x_2^2$. The gradient of the function is found to be zero at $x_1^* = 0.11734, x_2^* = 0.23467$. Verify that this point is also an isolated local minimum. Is it a global minimum?

Exercise 2.19. Function $f(x) = (1 + \exp(x_2)) \cos x_1 - x_2 \exp(x_2)$ has infinitely many local maxima but not a single minimum. Prove this claim using an optimal condition.

Exercise 2.20. Find all locally optimal solutions of the following problem:

$$\begin{aligned} \text{Opt } f(x) &= x_1x_2^2 \\ \text{s.t. } x_1 + x_2 &= 10. \end{aligned}$$

Exercise 2.21. Consider function $f(x) = x_1(x_1 - 2) + x_2^2(x_2 - 1)^2$. Are $x^* = (1, 1)^T$ and $y^* = (1, 0)^T$ its isolated global minima?

Exercise 2.22 (Equivalence of Norms and Unit Balls). On $\mathbb{R}^n$, recall the norms $\|x\|_1 = \sum_i |x_i|$, $\|x\|_2 = (\sum_i x_i^2)^{1/2}$, and $\|x\|_\infty = \max_i |x_i|$.

- (i) Prove the chain of inequalities $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ for every $x \in \mathbb{R}^n$, and show that each inequality is tight (attained by some nonzero x).
- (ii) Find the smallest constants c_1, c_2 (as functions of n) for which $\|x\|_1 \leq c_1\|x\|_2$ and $\|x\|_2 \leq c_2\|x\|_\infty$ hold for all x .
- (iii) Sketch the three unit balls in $\mathbb{R}^2$ and confirm that a smaller norm value corresponds to a larger ball. Give a point x with $\|x\|_\infty \leq 1$ but $\|x\|_1 > 1$.

Exercise 2.23 (Operator Norm of a Bounded Linear Map). Let $A \in \mathbb{R}^{m \times n}$ and regard $T(x) = Ax$ as a linear map from $(\mathbb{R}^n, \|\cdot\|_2)$ to $(\mathbb{R}^m, \|\cdot\|_2)$.

- (i) Show that the operator norm $\|T\|_{\text{op}} = \sup_{\|x\|_2 \leq 1} \|Ax\|_2$ equals the largest singular value of A .
- (ii) Let $X = Y = C[0, 1]$ with the supremum norm and let $(Tx)(t) = \int_0^1 k(t, \tau)x(\tau) d\tau$ for a continuous kernel k . Show that $\|T\|_{\text{op}} = \max_{t \in [0, 1]} \int_0^1 |k(t, \tau)| d\tau$.

Exercise 2.24 (Completeness and Cauchy Sequences). Let $X = C[0, 1]$ be the space of continuous functions on $[0, 1]$.

- (i) Equip X with the norm $\|x\|_1 = \int_0^1 |x(t)| dt$. Exhibit a Cauchy sequence in $(X, \|\cdot\|_1)$ that does not converge in X , and conclude that $(X, \|\cdot\|_1)$ is not a Banach space.
- (ii) Equip X instead with the supremum norm $\|x\|_\infty$. Explain why the same collection of functions is now complete. What structural property of uniform limits is responsible?

Exercise 2.25 (Coercivity and Attainment). Determine whether each functional attains a minimum over the indicated set. Justify your answer using coercivity, lower semicontinuity, and compactness as appropriate.

- (i) $f(x) = x_1^2 + x_2^2 - 4x_1$ over $\mathbb{R}^2$.
- (ii) $f(x) = e^{-x_1} + x_2^2$ over $\mathbb{R}^2$.
- (iii) $f(x) = \int_0^1 x(t) dt$ over the closed unit ball of $(C[0, 1], \|\cdot\|_\infty)$.

Exercise 2.26 (Gâteaux versus Fréchet Differentiability). Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = \frac{x^2 y}{x^4 + y^2}$ for $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$.

- (i) Show that the Gâteaux differential $\delta f(0; h)$ exists for every direction $h \in \mathbb{R}^2$ and compute it.
- (ii) Show that f is not continuous at the origin, and conclude that f is not Fréchet differentiable there.
- (iii) Explain in one or two sentences why this example does not contradict the fact that continuous partial derivatives imply Fréchet differentiability.

Exercise 2.27 (Tangent and Normal Cones). Let $D = \{x \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0\}$ be the nonnegative orthant.

- (i) Compute the tangent cone $T_D(x^*)$ and the normal cone $N_D(x^*)$ at $x^* = (0, 0)$, at $x^* = (1, 0)$, and at an interior point $x^* = (1, 1)$.
- (ii) For the objective $f(x) = (x_1 - 2)^2 + (x_2 + 1)^2$, use the condition $-\nabla f(x^*) \in N_D(x^*)$ to find the minimizer of f over D , and verify it geometrically.

Exercise 2.28 (Conjugate Functions). Compute the convex conjugate $f^*(\lambda) = \sup_x (\lambda^\top x - f(x))$ for each of the following, stating its effective domain.

- (i) $f(x) = \frac{1}{2} x^\top Q x$ on $\mathbb{R}^n$, with Q symmetric positive definite.
- (ii) $f(x) = e^x$ on $\mathbb{R}$.
- (iii) $f(x) = \delta_C(x)$, the indicator of the closed unit ball $C = \{x : \|x\|_2 \leq 1\}$.

Exercise 2.29 (Fenchel Duality and Weak Duality). Consider the primal problem $\min_{x \in \mathbb{R}^n} \frac{1}{2} \|x\|_2^2$ subject to $Ax = b$, where $A \in \mathbb{R}^{m \times n}$ has full row rank and $b \in \mathbb{R}^m$.

- (i) Encode the constraint with an indicator function and derive the Fenchel dual using the conjugates computed above.
- (ii) Solve the dual explicitly and verify that strong duality holds, i.e., the optimal primal and dual values coincide.

(iii) Recover the primal minimizer x^* from the dual solution.

Exercise 2.30 (Gâteaux Differential of the Max Norm on $C[0, 1]$). Let $f(x) = \|x\| = \max_{0 \leq t \leq 1} |x(t)|$ on $C[0, 1]$. Determine a class of functions $D \subset C[0, 1]$ such that for $x \in D$ the Gâteaux differential $\delta f(x; h)$ exists and is linear in h . *Hint:* Show that if x attains a unique maximum of $|x(t)|$ at $t_o \in (0, 1)$ and $h \in C[0, 1]$, then $\lim_{\alpha \rightarrow 0} \frac{1}{\alpha} \{ \|x + \alpha h\| - |x(t_o) + \alpha h(t_o)| \} = 0$.

Exercise 2.31 (Gâteaux Differentials of Max and L^1 Functionals). (i) Repeat the previous exercise for $f(x) = \max_{0 \leq t \leq 1} x(t)$.

(ii) Do the same for $f(x) = \int_0^1 |x(t)| dt$.

Exercise 2.32 (Gâteaux Differentials and Derivatives of Transformations). Obtain the Gâteaux differentials and Gâteaux derivatives of the following transformations, indicating in each case whether the Gâteaux derivative is a linear operator.

(i) $T : C[0, 1] \rightarrow C[0, 1], T(x)(t) = x(t)^2$.

(ii) $T : C[0, 1] \rightarrow \mathbb{R}, T(x) = x(\frac{1}{2})^2$.

(iii) $T : L^2[0, 1] \rightarrow L^2[0, 1], T(x)(t) = \int_0^t x(s)^2 ds$.

(iv) $T : L^2[0, 1] \rightarrow \mathbb{R}, T(x) = \int_0^1 x(s)^2 ds$.

Exercise 2.33 (Sum of Subspaces). Let M and N be subspaces of a vector space. Show that $[M \cup N] = M + N$, where $[\cdot]$ denotes the subspace generated by a set and $+$ denotes the sum of subspaces.

Exercise 2.34 (Best Approximation by Independent Vectors). Let $(X, \|\cdot\|)$ be a normed linear space and $x_1, \dots, x_n$ linearly independent vectors in X . For a fixed $y \in X$, show that there exist coefficients $a_1, \dots, a_n$ minimizing $\|y - a_1x_1 - \dots - a_nx_n\|$. *Hint:* Restrict the search to a closed and bounded finite-dimensional set and apply the Weierstrass theorem. Then, for $X = L^2[0, 1]$, $n = 2$, $x_1(t) = 1$, $x_2(t) = t$, $y(t) = t^3$ on $[0, 1]$, obtain the optimal a_1, a_2 ; and discuss the much more involved case $X = C[0, 1]$ with the Chebyshev (maximum) norm.

Exercise 2.35 (Strictly Normed Spaces). A normed space is *strictly (strongly) normed* if $\|x + y\| = \|x\| + \|y\|$ implies $y = \theta$ or $x = \alpha y$ for some scalar α .

(i) Show that $L^p[0, 1]$ is strictly normed for $1 < p < \infty$.

(ii) Show that if X is strictly normed, the best approximation of the previous exercise is unique.

Exercise 2.36 (Continuity of the Max Functional). Let $f : C[0, 2] \rightarrow \mathbb{R}$ be defined by $f(x) = \max_{0 \leq t \leq 2} x(t)$ (supremum norm on $C[0, 2]$).

(i) Show that f is continuous.

(ii) Is f uniformly continuous?

Exercise 2.37 (Incompleteness of a Sequence Space). Let ℓ'_p consist of all finite ordered tuples $(\xi_1, \dots, \xi_k)$ (k a natural number, $\xi_i \in \mathbb{R}$), with metric $\rho(x, y) = (\sum_{i=1}^{k_1} |\xi_i - \nu_i|^p + \sum_{j=k_1+1}^{k_2} |\nu_j|^p)^{1/p}$, $1 \leq p < \infty$, for $x = (\xi_1, \dots, \xi_{k_1})$, $y = (\nu_1, \dots, \nu_{k_2})$, $k_2 \geq k_1$. Show that ℓ'_p is not a complete metric space. *Hint:* Consider $x_n = (1, \frac{1}{2}, \dots, \frac{1}{2^{n-1}})$.

Exercise 2.38 (Nonseparability of $L_p(-\infty, \infty)$). The space $L_p[a, b]$ is separable for $1 \leq p < \infty$ on a finite interval. This fails on $(-\infty, \infty)$ with the norm $\|x\| = \left(\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^p dt\right)^{1/p}$. Construct a counterexample showing that $L_2(-\infty, \infty)$ is not separable. *Hint:* Use the continuum $\{\sin \alpha t : \alpha \in (-\infty, \infty)\}$.

Exercise 2.39 (Fixed Point of a Nonlinear Map). Consider the fixed-point equation $x(t) = \frac{1}{2}t^3 + \alpha \sin \pi x(t)$ on $t \in [-2, 2]$, with $x \in C[-2, 2]$ and $\alpha > 0$. For what values of α does a *unique* continuous solution exist on $[-2, 2]$? Prove existence and uniqueness for those α . *Hint:* Apply a contraction-mapping argument on a suitable subset of $C[-2, 2]$.

Exercise 2.40 (Affine Iteration and Contraction). Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $T(x) = Ax + b$, with $A = (a_{ij})$ and $b \in \mathbb{R}^n$.

- (i) Under what conditions on A, b is T a contraction in the Euclidean norm?
- (ii) Same question for the maximum norm $\|x\| = \max_i |x_i|$; are the conditions more or less restrictive than in (i)?
- (iii) For the iteration $x_{(i+1)} = Ax_{(i)} + b$, deduce conditions for convergence to a fixed point of T .
- (iv) Verify the condition in (iii) and compute the fixed point (to 5 places) for $A = \begin{pmatrix} 0.5 & 0.2 & 0.2 \\ 0.6 & 0.2 & 0.1 \\ 0.2 & 0.7 & 0 \end{pmatrix}$, $b = (1, 2, 3)^\top$, starting from $x_{(0)} = \theta$ and from $x_{(0)} = (-1, 1, -1)^\top$.

Exercise 2.41 (Fredholm Integral Equation via Contraction). Consider $x(t) + \lambda \int_0^2 (t - s)x(s) ds = t^3$, $0 \leq t \leq 2$, with parameter λ .

- (i) On $L^2[0, 2]$, for what values of λ does it admit a unique solution?
- (ii) Is the solution continuous, i.e. in $C[0, 2]$?
- (iii) On $L^1[0, 2]$, for what λ does Banach's contraction mapping theorem give a unique solution, and would it differ from the one in (i)?

Exercise 2.42 (Infinite Linear System on ℓ^1). Consider $\lambda \sum_{m=1}^{\infty} a_{nm}x_m = 3^{-2n} + x_n$, $n = 1, 2, \dots$, on ℓ^1 , with parameter λ and real a_{nm} .

- (i) Under what conditions on λ and $\{a_{nm}\}$ does this admit a unique solution $x^\circ \in \ell^1$? (Use the contraction mapping theorem.)
- (ii) For $a_{nm} = \frac{1}{2n^2}$ ($m = n$) and $a_{nm} = \frac{1}{4m^2n^2}$ ($m \neq n$), obtain the conditions on λ for which $T(x)_n = \lambda \sum_m a_{nm}x_m - 3^{-2n}$ is (i) a contraction, (ii) nonexpansive.

Exercise 2.43 (Contractive Maps on a Compact Set). Let X be a Banach space and S a compact subset of X . Let T be a *contractive* (not necessarily contraction) mapping of S into itself, i.e. $\|Tx - Ty\| < \|x - y\|$ for $x \neq y$. Show that T has a unique fixed point, obtainable by the iteration $x_{(n+1)} = T(x_{(n)})$ from any $x_{(0)} \in S$.

Exercise 2.44 (Existence via Weierstrass). Determine whether each problem attains its minimum. Justify using the Weierstrass theorem or exhibit a minimizing sequence with no feasible limit.

- (i) $\min e^x$ over $x \in \mathbb{R}$.

- (ii) $\min x_1^2 + x_2^2$ over $\{x \in \mathbb{R}^2 : x_1 x_2 \geq 1, x_1 > 0\}$.
- (iii) $\min -x_1 - x_2$ over $\{x \in \mathbb{R}^2 : \|x\|_\infty \leq 1\}$.
- (iv) $\min 1/x$ over $(0, 1]$.

Exercise 2.45 (Coercivity). A continuous function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is coercive if $f(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$.

- (i) Prove that a continuous coercive function attains its minimum over any nonempty closed set $D \subset \mathbb{R}^n$.
- (ii) Show that $f(x_1, x_2) = e^{x_1} + y_2^2$ with $y_2 := x_2 - x_1$ is not coercive, and determine whether it attains its infimum over $\mathbb{R}^2$.
- (iii) Give an example showing that coercivity cannot be dropped in (i) even when f is bounded below.

Exercise 2.46 (Lower Semicontinuity). (i) Show that $f(x) = 0$ for $x \neq 0$ and $f(0) = -1$ is lower semicontinuous on $\mathbb{R}$ and attains its minimum on $[-1, 1]$, although it is not continuous.

- (ii) Show that $g(x) = |x|$ for $x \neq 0$ and $g(0) = 1$ is not lower semicontinuous, and exhibit a compact set (for example $[-1, 1]$) on which g fails to attain its infimum.
- (iii) Prove that the pointwise supremum of a family of continuous functions is lower semicontinuous.

Exercise 2.47 (Local versus Global Minima). (i) Construct a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ with exactly two local minimizers, neither of which is global.

- (ii) Show that for a convex function on $\mathbb{R}^n$, every local minimizer is global.
- (iii) Let $f(x) = x^2(1 + \sin^2(1/x))$ for $x \neq 0$ and $f(0) = 0$. Show that $x = 0$ is the unique global minimizer even though f oscillates in every neighborhood of it.

Chapter 3

Classical Optimality Conditions

This chapter develops the analytical foundations of constrained optimization, with emphasis on first- and second-order optimality conditions for nonlinear programs involving equality and inequality constraints. Beginning with the geometric notion of feasibility and local optimality, we derive the Fritz–John necessary conditions using theorems of the alternative and clarify the role of constraint qualifications in recovering the Karush–Kuhn–Tucker (KKT) framework. Particular attention is paid to the Mangasarian–Fromovitz constraint qualification and its function in excluding degenerate multiplier configurations.

The chapter then advances to second-order analysis, introducing critical directions and second-order tangent behavior as the appropriate tools for characterizing curvature of the objective along feasible directions. We present both second-order necessary and sufficient conditions in Fritz–John form, highlighting their inherently directionwise nature and the subtle interaction between active constraints and curvature. The distinction between local, strict, and isolated local optima is examined in detail, with illustrative examples demonstrating how first- and second-order conditions may hold even when a point fails to be locally optimal.

Throughout, the chapter emphasizes geometric intuition, structural limitations of derivative-based conditions, and the precise circumstances under which second-order conditions guarantee isolation of a local optimum. These results provide the theoretical backbone for modern nonlinear programming methods and motivate algorithmic approaches such as sequential quadratic programming.

3.1 ■ Constrained Optimization and Optimality Concepts

We consider a general constrained optimization problem in which the feasible set is specified algebraically by a collection of equality and inequality constraints:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \\ & g_j(x) \leq 0, \quad j \in \mathcal{Q}. \end{aligned} \tag{3.1}$$

Here, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective function to be minimized, $h_i : \mathbb{R}^n \rightarrow \mathbb{R}$ represent equality constraints indexed by the finite set $\mathcal{P}$, and $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ represent inequality constraints indexed by the finite set $\mathcal{Q}$. The constraints carve out the region of $\mathbb{R}^n$ over which the objective is allowed to be evaluated, so that optimization is no longer a search over all of $\mathbb{R}^n$ but only over this

admissible region. We collect it in the *feasible set*

$$\mathcal{F} = \{x \in \mathbb{R}^n \mid h_i(x) = 0, i \in \mathcal{P}, g_j(x) \leq 0, j \in \mathcal{Q}\}.$$

Any point $x \in \mathcal{F}$ is called a *feasible point*, while points that violate at least one constraint are *infeasible*. When at least one of the functions f , h_i , or g_j is nonlinear, problem (3.1) is called a *nonlinear program*. The central difficulty, and the theme of this chapter, is that optimality can no longer be read off from ∇f alone: the shape of $\mathcal{F}$ near a candidate point enters decisively.

3.1.1 ■ Optimality Notions

We now introduce standard notions of optimality for constrained optimization problems.

Global minimum. A feasible point $x^* \in \mathcal{F}$ is a *global minimum* (or *globally optimal solution*) of (3.1) if $f(x^*) \leq f(x)$ for all $x \in \mathcal{F}$, and the value $f(x^*)$ is then called the *global optimal value*.

Local minimum. A feasible point $x^* \in \mathcal{F}$ is a *local minimum* if $f(x^*) \leq f(x)$ holds only for x in some feasible neighborhood $\mathcal{F} \cap N(x^*)$; it is a *strict local minimum* if the inequality is strict, $f(x^*) < f(x)$ for all $x \in \mathcal{F} \cap N(x^*)$ with $x \neq x^*$, and an *isolated local minimum* if x^* is the only local minimum of the problem in some feasible neighborhood of itself.

These three notions are genuinely distinct and become progressively stronger. A local minimum may sit on a flat plateau of equally good neighbors. A strict local minimum is the unique *best* point nearby, but competing local minima may still accumulate at it: for $f(x) = x^4(2 + \sin(1/x))$, extended by $f(0) = 0$, the origin is a strict local minimum, yet every neighborhood of the origin contains further local minima. An isolated local minimum excludes even that. The second-order sufficient conditions of this chapter certify strict local minimality directly, and, as the quadratic-growth estimate in the proof of Theorem 3.20 shows, isolation as well.

Every global minimum is also a local minimum, but the converse fails in general. The essential feature of the constrained setting is that local optimality is a *local geometric* property: it depends only on how f behaves along directions that keep x inside $\mathcal{F}$ near x^* , never on the global landscape of f over all of $\mathbb{R}^n$. Making this notion of “admissible direction” precise is the task of the next subsections.

3.1.2 ■ Regularity Conditions

A strict hierarchy of regularity

The analysis of constrained optimization problems relies on different levels of smoothness of the objective and constraint functions. These notions form a strict hierarchy of regularity:

$$f \in C^1(U) \implies f \text{ is differentiable on } U \implies f \text{ is continuous on } U,$$

where the converses do not generally hold. Each level of regularity enables progressively stronger analytical tools.

Continuity is the weakest regularity notion and ensures that small perturbations of the argument produce small changes in the function value.

Definition 3.1 (Continuity). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be *continuous at a point* $x \in \mathbb{R}^n$ if $\lim_{y \rightarrow x} f(y) = f(x)$, that is, for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|y - x\| < \delta \implies$

$|f(y) - f(x)| < \varepsilon$. The function f is said to be continuous on a set $U \subset \mathbb{R}^n$ if it is continuous at every point in U .

Differentiability strengthens continuity by requiring the existence of a linear approximation that locally captures the behavior of the function.

Definition 3.2 (Differentiability). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be differentiable at a point $x \in \mathbb{R}^n$ if there exists a vector $\nabla f(x) \in \mathbb{R}^n$ such that $\lim_{\|h\| \rightarrow 0} \frac{f(x+h) - f(x) - \nabla f(x)^\top h}{\|h\|} = 0$. The vector $\nabla f(x)$ is called the gradient of f at x .

Differentiability guarantees a first-order Taylor expansion at the point of interest, but does not impose any regularity on how the gradient behaves in a neighborhood of that point.

Continuous differentiability requires that the gradient itself varies smoothly.

Definition 3.3 (Continuous differentiability). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be continuously differentiable on an open set $U \subset \mathbb{R}^n$ if it is differentiable at every point $x \in U$ and the gradient mapping $x \mapsto \nabla f(x)$ is continuous on U . In this case, we write $f \in C^1(U)$.

Continuous differentiability ensures the stability of first-order approximations under small perturbations of the expansion point. In particular, it justifies the use of Taylor expansions with controlled remainder terms and is the standard smoothness assumption in first-order optimality theory.

Regularity of constrained optimization problems.

We now translate these smoothness notions into regularity conditions for constrained optimization problems.

Definition 3.4 (First-order regularity). The constrained optimization problem (3.1) is said to be first-order regular at a feasible point $x^* \in \mathcal{F}$ if the objective function f and all constraint functions h_i , $i \in \mathcal{P}$, and g_j , $j \in \mathcal{Q}$, are continuously differentiable in an open neighborhood of x^* .

First-order regularity ensures that gradients of the objective and constraint functions are well defined and vary continuously near x^* , allowing the formulation of feasible directions, constraint qualifications, and first-order optimality conditions such as the Fritz–John and Karush–Kuhn–Tucker conditions.

Definition 3.5 (Second-order regularity). The constrained optimization problem (3.1) is said to be second-order regular at a feasible point $x^* \in \mathcal{F}$ if the objective function f and all constraint functions h_i , $i \in \mathcal{P}$, and g_j , $j \in \mathcal{Q}$, are twice continuously differentiable in an open neighborhood of x^* .

Second-order regularity is required for the formulation of second-order necessary and sufficient optimality conditions, which distinguish local minima from local maxima and saddle points.

3.1.3 ■ Active and Inactive Constraints

To analyze local optimality conditions, it is essential to distinguish between constraints that are binding at a feasible point and those that are not.

Equality constraints as hard constraints. For every equality constraint $h_i(x) = 0, i \in \mathcal{P}$, feasibility requires that $h_i(x^*) = 0$ for any feasible point x^* . Hence, *all equality constraints are active at every feasible point*. Equality constraints must be satisfied exactly and impose hard geometric restrictions on the feasible set. In particular, feasible points lie on a lower-dimensional manifold in $\mathbb{R}^n$, and admissible local perturbations must remain tangent to this manifold.

Inequality constraints and the active set. Inequality constraints may or may not influence the local structure of the feasible set.

Definition 3.6 (Active inequality constraints). Let $x^* \in \mathcal{F}$. An inequality constraint $g_j(x) \leq 0$ is said to be *active at x^** if $g_j(x^*) = 0$, and *inactive* if $g_j(x^*) < 0$. The set of active inequality constraints at x^* is defined as

$$\mathcal{Q}(x^*) = \{j \in \mathcal{Q} \mid g_j(x^*) = 0\}.$$

If an inequality constraint is inactive at x^* , then by continuity there exists a neighborhood of x^* in which the constraint remains strictly satisfied. Such constraints therefore impose no restriction on sufficiently small feasible perturbations and play no role in local optimality analysis.

Theorem 3.7 (Characterization of Feasible Directions). Consider the constrained optimization problem (3.1), and let $x^* \in \mathcal{F}$ be a feasible point. If all constraint functions are differentiable at x^* , then a vector $d \in \mathbb{R}^n$ is a feasible direction at x^* only if

$$\begin{aligned} \nabla h_i(x^*)^\top d &= 0, & i \in \mathcal{P}, \\ \nabla g_j(x^*)^\top d &\leq 0, & j \in \mathcal{Q}(x^*), \end{aligned} \tag{3.2}$$

where $\mathcal{Q}(x^*) = \{j \in \mathcal{Q} \mid g_j(x^*) = 0\}$ denotes the set of active inequality constraints at x^* .

Proof. Let $d \in \mathbb{R}^n$ be a feasible direction at x^* . By definition, there exists $\bar{\alpha} > 0$ such that $x^* + \alpha d \in \mathcal{F}, \forall \alpha \in (0, \bar{\alpha}]$.

Equality constraints. For each $i \in \mathcal{P}$, feasibility implies $h_i(x^* + \alpha d) = 0 \forall \alpha \in (0, \bar{\alpha}]$. Using a first-order Taylor expansion of h_i at x^* ,

$$h_i(x^* + \alpha d) = h_i(x^*) + \alpha \nabla h_i(x^*)^\top d + o(\alpha).$$

Since $h_i(x^*) = 0$, dividing by α and letting $\alpha \downarrow 0$ yields $\nabla h_i(x^*)^\top d = 0$.

Inequality constraints. Consider an inequality constraint indexed by $j \in \mathcal{Q}$. If $j \notin \mathcal{Q}(x^*)$, then $g_j(x^*) < 0$ and, by continuity, $g_j(x^* + \alpha d) < 0$ for all sufficiently small $\alpha > 0$. Hence such constraints impose no restriction on d .

Now let $j \in \mathcal{Q}(x^*)$, so that $g_j(x^*) = 0$. Feasibility requires $g_j(x^* + \alpha d) \leq 0 \forall \alpha \in (0, \bar{\alpha}]$. Using a first-order Taylor expansion,

$$g_j(x^* + \alpha d) = \alpha \nabla g_j(x^*)^\top d + o(\alpha).$$

Dividing by α and letting $\alpha \downarrow 0$ gives $\nabla g_j(x^*)^\top d \leq 0$.

Combining the above conditions proves (3.2). $\square$

The set described by (3.2) is a *cone*, determined entirely by the equality constraints together with the *active* inequality constraints; inactive constraints drop out of the local analysis. This is

the payoff of the active-set viewpoint: optimality conditions never need to reference constraints that are slack at x^* . Note that Theorem 3.7 gives only an inclusion—every feasible direction satisfies the linearized conditions—and the reverse inclusion, which is what makes the linearized cone a faithful model of the local geometry, requires a constraint qualification. Section 3.3.2 returns to this point. Granting it, all first- and second-order optimality conditions in this chapter amount to a single demand: the objective must not decrease along any admissible direction.

3.2 ■ Method of Lagrange

We begin with constrained optimization problems involving *only equality constraints*. This classical setting leads to the method of Lagrange multipliers and provides the conceptual foundation for first-order optimality conditions in constrained optimization.

3.2.1 ■ Equality-constrained optimization

Consider the equality-constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \end{aligned} \tag{3.3}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective function and $h_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are equality constraint functions indexed by the finite set $\mathcal{P} = \{1, \dots, p\}$.

The feasible set associated with (3.3) is

$$\mathcal{F} = \{x \in \mathbb{R}^n \mid h_i(x) = 0, \quad i \in \mathcal{P}\}.$$

Throughout this section, we assume that f is differentiable and that each constraint function h_i is continuously differentiable in a neighborhood of any point of interest.

The geometric picture is the following. At a feasible point $x \in \mathcal{F}$ each gradient $\nabla h_i(x)$ points *normal* to the surface $\{h_i = 0\}$, so a first-order perturbation d keeps us on the constraint surface exactly when it is orthogonal to every constraint gradient, that is, when $\nabla h_i(x)^\top d = 0$ for all $i \in \mathcal{P}$. These admissible perturbations form the tangent space of $\mathcal{F}$ at x . Now suppose x^* is a local minimizer. Moving along any tangent direction d cannot decrease f to first order, and since $-d$ is tangent whenever d is, the directional derivative $\nabla f(x^*)^\top d$ must in fact *vanish* on the whole tangent space. Thus $\nabla f(x^*)$ is orthogonal to every feasible direction, which forces it into the *normal* space of the constraint manifold, the space spanned by $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$. In words: at an optimum the objective can only “pull” in directions that the constraints forbid.

3.2.2 ■ First-order necessary condition

The preceding geometric argument yields the fundamental first-order necessary condition for equality-constrained optimization.

Theorem 3.8 (Euler–Lagrange necessary condition). *Consider problem (3.3). Suppose that f is differentiable and each constraint h_i , $i \in \mathcal{P}$, is continuously differentiable at a feasible point $x^* \in \mathcal{F}$. If x^* is a local minimizer and the gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent, then*

$$\nabla f(x^*) \in \text{span}\{\nabla h_1(x^*), \dots, \nabla h_p(x^*)\}.$$

Equivalently, the vectors $\nabla f(x^)$, $\nabla h_1(x^*)$, $\dots$, $\nabla h_p(x^*)$ are linearly dependent.*

Proof. Let $x^* \in \mathcal{F}$ be a local minimizer, so that there exists $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in \mathcal{F} \cap B_\delta(x^*)$. Let d be a feasible first-order perturbation at x^* : by definition there exists a differentiable curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathcal{F}$ with $\gamma(0) = x^*$ and $\dot{\gamma}(0) = d$. Since $h_i(\gamma(t)) = 0$ for all t , differentiation at $t = 0$ gives $\nabla h_i(x^*)^\top d = 0$ for $i \in \mathcal{P}$, so every feasible direction lies in $T_{\mathcal{F}}(x^*) = \{d : \nabla h_i(x^*)^\top d = 0, i \in \mathcal{P}\}$. Finally, defining $\phi(t) = f(\gamma(t))$, local optimality gives $\phi'(0) = 0$, hence

$$\nabla f(x^*)^\top d = 0 \quad \forall d \in T_{\mathcal{F}}(x^*). \quad (3.4)$$

Contradiction argument. Let $M := \text{span}\{\nabla h_1(x^*), \dots, \nabla h_p(x^*)\}$ and suppose, by contradiction, that $\nabla f(x^*) \notin M$. Like every subspace of $\mathbb{R}^n$, M has an orthogonal complement, and $\mathbb{R}^n = M \oplus M^\perp$; this decomposition requires no assumption on the constraint gradients. Because $\nabla f(x^*) \notin M$, its projection onto $M^\perp$ is nonzero, and we may write uniquely $\nabla f(x^*) = u + v$ with $u \in M$, $v \in M^\perp$ and $v \neq 0$.

Now choose $d := v \in M^\perp$. A vector lies in $M^\perp$ if and only if it is orthogonal to every constraint gradient, i.e. $\nabla h_i(x^*)^\top d = 0$ for $i \in \mathcal{P}$; therefore $d \in T_{\mathcal{F}}(x^*)$. This is the step at which the linear independence of the constraint gradients (LICQ) is indispensable: a vector satisfying the linearized conditions need not be the velocity of any feasible curve (if the gradients are dependent, $\mathcal{F}$ may even reduce to the single point x^* while $M^\perp \neq \{0\}$; see Figure 3.2(b)). Because the gradients are independent, the map $x \mapsto (h_1(x), \dots, h_p(x))$ has surjective derivative at x^* , the implicit function theorem produces an actual feasible curve γ with $\gamma(0) = x^*$ and $\dot{\gamma}(0) = d$, so the linearized direction d is realized by admissible motion and (3.4) applies to it. On the other hand, since $u \in M$ and $v \in M^\perp$ give $u^\top v = 0$,

$$\nabla f(x^*)^\top d = (u + v)^\top v = u^\top v + v^\top v = \|v\|^2 > 0.$$

We have thus constructed a feasible first-order direction $d \in T_{\mathcal{F}}(x^*)$ with $\nabla f(x^*)^\top d \neq 0$, which contradicts the first-order optimality condition (3.4) (see Figure 3.1).

Therefore our assumption was false, and we conclude that

$$\nabla f(x^*) \in M = \text{span}\{\nabla h_1(x^*), \dots, \nabla h_p(x^*)\},$$

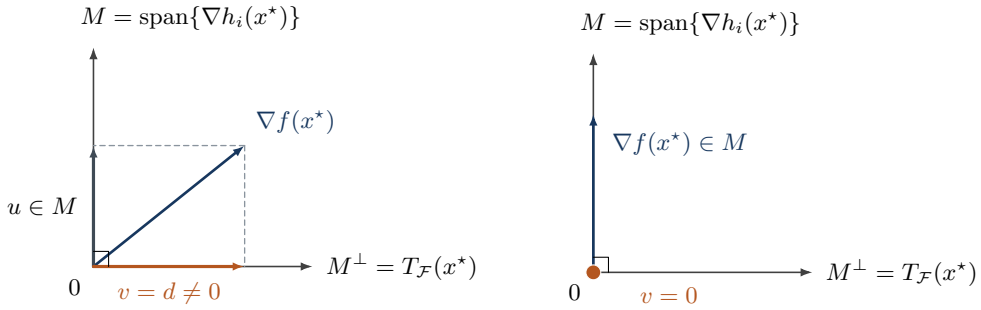
which establishes the Euler–Lagrange necessary condition. $\square$

The proof of Theorem 3.8 reveals that a key step relied on the fact that the constraint gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ form a linearly independent set. The decomposition of $\nabla f(x^*)$ into components in $M = \text{span}\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ and $M^\perp$ (Figure 3.1) is available for any set of gradients; what linear independence guarantees, through the implicit function theorem, is that every direction in $M^\perp$ is the velocity of an actual feasible curve, so that the tangential component is a genuine feasible direction. This property is not automatic; it must be imposed as an additional regularity assumption.

Definition 3.9 (Linear Independence Constraint Qualification (LICQ)). Let x^* be a feasible point of the equality-constrained problem (3.3). We say that the linear independence constraint qualification (LICQ) holds at x^* if the constraint gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent.

Remark 3.10 (Normal cone interpretation). The Euler–Lagrange condition can be derived directly from the normal-cone optimality principle. For the equality-constrained set $\mathcal{F} = \{x \in \mathbb{R}^n : h_i(x) = 0, i \in \mathcal{P}\}$, if LICQ holds at x^* , then the normal cone is given by

$$N_{\mathcal{F}}(x^*) = \text{span}\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}.$$



(a) Assume $\nabla f(x^*) \notin M$. Its tangential part $v \neq 0$ is a feasible direction d with $\nabla f(x^*)^\top d = \|v\|^2 > 0$, contradicting stationarity.

(b) At an optimum the tangential part vanishes, so $\nabla f(x^*) \in M = \text{span}\{\nabla h_i(x^*)\}$: the Euler–Lagrange condition.

Figure 3.1: Geometry of the proof of Theorem 3.8. Decompose $\nabla f(x^*) = u + v$ with $u \in M$ (normal space) and $v \in M^\perp$ (tangent space $T_{\mathcal{F}}(x^*)$). A nonzero tangential part v would be a feasible ascent/descent direction, contradicting first-order optimality; hence $v = 0$ and the gradient is a combination of the constraint gradients.

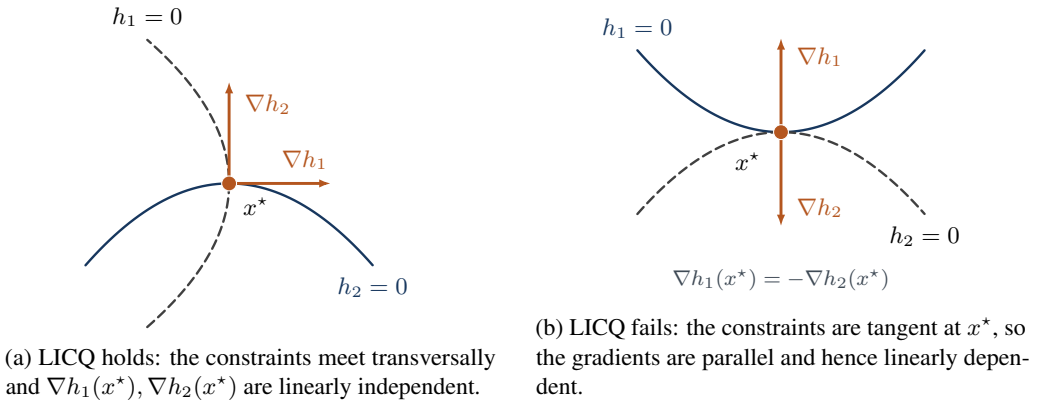


Figure 3.2: The linear independence constraint qualification (LICQ) at a feasible point x^* . The splitting $\mathbb{R}^n = M \oplus M^\perp$ with $M = \text{span}\{\nabla h_i(x^*)\}$ exists for any set of gradients; what LICQ adds, through the implicit function theorem, is that every $d \in M^\perp$ is the velocity of an actual feasible curve, so the tangential part used in the proof of Theorem 3.8 is a genuine feasible direction. When LICQ fails this link can break: in (b) the feasible set is the single point x^* , although $M^\perp$ is the whole x_1 -axis.

Since first-order optimality in cone form requires $-\nabla f(x^*) \in N_{\mathcal{F}}(x^*)$, we immediately obtain

$$\nabla f(x^*) \in \text{span}\{\nabla h_1(x^*), \dots, \nabla h_p(x^*)\},$$

which is exactly the Euler–Lagrange necessary condition.

Thus the Lagrange multiplier equation is simply the normal-cone inclusion expressed algebraically under LICQ.

Remark 3.11 (Interpretation and role of LICQ). LICQ is a geometric regularity condition on the feasible set. If the gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent, they define p independent

normal directions at x^* . Consequently, the feasible set locally behaves like a smooth $(n - p)$ -dimensional manifold embedded in $\mathbb{R}^n$.

In this case, the tangent cone is exactly the linearized constraint space: $T_{\mathcal{F}}(x^*) = \{d \in \mathbb{R}^n : \nabla h_i(x^*)^\top d = 0, i \in \mathcal{P}\}$, and the normal cone is $N_{\mathcal{F}}(x^*) = \text{span}\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$.

Thus LICQ guarantees that the first-order linear approximation of the constraints correctly captures the local geometry of the feasible set. Under this condition, the Euler–Lagrange theorem admits a clean geometric interpretation: at a local minimum, the gradient of the objective must lie in the normal space of the feasible manifold. Equivalently, there exist multipliers λ^* such that

$$\nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i^* \nabla h_i(x^*) = 0.$$

If LICQ fails, the local geometry of the feasible set may be degenerate. The constraint gradients may be linearly dependent, the tangent cone may not coincide with the linearized constraint space, and the Lagrange multiplier representation may fail to exist or fail to be unique.

In such cases, the appropriate first-order necessary condition is the *Fritz–John condition*, which asserts that there exist scalars $\nu_0 \geq 0$ and λ_i , not all zero, such that

$$\nu_0 \nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) = 0.$$

If $\nu_0 > 0$, one can normalize by dividing through by ν_0 and recover the standard Lagrange multiplier condition. If $\nu_0 = 0$, one obtains the so-called *abnormal case*, which can occur only when the constraint gradients are linearly dependent. Thus LICQ precisely excludes abnormal multipliers and guarantees the validity of the classical Lagrange multiplier representation.

3.2.3 ■ Lagrangian and Lagrange Multipliers

We now introduce the Lagrangian function, which provides a convenient algebraic representation of first-order optimality conditions for equality-constrained problems.

Definition 3.12 (Lagrangian). *The Lagrangian associated with problem (3.3) is defined as*

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i \in \mathcal{P}} \lambda_i h_i(x),$$

where $\lambda = (\lambda_i)_{i \in \mathcal{P}} \in \mathbb{R}^p$ is the vector of Lagrange multipliers.

The Lagrangian aggregates the objective and the constraints into a single scalar function. Its stationarity with respect to the decision variable x expresses the condition that the gradient of the objective be orthogonal to all feasible first-order perturbations.

Corollary 3.13 (Lagrange multiplier representation of the Euler–Lagrange condition). *Let x^* be a local minimizer of the equality-constrained problem (3.3). If the constraint gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent, then there exists a multiplier vector $\lambda^* \in \mathbb{R}^p$ such that*

$$\nabla_x \mathcal{L}(x^*, \lambda^*) = \nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i^* \nabla h_i(x^*) = 0. \quad (3.5)$$

Proof. By Theorem 3.8, $\nabla f(x^*)$ lies in the span of $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$. Since these vectors are linearly independent, this representation is exact, yielding coefficients λ_i^* such that (3.5) holds. $\square$

Corollary 3.13 shows that, under linear independence of the constraint gradients, the Euler–Lagrange necessary condition is equivalent to stationarity of the Lagrangian with respect to the primal variable.

The classical method of Lagrange multipliers consists of solving the system

$$\nabla_x \mathcal{L}(x, \lambda) = \nabla f(x) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x) = 0, \quad (3.6)$$

$$h_i(x) = 0, \quad i \in \mathcal{P}. \quad (3.7)$$

Definition 3.14 (Lagrange candidate). *A pair (x^*, λ^*) satisfying (3.6)–(3.7) is called a Lagrange candidate. Every local minimizer at which the constraint gradients are linearly independent must be a Lagrange candidate.*

The method of Lagrange multipliers provides *necessary* but not sufficient conditions for local optimality. In particular, a Lagrange candidate may correspond to a local maximum or a saddle point rather than a local minimum. Distinguishing among these cases requires second-order analysis based on the Hessian of the Lagrangian restricted to feasible directions.

Finally, if the constraint gradients are linearly dependent at a local minimizer, the Lagrangian stationarity condition (3.5) may fail to admit a solution. This degeneracy motivates the introduction of more general first-order conditions, such as the Fritz–John conditions, and the use of constraint qualifications to recover Karush–Kuhn–Tucker optimality conditions in the presence of inequality constraints.

Example 3.15 Degenerate vs. regular constraint at a single feasible point.

Consider the problem $\min f(x) = x$ under two different equality constraints.

Case (A): $h(x) = x^2 = 0$.

The constraint implies $x = 0$, so the feasible set is $\mathcal{F} = \{0\}$ and $x^* = 0$ is trivially a global (hence local) minimizer. The Lagrangian is $\mathcal{L}(x, \lambda) = x + \lambda x^2$. Its derivative is $\frac{d}{dx} \mathcal{L}(x, \lambda) = 1 + 2\lambda x$. Evaluating at $x^* = 0$ gives $1 + 2\lambda \cdot 0 = 1$. Thus stationarity would require $1 = 0$, which is impossible. Equivalently, in gradient form, $\nabla f(0) + \lambda \nabla h(0) = 0$ reduces to $1 + \lambda \cdot 0 = 0$, which has no solution.

The failure occurs because $\nabla h(0) = 0$. The constraint gradient vanishes, violating LICQ. The linearized constraint imposes no restriction on directions, even though the true feasible set consists of a single point. Hence the classical Lagrange multiplier equation fails despite optimality.

Fix via Fritz–John Conditions The Fritz–John condition asserts that there exist scalars $\nu_0 \geq 0$ and λ , not both zero, such that $\nu_0 \nabla f(0) + \lambda \nabla h(0) = 0$. Since $\nabla f(0) = 1$ and $\nabla h(0) = 0$, this reduces to $\nu_0 = 0$. Thus any pair (ν_0, λ) with $\nu_0 = 0$ and $\lambda \neq 0$ satisfies the condition.

This is the *abnormal case*, characterized by $\nu_0 = 0$. The objective gradient plays no role in the stationarity equation, which reduces to $\lambda \nabla h(0) = 0$ and holds automati-

ically. Thus the Fritz–John conditions remain valid as necessary conditions at the optimum even though no Lagrange multiplier exists. They are of course not sufficient: here they hold with no information from the objective at all. The degeneracy is absorbed into the additional scalar ν_0 .

Case (B): $h(x) = x = 0$.

Again the feasible set is $\mathcal{F} = \{0\}$, so $x^* = 0$ is optimal. The Lagrangian is $\mathcal{L}(x, \lambda) = x + \lambda x$, and its derivative is $1 + \lambda$. Evaluating at $x^* = 0$ yields $1 + \lambda = 0$, so $\lambda^* = -1$. In gradient form, $\nabla f(0) + \lambda^* \nabla h(0) = 0$ reduces to $1 + \lambda^* \cdot 1 = 0$, which is satisfied. Here $\nabla h(0) = 1 \neq 0$, so LICQ holds. The Fritz–John condition reduces to the classical Lagrange multiplier equation with $\nu_0 > 0$, and after normalization we recover the usual multiplier.

Both problems have the same feasible set $\{0\}$ and the same objective. The difference lies entirely in the *local geometry of the constraint*. In Case (A), the constraint is degenerate: its gradient vanishes, the linearization is useless, and only an abnormal Fritz–John multiplier exists. In Case (B), the constraint is regular: LICQ holds, the feasible set is locally a smooth 0-dimensional manifold, and the classical multiplier exists.

This example illustrates that Lagrange multipliers depend not only on the feasible set itself but also on how that set is represented by the constraint functions. $\square$

Example 3.16 Redundant nonlinear constraints and non-unique multipliers.

Consider the problem

$$\min_{x \in \mathbb{R}^2} f(x) = x_1 \quad \text{s.t.} \quad h_1(x) = x_1^2 + x_2^2 - 1 = 0, \quad h_2(x) = 2x_1^2 + 2x_2^2 - 2 = 0.$$

Both constraints describe the unit circle $\mathcal{F} = \{x \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}$. Thus the feasible set is the unit circle. The unique minimizer of $f(x) = x_1$ over the circle is $x^* = (-1, 0)$.

Failure of LICQ. The gradients are $\nabla h_1(x) = 2(x_1, x_2)$ and $\nabla h_2(x) = 4(x_1, x_2)$. At $x^* = (-1, 0)$ they become $\nabla h_1(x^*) = (-2, 0)$ and $\nabla h_2(x^*) = (-4, 0)$, which are linearly dependent. Hence LICQ fails.

Normal cone. Since the feasible set is a smooth curve, its normal cone at x^* is the span of the outward normal, namely $N_{\mathcal{F}}(x^*) = \text{span}\{(-1, 0)\}$.

Lagrange multipliers. The Lagrangian is

$$\mathcal{L}(x, \lambda_1, \lambda_2) = x_1 + \lambda_1(x_1^2 + x_2^2 - 1) + \lambda_2(2x_1^2 + 2x_2^2 - 2).$$

The stationarity condition is $\nabla f(x) + \lambda_1 \nabla h_1(x) + \lambda_2 \nabla h_2(x) = 0$. At x^* this becomes $(1, 0) + \lambda_1(-2, 0) + \lambda_2(-4, 0) = 0$. Equivalently, $1 - 2\lambda_1 - 4\lambda_2 = 0$. This single linear equation in (λ_1, λ_2) has infinitely many solutions. Thus multipliers exist but are not unique.

The feasible set is perfectly regular (a smooth manifold), and the normal cone is one-dimensional. However, because the constraints are redundant, their gradients are linearly dependent, so LICQ fails and the multiplier vector is not unique. This example illustrates that LICQ is not needed for optimality itself, but it is essential for obtaining a unique multiplier representation. $\square$

3.2.4 ■ Second-Order Sufficient Conditions

The first-order Euler–Lagrange condition characterizes candidate solutions by requiring stationarity of the objective function along feasible directions. However, as in unconstrained optimization, first-order conditions alone do not distinguish local minima from local maxima or saddle points. This distinction requires second-order information.

Throughout this subsection, we assume that the objective function f and the constraint functions $h_i, i \in \mathcal{P}$, are *twice continuously differentiable* in a neighborhood of a feasible point x^* .

Tangent space

Let $x^* \in \mathcal{F}$ be a feasible point at which the gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent. The tangent space of the feasible set at x^* is given by

$$\mathcal{T}(x^*) = \{d \in \mathbb{R}^n \mid \nabla h_i(x^*)^\top d = 0, i \in \mathcal{P}\}.$$

Let $N \in \mathbb{R}^{n \times (n-p)}$ be a matrix whose columns form a basis of $\mathcal{T}(x^*)$. Any feasible first-order perturbation can be written as $d = N\xi$ for some $\xi \in \mathbb{R}^{n-p}$.

Example 3.17 Tangent basis for a nonlinear constraint.

Consider the equality-constrained set

$$\mathcal{F} = \{(x_1, x_2) \in \mathbb{R}^2 \mid h(x) = x_1^2 + x_2^2 - 1 = 0\},$$

and the feasible point $x^* = (1, 0)$.

The gradient of the constraint is $\nabla h(x) = (2x_1, 2x_2)^\top$, and hence $\nabla h(x^*) = (2, 0)^\top$.

A vector $d = (d_1, d_2)^\top$ belongs to the tangent space $\mathcal{T}(x^*)$ if and only if $\nabla h(x^*)^\top d = 2d_1 = 0$. Thus $\mathcal{T}(x^*) = \{(0, d_2)^\top \mid d_2 \in \mathbb{R}\}$.

Choosing the basis vector $n_1 = (0, 1)^\top$, we obtain

$$N = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Every feasible first-order perturbation can therefore be written as $d = N\xi$ with $\xi \in \mathbb{R}$.

This example illustrates that N depends on the local geometry of the constraint manifold through the gradient $\nabla h(x^*)$, even though the constraint itself is nonlinear. $\square$

Example 3.18 Nonlinear constraint in $\mathbb{R}^3$.

Consider the nonlinear constraint $h(x) = x_1^2 + x_2^2 + x_3 - 1 = 0$ and the feasible point $x^* = (0, 0, 1)$. The gradient at x^* is $\nabla h(x^*) = (0, 0, 1)^\top$. A vector $d = (d_1, d_2, d_3)^\top$ belongs to the tangent space $\mathcal{T}(x^*)$ if and only if $\nabla h(x^*)^\top d = d_3 = 0$. Hence,

$$\mathcal{T}(x^*) = \{d \in \mathbb{R}^3 \mid d_3 = 0\} = \text{span}\{(1, 0, 0)^\top, (0, 1, 0)^\top\}.$$

Choosing

$$N = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix},$$

every tangent direction admits the representation $d = N\xi$ with $\xi \in \mathbb{R}^2$. $\square$

Lagrangian and second-order expansion

Suppose that x^* satisfies the Euler–Lagrange condition. Then there exist multipliers $\lambda_i \in \mathbb{R}$, $i \in \mathcal{P}$, such that

$$\nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) = 0. \quad (3.8)$$

Define the *Lagrangian* associated with the equality-constrained problem by

$$\mathcal{L}(x, \lambda) = f(x) + \sum_{i \in \mathcal{P}} \lambda_i h_i(x).$$

While the first-order condition characterizes stationarity along feasible directions, second-order optimality depends on how the objective curves along the constraint manifold. Because feasible perturbations must satisfy the constraints to second order, the appropriate curvature object is not $\nabla^2 f(x^*)$ alone, but the Hessian of the Lagrangian,

$$\nabla_{xx}^2 \mathcal{L}(x^*, \lambda) = \nabla^2 f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla^2 h_i(x^*).$$

This matrix governs the second-order variation of the objective along feasible directions, as made precise below.

Second-order necessary condition

We first state a second-order condition that must hold at any local minimum. This condition rules out directions of negative curvature along the constraint manifold but does not, by itself, guarantee local optimality.

Theorem 3.19 (Second-Order Necessary Condition for Equality Constraints). *Consider the equality-constrained optimization problem (3.3). Suppose that:*

- (1) *the objective function f and the constraint functions h_i , $i \in \mathcal{P}$, are twice continuously differentiable in a neighborhood of a feasible point x^* ,*
- (2) *the gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent,*
- (3) *there exist multipliers $\lambda_i \in \mathbb{R}$ such that the stationarity condition (3.8) holds.*

If x^ is a local minimum of the equality-constrained problem, then*

$$d^\top \left[\nabla^2 f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla^2 h_i(x^*) \right] d \geq 0 \quad \forall d \in \mathcal{T}(x^*). \quad (3.9)$$

Equivalently, the reduced Hessian

$$N^\top \left[\nabla^2 f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla^2 h_i(x^*) \right] N$$

is positive semidefinite.

Proof. Let $d \in \mathcal{T}(x^*)$ be arbitrary. Since the gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent and the constraints are twice continuously differentiable, the feasible set $\mathcal{F}$ is locally a smooth manifold. Hence there exists a twice continuously differentiable curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathcal{F}$ such that $\gamma(0) = x^*$ and $\dot{\gamma}(0) = d$.

Define $\phi(t) = f(\gamma(t))$. Because x^* is a local minimum and $\gamma(t)$ is feasible for all t , the function ϕ attains a local minimum at $t = 0$. Therefore $\phi'(0) = 0$ and $\phi''(0) \geq 0$.

The first derivative satisfies

$$\phi'(0) = \nabla f(x^*)^\top d = - \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*)^\top d = 0,$$

where the last equality follows from $d \in \mathcal{T}(x^*)$.

To compute the second derivative, note that

$$\phi''(0) = d^\top \nabla^2 f(x^*) d + \nabla f(x^*)^\top \ddot{\gamma}(0).$$

Since $\gamma(t)$ is feasible, we have $h_i(\gamma(t)) = 0$ for all t , and differentiating twice yields

$$\nabla h_i(x^*)^\top \ddot{\gamma}(0) + d^\top \nabla^2 h_i(x^*) d = 0, \quad i \in \mathcal{P}.$$

Multiplying by λ_i and summing over i gives

$$\nabla f(x^*)^\top \ddot{\gamma}(0) = - \sum_{i \in \mathcal{P}} \lambda_i d^\top \nabla^2 h_i(x^*) d.$$

Substituting into the expression for $\phi''(0)$ yields

$$\phi''(0) = d^\top \left[\nabla^2 f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla^2 h_i(x^*) \right] d.$$

Since $\phi''(0) \geq 0$, inequality (3.9) follows. $\square$

Second-order sufficient condition

We now strengthen the necessary condition to obtain a sufficient condition for local optimality.

Theorem 3.20 (Second-Order Sufficient Condition for Equality Constraints). *Consider the equality-constrained optimization problem (3.3). Suppose that:*

- (1) *the objective function f and the constraint functions h_i , $i \in \mathcal{P}$, are twice continuously differentiable in a neighborhood of a feasible point x^* ,*
- (2) *the gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent,*
- (3) *there exist multipliers $\lambda_i \in \mathbb{R}$ satisfying (3.8),*
- (4) *the reduced Hessian*

$$N^\top \left[\nabla^2 f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla^2 h_i(x^*) \right] N \quad (3.10)$$

is positive definite.

Then x^ is an isolated local minimum of the equality-constrained problem.*

Proof. Write $\mathcal{L}(x, \lambda) = f(x) + \sum_{i \in \mathcal{P}} \lambda_i h_i(x)$ for the Lagrangian with the multipliers supplied by hypothesis (3), so that $\nabla_x \mathcal{L}(x^*, \lambda) = 0$ by (3.8). The key point is that $\mathcal{L}(\cdot, \lambda)$ agrees with f on the feasible set, because every h_i vanishes there; the multiplier terms therefore absorb the curvature of the constraint surfaces, which a direct expansion of f alone would miss.

Suppose, for contradiction, that x^* is not a strict local minimum. Then there is a sequence $\{x_k\} \subset \mathcal{F}$ with $x_k \neq x^*$, $x_k \rightarrow x^*$ and $f(x_k) \leq f(x^*)$ for every k . Put $t_k := \|x_k - x^*\| \downarrow 0$ and $d_k := (x_k - x^*)/t_k$. Passing to a subsequence, $d_k \rightarrow d$ with $\|d\| = 1$.

The limit direction is tangent. For each $i \in \mathcal{P}$, feasibility of x_k and of x^* gives $h_i(x_k) = h_i(x^*) = 0$, so a first-order expansion yields $0 = h_i(x^* + t_k d_k) = t_k \nabla h_i(x^*)^\top d_k + o(t_k)$. Dividing by t_k and letting $k \rightarrow \infty$ gives $\nabla h_i(x^*)^\top d = 0$, that is, $d \in \mathcal{T}(x^*) \setminus \{0\}$.

Passing to the second order. Since x_k and x^* are both feasible, $\mathcal{L}(x_k, \lambda) = f(x_k)$ and $\mathcal{L}(x^*, \lambda) = f(x^*)$. A second-order expansion of $\mathcal{L}(\cdot, \lambda)$ at x^* , using $\nabla_x \mathcal{L}(x^*, \lambda) = 0$, therefore gives

$$0 \geq f(x_k) - f(x^*) = \mathcal{L}(x_k, \lambda) - \mathcal{L}(x^*, \lambda) = \frac{1}{2} t_k^2 d_k^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda) d_k + o(t_k^2).$$

Dividing by $t_k^2 > 0$ and letting $k \rightarrow \infty$ yields $d^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda) d \leq 0$. But d is a nonzero tangent direction, so $d = N\xi$ for some $\xi \neq 0$ and the reduced Hessian (3.10) is positive definite, whence $d^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda) d = \xi^\top N^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda) N \xi > 0$. This contradiction shows that x^* is a strict local minimum.

The same computation run quantitatively gives more: with $c := \frac{1}{2} \lambda_{\min}(N^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda) N) > 0$ one obtains the *quadratic growth* estimate $f(x) \geq f(x^*) + \frac{c}{2} \|x - x^*\|^2$ for all feasible x near

x^* . In particular x^* is the only local minimum of the problem in a neighborhood of itself, that is, an isolated local minimum. $\square$

Remark 3.21. The matrix in (3.10) is known as the *reduced Hessian*. Its positive definiteness ensures strictly positive second-order variation of the objective along all feasible directions, generalizing the classical unconstrained second-order sufficient condition.

Example 3.22 Second-order sufficiency for an equality-constrained problem.

Consider the equality-constrained problem

$$\min_{x \in \mathbb{R}^3} f(x) = x_1^2 + x_2^2 + x_3^2 \quad \text{s.t.} \quad h(x) = x_1 + x_2 + x_3 - 1 = 0.$$

The feasible set is the affine plane $\mathcal{F} = \{x \in \mathbb{R}^3 \mid x_1 + x_2 + x_3 = 1\}$. The gradients are $\nabla f(x) = (2x_1, 2x_2, 2x_3)^\top$ and $\nabla h(x) = (1, 1, 1)^\top$. The Euler–Lagrange condition requires $\nabla f(x^*) + \lambda \nabla h(x^*) = 0$ for some $\lambda \in \mathbb{R}$, which yields $x_1^* = x_2^* = x_3^* = \frac{1}{3}$ and $\lambda^* = -\frac{2}{3}$.

The tangent space at x^* is $\mathcal{T}(x^*) = \{d \in \mathbb{R}^3 \mid d_1 + d_2 + d_3 = 0\}$. A convenient basis is $n_1 = (1, 0, -1)^\top$ and $n_2 = (0, 1, -1)^\top$, hence

$$N = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}.$$

The Hessians satisfy $\nabla^2 f(x) = 2I_3$ and $\nabla^2 h(x) = 0$, so $\nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) = 2I_3$. The reduced Hessian is

$$N^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) N = 2N^\top N = 2 \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its eigenvalues are 2 and 6, both strictly positive, so the reduced Hessian is positive definite on $\mathcal{T}(x^*)$. All assumptions of the second-order sufficient condition hold. Therefore, $x^* = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ is an *isolated local minimum* of the equality-constrained problem. $\square$

Example 3.23 Minimization on a nonlinear manifold.

Consider the equality-constrained optimization problem

$$\min_{(x,y) \in \mathbb{R}^2} f(x, y) = x^2 + y^2 \quad \text{s.t.} \quad h(x, y) = x^2 y - 1 = 0.$$

The feasible set is the nonlinear curve $\mathcal{F} = \{(x, y) \in \mathbb{R}^2 \mid x^2 y = 1\}$, which consists of two symmetric branches corresponding to $x > 0$ and $x < 0$. Since $x = 0$ is infeasible, the constraint implies $y = \frac{1}{x^2}$ for $x \neq 0$.

First-order conditions. The Lagrangian is $\mathcal{L}(x, y, \lambda) = x^2 + y^2 + \lambda(x^2y - 1)$. Stationarity gives $\partial_x \mathcal{L} = 2x + 2\lambda xy = 0$, $\partial_y \mathcal{L} = 2y + \lambda x^2 = 0$, together with feasibility $x^2y = 1$.

Since $x \neq 0$, the first equation yields $2x(1 + \lambda y) = 0$, hence $1 + \lambda y = 0$ and therefore $\lambda = -\frac{1}{y}$. Substituting into the second condition gives $2y - \frac{x^2}{y} = 0$, equivalently $2y^2 = x^2$. Combining with $x^2y = 1$ yields $x^2 = 2^{1/3}$ and $y = 2^{-1/3}$.

Thus the Lagrange candidates are $(x^*, y^*) = (\pm 2^{1/6}, 2^{-1/3})$ with multiplier $\lambda^* = -2^{1/3}$.

Regularity and tangent space. We verify the linear independence constraint qualification. Since $\nabla h(x, y) = (2xy, x^2)$, we have $\nabla h(x^*, y^*) = (2x^*y^*, (x^*)^2)$, which is nonzero because $x^* \neq 0$. Hence the equality constraint gradient is linearly independent.

The tangent space is $T(x^*, y^*) = \{d \in \mathbb{R}^2 : \nabla h(x^*, y^*)^\top d = 0\}$. Writing $d = (d_1, d_2)^\top$, the tangent condition becomes $2x^*y^*d_1 + (x^*)^2d_2 = 0$. Since $(x^*)^2 = 2^{1/3}$ and $y^* = 2^{-1/3}$, we have $2x^*y^* = 2 \cdot 2^{1/6} \cdot 2^{-1/3} = 2^{5/6} \text{sign}(x^*)$, and therefore $d_2 = -\frac{2x^*y^*}{(x^*)^2}d_1 = -\frac{2^{5/6}}{2^{1/3}} \text{sign}(x^*)d_1 = -2^{1/2} \text{sign}(x^*)d_1$. (As a check, differentiating the constraint in the form $y = 1/x^2$ gives the same slope $dy/dx = -2/x^3 = -\sqrt{2}$ at $x^* = 2^{1/6}$.)

Thus $T(x^*, y^*)$ is one-dimensional, spanned by $n = (1, -2^{1/2} \text{sign}(x^*))^\top$.

Second-order sufficient condition. The Hessian of the Lagrangian is $\nabla_{xx}^2 \mathcal{L} = \nabla^2 f + \lambda \nabla^2 h$. Since $\nabla^2 f = 2I_2$ and $\nabla^2 h(x, y) = \begin{pmatrix} 2y & 2x \\ 2x & 0 \end{pmatrix}$, we obtain

$$\nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) = 2I_2 + \lambda^* \begin{pmatrix} 2y^* & 2x^* \\ 2x^* & 0 \end{pmatrix}.$$

Substituting $x^* = 2^{1/6} \text{sign}(x^*)$, $y^* = 2^{-1/3}$, and $\lambda^* = -2^{1/3}$ (using $\lambda^* 2y^* = -2$ and $\lambda^* 2x^* = -2^{3/2} \text{sign}(x^*)$) gives

$$\nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) = \begin{pmatrix} 0 & -2^{3/2} \text{sign}(x^*) \\ -2^{3/2} \text{sign}(x^*) & 2 \end{pmatrix},$$

so that, with $n = (1, -2^{1/2} \text{sign}(x^*))^\top$,

$$n^\top \nabla_{xx}^2 \mathcal{L}(x^*, y^*, \lambda^*) n = 2(2^{1/2})^2 (2^{3/2}) + 2(2^{1/2})^2 = 8 + 4 = 12 > 0.$$

Since the tangent space is one-dimensional, the reduced Hessian is positive definite. By the second-order sufficient condition for equality constraints, each candidate point is an isolated local minimum. $\square$

3.2.5 ■ Quadratic Penalty and Augmented Lagrangian Methods

The Lagrange multiplier theorem can be derived by approximating the equality-constrained problem with a sequence of unconstrained problems. This viewpoint provides both conceptual insight and a foundation for numerical algorithms.

Quadratic penalty method

We return to the equality-constrained problem (3.3). Recall that the feasible set is $\mathcal{F} = \{x \in \mathbb{R}^n \mid h_i(x) = 0, i \in \mathcal{P}\}$.

Let $x^* \in \mathcal{F}$ be a local minimizer of (3.3) at which LICQ holds; that is, the gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent. Consequently, near x^* the feasible set $\mathcal{F}$ is a smooth $(n - p)$ -dimensional manifold and

$$T_{\mathcal{F}}(x^*) = \{d \in \mathbb{R}^n : \nabla h_i(x^*)^\top d = 0, i \in \mathcal{P}\}.$$

For $k > 0$, define the quadratic penalty function

$$F^k(x) = f(x) + \frac{k}{2} \sum_{i \in \mathcal{P}} h_i(x)^2 = f(x) + \frac{k}{2} \|h(x)\|^2. \quad (3.11)$$

Instead of solving (3.3), we consider the unconstrained problem $\min_{x \in \mathbb{R}^n} F^k(x)$.

First-order stationarity. Let x^k be a local minimizer of F^k . Since the problem is unconstrained, first-order optimality gives

$$\nabla F^k(x^k) = \nabla f(x^k) + k \sum_{i \in \mathcal{P}} h_i(x^k) \nabla h_i(x^k) = 0. \quad (3.12)$$

Define

$$\lambda_i^k := k h_i(x^k), \quad i \in \mathcal{P}, \quad (3.13)$$

and let $\lambda^k = (\lambda_i^k)_{i \in \mathcal{P}}$. Then (3.12) becomes

$$\nabla f(x^k) + \sum_{i \in \mathcal{P}} \lambda_i^k \nabla h_i(x^k) = 0, \quad (3.14)$$

which has exactly the same algebraic structure as the Euler–Lagrange equation derived earlier.

Limit as $k \rightarrow \infty$. Suppose that $x^k \rightarrow x^*$ as $k \rightarrow \infty$. Because the penalty term forces constraint violations to be costly, one has $h(x^k) \rightarrow 0$, so x^* is feasible.

Passing to the limit in (3.14) and using continuity of ∇f and ∇h_i yields

$$\nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i^* \nabla h_i(x^*) = 0,$$

where

$$\lambda_i^* = \lim_{k \rightarrow \infty} k h_i(x^k).$$

Under LICQ, the multiplier vector λ^* is uniquely determined. Thus the Euler–Lagrange condition derived earlier arises as the limit of unconstrained stationarity conditions applied to the penalized problems.

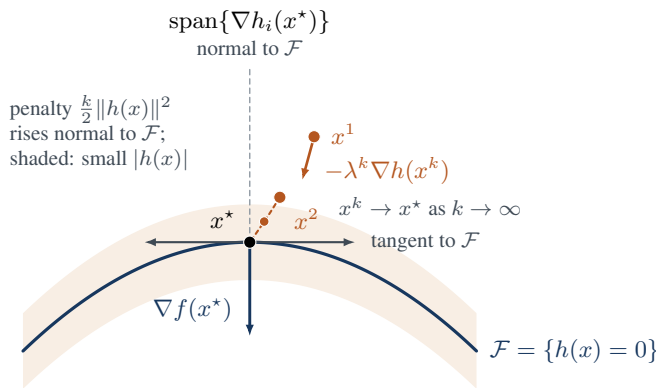


Figure 3.3: Geometry of the quadratic penalty method. The penalty $\frac{k}{2}\|h(x)\|^2$ builds curvature normal to the feasible manifold $\mathcal{F} = \{h(x) = 0\}$ that grows with k , so the unconstrained minimizers x^k of the penalized objective are pushed toward $\mathcal{F}$ and converge to x^* . The scaled residuals $\lambda^k = k h(x^k)$ converge to the multipliers λ^* , and the restoring force $-\lambda^k \nabla h(x^k)$ of the penalty acts along the normal; in the limit the objective gradient is purely normal, i.e. $\nabla f(x^*) \in \text{span}\{\nabla h_i(x^*)\}$.

Second-order limit. The Hessian of F^k is

$$\nabla^2 F^k(x) = \nabla^2 f(x) + k \sum_{i \in \mathcal{P}} \left(\nabla h_i(x) \nabla h_i(x)^\top + h_i(x) \nabla^2 h_i(x) \right).$$

If x^k is a local minimizer of F^k , then $d^\top \nabla^2 F^k(x^k) d \geq 0$ for all $d \in \mathbb{R}^n$.

Restricting to directions $d \in T_{\mathcal{F}}(x^*)$ and passing to the limit, using $k h_i(x^k) \rightarrow \lambda_i^*$, we obtain

$$d^\top \left(\nabla^2 f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i^* \nabla^2 h_i(x^*) \right) d \geq 0 \quad \forall d \in T_{\mathcal{F}}(x^*),$$

which is precisely the second-order Lagrange multiplier necessary condition established earlier.

Geometric interpretation. The quadratic penalty term $\frac{k}{2}\|h(x)\|^2$ introduces strong curvature in directions normal to the manifold $\mathcal{F}$. As k increases, infeasible directions become increasingly expensive, forcing minimizers toward $\mathcal{F}$. The scaled residuals $\lambda_i^k = k h_i(x^k)$ converge to the multipliers λ_i^* , which represent the normal components balancing the objective gradient.

Thus the classical Euler–Lagrange condition

$$\nabla f(x^*) \in \text{span}\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$$

emerges naturally from unconstrained optimality conditions applied to the penalized problems. Figure 3.3 summarizes this geometry.

Numerical implications of penalty and augmented Lagrangian methods

The quadratic penalty and augmented Lagrangian constructions are not merely analytical devices for recovering the Euler–Lagrange conditions. They provide a systematic bridge between the geometric optimality theory developed above and practical numerical algorithms for solving equality-constrained problems.

Consider the equality-constrained problem (3.3). The quadratic penalty method replaces it by the unconstrained problems

$$\min_{x \in \mathbb{R}^n} F^k(x) = f(x) + \frac{k}{2} \|h(x)\|^2.$$

In this way, constrained optimization is embedded into a sequence of unconstrained problems, each of which may be treated using standard methods from smooth unconstrained optimization, such as gradient or Newton-type schemes.

The first-order optimality condition for a local minimizer x^k of F^k is

$$\nabla f(x^k) + k \nabla h(x^k) h(x^k) = 0.$$

Introducing $\lambda^k = k h(x^k)$ transforms this condition into

$$\nabla f(x^k) + \nabla h(x^k) \lambda^k = 0,$$

which formally resembles the Lagrange multiplier equation. Thus the multiplier arises numerically as a scaled constraint violation. When $x^k \rightarrow x^*$ and $k \rightarrow \infty$, the limit relation $\lambda^* = \lim_{k \rightarrow \infty} \lambda^k$ recovers the Euler–Lagrange condition established earlier. The Lagrange multiplier therefore admits a natural computational interpretation as the asymptotic penalty weight required to enforce feasibility.

However, the quadratic penalty approach exhibits a fundamental numerical difficulty. The Hessian of the penalized objective is

$$\nabla^2 F^k(x) = \nabla^2 f(x) + k \nabla h(x) \nabla h(x)^\top + k \sum_{i=1}^p h_i(x) \nabla^2 h_i(x).$$

Near the feasible set, the dominant additional curvature term is $k \nabla h(x) \nabla h(x)^\top$. This term acts in directions normal to the constraint manifold, while tangential curvature is governed primarily by the Hessian of the Lagrangian. As k increases, curvature in the normal directions grows proportionally to k , whereas tangential curvature remains moderate. The resulting anisotropy leads to severe ill-conditioning, which degrades the performance of Newton and quasi-Newton methods. Thus pure penalty methods require increasingly large penalty parameters to enforce feasibility, but large penalties produce numerically stiff problems.

The augmented Lagrangian method resolves this difficulty by combining multiplier information with moderate penalization. For fixed $k > 0$, define

$$\mathcal{L}_k(x, \lambda) = f(x) + \lambda^\top h(x) + \frac{k}{2} \|h(x)\|^2.$$

Instead of sending $k \rightarrow \infty$, one alternates between minimizing $\mathcal{L}_k(x, \lambda^j)$ with respect to x and updating the multiplier according to

$$\lambda^{j+1} = \lambda^j + k h(x^{j+1}).$$

This update mirrors the asymptotic relation $\lambda^* = \lim_{k \rightarrow \infty} k h(x^k)$, but enforces it dynamically at finite penalty parameter.

At a feasible point, where $h(x) = 0$, the stationarity condition

$$\nabla f(x) + \nabla h(x) \lambda + k \nabla h(x) h(x) = 0$$

reduces to the classical multiplier equation. Thus the augmented Lagrangian preserves the normal-cone characterization $-\nabla f(x^*) \in N_{\mathcal{F}}(x^*)$ derived earlier.

From a second-order perspective, the Hessian of the augmented Lagrangian satisfies

$$\nabla_{xx}^2 \mathcal{L}_k(x, \lambda) = \nabla_{xx}^2 \mathcal{L}(x, \lambda) + k \nabla h(x) \nabla h(x)^\top + o(1).$$

Along directions tangent to the feasible manifold, the curvature approximates the reduced Hessian $N^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) N$ appearing in the second-order sufficient condition. Along normal directions, the additional quadratic term provides regularization. This balanced curvature structure explains the superior numerical behavior of augmented Lagrangian methods compared with pure penalty schemes.

Example 3.24 Penalty and augmented Lagrangian on a nonlinear constraint.

Consider the equality-constrained problem

$$\min_{x \in \mathbb{R}^2} f(x) = \frac{1}{2}(x_1^2 + x_2^2) \quad \text{subject to} \quad h(x) = x_1 x_2 - 1 = 0.$$

The feasible set is the hyperbola $\{x_1 x_2 = 1\}$, so the problem asks for the point of this hyperbola closest to the origin. The constraint is genuinely nonlinear, which is what makes the penalty geometry nontrivial.

Exact solution and multiplier. The symmetry $x \mapsto (x_2, x_1)$ places the minimizer on the line $x_1 = x_2$; intersecting with $x_1 x_2 = 1$ gives the first-quadrant solution $x^* = (1, 1)$, with $f(x^*) = 1$. Since $\nabla f(x) = (x_1, x_2)$ and $\nabla h(x) = (x_2, x_1)$, both gradients equal $(1, 1)$ at x^* , so the stationarity condition $\nabla f(x^*) + \lambda^* \nabla h(x^*) = 0$ yields $\lambda^* = -1$.

Quadratic penalty method. For $k > 0$ the penalized objective is

$$F^k(x) = \frac{1}{2}(x_1^2 + x_2^2) + \frac{k}{2}(x_1 x_2 - 1)^2,$$

with stationarity equations $x_1 + k(x_1 x_2 - 1)x_2 = 0$ and $x_2 + k(x_1 x_2 - 1)x_1 = 0$. Subtracting the two equations factors as

$$(x_1 - x_2) [1 - k(x_1 x_2 - 1)] = 0.$$

The bracket cannot vanish at a real minimizer: it would require $x_1 + x_2 = 0$ and hence $x_1 x_2 = -x_1^2 < 0$, incompatible with the resulting $x_1 x_2 = 1 + \frac{1}{k} > 0$. Thus every stationary point satisfies $x_1 = x_2 =: t$, and along this line

$$\phi(t) = t^2 + \frac{k}{2}(t^2 - 1)^2, \quad \phi'(t) = 2t[1 + k(t^2 - 1)].$$

For $k > 1$ the nonzero root gives the penalized minimizer

$$x^k = \left(\sqrt{1 - \frac{1}{k}}, \sqrt{1 - \frac{1}{k}} \right) \xrightarrow[k \rightarrow \infty]{} (1, 1) = x^*,$$

whose error decays at the linear rate $\|x^k - x^*\| = \sqrt{2}(1 - \sqrt{1 - 1/k}) = O(1/k)$. The scaled residual reproduces the multiplier *exactly* at every finite k :

$$h(x^k) = \left(1 - \frac{1}{k}\right) - 1 = -\frac{1}{k}, \quad \lambda^k = k h(x^k) = -1 = \lambda^*.$$

Ill-conditioning. The Hessian of the penalized objective is

$$\nabla^2 F^k(x) = \begin{pmatrix} 1 + kx_2^2 & k(2x_1x_2 - 1) \\ k(2x_1x_2 - 1) & 1 + kx_1^2 \end{pmatrix},$$

and at x^k , where $x_1x_2 = 1 - \frac{1}{k}$, it becomes

$$\nabla^2 F^k(x^k) = \begin{pmatrix} k & k-2 \\ k-2 & k \end{pmatrix},$$

with eigenvalue $2k - 2$ along the normal direction $(1, 1) \parallel \nabla h(x^k)$ and eigenvalue 2 along the tangent direction $(1, -1)$. Hence

$$\kappa(\nabla^2 F^k(x^k)) = \frac{2k-2}{2} = k-1,$$

which diverges linearly in k : the curvature normal to the constraint stiffens like k while the tangential curvature stays fixed at 2. This is exactly the anisotropy anticipated above and depicted in Figure 3.3, and it is why pure penalty methods become numerically stiff as k grows.

Augmented Lagrangian method. Now keep a moderate parameter, say $k = 2$, and use

$$\mathcal{L}_k(x, \lambda) = \frac{1}{2}(x_1^2 + x_2^2) + \lambda(x_1x_2 - 1) + \frac{k}{2}(x_1x_2 - 1)^2.$$

On the symmetric line $x_1 = x_2 = t$ stationarity reads $2t[1 + \lambda + k(t^2 - 1)] = 0$. Starting from $\lambda^0 = 0$, the first inner minimization is the pure-penalty solve above, returning $x^1 = x^k$ with $h(x^1) = -\frac{1}{k}$. The multiplier update already reaches λ^* :

$$\lambda^1 = \lambda^0 + k h(x^1) = 0 + k\left(-\frac{1}{k}\right) = -1.$$

With $\lambda = \lambda^1 = -1$ the bracket collapses to $k(t^2 - 1)$, whose root is $t = 1$, so the second inner minimization returns the *exact* solution

$$x^2 = (1, 1) = x^*, \quad h(x^2) = 0, \quad \lambda^2 = \lambda^1 = -1.$$

The iteration has converged. Thus the augmented Lagrangian recovers the exact primal–dual pair (x^*, λ^*) at a fixed, moderate penalty parameter after a single multiplier update, whereas the pure penalty method reaches x^* only as $k \rightarrow \infty$ and at the cost of a Hessian whose condition number $k - 1$ diverges. $\square$

3.3 ■ Fritz–John and Mangasarian–Fromovitz Conditions

This section extends the classical Lagrange multiplier method from equality-constrained optimization to the general setting with inequality constraints. The geometric framework shifts from tangent spaces to tangent cones, and first-order optimality is expressed as a normal cone inclusion. The existence of multipliers is then derived through theorems of alternatives, revealing multiplier systems as algebraic representations of cone polarity. This approach leads naturally to the Fritz–John conditions in full generality and, under appropriate constraint qualifications

such as the Mangasarian–Fromovitz condition, to the normalized Karush–Kuhn–Tucker (KKT) conditions. The following brief historical account places these developments in context.

Fritz–John conditions (1948). The earliest general first-order necessary conditions for constrained optimization were developed by Fritz John in 1948 [60]. John generalized the classical method of Lagrange multipliers to optimization problems involving both equality and inequality constraints, without imposing any regularity assumptions on the constraint set. The resulting Fritz–John conditions assert that, at a local minimum, there exist multipliers associated with the objective function and the constraints, not all zero, that satisfy a stationarity condition together with complementary slackness.

A distinctive feature of John’s formulation is the presence of an additional multiplier associated with the objective function, which is allowed to vanish. While this makes the Fritz–John conditions universally applicable, it also allows degenerate situations in which the objective function does not appear in the stationarity condition. Such cases later became known as *abnormal multipliers* and motivated the development of constraint qualifications.

Karush–Kuhn–Tucker conditions (1951). The conditions now known as the Karush–Kuhn–Tucker (KKT) conditions emerged in the early development of nonlinear programming. An early version appeared in the 1939 master’s thesis of William Karush [67], but the framework was independently and systematically developed by Kuhn and Tucker in 1951 [73], motivated in part by applications to economics and game theory.

The KKT conditions resemble the Fritz–John conditions but impose a normalization that fixes the multiplier associated with the objective function to one. This normalization eliminates abnormal multipliers and yields a clean stationarity condition involving only the gradients of the objective function and the active constraints. However, the validity of this normalization depends on additional regularity assumptions on the constraint set, which were not yet formalized in a systematic way at the time.

Constraint qualifications and Slater’s condition. The need to characterize precisely when abnormal multipliers are excluded led to the systematic study of constraint qualifications in the 1960s. A particularly influential contribution was made by Mangasarian and Fromovitz in 1967 [82], who introduced what is now known as the Mangasarian–Fromovitz Constraint Qualification (MFCQ). MFCQ requires linear independence of the equality constraint gradients and the existence of a feasible direction that strictly decreases all active inequality constraints. This condition ensures that the linearized feasible cone has nonempty interior relative to the equality subspace, thereby ruling out abnormal Fritz–John multipliers. Under MFCQ, the multiplier associated with the objective function is strictly positive, and the Fritz–John conditions reduce exactly to the KKT conditions.

In the special case of convex optimization, an even more transparent interiority condition had already appeared in the work of Slater in 1950 [110]. Slater’s condition requires the existence of a strictly feasible point satisfying all inequality constraints with strict inequality while meeting the equalities. In convex problems, Slater’s condition guarantees the validity of the KKT conditions and, moreover, ensures strong duality. From a geometric viewpoint, Slater’s condition implies that the feasible set has interior relative to the equality manifold, and thus automatically satisfies MFCQ at every feasible point. In this sense, Slater’s condition can be viewed as the convex-analytic precursor to modern constraint qualifications.

From a modern perspective, these developments establish a clear logical hierarchy in first-order optimality theory. The Fritz–John conditions provide the most general necessary conditions, valid without regularity assumptions. Constraint qualifications such as MFCQ identify settings in

which degeneracy is excluded. In convex programming, Slater’s condition guarantees both the KKT conditions and strong duality. Under such regularity assumptions, the Fritz–John conditions reduce to the KKT conditions, which form the cornerstone of modern nonlinear programming [10, 15, 90].

3.3.1 ■ Theorems of Alternatives: Gordan, Farkas, and Motzkin

We now present a family of fundamental results from linear algebra and convex analysis known as *theorems of alternatives*. These results assert that, given a pair of linear systems of a certain form, *exactly one* of the two systems admits a solution. Such results formalize a basic geometric dichotomy: either a feasible direction exists, or there is a dual certificate proving that no such direction can exist.

The classical theorems of alternatives form a strict hierarchy in expressive power:

Motzkin’s Transposition Theorem $\supset$ Farkas’ Lemma $\supset$ Gordan’s Theorem.

All three results are equivalent in finite dimensions, but they differ in how they treat equalities, weak inequalities, and strict inequalities. This distinction explains why different theorems appear naturally in different parts of constrained optimization theory.

Gordan’s theorem

We begin with the most restrictive but conceptually transparent result.

Theorem 3.25 (Gordan’s Theorem of the Alternative). *Let $B \in \mathbb{R}^{q \times n}$. Exactly one of the following two systems has a solution:*

- (1) *There exists $d \in \mathbb{R}^n$ such that $Bd < 0$.*
- (2) *There exists $\mu \in \mathbb{R}_{\geq 0}^q$, $\mu \neq 0$, such that $B^\top \mu = 0$.*

The two systems are mutually exclusive.

Proof. We show that exactly one of the two systems can hold.

Assume first that there exists $d \in \mathbb{R}^n$ such that $Bd < 0$. Suppose, for contradiction, that there also exists $\mu \in \mathbb{R}_{\geq 0}^q$, $\mu \neq 0$, such that $B^\top \mu = 0$. Taking the inner product with d yields $0 = \mu^\top (Bd) < 0$, since $\mu \geq 0$, $\mu \neq 0$, and $Bd < 0$, which is a contradiction. Hence, the second system cannot hold if the first does.

Conversely, assume that there is no $d \in \mathbb{R}^n$ such that $Bd < 0$. Equivalently, the range $R := \{Bd \mid d \in \mathbb{R}^n\}$, which is a linear subspace of $\mathbb{R}^q$, does not meet the open convex cone $U := \{v \in \mathbb{R}^q : v < 0\}$. Since U is open, convex and nonempty and R is a subspace disjoint from it, the separating hyperplane theorem provides a nonzero $\mu \in \mathbb{R}^q$ and a scalar α with

$$\mu^\top r \geq \alpha \geq \mu^\top v \quad \forall r \in R, \forall v \in U.$$

Because R is a subspace, the left inequality forces $\mu^\top r = 0$ for every $r \in R$, that is, $\mu^\top Bd = 0$ for all d , hence $B^\top \mu = 0$; it also forces $\alpha \leq 0$. For the right inequality, let $v = -\varepsilon e_k$ with e_k the k th standard basis vector and $\varepsilon > 0$; letting the remaining components of v tend to 0^- gives $-\varepsilon \mu_k \leq \alpha \leq 0$ for every $\varepsilon > 0$, whence $\mu_k \geq 0$. Thus $\mu \in \mathbb{R}_{\geq 0}^q$ is nonzero and satisfies $B^\top \mu = 0$, which is the second system. Therefore exactly one of the two systems is solvable. $\square$ Gordan’s theorem concerns *homogeneous* systems and *strict* inequalities. Geometrically, it

asserts that either there exists a direction that strictly decreases all components of Bd , or there exists a nontrivial nonnegative linear combination of the rows of B that annihilates all such directions. This theorem is the analytical backbone of strict feasibility conditions and appears naturally in constraint qualifications such as the Mangasarian–Fromovitz condition.

Example 3.26 Illustration of Gordan's Theorem.

We present two simple examples showing that exactly one of the two systems in Theorem 3.25 holds.

Case 1: Existence of a strictly feasible direction. Let $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$.

Consider the vector

$$d = \begin{pmatrix} -1 \\ -1 \end{pmatrix}.$$

Then

$$Bd = \begin{pmatrix} -1 \\ -1 \end{pmatrix} < 0,$$

so system (1) of Theorem 3.25 is satisfied. We now check system (2). The condition $B^\top \mu = 0$ becomes

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \mu_1 \geq 0, \mu_2 \geq 0, \quad \mu_1 = 0, \mu_2 = 0.$$

The only solution is $\mu = 0$, which is excluded. Hence system (2) has no solution, as predicted by Gordan's theorem.

Case 2: Existence of a nontrivial dual certificate. Let $B = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix} \in$

$\mathbb{R}^{2 \times 2}$. For any $d = (d_1, d_2)^\top \in \mathbb{R}^2$, we have $Bd = \begin{pmatrix} d_1 \\ -d_1 \end{pmatrix}$. It is impossible for both components to be strictly negative, so there exists no d such that $Bd < 0$. Thus system (1) has no solution.

Now consider system (2). Choose

$$\mu = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \in \mathbb{R}_{\geq 0}^2, \quad \mu \neq 0.$$

Then

$$B^\top \mu = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus system (2) is satisfied.

In the first case, there exists a direction that strictly decreases all linear forms defined by the rows of B . In the second case, no such direction exists, and this infeasibility is certified by a nontrivial nonnegative combination of the rows of B that annihilates all directions. $\square$

Farkas' Lemma

Farkas' Lemma generalizes Gordan's theorem by allowing equality constraints and weak inequalities. This additional flexibility makes it the central algebraic tool behind first-order optimality conditions.

Theorem 3.27 (Farkas' Lemma (Conic Form)). *Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{q \times n}$, and $w \in \mathbb{R}^n$ be given. Exactly one of the following two systems has a solution:*

(1) *There exists $d \in \mathbb{R}^n$ such that*

$$Ad = 0, \quad Bd \leq 0, \quad w^\top d < 0. \quad (3.15)$$

(2) *There exist multipliers $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}_{\geq 0}^q$ such that*

$$w + A^\top \lambda + B^\top \mu = 0. \quad (3.16)$$

Moreover, the two systems are mutually exclusive.

Remark 3.28 (Orientation of the multipliers). The sign in (3.16) is the one that matters in optimization. Written as $-w = A^\top \lambda + B^\top \mu$ it says that $-w$ lies in the polar of the cone $\{d : Ad = 0, Bd \leq 0\}$; written as in (3.16) it is already the stationarity equation of a Lagrangian, with $w = \nabla f(x^*)$, the rows of A the equality-constraint gradients and the rows of B the active inequality-constraint gradients. Reversing the sign of μ here would reverse the sign convention of every inequality multiplier in the chapter.

Proof. Define the cone $K = \{d \in \mathbb{R}^n \mid Ad = 0, Bd \leq 0\}$. The set K is a closed convex cone, obtained as the intersection of a linear subspace and a polyhedral cone.

Suppose first that multipliers $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}_{\geq 0}^q$ satisfy (3.16), i.e. $w = -A^\top \lambda - B^\top \mu$. Then for any $d \in K$,

$$w^\top d = -\lambda^\top (Ad) - \mu^\top (Bd) = 0 - \mu^\top (Bd) \geq 0,$$

since $Bd \leq 0$ and $\mu \geq 0$ give $\mu^\top (Bd) \leq 0$. Thus no $d \in K$ can satisfy $w^\top d < 0$, so the first system is infeasible.

Conversely, assume that no $d \in K$ satisfies $w^\top d < 0$. Then $w^\top d \geq 0 \forall d \in K$, which is equivalent to $(-w)^\top d \leq 0 \forall d \in K$. Hence $-w \in K^\circ$, where $K^\circ = \{y \in \mathbb{R}^n : y^\top d \leq 0 \forall d \in K\}$ denotes the polar cone of K .

Since $K = \{d : Ad = 0\} \cap \{d : Bd \leq 0\}$, the polar-of-intersection identity yields

$$K^\circ = \text{span}\{A^\top\} + \text{cone}\{B^\top\}.$$

Therefore there exist multipliers $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}_{\geq 0}^q$ such that $-w = A^\top \lambda + B^\top \mu$, which is exactly (3.16). (Note that λ may be re-signed freely, being unrestricted, but μ may not, so the sign in (3.16) cannot be absorbed into the multipliers.) $\square$

Farkas' Lemma therefore states that either there exists a direction in the cone K that strictly decreases the linear functional $w^\top d$, or $-w$ lies in the cone generated by the constraint normals with nonnegative multipliers. Geometrically, this expresses the polarity relation $K \cap (-K^\circ) = \{0\}$. The lemma is precisely the mechanism that converts the absence of feasible descent directions into the existence of Lagrange multipliers. It underlies the Fritz–John conditions and, under suitable constraint qualifications, the Karush–Kuhn–Tucker conditions.

Gordan's theorem is not literally a special case of the form above, because it concerns a system of *strict* inequalities with no distinguished objective row. Both are specializations of Motzkin's transposition theorem, stated next, which is why that theorem sits at the top of the hierarchy.

Motzkin's transposition theorem

The most general theorem of alternatives is due to Motzkin and allows equalities, weak inequalities and strict inequalities to appear in the same homogeneous system. It is the tool that produces Fritz–John multipliers.

Theorem 3.29 (Motzkin's Transposition Theorem). *Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{q \times n}$ and $C \in \mathbb{R}^{r \times n}$ be given, with $r \geq 1$. Exactly one of the following systems has a solution:*

(1) *There exists $d \in \mathbb{R}^n$ such that*

$$Ad = 0, \quad Bd \leq 0, \quad Cd < 0. \quad (3.17)$$

(2) *There exist multipliers $\lambda \in \mathbb{R}^m$, $\mu \in \mathbb{R}_{\geq 0}^q$ and $\nu \in \mathbb{R}_{\geq 0}^r$ with $\nu \neq 0$ such that*

$$A^\top \lambda + B^\top \mu + C^\top \nu = 0. \quad (3.18)$$

Proof. *The two systems exclude each other:* Suppose d solves (3.17) and (λ, μ, ν) solves (3.18). Pairing (3.18) with d gives

$$0 = \lambda^\top (Ad) + \mu^\top (Bd) + \nu^\top (Cd) = 0 + \mu^\top (Bd) + \nu^\top (Cd).$$

Here $\mu^\top (Bd) \leq 0$ because $\mu \geq 0$ and $Bd \leq 0$, while $\nu^\top (Cd) < 0$ because $\nu \geq 0$ is nonzero and $Cd < 0$ strictly. The right-hand side is therefore strictly negative, a contradiction.

At least one system holds. Assume (3.17) has no solution. Because the system is homogeneous, a solution could be rescaled by any $t > 0$, so (3.17) is solvable if and only if the system

$$Ad = 0, \quad Bd \leq 0, \quad Cd \leq -\mathbf{1}$$

is solvable, where $\mathbf{1} \in \mathbb{R}^r$ is the vector of ones: given a solution of (3.17), scale d so that every component of Cd is at most -1 ; conversely any solution of the displayed system already solves (3.17).

Consider the closed convex cone $K := \{(d, t) \in \mathbb{R}^n \times \mathbb{R} : Ad = 0, Bd \leq 0, Cd + t\mathbf{1} \leq 0, t \geq 0\}$ and the linear functional $(d, t) \mapsto -t$ on it. Infeasibility of the displayed system says precisely that $-t \geq 0$, i.e. $t \leq 0$, on K , hence $t = 0$ for every $(d, t) \in K$. Applying Farkas' Lemma (Theorem 3.27) to the cone K and the vector $w := (0, -1)$, for which no element of K achieves $w^\top (d, t) < 0$, yields multipliers $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}_{\geq 0}^q, \nu \in \mathbb{R}_{\geq 0}^r, \sigma \geq 0$ with

$$(0, -1) + (A^\top \lambda + B^\top \mu + C^\top \nu, \mathbf{1}^\top \nu - \sigma) = 0.$$

The first block gives (3.18), and the second gives $\mathbf{1}^\top \nu = 1 + \sigma \geq 1$, so $\nu \neq 0$. $\square$

Remark 3.30 (Specializations). Taking B absent recovers the mixed equality/strict-inequality alternative used below to derive the Fritz–John conditions. Taking A and B absent and $C = B_0$ recovers Gordan's Theorem 3.25: either $B_0 d < 0$ for some d , or $B_0^\top \nu = 0$ for some $\nu \geq 0$, $\nu \neq 0$. Taking C to be the single row $w^\top$ recovers Farkas' Lemma 3.27, since (3.18) then reads $A^\top \lambda + B^\top \mu + \nu w = 0$ with a scalar $\nu > 0$, which after division by ν is (3.16).

Motzkin's theorem strictly generalizes Farkas' Lemma by permitting strict and weak inequalities to coexist. Farkas' Lemma is obtained as a special case by eliminating strict inequalities, while Gordan's theorem arises as a homogeneous specialization. Although Motzkin's theorem is rarely stated explicitly in basic optimization texts, it provides the most general separation principle underlying all linear constraint systems.

The three results play complementary roles in optimality theory:

- *Gordan’s theorem* captures *strict* feasibility and is the engine behind constraint qualifications such as the Mangasarian–Fromovitz condition;
- *Farkas’ Lemma* converts the absence of feasible descent directions into the *existence of multipliers*, and is the algebraic heart of KKT theory;
- *Motzkin’s theorem* unifies strict and weak feasibility in a single statement, subsuming the other two.

Together they supply the algebraic foundation on which every first-order optimality condition in this chapter rests: each such condition is, at bottom, a statement that “no improving feasible direction exists,” and a theorem of the alternative is exactly what turns that non-existence into the multipliers we can compute with.

3.3.2 ■ Theorems of Alternatives and Polar Cones

The theorems of alternatives presented in Section 3.3.1 admit a unified geometric interpretation through cone polarity. Each theorem expresses a fundamental incompatibility: a closed convex cone and its polar cannot simultaneously contain mutually conflicting nontrivial elements. This polarity principle is the geometric backbone behind multiplier existence and normal-cone optimality conditions.

Polar cone and incompatibility principle. Let $K \subset \mathbb{R}^n$ be a closed convex cone. Its polar cone is defined by

$$K^\circ = \{y \in \mathbb{R}^n : y^\top x \leq 0 \text{ for all } x \in K\}. \quad (3.19)$$

The bipolar theorem yields $K^{\circ\circ} = K$. By definition, $x \in K$ and $y \in K^\circ$ imply $x^\top y \leq 0$. Consequently, $K \cap (-K^\circ) = \{0\}$. Thus a nonzero vector cannot belong simultaneously to a cone and to the negative of its polar. All classical theorems of alternatives are refined expressions of this incompatibility principle.

Gordan’s theorem as strict polarity

Consider Theorem 3.25. Let $K := \{Bd : d \in \mathbb{R}^n\} \subset \mathbb{R}^q$ denote the image cone generated by the rows of B . The system $Bd < 0$ asks whether K intersects the interior of the negative orthant $-\mathbb{R}_+^q$. If such a d exists, then $K \cap \text{int}(-\mathbb{R}_+^q) \neq \emptyset$. If not, the separating hyperplane theorem implies the existence of a nonzero vector $\mu \geq 0$ such that $\mu^\top (Bd) = 0 \forall d$, equivalently, $B^\top \mu = 0$. Since $\mu \geq 0$ belongs to $\mathbb{R}_+^q$, this means $\mathbb{R}_+^q \cap K^\circ \neq \{0\}$.

Thus Gordan’s theorem may be expressed in polarity form as

$$K \cap \text{int}(-\mathbb{R}_+^q) \neq \emptyset \iff \mathbb{R}_+^q \cap K^\circ = \{0\}.$$

Geometrically: either a strict descent direction exists, or the polar cone contains a nontrivial nonnegative annihilator, but not both.

Farkas’ lemma via polar of an intersection

The conic version of Farkas’ lemma (Theorem 3.27) fits exactly into the polar-cone framework. Define the cone $K = \{d \in \mathbb{R}^n : Ad = 0, Bd \leq 0\}$. This cone is the intersection of a linear subspace and a polyhedral cone: $K = \ker A \cap \{d : Bd \leq 0\}$.

The polar-of-intersection identity gives

$$K^\circ = (\ker A)^\circ + (\{Bd \leq 0\})^\circ. \quad (3.20)$$

Since

$$(\ker A)^\circ = \text{span}(A^\top), \quad (\{Bd \leq 0\})^\circ = \text{cone}(B^\top),$$

we obtain

$$K^\circ = \{A^\top \lambda + B^\top \mu : \lambda \in \mathbb{R}^m, \mu \geq 0\}.$$

Now consider a vector $w \in \mathbb{R}^n$. The absence of a feasible direction satisfying $Ad = 0, Bd \leq 0, w^\top d < 0$ is equivalent to $w^\top d \geq 0 \forall d \in K$, that is, $-w \in K^\circ$.

Using (3.20), this means there exist multipliers $\lambda \in \mathbb{R}^m$ and $\mu \geq 0$ such that $-w = A^\top \lambda + B^\top \mu$, equivalently $w + A^\top \lambda + B^\top \mu = 0$.

Thus Farkas' lemma is precisely the statement

$$\text{either } \exists d \in K \text{ with } w^\top d < 0, \quad \text{or } -w \in K^\circ,$$

but not both. It is a direct algebraic restatement of cone polarity.

Motzkin's theorem and multiplier structure

Motzkin's transposition theorem generalizes both Gordan and Farkas by allowing mixed systems. In cone language, if $K = \ker A \cap \{Bd \leq 0\}$, then

$$K^\circ = \text{span}(A^\top) + \text{cone}(B^\top).$$

The multipliers appearing in nonlinear programming are therefore not arbitrary: equality constraints contribute unrestricted multipliers (a linear span), while inequality constraints contribute nonnegative multipliers (a convex cone). This decomposition is nothing more than the polar-of-intersection identity applied to the linearized feasible cone.

Connection to normal cones and optimality. In Chapter 2, first-order optimality was expressed as $-\nabla f(x^*) \in N_{\mathcal{F}}(x^*)$. When the feasible set is described by linearized constraints, its tangent cone is

$$T_{\mathcal{F}}(x^*) = \{d : \nabla h_i(x^*)^\top d = 0, \nabla g_j(x^*)^\top d \leq 0\}.$$

The normal cone is the polar of this tangent cone, $N_{\mathcal{F}}(x^*) = T_{\mathcal{F}}(x^*)^\circ$.

Applying the polar-of-intersection identity recovers exactly the multiplier representation

$$-\nabla f(x^*) = \sum_i \lambda_i \nabla h_i(x^*) + \sum_j \mu_j \nabla g_j(x^*), \quad \mu_j \geq 0.$$

Thus the Fritz–John and KKT multiplier systems are not ad hoc constructions; they are algebraic manifestations of cone polarity. The classical theorems of alternatives are precisely the separation results that guarantee the existence of such multipliers.

3.3.3 ■ First-Order Necessary Conditions for General Constrained Programs

We now state a general first-order necessary condition for optimality for constrained optimization problems of the form (3.1). This condition applies to problems with both equality and inequality constraints and requires only first-order differentiability.

Theorem 3.31 (First-Order Necessary Condition (Fritz–John / Mangasarian–Fromovitz)). *Consider the constrained optimization problem (3.1). Assume that the objective function f and*

the inequality constraint functions g_j , $j \in \mathcal{Q}$, are differentiable at a feasible point $x^* \in \mathcal{F}$, and that the equality constraint functions h_i , $i \in \mathcal{P}$, are continuously differentiable at x^* .

If x^* is a local minimum, then there exist scalars

$$u_0 \geq 0, \quad u_j \geq 0 \quad (j \in \mathcal{Q}), \quad \lambda_i \in \mathbb{R} \quad (i \in \mathcal{P}),$$

not all zero, such that

$$u_0 \nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{Q}} u_j \nabla g_j(x^*) = 0, \quad (3.21)$$

together with the complementary slackness conditions

$$u_j g_j(x^*) = 0, \quad j \in \mathcal{Q}. \quad (3.22)$$

Proof. The proof is based on feasible directions and a theorem of alternatives.

Let $x^* \in \mathcal{F}$ be a local minimum of problem (3.1). By definition, there exists a neighborhood $N(x^*)$ such that $f(x^*) \leq f(x)$, $\forall x \in \mathcal{F} \cap N(x^*)$.

Step 1: a degenerate case handled directly. If the equality-constraint gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly dependent, there are scalars λ_i , not all zero, with $\sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) = 0$. Taking $u_0 = 0$ and $u_j = 0$ for all j then satisfies (3.21) with a nonzero multiplier vector, and (3.22) holds trivially. This is the *abnormal* branch of the Fritz–John conditions, and it costs nothing. So assume from now on that $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ is linearly independent.

Step 2: no strictly improving admissible direction exists. We claim there is no $d \in \mathbb{R}^n$ with

$$\nabla h_i(x^*)^\top d = 0 \quad (i \in \mathcal{P}), \quad \nabla g_j(x^*)^\top d < 0 \quad (j \in \mathcal{Q}(x^*)), \quad \nabla f(x^*)^\top d < 0. \quad (3.23)$$

It is worth emphasising why this claim, and not the linearized cone, is the right object: Theorem 3.7 only shows that feasible directions *lie inside* the linearized cone, so nonnegativity of $\nabla f(x^*)^\top d$ on feasible directions says nothing about the rest of that cone. The *strict* inequalities in (3.23) are what allow a genuine feasible arc to be constructed.

Suppose such a d existed. Since the equality gradients are linearly independent, the map $x \mapsto (h_1(x), \dots, h_p(x))$ has surjective derivative at x^* , so the implicit function theorem yields a C^1 curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathbb{R}^n$ with $\gamma(0) = x^*$, $\dot{\gamma}(0) = d$ and $h_i(\gamma(t)) = 0$ for every i and all small t . For $j \in \mathcal{Q}(x^*)$,

$$g_j(\gamma(t)) = g_j(x^*) + t \nabla g_j(x^*)^\top d + o(t) = t \nabla g_j(x^*)^\top d + o(t) < 0$$

for all small $t > 0$, while $g_j(\gamma(t)) < 0$ for inactive j by continuity. Hence $\gamma(t) \in \mathcal{F}$ for all small $t > 0$. But then

$$f(\gamma(t)) = f(x^*) + t \nabla f(x^*)^\top d + o(t) < f(x^*)$$

for all small $t > 0$, contradicting local optimality of x^* . This proves the claim.

Step 3: multipliers from Motzkin's theorem. System (3.23) is exactly system (1) of Theorem 3.29 with A the matrix whose rows are $\nabla h_i(x^*)^\top$, $i \in \mathcal{P}$, no weak-inequality block, and C the matrix whose rows are $\nabla g_j(x^*)^\top$, $j \in \mathcal{Q}(x^*)$, together with the extra row $\nabla f(x^*)^\top$. Since

that system is infeasible, Motzkin's theorem supplies $\lambda \in \mathbb{R}^p$ and nonnegative multipliers, not all of the strict block vanishing, such that

$$u_0 \nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{Q}(x^*)} u_j \nabla g_j(x^*) = 0, \quad (3.24)$$

with $u_0 \geq 0$, $u_j \geq 0$ and $(u_0, u_{\mathcal{Q}(x^*)}) \neq 0$. In particular the full multiplier vector is nonzero, as required.

Complementary slackness. For any $j \notin \mathcal{Q}(x^*)$, we have $g_j(x^*) < 0$. Such constraints do not restrict feasible directions and therefore cannot appear in the conic representation (3.24). Hence, $u_j = 0$, $j \notin \mathcal{Q}(x^*)$, which implies the complementary slackness conditions $u_j g_j(x^*) = 0$, $j \in \mathcal{Q}$.

The existence of multipliers satisfying (3.24) together with complementary slackness establishes the Fritz–John optimality conditions (3.21)–(3.22), completing the proof. $\square$

Example 3.32 Abnormal Fritz–John point.

Consider the constrained problem

$$\min_{x \in \mathbb{R}^2} f(x) = x_1 \quad \text{s.t.} \quad g(x) := x_1^2 + x_2^2 \leq 0.$$

The feasible set is $\mathcal{F} = \{(0, 0)\}$, so $x^* = (0, 0)$ is trivially a local (and global) minimum. The gradients at x^* are

$$\nabla f(0, 0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \nabla g(0, 0) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The KKT stationarity condition requires $\nabla f(x^*) + \mu \nabla g(x^*) = 0$, $\mu \geq 0$, which becomes

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ 0 \end{pmatrix} = 0.$$

This equation has no solution for any $\mu \geq 0$. Hence the KKT conditions fail.

However, the Fritz–John conditions allow a scalar multiplier $\nu_0 \geq 0$:

$$\nu_0 \nabla f(x^*) + \nu_1 \nabla g(x^*) = 0.$$

Choosing $\nu_0 = 0$ and $\nu_1 = 1$ we obtain $0 \cdot \nabla f(x^*) + 1 \cdot \nabla g(x^*) = 0$.

Thus the Fritz–John conditions are satisfied in the *abnormal case* $\nu_0 = 0$. The objective gradient plays no role in stationarity. This abnormality occurs because $\nabla g(x^*) = 0$, so the linearized feasible cone is the entire space, and no constraint qualification holds. $\square$

Remark 3.33 (Derivation via the normal cone). The Fritz–John conditions follow directly from the first-order optimality condition expressed in normal cone form.

Let

$$\mathcal{F} = \{x : h_i(x) = 0, i \in \mathcal{P}, g_j(x) \leq 0, j \in \mathcal{Q}\} \quad (3.25)$$

denote the feasible set and let

$$\mathcal{A}(x^*) = \{j \in \mathcal{Q} : g_j(x^*) = 0\} \quad (3.26)$$

be the active inequality index set.

By Theorem 2.135, if x^* is a local minimum, then

$$-\nabla f(x^*) \in N_{\mathcal{F}}(x^*), \quad (3.27)$$

where $N_{\mathcal{F}}(x^*)$ denotes the normal cone to $\mathcal{F}$.

Under differentiability, the tangent cone admits the linearized form

$$T_{\mathcal{F}}(x^*) = \{d : \nabla h_i(x^*)^\top d = 0, \nabla g_j(x^*)^\top d \leq 0 \text{ for } j \in \mathcal{A}(x^*)\}. \quad (3.28)$$

Since the normal cone is the polar of the tangent cone,

$$N_{\mathcal{F}}(x^*) = T_{\mathcal{F}}(x^*)^\circ, \quad (3.29)$$

Farkas' lemma implies the representation

$$N_{\mathcal{F}}(x^*) = \left\{ \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{A}(x^*)} \mu_j \nabla g_j(x^*) : \lambda_i \in \mathbb{R}, \mu_j \geq 0 \right\}. \quad (3.30)$$

Combining (3.27) and (3.30), we obtain

$$-\nabla f(x^*) = \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{A}(x^*)} \mu_j \nabla g_j(x^*), \quad \mu_j \geq 0. \quad (3.31)$$

Equation (3.31) is precisely the multiplier form of the first-order necessary condition. Allowing an arbitrary nonnegative scaling factor produces the homogeneous Fritz–John system.

Definition 3.34 (Mangasarian–Fromovitz Constraint Qualification (MFCQ)). Consider the constrained optimization problem (3.1), and let $x^* \in \mathcal{F}$ be a feasible point. Denote the active set of inequality constraints at x^* by $\mathcal{Q}(x^*) = \{j \in \mathcal{Q} \mid g_j(x^*) = 0\}$.

The Mangasarian–Fromovitz Constraint Qualification (MFCQ) is said to hold at x^* if the following conditions are satisfied:

- (1) The gradients of the equality constraints are linearly independent at x^* ; that is, the set $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ is linearly independent.
- (2) There exists a vector $d \in \mathbb{R}^n$ such that

$$\nabla h_i(x^*)^\top d = 0, \quad i \in \mathcal{P}, \quad \nabla g_j(x^*)^\top d < 0, \quad j \in \mathcal{Q}(x^*).$$

Example 3.35 MFCQ holds and KKT applies.

Consider

$$\min_{x \in \mathbb{R}^2} f(x) = x_1^2 + x_2^2 \quad \text{s.t.} \quad g(x) := x_1 \leq 0.$$

The unique local minimum is $x^* = (0, 0)$. The active set is $\mathcal{Q}(x^*) = \{1\}$, and $\nabla g(x^*) = (1, 0)$. Choosing the direction $d = (-1, 0)$ yields $\nabla g(x^*)^\top d = -1 < 0$, so the Mangasarian–Fromovitz Constraint Qualification holds. The KKT conditions therefore apply, yielding

$$\nabla f(x^*) + \mu \nabla g(x^*) = 0 \quad \Rightarrow \quad \mu = 0.$$

This example illustrates a regular boundary point where MFCQ guarantees the existence of Lagrange multipliers and rules out abnormal Fritz–John solutions. $\square$

Theorem 3.36 (Karush–Kuhn–Tucker Theorem under the Mangasarian–Fromovitz Constraint Qualification). *Consider the constrained optimization problem (3.1). Assume that the objective function f and the constraint functions h_i , $i \in \mathcal{P}$, and g_j , $j \in \mathcal{Q}$, are continuously differentiable at a feasible point $x^* \in \mathcal{F}$.*

If x^ is a local minimum and the Mangasarian–Fromovitz Constraint Qualification (MFCQ) holds at x^* , then there exist Lagrange multipliers $\lambda_i \in \mathbb{R}$ ($i \in \mathcal{P}$) and $\mu_j \geq 0$ ($j \in \mathcal{Q}$) such that the following Karush–Kuhn–Tucker conditions are satisfied:*

$$\nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{Q}} \mu_j \nabla g_j(x^*) = 0, \quad (3.32)$$

$$h_i(x^*) = 0, \quad i \in \mathcal{P}, \quad (3.33)$$

$$g_j(x^*) \leq 0, \quad j \in \mathcal{Q}, \quad (3.34)$$

$$\mu_j g_j(x^*) = 0, \quad j \in \mathcal{Q}. \quad (3.35)$$

Proof. Since x^* is a local minimum and all functions are continuously differentiable at x^* , the Fritz–John conditions (Theorem 3.31) apply. Hence, there exist multipliers

$$u_0 \geq 0, \quad \lambda_i \in \mathbb{R}, \quad i \in \mathcal{P}, \quad u_j \geq 0, \quad j \in \mathcal{Q},$$

not all zero, such that

$$u_0 \nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{Q}} u_j \nabla g_j(x^*) = 0, \quad (3.36)$$

together with the complementary slackness conditions $u_j g_j(x^*) = 0$, $j \in \mathcal{Q}$.

Reduction to active constraints. By complementary slackness, $u_j = 0$ for all $j \notin \mathcal{Q}(x^*)$, where $\mathcal{Q}(x^*) = \{j \in \mathcal{Q} \mid g_j(x^*) = 0\}$ denotes the active set. Thus (3.36) may be written as

$$u_0 \nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{Q}(x^*)} u_j \nabla g_j(x^*) = 0. \quad (3.37)$$

Exclusion of the abnormal case $u_0 = 0$. Assume, for contradiction, that $u_0 = 0$. Then (3.37) reduces to

$$\sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{Q}(x^*)} u_j \nabla g_j(x^*) = 0, \quad u_j \geq 0. \quad (3.38)$$

Since the gradients $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent (by MFCQ), and MFCQ guarantees the existence of a direction $d \in \mathbb{R}^n$ such that

$$\nabla h_i(x^*)^\top d = 0, \quad i \in \mathcal{P}, \quad \nabla g_j(x^*)^\top d < 0, \quad j \in \mathcal{Q}(x^*),$$

we may take the inner product of (3.38) with d to obtain

$$0 = \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*)^\top d + \sum_{j \in \mathcal{Q}(x^*)} u_j \nabla g_j(x^*)^\top d = \sum_{j \in \mathcal{Q}(x^*)} u_j \nabla g_j(x^*)^\top d.$$

Since $\nabla g_j(x^*)^\top d < 0$ for all $j \in \mathcal{Q}(x^*)$ and $u_j \geq 0$, the above equality can hold only if $u_j = 0 \forall j \in \mathcal{Q}(x^*)$. Substituting into (3.38) yields $\sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) = 0$. By linear independence of the equality constraint gradients, this implies $\lambda_i = 0$ for all $i \in \mathcal{P}$.

Thus all multipliers vanish, contradicting the Fritz–John condition that the multipliers are not all zero. Hence $u_0 > 0$.

Normalization and KKT conditions. Since $u_0 > 0$, we may normalize the Fritz–John multipliers by dividing (3.37) by u_0 and define

$$\mu_j = \frac{u_j}{u_0}, \quad j \in \mathcal{Q}, \quad \lambda_i = \frac{\lambda_i}{u_0}, \quad i \in \mathcal{P}.$$

This yields the stationarity condition

$$\nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{Q}} \mu_j \nabla g_j(x^*) = 0,$$

together with feasibility and complementary slackness,

$$h_i(x^*) = 0, \quad g_j(x^*) \leq 0, \quad \mu_j g_j(x^*) = 0, \quad j \in \mathcal{Q}.$$

These are precisely the Karush–Kuhn–Tucker conditions.

Under the Mangasarian–Fromovitz Constraint Qualification, every local minimum satisfies the Karush–Kuhn–Tucker conditions. $\square$

Example 3.37 Illustrative KKT example under MFCQ with indexed constraints.

Consider the constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & f(x) = e^{x_1} + (x_2 - 1)^2 + x_3^2 \\ \text{s.t.} \quad & h_1(x) := x_1 + x_2 - 1 = 0, \\ & g_1(x) := -x_3 \leq 0. \end{aligned}$$

Here $\mathcal{P} = \{1\}$ and $\mathcal{Q} = \{1\}$. The feasible set is the vertical half-plane $\mathcal{F} = \{x \in \mathbb{R}^3 \mid x_1 + x_2 = 1, x_3 \geq 0\}$.

Restricting the objective to the equality constraint by writing $x_2 = 1 - x_1$ gives $f(x) = e^{x_1} + x_1^2 + x_3^2$. The variables x_1 and x_3 decouple. Minimization in x_3 yields $x_3^* = 0$, at which the inequality constraint is active. The remaining one-dimensional problem is $\min_{x_1 \in \mathbb{R}} \psi(x_1) = e^{x_1} + x_1^2$, whose derivative $\psi'(x_1) = e^{x_1} + 2x_1$ has

a unique root $x_1^* < 0$. Define $x^* = (x_1^*, 1 - x_1^*, 0)$.

At x^* , the active inequality set is $\mathcal{Q}(x^*) = \{1\}$. The gradients at x^* are

$$\nabla f(x^*) = (e^{x_1^*}, -2x_1^*, 0)^\top, \quad \nabla h_1(x^*) = (1, 1, 0)^\top, \quad \nabla g_1(x^*) = (0, 0, -1)^\top.$$

The Mangasarian–Fromovitz Constraint Qualification holds at x^* . Indeed, $\nabla h_1(x^*) \neq 0$, and the direction $d = (0, 0, 1)^\top$ satisfies $\nabla h_1(x^*)^\top d = 0$ and $\nabla g_1(x^*)^\top d = -1 < 0$.

By Theorem 3.36, there exist multipliers $\lambda_1 \in \mathbb{R}$ and $\mu_1 \geq 0$ such that the KKT conditions hold:

$$\nabla f(x^*) + \lambda_1 \nabla h_1(x^*) + \mu_1 \nabla g_1(x^*) = 0, \quad \mu_1 g_1(x^*) = 0.$$

Substituting the gradients yields

$$\begin{pmatrix} e^{x_1^*} \\ -2x_1^* \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \mu_1 \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

which implies, from the first and second rows, that $\lambda_1 = -e^{x_1^*} = 2x_1^*$ and $\mu_1 = 0$, the two expressions for λ_1 agreeing because $\psi'(x_1^*) = e^{x_1^*} + 2x_1^* = 0$.

All Karush–Kuhn–Tucker conditions are satisfied at x^* . This example illustrates how the Mangasarian–Fromovitz Constraint Qualification guarantees the existence of indexed Lagrange multipliers and ensures the validity of the KKT conditions in a genuinely three-dimensional, non-polynomial constrained optimization problem. $\square$

Example 3.38 Nonlinear degeneracy and failure of MFCQ.

Consider the constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & f(x) = x_3 \\ \text{s.t.} \quad & h(x) := x_1^2 + x_2^2 - x_3^2 = 0, \\ & g_1(x) := x_3 \leq 0, \\ & g_2(x) := x_1 + x_3 \leq 0, \\ & g_3(x) := x_2 + x_3 \leq 0. \end{aligned} \tag{3.39}$$

The equality constraint defines a double cone $x_3^2 = x_1^2 + x_2^2$, while the inequality constraint $x_3 \leq 0$ restricts attention to the lower nappe. The remaining inequalities further intersect the cone with tilted half-spaces. The feasible set is therefore a curved cone with a degenerate apex at the origin.

The gradients of the constraint functions are

$$\nabla h(x) = (2x_1, 2x_2, -2x_3), \quad \nabla g_1 = (0, 0, 1), \quad \nabla g_2 = (1, 0, 1), \quad \nabla g_3 = (0, 1, 1).$$

A regular boundary point. Consider the feasible point $x^{(1)} = (-1, -1, -\sqrt{2})$. The equality constraint holds, and *no* inequality is active, since

$$g_1(x^{(1)}) = -\sqrt{2} < 0, \quad g_2(x^{(1)}) = g_3(x^{(1)}) = -1 - \sqrt{2} < 0.$$

Thus $\mathcal{Q}(x^{(1)}) = \emptyset$. With no active inequalities, MFCQ reduces to linear independence of the equality-constraint gradient, and indeed

$$\nabla h(x^{(1)}) = (2x_1, 2x_2, -2x_3)|_{x^{(1)}} = (-2, -2, 2\sqrt{2}) \neq 0.$$

Hence MFCQ holds at $x^{(1)}$. If $x^{(1)}$ were a local minimum, the Karush–Kuhn–Tucker conditions would necessarily hold at this point.

A degenerate point where MFCQ fails. Now consider the point $x^{(2)} = (0, 0, 0)$. This point satisfies all constraints, and all inequality constraints are active: $\mathcal{Q}(x^{(2)}) = \{1, 2, 3\}$. The gradients at $x^{(2)}$ are

$$\nabla h(x^{(2)}) = (0, 0, 0), \quad \nabla g_1 = (0, 0, 1), \quad \nabla g_2 = (1, 0, 1), \quad \nabla g_3 = (0, 1, 1).$$

The MFCQ condition would require a direction $d \in \mathbb{R}^3$ satisfying

$$\nabla g_1^\top d < 0, \quad \nabla g_2^\top d < 0, \quad \nabla g_3^\top d < 0,$$

that is,

$$d_3 < 0, \quad d_1 + d_3 < 0, \quad d_2 + d_3 < 0.$$

This requirement is in fact *satisfiable*: the direction $d = (0, 0, -1)$ gives

$$\nabla g_1^\top d = \nabla g_2^\top d = \nabla g_3^\top d = -1 < 0,$$

and $\nabla h(x^{(2)})^\top d = 0$ trivially. What fails is the *other* MFCQ requirement, linear independence of the equality-constraint gradients, because the equality gradient *vanishes* at the apex,

$$\nabla h(x^{(2)}) = (2x_1, 2x_2, -2x_3)|_0 = (0, 0, 0).$$

A zero vector is never linearly independent, so MFCQ fails at $x^{(2)}$.

This example illustrates that the failure of the Mangasarian–Fromovitz Constraint Qualification here stems from degeneracy of the *equality* constraint: at the cone apex the surface $x_1^2 + x_2^2 = x_3^2$ is not a smooth manifold, and its gradient collapses to zero, so the linearization no longer captures the curved constraint. At such points the Karush–Kuhn–Tucker conditions are not guaranteed to hold, and only the weaker Fritz–John necessary conditions apply. $\square$

Remark 3.39 (Geometric meaning of MFCQ, recovery of KKT, and relation to Slater).

The Mangasarian–Fromovitz Constraint Qualification (MFCQ) admits a natural geometric interpretation in terms of the linearized tangent cone (3.28) and the normal cone representation

(3.30). Recall that first-order optimality is equivalent to the normal cone inclusion (3.27), namely

$$-\nabla f(x^*) \in N_{\mathcal{F}}(x^*). \quad (3.40)$$

Using the explicit representation (3.30), this inclusion yields the homogeneous Fritz–John relation (3.31).

MFCQ requires that the gradients of the equality constraints $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent and that there exists a direction d such that

$$\nabla h_i(x^*)^\top d = 0 \quad \text{for all } i \in \mathcal{P}, \quad \nabla g_j(x^*)^\top d < 0 \quad \text{for all } j \in \mathcal{A}(x^*).$$

The first condition ensures that the linearized equality constraints define a smooth tangent subspace. The second condition asserts that, within this subspace, there exists a direction that strictly moves into the interior of the feasible region by decreasing every active inequality constraint.

Geometrically, this means that the linearized tangent cone (3.28) has nonempty interior relative to the equality constraint subspace. The active inequalities do not “pinch” the feasible directions into a lower-dimensional cone or collapse them to the origin. There is genuine local thickness of feasible directions along the equality surface.

In dual terms, this interiority condition implies that the active constraint normals cannot positively span a cone containing the origin. Equivalently, using the normal cone representation (3.30), the relation

$$\sum_{i \in \mathcal{P}} \lambda_i \nabla h_i(x^*) + \sum_{j \in \mathcal{A}(x^*)} \mu_j \nabla g_j(x^*) = 0, \quad \mu_j \geq 0,$$

can hold only when $\lambda_i = 0$ and $\mu_j = 0$. That is, no nontrivial conic combination of active constraint gradients can vanish.

Returning now to the homogeneous Fritz–John relation (3.31), MFCQ therefore excludes the abnormal case in which the multiplier of $\nabla f(x^*)$ is zero. The objective multiplier must be strictly positive. Because the multiplier system is homogeneous, it can be normalized so that the coefficient of $\nabla f(x^*)$ equals one. After normalization, the stationarity condition reduces precisely to the classical Karush–Kuhn–Tucker condition.

In convex optimization, Slater’s condition provides a global interiority requirement that guarantees MFCQ automatically. If there exists a strictly feasible point satisfying $g_j(x) < 0$ for all inequality constraints and the equalities are affine, then at every feasible point the linearized cone has interior directions relative to the equality subspace. Thus Slater’s condition ensures that the normal cone is generated only by active constraint normals and that the objective gradient must participate nontrivially in the supporting hyperplane condition (3.40). In this way, KKT conditions in convex programming can be understood as the normalized form of the normal cone optimality condition under an interiority assumption.

3.3.4 ■ Second-Order Conditions for General Constrained Programs

We now turn to second-order optimality conditions for constrained optimization problems of the form (3.1), allowing both equality and inequality constraints. In contrast to the equality-constrained case, second-order conditions for general nonlinear programs must account for the interaction between curvature, active constraints, and the geometry of feasible directions.

Throughout this subsection, we assume that the objective function f and all constraint functions h_i , $i \in \mathcal{P}$, and g_j , $j \in \mathcal{Q}$ are twice continuously differentiable in a neighborhood of a feasible point $x^* \in \mathcal{F}$.

Second-order behavior is governed by directions that are feasible to first order and along which the objective does not decrease at first order.

Definition 3.40 (Critical direction set). *The set of critical directions at x^* is*

$$D(x^*) = \left\{ d \in \mathbb{R}^n \left| \begin{array}{ll} \nabla f(x^*)^\top d \leq 0, \\ \nabla h_i(x^*)^\top d = 0, & i \in \mathcal{P}, \\ \nabla g_j(x^*)^\top d \leq 0, & j \in \mathcal{Q}(x^*) \end{array} \right. \right\}. \quad (3.41)$$

The set $D(x^*)$ is a closed convex cone and is always nonempty, since it contains $d = 0$. Intuitively, $D(x^*)$ collects the directions that remain feasible to first order and along which the objective is *first-order flat*. These are exactly the directions the first-order conditions cannot decide: along any direction with $\nabla f(x^*)^\top d > 0$ the objective already increases, so no violation of optimality can occur there. The verdict must instead come from curvature, and this is why second-order conditions are stated *only* on $D(x^*)$. A useful mental image is that $D(x^*)$ is the “blind spot” of the first-order theory, and the Hessian of the Lagrangian is the instrument that sees into it.

Second-order conditions require a Lagrangian formulation that remains valid even when constraint qualifications fail.

Definition 3.41 (Fritz–John Lagrangian).

$$\mathcal{L}(x, u_0, \lambda, u) = u_0 f(x) + \sum_{i \in \mathcal{P}} \lambda_i h_i(x) + \sum_{j \in \mathcal{Q}} u_j g_j(x),$$

where

$$u_0 \geq 0, \quad u_j \geq 0 \ (j \in \mathcal{Q}), \quad \lambda_i \in \mathbb{R}.$$

The argument order is (x, u_0, λ, u) : the objective weight first, then the equality multipliers, then the inequality multipliers. When $u_0 = 1$ the objective weight is suppressed and we write $\mathcal{L}(x, \lambda, \mu)$ for the ordinary Lagrangian of Section 3.2; this is the convention used from Theorem 3.43 onward, where a constraint qualification makes the normalization $u_0 = 1$ legitimate.

The coefficient u_0 corresponds to the objective function and may vanish in degenerate (abnormal) cases. Constraint qualifications such as MFCQ exclude this possibility and reduce the theory to the standard KKT framework.

Theorem 3.42 (Second-Order Necessary Condition for a Local Minimum). *Let $x^* \in \mathcal{F}$ be a local minimum of problem (3.1), and let $d \in D(x^*)$ be a critical direction that is derivable, meaning that some twice continuously differentiable curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathcal{F}$ satisfies $\gamma(0) = x^*$ and $\dot{\gamma}(0) = d$. (Every critical direction is derivable when MFCQ or LICQ holds at x^* ; without a constraint qualification the linearized cone can contain directions that no feasible curve realizes, and the conclusion below may fail for those.) Then there exist multipliers*

$$u_0 \geq 0, \quad u_j \geq 0 \ (j \in \mathcal{Q}), \quad \lambda_i \in \mathbb{R} \ (i \in \mathcal{P}),$$

not all zero, such that

$$\nabla_x \mathcal{L}(x^*, u_0, \lambda, u) = 0, \quad (3.42)$$

$$d^\top \nabla_{xx}^2 \mathcal{L}(x^*, u_0, \lambda, u) d \geq 0, \quad (3.43)$$

$$u_j g_j(x^*) = 0, \quad j \in \mathcal{Q}, \quad (3.44)$$

$$u_0 \nabla f(x^*)^\top d = 0, \quad (3.45)$$

$$u_j \nabla g_j(x^*)^\top d = 0, \quad j \in \mathcal{Q}. \quad (3.46)$$

Proof. Fix an arbitrary direction $d \in D(x^*)$.

First-order optimality and critical directions. Since x^* is a local minimum of problem (3.1), the Fritz–John necessary conditions apply (Theorem 3.31). Hence, there exist multipliers

$$u_0 \geq 0, \quad u_j \geq 0 \ (j \in \mathcal{Q}), \quad \lambda_i \in \mathbb{R} \ (i \in \mathcal{P}),$$

not all zero, such that the stationarity and complementary slackness conditions (3.42) and (3.44) hold.

By definition of the critical cone $D(x^*)$, the direction d satisfies

$$\nabla f(x^*)^\top d \leq 0, \quad \nabla h_i(x^*)^\top d = 0 \ (i \in \mathcal{P}), \quad \nabla g_j(x^*)^\top d \leq 0 \ (j \in \mathcal{Q}(x^*)).$$

Thus, d is a feasible first-order direction along which the objective does not increase to first order.

Reduction to first-order binding constraints. Define the index set of constraints that are *first-order binding along d* :

$$\mathcal{I}(d) := \{j \in \mathcal{Q}(x^*) \mid \nabla g_j(x^*)^\top d = 0\}.$$

Only constraints in $\mathcal{I}(d)$ can affect feasibility at second order. Constraints with $\nabla g_j(x^*)^\top d < 0$ become strictly inactive for small $t > 0$ and hence do not influence second-order behavior.

The orthogonality conditions

$$u_0 \nabla f(x^*)^\top d = 0, \quad u_j \nabla g_j(x^*)^\top d = 0 \ (j \in \mathcal{Q})$$

express precisely that multipliers may be nonzero only for first-order binding terms.

Second-order expansion along feasible curves. By hypothesis d is derivable, so a twice continuously differentiable curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathcal{F}$ exists with $\gamma(0) = x^*$, $\dot{\gamma}(0) = d$. This is exactly the point at which derivability is needed; the linearized description of $D(x^*)$ alone would not supply such a curve. Applying Taylor expansions along $\gamma(t)$ yields

$$f(\gamma(t)) = f(x^*) + t \nabla f(x^*)^\top d + \frac{t^2}{2} d^\top \nabla^2 f(x^*) d + o(t^2),$$

$$g_j(\gamma(t)) = g_j(x^*) + t \nabla g_j(x^*)^\top d + \frac{t^2}{2} d^\top \nabla^2 g_j(x^*) d + o(t^2).$$

For all $j \notin \mathcal{I}(d)$, the first-order term $\nabla g_j(x^*)^\top d < 0$ ensures feasibility for sufficiently small $t > 0$. For $j \in \mathcal{I}(d)$, feasibility depends on the second-order terms.

Second-order infeasibility of descent directions. Suppose, for contradiction, that there exist multipliers satisfying (3.42) and (3.44) such that $d^\top \nabla_{xx}^2 \mathcal{L}(x^*, u_0, \lambda, u) d < 0$. Then combining the second-order expansions with complementary slackness yields

$$\mathcal{L}(\gamma(t), u_0, \lambda, u) < \mathcal{L}(x^*, u_0, \lambda, u) \quad \text{for all sufficiently small } t > 0.$$

Since $u_0 \geq 0$ and all constraints are satisfied along $\gamma(t)$, this strict decrease in the Lagrangian implies $f(\gamma(t)) < f(x^*)$ for sufficiently small $t > 0$, contradicting that x^* is a local minimum.

The abnormal case. The contradiction just derived uses $u_0 > 0$, since only then does a strict decrease of $\mathcal{L}$ along γ force a strict decrease of f . When $u_0 = 0$ the curvature inequality must instead be obtained from a second-order theorem of the alternative applied to the system encoding strict descent and feasibility at second order; see [10, 15] for that argument, which is directionwise in exactly the same sense as the statement above. Under MFCQ the abnormal case cannot occur, and Theorem 3.43 below gives the conclusion in its familiar normalized form.

Since the direction $d \in D(x^*)$ was arbitrary, the second-order curvature inequality (3.43) holds for every critical direction. This establishes the second-order necessary condition. $\square$

Theorem 3.43 (Second-Order Necessary Condition under MFCQ). *Let x^* be a local minimum of problem (3.1). Assume:*

- (1) *the objective function f and the constraint functions h_i , $i \in \mathcal{P}$, and g_j , $j \in \mathcal{Q}$, are twice continuously differentiable in a neighborhood of x^* ;*
- (2) *the Mangasarian–Fromovitz Constraint Qualification (MFCQ) holds at x^* .*

Then there exist Lagrange multipliers $\lambda^ \in \mathbb{R}^{|\mathcal{P}|}$, $\mu^* \in \mathbb{R}_{\geq 0}^{|\mathcal{Q}|}$ satisfying the Karush–Kuhn–Tucker conditions at x^* such that*

$$d^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*, \mu^*) d \geq 0 \quad \text{for all } d \in \mathcal{C}(x^*), \quad (3.47)$$

where $\mathcal{C}(x^*)$ denotes the critical cone at x^* .

Proof. Since x^* is a local minimum and MFCQ holds at x^* , the Karush–Kuhn–Tucker theorem (Theorem 3.36) guarantees the existence of multipliers (λ^*, μ^*) with $\mu^* \geq 0$ such that

$$\begin{aligned} \nabla f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i^* \nabla h_i(x^*) + \sum_{j \in \mathcal{Q}} \mu_j^* \nabla g_j(x^*) &= 0, \\ \mu_j^* g_j(x^*) &= 0, \quad j \in \mathcal{Q}. \end{aligned}$$

Let $d \in \mathcal{C}(x^*)$ be arbitrary. By definition of the critical cone, d satisfies $\nabla h_i(x^*)^\top d = 0$ ($i \in \mathcal{P}$),

$$\nabla g_j(x^*)^\top d = 0 \quad (j \in \mathcal{A}^+), \quad \nabla g_j(x^*)^\top d \leq 0 \quad (j \in \mathcal{A}^0),$$

where

$$\mathcal{A}^+ := \{j \in \mathcal{Q} \mid \mu_j^* > 0\}, \quad \mathcal{A}^0 := \{j \in \mathcal{Q} \mid \mu_j^* = 0, g_j(x^*) = 0\}.$$

Because f , h_i , and g_j are twice continuously differentiable, there exists a twice continuously differentiable feasible curve $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathcal{F}$ such that

$$\gamma(0) = x^*, \quad \dot{\gamma}(0) = d.$$

Such a curve exists because MFCQ ensures a well-defined first-order tangent structure for the feasible set.

Define $\phi(t) := f(\gamma(t))$. Since x^* is a local minimum, $\phi(t) \geq \phi(0)$ for all sufficiently small t , implying

$$\phi'(0) = 0, \quad \phi''(0) \geq 0.$$

Using the chain rule and the KKT stationarity condition, we obtain

$$\phi'(0) = \nabla f(x^*)^\top d = - \sum_{i \in \mathcal{P}} \lambda_i^* \nabla h_i(x^*)^\top d - \sum_{j \in \mathcal{Q}} \mu_j^* \nabla g_j(x^*)^\top d = 0,$$

where the equality follows from feasibility of d and complementary slackness.

Differentiating a second time and using the feasibility of $\gamma(t)$ yields

$$\phi''(0) = d^\top \left(\nabla^2 f(x^*) + \sum_{i \in \mathcal{P}} \lambda_i^* \nabla^2 h_i(x^*) + \sum_{j \in \mathcal{Q}} \mu_j^* \nabla^2 g_j(x^*) \right) d = d^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*, \mu^*) d.$$

Since $\phi''(0) \geq 0$ for every $d \in \mathcal{C}(x^*)$, the inequality (3.47) follows. This completes the proof. $\square$

Example 3.44 Second-order sufficiency under MFCQ for an inequality-constrained problem.

Consider the constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & f(x) := -x_1 x_2 x_3 \\ \text{s.t.} \quad & g(x) := x_1^2 + 2x_2^2 + 4x_3^2 - 48 \leq 0. \end{aligned} \tag{3.48}$$

Let $x^* = (-4, -2\sqrt{2}, 2)$. We verify that x^* is an *isolated local minimum* using the second-order sufficient condition under MFCQ.

Feasibility and activity. A direct computation shows

$$(x_1^*)^2 + 2(x_2^*)^2 + 4(x_3^*)^2 = (-4)^2 + 2(-2\sqrt{2})^2 + 4(2)^2 = 16 + 16 + 16 = 48,$$

so $g(x^*) = 48 - 48 = 0$ and the inequality constraint is active at x^* . Hence, $\mathcal{Q}(x^*) = \{1\}$.

Gradients and MFCQ. The gradients of the objective and constraint are

$$\nabla f(x) = \begin{pmatrix} -x_2 x_3 \\ -x_1 x_3 \\ -x_1 x_2 \end{pmatrix}, \quad \nabla g(x) = \begin{pmatrix} 2x_1 \\ 4x_2 \\ 8x_3 \end{pmatrix}.$$

Evaluating at x^* gives

$$\nabla f(x^*) = \begin{pmatrix} 4\sqrt{2} \\ 8 \\ -8\sqrt{2} \end{pmatrix}, \quad \nabla g(x^*) = \begin{pmatrix} -8 \\ -8\sqrt{2} \\ 16 \end{pmatrix}.$$

Since $\nabla g(x^*) \neq 0$, the Mangasarian–Fromovitz Constraint Qualification holds at x^* . Therefore, KKT conditions are necessary for local optimality.

First-order (KKT) stationarity. We seek $\mu \geq 0$ such that $\nabla f(x^*) + \mu \nabla g(x^*) = 0$. Solving componentwise yields $\mu = \frac{\sqrt{2}}{2}$.

Critical directions. The critical cone at x^* is $D(x^*) = \{d \in \mathbb{R}^3 : \nabla g(x^*)^\top d = 0\}$. That is,

$$-8d_1 - 8\sqrt{2}d_2 + 16d_3 = 0, \quad \text{or equivalently} \quad d_1 + \sqrt{2}d_2 - 2d_3 = 0.$$

Hessian of the Lagrangian. The Hessian of the objective is

$$\nabla^2 f(x) = \begin{pmatrix} 0 & -x_3 & -x_2 \\ -x_3 & 0 & -x_1 \\ -x_2 & -x_1 & 0 \end{pmatrix},$$

and

$$\nabla^2 g(x) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$$

Thus, the Hessian of the Lagrangian $\mathcal{L}(x, \mu) = f(x) + \mu g(x)$ at (x^*, μ) is

$$\nabla_{xx}^2 \mathcal{L}(x^*, \mu) = \nabla^2 f(x^*) + \mu \nabla^2 g(x^*) = \begin{pmatrix} \sqrt{2} & -2 & 2\sqrt{2} \\ -2 & 2\sqrt{2} & 4 \\ 2\sqrt{2} & 4 & 4\sqrt{2} \end{pmatrix},$$

where $\mu \nabla^2 g(x^*) = \frac{\sqrt{2}}{2} \text{diag}(2, 4, 8) = \text{diag}(\sqrt{2}, 2\sqrt{2}, 4\sqrt{2})$ was added to $\nabla^2 f(x^*)$.

Restriction to the critical cone. Using the constraint $d_1 = -\sqrt{2}d_2 + 2d_3$, we eliminate d_1 and express the quadratic form in (d_2, d_3) :

$$d^\top \nabla_{xx}^2 \mathcal{L}(x^*, \mu) d = \begin{pmatrix} d_2 & d_3 \end{pmatrix} \begin{pmatrix} 8\sqrt{2} & -8 \\ -8 & 16\sqrt{2} \end{pmatrix} \begin{pmatrix} d_2 \\ d_3 \end{pmatrix}.$$

This 2×2 matrix has trace $24\sqrt{2} > 0$ and determinant $8\sqrt{2} \cdot 16\sqrt{2} - (-8)^2 = 256 - 64 = 192 > 0$, so both eigenvalues are strictly positive. Hence,

$$d^\top \nabla_{xx}^2 \mathcal{L}(x^*, \mu) d > 0 \quad \forall d \in D(x^*) \setminus \{0\}.$$

All assumptions of the second-order sufficient condition under MFCQ are satisfied. Therefore, $x^* = (-4, -2\sqrt{2}, 2)$ is an *isolated local minimum* of problem (3.48). (In fact, symmetry shows it is one of several global minima.) $\square$

3.4 ■ Further Reading

The study of optimization has long predated its modern mathematical formulation. Philosophical inquiries into “best” arrangements have appeared in ancient Greek thought. For example, Socrates sought general principles explaining how social and natural systems might be organized optimally. A decisive mathematical transition occurred in the seventeenth and eighteenth centuries, when optimization problems emerged naturally from mechanics. Foundational contributions by Galileo, Newton, Leibniz, Euler, Lagrange, and the Bernoulli brothers addressed questions such as maximal projectile range, minimal resistance shapes, and paths of shortest descent. These problems gave rise to the *calculus of variations*, whose central aim is to extremize a functional depending on an entire trajectory rather than on finitely many parameters; see, e.g., [77, 6].

Some constrained optimization problems are even older. Classical *isoperimetric problems*, studied by Zenodorus (ca. 100 BC) and Archimedes, ask which closed curve of fixed length encloses the largest area. These problems already involve implicit constraints and foreshadow modern optimality theory (Encyclopaedia Britannica, 1950). The modern theory of finite-dimensional constrained optimization emerged in the twentieth century, driven by applications in economics, operations research, and engineering. Pioneering work by von Neumann [121] and Kantorovich [65] demonstrated how economic planning and resource allocation problems could be modeled as optimization problems with inequality constraints, particularly linear ones.

First-order optimality conditions for inequality-constrained problems were systematically studied by Karush in his 1939 M.Sc. thesis, written in the context of the Chicago school of calculus of variations; see Kuhn [73]. The most general first-order necessary condition, now known as the *Fritz–John condition*, was established by Fritz John [60]. Its strengthened form, guaranteeing a nonzero multiplier on the objective function under mild regularity assumptions, was proved by Mangasarian and Fromovitz [82], introducing what is now called the *Mangasarian–Fromovitz Constraint Qualification (MFCQ)*.

By the late 1960s, the classical theory of optimality conditions had largely stabilized. Comprehensive surveys are provided by Mangasarian [82], and systematic treatments of constraint qualifications appear in Peterson [93]. Standard references on nonlinear programming include Avriel [8] and Bazaraa, Sherali, and Shetty [10]. Extensions to nonsmooth optimization require generalized derivatives. In the convex case, the subdifferential provides an adequate replacement for the gradient, while in the nonconvex case Clarke’s generalized gradient plays an analogous role; see Clarke [35] and Penot [92]. These developments highlight both the power and the limitations of derivative-based optimality theory.

3.5 ■ Exercises

Exercise 3.1 (Local vs. isolated extrema under inequality constraints). Consider the optimization problem $\max_{x \in \mathbb{R}^3} f(x) := \frac{x_1}{x_3}$ subject to

$$\begin{aligned} x_1 + x_2 - 1 &\leq 0, \\ -x_1 - x_2 x_3 + 1 &\leq 0, \\ x_1 &\geq 0, \quad x_2 \geq 0, \quad x_3 \geq \frac{1}{2}. \end{aligned}$$

- (i) Verify that the point $x^* = (1, 0, 1)$ is feasible. Identify explicitly which constraints are active at x^* .
- (ii) Compute the gradients of the objective function and of all active constraints at x^* . Determine the active set $\mathcal{Q}(x^*)$.

- (iii) Verify that x^* satisfies the first-order Fritz–John necessary conditions. Explain why these conditions alone do not guarantee local optimality.
- (iv) Show that x^* also satisfies all second-order *necessary* conditions discussed in this chapter, when evaluated along all critical directions.
- (v) Consider the feasible curve $\gamma(\alpha) = (1, 0, 1 - \alpha)^\top$, $\alpha \geq 0$. Show that $\gamma(\alpha)$ remains feasible for sufficiently small $\alpha > 0$, and prove that $f(\gamma(\alpha)) > f(x^*)$.
- (vi) Conclude that x^* is *not* a local maximizer, despite satisfying first- and second-order necessary conditions. Briefly explain the geometric reason for this failure.

Exercise 3.2 (Isolated local maximum via vanishing critical directions). Consider the same optimization problem as in the previous exercise and define $y^* = (1, 0, \frac{1}{2})$.

- (i) Verify that y^* is feasible and determine the active constraint set $\mathcal{Q}(y^*)$.
- (ii) Compute the set of critical directions $D(y^*)$ by imposing the first-order feasibility and stationarity conditions.
- (iii) Show that $D(y^*) = \{0\}$.
- (iv) Explain why the absence of any nonzero critical direction implies that y^* is an *isolated local maximum*. Relate your argument to the second-order sufficient condition for constrained optimization.

Exercise 3.3 (Unconstrained isolated local minimum). Consider the unconstrained optimization problem $\min_{x \in \mathbb{R}^3} f(x)$, where $f(x) = x_1^4 + x_1^3 - x_1 + x_2^4 - x_2^2 + x_2 + x_3^2 - x_3 + x_1 x_2 x_3$.

- (i) Compute $\nabla f(x)$ and verify that the point $x^* = (0.571, -0.940, 0.768)$ satisfies $\nabla f(x^*) = 0$.
- (ii) Compute the Hessian matrix $\nabla^2 f(x)$.
- (iii) Evaluate $\nabla^2 f(x^*)$ and show that it is positive definite.
- (iv) Conclude that x^* is an *isolated local minimum* of f . Explain why positive definiteness of the Hessian excludes the existence of any other local minimum in a neighborhood of x^* .

Exercise 3.4 (Failure of Lagrange multipliers under degeneracy). Consider the equality-constrained problem

$$\min_{x \in \mathbb{R}} f(x) = x \quad \text{s.t.} \quad h(x) = x^2 = 0.$$

- (i) Identify the feasible set and show that the unique feasible point $x^* = 0$ is a global minimizer.
- (ii) Compute $\nabla f(x^*)$ and $\nabla h(x^*)$.
- (iii) Show that the Lagrange multiplier equation $\nabla f(x^*) + \lambda \nabla h(x^*) = 0$ has no solution.
- (iv) Explain why this does not contradict first-order optimality. Identify the constraint qualification that fails and interpret this failure geometrically.

Exercise 3.5 (Second-order conditions without constraint qualifications). Consider the inequality-constrained problem

$$\min_{x \in \mathbb{R}} f(x) = x^2 \quad \text{s.t.} \quad g(x) = x^3 \leq 0.$$

- (i) Show that $x^* = 0$ is feasible and is a local minimum.
- (ii) Compute $\nabla g(x^*)$ and verify that the Mangasarian–Fromovitz constraint qualification fails at x^* .
- (iii) Determine whether the Karush–Kuhn–Tucker conditions hold at x^* .
- (iv) Verify explicitly that the Fritz–John conditions are satisfied.
- (v) Decide whether x^* is an isolated local minimum and explain how this example illustrates the distinction between optimality and regularity.

Exercise 3.6 (Equality constraints rewritten as inequalities). Consider the equality-constrained problem

$$\min f(x) \quad \text{s.t.} \quad h(x) = 0,$$

and rewrite it equivalently as $h(x) \leq 0, -h(x) \leq 0$.

- (i) Write down the Fritz–John conditions for the reformulated problem.
- (ii) Show that all multipliers may vanish at a local optimum.
- (iii) Explain why this degeneracy cannot occur when the Mangasarian–Fromovitz constraint qualification holds for the original problem.

Exercise 3.7 (Linear equality constraints: geometry and multipliers). Consider the equality-constrained problem

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad Ax = b,$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Assume that the feasible set $\mathcal{F} = \{x \in \mathbb{R}^n : Ax = b\}$ is nonempty.

- (i) Show that $\mathcal{F}$ is an affine subspace of $\mathbb{R}^n$. Determine its tangent cone $T_{\mathcal{F}}(x^*)$ at any feasible point x^* .
- (ii) Show that the normal cone at x^* is $N_{\mathcal{F}}(x^*) = \text{range}(A^\top)$.
- (iii) Suppose that x^* is a local minimizer and f is differentiable. Show that a necessary condition for optimality is $\nabla f(x^*) \in \text{range}(A^\top)$. Conclude that there exists $\lambda \in \mathbb{R}^m$ such that $\nabla f(x^*) + A^\top \lambda = 0$.
- (iv) Show that the multiplier λ is unique if and only if A has full row rank.

Exercise 3.8 (Lagrange multipliers and second-order sufficiency: a worked case). Consider the equality-constrained problem

$$\min_{x \in \mathbb{R}^2} f(x) = x_1^2 + x_2^2 \quad \text{s.t.} \quad h(x) = x_1 + 2x_2 - 5 = 0.$$

- (i) Write the Lagrangian and solve the stationarity equations $\nabla f(x) + \lambda \nabla h(x) = 0$ together with feasibility. Show that the unique Lagrange candidate is $x^* = (1, 2)$ with $\lambda^* = -2$.
- (ii) Determine a basis matrix N for the tangent space $\mathcal{T}(x^*) = \{d : \nabla h(x^*)^\top d = 0\}$ and compute the reduced Hessian $N^\top \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) N$. Conclude that x^* is an isolated local minimum.

- (iii) Replace the right-hand side 5 by a parameter b and let $V(b)$ denote the optimal value. Show that $V(b) = b^2/5$ and verify the *sensitivity relation* $V'(b) = -\lambda^*(b)$. Interpret λ^* as the marginal cost of tightening the constraint.

Exercise 3.9 (KKT conditions for a projection problem). A point $a = (3, 4)$ is to be projected onto the half-plane $\{x \in \mathbb{R}^2 : x_1 \leq 1\}$:

$$\min_{x \in \mathbb{R}^2} f(x) = \frac{1}{2} \|x - a\|^2 \quad \text{s.t.} \quad g(x) = x_1 - 1 \leq 0.$$

- (i) Write the KKT system (stationarity, primal feasibility, dual feasibility $\mu \geq 0$, and complementary slackness).
- (ii) Show that the constraint must be active at the optimum, and solve the KKT system to obtain $x^* = (1, 4)$ with $\mu^* = 2$.
- (iii) Verify MFCQ at x^* and confirm that the second-order sufficient condition holds (here trivially, since f is strongly convex).
- (iv) Give the geometric interpretation of the multiplier μ^* as the distance by which a overshoots the half-plane.

Exercise 3.10 (MFCQ holds where LICQ fails). Consider the feasible set defined by $g_1(x) = -x_2 \leq 0$ and $g_2(x) = x_1^2 - x_2 \leq 0$, and the point $x^* = (0, 0)$.

- (i) Show that both constraints are active at x^* and compute $\nabla g_1(x^*)$ and $\nabla g_2(x^*)$.
- (ii) Show that these gradients are linearly dependent, so that LICQ *fails* at x^* .
- (iii) Exhibit a direction d with $\nabla g_j(x^*)^\top d < 0$ for $j = 1, 2$, and conclude that MFCQ *holds*.
- (iv) Explain, in one or two sentences, why this shows MFCQ is strictly weaker than LICQ, and why a KKT point at x^* may nonetheless have non-unique multipliers.

Exercise 3.11 (A theorem-of-the-alternative certificate). Let

$$B = \begin{pmatrix} 1 & 1 \\ -1 & 2 \\ 0 & -1 \end{pmatrix}.$$

- (i) Decide whether there exists $d \in \mathbb{R}^2$ with $Bd < 0$ (componentwise). If so, produce one such d .
- (ii) Decide whether there exists $\mu \in \mathbb{R}_{\geq 0}^3$, $\mu \neq 0$, with $B^\top \mu = 0$. If so, produce one such μ .
- (iii) Reconcile your findings with Gordan's theorem (Theorem 3.25), which asserts that *exactly one* of the two systems is solvable.

Exercise 3.12 (Quadratic penalty method recovers the multiplier). Consider the scalar problem

$$\min_{x \in \mathbb{R}} f(x) = \frac{1}{2} x^2 \quad \text{s.t.} \quad h(x) = x - 1 = 0,$$

and its quadratic penalty approximation $F^k(x) = \frac{1}{2} x^2 + \frac{k}{2} (x - 1)^2$ for $k > 0$.

- (i) Compute the unconstrained minimizer x^k of F^k and show that $x^k = \frac{k}{k+1} \rightarrow 1$ as $k \rightarrow \infty$, so feasibility is recovered only in the limit.

- (ii) Define $\lambda^k = k h(x^k)$ and show that $\lambda^k = -\frac{k}{k+1} \rightarrow -1$, which equals the exact Lagrange multiplier at $x^* = 1$.
- (iii) Compute $F^{k''}(x)$ and explain why the penalty problem becomes increasingly ill-conditioned as $k \rightarrow \infty$. Describe how the augmented Lagrangian method avoids driving $k \rightarrow \infty$.

Exercise 3.13 (Necessary is not sufficient: a constrained saddle). Consider

$$\min_{x \in \mathbb{R}^2} f(x) = x_1^2 - x_2^2 \quad \text{s.t.} \quad h(x) = x_1 - 1 = 0.$$

- (i) Find all Lagrange candidates and the associated multiplier.
- (ii) Evaluate the reduced Hessian on the tangent space and show that the second-order *necessary* condition for a minimum *fails*.
- (iii) Conclude that the candidate is not a local minimum, and identify what kind of stationary point it is when the objective is restricted to the feasible line. Contrast this with a point satisfying the second-order *sufficient* condition.

Part II

Convex Optimization

Chapter 4

Basic Convex Optimization

A single idea runs through this chapter: *convexity turns local information into global conclusions*. For a general optimization problem, knowing that a point is better than all of its neighbors tells us almost nothing about the rest of the landscape; for a convex problem, the same local certificate proves global optimality. This one property is what separates the problems we can solve reliably and at scale from those we cannot, and it is the reason convex optimization sits at the heart of modern estimation, control, machine learning, and communications.

We develop the geometry, analysis, and duality principles that make this possible. We begin with convex sets and convex functions, then use that structure to explain why convex programs admit global optimality certificates, stable solution sets, and efficient algorithms. A second theme is *convexification*: rewriting or relaxing a nonconvex problem so that these tools apply. Here we distinguish exact reformulations—epigraph methods, changes of variables, eigenvalue representations, and fractional transformations—from relaxations based on lifting, convex envelopes, and Lagrangian duality. The two themes meet in the machinery of saddle points, Slater’s condition, and the Karush–Kuhn–Tucker (KKT) conditions, which we then use to read off concrete solutions. Examples such as water-filling and eigenvalue optimization show how the abstract theory produces working answers in communications, control, and resource allocation.

4.1 ■ Convex Optimization: Basic Concepts and Geometry

Convex optimization problems are optimization problems whose objectives and feasible regions are governed by convexity. A generic convex problem can be written in the form

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i \in \mathcal{Q} := \{1, 2, \dots, q\}, \end{aligned} \tag{4.1}$$

where the objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and the constraint functions $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are assumed to be convex.

The theory rests on three connected objects: convex sets, which describe feasible regions; convex functions, which describe objectives and constraints; and convex programs, which combine the two into problems with strong global optimality guarantees. The geometry of convex sets gives the quickest intuition for why local information can certify global conclusions.

The formulation (4.1) uses only inequalities. Algebraically, an equality $h(x) = 0$ can be written as $h(x) \leq 0$ and $-h(x) \leq 0$; in a convex program this preserves convexity only when h is affine. Equality constraints are therefore usually kept separate, especially when stating constraint qualifications and optimality conditions.

4.1.1 ■ Convex Sets

Convex sets provide the geometric foundation of convex optimization. Their simple defining property leads to strong closure properties and powerful separation results that underpin duality, optimality conditions, and algorithmic design.

Definition 4.1 (Convex Set). A set $C \subseteq \mathbb{R}^n$ is said to be convex if for any $x, y \in C$ and any scalar $\lambda \in [0, 1]$, the point $z := \lambda x + (1 - \lambda)y$ also belongs to C .

Geometrically, convexity means that the line segment connecting any two points of C remains in C ; see Figure 4.1. This simple property is the source of many analytical and computational advantages of convex optimization.

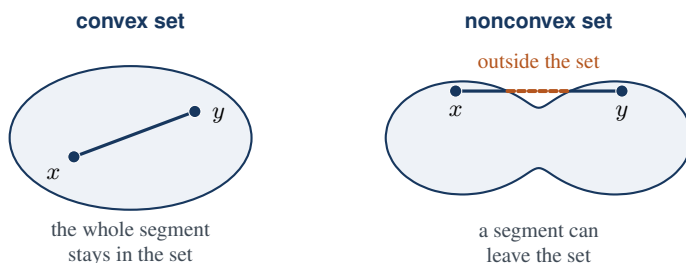


Figure 4.1: Convexity is a line-segment property. The feasible-set results in this chapter repeatedly use the fact that interpolation between feasible points stays feasible.

Examples of Convex Sets

A handful of canonical sets recur throughout optimization, and verifying their convexity is worth doing once carefully, because every verification follows the same three-step script: take two points of the set, form the convex combination $z = \lambda x + (1 - \lambda)y$, and check that the defining condition survives. The sets below are the atoms from which essentially every feasible region in this book is assembled.

Example 4.2 Affine sets and half-spaces.

The solution set of a linear system, $F := \{x \in \mathbb{R}^n : Ax = b\}$ with $A \in \mathbb{R}^{m \times n}$, is convex: if $Ax = b$ and $Ay = b$, then linearity gives $A(\lambda x + (1 - \lambda)y) = \lambda b + (1 - \lambda)b = b$, so the combination again lies in F . Relaxing the equality to an inequality changes nothing in the argument: for a half-space $H := \{x : a^\top x \leq b\}$ we have $a^\top(\lambda x + (1 - \lambda)y) = \lambda a^\top x + (1 - \lambda)a^\top y \leq \lambda b + (1 - \lambda)b = b$, so H is convex as well. Half-spaces are the fundamental building blocks of polyhedral sets, any finite intersection of them is a polyhedron, and the intersection property established below guarantees that convexity is inherited by the whole. $\square$

Example 4.3 Euclidean balls and ellipsoids.

For the ball $B(x_0, r) := \{x : \|x - x_0\|_2 \leq r\}$, writing a convex combination as $\lambda(x - x_0) + (1 - \lambda)(y - x_0)$ and applying the triangle inequality gives

$$\|\lambda(x - x_0) + (1 - \lambda)(y - x_0)\|_2 \leq \lambda\|x - x_0\|_2 + (1 - \lambda)\|y - x_0\|_2 \leq r,$$

so the combination stays within distance r of x_0 and $B(x_0, r)$ is convex. More generally, an ellipsoid $E := \{x : (x - x_0)^\top P^{-1}(x - x_0) \leq 1\}$ with $P \succ 0$ is convex, because it is the preimage of the unit ball under the affine map $x \mapsto P^{-1/2}(x - x_0)$. This illustrates a theme we exploit repeatedly: rather than re-verify the definition, one recognizes a set as the image or preimage of a set already known to be convex. $\square$

Example 4.4 Convex cones.

A set K is a *cone* if $x \in K$ implies $\alpha x \in K$ for all $\alpha \geq 0$, and a *convex cone* if it is in addition convex. For such a set the two properties fuse into a single closure rule: any conic combination $\alpha x + \beta y$ with $\alpha, \beta \geq 0$ again lies in K . Indeed αx and βy belong to K by the cone property, and for $\alpha + \beta > 0$ their convex combination

$$\frac{\alpha}{\alpha + \beta}(\alpha + \beta)x + \frac{\beta}{\alpha + \beta}(\alpha + \beta)y = \alpha x + \beta y$$

belongs to K by convexity (the case $\alpha = \beta = 0$ is trivial). The nonnegative orthant $\mathbb{R}_+^n$, the second-order (Lorentz) cone, and the cone of positive semidefinite matrices are the three cones on which modern conic optimization is built. $\square$

Example 4.5 Convex hulls.

The *convex hull* of a set S collects all of its finite convex combinations,

$$\text{conv}(S) = \left\{ \sum_{i=1}^k \lambda_i x_i \mid x_i \in S, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1 \right\}.$$

It is convex, because a convex combination of convex combinations is again a convex combination, and it is the *smallest* convex set containing S : any convex set that contains S must contain all finite convex combinations of its points, hence all of $\text{conv}(S)$. The convex hull is the canonical way to “fill in” a set to the nearest convex one, a construction we return to when relaxing combinatorial constraints. $\square$

Example 4.6 Descent cones.

For a differentiable f , the set of first-order descent directions at x , $S := \{d : \nabla f(x)^\top d < 0\}$, is an open half-space whose bounding hyperplane passes through the origin, and hence an *open* convex cone: it is convex as a half-space, and scaling any $d \in S$ by $\alpha > 0$ preserves the sign, since $\nabla f(x)^\top (\alpha d) = \alpha \nabla f(x)^\top d < 0$. The

origin itself is excluded (there $\nabla f(x)^\top d = 0$), so S satisfies the cone property only for $\alpha > 0$; its closure $\{d : \nabla f(x)^\top d \leq 0\}$ is a convex cone in the strict sense of the definition above. That these directions form a full cone rather than a scattered collection is precisely what lets descent algorithms select a search direction by a convex feasibility test at each step. $\square$

Properties of Convex Sets

Theorem 4.7 (Intersection Property). *If $\{C_i\}_{i \in \mathcal{P}}$ is a collection of convex sets, then their intersection $C := \bigcap_{i \in \mathcal{P}} C_i$ is convex.*

Proof. Let $x, y \in C$. Then $x, y \in C_i$ for all $i \in \mathcal{P}$. Since each C_i is convex, $\lambda x + (1 - \lambda)y \in C_i \forall i \in \mathcal{P}$. Hence $\lambda x + (1 - \lambda)y \in C$. $\square$

Proposition 4.8 (Linear Images and Preimages). *Let A be a linear map.*

- *If C is convex, then $A(C)$ is convex.*
- *If D is convex, then $A^{-1}(D)$ is convex.*

Proof. Both claims follow directly from linearity of A and preservation of convex combinations. $\square$

Convexity is likewise preserved under Cartesian products: if C_1 and C_2 are convex, so is $C_1 \times C_2$. Taken together, intersection, linear images and preimages, and products let us certify convexity compositionally, working outward from primitive sets. The topological operations of closure and interior also respect convexity, so we may pass freely between a convex set and its boundary or interior without leaving the convex world, a convenience we exploit repeatedly when manipulating feasible sets.

Proposition 4.9 (Closure and interior). *Let $C \subseteq \mathbb{R}^n$ be convex.*

- (1) *The closure $\overline{C}$ is convex.*
- (2) *If $\text{int}(C) \neq \emptyset$, then $\text{int}(C)$ is convex.*

Proof. (1) Let $x, y \in \overline{C}$ and fix $\theta \in [0, 1]$. By definition of closure, there exist sequences $\{x_k\} \subset C$ and $\{y_k\} \subset C$ such that $x_k \rightarrow x$ and $y_k \rightarrow y$. Since C is convex, $z_k := \theta x_k + (1 - \theta)y_k \in C$ for all k . By continuity of addition and scalar multiplication, $z_k \rightarrow \theta x + (1 - \theta)y$. Thus $\theta x + (1 - \theta)y$ is a limit of points in C , hence belongs to $\overline{C}$. Therefore $\overline{C}$ is convex.

(2) Let $x, y \in \text{int}(C)$ and fix $\theta \in [0, 1]$. By definition of interior, there exist radii $r_x > 0$ and $r_y > 0$ such that $B(x, r_x) \subset C$ and $B(y, r_y) \subset C$. Let $r := \min\{r_x, r_y\}$. For any u with $\|u\| < r$ we have $x + u \in C$ and $y + u \in C$. Since C is convex,

$$\theta(x + u) + (1 - \theta)(y + u) = \theta x + (1 - \theta)y + u \in C.$$

Thus $B(\theta x + (1 - \theta)y, r) \subset C$, which shows that $\theta x + (1 - \theta)y \in \text{int}(C)$. Hence $\text{int}(C)$ is convex. $\square$

Extreme Points, Carathéodory's Theorem, and Structural Operations

Definition 4.10 (Extreme point). Let $C \subseteq \mathbb{R}^n$ be convex. A point $x \in C$ is called an extreme point of C if whenever

$$x = \theta y + (1 - \theta)z, \quad y, z \in C, \quad \theta \in (0, 1),$$

it follows that $y = z = x$.

Extreme points are precisely the points of C that cannot be written as nontrivial convex combinations of other points of C ; geometrically, they are the “corners” of the set. In a polyhedron they coincide with the vertices, and this is exactly why linear programs can be solved by moving from vertex to vertex: a linear objective is minimized at an extreme point, so the search reduces to a finite set. More generally, a compact convex set is the convex hull of its extreme points (the Krein–Milman theorem), so these few points encode the entire set.

The next results catalogue the operations that build new convex sets from old ones. Together with the intersection property, they form an algebra of convexity: once a handful of primitive sets are known to be convex, convexity of a complicated feasible region can be verified by inspection rather than by returning to the definition each time.

Proposition 4.11 (Minkowski sum). If $C_1, C_2 \subseteq \mathbb{R}^n$ are convex, then their Minkowski sum

$$C_1 + C_2 := \{x_1 + x_2 : x_1 \in C_1, x_2 \in C_2\}$$

is convex.

Proof. Let $u = x_1 + x_2$ and $v = y_1 + y_2$ with $x_i, y_i \in C_i$. For $\theta \in [0, 1]$,

$$\theta u + (1 - \theta)v = (\theta x_1 + (1 - \theta)y_1) + (\theta x_2 + (1 - \theta)y_2).$$

Since C_1 and C_2 are convex, $\theta x_i + (1 - \theta)y_i \in C_i$. Hence $\theta u + (1 - \theta)v \in C_1 + C_2$. $\square$

Recall that the convex hull $\text{conv}(S)$ consists of all finite convex combinations $\sum_{i=1}^m \lambda_i x_i$ of points $x_i \in S$. The definition places no bound on the number m of points involved, which raises a natural question: to represent a given point of the hull, how many points of S do we actually need? Carathéodory's theorem answers that the ambient dimension alone sets the ceiling, at most $n + 1$ points suffice, no matter how large or complicated S is.

Theorem 4.12 (Carathéodory). Let $S \subseteq \mathbb{R}^n$ and let $C = \text{conv}(S)$. Then every $x \in C$ can be written as a convex combination of at most $n + 1$ points of S .

Proof. Let $x \in C$. By definition of the convex hull, there exist points $x_1, \dots, x_m \in S$ and coefficients $\lambda_i \geq 0$ with $\sum_{i=1}^m \lambda_i = 1$ such that $x = \sum_{i=1}^m \lambda_i x_i$. If $m \leq n + 1$, we are done. Suppose $m > n + 1$.

Affine dependence. Since $m > n + 1$, the vectors $x_1 - x_m, \dots, x_{m-1} - x_m$ form a collection of $m - 1 > n$ vectors in $\mathbb{R}^n$. Hence they are linearly dependent. Therefore, there exist scalars $\beta_1, \dots, \beta_{m-1}$, not all zero, such that $\sum_{i=1}^{m-1} \beta_i (x_i - x_m) = 0$. Expanding gives $\sum_{i=1}^{m-1} \beta_i x_i = \left(\sum_{i=1}^{m-1} \beta_i \right) x_m$. Define $\alpha_i = \beta_i$ for $i = 1, \dots, m - 1$ and $\alpha_m = -\sum_{i=1}^{m-1} \beta_i$. Then not all α_i are zero and we obtain $\sum_{i=1}^m \alpha_i x_i = 0$ and $\sum_{i=1}^m \alpha_i = 0$. Thus $\{x_i\}$ are affinely dependent.

Perturbation of coefficients. For $\varepsilon \in \mathbb{R}$ define $\lambda_i(\varepsilon) = \lambda_i + \varepsilon\alpha_i$. Because $\sum_{i=1}^m \alpha_i = 0$, we have $\sum_{i=1}^m \lambda_i(\varepsilon) = \sum_{i=1}^m \lambda_i + \varepsilon \sum_{i=1}^m \alpha_i = 1$. Moreover, $\sum_{i=1}^m \lambda_i(\varepsilon)x_i = \sum_{i=1}^m \lambda_i x_i + \varepsilon \sum_{i=1}^m \alpha_i x_i = x$. Thus the representation of x remains valid for all ε .

Eliminating one coefficient. We now choose ε so that all coefficients remain nonnegative but at least one becomes zero. Since $\sum_{i=1}^m \alpha_i = 0$ and not all α_i vanish, there must exist indices with $\alpha_i > 0$ and others with $\alpha_j < 0$. To preserve convexity we require $\lambda_i(\varepsilon) = \lambda_i + \varepsilon\alpha_i \geq 0$ for all i . If $\alpha_i > 0$, this inequality gives $\varepsilon \geq -\lambda_i/\alpha_i$. If $\alpha_i < 0$, it gives $\varepsilon \leq -\lambda_i/\alpha_i$. Hence all constraints together define an interval $\varepsilon^- \leq \varepsilon \leq \varepsilon^+$, where $\varepsilon^- = \max_{\alpha_i > 0}(-\lambda_i/\alpha_i)$ and $\varepsilon^+ = \min_{\alpha_i < 0}(-\lambda_i/\alpha_i)$. Since $\varepsilon = 0$ satisfies all constraints, the interval is nonempty and $\varepsilon^- \leq 0 \leq \varepsilon^+$ (the inequalities are strict unless some λ_i already vanishes, in which case that point may simply be discarded). Choose $\varepsilon = \varepsilon^+$ (or $\varepsilon = \varepsilon^-$). By definition of ε^+ , there exists an index k such that $\varepsilon^+ = -\lambda_k/\alpha_k$. For this index, $\lambda_k(\varepsilon^+) = \lambda_k + \varepsilon^+\alpha_k = \lambda_k - (\lambda_k/\alpha_k)\alpha_k = 0$. For all other indices, $\lambda_i(\varepsilon^+) \geq 0$. Discarding the point x_k , whose coefficient is now zero, gives a convex representation of x using at most $m - 1$ points of S .

Finite termination. Each reduction step decreases the number of nonzero coefficients. Since m is finite, after finitely many repetitions we obtain a representation with at most $n + 1$ points. This completes the proof. $\square$

The bound $n + 1$ is not a technicality: it guarantees that convex combinations can always be taken *sparse*. In statistics and machine learning this means any point in the convex hull of a data set, for instance a mixture distribution or an averaged prediction, can be reproduced using only $n + 1$ of the samples, and in optimization it underlies the fact that basic feasible solutions of a linear program activate at most n constraints.

Supporting Hyperplanes

Supporting hyperplanes formalize the idea that, at any boundary point of a convex set, one can place a hyperplane that “touches” the set without cutting through it. This result is purely geometric and underlies first-order optimality conditions in convex optimization.

Theorem 4.13 (Supporting Hyperplane Theorem). *Let $C \subset \mathbb{R}^n$ be a nonempty convex set, and let z be a boundary point of C . Then there exists a nonzero vector $a \in \mathbb{R}^n$ satisfying $a^\top x \leq a^\top z \forall x \in C$.*

Equivalently, the hyperplane $H := \{x \in \mathbb{R}^n : a^\top x = a^\top z\}$ contains the point z and places the entire set C in the closed half-space $\{x : a^\top x \leq a^\top z\}$. Such a hyperplane is called a *supporting hyperplane* of C at z .

Proof. We first replace C by its closure $\overline{C}$, which is legitimate because convexity is preserved under closure (Proposition 4.9) and any supporting hyperplane for $\overline{C}$ at z also supports the smaller set C . The point z still fails to be interior: for a convex set, $\text{int}(\overline{C}) = \text{int}(C)$ when the latter is nonempty, and both interiors are empty otherwise, so in either case $z \notin \text{int}(\overline{C})$ and hence $z \in \partial\overline{C}$.

This is the order in which the two steps must be taken. Passing to the closure *after* choosing an exterior sequence would be circular, since a point outside C need not lie outside $\overline{C}$, and the separation theorem below requires the latter.

Because z is a boundary point of the closed convex set $\overline{C}$, there exists a sequence $\{z_k\}_{k=1}^\infty \subset \mathbb{R}^n \setminus \overline{C}$ such that $z_k \rightarrow z$ as $k \rightarrow \infty$. For each k , the singleton $\{z_k\}$ and the closed convex set $\overline{C}$ are nonempty, convex, and disjoint.

By the strong separation theorem for a closed convex set and a point not belonging to it (a special case of the Hahn–Banach separation theorem), there exist, for each k , a nonzero vector $a_k \in \mathbb{R}^n$ and a scalar $b_k \in \mathbb{R}$ such that

$$a_k^\top x \leq b_k < a_k^\top z_k \quad \forall x \in C. \quad (4.2)$$

We may normalize each a_k so that $\|a_k\|_2 = 1$. Since the unit sphere in $\mathbb{R}^n$ is compact, there exists a subsequence $\{a_{k_j}\}$ that converges to some vector $a \in \mathbb{R}^n$ with $\|a\|_2 = 1$, and hence $a \neq 0$.

Fix an arbitrary point $x \in C$. From (4.2), for each index j , we have the chain of inequalities $a_{k_j}^\top x \leq b_{k_j} < a_{k_j}^\top z_{k_j}$. The left-hand inequality implies that the real sequence $\{a_{k_j}^\top x\}_{j=1}^\infty$ is bounded above by b_{k_j} for every j . Since $a_{k_j} \rightarrow a$ in $\mathbb{R}^n$, continuity of the inner product yields $\lim_{j \rightarrow \infty} a_{k_j}^\top x = a^\top x$. Applying the $\liminf$ operator to the inequality $a_{k_j}^\top x \leq b_{k_j}$ and using the fact that limits preserve inequalities under $\liminf$, we obtain $a^\top x = \lim_{j \rightarrow \infty} a_{k_j}^\top x \leq \liminf_{j \rightarrow \infty} b_{k_j}$.

Similarly, the strict inequality $b_{k_j} < a_{k_j}^\top z_{k_j}$ implies that $\{b_{k_j}\}$ is bounded above by the sequence $\{a_{k_j}^\top z_{k_j}\}$. Since $a_{k_j} \rightarrow a$ and $z_{k_j} \rightarrow z$, continuity of the inner product again gives $\lim_{j \rightarrow \infty} a_{k_j}^\top z_{k_j} = a^\top z$. Taking limits in the inequality $b_{k_j} < a_{k_j}^\top z_{k_j}$ and using the definition of $\liminf$, we conclude that

$$\liminf_{j \rightarrow \infty} b_{k_j} \leq \lim_{j \rightarrow \infty} a_{k_j}^\top z_{k_j} = a^\top z.$$

Combining these inequalities yields $a^\top x \leq \liminf_{j \rightarrow \infty} b_{k_j} \leq a^\top z$. Because $x \in C$ was chosen arbitrarily, this inequality holds for all $x \in C$, and therefore $a^\top x \leq a^\top z \forall x \in C$. This establishes the existence of a supporting hyperplane of C at the point z with normal vector a .

□

Separating Hyperplanes

Separating hyperplanes describe the relative geometry of two convex sets. They formalize the intuition that two disjoint convex sets can always be *separated* by a hyperplane, meaning that the sets lie on opposite sides of the hyperplane (possibly touching it).

Theorem 4.14 (Separating Hyperplane Theorem). *Let C and D be two nonempty, disjoint convex sets in $\mathbb{R}^n$. Then there exist a nonzero vector $a \in \mathbb{R}^n$ and a scalar $b \in \mathbb{R}$ such that*

$$a^\top x \leq b \quad \forall x \in C, \quad a^\top x \geq b \quad \forall x \in D.$$

Proof. Define the Minkowski difference $E := C - D := \{x - y : x \in C, y \in D\}$.

We first check that E is convex. Let $u_1 = x_1 - y_1$ and $u_2 = x_2 - y_2$ be arbitrary elements of E , with $x_1, x_2 \in C$ and $y_1, y_2 \in D$. For any $\lambda \in [0, 1]$, we have

$$\lambda u_1 + (1 - \lambda)u_2 = (\lambda x_1 + (1 - \lambda)x_2) - (\lambda y_1 + (1 - \lambda)y_2).$$

Since C and D are convex, $\lambda x_1 + (1 - \lambda)x_2 \in C$ and $\lambda y_1 + (1 - \lambda)y_2 \in D$, which implies $\lambda u_1 + (1 - \lambda)u_2 \in E$. Hence E is convex.

Next, the origin does not belong to E . If $0 \in E$, then there exist $x \in C$ and $y \in D$ such that $x - y = 0$, i.e., $x = y$. This implies $C \cap D \neq \emptyset$, contradicting the assumption that C and D are disjoint. Therefore, $0 \notin E$.

A fortiori the origin is not an interior point of E : since $0 \notin E$, it follows immediately that $0 \notin \text{int}(E)$. In particular, there exists no open ball centered at the origin that is contained in E .

This puts a supporting hyperplane at the origin within reach. Because E is a convex set and $0 \notin \text{int}(E)$, the Supporting Hyperplane Theorem applies. Hence, there exists a nonzero vector $a \in \mathbb{R}^n$ such that $a^\top u \leq 0 \forall u \in E$.

Separation of C and D now follows. For arbitrary $x \in C$ and $y \in D$, the vector $u := x - y$ belongs to E , and thus $a^\top(x - y) \leq 0$, which implies $a^\top x \leq a^\top y \forall x \in C, y \in D$.

Define $b := \sup_{x \in C} a^\top x \in [-\infty, +\infty]$. Then, by definition, $a^\top x \leq b \forall x \in C$. Moreover, the inequality $a^\top x \leq a^\top y$ for all $x \in C$ and $y \in D$ implies $a^\top y \geq b \forall y \in D$.

Therefore, the hyperplane $H := \{x \in \mathbb{R}^n : a^\top x = b\}$ separates the sets C and D , completing the proof. $\square$

4.1.2 ■ Convex Functions

Convex functions turn local information into global information. Gradients, subgradients, and supporting hyperplanes become global certificates rather than purely local approximations. This is why convex objectives avoid spurious local minima and why first-order conditions can be used as optimality conditions.

Let $C \subseteq \mathbb{R}^n$ be a convex set, and let $f : C \rightarrow \mathbb{R}$ be a real-valued function.

Definition 4.15 (Convex Function). *The function f is said to be convex on C if for all $x, y \in C$ and all $\lambda \in [0, 1]$,*

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y). \quad (4.3)$$

Inequality (4.3) says that linear interpolation in the domain upper-bounds the function value in the range. Equivalently, the graph of f lies below every chord connecting $(x, f(x))$ and $(y, f(y))$.

To formalize this geometric viewpoint, we introduce the notion of the epigraph.

Definition 4.16 (Epigraph). *The epigraph of a function $f : C \rightarrow \mathbb{R}$ is the set*

$$\text{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} \mid x \in C, t \geq f(x)\}.$$

The epigraph consists of all points on or above the graph of f , turning a function into a set in one higher dimension.

Theorem 4.17 (Epigraph Characterization of Convexity). *A function $f : C \rightarrow \mathbb{R}$ is convex on a convex set C if and only if its epigraph $\text{epi}(f)$ is a convex subset of $\mathbb{R}^{n+1}$.*

This characterization establishes a precise equivalence between convex functions and convex sets; Figure 4.2 gives the geometric picture. It is central in duality theory, constrained optimization, and variational analysis, where problems are often studied through epigraphs.

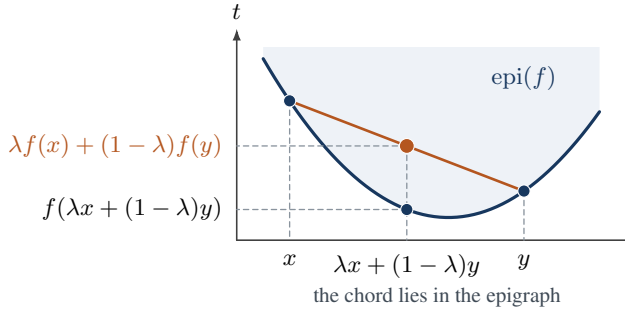


Figure 4.2: For a convex function, every chord between graph points lies above the graph. Equivalently, the epigraph is a convex set in one higher dimension.

Proof.

We prove both directions.

($\Rightarrow$) **Convex function $\Rightarrow$ convex epigraph.** Assume that f is convex on the convex set C . Let $(x_1, t_1), (x_2, t_2) \in \text{epi}(f)$. By definition of the epigraph, we have $t_1 \geq f(x_1)$ and $t_2 \geq f(x_2)$. For $\lambda \in [0, 1]$, consider their convex combination

$$(x_\lambda, t_\lambda) = \lambda(x_1, t_1) + (1 - \lambda)(x_2, t_2) = (\lambda x_1 + (1 - \lambda)x_2, \lambda t_1 + (1 - \lambda)t_2).$$

Because C is convex, $x_\lambda = \lambda x_1 + (1 - \lambda)x_2 \in C$. By convexity of f ,

$$f(x_\lambda) = f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2).$$

Since $t_i \geq f(x_i)$ for $i = 1, 2$, it follows that

$$\lambda f(x_1) + (1 - \lambda)f(x_2) \leq \lambda t_1 + (1 - \lambda)t_2 = t_\lambda.$$

Hence $t_\lambda \geq f(x_\lambda)$, which means $(x_\lambda, t_\lambda) \in \text{epi}(f)$. Therefore $\text{epi}(f)$ is convex.

($\Leftarrow$) **Convex epigraph $\Rightarrow$ convex function.** Assume that $\text{epi}(f)$ is convex. For any $x_1, x_2 \in C$, the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ belong to $\text{epi}(f)$. Convexity of the epigraph implies that for every $\lambda \in [0, 1]$, the convex combination $(\lambda x_1 + (1 - \lambda)x_2, \lambda f(x_1) + (1 - \lambda)f(x_2))$ also lies in $\text{epi}(f)$. By definition of the epigraph, this means

$$\lambda f(x_1) + (1 - \lambda)f(x_2) \geq f(\lambda x_1 + (1 - \lambda)x_2).$$

Rewriting yields

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2),$$

which is exactly the definition of convexity.

Thus f is convex on C if and only if $\text{epi}(f)$ is a convex subset of $\mathbb{R}^{n+1}$.

□

Example 4.18 Convexity preserved under partial minimization.

Let $C \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be convex and let $f : C \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex in the joint

variables (x, y) . For each $x \in \mathbb{R}^n$, define $C_x := \{y \in \mathbb{R}^m : (x, y) \in C\}$, and define the marginal function $g(x) := \inf_{y \in C_x} f(x, y)$, with the convention that $g(x) = +\infty$ if $C_x = \emptyset$. Then g is convex on its effective domain.

Proof (epigraph argument). We express the epigraph of g as

$$\text{epi}(g) = \{(x, t) \in \mathbb{R}^{n+1} \mid g(x) \leq t\}.$$

By definition of g , $g(x) \leq t$ means $\inf_{y \in C_x} f(x, y) \leq t$. Provided the infimum is attained, or more generally away from boundary points where it is approached but not attained, this is equivalent to the existence of some $y \in C_x$ with $f(x, y) \leq t$, so that

$$\text{epi}(g) = \{(x, t) \mid \exists y \text{ with } (x, y) \in C \text{ and } f(x, y) \leq t\}$$

up to such boundary points. The geometric argument below uses this identification; the *Direct verification* that follows then establishes convexity unconditionally, covering the non-attained case as well.

Let $S = \{(x, y, t) \in \mathbb{R}^{n+m+1} \mid (x, y) \in C, f(x, y) \leq t\}$. Then S is the epigraph of f restricted to C , hence convex because f is convex and C is convex. Moreover,

$$\text{epi}(g) = \{(x, t) \mid \exists y \text{ such that } (x, y, t) \in S\}.$$

To show that $\text{epi}(g)$ is convex, take $(x_1, t_1), (x_2, t_2) \in \text{epi}(g)$. Then there exist y_1, y_2 such that $(x_1, y_1, t_1), (x_2, y_2, t_2) \in S$. Thus $(x_i, y_i) \in C$ and $f(x_i, y_i) \leq t_i$ for $i = 1, 2$. For $\lambda \in [0, 1]$, convexity of S implies that

$$(x_\lambda, y_\lambda, t_\lambda) := (\lambda x_1 + (1 - \lambda)x_2, \lambda y_1 + (1 - \lambda)y_2, \lambda t_1 + (1 - \lambda)t_2) \in S.$$

Hence $(x_\lambda, y_\lambda) \in C$ and $f(x_\lambda, y_\lambda) \leq t_\lambda$. By definition of g ,

$$g(x_\lambda) = \inf_{y: (x_\lambda, y) \in C} f(x_\lambda, y) \leq f(x_\lambda, y_\lambda) \leq t_\lambda.$$

Therefore $(x_\lambda, t_\lambda) \in \text{epi}(g)$. □

Direct verification (alternative argument). Let x_1, x_2 belong to the effective domain of g and let $\theta \in [0, 1]$. Fix $\varepsilon > 0$. By definition of the infimum, there exist $y_1 \in C_{x_1}$ and $y_2 \in C_{x_2}$ such that $f(x_i, y_i) \leq g(x_i) + \varepsilon$ for $i = 1, 2$. Since C is convex, $(\theta x_1 + (1 - \theta)x_2, \theta y_1 + (1 - \theta)y_2) \in C$. By convexity of f ,

$$f(\theta x_1 + (1 - \theta)x_2, \theta y_1 + (1 - \theta)y_2) \leq \theta f(x_1, y_1) + (1 - \theta)f(x_2, y_2).$$

By definition of g as an infimum over y , $g(\theta x_1 + (1 - \theta)x_2) \leq f(\theta x_1 + (1 - \theta)x_2, \theta y_1 + (1 - \theta)y_2)$. Combining the inequalities gives $g(\theta x_1 + (1 - \theta)x_2) \leq \theta g(x_1) + (1 - \theta)g(x_2) + \varepsilon$. Since $\varepsilon > 0$ is arbitrary, we conclude

$$g(\theta x_1 + (1 - \theta)x_2) \leq \theta g(x_1) + (1 - \theta)g(x_2).$$

□

□

Example 4.19 Jensen's inequality via epigraph geometry.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex and let X be an $\mathbb{R}^n$ -valued random variable with finite $\mathbb{E}[X]$ and $\mathbb{E}[f(X)]$. Then $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$.

Proof. Since f is convex, its epigraph $\text{epi}(f) = \{(x, t) \in \mathbb{R}^{n+1} : t \geq f(x)\}$ is convex. For every outcome ω , we have $(X(\omega), f(X(\omega))) \in \text{epi}(f)$, so the random vector $(X, f(X))$ takes values in the convex set $\text{epi}(f)$. Since f is finite and convex it is continuous, so $\text{epi}(f)$ is closed; the expectation of a random vector supported in a closed convex set again lies in that set. Hence $\mathbb{E}[(X, f(X))] = (\mathbb{E}[X], \mathbb{E}[f(X)]) \in \text{epi}(f)$. By definition of the epigraph, this membership implies $f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$. $\square$

Proposition 4.20 (Support function). Let $C \subseteq \mathbb{R}^n$ be nonempty and convex. Define the support function $\sigma_C(a) := \sup_{x \in C} a^\top x$, $a \in \mathbb{R}^n$. Then σ_C is convex in a .

Proof. Let $a_1, a_2 \in \mathbb{R}^n$ and $\theta \in [0, 1]$. For any $x \in C$, $(\theta a_1 + (1-\theta)a_2)^\top x = \theta a_1^\top x + (1-\theta)a_2^\top x$. Taking the supremum over $x \in C$ gives

$$\sigma_C(\theta a_1 + (1-\theta)a_2) = \sup_{x \in C} [\theta a_1^\top x + (1-\theta)a_2^\top x] \leq \theta \sigma_C(a_1) + (1-\theta)\sigma_C(a_2).$$

Thus σ_C is convex. $\square$

The support function completely characterizes closed convex sets: two closed convex sets are equal if and only if their support functions coincide. It plays a central role in duality and separation theory.

Definition 4.21 (Strictly Convex Function). The function f is strictly convex on C if for all $x \neq y$ in C and all $\lambda \in (0, 1)$,

$$f(\lambda x + (1-\lambda)y) < \lambda f(x) + (1-\lambda)f(y).$$

Strict convexity excludes flat directions: every chord of the graph lies *strictly* above the function except at its endpoints. As a consequence, strictly convex functions admit at most one minimizer on a convex domain, a property that is particularly important in estimation, control, and learning applications where uniqueness of solutions is desirable.

Convexity alone already implies strong regularity. Even in the absence of differentiability, convex functions enjoy continuity and well-behaved local structure on the interior of their domains.

Theorem 4.22 (Continuity on the Interior). If f is convex on a convex set C , then f is continuous on $\text{int}(C)$.

Proof. Let $x_0 \in \text{int}(C)$. By definition of interior, there exists $\epsilon > 0$ such that the closed ball $B_\epsilon(x_0) \subset C$. *Step 1:* f is bounded on $B_\epsilon(x_0)$. A convex function is bounded above on a compact subset of the interior of its domain, since such a set is contained in a simplex whose vertices lie in C and convexity bounds f on the simplex by the largest vertex value; let $M := \sup_{y \in B_\epsilon(x_0)} f(y) < \infty$. A lower bound follows by reflection: for $y \in B_\epsilon(x_0)$ the reflected point $2x_0 - y$ also lies

in $B_\epsilon(x_0)$, and midpoint convexity at x_0 gives $f(x_0) \leq \frac{1}{2}f(y) + \frac{1}{2}f(2x_0 - y) \leq \frac{1}{2}f(y) + \frac{1}{2}M$, whence $f(y) \geq 2f(x_0) - M$. So $|f|$ is bounded on $B_\epsilon(x_0)$ by some constant, again denoted M .

Step 2: a local Lipschitz bound. Let $y \in B_{\epsilon/2}(x_0)$ with $y \neq x_0$, put $r := \|y - x_0\| \leq \epsilon/2$ and extend the segment from x_0 through y to the point $z := x_0 + \frac{\epsilon}{2r}(y - x_0)$, which satisfies $\|z - x_0\| = \epsilon/2$ and hence $z \in B_\epsilon(x_0)$. Writing y as the convex combination $y = (1 - \theta)x_0 + \theta z$ with $\theta = 2r/\epsilon \in (0, 1]$, convexity gives

$$f(y) - f(x_0) \leq \theta(f(z) - f(x_0)) \leq \frac{2r}{\epsilon} 2M.$$

Applying the same bound to the reflected point $2x_0 - y$ and using convexity once more yields $f(x_0) - f(y) \leq \frac{2r}{\epsilon} 2M$ as well. Together, $|f(y) - f(x_0)| \leq \frac{4M}{\epsilon} \|y - x_0\|$.

Thus f is locally Lipschitz in a neighborhood of x_0 , and hence continuous at x_0 . Since $x_0 \in \text{int}(C)$ was arbitrary, f is continuous on $\text{int}(C)$. $\square$

This result ensures that convex functions cannot exhibit abrupt oscillations or discontinuities in the interior of their domain. In particular, small changes in the argument lead to small changes in function values, which is essential for the stability of optimization algorithms.

Example 4.23 Failure of continuity at the boundary.

Let

$$C := (0, 1] \subset \mathbb{R}, \quad f(x) := \frac{1}{x}.$$

The function f is convex on C , since $f''(x) = \frac{2}{x^3} > 0 \ \forall x \in (0, 1]$. However, f cannot be extended continuously to the boundary point $x = 0 \notin C$. Indeed, $\lim_{x \downarrow 0} f(x) = +\infty$. Thus, although f is continuous on the interior $\text{int}(C) = (0, 1)$, it cannot be extended continuously to the boundary of C .

This example illustrates that convexity alone does not control behavior at boundary points, and that the continuity result in Theorem 4.22 is sharp. $\square$

Example 4.24 A convex function discontinuous at a point of its domain.

The previous example concerns a boundary point outside C . Discontinuity can also occur at a boundary point that *belongs* to C . On $C := [0, 1]$ define

$$f(x) := \begin{cases} 1, & x = 0, \\ 0, & x \in (0, 1]. \end{cases}$$

For $x \in (0, 1]$ and $\lambda \in [0, 1]$ we have $f(\lambda \cdot 0 + (1 - \lambda)x) = f((1 - \lambda)x) = 0 \leq \lambda$ whenever $\lambda < 1$, and the case $\lambda = 1$ is an equality; convexity on the rest of C is immediate since f is constant there. Thus f is convex on C , yet it is discontinuous at $0 \in C$. Interior continuity is all that Theorem 4.22 promises, and it is all that is true. $\square$

Remark 4.25. Convex functions may fail to be continuous on the boundary of C . However, interior continuity is sufficient for most optimization arguments, including the derivation of

optimality conditions and the convergence analysis of iterative methods.

In subsequent sections, we refine the notion of convexity by introducing first-order and second-order characterizations, including gradient inequalities, subgradients, and Hessian conditions. These analytical tools make precise the connection between the geometric definition given above and practical methods for solving convex optimization problems.

First-Order Characterization of Convexity

When the function f is differentiable, convexity admits a particularly powerful and intuitive characterization in terms of first-order information. This characterization reveals that for convex functions, local linear approximations provide global underestimates of the function, thereby linking geometry, analysis, and optimization in a precise way.

Theorem 4.26 (First-Order Condition for Convexity). *Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable. Then f is convex if and only if*

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) \quad \text{for all } x, y \in \mathbb{R}^n. \quad (4.4)$$

This inequality is often called the *gradient inequality*. It asserts that the function always lies above its first-order Taylor expansion at any point. Unlike general smooth functions, for which Taylor expansions are only locally accurate, convex functions enjoy a global linear lower bound at every point.

Inequality (4.4) states that the tangent hyperplane to the graph of f at x lies below the graph of f everywhere. In geometric terms, the graph of a convex function has the property that no tangent plane can cut through it. This global supporting-hyperplane property is the functional analog of supporting hyperplanes for convex sets and forms the basis of many optimality conditions.

Proof. ($\Rightarrow$) Assume f is convex. Fix arbitrary points $x, y \in \mathbb{R}^n$ and let $\lambda \in (0, 1)$. By convexity,

$$f(y + \lambda(x - y)) \leq (1 - \lambda)f(y) + \lambda f(x).$$

Rewriting this inequality yields

$$f(y + \lambda(x - y)) - f(y) \leq \lambda(f(x) - f(y)).$$

Dividing both sides by $\lambda > 0$ gives

$$\frac{f(y + \lambda(x - y)) - f(y)}{\lambda} \leq f(x) - f(y).$$

Taking the limit as $\lambda \rightarrow 0^+$ and using differentiability of f at y , we obtain $\nabla f(y)^\top (x - y) \leq f(x) - f(y)$, which, after rearranging terms, is equivalent to (4.4).

($\Leftarrow$) Assume that (4.4) holds for all $x, y \in \mathbb{R}^n$. Let $x, y \in \mathbb{R}^n$ and let $\lambda \in [0, 1]$. Define $x' := \lambda x + (1 - \lambda)y$. Applying (4.4) at the point x' with $y = x$ and $y = y$, respectively, yields

$$\nabla f(x')^\top (x - x') \leq f(x) - f(x'), \quad \nabla f(x')^\top (y - x') \leq f(y) - f(x').$$

Multiplying the first inequality by λ and the second by $(1 - \lambda)$ and summing, we obtain

$$\nabla f(x')^\top (\lambda(x - x') + (1 - \lambda)(y - x')) \leq \lambda f(x) + (1 - \lambda)f(y) - f(x').$$

Since $\lambda(x - x') + (1 - \lambda)(y - x') = 0$, the left-hand side vanishes, and we are left with $f(x') \leq \lambda f(x) + (1 - \lambda)f(y)$, which is precisely the convexity inequality (4.3). $\square$

Theorem 4.27 (Monotone Gradient Characterization). *Let $f \in C^1(\mathbb{R}^n)$. The following are equivalent:*

- (i) f is convex;
- (ii) the gradient mapping ∇f is monotone, that is,

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq 0 \quad \forall x, y \in \mathbb{R}^n. \quad (4.5)$$

Proof. Assume first that f is convex. Applying the first-order inequality (4.4) with (x, y) and then with (y, x) yields $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$, $f(x) \geq f(y) + \nabla f(y)^\top (x - y)$. Adding the two inequalities gives

$$0 \geq \nabla f(x)^\top (y - x) + \nabla f(y)^\top (x - y) = -(\nabla f(x) - \nabla f(y))^\top (x - y),$$

which is equivalent to (4.5).

Conversely, suppose (4.5) holds. Fix x, y and define $\phi(t) = f(x + t(y - x))$ for $t \in [0, 1]$. Then $\phi'(t) = \nabla f(x + t(y - x))^\top (y - x)$. Monotonicity of ∇f implies that $\phi'(t)$ is nondecreasing in t . Integrating from 0 to 1 yields

$$f(y) - f(x) = \int_0^1 \phi'(t) dt \geq \phi'(0) = \nabla f(x)^\top (y - x),$$

which is precisely (4.4). $\square$

Second-Order Characterization

When second derivatives exist, convexity can be characterized locally through curvature properties encoded in the Hessian matrix.

Theorem 4.28 (Hessian Test for Convexity). *Let $f \in C^2(\mathbb{R}^n)$. Then f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in \mathbb{R}^n$.*

Proof. ($\Rightarrow$) Assume f is convex. Fix $x \in \mathbb{R}^n$ and $v \in \mathbb{R}^n$. Define the univariate function $\phi(t) := f(x + tv)$. Since f is convex, its restriction to any line is convex, so ϕ is convex on $\mathbb{R}$. Because $f \in C^2$, we have

$$\phi''(t) = v^\top \nabla^2 f(x + tv)v.$$

Convexity of ϕ implies $\phi''(t) \geq 0$ for all t , and in particular $v^\top \nabla^2 f(x)v \geq 0$. Since v was arbitrary, $\nabla^2 f(x)$ is positive semidefinite.

($\Leftarrow$) Assume $\nabla^2 f(x) \succeq 0$ for all x . Fix $x, y \in \mathbb{R}^n$ and define $\phi(t) := f(x + t(y - x))$, $t \in [0, 1]$. Then

$$\phi''(t) = (y - x)^\top \nabla^2 f(x + t(y - x))(y - x) \geq 0,$$

so ϕ is convex on $[0, 1]$. Convexity of ϕ yields $\phi(1) \geq \phi(0) + \phi'(0)$, that is,

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x).$$

By Theorem 4.26, this first-order condition is equivalent to convexity of f . Hence f is convex. $\square$

Positive semidefiniteness of the Hessian means that the second directional derivative satisfies

$$\left. \frac{d^2}{dt^2} f(x + tv) \right|_{t=0} = v^\top \nabla^2 f(x) v \geq 0 \quad \forall v,$$

so the function does not bend downward in any direction. Convexity is thus a purely local curvature condition when f is twice continuously differentiable.

Strict Convexity

We begin with the definition.

Definition 4.29 (Strict convexity). A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly convex if

$$f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y), \quad \forall x \neq y, \theta \in (0, 1).$$

Strict convexity can be characterized in several equivalent first-order ways.

Proposition 4.30 (First-order characterizations). Let $f \in C^1(\mathbb{R}^n)$. The following statements are equivalent:

- (i) f is strictly convex;
- (ii) $f(y) > f(x) + \nabla f(x)^\top (y - x), \forall x \neq y$;
- (iii) The gradient mapping is strictly monotone:

$$(\nabla f(x) - \nabla f(y))^\top (x - y) > 0, \quad \forall x \neq y.$$

Proof. For differentiable convex functions, the supporting hyperplane inequality $f(y) \geq f(x) + \nabla f(x)^\top (y - x)$ holds for all x, y . If f is strictly convex and equality holds in the above inequality for some $x \neq y$, then f must be affine on the segment $[x, y]$, contradicting strict convexity. Hence (i) $\Rightarrow$ (ii).

To show (ii) $\Rightarrow$ (iii), write (ii) once for (x, y) and once for (y, x) and add:

$$f(y) > f(x) + \nabla f(x)^\top (y - x), \quad f(x) > f(y) + \nabla f(y)^\top (x - y).$$

Adding yields $(\nabla f(x) - \nabla f(y))^\top (x - y) > 0$.

For (iii) $\Rightarrow$ (ii), fix $x \neq y$ and set $d = y - x$. By strict monotonicity applied to x and $x + td$ for $t \in (0, 1]$, $\nabla f(x + td)^\top d > \nabla f(x)^\top d$. Integrating along the segment,

$$f(y) - f(x) = \int_0^1 \nabla f(x + td)^\top d \, dt > \nabla f(x)^\top (y - x),$$

which is exactly (ii).

It remains to close the cycle with (ii) $\Rightarrow$ (i). Let $x \neq y$ and $\theta \in (0, 1)$, and set $z := \theta x + (1 - \theta)y$, so that $z \neq x$ and $z \neq y$. Applying (ii) at the base point z to each of x and y gives

$$f(x) > f(z) + \nabla f(z)^\top (x - z), \quad f(y) > f(z) + \nabla f(z)^\top (y - z).$$

Multiplying the first by $\theta > 0$, the second by $1 - \theta > 0$ and adding, the gradient terms combine into $\nabla f(z)^\top (\theta x + (1 - \theta)y - z) = 0$, leaving $\theta f(x) + (1 - \theta)f(y) > f(z)$, which is strict convexity. $\square$

We now turn to second-order conditions.

Proposition 4.31 (Second-order sufficient condition for strict convexity). *Let $f \in C^2(\mathbb{R}^n)$. If $\nabla^2 f(x) \succ 0$ for all $x \in \mathbb{R}^n$, then f is strictly convex.*

Proof. Fix $x \in \mathbb{R}^n$ and a nonzero direction $v \in \mathbb{R}^n$, and define $\phi(t) := f(x + tv)$. Then $\phi''(t) = v^\top \nabla^2 f(x + tv) v$. Since $\nabla^2 f(x) \succ 0$ for all x , we have $v^\top \nabla^2 f(x + tv) v > 0$ for all $v \neq 0$ and all t . Hence ϕ is strictly convex as a function of one variable.

For $x \neq y$, write $y = x + v$ with $v \neq 0$. Strict convexity of ϕ on $[0, 1]$ yields $f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$ for all $\theta \in (0, 1)$, proving that f is strictly convex. $\square$

Remark. Positive definiteness of the Hessian is sufficient for strict convexity but not necessary. Strict convexity implies only $\nabla^2 f(x) \succeq 0$ for all x , and the Hessian may be singular at some points.

Example 4.32 Strict convexity with singular Hessian.

Consider $f(x) = x^4$ on $\mathbb{R}$. Then $f''(x) = 12x^2 \geq 0$ for all x , so f is convex. Moreover, for $x \neq y$ and $\theta \in (0, 1)$, $(\theta x + (1 - \theta)y)^4 < \theta x^4 + (1 - \theta)y^4$, so f is strictly convex. However, $f''(0) = 0$, so the Hessian is only positive semidefinite at $x = 0$. $\square$

4.1.3 ■ Strong Convexity

A uniform lower bound on the curvature, $\nabla^2 f(x) \succeq mI$ for some $m > 0$ and for all x , characterizes *strong convexity*.

Definition 4.33 (Strong convexity). *A differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is said to be m -strongly convex ($m > 0$) if for all $x, y \in \mathbb{R}^n$,*

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|^2.$$

This inequality strengthens the first-order condition for convexity by requiring a uniform quadratic growth away from every supporting hyperplane. Equivalently, the function $x \mapsto f(x) - \frac{m}{2} \|x\|^2$ is convex.

Proposition 4.34 (Second-order characterization). *If $f \in C^2(\mathbb{R}^n)$, then f is m -strongly convex if and only if*

$$\nabla^2 f(x) \succeq mI \quad \text{for all } x \in \mathbb{R}^n.$$

Strong convexity implies strict convexity and, moreover, the gradient mapping is strongly monotone:

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq m \|x - y\|^2, \quad \forall x, y.$$

In particular, a strongly convex function has a unique minimizer, and the function grows at least quadratically away from it.

The distinction between convexity, strict convexity, and strong convexity is one of *degree*, and it is worth keeping the hierarchy straight. Convexity forbids downward curvature ($\nabla^2 f \succeq 0$); strict convexity forbids flat segments and so buys uniqueness of the minimizer; strong convexity forbids the curvature from vanishing anywhere ($\nabla^2 f \succeq mI$) and so buys a *quantitative* handle on how fast the function rises. That quantitative handle is what algorithms consume. The quadratic lower bound implies the error bound $\frac{m}{2} \|x - x^*\|^2 \leq f(x) - f(x^*)$, so a small optimality gap certifies a small distance to the optimum: the problem is well conditioned and noise in the objective does not blur the location of the solution. When f is also L -smooth ($\nabla^2 f \preceq LI$), the ratio $\kappa = L/m$ is the *condition number*, and gradient descent converges linearly at a rate governed by κ . Thus m is not a cosmetic constant but the very quantity that separates fast, reliable optimization from slow, ill-conditioned optimization.

Characterization via one-dimensional restrictions

A convenient way to characterize convexity is to restrict f to lines. For $x \in \mathbb{R}^n$ and $v \in \mathbb{R}^n$, define the scalar function $\phi(t) := f(x + tv)$, $t \in \mathbb{R}$.

Proposition 4.35 (Line characterization of convexity). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be given. Then f is convex if and only if, for every x and v , the function $\phi(t) = f(x + tv)$ is convex in t .*

Proof. If f is convex, then for any t_1, t_2 and $\theta \in (0, 1)$,

$$\phi(\theta t_1 + (1 - \theta)t_2) = f(x + (\theta t_1 + (1 - \theta)t_2)v) = f(\theta(x + t_1 v) + (1 - \theta)(x + t_2 v))$$

is bounded above by

$$\theta f(x + t_1 v) + (1 - \theta)f(x + t_2 v) = \theta \phi(t_1) + (1 - \theta)\phi(t_2),$$

so ϕ is convex.

Conversely, let $x, y \in \mathbb{R}^n$ and set $v = y - x$. Then for $\theta \in (0, 1)$,

$$f(\theta x + (1 - \theta)y) = f(x + \theta(y - x)) = \phi(\theta).$$

Convexity of ϕ on $[0, 1]$ gives

$$\phi(\theta) \leq \theta \phi(1) + (1 - \theta)\phi(0) = \theta f(y) + (1 - \theta)f(x),$$

which is precisely convexity of f . $\square$

If $f \in C^2(\mathbb{R}^n)$, this characterization yields the second-order condition immediately: $\phi''(t) = v^\top \nabla^2 f(x + tv) v$. Thus f is convex if and only if $v^\top \nabla^2 f(z) v \geq 0$ for all z and all v , that is, $\nabla^2 f(z) \succeq 0$ everywhere.

Remark 4.36 (Extension beyond $\mathbb{R}^n$). The line-restriction characterization of convexity does not rely on the finite-dimensional structure of $\mathbb{R}^n$ and extends verbatim to arbitrary real vector spaces.

Let E be a real vector space and let $f : E \rightarrow \mathbb{R} \cup \{+\infty\}$. Then f is convex if and only if, for every $x \in E$ and every direction $v \in E$, the scalar function $\phi(t) := f(x + tv)$, $t \in \mathbb{R}$, is convex on its domain.

The proof is identical to the finite-dimensional case, since convexity is defined purely in terms of affine combinations and therefore depends only on the vector space structure, not on coordinates or dimension. Consequently, the same principle applies in Banach and Hilbert spaces for convex functionals, in spaces of measurable functions, in matrix spaces such as $\mathbb{S}_{++}^n$, and more generally in operator spaces arising in functional analysis. In all such settings, convexity of f is equivalent to convexity along every affine one-dimensional slice.

When appropriate second-order derivatives exist (for example, Fréchet derivatives in Banach spaces or Hessians in finite dimensions), the line characterization yields the generalized second-order test $\frac{d^2}{dt^2} f(x + tv) \geq 0$ for all x, v , which corresponds to positivity of the associated second directional derivative or quadratic form.

Example 4.37 Convexity of $-\log \det X$ on $\mathbb{S}_{++}^n$.

Let $\mathbb{S}_{++}^n$ denote the set of symmetric positive definite matrices. Define $f(X) := -\log \det X$ for $X \in \mathbb{S}_{++}^n$. Then f is convex (indeed strictly convex) on $\mathbb{S}_{++}^n$.

Proof. Since $\mathbb{S}_{++}^n$ is an open convex set, it suffices to verify convexity along every affine line. Fix $X \in \mathbb{S}_{++}^n$ and a symmetric matrix V . For all t such that $X + tV \succ 0$, define $\phi(t) := -\log \det(X + tV)$. Because $\mathbb{S}_{++}^n$ is open, there exists an open interval of t around 0 for which $X + tV$ remains positive definite.

Using the identity $\frac{d}{dt} \log \det Y(t) = \text{tr}(Y(t)^{-1}Y'(t))$, with $Y(t) = X + tV$ and $Y'(t) = V$, we obtain $\phi'(t) = -\text{tr}((X + tV)^{-1}V)$. To compute the second derivative, recall that $\frac{d}{dt} Y(t)^{-1} = -Y(t)^{-1}Y'(t)Y(t)^{-1}$. Hence

$$\frac{d}{dt}(X + tV)^{-1} = -(X + tV)^{-1}V(X + tV)^{-1}.$$

Differentiating once more gives

$$\phi''(t) = \text{tr}((X + tV)^{-1}V(X + tV)^{-1}V).$$

Let $W := (X + tV)^{-1/2}V(X + tV)^{-1/2}$, which is symmetric. By the cyclic property of the trace (note $(X + tV)^{-1}V(X + tV)^{-1}V$ is only *similar* to W^2 , but has the same trace),

$$\phi''(t) = \text{tr}((X + tV)^{-1}V(X + tV)^{-1}V) = \text{tr}(W^2) = \|W\|_F^2 \geq 0.$$

Thus ϕ is convex in t for every choice of X and V , and therefore f is convex on $\mathbb{S}_{++}^n$. Moreover, $\phi''(t) = 0$ only when $V = 0$, so f is strictly convex. $\square$

Example 4.38 Convexity of the log–integral–exponential functional.

Let $(\Omega, \mathcal{F}, \rho)$ be a σ -finite measure space and let f belong to a linear space of measurable functions such that $\int e^{f(x)} d\rho(x) < \infty$. (σ -finiteness is what the worked case below requires, since it takes ρ to be Lebesgue measure on $\mathbb{R}^n$; only finiteness

of the integral is used in the proof.) Define $\mathcal{T}(f) := \log \int e^{f(x)} d\rho(x)$. Then $\mathcal{T}$ is convex on its domain.

Proof. Since convexity of a functional on a linear space can be verified along affine lines, fix f and a direction h , and define $\phi(t) := \mathcal{T}(f+th) = \log \int e^{f(x)+th(x)} d\rho(x)$ for those t where the integral is finite.

Let $Z(t) = \int e^{f(x)+th(x)} d\rho(x)$. Then $\phi(t) = \log Z(t)$ and $\phi'(t) = Z'(t)/Z(t)$. Differentiating under the integral sign gives $Z'(t) = \int h(x)e^{f(x)+th(x)} d\rho(x)$, hence

$$\phi'(t) = \frac{\int h(x)e^{f(x)+th(x)} d\rho(x)}{\int e^{f(x)+th(x)} d\rho(x)}.$$

Introduce the probability measure

$$d\nu_t(x) = \frac{e^{f(x)+th(x)}}{Z(t)} d\rho(x).$$

Then

$$\phi'(t) = \int h(x) d\nu_t(x) = \mathbb{E}_{\nu_t}[h].$$

Differentiating once more,

$$\phi''(t) = \frac{Z''(t)Z(t) - (Z'(t))^2}{Z(t)^2}.$$

Since $Z''(t) = \int h(x)^2 e^{f(x)+th(x)} d\rho(x)$, a direct computation yields

$$\phi''(t) = \int h(x)^2 d\nu_t(x) - \left(\int h(x) d\nu_t(x) \right)^2.$$

Thus

$$\phi''(t) = \text{Var}_{\nu_t}(h) \geq 0.$$

Since the variance of any random variable is nonnegative, ϕ is convex in t for every direction h . Therefore $\mathcal{T}$ is convex.

Let $\Omega = \mathbb{R}^n$ with Lebesgue measure and consider $f(x) = -\frac{1}{2}x^\top Qx$ with $Q \succ 0$. Since Q is symmetric positive definite, there exists an orthogonal matrix U and a diagonal matrix $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ with $\lambda_i > 0$ such that $Q = U^\top \Lambda U$. Changing the variables $y = Ux$ (so $dx = dy$ and $x^\top Qx = y^\top \Lambda y = \sum_{i=1}^n \lambda_i y_i^2$), the integral factorizes as

$$\int_{\mathbb{R}^n} \exp\left(-\frac{1}{2}x^\top Qx\right) dx = \prod_{i=1}^n \int_{\mathbb{R}} \exp\left(-\frac{1}{2}\lambda_i y_i^2\right) dy_i.$$

Using the Gaussian identity $\int_{\mathbb{R}} e^{-\frac{1}{2}\lambda y^2} dy = \sqrt{2\pi/\lambda}$ for $\lambda > 0$, we obtain

$$\int_{\mathbb{R}^n} \exp\left(-\frac{1}{2}x^\top Qx\right) dx = (2\pi)^{n/2} \prod_{i=1}^n \lambda_i^{-1/2}.$$

Since $\det Q = \prod_{i=1}^n \lambda_i$, this yields

$$\int_{\mathbb{R}^n} \exp\left(-\frac{1}{2}x^\top Qx\right) dx = (2\pi)^{n/2} (\det Q)^{-1/2}.$$

Hence the log-integral-exponential functional $\mathcal{T}(f) = \log \int_{\mathbb{R}^n} e^{f(x)} dx$ reduces to

$$\mathcal{T}(f) = \frac{n}{2} \log(2\pi) - \frac{1}{2} \log \det Q.$$

Thus, in the purely quadratic case, the functional depends only on the curvature matrix Q through its determinant.

More generally, let $f(x) = -\frac{1}{2}x^\top Qx + b^\top x$ with $Q \succ 0$. Completing the square gives

$$-\frac{1}{2}x^\top Qx + b^\top x = -\frac{1}{2}(x - Q^{-1}b)^\top Q(x - Q^{-1}b) + \frac{1}{2}b^\top Q^{-1}b.$$

Therefore

$$\int_{\mathbb{R}^n} e^{f(x)} dx = e^{\frac{1}{2}b^\top Q^{-1}b} \int_{\mathbb{R}^n} \exp\left(-\frac{1}{2}(x - Q^{-1}b)^\top Q(x - Q^{-1}b)\right) dx.$$

By translation invariance of Lebesgue measure, the remaining integral equals the centered Gaussian integral computed above, and hence

$$\int_{\mathbb{R}^n} e^{f(x)} dx = e^{\frac{1}{2}b^\top Q^{-1}b} (2\pi)^{n/2} (\det Q)^{-1/2}.$$

Taking logarithms gives

$$\mathcal{T}(f) = \frac{1}{2}b^\top Q^{-1}b + \frac{n}{2} \log(2\pi) - \frac{1}{2} \log \det Q.$$

Finally, for the scaled functional $\mathcal{T}_\beta(f) = \frac{1}{\beta} \log \int_{\mathbb{R}^n} e^{\beta f(x)} dx$, the same computation with Q replaced by βQ and b by βb yields

$$\int_{\mathbb{R}^n} e^{\beta f(x)} dx = e^{\frac{\beta}{2}b^\top Q^{-1}b} (2\pi)^{n/2} \beta^{-n/2} (\det Q)^{-1/2}.$$

Taking logarithms and dividing by β gives the exact expression

$$\mathcal{T}_\beta(f) = \frac{1}{2}b^\top Q^{-1}b - \frac{1}{2\beta} \log \det Q + \frac{n}{2\beta} \log \left(\frac{2\pi}{\beta} \right).$$

In particular, as $\beta \rightarrow \infty$, the logarithmic terms vanish and $\mathcal{T}_\beta(f) \rightarrow \frac{1}{2}b^\top Q^{-1}b = \sup_x f(x)$, while as $\beta \rightarrow 0$, the dimensional term $\frac{n}{2\beta} \log\left(\frac{2\pi}{\beta}\right)$ dominates and the functional approaches the mean-value regime.

These quantities carry thermodynamic and probabilistic meaning.

Define the partition function

$$Z(\beta) = \int_{\Omega} e^{\beta f(x)} d\rho(x), \quad \beta > 0.$$

The function $\log Z(\beta)$ is the cumulant generating function of f under the base measure ρ . The scaled log-partition functional

$$\mathcal{T}_\beta(f) = \frac{1}{\beta} \log Z(\beta)$$

is the associated free energy at inverse temperature β . Introduce the Gibbs probability measure

$$d\nu_\beta(x) = \frac{e^{\beta f(x)}}{Z(\beta)} d\rho(x).$$

For any integrable function g , we then have $\mathbb{E}_{\nu_\beta}[g] = \frac{\int g(x) e^{\beta f(x)} d\rho(x)}{\int e^{\beta f(x)} d\rho(x)}$. Differentiating under the integral sign gives $Z'(\beta) = \int f(x) e^{\beta f(x)} d\rho(x)$, and therefore

$$\frac{d}{d\beta} \log Z(\beta) = \frac{Z'(\beta)}{Z(\beta)} = \mathbb{E}_{\nu_\beta}[f].$$

Thus the derivative of $\log Z(\beta)$ equals the Gibbs expectation of f . Differentiating once more yields

$$\frac{d^2}{d\beta^2} \log Z(\beta) = \text{Var}_{\nu_\beta}(f) = \mathbb{E}_{\nu_\beta}[f^2] - (\mathbb{E}_{\nu_\beta}[f])^2 \geq 0.$$

Hence $\log Z(\beta)$ is convex in β , with curvature equal to the Gibbs variance of f .

□

4.2 ■ Introduction to Convex Optimization

Convex optimization concerns a special class of optimization problems in which both the objective function and the feasible set possess convexity properties. These properties impose a strong global structure on the problem and lead to powerful theoretical guarantees that sharply distinguish convex programs from general, nonconvex optimization problems.

Throughout this section, we consider optimization problems of the form $\min_{x \in \mathcal{F}} f(x)$, where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function and the feasible set $\mathcal{F} \subseteq \mathbb{R}^n$ is defined by convex inequality constraints. We assume that $\mathcal{F}$ is nonempty.

4.2.1 ■ Convexity of the Feasible Set

Theorem 4.39 (Convexity of the Feasible Set). *Let $\mathcal{F} := \{x \in \mathbb{R}^n : g_i(x) \leq 0, i \in \mathcal{Q}\}$, where each $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function. Then the feasible set $\mathcal{F}$ is a convex set.*

Proof. Fix any index $i \in \mathcal{Q}$. Since g_i is convex, its sublevel set $\{x \in \mathbb{R}^n : g_i(x) \leq 0\}$ is convex by the definition of convexity in (4.3). Indeed, for any x, y in this set and any $\lambda \in [0, 1]$, convexity implies

$$g_i(\lambda x + (1 - \lambda)y) \leq \lambda g_i(x) + (1 - \lambda)g_i(y) \leq 0.$$

Thus $\lambda x + (1 - \lambda)y$ also belongs to the set.

The feasible set $\mathcal{F}$ is the intersection of these convex sublevel sets over all $i \in \mathcal{Q}$. Since arbitrary intersections of convex sets are convex, it follows that $\mathcal{F}$ is convex. $\square$

Convexity of the feasible set ensures that any two feasible points can be joined by a line segment that remains entirely feasible. This geometric property is crucial for both theoretical analysis and the design of optimization algorithms, as it allows global conclusions to be drawn from local behavior.

4.2.2 ■ Local Optimality Implies Global Optimality

One of the most striking and useful consequences of convexity is that local optimality automatically implies global optimality.

Theorem 4.40 (Local Minima Are Global). *For a convex optimization problem, every local minimum is also a global minimum.*

Proof. Let $x^* \in \mathcal{F}$ be a local minimum of f over $\mathcal{F}$, and let $y \in \mathcal{F}$ be arbitrary. Because $\mathcal{F}$ is convex by Theorem 4.39, for any $\lambda \in (0, 1)$ the point $x_\lambda := \lambda x^* + (1 - \lambda)y$ also belongs to $\mathcal{F}$.

By convexity of f (cf. (4.3)), $f(x_\lambda) \leq \lambda f(x^*) + (1 - \lambda)f(y)$. Since x^* is a local minimum, there exists a neighborhood of x^* within which $f(x) \geq f(x^*)$ for all feasible x . For λ sufficiently close to 1, the point x_λ lies in this neighborhood, and hence $f(x_\lambda) \geq f(x^*)$. Combining the two inequalities yields $\lambda f(x^*) + (1 - \lambda)f(y) \geq f(x^*)$, which simplifies to $(1 - \lambda)f(y) \geq (1 - \lambda)f(x^*)$. Since $1 - \lambda > 0$, we conclude that $f(y) \geq f(x^*)$. Because $y \in \mathcal{F}$ was arbitrary, x^* is a global minimum. $\square$

Remark 4.41. This result is one of the defining advantages of convex optimization: there are no spurious local minima. In contrast, nonconvex optimization problems may exhibit many local minima that are not globally optimal.

Theorem 4.42 (Uniqueness of the Minimizer). *Let $\mathcal{F}$ be convex and let f be strictly convex on $\mathcal{F}$. Then f has at most one global minimizer over $\mathcal{F}$.*

Proof. Suppose there exist two distinct minimizers $x_1 \neq x_2$ in $\mathcal{F}$ with $f(x_1) = f(x_2) = \inf_{x \in \mathcal{F}} f(x) =: f^*$. Fix $\lambda \in (0, 1)$. Convexity of $\mathcal{F}$ places $x_\lambda := \lambda x_1 + (1 - \lambda)x_2$ in $\mathcal{F}$, so it competes for the minimum, while strict convexity gives $f(x_\lambda) < \lambda f(x_1) + (1 - \lambda)f(x_2) = f^*$. This contradicts the definition of f^* as the infimum over $\mathcal{F}$. Hence the minimizer is unique. $\square$

4.3 ■ Constraint Qualifications and Slater's Condition

First-order and saddle-point optimality conditions require some regularity of the constraints. Such assumptions are called *constraint qualifications*. For convex programs, Slater's condition is the workhorse: it gives strong duality, guarantees Lagrange multipliers, and lets KKT conditions characterize global optimality.

4.3.1 ■ Slater's Condition

Consider the convex optimization problem (4.1), where f and $g_i, i \in \mathcal{Q}$, are convex functions.

Definition 4.43 (Slater's Condition). *The convex program (4.1) is said to satisfy Slater's condition if there exists a point $\bar{x} \in \mathbb{R}^n$ such that $g_i(\bar{x}) < 0$ for all $i \in \mathcal{Q}$. Such a point $\bar{x}$ is called a strictly feasible point.*

Slater's condition requires that all inequality constraints be satisfied strictly at some point. It implies that the feasible set $\mathcal{F} := \{x \in \mathbb{R}^n : g_i(x) \leq 0, i \in \mathcal{Q}\}$ has a nonempty interior, since a strictly feasible point is interior to $\mathcal{F}$ by continuity. The converse does not hold: for the convex constraint $g \equiv 0$ the feasible set is all of $\mathbb{R}^n$, so its interior is nonempty, yet no point is strictly feasible. Slater's condition is thus a statement about the *representation* of $\mathcal{F}$ by the functions g_i , not only about its geometry. This interiority rules out degenerate configurations in which feasibility is confined to the boundary of the constraint set.

Equality constraints can never satisfy Slater's condition, since they cannot be satisfied strictly. Consequently, Slater's condition is imposed only on inequality constraints. In the presence of equality constraints, strict feasibility is understood relative to the affine subspace defined by those equalities.

4.3.2 ■ Geometric Characterization

Let $x^* \in \mathcal{F}$ be a feasible point and define the active constraint set at x^* by

$$\mathcal{Q}(x^*) := \{i \in \mathcal{Q} : g_i(x^*) = 0\}.$$

Theorem 4.44 (Directional Characterization of Slater's Condition). *Assume that each g_i is differentiable. Slater's condition holds if and only if, for every boundary point $x^* \in \mathcal{F}$, there exists a direction $d \in \mathbb{R}^n$ such that $\nabla g_i(x^*)^\top d < 0, \forall i \in \mathcal{Q}(x^*)$.*

Proof. Assume first that Slater's condition holds, and let $\bar{x}$ be a strictly feasible point. Fix any boundary point $x^* \in \mathcal{F}$ and define $d := \bar{x} - x^*$. For any $i \in \mathcal{Q}(x^*)$, we have $g_i(x^*) = 0$ and $g_i(\bar{x}) < 0$. By convexity of g_i ,

$$g_i(x^* + \alpha d) = g_i((1 - \alpha)x^* + \alpha\bar{x}) \leq (1 - \alpha)g_i(x^*) + \alpha g_i(\bar{x}) = \alpha g_i(\bar{x}) < 0$$

for all sufficiently small $\alpha > 0$. Dividing by α and letting $\alpha \rightarrow 0^+$ yields $\nabla g_i(x^*)^\top d < 0, \forall i \in \mathcal{Q}(x^*)$.

Conversely, suppose that the stated directional condition holds at every boundary point. Fix any boundary point $x^* \in \mathcal{F}$. By assumption, there exists a direction d such that $\nabla g_i(x^*)^\top d < 0 \forall i \in \mathcal{Q}(x^*)$. By differentiability and continuity of each g_i , there exists $\epsilon > 0$ such that

$$g_i(x^* + \alpha d) < 0 \quad \forall i \in \mathcal{Q}(x^*), \quad \forall \alpha \in (0, \epsilon).$$

For constraints $i \notin \mathcal{Q}(x^*)$, we already have $g_i(x^*) < 0$, and by continuity this inequality remains strict for sufficiently small $\alpha > 0$. Hence $x^* + \alpha d$ is strictly feasible, and Slater's condition holds. $\square$

Geometrically, Theorem 4.44 says that every boundary point admits a direction pointing strictly into $\mathcal{F}$; see Figure 4.3.

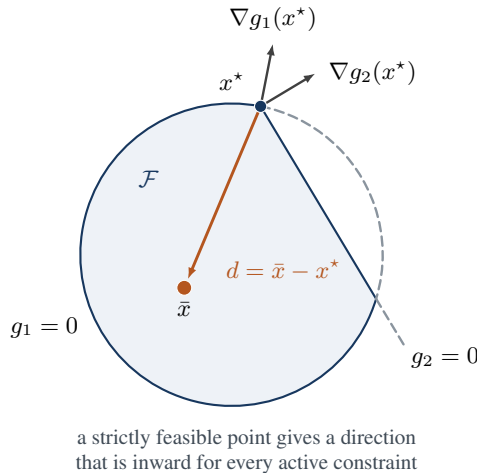


Figure 4.3: Slater's condition supplies a point $\bar{x}$ in the strict interior. From any boundary point x^* , the direction $d = \bar{x} - x^*$ satisfies $\nabla g_i(x^*)^\top d < 0$ for all active constraints, which is the geometry behind the KKT multiplier theorem.

Corollary 4.45. *If, for every boundary point $x^* \in \mathcal{F}$, the set of gradients $\{\nabla g_i(x^*) : i \in \mathcal{Q}(x^*)\}$ is linearly independent, then Slater's condition holds.*

Proof. Fix any boundary point $x^* \in \mathcal{F}$. Linear independence of the gradients implies that there exists no nontrivial vector of coefficients $\lambda_i \geq 0$, not all zero, such that $\sum_{i \in \mathcal{Q}(x^*)} \lambda_i \nabla g_i(x^*) = 0$. By Gordan's theorem of alternatives, this is equivalent to the existence of a vector $d \in \mathbb{R}^n$ satisfying $\nabla g_i(x^*)^\top d < 0 \forall i \in \mathcal{Q}(x^*)$. The conclusion then follows directly from Theorem 4.44. $\square$

4.3.3 ■ Minimal Index Sets and Constraint Reduction

Define the *minimal active index set* $\mathcal{Q}^= := \{i \in \mathcal{Q} : g_i(x) = 0 \text{ for all } x \in \mathcal{F}\}$. Constraints in $\mathcal{Q}^=$ are always active and cannot be satisfied strictly. Define the complementary index set $\mathcal{Q}^< := \mathcal{Q} \setminus \mathcal{Q}^=$.

Proposition 4.46. *The feasible set admits the representation*

$$\mathcal{F} = \{x : g_i(x) \leq 0, i \in \mathcal{Q}^<\} \cap \{x : g_i(x) = 0, i \in \mathcal{Q}^=\}.$$

Proof. By definition of $\mathcal{Q}^=$, each constraint indexed by $i \in \mathcal{Q}^=$ is active at every feasible point and hence acts as an equality constraint. Conversely, all constraints in $\mathcal{Q}^<$ are inactive at least at

one feasible point and therefore may admit strict satisfaction. The stated representation follows by partitioning the constraint set accordingly. $\square$

Proposition 4.47. *Let $F = \{x \in \mathbb{R}^n : g_i(x) \leq 0, i \in \mathcal{Q}\}$ be nonempty, where each g_i is convex. Define $\mathcal{Q}^= = \{i \in \mathcal{Q} : g_i(x) = 0 \text{ for all } x \in F\}$. Then Slater's condition holds if and only if $\mathcal{Q}^= = \emptyset$.*

Proof. Recall that Slater's condition requires the existence of $\bar{x} \in F$ such that $g_i(\bar{x}) < 0$ for all $i \in \mathcal{Q}$.

$(\Leftarrow)$ Assume $\mathcal{Q}^= = \emptyset$. By definition, for every $i \in \mathcal{Q}$ there exists $x^{(i)} \in F$ such that $g_i(x^{(i)}) < 0$. Since F is convex, any convex combination $\bar{x} = \sum_{i \in \mathcal{Q}} \alpha_i x^{(i)}$ with $\alpha_i > 0$ and $\sum_i \alpha_i = 1$ also belongs to F . By convexity of each g_i , $g_i(\bar{x}) \leq \sum_{j \in \mathcal{Q}} \alpha_j g_i(x^{(j)})$. Because $g_i(x^{(i)}) < 0$ and all other terms are ≤ 0 , the right-hand side is strictly negative, hence $g_i(\bar{x}) < 0$ for all i . Thus Slater's condition holds.

$(\Rightarrow)$ Assume Slater's condition holds. Then there exists $\bar{x} \in F$ with $g_i(\bar{x}) < 0$ for all $i \in \mathcal{Q}$. Therefore no index i can satisfy $g_i(x) = 0$ for all $x \in F$. Hence $\mathcal{Q}^= = \emptyset$. $\square$

Remark 4.48. Slater's condition holds if and only if $\mathcal{Q}^= = \emptyset$. In this sense, Slater's condition excludes degenerate inequality constraints that force feasibility to lie entirely on the boundary of the constraint set. Equality constraints are handled separately and do not obstruct Slater's condition when strict feasibility holds relative to their affine hull.

4.4 ■ Basic Convex Program Characterization

We now present a fundamental characterization of optimal solutions to convex programs in terms of saddle points of the Lagrangian. This result provides the conceptual bridge between primal optimality, dual variables, and KKT conditions. Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \end{aligned} \tag{4.6}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are convex functions.

The associated Lagrangian is defined by

$$\mathcal{L}(x, \mu) := f(x) + \sum_{i=1}^m \mu_i g_i(x), \quad \mu \in \mathbb{R}_+^m. \tag{4.7}$$

Theorem 4.49 (Saddle-Point Characterization of Optimality). *Assume that Slater's condition holds for (4.6). A point $x^* \in \mathbb{R}^n$ is a primal optimal solution of (4.6) if and only if there exists a vector of Lagrange multipliers $\mu^* \in \mathbb{R}_+^m$ such that (x^*, μ^*) is a saddle point of the Lagrangian, i.e.,*

$$\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*), \quad \forall x \in \mathbb{R}^n, \forall \mu \in \mathbb{R}_+^m. \tag{4.8}$$

Proof. We prove necessity and sufficiency.

Necessity. Assume x^* is optimal for (4.6). Define two sets in $\mathbb{R}^{m+1}$ (with componentwise order):

$$C_1 := \left\{ y \in \mathbb{R}^{m+1} : \exists x \in \mathbb{R}^n \text{ such that } y \geq \begin{bmatrix} f(x) \\ g_1(x) \\ \vdots \\ g_m(x) \end{bmatrix} \right\},$$

$$C_2 := \{ y \in \mathbb{R}^{m+1} : y_0 < f(x^*), y_i \leq 0 \ (i = 1, \dots, m) \}.$$

The strict inequality in the first coordinate is essential: with $y_0 \leq f(x^*)$ the two sets would share the point $(f(x^*), 0, \dots, 0)$ whenever x^* is feasible, and no separation argument could be applied to them.

The set C_2 is convex, being the intersection of an open half-space with finitely many closed ones, and it is nonempty with nonempty interior. The set C_1 is convex because convex combinations preserve componentwise domination and f, g_i are convex.

The two sets are disjoint. Indeed, a point $y \in C_1 \cap C_2$ would satisfy $y_0 < f(x^*)$ and $y_i \leq 0$ while dominating $(f(x), g_1(x), \dots, g_m(x))$ for some x ; that x would then obey $f(x) \leq y_0 < f(x^*)$ and $g_i(x) \leq y_i \leq 0$, i.e. it would be feasible with strictly smaller objective value, contradicting optimality of x^* .

Two disjoint nonempty convex sets in $\mathbb{R}^{m+1}$ can be separated, so there exist $(a_0, a) \neq 0$ and α such that

$$a_0 y_0 + a^\top y \geq \alpha \quad \forall (y_0, y) \in C_1, \quad a_0 y_0 + a^\top y \leq \alpha \quad \forall (y_0, y) \in C_2.$$

The point $(f(x^*), 0, \dots, 0)$ lies in the closure of C_2 , so letting $y \rightarrow (f(x^*), 0, \dots, 0)$ within C_2 in the second inequality gives $a_0 f(x^*) \leq \alpha$. Evaluating the first inequality at $(f(x), g_1(x), \dots, g_m(x)) \in C_1$ yields

$$a_0 f(x) + \sum_{i=1}^m a_i g_i(x) \geq \alpha, \quad \forall x \in \mathbb{R}^n.$$

Combining,

$$a_0 f(x) + \sum_{i=1}^m a_i g_i(x) \geq a_0 f(x^*), \quad \forall x. \quad (4.9)$$

The upward closed structure of C_1 forces $a_0 \geq 0$ and $a_i \geq 0$ for every i : increasing any coordinate of a point of C_1 keeps it in C_1 , so a negative coefficient would drive the left-hand side of the first inequality to $-\infty$.

Slater's condition now rules out $a_0 = 0$. Suppose $a_0 = 0$; then $(a_1, \dots, a_m) \neq 0$ and (4.9) reads $\sum_{i=1}^m a_i g_i(x) \geq 0$ for every x . Evaluating this at a strictly feasible point $\bar{x}$, where $g_i(\bar{x}) < 0$ for all i , gives $\sum_i a_i g_i(\bar{x}) < 0$, since the a_i are nonnegative and not all zero. This contradiction shows $a_0 > 0$. We may therefore define $\mu_i^* := \frac{a_i}{a_0} \geq 0$. Dividing (4.9) by a_0 gives

$$f(x^*) \leq f(x) + \sum_{i=1}^m \mu_i^* g_i(x) = \mathcal{L}(x, \mu^*), \quad \forall x.$$

Evaluating this inequality at $x = x^*$ yields

$$f(x^*) \leq f(x^*) + \sum_{i=1}^m \mu_i^* g_i(x^*).$$

Since $g_i(x^*) \leq 0$ and $\mu_i^* \geq 0$, the sum $\sum_i \mu_i^* g_i(x^*)$ is nonpositive. Therefore equality must hold: $\sum_{i=1}^m \mu_i^* g_i(x^*) = 0$. This is the complementary slackness condition, and it implies $\mathcal{L}(x^*, \mu^*) = f(x^*)$. Consequently, $\mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*) \forall x$, and for any $\mu \geq 0$,

$$\mathcal{L}(x^*, \mu) = f(x^*) + \sum_{i=1}^m \mu_i g_i(x^*) \leq f(x^*) = \mathcal{L}(x^*, \mu^*).$$

Thus

$$\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*), \quad \forall x, \forall \mu \geq 0,$$

so (x^*, μ^*) is a saddle point of the Lagrangian. $\square$

Sufficiency. Conversely, assume there exists $\mu^* \geq 0$ such that the saddle condition (4.8) holds:

$$\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*), \quad \forall x, \forall \mu \geq 0.$$

We first show that x^* is feasible. For any $\mu \geq 0$,

$$\mathcal{L}(x^*, \mu) = f(x^*) + \sum_{i=1}^m \mu_i g_i(x^*) \leq \mathcal{L}(x^*, \mu^*).$$

Since $\mathcal{L}(x^*, \mu^*)$ is finite and the inequality holds for all $\mu \geq 0$, the only possibility is that $g_i(x^*) \leq 0 \forall i$. Otherwise, if $g_j(x^*) > 0$ for some j , choosing $\mu_j \rightarrow +\infty$ would violate the inequality. Hence x^* is feasible. Now let x be any feasible point. Since $g_i(x) \leq 0$ and $\mu_i^* \geq 0$, we have

$$\mathcal{L}(x, \mu^*) = f(x) + \sum_{i=1}^m \mu_i^* g_i(x) \leq f(x).$$

Using the saddle inequality, $f(x^*) = \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*) \leq f(x)$. Therefore $f(x^*) \leq f(x)$ for every feasible x , so x^* minimizes f over the feasible set. Hence x^* is optimal.

$\square$

Remark 4.50. Note that Theorem 4.49 does not assume differentiability of f or g_i . Only convexity and Slater's condition are used, together with the separating hyperplane theorem in $\mathbb{R}^{m+1}$. Differentiability becomes relevant later when one derives first-order KKT conditions in terms of gradients.

Corollary 4.51 (Complementarity Conditions). Assume Slater's condition holds for the convex program (4.6). If (x^*, μ^*) is a saddle point of the Lagrangian, i.e., (4.8) holds, then the following complementarity conditions are satisfied:

$$\mu_i^* g_i(x^*) = 0, \quad i = 1, \dots, m. \quad (4.10)$$

Proof. Since (x^*, μ^*) is a saddle point, (4.8) implies $\mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*)$ for all x , and $\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*)$ for all $\mu \geq 0$.

From the second inequality it follows, as shown in the sufficiency part of the theorem, that x^* is feasible; hence $g_i(x^*) \leq 0$ for all i . Taking $x = x^*$ in the first inequality yields $\mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x^*, \mu)$, so equality holds. Consequently $\mathcal{L}(x^*, \mu^*) = f(x^*)$. By definition of the Lagrangian, $\mathcal{L}(x^*, \mu^*) = f(x^*) + \sum_{i=1}^m \mu_i^* g_i(x^*)$, and therefore $f(x^*) + \sum_{i=1}^m \mu_i^* g_i(x^*) = f(x^*)$. This implies $\sum_{i=1}^m \mu_i^* g_i(x^*) = 0$.

Since each $\mu_i^* \geq 0$ and each $g_i(x^*) \leq 0$, every term $\mu_i^* g_i(x^*)$ is nonpositive. A sum of nonpositive terms can equal zero only if each term vanishes individually, so $\mu_i^* g_i(x^*) = 0$ for all i . $\square$

Remark 4.52. The complementarity conditions (4.10) express a precise duality balance: each constraint is either inactive ($g_i(x^*) < 0$) with zero multiplier, or active ($g_i(x^*) = 0$) with possibly nonzero multiplier. Importantly, this result follows purely from convexity and the saddle-point property and does not require differentiability of the objective or constraint functions.

Theorem 4.53 (Karush–Kuhn–Tucker (KKT) Conditions). *Assume that the objective function f and the constraint functions g_i , $i = 1, \dots, m$, are continuously differentiable. Consider the convex optimization problem (4.6), and suppose that Slater’s condition holds. Then a feasible point $x^* \in \mathbb{R}^n$ is optimal if and only if there exists a vector of Lagrange multipliers $\mu^* = (\mu_1^*, \dots, \mu_m^*) \in \mathbb{R}_+^m$ such that*

$$(\text{Stationarity}) \quad \nabla f(x^*) + \sum_{i=1}^m \mu_i^* \nabla g_i(x^*) = 0, \quad (4.11)$$

$$(\text{Primal feasibility}) \quad g_i(x^*) \leq 0, \quad i = 1, \dots, m, \quad (4.12)$$

$$(\text{Dual feasibility}) \quad \mu_i^* \geq 0, \quad i = 1, \dots, m, \quad (4.13)$$

$$(\text{Complementary slackness}) \quad \mu_i^* g_i(x^*) = 0, \quad i = 1, \dots, m. \quad (4.14)$$

It is worth pausing on what the four conditions say together, because each plays a distinct role. Stationarity is a *force balance*: at the optimum the gradient of the objective is cancelled by a nonnegative combination of the active constraint gradients, so no feasible direction decreases f . Primal feasibility keeps x^* inside the feasible set; dual feasibility, $\mu_i^* \geq 0$, records that each constraint can only *push back*, never pull, since it is a one-sided inequality. Complementary slackness is the switch that couples the two: whenever a constraint is inactive ($g_i(x^*) < 0$) its multiplier must vanish, and only active constraints ($g_i(x^*) = 0$) may carry weight. The multipliers μ_i^* are thus more than bookkeeping, as the sensitivity analysis later in this chapter makes precise, μ_i^* is the *shadow price* of the i -th constraint, the rate at which the optimal value would improve if that constraint were relaxed. For convex programs under Slater’s condition these algebraic conditions are not merely necessary but sufficient, so solving the KKT system is equivalent to solving the optimization problem.

Proof. We prove necessity and sufficiency.

Necessity. Assume x^* is optimal for (4.6). Since Slater’s condition holds, Theorem 4.49 implies that there exists $\mu^* \in \mathbb{R}_+^m$ such that

$$\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*), \quad \forall x \in \mathbb{R}^n, \forall \mu \in \mathbb{R}_+^m, \quad (4.15)$$

where $\mathcal{L}(x, \mu) = f(x) + \sum_{i=1}^m \mu_i g_i(x)$. Primal feasibility (4.12) holds by assumption (x^* is feasible). Dual feasibility (4.13) holds by construction ($\mu^* \geq 0$).

Complementary slackness (4.14) follows from (4.15) as in Corollary 4.51. Feasibility gives $g_i(x^*) \leq 0$, while $\mu^* \geq 0$ gives $\sum_{i=1}^m \mu_i^* g_i(x^*) \leq 0$. Taking $\mu = 0$ in the left saddle inequality gives $f(x^*) \leq \mathcal{L}(x^*, \mu^*)$, so

$$0 \leq \sum_{i=1}^m \mu_i^* g_i(x^*) \leq 0.$$

Each term is nonpositive, hence each term must vanish: $\mu_i^* g_i(x^*) = 0$ for all i .

It remains to prove stationarity (4.11). From the right inequality in (4.15), we have $\mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*)$ for all $x \in \mathbb{R}^n$, so x^* is a *global minimizer* of the differentiable function $x \mapsto \mathcal{L}(x, \mu^*)$ over $\mathbb{R}^n$.

We now show that any global minimizer of a differentiable function on $\mathbb{R}^n$ must satisfy zero gradient. Consider the directional function $\varphi(t) := \mathcal{L}(x^* + td, \mu^*)$, $t \in \mathbb{R}$, for an arbitrary direction $d \in \mathbb{R}^n$. Since x^* minimizes $\mathcal{L}(\cdot, \mu^*)$ globally, $t = 0$ minimizes φ over $\mathbb{R}$, hence $\varphi'(0) = 0$. By the chain rule, $\varphi'(0) = \nabla_x \mathcal{L}(x^*, \mu^*)^\top d$. Since d is arbitrary, this implies $\nabla_x \mathcal{L}(x^*, \mu^*) = 0$, i.e., $\nabla f(x^*) + \sum_{i=1}^m \mu_i^* \nabla g_i(x^*) = 0$, which is (4.11). This completes the necessity proof.

Sufficiency. Conversely, assume that there exist (x^*, μ^*) satisfying (4.11)–(4.14). We show that x^* is optimal. Fix any $x \in \mathbb{R}^n$. By convexity of f and differentiability, the first-order condition for convexity implies $f(x) \geq f(x^*) + \nabla f(x^*)^\top (x - x^*)$. Similarly, for each i , $g_i(x) \geq g_i(x^*) + \nabla g_i(x^*)^\top (x - x^*)$. Multiply the latter inequalities by $\mu_i^* \geq 0$ and sum over i :

$$\sum_{i=1}^m \mu_i^* g_i(x) \geq \sum_{i=1}^m \mu_i^* g_i(x^*) + \left(\sum_{i=1}^m \mu_i^* \nabla g_i(x^*) \right)^\top (x - x^*).$$

Adding the inequality for $f(x)$ yields

$$\begin{aligned} \mathcal{L}(x, \mu^*) &= f(x) + \sum_{i=1}^m \mu_i^* g_i(x) \\ &\geq f(x^*) + \sum_{i=1}^m \mu_i^* g_i(x^*) + \left(\nabla f(x^*) + \sum_{i=1}^m \mu_i^* \nabla g_i(x^*) \right)^\top (x - x^*). \end{aligned}$$

By stationarity (4.11), the last inner-product term is zero, so $\mathcal{L}(x, \mu^*) \geq \mathcal{L}(x^*, \mu^*)$ for all $x \in \mathbb{R}^n$, which is the right saddle inequality.

To prove the left saddle inequality, fix any $\mu \geq 0$. Since x^* is feasible, $g_i(x^*) \leq 0$, we have

$$\mathcal{L}(x^*, \mu) = f(x^*) + \sum_{i=1}^m \mu_i g_i(x^*) \leq f(x^*).$$

On the other hand, complementary slackness (4.14) implies $\sum_i \mu_i^* g_i(x^*) = 0$, hence

$$\mathcal{L}(x^*, \mu^*) = f(x^*) + \sum_{i=1}^m \mu_i^* g_i(x^*) = f(x^*).$$

Therefore, for all $\mu \geq 0$, $\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*)$. Combining the two parts gives

$$\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*), \quad \forall x \in \mathbb{R}^n, \forall \mu \geq 0,$$

so (x^*, μ^*) is a saddle point of the Lagrangian. By the saddle-point characterization (Theorem 4.49), x^* is optimal for (4.6). This completes the proof. $\square$

Remark 4.54. The saddle-point characterization (Theorem 4.49) and complementary slackness can be derived without differentiability, using only convexity, Slater's condition, and separation. Differentiability is invoked in Theorem 4.53 only to express stationarity in the gradient form (4.11) and to use the first-order characterization of convex functions in the sufficiency direction.

Remark 4.55 (Stationarity over the active set). Let $\mathcal{Q}(x^*) := \{i \in \{1, \dots, m\} : g_i(x^*) = 0\}$ denote the set of active constraints at x^* . By the complementarity conditions, $\mu_i^* g_i(x^*) = 0$ for all i , so in particular $\mu_i^* = 0$ whenever $g_i(x^*) < 0$. Hence the stationarity condition

$$\nabla f(x^*) + \sum_{i=1}^m \mu_i^* \nabla g_i(x^*) = 0$$

can equivalently be written as

$$\nabla f(x^*) + \sum_{i \in \mathcal{Q}(x^*)} \mu_i^* \nabla g_i(x^*) = 0.$$

Thus only gradients of active constraints contribute to the stationarity condition. Inactive constraints play no role in the first-order balance.

Example 4.56 Quadratic Program with a Single Inequality Constraint.

Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & \frac{1}{2}(x_1^2 + x_2^2 + x_3^2) \\ \text{subject to} \quad & x_1 + x_2 + x_3 \leq -3. \end{aligned} \tag{4.16}$$

The objective function is strictly convex, with Hessian equal to the identity matrix. The constraint function $g(x) := x_1 + x_2 + x_3 + 3$ is affine and hence convex. The point $\bar{x} = (-2, -1, -1)$ satisfies $g(\bar{x}) < 0$, so Slater's condition holds. Since the problem is convex and Slater's condition holds, optimality is fully characterized by the Karush–Kuhn–Tucker conditions (4.11)–(4.14).

The Lagrangian is

$$\mathcal{L}(x, \mu) = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2) + \mu(x_1 + x_2 + x_3 + 3), \quad \mu \geq 0.$$

Stationarity (4.11) yields

$$\nabla_x \mathcal{L}(x^*, \mu^*) = \begin{bmatrix} x_1^* \\ x_2^* \\ x_3^* \end{bmatrix} + \mu^* \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0,$$

which implies $x_1^* = x_2^* = x_3^* = -\mu^*$. *Primal feasibility* (4.12) requires $x_1^* + x_2^* + x_3^* \leq -3$. *Dual feasibility* (4.13) requires $\mu^* \geq 0$. *Complementary slackness* (4.14) gives $\mu^*(x_1^* + x_2^* + x_3^* + 3) = 0$.

Substituting the stationarity condition into the constraint yields $-3\mu^* \leq -3$, so $\mu^* \geq 1$. Complementary slackness forces the constraint to be active, hence $x_1^* + x_2^* + x_3^* = -3$. Using $x_i^* = -\mu^*$, we obtain $-3\mu^* = -3$, and therefore $\mu^* = 1$, $x_1^* = x_2^* = x_3^* = -1$. The unique optimal solution of (4.16) is $x^* = (-1, -1, -1)$, $\mu^* = 1$. Uniqueness follows from strict convexity of the objective function. $\square$

Example 4.57 Exponential Program with Inequality Constraints.

Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & e^{x_1 - x_2} \\ \text{subject to} \quad & e^{x_1} + e^{x_2} \leq 20, \\ & x_1 \geq 0. \end{aligned} \tag{4.17}$$

The objective function $f(x) = e^{x_1 - x_2}$ is convex, since it is the exponential of an affine function. The constraint functions $g_1(x) := e^{x_1} + e^{x_2} - 20$ and $g_2(x) := -x_1$ are convex, and the point $\bar{x} = (0, 0)$ satisfies $g_1(\bar{x}) = 2 - 20 < 0$ and $g_2(\bar{x}) = 0$, so Slater's condition holds. Since the problem is convex and Slater's condition holds, the KKT conditions are necessary and sufficient for optimality. The Lagrangian is

$$\mathcal{L}(x, \mu) = e^{x_1 - x_2} + \mu_1(e^{x_1} + e^{x_2} - 20) + \mu_2(-x_1), \quad \mu_1, \mu_2 \geq 0.$$

Stationarity yields

$$\nabla_x \mathcal{L}(x^*, \mu^*) = \begin{bmatrix} e^{x_1^* - x_2^*} + \mu_1^* e^{x_1^*} - \mu_2^* \\ -e^{x_1^* - x_2^*} + \mu_1^* e^{x_2^*} \end{bmatrix} = 0.$$

Primal feasibility requires $e^{x_1^*} + e^{x_2^*} \leq 20$, $x_1^* \geq 0$. *Dual feasibility* requires $\mu_1^*, \mu_2^* \geq 0$. *Complementary slackness* gives $\mu_1^*(e^{x_1^*} + e^{x_2^*} - 20) = 0$ and $\mu_2^* x_1^* = 0$.

If both constraints are inactive, then $\mu_1^* = \mu_2^* = 0$, which would imply $e^{x_1^* - x_2^*} = 0$, a contradiction. If only the exponential constraint is active, stationarity again leads to a contradiction. Hence both constraints must be active, yielding $x_1^* = 0$, $e^{x_1^*} + e^{x_2^*} = 20$. Therefore $x_2^* = \ln 19$. The optimal solution of (4.17) is $x^* = (0, \ln 19)$. $\square$

Example 4.58 Quadratic Program with Two Convex Inequality Constraints.

Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & x_1^2 + x_2^2 - 4x_1 - 4x_2 \\ \text{subject to} \quad & x_1^2 - x_2 \leq 0, \\ & x_1 + x_2 \leq 2. \end{aligned} \tag{4.18}$$

The objective function is strictly convex, with Hessian equal to $2I$. The constraint functions $g_1(x) := x_1^2 - x_2$ and $g_2(x) := x_1 + x_2 - 2$ are convex. Slater's condition holds since there exist feasible points with strict inequalities.

The Lagrangian is

$$\mathcal{L}(x, \mu) = x_1^2 + x_2^2 - 4x_1 - 4x_2 + \mu_1(x_1^2 - x_2) + \mu_2(x_1 + x_2 - 2), \quad \mu_1, \mu_2 \geq 0.$$

Stationarity yields

$$\nabla_x \mathcal{L}(x^*, \mu^*) = \begin{bmatrix} 2x_1^* - 4 + 2\mu_1^* x_1^* + \mu_2^* \\ 2x_2^* - 4 - \mu_1^* + \mu_2^* \end{bmatrix} = 0.$$

Primal feasibility requires $x_1^{*2} - x_2^* \leq 0$, $x_1^* + x_2^* \leq 2$. *Dual feasibility* requires $\mu_1^*, \mu_2^* \geq 0$. *Complementary slackness* gives $\mu_1^*(x_1^{*2} - x_2^*) = 0$ and $\mu_2^*(x_1^* + x_2^* - 2) = 0$.

If neither constraint is active, then $x^* = (2, 2)$, which is infeasible. Activating only the nonlinear constraint leads to a solution violating $x_1 + x_2 \leq 2$. Taking $\mu_1^* = 0$ and activating the linear constraint yields $x_1^* = x_2^* = 1$, $\mu_2^* = 2$, which satisfies all KKT conditions. Note that at $x^* = (1, 1)$ the nonlinear constraint is in fact also tight, $g_1(x^*) = 1 - 1 = 0$, so both constraints hold with equality; this is a *weakly active* (degenerate) case, admissible because complementary slackness requires only $\mu_1^* g_1(x^*) = 0$, which holds here through $\mu_1^* = 0$. The unique optimal solution of (4.18) is $x^* = (1, 1)$, $\mu_1^* = 0$, $\mu_2^* = 2$. Uniqueness follows from strict convexity of the objective function. $\square$

4.4.1 ■ Sufficiency of KKT Conditions

The previous theorems used Slater's condition to guarantee that KKT points *exist* at an optimum. It is important to separate that existence question from the *sufficiency* of the conditions: once a KKT point is in hand, convexity alone certifies its optimality, whether or not Slater's condition holds. We first make this concrete with an example in which Slater fails yet the KKT conditions still pinpoint the solution, and then prove the general sufficiency statement.

To illustrate, consider the convex problem

$$\min_{x \in \mathbb{R}^2} f(x) = x_1^2 + x_2^2 \quad \text{subject to} \quad g(x) = (x_1 - x_2)^2 \leq 0.$$

The constraint $(x_1 - x_2)^2 \leq 0$ implies $x_1 = x_2$. Hence the feasible set is the affine subspace $\mathcal{F} = \{(t, t) : t \in \mathbb{R}\}$, a one-dimensional line in $\mathbb{R}^2$. Restricting the objective to $\mathcal{F}$ yields $f(t, t) = 2t^2$. The unique minimizer is therefore $x^* = (0, 0)$.

There exists no strictly feasible point because $(x_1 - x_2)^2 < 0$ has no solution. Thus Slater's condition does not hold. The Lagrangian is

$$\mathcal{L}(x, \mu) = x_1^2 + x_2^2 + \mu(x_1 - x_2)^2, \quad \mu \geq 0.$$

The gradients are

$$\nabla f(x) = \begin{pmatrix} 2x_1 \\ 2x_2 \end{pmatrix}, \quad \nabla g(x) = \begin{pmatrix} 2(x_1 - x_2) \\ -2(x_1 - x_2) \end{pmatrix}.$$

Stationarity requires

$$\nabla f(x^*) + \mu^* \nabla g(x^*) = 0.$$

At $x^* = (0, 0)$ we have $\nabla f(x^*) = 0$ and $\nabla g(x^*) = 0$, so the stationarity condition holds for every $\mu^* \geq 0$. Primal feasibility holds since $g(x^*) = 0$. Dual feasibility requires $\mu^* \geq 0$. Complementary holds because $\mu^* g(x^*) = 0$. Hence (x^*, μ^*) satisfies the KKT conditions for every $\mu^* \geq 0$.

Although Slater's condition fails because the feasible region lies entirely on the boundary of the constraint, the KKT conditions hold and remain sufficient for optimality. This example shows that Slater's condition is a regularity condition ensuring necessity of the KKT conditions and strong duality, but it is not required for their sufficiency once a saddle point exists.

Theorem 4.59 (Sufficiency of KKT Conditions). *Consider the convex optimization problem*

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i \in \mathcal{Q}, \end{aligned} \tag{4.19}$$

where f and g_i , $i \in \mathcal{Q}$, are convex and differentiable. If there exist a feasible point x^* and multipliers $\mu^* \geq 0$ such that the KKT conditions hold at x^* , then x^* is a global minimizer of (4.19).

Proof. The stationarity condition of the KKT system yields

$$\nabla f(x^*) + \sum_{i \in \mathcal{Q}(x^*)} \mu_i^* \nabla g_i(x^*) = 0, \tag{4.20}$$

where $\mathcal{Q}(x^*) := \{i \in \mathcal{Q} : g_i(x^*) = 0\}$ denotes the set of active constraints at x^* .

Let x be any feasible point. Taking the inner product of (4.20) with $(x - x^*)$ gives

$$0 = \nabla f(x^*)^\top (x - x^*) + \sum_{i \in \mathcal{Q}(x^*)} \mu_i^* \nabla g_i(x^*)^\top (x - x^*).$$

By convexity of f and g_i , the first-order inequality implies $\nabla f(x^*)^\top (x - x^*) \leq f(x) - f(x^*)$, and, for each $i \in \mathcal{Q}(x^*)$, $\nabla g_i(x^*)^\top (x - x^*) \leq g_i(x) - g_i(x^*)$. Substituting these inequalities yields

$$0 \leq f(x) - f(x^*) + \sum_{i \in \mathcal{Q}(x^*)} \mu_i^* (g_i(x) - g_i(x^*)).$$

Since $g_i(x^*) = 0$ for all active constraints and $g_i(x) \leq 0$ for all feasible x , each term in the summation is nonpositive. Consequently, $0 \leq f(x) - f(x^*)$, which implies $f(x^*) \leq f(x)$ for all feasible x .

Therefore, x^* is a global optimal solution of (4.19). $\square$

4.4.2 ■ Case Study: Water-Filling and the KKT Conditions

We illustrate the Karush–Kuhn–Tucker (KKT) conditions (Theorem 4.53, conditions (4.11)–(4.14)) through the classical *water-filling problem*, which arises in power allocation for parallel channels and, more broadly, in any resource allocation problem with a single coupling budget constraint. The canonical instance is the capacity of a set of parallel Gaussian channels: one must split a fixed power budget across channels of differing noise levels to maximize total rate. The optimal policy turns out to have a strikingly physical form, pour power like water into a landscape whose floor heights are set by the noise levels, so that the good channels fill up to a common surface and the hopelessly noisy ones are left dry. Our aim is to show how this picture falls out of the KKT conditions rather than being imposed by hand: stationarity produces the common level, and complementary slackness decides which channels stay dry.

Let $a_i > 0$, $i = 1, \dots, n$, be given constants. Consider

$$\begin{aligned} & \max_{x \in \mathbb{R}^n} \quad \sum_{i=1}^n \log(x_i + a_i) \\ & \text{subject to} \quad \sum_{i=1}^n x_i \leq P, \\ & \quad \quad \quad x_i \geq 0, \quad i = 1, \dots, n, \end{aligned} \tag{4.21}$$

where $P > 0$ denotes a fixed total resource budget.

The objective function is strictly concave on $\mathbb{R}_+^n$ since each $\log(x_i + a_i)$ is strictly concave and strictly increasing in x_i . The feasible set is convex because it is defined by linear inequalities. Hence, problem (4.21) is a convex optimization problem (equivalently, a concave maximization problem).

Slater's condition holds. Indeed, the point $x_i = \frac{P}{2n}$, $i = 1, \dots, n$, satisfies

$$\sum_{i=1}^n x_i = \frac{P}{2} < P, \quad x_i > 0 \quad \forall i.$$

Thus all inequality constraints are satisfied strictly. By Slater's theorem, the KKT conditions are both necessary and sufficient for optimality.

Introduce a multiplier $\mu \geq 0$ associated with the total resource constraint and multipliers $\nu_i \geq 0$, $i = 1, \dots, n$, associated with the nonnegativity constraints. The Lagrangian is

$$\mathcal{L}(x, \mu, \nu) = \sum_{i=1}^n \log(x_i + a_i) - \mu \left(\sum_{i=1}^n x_i - P \right) + \sum_{i=1}^n \nu_i x_i.$$

The nonnegativity constraint enters as $-x_i \leq 0$, so for a maximization problem its multiplier contributes $-\nu_i(-x_i) = +\nu_i x_i$; the resulting sign on ν_i is what decides which channels are left dry.

An optimal primal–dual triple (x^*, μ^*, ν^*) must satisfy:

Primal feasibility:

$$\sum_{i=1}^n x_i^* \leq P, \quad x_i^* \geq 0.$$

Dual feasibility: $\mu^* \geq 0$ and $\nu_i^* \geq 0$.

Stationarity:

$$\frac{1}{x_i^* + a_i} - \mu^* + \nu_i^* = 0, \quad i = 1, \dots, n. \tag{4.22}$$

Complementary slackness:

$$\nu_i^* x_i^* = 0, \quad i = 1, \dots, n, \quad \mu^* \left(\sum_{i=1}^n x_i^* - P \right) = 0. \tag{4.23}$$

The resource constraint is active at the optimum. Since each $\log(x_i + a_i)$ is strictly increasing in x_i , any feasible point with $\sum_{i=1}^n x_i < P$ can be strictly improved by increasing some x_i while remaining feasible. Therefore, $\sum_{i=1}^n x_i^* = P$, and by complementary slackness we must have $\mu^* > 0$.

It remains to analyse the components one at a time. Fix $i \in \{1, \dots, n\}$.

Case 1: $x_i^* > 0$. Complementary slackness implies $\nu_i^* = 0$. From stationarity, $\frac{1}{x_i^* + a_i} = \mu^*$, so $x_i^* = \frac{1}{\mu^*} - a_i$. Since $x_i^* > 0$, this requires $\frac{1}{\mu^*} > a_i$.

Case 2: $x_i^* = 0$. Stationarity gives $\nu_i^* = \mu^* - \frac{1}{a_i}$. Dual feasibility $\nu_i^* \geq 0$ implies

$$\mu^* \geq \frac{1}{a_i}, \quad \text{equivalently} \quad \frac{1}{\mu^*} \leq a_i.$$

Combining the two cases yields the equivalence

$$x_i^* > 0 \iff \frac{1}{\mu^*} > a_i,$$

and hence

$$x_i^* = \max \left\{ 0, \frac{1}{\mu^*} - a_i \right\}, \quad i = 1, \dots, n. \quad (4.24)$$

The multiplier μ^* is pinned down by the budget. Substituting (4.24) into the active resource constraint gives

$$\sum_{i=1}^n \max \left\{ 0, \frac{1}{\mu^*} - a_i \right\} = P.$$

The left-hand side is continuous and strictly decreasing in $\mu^* > 0$, with limit $+\infty$ as $\mu^* \downarrow 0$ and limit 0 as $\mu^* \rightarrow \infty$. Hence a unique μ^* exists.

The solution now has a transparent physical reading. Each constant a_i represents a baseline height, and $1/\mu^*$ represents a common water level. Resources are allocated so that all active components reach the same level $1/\mu^*$, while components with $a_i \geq 1/\mu^*$ receive zero allocation. The active set is determined precisely by complementary slackness.

Theorem 4.60 (Optimality and Uniqueness of the Water-Filling Solution). *Consider the water-filling problem (4.21). Let x^* be defined by (4.24), where $\mu^* > 0$ satisfies $\sum_{i=1}^n x_i^* = P$. Then x^* is the unique global optimal solution of problem (4.21).*

Figure 4.4 illustrates the structure of the KKT conditions (4.22)–(4.23). The gray bars represent the baseline levels a_i , while the horizontal blue line represents the common level $1/\mu^*$. For indices with $x_i^* > 0$, complementary slackness (4.23) implies $\nu_i^* = 0$, and stationarity (4.22) gives $\frac{1}{x_i^* + a_i} = \mu^*$, i.e., $x_i^* + a_i = 1/\mu^*$. Thus all active components are filled to the same height. For indices with $x_i^* = 0$, dual feasibility requires $\frac{1}{\mu^*} \leq a_i$, so these bars rise to or above the water level and receive no allocation. The total volume of water equals P by the active resource constraint, which determines μ^* uniquely through (4.24). The figure therefore visualizes stationarity as level equalization, complementary slackness as the active-set condition, and primal feasibility as conservation of total water.

4.5 ■ Duality and the Dual Function

Duality provides an alternative perspective on constrained optimization: instead of forbidding constraint violations outright, we *price* them. Each constraint $g_i(x) \leq 0$ is attached a nonnegative price μ_i , and violations are charged rather than banned, turning the constrained problem into a family of unconstrained ones indexed by the prices. Minimizing this penalized objective over x gives, for every price vector, a number that can never exceed the true optimum, a valid lower bound. The dual problem then searches for the prices that make this bound as tight as possible.

Two features make this construction powerful. First, the best bound is obtained by solving a *concave* maximization problem, regardless of whether the primal is convex, so the dual is

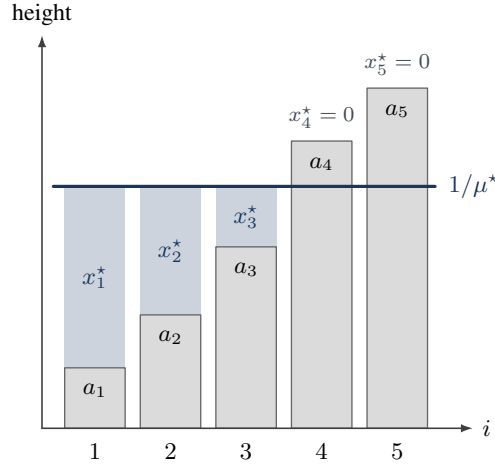


Figure 4.4: Water-filling interpretation of the KKT solution. Gray bars represent baseline levels a_i . The horizontal blue line is the water level $1/\mu^*$. Water (blue region) is allocated only to indices with $a_i < 1/\mu^*$, corresponding to $x_i^* > 0$. Bars above the water level receive no allocation. Complementary slackness determines the active set.

always tractable in principle and always furnishes a certificate of optimality. Second, the optimal prices are exactly the Lagrange multipliers of the KKT system and carry an economic meaning as marginal values, or shadow prices, of the resources encoded by the constraints. These ideas underpin optimality conditions, sensitivity analysis, and a large family of decomposition and dual-ascent algorithms.

Throughout this section, consider the convex optimization problem (4.1) and its associated Lagrangian

$$\mathcal{L}(x, \mu) = f(x) + \sum_{i \in \mathcal{Q}} \mu_i g_i(x), \quad \mu \geq 0,$$

where $\mu \in \mathbb{R}_+^{|\mathcal{Q}|}$ denotes the vector of Lagrange multipliers.

4.5.1 ■ The Dual Function and Dual Problem

Definition 4.61 (Dual Function). The dual function associated with the convex program (4.1) is defined by

$$d(\mu) := \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu), \quad \mu \geq 0, \quad (4.25)$$

where $\mathcal{L}(x, \mu)$ denotes the Lagrangian of the primal problem.

For each fixed $\mu \geq 0$, the value $d(\mu)$ provides a lower bound on the optimal value of the primal problem, i.e., $d(\mu) \leq f^*$, where f^* denotes the optimal primal value. This property is known as *weak duality*.

Definition 4.62 (Dual Problem). The dual problem corresponding to (4.1) is defined as

$$\sup_{\mu \geq 0} d(\mu). \quad (4.26)$$

The optimal value of the dual problem is denoted by $d^* := \sup_{\mu \geq 0} d(\mu)$.

4.5.2 ■ Properties of Dual Functions

Theorem 4.63 (Weak Duality). *For any dual feasible point $\mu \geq 0$, the dual function satisfies $d(\mu) \leq f^*$, where f^* is the optimal value of the primal problem (4.1).*

Proof. Let $\mu \geq 0$ and let x^* be any primal feasible point, i.e., $g_i(x^*) \leq 0$ for all $i \in \mathcal{Q}$. Then

$$\mathcal{L}(x^*, \mu) = f(x^*) + \sum_{i \in \mathcal{Q}} \mu_i g_i(x^*) \leq f(x^*),$$

since $\mu_i \geq 0$ and $g_i(x^*) \leq 0$. Taking the infimum over x yields

$$d(\mu) = \inf_x \mathcal{L}(x, \mu) \leq \mathcal{L}(x^*, \mu) \leq f(x^*).$$

Since this holds for any feasible x^* , it follows that $d(\mu) \leq f^*$. $\square$

A key structural property of the dual function is concavity, which holds independently of whether the primal problem is convex.

Theorem 4.64 (Concavity of the Dual Function). *The dual function $d(\mu)$ is concave on $\mathbb{R}_+^{|\mathcal{Q}|}$.*

Proof. Let $\mu^1, \mu^2 \geq 0$ and let $\lambda \in [0, 1]$. Define $\mu := \lambda\mu^1 + (1 - \lambda)\mu^2$. By definition of the dual function,

$$\begin{aligned} d(\mu) &= \inf_{x \in \mathbb{R}^n} \left(f(x) + \sum_{i \in \mathcal{Q}} \mu_i g_i(x) \right) \\ &= \inf_{x \in \mathbb{R}^n} \left(\lambda \mathcal{L}(x, \mu^1) + (1 - \lambda) \mathcal{L}(x, \mu^2) \right). \end{aligned}$$

Since the infimum of a convex combination of functions is greater than or equal to the same convex combination of their infima, we obtain

$$d(\mu) \geq \lambda \inf_x \mathcal{L}(x, \mu^1) + (1 - \lambda) \inf_x \mathcal{L}(x, \mu^2) = \lambda d(\mu^1) + (1 - \lambda) d(\mu^2).$$

Therefore, $d(\mu)$ is concave. $\square$

Remark 4.65. Although the primal problem (4.1) is a minimization problem, the dual problem (4.26) is always a concave maximization problem, even when the primal problem is nonconvex. This structural asymmetry underlies the power of dual methods in optimization.

Example 4.66 Quadratic Example Revisited.

We revisit the quadratic program introduced earlier in order to illustrate the construction of the dual function and the recovery of the primal optimizer from the dual solution. Consider

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & \frac{1}{2} (x_1^2 + x_2^2 + x_3^2) \\ \text{s.t.} \quad & x_1 + x_2 + x_3 \leq -3. \end{aligned}$$

Introducing a Lagrange multiplier $\mu \geq 0$ for the inequality constraint, the Lagrangian is

$$\mathcal{L}(x, \mu) = \frac{1}{2}\|x\|^2 + \mu(x_1 + x_2 + x_3 + 3).$$

The associated dual function is $d(\mu) = \inf_{x \in \mathbb{R}^3} \mathcal{L}(x, \mu)$.

We first minimize over the primal variables. Since the Lagrangian is strictly convex in x , the infimum is attained at the unique stationary point. Taking first-order conditions,

$$\frac{\partial \mathcal{L}}{\partial x_i} = x_i + \mu = 0, \quad i = 1, 2, 3,$$

which yields $x_1 = x_2 = x_3 = -\mu$.

Substituting this expression back into the Lagrangian gives the dual function

$$\begin{aligned} d(\mu) &= \frac{1}{2} \cdot 3\mu^2 + \mu(-3\mu + 3) \\ &= -\frac{3}{2}\mu^2 + 3\mu. \end{aligned}$$

It remains to maximize the dual and recover the primal optimizer. The dual problem is $\max_{\mu \geq 0} d(\mu)$. Since $d(\mu)$ is a concave quadratic function, the maximizer satisfies $\frac{d}{d\mu} d(\mu) = -3\mu + 3 = 0$, yielding $\mu^* = 1$. Recovering the primal solution from the minimizer of the Lagrangian, we obtain $x^* = (-1, -1, -1)$, which coincides with the solution obtained earlier and satisfies the constraint with equality. Finally, the optimal dual value $d(\mu^*) = -\frac{3}{2} + 3 = \frac{3}{2}$ equals the primal optimum $f^* = \frac{1}{2}(1 + 1 + 1) = \frac{3}{2}$, so strong duality holds and the duality gap is zero. $\square$

4.5.3 - Geometric Interpretation of the Dual Problem

The algebra of the dual function conceals a simple picture, and it is worth seeing it once because it makes strong duality and complementarity almost visual. The idea is to plot the objective and constraint values jointly: map each x to the point $(f(x), g_1(x), \dots, g_m(x)) \in \mathbb{R}^{m+1}$ and study the resulting region. Weak duality becomes “a hyperplane lies below the region,” strong duality becomes “a supporting hyperplane touches it at the optimum,” and complementary slackness becomes “the hyperplane is tangent only along the active coordinates.” We now make this precise.

The dual problem admits a natural geometric interpretation in $\mathbb{R}^{m+1}$ with coordinates $(y_0, y_1, \dots, y_m)$. Instead of considering only the image of F , it is convenient to define the *upper feasible image*

$$\mathcal{Y} := \{y \in \mathbb{R}^{m+1} : \exists x \in \mathbb{R}^n \text{ such that } y_0 \geq f(x), y_i \geq g_i(x), i = 1, \dots, m\}.$$

Thus $\mathcal{Y}$ consists of all points that dominate $(f(x), g_1(x), \dots, g_m(x))$ componentwise. When f and the g_i are convex, the set $\mathcal{Y}$ is convex and upward closed in the sense that $y \in \mathcal{Y}$ and $\tilde{y} \geq y$ imply $\tilde{y} \in \mathcal{Y}$.

The primal feasible region corresponds to the intersection of $\mathcal{Y}$ with the half-space $\{y_i \leq 0, i = 1, \dots, m\}$. The primal optimal value is

$$f^* = \inf\{y_0 : (y_0, 0, \dots, 0) \in \mathcal{Y}\},$$

that is, the smallest vertical coordinate among points of $\mathcal{Y}$ lying above the feasible slice.

For a fixed multiplier $\mu \geq 0$, the Lagrangian defines affine hyperplanes

$$H_{\mu,c} = \left\{ y \in \mathbb{R}^{m+1} : y_0 + \sum_{i=1}^m \mu_i y_i = c \right\},$$

with normal vector $(1, \mu)$. The dual function is

$$d(\mu) = \inf_{y \in \mathcal{Y}} \left(y_0 + \sum_{i=1}^m \mu_i y_i \right),$$

so $d(\mu)$ is the largest intercept c for which the hyperplane $H_{\mu,c}$ lies below $\mathcal{Y}$ in the sense that $y_0 + \sum_{i=1}^m \mu_i y_i \geq c \forall y \in \mathcal{Y}$.

Thus each $\mu \geq 0$ specifies a slope, and the dual function selects the tightest supporting hyperplane of that slope. The dual problem maximizes this intercept over all $\mu \geq 0$, thereby choosing the supporting hyperplane with the largest vertical intercept.

Strong duality corresponds geometrically to the existence of a supporting hyperplane to $\mathcal{Y}$ that touches the boundary of $\mathcal{Y}$ at a point with $y_i = 0$ for all $i = 1, \dots, m$ and whose intercept equals f^* . Complementarity reflects the fact that the hyperplane is tangent only along those coordinates where the constraint is active.

Example 4.67 Geometric Interpretation with One Constraint.

Consider the convex problem

$$\min_{x \in \mathbb{R}} f(x) = x^2 \quad \text{subject to} \quad g(x) = x - 1 \leq 0.$$

Define the upper feasible image

$$\mathcal{Y} := \{(y_1, y_0) \in \mathbb{R}^2 : \exists x \in \mathbb{R} \text{ such that } y_1 \geq g(x), y_0 \geq f(x)\}.$$

The boundary of $\mathcal{Y}$ is given by the curve $F(x) = (g(x), f(x)) = (x - 1, x^2)$.

Primal geometry. The feasible set is $x \leq 1$, corresponding to the slice $\mathcal{Y} \cap \{y_1 \leq 0\}$. The primal problem minimizes the vertical coordinate y_0 over this slice. Since $f(x) = x^2$ attains its global minimum at $x^* = 0$ and $0 \leq 1$, the constraint is inactive at the solution. Thus $F(x^*) = (-1, 0)$ and the optimal value is $f^* = 0$.

Dual hyperplanes. For each multiplier $\mu \geq 0$, consider affine lines $y_0 = c - \mu y_1$. The dual function is

$$d(\mu) = \inf_{(y_1, y_0) \in \mathcal{Y}} (y_0 + \mu y_1),$$

so $d(\mu)$ is the largest intercept c such that $y_0 \geq c - \mu y_1$ for all $(y_1, y_0) \in \mathcal{Y}$. Geometrically, we tilt a line with slope $-\mu$ until it becomes supporting to the convex set $\mathcal{Y}$.

Inactive constraint (see Figure 4.5). Because $x^* = 0$ lies strictly inside the feasible region, complementarity gives $\mu^* = 0$. The optimal supporting hyperplane is horizontal, $y_0 = f^* = 0$, touching $\mathcal{Y}$ at $F(x^*)$. Any $\mu > 0$ produces a strictly smaller

intercept. Thus when the constraint is inactive, the dual supporting hyperplane is not tilted.

Active constraint (see Figure 4.6). If instead the constraint is $g(x) = x + 1 \leq 0$, the feasible region becomes $x \leq -1$, and the minimizer occurs at the boundary $x^* = -1$. The constraint is active, so $\mu^* > 0$. Stationarity reads $2x^* + \mu^* = 0$, hence $\mu^* = 2$. The optimal supporting hyperplane is now tilted, $y_0 = 1 - 2y_1$, achieving intercept $f^* = 1$. Other multipliers, such as $\mu = 3$, yield supporting hyperplanes with strictly smaller intercepts. When the constraint binds, the slope adjusts to match the boundary geometry.

Degenerate feasible set (see Figure 4.7). If the constraint collapses to $|x| \leq 0$, the feasible set reduces to the single point $x^* = 0$. The intersection $\mathcal{Y} \cap \{y_1 \leq 0\}$ then consists only of $(0, 0)$. A supporting hyperplane still exists by convexity, but Slater's condition fails and the multiplier need not be unique. The geometry degenerates because the feasible slice has empty interior.

In all cases the dual problem selects the supporting hyperplane with the largest vertical intercept. When the constraint is inactive, $\mu^* = 0$ and the supporting hyperplane is horizontal. When the constraint is active, $\mu^* > 0$ and the hyperplane tilts to support the boundary. When the feasible set collapses, a supporting hyperplane still exists by convexity, but the multiplier loses uniqueness.

□

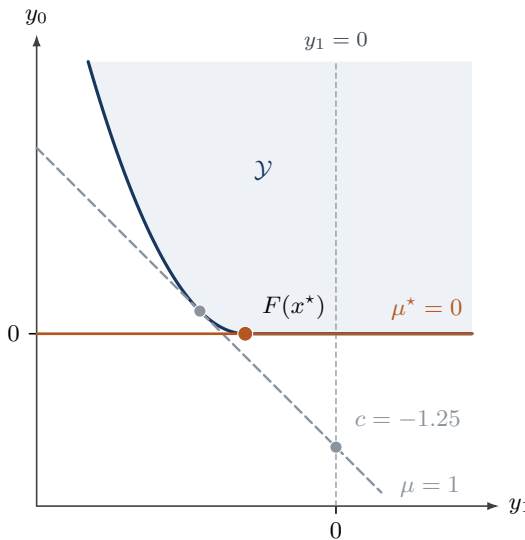


Figure 4.5: Dual supporting hyperplanes with $\mu \geq 0$ for the inactive constraint $x - 1 \leq 0$; the shaded region is the upper image $\mathcal{Y}$. The solid line ($\mu^* = 0$) supports $\mathcal{Y}$ at $F(x^*) = (-1, 0)$ and achieves the maximal intercept $f^* = 0$. The dashed line ($\mu = 1$) is a true supporting line that touches $\mathcal{Y}$ at $(-1.5, 0.25)$ but has the strictly smaller intercept -1.25 .

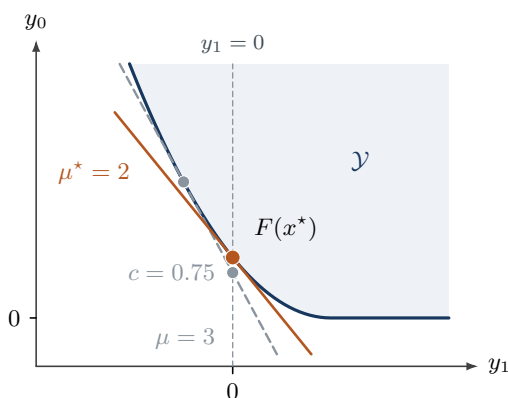


Figure 4.6: Supporting hyperplanes for $\mu \geq 0$ when the constraint $x + 1 \leq 0$ is active; the shaded region is the upper image $\mathcal{Y}$. The solid line ($\mu^* = 2$) supports $\mathcal{Y}$ at $F(x^*) = (0, 1)$ and achieves the maximal intercept $f^* = 1$. The dashed line ($\mu = 3$) touches the boundary of $\mathcal{Y}$ at $(-0.5, 2.25)$ and has smaller intercept 0.75.

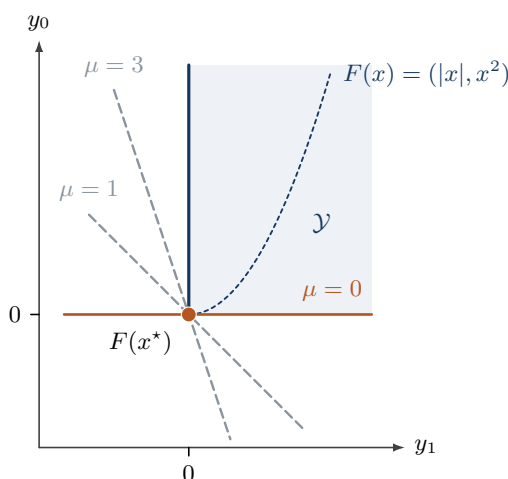


Figure 4.7: Degenerate problem $\min x^2$ subject to $|x| \leq 0$. The image set satisfies $y_0 = y_1^2$, $y_1 \geq 0$, and the upper image $\mathcal{Y}$ (shaded) is the quadrant $y_1 \geq 0$, $y_0 \geq 0$. The constraint $y_1 \leq 0$ forces $y_1 = 0$, so the feasible image reduces to the single point $(0, 0)$. The horizontal line $y_0 = 0$ is a supporting hyperplane; so is every line $y_0 = -\mu y_1$ with $\mu \geq 0$, which all attain the intercept $f^* = 0$, so the multiplier is not unique.

4.5.4 ■ Case Study: Duality in Linear Programming

Linear programming is the setting where duality first appeared and where it is cleanest, so it is the right place to see the general machinery pay off. Everything specializes sharply: the dual is again a linear program, weak duality is a one-line inequality, and, because polyhedra need no smoothness or curvature, strong duality holds under mere feasibility and boundedness, with no Slater condition required. The symmetry is worth remembering as a template: minimizing over primal variables against m constraints becomes maximizing over m dual variables against n

constraints, and the two optimal values meet exactly. We now specialize the general Lagrangian framework and recover the classical LP dual as a special case.

Consider the linear program

$$\begin{aligned} & \max_{x \in \mathbb{R}^n} && c^\top x \\ & \text{subject to} && Ax \leq b, \\ & && x \geq 0, \end{aligned} \tag{4.27}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$ are given. To place (4.27) into the standard convex minimization framework, we rewrite it equivalently as

$$\begin{aligned} & \min_{x \in \mathbb{R}^n} && -c^\top x \\ & \text{subject to} && Ax - b \leq 0, \\ & && -x \leq 0. \end{aligned} \tag{4.28}$$

This is a convex optimization problem with linear objective and linear inequality constraints.

Introduce Lagrange multipliers $\lambda_1 \in \mathbb{R}_+^m$ associated with the constraints $Ax \leq b$ and $\lambda_2 \in \mathbb{R}_+^n$ associated with the constraints $x \geq 0$. The Lagrangian is

$$\begin{aligned} \mathcal{L}(x, \lambda_1, \lambda_2) &= -c^\top x + \lambda_1^\top (Ax - b) - \lambda_2^\top x \\ &= (-c + A^\top \lambda_1 - \lambda_2)^\top x - \lambda_1^\top b. \end{aligned} \tag{4.29}$$

The dual function is

$$d(\lambda_1, \lambda_2) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda_1, \lambda_2).$$

From (4.29), the infimum over x is finite if and only if the linear coefficient of x vanishes, that is, $-c + A^\top \lambda_1 - \lambda_2 = 0$. If this condition is satisfied, then $d(\lambda_1, \lambda_2) = -\lambda_1^\top b$; otherwise, $d(\lambda_1, \lambda_2) = -\infty$.

The dual problem therefore becomes

$$\max_{\lambda_1 \geq 0, \lambda_2 \geq 0} d(\lambda_1, \lambda_2) \quad \text{subject to} \quad -c + A^\top \lambda_1 - \lambda_2 = 0.$$

Eliminating λ_2 yields the equivalent formulation

$$\begin{aligned} & \min_{\lambda_1 \geq 0} && \lambda_1^\top b \\ & \text{subject to} && A^\top \lambda_1 \geq c. \end{aligned} \tag{4.30}$$

This is precisely the classical dual linear program associated with (4.27).

For linear programs, weak duality holds automatically: for any primal feasible x and dual feasible λ_1 , $c^\top x \leq \lambda_1^\top b$. Moreover, if either the primal or the dual problem is feasible and bounded, then *strong duality* holds:

$$\max_x c^\top x = \min_{\lambda_1 \geq 0} \lambda_1^\top b.$$

At optimality, primal–dual optimal solutions (x^*, λ_1^*) satisfy the complementary slackness conditions

$$\lambda_{1,i}^* (A_i x^* - b_i) = 0, \quad i = 1, \dots, m, \quad x_j^* (A_{j,\cdot}^\top \lambda_1^* - c_j) = 0, \quad j = 1, \dots, n,$$

which fully characterize optimal solutions of linear programs.

Theorem 4.68 (Linear Programming Duality and Optimality). *Consider the primal linear program*

$$\begin{aligned} & \max_{x \in \mathbb{R}^n} && c^\top x \\ & \text{subject to} && Ax \leq b, \\ & && x \geq 0, \end{aligned} \tag{4.31}$$

and its dual

$$\begin{aligned} & \min_{\lambda \in \mathbb{R}^m} && b^\top \lambda \\ & \text{subject to} && A^\top \lambda \geq c, \\ & && \lambda \geq 0. \end{aligned} \tag{4.32}$$

Then the following statements hold.

- (i) Weak duality. For any primal feasible point x and any dual feasible multiplier λ , one has $c^\top x \leq b^\top \lambda$.
- (ii) Strong duality. If either (4.31) or (4.32) is feasible and has a finite optimal value, then both problems admit optimal solutions and their optimal values coincide:

$$\max\{c^\top x : Ax \leq b, x \geq 0\} = \min\{b^\top \lambda : A^\top \lambda \geq c, \lambda \geq 0\}.$$

- (iii) Existence of optimal solutions. If the primal problem (4.31) is feasible and bounded above, then (4.31) attains its maximum and (4.32) attains its minimum.
- (iv) Optimality characterization (complementary slackness). A pair (x^*, λ^*) is primal–dual optimal if and only if it is primal feasible and dual feasible, and satisfies the complementary slackness conditions

$$\lambda_i^* (b - Ax^*)_i = 0, \quad i = 1, \dots, m, \tag{4.33}$$

$$x_j^* ((A^\top \lambda^* - c)_j) = 0, \quad j = 1, \dots, n. \tag{4.34}$$

- (v) KKT special case. The conditions in item (iv) are precisely the KKT conditions for linear programming; they are necessary and sufficient for optimality, without requiring differentiability or any constraint qualification beyond feasibility.

Proof. *Weak duality.* Let x be primal feasible and λ be dual feasible. Since $A^\top \lambda \geq c$ and $x \geq 0$, we have $c^\top x \leq (A^\top \lambda)^\top x = \lambda^\top (Ax)$. Since $Ax \leq b$ and $\lambda \geq 0$, it follows that $\lambda^\top (Ax) \leq \lambda^\top b = b^\top \lambda$, and hence $c^\top x \leq b^\top \lambda$.

Strong duality and existence. Linear programs are convex (indeed polyhedral) optimization problems. Assume that one of the two problems is feasible and has a finite optimal value. Then the feasible region of that problem is a nonempty polyhedron on which the objective is linear and bounded in the optimizing direction. It follows that the optimal value is attained at an extreme point of the feasible polyhedron. Moreover, by the linear programming strong duality theorem (equivalently, a separating hyperplane theorem/Farkas' lemma argument), the duality gap is zero: the primal and dual optimal values coincide, and an optimal solution exists for the other problem as well. This establishes items (ii) and (iii).

Optimality characterization. Assume first that (x^*, λ^*) is primal–dual optimal. By weak duality, $c^\top x^* \leq b^\top \lambda^*$, and by strong duality the two sides are equal. Consider the nonnegative vector $b - Ax^* \geq 0$ and the nonnegative vector $A^\top \lambda^* - c \geq 0$. Using feasibility and the identity

$$b^\top \lambda^* - c^\top x^* = \lambda^{*\top} (b - Ax^*) + x^{*\top} (A^\top \lambda^* - c),$$

we obtain a sum of two nonnegative terms equal to zero, hence both terms must be zero: $\lambda^{*\top}(b - Ax^*) = 0$ and $x^{*\top}(A^\top \lambda^* - c) = 0$. Since each term is a sum of nonnegative summands, each summand must be zero, which yields (4.33)–(4.34).

Conversely, suppose x^* and λ^* are feasible and satisfy (4.33)–(4.34). Then

$$b^\top \lambda^* - c^\top x^* = \lambda^{*\top}(b - Ax^*) + x^{*\top}(A^\top \lambda^* - c) = 0,$$

so $c^\top x^* = b^\top \lambda^*$. By weak duality, $c^\top x^*$ is at most any dual feasible value and $b^\top \lambda^*$ is at least any primal feasible value; hence both are optimal and (x^*, λ^*) is primal–dual optimal.

KKT specialization. For linear programs, the Lagrangian stationarity condition coincides with $A^\top \lambda \geq c$ together with (4.34), and complementary slackness is exactly (4.33)–(4.34). Therefore, the conditions in item (iv) are the KKT conditions, and by item (iv) they are necessary and sufficient for optimality. $\square$

4.5.5 ■ Case Study: Support Vector Machines

Support Vector Machines (SVMs) constitute one of the most illuminating finite-dimensional examples of convex optimization. They transform the geometric problem of separating two classes of points into a strictly convex quadratic program whose structure reveals deep connections between convex sets, supporting hyperplanes, Lagrangian duality, and optimality conditions.

From a geometric perspective, the SVM constructs a separating hyperplane between two finite point sets by maximizing the margin, thereby selecting the most “robust” separator among infinitely many feasible ones. From an analytic perspective, the problem is a convex program for which Slater’s condition holds under linear separability, guaranteeing strong duality and a complete saddle-point characterization. From a dual viewpoint, the classifier admits a finite expansion in terms of a small subset of datapoints, the support vectors, revealing how extreme constraints determine the solution.

Let $\{(x_i, y_i)\}_{i=1}^N$ be labeled data with $x_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$. Assume that the data are linearly separable.

The maximum-margin separating hyperplane problem is

$$\begin{aligned} \min_{w, b} \quad & \frac{1}{2} \|w\|^2 \\ \text{s.t.} \quad & y_i(w^\top x_i + b) \geq 1, \quad i = 1, \dots, N. \end{aligned} \tag{4.35}$$

The objective is strictly convex in w , and the constraints are affine. Hence the feasible region is convex and the problem is a convex program. If the data are strictly separable, Slater’s condition holds and strong duality applies.

Theorem 4.69 (Dual Formulation of the Hard-Margin SVM). *Assume that the data are linearly separable so that (4.35) satisfies Slater’s condition. Then strong duality holds and the dual problem is*

$$\begin{aligned} \max_{\alpha \in \mathbb{R}^N} \quad & \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j x_i^\top x_j \\ \text{s.t.} \quad & \alpha_i \geq 0, \quad i = 1, \dots, N, \\ & \sum_{i=1}^N \alpha_i y_i = 0. \end{aligned} \tag{4.36}$$

Moreover, any optimal solution α^* of (4.36) determines the primal optimizer through

$$w^* = \sum_{i=1}^N \alpha_i^* y_i x_i, \quad (4.37)$$

and b^* can be recovered from any index k with $\alpha_k^* > 0$ by $b^* = y_k - w^{*\top} x_k$.

Proof. Since the data are linearly separable, there exists $(\bar{w}, \bar{b})$ such that $y_i(\bar{w}^\top x_i + \bar{b}) > 1$ for all $i = 1, \dots, N$. Hence Slater's condition holds for the convex program (4.35). Because the objective is convex and the constraints are affine, strong duality follows.

Introduce Lagrange multipliers $\alpha_i \geq 0$ for the constraints $1 - y_i(w^\top x_i + b) \leq 0$. The Lagrangian is

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 + \sum_{i=1}^N \alpha_i (1 - y_i(w^\top x_i + b)).$$

Expanding gives

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 + \sum_{i=1}^N \alpha_i - \sum_{i=1}^N \alpha_i y_i x_i^\top w - b \sum_{i=1}^N \alpha_i y_i.$$

The dual function is $g(\alpha) = \inf_{w,b} \mathcal{L}(w, b, \alpha)$. Since the dependence on b is linear, the infimum over b is finite if and only if $\sum_{i=1}^N \alpha_i y_i = 0$; otherwise $g(\alpha) = -\infty$. Hence the dual feasible set must satisfy $\sum_{i=1}^N \alpha_i y_i = 0$.

Under this condition the Lagrangian reduces to

$$\frac{1}{2} \|w\|^2 - \left(\sum_{i=1}^N \alpha_i y_i x_i \right)^\top w + \sum_{i=1}^N \alpha_i.$$

Define $s(\alpha) = \sum_{i=1}^N \alpha_i y_i x_i$. Then $\frac{1}{2} \|w\|^2 - s(\alpha)^\top w = \frac{1}{2} \|w - s(\alpha)\|^2 - \frac{1}{2} \|s(\alpha)\|^2$. The infimum over w is attained at $w = s(\alpha)$ and equals $\sum_{i=1}^N \alpha_i - \frac{1}{2} \|s(\alpha)\|^2$.

Since $\|s(\alpha)\|^2 = \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j x_i^\top x_j$, the dual function becomes

$$g(\alpha) = \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j x_i^\top x_j,$$

with constraints $\alpha_i \geq 0$ and $\sum_{i=1}^N \alpha_i y_i = 0$.

The quadratic term is associated with the Gram matrix $(y_i y_j x_i^\top x_j)_{i,j}$, which is positive semidefinite. Hence the quadratic form is convex and g is concave. The feasible set is convex; therefore the dual problem (4.36) is a concave maximization problem over a convex set.

By strong duality, there exists a saddle point (w^*, b^*, α^*) satisfying the KKT conditions. From the minimizing argument of the completed square, $w^* = s(\alpha^*) = \sum_{i=1}^N \alpha_i^* y_i x_i$, which proves (4.37).

Complementary slackness gives $\alpha_i^* (y_i(w^{*\top} x_i + b^*) - 1) = 0$ for all i . If $\alpha_k^* > 0$ for some k , then $y_k(w^{*\top} x_k + b^*) = 1$, which implies $b^* = y_k - w^{*\top} x_k$. Such an index exists because at optimality at least one constraint from each class must be active; otherwise the margin could be increased. This completes the proof. $\square$

Theorem 4.70 (KKT Characterization and Support Vectors). *Under the assumptions of Theorem 4.69, a triple (w^*, b^*, α^*) is optimal if and only if*

$$\begin{aligned} w^* &= \sum_{i=1}^N \alpha_i^* y_i x_i, & \sum_{i=1}^N \alpha_i^* y_i &= 0, & \alpha_i^* &\geq 0, \\ y_i(w^{*\top} x_i + b^*) &\geq 1, & \alpha_i^* (y_i(w^{*\top} x_i + b^*) - 1) &= 0 \quad \forall i. \end{aligned}$$

In particular,

$$\alpha_i^* > 0 \iff y_i(w^{*\top} x_i + b^*) = 1.$$

Proof. These conditions are precisely the Karush–Kuhn–Tucker conditions for the convex program (4.35). Since the problem is convex and Slater’s condition holds, they are necessary and sufficient. Complementary slackness yields the final equivalence. $\square$

The representation $w^* = \sum_i \alpha_i^* y_i x_i$ shows that the optimal hyperplane lies in the span of the training points. Only indices with $\alpha_i^* > 0$ contribute; these are the support vectors and lie exactly on the margin. All other datapoints are inactive in the dual and do not affect the classifier.

The dual objective depends on the data only through inner products $x_i^\top x_j$. Replacing these by a positive semidefinite kernel $K(x_i, x_j)$ yields a nonlinear classifier without altering convexity. Thus kernelization is a direct consequence of duality.

4.5.6 ■ Game-Theoretic Interpretation of Lagrangian Duality

A central object in constrained optimization is the *Lagrangian*, defined for the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g_i(x) \leq 0, \quad i = 1, \dots, m, \end{aligned} \tag{4.38}$$

by

$$\mathcal{L}(x, \mu) := f(x) + \sum_{i=1}^m \mu_i g_i(x), \quad \mu \in \mathbb{R}_+^m.$$

A fundamental question concerns the relationship between the two optimization orders

$$\sup_{\mu \geq 0} \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu) \quad \text{and} \quad \inf_{x \in \mathbb{R}^n} \sup_{\mu \geq 0} \mathcal{L}(x, \mu).$$

Consider first the sup–inf order, which produces the dual problem. The quantity $\sup_{\mu \geq 0} \inf_x \mathcal{L}(x, \mu)$ defines the *dual problem*. For each fixed $\mu \geq 0$, the inner minimization over x yields the *dual function*

$$d(\mu) := \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu). \tag{4.39}$$

The dual function is concave in μ , since it is the pointwise infimum of functions that are affine in μ . Maximizing $d(\mu)$ over $\mu \geq 0$ yields the tightest lower bound on the optimal value of the primal problem.

Reversing the order to inf–sup recovers the primal problem. For any fixed $x \in \mathbb{R}^n$, we have

$$\sup_{\mu \geq 0} \mathcal{L}(x, \mu) = \begin{cases} f(x), & \text{if } g_i(x) \leq 0 \quad \forall i, \\ +\infty, & \text{otherwise.} \end{cases}$$

Indeed, if any constraint $g_i(x) > 0$ is violated, the corresponding multiplier μ_i can be made arbitrarily large, driving the Lagrangian to $+\infty$.

Consequently, $\inf_x \sup_{\mu \geq 0} \mathcal{L}(x, \mu) = \inf \{f(x) \mid g_i(x) \leq 0, i = 1, \dots, m\}$, which coincides exactly with the original *primal problem* (4.38). Thus, the inf–sup formulation enforces feasibility by assigning infinite cost to constraint violations.

Saddle points and strong duality.

A pair (x^*, μ^*) is called a *saddle point* of the Lagrangian if

$$\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*), \quad \forall x \in \mathbb{R}^n, \forall \mu \geq 0. \quad (4.40)$$

The existence of a saddle point implies equality of the two optimization orders:

$$\sup_{\mu \geq 0} \inf_x \mathcal{L}(x, \mu) = \inf_x \sup_{\mu \geq 0} \mathcal{L}(x, \mu),$$

and hence equality of the primal and dual optimal values.

Theorem 4.71 (Strong Duality and Saddle-Point Existence). *Consider the convex optimization problem (4.38), where f and g_i , $i = 1, \dots, m$, are convex functions. Suppose that Slater's condition holds, i.e., there exists a point $\bar{x} \in \mathbb{R}^n$ such that $g_i(\bar{x}) < 0$, $i = 1, \dots, m$. Then:*

- (i) *strong duality holds: $\inf_x \sup_{\mu \geq 0} \mathcal{L}(x, \mu) = \sup_{\mu \geq 0} \inf_x \mathcal{L}(x, \mu)$;*
- (ii) *the dual supremum is attained at some $\mu^* \geq 0$; and if in addition the primal infimum is attained at some x^* , then (x^*, μ^*) is a saddle point of the Lagrangian satisfying (4.40).*

Remark 4.72. Attainment of the primal infimum has to be assumed separately in (ii). Slater's condition closes the duality gap, but it does not by itself produce a primal minimizer: the problem $\min_x e^x$ subject to $-x \leq 0$ has $p^* = 1$ attained, whereas $\min_x e^{-x}$ over the same feasible set has $p^* = 0$ and no minimizer, so no pair (x^*, μ^*) can satisfy the right-hand inequality in (4.40).

Proof. Let $p^* := \inf \{f(x) \mid g_i(x) \leq 0, i = 1, \dots, m\}$ denote the primal optimal value, and define the dual function $d(\mu) := \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu)$, $\mu \geq 0$, with dual optimal value $d^* := \sup_{\mu \geq 0} d(\mu)$. For any feasible x and any $\mu \geq 0$, we have

$$\mathcal{L}(x, \mu) = f(x) + \sum_{i=1}^m \mu_i g_i(x) \leq f(x),$$

since $g_i(x) \leq 0$ and $\mu_i \geq 0$. Taking the infimum over x and then the supremum over μ yields the weak duality bound $d^* \leq p^*$.

To show the reverse inequality, define the *perturbation set*

$$\mathcal{A} := \{(u, t) \in \mathbb{R}^m \times \mathbb{R} \mid \exists x \in \mathbb{R}^n \text{ with } g_i(x) \leq u_i \ (i = 1, \dots, m) \text{ and } f(x) \leq t\}.$$

Convexity of f and of each g_i makes $\mathcal{A}$ convex, and it is by construction *upward closed*: enlarging any coordinate of a point of $\mathcal{A}$ keeps it in $\mathcal{A}$. Note that $\mathcal{A}$, unlike a set built by subtracting a free scalar from $f(x)$, does not contain every value in its last coordinate; this is what makes a separation argument possible at all.

Put $\mathcal{B} := \{(0, t) : t < p^*\}$, which is convex and nonempty. The two sets are disjoint: a common point would supply an x with $g_i(x) \leq 0$ for every i and $f(x) \leq t < p^*$, contradicting the definition of p^* . By the separating hyperplane theorem there exist $(\lambda, \lambda_0) \neq 0$ and $\beta \in \mathbb{R}$ with

$$\lambda^\top u + \lambda_0 t \geq \beta \quad \forall (u, t) \in \mathcal{A}, \quad \lambda^\top u + \lambda_0 t \leq \beta \quad \forall (u, t) \in \mathcal{B}. \quad (4.41)$$

Upward closedness of $\mathcal{A}$ forces $\lambda \geq 0$ and $\lambda_0 \geq 0$, since a negative coefficient would send the left-hand side of the first inequality to $-\infty$. Taking $(0, t) \in \mathcal{B}$ and letting $t \uparrow p^*$ in the second inequality gives $\lambda_0 p^* \leq \beta$. Taking $(g(x), f(x)) \in \mathcal{A}$ in the first gives

$$\lambda_0 f(x) + \sum_{i=1}^m \lambda_i g_i(x) \geq \beta \geq \lambda_0 p^* \quad \forall x \in \mathbb{R}^n. \quad (4.42)$$

Slater's condition rules out $\lambda_0 = 0$. Indeed, if $\lambda_0 = 0$ then $\lambda \neq 0$, and evaluating (4.42) at the strictly feasible point $\bar{x}$ gives $\sum_i \lambda_i g_i(\bar{x}) \geq 0$; but every $g_i(\bar{x}) < 0$ and $\lambda \geq 0$ is nonzero, so the left-hand side is strictly negative, a contradiction. Hence $\lambda_0 > 0$.

Dividing (4.42) by λ_0 and setting $\mu_i^* := \lambda_i / \lambda_0 \geq 0$ yields $f(x) + \sum_{i=1}^m \mu_i^* g_i(x) \geq p^*$ for all x . Taking the infimum over x gives $d(\mu^*) \geq p^*$, and therefore $d^* \geq p^*$. Combined with weak duality, this shows $d^* = p^*$, which is (i), and the supremum defining d^* is attained at μ^* .

Finally, suppose the primal infimum is attained at some x^* , so that x^* is feasible and $f(x^*) = p^*$. Then

$$p^* = d(\mu^*) = \inf_x \mathcal{L}(x, \mu^*) \leq \mathcal{L}(x^*, \mu^*) = f(x^*) + \sum_{i=1}^m \mu_i^* g_i(x^*) \leq f(x^*) = p^*,$$

where the last inequality uses $g_i(x^*) \leq 0$ and $\mu_i^* \geq 0$. Every inequality in this chain is therefore an equality. Two consequences follow: complementary slackness $\sum_i \mu_i^* g_i(x^*) = 0$, and the fact that x^* attains $\inf_x \mathcal{L}(x, \mu^*)$, i.e. $\mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*)$ for all x . For any $\mu \geq 0$, feasibility of x^* gives

$$\mathcal{L}(x^*, \mu) = f(x^*) + \sum_{i=1}^m \mu_i g_i(x^*) \leq f(x^*) = \mathcal{L}(x^*, \mu^*).$$

Hence (x^*, μ^*) is a saddle point of the Lagrangian. $\square$

Game-Theoretic Interpretation and Minimax Theory

The Lagrangian formulation admits a natural interpretation as a two-player zero-sum game. The *primal player* selects a strategy $x \in \mathbb{R}^n$ with the objective of minimizing the payoff $\mathcal{L}(x, \mu)$, while the *dual player* selects a strategy $\mu \in \mathbb{R}_+^m$ with the objective of maximizing the same payoff. The value of the game therefore depends on the order in which the players act.

The quantity $\inf_{x \in \mathbb{R}^n} \sup_{\mu \geq 0} \mathcal{L}(x, \mu)$ corresponds to the outcome when the primal player moves first and the dual player responds optimally. As shown earlier, this value coincides with the optimal value of the primal constrained optimization problem, since infeasible choices of x are penalized by arbitrarily large multipliers.

Conversely, the quantity $\sup_{\mu \geq 0} \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu)$ corresponds to the outcome when the dual player moves first and the primal player responds optimally. This value defines the dual problem and provides the best lower bound on the primal optimal value via weak duality.

In general, for arbitrary payoff functions, these two values need not coincide. This asymmetry reflects the familiar phenomenon in game theory that the order of play may affect the outcome. A fundamental question, therefore, is under what conditions the minimax equality

$$\inf_x \sup_{\mu \geq 0} \mathcal{L}(x, \mu) = \sup_{\mu \geq 0} \inf_x \mathcal{L}(x, \mu)$$

holds.

Classical minimax theorems in game theory, such as von Neumann's minimax theorem for finite games and Sion's minimax theorem for infinite-dimensional games, provide sufficient

conditions for such an equality. These results typically require convexity (or linearity) in one player's strategy, concavity in the other's strategy, and appropriate compactness or interiority assumptions.

In the context of convex optimization, the Lagrangian $\mathcal{L}(x, \mu)$ is convex in x for fixed μ and affine (hence concave) in μ for fixed x . Slater's condition plays the role of a regularity assumption that substitutes for compactness and ensures that the feasible set has nonempty interior. Under these conditions, the hypotheses of a minimax theorem are effectively satisfied.

Strong duality can therefore be interpreted as a *minimax theorem* for the Lagrangian game: the order of play between the primal and dual players becomes immaterial, and the game admits a saddle-point equilibrium. That is, there exists a pair (x^*, μ^*) such that

$$\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*), \quad \forall x, \forall \mu \geq 0.$$

From this perspective, the Karush–Kuhn–Tucker conditions characterize Nash equilibria of the associated zero-sum game. The primal variables represent the strategy of the minimizing player, the Lagrange multipliers represent the strategy of the maximizing player, and complementary slackness determines which constraints (or actions) are active at equilibrium.

4.5.7 ■ Perturbations, Shadow Prices, and Sensitivity Analysis

So far the multipliers μ^* have appeared as auxiliary variables that make the KKT conditions close. We now show they have a concrete meaning of independent value: each μ_i^* is the *price* of the i th constraint, the rate at which the optimal value would improve if that constraint were loosened by one unit. This is why a manager reads the multipliers off an optimal solution to learn which resources are binding and how much an extra unit of each is worth, without re-solving the problem. We make the statement precise by studying how the optimal value reacts to perturbing the right-hand sides of the constraints.

We develop a unified view of *shadow prices* and *sensitivity analysis* by first establishing the perturbation theory for general convex optimization problems and then specializing the results to linear programs. This approach highlights linear-programming sensitivity formulas as direct consequences of convex duality.

Perturbations in Convex Optimization

Consider the convex optimization problem

$$\begin{aligned} \min_x \quad & f(x) \\ \text{s.t.} \quad & g_j(x) \leq 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{4.43}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ are convex functions.

For $\theta \in \mathbb{R}^{|\mathcal{Q}|}$, define the perturbed optimal value function

$$\varphi(\theta) := \inf \{f(x) \mid g_j(x) \leq \theta_j, \quad j \in \mathcal{Q}\}, \tag{4.44}$$

so that the original problem corresponds to $\theta = 0$.

Assume that the unperturbed problem satisfies *Slater's condition*. Then strong duality holds, and there exists an optimal vector of KKT multipliers $\mu^* = (\mu_j^*)_{j \in \mathcal{Q}}$.

Fix an index $i \in \mathcal{Q}$ and consider perturbations of only the i th constraint, $\theta = \theta_i e^i$, $\theta_i \geq 0$, where e^i is the i th canonical basis vector.

Theorem 4.73 (Shadow-Price Interpretation of Multipliers). *Under the above assumptions the value function φ is convex, and the following hold.*

(i) For every optimal multiplier vector μ^* ,

$$\varphi(\theta) \geq \varphi(0) - (\mu^*)^\top \theta \quad \forall \theta, \quad (4.45)$$

that is, $-\mu^* \in \partial\varphi(0)$.

(ii) Conversely, every subgradient of φ at 0 has the form $-\mu$ for an optimal multiplier vector μ ; thus $\partial\varphi(0)$ is exactly the negative of the set of optimal dual solutions.

(iii) Consequently the one-sided directional derivative satisfies

$$\frac{d}{d\theta_i^+} \varphi(0) = - \min \{ \mu_i : \mu \text{ optimal for the dual} \}. \quad (4.46)$$

In particular, if the optimal multiplier vector μ^* is unique, then φ is differentiable at 0 and

$$\frac{\partial\varphi}{\partial\theta_i}(0) = -\mu_i^*.$$

Remark 4.74. Uniqueness cannot be dropped from the last statement. When several optimal multipliers exist, φ has a kink at $\theta = 0$ and the left and right derivatives differ; each optimal μ^* then supplies a supporting slope rather than *the* slope, and only the extreme optimal multiplier realizes the one-sided derivative.

Proof. For $\theta = \theta_i e^i$, the perturbed problem takes the form

$$\varphi(\theta_i e^i) = \inf \{ f(x) \mid g_j(x) \leq 0, j \neq i, g_i(x) \leq \theta_i \}.$$

(i) Every optimal multiplier gives a supporting slope. Let μ^* be optimal for the dual, so that $d(\mu^*) = \varphi(0)$ by strong duality. Fix θ and let x be feasible for the perturbed problem, i.e. $g_j(x) \leq \theta_j$ for every j . Then

$$f(x) \geq f(x) + \sum_j \mu_j^* g_j(x) - \sum_j \mu_j^* \theta_j \geq d(\mu^*) - (\mu^*)^\top \theta = \varphi(0) - (\mu^*)^\top \theta,$$

where the first inequality uses $g_j(x) \leq \theta_j$ and $\mu_j^* \geq 0$. Taking the infimum over such x gives (4.45), i.e. $-\mu^* \in \partial\varphi(0)$.

(ii) Every subgradient comes from an optimal multiplier. Let $s \in \partial\varphi(0)$ and set $\mu := -s$. First, $\mu \geq 0$: relaxing a constraint cannot increase the optimal value, so φ is nonincreasing in each coordinate, forcing $s \leq 0$. Second, μ is dual optimal. Indeed, for any $x \in \mathbb{R}^n$ put $\theta := g(x)$; then x is feasible for the θ -problem, so $\varphi(\theta) \leq f(x)$, and the subgradient inequality $\varphi(\theta) \geq \varphi(0) + s^\top \theta$ gives

$$f(x) + \mu^\top g(x) \geq \varphi(\theta) + \mu^\top \theta \geq \varphi(0) + s^\top \theta + \mu^\top \theta = \varphi(0),$$

since $s = -\mu$. Taking the infimum over x yields $d(\mu) \geq \varphi(0)$, while weak duality gives $d(\mu) \leq \varphi(0)$; hence $d(\mu) = \varphi(0)$ and μ is dual optimal.

(iii) The directional derivative. For a convex function the one-sided directional derivative is the support function of the subdifferential, $\varphi'(0; e^i) = \max \{ s_i : s \in \partial\varphi(0) \}$. By (i) and (ii) this maximum is taken over $s = -\mu$ with μ dual optimal, which gives (4.46). If the optimal multiplier is unique then $\partial\varphi(0)$ is the singleton $\{-\mu^*\}$, so φ is differentiable at 0 with gradient $-\mu^*$. $\square$

The multiplier μ_i^* therefore measures the *marginal decrease in optimal value* resulting from relaxing the i th constraint. For small perturbations, $\varphi(e^i) - \varphi(0) \approx -\mu_i^*$, justifying the interpretation of μ_i^* as a *shadow price*.

Linear Programming Case Study

We now recover classical linear-programming sensitivity results as direct corollaries of the convex perturbation framework. Consider the linear program

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^\top x \\ \text{s.t.} \quad & Ax \geq b, \\ & x \geq 0. \end{aligned} \tag{4.47}$$

Its dual is

$$\begin{aligned} \max_{u \in \mathbb{R}^m} \quad & b^\top u \\ \text{s.t.} \quad & A^\top u \leq c, \\ & u \geq 0. \end{aligned} \tag{4.48}$$

Assume feasibility and boundedness. Strong duality yields optimal solutions (x^*, u^*) satisfying $c^\top x^* = b^\top u^* = f^\circ$.

Sensitivity and Shadow-Price Formulas

Viewing the LP as a convex program with affine data, Theorem 4.73 immediately yields the classical sensitivity identities.

Theorem 4.75 (Shadow-Price Formulas for Linear Programs). *Let (x^*, u^*) be an optimal primal–dual pair. Then the optimal value f° satisfies*

$$\frac{\partial f^\circ}{\partial c_i} = x_i^*, \quad i = 1, \dots, n, \tag{4.49}$$

$$\frac{\partial f^\circ}{\partial b_j} = u_j^*, \quad j = 1, \dots, m, \tag{4.50}$$

$$\frac{\partial f^\circ}{\partial a_{ij}} = -u_i^* x_j^*, \quad i = 1, \dots, m, \quad j = 1, \dots, n. \tag{4.51}$$

Proof. We derive each sensitivity formula by viewing the linear program as a special case of the convex perturbation framework established earlier.

Sensitivity with respect to b . Write the j th constraint in the standard form $g_j(x) = b_j - A_j x \leq 0$ used in (4.44). Increasing b_j *tightens* this constraint, so a perturbation $b \mapsto b + \theta e^j$ with $\theta \geq 0$ corresponds to the relaxation parameter $\theta_j = -\theta$. Theorem 4.73 gives $\partial f^\circ / \partial \theta_j = -u_j^*$, and the chain rule with $\theta_j = -\theta$ therefore yields

$$\frac{\partial f^\circ}{\partial b_j} = -\frac{\partial f^\circ}{\partial \theta_j} = u_j^*.$$

The sign is the expected one: $u^* \geq 0$ and tightening a constraint cannot decrease the minimum. It also matches the dual expression $f^\circ = b^\top u^*$ directly.

Sensitivity with respect to c . Now consider a perturbation of the objective coefficient c_i : $c \mapsto c + \theta e^i$. The perturbed objective is $(c + \theta e^i)^\top x = c^\top x + \theta x_i$. Let x^* be an optimal solution of the unperturbed problem. Since the feasible set is unchanged, we have

$$f^\circ(\theta) = \min_x \{c^\top x + \theta x_i\} \leq c^\top x^* + \theta x_i^* = f^\circ + \theta x_i^*.$$

Conversely, optimality of x^* implies no feasible x can achieve a smaller first-order variation. Hence,

$$\lim_{\theta \downarrow 0} \frac{f^\circ(\theta) - f^\circ}{\theta} = x_i^*,$$

which yields

$$\frac{\partial f^\circ}{\partial c_i} = x_i^*.$$

Sensitivity with respect to A . Finally, perturb a single constraint coefficient, $a_{ij} \mapsto a_{ij} + \theta$. In the standard form the i th constraint becomes $b_i - A_i x - \theta x_j \leq 0$, which is a relaxation by the amount $\theta_i = \theta x_j^*$ evaluated at the optimum. Applying Theorem 4.73,

$$\frac{\partial f^\circ}{\partial a_{ij}} = \frac{\partial f^\circ}{\partial \theta_i} \cdot x_j^* = -u_i^* x_j^*.$$

A quick scalar check confirms the sign: for $\min cx$ subject to $ax \geq b$, $x \geq 0$ with $a, b, c > 0$, one has $x^* = b/a$, $u^* = c/a$ and $f^\circ = cb/a$, so $\partial f^\circ / \partial a = -cb/a^2 = -u^* x^*$.

Combining the three cases establishes (4.49)–(4.51). $\square$

Economic Interpretation

The relation (4.49) shows that the optimal activity level x_i^* measures the marginal sensitivity of the objective value with respect to the cost coefficient of activity i : a small increase in its unit cost changes the optimal value proportionally to x_i^* . The identity (4.50) establishes that the dual variable u_j^* represents the shadow price of requirement j : raising the required level b_j by one unit raises the optimal cost at rate u_j^* , and conversely relaxing the requirement lowers it at the same rate. Finally, (4.51) captures the joint interaction between technology and resource valuation: making activity j more productive toward requirement i (a larger a_{ij}) lowers the optimal cost in proportion to both the shadow value of the requirement, u_i^* , and the optimal activity level, x_j^* .

4.6 ■ Transformation of Nonconvex Problems into Convex Problems

Many optimization problems in control, learning, and engineering are nonconvex as first written. Convexification asks whether the nonconvexity is essential or only an artifact of the formulation. Some transformations are exact and preserve global optimizers; others are relaxations that produce bounds or certificates. This section works through both.

We consider optimization problems of the form

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad g_i(x) \leq 0, \quad i = 1, \dots, m, \quad (4.52)$$

where the objective function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ and the constraint functions $g_i: \mathbb{R}^n \rightarrow \mathbb{R}$ may be nonconvex.

Definition 4.76 (Equivalent reformulation). A transformation of (4.52) is called an equivalent reformulation if it preserves the set of globally optimal solutions.

Definition 4.77 (Convex relaxation). A transformation is called a convex relaxation if it produces a convex optimization problem whose feasible set contains that of the original problem. The optimal value of the relaxed problem provides a bound on the original one.

4.6.1 ■ Epigraph Transformations

Definition 4.78 (Epigraph). Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, its epigraph is defined as $\text{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}$.

A fundamental result in convex analysis states that f is convex if and only if $\text{epi}(f)$ is a convex set.

The unconstrained problem $\min_{x \in \mathbb{R}^n} f(x)$ can be rewritten as

$$\min_{x, t} t \quad \text{subject to} \quad f(x) \leq t. \quad (4.53)$$

If f is convex, then (4.53) is a convex optimization problem equivalent to the original one.

Example 4.79 Minimax objectives and epigraph reformulation.

Consider the optimization problem

$$\min_{x \in \mathbb{R}^n} \max_{j=1, \dots, p} g_j(x), \quad (4.54)$$

where each function $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex.

Introducing an epigraph variable $t \in \mathbb{R}$, problem (4.54) can be equivalently rewritten as

$$\begin{aligned} \min_{x, t} \quad & t \\ \text{subject to} \quad & g_j(x) \leq t, \quad j = 1, \dots, p. \end{aligned} \quad (4.55)$$

The reformulated problem is a convex optimization problem, since the objective is linear and each constraint defines a convex sublevel set. This epigraph representation is fundamental in robust optimization and worst-case control. $\square$

Example 4.80 ℓ_∞ -norm objectives and epigraph reformulation.

Consider the optimization problem

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|_\infty, \quad (4.56)$$

where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

By definition of the infinity norm, $\|Ax - b\|_\infty = \max_{i=1, \dots, m} |(Ax - b)_i|$. Introducing

an epigraph variable $t \in \mathbb{R}$, problem (4.56) is equivalent to

$$\begin{aligned} \min_{x,t} \quad & t \\ \text{subject to} \quad & -t \leq (Ax - b)_i \leq t, \quad i = 1, \dots, m. \end{aligned} \quad (4.57)$$

This formulation is a linear program, since both the objective and constraints are affine. The epigraph transformation converts the nonsmooth infinity-norm objective into a set of linear inequality constraints. $\square$

Example 4.81 Maximum eigenvalue minimization.

Consider the optimization problem

$$\min_{x \in \mathbb{R}^n} \lambda_{\max}(A(x)), \quad (4.58)$$

where $A(x) := A_0 + \sum_{i=1}^n x_i A_i$, and $A_0, A_1, \dots, A_n \in \mathbb{S}^m$ are given symmetric matrices. At first glance, problem (4.58) appears highly nonlinear due to the eigenvalue operator.

Introduce an epigraph variable $t \in \mathbb{R}$. Using the variational characterization of the maximum eigenvalue, $\lambda_{\max}(A) \leq t$ if and only if $A \preceq tI$, problem (4.58) is equivalent to

$$\begin{aligned} \min_{x,t} \quad & t \\ \text{subject to} \quad & A_0 + \sum_{i=1}^n x_i A_i \preceq tI. \end{aligned} \quad (4.59)$$

The constraint $A_0 + \sum_{i=1}^n x_i A_i \preceq tI$ is a linear matrix inequality (LMI) in the variables (x, t) . The objective is linear. Therefore, problem (4.59) is a semidefinite program and hence a convex optimization problem.

Although the eigenvalue map $A \mapsto \lambda_{\max}(A)$ is nonlinear, it is a convex function on the space of symmetric matrices. The epigraph formulation exposes this convexity by replacing the eigenvalue inequality with an LMI. As a result, the original problem (4.58) admits an exact convex reformulation rather than a relaxation. $\square$

4.6.2 ■ Change of Variables

A powerful method for transforming nonconvex optimization problems into convex ones is an appropriate *change of variables*. While the original formulation may obscure convexity, a reparameterization can expose an equivalent problem with a convex objective and convex constraints. This approach is particularly effective when the nonconvexity arises from multiplicative structure or ratios of functions.

Example 4.82 Fractional objective and variable transformation.

Consider the optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & \frac{2x_1 + 3x_2 + 2}{x_1 + x_2 + 1} \\ \text{s.t.} \quad & A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq b, \end{aligned} \quad (4.60)$$

where $A \in \mathbb{R}^{m \times 2}$ and $b \in \mathbb{R}^m$ are given, and the denominator satisfies $x_1 + x_2 + 1 > 0$ on the feasible set.

Define the change of variables

$$y_1 := \frac{x_1}{x_1 + x_2 + 1}, \quad y_2 := \frac{x_2}{x_1 + x_2 + 1}, \quad z := \frac{1}{x_1 + x_2 + 1}. \quad (4.61)$$

Under this transformation, we have the identity

$$x_i = \frac{y_i}{z}, \quad i = 1, 2, \quad x_1 + x_2 + 1 = \frac{1}{z},$$

with $z > 0$. By construction the new variables obey the *normalization*

$$y_1 + y_2 + z = z(x_1 + x_2 + 1) = 1,$$

which must be carried into the reformulation to fix the scaling: without it the objective and constraints below are homogeneous in (y_1, y_2, z) , and the recovered point $x = (y_1/z, y_2/z)$ would no longer determine the objective value.

Substituting into the objective function yields the linear expression

$$\frac{2x_1 + 3x_2 + 2}{x_1 + x_2 + 1} = 2y_1 + 3y_2 + 2z.$$

The linear constraints transform as

$$A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \leq b \iff A \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \leq bz.$$

Therefore, problem (4.60) is equivalent to the linear optimization problem

$$\begin{aligned} \min_{(y_1, y_2, z)} \quad & 2y_1 + 3y_2 + 2z \\ \text{s.t.} \quad & A \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} - bz \leq 0, \\ & y_1 + y_2 + z = 1, \\ & z > 0. \end{aligned} \quad (4.62)$$

The feasible set of (4.62) is convex, and the objective function is linear. □

The transformation in Example 4.82 converts a fractional program into an equivalent linear program by lifting the variables into a higher-dimensional space. This reformulation is exact: every feasible solution of (4.60) corresponds uniquely to a feasible solution of (4.62), and vice versa. Such transformations form the basis of classical fractional programming methods and play an important role in convexification techniques.

Example 4.83 Logarithmic change of variables and convexification.

Consider the optimization problem

$$\min_{d \in \mathbb{R}_{++}^n} \sum_{i,j} M_{ij} \frac{d_i^2}{d_j^2}, \quad (4.63)$$

where $M_{ij} \geq 0$ are given coefficients and $\mathbb{R}_{++}^n := \{d \in \mathbb{R}^n : d_i > 0 \forall i\}$.

Problem (4.63) is nonconvex in the variables d_i , due to the presence of multiplicative ratios.

Introduce the logarithmic change of variables

$$x_i := \log d_i, \quad i = 1, \dots, n. \quad (4.64)$$

This transformation is bijective between $\mathbb{R}_{++}^n$ and $\mathbb{R}^n$.

Under (4.64), each ratio becomes $\frac{d_i^2}{d_j^2} = e^{2(x_i - x_j)}$. Hence, the objective function can be written as $\sum_{i,j} M_{ij} e^{2(x_i - x_j)}$. Taking the logarithm of the objective (which preserves minimizers since log is strictly increasing) yields the equivalent problem

$$\min_{x \in \mathbb{R}^n} f(x) := \log \left(\sum_{i,j} M_{ij} e^{2(x_i - x_j)} \right). \quad (4.65)$$

The function f in (4.65) is convex in x , since it is a log-sum-exp function of affine mappings of x . $\square$

Geometric Programming

Geometric programming (GP) is a class of optimization problems that admit an exact convex reformulation via a logarithmic change of variables. The intuition is simple: the objects that appear in a GP are products of powers of the variables, and the logarithm turns products into sums and powers into products with the exponents. A monomial, being a single such product, becomes an *affine* function of the log-variables, and a posynomial, being a sum of monomials, becomes a *log-sum-exp*, the canonical smooth convex function. Nonconvexity that looks essential in the original multiplicative form is therefore only an artifact of the coordinates; passing to logarithms exposes an exactly equivalent convex problem.

The two building blocks are the following. A *monomial* is a function $m : \mathbb{R}_{++}^n \rightarrow \mathbb{R}_{++}$ of the form $m(x) = c \prod_{i=1}^n x_i^{a_i}$, where $c > 0$, $x_i > 0$, and $a_i \in \mathbb{R}$. A *posynomial* is a finite sum of monomials, $p(x) = \sum_{k=1}^K c_k \prod_{i=1}^n x_i^{a_{ik}}$, $c_k > 0$. In general, posynomials are neither convex nor concave in the variable x .

A geometric program in standard form is then given by

$$\begin{aligned} \min_{x \in \mathbb{R}_{++}^n} \quad & p_0(x) \\ \text{subject to} \quad & p_i(x) \leq 1, \quad i = 1, \dots, m, \\ & m_j(x) = 1, \quad j = 1, \dots, r, \end{aligned} \quad (4.66)$$

where p_i are posynomials and m_j are monomials.

Now define the transformation $y := \log x$, that is, $y_i = \log x_i$, $i = 1, \dots, n$, which is a bijection between $\mathbb{R}_{++}^n$ and $\mathbb{R}^n$ with inverse $x = \exp(y)$. Applying it to the objective and constraints in (4.66), and taking logarithms of the objective and inequality constraints, yields the equivalent problem

$$\begin{aligned} \min_{y \in \mathbb{R}^n} \quad & \tilde{p}_0(y) := \log p_0(e^y) \\ \text{subject to} \quad & \tilde{p}_i(y) := \log p_i(e^y) \leq 0, \quad i = 1, \dots, m, \\ & \tilde{m}_j(y) := \log m_j(e^y) = 0, \quad j = 1, \dots, r. \end{aligned} \quad (4.67)$$

The transformed problem is convex. For each monomial m_j , $\tilde{m}_j(y) = \log c_j + \sum_{i=1}^n a_{ij}y_i$, which is affine in y . For each posynomial p_i ,

$$\tilde{p}_i(y) = \log \left(\sum_{k=1}^K \exp \left(\log c_{ik} + \sum_{l=1}^n a_{ilk}y_l \right) \right),$$

which is a log-sum-exp function and hence convex in y . Therefore, problem (4.67) is a convex optimization problem.

Because the logarithmic transformation is bijective, problems (4.66) and (4.67) are equivalent: an optimal solution x^* of the original problem corresponds to $y^* = \log x^*$ solving the transformed problem, and vice versa. Moreover, both problems have the same optimal value.

Fractional Programming

Fractional programming concerns optimization problems in which the objective function is expressed as a ratio of two functions. Such problems are generally nonconvex in their original formulation, even when the numerator is convex and the denominator is affine.

Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathcal{F}} \quad & \frac{f(x)}{g(x)} \\ \text{subject to} \quad & g(x) > 0, \end{aligned} \quad (4.68)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function, $g : \mathbb{R}^n \rightarrow \mathbb{R}_{++}$ is a positive affine function, and $\mathcal{F} \subseteq \mathbb{R}^n$ is a convex feasible set. Assume throughout that the denominator is affine and strictly positive on the feasible set, $g(x) = a^\top x + b$ with $a^\top x + b > 0$ for all $x \in \mathcal{F}$.

The Charnes–Cooper transformation now introduces the change of variables

$$y := \frac{x}{g(x)}, \quad t := \frac{1}{g(x)}. \quad (4.69)$$

This transformation is bijective on the domain $g(x) > 0$, with inverse mapping $x = \frac{y}{t}$, $t > 0$.

Substituting $x = y/t$ into (4.68) yields the equivalent problem

$$\begin{aligned} \min_{y,t} \quad & tf\left(\frac{y}{t}\right) \\ \text{subject to} \quad & a^\top y + bt = 1, \\ & y \in t\mathcal{F}, \\ & t > 0, \end{aligned} \tag{4.70}$$

where $t\mathcal{F} := \{tx \mid x \in \mathcal{F}\}$.

This problem is convex. If f is convex, then the function $(y, t) \mapsto tf\left(\frac{y}{t}\right)$ is the *perspective* of f and is jointly convex on the domain $t > 0$. The equality constraint $a^\top y + bt = 1$ is affine, and the set $t\mathcal{F}$ is convex whenever $\mathcal{F}$ is convex. Consequently, problem (4.70) is a convex optimization problem. Finally, the Charnes–Cooper transformation establishes a one-to-one correspondence between feasible points of (4.68) and feasible points of (4.70). Moreover, the two problems have identical optimal values, and optimal solutions are mapped through $x^* = y^*/t^*$.

Dinkelbach's Method for Fractional Programming

An alternative and widely used approach to fractional programming is *Dinkelbach's method*, which converts a fractional optimization problem into a sequence of parameterized convex optimization problems. In contrast to the Charnes–Cooper transformation, which produces a single equivalent convex reformulation, Dinkelbach's method yields an iterative procedure based on a fundamental equivalence between the original fractional problem and a family of difference-form problems.

Consider the fractional optimization problem

$$\min_{x \in \mathcal{F}} \frac{f(x)}{g(x)}, \tag{4.71}$$

where $\mathcal{F} \subseteq \mathbb{R}^n$ is a nonempty compact convex set, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a convex function, and $g : \mathbb{R}^n \rightarrow \mathbb{R}_{++}$ is a positive continuous function.

For each scalar parameter $\lambda \in \mathbb{R}$, define the auxiliary optimization problem

$$\phi(\lambda) := \min_{x \in \mathcal{F}} \{f(x) - \lambda g(x)\}. \tag{4.72}$$

For fixed λ , problem (4.72) is a convex optimization problem whenever f is convex and g is affine or convex.

The parametric family (4.72) encodes the original problem completely. Let $\lambda^* := \min_{x \in \mathcal{F}} \frac{f(x)}{g(x)}$. Then the following statements hold:

- (i) λ^* is the unique scalar satisfying

$$\phi(\lambda^*) = 0. \tag{4.73}$$

- (ii) A point $x^* \in \mathcal{F}$ is an optimal solution of (4.71) if and only if it minimizes (4.72) at $\lambda = \lambda^*$.

Thus, solving the original fractional problem is equivalent to finding the unique root of the scalar function $\phi(\lambda)$.

Motivated by this equivalence, Dinkelbach's method generates a sequence $\{x^{(k)}, \lambda^{(k)}\}$ according to

$$x^{(k)} \in \arg \min_{x \in \mathcal{F}} \{f(x) - \lambda^{(k)} g(x)\}, \tag{4.74}$$

$$\lambda^{(k+1)} := \frac{f(x^{(k)})}{g(x^{(k)})}. \tag{4.75}$$

The iteration is terminated when $|f(x^{(k)}) - \lambda^{(k)}g(x^{(k)})|$ falls below a prescribed tolerance.

The method has strong convergence guarantees. If f is convex, g is positive and affine, and $\mathcal{F}$ is compact, then the iterates $\{\lambda^{(k)}\}$ decrease monotonically to λ^* , and under mild regularity conditions the convergence is superlinear. Each step pays for this only with the solution of one convex problem of the form (4.72), so the nonconvexity induced by the ratio is dissolved into a short sequence of convex difference problems.

Two contrasts with the Charnes–Cooper transformation are worth drawing. Dinkelbach’s method reaches the exact optimum without introducing auxiliary variables or a lifted formulation, which makes it attractive when the feasible set $\mathcal{F}$ is complicated or when a change of variables would distort it; Charnes–Cooper, by contrast, produces a single convex program in one shot but at the cost of reparameterizing the domain. The choice between them is thus a trade-off between solving one larger reformulated problem and solving a handful of smaller ones.

4.6.3 ■ Decomposition

Many large-scale optimization problems possess internal structure that can be exploited to simplify analysis and computation. A common theme is *decomposition*: rewriting a problem so that it can be solved by addressing smaller subproblems, either sequentially or in parallel. Decomposition plays a central role in distributed optimization, network resource allocation, and large-scale learning.

We distinguish between *primal (sequential) decomposition*, which eliminates variables directly, and *dual decomposition*, which relaxes coupling constraints via Lagrange multipliers.

Primal (Sequential) Decomposition

Example 4.84 Sequential minimization.

Consider the optimization problem

$$\begin{aligned} \min_{x_1, x_2} \quad & f(x_1, x_2) \\ \text{s.t.} \quad & g(x_1) \leq 0, \\ & \tilde{g}(x_2) \leq 0. \end{aligned} \tag{4.76}$$

Define the *partial minimization* with respect to x_2 :

$$\tilde{f}(x_1) := \min_{x_2} f(x_1, x_2) \quad \text{s.t.} \quad \tilde{g}(x_2) \leq 0. \tag{4.77}$$

Provided the minimum in (4.77) is attained for each x_1 of interest, the original problem (4.76) is equivalent to

$$\min_{x_1} \tilde{f}(x_1) \quad \text{s.t.} \quad g(x_1) \leq 0. \tag{4.78}$$

Thus, the original problem is reduced to a sequence of smaller problems: an inner optimization over x_2 , followed by an outer optimization over x_1 . $\square$

Remark 4.85. If f is jointly convex in (x_1, x_2) and $\tilde{g}$ defines a convex feasible set, then the

partially minimized function $\tilde{f}$ is convex in x_1 . Hence, primal decomposition preserves convexity.

Dual Decomposition

Primal decomposition is effective when variables can be separated directly. However, many problems are coupled through shared constraints. In such cases, *dual decomposition* provides a powerful alternative by relaxing the coupling constraint using Lagrange multipliers.

Example 4.86 Dual decomposition via a coupling constraint.

Consider the convex optimization problem

$$\begin{aligned} \min_{\{x_i\}_{i=1}^N} \quad & \sum_{i=1}^N f_i(x_i) \\ \text{s.t.} \quad & \sum_{i=1}^N x_i \leq C, \end{aligned} \tag{4.79}$$

where each $f_i : \mathbb{R} \rightarrow \mathbb{R}$ is convex and $C > 0$ is a shared resource constraint.

Introduce a Lagrange multiplier $\lambda \geq 0$ associated with the coupling constraint. The Lagrangian is

$$\mathcal{L}(x_1, \dots, x_N, \lambda) = \sum_{i=1}^N f_i(x_i) + \lambda \left(\sum_{i=1}^N x_i - C \right).$$

For a fixed λ , the Lagrangian separates as

$$\mathcal{L}(x_1, \dots, x_N, \lambda) = \sum_{i=1}^N (f_i(x_i) + \lambda x_i) - \lambda C.$$

The dual function is therefore

$$q(\lambda) = \sum_{i=1}^N \underbrace{\inf_{x_i} (f_i(x_i) + \lambda x_i)}_{\text{independent subproblem}} - \lambda C.$$

The dual problem is

$$\max_{\lambda \geq 0} q(\lambda). \tag{4.80}$$

Each subproblem involves only a single variable x_i and can be solved independently. Coordination between subsystems occurs solely through the shared dual variable λ . $\square$

Remark 4.87. Dual decomposition converts a globally coupled primal problem into a collection of decoupled subproblems linked by a low-dimensional dual variable. This structure underlies many distributed algorithms, including price-based resource allocation, congestion control, and

network utility maximization.

Remark 4.88. When Slater’s condition holds for (4.79), strong duality guarantees that solving the dual problem (4.80) recovers the primal optimum. In this case, primal and dual decomposition are exact, not relaxations.

4.6.4 ■ Convex Relaxation

When an exact convex reformulation of a nonconvex problem is not available, convexification may instead proceed through *relaxation*: the feasible set or objective is enlarged to a convex superset or lowered to a convex underestimator. The resulting problem is convex but no longer equivalent to the original; for a minimization problem its optimal value is a *lower bound* on the true optimum. Such a bound is useful in two ways even when it is loose—it quantifies how far any candidate solution can be from optimal, and it drives the bounding step of branch-and-bound—while in the fortunate case that the relaxed solution turns out to be feasible for the original problem, the bound is attained and the relaxation is *exact*, delivering a genuine global optimizer.

The three subsections that follow organize the main strategies by *what* they relax: lifting enlarges the feasible description by dropping a rank constraint, convex envelopes replace the objective by its tightest convex minorant, and dual relaxations sidestep the primal entirely by optimizing the always-convex dual. All three are central to global optimization, approximation algorithms, and control, where exact solutions are often intractable.

Lifting and rank relaxation.

Quadratic and polynomial optimization problems can often be lifted to a higher-dimensional space in which objectives and constraints become linear. Nonconvexity is then isolated in rank constraints, which are subsequently dropped to obtain a convex relaxation.

Example 4.89 Quadratically constrained quadratic program.

Consider the nonconvex problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & x^\top Q x \\ \text{subject to} \quad & x^\top A_i x \leq b_i, \quad i = 1, \dots, m, \end{aligned} \tag{4.81}$$

where Q and A_i are symmetric but not necessarily positive semidefinite. It is a *quadratically constrained quadratic program* (QCQP). In general, such problems are nonconvex when either the objective matrix Q or one or more constraint matrices A_i fail to be positive semidefinite. As a consequence, the feasible set defined by the quadratic inequalities may be nonconvex, and the problem is, in general, NP-hard.

The lifting transformation $X = xx^\top$ maps the original vector variable $x \in \mathbb{R}^n$ into a matrix variable $X \in \mathbb{S}^n$, where $\mathbb{S}^n$ denotes the space of symmetric $n \times n$ matrices. Under this transformation, quadratic forms become linear: $x^\top Q x = \text{tr}(Qxx^\top) = \text{tr}(QX)$ and $x^\top A_i x = \text{tr}(A_i X)$. Thus, apart from the structural constraint $X = xx^\top$, the objective and constraints become affine functions of X .

The condition $X = xx^\top$ is equivalently characterized by the two constraints $X \succeq 0$ and $\text{rank}(X) = 1$. The positive semidefiniteness ensures that X can be factored as $X = YY^\top$ for some Y , while the rank-one constraint enforces that this factorization

corresponds to a single vector x rather than a higher-dimensional embedding. It is precisely the rank constraint that renders the lifted formulation nonconvex.

The *Shor relaxation* is obtained by dropping the rank constraint and retaining only the convex condition $X \succeq 0$. The resulting problem is a semidefinite program (SDP):

$$\begin{aligned} \min_{X \succeq 0} \quad & \text{tr}(QX) \\ \text{subject to} \quad & \text{tr}(A_i X) \leq b_i, \quad i = 1, \dots, m, \end{aligned}$$

which can be solved efficiently using interior-point or first-order methods.

Since the feasible set of the relaxed SDP strictly contains the set of rank-one matrices $xx^\top$, the optimal value of the relaxation provides a *lower bound* on the optimal value of the original QCQP. If the SDP solution happens to have rank one, then the relaxation is said to be *exact*, and the corresponding vector x can be recovered as a global optimizer of the original problem.

From a geometric perspective, the Shor relaxation replaces the nonconvex set $\{xx^\top : x \in \mathbb{R}^n\}$ with its convex hull intersected with the positive semidefinite cone. This relaxation therefore plays a role analogous to convex envelopes in finite dimensions: it yields the tightest convex approximation obtainable through lifting and linearization of quadratic forms.

Shor relaxations are central to global optimization and approximation algorithms for nonconvex quadratic problems. They form the basis of polynomial-time approximation schemes in combinatorial optimization (e.g., Max-Cut), provide certificates of optimality gaps, and serve as bounding subproblems within branch-and-bound frameworks. In optimal control and signal processing, related relaxations arise in stability analysis, Lyapunov synthesis, and robust control via linear matrix inequalities.

□

Lifting relaxes the *feasible description* by dropping a rank constraint. A complementary strategy leaves the feasible set alone and instead relaxes the *function* being optimized, replacing a nonconvex term by the tightest convex function beneath it. This is the idea of a convex envelope.

Convex envelopes.

Given a nonconvex function defined on a compact domain, its *convex envelope* is the tightest convex function that underestimates it on that domain. Replacing the original function by its convex envelope yields the strongest possible convex relaxation over the specified region.

Example 4.90 Bilinear term relaxation.

The bilinear function $f(x, y) = xy$ is neither convex nor concave on a rectangular domain unless one of the variables is fixed. Geometrically, its graph over $[\underline{x}, \bar{x}] \times [\underline{y}, \bar{y}]$ is a saddle surface, and local optimality conditions alone are insufficient to characterize global behavior.

The *convex envelope* of f over a given domain is defined as the largest convex function that underestimates f everywhere on that domain. Equivalently, it is the pointwise supremum of all convex functions that lie below f . Replacing a nonconvex function by its convex envelope yields the tightest possible convex relaxation of the original problem.

For a bilinear term on a rectangular domain, the convex envelope admits a closed polyhedral representation. The key observation is that the graph of $f(x, y) = xy$ over the rectangle is completely characterized by its values at the four extreme points

$$(\underline{x}, \underline{y}), (\underline{x}, \bar{y}), (\bar{x}, \underline{y}), (\bar{x}, \bar{y}).$$

Each of the McCormick inequalities corresponds to an affine function whose graph is a supporting hyperplane to the bilinear surface at two of these corner points. Taken together, the four inequalities describe the convex hull of the set

$$\{(x, y, z) \in \mathbb{R}^3 : z = xy, x \in [\underline{x}, \bar{x}], y \in [\underline{y}, \bar{y}]\}.$$

Introducing an auxiliary variable $z = xy$, the nonlinear equality constraint $z = xy$ is replaced by the linear constraints

$$\begin{aligned} z &\geq \underline{x}\underline{y} + \bar{x}\bar{y} - \bar{x}\underline{y}, \\ z &\geq \bar{x}\underline{y} + \underline{x}\bar{y} - \underline{x}\underline{y}, \\ z &\leq \underline{x}\underline{y} + \bar{x}\bar{y} - \underline{x}\bar{y}, \\ z &\leq \bar{x}\underline{y} + \underline{x}\bar{y} - \bar{x}\underline{y}. \end{aligned}$$

The resulting feasible set in (x, y, z) -space is a polyhedron that coincides with the convex envelope of the bilinear graph over the rectangular domain.

This relaxation is exact at the boundary points of the rectangle and becomes progressively tighter as the variable bounds are tightened. In global optimization algorithms such as branch-and-bound and spatial branch-and-cut, the domain is recursively partitioned, and the corresponding McCormick relaxations are updated at each node. In the limit, as the domain shrinks to a single point, the relaxation converges to the original bilinear constraint.

From a modeling perspective, McCormick relaxations provide a systematic mechanism for embedding nonconvex bilinear terms into mixed-integer linear or convex optimization frameworks, thereby enabling global solution methods while preserving tightness guarantees. $\square$

Lifting and convex envelopes both act on the primal description of the problem. The third strategy is more indirect: rather than convexifying the primal at all, it passes to the dual, which is convex automatically. Nonconvexity is not removed but *transferred* to the dual variables, where it becomes harmless.

Dual relaxations.

Lagrangian duality provides a systematic mechanism for convex relaxation by transferring nonconvexity from the primal variables to dual variables. Even when the primal problem is nonconvex, the dual problem is always convex and yields a lower bound on the primal optimal value.

Example 4.91 Nonconvex quadratic minimization.

The optimization problem

$$\min_{x \in \mathbb{R}^n} x^\top Q x \quad \text{subject to} \quad \|x\|_2^2 \leq 1$$

is a special case of a *trust-region subproblem*. When the matrix Q is indefinite, the objective function is nonconvex and admits directions of negative curvature. As a result, classical convexity-based arguments do not apply, and local optimality conditions alone are insufficient to characterize the global solution.

Despite this nonconvexity, the problem possesses a remarkable hidden convex structure. To reveal it, consider the Lagrangian

$$\mathcal{L}(x, \lambda) = x^\top Q x + \lambda(x^\top x - 1), \quad \lambda \geq 0.$$

For a fixed multiplier λ , the Lagrangian is a quadratic form in x :

$$\mathcal{L}(x, \lambda) = x^\top (Q + \lambda I) x - \lambda.$$

The dual function is obtained by minimizing $\mathcal{L}(x, \lambda)$ with respect to x . This minimum is finite if and only if $Q + \lambda I \succeq 0$, in which case the minimum is attained at $x = 0$ and equals $-\lambda$. If $Q + \lambda I$ is not positive semidefinite, the quadratic form is unbounded below and the dual function equals $-\infty$.

The dual problem therefore takes the form

$$\begin{aligned} \max_{\lambda \geq 0} \quad & -\lambda \\ \text{subject to} \quad & Q + \lambda I \succeq 0. \end{aligned}$$

The constraint $Q + \lambda I \succeq 0$ is equivalent to $\lambda \geq -\lambda_{\min}(Q)$, where $\lambda_{\min}(Q)$ denotes the smallest eigenvalue of Q . Hence the dual problem reduces to $\max_{\lambda \geq -\lambda_{\min}(Q)} (-\lambda)$, whose optimal solution is $\lambda^* = -\lambda_{\min}(Q)$ and $d^* = \lambda_{\min}(Q)$.

On the primal side, observe that minimizing $x^\top Q x$ over the unit ball is equivalent to minimizing it over the unit sphere whenever Q has at least one negative eigenvalue. Indeed, along the eigenvector $v_{\min}$ associated with $\lambda_{\min}(Q)$, $x^\top Q x = \lambda_{\min}(Q) \|x\|^2$, and the minimum is achieved at $x = v_{\min}$ with $\|x\| = 1$. Therefore, the primal optimal value satisfies $p^* = \lambda_{\min}(Q)$.

Since $p^* = d^*$, strong duality holds despite the nonconvexity of the primal problem. Moreover, the dual problem is convex: it consists of maximizing a linear function subject to a linear matrix inequality, or equivalently, minimizing the maximum eigenvalue of a shifted matrix. This example illustrates that certain nonconvex quadratic

problems admit exact convex dual formulations, and that eigenvalue optimization naturally emerges as the dual counterpart of trust-region-type constraints.

From a systems and control perspective, this phenomenon underlies many classical results. The appearance of eigenvalue constraints mirrors Lyapunov stability conditions, while the exactness of the relaxation explains why nonconvex energy minimization problems can often be solved reliably via convex optimization tools.

□

More generally, dual relaxations are exact for important problem classes such as trust-region subproblems and certain QCQPs, while providing tight bounds for broader classes of nonconvex problems.

Convex relaxations replace a difficult nonconvex feasible set with a convex set that contains it, or replace a nonconvex objective with a convex underestimator. The resulting solution provides a certificate, either as a bound or as an approximate solution, that can be refined or combined with other methods.

4.7 ■ Hahn-Banach Theorems

The Hahn–Banach theorem provides the fundamental analytic mechanism underlying geometric separation results for convex sets. In this section, we make this connection precise by showing how convex sets give rise to sublinear functionals and how Hahn–Banach extensions yield separating hyperplanes.

Definition 4.92 (Sublinear functional). *Let X be a real vector space. A function $p : X \rightarrow \mathbb{R}$ is called sublinear if it satisfies:*

- (i) Positive homogeneity: $p(\lambda x) = \lambda p(x), \forall x \in X, \forall \lambda \geq 0$,
- (ii) Subadditivity: $p(x + y) \leq p(x) + p(y), \forall x, y \in X$.

Sublinear functionals generalize norms by allowing asymmetry and degeneracy. They measure size or cost in a way that is compatible with convex combinations and scaling, but do not necessarily arise from inner products.

Example 4.93 Norms are sublinear.

Let $(X, \|\cdot\|)$ be a normed linear space. The norm function $p(x) := \|x\|$ is sublinear. Positive homogeneity follows from the definition of a norm, and subadditivity is precisely the triangle inequality. □

Example 4.94 Maximum of linear functionals.

Let $X = \mathbb{R}^n$ and let $\{a_1, \dots, a_m\} \subset \mathbb{R}^n$. Define $p(x) := \max_{1 \leq i \leq m} a_i^\top x$. Then p is sublinear. This follows from linearity of each map $x \mapsto a_i^\top x$ and the fact that the

pointwise maximum of linear functionals is sublinear. □

Example 4.95 Support function of a convex set.

Let $K \subset \mathbb{R}^n$ be a nonempty compact convex set. Define $p(x) := \sup_{y \in K} x^\top y$. Then p is sublinear. This function is known as the *support function* of K and plays a central role in convex geometry and duality theory. □

Why sublinear functionals encode convex sets. Let X be a real vector space and let $C \subset X$ be a nonempty convex set containing the origin. The geometry of C can be encoded by a function that measures how much C must be dilated in order to contain a given vector.

This leads to the *Minkowski functional* (or *gauge*) of C , defined by $p_C(x) := \inf\{\lambda > 0 : x \in \lambda C\}$.

The function p_C assigns to each $x \in X$ the smallest scaling factor required to include x in the set C . Thus, $p_C(x) \leq 1$ if and only if $x \in C$.

The structural properties of C are reflected directly in the algebraic properties of p_C . Convexity of C implies that p_C is subadditive, while containment of the origin implies positive homogeneity. Consequently, p_C is a sublinear functional.

Before pursuing this correspondence we record the elementary fact that sublinearity is nothing other than convexity together with positive homogeneity.

Proposition 4.96 (Sublinear = convex + positively homogeneous). *Let X be a real vector space and let $p : X \rightarrow \mathbb{R}$. The following are equivalent:*

- (1) p is sublinear;
- (2) p is convex and positively homogeneous.

Proof. ($1 \Rightarrow 2$) Assume that p is sublinear. Let $x, y \in X$ and $\theta \in [0, 1]$. By subadditivity and positive homogeneity,

$$p(\theta x + (1 - \theta)y) \leq p(\theta x) + p((1 - \theta)y) = \theta p(x) + (1 - \theta)p(y),$$

which shows that p is convex. Positive homogeneity holds by assumption.

($2 \Rightarrow 1$) Assume that p is convex and positively homogeneous. For any $x, y \in X$, convexity with $\theta = \frac{1}{2}$ gives

$$p\left(\frac{x+y}{2}\right) \leq \frac{1}{2}p(x) + \frac{1}{2}p(y).$$

Using positive homogeneity,

$$p(x + y) = 2p\left(\frac{x+y}{2}\right) \leq p(x) + p(y),$$

which establishes subadditivity. Positive homogeneity is given, hence p is sublinear. □

4.7.1 ■ Equivalence Between Convex Sets and Sublinear Functionals

The relationship between convex geometry and sublinear analysis is made precise through the notion of the Minkowski functional. In this sense, convex sets containing the origin and sublinear functionals provide dual representations of the same structure.

Definition 4.97 (Minkowski functional). Let X be a real vector space and let $C \subset X$ be a nonempty convex set containing the origin. The Minkowski functional (or gauge) of C is defined by $p_C(x) := \inf\{\lambda > 0 : x \in \lambda C\}, x \in X$.

Proposition 4.98 (Gauge representation of convex sets). Let $C \subset \mathbb{R}^n$ be a closed convex set with $0 \in \text{int}(C)$. Then the Minkowski functional p_C is finite, nonnegative and sublinear, and $C = \{x \in \mathbb{R}^n : p_C(x) \leq 1\}$.

Remark 4.99. Neither hypothesis is decorative. If C is the open unit ball then $p_C(x) = \|x\|_2$ and $\{p_C \leq 1\}$ is the closed ball, so the set identity fails; closedness is what repairs it. If 0 is merely in C rather than in its interior—say $C = \{(x, y) : y = 0, x \geq 0\} \subset \mathbb{R}^2$ —then p_C takes the value $+\infty$ off the ray and is no longer a real-valued functional.

Proof. Since C contains the origin and is convex, it is closed under convex combinations and positive scaling. Positive homogeneity of p_C follows directly from the definition. Subadditivity follows from convexity of C : if $x \in \lambda C$ and $y \in \mu C$, then $x + y \in (\lambda + \mu)C$, which implies $p_C(x + y) \leq \lambda + \mu$. Taking infima over all such λ, μ yields subadditivity.

Finiteness of p_C follows from $0 \in \text{int}(C)$: some ball $B_r(0)$ lies in C , so $x \in (\|x\|/r)C$ and $p_C(x) \leq \|x\|/r < \infty$. Nonnegativity is immediate from the definition, which takes an infimum over positive λ .

For the set identity, if $x \in C$ then $\lambda = 1$ is admissible and $p_C(x) \leq 1$. Conversely, suppose $p_C(x) \leq 1$ and pick $\lambda_k \downarrow p_C(x)$ with $x \in \lambda_k C$, i.e. $x/\lambda_k \in C$. If $p_C(x) > 0$ then $x/\lambda_k \rightarrow x/p_C(x)$, and closedness of C places $x/p_C(x)$ in C ; since $p_C(x) \leq 1$ and C is convex with $0 \in C$, the point $x = p_C(x) \cdot (x/p_C(x)) + (1 - p_C(x)) \cdot 0$ lies in C as well. If $p_C(x) = 0$ then $x/\lambda_k \in C$ with $\lambda_k \downarrow 0$, so $x = \lambda_k(x/\lambda_k) \rightarrow 0 \in C$, and convexity with $0 \in C$ again gives $x \in C$. This is precisely the step at which closedness is used. $\square$

Proposition 4.100 (Convex sets induced by sublinear functionals). Let $p : X \rightarrow \mathbb{R}$ be a sublinear functional. Define $C_p := \{x \in X : p(x) \leq 1\}$. Then C_p is a convex set containing the origin.

Proof. Since $p(0) = 0$, the origin belongs to C_p . Convexity follows from subadditivity and positive homogeneity: for any $x, y \in C_p$ and $\theta \in [0, 1]$,

$$p(\theta x + (1 - \theta)y) \leq \theta p(x) + (1 - \theta)p(y) \leq 1.$$

Thus $\theta x + (1 - \theta)y \in C_p$. $\square$

Propositions 4.98 and 4.100 establish a one-to-one correspondence between convex sets containing the origin and sublinear functionals, up to normalization. In this sense, convex geometry and sublinear analysis provide equivalent descriptions of the same underlying objects.

Theorem 4.101 (Equivalence between convex bodies and nonnegative sublinear functionals). The assignments $C \mapsto p_C$ and $p \mapsto C_p := \{x : p(x) \leq 1\}$ are mutually inverse bijections

between

- the closed convex sets $C \subset \mathbb{R}^n$ with $0 \in \text{int}(C)$, and
- the finite nonnegative sublinear functionals $p : \mathbb{R}^n \rightarrow [0, \infty)$.

That is, $C_{p_C} = C$ for every such C , and $p_{C_p} = p$ for every such p .

Remark 4.102. Both restrictions are needed. Dropping closedness breaks $C_{p_C} = C$, as the open unit ball shows. Dropping nonnegativity breaks $p_{C_p} = p$: on $\mathbb{R}$ the functional $p(x) = -x$ is sublinear, $C_p = [-1, \infty)$, and $p_{C_p}(x) = \max\{0, -x\}$, which differs from p at every $x > 0$. Nonnegative sublinear functionals are exactly the gauges, and it is with them that the correspondence is exact.

We now illustrate this correspondence with several canonical examples.

Example 4.103 Euclidean unit ball and norm.

Let $X = \mathbb{R}^n$ and define $C := \{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$. Then C is a closed, convex, and centrally symmetric set containing the origin. Its Minkowski functional is $p_C(x) = \inf\{\lambda > 0 : \|x\|_2 \leq \lambda\} = \|x\|_2$. Thus, the Euclidean norm is precisely the Minkowski functional of its unit ball. This example shows that norms arise as gauges of convex, balanced sets. $\square$

Example 4.104 Asymmetric convex set and sublinear gauge.

Let $X = \mathbb{R}^2$ and define $C := \{(x, y) \in \mathbb{R}^2 : x \leq 1, |y| \leq 1\}$. The set C is convex and contains the origin, but is not symmetric about the origin. Its Minkowski functional is given by

$$p_C(x, y) = \inf\{\lambda > 0 : (x, y) \in \lambda C\} = \max\{x, |y|\}.$$

The function p_C is sublinear but not symmetric, and therefore does not define a norm. This example illustrates that sublinear functionals strictly generalize norms and faithfully encode asymmetries in the geometry of the underlying convex set. $\square$

Example 4.105 Polyhedral gauge.

Let $X = \mathbb{R}^n$ and define $C := \{x \in \mathbb{R}^n : Ax \leq \mathbf{1}\}$, where $A \in \mathbb{R}^{m \times n}$ and inequalities are interpreted componentwise. Then C is a convex polyhedron containing the origin. The associated Minkowski functional satisfies

$$p_C(x) = \inf\{\lambda > 0 : Ax \leq \lambda \mathbf{1}\} = \max\left\{0, \max_{1 \leq i \leq m} (A_i x)\right\},$$

where A_i denotes the i th row of A . Hence, p_C is the pointwise maximum of finitely many linear functionals together with the zero functional, and is therefore sublinear. The zero term cannot be omitted: a gauge is nonnegative by construction, whereas

$\max_i (A_i x)$ is negative for any x with $Ax < 0$, such as $x = 0$ perturbed along a recession direction of C . $\square$

4.7.2 ■ Hahn–Banach Theorem (Extension Form)

Before stating the Hahn–Banach theorem, we present a concrete finite-dimensional example whose purpose is purely intuitive. In $\mathbb{R}^n$, every linear functional can be written explicitly as an inner product, which allows the extension guaranteed by Hahn–Banach to be visualized directly.

Example 4.106 Explicit Hahn–Banach extension in $\mathbb{R}^2$.

Let $X = \mathbb{R}^2$ equipped with the Euclidean norm $\|\mathbf{v}\|_2 = \sqrt{v_1^2 + v_2^2}$. Define the sublinear functional $p(\mathbf{v}) := \|\mathbf{v}\|_2$. The sublevel set $\{\mathbf{v} \in \mathbb{R}^2 : p(\mathbf{v}) \leq 1\}$ is the closed unit disk.

Let $M := \{(\alpha, 0) : \alpha \in \mathbb{R}\}$ be the x -axis, and define a linear functional $f : M \rightarrow \mathbb{R}$, $f(\alpha, 0) := \alpha$. For every $\mathbf{m} = (\alpha, 0) \in M$, we have $f(\mathbf{m}) = \alpha \leq |\alpha| = \|(\alpha, 0)\|_2 = p(\mathbf{m})$, so the domination condition $f \leq p$ holds on M .

Since X is finite-dimensional, every linear functional $F : \mathbb{R}^2 \rightarrow \mathbb{R}$ admits a representation as an inner product with a fixed vector. That is, there exists $\mathbf{y} \in \mathbb{R}^2$ such that $F(\mathbf{v}) = \mathbf{y}^\top \mathbf{v}$, $\forall \mathbf{v} \in \mathbb{R}^2$.

The extension requirement $F|_M = f$ uniquely determines $\mathbf{y}$. Indeed, for all $\alpha \in \mathbb{R}$, $\mathbf{y}^\top(\alpha, 0) = \alpha$, which implies $\mathbf{y} = (1, 0)$. Consequently, the extended functional is $F(x_1, x_2) = (1, 0)^\top(x_1, x_2) = x_1$.

This functional satisfies:

- (i) $F|_M = f$, since $F(\alpha, 0) = \alpha$,
- (ii) $F(\mathbf{v}) \leq \|\mathbf{v}\|_2$ for all $\mathbf{v} \in \mathbb{R}^2$, by the Cauchy–Schwarz inequality.

Geometric interpretation. The functional $F(\mathbf{v}) = x_1$ defines the supporting line $\{\mathbf{v} \in \mathbb{R}^2 : x_1 = 1\}$ to the unit disk at the boundary point $(1, 0)$. This example illustrates explicitly how the Hahn–Banach theorem extends a linear functional defined on a subspace to a global supporting hyperplane in a finite-dimensional space. $\square$

We now state the general extension theorem.

Theorem 4.107 (Hahn–Banach Theorem). *Let X be a real normed linear space and let $p : X \rightarrow \mathbb{R}$ be a continuous sublinear functional. Let $M \subset X$ be a linear subspace and $f : M \rightarrow \mathbb{R}$ a linear functional satisfying $f(m) \leq p(m) \forall m \in M$. Then there exists a linear functional $F : X \rightarrow \mathbb{R}$ extending f such that $F(x) \leq p(x) \forall x \in X$.*

The Hahn–Banach theorem asserts that any linear inequality compatible with a convex (sub-linear) constraint on a subspace can be extended globally. This extension principle is the analytic

core from which supporting hyperplane and separation theorems for convex sets follow.

4.7.3 ■ Separation of a Point and a Convex Set

We now show how Hahn–Banach yields a geometric separation theorem.

Theorem 4.108 (Separation by a hyperplane). *Let X be a real normed linear space and let $C \subset X$ be a nonempty closed convex set. If $x_0 \notin C$, then there exists a continuous linear functional $F \in X^*$ and a real number α such that $F(x_0) > \alpha \geq F(x) \forall x \in C$.*

Proof. The gauge machinery of the previous subsection requires a convex set that contains the origin in its interior, so the translation must be arranged with that in mind: translating by x_0 would produce a set missing the origin altogether, and its Minkowski functional would not be defined.

Since C is closed and $x_0 \notin C$, the distance $d := \inf_{c \in C} \|x_0 - c\|$ is strictly positive. Fix any $c_0 \in C$ and set

$$K := (C + B(0, d/2)) - c_0, \quad z := x_0 - c_0,$$

where $B(0, r)$ denotes the open ball of radius r . Then K is convex and open, and $0 \in K$ because $c_0 \in C$; in particular $0 \in \text{int}(K)$, so the Minkowski functional $p := p_K$ is finite and sublinear. Moreover $z \notin K$: any point of $C + B(0, d/2)$ is within $d/2$ of C , whereas x_0 is at distance d from C . Since K is open, $K = \{p < 1\}$, and therefore $p(z) \geq 1$.

Define f on the one-dimensional subspace $M := \text{span}\{z\}$ by $f(\lambda z) := \lambda p(z)$. Then $f \leq p$ on M : for $\lambda \geq 0$ positive homogeneity gives $f(\lambda z) = p(\lambda z)$, while for $\lambda < 0$ we have $f(\lambda z) < 0 \leq p(\lambda z)$. By the Hahn–Banach theorem f extends to a linear $F : X \rightarrow \mathbb{R}$ with $F \leq p$ everywhere. Because K contains a ball $B(0, r)$, we get $p(x) \leq \|x\|/r$ and hence F is bounded, so $F \in X^*$; also $F \neq 0$ since $F(z) = p(z) \geq 1$.

Finally we read off the separation. For $x \in C$ and $\|b\| < d/2$ the point $x + b - c_0$ lies in K , so $F(x - c_0) + F(b) \leq 1$. Taking the supremum over such b gives $F(x - c_0) \leq 1 - \frac{d}{2}\|F\|$ for every $x \in C$, whereas $F(z) = F(x_0 - c_0) \geq 1$. Since $\|F\| > 0$, setting $\alpha := F(c_0) + 1 - \frac{d}{2}\|F\|$ yields $F(x) \leq \alpha < F(x_0)$ for all $x \in C$, which is the asserted strict separation. $\square$

Consequently, in finite-dimensional convex optimization, every separating hyperplane takes the familiar form $\langle y, x \rangle = \alpha$, where $y \in \mathbb{R}^n$ is a normal vector to the hyperplane and $\alpha \in \mathbb{R}$ is a scalar offset.

Applying the separation theorem in $\mathbb{R}^n$, we obtain the following specialized result: if $C \subset \mathbb{R}^n$ is a nonempty closed convex set and $x_0 \notin C$, then there exist $y \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$ such that $\langle y, x_0 \rangle > \alpha \geq \langle y, x \rangle \forall x \in C$.

The hyperplane $\{x \in \mathbb{R}^n : \langle y, x \rangle = \alpha\}$ separates the point x_0 from the convex set C , with y acting as the outward normal. Thus, in finite dimensions, the abstract Hahn–Banach separation theorem reduces to the existence of a supporting or separating hyperplane expressed by an inner product.

Strict versus non-strict separation. The strict inequality $F(x_0) > \alpha \geq F(x) \forall x \in C$ expresses a *strict separation* between the point x_0 and the convex set C . Whether strict separation is possible depends on the geometric and topological properties of C and the location of x_0 .

If C is closed and convex and $x_0 \notin C$, then strict separation is always possible. Intuitively, closedness ensures that x_0 lies at a positive distance from C , allowing the separating hyperplane to be placed with a strict gap. Formally, the Hahn–Banach separation theorem guarantees the existence of a functional F such that $\sup_{x \in C} F(x) < F(x_0)$.

If C is convex but not closed, or if x_0 lies in the closure $\overline{C}$, then strict separation may fail. In this case, one can generally only guarantee the existence of $F \in X^*$ and $\alpha \in \mathbb{R}$ such that $F(x_0) \geq \alpha \geq F(x) \forall x \in C$, with equality possibly holding at x_0 or along a boundary sequence.

Geometrically, this corresponds to a supporting hyperplane that *touches* the closure of C at x_0 or asymptotically approaches it, leaving no room for a strict gap.

Boundary points. If $x_0 \in \partial C$, then separation is necessarily non-strict: $F(x_0) = \alpha \geq F(x) \forall x \in C$. In this case, F defines a *supporting hyperplane* to C at the boundary point x_0 .

4.7.4 ■ Separation of Two Convex Sets

Once separation between a point and a convex set is established, the result can be extended to separation between two convex sets. This generalization captures the fundamental geometric principle underlying duality and feasibility certification in convex analysis.

Theorem 4.109 asserts that two disjoint convex sets can be separated by a hyperplane. The linear functional F defines a hyperplane $\{x \in X : F(x) = \alpha\}$ such that C_1 lies entirely on one side of the hyperplane and C_2 lies on the other. This is referred to as *weak (or non-strict) separation*, since equality may occur for boundary points of the sets.

Theorem 4.109 (Separation of two convex sets). *Let X be a real normed linear space and let $C_1, C_2 \subset X$ be nonempty convex sets with $C_1 \cap C_2 = \emptyset$, and suppose that at least one of them has nonempty interior (which is automatic when $X = \mathbb{R}^n$ and the sets are full-dimensional). Then there exists a nonzero continuous linear functional $F \in X^*$ and a scalar $\alpha \in \mathbb{R}$ such that $F(x) \leq \alpha \leq F(y) \forall x \in C_1, \forall y \in C_2$.*

Remark 4.110. Some interiority hypothesis is genuinely needed in infinite dimensions: there exist disjoint nonempty convex subsets of an infinite-dimensional normed space that cannot be separated by any nonzero *continuous* linear functional. In $\mathbb{R}^n$ no such hypothesis is required, because disjoint convex sets can always be separated there.

The separation of a point from a convex set is a special case of Theorem 4.109. Indeed, by taking $C_2 = \{x_0\}$ and $C_1 = C$, the condition $C_1 \cap C_2 = \emptyset$ reduces to $x_0 \notin C$, and the conclusion recovers the point-set separation theorem.

A standard proof reduces the problem to separating the origin from a convex set. Define the difference set $C := C_1 - C_2 = \{x - y : x \in C_1, y \in C_2\}$. The set C is convex and does not contain the origin. If, say, C_1 has nonempty interior, then so does C , and a supporting hyperplane to C at 0 supplies a functional whose inequality rearranges to the stated separation between C_1 and C_2 . Note that the closed-set version proved above cannot be applied directly, since $C_1 - C_2$ need not be closed even when both sets are; this is exactly why the interiority hypothesis appears in the statement.

Strict separation. In general, the inequalities in Theorem 4.109 cannot be made strict. However, if additional regularity assumptions hold, such as one set being compact and the other closed, then there exist $\alpha_1 < \alpha_2$ such that $F(x) \leq \alpha_1 < \alpha_2 \leq F(y) \forall x \in C_1, \forall y \in C_2$. This corresponds to a hyperplane separating the two sets with a positive gap.

Finite-dimensional special case. If $X = \mathbb{R}^n$, then every continuous linear functional $F \in X^*$ can be written as $F(x) = \langle y, x \rangle$ for some vector $y \in \mathbb{R}^n$. In this case, separation is expressed by a hyperplane of the form $\langle y, x \rangle = \alpha$, with y acting as the normal direction.

4.8 ■ Further Reading

The material in this chapter draws on classical results in convex analysis, duality theory, and optimization, as well as important application-driven developments in communications and control. The following references provide authoritative treatments and complementary perspectives.

Saddle-point theory provides the unifying framework behind Lagrangian duality, KKT conditions, and constrained optimization. A classical and rigorous treatment can be found in [79], which develops saddle-point optimality conditions, constraint qualifications, and duality in both convex and nonconvex settings. Closely related geometric and variational perspectives are presented in [102, 119]. For a modern, optimization-focused exposition emphasizing duality and KKT conditions, see [26].

The water-filling solution is a canonical example illustrating dual variables, complementary slackness, and optimal resource allocation. It appears prominently in information theory and communications. Classical derivations are given in [36]. Convex-optimization interpretations of water-filling and its generalizations to network utility maximization are discussed in [26] and in the broader literature on network optimization.

Fractional programming problems arise naturally in energy efficiency, throughput optimization, and economic modeling. The foundational theory, including the Charnes–Cooper transformation and Dinkelbach’s method, is developed in the monograph [112, 128]. Algorithmic and parametric perspectives on fractional programming are also discussed in the literature on quasiconvex optimization and parametric programming.

Duality provides a systematic mechanism for convex relaxation of nonconvex problems, including quadratic and polynomial optimization. The role of duality in trust-region methods and quadratic programming is discussed in [90]. Semidefinite relaxation and duality-based approximation methods are treated in depth in [126]. These ideas also form the foundation of sum-of-squares and moment-based relaxations in polynomial optimization.

4.9 ■ Exercises

Exercise 4.1 (Convex Sets).

(i) Let $C \subset \mathbb{R}^n$ be a convex set, and let $x_1, \dots, x_k \in C$. Suppose $\theta_1, \dots, \theta_k \in \mathbb{R}$ satisfy $\theta_i > 0$ and $\theta_1 + \dots + \theta_k = 1$. Show that $\theta_1 x_1 + \dots + \theta_k x_k \in C$.

(ii) Let $C \subset \mathbb{R}^n$ be the solution set of the quadratic inequality

$$C = \{x \in \mathbb{R}^n \mid x^\top A x + b^\top x + c \leq 0\},$$

where $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$. Show that C is convex if A is positive semidefinite.

(iii) Show that the hyperbolic set

$$\{x \in \mathbb{R}_+^2 \mid x_1 x_2 \geq 1\}$$

is convex. As a generalization, show that the set

$$\left\{x \in \mathbb{R}_+^n \mid \prod_{i=1}^n x_i \geq 1\right\}$$

is convex.

Exercise 4.2 (Convex Functions).

(i) Let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an increasing convex function. Show that the composition $f(g(x))$ is a convex function.

(ii) Show that the log-sum-exp function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by

$$f(x) = \log \left(\sum_{i=1}^n e^{x_i} \right)$$

is convex.

(iii) Let $p < 1$, $p \neq 0$. Show that the function $f : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$ defined by

$$f(x) = \left(\sum_{i=1}^n x_i^p \right)^{1/p}$$

is concave.

(iv) Show that the function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $f(x) = \|Ax - b\|$, where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $\|\cdot\|$ is a norm on $\mathbb{R}^m$, is convex.

Exercise 4.3 (Integral Averaging of a Convex Function). Let $\Phi : [0, \infty) \rightarrow \mathbb{R}$ be a convex function, and define $f(t) := \frac{1}{t} \int_0^t \Phi(\alpha) d\alpha$, $t > 0$.

(i) Show that f is well defined for all $t > 0$.

(ii) Prove that f is convex on the interval $(0, \infty)$.

Exercise 4.4 (Failure of Convexity via the Hessian). Consider the function

$$f(\alpha, \beta) = \alpha^\alpha + \beta^\beta - \alpha^\beta - \beta^\alpha, \quad \alpha > 0, \beta > 0.$$

(i) Compute the second-order partial derivative $\frac{\partial^2 f}{\partial \alpha^2}$.

(ii) Show that

$$\frac{\partial^2 f}{\partial \alpha^2}(1, e) = 2 - e^2 < 0.$$

(iii) Conclude that the Hessian matrix of f is not positive semidefinite at $(\alpha, \beta) = (1, e)$.

(iv) Deduce that f is not convex in any neighborhood of $(1, e)$, and hence is not convex in the interior of the first quadrant.

Exercise 4.5 (Infimal Convolution and Convexity). Consider the function $g : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $g(x, y) = \|y\|_2 + \frac{1}{2} \|x - y\|_2^2$.

(i) Find $f(x) := \inf_{y \in \mathbb{R}^n} g(x, y)$.

(ii) Show that the function f is convex on $\mathbb{R}^n$.

Exercise 4.6 (Log-Sum-Exp Optimization and Duality). Let $x, a_i \in \mathbb{R}^n$ for $i = 1, \dots, m$, and let $b_i, y_i \in \mathbb{R}$. Let $p_i \in (0, 1]$, $i = 1, \dots, m$, be weighting factors. Consider the optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \log \left(\sum_{i=1}^m p_i e^{y_i} \right) \\ \text{s.t.} \quad & a_i^\top x + b_i = y_i, \quad i = 1, \dots, m. \end{aligned}$$

- (i) Show that the above problem is a convex optimization problem.
- (ii) Derive the dual problem.
- (iii) Determine whether the dual problem is concave, and justify your answer.

Exercise 4.7 (KKT Conditions). Use the Karush–Kuhn–Tucker (KKT) conditions to solve the following optimization problem and determine the optimal solution $(x_0, x_1, \dots, x_n)$:

$$\begin{aligned} \max_{x_0, x_1, \dots, x_n} \quad & \log(1 + x_0) + \sum_{i=1}^n \log x_i \\ \text{s.t.} \quad & x_0 + x_i \leq 1, \quad i = 1, \dots, n, \\ & x_0, x_1, \dots, x_n \geq 0. \end{aligned}$$

Exercise 4.8 (Entropy-Regularized Transport Duality). Let $\Sigma = \{\Sigma_{ij} \mid i = 1, \dots, n, j = 1, \dots, m\}$ and $C = \{C_{ij} \mid i = 1, \dots, n, j = 1, \dots, m\}$ be real $n \times m$ matrices. Define the inner product

$$\langle \Sigma, C \rangle = \sum_{i=1}^n \sum_{j=1}^m C_{ij} \Sigma_{ij}.$$

Let $a \in \mathbb{R}^n$, $b \in \mathbb{R}^m$, and denote by $\mathbf{1}_n$ and $\mathbf{1}_m$ the vectors of all ones in $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. Consider the optimization problem

$$\begin{aligned} \text{(P1)} \quad \min_{\Sigma} \quad & \langle \Sigma, C \rangle + \lambda \sum_{i=1}^n \sum_{j=1}^m \Sigma_{ij} \log \Sigma_{ij} \\ \text{s.t.} \quad & \Sigma \mathbf{1}_m = a, \\ & \Sigma^\top \mathbf{1}_n = b, \\ & \Sigma_{ij} \geq 0, \quad i = 1, \dots, n, j = 1, \dots, m, \end{aligned}$$

where $\lambda > 0$. Derive the dual problem associated with (P1).

Exercise 4.9 (Quasi-Concavity). Let $f : C \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a function defined on a convex set C . The function f is said to be *quasi-concave* if

$$f(\lambda x_1 + (1 - \lambda)x_2) \geq \min\{f(x_1), f(x_2)\}, \quad \forall x_1, x_2 \in C, \forall \lambda \in [0, 1].$$

- (i) Show that every concave function is quasi-concave.
- (ii) Let $f : C \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable. Show that f is quasi-concave if and only if

$$f(x_2) \leq f(x_1) \implies \nabla f(x_2)^\top (x_1 - x_2) \geq 0, \quad \forall x_1, x_2 \in C.$$

Exercise 4.10 (Bregman Divergence and Mirror Projections). Let $\mathcal{X}$ be a closed convex set and let $f : \mathcal{X} \rightarrow \mathbb{R}$ be a continuously differentiable convex function. The Bregman divergence associated with f is defined by

$$D_f(x, y) = f(x) - f(y) - \langle \nabla f(y), x - y \rangle.$$

An example is the negative entropy $f(x) = \sum_{i=1}^n x_i \log x_i$, where $x = (x_1, \dots, x_n)$ belongs to the n -dimensional probability simplex.

The Bregman projection of y onto $\mathcal{X}$ is defined as

$$\Pi_{\mathcal{X}}^f(y) = \arg \min_{x \in \mathcal{X}} D_f(x, y).$$

(i) Let $\hat{x} = \nabla f(x)$ and $\hat{y} = \nabla f(y)$. Let f^* denote the Fenchel conjugate of f . Show that $\nabla f^*(\hat{x}) = x$ and $D_f(x, y) = D_{f^*}(\hat{y}, \hat{x})$.

(ii) Show that for all $x, y, z \in \mathcal{X}$,

$$D_f(x, y) + D_f(z, x) - D_f(z, y) = (\nabla f(x) - \nabla f(y))^\top (x - z).$$

(iii) Assume that f is strictly convex. Let $\pi = \Pi_{\mathcal{X}}^f(y)$. Show that $(\nabla f(y) - \nabla f(\pi))^\top (w - \pi) \leq 0, \forall w \in \mathcal{X}$.

(iv) Let $\mathcal{X}$ be the probability simplex and let D be a convex cone such that $\mathcal{X} \cap D \neq \emptyset$. For $y \in D \setminus \mathcal{X}$ with $y > 0$, show that the Bregman projection under negative entropy satisfies

$$\Pi_{\mathcal{X} \cap D}^f(y) = \frac{y}{\|y\|_1}.$$

Exercise 4.11 (Lasso Problem and Its Dual (10 pts)). Consider the standard Lasso problem $\min_{x \in \mathbb{R}^n} \frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1$, where $x, y \in \mathbb{R}^n$ and $\lambda \geq 0$. This unconstrained problem can be equivalently rewritten as the constrained optimization problem

$$\begin{aligned} \min_{u, x} \quad & \frac{1}{2} \|y - u\|_2^2 + \lambda \|x\|_1 \\ \text{s.t.} \quad & u = Ax. \end{aligned}$$

(i) Show that the Lagrangian dual of the reformulated problem is given by

$$\begin{aligned} \min_{r \in \mathbb{R}^n} \quad & \frac{1}{2} \|y - r\|_2^2 \\ \text{s.t.} \quad & \|A^\top r\|_\infty \leq \lambda. \end{aligned}$$

(ii) Define the set $C := \{z \in \mathbb{R}^n \mid \|A^\top z\|_\infty \leq \lambda\}$, and let $\Pi_C : \mathbb{R}^n \rightarrow C$ denote the Euclidean projection onto C . Show that the optimal primal and dual solutions satisfy $u^* = y - \Pi_C(y)$ and $x^* = A^\dagger u^*$, where $A^\dagger$ denotes the Moore–Penrose pseudoinverse of A .

Exercise 4.12 (Augmented Lagrangian Function). Consider the following convex program (CP):

$$\begin{aligned} \min_{x \in X} \quad & f(x) \\ \text{s.t.} \quad & g(x) \leq 0, \end{aligned} \tag{4.82}$$

where X is a nonempty subset of $\mathbb{R}^n$, $f : X \rightarrow \mathbb{R}$ is a given function, and $g(x) = (g_1(x), \dots, g_r(x))$ with $g_j : X \rightarrow \mathbb{R}, j = 1, \dots, r$.

Let C_u be the perturbed constraint set $C_u = \{x \in X \mid g(x) \leq u\}$, $u \in \mathbb{R}^r$, and define

$$F(x, u) = \begin{cases} f(x), & \text{if } x \in C_u, \\ +\infty, & \text{otherwise.} \end{cases}$$

Then

$$p(u) = \inf_{x \in \mathbb{R}^n} F(x, u) = \inf_{x \in X, g(x) \leq u} f(x).$$

Now consider the modified function

$$F_c(x, u) = \begin{cases} f(x) + \frac{c}{2}\|u\|^2, & \text{if } x \in C_u, \\ +\infty, & \text{otherwise,} \end{cases}$$

where $c > 0$. Define

$$p_c(u) = \inf_{x \in \mathbb{R}^n} F_c(x, u) = \inf_{x \in X, g(x) \leq u} \left\{ f(x) + \frac{c}{2}\|u\|^2 \right\}, \quad u \geq 0.$$

The corresponding dual function is

$$q_c(\mu) = \inf_{x \in X, g(x) \leq u} \left\{ f(x) + \mu^T u + \frac{c}{2}\|u\|^2 \right\}, \quad \mu \in \mathbb{R}^r,$$

where the infimum is taken over both x and u .

(i) Show that for a fixed $x \in X$, the minimizer with respect to u is given componentwise by

$$g_j^+(x, \mu, c) := \max \left\{ -\frac{\mu_j}{c}, g_j(x) \right\}, \quad j = 1, \dots, r.$$

(ii) By substituting this expression into $q_c(\mu)$, show that

$$q_c(\mu) = \inf_{x \in X} \left\{ f(x) + \mu^T g^+(x, \mu, c) + \frac{c}{2}\|g^+(x, \mu, c)\|^2 \right\}.$$

(iii) Show that the expression inside braces is the *augmented Lagrangian function*.

Exercise 4.13 (Optimality Conditions). Find all locally optimal solutions of

$$\begin{aligned} \text{opt}_{x_1, x_2 \in \mathbb{R}} \quad & x_1 x_2^2 \\ \text{s.t.} \quad & x_1 + x_2 = 0. \end{aligned}$$

Using appropriate second-order optimality conditions, determine which of these are isolated local minima.

Exercise 4.14 (Slater's Condition). Consider the following program:

$$\begin{aligned} \min_{x \in \mathbb{R}^3} \quad & x_3 \\ \text{s.t.} \quad & (x_1 - 1)^2 + (x_2 - 1)^2 \leq 2, \\ & x_1 + x_2 + x_3^2 \leq 0, \\ & x_3 \leq 1. \end{aligned}$$

- (i) Check whether the constraints satisfy Slater's condition.
- (ii) Find the minimal index set of active constraints P° .
- (iii) Steve claims that $x^* = (0, 0, 0)^T$ is a globally optimal solution. Determine whether he is correct using an appropriate characterization of optimality. State clearly the characterization and any results you are using. Do not simplify the program.

Exercise 4.15 (Sufficiency of KKT). Consider the following convex program (CP):

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & f^i(x) \leq 0, \quad i \in \{1, \dots, n\}. \end{aligned}$$

Assume that $f, f^i : \mathbb{R}^n \rightarrow \mathbb{R}$ are convex and differentiable. Show that if the Karush–Kuhn–Tucker (KKT) conditions are satisfied at a feasible point x^* , then x^* is an optimal solution.

Exercise 4.16 (Linear Programming, 10pt). Show that the dual of the linear program (LP)

$$\begin{aligned} \min_{x_1 \geq 0, x_2 \geq 0} \quad & x_1 - x_2 \\ \text{s.t.} \quad & x_1 + 1 \leq 0, \\ & 1 - x_1 - x_2 \leq 0, \end{aligned}$$

is the linear program

$$\begin{aligned} \max_{\mu_1 \geq 0, \mu_2 \geq 0} \quad & \mu_1 + \mu_2 \\ \text{s.t.} \quad & -\mu_1 + \mu_2 - 1 \leq 0, \\ & \mu_2 + 1 \leq 0. \end{aligned}$$

Note that both primal and dual problems are infeasible.

Exercise 4.17 (Fenchel Duality). Consider the problem $\min_{x \in \mathbb{R}^n} f_1(x) + f_2(Ax)$, where $A \in \mathbb{R}^{m \times n}$, $f_1 : \mathbb{R}^n \rightarrow (-\infty, \infty]$, and $f_2 : \mathbb{R}^m \rightarrow (-\infty, \infty]$ are closed convex functions. Assume that there exists a feasible solution. We can convert the problem into an equivalent problem with decision variables $x_1 \in \mathbb{R}^n$ and $x_2 \in \mathbb{R}^m$:

$$\begin{aligned} \min_{x_1 \in \mathbb{R}^n, x_2 \in \mathbb{R}^m} \quad & f_1(x_1) + f_2(x_2) \\ \text{s.t.} \quad & x_2 = Ax_1. \end{aligned}$$

Show that the dual problem takes the following form: $\max_{\lambda \in \mathbb{R}^m} -f_1^*(A^T \lambda) - f_2^*(-\lambda)$, where f_1^* and f_2^* are the conjugate functions of f_1 and f_2 , respectively:

$$f_1^*(\lambda) = \sup_{x \in \mathbb{R}^n} \{\lambda^T x - f_1(x)\}, \quad \lambda \in \mathbb{R}^n,$$

$$f_2^*(\lambda) = \sup_{x \in \mathbb{R}^m} \{\lambda^T x - f_2(x)\}, \quad \lambda \in \mathbb{R}^m.$$

Exercise 4.18 (Separating Hyperplanes and Farkas' Lemma). Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Consider the two systems

$$(I) \quad \exists x \geq 0 \text{ with } Ax = b, \quad (II) \quad \exists y \in \mathbb{R}^m \text{ with } A^T y \leq 0, \quad b^T y > 0.$$

- (i) Show that (I) and (II) cannot hold simultaneously.
- (ii) Suppose (I) fails. The cone $K = \{Ax : x \geq 0\}$ is a nonempty closed convex cone (you may assume closedness), and $b \notin K$. Apply strict separation of the point b from the closed convex set K to produce a vector y certifying (II).
- (iii) Conclude *Farkas' lemma*: exactly one of (I), (II) holds.

Exercise 4.19 (Dual Cones and Self-Duality). For a set $K \subseteq \mathbb{R}^n$, define the *dual cone*

$$K^* := \{y \in \mathbb{R}^n : y^\top x \geq 0 \text{ for all } x \in K\}.$$

- (i) Show that K^* is a closed convex cone, regardless of whether K is convex, closed, or a cone.
- (ii) Show that the nonnegative orthant is self-dual, i.e. $(\mathbb{R}_+^n)^* = \mathbb{R}_+^n$.
- (iii) Show that the second-order (Lorentz) cone $\mathcal{Q} = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} : \|x\|_2 \leq t\}$ is self-dual under the inner product $\langle (x, t), (y, s) \rangle = x^\top y + ts$.
- (iv) Using the trace inner product $\langle X, Y \rangle = \text{tr}(XY)$ on symmetric matrices, show that the positive semidefinite cone $\mathbb{S}_+^n$ is self-dual.

Exercise 4.20 (Support Functions). Let $C \subseteq \mathbb{R}^n$ be nonempty, and recall the support function $\sigma_C(a) = \sup_{x \in C} a^\top x$.

- (i) Show that $\sigma_{C_1+C_2} = \sigma_{C_1} + \sigma_{C_2}$ and $\sigma_{\lambda C} = \lambda \sigma_C$ for every $\lambda \geq 0$.
- (ii) Let $B = \{x : \|x\| \leq 1\}$ be the unit ball of a norm $\|\cdot\|$. Show that $\sigma_B(a) = \|a\|_*$, the dual norm.
- (iii) For a polytope $C = \text{conv}\{v_1, \dots, v_k\}$, show that $\sigma_C(a) = \max_i a^\top v_i$.
- (iv) Prove that two nonempty closed convex sets are equal if and only if their support functions coincide. (Use a separating hyperplane.)

Exercise 4.21 (Perspective of a Convex Function). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex. Its *perspective* is

$$g(x, t) := t f\left(\frac{x}{t}\right), \quad (x, t) \text{ with } t > 0.$$

- (i) Show that g is convex on its domain. (Hint: express $\text{epi}(g)$ as the image of $\text{epi}(f)$ under a suitable linear map, or verify the convexity inequality directly.)
- (ii) Deduce that the quadratic-over-linear function x^2/t is convex on $\mathbb{R} \times \mathbb{R}_{++}$.
- (iii) Applying the perspective to $f(u) = u \log u$, deduce that the relative-entropy kernel $t \log(t/s)$ is convex on $\mathbb{R}_{++}^2$.

Exercise 4.22 (Strong Convexity, the PL Inequality, and Linear Convergence). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfy $mI \preceq \nabla^2 f(x) \preceq LI$ for all x (i.e., f is m -strongly convex and L -smooth), with minimizer x^* and $f^* = f(x^*)$.

- (i) Establish the two-sided bound

$$\frac{m}{2} \|x - x^*\|^2 \leq f(x) - f^* \leq \frac{L}{2} \|x - x^*\|^2.$$

- (ii) Prove the Polyak–Łojasiewicz inequality $\|\nabla f(x)\|^2 \geq 2m(f(x) - f^*)$. (Hint: minimize the strong-convexity lower bound $f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|^2$ over y .)
- (iii) For gradient descent $x^+ = x - \frac{1}{L} \nabla f(x)$, use L -smoothness to show $f(x^+) \leq f(x) - \frac{1}{2L} \|\nabla f(x)\|^2$, and combine with (ii) to obtain the linear rate $f(x^+) - f^* \leq (1 - \frac{m}{L})(f(x) - f^*)$. Interpret the ratio $\kappa = L/m$.

Exercise 4.23 (Weighted Water-Filling). Given weights $\alpha_i > 0$ and noise levels $n_i > 0$, consider the power allocation problem

$$\max_{x \geq 0} \sum_{i=1}^n \alpha_i \log \left(1 + \frac{x_i}{n_i} \right) \quad \text{subject to} \quad \sum_{i=1}^n x_i \leq P.$$

- (i) Show that the problem is convex and that Slater's condition holds.
- (ii) Writing the KKT conditions with multiplier $\mu \geq 0$ for the budget constraint and $\nu_i \geq 0$ for $x_i \geq 0$, show that the optimal allocation has the water-filling form

$$x_i^* = \left(\frac{\alpha_i}{\mu^*} - n_i \right)_+, \quad (\cdot)_+ := \max\{\cdot, 0\},$$

with $\mu^* > 0$.

- (iii) Explain how μ^* is determined from the budget, and identify the per-channel “water level” α_i/μ^* . Contrast the allocation with the unweighted case $\alpha_i \equiv 1$.

Exercise 4.24 (A Convex Problem with a Positive Duality Gap). Consider, on the domain $y > 0$,

$$\min_{x,y} e^{-x} \quad \text{subject to} \quad \frac{x^2}{y} \leq 0.$$

- (i) Show that the objective and the constraint function are convex on the domain, yet Slater's condition fails.
- (ii) Show that feasibility forces $x = 0$, so the primal optimal value is $p^* = 1$.
- (iii) Compute the dual function $d(\lambda) = \inf_{x,y>0} \{e^{-x} + \lambda x^2/y\}$ for $\lambda \geq 0$, and show $d(\lambda) = 0$; hence $d^* = 0$.
- (iv) Conclude that the duality gap is $p^* - d^* = 1$, and explain why this does not contradict strong duality.

Exercise 4.25 (Geometric Programming via the Log Transformation). A posynomial in $x \in \mathbb{R}_{++}^n$ is a finite sum $\sum_k c_k \prod_{i=1}^n x_i^{a_{ki}}$ with coefficients $c_k > 0$ and exponents $a_{ki} \in \mathbb{R}$. Consider the geometric program

$$\min_{x>0} f_0(x) \quad \text{subject to} \quad f_j(x) \leq 1, \quad j = 1, \dots, r,$$

where $f_0, f_1, \dots, f_r$ are posynomials.

- (i) Under the substitution $x_i = e^{z_i}$, show that each monomial becomes $e^{a_k^\top z + b_k}$ with $b_k = \log c_k$, so each posynomial becomes a sum of exponentials of affine functions of z .
- (ii) Taking logarithms of the objective and each constraint, show that the program becomes

$$\min_z \log \sum_k e^{a_{0k}^\top z + b_{0k}} \quad \text{s.t.} \quad \log \sum_k e^{a_{jk}^\top z + b_{jk}} \leq 0, \quad j = 1, \dots, r,$$

and explain why convexity of log-sum-exp makes this a convex program.

- (iii) Apply the transformation to $\min_{x>0} (x_1 x_2 x_3)^{-1}$ subject to $x_1 x_2 + x_1 x_3 + x_2 x_3 \leq 1$, and write the resulting convex problem in z .

Exercise 4.26 (SDP Relaxation of Two-Way Partitioning). Consider the Boolean quadratic problem

$$p^* = \min_{x \in \{-1,1\}^n} x^\top W x, \quad W \in \mathbb{S}^n.$$

- (i) Writing the constraints as $x_i^2 = 1$, form the Lagrangian with multipliers $\nu \in \mathbb{R}^n$ and show that the dual problem is

$$\max_{\nu} -\mathbf{1}^\top \nu \quad \text{subject to} \quad W + \text{diag}(\nu) \succeq 0.$$

- (ii) Show that this dual is the SDP dual of $\min_X \langle W, X \rangle$ subject to $X \succeq 0$, $X_{ii} = 1$, and that its optimal value d^* is a lower bound, $d^* \leq p^*$.
- (iii) Show that if the SDP admits a rank-one optimal solution $X^* = xx^\top$, then $x \in \{-1, 1\}^n$ solves the original problem and the relaxation is tight.

Exercise 4.27 (Carathéodory and Extreme Points of the Simplex). Let $\Delta_n = \{x \in \mathbb{R}_+^n : \mathbf{1}^\top x = 1\}$ denote the probability simplex, and let $e_1, \dots, e_n$ be the standard basis vectors.

- (i) Show that $\Delta_n = \text{conv}\{e_1, \dots, e_n\}$, so that every point of Δ_n is a convex combination of at most n vertices. Explain why the affine constraint $\mathbf{1}^\top x = 1$ sharpens the generic Carathéodory bound of $n + 1$ to n .
- (ii) Show that the extreme points of Δ_n are exactly $e_1, \dots, e_n$: verify each e_i is extreme, and that any x with two or more positive coordinates can be written as a nontrivial convex combination of two other points of Δ_n .
- (iii) Exhibit a point of Δ_3 whose every representation as a convex combination of the e_i uses all three vertices, showing the bound in (i) is tight.

Exercise 4.28 (Minimax Theorem and a Matrix Game). Let $A \in \mathbb{R}^{m \times n}$ be the payoff matrix of a two-player zero-sum game: the row player chooses a mixed strategy $x \in \Delta_m$, the column player chooses $y \in \Delta_n$, and the row player pays $x^\top A y$.

- (i) Show that $\max_{y \in \Delta_n} x^\top A y = \max_j (A^\top x)_j$, so the inner maximization is attained at a pure strategy (a vertex of Δ_n). Write the row player's problem $\min_{x \in \Delta_m} \max_{y \in \Delta_n} x^\top A y$ as a linear program.
- (ii) Using LP duality, show that the game has a value, $\min_{x \in \Delta_m} \max_{y \in \Delta_n} x^\top A y = \max_{y \in \Delta_n} \min_{x \in \Delta_m} x^\top A y$, and relate this to the existence of a saddle point of the bilinear payoff.
- (iii) Compute the value and a pair of optimal strategies for matching pennies, $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

Exercise 4.29 (Shadow Prices and Sensitivity in a Resource LP). A shop makes two products at unit profits $c = (3, 5)$, using $\max_{x \geq 0} 3x_1 + 5x_2$ subject to $x_1 \leq 4$, $2x_2 \leq 12$, and $3x_1 + 2x_2 \leq 18$, with right-hand side $b = (4, 12, 18)$.

- (i) Solve the LP and find the optimal multipliers $\mu^* \geq 0$ associated with the three constraints.
- (ii) Verify the shadow-price relation $\partial p^* / \partial b_i = \mu_i^*$ by perturbing each b_i and recomputing the optimal value, and explain why an inactive constraint has zero shadow price (complementary slackness).

- (iii) For the binding constraints, determine the range of b_i over which the shadow price μ_i^* stays constant (i.e., the optimal basis is unchanged), and interpret what happens to the marginal value of the resource beyond that range.

Exercise 4.30 (Fractional Programming: Charnes–Cooper and Dinkelbach). Let $X = \{x : Bx \leq d\}$ be a nonempty bounded polyhedron on which $e^\top x + f > 0$, and consider the linear-fractional program $\max_{x \in X} \frac{c^\top x + \alpha}{e^\top x + f}$.

- (i) (*Charnes–Cooper.*) With the substitution $y = x/(e^\top x + f)$ and $t = 1/(e^\top x + f)$, show that the program is equivalent to the linear program $\max_{y,t} c^\top y + \alpha t$ subject to $By \leq dt$, $e^\top y + ft = 1$, $t \geq 0$, and recover $x^* = y^*/t^*$.
- (ii) (*Dinkelbach.*) Define $F(\lambda) := \max_{x \in X} \{c^\top x + \alpha - \lambda(e^\top x + f)\}$. Show that F is convex, nonincreasing, and has a unique root λ^* , and that λ^* equals the optimal value of the fractional program with any maximizer of $F(\lambda^*)$ optimal for the original problem.
- (iii) Explain how Newton’s method applied to $F(\lambda) = 0$ yields the classical Dinkelbach iteration, and why each iterate requires solving only one linear program.

Exercise 4.31 (Dual Decomposition of a Separable Problem). Consider N subsystems coupled only through a shared resource,

$$\min_{x_1, \dots, x_N} \sum_{i=1}^N f_i(x_i) \quad \text{subject to} \quad \sum_{i=1}^N A_i x_i = b,$$

where each $f_i : \mathbb{R}^{n_i} \rightarrow \mathbb{R}$ is convex and $x_i \in X_i$ for convex sets X_i .

- (i) Form the Lagrangian with multiplier $\lambda \in \mathbb{R}^m$ for the coupling constraint, and show that it separates as $L(x, \lambda) = -\lambda^\top b + \sum_{i=1}^N (f_i(x_i) + \lambda^\top A_i x_i)$, so that the dual function decouples into N independent subproblems.
- (ii) Write the dual problem $\max_\lambda q(\lambda)$ and show q is concave. Assuming strong duality, describe the projected subgradient (dual ascent) update $\lambda \leftarrow \lambda + \gamma(\sum_i A_i x_i(\lambda) - b)$, where $x_i(\lambda)$ solves the i th subproblem.
- (iii) Interpret λ as a vector of resource prices and the update as a price-adjustment (tâtonnement) mechanism: prices rise on over-demanded resources and fall on under-used ones until supply meets demand. State a condition on f_i that guarantees each $x_i(\lambda)$ is unique, so the subproblems return single-valued responses.

Exercise 4.32 (Convex Hull as the Set of Convex Combinations). A convex combination of $x_1, \dots, x_n$ is a linear combination $\alpha_1 x_1 + \dots + \alpha_n x_n$ with $\alpha_i \geq 0$ and $\sum_i \alpha_i = 1$. Given a set S in a vector space, let K be the set of all convex combinations of finitely many points of S . Show that $K = \text{co}(S)$, the convex hull of S .

Chapter 5

Generalized Convex Optimization

This chapter develops a geometric framework for convex optimization beyond the classical Karush–Kuhn–Tucker setting. The central idea is to separate the constraints that define the feasible geometry from the inequalities that should carry dual multipliers. This separation leads to tangent-cone, normal-cone, and saddle-point optimality conditions that remain valid even when Slater’s condition fails.

The chapter proceeds in three steps. First, it derives cone-based first-order conditions for convex minimization. Second, it proves a restricted saddle-point theorem that dualizes only effective inequalities. Third, it shows how the restricted formulation reduces to the complete Lagrangian and classical KKT conditions under standard regularity assumptions. The theory is then illustrated through affine constraints, entropy optimization, convex optimal control, value-function sensitivity, and nonsmooth convex programs.

5.1 ■ Generalized Saddle-Point Theorem

Let $\mathcal{P} := \{1, \dots, p\}$, $\mathcal{Q} := \{1, \dots, q\}$. Consider the constrained optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \\ & g_j(x) \leq 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{5.1}$$

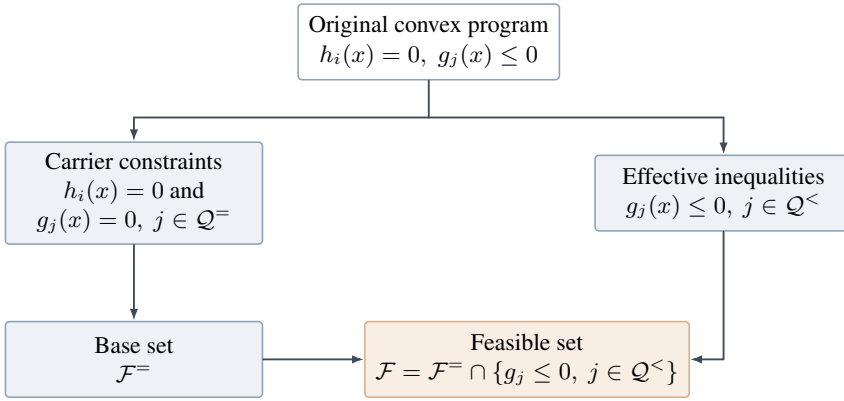
where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, $\{g_j\}_{j \in \mathcal{Q}}$ are convex inequality constraint functions, and $\{h_i\}_{i \in \mathcal{P}}$ define the equality constraints.

Throughout, we assume that the equality constraint set $\{x \in \mathbb{R}^n : h_i(x) = 0, i \in \mathcal{P}\}$ is a convex set. In general, even if each h_i is convex, the set $\{x : h_i(x) = 0\}$ need *not* be convex. Therefore, to ensure convexity of the feasible set, it is standard to restrict to equality constraints that are *affine*, i.e., $h_i(x) = a_i^\top x - b_i$, $i \in \mathcal{P}$, or, more generally, to functions whose zero level sets are convex. Affine functions are the most common and practically important case.

Under these assumptions, the feasible set of (5.1) is convex, and the problem is a convex optimization problem in the standard sense.

5.1.1 ■ Feasible-Set Decomposition

All equality-type constraints are incorporated into the geometry of the feasible set. Among the inequality constraints indexed by $\mathcal{Q}$, we distinguish those that are active at every feasible point.



geometry built into the domain is separated from the inequalities that keep dual multipliers

Figure 5.1: Feasible-set decomposition. Constraints that are always active become part of the base geometry, while the remaining inequalities define the effective dualized constraints.

$$\mathcal{Q}^= := \{j \in \mathcal{Q} \mid g_j(x) = 0 \text{ for all } x \in \mathcal{F}\}, \quad \mathcal{Q}^< := \mathcal{Q} \setminus \mathcal{Q}^=. \quad (5.2)$$

We define the *equality feasible set* (base set) as

$$\mathcal{F}^= := \{x \in \mathbb{R}^n \mid h_i(x) = 0, i \in \mathcal{P}, g_j(x) = 0, j \in \mathcal{Q}^=\}. \quad (5.3)$$

The *full feasible set* is the set of points satisfying *all* constraints,

$$\mathcal{F} := \{x \in \mathbb{R}^n \mid h_i(x) = 0, i \in \mathcal{P}, g_j(x) \leq 0, j \in \mathcal{Q}\}. \quad (5.4)$$

Equivalently, the full feasible set can be written as

$$\mathcal{F} = \mathcal{F}^= \cap \{x \in \mathbb{R}^n \mid g_j(x) \leq 0, j \in \mathcal{Q}^<\}. \quad (5.5)$$

5.1.2 ■ General Convex Optimization Problem

Consider the convex optimization problem

$$\min_{x \in C} f(x), \quad (5.6)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable, and $C \subset \mathbb{R}^n$ is a nonempty convex set.

Since both the objective function and the feasible set are convex, every local minimizer of (5.6) is also a global minimizer. Optimality can therefore be characterized using first-order information and the local geometry of the feasible set. The guiding intuition is simple: a point is optimal precisely when no admissible direction of motion decreases the objective. Making “admissible direction” precise is exactly what the tangent cone does.

Theorem 5.1 (First-Order Optimality Condition: Tangent-Cone Form). *A point $x^* \in C$ is optimal for (5.6) if and only if*

$$\nabla f(x^*)^\top d \geq 0 \quad \forall d \in T_C(x^*), \quad (5.7)$$

where $T_C(x^*)$ denotes the tangent (contingent) cone to C at x^* .

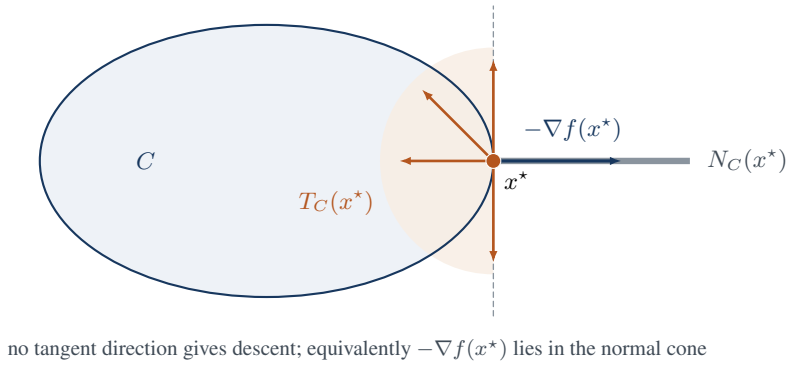


Figure 5.2: Tangent and normal cones at a boundary optimum, drawn at a point where the supporting line (dotted) is vertical. The tangent cone (shaded half-plane, translated to x^*) collects feasible infinitesimal motions; its polar, the normal cone, is the outward ray and contains $-\nabla f(x^*)$ at an optimum.

The tangent cone $T_C(x^*)$ captures all directions in which one can infinitesimally move from x^* while remaining feasible; formally,

$$T_C(x^*) := \{d \in \mathbb{R}^n \mid \exists \varepsilon_k \downarrow 0, \exists d_k \rightarrow d \text{ such that } x^* + \varepsilon_k d_k \in C \ \forall k\}. \quad (5.8)$$

Read in these terms, condition (5.7) says that the directional derivative $\nabla f(x^*)^\top d$ is nonnegative along every direction that is feasible to first order: there is no feasible descent direction at x^* . The condition also contains the classical unconstrained case as a limit. If $x^* \in \text{int}(C)$, then $T_C(x^*) = \mathbb{R}^n$, so the inequality must hold for d and $-d$ alike, forcing $\nabla f(x^*) = 0$.

It is often more convenient to state optimality through the polar object, the *normal cone* $N_C(x^*) := \{v \in \mathbb{R}^n \mid v^\top d \leq 0 \ \forall d \in T_C(x^*)\}$, whose elements make an obtuse angle with every feasible direction. Condition (5.7) is then the single inclusion

$$-\nabla f(x^*) \in N_C(x^*), \quad (5.9)$$

that is, the steepest-descent direction $-\nabla f(x^*)$ points “out of” the feasible set. This cone-based condition is the geometric seed from which Lagrangian duality, KKT conditions, and the generalized saddle-point theorems of this chapter all grow.

Proof. We prove the two directions separately.

Necessity. Assume that $x^* \in C$ is an optimal solution of $\min_{x \in C} f(x)$, where f is convex and differentiable. Let $d \in T_C(x^*)$ be an arbitrary tangent direction. By definition of the tangent cone, there exist sequences $\varepsilon_k \downarrow 0$ and $d_k \rightarrow d$ such that $x^* + \varepsilon_k d_k \in C \ \forall k$.

Since x^* is optimal over C , we have $f(x^* + \varepsilon_k d_k) \geq f(x^*) \ \forall k$. Dividing by $\varepsilon_k > 0$ yields $\frac{f(x^* + \varepsilon_k d_k) - f(x^*)}{\varepsilon_k} \geq 0$.

Because f is differentiable at x^* , the directional derivative exists and satisfies

$$\lim_{k \rightarrow \infty} \frac{f(x^* + \varepsilon_k d_k) - f(x^*)}{\varepsilon_k} = \nabla f(x^*)^\top d.$$

Taking the limit on both sides gives $\nabla f(x^*)^\top d \geq 0$.

Since $d \in T_C(x^*)$ was arbitrary, we conclude that $\nabla f(x^*)^\top d \geq 0 \ \forall d \in T_C(x^*)$.

Sufficiency. Now assume that $\nabla f(x^*)^\top d \geq 0 \forall d \in T_C(x^*)$. Let $x \in C$ be arbitrary and define $d := x - x^*$. Because C is convex, the line segment $x^* + \varepsilon(x - x^*) = (1 - \varepsilon)x^* + \varepsilon x$ belongs to C for all $\varepsilon \in [0, 1]$. Hence $d \in T_C(x^*)$.

By the assumed condition, $\nabla f(x^*)^\top (x - x^*) \geq 0$.

Since f is convex, it satisfies the first-order convexity inequality $f(x) \geq f(x^*) + \nabla f(x^*)^\top (x - x^*)$. Combining the two inequalities yields $f(x) \geq f(x^*)$.

Because $x \in C$ was arbitrary, this shows that x^* is a global minimizer of f over C .

□

5.2 ■ Saddle-Point Characterization without Slater's Condition

We consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \\ & g_j(x) \leq 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{5.10}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, $\{h_i\}_{i \in \mathcal{P}}$ are affine equality constraint functions, and $\{g_j\}_{j \in \mathcal{Q}}$ are convex inequality constraint functions.

Feasible-Set Decomposition

All equality constraints are incorporated into the geometry of the feasible set. Among the inequality constraints indexed by $\mathcal{Q}$, we distinguish those that are active at every feasible point.

Define the index sets

$$\mathcal{Q}^= := \{j \in \mathcal{Q} \mid g_j(x) = 0 \text{ for all } x \in \mathcal{F}\}, \quad \mathcal{Q}^< := \mathcal{Q} \setminus \mathcal{Q}^=. \tag{5.11}$$

Equivalently, for each $j \in \mathcal{Q}^<$ there exists at least one feasible point $x' \in \mathcal{F}$ such that $g_j(x') < 0$.

The *base feasible set* is defined by all equality constraints together with the inequality constraints that behave as equalities:

$$\mathcal{F}^= := \{x \in \mathbb{R}^n \mid h_i(x) = 0, \quad i \in \mathcal{P}, \quad g_j(x) = 0, \quad j \in \mathcal{Q}^=\}. \tag{5.12}$$

The set $\mathcal{F}^=$ is convex (typically affine).

The *full feasible set* of (5.1) is

$$\mathcal{F} = \mathcal{F}^= \cap \{x \in \mathbb{R}^n \mid g_j(x) \leq 0, \quad j \in \mathcal{Q}^<\}. \tag{5.13}$$

The constraints indexed by $\mathcal{Q}^<$ are the *effective inequality constraints*: these are the only inequalities that can be strictly slack somewhere on $\mathcal{F}$, and so they are the only ones that deserve a dual multiplier. The constraints in $\mathcal{Q}^=$ hold with equality everywhere on the feasible set and therefore belong to the *geometry* of $\mathcal{F}^=$ rather than to the list of dualized constraints. Crucially, this split is defined without any constraint qualification; in particular, Slater's condition is not assumed.

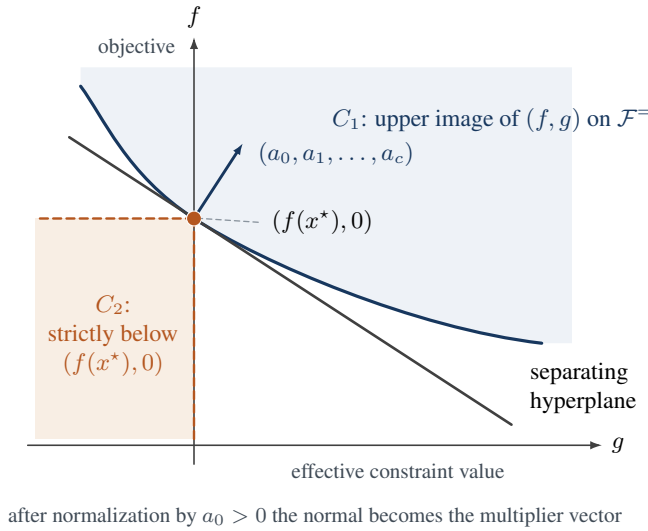


Figure 5.3: Separation proof for the restricted saddle theorem. Separating the upper image C_1 of the objective and effective constraints from the set C_2 of strictly improving points yields a supporting hyperplane, tangent to C_1 at $(f(x^*), 0)$, whose nonnegative normal $(a_0, a_1, \dots, a_c)$ is, once normalized by $a_0 > 0$, the multiplier vector. A slanted hyperplane corresponds to nonzero constraint multipliers $a_i/a_0 > 0$.

Following this principle, we introduce multipliers only for the effective inequalities and define the *restricted Lagrangian*

$$\mathcal{L}^c(x, \mu) := f(x) + \sum_{j \in Q^<} \mu_j g_j(x), \quad \mu \in \mathbb{R}_+^{|Q^<|}, \quad (5.14)$$

with the understanding that $x \in \mathcal{F}^=$.

Theorem 5.2 (Saddle-Point Characterization without Slater's Condition). *Consider the convex optimization problem (5.1). Let $\mathcal{F}^=$ denote the base feasible set and let $Q^<$ be the index set of effective inequality constraints, with $|Q^<| = c$. Assume that $\mathcal{F}^=$ is nonempty.*

A point $x^ \in \mathcal{F}^=$ is an optimal solution of (5.1) if and only if there exists a multiplier vector $\mu^* \in \mathbb{R}_+^c$ such that (x^*, μ^*) is a saddle point of the restricted Lagrangian $\mathcal{L}^c$ over $\mathcal{F}^= \times \mathbb{R}_+^c$, that is,*

$$\mathcal{L}^c(x^*, \mu) \leq \mathcal{L}^c(x^*, \mu^*) \leq \mathcal{L}^c(x, \mu^*), \quad \forall \mu \in \mathbb{R}_+^c, \forall x \in \mathcal{F}^=. \quad (5.15)$$

Proof. We prove necessity and sufficiency separately.

Necessity. Assume that $x^* \in \mathcal{F}^=$ is an optimal solution of (5.1). Without loss of generality, relabel the effective inequality constraints so that $Q^< = \{1, 2, \dots, c\}$.

Define the sets $C_1, C_2 \subset \mathbb{R}^{c+1}$ by

$$C_1 := \left\{ y \in \mathbb{R}^{c+1} \mid y \geq \begin{pmatrix} f(x) \\ g_1(x) \\ \vdots \\ g_c(x) \end{pmatrix} \text{ for some } x \in \mathcal{F}^\circ \right\}, \quad (5.16)$$

$$C_2 := \left\{ y \in \mathbb{R}^{c+1} \mid y < \begin{pmatrix} f(x^*) \\ 0 \\ \vdots \\ 0 \end{pmatrix} \right\}, \quad (5.17)$$

where inequalities are understood componentwise.

Since $\mathcal{F}^\circ$ is convex and the functions $f, g_1, \dots, g_c$ are convex, the set C_1 is convex. The set C_2 is convex and open. Moreover, $C_1 \cap C_2 = \emptyset$. Indeed, if $y \in C_1 \cap C_2$, then there exists $x \in \mathcal{F}^\circ$ such that

$$f(x) < f(x^*), \quad g_j(x) < 0, \quad j = 1, \dots, c,$$

which implies that x is feasible for (5.1) and yields a strictly smaller objective value, contradicting the optimality of x^* .

By the Hyperplane Separation Theorem, there exist a nonzero vector $a = (a_0, a_1, \dots, a_c) \in \mathbb{R}^{c+1}$ and a scalar $\alpha \in \mathbb{R}$ such that

$$\langle a, y^2 \rangle \leq \alpha \leq \langle a, y^1 \rangle \quad \forall y^1 \in C_1, \forall y^2 \in C_2. \quad (5.18)$$

Since C_2 is unbounded below in every coordinate, it follows that $a_i \geq 0, i = 0, 1, \dots, c$.

Choose

$$y^1 = \begin{pmatrix} f(x) \\ g_1(x) \\ \vdots \\ g_c(x) \end{pmatrix}, \quad x \in \mathcal{F}^\circ, \quad \bar{y} = \begin{pmatrix} f(x^*) \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

The point $\bar{y}$ itself lies on the boundary of C_2 rather than in C_2 , since C_2 was defined by strict inequalities; but $\bar{y}$ is a limit of points $y^2 \in C_2$, so letting $y^2 \rightarrow \bar{y}$ in (5.18) and using continuity of $y \mapsto \langle a, y \rangle$ gives $\langle a, \bar{y} \rangle \leq \alpha$. Combining with the right-hand inequality of (5.18) then yields

$$a_0 f(x^*) \leq a_0 f(x) + \sum_{j=1}^c a_j g_j(x), \quad \forall x \in \mathcal{F}^\circ. \quad (5.19)$$

We claim that $a_0 > 0$. Otherwise, if $a_0 = 0$, then at least one $a_j > 0$, which would imply $\sum_{j=1}^c a_j g_j(x) \geq 0 \forall x \in \mathcal{F}^\circ$, contradicting the definition of $\mathcal{Q}^<$, since for each $j \in \mathcal{Q}^<$ there exists $x' \in \mathcal{F}$ such that $g_j(x') < 0$ while $g_k(x') \leq 0$ for all $k \in \mathcal{Q}^<$.

Hence $a_0 > 0$. Define the multipliers $\mu_j^* := \frac{a_j}{a_0}, j = 1, \dots, c$.

Dividing (5.19) by a_0 yields

$$f(x^*) \leq f(x) + \sum_{j=1}^c \mu_j^* g_j(x), \quad \forall x \in \mathcal{F}^\circ, \quad (5.20)$$

that is,

$$\mathcal{L}^c(x^*, \mu^*) \leq \mathcal{L}^c(x, \mu^*), \quad \forall x \in \mathcal{F}^\circ.$$

Moreover, since x^* is feasible and $\mu \geq 0$,

$$\mathcal{L}^c(x^*, \mu) = f(x^*) + \sum_{j=1}^c \mu_j g_j(x^*) \leq f(x^*) = \mathcal{L}^c(x^*, \mu^*), \quad \forall \mu \geq 0.$$

Thus (x^*, μ^*) satisfies the saddle-point inequalities.

Sufficiency. Conversely, assume that there exists $\mu^* \in \mathbb{R}_+^c$ such that

$$\mathcal{L}^c(x^*, \mu) \leq \mathcal{L}^c(x^*, \mu^*) \leq \mathcal{L}^c(x, \mu^*), \quad \forall \mu \geq 0, \forall x \in \mathcal{F}^=.$$

First, we show that x^* is feasible for the inequality constraints. If $g_k(x^*) > 0$ for some $k \in \mathcal{Q}^<$, choose $\mu_k \rightarrow \infty$ and $\mu_j = 0$ for $j \neq k$, which violates the left inequality. Hence $g_j(x^*) \leq 0$ for all j .

The left inequality with $\mu = 0$, together with feasibility of x^* , gives

$$f(x^*) \leq \mathcal{L}^c(x^*, \mu^*) \leq f(x^*),$$

so $\mathcal{L}^c(x^*, \mu^*) = f(x^*)$. The right inequality then implies

$$f(x^*) \leq \mathcal{L}^c(x, \mu^*) \leq f(x)$$

for every feasible x , because $g_j(x) \leq 0$ and $\mu_j^* \geq 0$. Therefore x^* is optimal.

□

Remark 5.3 (Saddle Point and Complementary Slackness). A pair (x^*, μ^*) satisfying (5.15) is called a *saddle point* of the restricted Lagrangian $\mathcal{L}^c$ over $\mathcal{F}^= \times \mathbb{R}_+^c$.

(i) From the saddle-point inequalities and feasibility of x^* , it follows that

$$\mu_j^* g_j(x^*) = 0, \quad j \in \mathcal{Q}^<. \quad (5.21)$$

These are the *complementary slackness conditions*. They express the fact that only active inequality constraints influence the optimality conditions: if $g_j(x^*) < 0$ for some $j \in \mathcal{Q}^<$, then necessarily $\mu_j^* = 0$.

(ii) The restricted Lagrangian $\mathcal{L}^c$ can equivalently be replaced by the classical Lagrangian

$$\mathcal{L}(x, \mu) = f(x) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x),$$

without affecting the saddle-point characterization, provided that the multipliers satisfy $\mu_j = 0$ for all $j \in \mathcal{Q}^=$. Indeed, by definition of $\mathcal{Q}^=$, each constraint g_j with $j \in \mathcal{Q}^=$ satisfies $g_j(x) = 0$ for all $x \in \mathcal{F}^=$. Consequently, the corresponding Lagrange terms vanish identically on the domain $\mathcal{F}^=$, and the classical and restricted Lagrangians coincide on $\mathcal{F}^= \times \mathbb{R}_+^c$ once $\mu_j = 0$ is enforced for $j \in \mathcal{Q}^=$. These constraints therefore play a purely geometric role in defining the feasible set and do not contribute additional dual degrees of freedom.

(iii) The separating set used in the proof of Theorem 5.2 depends only on the feasible set $\mathcal{F}^=$ and not on the particular choice of an optimal solution. Consequently, the set of multipliers μ^* satisfying the saddle-point condition is the same for all optimal solutions of (5.1).

Corollary 5.4 (Saddle-Point Characterization under Slater's Condition). *Consider the convex optimization problem (5.1). Assume that the inequality constraints satisfy Slater's condition, i.e., there exists $\bar{x} \in \mathbb{R}^n$ with $h_i(\bar{x}) = 0$ for $i \in \mathcal{P}$ and $g_j(\bar{x}) < 0$ for $j \in \mathcal{Q}$.*

Then $\mathcal{Q}^= = \emptyset$ and $\mathcal{F}^= = \{x \in \mathbb{R}^n \mid h_i(x) = 0, i \in \mathcal{P}\}$. The restricted Lagrangian over $\mathcal{F}^=$ coincides with the classical inequality Lagrangian. A point $x^ \in \mathcal{F}^=$ is optimal if and only if there exists $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}|}$ such that*

$$\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*), \quad \forall \mu \in \mathbb{R}_+^{|\mathcal{Q}|}, \forall x \in \mathcal{F}^=.$$

With affine equality constraints and equality multipliers, this is precisely the classical KKT saddle-point condition.

Proof. Let $\bar{x}$ be a Slater point, so $g_j(\bar{x}) < 0$ for every $j \in \mathcal{Q}$. Then no inequality constraint can hold with equality at every feasible point, so $\mathcal{Q}^= = \emptyset$ and $\mathcal{Q}^< = \mathcal{Q}$. Consequently the base set is $\mathcal{F}^= = \{x : h_i(x) = 0, i \in \mathcal{P}\}$, and the restricted Lagrangian $\mathcal{L}^c$, which carries multipliers for all of $\mathcal{Q}^< = \mathcal{Q}$, coincides with the classical inequality Lagrangian $\mathcal{L}$. The stated equivalence is therefore Theorem 5.2 in the special case $\mathcal{Q}^= = \emptyset$. $\square$

5.2.1 ■ First-Order Optimality without Slater's Condition

We now derive a first-order characterization of optimality that does not rely on any constraint qualification, including Slater's condition. The result is a direct geometric consequence of the saddle-point characterization established earlier and expresses optimality in terms of normal and tangent cones. Recall that $\mathcal{F}^=$ denotes the base feasible set induced by all equality-type constraints, and that $\mathcal{Q}^<$ indexes the effective inequality constraints.

Theorem 5.5 (First-Order Optimality without Slater). *Consider the convex optimization problem (5.1). Assume that all functions are differentiable at a feasible point $x^* \in \mathcal{F}$. Then, x^* is an optimal solution of (5.1) if and only if there exists a multiplier vector $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that*

$$-\nabla f(x^*) - \sum_{j \in \mathcal{Q}^<} \mu_j^* \nabla g_j(x^*) \in N_{\mathcal{F}^=}(x^*), \quad (5.22)$$

where $N_{\mathcal{F}^=}(x^*)$ denotes the normal cone to $\mathcal{F}^=$ at x^* .

Using the polarity relation between normal and tangent cones, condition (5.22) is equivalent to

$$\left[\nabla f(x^*) + \sum_{j \in \mathcal{Q}^<} \mu_j^* \nabla g_j(x^*) \right]^\top d \geq 0 \quad \forall d \in T_{\mathcal{F}^=}(x^*), \quad (5.23)$$

where $T_{\mathcal{F}^=}(x^*)$ is the tangent cone to $\mathcal{F}^=$ at x^* .

Proof. We prove necessity and sufficiency separately.

Necessity. Assume that x^* is an optimal solution of (5.1). By the saddle-point characterization without Slater's condition (Theorem 5.2), there exists $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that

$$\mathcal{L}^c(x^*, \mu) \leq \mathcal{L}^c(x^*, \mu^*) \leq \mathcal{L}^c(x, \mu^*) \quad \forall \mu \geq 0, \forall x \in \mathcal{F}^=.$$

In particular, the right-hand inequality implies that x^* minimizes the convex function

$$x \mapsto \mathcal{L}^c(x, \mu^*) = f(x) + \sum_{j \in \mathcal{Q}^<} \mu_j^* g_j(x)$$

over the convex set $\mathcal{F}^=$.

Since $\mathcal{L}^c(\cdot, \mu^*)$ is differentiable at x^* , the first-order optimality condition for convex minimization over a convex set yields

$$\nabla_x \mathcal{L}^c(x^*, \mu^*)^\top d \geq 0 \quad \forall d \in T_{\mathcal{F}^=}(x^*).$$

Expanding the gradient gives

$$\left[\nabla f(x^*) + \sum_{j \in \mathcal{Q}^<} \mu_j^* \nabla g_j(x^*) \right]^\top d \geq 0 \quad \forall d \in T_{\mathcal{F}^=}(x^*),$$

which is precisely (5.23). By definition of the normal cone, this is equivalent to (5.22).

Sufficiency. Conversely, suppose there exists $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that (5.22) holds. Equivalently, (5.23) is satisfied.

Then, for every $x \in \mathcal{F}^=$, convexity of $\mathcal{L}^c(\cdot, \mu^*)$ implies

$$\mathcal{L}^c(x, \mu^*) \geq \mathcal{L}^c(x^*, \mu^*) + \nabla_x \mathcal{L}^c(x^*, \mu^*)^\top (x - x^*).$$

Since $x - x^* \in T_{\mathcal{F}^=}(x^*)$, the second term on the right-hand side is nonnegative by (5.23). Hence

$$\mathcal{L}^c(x, \mu^*) \geq \mathcal{L}^c(x^*, \mu^*) \quad \forall x \in \mathcal{F}^=.$$

Together with feasibility of x^* and nonnegativity of μ^* , this implies the saddle-point inequalities

$$\mathcal{L}^c(x^*, \mu) \leq \mathcal{L}^c(x^*, \mu^*) \leq \mathcal{L}^c(x, \mu^*),$$

for all $\mu \geq 0$ and all $x \in \mathcal{F}^=$. By Theorem 5.2, x^* is therefore an optimal solution of (5.1). $\square$

We now illustrate Theorem 5.5 through a simple example, highlighting how optimality is characterized via the normal cone to the base feasible set $\mathcal{F}^=$ even when classical KKT conditions fail.

Example 5.6 Singleton Feasible Set.

Consider the convex optimization problem

$$\min_{x \in \mathbb{R}} f(x) := -x \quad \text{s.t.} \quad g(x) := x^2 \leq 0.$$

The feasible set is $\mathcal{F} = \{0\}$, and the unique feasible point is $x^* = 0$. Since $g(x) = 0$ for all $x \in \mathcal{F}$, the inequality constraint acts as an equality, so $\mathcal{Q}^= = \{1\}$ and $\mathcal{Q}^< = \emptyset$. Consequently, $\mathcal{F}^= = \{0\}$.

The normal cone to $\mathcal{F}^=$ at x^* is $N_{\mathcal{F}^=}(0) = \mathbb{R}$. Moreover, $\nabla f(0) = -1$. Since $\mathcal{Q}^< =$

$\emptyset$, the first-order optimality condition (5.22) reduces to $-\nabla f(0) = 1 \in N_{\mathcal{F}^=}(0)$, which holds. Therefore, $x^* = 0$ is an optimal solution.

This example demonstrates that although no Lagrange multiplier can satisfy the classical KKT stationarity condition, the normal-cone characterization of Theorem 5.5 correctly certifies optimality. $\square$

The previous example is degenerate in that every inequality is a carrier constraint, so no effective inequality survives to receive a multiplier. The next example is genuinely mixed: it combines a carrier inequality, which is absorbed into the base feasible set, with an effective inequality, which is dualized through the restricted Lagrangian. It therefore exercises the full feasible-set decomposition $\mathcal{Q} = \mathcal{Q}^= \cup \mathcal{Q}^<$ in a setting where Slater's condition still fails.

Example 5.7 Carrier and Effective Inequalities in the Plane.

Consider the convex program

$$\min_{x \in \mathbb{R}^2} f(x) := x_2 \quad \text{s.t.} \quad g_1(x) := x_1^2 \leq 0, \quad g_2(x) := -x_2 \leq 0.$$

The constraint $g_1(x) \leq 0$ forces $x_1 = 0$, while $g_2(x) \leq 0$ requires $x_2 \geq 0$, so the feasible set is the nonnegative x_2 -axis, $\mathcal{F} = \{(0, t) : t \geq 0\}$. Because g_1 vanishes at every feasible point, it is a carrier inequality and $\mathcal{Q}^= = \{1\}$; by contrast g_2 is strictly negative whenever $x_2 > 0$, so it is effective and $\mathcal{Q}^< = \{2\}$. The base feasible set is the entire vertical axis,

$$\mathcal{F}^= = \{x \in \mathbb{R}^2 : g_1(x) = 0\} = \{x \in \mathbb{R}^2 : x_1 = 0\},$$

an affine line, and the full feasible set is recovered as $\mathcal{F} = \mathcal{F}^= \cap \{g_2(x) \leq 0\}$.

Slater's condition fails, since $x_1^2 \geq 0$ prevents any point from satisfying $g_1(x) < 0$; moreover $\nabla g_1(x) = (2x_1, 0)$ vanishes on the whole feasible set, so the classical constraint qualifications degenerate. Minimizing x_2 over $\mathcal{F}$ nonetheless has the obvious solution $x^* = (0, 0)$ with $f(x^*) = 0$.

We verify optimality through Theorem 5.5. The tangent cone to the affine set $\mathcal{F}^=$ at x^* is the axis direction $T_{\mathcal{F}^=}(x^*) = \{d \in \mathbb{R}^2 : d_1 = 0\}$, and its polar is the horizontal line $N_{\mathcal{F}^=}(x^*) = \{(s, 0) : s \in \mathbb{R}\}$. With $\nabla f(x^*) = (0, 1)$ and $\nabla g_2(x^*) = (0, -1)$, the first-order condition (5.22) requires a multiplier $\mu_2^* \geq 0$ with

$$-\nabla f(x^*) - \mu_2^* \nabla g_2(x^*) = (0, -1 + \mu_2^*) \in N_{\mathcal{F}^=}(x^*) = \{(s, 0) : s \in \mathbb{R}\}.$$

The membership holds precisely when the second component vanishes, i.e. $\mu_2^* = 1 \geq 0$. Equivalently, (x^*, μ_2^*) is a saddle point of the restricted Lagrangian $\mathcal{L}^c(x, \mu_2) = x_2 - \mu_2 x_2$ over $\mathcal{F}^= \times \mathbb{R}_+$: at $\mu_2^* = 1$ the map $x \mapsto \mathcal{L}^c(x, 1) \equiv 0$ is minimized by x^* on the axis, and $\mu_2 \mapsto \mathcal{L}^c(x^*, \mu_2) = 0$ is trivially maximized. The carrier inequality g_1 receives no multiplier and is represented entirely through the geometry of $\mathcal{F}^=$, whereas the effective inequality g_2 is dualized with $\mu_2^* = 1$. This illustrates how the decomposition sorts the two constraints into geometry and duality even though Slater's condition is violated. $\square$

Definition 5.8 (Cone of Directions of Constancy). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function and let $x^* \in \mathbb{R}^n$. The cone of directions of constancy of f at x^* is defined as

$$\mathcal{D}_j(x^*) := \{d \in \mathbb{R}^n \mid g_j(x^* + \alpha d) = g_j(x^*) \text{ for all } 0 < \alpha < \alpha_0 \text{ for some } \alpha_0 > 0\}.$$

To state the identity we must account for *all* the constraints that define $\mathcal{F}^=$, not only the carrier inequalities. Write

$$\mathcal{F}^= = \{x : c_k(x) = 0, k \in \mathcal{K}\}, \quad \{c_k\}_{k \in \mathcal{K}} := \{h_i\}_{i \in \mathcal{P}} \cup \{g_j\}_{j \in \mathcal{Q}^=},$$

and let $\mathcal{D}_k(x^*)$ denote the cone of directions of constancy of c_k at x^* , defined exactly as above.

Proposition 5.9 (Constancy–Tangent Cone Identity). Let $x^* \in \mathcal{F}^=$. Then the cone of directions along which x^* can be moved while staying in $\mathcal{F}^=$ is exactly $\bigcap_{k \in \mathcal{K}} \mathcal{D}_k(x^*)$, and consequently

$$T_{\mathcal{F}^=}(x^*) = \text{cl}\left(\bigcap_{k \in \mathcal{K}} \mathcal{D}_k(x^*)\right). \quad (5.24)$$

If $\mathcal{F}^=$ is affine or, more generally, polyhedral—the case that occurs throughout this chapter—the intersection is already closed and (5.24) holds without the closure. In all cases

$$N_{\mathcal{F}^=}(x^*) = \left(\bigcap_{k \in \mathcal{K}} \mathcal{D}_k(x^*)\right)^+, \quad (5.25)$$

where $(\cdot)^+$ denotes the polar cone, since a cone and its closure have the same polar.

Remark 5.10. Two features of the statement are forced. First, the intersection must run over all of $\mathcal{K}$: with a single affine equality $h(x) = x_1$ and no carrier inequalities, $\mathcal{Q}^= = \emptyset$ and an intersection over $\mathcal{Q}^=$ alone would be all of $\mathbb{R}^n$, whereas $T_{\mathcal{F}^=}(x^*) = \{d : d_1 = 0\}$. Second, the closure cannot be dropped for general convex c_k . Taking $c(x) = \text{dist}(x, D)^2$ for the closed unit disk $D \subset \mathbb{R}^2$ gives $\mathcal{F}^= = D$; at a boundary point x^* the constancy cone is the set of strictly inward directions $\{d : d^\top x^* < 0\} \cup \{0\}$, whose closure $\{d : d^\top x^* \leq 0\}$ is the tangent cone.

Proof. Write $\mathcal{D}(x^*) := \bigcap_{k \in \mathcal{K}} \mathcal{D}_k(x^*)$.

$\mathcal{D}(x^*)$ is exactly the cone of feasible directions. Let $d \in \mathcal{D}(x^*)$. For each $k \in \mathcal{K}$ there is $\alpha_k > 0$ with $c_k(x^* + \alpha d) = c_k(x^*) = 0$ for $\alpha \in (0, \alpha_k)$. Since $\mathcal{K}$ is finite, $\bar{\alpha} := \min_k \alpha_k > 0$ and every c_k vanishes at $x^* + \alpha d$ for $\alpha \in (0, \bar{\alpha})$, i.e. $x^* + \alpha d \in \mathcal{F}^=$. Conversely, if $x^* + \alpha d \in \mathcal{F}^=$ for all small $\alpha > 0$ then each c_k is identically zero, hence constant, along that segment, so $d \in \mathcal{D}(x^*)$. This proves the first assertion.

Passing to the tangent cone. For a convex set the tangent cone at a point is the closure of the cone of feasible directions: $T_{\mathcal{F}^=}(x^*) = \text{cl}(\text{cone}(\mathcal{F}^= - x^*))$, and by the previous paragraph $\text{cone}(\mathcal{F}^= - x^*) = \mathcal{D}(x^*)$. This gives (5.24). It also shows where the closure is needed: a tangent direction is only a *limit* of directions along which one can actually move, so the inclusion $T_{\mathcal{F}^=}(x^*) \subseteq \mathcal{D}(x^*)$ may fail, as the disk example in the preceding remark shows.

When $\mathcal{F}^=$ is polyhedral, $\mathcal{D}(x^*)$ is a finite intersection of half-spaces and subspaces, hence already closed, and the closure may be omitted.

Normal cone identity. A cone and its closure have the same polar, so $N_{\mathcal{F}^=}(x^*) = (T_{\mathcal{F}^=}(x^*))^\perp = \mathcal{D}(x^*)^\perp$, which is (5.25). $\square$

Example 5.11 Linear Equality Constraint.

Consider the convex function $g(x) = a^\top x - b$, where $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$, and the equality-constrained feasible set $\mathcal{F}^= = \{x \in \mathbb{R}^n \mid a^\top x = b\}$.

Let $x^* \in \mathcal{F}^=$ be arbitrary. For any direction $d \in \mathbb{R}^n$ and any scalar α , we have, using $a^\top x^* = b$, $g(x^* + \alpha d) = a^\top x^* + \alpha a^\top d - b = \alpha a^\top d$. Since $g(x^*) = 0$, it follows that $g(x^* + \alpha d) = g(x^*)$ for all α if and only if $a^\top d = 0$.

Therefore, the cone of directions of constancy is $\mathcal{D}_g(x^*) = \{d \in \mathbb{R}^n \mid a^\top d = 0\}$. Since $\mathcal{F}^=$ is an affine set, its tangent cone at x^* is $T_{\mathcal{F}^=}(x^*) = \{d \in \mathbb{R}^n \mid a^\top d = 0\}$. Thus $T_{\mathcal{F}^=}(x^*) = \mathcal{D}_g(x^*)$, which verifies the constancy-tangent cone identity in the affine case. $\square$

Example 5.12 Degenerate Convex Equality.

Consider the convex function $g(x) = x^2$, $x \in \mathbb{R}$, with equality constraint $g(x) = 0$. The feasible set is $\mathcal{F}^= = \{0\}$. Let $x^* = 0$. For any direction $d \in \mathbb{R}$, $g(0 + \alpha d) = \alpha^2 d^2$. Hence $g(0 + \alpha d) = g(0)$ for all sufficiently small $\alpha > 0 \iff d = 0$.

Therefore, $\mathcal{D}_g(0) = \{0\}$. Since $\mathcal{F}^=$ is a singleton, its tangent cone is $T_{\mathcal{F}^=}(0) = \{0\}$. Thus, $T_{\mathcal{F}^=}(0) = \mathcal{D}_g(0)$, illustrating how degeneracy collapses both the constancy and tangent cones. $\square$

5.2.2 ■ Complete Lagrangian Characterization

Consider the convex optimization problem (5.1). We define the *complete Lagrangian* as

$$\mathcal{L}(x, \mu, \nu) := f(x) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x) + \sum_{i \in \mathcal{P}} \nu_i h_i(x), \quad (5.26)$$

where $\mu \in \mathbb{R}_+^{|\mathcal{Q}|}$ and $\nu \in \mathbb{R}^{|\mathcal{P}|}$. No sign restriction is imposed on the equality multipliers ν , reflecting the two-sided nature of equality constraints.

Recall that the inequality constraint index set admits a decomposition $\mathcal{Q} = \mathcal{Q}^= \dot{\cup} \mathcal{Q}^<$, and that the base feasible set is defined as

$$\mathcal{F}^= := \{x \mid h_i(x) = 0, i \in \mathcal{P}, g_j(x) = 0, j \in \mathcal{Q}^=\}$$

By Theorem 5.5, a feasible point $x^* \in \mathcal{F}^=$ is an optimal solution of (5.1) if and only if there exists a multiplier vector $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that

$$-\nabla f(x^*) - \sum_{j \in \mathcal{Q}^<} \mu_j^* \nabla g_j(x^*) \in N_{\mathcal{F}^=}(x^*). \quad (5.27)$$

When the inequality constraints satisfy Slater's condition, no inequality constraint can be active at every feasible point; equivalently, $\mathcal{Q}^= = \emptyset$. In this case, the base feasible set is determined

solely by the equality constraints. For the complete Lagrangian formulation we take the standard convex-programming case in which the equality constraints are affine, $h_i(x) = a_i^\top x - b_i$, $i \in \mathcal{P}$. Writing A for the matrix with rows $a_i^\top$, the base set is $\mathcal{F}^\circ = \{x \mid Ax = b\}$. Its normal cone is generated exactly by the equality gradients:

$$N_{\mathcal{F}^\circ}(x^*) = \text{range}(A^\top) = \text{span}\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}.$$

Thus the normal-cone condition (5.27) can be written as the complete Lagrangian stationarity condition

$$\nabla_x \mathcal{L}(x^*, \mu^*, \nu^*) = 0. \quad (5.28)$$

If the equality gradients are linearly independent, the equality multiplier ν^* is unique. This rank condition is the *Linear Independence Constraint Qualification* (LICQ).

Lemma 5.13 (Normal Cone to an Affine Equality-Feasible Set). *Let $h_i(x) = a_i^\top x - b_i$, $i \in \mathcal{P}$, and define $\mathcal{F}^\circ := \{x \mid Ax = b\}$, where the rows of A are $a_i^\top$. For any $x^* \in \mathcal{F}^\circ$,*

$$T_{\mathcal{F}^\circ}(x^*) = \ker(A), \quad N_{\mathcal{F}^\circ}(x^*) = \text{range}(A^\top).$$

If the rows of A are linearly independent, then the multiplier representation $A^\top \nu$ of a normal vector is unique.

Proof. For an affine set, feasible infinitesimal motions are exactly the directions that preserve the affine equation: $A(x^* + \epsilon d) = b$ for all small ϵ if and only if $Ad = 0$. Hence $T_{\mathcal{F}^\circ}(x^*) = \ker(A)$. The normal cone is the polar of this tangent space, so $N_{\mathcal{F}^\circ}(x^*) = \ker(A)^\perp = \text{range}(A^\top)$. If the rows of A are linearly independent, then $A^\top \nu = A^\top \tilde{\nu}$ implies $\nu = \tilde{\nu}$, giving uniqueness. $\square$

Theorem 5.14 (Complete Lagrangian Characterization under Slater and LICQ). *Consider the convex optimization problem (5.1). Assume that:*

- *Slater's condition holds for the inequality constraints, and*
- *the equality constraints satisfy the Linear Independence Constraint Qualification (LICQ) at x^* , i.e., $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent.*

Suppose that all functions are differentiable.

Then, a feasible point x^ is an optimal solution of (5.1) if and only if there exist multipliers $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}|}$ and $\nu^* \in \mathbb{R}^{|\mathcal{P}|}$ such that the following conditions hold:*

(1) **Primal feasibility:**

$$h_i(x^*) = 0, \quad i \in \mathcal{P}, \quad g_j(x^*) \leq 0, \quad j \in \mathcal{Q}.$$

(2) **Stationarity:** $\nabla_x \mathcal{L}(x^*, \mu^*, \nu^*) = 0$.

(3) **Complementary slackness:** $\mu_j^* g_j(x^*) = 0, j \in \mathcal{Q}$.

(4) **Dual feasibility:** $\mu_j^* \geq 0, j \in \mathcal{Q}$.

Moreover, under LICQ the multiplier vector ν^ is unique. Equivalently, (x^*, μ^*, ν^*) is a saddle point of the complete Lagrangian $\mathcal{L}$ over $\mathbb{R}^n \times \mathbb{R}_+^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$.*

Proof. Slater's condition guarantees a strictly feasible point, so no inequality is active throughout $\mathcal{F}$; hence $\mathcal{Q}^= = \emptyset$, $\mathcal{Q}^< = \mathcal{Q}$, and the base set reduces to the affine equality set $\mathcal{F}^= = \{x : Ax = b\}$, where A has rows $a_i^\top = \nabla h_i^\top$.

By Theorem 5.5, the feasible point x^* is optimal if and only if there exists $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}|}$ with

$$-\nabla f(x^*) - \sum_{j \in \mathcal{Q}} \mu_j^* \nabla g_j(x^*) \in N_{\mathcal{F}^=}(x^*).$$

By Lemma 5.13, $N_{\mathcal{F}^=}(x^*) = \text{range}(A^\top) = \text{span}\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$, so the inclusion holds if and only if there exists $\nu^* \in \mathbb{R}^{|\mathcal{P}|}$ with $-\nabla f(x^*) - \sum_j \mu_j^* \nabla g_j(x^*) = \sum_i \nu_i^* \nabla h_i(x^*)$; that is, $\nabla_x \mathcal{L}(x^*, \mu^*, \nu^*) = 0$, which is the stationarity condition (2). Primal feasibility (1) holds because $x^* \in \mathcal{F}$, and dual feasibility (4) is the requirement $\mu^* \geq 0$ already present in Theorem 5.5. Complementary slackness (3) follows from the saddle-point characterization of Theorem 5.2: feasibility of x^* forces $\sum_j \mu_j^* g_j(x^*) = 0$, and since each term is nonpositive, every term vanishes.

For the converse, suppose (x^*, μ^*, ν^*) satisfies conditions (1)–(4). Because $\mathcal{L}(\cdot, \mu^*, \nu^*)$ is convex and its gradient vanishes at x^* , the point x^* minimizes it over $\mathbb{R}^n$; together with primal feasibility and complementary slackness this gives, for every feasible x ,

$$f(x^*) = \mathcal{L}(x^*, \mu^*, \nu^*) \leq \mathcal{L}(x, \mu^*, \nu^*) = f(x) + \sum_j \mu_j^* g_j(x) \leq f(x),$$

so x^* is optimal. This chain of equalities and inequalities is exactly the saddle-point property of (x^*, μ^*, ν^*) . Finally, if the gradients $\{\nabla h_i(x^*)\}$ are linearly independent (LICQ), then $A^\top \nu^* = A^\top \tilde{\nu}^*$ forces $\nu^* = \tilde{\nu}^*$, so the equality multiplier is unique. $\square$

Remark 5.15 (Role of Slater's Condition and LICQ). Slater's condition implies that no inequality constraint can be active at every feasible point; that is, $\mathcal{Q}^= = \emptyset$. Consequently, all inequality constraints are effective and generate dual variables, while the base feasible set is determined solely by the equality constraints.

LICQ ensures that the normal cone to the equality-feasible set admits the exact representation $N_{\mathcal{F}^=}(x^*) = \text{span}\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$, which makes the normal-cone optimality condition equivalent to the complete Lagrangian stationarity condition. Under these assumptions, the generalized saddle-point and normal-cone characterizations developed earlier reduce to the classical Karush–Kuhn–Tucker conditions.

Theorem 5.14 therefore follows as a special case of the more general theory established in the absence of constraint qualifications.

5.2.3 ■ Case Study: Convex Optimization with Affine Constraints under Slater's Condition

We revisit the convex optimization problem with affine equality constraints and convex inequality constraints, now assuming that Slater's condition holds for the inequality constraints. This setting illustrates how the generalized framework recovers the classical Karush–Kuhn–Tucker (KKT) conditions while preserving the geometric interpretation developed earlier.

Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & Ax = b, \\ & g_j(x) \leq 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{5.29}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and each $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable.

The equality constraints define the base feasible set $\mathcal{F}^= := \{x \in \mathbb{R}^n \mid Ax = b\}$, which is assumed to be nonempty.

Slater's Condition

We assume that Slater's condition holds for the inequality constraints; that is, there exists a point $\bar{x} \in \mathcal{F}^=$ such that $g_j(\bar{x}) < 0$ for all $j \in \mathcal{Q}$.

Under this assumption, no inequality constraint is active everywhere on $\mathcal{F}^=$, and therefore $\mathcal{Q}^= = \emptyset$ and $\mathcal{Q}^< = \mathcal{Q}$.

Geometry of the Feasible Set

We derive explicit expressions for the tangent and normal cones associated with the affine feasible set $\mathcal{F}^= := \{x \in \mathbb{R}^n \mid Ax = b\}$, where $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

Tangent cone. By definition, the tangent cone to $\mathcal{F}^=$ at $x^* \in \mathcal{F}^=$ is

$$T_{\mathcal{F}^=}(x^*) := \{d \in \mathbb{R}^n \mid \exists \varepsilon_k \downarrow 0, \exists d_k \rightarrow d \text{ such that } x^* + \varepsilon_k d_k \in \mathcal{F}^= \forall k\}.$$

For any $x^* \in \mathcal{F}^=$ and any feasible sequence $x^* + \varepsilon_k d_k \in \mathcal{F}^=$, we must have $A(x^* + \varepsilon_k d_k) = b$. Since $Ax^* = b$, this implies $Ad_k = 0 \forall k$. Passing to the limit yields $Ad = 0$.

Conversely, if $d \in \mathbb{R}^n$ satisfies $Ad = 0$, then $x^* + \varepsilon d \in \mathcal{F}^= \forall \varepsilon \in \mathbb{R}$, which shows that $d \in T_{\mathcal{F}^=}(x^*)$.

Therefore $T_{\mathcal{F}^=}(x^*) = \{d \in \mathbb{R}^n \mid Ad = 0\}$.

Normal cone. The normal cone to $\mathcal{F}^=$ at x^* is defined as the polar cone of the tangent cone:

$$N_{\mathcal{F}^=}(x^*) := \{v \in \mathbb{R}^n \mid v^\top d \leq 0 \quad \forall d \in T_{\mathcal{F}^=}(x^*)\}.$$

Using the expression for the tangent cone derived above, this becomes

$$N_{\mathcal{F}^=}(x^*) = \{v \in \mathbb{R}^n \mid v^\top d = 0 \quad \forall d \text{ such that } Ad = 0\}.$$

The set of vectors orthogonal to $\ker(A)$ is precisely the range of $A^\top$. Hence $N_{\mathcal{F}^=}(x^*) = \{A^\top \lambda \mid \lambda \in \mathbb{R}^m\}$.

For any $x^* \in \mathcal{F}^=$, the tangent and normal cones are therefore given by

$$\begin{aligned} T_{\mathcal{F}^=}(x^*) &= \{d \in \mathbb{R}^n \mid Ad = 0\}, \\ N_{\mathcal{F}^=}(x^*) &= \{A^\top \lambda \mid \lambda \in \mathbb{R}^m\}. \end{aligned}$$

First-Order Optimality

By Theorem 5.5, a feasible point $x^* \in \mathcal{F}^=$ is optimal for (5.29) if and only if there exists a multiplier vector $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}|}$ such that

$$-\nabla f(x^*) - \sum_{j \in \mathcal{Q}} \mu_j^* \nabla g_j(x^*) \in N_{\mathcal{F}^=}(x^*). \quad (5.30)$$

Equivalently, there exists $\lambda^* \in \mathbb{R}^m$ such that

$$\nabla f(x^*) + \sum_{j \in \mathcal{Q}} \mu_j^* \nabla g_j(x^*) + A^\top \lambda^* = 0, \quad (5.31)$$

together with the complementary slackness conditions $\mu_j^* \geq 0$ and $\mu_j^* g_j(x^*) = 0$ for $j \in \mathcal{Q}$.

Under Slater's condition, the normal-cone optimality condition of Theorem 5.5 reduces exactly to the classical KKT conditions. In this case, all inequality constraints are effective, and no carrier inequalities are present. The generalized framework therefore strictly extends the classical theory, recovering it as a special case when standard constraint qualifications hold.

Example 5.16 Entropy Optimization with Cost and Equality Constraints.

We consider entropy regularization with a linear cost under affine equality constraints, a formulation that appears in free energy minimization, information theory, and regularized optimal transport. Let $p = (p_1, \dots, p_n) \in \mathbb{R}^n$ denote a probability vector and $c = (c_1, \dots, c_n) \in \mathbb{R}^n$ a cost vector. Consider the convex program

$$\begin{aligned} \min_{p \in \mathbb{R}^n} \quad & f(p) := \sum_{i=1}^n p_i \log p_i + \sum_{i=1}^n c_i p_i \\ \text{s.t.} \quad & \sum_{i=1}^n p_i = 1, \\ & \sum_{i=1}^n a_i p_i = b, \\ & p_i \geq 0, \quad i = 1, \dots, n. \end{aligned} \tag{5.32}$$

The function $x \mapsto x \log x$ is strictly convex on $(0, \infty)$, hence f is strictly convex on the probability simplex. Therefore, if an optimizer exists, it is unique.

Affine constraint set. The equality constraints define the affine set

$$\mathcal{F}^= = \left\{ p \in \mathbb{R}^n \mid \sum_{i=1}^n p_i = 1, \sum_{i=1}^n a_i p_i = b \right\} = \{p \in \mathbb{R}^n \mid Ap = b'\}, \tag{5.33}$$

where

$$A = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_1 & a_2 & \cdots & a_n \end{pmatrix}, \quad b' = \begin{pmatrix} 1 \\ b \end{pmatrix}.$$

For any $p^* \in \mathcal{F}^=$, the tangent space to $\mathcal{F}^=$ is

$$T_{\mathcal{F}^=}(p^*) = \left\{ d \in \mathbb{R}^n \mid \sum_{i=1}^n d_i = 0, \sum_{i=1}^n a_i d_i = 0 \right\}, \tag{5.34}$$

and the normal space is

$$N_{\mathcal{F}^=}(p^*) = \{ \lambda_1 \mathbf{1} + \lambda_2 a \mid \lambda_1, \lambda_2 \in \mathbb{R} \}, \tag{5.35}$$

where $\mathbf{1} = (1, \dots, 1)$ and $a = (a_1, \dots, a_n)$.

Lagrangian and stationarity. Introduce multipliers $\lambda_1, \lambda_2 \in \mathbb{R}$ for the two equality constraints. The Lagrangian is

$$\mathcal{L}(p, \lambda_1, \lambda_2) = \sum_{i=1}^n p_i \log p_i + \sum_{i=1}^n c_i p_i + \lambda_1 \left(\sum_{i=1}^n p_i - 1 \right) + \lambda_2 \left(\sum_{i=1}^n a_i p_i - b \right). \tag{5.36}$$

For $p_i > 0$, the partial derivative with respect to p_i is

$$\frac{\partial \mathcal{L}}{\partial p_i} = 1 + \log p_i + c_i + \lambda_1 + \lambda_2 a_i. \quad (5.37)$$

At an interior optimum p^* , the first-order stationarity condition is

$$1 + \log p_i^* + c_i + \lambda_1 + \lambda_2 a_i = 0, \quad i = 1, \dots, n. \quad (5.38)$$

Equivalently,

$$\log p_i^* = -1 - c_i - \lambda_1 - \lambda_2 a_i, \quad i = 1, \dots, n. \quad (5.39)$$

Exponentiating both sides yields

$$p_i^* = \exp(-1 - \lambda_1) \exp(-c_i - \lambda_2 a_i), \quad i = 1, \dots, n. \quad (5.40)$$

Thus all coordinates share the same multiplicative factor $\exp(-1 - \lambda_1)$, while the relative weights are determined by $\exp(-c_i - \lambda_2 a_i)$.

Normalization. To determine the constant $\exp(-1 - \lambda_1)$, we impose the probability constraint $\sum_{i=1}^n p_i^* = 1$. Summing (5.40) over i gives

$$\sum_{i=1}^n p_i^* = \exp(-1 - \lambda_1) \sum_{i=1}^n \exp(-c_i - \lambda_2 a_i).$$

Since $\sum_{i=1}^n p_i^* = 1$, it follows that

$$\exp(-1 - \lambda_1) = \frac{1}{\sum_{k=1}^n \exp(-c_k - \lambda_2 a_k)}. \quad (5.41)$$

Substituting (5.41) into (5.40) yields

$$p_i^* = \frac{\exp(-c_i - \lambda_2 a_i)}{\sum_{k=1}^n \exp(-c_k - \lambda_2 a_k)}, \quad i = 1, \dots, n. \quad (5.42)$$

This automatically enforces $\sum_i p_i^* = 1$. The remaining multiplier λ_2 must be chosen so that the second equality constraint holds:

$$\sum_{i=1}^n a_i \frac{\exp(-c_i - \lambda_2 a_i)}{\sum_{k=1}^n \exp(-c_k - \lambda_2 a_k)} = b. \quad (5.43)$$

Equivalently, if we define the partition function,

$$Z(\lambda_2) := \sum_{k=1}^n \exp(-c_k - \lambda_2 a_k), \quad (5.44)$$

then (5.42) becomes

$$p_i^* = \frac{e^{-c_i - \lambda_2 a_i}}{Z(\lambda_2)}, \quad i = 1, \dots, n, \quad (5.45)$$

and the moment constraint is

$$\sum_{i=1}^n a_i p_i^* = b. \quad (5.46)$$

Verification of optimality. Let p^* be of the form (5.42) and suppose that it satisfies the constraint (5.43). Then p^* is feasible. Moreover, by construction, there exist multipliers $\lambda_1, \lambda_2 \in \mathbb{R}$ such that (5.38) holds for every i . Hence, $\nabla f(p^*) + \lambda_1 \mathbf{1} + \lambda_2 a = 0$, which is exactly the first-order optimality condition on the affine constraint set. Since f is convex and the feasible region is convex, any feasible point satisfying the KKT conditions is a global minimizer; and because f is strictly convex on the simplex, this minimizer is unique. $\square$

The following proposition records the conclusion of Example 5.16 as a formal statement.

Proposition 5.17 (Entropy–Cost Optimization under Affine Constraints). *Assume that the feasible set of (5.32) is nonempty. Then the problem admits a unique optimal solution p^* . Moreover, there exists $\lambda_2 \in \mathbb{R}$ such that*

$$p_i^* = \frac{e^{-c_i - \lambda_2 a_i}}{\sum_{k=1}^n e^{-c_k - \lambda_2 a_k}}, \quad i = 1, \dots, n, \quad (5.47)$$

and λ_2 is determined by the constraint $\sum_{i=1}^n a_i p_i^* = b$. Conversely, any vector of the form (5.47) satisfying the moment constraint is the unique optimal solution of (5.32).

Proof. The feasible set is the intersection of the probability simplex with the affine moment constraint; it is convex, compact, and nonempty by hypothesis. Since $p \mapsto \sum_i p_i \log p_i$ is continuous and strictly convex on the simplex, it attains its minimum at a unique point p^* , which establishes existence and uniqueness. Moreover p^* is interior ($p_i^* > 0$ for all i): the directional derivative of $x \log x$ tends to $-\infty$ as $x \downarrow 0$, so moving mass onto any coordinate with $p_i = 0$ strictly decreases the objective, and no boundary point can be optimal.

At the interior optimum the affine-constrained first-order condition $\nabla f(p^*) + \lambda_1 \mathbf{1} + \lambda_2 a = 0$ holds for some multipliers λ_1, λ_2 ; this is precisely (5.38), whose solution is the Gibbs form (5.47) after eliminating λ_1 through normalization, with λ_2 fixed by the moment constraint $\sum_i a_i p_i^* = b$. Conversely, any vector of the form (5.47) is strictly positive and sums to one, and if it also meets the moment constraint it satisfies the same stationarity condition on the affine feasible set; by convexity it is then the global minimizer, hence equals p^* . $\square$

The entropy example dualizes only equality constraints. The next example adds inequality constraints, so that complementary slackness becomes decisive: it selects the subset of constraints that are binding at the optimum. It is the Chebyshev-center problem, a linear program whose active constraints are the facets of a polyhedron that the largest inscribed ball touches.

Example 5.18 Chebyshev Center of a Polyhedron.

Let a polyhedron be described by m affine inequalities,

$$\mathcal{P} = \{x \in \mathbb{R}^n : a_j^\top x \leq b_j, j = 1, \dots, m\}, \quad \|a_j\|_2 = 1,$$

where the normals have been scaled to unit length, and assume $\mathcal{P}$ has nonempty interior. The *Chebyshev center* is the center of the largest Euclidean ball $\{x_c + u : \|u\|_2 \leq r\}$ contained in $\mathcal{P}$. A ball of radius r centered at x_c lies in the halfspace $\{a_j^\top x \leq b_j\}$ if and only if $\sup_{\|u\|_2 \leq r} a_j^\top (x_c + u) = a_j^\top x_c + r \leq b_j$, so the problem is the linear program

$$\max_{x_c \in \mathbb{R}^n, r \in \mathbb{R}} r \quad \text{s.t.} \quad a_j^\top x_c + r \leq b_j, \quad j = 1, \dots, m, \quad r \geq 0.$$

All constraints are affine, and any interior point x_0 of $\mathcal{P}$ together with a small $r > 0$ satisfies every inequality strictly; hence Slater's condition holds and the complete Lagrangian characterization of Theorem 5.14 applies.

Complete Lagrangian and stationarity. Writing the objective as the minimization of $-r$ and attaching multipliers $\lambda_j \geq 0$ to the halfspace constraints and $\eta \geq 0$ to $r \geq 0$, the complete Lagrangian is

$$\mathcal{L}(x_c, r, \lambda, \eta) = -r + \sum_{j=1}^m \lambda_j (a_j^\top x_c + r - b_j) - \eta r.$$

Stationarity with respect to x_c and r gives

$$\nabla_{x_c} \mathcal{L} = \sum_{j=1}^m \lambda_j a_j = 0, \quad \frac{\partial \mathcal{L}}{\partial r} = -1 + \sum_{j=1}^m \lambda_j - \eta = 0.$$

At the optimum the largest ball has $r^* > 0$, so $\eta^* = 0$ by complementary slackness, and the second equation reduces to $\sum_{j=1}^m \lambda_j^* = 1$. The optimal multipliers thus form a probability vector satisfying $\sum_j \lambda_j^* a_j = 0$: the touched facet normals are in convex-combination equilibrium, which is exactly the condition that the ball cannot expand or translate without leaving $\mathcal{P}$.

Active facets via complementary slackness. Complementary slackness reads $\lambda_j^* (a_j^\top x_c^* + r^* - b_j) = 0$. Hence $\lambda_j^* > 0$ forces $a_j^\top x_c^* + r^* = b_j$, meaning the inscribed ball is tangent to facet j ; every facet at positive distance, $a_j^\top x_c^* + r^* < b_j$, has $\lambda_j^* = 0$ and is absent from the equilibrium condition $\sum_j \lambda_j^* a_j = 0$. Since Slater's condition holds and the program is convex, these KKT conditions are necessary and sufficient, and by Theorem 5.30 strong duality holds, with (x_c^*, r^*, λ^*) a saddle point of the complete Lagrangian. This example shows how complementary slackness partitions the constraints into the binding facets, which alone determine the center and radius, and the inactive facets, which play no role in the solution. $\square$

The Chebyshev center is a minimax notion of “center.” A complementary, averaging notion is the *barycenter*, and in the space of probability measures its natural form is the Wasserstein

(optimal-transport) barycenter. Its discrete, fixed-support version is again a linear program whose complete Lagrangian dual consists of Kantorovich potentials, providing a substantial optimal-transport illustration of the theory.

Example 5.19 Wasserstein Barycenter of Discrete Measures.

Let $\mu^{(1)}, \dots, \mu^{(K)}$ be probability measures on $\mathbb{R}^d$ with mixing weights $\rho_k > 0$, $\sum_{k=1}^K \rho_k = 1$, where each $\mu^{(k)} = \sum_{j=1}^{m_k} b_j^{(k)} \delta_{y_j^{(k)}}$ is supported on points $y_j^{(k)}$ with masses $b_j^{(k)} > 0$, $\sum_j b_j^{(k)} = 1$. The 2-Wasserstein barycenter is the measure minimizing the weighted sum of squared transport distances $\sum_{k=1}^K \rho_k W_2^2(\mu, \mu^{(k)})$; it is the Fréchet mean of the $\mu^{(k)}$ under the Wasserstein metric. We fix a candidate support $\{z_1, \dots, z_n\} \subset \mathbb{R}^d$ for the barycenter and optimize over its masses $p \in \mathbb{R}^n$ and the transport plans $\pi^{(k)} \in \mathbb{R}^{n \times m_k}$, with squared-distance costs $C_{ij}^{(k)} = \|z_i - y_j^{(k)}\|_2^2$. The fixed-support barycenter is then the linear program

$$\min_{p, \{\pi^{(k)}\}} \sum_{k=1}^K \rho_k \sum_{i=1}^n \sum_{j=1}^{m_k} C_{ij}^{(k)} \pi_{ij}^{(k)}$$

subject to

$$\sum_{i=1}^n \pi_{ij}^{(k)} = b_j^{(k)}, \quad \sum_{j=1}^{m_k} \pi_{ij}^{(k)} = p_i, \quad \sum_{i=1}^n p_i = 1, \quad \pi_{ij}^{(k)} \geq 0, \quad p_i \geq 0.$$

The first family of constraints fixes the marginal of each plan to the data measure $\mu^{(k)}$, the second couples every plan to the common barycenter masses p , and the third normalizes p . (Optimizing the support locations z_i as well would make the problem nonconvex; with the support fixed it is a convex linear program.) All constraints are affine and, for instance, the independent couplings $\pi_{ij}^{(k)} = p_i b_j^{(k)}$ with p uniform are strictly positive and feasible, so Slater's condition holds and Theorem 5.14 applies.

Complete Lagrangian and Kantorovich potentials. Attach potentials $\psi_j^{(k)} \in \mathbb{R}$ to the data marginals, $\phi_i^{(k)} \in \mathbb{R}$ to the coupling constraints, and $\lambda \in \mathbb{R}$ to the normalization. The complete Lagrangian is

$$\begin{aligned} \mathcal{L} = & \sum_{k=1}^K \rho_k \langle C^{(k)}, \pi^{(k)} \rangle + \sum_{k=1}^K \sum_{j=1}^{m_k} \psi_j^{(k)} \left(b_j^{(k)} - \sum_i \pi_{ij}^{(k)} \right) \\ & + \sum_{k=1}^K \sum_{i=1}^n \phi_i^{(k)} \left(p_i - \sum_j \pi_{ij}^{(k)} \right) + \lambda \left(1 - \sum_i p_i \right). \end{aligned}$$

Because the program is linear with nonnegativity constraints on $\pi^{(k)}$ and p , minimizing $\mathcal{L}$ over the nonnegative orthant gives the stationarity (reduced-cost) conditions

$$\phi_i^{(k)} + \psi_j^{(k)} \leq \rho_k C_{ij}^{(k)} \quad \forall i, j, k, \quad \sum_{k=1}^K \phi_i^{(k)} \geq \lambda \quad \forall i,$$

each holding with equality on the coordinates that carry positive mass.

Complementary slackness. The two tightness conditions are exactly complementary slackness:

$$\pi_{ij}^{(k)\star} > 0 \implies \phi_i^{(k)\star} + \psi_j^{(k)\star} = \rho_k C_{ij}^{(k)}, \quad p_i^\star > 0 \implies \sum_{k=1}^K \phi_i^{(k)\star} = \lambda^\star.$$

The first is the classical optimal-transport support condition: mass flows only on pairs $(z_i, y_j^{(k)})$ where the potentials meet the cost. The second states that the barycenter places mass only on support points z_i that minimize the aggregated potential $\sum_k \phi_i^{(k)}$, whose common value is $\lambda^\star = \min_i \sum_k \phi_i^{(k)\star}$.

Dual problem and strong duality. Substituting the stationarity conditions collapses $\mathcal{L}$ to its constant part, yielding the complete dual problem

$$\max_{\phi, \psi, \lambda} \sum_{k=1}^K \sum_{j=1}^{m_k} b_j^{(k)} \psi_j^{(k)} + \lambda \quad \text{s.t.} \quad \phi_i^{(k)} + \psi_j^{(k)} \leq \rho_k C_{ij}^{(k)}, \quad \sum_{k=1}^K \phi_i^{(k)} \geq \lambda,$$

a Kantorovich-type dual coupling the K transport problems through the shared constraint $\sum_k \phi_i^{(k)} \geq \lambda$. Since Slater's condition holds and the program is convex, by Theorem 5.30 strong duality holds with no gap, and any optimal primal–dual pair is a saddle point of the complete Lagrangian. The example illustrates how the complete Lagrangian simultaneously produces the transport potentials, the coupling that ties the measures to a common barycenter, and the complementary-slackness rules that select the active transport pairs and the barycenter's support. $\square$

5.2.4 ■ Associated Dual Problems: Restricted and Complete Lagrangian Formulations

This subsection compares the two dual descriptions used throughout the chapter. The *restricted* formulation remains meaningful without Slater's condition because it dualizes only effective inequalities. The *complete* formulation is the classical Lagrangian dual; it becomes exact under Slater's condition and LICQ. The two formulations coincide when strict feasibility holds, but differ when some inequalities are really part of the feasible geometry.

Restricted Lagrangian Formulation

Let $\mathcal{Q}^<$ denote the index set of *effective* inequality constraints, and let $\mathcal{Q}^=$ denote the complementary index set of ineffective inequality constraints. Define the *base feasible set*

$$\mathcal{F}^= := \{x \in \mathbb{R}^n : h_i(x) = 0, i \in \mathcal{P}, g_j(x) = 0, j \in \mathcal{Q}^=\},$$

obtained by incorporating all equality constraints and the inequality constraints indexed by $\mathcal{Q}^=$ directly into the feasible region.

Definition 5.20 (Restricted Lagrangian). *The restricted Lagrangian associated with (5.1) is*

defined by

$$\mathcal{L}^c(x, \mu) = f(x) + \sum_{j \in \mathcal{Q}^<} \mu_j g_j(x), \quad x \in \mathcal{F}^=, \mu \in \mathbb{R}_+^c. \quad (5.48)$$

Unlike the complete Lagrangian, multipliers are introduced only for the inequality constraints indexed by $\mathcal{Q}^<$. Constraints indexed by $\mathcal{Q}^=$ are enforced through the geometry of $\mathcal{F}^=$ and do not give rise to independent dual variables.

Proposition 5.21 (Restricted Dual Function). *The restricted dual function associated with $\mathcal{L}^c$ is given by*

$$d(\mu) := \inf_{x \in \mathcal{F}^=} \left\{ f(x) + \sum_{j \in \mathcal{Q}^<} \mu_j g_j(x) \right\}, \quad \mu \in \mathbb{R}_+^c. \quad (5.49)$$

Moreover, $d(\mu)$ is a concave function on $\mathbb{R}_+^c$.

Proof. Concavity in μ has nothing to do with convexity in x : it comes from the fact that, for each fixed $x \in \mathcal{F}^=$, the map $\mu \mapsto f(x) + \sum_{j \in \mathcal{Q}^<} \mu_j g_j(x)$ is affine. The function d is therefore a pointwise infimum of affine functions of μ , and such an infimum is concave. Explicitly, for $\mu^1, \mu^2 \in \mathbb{R}_+^c$ and $\theta \in [0, 1]$,

$$d(\theta\mu^1 + (1-\theta)\mu^2) = \inf_{x \in \mathcal{F}^=} [\theta \mathcal{L}^c(x, \mu^1) + (1-\theta) \mathcal{L}^c(x, \mu^2)] \geq \theta d(\mu^1) + (1-\theta) d(\mu^2).$$

Note that no convexity of f , g_j or $\mathcal{F}^=$ is used; the dual function is concave even for nonconvex data. $\square$

Definition 5.22 (Restricted Dual Problem). *The restricted dual problem associated with (5.1) is*

$$\max_{\mu \in \mathbb{R}_+^c} d(\mu), \quad (5.50)$$

where $d(\mu)$ is defined in (5.49).

Theorem 5.23 (Weak Duality for the Restricted Formulation). *For any primal feasible point $x \in \mathcal{F}$ and any $\mu \in \mathbb{R}_+^c$,*

$$d(\mu) \leq \mathcal{L}^c(x, \mu) \leq f(x).$$

Consequently, the optimal value of the restricted dual problem provides a lower bound on the primal optimal value.

Proof. By definition of $d(\mu)$, $d(\mu) = \inf_{y \in \mathcal{F}^=} \mathcal{L}^c(y, \mu) \leq \mathcal{L}^c(x, \mu)$ for any $x \in \mathcal{F}^=$. If $x \in \mathcal{F}$ is primal feasible, then $g_j(x) \leq 0$ for all $j \in \mathcal{Q}^<$, and hence

$$\mathcal{L}^c(x, \mu) = f(x) + \sum_{j \in \mathcal{Q}^<} \mu_j g_j(x) \leq f(x),$$

since $\mu_j \geq 0$. $\square$

Theorem 5.24 (Strong Duality via Restricted Saddle Points). *Strong duality holds between the primal problem (5.1) and the restricted dual problem (5.50) if and only if there exists $(x^*, \mu^*) \in$*

$\mathcal{F}^= \times \mathbb{R}_+^c$ such that (x^*, μ^*) is a saddle point of the restricted Lagrangian $\mathcal{L}^c$, i.e.,

$$\mathcal{L}^c(x^*, \mu) \leq \mathcal{L}^c(x^*, \mu^*) \leq \mathcal{L}^c(x, \mu^*), \quad \forall x \in \mathcal{F}^=, \quad \forall \mu \in \mathbb{R}_+^c. \quad (5.51)$$

In this case,

$$\min_{x \in \mathcal{F}} f(x) = \max_{\mu \in \mathbb{R}_+^c} d(\mu),$$

and μ^* is dual optimal.

Proof. If (x^*, μ^*) is a saddle point of $\mathcal{L}^c$, then

$$d(\mu^*) = \inf_{x \in \mathcal{F}^=} \mathcal{L}^c(x, \mu^*) = \mathcal{L}^c(x^*, \mu^*).$$

By the right-hand inequality in (5.51) and primal feasibility of x^* , we have $\mathcal{L}^c(x^*, \mu^*) \leq f(x^*)$. Combining this with weak duality yields $d(\mu^*) = f(x^*)$, which establishes strong duality and dual optimality of μ^* .

Conversely, if strong duality holds and $f(x^*) = d(\mu^*)$, then (x^*, μ^*) satisfies the saddle-point inequalities (5.51) by standard convex duality arguments. $\square$

Complete Lagrangian Formulation

We now consider the classical Lagrangian formulation under standard regularity assumptions.

Assumption 5.25 (Slater and LICQ). Assume that:

- Slater's condition holds for the inequality constraints, i.e., there exists $\bar{x} \in \mathbb{R}^n$ such that

$$g_j(\bar{x}) < 0, \quad \forall j \in \mathcal{Q}, \quad h_i(\bar{x}) = 0, \quad \forall i \in \mathcal{P};$$

- the equality constraints satisfy the Linear Independence Constraint Qualification (LICQ) at an optimal solution x^* , i.e., $\{\nabla h_i(x^*)\}_{i \in \mathcal{P}}$ are linearly independent.

Definition 5.26 (Complete Lagrangian). Under Assumption 5.25, the complete Lagrangian associated with the convex optimization problem (5.1) is defined as

$$\mathcal{L}(x, \mu, \nu) = f(x) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x) + \sum_{i \in \mathcal{P}} \nu_i h_i(x), \quad \mu \in \mathbb{R}_+^{|\mathcal{Q}|}, \quad \nu \in \mathbb{R}^{|\mathcal{P}|}. \quad (5.52)$$

Proposition 5.27 (Complete Dual Function). The dual function associated with the complete Lagrangian is given by

$$d(\mu, \nu) := \inf_{x \in \mathbb{R}^n} \left\{ f(x) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x) + \sum_{i \in \mathcal{P}} \nu_i h_i(x) \right\}. \quad (5.53)$$

Moreover, $d(\mu, \nu)$ is concave on $\mathbb{R}_+^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$.

Proof. As in Proposition 5.21, concavity follows from affineness in the multipliers rather than from convexity in x . For each fixed $x \in \mathbb{R}^n$ the map

$$(\mu, \nu) \mapsto f(x) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x) + \sum_{i \in \mathcal{P}} \nu_i h_i(x)$$

is affine, and d is the pointwise infimum over x of this family; a pointwise infimum of affine functions is concave. Convexity of f and g_j and affinity of h_i are needed elsewhere—for strong duality—but not here. $\square$

Definition 5.28 (Complete Dual Problem). *The complete dual problem associated with (5.1) is*

$$\begin{aligned} \max_{\mu, \nu} \quad & d(\mu, \nu) \\ \text{s.t.} \quad & \mu \geq 0, \end{aligned} \tag{5.54}$$

where $d(\mu, \nu)$ is defined in (5.53).

Theorem 5.29 (Weak Duality for the Complete Formulation). *For any primal feasible point $x \in \mathcal{F}$ and any dual feasible (μ, ν) ,*

$$d(\mu, \nu) \leq \mathcal{L}(x, \mu, \nu) \leq f(x).$$

Proof. By definition of $d(\mu, \nu)$,

$$d(\mu, \nu) = \inf_{y \in \mathbb{R}^n} \mathcal{L}(y, \mu, \nu) \leq \mathcal{L}(x, \mu, \nu).$$

If x is primal feasible, then $g_j(x) \leq 0$ and $h_i(x) = 0$ for all j, i , and since $\mu \geq 0$,

$$\mathcal{L}(x, \mu, \nu) = f(x) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x) \leq f(x).$$

$\square$

Theorem 5.30 (Strong Duality and Saddle-Point Characterization). *Under Assumption 5.25, strong duality holds between the primal problem (5.1) and the complete dual problem (5.54), i.e.,*

$$\min_{x \in \mathcal{F}} f(x) = \max_{\mu \geq 0, \nu} d(\mu, \nu). \tag{5.55}$$

Moreover, there exists a primal–dual optimal triple (x^, μ^*, ν^*) such that $f(x^*) = d(\mu^*, \nu^*)$. This equality holds if and only if (x^*, μ^*, ν^*) is a saddle point of the complete Lagrangian, i.e.,*

$$\mathcal{L}(x^*, \mu, \nu) \leq \mathcal{L}(x^*, \mu^*, \nu^*) \leq \mathcal{L}(x, \mu^*, \nu^*), \quad \forall x \in \mathbb{R}^n, \forall \mu \geq 0, \forall \nu. \tag{5.56}$$

Proof. Let x^* be an optimal solution of the primal problem (5.1). Under Assumption 5.25, Theorem 5.14 provides multipliers $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}|}$ and $\nu^* \in \mathbb{R}^{|\mathcal{P}|}$ satisfying primal feasibility, stationarity $\nabla_x \mathcal{L}(x^*, \mu^*, \nu^*) = 0$, complementary slackness $\mu_j^* g_j(x^*) = 0$ for all $j \in \mathcal{Q}$, and dual feasibility $\mu^* \geq 0$. Because f and each g_j are convex and each h_i is affine, the map $x \mapsto \mathcal{L}(x, \mu^*, \nu^*)$ is convex, so the vanishing of its gradient at x^* implies that x^* is a global minimizer of $\mathcal{L}(\cdot, \mu^*, \nu^*)$ over $\mathbb{R}^n$. Consequently

$$d(\mu^*, \nu^*) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \mu^*, \nu^*) = \mathcal{L}(x^*, \mu^*, \nu^*) = f(x^*) + \sum_{j \in \mathcal{Q}} \mu_j^* g_j(x^*) + \sum_{i \in \mathcal{P}} \nu_i^* h_i(x^*) = f(x^*), \tag{5.57}$$

where the last equality uses $h_i(x^*) = 0$ together with the complementary slackness relations $\mu_j^* g_j(x^*) = 0$.

The identity (5.57) already yields strong duality. Indeed, weak duality (Theorem 5.29) gives $d(\mu, \nu) \leq f(x)$ for every primal feasible x and every dual feasible (μ, ν) , whence

$$\max_{\mu \geq 0, \nu} d(\mu, \nu) \leq \min_{x \in \mathcal{F}} f(x).$$

Since x^* is primal feasible and (μ^*, ν^*) is dual feasible, (5.57) forces this inequality to hold with equality, and it identifies x^* and (μ^*, ν^*) as maximizers and minimizers of the respective problems. Thus the extrema in (5.55) are attained by the triple (x^*, μ^*, ν^*) , and $f(x^*) = d(\mu^*, \nu^*)$.

It remains to show that the equality $f(x^*) = d(\mu^*, \nu^*)$ is equivalent to the saddle-point property (5.56). Suppose first that $f(x^*) = d(\mu^*, \nu^*)$, with x^* primal feasible and $\mu^* \geq 0$. Feasibility gives $\mathcal{L}(x^*, \mu^*, \nu^*) = f(x^*) + \sum_j \mu_j^* g_j(x^*) \leq f(x^*)$, while weak duality gives $f(x^*) = d(\mu^*, \nu^*) \leq \mathcal{L}(x^*, \mu^*, \nu^*)$; hence both inequalities are equalities. In particular $\sum_j \mu_j^* g_j(x^*) = 0$ and $d(\mu^*, \nu^*) = \mathcal{L}(x^*, \mu^*, \nu^*)$, so x^* minimizes $\mathcal{L}(\cdot, \mu^*, \nu^*)$ over $\mathbb{R}^n$; this is the right-hand inequality in (5.56). For the left-hand inequality, let $\mu \geq 0$ and ν be arbitrary. Using $g_j(x^*) \leq 0$ and $h_i(x^*) = 0$,

$$\mathcal{L}(x^*, \mu, \nu) = f(x^*) + \sum_{j \in \mathcal{Q}} \mu_j g_j(x^*) + \sum_{i \in \mathcal{P}} \nu_i h_i(x^*) \leq f(x^*) = \mathcal{L}(x^*, \mu^*, \nu^*),$$

which establishes the saddle-point inequalities (5.56). Conversely, if (x^*, μ^*, ν^*) is a saddle point, then the right-hand inequality shows that x^* minimizes $\mathcal{L}(\cdot, \mu^*, \nu^*)$, so $d(\mu^*, \nu^*) = \mathcal{L}(x^*, \mu^*, \nu^*)$; taking the supremum of the left-hand inequality over $\mu \geq 0$ and ν shows that this supremum is finite, which forces $g_j(x^*) \leq 0$ and $h_i(x^*) = 0$, and, comparing the choices (μ^*, ν^*) and $(0, \nu^*)$, yields $\sum_j \mu_j^* g_j(x^*) = 0$. Therefore $\mathcal{L}(x^*, \mu^*, \nu^*) = f(x^*)$, and combining the two displays gives $d(\mu^*, \nu^*) = f(x^*)$, as claimed. $\square$

Corollary 5.31 (Uniqueness of Equality Multipliers). *Under LICQ, the equality multiplier vector ν^* is unique. Consequently, the optimal value function is locally differentiable with respect to perturbations of the equality constraints.*

Proof. LICQ ensures linear independence of the gradients of the equality constraints at x^* , which implies uniqueness of the corresponding Lagrange multipliers by standard sensitivity analysis results. $\square$

Comparison with the Restricted Formulation. When Slater's condition holds, all inequality constraints are effective and the restricted and complete Lagrangian formulations coincide. When Slater's condition fails, the complete Lagrangian formulation may fail to admit a saddle point or a dual optimizer. In such cases, the restricted Lagrangian formulation provides a dual characterization that restores strong duality by introducing multipliers only for the effective inequality constraints, while absorbing the remaining constraints into the geometry of the feasible set.

The following example makes the complete Lagrangian and its dual entirely explicit for an equality-constrained quadratic program, where every object in Theorem 5.30 admits a closed form.

Example 5.32 Equality-Constrained Quadratic Program via the Complete Lagrangian.

Let $Q \in \mathbb{R}^{n \times n}$ be symmetric positive definite, let $A \in \mathbb{R}^{m \times n}$ have full row rank $m \leq n$, and let $c \in \mathbb{R}^n$, $b \in \mathbb{R}^m$. Consider the convex program

$$\min_{x \in \mathbb{R}^n} f(x) := \frac{1}{2} x^\top Q x - c^\top x \quad \text{s.t.} \quad Ax = b.$$

There are no inequality constraints, so Slater's condition holds vacuously, and LICQ is exactly the full-row-rank assumption on A ; Assumption 5.25 is therefore satisfied and Theorem 5.30 applies. The complete Lagrangian carries only the equality multiplier $\nu \in \mathbb{R}^m$,

$$\mathcal{L}(x, \nu) = \frac{1}{2} x^\top Q x - c^\top x + \nu^\top (Ax - b).$$

Dual function. Since $Q \succ 0$, the map $x \mapsto \mathcal{L}(x, \nu)$ is strictly convex, and its unconstrained minimizer is obtained from $\nabla_x \mathcal{L}(x, \nu) = Qx - c + A^\top \nu = 0$, namely $x(\nu) = Q^{-1}(c - A^\top \nu)$. Substituting $x(\nu)$ into $\mathcal{L}$ and simplifying yields the dual function

$$d(\nu) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \nu) = -\frac{1}{2} (c - A^\top \nu)^\top Q^{-1} (c - A^\top \nu) - b^\top \nu,$$

a concave quadratic in ν with Hessian $-AQ^{-1}A^\top \prec 0$.

Dual optimum and strong duality. The dual problem $\max_\nu d(\nu)$ is unconstrained, and its stationarity condition $\nabla_\nu d(\nu) = AQ^{-1}(c - A^\top \nu) - b = 0$ gives, using the invertibility of $AQ^{-1}A^\top$ (a consequence of $Q \succ 0$ and full row rank of A), $\nu^* = (AQ^{-1}A^\top)^{-1}(AQ^{-1}c - b)$. The corresponding primal point $x^* = Q^{-1}(c - A^\top \nu^*)$ is feasible, since

$$Ax^* = AQ^{-1}c - AQ^{-1}A^\top \nu^* = AQ^{-1}c - (AQ^{-1}c - b) = b,$$

and the pair (x^*, ν^*) solves the block KKT system

$$\begin{pmatrix} Q & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} x^* \\ \nu^* \end{pmatrix} = \begin{pmatrix} c \\ b \end{pmatrix}.$$

By Theorem 5.30 strong duality holds with no duality gap, $f(x^*) = d(\nu^*)$, and (x^*, ν^*) is a saddle point of $\mathcal{L}$. Geometrically, ν^* measures the sensitivity of the optimal cost to perturbations of the right-hand side b , anticipating the value-function interpretation of multipliers developed later in this chapter (Theorem 5.42). $\square$

5.2.5 ■ Case Study: Discrete-Time Convex Optimal Control

We illustrate how the Lagrangian duality framework developed earlier extends to dynamic optimization problems by considering a discrete-time convex optimal control problem. This case study makes explicit how strong duality, adjoint equations, and stagewise decomposition arise from the complete Lagrangian.

Problem 5.33 (Discrete-Time Convex Optimal Control). Let $T \geq 2$ be a finite horizon.

Consider the optimization problem

$$\begin{aligned} \min_{\{u_t\}_{t=1}^{T-1}} \quad & \sum_{t=1}^{T-1} f_t(u_t) \\ \text{s.t.} \quad & x_{t+1} = A_t x_t + B_t u_t, \quad t = 1, \dots, T-1, \\ & x_T = d, \\ & u_t \in U_t, \quad t = 1, \dots, T-1, \end{aligned} \quad (5.58)$$

where the initial state $x_1 \in \mathbb{R}^n$ is given. The state variables satisfy $x_t \in \mathbb{R}^n$, the control variables satisfy $u_t \in \mathbb{R}^m$, the matrices $A_t \in \mathbb{R}^{n \times n}$ and $B_t \in \mathbb{R}^{n \times m}$ are given, and $d \in \mathbb{R}^n$ is a prescribed terminal state.

Assumption 5.34 (Convexity and Slater Condition). For each $t = 1, \dots, T-1$, the function $f_t : \mathbb{R}^m \rightarrow \mathbb{R}$ is convex and the set $U_t \subset \mathbb{R}^m$ is convex. Moreover, Slater's condition holds: there exists a control sequence $\{\bar{u}_t\}_{t=1}^{T-1}$ with $\bar{u}_t \in \text{int } U_t$ such that the corresponding state trajectory $\{\bar{x}_t\}_{t=1}^T$, generated by (5.58), satisfies $\bar{x}_T = d$.

Under Assumption 5.34, problem (5.58) is a convex optimization problem for which strong duality holds.

Complete Lagrangian. Introduce Lagrange multipliers $\mu_t \in \mathbb{R}^n$ for the state transition constraints and a multiplier $\nu \in \mathbb{R}^n$ for the terminal constraint $x_T = d$. The complete Lagrangian associated with (5.58) is

$$\mathcal{L}(x, u, \mu, \nu) = \sum_{t=1}^{T-1} f_t(u_t) + \sum_{t=1}^{T-1} \langle \mu_t, x_{t+1} - A_t x_t - B_t u_t \rangle + \langle \nu, d - x_T \rangle. \quad (5.59)$$

Dual function. The dual function is defined by minimizing the Lagrangian over the primal variables, subject to $u_t \in U_t$ and with x_1 fixed:

$$d(\mu, \nu) := \inf_{\substack{x_2, \dots, x_T \in \mathbb{R}^n \\ u_t \in U_t}} \mathcal{L}(x, u, \mu, \nu). \quad (5.60)$$

Expanding the inner products in (5.59) yields

$$\mathcal{L} = \sum_{t=1}^{T-1} f_t(u_t) + \sum_{t=1}^{T-1} \langle \mu_t, x_{t+1} \rangle - \sum_{t=1}^{T-1} \langle \mu_t, A_t x_t \rangle - \sum_{t=1}^{T-1} \langle \mu_t, B_t u_t \rangle + \langle \nu, d \rangle - \langle \nu, x_T \rangle.$$

Using $\langle \mu_t, A_t x_t \rangle = \langle A_t^\top \mu_t, x_t \rangle$, we group terms according to the state variables.

The terminal state x_T appears only through

$$\langle \mu_{T-1}, x_T \rangle - \langle \nu, x_T \rangle = \langle \mu_{T-1} - \nu, x_T \rangle.$$

If $\mu_{T-1} \neq \nu$, the infimum over $x_T \in \mathbb{R}^n$ is $-\infty$. Hence, finiteness of the dual function requires

$$\mu_{T-1} = \nu. \quad (5.61)$$

For intermediate states x_t , $t = 2, \dots, T-1$, the corresponding terms are

$$\langle \mu_{t-1}, x_t \rangle - \langle A_t^\top \mu_t, x_t \rangle = \langle \mu_{t-1} - A_t^\top \mu_t, x_t \rangle.$$

Again, finiteness of the infimum over $x_t \in \mathbb{R}^n$ requires

$$\mu_{t-1} = A_t^\top \mu_t, \quad t = 2, \dots, T-1. \quad (5.62)$$

Equations (5.61) and (5.62) together yield the discrete-time adjoint equations

$$\mu_{t-1} = A_t^\top \mu_t, \quad t = 2, \dots, T-1, \quad \mu_{T-1} = \nu. \quad (5.63)$$

Assuming (5.63) holds, all dependence on $x_2, \dots, x_T$ vanishes and the Lagrangian reduces to

$$\mathcal{L}(x, u, \mu, \nu) = \langle \nu, d \rangle - \langle A_1^\top \mu_1, x_1 \rangle + \sum_{t=1}^{T-1} (f_t(u_t) - \langle \mu_t, B_t u_t \rangle), \quad (5.64)$$

where the term involving x_1 is constant. The dual function is obtained from (5.64) by the remaining minimization over the controls,

$$d(\mu, \nu) = \langle \nu, d \rangle - \langle A_1^\top \mu_1, x_1 \rangle + \inf_{u_t \in U_t, t=1, \dots, T-1} \sum_{t=1}^{T-1} (f_t(u_t) - \langle \mu_t, B_t u_t \rangle).$$

The control variables decouple across time, so this minimization separates as

$$\inf_{u_t \in U_t} \sum_{t=1}^{T-1} (f_t(u_t) - \langle \mu_t, B_t u_t \rangle) = \sum_{t=1}^{T-1} \inf_{u_t \in U_t} (f_t(u_t) - \langle \mu_t, B_t u_t \rangle).$$

For each t , the stagewise minimization problem is

$$\min_{u_t \in U_t} \{f_t(u_t) - \langle \mu_t, B_t u_t \rangle\}. \quad (5.65)$$

Let $\hat{u}_t(\mu_t)$ denote its (assumed unique) minimizer.

Substituting $\hat{u}_t(\mu_t)$ into (5.64) yields the dual function

$$d(\nu) = \sum_{t=1}^{T-1} [f_t(\hat{u}_t(\mu_t(\nu))) - \langle \mu_t(\nu), B_t \hat{u}_t(\mu_t(\nu)) \rangle] + \langle \nu, d \rangle - \langle A_1^\top \mu_1(\nu), x_1 \rangle, \quad (5.66)$$

where $\mu_t(\nu)$ is determined uniquely from ν via (5.63).

The associated dual problem is

$$\max_{\nu \in \mathbb{R}^n} d(\nu). \quad (5.67)$$

Theorem 5.35 (Dynamic Decomposition via the Complete Lagrangian). *Under Assumption 5.34, the following statements hold:*

- (1) *Strong duality holds between the primal problem (5.58) and the dual problem (5.67).*
- (2) *There exist multipliers (μ^*, ν^*) satisfying the adjoint equations (5.63).*
- (3) *The optimal control is obtained stagewise by solving (5.65) with $\mu_t = \mu_t^*$.*
- (4) *A feasible trajectory (x^*, u^*) is optimal if and only if (x^*, u^*, μ^*, ν^*) is a saddle point of the complete Lagrangian (5.59).*

Proof. Problem (5.58) is convex: each f_t is convex, each U_t is convex, and the dynamics and terminal constraint are affine in (x, u) . Under Assumption 5.34 a strictly feasible control sequence exists, so Slater's condition holds and, by the strong-duality and saddle-point theorem for convex programs (Theorem 5.30), the optimal primal value equals the optimal dual value and a saddle point of the complete Lagrangian (5.59) exists. This proves (1) and (4), the latter being exactly the statement that primal- and dual-optimal points are paired as a saddle point.

To identify the multipliers, recall the dual-function computation preceding the theorem. Grouping the terms of $\mathcal{L}$ by the state variables, the coefficient of each free state x_t ($t = 2, \dots, T$) is affine, so the infimum over $x_t \in \mathbb{R}^n$ is finite only if that coefficient vanishes; these vanishing conditions are precisely the adjoint equations $\mu_{t-1} = A_t^\top \mu_t$ with $\mu_{T-1} = \nu$ in (5.63). Any dual optimizer (μ^*, ν^*) keeps the dual function finite and therefore satisfies (5.63), giving (2). With the adjoint relations enforced, all dependence on the states cancels and the Lagrangian separates across time into the stagewise problems $\min_{u_t \in U_t} \{f_t(u_t) - \langle \mu_t^*, B_t u_t \rangle\}$ of (5.65); minimizing $\mathcal{L}$ over the primal variables at the saddle point thus reduces to solving these decoupled problems with $\mu_t = \mu_t^*$, which is (3). $\square$

5.2.6 ■ Extension to Output-Feedback Linear-Quadratic Regulator

We now extend the dynamic duality framework to a linear-quadratic regulator (LQR) with output penalties. The point of the exercise is not the answer, which is the familiar Riccati recursion, but the route: everything follows from forming the Lagrangian and eliminating the primal variables one block at a time. Penalizing the output rather than the full state changes only one term, and watching where that term lands in the adjoint equation is the quickest way to see what the Riccati recursion is actually doing.

Let the system dynamics be

$$x_{t+1} = A_t x_t + B_t u_t, \quad y_t = C_t x_t,$$

and consider the finite-horizon optimization problem

$$\begin{aligned} \min_{\{u_t\}_{t=1}^{T-1}} \quad & \sum_{t=1}^{T-1} \left(\frac{1}{2} u_t^\top R_t u_t + \frac{1}{2} y_t^\top Q_t y_t \right) \\ \text{s.t.} \quad & x_{t+1} = A_t x_t + B_t u_t, \\ & x_T = d, \end{aligned} \tag{5.68}$$

where $R_t \in \mathbb{R}^{m \times m}$ satisfies $R_t \succ 0$ and $Q_t \in \mathbb{R}^{p \times p}$ satisfies $Q_t \succeq 0$. Equivalently, the stage cost can be written as

$$\frac{1}{2} u_t^\top R_t u_t + \frac{1}{2} x_t^\top C_t^\top Q_t C_t x_t.$$

Problem (5.68) is a strictly convex optimization problem, and strong duality holds, so nothing is lost by working with the dual.

We therefore begin by attaching a multiplier to every constraint. Introduce multipliers $\mu_t \in \mathbb{R}^n$ for the state transition constraints and $\nu \in \mathbb{R}^n$ for the terminal constraint $x_T = d$. The complete Lagrangian is

$$\begin{aligned} \mathcal{L}(x, u, \mu, \nu) = & \sum_{t=1}^{T-1} \left(\frac{1}{2} u_t^\top R_t u_t + \frac{1}{2} x_t^\top C_t^\top Q_t C_t x_t \right) \\ & + \sum_{t=1}^{T-1} \langle \mu_t, x_{t+1} - A_t x_t - B_t u_t \rangle + \langle \nu, d - x_T \rangle. \end{aligned} \tag{5.69}$$

The dual function is obtained by minimizing $\mathcal{L}$ over the primal variables, and it is convenient to do this one block at a time, states first. Collecting terms involving x_t for $t = 2, \dots, T-1$ yields

$$\frac{1}{2}x_t^\top C_t^\top Q_t C_t x_t + \langle \mu_{t-1}, x_t \rangle - \langle A_t^\top \mu_t, x_t \rangle.$$

Minimization over $x_t \in \mathbb{R}^n$ requires the first-order condition

$$C_t^\top Q_t C_t x_t + \mu_{t-1} - A_t^\top \mu_t = 0,$$

which gives the modified adjoint recursion

$$\mu_{t-1} = A_t^\top \mu_t - C_t^\top Q_t C_t x_t, \quad t = 2, \dots, T-1. \quad (5.70)$$

For the terminal state x_T , as before, finiteness of the dual function implies $\mu_{T-1} = \nu$. Note where the output weighting has gone: it enters the adjoint recursion only through the matrix $C_t^\top Q_t C_t$, so penalizing the output is indistinguishable from penalizing the state with that particular weight.

Minimizing over the controls is simpler, because they are unconstrained and appear quadratically. The terms involving u_t are $\frac{1}{2}u_t^\top R_t u_t - \langle \mu_t, B_t u_t \rangle$. Taking the gradient with respect to u_t yields $R_t u_t - B_t^\top \mu_t = 0$, and therefore the optimal control is

$$u_t^* = R_t^{-1} B_t^\top \mu_t. \quad (5.71)$$

So far the optimality system couples a forward state recursion to a backward costate recursion, which is not yet a usable control law. The step that closes the loop is the observation that, at optimality, the costate is linear in the state,

$$\mu_t = -P_{t+1} x_{t+1}, \quad (5.72)$$

for a matrix sequence $\{P_t\}$ still to be determined. The minus sign is dictated by the sign convention adopted for the multipliers: μ_t was attached to $x_{t+1} - A_t x_t - B_t u_t$, and with $u_t^* = R_t^{-1} B_t^\top \mu_t$ the positive-sign ansatz would produce $u_t^* = (R_t - B_t^\top P_{t+1} B_t)^{-1} B_t^\top P_{t+1} A_t x_t$, which has both the wrong sign and the wrong curvature term relative to (5.74). Substituting this relation, together with the dynamics and (5.71), into (5.70) yields the backward Riccati recursion

$$\begin{aligned} P_t &= C_t^\top Q_t C_t + A_t^\top P_{t+1} A_t \\ &\quad - A_t^\top P_{t+1} B_t (R_t + B_t^\top P_{t+1} B_t)^{-1} B_t^\top P_{t+1} A_t, \end{aligned} \quad (5.73)$$

with terminal condition $P_T = 0$ (or $P_T = C_T^\top Q_T C_T$ if a terminal output penalty is included). The matrix P_t is thus generated backward in time, before any state is known, which is what makes the resulting policy implementable online.

Remark 5.36 (Which terminal condition goes with which problem). The choice $P_T = 0$ belongs to the *free-endpoint* variant of (5.68), in which the hard constraint $x_T = d$ is dropped, or replaced by a terminal penalty $\frac{1}{2}x_T^\top P_T x_T$. It is not compatible with the hard terminal constraint as posed: there $\mu_{T-1} = \nu$ is a *free* multiplier chosen to steer the state exactly onto d , whereas $P_T = 0$ together with (5.72) would force $\nu = 0$, i.e. no terminal constraint at all. With $x_T = d$ retained, the costate acquires an affine part, $\mu_t = -P_{t+1} x_{t+1} + r_{t+1}$, and the optimal control becomes an affine feedback law whose offset is driven by ν ; the Riccati recursion (5.73) for the quadratic part is unchanged.

Combining the linear costate relation with (5.71) gives the optimal policy as a state-feedback law,

$$u_t^* = - (R_t + B_t^\top P_{t+1} B_t)^{-1} B_t^\top P_{t+1} A_t x_t^*. \quad (5.74)$$

Theorem 5.37 (Optimal Control and Riccati Characterization for Output-Weighted LQR).

Consider the output-weighted linear-quadratic regulator problem (5.68). There exists a unique sequence of symmetric matrices $\{P_t\}_{t=1}^T$, $P_t \succeq 0$, satisfying the backward Riccati recursion (5.73), with terminal condition $P_T = 0$ (or $P_T = C_T^\top Q_T C_T$ if a terminal output penalty is included).

Furthermore, the optimal control sequence $\{u_t^*\}_{t=1}^{T-1}$ is given by the state-feedback law (5.74). The associated costate variables satisfy $\mu_t = -P_{t+1}x_{t+1}^*$, and are consistent with the adjoint equations (5.70). The optimal solution of (5.68) is completely characterized by the Riccati equation (5.73) and the feedback law (5.74).

Proof. Because $R_t \succ 0$ and $Q_t \succeq 0$, the stage cost is strictly convex in u_t and convex in x_t , while the dynamics and terminal condition are affine. The objective is therefore strictly convex on the affine feasible manifold, so the optimal control sequence $\{u_t^*\}$ exists and is unique. The associated optimality system consists of the dynamics, the adjoint recursion (5.70) with $\mu_{T-1} = \nu$, and the stationarity condition (5.71).

Posit the linear costate relation (5.72), $\mu_t = -P_{t+1}x_{t+1}$. Substituting (5.71) and the dynamics into the adjoint recursion (5.70) and matching the coefficient of x_t yields the backward recursion (5.73). Since $R_t \succ 0$ and $P_{t+1} \succeq 0$, the matrix $R_t + B_t^\top P_{t+1} B_t$ is positive definite and hence invertible, so each step of the recursion is well defined. Starting from the terminal value $P_T \succeq 0$ and proceeding backward, induction produces a unique sequence of symmetric matrices $\{P_t\}$. Each P_t is positive semidefinite because the optimal cost-to-go from state x_t is the convex quadratic $V_t(x_t) = \frac{1}{2}x_t^\top P_t x_t$, whose Hessian P_t is therefore $\succeq 0$.

Finally, eliminating the costate through $u_t^* = R_t^{-1}B_t^\top \mu_t = -R_t^{-1}B_t^\top P_{t+1}x_{t+1}$ and the dynamics $x_{t+1} = A_t x_t + B_t u_t$ and solving the resulting linear relation for u_t gives $(R_t + B_t^\top P_{t+1} B_t)u_t^* = -B_t^\top P_{t+1} A_t x_t^*$, which is the state-feedback law (5.74). Uniqueness of $\{P_t\}$ and of $\{u_t^*\}$ completes the characterization. $\square$

Theorem 5.38 (Dual Optimal Control Formulation of LQR). The Lagrangian dual of the output-feedback LQR problem (5.68) is the following concave maximization problem:

$$\begin{aligned} \max_{\{\mu_t\}_{t=1}^{T-1}, \nu} \quad & \langle \nu, d \rangle - \langle \mu_1, A_1 x_1 \rangle - \sum_{t=1}^{T-1} \frac{1}{2} \mu_t^\top B_t R_t^{-1} B_t^\top \mu_t \\ \text{s.t.} \quad & \mu_{t-1} = A_t^\top \mu_t - C_t^\top Q_t C_t x_t, \quad t = 2, \dots, T-1, \\ & \mu_{T-1} = \nu, \end{aligned} \tag{5.75}$$

where the state trajectory $\{x_t\}$ is implicitly defined by the adjoint recursion. Strong duality holds, and the optimal primal control is recovered via $u_t^* = R_t^{-1}B_t^\top \mu_t^*$.

Proof. We derive the dual problem by explicit minimization of the complete Lagrangian associated with the output-weighted LQR problem (5.68).

Recall that the complete Lagrangian is

$$\mathcal{L}(x, u, \mu, \nu) = \sum_{t=1}^{T-1} \left(\frac{1}{2} u_t^\top R_t u_t + \frac{1}{2} x_t^\top C_t^\top Q_t C_t x_t \right) + \sum_{t=1}^{T-1} \langle \mu_t, x_{t+1} - A_t x_t - B_t u_t \rangle + \langle \nu, d - x_T \rangle,$$

where x_1 is fixed.

The dual function is defined as

$$d(\mu, \nu) := \inf_{\substack{x_2, \dots, x_T \in \mathbb{R}^n \\ u_t \in \mathbb{R}^m}} \mathcal{L}(x, u, \mu, \nu).$$

Expanding the inner products and grouping terms according to the state variables, the dependence on the terminal state x_T appears only through $\langle \mu_{T-1} - \nu, x_T \rangle$. Since this expression is affine in x_T , the infimum over $x_T \in \mathbb{R}^n$ is finite if and only if $\mu_{T-1} = \nu$.

For each intermediate state x_t , $t = 2, \dots, T-1$, the corresponding terms are

$$\frac{1}{2} x_t^\top C_t^\top Q_t C_t x_t + \langle \mu_{t-1} - A_t^\top \mu_t, x_t \rangle.$$

This expression is strictly convex in x_t . Its infimum is finite and attained if and only if the first-order optimality condition holds, yielding

$$\mu_{t-1} = A_t^\top \mu_t - C_t^\top Q_t C_t x_t, \quad t = 2, \dots, T-1.$$

Together with $\mu_{T-1} = \nu$, these relations define the adjoint (costate) recursion in (5.75). When these conditions are satisfied, all dependence on $x_2, \dots, x_T$ is eliminated from the dual function.

The remaining terms involving the control variables are $\frac{1}{2} u_t^\top R_t u_t - \langle \mu_t, B_t u_t \rangle$, $t = 1, \dots, T-1$. Since $R_t \succ 0$, each term is strictly convex in u_t and admits a unique minimizer. The first-order optimality condition gives $R_t u_t = B_t^\top \mu_t$, so $u_t^* = R_t^{-1} B_t^\top \mu_t$. Substituting u_t^* back into the Lagrangian yields

$$\inf_{u_t} \left(\frac{1}{2} u_t^\top R_t u_t - \langle \mu_t, B_t u_t \rangle \right) = -\frac{1}{2} \mu_t^\top B_t R_t^{-1} B_t^\top \mu_t.$$

Collecting all remaining terms, the dual function reduces to

$$d(\mu, \nu) = \langle \nu, d \rangle - \langle \mu_1, A_1 x_1 \rangle - \sum_{t=1}^{T-1} \frac{1}{2} \mu_t^\top B_t R_t^{-1} B_t^\top \mu_t,$$

provided the adjoint relations are satisfied; otherwise $d(\mu, \nu) = -\infty$. Maximizing this concave function subject to the adjoint constraints yields the dual problem (5.75).

Under the standing convexity assumptions and Slater's condition for the primal problem, strong duality holds. Hence, there exists an optimal dual solution (μ^*, ν^*) attaining the primal optimal value. The corresponding primal control recovered via $u_t^* = R_t^{-1} B_t^\top \mu_t^*$ is optimal for the primal problem. $\square$

5.2.7 ■ Parametric Value Function and Complete Lagrangian Optimality

Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & g_j(x) \leq \theta_j, \quad j \in \mathcal{Q}, \\ & h_i(x) = \delta_i, \quad i \in \mathcal{P}, \end{aligned} \tag{5.76}$$

where f and g_j are convex functions and h_i are affine. The vectors $\theta \in \mathbb{R}^{|\mathcal{Q}|}$ and $\delta \in \mathbb{R}^{|\mathcal{P}|}$ parameterize the right-hand sides of the inequality and equality constraints, respectively.

Rather than fixing the constraint levels a priori, we regard (5.76) as a member of a family of optimization problems indexed by the parameters (θ, δ) . This parametric perspective allows us to study how the optimal cost varies as the constraints are tightened or relaxed. For given (θ, δ) , define the feasible set

$$\mathcal{F}(\theta, \delta) = \{x \in \mathbb{R}^n : g(x) \leq \theta, h(x) = \delta\}. \tag{5.77}$$

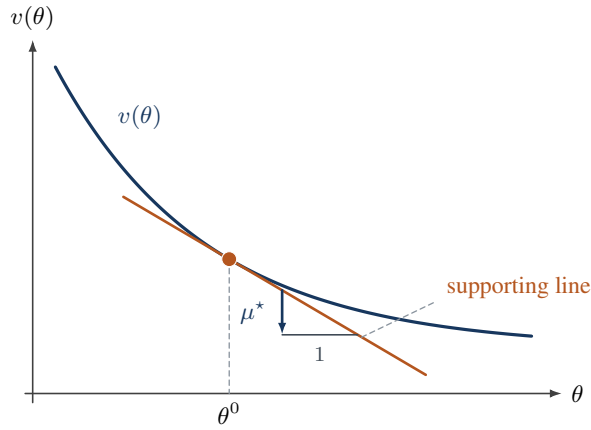


Figure 5.4: Sensitivity interpretation of KKT multipliers. Perturbing the right-hand side traces out the convex value function $v(\theta)$, which is nonincreasing because a larger θ enlarges the feasible set. It lies above each of its supporting lines and touches one at the nominal level θ^0 . The optimal multiplier μ^* is the negative slope of that supporting line, read off as the drop over a unit step in θ .

The set $\mathcal{F}(\theta, \delta)$ collects all decision variables x that satisfy the inequality constraints with upper bounds θ and the equality constraints with right-hand side δ . When θ increases componentwise, the feasible set expands, while changes in δ translate the affine constraint manifold.

The associated *optimal value function* is defined by

$$v(\theta, \delta) = \begin{cases} \inf_{x \in \mathcal{F}(\theta, \delta)} f(x), & \mathcal{F}(\theta, \delta) \neq \emptyset, \\ +\infty, & \text{otherwise.} \end{cases} \quad (5.78)$$

Thus, $v(\theta, \delta)$ represents the minimum achievable objective value under constraint levels (θ, δ) . Assigning the value $+\infty$ to infeasible parameter pairs allows v to be treated as an extended-real-valued convex function on $\mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$.

The original optimization problem corresponds to the nominal parameter choice $(\theta, \delta) = (0, 0)$. In this sense, $v(\theta, \delta)$ provides a global description of the problem's sensitivity to perturbations in both inequality and equality constraints and serves as the central object linking dual variables, Lagrange multipliers, and first-order optimality conditions.

Lemma 5.39 (Convexity of the value function). *The function $v : \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|} \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex.*

Proof. Let (θ^1, δ^1) and (θ^2, δ^2) be given and $\alpha \in (0, 1)$. Set

$$(\theta, \delta) = \alpha(\theta^1, \delta^1) + (1 - \alpha)(\theta^2, \delta^2).$$

For any $x^1 \in \mathcal{F}(\theta^1, \delta^1)$ and $x^2 \in \mathcal{F}(\theta^2, \delta^2)$, convexity of g_j and affinity of h_i imply $g(\alpha x^1 + (1 - \alpha)x^2) \leq \theta$ and $h(\alpha x^1 + (1 - \alpha)x^2) = \delta$. Hence $\alpha x^1 + (1 - \alpha)x^2 \in \mathcal{F}(\theta, \delta)$. By convexity of f ,

$$f(\alpha x^1 + (1 - \alpha)x^2) \leq \alpha f(x^1) + (1 - \alpha)f(x^2).$$

Taking infima over x^1 and x^2 yields

$$v(\theta, \delta) \leq \alpha v(\theta^1, \delta^1) + (1 - \alpha)v(\theta^2, \delta^2),$$

proving convexity. $\square$

Define the complete Lagrangian

$$\mathcal{L}(x, \mu, \nu) = f(x) + \langle \mu, g(x) - \theta \rangle + \langle \nu, h(x) - \delta \rangle, \quad (5.79)$$

with multipliers $\mu \in \mathbb{R}_+^{|\mathcal{Q}|}$ and $\nu \in \mathbb{R}^{|\mathcal{P}|}$. Fix (θ^0, δ^0) and define the extended dual function

$$D(\mu, \nu) = \begin{cases} \inf_{x \in \mathbb{R}^n} [f(x) + \langle \mu, g(x) - \theta^0 \rangle + \langle \nu, h(x) - \delta^0 \rangle], & \mu \geq 0, \\ -\infty, & \text{otherwise.} \end{cases} \quad (5.80)$$

Fenchel conjugate of the value function.

Since the optimal value function $v : \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|} \rightarrow \mathbb{R} \cup \{+\infty\}$ is convex, we may associate with it its *Fenchel conjugate* $v^* : \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|} \rightarrow \mathbb{R} \cup \{+\infty\}$, defined by

$$v^*(s, u) := \sup_{(\theta, \delta)} \left\{ \langle s, \theta \rangle + \langle u, \delta \rangle - v(\theta, \delta) \right\}, \quad (5.81)$$

where $(s, u) \in \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$.

Equivalently, $v^*(s, u)$ is the support function of the epigraph of v and encodes all affine lower bounds of v . In particular, (s, u) belongs to the subdifferential $\partial v(\theta^0, \delta^0)$ if and only if

$$v(\theta^0, \delta^0) + v^*(s, u) = \langle s, \theta^0 \rangle + \langle u, \delta^0 \rangle, \quad (5.82)$$

which is the Fenchel–Young equality.

Lemma 5.40 (Conjugate representation of the dual function). *Let v^* denote the Fenchel conjugate of v . Then*

$$D(\mu, \nu) = -v^*(-\mu, -\nu) - \langle \mu, \theta^0 \rangle - \langle \nu, \delta^0 \rangle. \quad (5.83)$$

Proof. For $\mu \geq 0$ we use the identities

$$\langle \mu, g(x) - \theta^0 \rangle = \inf_{\theta \geq g(x)} \langle \mu, \theta - \theta^0 \rangle, \quad \langle \nu, h(x) - \delta^0 \rangle = \inf_{\delta = h(x)} \langle \nu, \delta - \delta^0 \rangle.$$

Substituting and exchanging infima,

$$D(\mu, \nu) = \inf_{(\theta, \delta)} [v(\theta, \delta) + \langle \mu, \theta - \theta^0 \rangle + \langle \nu, \delta - \delta^0 \rangle].$$

By definition of the Fenchel conjugate,

$$\inf_{(\theta, \delta)} [v(\theta, \delta) + \langle \mu, \theta \rangle + \langle \nu, \delta \rangle] = -v^*(-\mu, -\nu),$$

which yields the stated formula. $\square$

Subdifferential and sensitivity of the value function.

Let $v : \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|} \rightarrow \mathbb{R} \cup \{+\infty\}$ be the optimal value function associated with the parametric problem. Since v is convex (Lemma 5.39), it admits a *subdifferential* at every point.

Definition 5.41 (Subdifferential). The subdifferential of v at (θ^0, δ^0) , denoted $\partial v(\theta^0, \delta^0)$, is the set of all vectors $(s, u) \in \mathbb{R}^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$ such that

$$v(\theta, \delta) \geq v(\theta^0, \delta^0) + \langle s, \theta - \theta^0 \rangle + \langle u, \delta - \delta^0 \rangle \quad \text{for all } (\theta, \delta). \quad (5.84)$$

Each element of $\partial v(\theta^0, \delta^0)$ is called a *subgradient*. Geometrically, subgradients correspond to supporting hyperplanes of the epigraph of v at (θ^0, δ^0) .

Theorem 5.42 (Value-function characterization of KKT multipliers). Assume Slater's condition holds for the inequality constraints. Then

$$\partial v(\theta^0, \delta^0) = -\left\{ (\mu, \nu) : D(\mu, \nu) = v(\theta^0, \delta^0) \right\}. \quad (5.85)$$

Proof. By Definition 5.41, a vector $(s, u) \in \partial v(\theta^0, \delta^0)$ if and only if

$$v(\theta, \delta) \geq v(\theta^0, \delta^0) + \langle s, \theta - \theta^0 \rangle + \langle u, \delta - \delta^0 \rangle \quad \text{for all } (\theta, \delta).$$

For convex functions, this condition is equivalent to the Fenchel–Young equality

$$v(\theta^0, \delta^0) + v^*(s, u) = \langle s, \theta^0 \rangle + \langle u, \delta^0 \rangle.$$

Setting $(s, u) = (-\mu, -\nu)$ and invoking Lemma 5.40, the above equality holds if and only if $D(\mu, \nu) = v(\theta^0, \delta^0)$. Under Slater's condition, strong duality guarantees the existence of such multipliers, completing the proof. $\square$

Corollary 5.43 (Alignment with complete Lagrangian optimality). Assume Slater's condition, LICQ at x^* , and differentiability of all functions, and let (μ^*, ν^*) be KKT multipliers. Then $(-\mu^*, -\nu^*) \in \partial v(0, 0)$. If in addition the KKT multiplier pair is unique, then $\partial v(0, 0)$ is the singleton $\{(-\mu^*, -\nu^*)\}$, so v is differentiable at $(0, 0)$ with $\nabla v(0, 0) = (-\mu^*, -\nu^*)$. Uniqueness cannot be dropped: the LICQ of Assumption 5.25 constrains only the equality gradients and pins down ν^* , whereas μ^* may still range over a set, in which case v has a kink at the origin and only the subdifferential statement survives. Equivalently, (x^*, μ^*, ν^*) is a saddle point of the complete Lagrangian $\mathcal{L}$ over $\mathbb{R}^n \times \mathbb{R}_+^{|\mathcal{Q}|} \times \mathbb{R}^{|\mathcal{P}|}$.

5.3 ■ Introduction to Non-Smooth Convex Programs

Optimality of a feasible point in a convex program can be characterized using saddle-point conditions regardless of whether the objective and constraint functions are differentiable. An alternative and more general approach relies on the notions of *subgradients* and *subdifferentials*, which extend the concept of gradients to nonsmooth convex functions.

5.3.1 ■ Subgradients and Subdifferentials

Definition 5.44 (Subgradient). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function and let $x^* \in \mathbb{R}^n$. A vector $h \in \mathbb{R}^n$ is called a *subgradient* of f at x^* if

$$f(x) - f(x^*) \geq h^\top (x - x^*) \quad \forall x \in \mathbb{R}^n. \quad (5.86)$$

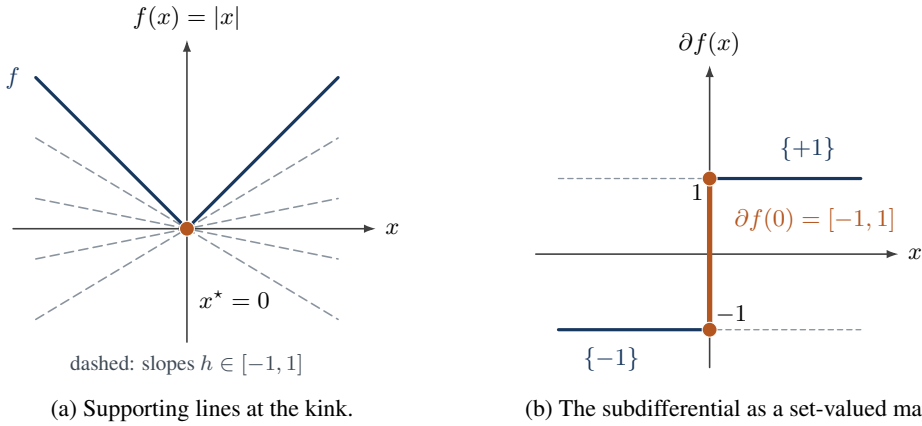


Figure 5.5: Subgradients and the subdifferential of $f(x) = |x|$. (a) A vector h is a subgradient at x^* exactly when the affine function $x \mapsto f(x^*) + h(x - x^*)$ lies below the graph; at the kink an entire fan of such lines exists, one for each slope in $[-1, 1]$. (b) The resulting set-valued map $x \mapsto \partial f(x)$ is the singleton $\{-1\}$ for $x < 0$ and $\{+1\}$ for $x > 0$, and jumps through the whole interval $[-1, 1]$ at the single point of nondifferentiability. Where f is differentiable the subdifferential collapses to the gradient.

Definition 5.45 (Subdifferential). The set of all subgradients of f at x^* is called the subdifferential of f at x^* and is denoted by $\partial f(x^*)$.

Geometrically, $\partial f(x^*)$ consists of the slopes of all supporting hyperplanes to the graph of f at x^* ; equivalently, h is a subgradient exactly when the affine function $x \mapsto f(x^*) + h^\top(x - x^*)$ underestimates f everywhere and touches it at x^* . If f is convex and differentiable at x^* , only one such hyperplane exists and the subdifferential reduces to a singleton: $\partial f(x^*) = \{\nabla f(x^*)\}$. Thus, the concept of subgradients strictly generalizes classical gradients. At a kink, by contrast, a whole family of supporting hyperplanes fits underneath the graph, and $\partial f(x^*)$ records their slopes as a set. This is the picture drawn in Figure 5.5 for the absolute value.

Example 5.46 Absolute Value Function.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = |x|$.

The function f is convex but not differentiable at $x^* = 0$. For $x^* = 0$, a vector $h \in \mathbb{R}$ is a subgradient if and only if $|x| - |0| \geq h(x - 0) \forall x \in \mathbb{R}$.

This inequality holds precisely when $h \in [-1, 1]$. Hence, $\partial f(0) = [-1, 1]$. Geometrically, each $h \in [-1, 1]$ corresponds to the slope of a supporting line to the graph of $|x|$ at the origin. For $x^* \neq 0$, the function is differentiable and $\partial f(x^*) = \{\text{sign}(x^*)\}$, as Figure 5.5(b) shows. $\square$

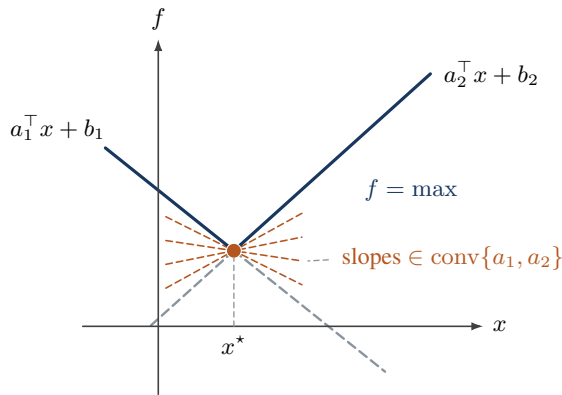


Figure 5.6: Subdifferential of a maximum of two affine functions. Away from the kink exactly one piece is active and ∂f is the single gradient of that piece. At the kink x^* both are active, every slope between a_1 and a_2 supports the graph (dashed), and $\partial f(x^*) = \text{conv}\{a_1, a_2\}$: the subdifferential is the convex hull of the gradients of the active pieces.

Example 5.47 Maximum of Affine Functions.

Let $f(x) := \max\{a_1^\top x + b_1, a_2^\top x + b_2\}$, where $a_1, a_2 \in \mathbb{R}^n$ and $b_1, b_2 \in \mathbb{R}$.

The function f is convex but generally nondifferentiable at points where both affine functions are active. If $a_1^\top x^* + b_1 = a_2^\top x^* + b_2$, then the subdifferential is $\partial f(x^*) = \text{conv}\{a_1, a_2\}$, the convex hull of the active gradients.

If only one affine function is active at x^* , say $f(x^*) = a_1^\top x^* + b_1$, then $\partial f(x^*) = \{a_1\}$. Figure 5.6 illustrates both cases.

Thus, the subdifferential aggregates the gradients of all active components. $\square$

Example 5.48 Indicator Function of a Convex Set.

Let $C \subset \mathbb{R}^n$ be a closed convex set and define the indicator function

$$\delta_C(x) := \begin{cases} 0, & x \in C, \\ +\infty, & x \notin C. \end{cases}$$

The function δ_C is convex. Fix a point $x^* \in C$. By definition, a vector $v \in \mathbb{R}^n$ belongs to the subdifferential $\partial \delta_C(x^*)$ if and only if

$$\delta_C(x) - \delta_C(x^*) \geq v^\top (x - x^*) \quad \forall x \in \mathbb{R}^n. \quad (5.87)$$

Since $\delta_C(x^*) = 0$, we distinguish two cases. If $x \notin C$, then $\delta_C(x) = +\infty$ and (5.87) holds automatically. If $x \in C$, then $\delta_C(x) = 0$ and (5.87) reduces to $0 \geq v^\top (x - x^*) \quad \forall x \in C$.

Therefore,

$$\partial\delta_C(x^*) = \{v \in \mathbb{R}^n \mid v^\top(x - x^*) \leq 0 \ \forall x \in C\},$$

which is precisely the normal cone to C at x^* : $\partial\delta_C(x^*) = N_C(x^*)$.

This example shows that normal cones arise naturally as subdifferentials of indicator functions of convex sets, providing a direct link between subgradient theory and geometric optimality conditions. $\square$

Example 5.49 Negative Entropy.

Let $f : \mathbb{R}_+^n \rightarrow \mathbb{R}$ be the negative entropy $f(p) = \sum_{i=1}^n p_i \log p_i$.

On the interior of the positive orthant, f is differentiable, with $\partial f(p) = \{\nabla f(p)\}$ and $\nabla f(p)_i = 1 + \log p_i$.

At a boundary point where $p_i = 0$ for some i , the subdifferential is in fact *empty*: any subgradient v would require $q_i \log q_i \geq v_i q_i$, i.e. $\log q_i \geq v_i$, for all $q_i > 0$, which is impossible since $\log q_i \rightarrow -\infty$ as $q_i \downarrow 0$. Thus $\partial f(p) = \emptyset$. This vertical tangent of $x \log x$ at the origin, the very infinite slope that precludes a finite affine minorant, is why entropy-based problems are posed on the open orthant (or regularized) so that subgradients remain available. $\square$

Theorem 5.50 (Existence of Subgradients). *Let $C \subset \mathbb{R}^n$ be convex and let $f : C \rightarrow \mathbb{R}$ be convex, and understand the subdifferential relative to C ,*

$$\partial f(x^*) := \{h \in \mathbb{R}^n : f(x) - f(x^*) \geq h^\top(x - x^*) \ \forall x \in C\}.$$

Then $\partial f(x^)$ is nonempty for every $x^* \in \text{int}(C)$, and it is a compact convex set. When $C = \mathbb{R}^n$ this is the subdifferential of Definition 5.41.*

Proof. Fix a point $x^* \in \text{int}(C)$. Consider the epigraph of f ,

$$\text{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} \mid t \geq f(x)\},$$

which is a closed convex subset of $\mathbb{R}^{n+1}$ since f is convex.

Because $x^* \in \text{int}(C)$, the point $(x^*, f(x^*))$ lies on the boundary of $\text{epi}(f)$, and there exist points strictly below the epigraph in a neighborhood of x^* . By the supporting hyperplane theorem there is a nonzero normal $(\tilde{h}, \beta) \in \mathbb{R}^{n+1}$ supporting $\text{epi}(f)$ at $(x^*, f(x^*))$. Necessarily $\beta < 0$: taking $t \rightarrow +\infty$ in the supporting inequality rules out $\beta > 0$, and $\beta = 0$ would make the hyperplane vertical, which is impossible because $x^* \in \text{int}(C)$ and f is bounded on a neighbourhood of x^* . Rescaling so that $\beta = -1$ and writing $h := \tilde{h}$, we obtain

$$(h, -1)^\top((x, t) - (x^*, f(x^*))) \leq 0 \quad \forall (x, t) \in \text{epi}(f).$$

Expanding the inequality yields $h^\top(x - x^*) - (t - f(x^*)) \leq 0$ for all $t \geq f(x)$, which, taking $t = f(x)$, implies $f(x) - f(x^*) \geq h^\top(x - x^*)$ for all $x \in C$.

Thus, $h \in \partial f(x^*)$, showing that the subdifferential is nonempty.

Next, we show that $\partial f(x^*)$ is convex. Let $h_1, h_2 \in \partial f(x^*)$ and let $\lambda \in [0, 1]$. Then for all x ,

$$f(x) - f(x^*) \geq h_1^\top(x - x^*), \quad f(x) - f(x^*) \geq h_2^\top(x - x^*).$$

Taking a convex combination gives

$$f(x) - f(x^*) \geq (\lambda h_1 + (1 - \lambda)h_2)^\top (x - x^*),$$

so $\lambda h_1 + (1 - \lambda)h_2 \in \partial f(x^*)$. Hence, $\partial f(x^*)$ is convex.

Finally, we show compactness. Since $x^* \in \text{int}(C)$, there exists $\varepsilon > 0$ such that $B_\varepsilon(x^*) \subset C$. For any $h \in \partial f(x^*)$ and any u with $\|u\| = 1$, applying the subgradient inequality at $x^* + \varepsilon u$ gives $f(x^* + \varepsilon u) - f(x^*) \geq \varepsilon h^\top u$. Thus, $h^\top u \leq \frac{f(x^* + \varepsilon u) - f(x^*)}{\varepsilon}$. Since the right-hand side is finite and uniformly bounded over the unit sphere, the set $\partial f(x^*)$ is bounded.

Because $\partial f(x^*)$ is closed (as an intersection of closed half-spaces) and bounded, it is compact. This completes the proof. $\square$

Theorem 5.51 (Basic Subdifferential Calculus). *Let $f, f_1, f_2 : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex functions, and let $\alpha \geq 0$ be a scalar. Then, for every $x^* \in \mathbb{R}^n$:*

(1) **Positive scaling:** $\partial(\alpha f)(x^*) = \alpha \partial f(x^*)$.

(2) **Additivity:**

$$\partial(f_1 + f_2)(x^*) = \partial f_1(x^*) + \partial f_2(x^*),$$

where the sum on the right-hand side is the Minkowski sum of sets.

Proof. *Positive scaling.* If $\alpha = 0$ both sides equal $\{0\}$. For $\alpha > 0$, $h \in \partial f(x^*)$ means $f(x) - f(x^*) \geq h^\top (x - x^*)$ for all x , which after multiplying by $\alpha > 0$ is equivalent to $(\alpha f)(x) - (\alpha f)(x^*) \geq (\alpha h)^\top (x - x^*)$, i.e. $\alpha h \in \partial(\alpha f)(x^*)$. Hence $\partial(\alpha f)(x^*) = \alpha \partial f(x^*)$.

Additivity, inclusion “ $\supseteq$ ”. Let $h_k \in \partial f_k(x^*)$ for $k = 1, 2$. Adding the two subgradient inequalities gives $(f_1 + f_2)(x) - (f_1 + f_2)(x^*) \geq (h_1 + h_2)^\top (x - x^*)$ for all x , so $h_1 + h_2 \in \partial(f_1 + f_2)(x^*)$; thus $\partial f_1(x^*) + \partial f_2(x^*) \subseteq \partial(f_1 + f_2)(x^*)$.

Additivity, inclusion “ $\subseteq$ ”. For a finite convex function g , the one-sided directional derivative $g'(x^*; d) = \lim_{t \downarrow 0} \frac{g(x^* + td) - g(x^*)}{t}$ is the support function of the subdifferential,

$$g'(x^*; d) = \max_{h \in \partial g(x^*)} h^\top d, \quad \forall d \in \mathbb{R}^n,$$

which is well defined because $\partial g(x^*)$ is nonempty, compact, and convex (Theorem 5.50). Directional derivatives are additive, $(f_1 + f_2)'(x^*; d) = f_1'(x^*; d) + f_2'(x^*; d)$, and the support function of a Minkowski sum is the sum of the support functions. Therefore the sets $\partial(f_1 + f_2)(x^*)$ and $\partial f_1(x^*) + \partial f_2(x^*)$ have the same support function on all of $\mathbb{R}^n$. Two nonempty compact convex sets with identical support functions coincide, which gives the reverse inclusion and hence equality. $\square$

5.3.2 ■ Non-Smooth Convex Optimality

Theorem 5.52 (Equivalent First-Order Optimality Conditions). *Let $C \subset \mathbb{R}^n$ be a convex set and let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. For a point $x^* \in C$, the following statements are equivalent:*

(i) x^* is an optimal solution of $\min_{x \in C} f(x)$.

(ii) There exists a subgradient $h \in \partial f(x^*)$ such that

$$h^\top(x - x^*) \geq 0, \quad \forall x \in C. \quad (5.88)$$

(iii) There exists $h \in \partial f(x^*)$ such that

$$-h \in N_C(x^*). \quad (5.89)$$

(iv) The following set inclusion holds:

$$0 \in \partial f(x^*) + N_C(x^*). \quad (5.90)$$

Proof. (i) $\Rightarrow$ (iv) Optimality of x^* over C says exactly that x^* minimizes the extended-valued convex function $f + \delta_C$ over all of $\mathbb{R}^n$, which in turn is equivalent to $0 \in \partial(f + \delta_C)(x^*)$. Because f is finite-valued on $\mathbb{R}^n$, hence continuous, the subdifferential sum rule applies without further qualification and gives

$$\partial(f + \delta_C)(x^*) = \partial f(x^*) + \partial \delta_C(x^*) = \partial f(x^*) + N_C(x^*),$$

the last equality being the indicator-function example above. Therefore $0 \in \partial f(x^*) + N_C(x^*)$, which is (5.90).

It is worth noting why this route is necessary: optimality does *not* imply that an arbitrarily chosen $h \in \partial f(x^*)$ satisfies (5.88). At a kink the subdifferential is a set, and only *some* of its elements are normal to C ; the sum rule is what produces one of them.

(ii) $\Leftrightarrow$ (iii) By definition of the normal cone,

$$-h \in N_C(x^*) \iff h^\top(x - x^*) \geq 0, \quad \forall x \in C.$$

(iii) $\Leftrightarrow$ (iv) Condition (5.89) holds if and only if there exist $h \in \partial f(x^*)$ and $w \in N_C(x^*)$ with $h + w = 0$, which is exactly (5.90). Combined with the implication proved first, this closes the cycle (i) $\Rightarrow$ (iv) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (ii).

(ii) $\Rightarrow$ (i) If (5.88) holds, then using the subgradient inequality, $f(x) \geq f(x^*) + h^\top(x - x^*) \geq f(x^*)$, $\forall x \in C$, so x^* is optimal. $\square$

Example 5.53 Soft thresholding: why set-valuedness matters.

The previous examples computed subdifferentials. This one puts the machinery to work and shows that the set-valuedness at a kink is not a technical nuisance but the source of a genuinely useful effect. Fix $a \in \mathbb{R}$ and $\lambda > 0$ and consider the scalar problem

$$\min_{x \in \mathbb{R}} F(x), \quad F(x) := \frac{1}{2}(x - a)^2 + \lambda|x|, \quad (5.91)$$

which is the one-dimensional model of ℓ^1 -regularized least squares. The objective is convex, and it is nondifferentiable at exactly one point.

Three properties of subgradients do all the work. First, the feasible set is all of $\mathbb{R}$, so $N_{\mathbb{R}}(x^*) = \{0\}$ and part (iv) of Theorem 5.52 reduces to the inclusion $0 \in \partial F(x^*)$. Second, because the quadratic term is differentiable, the subdifferential of the sum

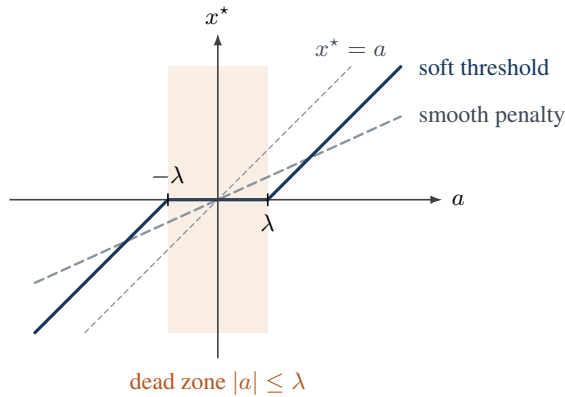


Figure 5.7: Minimizer of $\frac{1}{2}(x - a)^2 + \lambda|x|$ as a function of a . Because $\partial|\cdot|(0)$ is the whole interval $[-1, 1]$ rather than a single number, the stationarity inclusion $0 \in \partial F(x^*)$ is satisfied at $x^* = 0$ for every a in the shaded dead zone $|a| \leq \lambda$. Replacing $\lambda|x|$ by a differentiable penalty gives the dashed line, which is never exactly zero: set-valuedness at the kink is what produces sparsity. The dotted line is the unpenalized solution $x^* = a$.

splits as $\partial F(x) = (x - a) + \lambda \partial|x|$. Third, $\partial|x| = \{\text{sign}(x)\}$ away from the origin and $\partial|0| = [-1, 1]$. Note that the inclusion is a genuine optimality condition rather than mere stationarity: by Theorem 5.52 it characterizes *global* minimizers, which is exactly what convexity buys.

Solving the inclusion case by case is now immediate. If $x^* > 0$ then $0 = x^* - a + \lambda$, so $x^* = a - \lambda$, which is consistent with $x^* > 0$ precisely when $a > \lambda$. If $x^* < 0$ then $x^* = a + \lambda$, consistent when $a < -\lambda$. If $x^* = 0$ the inclusion reads $0 \in -a + \lambda[-1, 1]$, that is, $|a| \leq \lambda$. The three cases are mutually exclusive and cover the line, so the minimizer is unique and given by the *soft-thresholding* operator

$$x^* = \text{sign}(a) \max\{|a| - \lambda, 0\}, \quad (5.92)$$

plotted in Figure 5.7.

The point worth keeping is the third case. At a differentiable point the stationarity condition is one equation in one unknown and pins x^* down; at the kink it is an *inclusion* in an interval, and a single candidate $x^* = 0$ can therefore satisfy it for a whole range of data a . That range is the dead zone $|a| \leq \lambda$, on which the solution is exactly zero rather than merely small. Replacing $\lambda|x|$ by the differentiable penalty λx^2 makes the subdifferential a singleton everywhere and gives $x^* = a/(1 + 2\lambda)$, which vanishes only when a does. Sparsity is thus produced by the nonsmoothness, and the subdifferential is what makes it computable. $\square$

Theorem 5.54 (First-Order Optimality without Slater, Nonsmooth Case). Consider the convex optimization problem (5.1). Assume that all functions are convex (not necessarily differentiable), and let $\mathcal{F}^\circ$ denote the base feasible set. Let $x^* \in \mathcal{F}^\circ$ be feasible.

Then x^* is an optimal solution if and only if there exists a multiplier vector $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ and

subgradients $h \in \partial f(x^*)$ and $h_j \in \partial g_j(x^*)$, $j \in \mathcal{Q}^<$, such that

$$-h - \sum_{j \in \mathcal{Q}^<} \mu_j^* h_j \in N_{\mathcal{F}^=}(x^*). \quad (5.93)$$

Proof. By Theorem 5.2, x^* is optimal if and only if there exists $\mu^* \in \mathbb{R}_+^{|\mathcal{Q}^<|}$ such that x^* minimizes the restricted Lagrangian

$$\mathcal{L}^c(x, \mu^*) = f(x) + \sum_{j \in \mathcal{Q}^<} \mu_j^* g_j(x)$$

over $\mathcal{F}^=$.

Since $\mathcal{L}^c(\cdot, \mu^*)$ is convex and $\mathcal{F}^=$ is convex, Theorem 5.52 applies. Thus, x^* minimizes $\mathcal{L}^c(\cdot, \mu^*)$ over $\mathcal{F}^=$ if and only if there exists $g \in \partial_x \mathcal{L}^c(x^*, \mu^*)$ such that $g^\top(x - x^*) \geq 0, \forall x \in \mathcal{F}^=$.

The subdifferential sum rule for convex functions yields

$$\partial_x \mathcal{L}^c(x^*, \mu^*) = \partial f(x^*) + \sum_{j \in \mathcal{Q}^<} \mu_j^* \partial g_j(x^*).$$

Hence, there exist $h \in \partial f(x^*)$ and $h_j \in \partial g_j(x^*)$ such that $g = h + \sum_{j \in \mathcal{Q}^<} \mu_j^* h_j$. The inequality $g^\top(x - x^*) \geq 0, \forall x \in \mathcal{F}^=$ is equivalent to $-g \in N_{\mathcal{F}^=}(x^*)$, which proves (5.93). $\square$

Remark 5.55. If all functions are differentiable at x^* , then each subdifferential reduces to a singleton containing the corresponding gradient. In this case, the condition (5.93) reduces to the normal-cone optimality condition stated in Theorem 5.5.

No constraint qualification, including Slater's condition, is required for this result. Inequality constraints that act as equalities on the feasible set are absorbed into the base set $\mathcal{F}^=$, while inactive inequality constraints do not contribute to the stationarity condition.

The normal-cone formulation emphasizes the geometric nature of optimality. Stationarity means that there exists no feasible descent direction at x^* , or equivalently, that the negative subgradient of the objective lies in the normal cone to $\mathcal{F}^=$, which is the polar of the tangent cone $T_{\mathcal{F}^=}(x^*)$.

Corollary 5.56 (Non-Smooth Karush–Kuhn–Tucker Conditions). Consider the convex optimization problem (5.1). Assume that all functions are convex (not necessarily differentiable) and that the inequality constraints satisfy Slater's condition, so that $\mathcal{Q}^= = \emptyset$ and the base feasible set reduces to the equality manifold $\mathcal{F}^= = \{x : h_i(x) = 0, i \in \mathcal{P}\}$.

Then a feasible point x^* is an optimal solution if and only if there exist multipliers $\mu_j^* \geq 0$, $j \in \mathcal{Q}$, such that

$$0 \in \partial f(x^*) + \sum_{j \in \mathcal{Q}} \mu_j^* \partial g_j(x^*) + N_{\mathcal{F}^=}(x^*). \quad (5.94)$$

In the purely inequality-constrained case $\mathcal{P} = \emptyset$ one has $\mathcal{F}^= = \mathbb{R}^n$ and $N_{\mathcal{F}^=}(x^*) = \{0\}$, so the last term drops out and (5.94) becomes the familiar inclusion $0 \in \partial f(x^*) + \sum_j \mu_j^* \partial g_j(x^*)$. With affine equalities, $N_{\mathcal{F}^=}(x^*) = \text{range}(A^\top)$ and the term contributes exactly the equality multipliers ν^* .

Proof. Under Slater's condition no inequality is active throughout $\mathcal{F}$, so $\mathcal{Q}^= = \emptyset$, $\mathcal{Q}^< = \mathcal{Q}$, and the base set is the equality manifold $\mathcal{F}^=$. Applying Theorem 5.54, x^* is optimal if and only if there

exist $\mu^* \in \mathbb{R}_+^{|Q|}$ and subgradients $h \in \partial f(x^*)$, $h_j \in \partial g_j(x^*)$ with $-h - \sum_{j \in Q} \mu_j^* h_j \in N_{\mathcal{F}^=}(x^*)$, that is, $0 \in h + \sum_j \mu_j^* h_j + N_{\mathcal{F}^=}(x^*)$. By the subdifferential sum rule (Theorem 5.51) this is exactly (5.94). If $\mathcal{P} = \emptyset$ then $\mathcal{F}^= = \mathbb{R}^n$ and $N_{\mathcal{F}^=}(x^*) = \{0\}$, giving the reduced form. $\square$

Example 5.57 ℓ^1 - ℓ^∞ Norms and Non-Smooth KKT Conditions.

Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \|x\|_1 := \sum_{i=1}^n |x_i| \\ \text{s.t.} \quad & \|x\|_\infty := \max_{1 \leq i \leq n} |x_i| \leq 1. \end{aligned} \tag{5.95}$$

The feasible set is the ℓ^∞ unit ball $C := \{x \in \mathbb{R}^n \mid \|x\|_\infty \leq 1\}$, which is compact, convex, and has nonempty interior. Hence Slater's condition holds. Since $\|x\|_1 \geq 0$ for all x and $\|0\|_1 = 0$, the unique optimal solution is $x^* = 0$. The subdifferential of the ℓ^1 norm at the origin is given by $\partial\|x\|_1(0) = \{v \in \mathbb{R}^n \mid \|v\|_\infty \leq 1\}$.

The inequality constraint can be written as $g(x) := \|x\|_\infty - 1 \leq 0$, whose subdifferential at $x^* = 0$ is $\partial g(0) = \partial\|x\|_\infty(0) = \text{conv}\{\pm e_1, \dots, \pm e_n\}$, where $\{e_i\}$ denotes the standard basis of $\mathbb{R}^n$.

Since $x^* = 0$ lies in the interior of the feasible set C , the normal cone satisfies $N_C(0) = \{0\}$.

By the non-smooth Karush–Kuhn–Tucker conditions, optimality requires the existence of a multiplier $\mu^* \geq 0$ such that $0 \in \partial\|x\|_1(0) + \mu^* \partial g(0)$. Choosing $\mu^* = 0$ yields $0 \in \partial\|x\|_1(0)$, which holds. Therefore, $x^* = 0$ satisfies the non-smooth KKT conditions and is the optimal solution of (5.95). This example illustrates the dual geometric relationship between the ℓ^1 and ℓ^∞ norms: the subdifferential of $\|\cdot\|_1$ is an ℓ^∞ ball, while the subdifferential of $\|\cdot\|_\infty$ is an ℓ^1 polytope. $\square$

5.4 - Further Reading

A comprehensive and unifying framework for convex analysis, including faithfully convex functions and duality, is developed in [119]. Cone-based formulations of feasibility, consistency, and optimality conditions are presented in [14, 13]. Theorems of the alternative and their role in optimality and duality are treated in [81]. Modern treatments of Lagrangian duality, saddle-point theory, and constraint qualifications are given in [15, 18]. A general variational framework for optimality conditions beyond standard constraint qualifications is presented in [43]. Subdifferential calculus and non-smooth convex programming are treated in [8, 10, 43], together with numerical methods such as penalty, barrier, SUMT, and augmented Lagrangian methods; see also [15].

5.5 - Exercises

Exercise 5.1 (Locally Flat Surfaces for Faithfully Convex Functions). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable *faithfully convex* function of the form $f(x) = \varphi(Ax + b) + a^T x + \alpha$, where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $a \in \mathbb{R}^n$, $\alpha \in \mathbb{R}$, and $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}$ is strictly convex.

Recall that f is said to have a *locally flat surface* (LFS) at $x^* \in \mathbb{R}^n$ if $\mathcal{N}(\nabla f(x^*)) = \mathcal{D}^\circ f(x^*)$, where $\mathcal{D}^\circ f(x^*)$ denotes the cone of directions along which f is locally constant at x^* .

- (i) Show that $\nabla f(x) = A^T \nabla \varphi(Ax + b) + a$.
- (ii) Prove that a direction $d \in \mathbb{R}^n$ satisfies $f(x^* + td) = f(x^*)$ for all sufficiently small t if and only if $Ad = 0$ and $a^T d = 0$.
- (iii) Show that if $\nabla f(x^*) = 0$, then f has the LFS property at x^* if and only if f is constant.
- (iv) Suppose $\nabla f(x^*) \neq 0$. Prove that f has the LFS property at x^* if and only if

$$\text{rank} \begin{bmatrix} A \\ a^T \end{bmatrix} = 1.$$

- (v) Briefly interpret the rank condition above in terms of the geometry of the level sets of f near x^* .

Exercise 5.2 (Parametric Linear Program and Optimal Value Function). Consider the linear programming problem parameterized by $\theta \in \mathbb{R}$ and $\delta \in \mathbb{R}$:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & \langle c, x \rangle \\ \text{s.t.} \quad & \langle a, x \rangle \geq \theta, \\ & 0 \leq x_j \leq \delta, \quad j = 1, \dots, n, \end{aligned} \tag{5.96}$$

where the vectors $c \in \mathbb{R}^n$ and $a \in \mathbb{R}^n$ are fixed, and the box constraint $0 \leq x \leq \delta$ is understood componentwise. Assume throughout that $\delta \geq 0$.

- (i) Determine for which values of (θ, δ) the feasible set of (5.96) is nonempty. In particular, compute the maximum value of $\langle a, x \rangle$ over the box $0 \leq x \leq \delta$, and characterize the largest admissible θ as a function of δ .
- (ii) Show that if the problem is feasible, then an optimal solution can be chosen at an extreme point of the feasible polyhedron. Describe the structure of these extreme points.
- (iii) Assume the problem is feasible. Show that an optimal solution is obtained by setting

$$x_j = \begin{cases} 0, & c_j > 0, \\ \delta, & c_j < 0, \end{cases}$$

except possibly for a subset of indices that must be adjusted in order to satisfy the constraint $\langle a, x \rangle \geq \theta$.

- (iv) Define the optimal value function

$$v(\theta, \delta) = \inf \{ \langle c, x \rangle : \langle a, x \rangle \geq \theta, 0 \leq x_j \leq \delta, j = 1, \dots, n \}.$$

Show that $v(\theta, \delta)$ is a convex, piecewise linear function of (θ, δ) .

- (v) Assume the problem is feasible and bounded. Interpret the partial derivatives (or subgradients) of $v(\theta, \delta)$ with respect to θ and δ in terms of the Lagrange multipliers associated with the constraints $\langle a, x \rangle \geq \theta$ and $x_j \leq \delta$.

- (vi) Analyze explicitly the case $a = \mathbf{1}$ (the all-ones vector). Give an explicit formula for $v(\theta, \delta)$ and describe how the set of active constraints depends on θ .

Exercise 5.3 (Failure of Slater's Condition in a Degenerate Convex Program). Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & x_2 \\ \text{s.t.} \quad & x_2 \geq x_1^2, \\ & x_2 \leq 0. \end{aligned} \tag{5.97}$$

- (i) Show that the feasible set of (5.97) is nonempty and convex, and identify it explicitly.
- (ii) Show that Slater's condition does not hold for this problem. In particular, show that there is no point satisfying both inequality constraints strictly.
- (iii) Solve (5.97) and determine the unique optimal solution and optimal value.
- (iv) Describe the geometry of the feasible set and explain why it has empty interior relative to $\mathbb{R}^2$. Relate this to the notion of constraint degeneracy.
- (v) Form the Lagrangian and derive the KKT conditions. Determine whether Lagrange multipliers exist that satisfy the KKT system.
- (vi) Derive the Lagrange dual problem explicitly. Show that strong duality holds despite the failure of Slater's condition.

Exercise 5.4 (Lasso Problem). Consider the standard Lasso problem: $\min_{x \in \mathbb{R}^n} \frac{1}{2} \|y - Ax\|_2^2 + \lambda \|x\|_1$, where $x, y \in \mathbb{R}^n$ and $\lambda \geq 0$.

Show that for the case $A = I$, the solution of the Lasso problem is $x^* = S_\lambda(y)$, where S_λ is the soft-thresholding operator defined componentwise by

$$[S_\lambda(y)]_i = \begin{cases} y_i - \lambda, & \text{if } y_i > \lambda, \\ 0, & \text{if } -\lambda \leq y_i \leq \lambda, \\ y_i + \lambda, & \text{if } y_i < -\lambda. \end{cases}$$

Here, $[S_\lambda(y)]_i$ denotes the i -th component of the vector $S_\lambda(y)$.

Hint: You need to compute the subgradients of the objective function.

Exercise 5.5 (Resource Allocation). Let $A > 0$, $s_i > 0$ for $i = 1, 2, \dots, n$, and $\mathcal{N} = \{1, 2, \dots, n\}$. Consider the optimization problem

$$\begin{aligned} \max_{x_i, i \in \mathcal{N}} \quad & \sum_{i=1}^n \frac{p_i x_i}{s_i + x_i} \\ \text{s.t.} \quad & \sum_{i=1}^n x_i = A, \\ & x_i \geq 0, \quad i = 1, \dots, n. \end{aligned}$$

Assume that

$$\frac{p_1}{s_1} > \frac{p_2}{s_2} > \dots > \frac{p_n}{s_n}.$$

Show that there exists a scalar λ^* such that the optimal solution is given by

$$x_i^* = \begin{cases} \sqrt{\frac{s_i p_i}{\lambda^*}} - s_i, & \text{for } i = 1, \dots, m^*, \\ 0, & \text{for } i = m^* + 1, \dots, n, \end{cases}$$

where m^* is the largest index m for which $\frac{p_m}{s_m} \geq \lambda^*$.

Exercise 5.6 (Carrier Inequalities and the Restricted Lagrangian). Consider the convex program

$$\min_{x \in \mathbb{R}^2} f(x) = x_1^2 + x_2^2 \quad \text{s.t.} \quad g_1(x) = -x_1 \leq 0, \quad g_2(x) = x_1 \leq 0.$$

- (i) Show that the feasible set is $\mathcal{F} = \{x \in \mathbb{R}^2 : x_1 = 0\}$ and that Slater's condition fails, because no point satisfies both inequalities strictly.
- (ii) Identify the index sets $\mathcal{Q}^=$ and $\mathcal{Q}^<$ of carrier and effective inequalities, and write down the base feasible set $\mathcal{F}^=$.
- (iii) Compute the tangent cone $T_{\mathcal{F}}(x^*)$ and the normal cone $N_{\mathcal{F}}(x^*)$ at the optimal point $x^* = (0, 0)$.
- (iv) Using $\mathcal{Q}^< = \emptyset$, verify the first-order optimality condition $-\nabla f(x^*) \in N_{\mathcal{F}}(x^*)$ of Theorem 5.5, even though no inequality multiplier is involved.
- (v) Now form the *complete* Lagrangian $\mathcal{L}(x, \mu) = f(x) + \mu_1 g_1(x) + \mu_2 g_2(x)$ and show that the KKT stationarity condition is satisfied by the entire family $\mu_1 = \mu_2 = \tau$, $\tau \geq 0$. Explain why the multipliers are non-unique here, and relate this to the fact that g_1, g_2 are carrier constraints absorbed into $\mathcal{F}^=$.

Exercise 5.7 (Tangent and Normal Cones of a Polyhedron). Let $\mathcal{P} = \{x \in \mathbb{R}^n : a_i^\top x \leq b_i, i = 1, \dots, m\}$ be a polyhedron, let $x^* \in \mathcal{P}$, and let $I(x^*) = \{i : a_i^\top x^* = b_i\}$ be the set of active constraints.

- (i) Show that the tangent cone is $T_{\mathcal{P}}(x^*) = \{d \in \mathbb{R}^n : a_i^\top d \leq 0, i \in I(x^*)\}$.
- (ii) Show that the normal cone is the finitely generated cone $N_{\mathcal{P}}(x^*) = \text{cone}\{a_i : i \in I(x^*)\} = \{\sum_{i \in I(x^*)} \mu_i a_i : \mu_i \geq 0\}$.
- (iii) For $n = 2$ and $\mathcal{P} = \{x : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 \leq 1\}$, compute both cones at the vertex $x^* = (0, 0)$ and at the edge point $x^* = (\frac{1}{2}, 0)$, and sketch them.
- (iv) Deduce that x^* minimizes a differentiable convex f over $\mathcal{P}$ if and only if $-\nabla f(x^*) = \sum_{i \in I(x^*)} \mu_i a_i$ for some $\mu_i \geq 0$, and identify this as the KKT condition for linear inequality constraints.

Exercise 5.8 (Maximum Entropy, the Gibbs Distribution, and Temperature). Fix values $a_1 < a_2 < \dots < a_n$ and a target $b \in (a_1, a_n)$, and consider

$$\min_{p \in \mathbb{R}_+^n} \sum_{i=1}^n p_i \log p_i \quad \text{s.t.} \quad \sum_{i=1}^n p_i = 1, \quad \sum_{i=1}^n a_i p_i = b.$$

This is problem (5.32) with cost $c \equiv 0$; write β for the multiplier of the moment constraint.

- (i) Using the stationarity conditions, show that the optimizer is the Gibbs distribution $p_i^*(\beta) = e^{-\beta a_i} / Z(\beta)$ with $Z(\beta) = \sum_k e^{-\beta a_k}$, so that β plays the role of an inverse temperature.
- (ii) Define the moment function $b(\beta) = \sum_i a_i p_i^*(\beta)$. Show that $b'(\beta) = -\text{Var}_{p^*(\beta)}(a) \leq 0$, with strict inequality whenever the a_i are not all equal.
- (iii) Conclude that $\beta \mapsto b(\beta)$ is a strictly decreasing bijection from $\mathbb{R}$ onto (a_1, a_n) , so the multiplier β is uniquely determined by the target moment b .
- (iv) Interpret the limits $\beta \rightarrow +\infty$, $\beta \rightarrow -\infty$, and $\beta = 0$ in terms of the concentration of p^* on the smallest, largest, and uniform outcomes, respectively.

Exercise 5.9 (Discrete-Time Adjoint Equations from the Lagrangian). Consider the equality-constrained control problem

$$\min_{\{u_t\}_{t=1}^{T-1}} \sum_{t=1}^{T-1} f_t(u_t) \quad \text{s.t.} \quad x_{t+1} = A_t x_t + B_t u_t, \quad t = 1, \dots, T-1, \quad x_T = d,$$

with x_1 given and each f_t convex and differentiable.

- (i) Introduce multipliers μ_t for the dynamics constraints $x_{t+1} - A_t x_t - B_t u_t = 0$ and ν for the terminal constraint $x_T - d = 0$, and show that finiteness of the dual function forces the adjoint recursion $\mu_{t-1} = A_t^\top \mu_t$ for $t = 2, \dots, T-1$ with terminal condition $\mu_{T-1} = -\nu$.
- (ii) Show that the stagewise minimization decouples into $\min_{u_t} \{f_t(u_t) - \langle \mu_t, B_t u_t \rangle\}$, with stationarity condition $\nabla f_t(u_t^*) = B_t^\top \mu_t$.
- (iii) Specialize to $f_t(u_t) = \frac{1}{2} \|u_t\|_2^2$ and give u_t^* explicitly in terms of the costate μ_t .
- (iv) Explain in one or two sentences why, in this convex setting, the adjoint equations arise purely from requiring the dual function to be finite, with no appeal to the Pontryagin maximum principle.

Chapter 6

Vector Space Convex Optimization

Everything so far has taken the decision variable to be a vector in $\mathbb{R}^n$ and the constraints to be finitely many scalar inequalities. Many problems do not come that way. A control signal is a function of time, a covariance is a matrix required to be positive semidefinite, a probability density is constrained at a continuum of points. This chapter develops convex optimization in vector spaces, where the decision variable may be any element of a real vector space and “ $\leq$ ” is interpreted through a convex cone. Two ideas do the organizing work: duality and saddle points. Linear functionals, dual spaces, and cones supply the language, and the reward is that one set of arguments covers finite-dimensional programs, semidefinite programs, integral constraints, optimal control, and variational problems in function spaces at once.

The central result is that cone-convex programs admit a saddle-point characterization and satisfy strong duality under a Slater-type regularity condition, exactly as in Chapter 4. Nothing about finite dimension was ever essential; what mattered was convexity and the existence of an interior point. When the data are in addition Fréchet differentiable, the saddle-point statement specializes to a first-order optimality system, and the Karush–Kuhn–Tucker conditions reappear, here as a consequence of convex duality rather than of any algebraic rank assumption on the constraint gradients.

6.1 ■ Linear Functionals and Dual Spaces

We begin with the functional-analytic vocabulary needed for convex duality: linear functionals, boundedness, and dual spaces. The examples keep the abstract definitions tied to optimization and control.

Linear functionals. Let Z be a real vector space. A functional $f : Z \rightarrow \mathbb{R}$ is called *linear* if for all $z_1, z_2 \in Z$ and all scalars $\alpha, \beta \in \mathbb{R}$, $f(\alpha z_1 + \beta z_2) = \alpha f(z_1) + \beta f(z_2)$. Linear functionals represent measurements, costs, or constraints that depend linearly on the decision variable.

Example 6.1 Finite-dimensional linear functionals.

Let $Z = \mathbb{R}^n$ and $z = (z_1, \dots, z_n)$. Any functional of the form $f(z) = \sum_{i=1}^n \eta_i z_i$, $\eta \in \mathbb{R}^n$, is linear. Conversely, every linear functional on $\mathbb{R}^n$ admits such a representation. Thus, linear functionals on $\mathbb{R}^n$ are in one-to-one correspon-

dence with vectors in $\mathbb{R}^n$. □

Example 6.2 Evaluation functional.

Let $Z = C[0, 1]$, the space of continuous functions on $[0, 1]$. The functional

$$f(x) = x\left(\frac{1}{2}\right)$$

is linear. This example shows that linear functionals need not involve integrals or sums; point evaluation is a linear operation. □

Example 6.3 Integral functional.

Let $Z = L^2[0, 1]$. For a fixed function $y \in L^2[0, 1]$, define $f(x) = \int_0^1 x(t)y(t) dt$. This functional is linear in x and arises naturally in optimal control and variational problems. □

Continuity and boundedness. When Z is equipped with a norm, not every linear functional is suitable for analysis or optimization. We therefore focus on linear functionals that are continuous.

A linear functional f on a normed space Z is said to be *bounded* if there exists a constant $M \geq 0$ such that $|f(z)| \leq M\|z\|$ for all $z \in Z$. The smallest such constant is called the *operator norm* of f and is denoted $\|f\| := \sup_{\|z\| \leq 1} |f(z)|$.

A fundamental and useful fact is that for linear functionals, boundedness and continuity are equivalent: a linear functional is continuous if and only if it is bounded.

Example 6.4 Boundedness via Cauchy–Schwarz.

Let $Z = L^2[0, 1]$ and $f(x) = \int_0^1 x(t)y(t) dt$ with $y \in L^2[0, 1]$. By the Cauchy–Schwarz inequality, $|f(x)| \leq \|x\|_2 \|y\|_2$. Hence f is bounded with $\|f\| \leq \|y\|_2$; the choice $x = y/\|y\|_2$ attains the bound, so in fact $\|f\| = \|y\|_2$. □

Dual spaces. The set of all bounded linear functionals on a normed space Z forms a vector space, called the *dual space* of Z and denoted by Z^* . Equipped with the operator norm $\|z^*\| := \sup_{\|z\| \leq 1} |z^*(z)|$, the dual space Z^* is itself a normed space. Moreover, Z^* is always complete, even when Z is not; hence Z^* is a Banach space.

Dual spaces provide the natural setting for Lagrange multipliers, adjoint variables, and sensitivity information in optimization. In particular, first- order optimality conditions are expressed through elements of the dual space.

Example 6.5 Dual of $\mathbb{R}^n$.

The dual space $(\mathbb{R}^n)^*$ can be identified with $\mathbb{R}^n$ itself. Under this identification, every linear functional $z^* \in (\mathbb{R}^n)^*$ acts on $z \in \mathbb{R}^n$ via the Euclidean inner product $\langle z, z^* \rangle = z^\top z^*$. Thus, in finite-dimensional spaces, dual variables can be represented as vectors, and no distinction is made between primal and dual spaces at the level of sets. $\square$

Example 6.6 Gradients as dual elements in finite dimensions.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at x . The directional derivative of f at x in the direction d is $\delta f(x; d) = \nabla f(x)^\top d$. For fixed x , the mapping $d \mapsto \delta f(x; d)$ is linear and continuous in d . Hence, $\nabla f(x) \in (\mathbb{R}^n)^*$. This observation explains why gradients appear naturally as multipliers in first-order optimality conditions. $\square$

Example 6.7 Gradients in Hilbert spaces.

Let Z be a real Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\|z\| = \langle z, z \rangle^{1/2}$. Let $f : Z \rightarrow \mathbb{R}$ be Fréchet differentiable at a point $x \in Z$.

By definition, Fréchet differentiability means that there exists a bounded linear functional $Df(x) \in Z^*$ such that $\lim_{\|d\| \rightarrow 0} \frac{f(x+d) - f(x) - Df(x)[d]}{\|d\|} = 0$. Since $Df(x)$ is a bounded linear functional on the Hilbert space Z , the Riesz representation theorem guarantees the existence of a unique element $\nabla f(x) \in Z$ satisfying $Df(x)[d] = \langle \nabla f(x), d \rangle, \forall d \in Z$. Equivalently, the directional derivative of f at x in the direction d can be written as $\delta f(x; d) = \langle \nabla f(x), d \rangle, \forall d \in Z$.

In a Hilbert space, the gradient $\nabla f(x)$ is therefore the Riesz representative of the derivative $Df(x) \in Z^*$. This canonical identification does not depend on coordinates, and the condition $\nabla f(x) = 0$ is equivalent to the vanishing of the derivative $Df(x)$ and therefore characterizes unconstrained minimizers of convex functionals on Hilbert spaces.

Consider the quadratic functional $f(z) = \frac{1}{2} \|z\|^2, z \in Z$. A direct computation yields $\delta f(z; d) = \langle z, d \rangle$, so that $\nabla f(z) = z$. The quadratic case makes explicit the dual role of a gradient: it is a vector in Z and, through the inner product, a bounded linear functional on Z . $\square$

Example 6.8 Integral functionals and duality.

Let $Z = L^2[a, b]$ and define $f(x) = \frac{1}{2} \int_a^b x(t)^2 dt$. The Fréchet derivative at x is the bounded linear functional $d \mapsto \int_a^b x(t)d(t) dt$; equivalently, the directional derivative is $\delta f(x; d) = \int_a^b x(t)d(t) dt$. Thus, the gradient of f at x is the function x itself, viewed as an element of the dual space $(L^2[a, b])^* \cong L^2[a, b]$. This identification is central in optimal control and calculus of variations. $\square$

ℓ^p and L^p spaces. For $1 \leq p < \infty$, the space ℓ^p consists of sequences $x = \{x_i\}_{i=1}^\infty$ satisfying $\sum_{i=1}^\infty |x_i|^p < \infty$, with norm $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$. The space ℓ^∞ consists of bounded sequences with the supremum norm.

Similarly, for an interval $[a, b]$, the space $L^p[a, b]$ consists of measurable functions satisfying $\int_a^b |x(t)|^p dt < \infty$, with $L^\infty[a, b]$ defined via the essential supremum.

Hölder's inequality and duality. Let $1 \leq p < \infty$ and define the conjugate exponent q by $1/p + 1/q = 1$. Hölder's inequality states that for $x \in \ell^p$ and $y \in \ell^q$, $\sum_{i=1}^\infty |x_i y_i| \leq \|x\|_p \|y\|_q$, and similarly for $x \in L^p[a, b]$ and $y \in L^q[a, b]$, $\int_a^b |x(t)y(t)| dt \leq \|x\|_p \|y\|_q$.

Example 6.9 Dual of ℓ^p .

For $1 < p < \infty$, every bounded linear functional on ℓ^p admits the representation $f(x) = \sum_{i=1}^\infty x_i y_i$ for a unique $y \in \ell^q$, and $\|f\| = \|y\|_q$. Thus, $(\ell^p)^* \cong \ell^q$. $\square$

Example 6.10 Dual of L^p .

For $1 < p < \infty$, every bounded linear functional on $L^p[a, b]$ can be written as $f(x) = \int_a^b x(t)y(t) dt$ for a unique $y \in L^q[a, b]$. This representation is fundamental in infinite-dimensional optimization, where multipliers are functions rather than vectors. $\square$

6.2 ■ Cones

A cone defines what “positive” means in a vector space. Once positivity is specified, inequalities, convexity, and multipliers can be formulated without coordinates.

6.2.1 ■ Positive and Negative Cones

Let X be a real vector space.

Definition 6.11 (Positive cone and induced order). Let $P \subset X$ be a convex cone, i.e.,

$$x, y \in P, \alpha, \beta \geq 0 \implies \alpha x + \beta y \in P.$$

For $x, y \in X$, we define

$$x \succeq y \iff x - y \in P.$$

The cone P is called the positive cone in X . The cone $N := -P$ is called the negative cone, and we write $y \preceq x$ if $y - x \in N$.

Example 6.12 Positive orthant in $\mathbb{R}^n$.

In $X = \mathbb{R}^n$, the cone $P := \{x = (\xi_1, \dots, \xi_n) \in \mathbb{R}^n : \xi_i \geq 0, i = 1, \dots, n\}$ defines the standard componentwise partial order on $\mathbb{R}^n$. $\square$

Example 6.13 Function spaces.

Let X be a space of real-valued functions on an interval $[t_1, t_2]$. A natural positive cone is $P := \{f \in X : f(t) \geq 0 \text{ for all } t \in [t_1, t_2]\}$. $\square$

The induced relation $\succeq$ is reflexive and transitive. Indeed, $x \succeq x$ since $0 \in P$, and if $x \succeq y$ and $y \succeq z$ then $x - z = (x - y) + (y - z) \in P$.

6.2.2 ■ Closed Cones and Interior Points

In normed spaces, it is often important that the positive cone be closed.

Definition 6.14 (Strict positivity). Let X be a normed space and $P \subset X$ a positive cone. We write $x \succ 0$ if x belongs to the interior of P .

The existence of interior points is crucial for separation arguments.

Example 6.15 Lack of interior in L^1 .

Let $X = L^1[t_1, t_2]$ and let P be the cone of nonnegative functions. Then P has empty interior. Indeed, given any $f \geq 0$ and any $\varepsilon > 0$, pick a subset $A \subset [t_1, t_2]$ of positive but small measure and subtract a tall spike supported on A : the result is within ε of f in the L^1 -norm yet negative on A , so no L^1 -ball around f lies in P . $\square$

Example 6.16 Interior points in $C[t_1, t_2]$.

Let $X = C[t_1, t_2]$ equipped with the uniform norm, and let P be the cone of nonnegative continuous functions. In this case, P possesses interior points: for instance, the constant function 1 is interior, since any g with $\|g - 1\|_\infty < 1$ satisfies $g(t) > 0$ for all t , hence $g \in P$. $\square$

6.2.3 ■ Dual Cones

Let X be a normed space with dual space X^* .

Definition 6.17 (Dual cone). Given a positive cone $P \subset X$, the dual cone $P^\oplus \subset X^*$ is defined by

$$P^\oplus := \{x^* \in X^* : \langle x, x^* \rangle \geq 0 \text{ for all } x \in P\}.$$

Example 6.18 Finite-dimensional case.

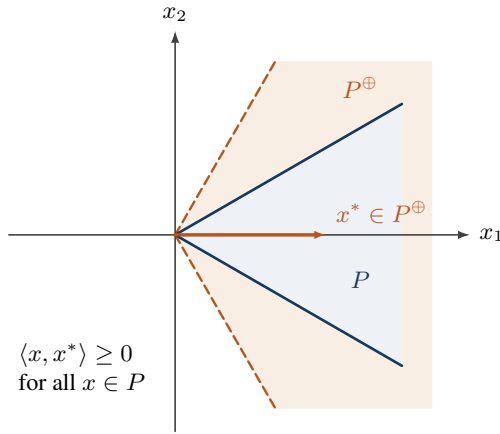


Figure 6.1: The dual cone $P^\oplus$ collects all continuous linear functionals that are nonnegative on the primal cone P . If P is symmetric about its axis with half-angle α , then $P^\oplus$ has half-angle $90^\circ - \alpha$ about the same axis, so each boundary ray of $P^\oplus$ is orthogonal to the opposite boundary ray of P . A narrow primal cone therefore has a wide dual cone; here $\alpha = 30^\circ$ and $P^\oplus$ opens at 60° .

If $X = \mathbb{R}^n$ and P is the positive orthant, then $P^\oplus$ consists of all linear functionals represented by vectors with nonnegative components. $\square$

Example 6.19 Continuous functions.

If $X = C[t_1, t_2]$ and P is the cone of nonnegative continuous functions, then $P^\oplus$ consists of all linear functionals represented by functions of bounded variation that are nondecreasing on $[t_1, t_2]$. $\square$

The dual cone $P^\oplus$ is always closed, even when P is not.

6.2.4 ■ Characterization via Duality

Proposition 6.20 (Dual characterization of positivity). *Let X be a normed space and let $P \subset X$ be a closed positive cone. If $x \in X$ satisfies $\langle x, x^* \rangle \geq 0$ for all $x^* \in P^\oplus$, then $x \in P$.*

Proof. Assume $x \notin P$. Since P is closed and convex, the separating hyperplane theorem implies that there exists $x^* \in X^*$ such that $\langle x, x^* \rangle < \langle p, x^* \rangle$ for all $p \in P$. Because P is a cone, $\langle p, x^* \rangle$ cannot be negative; otherwise scaling p would violate the inequality. Hence $x^* \in P^\oplus$. Moreover, $\inf_{p \in P} \langle p, x^* \rangle = 0$, which implies $\langle x, x^* \rangle < 0$, contradicting the assumption. $\square$

6.2.5 ■ Convex Mappings with Respect to a Cone

The same order relation allows convexity to be defined for vector-valued mappings.

Definition 6.21 (Cone-convex mapping). Let X and Z be vector spaces, and let $P \subset Z$ be a specified positive cone. A mapping $G : X \rightarrow Z$ is said to be convex with respect to P if:

- (1) its domain $\Omega \subset X$ is convex, and
- (2) for all $x_1, x_2 \in \Omega$ and all $\alpha \in (0, 1)$,

$$G(\alpha x_1 + (1 - \alpha)x_2) \preceq \alpha G(x_1) + (1 - \alpha)G(x_2).$$

Remark 6.22. Convexity of a mapping is not an intrinsic property of the function alone; it depends on the choice of the positive cone in the range space.

The following examples show how scalar convexity, componentwise inequalities, matrix inequalities, and integral constraints are all cone-convexity with different range cones.

Example 6.23 Scalar-valued convex functions.

Let $Z = \mathbb{R}$ and let $P = \mathbb{R}_+$. A mapping $G : X \rightarrow \mathbb{R}$ is convex with respect to P if and only if

$$G(\alpha x_1 + (1 - \alpha)x_2) \leq \alpha G(x_1) + (1 - \alpha)G(x_2),$$

which is precisely the standard definition of convexity for real-valued functions. Thus, scalar convex functions are exactly cone-convex mappings with respect to the nonnegative real cone. $\square$

Example 6.24 Componentwise convex inequality constraints.

Let $X = \mathbb{R}^n$, $Z = \mathbb{R}^m$, and $P = \mathbb{R}_+^m$. Define

$$G(x) = \begin{pmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{pmatrix},$$

where each $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$. Then G is convex with respect to P if and only if each component g_i is convex. The constraint $G(x) \preceq 0$ is exactly the system $g_i(x) \leq 0$, $i = 1, \dots, m$. $\square$

Example 6.25 Matrix-valued convex mappings and LMIs.

Let $Z = \mathbb{S}^n$, the space of symmetric $n \times n$ matrices, equipped with the cone of positive semidefinite matrices $\mathbb{S}_+^n$. Define $G(x) = A_0 + \sum_{i=1}^k x_i A_i$, $A_i \in \mathbb{S}^n$. Then G is affine and hence convex with respect to $\mathbb{S}_+^n$. The constraint $G(x) \preceq 0$ is a linear matrix inequality (LMI), which forms the basic feasibility condition in semidefinite programming. $\square$

Example 6.26 Nonlinear matrix-valued convex mappings.

Let $Z = \mathbb{S}^2$ with cone $\mathbb{S}_+^2$, and define

$$G(x) = \begin{pmatrix} x & 0 \\ 0 & x^2 \end{pmatrix}, \quad x \in \mathbb{R}.$$

Because $G(x)$ is diagonal, the Löwner order reduces to the componentwise order on the diagonal entries. The difference

$$\alpha G(x_1) + (1-\alpha)G(x_2) - G(\alpha x_1 + (1-\alpha)x_2) = \text{diag}\left(0, \alpha x_1^2 + (1-\alpha)x_2^2 - (\alpha x_1 + (1-\alpha)x_2)^2\right)$$

is positive semidefinite, since $x \mapsto x^2$ is convex. Hence G is convex with respect to $\mathbb{S}_+^2$, showing that cone-convexity is not limited to affine mappings. $\square$

Example 6.27 Integral inequality constraints.

Let $X = L^2[0, T]$, $Z = \mathbb{R}$, and $P = \mathbb{R}_+$. Define $G(u) = \int_0^T g(u(t)) dt - b$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is convex and $b \in \mathbb{R}$ is fixed. Then G is convex with respect to P , so integral constraints fit directly into the cone-convex framework. $\square$

Example 6.28 Vector-valued integral constraints.

Let $X = L^2[0, T]$, $Z = \mathbb{R}^m$, and $P = \mathbb{R}_+^m$. Define

$$G(u) = \begin{pmatrix} \int_0^T g_1(u(t)) dt \\ \vdots \\ \int_0^T g_m(u(t)) dt \end{pmatrix},$$

where each g_i is convex. Then G is convex with respect to P , and the constraint $G(u) \preceq 0$ represents multiple integral inequalities. $\square$

Example 6.29 Same mapping, different cone, different convexity.

Let $G : \mathbb{R} \rightarrow \mathbb{R}^2$ be defined by $G(x) = (x, -x^2)$. We show that the convexity of G depends on the choice of the positive cone in the range space.

Case 1: $P = \mathbb{R}_+^2$. Recall that G is convex with respect to P if and only if, for all $x_1, x_2 \in \mathbb{R}$ and all $\alpha \in (0, 1)$,

$$G(\alpha x_1 + (1-\alpha)x_2) \preceq \alpha G(x_1) + (1-\alpha)G(x_2),$$

where $\preceq$ denotes the componentwise order induced by $\mathbb{R}_+^2$.

Evaluating both sides, we obtain

$$G(\alpha x_1 + (1 - \alpha)x_2) = (\alpha x_1 + (1 - \alpha)x_2, -(\alpha x_1 + (1 - \alpha)x_2)^2),$$

and

$$\alpha G(x_1) + (1 - \alpha)G(x_2) = (\alpha x_1 + (1 - \alpha)x_2, -\alpha x_1^2 - (1 - \alpha)x_2^2).$$

The first components are equal. For the second component, the required inequality is

$$-(\alpha x_1 + (1 - \alpha)x_2)^2 \leq -\alpha x_1^2 - (1 - \alpha)x_2^2,$$

which is equivalent to

$$(\alpha x_1 + (1 - \alpha)x_2)^2 \geq \alpha x_1^2 + (1 - \alpha)x_2^2.$$

This inequality is violated for any $x_1 \neq x_2$, since the square function is strictly convex. For example, taking $x_1 = -1$, $x_2 = 1$, and $\alpha = \frac{1}{2}$, we obtain

$$(\frac{1}{2}(-1) + \frac{1}{2}(1))^2 = 0 < 1 = \frac{1}{2}(1) + \frac{1}{2}(1).$$

Hence the defining inequality for cone-convexity fails, and G is *not* convex with respect to $\mathbb{R}_+^2$.

Case 2: $P = \{(u, v) \in \mathbb{R}^2 \mid u \geq 0\}$.

In this case, the partial order induced by P depends only on the first component. That is,

$$(u_1, v_1) \preceq (u_2, v_2) \iff u_1 \leq u_2,$$

with no restriction on the second component. For all $x_1, x_2 \in \mathbb{R}$ and $\alpha \in (0, 1)$, we again compute

$$G(\alpha x_1 + (1 - \alpha)x_2) = (\alpha x_1 + (1 - \alpha)x_2, -(\alpha x_1 + (1 - \alpha)x_2)^2),$$

and

$$\alpha G(x_1) + (1 - \alpha)G(x_2) = (\alpha x_1 + (1 - \alpha)x_2, -\alpha x_1^2 - (1 - \alpha)x_2^2).$$

Since the first components are identical, we have

$$G(\alpha x_1 + (1 - \alpha)x_2) \preceq \alpha G(x_1) + (1 - \alpha)G(x_2)$$

for all x_1, x_2 and $\alpha \in (0, 1)$, regardless of the behavior of the second component.

Thus G is convex with respect to the cone $P = \{(u, v) \mid u \geq 0\}$. □

6.3 ■ General Convex Programming in Vector Spaces

We now put the pieces together. A cone in the constraint space turns $G(x) \preceq 0$ into a generalized inequality, and the dual cone supplies the right multiplier space for the Lagrangian.

6.3.1 ■ Problem Statement

Let X be a real vector space and let $\Omega \subset X$ be a nonempty convex set. We consider optimization problems of the form

$$\begin{aligned} \min_{x \in \Omega} \quad & f(x) \\ \text{subject to} \quad & G(x) \preceq 0, \end{aligned} \tag{6.1}$$

where the objective function $f : \Omega \rightarrow \mathbb{R}$ is a convex functional and the constraint mapping $G : \Omega \rightarrow Z$ takes values in a normed vector space Z .

To interpret the vector inequality in (6.1), equip Z with a closed convex positive cone $P \subset Z$. It induces the order

$$z_1 \succeq z_2 \iff z_1 - z_2 \in P. \tag{6.2}$$

Thus $G(x) \preceq 0$ means $G(x) \in -P$. Componentwise inequalities in $\mathbb{R}^m$, matrix inequalities, and function-space inequalities are all instances of this construction.

Cone-Convex Constraint Mappings

The constraint mapping $G : \Omega \rightarrow Z$ is assumed to be convex with respect to the cone P . Specifically, for all $x_1, x_2 \in \Omega$ and all $\alpha \in (0, 1)$, the mapping satisfies

$$G(\alpha x_1 + (1 - \alpha)x_2) \preceq \alpha G(x_1) + (1 - \alpha)G(x_2). \tag{6.3}$$

This is ordinary convexity with scalar inequalities replaced by the cone order. It keeps the feasible set convex and makes the constraint compatible with separation arguments.

Dual Cone and Lagrangian

Let Z^* denote the dual space of Z , consisting of all continuous linear functionals on Z . Associated with the positive cone $P \subset Z$ is the dual cone

$$P^\oplus := \{z^* \in Z^* \mid \langle z, z^* \rangle \geq 0 \text{ for all } z \in P\}. \tag{6.4}$$

Elements of $P^\oplus$ are the order-preserving linear functionals: they assign nonnegative values to every positive direction.

The Lagrangian associated with the general convex program (6.1) is defined as

$$\mathcal{L}(x, z^*) := f(x) + \langle G(x), z^* \rangle, \quad z^* \in P^\oplus. \tag{6.5}$$

This is the classical Lagrangian with nonnegative scalar multipliers replaced by dual-cone elements. The pairing $\langle G(x), z^* \rangle$ converts the vector-valued constraint into a scalar penalty.

Slater Condition

Lagrange multipliers require a nondegenerate feasible set relative to the cone order.

Definition 6.30 (Slater Condition). *Problem (6.1) is said to satisfy Slater's condition if there exists a point $\bar{x} \in \Omega$ such that*

$$G(\bar{x}) \prec 0, \tag{6.6}$$

that is, $G(\bar{x})$ lies in the interior of the negative cone $-P$.

Slater's condition is strict feasibility. Geometrically, it gives the separation argument room to produce a nontrivial supporting hyperplane, which becomes the Lagrange multiplier.

Illustrative Examples

Three examples show how the same template changes only through the choice of range space and cone.

Example 6.31 Finite-dimensional inequality-constrained convex program.

Let $X = \mathbb{R}^n$ and let $\Omega \subset \mathbb{R}^n$ be a convex set. Let the objective function $f : \Omega \rightarrow \mathbb{R}$ be convex, and define the constraint mapping $G : \Omega \rightarrow \mathbb{R}^m$ componentwise by $G(x) = (g_1(x), \dots, g_m(x))$, where each component function $g_i : \Omega \rightarrow \mathbb{R}$ is convex.

Equip $Z = \mathbb{R}^m$ with the positive orthant $P := \mathbb{R}_+^m = \{z \in \mathbb{R}^m : z_i \geq 0, i = 1, \dots, m\}$. The cone P induces the standard componentwise partial order on $\mathbb{R}^m$. Under this order, the vector inequality $G(x) \preceq 0$ is equivalent to the system of scalar constraints $g_i(x) \leq 0, i = 1, \dots, m$.

Convexity of each g_i implies that the mapping G is convex with respect to the cone P in the sense of (6.3). As a result, the general convex programming problem (6.1) reduces exactly to the classical finite-dimensional convex optimization problem with inequality constraints.

In this setting, the dual space satisfies $Z^* \cong \mathbb{R}^m$ under the standard inner product, and the dual cone $P^\oplus$ coincides with $\mathbb{R}_+^m$. The requirement that dual variables belong to $P^\oplus$ therefore recovers the familiar nonnegativity condition on Lagrange multipliers. The Lagrangian (6.5) takes the form $\mathcal{L}(x, \lambda) = f(x) + \sum_{i=1}^m \lambda_i g_i(x)$, $\lambda_i \geq 0$, recovering the usual Lagrange multipliers as a special case. $\square$

Example 6.32 Integral inequality constraint in a function space.

Let $X = L^2([0, T])$ and let $\Omega \subset L^2([0, T])$ be a convex set of admissible functions. Consider the objective functional $f(x) := \int_0^T \ell(x(t)) dt$, where $\ell : \mathbb{R} \rightarrow \mathbb{R}$ is convex. Define the constraint mapping $G : \Omega \rightarrow \mathbb{R}$ by $G(x) := \int_0^T x(t) dt - b$, where $b \in \mathbb{R}$ is a prescribed constant.

Let $Z = \mathbb{R}$ equipped with the positive cone $P = \mathbb{R}_+$. The constraint $G(x) \preceq 0$ then corresponds to the integral inequality $\int_0^T x(t) dt \leq b$. Since G is affine, it is convex with respect to P , and the resulting problem fits the general convex programming formulation (6.1).

Slater's condition holds provided there exists a function $\bar{x} \in \Omega$ such that $\int_0^T \bar{x}(t) dt < b$. In this case, strong duality applies. The dual space Z^* is again identified with $\mathbb{R}$, and the dual cone $P^\oplus$ is $\mathbb{R}_+$. The associated dual variable is therefore a scalar multiplier $\lambda \geq 0$, and the Lagrangian becomes

$$\mathcal{L}(x, \lambda) = \int_0^T \ell(x(t)) dt + \lambda \left(\int_0^T x(t) dt - b \right).$$

The multiplier has the same algebraic form as in finite dimensions, even though the decision variable is a function. $\square$

Example 6.33 Matrix inequality constraints and semidefinite programming.

Let $X = \mathbb{R}^n$ and let $\Omega \subset \mathbb{R}^n$ be convex. Suppose that the constraint mapping $G : \Omega \rightarrow \mathbb{S}^m$ takes values in the space $\mathbb{S}^m$ of real symmetric $m \times m$ matrices. Equip $Z = \mathbb{S}^m$ with the positive semidefinite cone $P := \mathbb{S}_+^m = \{M \in \mathbb{S}^m : M \succeq 0\}$.

The constraint $G(x) \preceq 0$ is then interpreted in the Löwner order, meaning that $G(x)$ is negative semidefinite. When f is linear and G is affine (so that $G(x) \preceq 0$ is an LMI), (6.1) is a semidefinite program; for a general G that is merely convex with respect to $\mathbb{S}_+^m$ one obtains a convex conic (“nonlinear semidefinite”) program over the same cone.

The dual space of $\mathbb{S}^m$ is naturally identified with $\mathbb{S}^m$ itself via the trace pairing $\langle M, Y \rangle := \text{trace}(MY)$. Under this identification, the dual cone $P^\oplus$ coincides with $\mathbb{S}_+^m$. Thus, the dual variables are themselves positive semidefinite matrices, and the Lagrangian takes the form $\mathcal{L}(x, Y) = f(x) + \text{trace}(G(x)Y)$, $Y \succeq 0$.

Here Slater’s condition asks for a point $\bar{x} \in \Omega$ with $G(\bar{x})$ negative definite. Under this condition, the same saddle-point and strong-duality theorem applies to semidefinite programs. $\square$

6.3.2 ■ Saddle-Point Characterization and Strong Duality

The following theorem is the analytic heart of the chapter. It says that, under convexity and a single strict-feasibility (Slater) assumption, solving the constrained program is equivalent to finding a saddle point of the Lagrangian: the primal player chooses x to make $\mathcal{L}$ small, the dual player chooses $z^* \in P^\oplus$ to make it large, and at optimality neither can improve. The same statement simultaneously delivers the multiplier, complementary slackness, and strong duality, and its proof uses nothing beyond convexity and the separating hyperplane theorem, no differentiability and no rank condition.

Theorem 6.34 (Saddle-Point and Strong Duality). *Assume that f and G are convex and that Slater’s condition (6.6) holds. A point $x^* \in \Omega$ is an optimal solution of (6.1) if and only if there exists $z^{**} \in P^\oplus$ such that*

$$\mathcal{L}(x^*, z^*) \leq \mathcal{L}(x^*, z^{**}) \leq \mathcal{L}(x, z^{**}), \quad \forall x \in \Omega, \forall z^* \in P^\oplus. \quad (6.7)$$

Moreover, strong duality holds:

$$\inf_{x \in \Omega} \sup_{z^* \in P^\oplus} \mathcal{L}(x, z^*) = \sup_{z^* \in P^\oplus} \inf_{x \in \Omega} \mathcal{L}(x, z^*). \quad (6.8)$$

Proof. We prove the equivalence in two directions and then derive strong duality as a consequence. Throughout the proof, recall that the Lagrangian is defined by $\mathcal{L}(x, z^*) = f(x) + \langle G(x), z^* \rangle$, $z^* \in P^\oplus$.

($\Leftarrow$) Existence of a saddle point implies primal optimality. Assume that there exists a pair $(x^*, z^{**}) \in \Omega \times P^\oplus$ satisfying the saddle-point inequalities (6.7), namely,

$$\mathcal{L}(x^*, z^*) \leq \mathcal{L}(x^*, z^{**}) \leq \mathcal{L}(x, z^{**}), \quad \forall x \in \Omega, \forall z^* \in P^\oplus.$$

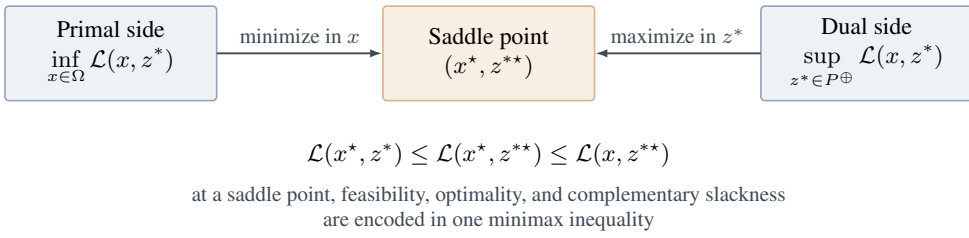


Figure 6.2: The saddle inequality couples the primal minimization and the dual maximization. Strong duality says the two orders of play have the same value.

We first show that x^* must be feasible. Taking $z^* = 0$ in the left inequality gives

$$f(x^*) = \mathcal{L}(x^*, 0) \leq \mathcal{L}(x^*, z^{**}) = f(x^*) + \langle G(x^*), z^{**} \rangle,$$

which implies

$$\langle G(x^*), z^{**} \rangle \geq 0. \quad (6.9)$$

Suppose, for contradiction, that $G(x^*) \not\leq 0$, i.e. $G(x^*) \notin -P$. Since $-P$ is closed and convex, the separating hyperplane theorem provides a continuous linear functional $w^* \in Z^*$ with $\langle G(x^*), w^* \rangle > \sup_{z \in -P} \langle z, w^* \rangle$. The supremum on the right is 0, because $-P$ is a cone containing the origin, and finiteness of the supremum forces $\langle p, w^* \rangle \geq 0$ for every $p \in P$, i.e. $w^* \in P^\oplus$. Thus $\langle G(x^*), w^* \rangle > 0$ with $w^* \in P^\oplus$. Substituting the admissible multipliers $t w^* \in P^\oplus$, $t > 0$, into the left saddle-point inequality gives

$$f(x^*) + t \langle G(x^*), w^* \rangle = \mathcal{L}(x^*, t w^*) \leq \mathcal{L}(x^*, z^{**}),$$

whose left side tends to $+\infty$ as $t \rightarrow \infty$ while the right side is a fixed real number. This contradiction shows $G(x^*) \leq 0$, so x^* is feasible.

Next, since $G(x^*) \leq 0$ and $z^{**} \in P^\oplus$, we also have

$$\langle G(x^*), z^{**} \rangle \leq 0. \quad (6.10)$$

Combining (6.9) and (6.10) yields $\langle G(x^*), z^{**} \rangle = 0$.

Now let $x \in \Omega$ be any feasible point, i.e. $G(x) \leq 0$. From the right inequality in (6.7),

$$\mathcal{L}(x^*, z^{**}) \leq \mathcal{L}(x, z^{**}) = f(x) + \langle G(x), z^{**} \rangle.$$

Since $G(x) \leq 0$ and $z^{**} \in P^\oplus$, the inner product on the right-hand side is nonpositive. Using $\mathcal{L}(x^*, z^{**}) = f(x^*)$, we conclude $f(x^*) \leq f(x)$. Because x was arbitrary among feasible points, x^* is a primal optimal solution of (6.1).

($\Rightarrow$) Primal optimality implies existence of a saddle point. Assume now that $x^* \in \Omega$ is a primal optimal solution of (6.1). Define the optimal value $\mu^* := f(x^*)$.

The key idea is to encode the primal problem geometrically in a higher-dimensional space so that convex separation can be applied. Consider the product space $\mathbb{R} \times Z$ and define the sets

$$\mathcal{A} := \{(r, z) \in \mathbb{R} \times Z \mid \exists x \in \Omega \text{ such that } r \geq f(x), z \succeq G(x)\},$$

and

$$\mathcal{B} := \{(r, z) \in \mathbb{R} \times Z \mid r \leq \mu^*, z \preceq 0\}.$$

The set $\mathcal{A}$ is convex because it is the image of the convex set Ω under the convex mapping $x \mapsto (f(x), G(x))$ followed by upward closure with respect to the cone order. The set $\mathcal{B}$ is convex because it is defined by linear inequalities.

By definition of μ^* , there exists no point that simultaneously satisfies $r \geq f(x)$ with $r \leq \mu^*$ and $z \succeq G(x)$ with $z \preceq 0$. Hence, $\mathcal{A} \cap \mathcal{B} = \emptyset$.

Slater's condition ensures the existence of a strictly feasible point $\bar{x} \in \Omega$ such that $G(\bar{x}) \prec 0$. This implies that $\mathcal{B}$ contains interior points in $\mathbb{R} \times Z$, which prevents degenerate (vertical) separating hyperplanes and guarantees the applicability of the separating hyperplane theorem.

By the separating hyperplane theorem, there exists a nonzero continuous linear functional $(\alpha, z^{**}) \in \mathbb{R} \times Z^*$ such that

$$\alpha r + \langle z, z^{**} \rangle \geq \alpha r' + \langle z', z^{**} \rangle \quad (6.11)$$

for all $(r, z) \in \mathcal{A}$ and $(r', z') \in \mathcal{B}$.

We now analyze the structure of the separating functional. From the definition of $\mathcal{B}$, inequality (6.11) implies $\alpha \geq 0$ and $\langle z, z^{**} \rangle \geq 0$ for all $z \in P$. Hence $z^{**} \in P^\oplus$.

Moreover $\alpha \neq 0$, and here is where Slater's condition is used. Suppose $\alpha = 0$; then $(\alpha, z^{**}) \neq 0$ forces $z^{**} \neq 0$. Taking $(r, z) = (f(\bar{x}), G(\bar{x})) \in \mathcal{A}$ for the Slater point $\bar{x}$ and $(r', z') = (\mu^*, 0) \in \mathcal{B}$ in (6.11) gives $\langle G(\bar{x}), z^{**} \rangle \geq 0$. But $G(\bar{x}) \prec 0$ means $G(\bar{x})$ lies in the interior of $-P$, so $G(\bar{x}) + \varepsilon v \in -P$ for every v of unit norm and all small $\varepsilon > 0$; applying $z^{**} \in P^\oplus$ to $-(G(\bar{x}) + \varepsilon v) \in P$ yields $\langle G(\bar{x}), z^{**} \rangle \leq -\varepsilon \langle v, z^{**} \rangle$ for all such v , and choosing v with $\langle v, z^{**} \rangle > 0$ gives $\langle G(\bar{x}), z^{**} \rangle < 0$, a contradiction. Hence $\alpha > 0$, and we normalize $\alpha = 1$.

With this normalization, inequality (6.11) implies $f(x) + \langle G(x), z^{**} \rangle \geq \mu^*, \forall x \in \Omega$, or equivalently,

$$\mathcal{L}(x, z^{**}) \geq f(x^*), \quad \forall x \in \Omega. \quad (6.12)$$

Finally, for any $z^* \in P^\oplus$, feasibility of x^* implies

$$\mathcal{L}(x^*, z^*) = f(x^*) + \langle G(x^*), z^* \rangle \leq f(x^*) = \mathcal{L}(x^*, z^{**}),$$

while (6.12) gives $\mathcal{L}(x^*, z^{**}) \leq \mathcal{L}(x, z^{**}), \forall x \in \Omega$. Thus (x^*, z^{**}) satisfies the saddle-point inequalities (6.7).

Strong duality. The equivalence established above shows that primal optimal solutions exist if and only if saddle points exist. For any saddle point (x^*, z^{**}) ,

$$\sup_{z^* \in P^\oplus} \inf_{x \in \Omega} \mathcal{L}(x, z^*) = \mathcal{L}(x^*, z^{**}) = \inf_{x \in \Omega} \sup_{z^* \in P^\oplus} \mathcal{L}(x, z^*),$$

which establishes the minimax equality (6.8). $\square$

Corollary 6.35 (Complementary Slackness). *Suppose that the assumptions of Theorem 6.34 hold, and let $(x^*, z^{**}) \in \Omega \times P^\oplus$ be a saddle point of the Lagrangian $\mathcal{L}(x, z^*) = f(x) + \langle G(x), z^* \rangle$. Then the primal and dual variables satisfy the complementary slackness condition*

$$\langle G(x^*), z^{**} \rangle = 0. \quad (6.13)$$

Proof. Since (x^*, z^{**}) is a saddle point, x^* is a feasible point of the primal problem, and $z^{**} \in P^\oplus$ is dual feasible. Hence $G(x^*) \preceq 0$ and, by definition of the dual cone,

$$\langle G(x^*), z^{**} \rangle \leq 0. \quad (6.14)$$

On the other hand, the left-hand inequality in the saddle-point condition (6.7) holds for all $z^* \in P^\oplus$. In particular, evaluating it at $z^* = 0$ yields $\mathcal{L}(x^*, 0) \leq \mathcal{L}(x^*, z^{**})$, or equivalently,

$$f(x^*) \leq f(x^*) + \langle G(x^*), z^{**} \rangle. \quad (6.15)$$

This implies $\langle G(x^*), z^{**} \rangle \geq 0$.

Combining (6.14) and (6.15) establishes (6.13). $\square$

6.3.3 ■ Connection to Sion's Minimax Theorem and Zero-Sum Games

The saddle-point characterization is a minimax principle. It is closely related to Sion's minimax theorem and has a zero-sum game interpretation.

Sion's minimax theorem applies to a function $\Phi : X \times Y \rightarrow \mathbb{R}$ defined on convex sets X and Y , provided that $\Phi(\cdot, y)$ is convex and lower semicontinuous for each fixed y , $\Phi(x, \cdot)$ is concave and upper semicontinuous for each fixed x , and at least one of the sets is compact. Under these conditions, Sion's theorem asserts that

$$\inf_{x \in X} \sup_{y \in Y} \Phi(x, y) = \sup_{y \in Y} \inf_{x \in X} \Phi(x, y).$$

In the present setting, we consider the function $\Phi(x, z^*) := \mathcal{L}(x, z^*) = f(x) + \langle G(x), z^* \rangle$, defined on the product set $\Omega \times P^\oplus$. For each fixed $z^* \in P^\oplus$, the function $x \mapsto \mathcal{L}(x, z^*)$ is convex, since it is the sum of a convex functional and a linear functional. For each fixed $x \in \Omega$, the function $z^* \mapsto \mathcal{L}(x, z^*)$ is affine and hence concave over the convex set $P^\oplus$.

Thus the Lagrangian has the convex–concave structure required by Sion's theorem. The missing hypothesis is usually compactness: neither Ω nor $P^\oplus$ is compact in many optimization problems, especially in function spaces. Slater's condition and the separating hyperplane argument replace that compactness assumption in Theorem 6.34.

Zero-sum game interpretation. The Lagrangian also defines a zero-sum game. The primal player chooses $x \in \Omega$ to minimize $\mathcal{L}(x, z^*)$; the dual player chooses $z^* \in P^\oplus$ to maximize it: $(x, z^*) \mapsto \mathcal{L}(x, z^*)$.

The dual player penalizes constraint violations through an order-preserving linear functional; the restriction $z^* \in P^\oplus$ keeps those penalties consistent with the cone order.

A saddle point (x^*, z^{**}) of the Lagrangian corresponds exactly to a Nash equilibrium of this zero-sum game. The inequalities $\mathcal{L}(x^*, z^*) \leq \mathcal{L}(x^*, z^{**}) \leq \mathcal{L}(x, z^{**})$ state that neither player can improve by deviating unilaterally.

The minimax equality (6.8) expresses the equivalence between the value of the game when the primal player moves first and the value when the dual player moves first. Strong duality is therefore the existence of a game value and a pure-strategy equilibrium.

6.3.4 ■ Primal Problem

The saddle-point theorem immediately gives a primal-side Lagrange multiplier representation of the optimal value.

Corollary 6.36 (Lagrange Multiplier Theorem (Primal Form)). *Let all assumptions of Theorem 6.34 hold, and define the optimal value*

$$\mu_0 := \inf\{f(x) : x \in \Omega, G(x) \preceq 0\}. \quad (6.16)$$

Then there exists an element $z_0^* \in P^\oplus$ such that

$$\mu_0 = \inf_{x \in \Omega} \{f(x) + \langle G(x), z_0^* \rangle\}. \quad (6.17)$$

Moreover, if the infimum in (6.16) is attained at some $x_0 \in \Omega$ with $G(x_0) \preceq 0$, then x_0 also attains the infimum in (6.17) and satisfies the complementary slackness condition

$$\langle G(x_0), z_0^* \rangle = 0. \quad (6.18)$$

Proof. By Theorem 6.34, there exists a saddle point $(x^*, z_0^*) \in \Omega \times P^\oplus$ of the Lagrangian such that

$$\mathcal{L}(x^*, z^*) \leq \mathcal{L}(x^*, z_0^*) \leq \mathcal{L}(x, z_0^*) \quad \forall x \in \Omega, \forall z^* \in P^\oplus.$$

Restricting this inequality to the primal variable yields

$$\mathcal{L}(x, z_0^*) \geq \mathcal{L}(x^*, z_0^*) = f(x^*) = \mu_0, \quad \forall x \in \Omega.$$

Taking the infimum over $x \in \Omega$ gives

$$\inf_{x \in \Omega} \{f(x) + \langle G(x), z_0^* \rangle\} = \mu_0,$$

which proves (6.17).

If x_0 attains the infimum in (6.16), then feasibility implies $f(x_0) + \langle G(x_0), z_0^* \rangle \leq f(x_0) = \mu_0$. Combined with (6.17), this yields $\langle G(x_0), z_0^* \rangle = 0$, establishing complementary slackness. $\square$

6.3.5 ■ Dual Problem

Projecting the saddle-point inequality onto the dual variable gives the dual optimization problem.

Corollary 6.37 (Dual Problem and Dual Optimality). *Let all assumptions of Theorem 6.34 hold, and define the dual function*

$$\psi(z^*) := \inf_{x \in \Omega} \{f(x) + \langle G(x), z^* \rangle\}, \quad z^* \in P^\oplus. \quad (6.19)$$

Then the dual problem is

$$\sup_{z^* \in P^\oplus} \psi(z^*). \quad (6.20)$$

Moreover, there exists $z^{**} \in P^\oplus$ such that

$$\mu_0 = \sup_{z^* \in P^\oplus} \psi(z^*) = \psi(z^{**}), \quad (6.21)$$

and any such z^{**} is a dual optimal solution.

Proof. By Theorem 6.34, there exists a saddle point $(x^*, z^{**}) \in \Omega \times P^\oplus$ satisfying

$$\mathcal{L}(x^*, z^*) \leq \mathcal{L}(x^*, z^{**}) \leq \mathcal{L}(x, z^{**}), \quad \forall x \in \Omega, \forall z^* \in P^\oplus.$$

Restricting this inequality to the dual variable yields

$$\mathcal{L}(x^*, z^*) \leq \mathcal{L}(x^*, z^{**}), \quad \forall z^* \in P^\oplus.$$

Since $\psi(z^*) = \inf_{x \in \Omega} \mathcal{L}(x, z^*) \leq \mathcal{L}(x^*, z^*)$, we obtain

$$\psi(z^*) \leq \mathcal{L}(x^*, z^{**}) = f(x^*) = \mu_0, \quad \forall z^* \in P^\oplus.$$

Thus μ_0 is an upper bound on the dual objective.

On the other hand, evaluating the dual function at z^{**} gives

$$\psi(z^{**}) = \inf_{x \in \Omega} \mathcal{L}(x, z^{**}) = \mathcal{L}(x^*, z^{**}) = \mu_0,$$

where the second equality follows from the primal-side inequality of the saddle point. Hence z^{**} attains the supremum in (6.20), and (6.21) follows. $\square$

The next properties hold independently of optimality and explain the shape of the dual problem.

Lemma 6.38 (Concavity of the Dual Function). *The dual function $\psi : P^\oplus \rightarrow \mathbb{R} \cup \{-\infty\}$ defined by (6.19) is concave.*

Proof. For each fixed $x \in \Omega$, the mapping $z^* \mapsto f(x) + \langle G(x), z^* \rangle$ is affine in z^* . The dual function ψ is defined as the pointwise infimum over $x \in \Omega$ of these affine functions. Since the infimum of affine functions is concave, ψ is concave on $P^\oplus$. $\square$

Lemma 6.39 (Weak Duality). *For every feasible primal point $x \in \Omega$ with $G(x) \preceq 0$ and every dual feasible point $z^* \in P^\oplus$, the inequality*

$$\psi(z^*) \leq f(x) \tag{6.22}$$

holds. In particular, $\sup_{z^ \in P^\oplus} \psi(z^*) \leq \mu_0$.*

Proof. Let $x \in \Omega$ satisfy $G(x) \preceq 0$, and let $z^* \in P^\oplus$. By definition of the dual function,

$$\psi(z^*) = \inf_{y \in \Omega} \{f(y) + \langle G(y), z^* \rangle\} \leq f(x) + \langle G(x), z^* \rangle.$$

Since $z^* \in P^\oplus$ and $G(x) \preceq 0$, we have $\langle G(x), z^* \rangle \leq 0$, which yields (6.22). Taking the supremum over $z^* \in P^\oplus$ proves the second claim. $\square$

Lemma 6.40 (Upper Envelope Interpretation). *The dual problem (6.20) seeks the tightest lower bound on the primal objective among all Lagrangian relaxations. That is,*

$$\sup_{z^* \in P^\oplus} \psi(z^*) = \sup_{z^* \in P^\oplus} \inf_{x \in \Omega} \{f(x) + \langle G(x), z^* \rangle\}.$$

Proof. For each $z^* \in P^\oplus$, the function $x \mapsto f(x) + \langle G(x), z^* \rangle$ defines a relaxation of the primal problem in which constraint violations are penalized linearly. The dual function $\psi(z^*)$ is the optimal value of this relaxed problem. Maximizing ψ over $P^\oplus$ therefore selects the strongest such relaxation, i.e., the largest lower bound on the primal optimal value. $\square$

6.3.6 ■ Dealing with Equality Constraints

So far all constraints have been inequalities. Now suppose the problem also has a linear equality constraint

$$H(x) = 0, \quad (6.23)$$

where $H : X \rightarrow Y$ is a continuous *affine* mapping into another vector space Y ; in finite dimensions one may think of $H(x) = Ax - b$, which is affine rather than linear unless $b = 0$. Affineness is what the arguments below actually use: it makes the set $\{x : H(x) = 0\}$ convex and the map $x \mapsto H(x)$ compatible with convex combinations.

Formally, the equality constraint (6.23) can be written as the pair of convex inequalities $H(x) \preceq 0$ and $-H(x) \preceq 0$, but this destroys the regularity condition: there is no point x_1 at which $H(x_1) \prec 0$ and $-H(x_1) \prec 0$ hold simultaneously. Slater's condition therefore fails automatically.

The obstruction is geometric: $H(x) = 0$ lies on the boundary in both directions, so there is no strictly feasible interior point for the paired inequalities.

Equality constraints must therefore be treated separately. Consider

$$\begin{aligned} \min_{x \in \Omega} \quad & f(x) \\ \text{s.t.} \quad & G(x) \preceq 0, \\ & H(x) = 0, \end{aligned} \quad (6.24)$$

where $G : \Omega \rightarrow Z$ is convex with respect to a cone $P \subset Z$ with nonempty interior, and $H : X \rightarrow Y$ is a continuous affine mapping.

In this setting, the appropriate Lagrangian is defined by

$$\mathcal{L}(x, z^*, \lambda) = f(x) + \langle G(x), z^* \rangle + \langle H(x), \lambda \rangle, \quad (6.25)$$

where $z^* \in P^\oplus$ is the multiplier associated with the inequality constraints, and $\lambda \in Y^*$ is an unrestricted multiplier associated with the equality constraint.

Equality multipliers are unrestricted because equalities impose two-sided restrictions.

Regularity and duality. A suitable regularity condition for the mixed problem (6.24) is the existence of a point $\bar{x} \in \Omega$ with $G(\bar{x}) \prec 0$ and $H(\bar{x}) = 0$. Under this condition, the same separation argument gives saddle points, strong duality, and, under differentiability, KKT conditions for the mixed problem.

Theorem 6.41 (Saddle-Point and Strong Duality with Equality Constraints).

Let X be a real vector space and $\Omega \subset X$ a nonempty convex set. Let $f : \Omega \rightarrow \mathbb{R}$ be a convex functional. Let $G : \Omega \rightarrow Z$ be convex with respect to a closed convex cone $P \subset Z$ with nonempty interior, and let $H : X \rightarrow Y$ be a continuous affine mapping.

Consider the mixed constrained problem

$$\begin{aligned} \min_{x \in \Omega} \quad & f(x) \\ \text{s.t.} \quad & G(x) \preceq 0, \\ & H(x) = 0. \end{aligned} \quad (6.26)$$

Assume that there exists a point $\bar{x} \in \Omega$ such that

$$G(\bar{x}) \prec 0, \quad H(\bar{x}) = 0, \quad (6.27)$$

(i.e., Slater's condition holds for the inequality constraints and the equality constraints are feasible).

Define the Lagrangian

$$\mathcal{L}(x, z^*, \lambda) = f(x) + \langle G(x), z^* \rangle + \langle H(x), \lambda \rangle, \quad z^* \in P^\oplus, \lambda \in Y^*. \quad (6.28)$$

Then a point $x^* \in \Omega$ is an optimal solution of (6.26) if and only if there exist multipliers $(z^{**}, \lambda^*) \in P^\oplus \times Y^*$ such that

$$\mathcal{L}(x^*, z^*, \lambda) \leq \mathcal{L}(x^*, z^{**}, \lambda^*) \leq \mathcal{L}(x, z^{**}, \lambda^*), \quad \forall x \in \Omega, \forall z^* \in P^\oplus, \forall \lambda \in Y^*. \quad (6.29)$$

Moreover, strong duality holds:

$$\inf_{x \in \Omega} \sup_{\substack{z^* \in P^\oplus \\ \lambda \in Y^*}} \mathcal{L}(x, z^*, \lambda) = \sup_{\substack{z^* \in P^\oplus \\ \lambda \in Y^*}} \inf_{x \in \Omega} \mathcal{L}(x, z^*, \lambda). \quad (6.30)$$

Proof. The argument parallels Theorem 6.34, with an extra coordinate for the equality constraint.

($\Leftarrow$) A saddle point is primal optimal. Suppose (x^*, z^{**}, λ^*) satisfies (6.29). The left inequality $\mathcal{L}(x^*, z^*, \lambda) \leq \mathcal{L}(x^*, z^{**}, \lambda^*)$ holds for all $z^* \in P^\oplus$ and all $\lambda \in Y^*$. Since λ ranges over the whole space Y^* and enters through the linear term $\langle H(x^*), \lambda \rangle$, boundedness forces $H(x^*) = 0$. Holding $\lambda = \lambda^*$ and letting z^* range over $P^\oplus$, the same argument as in Theorem 6.34 yields $G(x^*) \leq 0$ and the complementary slackness $\langle G(x^*), z^{**} \rangle = 0$; hence $\mathcal{L}(x^*, z^{**}, \lambda^*) = f(x^*)$ and x^* is feasible. For any feasible x (so $G(x) \leq 0, H(x) = 0$), the right inequality gives

$$f(x^*) = \mathcal{L}(x^*, z^{**}, \lambda^*) \leq \mathcal{L}(x, z^{**}, \lambda^*) = f(x) + \langle G(x), z^{**} \rangle + \langle H(x), \lambda^* \rangle \leq f(x),$$

because $\langle G(x), z^{**} \rangle \leq 0$ and $\langle H(x), \lambda^* \rangle = 0$. Thus x^* is optimal.

($\Rightarrow$) Primal optimality gives a saddle point. Let x^* be optimal with value $\mu^* = f(x^*)$, and work in the product space $\mathbb{R} \times Z \times Y$. Define

$$\mathcal{A} = \{(r, z, y) : \exists x \in \Omega, r \geq f(x), z \succeq G(x), y = H(x)\}, \quad \mathcal{B} = \{(r, 0, 0) : r < \mu^*\}.$$

Both are convex: $\mathcal{A}$ is the image of the convex set Ω under the convex/affine map $x \mapsto (f(x), G(x), H(x))$, closed upward in r and in the cone order on z , with $y = H(x)$ affine; $\mathcal{B}$ is a half-line. They are disjoint, for a point in $\mathcal{A} \cap \mathcal{B}$ would furnish $x \in \Omega$ with $G(x) \leq 0$, $H(x) = 0$, and $f(x) \leq r < \mu^*$, contradicting optimality. The Slater point $\bar{x}$ (with $G(\bar{x}) \prec 0$, $H(\bar{x}) = 0$) supplies interior directions, so the separating hyperplane theorem provides a nonzero continuous functional $(\alpha, z^{**}, \lambda^*)$ with

$$\alpha r + \langle z, z^{**} \rangle + \langle y, \lambda^* \rangle \geq \alpha r' \quad \forall (r, z, y) \in \mathcal{A}, (r', 0, 0) \in \mathcal{B}.$$

Testing against $\mathcal{B}$ forces $\alpha \geq 0$, and testing the cone directions of z forces $z^{**} \in P^\oplus$; the y -component carries no sign restriction because $\mathcal{B}$ fixes $y = 0$, so $\lambda^* \in Y^*$ is unrestricted. Strict feasibility rules out $\alpha = 0$, so we normalize $\alpha = 1$. The resulting inequality reads $f(x) + \langle G(x), z^{**} \rangle + \langle H(x), \lambda^* \rangle \geq \mu^*$ for all $x \in \Omega$, i.e. $\mathcal{L}(x, z^{**}, \lambda^*) \geq f(x^*)$. Together with $\mathcal{L}(x^*, z^*, \lambda) \leq f(x^*) = \mathcal{L}(x^*, z^{**}, \lambda^*)$, which holds for all (z^*, λ) by feasibility of x^* , this is the saddle-point property (6.29).

Strong duality. At any saddle point, $\sup_{z^*, \lambda} \inf_x \mathcal{L} = \mathcal{L}(x^*, z^{**}, \lambda^*) = \inf_x \sup_{z^*, \lambda} \mathcal{L}$, which is (6.30). $\square$

Corollary 6.42 (Lagrange Multiplier Theorem (Primal Form with Equality Constraints)).

Let all assumptions of Theorem 6.41 hold, and define the optimal value

$$\mu_0 := \inf\{f(x) : x \in \Omega, G(x) \preceq 0, H(x) = 0\}. \quad (6.31)$$

Assume that μ_0 is finite.

Then there exist multipliers $z_0^* \in P^\oplus$ and $\lambda_0 \in Y^*$ such that

$$\mu_0 = \inf_{x \in \Omega} \{f(x) + \langle G(x), z_0^* \rangle + \langle H(x), \lambda_0 \rangle\}. \quad (6.32)$$

Moreover, if the infimum in (6.31) is attained at some $x_0 \in \Omega$ with $G(x_0) \preceq 0$ and $H(x_0) = 0$, then x_0 also attains the infimum in (6.32) and satisfies the complementary slackness condition

$$\langle G(x_0), z_0^* \rangle = 0. \quad (6.33)$$

Proof. By Theorem 6.41 there is a saddle point (x^*, z_0^*, λ_0) of $\mathcal{L}$. Its right inequality $\mathcal{L}(x^*, z_0^*, \lambda_0) \leq \mathcal{L}(x, z_0^*, \lambda_0)$ for all $x \in \Omega$, together with $\mathcal{L}(x^*, z_0^*, \lambda_0) = f(x^*) = \mu_0$ established in the theorem, gives $\inf_{x \in \Omega} \{f(x) + \langle G(x), z_0^* \rangle + \langle H(x), \lambda_0 \rangle\} = \mu_0$, which is (6.32). If the primal infimum is attained at a feasible x_0 (so $G(x_0) \preceq 0$, $H(x_0) = 0$), then $f(x_0) + \langle G(x_0), z_0^* \rangle + \langle H(x_0), \lambda_0 \rangle = f(x_0) + \langle G(x_0), z_0^* \rangle \leq f(x_0) = \mu_0$, while (6.32) bounds the left side below by μ_0 . Hence $\langle G(x_0), z_0^* \rangle = 0$ and x_0 attains the Lagrangian infimum. $\square$

6.3.7 ■ First-Order Optimality Conditions

First-Order Optimality for Optimization over Cones

First consider the cone-constrained problem

$$\begin{aligned} \min_{x \in X} \quad & f(x) \\ \text{s.t.} \quad & x \in P, \end{aligned} \quad (6.34)$$

where X is a real normed vector space, $P \subset X$ is a convex cone, and $f : X \rightarrow \mathbb{R}$ is a convex functional. The cone P models one-sided constraints such as nonnegativity, order constraints, or tangent cones.

For differentiable convex f , optimality over P can be read from directional derivatives.

Lemma 6.43 (First-Order Optimality over a Cone).

Let X be a real normed vector space and let $P \subset X$ be a convex cone. Let $f : X \rightarrow \mathbb{R}$ be a Fréchet differentiable convex functional. A point $x_0 \in P$ minimizes f over P if and only if

$$\delta f(x_0; x) \geq 0, \quad \forall x \in P, \quad (6.35)$$

$$\delta f(x_0; x_0) = 0. \quad (6.36)$$

Proof. Necessity. Assume that $x_0 \in P$ minimizes f over P . Since P is convex, for any $x \in P$ and any $\alpha \geq 0$ sufficiently small, the point $x_0 + \alpha(x - x_0) \in P$ is feasible. Therefore, the function $\alpha \mapsto f(x_0 + \alpha(x - x_0))$ attains a minimum at $\alpha = 0$, which implies

$$\left. \frac{d}{d\alpha} f(x_0 + \alpha(x - x_0)) \right|_{\alpha=0} \geq 0.$$

By definition of the Fréchet directional derivative, this yields

$$\delta f(x_0; x - x_0) \geq 0, \quad \forall x \in P. \quad (6.37)$$

Choosing $x = \frac{1}{2}x_0 \in P$ in (6.37) gives $\delta f(x_0; -\frac{1}{2}x_0) \geq 0$, which implies

$$\delta f(x_0; x_0) \leq 0. \quad (6.38)$$

Similarly, choosing $x = 2x_0 \in P$ yields $\delta f(x_0; x_0) \geq 0$. Together, these inequalities imply (6.36). Substituting (6.36) back into (6.37) establishes (6.35).

Sufficiency. Assume that (6.35) and (6.36) hold. By convexity of f , for any $x \in P$ and any $0 < \alpha < 1$,

$$f(x_0 + \alpha(x - x_0)) \leq f(x_0) + \alpha(f(x) - f(x_0)).$$

Rearranging yields

$$f(x) - f(x_0) \geq \frac{1}{\alpha} [f(x_0 + \alpha(x - x_0)) - f(x_0)].$$

Taking the limit as $\alpha \downarrow 0$ and using Fréchet differentiability, we obtain

$$f(x) - f(x_0) \geq \delta f(x_0; x - x_0).$$

Using (6.35) and (6.36), we conclude $f(x) \geq f(x_0), \forall x \in P$, and hence x_0 minimizes f over P . $\square$

Lemma 6.43 says that the derivative must be nonnegative in all cone directions and zero along the radial direction x_0 . Applied to a Lagrangian, this is the differential source of complementary slackness and KKT stationarity.

First-Order Optimality Conditions for Mixed Constraints.

Return to the mixed problem (6.26). Convexity and Slater regularity already give multipliers and a saddle point. If the data are Fréchet differentiable, minimizing the Lagrangian in the primal variable is equivalent to vanishing derivative, yielding the abstract KKT system.

Corollary 6.44 (Lagrange Multiplier Theorem with First-Order Optimality). *Let all assumptions of Theorem 6.41 hold. Assume in addition that $f : X \rightarrow \mathbb{R}$, $G : X \rightarrow Z$ and $H : X \rightarrow Y$ are Fréchet differentiable at a point $x_0 \in \Omega$, and that x_0 attains the optimal value $\mu_0 = \inf\{f(x) : x \in \Omega, G(x) \preceq 0, H(x) = 0\}$.*

Then there exist multipliers $z_0^ \in P^\oplus$ and $\lambda_0 \in Y^*$ such that the following conditions hold:*

(1) **Primal feasibility:**

$$G(x_0) \preceq 0, \quad H(x_0) = 0. \quad (6.39)$$

(2) **Dual feasibility:**

$$z_0^* \in P^\oplus. \quad (6.40)$$

(3) **Complementary slackness:**

$$\langle G(x_0), z_0^* \rangle = 0. \quad (6.41)$$

(4) **First-order stationarity:**

$$Df(x_0) + DG(x_0)^* z_0^* + DH(x_0)^* \lambda_0 = 0 \quad \text{in } X^*. \quad (6.42)$$

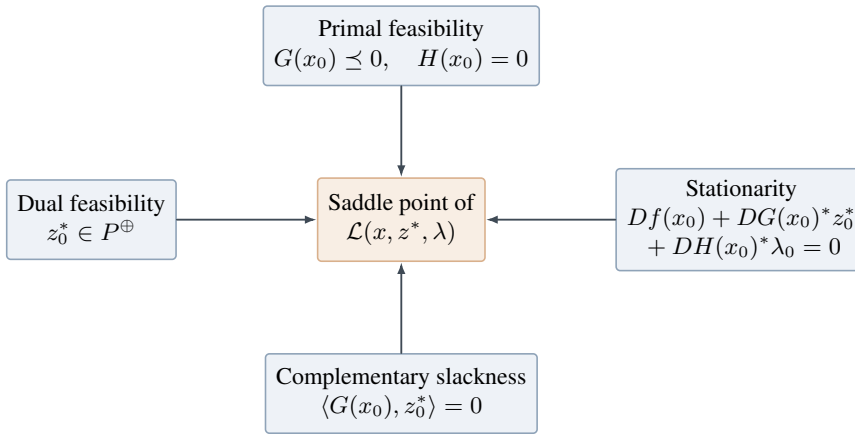


Figure 6.3: The differentiable KKT system is the saddle-point condition written in four complementary pieces: primal feasibility, dual feasibility, complementary slackness, and stationarity.

Proof. By Corollary 6.42 there exist multipliers $z_0^* \in P^\oplus$ and $\lambda_0 \in Y^*$ such that x_0 attains $\inf_{x \in \Omega} \mathcal{L}(x, z_0^*, \lambda_0)$ and $\langle G(x_0), z_0^* \rangle = 0$. Conditions (1)–(3) are then immediate: primal feasibility holds because x_0 attains μ_0 , dual feasibility is $z_0^* \in P^\oplus$, and complementary slackness is the identity just quoted.

It remains to establish stationarity (4). The functional $x \mapsto \mathcal{L}(x, z_0^*, \lambda_0) = f(x) + \langle G(x), z_0^* \rangle + \langle H(x), \lambda_0 \rangle$ is convex, and it is Fréchet differentiable at x_0 because f, G, H are; by the chain rule its derivative is the element of X^*

$$DL(x_0) = Df(x_0) + z_0^* \circ DG(x_0) + \lambda_0 \circ DH(x_0) = Df(x_0) + DG(x_0)^* z_0^* + DH(x_0)^* \lambda_0,$$

where $DG(x_0)^*$ and $DH(x_0)^*$ are the adjoints of the derivative maps. Since x_0 minimizes the convex differentiable functional $\mathcal{L}(\cdot, z_0^*, \lambda_0)$ over Ω at an interior point, the derivative must vanish, $DL(x_0) = 0$, which is (6.42). (Differentiability is used only here, to turn minimization of the Lagrangian into a stationarity equation; existence of the multipliers and strong duality already followed from convexity and Slater regularity.) $\square$

Role of differentiability. Corollary 6.44 refines the primal Lagrange multiplier theorem by expressing optimality through first-order conditions. Differentiability is not required for the existence of Lagrange multipliers or for strong duality; those results follow purely from convexity and Slater-type regularity. The role of differentiability is instead to convert global minimization of the Lagrangian into a stationarity condition in the dual space X^* .

Absence of constraint qualifications. No linear independence or rank condition on the constraint mappings is assumed. In particular, linear independence constraint qualifications such as LICQ are not required. The multipliers arise from convex separation arguments rather than from solvability of systems of derivative equations. Constraint qualifications become relevant only when convexity is absent, or when one seeks uniqueness, stability, or sensitivity properties of the multipliers beyond existence.

Geometric meaning of complementary slackness. The complementary slackness condition $\langle G(x_0), z_0^* \rangle = 0$ has a direct geometric interpretation. It expresses orthogonality between the dual

multiplier z_0^* and the active face of the cone P at the point $G(x_0)$. Equivalently, it characterizes the supporting hyperplane that separates the epigraph of the Lagrangian from the feasible region in the product space $X \times Z$.

Equality versus inequality constraints. Equality constraints and inequality constraints play fundamentally different roles in the dual formulation. Equality constraints give rise to unrestricted multipliers in the dual space, reflecting the two-sided nature of equality restrictions. In contrast, inequality constraints induce cone-restricted multipliers and lead to complementary slackness conditions. This distinction is intrinsic and cannot be removed by reformulating equalities as pairs of inequalities without destroying regularity.

Connection to classical KKT conditions. In finite-dimensional settings with smooth data, the conditions in Corollary 6.44 reduce to the classical Karush–Kuhn–Tucker system. From this viewpoint, KKT theory is a differentiable specialization of convex duality rather than its foundation. The saddle-point framework provides the conceptual basis, while differentiability merely supplies a convenient representation.

Interpretation as a saddle point. The first-order stationarity condition $Df(x_0) + DG(x_0)^*z_0^* + DH(x_0)^*\lambda_0 = 0$ characterizes x_0 as the primal component of a saddle point of the Lagrangian. Primal feasibility, dual feasibility, complementary slackness, and stationarity are thus unified as consequences of the saddle-point property, rather than as independent optimality requirements.

6.4 ■ Case Study: Optimal Control with Terminal Inequality Constraints

Consider the linear time-varying system

$$\dot{x}(t) = A(t)x(t) + b(t)u(t), \quad t \in [t_0, t_1], \quad (6.43)$$

where $x(t) \in \mathbb{R}^n$, the input $u(t) \in \mathbb{R}$ is scalar, and the initial state $x(t_0)$ is given.

The objective is to minimize the quadratic control effort

$$J(u) = \frac{1}{2} \int_{t_0}^{t_1} u^2(t) dt, \quad (6.44)$$

subject to the terminal inequality constraint

$$x(t_1) \succeq c, \quad (6.45)$$

where the inequality is componentwise.

Let $\Phi(t, s)$ denote the fundamental solution matrix of the homogeneous system $\dot{x}(t) = A(t)x(t)$. Under continuity assumptions on $A(t)$ and $b(t)$, the solution of (6.43) at time t_1 can be expressed as

$$x(t_1) = \Phi(t_1, t_0)x(t_0) + \int_{t_0}^{t_1} \Phi(t_1, t)b(t)u(t) dt. \quad (6.46)$$

Define the linear operator $K : L^2[t_0, t_1] \rightarrow \mathbb{R}^n$ by

$$Ku := \int_{t_0}^{t_1} \Phi(t_1, t)b(t)u(t) dt, \quad (6.47)$$

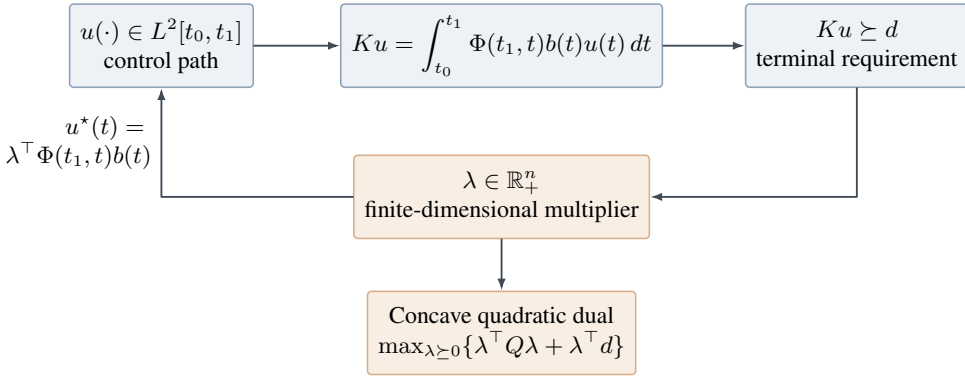


Figure 6.4: The terminal constraint depends on the control only through the finite-dimensional operator K . Duality therefore reduces the control problem to an optimization over $\lambda \in \mathbb{R}_+^n$.

and set

$$d := c - \Phi(t_1, t_0)x(t_0). \quad (6.48)$$

The terminal constraint is therefore equivalent to

$$Ku \succeq d. \quad (6.49)$$

The optimal control problem can now be written as the convex optimization problem

$$\begin{aligned} \min_{u \in L^2[t_0, t_1]} \quad & \frac{1}{2} \int_{t_0}^{t_1} u^2(t) dt \\ \text{subject to} \quad & Ku \succeq d. \end{aligned} \quad (6.50)$$

The objective is strictly convex on $L^2[t_0, t_1]$, while K is linear and takes values in the finite-dimensional ordered space $(\mathbb{R}^n, \mathbb{R}_+^n)$.

Introduce a Lagrange multiplier $\lambda \in \mathbb{R}_+^n$ associated with the constraint (6.49). The Lagrangian is

$$\mathcal{L}(u, \lambda) = \frac{1}{2} \int_{t_0}^{t_1} u^2(t) dt + \lambda^\top (d - Ku). \quad (6.51)$$

The dual function is defined by

$$\psi(\lambda) = \inf_{u \in L^2[t_0, t_1]} \mathcal{L}(u, \lambda), \quad \lambda \in \mathbb{R}_+^n. \quad (6.52)$$

Expanding the Lagrangian yields

$$\mathcal{L}(u, \lambda) = \int_{t_0}^{t_1} \left(\frac{1}{2} u^2(t) - \lambda^\top \Phi(t_1, t) b(t) u(t) \right) dt + \lambda^\top d.$$

For each t , the integrand is a strictly convex quadratic function of $u(t)$. The minimizing control is therefore given pointwise by

$$u^*(t) = \lambda^\top \Phi(t_1, t) b(t). \quad (6.53)$$

Substituting (6.53) into (6.51) yields

$$\mu_0 = \max_{\lambda \geq 0} \{ \lambda^\top Q \lambda + \lambda^\top d \}, \quad (6.54)$$

where

$$Q = -\frac{1}{2} \int_{t_0}^{t_1} \Phi(t_1, t) b(t) b^\top(t) \Phi^\top(t_1, t) dt. \quad (6.55)$$

The matrix Q is negative semidefinite, and the dual problem (6.54) is a concave quadratic maximization problem in $\mathbb{R}^n$.

Let λ^* denote a maximizer of (6.54). Then the corresponding optimal control is given by

$$u^*(t) = (\lambda^*)^\top \Phi(t_1, t) b(t). \quad (6.56)$$

The multiplier λ^* is the shadow price of the terminal target: it is the gradient of the optimal value μ_0 with respect to c and measures the marginal cost of tightening the terminal inequality.

Remark 6.45. Convex duality reduces an infinite-dimensional control problem to a finite-dimensional concave quadratic maximization. The derivation uses saddle-point theory rather than Pontryagin's Maximum Principle.

6.5 ■ Case Study: Optimal Extraction of an Exhaustible Resource

A firm owns an oil deposit of size $\alpha > 0$ and chooses an extraction rate $x : [0, \infty) \rightarrow \mathbb{R}_+$, where $x(t)$ is the extraction rate at time t . The stock constraint is

$$\int_0^\infty x(t) dt \leq \alpha. \quad (6.57)$$

Instantaneous discounted profit is $F(x(t))v(t)$, where

- $F : [0, \infty) \rightarrow \mathbb{R}$ is the profit function,
- $v : [0, \infty) \rightarrow \mathbb{R}_+$ is a discount factor.

The objective is

$$\max_{x(\cdot)} \int_0^\infty F(x(t)) v(t) dt, \quad (6.58)$$

subject to (6.57) and $x(t) \geq 0$ for all t .

Assume the following.

Profit function. The function F satisfies:

- (1) $F(0) = 0$,
- (2) F is strictly increasing and strictly concave on $[0, \infty)$,
- (3) F is continuously differentiable, with derivative denoted F_x .

Consequently, F_x is continuous and strictly decreasing, and therefore admits a continuous, strictly decreasing inverse $(F_x)^{-1}$ on its range.

Discount factor. The discount factor $v(t)$ is continuous, strictly positive, strictly decreasing, satisfies $v(t) \rightarrow 0$ as $t \rightarrow \infty$, and is integrable on $[0, \infty)$.

6.5.1 ■ Finite-horizon restriction

First restrict attention to continuous extraction policies on a finite horizon $[0, T]$:

$$\begin{aligned} \max_{x(\cdot)} \quad & \int_0^T F(x(t)) v(t) dt \\ \text{s.t.} \quad & \int_0^T x(t) dt \leq \alpha, \\ & x(t) \geq 0 \quad \text{for } t \in [0, T], \\ & x \text{ continuous on } [0, T]. \end{aligned} \tag{6.59}$$

Introduce a Lagrange multiplier $\lambda \geq 0$ associated with the resource constraint. The Lagrangian is

$$\mathcal{L}(x, \lambda) = \int_0^T F(x(t)) v(t) dt - \lambda \left(\int_0^T x(t) dt - \alpha \right). \tag{6.60}$$

Since the objective is concave and the constraints are convex, optimal solutions correspond to saddle points of the Lagrangian. For a *maximization* primal the two orders are

$$\max_{x \geq 0} \min_{\lambda \geq 0} \mathcal{L}(x, \lambda) \quad (\text{primal}), \quad \min_{\lambda \geq 0} \max_{x \geq 0} \mathcal{L}(x, \lambda) \quad (\text{dual}),$$

and they agree at a saddle point. The inner minimization over $\lambda \geq 0$ in the primal order returns the objective when $\int_0^T x dt \leq \alpha$ and $-\infty$ otherwise, which is exactly how the constraint is enforced.

Let (x_0, λ_0) denote an optimal primal–dual pair. Introducing a pointwise multiplier $\mu(t) \geq 0$ for the constraint $x(t) \geq 0$, the Karush–Kuhn–Tucker conditions are:

Primal feasibility

$$x_0(t) \geq 0, \quad \int_0^T x_0(t) dt \leq \alpha.$$

Dual feasibility

$$\lambda_0 \geq 0, \quad \mu(t) \geq 0 \quad \text{for all } t \in [0, T].$$

Complementary slackness

$$\lambda_0 \left(\int_0^T x_0(t) dt - \alpha \right) = 0, \quad \mu(t) x_0(t) = 0 \quad \forall t. \tag{6.61}$$

Stationarity

$$F_x(x_0(t)) v(t) - \lambda_0 + \mu(t) = 0 \quad \forall t \in [0, T]. \tag{6.62}$$

Because F is increasing and $v(t) > 0$, the resource constraint binds at optimality, so $\int_0^T x_0(t) dt = \alpha$ and $\lambda_0 > 0$.

6.5.2 ■ Characterization of the optimal extraction path

From (6.62) and (6.61), two cases arise.

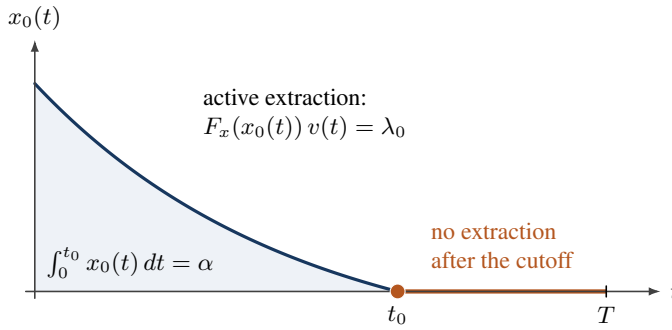


Figure 6.5: The optimal extraction path front-loads production and stops at the endogenous cutoff time t_0 . The shaded area is the total available stock α . The curve is drawn for $F(x) = \log(1+x)$ and $v(t) = e^{-rt}$, for which $x_0(t) = e^{r(t_0-t)} - 1$ on $[0, t_0]$.

Active extraction. If $x_0(t) > 0$, then $\mu(t) = 0$ and $F_x(x_0(t)) v(t) = \lambda_0$, which implies

$$x_0(t) = (F_x)^{-1} \left(\frac{\lambda_0}{v(t)} \right). \quad (6.63)$$

No extraction. If $x_0(t) = 0$, then $\mu(t) = \lambda_0 - F_x(0)v(t) \geq 0$, so extraction ceases when $\lambda_0 \geq F_x(0)v(t)$.

Since $v(t)$ is strictly decreasing, there exists a cutoff time $t_0 \in (0, T]$ such that $\lambda_0 = F_x(0)v(t_0)$. Consequently, the optimal extraction policy takes the form

$$x_0(t) = \begin{cases} (F_x)^{-1} \left(\frac{F_x(0)v(t_0)}{v(t)} \right), & 0 \leq t \leq t_0, \\ 0, & t_0 \leq t \leq T. \end{cases} \quad (6.64)$$

The cutoff time t_0 is uniquely determined by the binding resource constraint:

$$\int_0^{t_0} (F_x)^{-1} \left(\frac{F_x(0)v(t_0)}{v(t)} \right) dt = \alpha.$$

Define

$$J(t_0) = \int_0^{t_0} (F_x)^{-1} \left(\frac{F_x(0)v(t_0)}{v(t)} \right) dt.$$

The function J is continuous and *strictly increasing*: raising t_0 lowers $v(t_0)$, hence lowers the argument $F_x(0)v(t_0)/v(t)$, and since $(F_x)^{-1}$ is strictly decreasing this raises the integrand pointwise while also lengthening the interval of integration. Strict monotonicity is what delivers uniqueness; continuity and the limits give existence.

On the infinite horizon, $J(0^+) = 0$ and $J(t_0) \rightarrow \infty$ as $t_0 \rightarrow \infty$, so $J(t_0) = \alpha$ has exactly one solution for every $\alpha > 0$. On the finite horizon $[0, T]$ the cutoff is confined to $(0, T]$, so a solution with $t_0 < T$ exists precisely when $\alpha < J(T)$; for $\alpha \geq J(T)$ the constraint does not force extraction to stop before T and the cutoff is $t_0 = T$.

6.5.3 ■ Extension to the infinite horizon

Any admissible plan with $\int_0^\infty y(t) dt = \alpha$ can be approximated in discounted profit by a continuous compactly supported plan. Because $v(t) \rightarrow 0$, truncating far in the future has arbitrarily small value loss. Thus (6.64) is optimal for the infinite-horizon problem.

The optimal policy equalizes discounted marginal profit: $F_x(x_0(t))v(t) = \lambda_0$ for all t with $x_0(t) > 0$. Extraction is front-loaded, declining over time, and ceases at a finite endogenous exhaustion time t_0 .

The multiplier λ_0 admits a shadow-price interpretation:

$$\lambda_0 = \frac{d}{d\alpha} \left(\sup_{x(\cdot)} \int_0^\infty F(x(t))v(t) dt \right),$$

representing the marginal value of an additional unit of oil in the ground.

6.6 ■ Further Reading

This chapter follows the vector-space perspective in Luenberger [79]. That viewpoint treats finite-dimensional convex programs, semidefinite programs [126], conic programs [4], optimal control problems, and PDE-constrained optimization as instances of one duality theory. In this setting, finite-dimensional optimization is a special case rather than the foundation. For more background on vector spaces, dual spaces, and their role in optimization, see [79].

6.7 ■ Exercises

Exercise 6.1 (Optimization in Function Spaces). Find a continuous function $x(\cdot)$ on the interval $[-1, 1]$ that maximizes

$$\int_{-1}^1 t^3 x(t) dt$$

subject to the constraint

$$\int_{-1}^1 |x(t)|^3 dt \leq 2.$$

Hint: Use Hölder's inequality on a suitably chosen normed linear space.

Exercise 6.2 (Dual Optimization Problem). Consider the same setting as in the previous exercise, but now minimize

$$\int_{-1}^1 t^3 x(t) dt$$

subject to

$$\int_{-1}^1 |x(t)|^3 dt \leq 2.$$

What is the optimal minimizing solution?

Exercise 6.3 (ℓ^p Spaces). Suppose $x = \{\xi_1, \xi_2, \dots\} \in \ell^{p_0}$ for some $1 \leq p_0 \leq \infty$.

(i) Show that $x \in \ell^p$ for all $p \geq p_0$.

(ii) Show that

$$\|x\|_\infty := \sup_i |\xi_i| = \lim_{p \rightarrow \infty} \|x\|_p.$$

For terminology and notation, see Section 2.10 of [79].

Exercise 6.4 (Bounded Linear Functionals on L^2). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $L_2(\Omega, \mathbb{P}; [0, 1])$ denote the Hilbert space of scalar second-order random variables defined on $[0, 1] \times \Omega$. Consider the mappings

$$g : L_2(\Omega, \mathbb{P}; [0, 1]) \rightarrow L_2(\Omega, \mathbb{P}), \quad f : L_2(\Omega, \mathbb{P}) \rightarrow \mathbb{R},$$

defined by

$$g(x) = \int_0^1 b(t)x(t, \omega) dt, \quad b \in L_2[0, 1],$$

and

$$f(y) = \mathbb{E}[y].$$

Let $h = f \circ g : L_2(\Omega, \mathbb{P}; [0, 1]) \rightarrow \mathbb{R}$. Show that both f and h are bounded linear functionals. *Hint:* See Chapter 5.2 of [79].

Exercise 6.5 (Dual Space of $C[0, 1]$). Let $X = C[0, 1]$ equipped with the supremum norm, and let $f \in X^*$ be defined by

$$f(x) = x\left(\frac{1}{4}\right) + 3x\left(\frac{1}{2}\right) + 2 \int_0^1 t x(t) dt.$$

Compute the operator norm $\|f\|$.

Exercise 6.6 (Interior points of the nonnegative cone in function spaces). Let $[a, b] \subset \mathbb{R}$ be a compact interval and let $P := \{f \mid f \geq 0\}$, where nonnegativity is interpreted according to the ambient function space.

- (1) Let $1 \leq p < \infty$ and consider the space $L_p[a, b]$ endowed with the L_p norm. Define $P := \{f \in L_p[a, b] \mid f(t) \geq 0 \text{ a.e. on } [a, b]\}$. Prove that P has *empty interior* in $L_p[a, b]$.
- (2) Consider either $C[a, b]$ endowed with the uniform norm $\|\cdot\|_\infty$ or $L_\infty[a, b]$ endowed with the essential supremum norm. Define $P := \{f \mid f(t) \geq 0 \text{ for all } t \in [a, b]\}$. Prove that P has *nonempty interior* in these spaces.

Exercise 6.7 (Minimum-norm duality via first-order optimality). Let X be a real normed linear space with dual X^* , and let $M \subset X$ be a closed linear subspace. Fix $x \in X$ and consider the primal optimization problem

$$\inf_{m \in M} \|x - m\|. \quad (6.65)$$

- (1) Show that problem (6.65) is a convex optimization problem and that the feasible set M may be represented as the solution set of a linear equality constraint $H(m) = 0$, for a suitable continuous linear mapping $H : X \rightarrow Y$.
- (2) Assume that the infimum in (6.65) is attained at some $m_0 \in M$, that $x - m_0 \neq 0$, and that the norm of X is Fréchet differentiable at $x - m_0$ (equivalently, $x - m_0$ is a point of smoothness of the norm, so it has a unique norming functional). Show that the objective function $f(m) := \|x - m\|$ is Fréchet differentiable at m_0 and compute its derivative.
- (3) Apply Corollary 6.44 to the problem (6.65) and show that there exists a Lagrange multiplier $\lambda_0 \in Y^*$ such that $Df(m_0) + DH(m_0)^* \lambda_0 = 0$.
- (4) Writing $x_0^* := -Df(m_0)$, deduce that $x_0^* \in M^\perp$ and that $\|x_0^*\| = 1$, where

$$M^\perp := \{x^* \in X^* \mid \langle m, x^* \rangle = 0 \text{ for all } m \in M\}.$$

(5) Show that $\|x - m_0\| = \langle x, x_0^* \rangle$, and conclude that

$$\inf_{m \in M} \|x - m\| = \max_{\substack{x^* \in M^\perp \\ \|x^*\| \leq 1}} \langle x, x^* \rangle.$$

Remark. This exercise shows that the minimum-norm duality theorem arises as a direct consequence of first-order optimality for a convex equality-constrained problem, with the dual functional appearing naturally as a Lagrange multiplier rather than as an independently postulated dual variable.

Exercise 6.8 (Self-Duality of the Positive Semidefinite Cone). Equip the space $\mathbb{S}^n$ of real symmetric matrices with the trace inner product $\langle X, Y \rangle = \text{trace}(XY)$, and let $\mathbb{S}_+^n$ denote the cone of positive semidefinite matrices.

- (i) Show that if $X \succeq 0$ and $Y \succeq 0$, then $\text{trace}(XY) \geq 0$. *Hint:* write $X = \sum_k \lambda_k v_k v_k^\top$ with $\lambda_k \geq 0$ and use $\text{trace}(v_k v_k^\top Y) = v_k^\top Y v_k$.
- (ii) Conversely, suppose $Y \in \mathbb{S}^n$ satisfies $\text{trace}(XY) \geq 0$ for all $X \succeq 0$. By testing against the rank-one matrices $X = vv^\top$, show that $v^\top Y v \geq 0$ for all v , hence $Y \succeq 0$.
- (iii) Conclude that $(\mathbb{S}_+^n)^\oplus = \mathbb{S}_+^n$: the semidefinite cone is self-dual. Explain why this makes the dual variable of a semidefinite program itself a positive semidefinite matrix, as in Example 6.33.

Exercise 6.9 (The Second-Order Cone and Its Dual). Let $K = \{(x, t) \in \mathbb{R}^n \times \mathbb{R} : \|x\|_2 \leq t\}$ be the second-order (Lorentz) cone, and equip $\mathbb{R}^{n+1}$ with the standard inner product $\langle (x, t), (y, s) \rangle = x^\top y + ts$.

- (i) Show that K is a closed convex cone with nonempty interior.
- (ii) Using the Cauchy–Schwarz inequality, show that for $(y, s) \in K$ and $(x, t) \in K$ one has $x^\top y + ts \geq 0$, so that $K \subseteq K^\oplus$.
- (iii) Conversely, suppose (y, s) satisfies $x^\top y + ts \geq 0$ for all $(x, t) \in K$. By choosing $t = \|x\|_2$ and $x = -\frac{s}{\|y\|_2} y$ (when $y \neq 0$), show that $\|y\|_2 \leq s$, hence $(y, s) \in K$. Conclude that K is self-dual.
- (iv) Interpret the self-duality in the language of generalized inequalities: the constraint $(x, t) \in K$ and its multiplier both live in the same cone.

Exercise 6.10 (Minimum-Energy Control to a Terminal State). Reconsider the control setting of Section 6.4, but with the terminal *equality* constraint $Ku = d$ in place of the inequality, where $K : L^2[t_0, t_1] \rightarrow \mathbb{R}^n$ is the reachability operator $Ku = \int_{t_0}^{t_1} \Phi(t_1, t)b(t)u(t) dt$:

$$\min_{u \in L^2[t_0, t_1]} \frac{1}{2} \int_{t_0}^{t_1} u^2(t) dt \quad \text{s.t.} \quad Ku = d.$$

- (i) Form the Lagrangian $\mathcal{L}(u, \lambda) = \frac{1}{2} \int u^2 + \lambda^\top (d - Ku)$ with unrestricted $\lambda \in \mathbb{R}^n$, and show that the inner minimization gives $u^*(t) = \lambda^\top \Phi(t_1, t)b(t)$.
- (ii) Substituting back, show that the dual problem is the unconstrained concave quadratic $\max_{\lambda \in \mathbb{R}^n} \{\lambda^\top d - \frac{1}{2} \lambda^\top W \lambda\}$, where $W = \int_{t_0}^{t_1} \Phi(t_1, t)b(t)b^\top(t)\Phi^\top(t_1, t) dt$ is the controllability Gramian.

- (iii) Assuming $W \succ 0$, solve the dual to obtain $\lambda^* = W^{-1}d$ and the minimum-energy control $u^*(t) = d^T W^{-1} \Phi(t_1, t) b(t)$.
- (iv) Show that the optimal cost equals $\frac{1}{2} d^T W^{-1} d$, and explain why $W \succ 0$ (controllability) is exactly the regularity condition that makes the equality-constrained problem solvable.

Exercise 6.11 (Hilbert–Schmidt Functional and Riesz Representation). Let H be the Hilbert space of Lebesgue-measurable functions on $[0, \infty)$ with inner product $(x, y) = \int_0^\infty e^{-2t} x(t) y(t) dt$. Define a functional f on H by $f(x) = \int_0^\infty e^{-2t} dt \int_0^t K(t, s) x(s) ds$, where $K(t, s)$ is a square-integrable (Hilbert–Schmidt) kernel on $[0, \infty) \times [0, \infty)$, i.e. $\int_0^\infty \int_0^\infty |K(t, s)|^2 dt ds < \infty$.

- (i) Show that f is linear and continuous.
- (ii) Determine an element $y \in H$ such that $f(x) = (x, y)$. Is the solution unique?

Exercise 6.12 (A Functional on $L^4[0, 1]$). Define a functional f on $L^4[0, 1]$ by $f(x) = \int_0^1 dt \int_0^t K(t, s) x(s) ds$.

- (i) Obtain tight conditions on the kernel $K(\cdot, \cdot)$ such that f is linear and continuous.
- (ii) Assuming K satisfies the condition(s) in (i), determine $\|f\|$ in terms of K .
- (iii) Compute $\|f\|$ when $K(t, s) = t - s$.

Note: $L^4[0, 1]$ is not a Hilbert space.

Exercise 6.13 (Dependent Linear Functionals via Hahn–Banach). Let $g_1, \dots, g_n$ be linearly independent linear functionals on a vector space X , and let f be a linear functional on X such that $g_i(x) = 0$ for all $i = 1, \dots, n$ implies $f(x) = 0$. Show that there exist constants $\lambda_1, \dots, \lambda_n$ with $f = \sum_{i=1}^n \lambda_i g_i$. *Hint:* Use the Hahn–Banach theorem (extension form).

Exercise 6.14 (Aligned and Orthogonal Functionals on c_0). Let c_0 be the space of real sequences converging to zero, with the max norm, and let $x = \{\xi_i\} \in c_0$ with $\xi_1 = -1$ and $\xi_i = 2/i$ for $i \geq 2$. A functional f is *aligned* with x if $f(x) = \|f\| \|x\|$, and g is *orthogonal* to x if $g(x) = 0$.

- (i) Find a linear functional f on c_0 of unit norm aligned with x . Is it unique?
- (ii) Find $g \in c_0^*$ of unit norm orthogonal to x . Is it unique?

Exercise 6.15 (Aligned and Orthogonal Functionals on $C[0, 3]$). Let $x \in C[0, 3]$ be

$$x(t) = \begin{cases} 1 & 0 \leq t \leq 1/2, \\ 2 - 2t & 1/2 \leq t \leq 3/2, \\ 2t - 4 & 3/2 \leq t \leq 5/2, \\ 1 & 5/2 \leq t \leq 3. \end{cases}$$

With $X = C[0, 3]$, find $x^* \in X^*$ and $y^* \in X^*$, both of unit norm, such that x^* is aligned with x and y^* is orthogonal to x . Are the solutions unique?

Exercise 6.16 (A Constrained Minimum-Norm Problem). Find a continuous function x_o that minimizes $\int_0^2 |x(t)|^3 dt$ subject to $\int_0^2 t^2 x(t) dt = 2$. Identify the space(s) in which you work, and indicate whether the solution is unique.

Exercise 6.17 (Minimum-Time L^∞ Problem). Find a function x with $\|x\|_\infty \leq 1$ and $\int_0^T x(t) dt = -3$, $\int_0^T t x(t) dt = 0$, for the smallest possible $T > 0$. (Find the optimal T_o and a corresponding $x(t)$ on $[0, T_o]$.) *Hint:* First minimize $\|x\|_\infty$ subject to the integral constraints with T fixed (call it x_T), then find the smallest T with $\|x_T\|_\infty \leq 1$.

Exercise 6.18 (Two Coupled Minimum-Norm Constraints). Find x and y minimizing $\int_0^1 [x^2(t) + y^2(t)]^{1/2} dt$ subject to $\int_0^1 (1-t)x(t) dt = 1$ and $\int_0^1 (1-t)y(t) dt = \frac{3}{2}$. *Note:* Part of the problem is to identify the space in which to seek the solution.

Exercise 6.19 (Weak Convergence, Norm Convergence, and Cesàro Means). Let X be a Hilbert space and $\{x_n\}$ converge weakly to $x^o \in X$.

- (i) Show the convergence is strong (in norm) if in addition $\|x_n\| \rightarrow \|x^o\|$.
- (ii) Show one can extract a subsequence $\{x_{n_k}\}$ whose arithmetic means $y_m = \frac{1}{m} \sum_{k=1}^m x_{n_k}$ converge strongly to x^o .

Exercise 6.20 (Weakly and Weak* Continuous Functionals). Let X be a real normed linear space with dual X^* .

- (i) Show a linear functional f on X is weakly continuous iff $f(x) = \langle x, x^* \rangle$ for some $x^* \in X^*$.
- (ii) Show a linear functional g on X^* is weak* continuous iff $g(x^*) = \langle x, x^* \rangle$ for some $x \in X$.

Hint: Use Exercise 6.13.

Exercise 6.21 (Neumann Series). Let X be a Banach space and Q a bounded linear operator on X with $\|Q\| < 1$. Show that $(I - Q)$ is invertible and $(I - Q)^{-1} = \sum_{n=0}^{\infty} Q^n$ (the *Neumann series*). *Hint:* Use the Banach contraction mapping theorem.

Exercise 6.22 (Adjoint of an Integral Operator on L^p). Let $X = L^p[0, 2]$, $1 < p < \infty$, $Y = L^q[0, 2]$ with $\frac{1}{p} + \frac{1}{q} = 1$, and $A \in B(X, Y)$ defined by $A(x) = \int_0^2 K(t, s)x(s) ds$, where $\int_0^2 \int_0^2 |K(t, s)|^q dt ds < \infty$. Find A^* , the adjoint of A .

Exercise 6.23 (Adjoint of a Conditional-Expectation-Type Operator). Let x, y be scalar second-order random variables on $(\Omega, \mathcal{F}, \mathcal{P})$ with joint density $f_{x,y}(\xi, \eta)$ and marginals $f_1(\xi) = \int f_{x,y}(\xi, \eta) d\eta$, $f_2(\eta) = \int f_{x,y}(\xi, \eta) d\xi$. Let H_i be the space of measurable γ with $\|\gamma\|_i = \{\int |\gamma(s)|^2 f_i(s) ds\}^{1/2} < \infty$, $i = 1, 2$, and define $A : H_1 \rightarrow H_2$ by $A(\gamma)(\eta) = \int \gamma(\xi) [f_{x,y}(\xi, \eta)/f_2(\eta)] d\xi$.

- (i) Show that A is linear and bounded.
- (ii) Find the norm of A .
- (iii) Obtain the adjoint of A for the inner product $(\gamma, \beta)_i = \int \gamma(\xi)\beta(\xi)f_i(\xi) d\xi$, $i = 1, 2$.

Exercise 6.24 (Norms of A^*A and AA^*). Let H, G be real Hilbert spaces and $A : H \rightarrow G$ a bounded linear operator with adjoint A^* . Prove or disprove:

- (i) the operator norms of A^*A and AA^* coincide;
- (ii) the operator norm of A^*A equals $\|A\|^2$.

Exercise 6.25 (Pseudoinverse of a Matrix Operator). Find the pseudoinverse of the operator A on $\mathbb{R}^3$ with matrix $A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix}$.

Exercise 6.26 (Norm Optimization via Hölder's Inequality). (i) Obtain a continuous $x(\cdot)$ on $[-1, 1]$ maximizing $\int_{-1}^1 t^3 x(t) dt$ subject to $\int_{-1}^1 |x(t)|^3 dt \leq 2$.

(ii) Consider the same integral but to be *minimized*; what is the optimal solution?

Hint: Apply Hölder's inequality in a suitable normed space.

Exercise 6.27 (Minimization on the Unit Sphere of L^2). Find a function $x \in L^2[0, 3]$ of unit norm ($\int_0^3 |x(t)|^2 dt = 1$) that minimizes the functional $f(x) = \int_0^3 x(t) \cos \pi t dt$. Is the solution unique?

Exercise 6.28 (An Inner Product on Complex Matrices). Let $\mathbf{H}$ be the space of $m \times m$ complex matrices with the usual operations, and define $(A, B) = \text{Trace}(A'QB)$, where A' is the transpose of A , $\bar{B}$ the complex conjugate of B , and Q Hermitian positive definite. Prove that (A, B) is an inner product on $\mathbf{H}$.

Exercise 6.29 (Candidate Inner Products on $C[1, 4]$). Let $X = C[1, 4]$. For each of the following, determine whether it defines an inner product on X , justifying your answer:

(i) $(x, y) = \int_1^4 \int_1^4 s^2 x(s) t^2 y(t) ds dt$;

(ii) $(x, y) = \int_1^4 t^3 x(t) y(t) dt$.

Exercise 6.30 (Best Affine Approximation of t^n). Seek an affine function $m(t) = a + bt$ minimizing $F(m) = \int_{-1}^1 [t^n - m(t)]^2 dt$, $n > 1$ an integer.

(i) Formulate this as a projection problem in a Hilbert space H , identifying H and the subspace M .

(ii) Obtain the solution, parametrized by n . Is it unique?

(iii) What is the minimum value of F ?

Repeat with the weighted functional $F(m) = \int_{-1}^1 t^2 [t^n - m(t)]^2 dt$.

Exercise 6.31 (Minimum-Norm Problem with Two Moment Constraints). Obtain the minimum-norm solution of the problem of minimizing $\int_1^2 t^{-1} x^2(t) dt$ subject to $\int_1^2 x(t) t^{-2/3} dt = 1$ and $\int_1^2 x(t) t^{-1/3} dt = -1$. Is the solution unique?

Exercise 6.32 (Projection Theorem for a Quadratic Program). Using the projection theorem, solve $\min x^\top Qx$ subject to $Ax = b$, where $x \in \mathbb{R}^n$, Q is symmetric positive definite, $A \in \mathbb{R}^{m \times n}$ ($m < n$) has full row rank, and $b \in \mathbb{R}^m$. Obtain a closed-form expression in terms of A, Q, b .

Exercise 6.33 (Constrained Best Polynomial Approximation). Given $x \in L^2[a, b]$ on a finite interval, seek a polynomial p of degree $\leq n$ minimizing $\int_a^b |x(t) - p(t)|^2 dt$ subject to $\int_a^b p(t) dt = 0$.

(i) Show the problem admits a unique solution.

(ii) Show it can be solved by first finding the unconstrained minimizer q° of $\int_a^b |x(t) - q(t)|^2 dt$ (degree $\leq n$), and then projecting q° onto the constraint.

Exercise 6.34 (Minimum-Norm Vector under Inequality Constraints). Find the vector x of minimum norm satisfying $(x, y_i) \geq c_i$, $i = 1, \dots, n$, with y_i linearly independent and c_i given.

- (i) Show the problem admits a unique solution.
- (ii) Show that $x = \sum_i a_i y_i$ solves it iff the vector a satisfies $G^\top a \geq c$, $a \geq \theta$, and $a_i = 0$ whenever $(x, y_i) > c_i$, where G is the Gram matrix of $\{y_1, \dots, y_n\}$.

Exercise 6.35 (Tight Frames in a Hilbert Space). Let $\{\xi_j\}_{j \in J}$ be a family in a Hilbert space H . It is a *frame* if $A\|f\|^2 \leq \sum_j |\langle f, \xi_j \rangle|^2 \leq B\|f\|^2$ for all f , with $0 < A \leq B < \infty$; it is *tight* if $A = B$.

- (i) Show that for a tight frame, $f = A^{-1} \sum_j \langle f, \xi_j \rangle \xi_j$.
- (ii) In $H = \mathbb{C}^2$, show that $\xi_1 = (0, 1)^\top$, $\xi_2 = (-\frac{\sqrt{3}}{2}, -\frac{1}{2})^\top$, $\xi_3 = (\frac{\sqrt{3}}{2}, -\frac{1}{2})^\top$ form a tight frame; find the frame bound A .
- (iii) Show that if a tight frame has $A = 1$ and $\|\xi_j\| = 1$ for all j , then $\{\xi_j\}$ is an orthonormal basis.

Exercise 6.36 (Minimum-Norm Functional with Side Conditions). Let $X = C[-1, 1]$ with the maximum norm. Find a linear functional f on X of minimum norm satisfying $f(x_1) = 1$, $f(x_2) = 2$, where $x_1(t) = 2t^2 - t$ and $x_2(t) = -t^2 + t + 1$.

- (i) Is the problem feasible? Justify.
- (ii) If so, obtain the solution.
- (iii) Repeat (i)–(ii) with $X = L^2[-1, 1]$.
- (iv) Would (iii) change if X were $C[-1, 1]$ with the L^2 norm?
- (v) For the solution x^* of (iii), describe all $x \in L^2[-1, 1]$ orthogonal to x^* , i.e. $\langle x, x^* \rangle = 0$.

Exercise 6.37 (Fréchet Derivative and Minimization of a Functional). Let $J : L^2[0, 1] \rightarrow \mathbb{R}$, $J(x) = \int_0^1 [n(t) + x(t)] p(t; x) dt$, where $p(t; x) = m(t) + \int_0^t k(t, s) x(s) ds$, with $m, n \in L^2[0, 1]$ and $k \in L^2([0, 1] \times [0, 1])$.

- (i) Obtain the Fréchet derivative of J , identifying the space it belongs to and verifying it is indeed the Fréchet derivative.
- (ii) Repeat (i) with $L^2[0, 1]$ replaced by $C[0, 1]$ (maximum norm).
- (iii) Working in $L^2[0, 1]$, give conditions on n, m, k for J to admit a unique global minimum, and obtain (or describe an iterative scheme with justified convergence for) the solution.
- (iv) Apply the scheme in (iii) to $k(t, s) = 1 + s$, $n(t) \equiv 0$, $m(t) = \frac{5}{6}t^3 + t^2 - \frac{1}{2}t - \frac{1}{2}$.

Exercise 6.38 (Dual of a Constrained Minimum-Norm Problem and Sensitivity). Let H be a real Hilbert space and $p, q \in H$ with $\|p\| = \|q\| = 2$ and $(p, q) = -1$. Consider minimizing $J(x) = (p, x)$ over $x \in H$ subject to $(x, x) \leq \frac{17}{2}$ and $(q, x) \leq \frac{1}{2}$.

- (i) Formulate and solve the dual problem.
- (ii) What is the minimum value of the primal?
- (iii) Replace the second constraint by $(q, x) \leq c$ ($c > 0$), with primal optimal value $J_o(c)$; compute $\frac{dJ_o(c)}{dc}$ at $c = 0.5$ without an explicit $J_o(c)$, justifying the reasoning.

- (iv) In $H = L^2[-1, 1]$ with $p(t) = \sqrt{6}t$, $q(t) = -\frac{\sqrt{6}}{4}t + r(t)$ (choose r so that $\|q\| = 2$, $(p, q) = -1$), obtain $r(t)$ and an explicit primal solution x° , and verify $J(x^\circ)$ matches part (ii).

Exercise 6.39 (Nonattainment in Function Space). On the space $C[0, 1]$ with the supremum norm, consider

$$J(x) = \int_0^1 (2t - 1)x(t) dt \quad \text{over } D = \{x \in C[0, 1] : \|x\|_\infty \leq 1\}.$$

- (i) Compute $\inf_{x \in D} J(x)$ by choosing $x(t) = \pm 1$ to oppose the sign of the weight $2t - 1$ where advantageous.
- (ii) Show that the infimum is not attained by any continuous function.
- (iii) Explain which hypothesis of the Weierstrass theorem fails, and relate your finding to the failure of compactness of closed, bounded sets in infinite-dimensional normed spaces.

Part III

Algorithms

Chapter 7

Algorithms for Unconstrained Problems

This chapter develops a unified and rigorous treatment of modern optimization methods for unconstrained problems. Beginning with first-order methods, the chapter analyzes gradient descent under exact and perturbed gradients, emphasizing stability, descent properties, and convergence under deterministic and martingale noise. Stochastic gradient and Robbins–Monro algorithms are formulated within the general stochastic approximation framework, and their convergence is established using Lyapunov techniques and the ODE method.

The exposition then turns to second-order methods. Newton’s method is derived from local quadratic modeling, and its global and local convergence behavior is analyzed in detail. Under strong convexity and Lipschitz Hessian assumptions, the method exhibits a damped global descent phase followed by a local quadratic convergence regime. The Newton decrement is interpreted as a natural measure of optimality and as an accurate predictor of local objective decrease.

Throughout, the chapter emphasizes a dynamical-systems viewpoint: optimization algorithms are interpreted as discrete-time approximations of continuous flows. This perspective unifies deterministic descent, stochastic approximation, fixed-point iterations, and second-order methods within a common analytical framework based on Lyapunov stability, contraction principles, and martingale convergence theory.

7.1 ■ Numerical Algorithms for Unconstrained Optimization

We consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x), \tag{7.1}$$

where the objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is assumed to be continuously differentiable.

Unconstrained smooth optimization constitutes one of the central paradigms of optimization theory and arises throughout engineering, control, economics, and machine learning. Except in special cases such as quadratic objectives or separable structures, problem (7.1) does not admit a closed-form solution. Consequently, one relies on numerical algorithms that iteratively generate approximations to minimizers.

7.1.1 ■ Iterative Descent Framework

A cornerstone of smooth optimization is the first-order necessary condition for local optimality: if x^* is a local minimizer of f , then $\nabla f(x^*) = 0$.

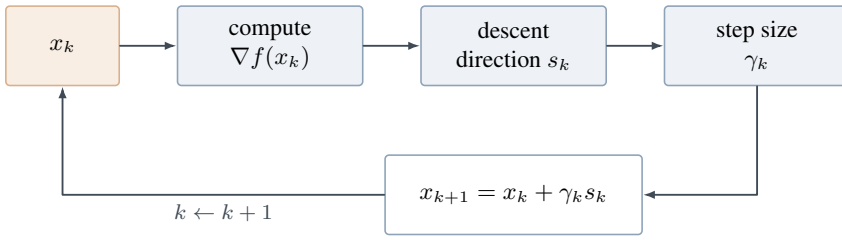


Figure 7.1: A descent algorithm alternates between local information, a search direction, a step-size rule, and an iterate update.

Definition 7.1 (Stationary point). A point $x^* \in \mathbb{R}^n$ is a stationary point of f if $\nabla f(x^*) = 0$.

In the absence of convexity, a stationary point may be a local minimum, a local maximum, or a saddle point, so stationarity is necessary but not sufficient for optimality. When $f \in C^2(\mathbb{R}^n)$, second-order information refines the classification: $\nabla^2 f(x^*) \succ 0$ makes x^* a strict local minimizer, $\nabla^2 f(x^*) \prec 0$ a strict local maximizer, and indefinite $\nabla^2 f(x^*)$ a saddle point. Nevertheless, most first-order numerical algorithms are constructed so that any limit point of the generated sequence, when it exists, satisfies the stationarity condition.

Since problem (7.1) rarely admits a closed-form solution, numerical methods generate a sequence $\{x_k\}_{k \geq 0} \subset \mathbb{R}^n$ intended to approach a stationary point of f . A broad and unifying representation of such methods is the iteration

$$x_{k+1} = x_k + \gamma_k s_k, \quad k = 0, 1, 2, \dots, \quad (7.2)$$

where $s_k \in \mathbb{R}^n$ is a search direction and $\gamma_k > 0$ is a step size. The scheme (7.2) is the generic form of an *iterative descent method*.

This formulation separates the algorithmic design into two fundamental components: the choice of direction s_k , which determines how local information about the objective is used, and the step-size rule γ_k , which controls stability and progress along that direction. The guiding principle of descent methods is that each iteration produces a sufficient decrease in the objective value, thereby driving the sequence toward a stationary point.

7.1.2 ▀ Descent Directions

The essential structural requirement for a descent-based method is that the search direction yields a local decrease of the objective function.

Definition 7.2 (Descent direction). A vector $s \in \mathbb{R}^n$ is called a descent direction for f at a point x if $\nabla f(x)^\top s < 0$.

This condition admits a clear geometric interpretation. The gradient $\nabla f(x)$ identifies the direction of steepest local increase of f under the Euclidean norm. The directional derivative of f at x along s is $Df(x)[s] = \nabla f(x)^\top s$. Hence, the inequality $\nabla f(x)^\top s < 0$ means that f decreases locally in the direction s .

The descent property follows directly from the first-order Taylor expansion:

$$f(x + \gamma s) = f(x) + \gamma \nabla f(x)^\top s + o(\gamma), \quad \gamma \rightarrow 0.$$

If $\nabla f(x)^\top s < 0$, then there exists $\bar{\gamma} > 0$ such that $f(x + \gamma s) < f(x)$, $\forall \gamma \in (0, \bar{\gamma})$. Thus, descent directions guarantee strict local decrease for sufficiently small step sizes. If $\nabla f(x) = 0$, then

$\nabla f(x)^\top s = 0$ for all $s \in \mathbb{R}^n$, so no descent direction exists. This explains why first-order methods naturally terminate at stationary points.

7.1.3 ■ Choice of Descent Direction

Different numerical methods arise from different choices of the search direction s_k , reflecting how much local information about the objective function is exploited.

Steepest Descent Method

The steepest descent (or gradient) method selects

$$s_k = -\nabla f(x_k). \quad (7.3)$$

The gradient points in the direction of steepest local increase of the objective; therefore its negative corresponds to the direction of steepest local decrease under the Euclidean norm.

Among all unit-norm directions d , the direction minimizing the directional derivative $\nabla f(x_k)^\top d$ is $d = -\frac{\nabla f(x_k)}{\|\nabla f(x_k)\|}$. Thus, the steepest descent direction is locally optimal in a first-order sense. Despite its simplicity, the method may converge slowly when the level sets of f are highly elongated, i.e., when the problem is ill-conditioned.

Newton's Method

Newton's method incorporates second-order information through the Hessian matrix. The search direction is defined as

$$s_k = -(\nabla^2 f(x_k))^{-1} \nabla f(x_k), \quad (7.4)$$

assuming that the Hessian $\nabla^2 f(x_k)$ is symmetric and positive definite.

This direction is the unique minimizer of the second-order Taylor model of f at x_k ; the derivation, together with the damping needed to make the method reliable away from a minimizer, is deferred to Section 7.5.

When f is twice continuously differentiable and strongly convex in a neighborhood of a minimizer, Newton's method exhibits quadratic local convergence. However, if the Hessian is indefinite or poorly conditioned, the method may fail or require damping or line-search strategies to ensure global convergence.

Quasi-Newton Methods

Quasi-Newton methods approximate the Newton direction without explicitly computing or inverting the Hessian. The search direction takes the form

$$s_k = -D_k \nabla f(x_k), \quad (7.5)$$

where D_k is a symmetric positive definite matrix intended to approximate $(\nabla^2 f(x_k))^{-1}$.

The matrix D_k is updated iteratively using gradient differences so as to satisfy secant conditions that capture curvature information. Well-known quasi-Newton schemes include BFGS and limited-memory BFGS (L-BFGS). Under suitable assumptions, these methods achieve superlinear local convergence while maintaining moderate computational cost.

The secant condition requires that the updated inverse-Hessian approximation D_{k+1} reproduce the observed curvature along the last step, i.e. $D_{k+1}q_k = p_k$, where $p_k = x_{k+1} - x_k$ is the step actually taken and $q_k = \nabla f(x_{k+1}) - \nabla f(x_k)$ the corresponding gradient change. (The step $p_k = \gamma_k s_k$ is a positive multiple of the search direction s_k of (7.5); the two are kept distinct

here because the secant condition refers to the step, not the direction.) The BFGS update is the rank-two correction that satisfies this condition while preserving symmetry and positive definiteness whenever the curvature condition $p_k^\top q_k > 0$ holds; the latter is guaranteed by a line search satisfying the Wolfe conditions.

ALGORITHM 7.1. BFGS Quasi-Newton Method.

Require: initial point x_0 , initial inverse-Hessian estimate $D_0 \succ 0$ (e.g. $D_0 = I$), tolerance $\varepsilon > 0$.

- 1: **for** $k = 0, 1, 2, \dots$ **do**
- 2: compute the gradient $g_k = \nabla f(x_k)$;
- 3: **if** $\|g_k\| \leq \varepsilon$ **then**
- 4: **stop** and return x_k ;
- 5: **end if**
- 6: compute the search direction $s_k = -D_k g_k$;
- 7: choose a step size $\gamma_k > 0$ satisfying the Wolfe conditions by line search;
- 8: update the iterate $x_{k+1} = x_k + \gamma_k s_k$;
- 9: set $p_k = x_{k+1} - x_k$ and $q_k = \nabla f(x_{k+1}) - g_k$;
- 10: compute $\rho_k = 1/(q_k^\top p_k)$;
- 11: update the inverse-Hessian approximation

$$D_{k+1} = (I - \rho_k p_k q_k^\top) D_k (I - \rho_k q_k p_k^\top) + \rho_k p_k p_k^\top;$$

12: **end for**

7.1.4 ■ Step Size Rules

The scalar $\gamma_k > 0$ in the update $x_{k+1} = x_k + \gamma_k s_k$ is called the *step size* (or learning rate). Its selection plays a decisive role in determining both convergence and efficiency.

Constant Step Size. A constant step size $\gamma_k = \gamma$ is simple to implement. If γ is too large, the iterates may oscillate or diverge; if too small, convergence may be excessively slow. Constant step sizes are therefore most effective when the problem is well conditioned or when local convergence is of primary interest.

Exact and Bounded Line Search. An alternative is to select γ_k by minimizing the objective along the search direction, $\gamma_k = \arg \min_{\gamma \geq 0} f(x_k + \gamma s_k)$. This is known as an exact line search. In practice, one often restricts the search to a bounded interval $\gamma \in [0, \bar{\gamma}]$, leading to bounded line-search methods. Although effective for classical smooth problems, such procedures may be computationally burdensome in large-scale settings.

Diminishing Step Size. A third important class of rules consists of diminishing step sizes satisfying

$$\gamma_k \rightarrow 0, \quad \sum_{k=0}^{\infty} \gamma_k = \infty. \quad (7.6)$$

The first condition ensures increasingly conservative steps, while the second prevents premature stagnation.

To see why these conditions exclude convergence to nonstationary points, suppose that $x_k \rightarrow \bar{x}$ with $\nabla f(\bar{x}) \neq 0$. For sufficiently large indices $m > n$, we have $x_m \approx x_n \approx \bar{x}$, and therefore

$$x_m \approx x_n - \left(\sum_{k=n}^{m-1} \gamma_k \right) \nabla f(\bar{x}).$$

Since $\sum_{k=n}^{m-1} \gamma_k$ can be made arbitrarily large by (7.6), this contradicts convergence. Hence, any accumulation point must be stationary.

7.1.5 ■ Algorithmic Summary

The direction and step-size choices above combine into the generic descent template of Algorithm 7.2, which underlies steepest descent, Newton, and quasi-Newton methods alike: only the rule for s_k changes. Algorithm 7.3 instantiates the template for steepest descent with an Armijo backtracking line search, the most common practical variant.

ALGORITHM 7.2. Generic Iterative Descent Method.

Require: initial point $x_0 \in \mathbb{R}^n$, tolerance $\varepsilon > 0$.

- 1: **for** $k = 0, 1, 2, \dots$ **do**
- 2: compute the gradient $\nabla f(x_k)$;
- 3: **if** $\|\nabla f(x_k)\| \leq \varepsilon$ **then**
- 4: **stop** and return x_k ;
- 5: **end if**
- 6: choose a descent direction s_k with $\nabla f(x_k)^\top s_k < 0$ ▷ e.g. $-\nabla f(x_k)$, Newton, or quasi-Newton
- 7: choose a step size $\gamma_k > 0$ ▷ constant, line search, or diminishing
- 8: update $x_{k+1} = x_k + \gamma_k s_k$;
- 9: **end for**

ALGORITHM 7.3. Gradient Descent with Backtracking (Armijo) Line Search.

Require: initial point x_0 , tolerance $\varepsilon > 0$, parameters $\sigma \in (0, 1)$, $\beta \in (0, 1)$, initial trial step $\bar{\gamma} > 0$.

- 1: **for** $k = 0, 1, 2, \dots$ **do**
- 2: compute $g_k = \nabla f(x_k)$;
- 3: **if** $\|g_k\| \leq \varepsilon$ **then**
- 4: **stop** and return x_k ;
- 5: **end if**
- 6: set the direction $s_k = -g_k$ and the trial step $\gamma_k = \bar{\gamma}$;
- 7: **while** $f(x_k + \gamma_k s_k) > f(x_k) - \sigma \gamma_k \|g_k\|^2$ **do** ▷ Armijo sufficient-decrease test

```

8:       $\gamma_k \leftarrow \beta \gamma_k$ ;
9:  end while
10:  update  $x_{k+1} = x_k + \gamma_k s_k$ ;
11: end for

```

7.2 • Convergence Results

We now investigate the convergence properties of the general descent iteration (7.2).

A first-order Taylor expansion of f around x_k yields

$$f(x_{k+1}) = f(x_k + \gamma_k s_k) = f(x_k) + \gamma_k \nabla f(x_k)^\top s_k + o(\gamma_k), \quad \gamma_k \rightarrow 0,$$

where $o(\gamma_k)$ denotes higher-order terms satisfying $o(\gamma_k)/\gamma_k \rightarrow 0$.

This expansion makes explicit the local mechanism governing descent: the term $\nabla f(x_k)^\top s_k$ determines whether the objective decreases to first order, while the higher-order terms must be controlled through regularity assumptions.

Fundamental Question. Under what conditions does every limit point of the sequence generated by (7.2) correspond to a stationary point of f ?

The Taylor expansion alone does not guarantee convergence to stationarity. A potential failure mode arises if $\nabla f(x_k)^\top s_k \rightarrow 0$ while $\nabla f(x_k) \not\rightarrow 0$, that is, if the search direction becomes asymptotically orthogonal to the gradient. In such a situation, the algorithm may cease to make progress even though the gradient remains nonzero, leading to stagnation at a nonstationary point.

To exclude this pathological behavior, one imposes quantitative conditions ensuring that the search direction maintains a uniformly favorable angle with respect to the negative gradient and does not grow excessively in magnitude.

Assumption 7.3 (Uniform descent conditions). *There exist constants $C_1 > 0$ and $C_2 > 0$ such that, for all k ,*

$$C_1 \|\nabla f(x_k)\|^2 \leq -\nabla f(x_k)^\top s_k, \quad (7.7)$$

$$\|s_k\| \leq C_2 \|\nabla f(x_k)\|. \quad (7.8)$$

Condition (7.7) ensures a uniform amount of descent proportional to the squared gradient norm. It rules out directions that become nearly orthogonal to the gradient and guarantees that the decrease in the objective is quantitatively significant whenever the gradient is nonzero.

Condition (7.8) controls the size of the search direction relative to the gradient. This prevents the step $\gamma_k s_k$ from being dominated by excessively large directions that could invalidate the local Taylor approximation or destabilize the iteration. These conditions ensure that the descent mechanism remains effective and well scaled throughout the iterative process.

For the steepest descent method, where $s_k = -\nabla f(x_k)$, both inequalities in Assumption 7.3 hold trivially with $C_1 = C_2 = 1$. Under suitable regularity assumptions, such as positive definiteness and boundedness of the Hessian, Newton and quasi-Newton directions also satisfy these conditions, thereby fitting into the same convergence framework.

7.2.1 ■ Convergence for a Constant Step Size

We analyze the convergence of the iteration

$$x_{k+1} = x_k + \gamma s_k, \quad (7.9)$$

with a constant step size $\gamma > 0$.

Assumption 7.4 (Lipschitz continuity of the gradient). *There exists a constant $L > 0$ such that*

$$\|\nabla f(x) - \nabla f(\bar{x})\| \leq L\|x - \bar{x}\|, \quad \forall x, \bar{x} \in \mathbb{R}^n.$$

Lemma 7.5 (Quadratic upper bound under Lipschitz gradient). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable and suppose its gradient is L -Lipschitz, i.e.,*

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

Then, for all $x, z \in \mathbb{R}^n$,

$$f(x + z) \leq f(x) + z^\top \nabla f(x) + \frac{L}{2} \|z\|^2. \quad (7.10)$$

Proof. Fix $x, z \in \mathbb{R}^n$ and define the scalar function $g(\xi) := f(x + \xi z)$, $\xi \in [0, 1]$. Since f is continuously differentiable, the chain rule gives $g'(\xi) = \nabla f(x + \xi z)^\top z$.

By the fundamental theorem of calculus,

$$f(x + z) - f(x) = g(1) - g(0) = \int_0^1 g'(\xi) d\xi = \int_0^1 z^\top \nabla f(x + \xi z) d\xi. \quad (7.11)$$

We now add and subtract $\nabla f(x)$ inside the integrand:

$$\begin{aligned} f(x + z) - f(x) &= \int_0^1 z^\top \nabla f(x) d\xi + \int_0^1 z^\top (\nabla f(x + \xi z) - \nabla f(x)) d\xi \\ &= z^\top \nabla f(x) + \int_0^1 z^\top (\nabla f(x + \xi z) - \nabla f(x)) d\xi. \end{aligned} \quad (7.12)$$

To estimate the second term, apply the Cauchy–Schwarz inequality:

$$z^\top (\nabla f(x + \xi z) - \nabla f(x)) \leq \|z\| \|\nabla f(x + \xi z) - \nabla f(x)\|.$$

By the Lipschitz continuity of the gradient,

$$\|\nabla f(x + \xi z) - \nabla f(x)\| \leq L\|x + \xi z - x\| = L\xi\|z\|.$$

Therefore,

$$z^\top (\nabla f(x + \xi z) - \nabla f(x)) \leq L\xi\|z\|^2.$$

Substituting this estimate into (7.12) gives

$$\begin{aligned} f(x + z) - f(x) &\leq z^\top \nabla f(x) + \int_0^1 L\xi\|z\|^2 d\xi \\ &= z^\top \nabla f(x) + L\|z\|^2 \int_0^1 \xi d\xi. \end{aligned}$$

Since $\int_0^1 \xi \, d\xi = \frac{1}{2}$, we obtain

$$f(x+z) - f(x) \leq z^\top \nabla f(x) + \frac{L}{2} \|z\|^2,$$

which establishes (7.10). $\square$

Proposition 7.6 (Convergence with constant step size). *Let $\{x_k\}$ be generated by (7.9) and suppose Assumptions 7.4 and 7.3 hold. If*

$$0 < \gamma < \frac{2C_1}{LC_2^2}, \quad (7.13)$$

then either $f(x_k) \rightarrow -\infty$, or $f(x_k)$ converges to a finite value and $\lim_{k \rightarrow \infty} \nabla f(x_k) = 0$. Moreover, every limit point of $\{x_k\}$ is a stationary point of f .

Proof. Applying Lemma 7.5 with $x = x_k$ and $z = \gamma s_k$, we obtain

$$f(x_{k+1}) - f(x_k) \leq \gamma \nabla f(x_k)^\top s_k + \frac{L}{2} \gamma^2 \|s_k\|^2. \quad (7.14)$$

Using the descent conditions in Assumption 7.3,

$$-\nabla f(x_k)^\top s_k \geq C_1 \|\nabla f(x_k)\|^2, \quad \|s_k\|^2 \leq C_2^2 \|\nabla f(x_k)\|^2.$$

Substituting these bounds into (7.14) yields

$$\begin{aligned} f(x_{k+1}) - f(x_k) &\leq -\gamma C_1 \|\nabla f(x_k)\|^2 + \frac{L}{2} \gamma^2 C_2^2 \|\nabla f(x_k)\|^2 \\ &= -\gamma \left(C_1 - \frac{L}{2} \gamma C_2^2 \right) \|\nabla f(x_k)\|^2. \end{aligned} \quad (7.15)$$

Equivalently,

$$f(x_k) - f(x_{k+1}) \geq \gamma \left(C_1 - \frac{L}{2} \gamma C_2^2 \right) \|\nabla f(x_k)\|^2. \quad (7.16)$$

Condition (7.13) guarantees that

$$\alpha := \gamma \left(C_1 - \frac{L}{2} \gamma C_2^2 \right) > 0.$$

Hence

$$f(x_{k+1}) \leq f(x_k) - \alpha \|\nabla f(x_k)\|^2, \quad (7.17)$$

which shows that $\{f(x_k)\}$ is nonincreasing.

If $f(x_k) \rightarrow -\infty$, the conclusion holds. Suppose instead that $f(x_k)$ is bounded below. Then, since it is monotone, $f(x_k)$ converges to a finite limit.

Summing (7.16) from $k = 0$ to $T - 1$ gives

$$\sum_{k=0}^{T-1} \alpha \|\nabla f(x_k)\|^2 \leq f(x_0) - f(x_T).$$

Letting $T \rightarrow \infty$ and using boundedness of $f(x_T)$, we obtain

$$\sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 < \infty.$$

Therefore,

$$\lim_{k \rightarrow \infty} \|\nabla f(x_k)\| = 0.$$

Finally, let $x_{k_j} \rightarrow \bar{x}$ be a convergent subsequence. Since $\nabla f(x_{k_j}) \rightarrow 0$ and ∇f is continuous,

$$\nabla f(\bar{x}) = \lim_{j \rightarrow \infty} \nabla f(x_{k_j}) = 0.$$

Hence every limit point $\bar{x}$ is stationary, completing the proof. $\square$

7.2.2 ■ Convergence for a Diminishing Step Size

We now return to the iteration (7.2) with the step size allowed to vary, and ask what replaces the fixed admissible range (7.13) when γ_k is driven to zero.

Proposition 7.7 (Convergence with diminishing step size). *Suppose Assumptions 7.4 and 7.3 hold, and that*

$$\gamma_k \rightarrow 0, \quad \sum_{k=0}^{\infty} \gamma_k = \infty. \quad (7.18)$$

Then either $f(x_k) \rightarrow -\infty$, or $f(x_k)$ converges to a finite value and $\lim_{k \rightarrow \infty} \nabla f(x_k) = 0$. Every limit point of $\{x_k\}$ is a stationary point of f .

Proof. Applying the quadratic upper bound (7.10) with $x = x_k$ and $z = \gamma_k s_k$ yields

$$f(x_{k+1}) - f(x_k) \leq \gamma_k \nabla f(x_k)^\top s_k + \frac{L}{2} \gamma_k^2 \|s_k\|^2.$$

Using the descent conditions

$$-\nabla f(x_k)^\top s_k \geq C_1 \|\nabla f(x_k)\|^2, \quad \|s_k\|^2 \leq C_2^2 \|\nabla f(x_k)\|^2,$$

we obtain

$$\begin{aligned} f(x_{k+1}) - f(x_k) &\leq -\gamma_k C_1 \|\nabla f(x_k)\|^2 + \frac{L}{2} \gamma_k^2 C_2^2 \|\nabla f(x_k)\|^2 \\ &= -\gamma_k \left(C_1 - \frac{L}{2} \gamma_k C_2^2 \right) \|\nabla f(x_k)\|^2. \end{aligned} \quad (7.19)$$

Since $\gamma_k \rightarrow 0$, there exists $\bar{k}$ such that for all $k \geq \bar{k}$, $\gamma_k \leq \frac{C_1}{LC_2^2}$. Hence, for $k \geq \bar{k}$,

$$C_1 - \frac{L}{2} \gamma_k C_2^2 \geq \frac{C_1}{2}.$$

Substituting into (7.19) gives

$$f(x_{k+1}) \leq f(x_k) - \frac{C_1}{2} \gamma_k \|\nabla f(x_k)\|^2, \quad k \geq \bar{k}. \quad (7.20)$$

Thus, for sufficiently large k , the objective decreases by an amount proportional to $\gamma_k \|\nabla f(x_k)\|^2$. In particular, $\{f(x_k)\}$ is eventually nonincreasing. If $f(x_k) \rightarrow -\infty$, the conclusion holds. Assume instead that $f(x_k)$ is bounded below. Since it is eventually monotone decreasing, it converges to a finite limit.

Summing (7.20) from $k = \bar{k}$ to $T - 1$ yields

$$\frac{C_1}{2} \sum_{k=\bar{k}}^{T-1} \gamma_k \|\nabla f(x_k)\|^2 \leq f(x_{\bar{k}}) - f(x_T).$$

Letting $T \rightarrow \infty$ and using boundedness of $f(x_T)$, we obtain

$$\sum_{k=\bar{k}}^{\infty} \gamma_k \|\nabla f(x_k)\|^2 < \infty.$$

We now upgrade this summability to $\|\nabla f(x_k)\| \rightarrow 0$ in two steps. Note that one cannot argue by simply extracting a subsequence along which the gradient stays bounded away from zero: divergence of $\sum_k \gamma_k$ says nothing about the sum of γ_k along an arbitrary subsequence. (For $\gamma_k = 1/k$ the full series diverges, while $\sum_j \gamma_{2^j}$ converges.) The Lipschitz continuity of ∇f is what supplies the missing control.

Step 1: $\liminf_k \|\nabla f(x_k)\| = 0$. If not, there are $\varepsilon > 0$ and $K \geq \bar{k}$ with $\|\nabla f(x_k)\| \geq \varepsilon$ for every $k \geq K$. Then $\sum_{k \geq K} \gamma_k \|\nabla f(x_k)\|^2 \geq \varepsilon^2 \sum_{k \geq K} \gamma_k = \infty$, contradicting the summability just established. Here the lower bound holds for all large k , not merely along a subsequence, so divergence of $\sum_k \gamma_k$ may legitimately be invoked.

Step 2: the $\limsup$ equals the $\liminf$. Suppose $\limsup_k \|\nabla f(x_k)\| > 0$ and fix $\varepsilon > 0$ with $2\varepsilon < \limsup_k \|\nabla f(x_k)\|$. By Step 1 the sequence $\|\nabla f(x_k)\|$ drops below ε infinitely often and rises above 2ε infinitely often, so we may select infinitely many disjoint index blocks $n_i < m_i$ such that

$$\|\nabla f(x_{n_i})\| < \varepsilon, \quad \|\nabla f(x_{m_i})\| > 2\varepsilon, \quad \varepsilon \leq \|\nabla f(x_k)\| \leq 2\varepsilon \text{ for } n_i < k < m_i,$$

by taking m_i to be the first crossing above 2ε after n_i and n_i the last crossing below ε before it.

Across such a block, the triangle inequality, Assumption 7.4 and (7.8) give

$$\varepsilon < \|\nabla f(x_{m_i})\| - \|\nabla f(x_{n_i})\| \leq \|\nabla f(x_{m_i}) - \nabla f(x_{n_i})\| \leq L \sum_{k=n_i}^{m_i-1} \gamma_k \|s_k\| \leq LC_2 \sum_{k=n_i}^{m_i-1} \gamma_k \|\nabla f(x_k)\|.$$

Since $\gamma_k \rightarrow 0$, for all large i the gradient norm is at most 3ε on the whole block, whence $\sum_{k=n_i}^{m_i-1} \gamma_k \geq 1/(3LC_2)$. Because $\|\nabla f(x_k)\| \geq \varepsilon$ throughout the block,

$$\sum_{k=n_i}^{m_i-1} \gamma_k \|\nabla f(x_k)\|^2 \geq \frac{\varepsilon^2}{3LC_2} > 0,$$

a fixed positive amount contributed by each of infinitely many disjoint blocks. This contradicts $\sum_{k \geq \bar{k}} \gamma_k \|\nabla f(x_k)\|^2 < \infty$. Hence $\limsup_k \|\nabla f(x_k)\| = 0$, that is, $\lim_{k \rightarrow \infty} \|\nabla f(x_k)\| = 0$.

Finally, let $x_{k_j} \rightarrow \bar{x}$ along a convergent subsequence. By continuity of ∇f ,

$$\nabla f(\bar{x}) = \lim_{k \rightarrow \infty} \nabla f(x_{k_j}) = 0,$$

so every limit point is stationary. $\square$

7.2.3 ■ Gradient Descent and Stationary Points

Propositions 7.6 and 7.7 were stated for a general search direction s_k satisfying the uniform descent conditions. We now specialize them to steepest descent, $s_k = -\nabla f(x_k)$, and record the

resulting statements in the form used throughout the rest of the chapter. Nothing here assumes convexity: the conclusions are exactly those available for a smooth objective with no further structure, and they will serve as the common starting point for the convex analysis of Section 7.2.4 and for the nonconvex analysis of Section 7.3.

Throughout, f is the objective of problem (7.1) and x^* denotes a stationary point in the sense of Definition 7.1. We consider the gradient descent recursion

$$x_{k+1} = x_k - \gamma \nabla f(x_k), \quad k = 0, 1, 2, \dots, \quad (7.21)$$

with constant step size $\gamma > 0$.

Steepest descent satisfies the uniform descent conditions of Assumption 7.3 with $C_1 = C_2 = 1$, as noted when that assumption was introduced. The admissible range $0 < \gamma < 2C_1/(LC_2^2)$ of (7.13) therefore collapses to the single condition recorded next, and the per-step decrease constant $\alpha = \gamma(C_1 - \frac{L}{2}\gamma C_2^2)$ becomes $\alpha = \gamma(1 - L\gamma/2)$.

Assumption 7.8 (Step size). *The step size satisfies $0 < \gamma < \frac{2}{L}$.*

The next two results are Lemma 7.5 and Proposition 7.6 written out in this special case. We restate them because these particular constants recur in every later argument.

Lemma 7.9 (Descent inequality). *Under Assumption 7.4, the iterates generated by (7.21) satisfy*

$$f(x_{k+1}) \leq f(x_k) - \alpha \|\nabla f(x_k)\|^2, \quad (7.22)$$

where $\alpha := \gamma(1 - \frac{L\gamma}{2}) > 0$.

Proof. By the descent lemma,

$$f(x_{k+1}) \leq f(x_k) + \langle \nabla f(x_k), x_{k+1} - x_k \rangle + \frac{L}{2} \|x_{k+1} - x_k\|^2.$$

Using the update $x_{k+1} - x_k = -\gamma \nabla f(x_k)$, we obtain $\langle \nabla f(x_k), x_{k+1} - x_k \rangle = -\gamma \|\nabla f(x_k)\|^2$ and $\|x_{k+1} - x_k\|^2 = \gamma^2 \|\nabla f(x_k)\|^2$. Substituting gives

$$f(x_{k+1}) \leq f(x_k) - \gamma \|\nabla f(x_k)\|^2 + \frac{L}{2} \gamma^2 \|\nabla f(x_k)\|^2,$$

which yields (7.22). $\square$

Proposition 7.10 (Vanishing gradient and stationarity of limit points). *Suppose that f is bounded below, i.e., there exists $f_{\inf} \in \mathbb{R}$ such that $f(x) \geq f_{\inf}$, $\forall x \in \mathbb{R}^n$. Then the sequence $\{x_k\}$ generated by (7.21) satisfies*

$$\sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 < \infty, \quad (7.23)$$

$$\|\nabla f(x_k)\| \rightarrow 0. \quad (7.24)$$

Moreover, every accumulation point of $\{x_k\}$ is a stationary point.

Proof. Summing the descent inequality (7.22) from $k = 0$ to $T - 1$ yields

$$\sum_{k=0}^{T-1} (f(x_{k+1}) - f(x_k)) \leq -\alpha \sum_{k=0}^{T-1} \|\nabla f(x_k)\|^2.$$

The left-hand side telescopes:

$$\sum_{k=0}^{T-1} (f(x_{k+1}) - f(x_k)) = f(x_T) - f(x_0).$$

Hence,

$$f(x_T) - f(x_0) \leq -\alpha \sum_{k=0}^{T-1} \|\nabla f(x_k)\|^2. \quad (7.25)$$

Rearranging gives

$$\alpha \sum_{k=0}^{T-1} \|\nabla f(x_k)\|^2 \leq f(x_0) - f(x_T). \quad (7.26)$$

Since $f(x_T) \geq f_{\inf}$, we obtain

$$\alpha \sum_{k=0}^{T-1} \|\nabla f(x_k)\|^2 \leq f(x_0) - f_{\inf}.$$

Letting $T \rightarrow \infty$ yields

$$\sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 \leq \frac{f(x_0) - f_{\inf}}{\alpha} < \infty,$$

which proves (7.23).

We now show that $\|\nabla f(x_k)\| \rightarrow 0$. Suppose, by contradiction, that this is not the case. Then there exists $\varepsilon > 0$ and an infinite subsequence $\{k_j\}$ such that $\|\nabla f(x_{k_j})\| \geq \varepsilon$. Therefore,

$$\sum_{j=0}^{\infty} \|\nabla f(x_{k_j})\|^2 \geq \sum_{j=0}^{\infty} \varepsilon^2 = \infty,$$

which contradicts the summability established above. Hence, $\|\nabla f(x_k)\| \rightarrow 0$. Finally, let $x_{k_j} \rightarrow \bar{x}$ be any convergent subsequence. By continuity of ∇f , $\nabla f(\bar{x}) = \lim_{j \rightarrow \infty} \nabla f(x_{k_j}) = 0$, so $\bar{x}$ is a stationary point. $\square$

7.2.4 ■ Strong Convexity and Gradient Descent

Section 7.2.3 extracted everything that smoothness alone yields: the objective decreases monotonically, the gradients are square summable, and every accumulation point of the iterates is stationary. What it does not yield is convergence of the sequence itself, nor any rate. Two different kinds of additional structure repair this. The first, treated here, is convexity, which forces the stationary point to be unique and the convergence to be geometric. The second, treated in Section 7.3, dispenses with convexity altogether and replaces it by a geometric regularity condition on the objective near its critical points.

We now state and prove a precise convergence theorem for gradient descent under smoothness and strong convexity assumptions.

Definition 7.11 (Strong convexity and smoothness). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable.

(1) The function f is strongly convex with parameter $m > 0$ if

$$\nabla^2 f(x) \succeq mI, \quad \forall x \in \mathbb{R}^n. \quad (7.27)$$

(2) The function f is L -smooth if its gradient is L -Lipschitz continuous, that is,

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^n. \quad (7.28)$$

Strong convexity is equivalent to the quadratic lower bound

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{m}{2} \|y - x\|^2, \quad \forall x, y \in \mathbb{R}^n. \quad (7.29)$$

Under assumptions (7.27)–(7.28), the function f admits a unique minimizer x^* , characterized by $\nabla f(x^*) = 0$.

Theorem 7.12 (Linear convergence of gradient descent). *Let f be m -strongly convex and L -smooth. Consider the gradient descent iteration*

$$x_{k+1} = x_k - \gamma \nabla f(x_k), \quad (7.30)$$

with constant step size $\gamma > 0$. If

$$0 < \gamma < \frac{2m}{L^2}, \quad (7.31)$$

then the iterates converge linearly to the unique minimizer x^* . More precisely,

$$\|x_k - x^*\| \leq \rho^k \|x_0 - x^*\|, \quad (7.32)$$

where $\rho = \sqrt{1 - 2m\gamma + \gamma^2 L^2} \in (0, 1)$.

Proof. Subtracting x^* from both sides of (7.30) gives

$$x_{k+1} - x^* = x_k - x^* - \gamma(\nabla f(x_k) - \nabla f(x^*)).$$

Taking squared norms yields

$$\begin{aligned} \|x_{k+1} - x^*\|^2 &= \|x_k - x^*\|^2 + \gamma^2 \|\nabla f(x_k) - \nabla f(x^*)\|^2 \\ &\quad - 2\gamma(x_k - x^*)^\top (\nabla f(x_k) - \nabla f(x^*)). \end{aligned} \quad (7.33)$$

By L -smoothness,

$$\|\nabla f(x_k) - \nabla f(x^*)\| \leq L\|x_k - x^*\|.$$

By strong convexity,

$$(\nabla f(x_k) - \nabla f(x^*))^\top (x_k - x^*) \geq m\|x_k - x^*\|^2.$$

Substituting these bounds into (7.33) yields

$$\|x_{k+1} - x^*\|^2 \leq (1 - 2m\gamma + \gamma^2 L^2) \|x_k - x^*\|^2.$$

Define $q(\gamma) = 1 - 2m\gamma + \gamma^2 L^2$. Condition (7.31) guarantees that $q(\gamma) \in (0, 1)$. Let $\rho = \sqrt{q(\gamma)} \in (0, 1)$. Then

$$\|x_{k+1} - x^*\| \leq \rho \|x_k - x^*\|.$$

Iterating this inequality proves (7.32). $\square$

Proposition 7.13 (Optimal step size and conditioning). *Under the assumptions of Theorem 7.12, the squared contraction factor $q(\gamma) = 1 - 2m\gamma + \gamma^2 L^2$ is minimized at $\gamma^* = \frac{m}{L^2}$, where it takes the value $q(\gamma^*) = 1 - m^2/L^2$. In the quadratic case $f(x) = \frac{1}{2}x^\top Qx$, one has $m = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$. Hence the convergence rate depends explicitly on the ratio $\kappa(Q) = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}$, which is the condition number of Q : the guaranteed factor is $\rho = \sqrt{1 - \kappa(Q)^{-2}}$. As $\kappa(Q)$ increases, the contraction factor approaches one, and convergence becomes arbitrarily slow.*

7.2.5 ■ Case Study: Steepest Descent on a Quadratic Function

The quadratic objective is the one case in which every quantity appearing in the preceding two subsections can be computed in closed form, and it is worth working through in full. The iteration becomes a linear dynamical system, the smoothness and strong convexity constants are the extreme eigenvalues of a single matrix, the admissible step-size range and the optimal step size are explicit, and the contraction factor is exactly the quantity that Theorem 7.12 bounds. The case study therefore serves both as a check on the general theory and as the source of the standard intuition that the difficulty of a smooth problem is measured by the condition number of its Hessian.

We take as the object of study the gradient descent iteration

$$x_{k+1} = x_k - \gamma_k \nabla f(x_k), \quad k = 0, 1, 2, \dots, \quad (7.34)$$

where $\gamma_k > 0$ is the step size.

Before specializing to a quadratic objective it is convenient to fix the vocabulary of convergence rates, since the whole point of the case study is that all three of the following behaviours are realized by the same iteration for different step-size rules. Let x^* be a stationary point of f . The sequence $\{x_k\}$ is said to converge:

- (1) *Linearly* (or geometrically) if there exist constants $\rho \in (0, 1)$ and $C > 0$ such that

$$\|x_k - x^*\| \leq C\rho^k, \quad \forall k \geq 0. \quad (7.35)$$

- (2) *Sublinearly* if $\|x_k - x^*\| \rightarrow 0$ but no geometric rate holds, for example,

$$\|x_k - x^*\| = \mathcal{O}\left(\frac{1}{k}\right). \quad (7.36)$$

- (3) In *finite time* if there exists $T < \infty$ such that $x_k = x^*$ for all $k \geq T$.

Equivalent notions can be formulated in terms of objective values $f(x_k) - f(x^*)$ or gradient norms $\|\nabla f(x_k)\|$.

We now fix the objective. Consider the quadratic function

$$f(x) = \frac{1}{2}x^\top Qx, \quad (7.37)$$

where $Q \in \mathbb{R}^{n \times n}$ is symmetric positive definite. Then f is strongly convex, the unique minimizer is $x^* = 0$, and the gradient satisfies $\nabla f(x) = Qx$.

Substituting into (7.34) gives

$$x_{k+1} = (I - \gamma_k Q)x_k. \quad (7.38)$$

Thus the method reduces to a linear time-varying dynamical system.

Suppose first that the step size is held fixed, $\gamma_k = \gamma$. Then (7.38) yields

$$x_k = (I - \gamma Q)^k x_0. \quad (7.39)$$

Let $\{\lambda_i\}_{i=1}^n$ denote the eigenvalues of Q . Since $Q \succ 0$, we have $0 < \lambda_{\min}(Q) \leq \lambda_i \leq \lambda_{\max}(Q)$. The eigenvalues of $I - \gamma Q$ are $1 - \gamma\lambda_i$, and the spectral radius satisfies

$$\rho(I - \gamma Q) = \max_{1 \leq i \leq n} |1 - \gamma\lambda_i|.$$

Proposition 7.14. *The iteration (7.39) converges to zero for all initial conditions x_0 if and only if*

$$0 < \gamma < \frac{2}{\lambda_{\max}(Q)}. \quad (7.40)$$

In this case,

$$\|x_k\| \leq \rho^k \|x_0\|, \quad \rho = \rho(I - \gamma Q) < 1,$$

and convergence is linear.

Proof. Convergence holds if and only if $\rho(I - \gamma Q) < 1$. Since $|1 - \gamma\lambda_i| < 1 \forall i$, this is equivalent to $-1 < 1 - \gamma\lambda_i < 1, \forall i$. The right inequality gives $\gamma\lambda_i > 0$, which holds for $\gamma > 0$. The left inequality yields $\gamma\lambda_i < 2$ for all i , hence $\gamma < 2/\lambda_{\max}(Q)$. This proves (7.40). $\square$

Proposition 7.14 leaves γ free within the stability range, and it is natural to ask which choice contracts fastest. The contraction factor is minimized by choosing

$$\gamma^* = \frac{2}{\lambda_{\min}(Q) + \lambda_{\max}(Q)},$$

which yields

$$\rho^* = \frac{\kappa(Q) - 1}{\kappa(Q) + 1}, \quad \kappa(Q) = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}.$$

As $\kappa(Q) \rightarrow \infty$, we have $\rho^* \rightarrow 1$, and convergence becomes arbitrarily slow.

Remark 7.15 (Two different “optimal” step sizes). It is worth reconciling this with Proposition 7.13, which reported $\gamma^* = m/L^2$ and $\rho = \sqrt{1 - \kappa^{-2}}$. The two answers optimize different objects. Proposition 7.13 minimizes the *upper bound* $q(\gamma) = 1 - 2m\gamma + \gamma^2 L^2$ produced by the generic strongly convex argument, whereas the present computation minimizes the *exact* contraction factor $\rho(I - \gamma Q)$ available for a quadratic. The exact factor is the smaller of the two: at $\kappa = 2$, for instance, $(\kappa - 1)/(\kappa + 1) = 1/3$ against $\sqrt{1 - \kappa^{-2}} \approx 0.87$. Both degrade to 1 as $\kappa \rightarrow \infty$, which is the conclusion that matters.

Thus the linear rate depends explicitly on the condition number of the Hessian. It is worth comparing this with Theorem 7.12, whose constants for the quadratic are $L = \lambda_{\max}(Q)$ and $m = \lambda_{\min}(Q)$. That theorem is valid but conservative: it admits only $0 < \gamma < 2m/L^2$ and bounds the contraction by $\sqrt{1 - 2m\gamma + \gamma^2 L^2}$, whereas Proposition 7.14 admits the far larger range $0 < \gamma < 2/\lambda_{\max}$ and identifies the contraction exactly. For $Q = \text{diag}(1, 10)$ the theorem permits only $\gamma < 0.02$ and gives $\rho \leq 0.995$ at $\gamma = 0.01$, while the exact analysis permits $\gamma < 0.2$ and attains $\rho^* = 9/11$ at $\gamma^* = 2/11$. The gap is the price of a proof that never looks at the eigenvectors of Q , only at the extreme eigenvalues. Figure 7.2 displays $\rho(\gamma)$ for $\kappa = 10$. The two branches $|1 - \gamma|$ and $|1 - \gamma\kappa|$ cross at γ^* , producing a kink: the optimum is not a smooth interior

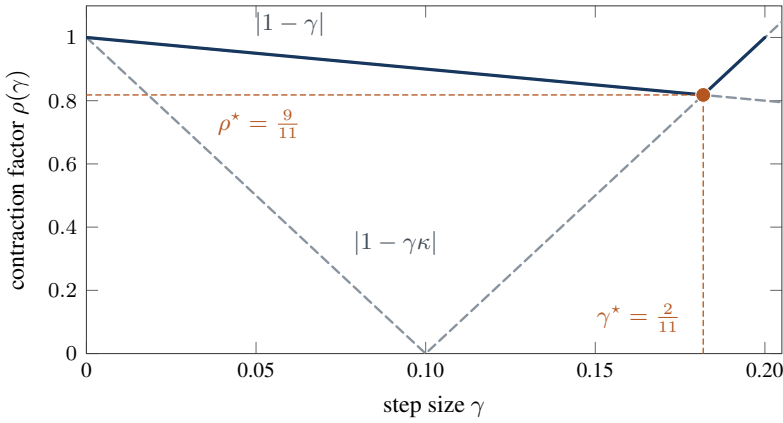


Figure 7.2: Contraction factor $\rho(\gamma) = \max\{|1 - \gamma|, |1 - \gamma\kappa|\}$ (solid) for $\kappa = 10$, with the two branches dashed. For $\gamma < \gamma^*$ the maximum is the slowly descending $|1 - \gamma|$; beyond γ^* it is the steeply ascending $|1 - \gamma\kappa|$. They meet at $\gamma^* = 2/11$, where ρ attains its minimum $\rho^* = 9/11$. Stability is lost at $\gamma = 2/\lambda_{\max} = 0.2$.

minimum but the point at which the slowest and the fastest coordinate contract at the same rate. The curve also shows how unforgiving the choice is on the right, where ρ climbs to one over the short interval $(\gamma^*, 2/\lambda_{\max})$.

Suppose now that the step size is allowed to vanish, $\gamma_k \rightarrow 0$, for example $\gamma_k = c/k$. Then the iteration (7.38) becomes time-varying. In this regime the spectral radius argument no longer yields a uniform contraction factor. Instead one typically obtains sublinear convergence rates, such as

$$f(x_k) - f(x^*) = \mathcal{O}\left(\frac{1}{k}\right), \quad \|\nabla f(x_k)\| = \mathcal{O}\left(\frac{1}{\sqrt{k}}\right).$$

These rates coincide with those established in the general smooth optimization framework without strong convexity. The comparison is worth making concrete. Figure 7.3 runs both rules on $Q = \text{diag}(1, 10)$ from $x_0 = (1, 1)^\top$ and plots the normalized objective gap $f(x_k)/f(x_0)$ on a logarithmic scale. The constant step $\gamma^* = 2/11$ produces a straight line of slope $2 \log_{10} \rho^*$, reaching 10^{-7} within forty iterations, whereas the diminishing rule $\gamma_k = 0.18/(k+1)$ drops quickly during the first few iterations and then stalls near 2×10^{-2} . The reason is visible in the step-size rule itself: the total movement still available after step k is governed by $\sum_{s \geq k} \gamma_s$, which diverges, so the iterates do converge, but only at the logarithmic pace of the harmonic tail. Diminishing step sizes buy robustness, in the sense that no knowledge of L is needed to guarantee stability, and they pay for it with the rate. They become the rule of choice only when a constant step is untenable, as in the noisy setting of Section 7.4.

Remark 7.16. The quadratic objective provides an exact realization of the general theory. Lipschitz continuity of the gradient corresponds to $\|\nabla^2 f\| \leq L$, while strong convexity corresponds to $\nabla^2 f \succeq mI$. In this setting the quadratic upper and lower bounds become exact, and the distinction between linear and sublinear convergence is completely characterized by the spectral properties of Q .

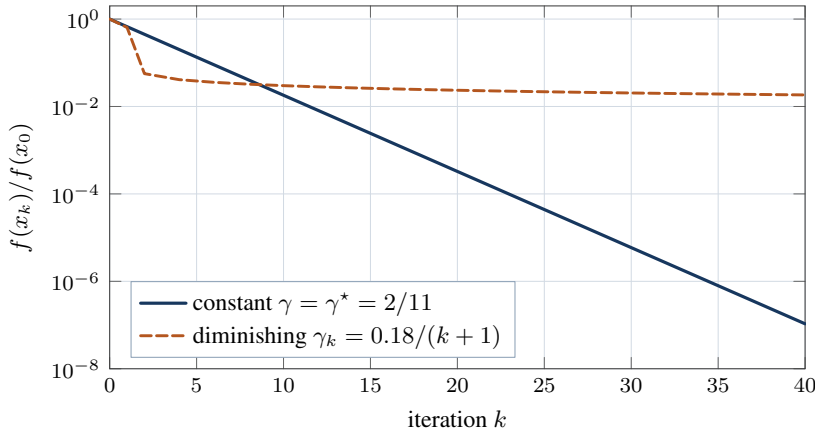


Figure 7.3: Normalized objective gap for steepest descent on $f(x) = \frac{1}{2}x^\top \text{diag}(1, 10)x$ from $x_0 = (1, 1)^\top$. The constant step γ^* decays geometrically; the diminishing rule $\gamma_k = 0.18/(k+1)$ makes rapid initial progress and then stalls, converging only at the sublinear rate dictated by the harmonic tail $\sum_{s \geq k} \gamma_s$.

Example 7.17 Effect of conditioning on steepest descent.

Consider $f(x) = \frac{1}{2}x^\top Qx$ on $\mathbb{R}^2$ with $Q = \text{diag}(1, \kappa)$, $\kappa \geq 1$, so that $\lambda_{\min} = 1$, $\lambda_{\max} = \kappa$, and the condition number is $\kappa(Q) = \kappa$. We run steepest descent from $x_0 = (1, 1)^\top$ with a constant step γ . We derive the exact convergence behaviour step by step.

Step 1 (decoupling the coordinates). Since $\nabla f(x) = Qx$ and Q is diagonal, the iteration $x_{k+1} = (I - \gamma Q)x_k$ acts independently on each coordinate, $x_k^{(1)} = (1 - \gamma)^k x_0^{(1)}$ and $x_k^{(2)} = (1 - \gamma\kappa)^k x_0^{(2)}$. Thus $\|x_k\| = [(1 - \gamma)^{2k} + (1 - \gamma\kappa)^{2k}]^{1/2}$, and the asymptotic decay rate is the *contraction factor* $\rho(\gamma) = \max\{|1 - \gamma|, |1 - \gamma\kappa|\}$.

Step 2 (optimal step size). We minimise $\rho(\gamma)$ over $\gamma > 0$. The term $|1 - \gamma|$ increases and $|1 - \gamma\kappa|$ decreases in a way that the maximum is smallest when the two are equal with opposite signs, $1 - \gamma = -(1 - \gamma\kappa)$, giving

$$\gamma^* = \frac{2}{1 + \kappa}, \quad \rho^* = 1 - \gamma^* = \frac{\kappa - 1}{\kappa + 1}.$$

Both coordinate factors then have the *same magnitude* $|1 - \gamma^*| = |1 - \gamma^*\kappa| = \rho^*$, so that $\|x_k\| = \rho^{*k} \|x_0\|$ *exactly*.

Step 3 (well-conditioned case). If $\kappa = 1$ then $\gamma^* = 1$ and $1 - \gamma^*\lambda_i = 0$ for both coordinates, so $x_1 = 0$: a single step reaches the minimiser, as expected for circular level sets.

Step 4 (iteration complexity). To reach relative accuracy ε we need $\rho^{*k} \leq \varepsilon$, i.e.

$$k \geq \frac{\ln(1/\varepsilon)}{\ln(1/\rho^*)} \approx \frac{\ln(1/\varepsilon)}{2} \kappa \quad \text{for large } \kappa,$$

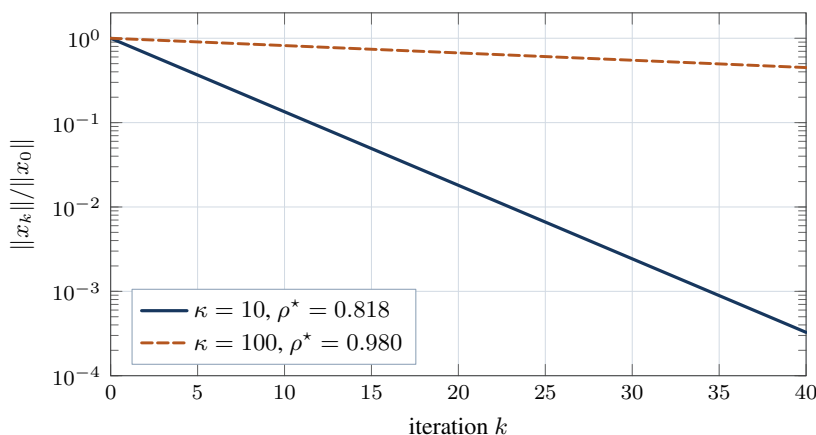


Figure 7.4: Relative error of steepest descent on the quadratic $f(x) = \frac{1}{2}x^\top \text{diag}(1, \kappa)x$ with optimal step size. The rate degrades from $\rho^* = 0.818$ ($\kappa = 10$) to $\rho^* = 0.980$ ($\kappa = 100$) as conditioning worsens.

using $\ln(1/\rho^*) = \ln \frac{\kappa+1}{\kappa-1} \approx 2/\kappa$. The iteration count therefore grows *linearly in* κ . Substituting $\varepsilon = 10^{-3}$:

κ	γ^*	ρ^*	k for $\ x_k\ /\ x_0\ \leq 10^{-3}$
1	1	0	1
10	0.1818	0.818	35
100	0.0198	0.980	346

Step 5 (why it zig-zags). For $\kappa = 10$, $\gamma^* = 2/11$, the two factors are $1 - \gamma^* = +\frac{9}{11}$ and $1 - \gamma^*\kappa = -\frac{9}{11}$. Hence $x_k^{(1)} = (\frac{9}{11})^k$ decays monotonically while $x_k^{(2)} = (-\frac{9}{11})^k$ alternates sign. The iterates therefore bounce back and forth across the narrow valley (Figure 7.5) instead of moving straight to the minimiser, which is the geometric reason for the slow rate.

Figure 7.4 plots the relative error $\|x_k\|/\|x_0\| = \rho^{*k}$ on a logarithmic scale: each curve is a straight line of slope $\log_{10} \rho^*$, confirming the linear rate of Proposition 7.14, with the slope flattening dramatically as conditioning worsens. $\square$

7.3 ■ Nonconvex Gradient Descent and the Łojasiewicz Framework

The previous section followed the two branches that open up once the gradients are known to vanish. Along the first, strong convexity pinned the iterates to a unique minimizer and delivered a geometric rate. We now take the second, and ask what can be salvaged when convexity is dropped entirely.

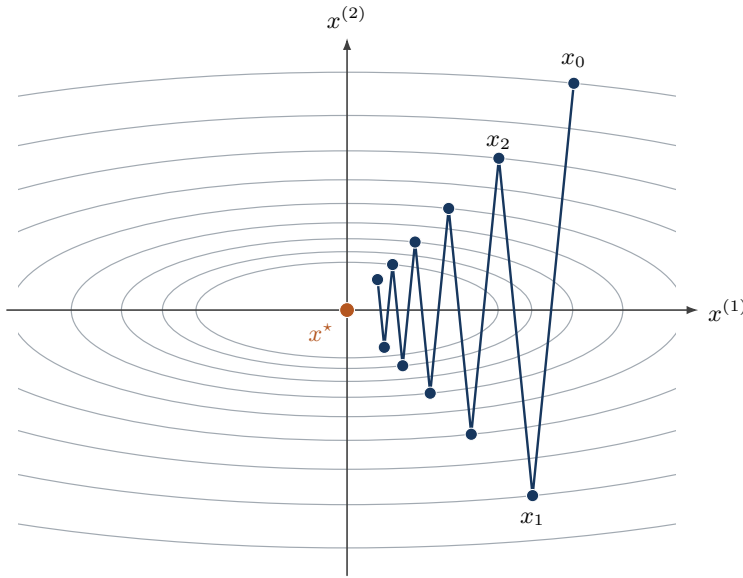


Figure 7.5: Steepest-descent iterates on $f(x) = \frac{1}{2}(x_1^2 + 10x_2^2)$ with the optimal step $\gamma^* = 2/11$. The second coordinate alternates sign, so the path zig-zags across the elongated valley (gray level sets, one through each of $x_0, \dots, x_8$) rather than heading straight to x^* ; this is the geometric cause of the slow linear rate.

The obstacles are structural. Critical points may be local minima, saddle points or local maxima, and stationarity by itself distinguishes none of them. The objective may have flat regions in which the gradient is arbitrarily small far from any minimizer. Above all, Proposition 7.10 establishes that gradient descent drives the gradient to zero and that all accumulation points are stationary, but it does not guarantee convergence of the entire sequence $\{x_k\}$: the iterates may oscillate among several stationary points, or drift along a flat valley without settling anywhere.

The gap is one of *length rather than limit points*. Vanishing gradients do not preclude a trajectory of infinite total length, and a sequence of infinite length need not converge even when all of its accumulation points are stationary. What is missing is a quantitative link between how small the gradient is and how close the function value is to its critical value, a condition forbidding the objective from being too flat near a critical point. That role is played by the Łojasiewicz gradient inequality, which we introduce next and which requires no convexity whatsoever.

7.3.1 ■ Łojasiewicz Gradient Inequality

Definition 7.18 (Łojasiewicz gradient inequality). Let x^* be a stationary point of f . We say that f satisfies the Łojasiewicz gradient inequality at x^* if there exist constants $\theta \in [0, 1)$, $C > 0$, and a neighborhood U of x^* such that

$$|f(x) - f(x^*)|^{1-\theta} \leq C \|\nabla f(x)\|, \quad \forall x \in U. \quad (7.41)$$

Equivalently, there exists $c > 0$ such that

$$\|\nabla f(x)\| \geq c|f(x) - f(x^*)|^\theta, \quad x \in U.$$

The inequality establishes a quantitative relationship between the energy gap $\Delta(x) := f(x) - f(x^*)$ and the gradient magnitude $\|\nabla f(x)\|$. It prevents the function from becoming arbitrarily flat near a critical point unless the objective value is also close to optimal. The exponent θ characterizes the local geometry: small values correspond to sharp minima, while values close to 1 correspond to flat landscapes.

Example 7.19 Strongly convex quadratic: $\theta = 1/2$.

Let $f(x) = \frac{1}{2}x^\top Qx$, $Q \succ 0$, $x^* = 0$. Then, $\nabla f(x) = Qx$. Using spectral bounds, $f(x) \leq \frac{\lambda_{\max}}{2}\|x\|^2$, $\|\nabla f(x)\| \geq \lambda_{\min}\|x\|$, so that $\|x\| \geq \sqrt{2/\lambda_{\max}} f(x)^{1/2}$. Hence,

$$\|\nabla f(x)\| \geq \lambda_{\min}\|x\| \geq \lambda_{\min} \sqrt{\frac{2}{\lambda_{\max}}} f(x)^{1/2}.$$

Thus the Łojasiewicz inequality holds with $\theta = 1/2$. This corresponds to the strongly convex case and yields linear convergence. $\square$

Example 7.20 Flat polynomial minimum: $\theta = \frac{2p-1}{2p}$.

Let $f(x) = x^{2p}$, $p \geq 1$, $x^* = 0$. Then, $\nabla f(x) = 2px^{2p-1}$. We compute $|f(x)| = |x|^{2p}$, $\|\nabla f(x)\| = 2p|x|^{2p-1}$. Thus, $\|\nabla f(x)\| = 2p|f(x)|^{\frac{2p-1}{2p}}$. Hence, the inequality holds with $\theta = \frac{2p-1}{2p} \in [1/2, 1)$, with $\theta = 1/2$ recovered at $p = 1$ (the strongly convex quadratic $f = x^2$). As p increases, the function becomes flatter near the minimum and $\theta \rightarrow 1$, leading to slower convergence. $\square$

Example 7.21 Sharp minimum: $\theta = 0$ (intuition).

Consider $f(x) = |x|$. Then, for $x \neq 0$, $|f(x) - f(0)| = |x|$, $\|\nabla f(x)\| = 1$. Hence, $\|\nabla f(x)\| \geq c$, which corresponds formally to $\theta = 0$. This represents an extreme case of a sharp minimum, where convergence occurs in finite time in continuous-time dynamics. $\square$

Example 7.22 Saddle point still satisfies Łojasiewicz.

Let $f(x, y) = x^2 - y^2$, $(x^*, y^*) = (0, 0)$. Then $\nabla f(x, y) = (2x, -2y)$. We have $|f(x, y)| = |x^2 - y^2| \leq x^2 + y^2$, and $\|\nabla f(x, y)\| = 2\sqrt{x^2 + y^2}$. Thus, $|f(x, y)|^{1/2} \leq \sqrt{x^2 + y^2} = \frac{1}{2}\|\nabla f(x, y)\|$. Hence the inequality holds locally with $\theta = 1/2$. This shows that the Łojasiewicz inequality applies to all critical points, not only minima. $\square$

The preceding examples are analytical. It is instructive to see numerically that the exponent is not a free parameter to be chosen for convenience: for a given objective there is a threshold below

Table 7.1: Verification of the Łojasiewicz inequality for $f(x) = x^4$. The ratio at the correct exponent $\theta = 3/4$ is bounded away from zero, so (7.41) holds with $c = 4$; at $\theta = 1/2$ the ratio tends to zero, so no positive constant exists and the inequality fails.

x	$\Delta(x) = x^4$	$ f'(x) $	$ f'(x) /\Delta^{3/4}$	$ f'(x) /\Delta^{1/2}$
1	1	4	4	4
0.5	6.25×10^{-2}	5×10^{-1}	4	2
0.2	1.6×10^{-3}	3.2×10^{-2}	4	0.8
0.1	1×10^{-4}	4×10^{-3}	4	0.4
0.05	6.25×10^{-6}	5×10^{-4}	4	0.2
0.01	1×10^{-8}	4×10^{-6}	4	0.04

which no constant C can work.

Example 7.23 Numerical check of the exponent.

Take $f(x) = x^4$ with $x^* = 0$, so that $\Delta(x) = x^4$ and $|f'(x)| = 4|x|^3$. The analytical exponent is $\theta = 3/4$, and indeed $|f'(x)| = 4\Delta(x)^{3/4}$ identically. Table 7.1 evaluates the two candidate ratios $|f'(x)|/\Delta(x)^\theta$ for $\theta = 3/4$ and for the strongly convex value $\theta = 1/2$ as x approaches the minimizer.

The fourth column is constant, confirming (7.41) with $c = 4$ and $\theta = 3/4$ on all of $\mathbb{R}$. The fifth column decays like $4|x|$, so for $\theta = 1/2$ no positive constant survives in any neighborhood of the origin: the quartic is genuinely flatter at its minimum than a quadratic. Since the exponent controls the convergence rate, as Section 7.3.3 makes precise, this degeneracy is not a technicality but the numerical signature of slow convergence. $\square$

7.3.2 ■ Convergence of the Entire Sequence

We now establish global convergence of the gradient descent iterates under the Łojasiewicz gradient inequality. The result shows that boundedness together with a local geometric regularity condition excludes oscillatory behavior among multiple stationary points.

Theorem 7.24. *Suppose Assumptions 7.4 and 7.8 hold. Assume further that*

- (i) *the sequence $\{x_k\}$ generated by (7.21) is bounded;*
- (ii) *the function f satisfies the Łojasiewicz gradient inequality at every stationary point.*

Then the sequence $\{x_k\}$ converges to a single stationary point x^ of f .*

Proof. From the descent inequality (7.22), $f(x_{k+1}) \leq f(x_k) - \alpha \|\nabla f(x_k)\|^2$, so $\{f(x_k)\}$ is monotonically decreasing. Since $\{x_k\}$ is bounded and f is continuous, f is bounded below on the trajectory, and therefore there exists $f_\infty \in \mathbb{R}$ such that $f(x_k) \downarrow f_\infty$.

Summing the descent inequality yields

$$\sum_{k=0}^{\infty} \|\nabla f(x_k)\|^2 \leq \frac{1}{\alpha} (f(x_0) - f_\infty) < \infty,$$

which implies $\|\nabla f(x_k)\| \rightarrow 0$.

Since $\{x_k\}$ is bounded, it admits at least one accumulation point. Let x^* be any limit point and let $x_{k_j} \rightarrow x^*$. By continuity of ∇f , $\nabla f(x^*) = \lim_{j \rightarrow \infty} \nabla f(x_{k_j}) = 0$, so x^* is a stationary point. Moreover, by continuity of f , $f(x^*) = \lim_{j \rightarrow \infty} f(x_{k_j}) = f_\infty$.

Define the energy gap $\Delta_k := f(x_k) - f(x^*) = f(x_k) - f_\infty \geq 0$, which decreases monotonically to 0.

The Łojasiewicz neighborhood. By hypothesis (ii) there are $\rho > 0$, $c > 0$ and $\theta \in [0, 1)$ such that

$$\|\nabla f(x)\| \geq c(f(x) - f_\infty)^\theta \quad \text{whenever } \|x - x^*\| < \rho \text{ and } f(x) > f_\infty. \quad (7.42)$$

It is tempting to assert at this point that the whole tail of $\{x_k\}$ lies in this ball because a subsequence converges to x^* ; that would be circular, since staying near x^* is precisely what remains to be shown. The correct route is to use the descent property to *bound the remaining path length* before the trajectory can escape.

A telescoping length estimate. Introduce the concave increasing function $\varphi(s) := \frac{s^{1-\theta}}{c(1-\theta)}$ on $[0, \infty)$, so that $\varphi'(s) = s^{-\theta}/c$. Suppose x_k lies in the ball of (7.42) and $\Delta_k > 0$. Concavity of φ gives

$$\varphi(\Delta_k) - \varphi(\Delta_{k+1}) \geq \varphi'(\Delta_k)(\Delta_k - \Delta_{k+1}) = \frac{\Delta_k - \Delta_{k+1}}{c \Delta_k^\theta}.$$

The descent inequality (7.22) bounds the numerator below by $\alpha \|\nabla f(x_k)\|^2$, while (7.42) bounds the denominator above by $\|\nabla f(x_k)\|$. Hence

$$\varphi(\Delta_k) - \varphi(\Delta_{k+1}) \geq \frac{\alpha \|\nabla f(x_k)\|^2}{\|\nabla f(x_k)\|} = \alpha \|\nabla f(x_k)\| = \frac{\alpha}{\gamma} \|x_{k+1} - x_k\|, \quad (7.43)$$

using $\|x_{k+1} - x_k\| = \gamma \|\nabla f(x_k)\|$. Each step of the trajectory is therefore paid for by a drop in $\varphi(\Delta_k)$, and $\varphi(\Delta_k)$ is bounded and decreasing.

The trajectory cannot escape. Fix k_0 so large that

$$\|x_{k_0} - x^*\| + \frac{\gamma}{\alpha} \varphi(\Delta_{k_0}) < \rho, \quad (7.44)$$

which is possible because some subsequence of $\{x_k\}$ converges to x^* and $\Delta_k \downarrow 0$, so $\varphi(\Delta_k) \downarrow 0$. We claim $\|x_k - x^*\| < \rho$ for every $k \geq k_0$. Indeed, suppose this holds for $k_0, \dots, k$. Then (7.43) applies at each of those indices and telescopes to

$$\sum_{j=k_0}^k \|x_{j+1} - x_j\| \leq \frac{\gamma}{\alpha} (\varphi(\Delta_{k_0}) - \varphi(\Delta_{k+1})) \leq \frac{\gamma}{\alpha} \varphi(\Delta_{k_0}),$$

whence, by the triangle inequality and (7.44),

$$\|x_{k+1} - x^*\| \leq \|x_{k_0} - x^*\| + \sum_{j=k_0}^k \|x_{j+1} - x_j\| \leq \|x_{k_0} - x^*\| + \frac{\gamma}{\alpha} \varphi(\Delta_{k_0}) < \rho.$$

The claim follows by induction. (If $\Delta_k = 0$ for some k , the descent inequality forces $\nabla f(x_k) = 0$ and the iteration is stationary from then on, so the conclusion is immediate.)

Finite length. The induction just performed shows in addition that the bound holds for every k , so letting $k \rightarrow \infty$,

$$\sum_{k=k_0}^{\infty} \|x_{k+1} - x_k\| \leq \frac{\gamma}{\alpha} \varphi(\Delta_{k_0}) < \infty. \quad (7.45)$$

Therefore the sequence $\{x_k\}$ has finite length and is Cauchy. Since $\mathbb{R}^n$ is complete, there exists $\bar{x}$ such that $x_k \rightarrow \bar{x}$. By continuity of ∇f and the fact that $\|\nabla f(x_k)\| \rightarrow 0$, $\nabla f(\bar{x}) = 0$, so $\bar{x}$ is a stationary point. Finally, since every accumulation point must coincide with $\bar{x}$, the entire sequence converges to the same limit. $\square$

The key mechanism behind the proof is the finite-length property $\sum_{k=0}^{\infty} \|x_{k+1} - x_k\| < \infty$. Without the Łojasiewicz inequality, gradient descent guarantees only that gradients vanish asymptotically, and the sequence may wander along flat regions of stationary points. The Łojasiewicz condition prevents such behavior by enforcing a quantitative relation between objective decrease and gradient magnitude, thereby converting asymptotic stationarity into full convergence.

The following numerical example exhibits all three features of Theorem 7.24 at once: a genuinely nonconvex objective with several stationary points, convergence of each trajectory to a single one of them, and the finite length that makes this possible.

Example 7.25 Double-well objective.

Consider the nonconvex function

$$f(x) = (x_1^2 - 1)^2 + x_2^2, \quad x = (x_1, x_2)^\top \in \mathbb{R}^2, \quad (7.46)$$

with gradient $\nabla f(x) = (4x_1(x_1^2 - 1), 2x_2)^\top$. The stationary points are the two global minimizers $(\pm 1, 0)$, with $f = 0$, and the saddle $(0, 0)$, with $f = 1$; the Hessian at the saddle is $\text{diag}(-4, 2)$, so it is a strict saddle. The function is a polynomial, hence real analytic, and therefore satisfies the Łojasiewicz inequality at each of its critical points; at the two minimizers the Hessian is $\text{diag}(8, 2) \succ 0$ and the exponent is $\theta = 1/2$.

We run gradient descent with the constant step $\gamma = 0.05$ from three initializations. Figure 7.6 shows the trajectories, and Table 7.2 reports the cumulative path length $\Lambda_T := \sum_{k=0}^{T-1} \|x_{k+1} - x_k\|$.

Three points deserve emphasis. First, the limit is genuinely initialization-dependent: the same algorithm with the same step size converges to $(1, 0)$ or to $(-1, 0)$ according to which basin contains x_0 . Nothing in Theorem 7.24 claims otherwise: it asserts convergence to a stationary point, not to a global minimizer. Second, each length sequence Λ_T saturates, and the increments beyond $T = 100$ are below 10^{-3} ; the trajectory is rectifiable, which is exactly the mechanism that upgrades $\|\nabla f(x_k)\| \rightarrow 0$ to $x_k \rightarrow \bar{x}$. Third, the third trajectory illustrates the role of the saddle. Starting at $x_0 = (0.05, 0.6)$, the first coordinate grows only from 0.05 to 0.30 over ten iterations, because $\|\nabla f\|$ is small near the saddle; thereafter it escapes along the unstable eigendirection and reaches the well. Gradient descent is not attracted to the strict saddle here, but it is slowed by it, the Łojasiewicz inequality holds at $(0, 0)$ with $\theta = 1/2$ and therefore guarantees escape in a controlled number of steps rather than stagnation. $\square$

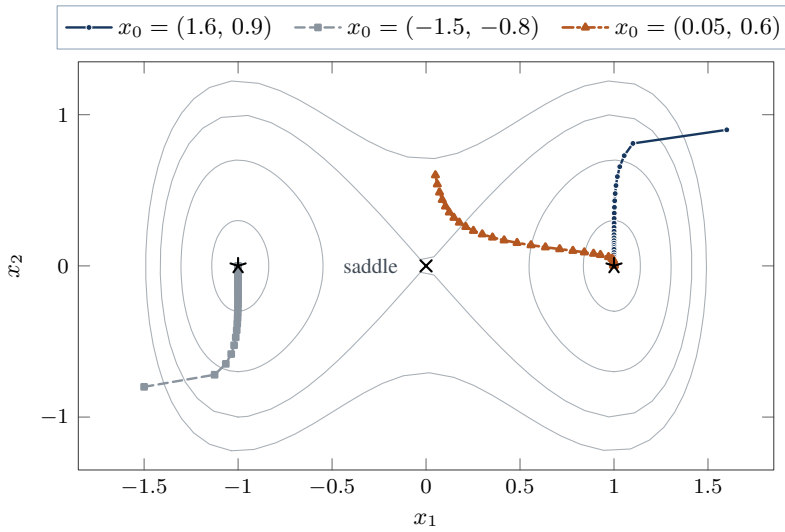


Figure 7.6: Gradient descent with $\gamma = 0.05$ on $f(x) = (x_1^2 - 1)^2 + x_2^2$. Stars mark the two minimizers $(\pm 1, 0)$, the cross marks the saddle $(0, 0)$; gray curves are the level sets $f = 0.09, 0.49, 1, 1.5$, the level $f = 1$ crossing itself at the saddle. Each trajectory converges to a single stationary point, and the limit depends on the initialization. The dash-dotted trajectory starts close to the saddle and lingers there before being expelled along the unstable direction.

Table 7.2: Cumulative path length $\Lambda_T = \sum_{k < T} \|x_{k+1} - x_k\|$ for the three trajectories of Example 7.25. Each series saturates, which is the finite-length property underlying Theorem 7.24.

x_0	Λ_5	Λ_{10}	Λ_{20}	Λ_{40}	Λ_{100}	Λ_{400}	limit
$(1.6, 0.9)$	0.8037	1.0216	1.2260	1.3221	1.3353	1.3354	$(1, 0)$
$(-1.5, -0.8)$	0.6614	0.8553	1.0370	1.1224	1.1342	1.1342	$(-1, 0)$
$(0.05, 0.6)$	0.2581	0.4895	1.1342	1.2428	1.2516	1.2516	$(1, 0)$

7.3.3 ■ Convergence Rates

We now make the dependence of the convergence rate on the Łojasiewicz exponent precise. Let x^* denote the limit point established in Theorem 7.24, and let $\theta \in [0, 1)$ be the corresponding Łojasiewicz exponent. Define the energy gap $\Delta_k := f(x_k) - f(x^*)$. For all sufficiently large k , the iterates lie in a neighborhood of x^* where the Łojasiewicz gradient inequality holds. Combining the descent inequality

$$\Delta_k - \Delta_{k+1} \geq \alpha \|\nabla f(x_k)\|^2$$

with $\|\nabla f(x_k)\| \geq c\Delta_k^\theta$, we obtain the fundamental discrete inequality

$$\Delta_k - \Delta_{k+1} \geq C\Delta_k^{2\theta}, \quad C := \alpha c^2 > 0. \quad (7.47)$$

The asymptotic behavior of $\{\Delta_k\}$ is governed by the exponent 2θ .

Case $\theta = 0$. When $\theta = 0$, the Łojasiewicz inequality implies $\|\nabla f(x)\| \geq c > 0$ whenever $f(x) > f(x^*)$. Substituting into (7.47) yields $\Delta_k - \Delta_{k+1} \geq C$, for some constant $C > 0$ as long

as $\Delta_k > 0$. Since $\Delta_k \geq 0$ and decreases by at least a fixed amount, it must reach zero in finitely many steps. Thus convergence occurs in finite time. This corresponds to a sharp critical point.

Case $0 < \theta \leq \frac{1}{2}$. In this regime, $2\theta \leq 1$. Since $\Delta_k \rightarrow 0$, there exists k_0 such that for all $k \geq k_0$, $\Delta_k \leq 1$. Hence, $\Delta_k^{2\theta} \geq \Delta_k$, $k \geq k_0$. Substituting into (7.47) gives $\Delta_k - \Delta_{k+1} \geq C\Delta_k$ for $k \geq k_0$, which implies $\Delta_{k+1} \leq (1 - C)\Delta_k$. Since $C \in (0, 1)$ (possibly after shrinking C), we obtain a contraction. Iterating yields $\Delta_k \leq \Delta_{k_0}(1 - C)^{k-k_0}$. Thus there exist constants $C_1 > 0$ and $\rho \in (0, 1)$ such that $f(x_k) - f(x^*) \leq C_1\rho^k$. This establishes linear convergence of the function values.

To transfer this rate from function values to iterates, use the length bound (7.45) rather than a Lipschitz estimate: bounding $\|x_{k+1} - x_k\|$ by $\|x_k - x^*\|$ would presuppose the very decay we are trying to establish. Applying (7.45) from index k onwards and letting the tail converge to the limit $\bar{x}$ gives

$$\|x_k - \bar{x}\| \leq \sum_{j=k}^{\infty} \|x_{j+1} - x_j\| \leq \frac{\gamma}{\alpha} \varphi(\Delta_k) = \frac{\gamma}{\alpha} \frac{\Delta_k^{1-\theta}}{c(1-\theta)}.$$

With $\Delta_k \leq \Delta_{k_0}(1 - C)^{k-k_0}$ and $1 - \theta \geq \frac{1}{2}$, the right-hand side decays geometrically, so $\|x_k - \bar{x}\| \leq C_2\rho_1^k$ for some $\rho_1 \in (0, 1)$: the iterates converge linearly as well, at the rate $\rho^{1-\theta}$ inherited from the function values.

The borderline case $\theta = \frac{1}{2}$ corresponds to the strongly convex situation, where the Łojasiewicz inequality reduces to a quadratic growth condition.

Case $\frac{1}{2} < \theta < 1$. In this case, $2\theta > 1$. The inequality (7.47) can be compared to the differential equation $\dot{\Delta} = -C\Delta^{2\theta}$. Solving this equation yields $\Delta(k) = (\Delta(0)^{1-2\theta} + C(2\theta - 1)k)^{-\frac{1}{2\theta-1}}$, which decays polynomially.

A standard comparison argument for difference inequalities shows that $\Delta_k = O\left(k^{-\frac{1}{2\theta-1}}\right)$. Thus the convergence rate is sublinear.

Example 7.26 Numerical verification of the predicted rates.

The three regimes above are directly observable. Consider the family $f_p(x) = x^{2p}$, $p = 1, 2, 3$, on $\mathbb{R}$, for which Example 7.23 and its predecessor give the exponents $\theta_p = (2p - 1)/(2p)$, that is $\theta_1 = 1/2$, $\theta_2 = 3/4$, $\theta_3 = 5/6$. The theory therefore predicts linear convergence for $p = 1$ and, for $p \geq 2$, the polynomial rate

$$\Delta_k = O\left(k^{-\frac{1}{2\theta_p-1}}\right) = O\left(k^{-\frac{p}{p-1}}\right),$$

namely $O(k^{-2})$ for $p = 2$ and $O(k^{-3/2})$ for $p = 3$. We run $x_{k+1} = x_k - \gamma f'_p(x_k)$ with $x_0 = 1$ and $\gamma = 0.1$ for all three, and measure the empirical decay exponent as the slope of $\log \Delta_k$ against $\log k$ over $k \in [10^3, 4 \times 10^3]$.

Table 7.3 and Figure 7.7 confirm the predictions to two decimal places. For $p = 1$ the iteration is $x_{k+1} = 0.8x_k$, so $\Delta_k = 0.64^k$ exactly and the curve leaves the plotting window within sixty iterations. For $p = 2$ and $p = 3$ the curves are straight on the log-log axes, with slopes -1.99 and -1.49 against the predicted -2 and $-3/2$. The practical consequence is stark: reaching $\Delta_k \leq 10^{-6}$ takes 31 iterations for the

Table 7.3: Predicted and measured decay of $\Delta_k = f_p(x_k)$ for gradient descent on $f_p(x) = x^{2p}$ with $\gamma = 0.1$ and $x_0 = 1$. The measured exponent is the log–log slope over $k \in [10^3, 4 \times 10^3]$.

p	θ_p	predicted rate	measured exponent	Δ_{100}	k to reach 10^{-6}
1	1/2	linear, $\rho = 0.64$	—	$< 10^{-16}$	31
2	3/4	k^{-2}	−1.993	1.32×10^{-4}	1244
3	5/6	$k^{-3/2}$	−1.486	2.16×10^{-4}	$> 4 \times 10^3$

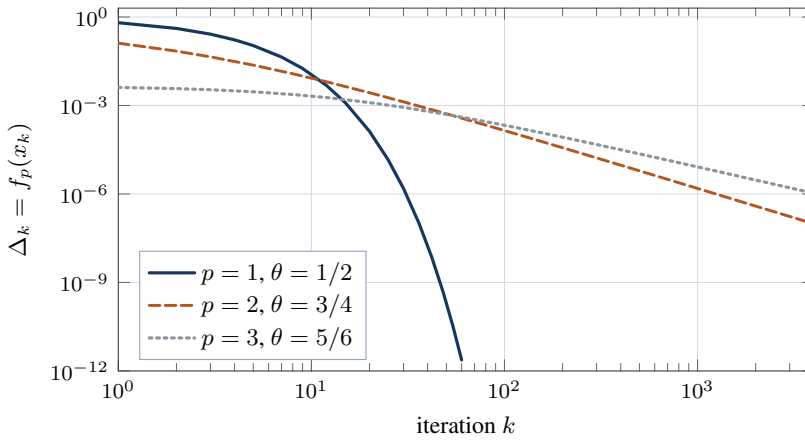


Figure 7.7: Objective gap Δ_k for gradient descent on $f_p(x) = x^{2p}$, $x_0 = 1$, $\gamma = 0.1$, on log–log axes. The case $p = 1$ ($\theta = 1/2$) decays geometrically and so falls off any polynomial reference line. The cases $p = 2, 3$ are straight lines of slope -1.99 and -1.49 , matching the predicted $-1/(2\theta - 1)$ equal to -2 and $-3/2$.

quadratic, forty times as many for the quartic, and more than four thousand for the sextic. The single number responsible is θ . □

The exponent θ quantifies the local geometry of the objective near the limit point. Small values of θ correspond to sharp minima and fast convergence, while values of θ close to one correspond to flatter geometries and slower rates. Convexity is not required; instead, the Łojasiewicz exponent plays the role of a curvature surrogate, encoding how rapidly the gradient vanishes relative to suboptimality of the function. This determines both the qualitative and quantitative behavior of gradient descent in the nonconvex regime.

7.4 ■ Gradient Methods with Noise or Errors

In many practical optimization problems, the gradient $\nabla f(x_k)$ is not available in exact form. Instead, the algorithm has access to an approximation that can be written as $\nabla f(x_k) + w_k$, where $w_k \in \mathbb{R}^n$ denotes an error or noise term. Such perturbations arise naturally in a variety of settings. In deterministic numerical computation, round-off errors and finite-precision arithmetic introduce discrepancies between the true and computed gradient. In problems involving discretization, such as PDE-constrained optimization, numerical approximation of derivatives generates additional errors. In data-driven and large-scale learning problems, gradients are often estimated from

samples, leading to stochastic fluctuations due to sampling variability.

When the gradient is perturbed in this way, the update rule becomes

$$x_{k+1} = x_k - \gamma_k (\nabla f(x_k) + w_k), \quad (7.48)$$

where $\gamma_k > 0$ is the step size. The iteration (7.48) is commonly referred to as a *perturbed gradient method* or, in stochastic settings, *stochastic gradient descent*.

Without additional assumptions on the noise sequence $\{w_k\}$, convergence cannot, in general, be guaranteed. If the perturbations are arbitrary, adversarial, or unbounded, the iterates may diverge or fail to approach a stationary point. A meaningful convergence analysis therefore requires structural assumptions on the noise, such as boundedness, vanishing magnitude, martingale-difference structure, or bounded variance. These structured noise models reflect the mechanisms generating the perturbations and allow one to establish convergence guarantees under appropriate step-size conditions.

7.4.1 ■ Representative Noise Models

We now describe several representative perturbation models that arise naturally in deterministic and stochastic optimization and illustrate their qualitative effect on convergence.

1. Error small relative to the gradient Suppose that the perturbation satisfies $\|w_k\| \leq \varepsilon_k \|\nabla f(x_k)\|$, $0 \leq \varepsilon_k < 1$, for all k . Consider the perturbed search direction $s_k = -\nabla f(x_k) - w_k$. Then

$$\begin{aligned} \nabla f(x_k)^\top s_k &= -\|\nabla f(x_k)\|^2 - \nabla f(x_k)^\top w_k \\ &\leq -\|\nabla f(x_k)\|^2 + \|\nabla f(x_k)\| \|w_k\| \\ &\leq -(1 - \varepsilon_k) \|\nabla f(x_k)\|^2. \end{aligned}$$

Hence, whenever $\varepsilon_k < 1$ and $\nabla f(x_k) \neq 0$, the direction s_k remains a descent direction. In particular, relative errors that are uniformly bounded away from one do not destroy the descent property, and the qualitative behavior of gradient descent is preserved.

2. Uniformly bounded error Assume that the perturbation is uniformly bounded: $\|w_k\| \leq \delta, \forall k$, for some constant $\delta > 0$.

If the noise is time-invariant, that is, $w_k = w$ for all k , then the update

$$x_{k+1} = x_k - \gamma_k (\nabla f(x_k) + w)$$

is equivalent to applying gradient descent to the perturbed objective $\tilde{f}(x) = f(x) + w^\top x$. A stationary point $\tilde{x}$ of $\tilde{f}$ satisfies $\nabla \tilde{f}(\tilde{x}) = -w$. Thus the method converges to the minimizer of a linearly perturbed objective rather than to a true stationary point of f .

If w_k varies with time but remains bounded, exact convergence to a stationary point of f cannot generally be expected. Instead, the iterates typically enter and remain within a neighborhood of the stationary set characterized by $\|\nabla f(x)\| \leq \delta$, so that the residual gradient is controlled by the noise magnitude.

3. Step-size-proportional error A particularly important and benign noise model assumes that

$$\|w_k\| \leq \gamma_k (q + p \|\nabla f(x_k)\|), \quad \forall k, \quad (7.49)$$

for some constants $p, q \geq 0$.

In this case the perturbation vanishes asymptotically whenever $\gamma_k \rightarrow 0$. Such errors naturally arise in stochastic approximation, incremental gradient methods, and online learning algorithms. Because the noise magnitude scales with the step size, the cumulative perturbation remains controlled, and under standard diminishing step-size conditions one can recover convergence to a stationary point of f .

7.4.2 ■ General Convergence Result with Errors

We now consider a perturbed descent scheme of the form

$$x_{k+1} = x_k + \gamma_k (s_k + w_k), \quad (7.50)$$

where $\gamma_k > 0$ is a step size, s_k is a search direction, and w_k is a perturbation term.

The directions s_k are assumed to satisfy the uniform descent conditions

$$C_1 \|\nabla f(x_k)\|^2 \leq -\nabla f(x_k)^\top s_k, \quad (7.51)$$

$$\|s_k\| \leq C_2 \|\nabla f(x_k)\|, \quad (7.52)$$

for some constants $C_1, C_2 > 0$. These conditions ensure that, in the absence of noise, the method possesses a uniform descent mechanism comparable to that of gradient-type algorithms.

We further assume that the perturbations satisfy the step-size-proportional bound

$$\|w_k\| \leq \gamma_k (q + p \|\nabla f(x_k)\|), \quad \forall k, \quad (7.53)$$

for some constants $p, q \geq 0$. Finally, the step sizes are required to satisfy

$$\sum_{k=0}^{\infty} \gamma_k = \infty, \quad \sum_{k=0}^{\infty} \gamma_k^2 < \infty. \quad (7.54)$$

The first condition in (7.54) ensures that the method does not stagnate prematurely, while the second guarantees that the cumulative effect of the vanishing perturbations remains finite.

Proposition 7.27 (Convergence under vanishing perturbations). *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable, and suppose that ∇f is Lipschitz continuous. Assume that the iteration (7.50) satisfies (7.51)–(7.54). Then one of the following alternatives holds: $f(x_k) \rightarrow -\infty$, or $\{f(x_k)\}$ converges to a finite value and $\lim_{k \rightarrow \infty} \|\nabla f(x_k)\| = 0$. In the latter case, every limit point of the sequence $\{x_k\}$ is a stationary point of f .*

Remark 7.28. The result shows that step-size-proportional perturbations are asymptotically negligible under the classical stochastic approximation conditions (7.54). The square-summability condition $\sum_k \gamma_k^2 < \infty$ ensures that the accumulated noise remains controlled, while the divergence of $\sum_k \gamma_k$ forces the iterates to continue making progress. These conditions are standard in the analysis of Robbins–Monro procedures and stochastic gradient methods.

Proof. [Proof of Proposition 7.27] Write $g_k := \|\nabla f(x_k)\|$. Note first that $\sum_k \gamma_k^2 < \infty$ forces $\gamma_k \rightarrow 0$.

A perturbed descent inequality. Apply Lemma 7.5 with $z = \gamma_k (s_k + w_k)$:

$$f(x_{k+1}) - f(x_k) \leq \gamma_k \nabla f(x_k)^\top s_k + \gamma_k \nabla f(x_k)^\top w_k + \frac{L}{2} \gamma_k^2 \|s_k + w_k\|^2.$$

The first term is at most $-\gamma_k C_1 g_k^2$ by (7.51). For the second, Cauchy–Schwarz and (7.53) give $\nabla f(x_k)^\top w_k \leq g_k \|w_k\| \leq \gamma_k (q g_k + p g_k^2)$, and $q g_k \leq \frac{1}{2} (q^2 + g_k^2)$. For the third, $\|s_k + w_k\|^2 \leq$

$2\|s_k\|^2 + 2\|w_k\|^2$ together with (7.52) and $(q + pg_k)^2 \leq 2q^2 + 2p^2g_k^2$ gives $\|s_k + w_k\|^2 \leq 2C_2^2g_k^2 + 4\gamma_k^2q^2 + 4\gamma_k^2p^2g_k^2$. Collecting,

$$f(x_{k+1}) \leq f(x_k) - \gamma_k \left[C_1 - \gamma_k \left(p + \frac{1}{2} + LC_2^2 + 2L\gamma_k^2p^2 \right) \right] g_k^2 + \beta_k, \quad \beta_k := \frac{1}{2}\gamma_k^2q^2 + 2L\gamma_k^4q^2. \quad (7.55)$$

Since $\gamma_k \rightarrow 0$, there is $\bar{k}$ beyond which the bracket exceeds $C_1/2$; and $\sum_k \beta_k < \infty$ because $\sum_k \gamma_k^2 < \infty$ and γ_k is bounded.

The dichotomy and summability. Put $V_k := f(x_k) + \sum_{j \geq k} \beta_j$, which is well defined for $k \geq \bar{k}$. By (7.55), $V_{k+1} \leq V_k - \frac{C_1}{2}\gamma_k g_k^2 \leq V_k$, so $\{V_k\}$ is nonincreasing. Either $V_k \rightarrow -\infty$, in which case $f(x_k) \rightarrow -\infty$ since the tail sum vanishes, or V_k converges to a finite limit and hence so does $f(x_k)$. In the latter case, summing the displayed inequality from $\bar{k}$ to $T - 1$ and letting $T \rightarrow \infty$ yields

$$\sum_{k \geq \bar{k}} \gamma_k g_k^2 < \infty. \quad (7.56)$$

From summability to $g_k \rightarrow 0$. This is exactly the situation of Proposition 7.7: (7.56) together with $\sum_k \gamma_k = \infty$ and the Lipschitz continuity of ∇f . The two-step argument given there—first $\liminf_k g_k = 0$, then a crossing argument bounding $\|\nabla f(x_m) - \nabla f(x_n)\|$ across blocks on which g_k stays above a threshold—applies verbatim, the only change being that the step $\|x_{k+1} - x_k\| \leq \gamma_k \|s_k + w_k\| \leq \gamma_k (C_2 + \gamma_k p) g_k + \gamma_k^2 q$ now carries the extra perturbation terms, which are summable and therefore do not affect the conclusion. Hence $g_k \rightarrow 0$.

Finally, if $x_{k_j} \rightarrow \bar{x}$ then $\nabla f(\bar{x}) = \lim_j \nabla f(x_{k_j}) = 0$ by continuity, so every limit point is stationary. $\square$

7.4.3 ■ Stochastic Gradient Noise and Robbins–Monro Approximation

A fundamentally important class of perturbations arises from *random sampling*. In this setting, the noise is neither deterministic nor adversarial, but is generated by random variables with zero mean and finite variance. This framework underlies stochastic gradient methods and classical stochastic approximation theory.

Stochastic Gradient Model

Consider an objective function defined as an expectation:

$$f(x) = \mathbb{E}_v [F(x, v)], \quad (7.57)$$

where $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ is continuously differentiable with respect to x , and v is a random vector defined on a probability space $(\Omega, \mathcal{A}, P)$.

Under standard regularity conditions ensuring interchange of differentiation and expectation (e.g., dominated convergence), the gradient of f is given by

$$\nabla f(x) = \mathbb{E}_v [\nabla_x F(x, v)]. \quad (7.58)$$

The exact gradient descent iteration therefore reads

$$x_{k+1} = x_k - \gamma_k \nabla f(x_k) = x_k - \gamma_k \mathbb{E}_v [\nabla_x F(x_k, v)].$$

In many applications, the expectation in (7.58) is computationally intractable. Instead, one draws an independent sample $v_k \sim P$ and defines the stochastic search direction $s_k = -\nabla_x F(x_k, v_k)$. This leads to the stochastic gradient iteration

$$x_{k+1} = x_k - \gamma_k \nabla_x F(x_k, v_k). \quad (7.59)$$

Noise Decomposition

The iteration (7.59) can be written as a perturbed gradient method. Indeed,

$$\begin{aligned} x_{k+1} &= x_k - \gamma_k \left(\mathbb{E}_v [\nabla_x F(x_k, v)] + \underbrace{\nabla_x F(x_k, v_k) - \mathbb{E}_v [\nabla_x F(x_k, v)]}_{\triangleq w_k} \right) \\ &= x_k - \gamma_k \nabla f(x_k) + \gamma_k w_k. \end{aligned} \quad (7.60)$$

The random vector

$$w_k = \nabla_x F(x_k, v_k) - \mathbb{E}_v [\nabla_x F(x_k, v)] \quad (7.61)$$

satisfies $\mathbb{E}(w_k \mid \mathcal{F}_k) = 0$, where $\mathcal{F}_k$ denotes the sigma-algebra generated by $\{x_0, v_0, \dots, v_{k-1}\}$. Thus, $\{w_k\}$ forms a *martingale difference sequence*. Moreover, if $\mathbb{E}(\|w_k\|^2 \mid \mathcal{F}_k) \leq \sigma^2$, the noise has uniformly bounded conditional variance.

Although the direction $-\nabla_x F(x_k, v_k)$ is not necessarily a descent direction for the deterministic objective f , it satisfies $\mathbb{E}[\nabla f(x_k)^\top s_k \mid \mathcal{F}_k] = -\|\nabla f(x_k)\|^2$, so the descent property holds *in conditional expectation*. This is the key structural feature enabling convergence analysis.

Robbins–Monro Stochastic Approximation

The stochastic gradient method is a special case of the classical Robbins–Monro stochastic approximation scheme. Consider the problem of finding a root of

$$h(x) = \mathbb{E}_v[g(x, v)] = 0, \quad (7.62)$$

where $g : \mathbb{R}^d \times \mathbb{R}^m \rightarrow \mathbb{R}^d$.

If the expectation were computable, one could apply the deterministic iteration

$$x_{k+1} = x_k - \gamma_k h(x_k).$$

When the distribution of v is unknown or the expectation is expensive to evaluate, it is replaced by a single sample:

$$x_{k+1} = x_k - \gamma_k g(x_k, v_k).$$

Writing

$$g(x_k, v_k) = h(x_k) + (g(x_k, v_k) - h(x_k)),$$

we obtain

$$x_{k+1} = x_k - \gamma_k (h(x_k) + w_k), \quad (7.63)$$

where w_k is a martingale difference noise term.

Algorithms of the form (7.63) are known as *Robbins–Monro stochastic approximation algorithms*. Under appropriate conditions on the step sizes, typically

$$\sum_{k=0}^{\infty} \gamma_k = \infty, \quad \sum_{k=0}^{\infty} \gamma_k^2 < \infty,$$

and suitable stability assumptions on h , the iterates converge almost surely to a solution of (7.62).

Stochastic gradient descent corresponds to the special case $h(x) = \nabla f(x)$, so that minimizing an expected objective is equivalent to solving the stochastic approximation problem $\nabla f(x) = 0$.

7.4.4 ■ ODE Method for Stochastic Approximation

We begin by stating the central result underlying the ODE method for stochastic approximation. This theorem formalizes the idea that a stochastic recursion tracks the trajectories of an associated deterministic ordinary differential equation and inherits its asymptotic behavior.

Theorem 7.29 (ODE method: convergence to the equilibrium set). *Consider the stochastic approximation recursion*

$$x_{k+1} = x_k + \gamma_k (h(x_k) + w_{k+1}), \quad x_0 \text{ given}, \quad (7.64)$$

where $h : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is continuous, $\{w_k\}$ is a martingale difference sequence with respect to the natural filtration, and $\{\gamma_k\}$ is a deterministic step-size sequence satisfying

$$\sum_{k=0}^{\infty} \gamma_k = \infty, \quad \sum_{k=0}^{\infty} \gamma_k^2 < \infty.$$

Assume further that the iterates $\{x_k\}$ are almost surely bounded and that the ordinary differential equation

$$\dot{z}(\tau) = h(z(\tau)) \quad (7.65)$$

admits a continuously differentiable Lyapunov function V such that $\langle \nabla V(x), h(x) \rangle \leq 0$, with equality if and only if $h(x) = 0$. Then, almost surely, every limit point of $\{x_k\}$ belongs to the invariant set $\mathcal{E} := \{x \in \mathbb{R}^d : h(x) = 0\}$, and the stochastic recursion converges to $\mathcal{E}$.

We state Theorem 7.29 here for orientation and give the precise hypotheses and a complete proof in Section 7.4.6, as Theorem 7.48.

The theorem provides a unifying framework for analyzing a wide class of stochastic approximation algorithms, including stochastic gradient methods and Robbins–Monro schemes. Its significance lies in the fact that the stochastic recursion (7.64) may be understood as a noisy, time-discretized approximation of the deterministic dynamical system (7.65) (Figure 7.8). The diminishing step sizes ensure that the discretization becomes increasingly accurate, while the square-summability condition guarantees that the cumulative effect of the stochastic noise remains finite almost surely.

To make this connection precise, one introduces the continuous-time scale $\tau_k := \sum_{s=0}^{k-1} \gamma_s$ and constructs a piecewise linear interpolation $\bar{x}(\tau)$ of the iterates by setting $\bar{x}(\tau_k) = x_k$ and interpolating linearly on each interval $[\tau_k, \tau_{k+1}]$. For each k , let $z^{(k)}(\cdot)$ denote the solution of the ODE (7.65) initialized at x_k at time τ_k . Under the stated assumptions, one can show that for any fixed time horizon $T > 0$,

$$\sup_{\tau \in [\tau_k, \tau_k + T]} \|\bar{x}(\tau) - z^{(k)}(\tau)\| \longrightarrow 0 \quad \text{almost surely.}$$

This tracking property (illustrated in Figure 7.9) formalizes the intuition that, on finite time windows, the stochastic iterates evolve like trajectories of the limiting ODE, with the noise becoming negligible as the step size decreases.

Once this asymptotic tracking is established, the long-term behavior of the stochastic recursion is governed by the qualitative dynamics of the ODE (7.65). In particular, if the ODE admits a Lyapunov function with the stated properties, then all ODE trajectories converge to the invariant set $\mathcal{E}$. The stochastic recursion inherits this convergence almost surely, since it remains arbitrarily close to ODE trajectories over arbitrarily long finite horizons.

The ODE method thus reduces the analysis of stochastic approximation algorithms to the study of deterministic dynamical systems. Stability, attractivity, and invariance properties of the

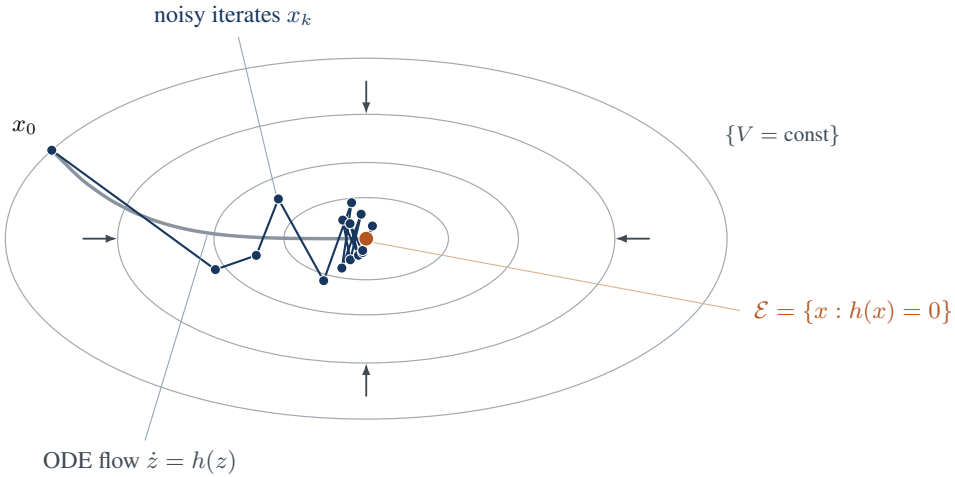


Figure 7.8: Phase-space picture of the ODE method (Theorem 7.29). The noisy iterates x_k (navy) shadow a trajectory of the limiting ODE $\dot{z} = h(z)$ (thick gray). The ODE trajectory descends the Lyapunov level sets $\{V = \text{const}\}$ (thin ellipses) monotonically, since $\langle \nabla V, h \rangle \leq 0$; individual iterates may step back up a level set because of the noise, but they follow the same descent on average and are absorbed into the invariant set $\mathcal{E} = \{h = 0\}$ (orange).

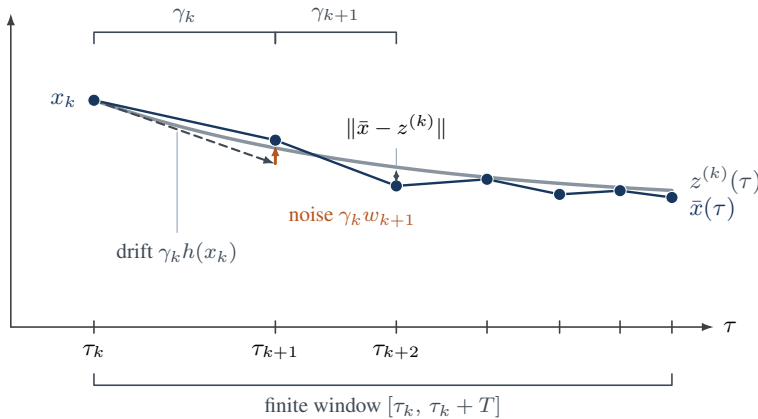


Figure 7.9: The tracking argument behind the proof of Theorem 7.29. On the continuous-time scale $\tau_k = \sum_{j < k} \gamma_j$, each iterate advances by a *drift* $\gamma_k h(x_k)$ (an Euler step along the ODE, dashed) plus a *martingale-noise* term $\gamma_k w_{k+1}$ (orange). The piecewise-linear interpolation $\bar{x}(\tau)$ (navy) is compared with the ODE solution $z^{(k)}(\tau)$ (gray) started at x_k . Because $\sum_k \gamma_k^2 < \infty$, the accumulated noise $\sum_j \gamma_j w_{j+1}$ converges almost surely, and a Gronwall estimate then forces the tracking gap $\sup_{\tau \in [\tau_k, \tau_k + T]} \|\bar{x}(\tau) - z^{(k)}(\tau)\| \rightarrow 0$.

ODE directly determine the almost sure limit set of the stochastic algorithm. The martingale noise affects transient behavior but does not alter the asymptotic outcome, provided it is properly scaled by diminishing step sizes.

At a conceptual level, the proof of Theorem 7.29 proceeds by first establishing the tracking property using martingale convergence arguments and discrete-time Gronwall inequalities, and

then invoking Lyapunov stability theory or invariance principles to characterize the limit set of the ODE. This two-stage structure, i.e., tracking followed by stability, lies at the heart of modern stochastic approximation theory.

7.4.5 ■ Mathematical Background and Preliminaries

We collect several probabilistic and analytical tools that will be used repeatedly in the convergence analysis of stochastic approximation algorithms.

Martingales

Definition 7.30 (Martingale). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\{\mathcal{F}_n\}_{n \geq 0}$ a filtration. A sequence of random vectors $\{Y_n\}_{n \geq 0}$ is called a martingale with respect to $\{\mathcal{F}_n\}$ if:

- (i) Y_n is $\mathcal{F}_n$ -measurable for all n ,
- (ii) $\mathbb{E}[\|Y_n\|] < \infty$ for all n ,
- (iii) $\mathbb{E}(Y_{n+1} \mid \mathcal{F}_n) = Y_n$ almost surely.

Example 7.31 Partial sums of martingale differences.

Let $\{w_{k+1}\}$ be adapted to $\{\mathcal{F}_{k+1}\}$ (so each w_{k+1} is $\mathcal{F}_{k+1}$ -measurable) and satisfy $\mathbb{E}(w_{k+1} \mid \mathcal{F}_k) = 0$ and $\mathbb{E}(\|w_{k+1}\|^2) < \infty$. Define $Y_k := \sum_{j=0}^{k-1} w_{j+1}$. Then $\{Y_k\}$ is a martingale with respect to $\{\mathcal{F}_k\}$.

This structure appears directly in stochastic approximation algorithms of the form $x_{k+1} = x_k + \gamma_k(h(x_k) + w_{k+1})$, where the cumulative noise term $\sum_{j=0}^{k-1} \gamma_j w_{j+1}$ is a scaled martingale. $\square$

Example 7.32 Sample noise in stochastic gradient descent.

Let $f(x) = \mathbb{E}_v[F(x, v)]$, so that $\nabla f(x) = \mathbb{E}_v[\nabla_x F(x, v)]$. Suppose v_k are i.i.d. samples and define $w_{k+1} = \nabla_x F(x_k, v_k) - \nabla f(x_k)$. Then $\mathbb{E}(w_{k+1} \mid \mathcal{F}_k) = 0$, so $\{w_{k+1}\}$ forms a martingale difference sequence. This is the canonical noise model underlying Robbins–Monro and stochastic gradient methods. $\square$

Martingale Convergence

Theorem 7.33 (Martingale Convergence Theorem). Let $\{Y_n\}$ be a martingale with respect to $\{\mathcal{F}_n\}$. Suppose that

$$\sum_{n=0}^{\infty} \mathbb{E}(\|Y_{n+1} - Y_n\|^2 \mid \mathcal{F}_n) < \infty \quad \text{almost surely.} \quad (7.66)$$

Then there exists a finite random vector Y_{∞} such that $Y_n \rightarrow Y_{\infty}$ almost surely.

Proof. Define the martingale difference sequence $D_{n+1} := Y_{n+1} - Y_n$, $\mathbb{E}(D_{n+1} \mid \mathcal{F}_n) = 0$. Then $Y_n = Y_0 + \sum_{k=0}^{n-1} D_{k+1}$. Condition (7.66) implies that $\sum_{n=0}^{\infty} \mathbb{E}(\|D_{n+1}\|^2) < \infty$. By Kolmogorov's convergence theorem for vector-valued martingale difference sequences, the series $\sum_n D_{n+1}$ converges almost surely. Hence Y_n converges almost surely to a finite random vector Y_{∞} . $\square$

Example 7.34 Almost sure convergence of accumulated noise.

Let $S_k := \sum_{j=0}^{k-1} \gamma_j w_{j+1}$, where $\{w_{k+1}\}$ is a martingale difference sequence with bounded conditional second moments, $\sup_j \mathbb{E}(\|w_{j+1}\|^2 \mid \mathcal{F}_j) \leq C < \infty$ a.s., and $\sum_{j=0}^{\infty} \gamma_j^2 < \infty$. Then

$$\sum_{j=0}^{\infty} \mathbb{E}(\|\gamma_j w_{j+1}\|^2 \mid \mathcal{F}_j) \leq C \sum_{j=0}^{\infty} \gamma_j^2 < \infty \quad \text{a.s.}$$

By Theorem 7.33, S_k converges almost surely. This fact is essential in showing that stochastic approximation iterates track the limiting ODE asymptotically. $\square$

Gronwall Inequalities

Gronwall inequalities are fundamental tools for bounding solutions of differential and difference inequalities.

Theorem 7.35 (Gronwall Inequality: Differential Form). Let $\phi : [0, T] \rightarrow \mathbb{R}$ be nonnegative and absolutely continuous, and suppose $\dot{\phi}(t) \leq L(t)\phi(t)$ for almost every $t \in [0, T]$, where $L(\cdot)$ is a nonnegative integrable function. Then

$$\phi(t) \leq \phi(0) \exp\left(\int_0^t L(\tau) d\tau\right), \quad \forall t \in [0, T].$$

In particular, for a constant $L \geq 0$ this reads $\phi(t) \leq \phi(0)e^{Lt}$.

Proof. Write $\Lambda(t) := \int_0^t L(\tau) d\tau$ and set $y(t) := \phi(t)e^{-\Lambda(t)}$, which is absolutely continuous. For almost every t ,

$$\dot{y}(t) = e^{-\Lambda(t)} (\dot{\phi}(t) - L(t)\phi(t)) \leq 0,$$

so y is nonincreasing and $y(t) \leq y(0) = \phi(0)$. Multiplying by $e^{\Lambda(t)}$ gives the stated bound. Note that the argument never divides by $\phi(t)$, so it remains valid where ϕ vanishes. $\square$

Theorem 7.36 (Gronwall Inequality: Integral Form). Let $\phi : [0, T] \rightarrow \mathbb{R}$ be bounded and nonnegative, and suppose

$$\phi(t) \leq B + \int_0^t L(\tau)\phi(\tau) d\tau, \quad \forall t \in [0, T],$$

for some constant $B \geq 0$ and nonnegative integrable function L . Then

$$\phi(t) \leq B \exp\left(\int_0^t L(\tau) d\tau\right), \quad \forall t \in [0, T].$$

Proof. Define $u(t) := B + \int_0^t L(\tau)\phi(\tau) d\tau$. Then $\phi(t) \leq u(t)$ and $\dot{u}(t) = L(t)\phi(t) \leq L(t)u(t)$. Applying Theorem 7.35 to u yields the desired bound. $\square$

Example 7.37 Stability of linear ODEs.

Consider the scalar ODE $\dot{\phi}(t) = a(t)\phi(t)$, $a(t) \leq L$, $\phi(0) \geq 0$. Then

$$\phi(t) = \phi(0) \exp\left(\int_0^t a(\tau) d\tau\right) \leq \phi(0)e^{Lt},$$

where the inequality uses $a(\tau) \leq L$ together with $\phi(0) \geq 0$. This is a direct application of the generalized Gronwall inequality (Theorem 7.35) and is used to bound the growth of error trajectories in ODE approximations. $\square$

Example 7.38 Perturbation of ODE solutions.

Let $\phi(t)$ and $\tilde{\phi}(t)$ satisfy $\dot{\phi}(t) = h(\phi(t))$ and $\dot{\tilde{\phi}}(t) = h(\tilde{\phi}(t)) + \delta(t)$, with $\|\delta(t)\| \leq \varepsilon$ and the same initial condition $\phi(0) = \tilde{\phi}(0)$. Then, $\|\phi(t) - \tilde{\phi}(t)\| \leq \frac{\varepsilon}{L}(e^{Lt} - 1) \leq \varepsilon t e^{Lt}$, where L is the Lipschitz constant of h . This estimate underlies the ODE method for stochastic approximation. $\square$

Discrete-Time Gronwall Inequality

We now state a discrete analogue that plays a central role in analyzing recursive algorithms.

Lemma 7.39 (Discrete-Time Gronwall Inequality). Let $K \geq 0$, and let $\{u_n\}_{n \geq 0}$ and $\{a_n\}_{n \geq 0}$ be nonnegative sequences satisfying

$$u_{n+1} \leq K + \sum_{m=0}^n a_m u_m, \quad \forall n \geq 0.$$

Then, for all $n \geq 0$,

$$u_n \leq K \exp\left(\sum_{m=0}^{n-1} a_m\right).$$

Proof. Define $S_n := \sum_{m=0}^{n-1} a_m$, $y_n := u_n e^{-S_n}$. Then

$$y_{n+1} = u_{n+1} e^{-S_{n+1}} \leq K e^{-S_{n+1}} + \sum_{m=0}^n a_m u_m e^{-S_{n+1}} = K e^{-S_{n+1}} + \sum_{m=0}^n a_m y_m e^{-(S_{n+1} - S_m)}.$$

Since $S_{n+1} - S_m \geq 0$, the sum is bounded by $\sum_{m=0}^n a_m y_m$. Iterating yields $y_n \leq K$, which implies the desired bound. $\square$

Remark 7.40. Lemma 7.39 is the discrete counterpart of the integral Gronwall inequality and is a fundamental tool for controlling error growth in numerical schemes, ODE approximations, and stochastic approximation algorithms.

Example 7.41 Bounding recursive error sequences.

Suppose a sequence $\{u_n\}$ satisfies $u_{n+1} \leq K + \sum_{m=0}^n a_m u_m$, $a_m \geq 0$. Such inequalities arise when comparing a numerical iterate to an exact ODE solution.

Lemma 7.39 implies

$$u_n \leq K \exp\left(\sum_{m=0}^{n-1} a_m\right),$$

providing a uniform bound on error growth. $\square$

Example 7.42 Tracking error in stochastic approximation.

In the ODE method, one encounters bounds of the form

$$\|z^{(k)}(\tau_u) - x_u\| \leq \sum_{j=k}^{u-1} \gamma_j L \|z^{(k)}(\tau_j) - x_j\| + \left\| \sum_{j=k}^{u-1} \gamma_j w_{j+1} \right\|.$$

Applying Lemma 7.39 allows one to control the tracking error uniformly over finite time windows. $\square$

7.4.6 ■ Stochastic Approximation Framework

We now present a standard convergence framework for stochastic approximation algorithms based on the ODE method. Consider the stochastic recursion

$$x_{k+1} = x_k + \gamma_k (h(x_k) + w_{k+1}), \quad x_0 \text{ given}, \quad (7.67)$$

where $\{x_k\} \subset \mathbb{R}^d$, $h : \mathbb{R}^d \rightarrow \mathbb{R}^d$, $\{\gamma_k\}$ is a step-size sequence, and $\{w_k\}$ is a noise process adapted to a filtration $\{\mathcal{F}_k\}$.

Assumption 7.43 (A1: Lipschitz continuity of h). *There exists $L > 0$ such that $\|h(x) - h(\bar{x})\| \leq L\|x - \bar{x}\|$, $\forall x, \bar{x} \in \mathbb{R}^d$.*

Assumption 7.44 (A2: Diminishing step sizes). *The step sizes satisfy*

$$\sum_{k=0}^{\infty} \gamma_k = \infty, \quad \sum_{k=0}^{\infty} \gamma_k^2 < \infty, \quad \gamma_k \rightarrow 0.$$

Assumption 7.45 (A3: Martingale difference noise). *There exists $K > 0$ such that, for all k ,*

$$\mathbb{E}(w_{k+1} \mid \mathcal{F}_k) = 0, \quad \mathbb{E}(\|w_{k+1}\|^2 \mid \mathcal{F}_k) \leq K(1 + \|x_k\|^2),$$

where $\mathcal{F}_k = \sigma(x_0, w_1, \dots, x_k, w_k)$.

Assumption 7.46 (A4: Bounded iterates). *$\sup_{k \geq 0} \|x_k\| < \infty$ almost surely.*

Assumption 7.47 (A5: Lyapunov stability). *There exists a continuously differentiable, positive, radially unbounded function $V : \mathbb{R}^d \rightarrow \mathbb{R}$ such that $\langle \nabla V(x), h(x) \rangle \leq 0, \forall x \in \mathbb{R}^d$, with strict inequality whenever $x \notin \mathcal{S}$, $\mathcal{S} := \{x : \nabla V(x) = 0\}$.*

Theorem 7.48 (ODE method: Lyapunov form). *Under assumptions (A1)–(A5), the sequence $\{x_k\}$ generated by (7.67) satisfies $V(x_k) \rightarrow V_\infty$ almost surely, for some random limit V_∞ , and every limit point of $\{x_k\}$ belongs almost surely to the internally chain transitive set of the ODE $\dot{z} = h(z)$. If, in addition, the ODE has a globally asymptotically stable equilibrium at the origin, then $x_k \rightarrow 0$ almost surely.*

Proof. The proof follows three connected ideas.

Continuous-time interpolation. Define the continuous-time scale $\tau_k := \sum_{j=0}^{k-1} \gamma_j$, $\tau_0 = 0$. By Assumption (A2), $\tau_k \rightarrow \infty$.

Define the piecewise linear interpolation $\bar{x}(\tau)$ of the iterates by

$$\bar{x}(\tau_k) = x_k, \quad \bar{x}(\tau) = x_k + \frac{\tau - \tau_k}{\gamma_k}(x_{k+1} - x_k), \quad \tau \in [\tau_k, \tau_{k+1}].$$

For each k , let $z^{(k)}(\tau)$ denote the solution of the ODE $\dot{z}(\tau) = h(z(\tau))$, with initial condition $z^{(k)}(\tau_k) = x_k$. The key idea is to show that $\bar{x}(\tau)$ asymptotically tracks the ODE trajectory $z^{(k)}(\tau)$ on finite time horizons, as illustrated in Figure 7.9.

Proof of the tracking property. Fix $T > 0$ and define $m := \inf\{u \geq k : \tau_u \geq \tau_k + T\}$.

For any $k \leq u \leq m$, the ODE solution satisfies

$$\begin{aligned} z^{(k)}(\tau_u) &= x_k + \int_{\tau_k}^{\tau_u} h(z^{(k)}(s)) ds \\ &= x_k + \sum_{j=k}^{u-1} \int_{\tau_j}^{\tau_{j+1}} h(z^{(k)}(s)) ds. \end{aligned}$$

The stochastic recursion yields

$$x_u = x_k + \sum_{j=k}^{u-1} \gamma_j h(x_j) + \sum_{j=k}^{u-1} \gamma_j w_{j+1}.$$

Subtracting the two expressions gives

$$\|z^{(k)}(\tau_u) - x_u\| \leq \sum_{j=k}^{u-1} \gamma_j \|h(z^{(k)}(\tau_j)) - h(x_j)\| \quad (7.68)$$

$$+ \left\| \sum_{j=k}^{u-1} \gamma_j w_{j+1} \right\| \quad (7.69)$$

$$+ \sum_{j=k}^{u-1} \int_{\tau_j}^{\tau_{j+1}} \|h(z^{(k)}(s)) - h(z^{(k)}(\tau_j))\| ds. \quad (7.70)$$

Term (7.68). By Assumption (A1), $\|h(z^{(k)}(\tau_j)) - h(x_j)\| \leq L\|z^{(k)}(\tau_j) - x_j\|$.

Term (7.69). Define $S_u := \sum_{j=0}^{u-1} \gamma_j w_{j+1}$. By Assumptions (A2)-(A4),

$$\sum_{k=0}^{\infty} \mathbb{E}(\|S_{k+1} - S_k\|^2 \mid \mathcal{F}_k) = \sum_{k=0}^{\infty} \gamma_k^2 \mathbb{E}(\|w_{k+1}\|^2 \mid \mathcal{F}_k) < \infty \quad \text{a.s.}$$

Hence, by the martingale convergence theorem, $S_u \rightarrow S_{\infty}$ almost surely. Therefore,

$$\sup_{k \leq u \leq m} \left\| \sum_{j=k}^{u-1} \gamma_j w_{j+1} \right\| \longrightarrow 0 \quad \text{a.s.}$$

Term (7.70). Using Lipschitz continuity of h , $\|h(z^{(k)}(s)) - h(z^{(k)}(\tau_j))\| \leq L \|z^{(k)}(s) - z^{(k)}(\tau_j)\|$. Since ODE solutions are locally Lipschitz on compact intervals and $\gamma_j \rightarrow 0$,

$$\sum_{j=k}^{u-1} \int_{\tau_j}^{\tau_{j+1}} \|h(z^{(k)}(s)) - h(z^{(k)}(\tau_j))\| ds = \mathcal{O} \left(\sum_{j=k}^{u-1} \gamma_j^2 \right) \longrightarrow 0.$$

Conclusion of the tracking argument. Combining all terms, we obtain

$$\|z^{(k)}(\tau_u) - x_u\| \leq \sum_{j=k}^{u-1} \gamma_j L \|z^{(k)}(\tau_j) - x_j\| + o(1).$$

Applying a discrete Gronwall inequality yields

$$\sup_{k \leq u \leq m} \|z^{(k)}(\tau_u) - x_u\| \longrightarrow 0 \quad \text{a.s.}$$

By linear interpolation,

$$\sup_{\tau \in [\tau_k, \tau_k + T]} \|z^{(k)}(\tau) - \bar{x}(\tau)\| \longrightarrow 0 \quad \text{a.s.}$$

which establishes the tracking property.

Lyapunov convergence. By Assumption (A5), $\frac{d}{d\tau} V(z(\tau)) = \langle \nabla V(z(\tau)), h(z(\tau)) \rangle \leq 0$, with strict inequality outside the invariant set $\mathcal{S} := \{x : \nabla V(x) = 0\}$. Thus, along the flow V is nonincreasing, and all ODE trajectories descend the Lyapunov level sets and approach $\mathcal{S}$; this is precisely the geometric mechanism depicted in Figure 7.8.

By the tracking property, $\bar{x}(\tau)$ remains asymptotically close to ODE trajectories on every finite horizon. Hence, $V(x_k) = V(\bar{x}(\tau_k)) \rightarrow V_{\infty}$ a.s. and all limit points of $\{x_k\}$ belong to $\mathcal{S}$.

Therefore, x_k converges almost surely to the invariant set of the ODE $\dot{z} = h(z)$, completing the proof. $\square$

7.4.7 ■ Results for Stochastic Gradient and Robbins–Monro Algorithms

The stochastic gradient iteration (7.59) and the fixed-point recursion (7.63) both fall within the general framework of stochastic approximation:

$$x_{k+1} = x_k + \gamma_k (h(x_k) + w_k), \quad (7.71)$$

where:

$$h(x) = \begin{cases} -\nabla f(x), & \text{(stochastic gradient)} \\ Hr - x, & \text{(fixed-point / Robbins–Monro)} \end{cases}$$

and $\{w_k\}$ is a martingale difference noise sequence adapted to a filtration $\{\mathcal{F}_k\}$.

This formulation allows us to apply the ODE method and martingale convergence results developed earlier.

Standing Assumptions

The following are Assumptions (A1)–(A4) of Section 7.4.6 written out for the particular drift $h = -\nabla f$:

(SG1) (Smoothness) f is continuously differentiable and ∇f is Lipschitz: $\|\nabla f(x) - \nabla f(\bar{x})\| \leq L\|x - \bar{x}\|$.

(SG2) (Step sizes)

$$\sum_{k=0}^{\infty} \gamma_k = \infty, \quad \sum_{k=0}^{\infty} \gamma_k^2 < \infty, \quad \gamma_k \rightarrow 0.$$

(SG3) (Martingale noise)

$$\mathbb{E}(w_k \mid \mathcal{F}_k) = 0, \quad \mathbb{E}(\|w_k\|^2 \mid \mathcal{F}_k) \leq C(1 + \|x_k\|^2).$$

(SG4) (Stability) $\sup_k \|x_k\| < \infty$ almost surely.

These assumptions are standard in stochastic approximation and are satisfied by a broad class of stochastic gradient and learning algorithms.

ODE Interpretation

Under the above assumptions, the stochastic recursion (7.71) asymptotically tracks the solution of the limiting ODE

$$\dot{z}(\tau) = h(z(\tau)). \quad (7.72)$$

In particular:

- For stochastic gradient descent, $\dot{z}(\tau) = -\nabla f(z(\tau))$.
- For Robbins–Monro fixed-point iterations, $\dot{z}(\tau) = Hz(\tau) - z(\tau)$.

Main Convergence Result

Theorem 7.49 (Almost sure convergence of stochastic gradient methods). *Under assumptions (SG1)–(SG4), the iterates generated by (7.59) satisfy:*

- Either $f(x_k) \rightarrow -\infty$, or $\{f(x_k)\}$ converges to a finite value almost surely.*
- $\lim_{k \rightarrow \infty} \nabla f(x_k) = 0$ almost surely.*
- Every limit point of $\{x_k\}$ is a stationary point of f .*

Proof. The stochastic gradient recursion can be written as

$$x_{k+1} = x_k - \gamma_k \nabla f(x_k) + \gamma_k w_k, \quad (7.73)$$

where $\{w_k\}$ is a martingale difference sequence satisfying

$$\mathbb{E}(w_k \mid \mathcal{F}_k) = 0, \quad \mathbb{E}(\|w_k\|^2 \mid \mathcal{F}_k) \leq C(1 + \|x_k\|^2).$$

Descent inequality. Since ∇f is Lipschitz continuous with constant L , the quadratic upper bound yields

$$f(x_{k+1}) \leq f(x_k) + \nabla f(x_k)^\top (x_{k+1} - x_k) + \frac{L}{2} \|x_{k+1} - x_k\|^2.$$

Substituting (7.73) gives

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) - \gamma_k \|\nabla f(x_k)\|^2 + \gamma_k \nabla f(x_k)^\top w_k \\ &\quad + \frac{L}{2} \gamma_k^2 \|\nabla f(x_k) + w_k\|^2. \end{aligned}$$

Conditional expectation. Taking conditional expectation with respect to $\mathcal{F}_k$ and using $\mathbb{E}(w_k \mid \mathcal{F}_k) = 0$,

$$\mathbb{E}[f(x_{k+1}) \mid \mathcal{F}_k] \leq f(x_k) - \gamma_k \|\nabla f(x_k)\|^2 + C' \gamma_k^2 (1 + \|\nabla f(x_k)\|^2),$$

for some constant $C' > 0$, where boundedness of $\{x_k\}$ has been used.

Almost sure summability. Summing over k and using $\sum_k \gamma_k^2 < \infty$, the Robbins–Siegmund supermartingale convergence theorem implies $\sum_{k=0}^{\infty} \gamma_k \|\nabla f(x_k)\|^2 < \infty$ a.s. Since $\sum_k \gamma_k = \infty$, this implies $\liminf_{k \rightarrow \infty} \|\nabla f(x_k)\| = 0$ a.s.

Full convergence of the gradient. Suppose, by contradiction, that $\limsup_{k \rightarrow \infty} \|\nabla f(x_k)\| > 0$. Then there exists $\varepsilon > 0$ and a subsequence $\{k_j\}$ such that $\|\nabla f(x_{k_j})\| \geq \varepsilon$ for all j . Since $\gamma_k > 0$ and $\sum_k \gamma_k = \infty$, the summability $\sum_k \gamma_k \|\nabla f(x_k)\|^2 < \infty$ would be violated. Hence, $\lim_{k \rightarrow \infty} \nabla f(x_k) = 0$ a.s.

Limit points. Let $\bar{x}$ be a limit point of $\{x_k\}$. By continuity of ∇f , $\nabla f(\bar{x}) = 0$, so every limit point is stationary.

Finally, since $f(x_{k+1}) \leq f(x_k) + \mathcal{O}(\gamma_k^2)$, and $\sum_k \gamma_k^2 < \infty$, either $f(x_k) \rightarrow -\infty$ or $\{f(x_k)\}$ converges to a finite value. $\square$

Consequences for Convex Objectives

Corollary 7.50 (Convergence for convex stochastic optimization). *If, in addition, f is convex, then the iterates x_k converge almost surely to the set of global minimizers of f . If f is strongly convex, the convergence is to the unique minimizer x^* .*

Proof. If f is convex, every stationary point is a global minimizer. Since all limit points of $\{x_k\}$ are stationary by Theorem 7.49, the result follows. If f is strongly convex, the minimizer is unique. Hence all convergent subsequences of $\{x_k\}$ share the same limit, implying $x_k \rightarrow x^*$ almost surely. $\square$

Convergence of Robbins–Monro Algorithms

Theorem 7.51 (Robbins–Monro convergence). *Consider the stochastic approximation scheme*

$$x_{k+1} = (1 - \gamma_k)x_k + \gamma_k(Hr_k + w_k),$$

where H is a contraction mapping with modulus $\rho < 1$ and $\{w_k\}$ is martingale difference noise. Under assumptions (SG2)–(SG4), $x_k \rightarrow x^*$ almost surely, where x^* is the unique solution of the fixed-point equation $Hr = x$.

Proof. Rewrite the recursion as

$$x_{k+1} - x^* = (1 - \gamma_k)(x_k - x^*) + \gamma_k(Hr_k - Hx^*) + \gamma_k w_k.$$

Using contraction of H , $\|Hr_k - Hx^*\| \leq \rho \|x_k - x^*\|$. Hence,

$$\|x_{k+1} - x^*\| \leq (1 - \gamma_k(1 - \rho))\|x_k - x^*\| + \gamma_k \|w_k\|.$$

Taking conditional expectations and applying the Robbins–Siegmund theorem, together with $\sum_k \gamma_k = \infty$ and $\sum_k \gamma_k^2 < \infty$, yields $x_k \rightarrow x^*$ almost surely. $\square$

Remark 7.52. These results show that stochastic gradient and Robbins–Monro algorithms asymptotically behave like their deterministic counterparts:

- stochastic gradient descent tracks the gradient flow $\dot{z} = -\nabla f(z)$,
- Robbins–Monro iterations track stable fixed-point dynamics.

Martingale noise is averaged out provided step sizes are diminishing and square-summable.

7.4.8 ■ Illustrative Examples

We now present concrete examples illustrating how classical stochastic algorithms fit into the stochastic approximation framework (7.67) and satisfy Assumptions (A1)–(A5). The prototype scheme analyzed throughout is summarized in Algorithm 7.4.

ALGORITHM 7.4. Stochastic Gradient / Robbins–Monro Method.

Require: initial point x_0 , step-size sequence $\{\gamma_k\}$ with $\gamma_k > 0$, $\sum_k \gamma_k = \infty$, $\sum_k \gamma_k^2 < \infty$.

- 1: **for** $k = 0, 1, 2, \dots$ **do**
- 2: draw a fresh sample ξ_{k+1} and obtain an unbiased estimate $g(x_k, \xi_{k+1})$ with $\mathbb{E}[g(x_k, \xi_{k+1}) \mid \mathcal{F}_k] = \nabla f(x_k)$;
- 3: update $x_{k+1} = x_k - \gamma_k g(x_k, \xi_{k+1})$;
- 4: **end for**

Equivalently $x_{k+1} = x_k + \gamma_k(h(x_k) + w_{k+1})$ with $h(x) = -\nabla f(x)$ and martingale-difference noise w_{k+1} .

Stochastic Gradient Descent for Strongly Convex Functions

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable, m -strongly convex, and with Lipschitz gradient. Suppose that instead of the true gradient, we observe a noisy estimate $g(x, \xi)$, where

ξ is a random sample and $\mathbb{E}[g(x, \xi)] = \nabla f(x)$. The stochastic gradient iteration is $x_{k+1} = x_k - \gamma_k g(x_k, \xi_{k+1})$. Writing

$$w_{k+1} = -(g(x_k, \xi_{k+1}) - \nabla f(x_k)),$$

the recursion becomes

$$x_{k+1} = x_k + \gamma_k (-\nabla f(x_k) + w_{k+1}),$$

which is of the form (7.67) with $h(x) = -\nabla f(x)$.

We now verify that the stochastic gradient recursion satisfies Assumptions (A1)–(A5) of the stochastic approximation framework. This allows us to invoke the ODE method and conclude almost sure convergence of the iterates. Throughout, we assume that f is continuously differentiable, its gradient is Lipschitz continuous, and f is strongly convex.

(A1) The map $h(x) = -\nabla f(x)$ is Lipschitz continuous because ∇f is Lipschitz by assumption. Hence $\|h(x) - h(\bar{x})\| = \|\nabla f(\bar{x}) - \nabla f(x)\| \leq L\|x - \bar{x}\|$.

(A2) The step-size conditions are satisfied for standard choices such as $\gamma_k = \frac{1}{k+1}$, since

$$\sum_{k=0}^{\infty} \gamma_k = \infty, \quad \sum_{k=0}^{\infty} \gamma_k^2 < \infty, \quad \gamma_k \rightarrow 0.$$

(A3) The noise sequence w_{k+1} is a martingale difference: $\mathbb{E}(w_{k+1} \mid \mathcal{F}_k) = 0$, and under standard moment assumptions $\mathbb{E}(\|w_{k+1}\|^2 \mid \mathcal{F}_k) \leq K(1 + \|x_k\|^2)$, so the conditional variance is controlled.

(A4) Since f is strongly convex and coercive, its sublevel sets are compact. The descent inequality $f(x_{k+1}) \leq f(x_k) + \mathcal{O}(\gamma_k^2)$ and the summability $\sum_k \gamma_k^2 < \infty$ imply that $\{f(x_k)\}$ remains bounded almost surely. Therefore $\{x_k\}$ remains in a compact sublevel set of f , and hence $\sup_k \|x_k\| < \infty$ a.s.

(A5) Let x^* denote the unique minimizer of f . Define the Lyapunov function $V(x) := f(x) - f(x^*)$. Then V is continuously differentiable, positive definite, and radially unbounded. Moreover,

$$\langle \nabla V(x), h(x) \rangle = \langle \nabla f(x), -\nabla f(x) \rangle = -\|\nabla f(x)\|^2 \leq 0,$$

with equality if and only if $\nabla f(x) = 0$, that is, $x = x^*$.

The limiting ODE associated with the recursion is $\dot{z} = -\nabla f(z)$, which is the gradient flow of f . Because f is strongly convex, this ODE has a unique globally asymptotically stable equilibrium at x^* . Therefore, by Theorem 7.48, $x_k \rightarrow x^*$ almost surely.

Example 7.53 Diminishing versus constant step size: rate and noise floor.

Take the quadratic $f(x) = \frac{1}{2}x^2$ with $\nabla f(x) = x$ and $x^* = 0$, and stochastic gradients $g(x_k, \xi_{k+1}) = x_k + w_{k+1}$ with $\mathbb{E}[w_{k+1} \mid \mathcal{F}_k] = 0$ and $\mathbb{E}[w_{k+1}^2 \mid \mathcal{F}_k] = \sigma^2$.

Step 1 (mean-square recursion). The iteration $x_{k+1} = x_k - \gamma_k g(x_k, \xi_{k+1}) = (1 - \gamma_k)x_k - \gamma_k w_{k+1}$. Squaring,

$$x_{k+1}^2 = (1 - \gamma_k)^2 x_k^2 - 2(1 - \gamma_k)\gamma_k x_k w_{k+1} + \gamma_k^2 w_{k+1}^2.$$

Taking the conditional expectation given $\mathcal{F}_k$, the cross term vanishes because $\mathbb{E}[w_{k+1} \mid \mathcal{F}_k] = 0$, while $\mathbb{E}[w_{k+1}^2 \mid \mathcal{F}_k] = \sigma^2$. Taking expectations and writing $a_k = \mathbb{E}[x_k^2]$ yields the *exact* deterministic recursion

$$a_{k+1} = (1 - \gamma_k)^2 a_k + \gamma_k^2 \sigma^2, \quad a_1 = 1, \quad \sigma^2 = 1,$$

with the recursion running for $k \geq 1$ (so that $\gamma_k = 1/k$ is defined). We compare the diminishing schedule $\gamma_k = 1/k$ with a constant step $\gamma_k = 0.1$.

Step 2 (constant step: noise floor). For $\gamma_k \equiv \gamma$ a fixed point a_∞ satisfies $a_\infty = (1 - \gamma)^2 a_\infty + \gamma^2 \sigma^2$, hence

$$a_\infty = \frac{\gamma^2 \sigma^2}{1 - (1 - \gamma)^2} = \frac{\gamma \sigma^2}{2 - \gamma}.$$

For $\gamma = 0.1$, $\sigma^2 = 1$ this gives $a_\infty = 0.1/1.9 \approx 5.26 \times 10^{-2}$: the variance injected at every step is never annealed away, so the error stalls at this floor.

Step 3 (diminishing step: $\Theta(1/k)$ rate). For $\gamma_k = 1/k$, substitute the ansatz $a_k \sim c/k$. Then $a_{k+1} - a_k \sim -c/k^2$, while the recursion gives

$$a_{k+1} - a_k = \left[(1 - \frac{1}{k})^2 - 1\right] a_k + \frac{\sigma^2}{k^2} = -\frac{2}{k} a_k + O\left(\frac{a_k}{k^2}\right) + \frac{\sigma^2}{k^2} \sim \frac{-2c + \sigma^2}{k^2}.$$

Matching the two expressions gives $-c = -2c + \sigma^2$, i.e. $c = \sigma^2$, so $a_k \sim \sigma^2/k$ and $k a_k \rightarrow \sigma^2 = 1$. The schedule satisfies the Robbins–Monro conditions $\sum_k \gamma_k = \infty$, $\sum_k \gamma_k^2 < \infty$, matching the almost-sure convergence established above.

k	$\gamma_k = 1/k$: a_k	$\gamma_k = 0.1$: a_k
1	1.00	1.00
5	2.50×10^{-1}	4.60×10^{-1}
10	1.11×10^{-1}	1.95×10^{-1}
20	5.26×10^{-2}	6.99×10^{-2}
50	2.04×10^{-2}	5.27×10^{-2}
100	1.01×10^{-2}	5.26×10^{-2}

These two schedules capture the central trade-off of stochastic approximation: the constant step contracts fast but stalls at the noise floor, whereas the diminishing step converges to the true minimiser, at the slower $\Theta(1/k)$ rate. Figure 7.10 plots both mean-square trajectories on a log–log scale against the $1/k$ reference line. $\square$

Example 7.54 Robbins–Monro Root-Finding.

Let $H : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a contraction mapping with modulus $\rho < 1$, and let x^* denote

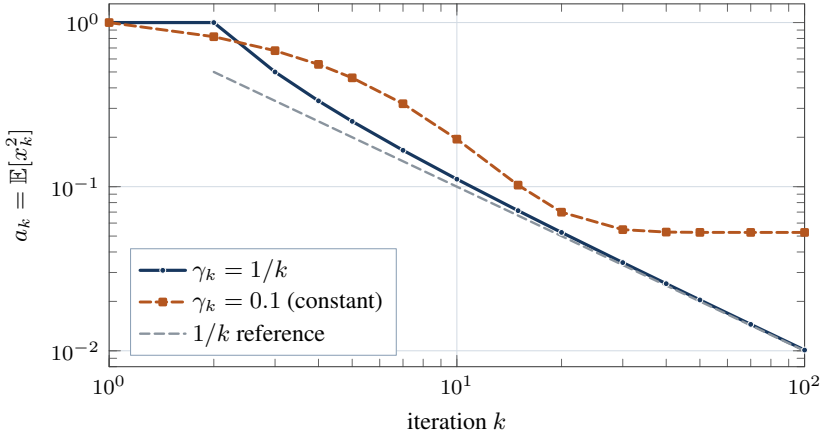


Figure 7.10: Mean-square error of stochastic gradient descent on $f(x) = \frac{1}{2}x^2$. The diminishing step size drives $a_k \rightarrow 0$ at rate $\Theta(1/k)$; the constant step stalls at the noise floor $a_\infty \approx 5.3 \times 10^{-2}$.

its unique fixed point, i.e., $H(x^*) = x^*$. Suppose that instead of observing $H(x)$ directly, we observe noisy evaluations of the form $\tilde{H}(x, \xi) = H(x) + \text{noise}$, where the noise has zero conditional mean.

The Robbins–Monro iteration is

$$x_{k+1} = x_k + \gamma_k (\tilde{H}(x_k, \xi_{k+1}) - x_k).$$

Rewriting,

$$x_{k+1} = x_k + \gamma_k (H(x_k) - x_k) + \gamma_k (\tilde{H}(x_k, \xi_{k+1}) - H(x_k)).$$

Defining $h(x) = H(x) - x$ and $w_{k+1} = \tilde{H}(x_k, \xi_{k+1}) - H(x_k)$, the recursion takes the stochastic approximation form

$$x_{k+1} = x_k + \gamma_k (h(x_k) + w_{k+1}).$$

The associated limiting ODE is $\dot{z} = H(z) - z$. To study its stability, consider the Lyapunov function $V(x) = \frac{1}{2}\|x - x^*\|^2$. Since H is a contraction, $\|H(x) - H(x^*)\| \leq \rho\|x - x^*\|$, and using $H(x^*) = x^*$ we compute

$$\langle x - x^*, h(x) \rangle = \langle x - x^*, H(x) - x \rangle = \langle x - x^*, H(x) - H(x^*) \rangle - \|x - x^*\|^2.$$

By Cauchy–Schwarz and the contraction property,

$$\langle x - x^*, H(x) - H(x^*) \rangle \leq \rho\|x - x^*\|^2,$$

and therefore

$$\langle x - x^*, h(x) \rangle \leq -(1 - \rho)\|x - x^*\|^2.$$

Thus along solutions of the ODE,

$$\frac{d}{d\tau} V(z(\tau)) = \langle z - x^*, H(z) - z \rangle \leq -(1 - \rho)\|z - x^*\|^2,$$

so x^* is globally asymptotically stable.

Under the standard stochastic approximation assumptions on the step sizes and noise, the Robbins–Monro iteration therefore converges almost surely to x^* . $\square$

Example 7.55 Linear Stochastic Approximation.

Consider the stochastic approximation scheme with drift $h(x) = -Ax$, where $A \in \mathbb{R}^{d \times d}$ is symmetric positive definite. The recursion becomes

$$x_{k+1} = x_k - \gamma_k Ax_k + \gamma_k w_{k+1}.$$

This fits the standard SA form $x_{k+1} = x_k + \gamma_k(h(x_k) + w_{k+1})$. The associated limiting ODE is $\dot{z} = -Az$. Since A is positive definite, its eigenvalues satisfy $\lambda_{\min}(A) > 0$. The solution of the ODE is $z(\tau) = e^{-A\tau}z(0)$, which converges exponentially fast to the origin, because $\|e^{-At}\| \leq e^{-\lambda_{\min}(A)t}$.

To verify Lyapunov stability, consider $V(x) = \frac{1}{2}x^\top x$. Then $\nabla V(x) = x$, and

$$\langle \nabla V(x), h(x) \rangle = \langle x, -Ax \rangle = -x^\top Ax \leq -\lambda_{\min}(A)\|x\|^2.$$

Thus $\dot{V}(z(\tau)) \leq -\lambda_{\min}(A)\|z(\tau)\|^2$, so the origin is globally asymptotically stable for the ODE, with exponential rate. Under the standard stochastic approximation assumptions (Lipschitz drift, diminishing step sizes, martingale noise, and bounded iterates), the discrete iterates track this stable ODE. Consequently, $x_k \rightarrow 0$ almost surely. $\square$

7.5 ■ Newton's Method

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice continuously differentiable. Newton's method is a second-order optimization algorithm derived from a local quadratic approximation of the objective function.

7.5.1 ■ Newton Direction and Quadratic Model

Given an iterate $x_k \in \mathbb{R}^n$, the *pure Newton iteration* is defined by

$$x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k), \quad (7.74)$$

provided the Hessian $\nabla^2 f(x_k)$ is invertible. The corresponding *Newton direction* is

$$d_k = -[\nabla^2 f(x_k)]^{-1} \nabla f(x_k). \quad (7.75)$$

The Newton direction arises as the unique minimizer of the quadratic Taylor model of f at x_k ,

$$m_k(d) := f(x_k) + \nabla f(x_k)^\top d + \frac{1}{2} d^\top \nabla^2 f(x_k) d. \quad (7.76)$$

Indeed, differentiating $m_k(d)$ with respect to d and setting the result equal to zero yields $\nabla m_k(d) = \nabla f(x_k) + \nabla^2 f(x_k)d = 0$, which gives $d = d_k$ whenever $\nabla^2 f(x_k)$ is nonsingular. If, in addition, $\nabla^2 f(x_k) \succ 0$, then d_k is the unique global minimizer of m_k .

Definition 7.56 (Newton decrement). The Newton decrement at x_k is defined as

$$\lambda(x_k) := \left(\nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k) \right)^{1/2}. \quad (7.77)$$

Evaluating the quadratic model (7.76) at the minimizer d_k yields

$$m_k(0) - m_k(d_k) = -\nabla f(x_k)^\top d_k - \frac{1}{2} d_k^\top \nabla^2 f(x_k) d_k.$$

Using $\nabla^2 f(x_k) d_k = -\nabla f(x_k)$, we obtain

$$m_k(0) - m_k(d_k) = \frac{1}{2} \nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k) = \frac{1}{2} \lambda(x_k)^2.$$

Thus $\lambda(x_k)^2/2$ represents the predicted decrease in the quadratic model.

Moreover,

$$\lambda(x_k)^2 = d_k^\top \nabla^2 f(x_k) d_k = \|d_k\|_{\nabla^2 f(x_k)}^2,$$

so the Newton decrement is precisely the norm of the Newton step in the local Hessian metric. When $\nabla^2 f(x_k) \succ 0$, the condition $\lambda(x_k) = 0$ is equivalent to $\nabla f(x_k) = 0$.

7.5.2 ■ Newton's Method with Backtracking Line Search

To ensure global convergence from arbitrary initial points, the pure Newton iteration is combined with a backtracking line search. This yields a damped Newton method that guarantees sufficient decrease at every iteration.

Let $\varepsilon > 0$, $\sigma \in (0, 1/2)$, and $\beta \in (0, 1)$. Given an initial point $x_0 \in \mathbb{R}^n$, generate the sequence $\{x_k\}$ as follows.

ALGORITHM 7.5. Newton's Method with Backtracking Line Search.

Require: initial point $x_0 \in \mathbb{R}^n$, tolerance $\varepsilon > 0$, parameters $\sigma \in (0, 1/2)$, $\beta \in (0, 1)$.

- 1: **for** $k = 0, 1, 2, \dots$ **do**
- 2: compute the Newton direction $d_k = -[\nabla^2 f(x_k)]^{-1} \nabla f(x_k) \triangleright \nabla^2 f(x_k)$ assumed nonsingular
- 3: compute the Newton decrement $\lambda(x_k)^2 = \nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$;
- 4: **if** $\frac{1}{2} \lambda(x_k)^2 \leq \varepsilon$ **then**
- 5: **stop** and return x_k $\triangleright$ small predicted decrease of the quadratic model
- 6: **end if**
- 7: set $\gamma_k = 1$;
- 8: **while** $f(x_k + \gamma_k d_k) > f(x_k) + \sigma \gamma_k \nabla f(x_k)^\top d_k$ **do** $\triangleright$ Armijo backtracking
- 9: $\gamma_k \leftarrow \beta \gamma_k$;
- 10: **end while**
- 11: update $x_{k+1} = x_k + \gamma_k d_k$;
- 12: **end for**

7.5.3 ■ Global and Local Convergence Analysis

Newton's method exhibits two distinct regimes: a *damped global phase*, ensuring descent from arbitrary initial points, and a *local quadratic phase* once the iterates enter a neighborhood of the minimizer.

Let $L_0 := \{x \in \mathbb{R}^n \mid f(x) \leq f(x_0)\}$ denote the level set determined by the initial point x_0 .

Assumption 7.57 (Regularity and curvature). *The following properties hold on L_0 :*

- (1) *The level set L_0 is closed and bounded.*
- (2) *f is strongly convex with parameter $m > 0$: $\nabla^2 f(x) \succeq mI, \forall x \in L_0$.*
- (3) *The Hessian is Lipschitz continuous with constant $L > 0$:*

$$\|\nabla^2 f(x) - \nabla^2 f(\bar{x})\| \leq L\|x - \bar{x}\|, \quad \forall x, \bar{x} \in L_0.$$

- (4) *The Hessian is uniformly bounded above: $\nabla^2 f(x) \preceq MI, \forall x \in L_0$.*

Under Assumption 7.57, the function f admits a unique minimizer $x^* \in L_0$.

Theorem 7.58 (Global convergence: damped Newton phase). *Under Assumption 7.57, there exists a constant $\eta \in (0, \frac{2m^2}{L})$ such that if $\|\nabla f(x_k)\| \geq \eta$, then the backtracking line search accepts a step size satisfying $\gamma_k \geq \beta \frac{m}{M}$, and the objective decrease satisfies*

$$f(x_{k+1}) - f(x_k) \leq -\delta, \quad \delta := \sigma\beta \frac{m}{M^2} \eta^2. \quad (7.78)$$

Consequently, this phase terminates after finitely many iterations.

Proof. Let $d_k = -[\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$ be the Newton direction. Since $\nabla^2 f(x) \preceq MI$ on L_0 , the standard quadratic upper bound gives, for all $\gamma \in [0, 1]$,

$$f(x_k + \gamma d_k) \leq f(x_k) + \gamma \nabla f(x_k)^\top d_k + \frac{\gamma^2 M}{2} \|d_k\|^2. \quad (7.79)$$

First observe

$$\nabla f(x_k)^\top d_k = -\nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k) = -\lambda(x_k)^2.$$

Strong convexity implies

$$\nabla^2 f(x_k) \succeq mI \quad \Rightarrow \quad \|d_k\| \leq \frac{1}{m} \|\nabla f(x_k)\|,$$

and also $\|d_k\|^2 \leq \frac{1}{m} \lambda(x_k)^2$. Substituting into (7.79) yields

$$f(x_k + \gamma d_k) \leq f(x_k) - \gamma \lambda(x_k)^2 + \frac{\gamma^2 M}{2m} \lambda(x_k)^2 = f(x_k) - \gamma \left(1 - \frac{\gamma M}{2m}\right) \lambda(x_k)^2.$$

Choosing $\gamma = \frac{m}{M}$ gives $f(x_k + \gamma d_k) \leq f(x_k) - \frac{m}{2M} \lambda(x_k)^2$.

Since $\sigma < \frac{1}{2}$, this implies the Armijo condition

$$f(x_k + \gamma d_k) \leq f(x_k) + \sigma \gamma \nabla f(x_k)^\top d_k.$$

Hence the backtracking line search accepts some $\gamma_k \geq \beta \frac{m}{M}$. Using

$$\lambda(x_k)^2 = \nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k) \geq \frac{1}{M} \|\nabla f(x_k)\|^2,$$

we obtain

$$f(x_{k+1}) - f(x_k) \leq -\sigma \gamma_k \lambda(x_k)^2 \leq -\sigma \beta \frac{m}{M^2} \|\nabla f(x_k)\|^2.$$

If $\|\nabla f(x_k)\| \geq \eta$, then (7.78) follows immediately. Since f is bounded below on L_0 , only finitely many decreases of magnitude at least δ can occur. Thus the damped phase terminates in finite time. $\square$

Theorem 7.59 (Local quadratic convergence). *Under Assumption 7.57, if $\|\nabla f(x_k)\| < \eta$, then the full Newton step ($\gamma_k = 1$) is accepted and*

$$\|\nabla f(x_{k+1})\| \leq \frac{L}{2m^2} \|\nabla f(x_k)\|^2. \quad (7.80)$$

Proof. The proof proceeds as follows.

Second-order expansion of the gradient. By the mean-value form of Taylor's theorem,

$$\nabla f(x_k + d_k) = \nabla f(x_k) + \int_0^1 \nabla^2 f(x_k + sd_k) d_k ds.$$

Add and subtract $\nabla^2 f(x_k) d_k$ inside the integral to obtain

$$\begin{aligned} \nabla f(x_k + d_k) &= \nabla f(x_k) + \nabla^2 f(x_k) d_k \\ &\quad + \int_0^1 (\nabla^2 f(x_k + sd_k) - \nabla^2 f(x_k)) d_k ds. \end{aligned}$$

Since d_k is the Newton direction, $\nabla f(x_k) + \nabla^2 f(x_k) d_k = 0$, and therefore

$$\nabla f(x_k + d_k) = \int_0^1 (\nabla^2 f(x_k + sd_k) - \nabla^2 f(x_k)) d_k ds.$$

Use of Lipschitz continuity of the Hessian. By Assumption 7.57, $\|\nabla^2 f(x) - \nabla^2 f(\bar{x})\| \leq L\|x - \bar{x}\|$. Hence, for each $s \in [0, 1]$, $\|\nabla^2 f(x_k + sd_k) - \nabla^2 f(x_k)\| \leq Ls\|d_k\|$.

Therefore,

$$\begin{aligned} \|\nabla f(x_k + d_k)\| &\leq \int_0^1 \|\nabla^2 f(x_k + sd_k) - \nabla^2 f(x_k)\| \|d_k\| ds \\ &\leq \int_0^1 Ls\|d_k\|^2 ds = \frac{L}{2} \|d_k\|^2. \end{aligned}$$

This yields $\|\nabla f(x_{k+1})\| \leq \frac{L}{2} \|d_k\|^2$.

Bounding the Newton direction. Strong convexity implies $\nabla^2 f(x_k) \succeq mI$, so $\|[\nabla^2 f(x_k)]^{-1}\| \leq \frac{1}{m}$. Hence,

$$\|d_k\| = \|[\nabla^2 f(x_k)]^{-1} \nabla f(x_k)\| \leq \frac{1}{m} \|\nabla f(x_k)\|.$$

Substituting into the previous inequality gives

$$\|\nabla f(x_{k+1})\| \leq \frac{L}{2m^2} \|\nabla f(x_k)\|^2,$$

which establishes (7.80).

Acceptance of the full Newton update. It remains to show that the full step $\gamma_k = 1$ satisfies the Armijo condition. The quadratic upper bound (7.79) is *not* adequate here: with $\gamma = 1$ it produces the coefficient $1 - \frac{M}{2m}$, which is negative as soon as $M > 2m$ and therefore certifies no decrease at all. What is needed is the sharper cubic bound supplied by the Lipschitz Hessian.

Integrating the Hessian twice and using $\|\nabla^2 f(x_k + sd_k) - \nabla^2 f(x_k)\| \leq Ls\|d_k\|$ gives

$$f(x_k + d_k) \leq f(x_k) + \nabla f(x_k)^\top d_k + \frac{1}{2} d_k^\top \nabla^2 f(x_k) d_k + \frac{L}{6} \|d_k\|^3.$$

For the Newton direction, $\nabla f(x_k)^\top d_k = -\lambda(x_k)^2$ and $d_k^\top \nabla^2 f(x_k) d_k = \nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k) = \lambda(x_k)^2$, so the first two curvature terms combine to $-\frac{1}{2}\lambda(x_k)^2$:

$$f(x_k + d_k) \leq f(x_k) - \frac{1}{2}\lambda(x_k)^2 + \frac{L}{6} \|d_k\|^3. \quad (7.81)$$

Since $\nabla^2 f(x_k) \succeq mI$ we have $\|d_k\|^2 \leq \frac{1}{m}\lambda(x_k)^2$, hence $\|d_k\|^3 \leq m^{-3/2}\lambda(x_k)^3$, and (7.81) becomes

$$f(x_k + d_k) \leq f(x_k) - \lambda(x_k)^2 \left(\frac{1}{2} - \frac{L}{6m^{3/2}} \lambda(x_k) \right).$$

The Armijo test at $\gamma_k = 1$ asks for $f(x_k + d_k) \leq f(x_k) - \sigma\lambda(x_k)^2$, which therefore holds as soon as

$$\frac{L}{6m^{3/2}} \lambda(x_k) \leq \frac{1}{2} - \sigma, \quad \text{i.e.} \quad \lambda(x_k) \leq \frac{3(1 - 2\sigma)m^{3/2}}{L}.$$

Because $\lambda(x_k) \leq \|\nabla f(x_k)\|/\sqrt{m}$, it suffices that $\|\nabla f(x_k)\| \leq 3(1 - 2\sigma)m^2/L$. Accordingly we fix the threshold of Theorem 7.58 to be

$$0 < \eta \leq \min \left\{ \frac{2m^2}{L}, \frac{3(1 - 2\sigma)m^2}{L} \right\}, \quad (7.82)$$

which is positive because $\sigma < \frac{1}{2}$. With this choice, whenever $\|\nabla f(x_k)\| < \eta$ the unit step passes the Armijo test on its first trial, so $\gamma_k = 1$ and the quadratic estimate (7.80) applies. $\square$

Corollary 7.60 (Quadratic convergence rate). Let $c := L/(2m^2)$. If x_k lies in the local convergence region, then for all $\ell \geq 1$,

$$\|\nabla f(x_{k+\ell})\| \leq \frac{1}{c} (c \|\nabla f(x_k)\|)^{2^\ell}.$$

Proof. From Theorem 7.59, once $\|\nabla f(x_k)\| < \eta$, we have

$$\|\nabla f(x_{j+1})\| \leq c \|\nabla f(x_j)\|^2, \quad c := \frac{L}{2m^2}. \quad (7.83)$$

Applying (7.83) repeatedly yields $\|\nabla f(x_{k+1})\| \leq c\|\nabla f(x_k)\|^2$,

$$\|\nabla f(x_{k+2})\| \leq c\|\nabla f(x_{k+1})\|^2 \leq c(c\|\nabla f(x_k)\|^2)^2 = c^3\|\nabla f(x_k)\|^4,$$

and, by induction,

$$\|\nabla f(x_{k+\ell})\| \leq c^{(2^\ell-1)}\|\nabla f(x_k)\|^{2^\ell}.$$

Rewriting the right-hand side,

$$c^{(2^\ell-1)}\|\nabla f(x_k)\|^{2^\ell} = \frac{1}{c} (c\|\nabla f(x_k)\|)^{2^\ell},$$

which is the stated bound. The factor $1/c$ cannot simply be dropped: when $c < 1$ it exceeds one, so discarding it would claim a *stronger* inequality than the recursion delivers. What matters for the rate is that $c\|\nabla f(x_k)\| < 1$ in the local region, by the choice of η in (7.82), so the bracketed quantity is raised to a doubly exponentially growing power and the convergence is quadratic. $\square$

Theorem 7.61 (Complete convergence result). *Under Assumption 7.57, Newton's method with backtracking line search:*

- (i) *generates a sequence $\{x_k\} \subset L_0$,*
- (ii) *decreases the objective globally until entering a neighborhood of the minimizer,*
- (iii) *converges quadratically to the unique minimizer x^* .*

Moreover,

$$\lambda(x_k) \rightarrow 0, \quad \frac{\lambda(x_k)^2}{2} \text{ accurately predicts the local objective decrease.}$$

Proof. The proof follows the evolution from global descent to the local quadratic regime.

Invariance of the level set. By the Armijo condition, $f(x_{k+1}) \leq f(x_k)$, so the sequence $\{f(x_k)\}$ is nonincreasing. Hence $x_k \in L_0$, $\forall k$. Since L_0 is closed and bounded (Assumption 7.57), the sequence $\{x_k\}$ is bounded.

Global descent phase. By Theorem 7.58, whenever $\|\nabla f(x_k)\| \geq \eta$, the decrease satisfies $f(x_{k+1}) - f(x_k) \leq -\delta$, for some fixed $\delta > 0$. Since f is bounded below on L_0 , only finitely many such decreases can occur. Therefore there exists k_0 such that $\|\nabla f(x_k)\| < \eta \forall k \geq k_0$.

Thus the sequence eventually enters the local convergence region.

Local quadratic convergence. For all $k \geq k_0$, Theorem 7.59 applies, yielding

$$\|\nabla f(x_{k+1})\| \leq \frac{L}{2m^2} \|\nabla f(x_k)\|^2.$$

Hence $\|\nabla f(x_k)\| \rightarrow 0$ quadratically.

Convergence of the iterates. Strong convexity implies the gradient inequality $\|\nabla f(x)\| \geq m\|x - x^*\|$. Indeed,

$$\langle \nabla f(x) - \nabla f(x^*), x - x^* \rangle \geq m\|x - x^*\|^2,$$

and since $\nabla f(x^*) = 0$,

$$\|\nabla f(x)\| \|x - x^*\| \geq m\|x - x^*\|^2.$$

Canceling $\|x - x^*\| \neq 0$ gives $\|x - x^*\| \leq \frac{1}{m}\|\nabla f(x)\|$. Thus

$$\|x_k - x^*\| \leq \frac{1}{m}\|\nabla f(x_k)\| \rightarrow 0,$$

so $x_k \rightarrow x^*$.

Uniqueness of the minimizer follows from strong convexity.

Behavior of the Newton decrement. By definition,

$$\lambda(x_k)^2 = \nabla f(x_k)^\top [\nabla^2 f(x_k)]^{-1} \nabla f(x_k).$$

Since

$$\frac{1}{M}\|\nabla f(x_k)\|^2 \leq \lambda(x_k)^2 \leq \frac{1}{m}\|\nabla f(x_k)\|^2,$$

we obtain $\lambda(x_k) \rightarrow 0$. Moreover, in the local region, $f(x_k) - f(x_{k+1}) = \frac{1}{2}\lambda(x_k)^2 + \mathcal{O}(\lambda(x_k)^3)$, and the quadratic model prediction $\lambda(x_k)^2/2$ asymptotically matches the true objective decrease.

□

Example 7.62 Quadratic versus linear convergence: Newton against gradient descent.

Consider the strictly convex scalar objective

$$f(x) = x - \ln x, \quad x > 0, \quad f'(x) = 1 - \frac{1}{x}, \quad f''(x) = \frac{1}{x^2},$$

with unique minimiser $x^* = 1$ (since $f'(1) = 0$) and $f''(1) = 1$. We compare Newton's method and gradient descent, deriving the error recursion of each.

Newton's method. The Newton step is

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} = x_k - \left(1 - \frac{1}{x_k}\right)x_k^2 = x_k - (x_k^2 - x_k) = 2x_k - x_k^2.$$

Introduce the error $e_k = x_k - 1$. Then

$$e_{k+1} = x_{k+1} - 1 = 2x_k - x_k^2 - 1 = -(x_k - 1)^2 = -e_k^2, \quad \text{so} \quad |e_{k+1}| = |e_k|^2.$$

This is *exact* quadratic convergence with constant 1: the number of correct digits doubles at each step, matching Theorem 7.59.

Gradient descent. With constant step γ , $x_{k+1} = x_k - \gamma(1 - \frac{1}{x_k})$. Writing $1 - \frac{1}{x_k} = \frac{e_k}{1 + e_k}$ and linearising near $x^* = 1$,

$$e_{k+1} = e_k - \gamma \frac{e_k}{1 + e_k} = (1 - \gamma)e_k + O(e_k^2).$$

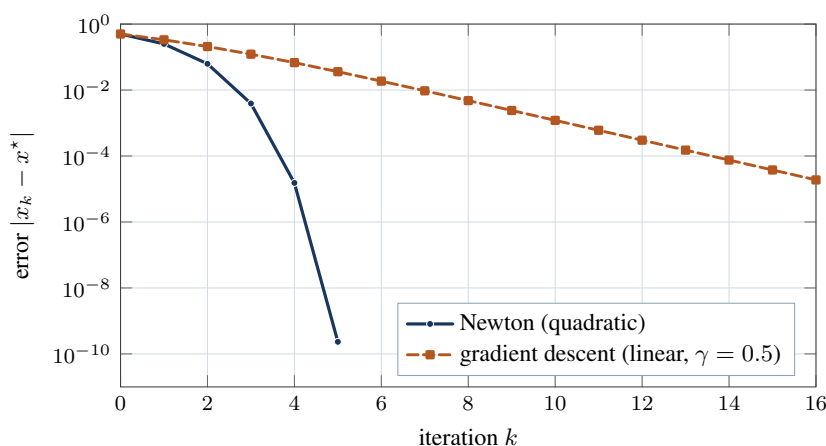


Figure 7.11: Error of Newton’s method versus gradient descent on $f(x) = x - \ln x$, $x^* = 1$, from $x_0 = 1.5$. On the logarithmic scale gradient descent is a straight line (constant factor $|1 - \gamma| = 0.5$ per step), while Newton bends sharply downward as the error is squared each iteration.

Thus the error contracts by the constant factor $|1 - \gamma|$ per step (here $f''(x^*) = 1$): *linear* convergence, adding a fixed number of digits per iteration. We take $\gamma = 0.5$, giving asymptotic factor 0.5. Starting both methods from $x_0 = 1.5$:

k	Newton $ e_k $	Gradient descent $ e_k $
0	5.00×10^{-1}	5.00×10^{-1}
1	2.50×10^{-1}	3.33×10^{-1}
2	6.25×10^{-2}	2.08×10^{-1}
3	3.91×10^{-3}	1.22×10^{-1}
4	1.53×10^{-5}	6.77×10^{-2}
5	2.33×10^{-10}	3.60×10^{-2}

The Newton errors follow $0.5 \rightarrow 0.5^2 \rightarrow 0.5^4 \rightarrow 0.5^8 \rightarrow \dots$, each is the square of the previous, whereas gradient descent multiplies by ≈ 0.5 each step. Reaching accuracy 10^{-10} takes Newton about 5 iterations but gradient descent about 33 (Figure 7.11). The price of Newton’s speed is the per-iteration cost of forming and inverting the Hessian, and the need for damping (Algorithm 7.5) to secure this fast local behaviour from arbitrary starting points: from x_0 far from 1 the undamped map $2x - x^2$ can leave the domain $(0, \infty)$. $\square$

7.6 ■ Further Reading

This chapter develops foundational methods for deterministic and stochastic optimization. The following references provide deeper treatments and modern applications. First-order methods form the backbone of modern deep learning. For optimization perspectives on neural networks, see [53, 25]. For large-scale convex and nonconvex first-order theory, see [88, 11]. Adaptive and

stochastic gradient methods widely used in practice (e.g., SGD, momentum, Adam) are discussed in [71]. Gradient-based algorithms for matrix optimization, including low-rank factorization and semidefinite problems, are treated in [1, 41]. For convex relaxations and matrix completion, see [31]. Applications to control and signal processing appear in [26].

Classical foundations of stochastic approximation originate in [100]. Comprehensive modern treatments are given in [75, 24]. The ODE method and martingale techniques are central tools in analyzing convergence. Stochastic approximation was originally developed for root-finding problems of the form $h(x) = 0$. Connections to fixed-point iterations and monotone operators are developed in [72]. In reinforcement learning, temporal-difference and Q-learning algorithms can be viewed as stochastic approximation schemes. For foundational treatments, see [114, 122, 117]. These works interpret learning dynamics as noisy approximations to contraction mappings or Bellman fixed points.

7.7 ■ Exercises

Exercise 7.1 (Total Variation Image Denoising). In this exercise, we consider total variation denoising for images, which aims to reduce noise while preserving edges. Let $Y \in \mathbb{R}^{n \times m}$ be a noisy image, where each element of Y is the pixel value of an image of size $n \times m$. From the two-dimensional image data Y , extract each column vector, say $y \in \mathbb{R}^n$, and solve the following optimization problem column by column:

$$\min_{x \in \mathbb{R}^n} \|x - y\|_2^2 + \lambda \sum_{i=1}^{n-1} |x_{i+1} - x_i|. \quad (7.84)$$

The first term measures fidelity to the data, while the second term is the total variation regularization. Develop an algorithm to solve this optimization problem.

Exercise 7.2 (Regularized Fictitious Play and Stochastic Approximation). Consider a finite two-player game. Let player 1 have action set $\mathcal{A} = \{1, \dots, m\}$ and payoff matrix $U \in \mathbb{R}^{m \times n}$. Let $x_n \in \Delta_m$ denote player 1's mixed strategy at time n , and let $y_n \in \Delta_n$ denote the empirical frequency of play of player 2. Classical fictitious play updates the empirical belief via

$$y_{n+1} = y_n + \varepsilon_n(e_{a_n} - y_n),$$

where e_{a_n} is the observed pure action of player 2 and $\{\varepsilon_n\}$ is a step-size sequence.

We now introduce an entropy-regularized best response. Given belief y , define the regularized payoff

$$\Phi(x; y) = x^\top U y + \tau H(x), \quad H(x) = - \sum_{i=1}^m x_i \log x_i,$$

where $\tau > 0$ is a regularization parameter.

The regularized best response of player 1 is

$$\beta_\tau(y) = \arg \max_{x \in \Delta_m} \{x^\top U y + \tau H(x)\}.$$

(a) Show that $\beta_\tau(y)$ has the softmax form

$$(\beta_\tau(y))_i = \frac{\exp((Uy)_i/\tau)}{\sum_{k=1}^m \exp((Uy)_k/\tau)}.$$

(b) Consider the stochastic recursion

$$x_{n+1} = x_n + \varepsilon_n(\beta_\tau(y_n) - x_n + M_{n+1}),$$

where M_{n+1} is martingale difference noise representing sampling fluctuations. Show that this recursion fits into the stochastic approximation framework

$$x_{n+1} = x_n + \varepsilon_n(h(x_n) + M_{n+1}),$$

and identify $h(x)$.

(c) Derive the limiting ODE associated with this recursion.

(d) Show that the ODE can be written as

$$\dot{x} = \beta_\tau(y) - x,$$

and interpret this as a smooth best-response dynamic.

(e) What happens in the limits (i) $\tau \rightarrow 0$ and (ii) $\tau \rightarrow \infty$? Explain the behavioral interpretation.

Exercise 7.3 (Descent Directions and Steepest Descent). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable and $\nabla f(x) \neq 0$.

- (i) Show that d is a descent direction at x (i.e. $f(x + \alpha d) < f(x)$ for all sufficiently small $\alpha > 0$) if and only if $\langle \nabla f(x), d \rangle < 0$.
- (ii) Show that, among all unit vectors in the Euclidean norm, the minimizer of the directional derivative $\langle \nabla f(x), d \rangle$ is $d = -\nabla f(x) / \|\nabla f(x)\|$ (the steepest-descent direction).
- (iii) For a symmetric positive-definite P , show that the steepest-descent direction in the norm $\|z\|_P = (z^\top P z)^{1/2}$ is proportional to $-P^{-1}\nabla f(x)$, and identify the choice $P = \nabla^2 f(x)$ with the Newton direction.

Exercise 7.4 (Descent Lemma and the Sublinear Rate). Let f be convex with L -Lipschitz gradient.

- (i) Prove the descent lemma $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2$.
- (ii) For gradient descent $x_{k+1} = x_k - \frac{1}{L}\nabla f(x_k)$, deduce the sufficient decrease $f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2$.
- (iii) Using convexity, obtain the $O(1/k)$ rate $f(x_k) - f^* \leq \frac{L\|x_0 - x^*\|^2}{2k}$.

Exercise 7.5 (Linear Convergence under Strong Convexity). Let f be m -strongly convex with L -Lipschitz gradient, and let $\kappa = L/m$ be the condition number.

- (i) Show that gradient descent with constant step $1/L$ satisfies $\|x_{k+1} - x^*\|^2 \leq (1 - \frac{1}{\kappa})\|x_k - x^*\|^2$.
- (ii) Show that the constant step minimizing the worst-case contraction is $\alpha^* = 2/(L + m)$, with rate factor $(\kappa - 1)/(\kappa + 1)$. Comment on the behavior as $\kappa \rightarrow \infty$.

Exercise 7.6 (Steepest Descent on a Quadratic). Let $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ with $Q \succ 0$, eigenvalues in $[m, L]$, and $\kappa = L/m$.

- (i) For steepest descent with exact line search, compute the step $\alpha_k = \frac{\|g_k\|^2}{g_k^\top Q g_k}$, where $g_k = \nabla f(x_k)$.
- (ii) Establish the Kantorovich-type bound $f(x_{k+1}) - f^* \leq \left(\frac{\kappa-1}{\kappa+1}\right)^2 (f(x_k) - f^*)$.
- (iii) For $Q = \text{diag}(1, \kappa)$, describe the characteristic zig-zag path and its dependence on κ .

Exercise 7.7 (Backtracking (Armijo) Line Search). Consider gradient descent with the Armijo condition $f(x - \alpha \nabla f(x)) \leq f(x) - c\alpha \|\nabla f(x)\|^2$, $c \in (0, 1)$, using backtracking $\alpha \leftarrow \beta\alpha$, $\beta \in (0, 1)$.

- (i) For L -smooth f , show every $\alpha \leq 2(1 - c)/L$ satisfies the Armijo condition; conclude that backtracking terminates in finitely many steps.
- (ii) Lower-bound the accepted step and show that each iteration decreases f by at least a fixed multiple of $\|\nabla f(x)\|^2$.

Exercise 7.8 (Newton's Method: Quadratic Convergence and Affine Invariance). Let f be twice continuously differentiable with $\nabla^2 f(x^*) \succ 0$ and L_H -Lipschitz Hessian.

- (i) Show that the pure Newton iteration converges locally quadratically: $\|x_{k+1} - x^*\| \leq C \|x_k - x^*\|^2$ for x_k near x^* ; identify C .
- (ii) Show Newton's method is affine invariant: if $g(y) = f(Ay)$ with A invertible, the Newton iterates satisfy $x_k = Ay_k$ whenever $x_0 = Ay_0$. Contrast with gradient descent.

Exercise 7.9 (The Newton Decrement). For f with $\nabla^2 f(x) \succ 0$, define the Newton decrement $\lambda(x) = (\nabla f(x)^\top \nabla^2 f(x)^{-1} \nabla f(x))^{1/2}$ and the Newton step $\Delta x_{\text{nt}} = -\nabla^2 f(x)^{-1} \nabla f(x)$.

- (i) Show $\lambda(x)^2 = -\langle \nabla f(x), \Delta x_{\text{nt}} \rangle = \Delta x_{\text{nt}}^\top \nabla^2 f(x) \Delta x_{\text{nt}}$.
- (ii) Show that $\frac{1}{2}\lambda(x)^2$ estimates $f(x) - f^*$ through the local quadratic model, and justify the stopping rule $\frac{1}{2}\lambda(x)^2 \leq \epsilon$.
- (iii) Verify that $\lambda(x)$ is affine invariant.

Exercise 7.10 (Quasi-Newton and the Secant Equation). Let $s_k = x_{k+1} - x_k$ and $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$.

- (i) State the secant equation $B_{k+1}s_k = y_k$ and explain why it is the natural quasi-Newton condition.
- (ii) Show that the BFGS update preserves symmetry and positive definiteness of B_k provided the curvature condition $s_k^\top y_k > 0$ holds.
- (iii) Show that the Wolfe conditions guarantee $s_k^\top y_k > 0$.

Exercise 7.11 (Polyak–Łojasiewicz Inequality and Linear Convergence). Say f (L -smooth) satisfies the Polyak–Łojasiewicz (PL) inequality $\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu(f(x) - f^*)$ for some $\mu > 0$.

- (i) Show that gradient descent with step $1/L$ converges linearly, $f(x_k) - f^* \leq (1 - \mu/L)^k (f(x_0) - f^*)$, *without* assuming convexity.
- (ii) Show that m -strong convexity implies PL with $\mu = m$, and exhibit a *nonconvex* function that nonetheless satisfies PL.

Exercise 7.12 (Step-Size Schedules for Stochastic Approximation). Consider step sizes $\varepsilon_n = c/n^p$ with $c > 0$.

- (i) Determine all p for which the Robbins–Monro conditions $\sum_n \varepsilon_n = \infty$ and $\sum_n \varepsilon_n^2 < \infty$ both hold.
- (ii) For a constant step $\varepsilon_n \equiv \varepsilon$, explain (e.g. via a scalar linear recursion with additive noise) why the iterates settle around a nonzero *noise floor* of order ε , and describe the bias–variance trade-off between p close to 1 and p close to $1/2$.

Exercise 7.13 (ODE Method: Convergence of a Mean-Estimation Recursion). Consider $X_{n+1} = X_n + \varepsilon_n(a - X_n + M_n)$, where $a \in \mathbb{R}^d$, $\{M_n\}$ is a martingale-difference sequence, and $\{\varepsilon_n\}$ satisfies the Robbins–Monro conditions.

- (i) Identify $h(x)$ and the limiting ODE $\dot{X} = h(X)$, and find the invariant set $\mathcal{E} = \{x : h(x) = 0\}$.
- (ii) Exhibit a Lyapunov function V with $\langle \nabla V(x), h(x) \rangle \leq 0$, equality iff $h(x) = 0$, and apply Theorem 7.29 to conclude $X_n \rightarrow a$ almost surely.
- (iii) Sketch the phase portrait and the finite-window tracking picture, in analogy with Figures 7.8 and 7.9.

Exercise 7.14 (Mean-Square Rate of Stochastic Gradient Descent). Let f be m -strongly convex and L -smooth, and consider $x_{n+1} = x_n - \varepsilon_n g(x_n, \xi_{n+1})$ with $\mathbb{E}[g(x, \xi) \mid \mathcal{F}] = \nabla f(x)$ and bounded variance $\mathbb{E}\|g(x, \xi) - \nabla f(x)\|^2 \leq \sigma^2$.

- (i) With $\varepsilon_n = 1/(mn)$, show that $\mathbb{E}\|x_n - x^*\|^2 \leq C/n$ and identify C .
- (ii) With a constant step $\varepsilon_n \equiv \varepsilon$, show that $\mathbb{E}\|x_n - x^*\|^2$ converges linearly to a noise floor of order $\varepsilon\sigma^2/m$, and discuss the resulting step-size trade-off.

Chapter 8

Algorithms for Constrained Problems

This chapter introduces fundamental algorithmic frameworks for convex optimization with constraints. The focus is on geometric and variational principles that underpin modern constrained methods. We begin with feasible direction methods, formalizing optimality as the absence of feasible descent directions and connecting local geometry to the Karush–Kuhn–Tucker conditions. We then develop projection methods, highlighting the role of Euclidean projection and normal cones. Projection geometry recasts constrained optimization as a fixed-point problem, leading naturally to the projected gradient method and its convergence theory.

Barrier and homotopy methods are presented as smooth deformations of the KKT system, with the central path viewed as a continuation trajectory toward primal–dual optimality. We also study primal–dual gradient algorithms from a saddle-point perspective, interpreting them as schemes for monotone inclusions and variational inequalities. Throughout, the emphasis is on core structural ideas, including convexity, normal cones, fixed-point mappings, and saddle-point formulations, that form the conceptual foundation for advanced optimization methods.

8.1 ■ Feasible Direction Methods

Feasible direction methods are based on a fundamental geometric principle in constrained optimization: a feasible point is optimal if and only if there exists no feasible direction along which the objective can be strictly decreased. In convex programming, this principle admits a clean and exact first-order characterization.

8.1.1 ■ Primal Characterization of Optimality

Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{s.t.} \quad & h_i(x) = 0, \quad i \in \mathcal{P}, \\ & g_j(x) \leq 0, \quad j \in \mathcal{Q}, \end{aligned} \tag{8.1}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable, each g_j is convex and differentiable, and the equality feasible set $\{x : h_i(x) = 0, i \in \mathcal{P}\}$ is convex, as occurs in particular when each h_i is affine. Let $\mathcal{F}$ denote the feasible set and let $x^* \in \mathcal{F}$ be a feasible point. The feasible region is obtained by intersecting a convex equality manifold with convex inequality regions. Near a feasible point x^* , only the inequalities that are active influence the local geometry of $\mathcal{F}$.

We define the active inequality set by

$$\mathcal{Q}(x^*) := \{j \in \mathcal{Q} \mid g_j(x^*) = 0\}. \quad (8.2)$$

Inactive inequalities, being strictly satisfied, impose no local restriction on infinitesimal motion.

A vector $d \in \mathbb{R}^n$ is called a feasible direction at x^* if there exists $\gamma_0 > 0$ such that $x^* + \gamma d \in \mathcal{F}$ for all $\gamma \in (0, \gamma_0)$. Thus, d is feasible if one may move infinitesimally from x^* along d without leaving the feasible set. The collection of all such directions forms the feasible direction cone $\mathcal{D}(x^*)$.

Under differentiability, first-order Taylor expansion yields the local conditions that characterize feasible directions. Equality constraints must remain locally unchanged, so feasible directions must lie in the tangent space of the equality manifold:

$$\nabla h_i(x^*)^\top d = 0, \quad i \in \mathcal{P}. \quad (8.3)$$

For active inequalities, the first-order change cannot be positive, otherwise feasibility would be violated immediately. Hence

$$\nabla g_j(x^*)^\top d \leq 0, \quad j \in \mathcal{Q}(x^*). \quad (8.4)$$

These conditions describe the first-order tangent cone to the feasible set at x^* .

Among the active inequalities, some may remain locally constant along certain directions. For $j \in \mathcal{Q}(x^*)$, define the cone of directions of constancy

$$D_j^-(x^*) := \{d \in \mathbb{R}^n : g_j(x^* + \gamma d) = g_j(x^*) \text{ for all sufficiently small } \gamma > 0\}.$$

These are directions tangent to the boundary surface of constraint j . Let $\mathcal{Q}^= \subseteq \mathcal{Q}(x^*)$ denote the set of active inequalities that behave like equalities *locally at x^** , i.e. those j for which every feasible direction lies in $D_j^-(x^*)$, and define $\mathcal{Q}^< := \mathcal{Q}(x^*) \setminus \mathcal{Q}^=$. This is a *local* notion, attached to the point x^* , and should not be confused with the global set of carrier inequalities of the same name in Chapter 5, which collects the constraints active at *every* feasible point. The global set is contained in the local one at each feasible point, and the two coincide when the feasible set has no lower-dimensional faces beyond those forced by the carrier constraints. Constraints in $\mathcal{Q}^=$ restrict motion as equalities, while those in $\mathcal{Q}^<$ admit strict decrease while preserving feasibility.

If a feasible direction d satisfies $\nabla f(x^*)^\top d < 0$, then moving slightly along d strictly decreases the objective while remaining feasible. Such a direction is called a feasible descent direction. Optimality therefore requires that no such direction exist.

Theorem 8.1 (Primal characterization of optimality). *Under the above assumptions, a feasible point x^* is optimal for (8.1) if and only if there exists no direction $d \in \mathbb{R}^n$ satisfying*

$$\nabla f(x^*)^\top d < 0, \quad (8.5)$$

$$\nabla h_i(x^*)^\top d = 0, \quad i \in \mathcal{P}, \quad (8.6)$$

$$\nabla g_j(x^*)^\top d < 0, \quad j \in \mathcal{Q}^<, \quad (8.7)$$

$$d \in D_j^-(x^*), \quad j \in \mathcal{Q}^=. \quad (8.8)$$

Equivalently, the above system is inconsistent.

The theorem formalizes the geometric idea that at an optimal point, one cannot move tangentially to the equality manifold, remain tangent to the rigid active constraints, strictly relax the remaining active constraints, and simultaneously decrease the objective. In convex problems, this first-order condition is not merely necessary but also sufficient for global optimality.

In the regular case where Slater's condition holds, or more generally when $\mathcal{Q}^\circ = \emptyset$, the situation simplifies. All active inequalities admit strict decrease directions, and the optimality condition reduces to the inconsistency of

$$\nabla f(x^*)^\top d < 0, \quad \nabla g_j(x^*)^\top d < 0, \quad j \in \mathcal{Q}(x^*), \quad \nabla h_i(x^*)^\top d = 0, \quad i \in \mathcal{P}.$$

Thus, in the regular convex case, optimality is equivalent to the nonexistence of a feasible descent direction. This primal viewpoint forms the conceptual foundation for multiplier theory and duality, which may be derived from this inconsistency via separation arguments.

8.1.2 ■ Feasible Direction Methods

Consider the convex program (8.1). Assume that all functions are continuously differentiable and that Slater's condition holds. Under this regularity assumption the degenerate active set $\mathcal{Q}^\circ$ is empty, and the primal characterization of Theorem 8.1 reduces to a directional optimality condition equivalent to the Karush–Kuhn–Tucker (KKT) conditions.

Proposition 8.2 (Directional Optimality Condition). *A feasible point $x^* \in \mathcal{F}$ is optimal if and only if the system in the direction variable $d \in \mathbb{R}^n$*

$$\nabla f(x^*)^\top d < 0, \tag{8.9}$$

$$\nabla g_j(x^*)^\top d < 0, \quad j \in \mathcal{Q}(x^*), \tag{8.10}$$

$$\nabla h_i(x^*)^\top d = 0, \quad i \in \mathcal{P}, \tag{8.11}$$

is inconsistent. Equivalently, x^ is not optimal if and only if there exists a direction d satisfying (8.9)–(8.11).*

The proposition formalizes the geometric principle that at an optimal point there exists no feasible descent direction. If x^* is not optimal, any direction d satisfying (8.9)–(8.11) has two essential properties. First, it is a feasible direction, meaning that for sufficiently small $\gamma > 0$, the point $x^* + \gamma d$ remains feasible. Second, it is a descent direction for the objective, so that $f(x^* + \gamma d) < f(x^*)$ for all sufficiently small $\gamma > 0$. Thus, nonoptimality is characterized by the existence of a direction that simultaneously preserves feasibility and strictly reduces the objective value.

This observation motivates a class of iterative procedures known as *feasible direction methods*. Starting from a feasible initial point $x_0 \in \mathcal{F}$, these methods generate a sequence $\{x_k\} \subset \mathcal{F}$ such that $f(x_{k+1}) \leq f(x_k)$, $k = 0, 1, 2, \dots$. The algorithm repeatedly constructs an improving feasible direction and then performs a line search along the feasible portion of that direction.

Feasible and improvable directions. Let $x_k \in \mathcal{F}$. A vector $d \in \mathbb{R}^n$ is called a feasible direction at x_k if there exists $T > 0$ such that $x_k + td \in \mathcal{F}$, $0 < t \leq T$. A feasible direction is called improvable if there exists $\tau > 0$ such that $f(x_k + td) < f(x_k)$, $0 < t \leq \tau$. Once such a direction d_k is identified, the method solves the one-dimensional constrained minimization problem along the feasible ray $\{x_k + \gamma d_k : \gamma \geq 0\}$, often referred to as the *step-size problem*. If γ_k denotes the optimal step size, the next iterate is defined by $x_{k+1} = x_k + \gamma_k d_k$.

Feasible direction generators. The central computational task is the generation of a direction d_k that satisfies the first-order feasibility conditions and yields descent. A typical generator for

(8.1) is

$$\begin{aligned}
 & \min_{d, \sigma} \quad \nabla f(x_k)^\top d \\
 & \text{subject to} \quad \nabla g_j(x_k)^\top d + \sigma \leq 0, \quad j \in \mathcal{Q}(x_k), \\
 & \quad \quad \quad \nabla h_i(x_k)^\top d = 0, \quad i \in \mathcal{P}, \\
 & \quad \quad \quad \|d\| \leq 1, \\
 & \quad \quad \quad \sigma \geq 0.
 \end{aligned} \tag{8.12}$$

Here $\mathcal{Q}(x_k)$ denotes the active inequality set at x_k . The normalization $\|d\| \leq 1$ prevents trivial scaling of the direction.

To reduce zig-zagging near the boundary, one often replaces $\mathcal{Q}(x_k)$ by an ε -active set

$$\mathcal{Q}_\varepsilon(x_k) = \{j \in \mathcal{Q} : -\varepsilon \leq g_j(x_k) \leq 0\},$$

with $\varepsilon \downarrow 0$ as $k \rightarrow \infty$.

An alternative generator emphasizes maximal descent margin:

$$\begin{aligned}
 & \max_{d, \sigma} \quad \sigma \\
 & \text{subject to} \quad \nabla f(x_k)^\top d + \sigma \leq 0, \\
 & \quad \quad \quad \nabla g_j(x_k)^\top d + \sigma \leq 0, \quad j \in \mathcal{Q}(x_k), \\
 & \quad \quad \quad \nabla h_i(x_k)^\top d = 0, \quad i \in \mathcal{P}, \\
 & \quad \quad \quad \|d\| \leq 1, \quad \sigma \geq 0.
 \end{aligned} \tag{8.13}$$

The stopping test is the optimal value: if $\sigma_k = 0$, no direction achieves a strictly negative margin, the directional system (8.9)–(8.11) is inconsistent, and x_k is declared optimal. Testing $d_k = 0$ instead would be too strong, since at optimality the maximizer d_k need not be unique and need not be zero; it is the vanishing of the margin σ_k , not of the direction, that certifies optimality.

The step-size problem enforces all original constraints:

$$\begin{aligned}
 & \min_{\gamma \geq 0} \quad f(x_k + \gamma d_k) \\
 & \text{subject to} \quad g_j(x_k + \gamma d_k) \leq 0, \quad j \in \mathcal{Q}, \\
 & \quad \quad \quad h_i(x_k + \gamma d_k) = 0, \quad i \in \mathcal{P}.
 \end{aligned} \tag{8.14}$$

Feasibility is preserved at every iteration.

Theorem 8.3 (Global Convergence). *Suppose that the problem is convex and continuously differentiable, Slater's condition holds, each nonoptimal iterate admits a nonzero descent direction, and the step size is chosen so that $x_{k+1} = x_k + \gamma_k d_k$ remains feasible with $f(x_{k+1}) < f(x_k)$. Then $\{f(x_k)\}$ is monotonically nonincreasing, every accumulation point of $\{x_k\}$ is optimal, and if the optimal solution is unique, then $x_k \rightarrow x^*$.*

Proof. Monotonic decrease follows from descent along the feasible ray. Convexity ensures boundedness below by the optimal value. If an accumulation point were nonoptimal, the directional optimality condition would produce a uniform descent direction in its neighborhood, yielding a strict decrease bounded away from zero and contradicting boundedness. Hence all accumulation points are optimal. Uniqueness implies convergence of the full sequence. $\square$

Feasible direction methods are globally convergent and conceptually transparent. However, because they rely only on first-order information and enforce feasibility at every step, they typically

exhibit sublinear convergence and may display boundary zig-zagging behavior. Their theoretical importance lies in the clear geometric interpretation they provide of optimality and their direct connection to the KKT conditions.

ALGORITHM 8.1. Feasible Direction Method for Convex Programs with Mixed Constraints.

Require: Initial feasible point $x_0 \in \mathcal{F}$; tolerance $\varepsilon > 0$; maximum iterations $K_{\max}$.

Ensure: Approximate optimal solution x^* .

1: $k \leftarrow 0$

2: **while** $k < K_{\max}$ **do**

3: Compute the ε -active inequality set

$$\mathcal{Q}_\varepsilon(x_k) := \{j \in \mathcal{Q} : -\varepsilon \leq g_j(x_k) \leq 0\}.$$

4: Solve the feasible direction subproblem:

$$\begin{aligned} & \min_{d, \sigma} \quad \nabla f(x_k)^\top d \\ & \text{subject to} \quad \nabla g_j(x_k)^\top d + \sigma \leq 0, \quad j \in \mathcal{Q}_\varepsilon(x_k), \\ & \quad \quad \quad \nabla h_i(x_k)^\top d = 0, \quad i \in \mathcal{P}, \\ & \quad \quad \quad \|d\| \leq 1, \\ & \quad \quad \quad \sigma \geq 0. \end{aligned}$$

5: **if** the optimal value $\nabla f(x_k)^\top d$ is zero **then**

6: **terminate** and return x_k as optimal ▷ no strictly improving feasible direction exists

7: **end if**

8: Solve the step-size problem:

$$\begin{aligned} & \gamma_k \in \arg \min_{\gamma \geq 0} \quad f(x_k + \gamma d) \\ & \text{subject to} \quad g_j(x_k + \gamma d) \leq 0, \quad j \in \mathcal{Q}, \\ & \quad \quad \quad h_i(x_k + \gamma d) = 0, \quad i \in \mathcal{P}. \end{aligned}$$

9: Update the iterate: $x_{k+1} \leftarrow x_k + \gamma_k d$

10: $k \leftarrow k + 1$

11: **end while**

12: **return** x_k

Theorem 8.4 (Global Convergence of Feasible Direction Methods). *Suppose that:*

- (1) *the optimization problem (8.1) is convex and all functions are continuously differentiable;*
- (2) *Slater's condition holds;*
- (3) *at each iterate x_k , if x_k is not optimal, the feasible direction generator produces a nonzero direction d_k satisfying (8.9)–(8.11);*

(4) the step size γ_k is chosen so that $x_{k+1} = x_k + \gamma_k d_k$ remains feasible and $f(x_{k+1}) < f(x_k)$.

Then the sequence $\{x_k\}$ generated by Algorithm 8.1 satisfies:

- (1) the sequence $\{f(x_k)\}$ is monotonically nonincreasing;
- (2) every accumulation point of $\{x_k\}$ is an optimal solution of (8.1);
- (3) if the optimal solution is unique, then $x_k \rightarrow x^*$.

Proof. Monotonic decrease. By construction, x_{k+1} is obtained by minimizing f along the feasible ray $\{x_k + \gamma d_k : \gamma \geq 0\}$. Since d_k is an improvable feasible direction, there exists $\bar{\gamma} > 0$ such that $f(x_k + \gamma d_k) < f(x_k)$ for all $\gamma \in (0, \bar{\gamma}]$. Hence $f(x_{k+1}) \leq f(x_k)$, and the sequence $\{f(x_k)\}$ is monotonically nonincreasing.

Boundedness of the objective values. Because f is convex and $\mathcal{F}$ is nonempty, the optimal value $f^* := \inf_{x \in \mathcal{F}} f(x)$ is finite. Therefore, $\{f(x_k)\}$ is bounded below by f^* and converges to some limit $\bar{f} \geq f^*$.

Characterization of nonoptimal accumulation points. Assume, by contradiction, that $\bar{x}$ is an accumulation point of $\{x_k\}$ that is not optimal. By Proposition 8.2, there exists a direction d satisfying (8.9)–(8.11) at $\bar{x}$. By continuity of the gradients and feasibility conditions, the same system remains consistent in a neighborhood of $\bar{x}$. Hence, for all sufficiently large k , the feasible direction generator produces a nonzero descent direction d_k with a uniform descent property: there exists $\eta > 0$ such that $\nabla f(x_k)^\top d_k \leq -\eta$.

Strict decrease away from optimality. By the descent property and the step-size rule, there exists $\delta > 0$ such that $f(x_{k+1}) \leq f(x_k) - \delta$ for all sufficiently large k . This implies that $f(x_k) \rightarrow -\infty$, contradicting boundedness of $\{f(x_k)\}$.

Optimality of accumulation points. The contradiction shows that every accumulation point of $\{x_k\}$ must be optimal. This proves (ii).

Convergence under uniqueness. If the optimal solution x^* is unique, then the set of accumulation points is the singleton $\{x^*\}$, which implies $x_k \rightarrow x^*$. $\square$

Remark 8.5. The convergence result is global but generally does not provide a rate. Feasible direction methods typically exhibit sublinear convergence and may suffer from zig-zagging near the boundary of the feasible set.

Remark 8.6. The proof relies critically on convexity and Slater regularity. Without convexity, feasible direction methods may converge to nonoptimal stationary points.

8.2 ■ Projection Methods

Projection methods form one of the most fundamental algorithmic approaches for constrained convex optimization. Their central idea is simple yet powerful: one performs an unconstrained descent step and then restores feasibility by projecting back onto the constraint set. The geometry of convex sets ensures that this projection is well defined and admits strong variational properties that lead to clean convergence theory.

We consider the constrained convex optimization problem

$$\min_{x \in X} f(x), \tag{8.15}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $X \subset \mathbb{R}^n$ is the feasible set.

Assumption 8.7. The following conditions hold:

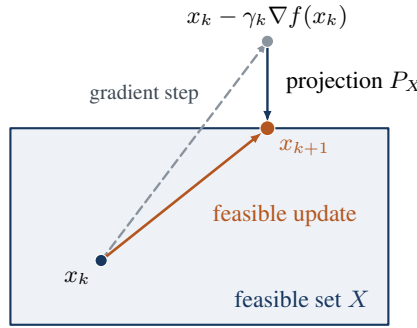


Figure 8.1: Projected methods separate an unconstrained descent proposal from the projection that restores feasibility.

- (1) f is convex and continuously differentiable on $\mathbb{R}^n$;
- (2) the feasible set X is nonempty, closed, and convex;
- (3) the optimal value $f^* := \inf_{x \in X} f(x)$ is finite.

These assumptions guarantee a well-posed convex optimization problem. Convexity of f ensures that every local minimizer is global. Closedness and convexity of X guarantee that the projection operator onto X is single-valued and continuous. Finiteness of the optimal value excludes pathological cases where the objective diverges to $-\infty$ over X .

Under Assumption 8.7, problem (8.15) admits at least one minimizing sequence. If X is additionally compact, existence of an optimal solution follows immediately from continuity of f . Even when X is unbounded, convexity allows one to derive strong first-order characterizations of optimality via normal cones and variational inequalities.

8.2.1 ■ Normal Cones and Projection Geometry

Projection methods are grounded in the geometry of closed convex sets. Their analysis rests on the interplay between Euclidean projection and the normal cone.

Definition 8.8 (Euclidean Projection). Let $X \subset \mathbb{R}^n$ be nonempty, closed, and convex. The Euclidean projection onto X is the mapping $P_X : \mathbb{R}^n \rightarrow X$ defined by $P_X(y) := \arg \min_{x \in X} \|x - y\|$.

Proposition 8.9 (Existence and Uniqueness). For every $y \in \mathbb{R}^n$, the projection $P_X(y)$ exists and is unique.

Proof. The function $x \mapsto \|x - y\|^2$ is strongly convex and continuous. Since X is closed and convex, the restriction of this function to X is strongly convex and lower semicontinuous. Therefore, it admits a unique minimizer on X . $\square$

Proposition 8.10 (Variational Characterization). Let $\bar{x} \in X$ and $y \in \mathbb{R}^n$. Then $\bar{x} = P_X(y)$ if and only if

$$(y - \bar{x})^\top (x - \bar{x}) \leq 0, \quad \forall x \in X. \quad (8.16)$$

Proof. The result follows from the first-order optimality condition for the convex problem $\min_{x \in X} \|x - y\|^2$. $\square$

Condition (8.16) shows that the projection error vector $y - \bar{x}$ forms a nonacute angle with every feasible displacement. This expresses the orthogonality of the projection in variational form.

Definition 8.11 (Normal Cone). Let $X \subset \mathbb{R}^n$ be closed and convex and let $z \in X$. The normal cone to X at z is

$$N_X(z) := \{v \in \mathbb{R}^n : v^\top(x - z) \leq 0 \forall x \in X\}.$$

The normal cone is a closed convex cone. It consists of all vectors that define supporting hyperplanes to X at z .

Theorem 8.12 (Projection-Normal Cone Equivalence). For $z \in X$ and $y \in \mathbb{R}^n$,

$$z = P_X(y) \iff y - z \in N_X(z).$$

Proof. The equivalence follows immediately by comparing (8.16) with the definition of $N_X(z)$. $\square$

This equivalence shows that projection subtracts a normal component from y in order to reach feasibility.

Lemma 8.13 (Scaling Invariance). If $z = P_X(z - v)$ for some $v \in \mathbb{R}^n$, then $z = P_X(z - \tau v)$, $\forall \tau > 0$.

Proof. The hypothesis is equivalent to $v \in N_X(z)$. Since $N_X(z)$ is a cone, $\tau v \in N_X(z)$ for all $\tau > 0$. Applying the projection-normal cone equivalence yields the claim. $\square$

This property reflects the conic structure of $N_X(z)$: once a direction lies in the normal cone, its magnitude does not affect the projection.

Example 8.14 Projection onto an Affine Subspace.

Let $X = \{x \in \mathbb{R}^n : Ax = b\}$, where $A \in \mathbb{R}^{m \times n}$ has full row rank. Then X is closed and convex. The projection is given explicitly by $P_X(y) = y - A^\top(AA^\top)^{-1}(Ay - b)$. The normal cone at any $z \in X$ is $N_X(z) = \{A^\top \lambda : \lambda \in \mathbb{R}^m\}$. Hence the projection error $y - P_X(y)$ lies in the range of $A^\top$, i.e., it is orthogonal to the tangent space of X . $\square$

Example 8.15 Projection onto the Nonnegative Orthant.

Let $X = \mathbb{R}_+^n = \{x \in \mathbb{R}^n : x_i \geq 0, i = 1, \dots, n\}$. The projection is given componentwise: $(P_X(y))_i = \max\{y_i, 0\}$. At a boundary point z , the normal cone consists of vectors v satisfying $v_i \leq 0$ whenever $z_i = 0$ and $v_i = 0$ whenever $z_i > 0$. The projection subtracts the negative components of y , which lie in this normal cone.



These examples illustrate how projection removes precisely the normal component of a point relative to the feasible set. In general convex geometry, this decomposition $y - P_X(y) \in N_X(P_X(y))$ provides the structural foundation for projected gradient methods and for the variational inequality formulation of constrained optimization problems.

8.2.2 ■ Projection Characterization of Optimality

The geometry of projection yields a precise and complete characterization of optimality for convex constrained problems.

Theorem 8.16 (Projection Characterization of Optimality). *Let $z \in X$. The following statements are equivalent:*

- (1) z is an optimal solution of $\min_{x \in X} f(x)$;
- (2) $0 \in \nabla f(z) + N_X(z)$;
- (3) $z = P_X(z - \nabla f(z))$.

Proof. (1) $\Rightarrow$ (2). Since f is convex and differentiable and X is convex, the first-order necessary and sufficient condition for optimality is the variational inequality $\nabla f(z)^\top (x - z) \geq 0, \forall x \in X$. This condition is equivalent to $-\nabla f(z) \in N_X(z)$, or equivalently, $0 \in \nabla f(z) + N_X(z)$.

(2) $\Rightarrow$ (3). If $-\nabla f(z) \in N_X(z)$, then by the projection-normal cone equivalence, $z = P_X(z - \nabla f(z))$.

(3) $\Rightarrow$ (2). Conversely, if $z = P_X(z - \nabla f(z))$, then by the projection characterization, $-\nabla f(z) \in N_X(z)$, which implies (2).

Since (1) and (2) are equivalent by convexity, all three statements are equivalent. $\square$

Fixed-Point Interpretation Define the *projected gradient mapping* $T(x) := P_X(x - \nabla f(x))$. Theorem 8.16 shows that z is optimal if and only if $T(z) = z$.

Thus the constrained minimization problem is equivalent to the fixed-point problem $x = T(x)$.

This equivalence has fundamental algorithmic consequences. Projected gradient methods can be viewed as fixed-point iterations for the mapping T . Nonexpansiveness of T is not automatic: P_X is always nonexpansive, but the inner map $x \mapsto x - \gamma \nabla f(x)$ is nonexpansive only when the step size is compatible with the curvature of f . For f convex and L -smooth this holds precisely for $\gamma \leq 2/L$, and with such a step $T_\gamma(x) := P_X(x - \gamma \nabla f(x))$ is a composition of nonexpansive maps and hence nonexpansive. The unit step used above corresponds to $L \leq 2$. Moreover, optimality is characterized by the inclusion $0 \in \nabla f(z) + N_X(z)$, which places constrained convex optimization within the general framework of monotone operator theory, where one seeks zeros of the sum of two monotone operators.

In this sense, projection geometry transforms constrained minimization into a problem of finding a fixed point or, equivalently, a zero of a monotone inclusion.

Example 8.17 Projection Characterization on a Halfspace.

Let $X = \{x \in \mathbb{R}^n : a^\top x \leq b\}$, $a \neq 0$. Then X is a nonempty, closed, convex halfspace. For any $y \in \mathbb{R}^n$, the Euclidean projection onto X is given by

$$P_X(y) = \begin{cases} y, & \text{if } a^\top y \leq b, \\ y - \frac{a^\top y - b}{\|a\|^2} a, & \text{if } a^\top y > b. \end{cases}$$

At any boundary point z satisfying $a^\top z = b$, the normal cone is $N_X(z) = \{\lambda a : \lambda \geq 0\}$. Interior points have $N_X(z) = \{0\}$. We now illustrate the projection characterization of optimality for the convex problem $\min_{x \in X} f(x)$, where f is convex and differentiable.

By Theorem 8.16, a point $z \in X$ is optimal if and only if $z = P_X(z - \nabla f(z))$.

There are two cases.

Case 1: Interior solution. If $a^\top z < b$, then z lies in the interior of X . Hence P_X acts as the identity locally. The fixed-point condition becomes $z = z - \nabla f(z)$, which implies $\nabla f(z) = 0$. Thus the optimality condition reduces to the unconstrained first-order condition.

Case 2: Boundary solution. Suppose $a^\top z = b$. Then $z = P_X(z - \nabla f(z))$ implies, using the explicit projection formula,

$$z = z - \nabla f(z) - \frac{a^\top (z - \nabla f(z)) - b}{\|a\|^2} a.$$

Since $a^\top z = b$, this simplifies to

$$0 = -\nabla f(z) + \frac{a^\top \nabla f(z)}{\|a\|^2} a.$$

Therefore,

$$\nabla f(z) = \frac{a^\top \nabla f(z)}{\|a\|^2} a.$$

Setting $\mu := -\frac{a^\top \nabla f(z)}{\|a\|^2}$, we obtain $\nabla f(z) = -\mu a$.

Since the projection formula applies only when $a^\top (z - \nabla f(z)) > b$, that is $a^\top \nabla f(z) < 0$, we must have $\mu \geq 0$. Hence $\nabla f(z) + \mu a = 0$, $\mu \geq 0$ and $a^\top z = b$, which is precisely the Karush–Kuhn–Tucker condition for $\min_{x \in X} f(x)$ with the active inequality constraint $a^\top x \leq b$ (multiplier $\mu \geq 0$, complementary slackness holding since $a^\top z = b$).

Thus the projection fixed-point condition $z = P_X(z - \nabla f(z))$ recovers both interior and boundary optimality conditions in a unified geometric form.

In particular, the projection subtracts exactly the outward normal component proportional to a , enforcing feasibility while preserving optimality. $\square$

8.2.3 ■ Fundamental Projection Inequality

The projection operator satisfies a fundamental variational inequality that lies at the heart of the convergence analysis of projection-based algorithms.

Lemma 8.18 (Projection Inequality). *Let $X \subset \mathbb{R}^n$ be nonempty, closed, and convex. For any $y \in \mathbb{R}^n$ and any $x \in X$,*

$$(P_X(y) - y)^\top (x - P_X(y)) \geq 0. \quad (8.17)$$

Proof. Let $\bar{x} := P_X(y)$. Since $\bar{x}$ minimizes the strongly convex function $\|x - y\|^2$ over X , the first-order optimality condition gives $(y - \bar{x})^\top (x - \bar{x}) \leq 0, \forall x \in X$. Multiplying by -1 yields $(\bar{x} - y)^\top (x - \bar{x}) \geq 0, \forall x \in X$, which is precisely (8.17). $\square$

Projection Norm Identity The projection inequality admits an equivalent and frequently used norm form. For any $x \in X$,

$$\|x - y\|^2 = \|x - P_X(y)\|^2 + \|P_X(y) - y\|^2 + 2(P_X(y) - y)^\top (x - P_X(y)). \quad (8.18)$$

Using (8.17), we obtain

$$\|x - y\|^2 \geq \|x - P_X(y)\|^2 + \|P_X(y) - y\|^2. \quad (8.19)$$

Inequality (8.19) is sometimes called the *generalized Pythagorean inequality*. It shows that the projection strictly reduces the distance to the feasible set unless y is already feasible.

Inequality (8.17) states that the projection error vector $P_X(y) - y$ lies in the normal cone to X at $P_X(y)$. Equivalently, this vector forms a nonacute angle with every feasible displacement $x - P_X(y)$.

Geometrically, the projection removes precisely the outward normal component of y relative to the set X . The decomposition $y - P_X(y) \in N_X(P_X(y))$ implies that the projection is orthogonal (in the variational sense) to the feasible region.

8.2.4 ■ Projected Gradient Method

Let $X \subset \mathbb{R}^n$ be nonempty, closed, and convex, and consider the problem $\min_{x \in X} f(x)$, where f is convex and continuously differentiable. Given an initial point $x_0 \in X$, the projected gradient method generates a sequence $\{x_k\}$ according to

$$x_{k+1} = P_X(x_k - \gamma_k \nabla f(x_k)), \quad (8.20)$$

where $\gamma_k > 0$ is the step size.

This iteration consists of two steps: first, a gradient descent step $x_k - \gamma_k \nabla f(x_k)$, which may leave the feasible set, and second, a projection onto X to restore feasibility. We assume throughout that ∇f is Lipschitz continuous on X with constant $L > 0$, i.e.,

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in X.$$

This is equivalent to the standard descent inequality

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|^2. \quad (8.21)$$

Lemma 8.19 (Descent Estimate). *Let f be convex with L -Lipschitz gradient. Let $x \in X$ and define $y = P_X(x - \gamma \nabla f(x))$, $\gamma > 0$. Then for any $z \in X$,*

$$\|y - z\|^2 \leq \|x - z\|^2 - 2\gamma(f(x) - f(z)) + \gamma^2 \|\nabla f(x)\|^2.$$

Proof. By nonexpansiveness of the projection, $\|y - z\| \leq \|x - \gamma \nabla f(x) - z\|$. Squaring both sides gives

$$\|y - z\|^2 \leq \|x - z\|^2 - 2\gamma \nabla f(x)^\top (x - z) + \gamma^2 \|\nabla f(x)\|^2.$$

By convexity, $f(x) - f(z) \leq \nabla f(x)^\top (x - z)$. Substituting yields the result. $\square$

This inequality shows that the projected gradient step reduces distance to any solution $z \in X^*$ up to a second-order error term.

Theorem 8.20 (Convergence of Projected Gradient). *Assume that*

- (1) f is convex and continuously differentiable,
- (2) ∇f is Lipschitz continuous with constant L ,
- (3) the solution set $X^* := \arg \min_{x \in X} f(x)$ is nonempty.

If the step size is constant and satisfies $0 < \gamma < \frac{2}{L}$, then the sequence $\{x_k\}$ generated by (8.20) converges to a point in X^ .*

Proof. Let $z \in X^*$. Since z is optimal, $z = P_X(z - \gamma \nabla f(z)) \forall \gamma > 0$.

Using nonexpansiveness of P_X ,

$$\|x_{k+1} - z\|^2 \leq \|x_k - \gamma \nabla f(x_k) - (z - \gamma \nabla f(z))\|^2.$$

Expanding gives

$$\|x_{k+1} - z\|^2 \leq \|x_k - z\|^2 - 2\gamma(\nabla f(x_k) - \nabla f(z))^\top (x_k - z) + \gamma^2 \|\nabla f(x_k) - \nabla f(z)\|^2.$$

By convexity and Lipschitz continuity,

$$(\nabla f(x_k) - \nabla f(z))^\top (x_k - z) \geq \frac{1}{L} \|\nabla f(x_k) - \nabla f(z)\|^2.$$

Hence

$$\|x_{k+1} - z\|^2 \leq \|x_k - z\|^2 - \gamma \left(\frac{2}{L} - \gamma \right) \|\nabla f(x_k) - \nabla f(z)\|^2.$$

Since $0 < \gamma < 2/L$, the coefficient is positive, so $\{\|x_k - z\|\}$ is monotonically nonincreasing. Summing over k yields

$$\sum_{k=0}^{\infty} \|\nabla f(x_k) - \nabla f(z)\|^2 < \infty,$$

and hence $\nabla f(x_k) \rightarrow \nabla f(z)$.

Boundedness of $\{x_k\}$ follows from Fejér monotonicity. Every accumulation point satisfies $x = P_X(x - \gamma \nabla f(x))$, hence is optimal. Convexity implies that the whole sequence converges to a point in X^* . $\square$

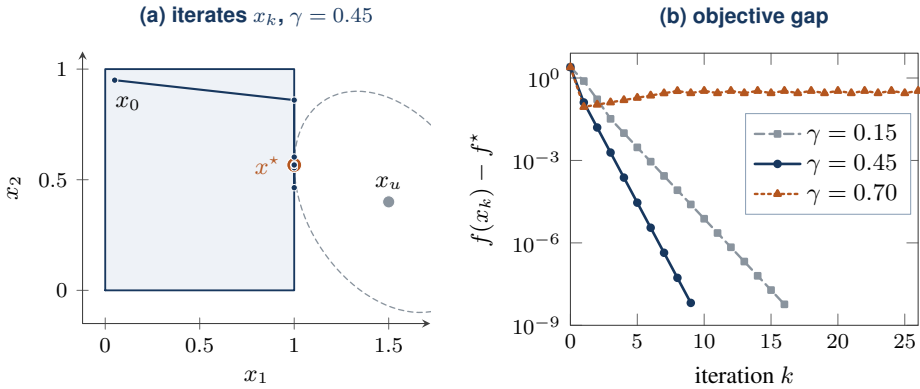


Figure 8.2: Projected gradient method on the box-constrained quadratic of Example 8.21. (a) The iterates immediately hit the active face $x_1 = 1$ and then converge along it, through damped oscillations, to $x^* = (1, \frac{17}{30})$, where the level set $f = f^*$ (dashed) is tangent to the face; the unconstrained minimizer $x_u = (1.5, 0.4)$ lies outside the feasible box. (b) With $L = 4$ the theoretical stability threshold is $2/L = 0.5$: the stepsizes $\gamma = 0.15$ and $\gamma = 0.45$ yield linear convergence, whereas $\gamma = 0.70 > 2/L$ produces a persistent two-cycle that never reaches x^* .

Example 8.21 Projected Gradient on a Box-Constrained Quadratic.

Consider the strongly convex quadratic

$$f(x) = \frac{1}{2}x^\top Qx - b^\top x, \quad Q = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 4.9 \\ 2.7 \end{pmatrix},$$

minimized over the box $X = [0, 1]^2$. The eigenvalues of Q are 2 and 4, so $\nabla f(x) = Qx - b$ is Lipschitz with constant $L = 4$ and the convergence threshold of Theorem 8.20 is $2/L = 0.5$. Projection onto the box is the componentwise clamp $P_X(x) = (\min\{\max\{x_1, 0\}, 1\}, \min\{\max\{x_2, 0\}, 1\})$.

The unconstrained minimizer $x_u = Q^{-1}b = (1.5, 0.4)$ lies *outside* X , so the constraint is active at the solution. Fixing the active face $x_1 = 1$ and minimizing over x_2 gives the constrained optimum $x^* = (1, \frac{17}{30}) \approx (1, 0.567)$ with $f^* = -3.8817$.

Starting from $x_0 = (0.05, 0.95)$, the iteration $x_{k+1} = P_X(x_k - \gamma \nabla f(x_k))$ was run for three constant stepsizes. As shown in Figure 8.2, the larger stepsizes $\gamma = 0.45$ and $\gamma = 0.70$ drive x_1 onto the face $x_1 = 1$ in a single step, while $\gamma = 0.15$ reaches it after a few iterations. On the active face the reduced iteration for x_2 is the clamped affine map $x_2 \mapsto (1 - 3\gamma)x_2 + 1.7\gamma$, with contraction factor $|1 - 3\gamma|$ and fixed point $x_2^* = \frac{17}{30}$. For $\gamma = 0.15$ (factor 0.55) the iterates converge monotonically; for $\gamma = 0.45$ (factor -0.35) they converge through damped oscillations, faster than $\gamma = 0.15$. In contrast $\gamma = 0.70 > 2/L$ makes $|1 - 3\gamma| = 1.1 > 1$, violating the stability condition, and the iterates settle into a persistent two-cycle, oscillating around x^* without ever converging, a concrete illustration of the sharpness of the bound $\gamma < 2/L$. $\square$

Convergence with bounded gradients.

The convergence theorem above assumes a Lipschitz gradient and a constant stepsize below $2/L$. We now establish convergence properties of the projected gradient method under the milder assumption that the gradient is merely uniformly bounded on the feasible set, which also covers diminishing-stepsize schedules. Assume throughout that $\|\nabla f(x)\| \leq G$, $\forall x \in X$, for some constant $G > 0$ (distinct from the Lipschitz constant L above).

Theorem 8.22. *Let Assumption 8.7 hold and suppose $\|\nabla f(x)\| \leq G$ for all $x \in X$. Let $\{x_k\}$ be generated by $x_{k+1} = P_X(x_k - \gamma_k \nabla f(x_k))$. Then:*

- (1) *If $\gamma_k \equiv \gamma > 0$ is constant, then $\liminf_{k \rightarrow \infty} f(x_k) \leq f^* + \frac{\gamma G^2}{2}$.*
- (2) *If the stepsizes satisfy*

$$\gamma_k > 0, \quad \sum_{k=0}^{\infty} \gamma_k = \infty, \quad \sum_{k=0}^{\infty} \gamma_k^2 < \infty,$$

then $\lim_{k \rightarrow \infty} f(x_k) = f^$.*

Proof. Let $x^* \in X$ be an optimal solution so that $f(x^*) = f^*$.

Fundamental inequality. From the standard descent estimate for projected gradient (with bounded gradients), we have for each k ,

$$f(x_{k+1}) \leq f(x_k) - \gamma_k \nabla f(x_k)^\top (x_k - x^*) + \frac{\gamma_k^2 G^2}{2}. \quad (8.22)$$

By convexity, $f(x_k) - f^* \leq \nabla f(x_k)^\top (x_k - x^*)$. Substituting into (8.22) gives

$$f(x_{k+1}) - f^* \leq f(x_k) - f^* - \gamma_k (f(x_k) - f^*) + \frac{\gamma_k^2 G^2}{2}. \quad (8.23)$$

Equivalently,

$$f(x_{k+1}) - f^* \leq (1 - \gamma_k)(f(x_k) - f^*) + \frac{\gamma_k^2 G^2}{2}.$$

Part (i): Constant stepsize. Assume $\gamma_k \equiv \gamma > 0$. Rearranging (8.23) gives

$$\gamma(f(x_k) - f^*) \leq (f(x_k) - f^*) - (f(x_{k+1}) - f^*) + \frac{\gamma^2 G^2}{2}.$$

Summing from $k = 0$ to $N - 1$ yields

$$\gamma \sum_{k=0}^{N-1} (f(x_k) - f^*) \leq f(x_0) - f^* + \frac{\gamma^2 G^2}{2} N.$$

Dividing by γN gives

$$\frac{1}{N} \sum_{k=0}^{N-1} (f(x_k) - f^*) \leq \frac{f(x_0) - f^*}{\gamma N} + \frac{\gamma G^2}{2}.$$

Letting $N \rightarrow \infty$ yields

$$\liminf_{k \rightarrow \infty} f(x_k) \leq f^* + \frac{\gamma G^2}{2}.$$

Part (ii): Diminishing stepsizes. Summing (8.23) gives

$$\sum_{k=0}^{N-1} \gamma_k (f(x_k) - f^*) \leq f(x_0) - f^* + \frac{G^2}{2} \sum_{k=0}^{N-1} \gamma_k^2.$$

Since $\sum \gamma_k^2 < \infty$, the right-hand side is bounded. Hence

$$\sum_{k=0}^{\infty} \gamma_k (f(x_k) - f^*) < \infty.$$

Because $\sum \gamma_k = \infty$, this is possible only if $\liminf_{k \rightarrow \infty} (f(x_k) - f^*) = 0$.

Let $\varepsilon > 0$. If $f(x_k) - f^* \geq \varepsilon$ for infinitely many k , the above sum would diverge, contradicting finiteness. Thus $\lim_{k \rightarrow \infty} f(x_k) = f^*$. $\square$

8.3 ■ Homotopy and Barrier Method

We consider the convex optimization problem

$$\begin{aligned} \min_x \quad & f(x) \\ \text{subject to} \quad & g_j(x) \leq 0, \quad j = 1, \dots, m, \\ & Ax = b, \end{aligned} \tag{8.24}$$

where f and g_j are convex and differentiable, and Slater's condition holds.

Theorem 8.23 (KKT Optimality for the Original Problem). *A point x^* is optimal for (8.24) if and only if there exist multipliers $\mu^* \succeq 0$ and λ^* such that*

$$\begin{aligned} Ax^* &= b, \\ g_j(x^*) &\leq 0, \quad j = 1, \dots, m, \\ \nabla f(x^*) + \sum_{j=1}^m \mu_j^* \nabla g_j(x^*) + A^T \lambda^* &= 0, \\ (\mu^*)^T g(x^*) &= 0. \end{aligned}$$

Proof. This is the classical Karush–Kuhn–Tucker theorem for convex optimization under Slater's condition. Convexity ensures sufficiency, while the constraint qualification guarantees existence of Lagrange multipliers. $\square$

8.3.1 ■ Barrier Approximation

Define the logarithmic barrier function

$$\phi(x) = - \sum_{j=1}^m \ln(-g_j(x)), \quad \text{dom } \phi = \{x \mid g_j(x) < 0\}.$$

For $t > 0$, consider the equality constrained barrier problem

$$\begin{aligned} \min_x \quad & tf(x) + \phi(x) \\ \text{subject to} \quad & Ax = b. \end{aligned} \tag{8.25}$$

Theorem 8.24 (Existence and Characterization of the Central Path). *Assume that for each $t > 0$, problem (8.25) admits a unique minimizer $x^*(t)$. Then $x^*(t)$ satisfies:*

(1) (Strict feasibility)

$$Ax^*(t) = b, \quad g_j(x^*(t)) < 0 \quad \forall j.$$

(2) (Stationarity) *There exists $\hat{\lambda}(t)$ such that $t\nabla f(x^*(t)) + \nabla\phi(x^*(t)) + A^T\hat{\lambda}(t) = 0$, where $\nabla\phi(x) = -\sum_{j=1}^m \frac{1}{g_j(x)} \nabla g_j(x)$.*

The set $\{x^*(t) \mid t > 0\}$ is called the central path.

Proof. Since the objective in (8.25) is convex and differentiable on $\text{dom } \phi$, first-order optimality conditions are both necessary and sufficient. The logarithmic barrier enforces strict feasibility automatically, while the equality constraint is handled by a Lagrange multiplier $\hat{\lambda}(t)$. $\square$

8.3.2 ■ Approximate KKT Conditions

Having defined the barrier problem and its central path, we now ask in what precise sense a central point approximates a solution of the original constrained problem. The following result answers this by showing that each central point satisfies a *perturbed KKT* system, in which exact complementary slackness $\mu_j g_j(x) = 0$ is replaced by a controlled residual of size $1/t$.

Theorem 8.25 (Barrier Solutions Satisfy Approximate KKT Conditions). *Let $x^*(t)$ be the minimizer of (8.25). Define*

$$\mu_j^*(t) := -\frac{1}{tg_j(x^*(t))}, \quad \lambda^*(t) := \frac{\hat{\lambda}(t)}{t}.$$

Then the triplet $(x^(t), \mu^*(t), \lambda^*(t))$ satisfies:*

$$\begin{aligned} Ax^*(t) &= b, \\ g_j(x^*(t)) &< 0, \\ \mu^*(t) &\succeq 0, \\ \nabla f(x^*(t)) + \sum_{j=1}^m \mu_j^*(t) \nabla g_j(x^*(t)) + A^T \lambda^*(t) &= 0, \\ (\mu^*(t))^T g(x^*(t)) &= -\frac{m}{t}. \end{aligned}$$

Proof. Substituting the definition of $\nabla\phi(x)$ into the stationarity condition of Theorem 8.24 and rescaling by $1/t$ yields the stated gradient condition. Since $g_j(x^*(t)) < 0$, we have $\mu_j^*(t) > 0$, ensuring dual feasibility. The final identity follows directly:

$$\sum_{j=1}^m \mu_j^*(t) g_j(x^*(t)) = \sum_{j=1}^m \left(-\frac{1}{tg_j(x^*(t))} \right) g_j(x^*(t)) = -\frac{m}{t}.$$

□

8.3.3 ■ Convergence to the True KKT Conditions

The approximate KKT conditions of the previous subsection carry an explicit error term of order $1/t$. It remains to verify that this error vanishes as the barrier parameter grows, so that central points converge to a genuine optimal solution. The following theorem makes this quantitative by bounding the suboptimality of the barrier minimizer through weak duality.

Theorem 8.26 (Convergence of Barrier Solutions). *Let f^* denote the optimal value of (8.24). Then*

$$f(x^*(t)) - \frac{m}{t} = q(\mu^*(t), \lambda^*(t)) \leq f^*,$$

and hence $f^ \leq f(x^*(t)) \leq f^* + \frac{m}{t}$. In particular, $\lim_{t \rightarrow \infty} f(x^*(t)) = f^*$, and the approximate complementary slackness condition converges to $(\mu^*)^T g(x^*) = 0$.*

Proof. The value $\mathcal{L}(x^*(t), \mu^*(t), \lambda^*(t))$ equals $f(x^*(t)) - m/t$ by direct computation. Since the dual function provides a lower bound on the primal optimum, the stated inequalities follow. Letting $t \rightarrow \infty$ yields convergence of both the objective value and the KKT system. □

The logarithmic barrier method replaces the hard complementary slackness condition $\mu^T g(x) = 0$ by the smooth approximation $\mu^T g(x) = -\frac{m}{t}$, which vanishes asymptotically. Interior-point methods can thus be interpreted as tracking a smooth path of approximate KKT points that converges to an exact solution of the original inequality constrained problem.

8.3.4 ■ Case Study: The Central Path of a Linear Program

We now specialize the general barrier and central-path framework to linear programming, where every ingredient becomes fully explicit. After setting up the logarithmic barrier, we compute its gradient and Hessian, derive the equation that characterizes the central path, and close with a worked two-dimensional example that traces the path numerically.

Consider the linear program

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & c^T x \\ \text{subject to} \quad & a_j^T x \leq b_j, \quad j = 1, \dots, m. \end{aligned} \tag{8.26}$$

Let $A \in \mathbb{R}^{m \times n}$ denote the matrix whose rows are a_j^T .

Theorem 8.27 (Logarithmic Barrier for Linear Programs). *The logarithmic barrier associated with (8.26) is $\phi(x) = -\sum_{j=1}^m \ln(b_j - a_j^T x)$, with domain $\text{dom } \phi = \{x \in \mathbb{R}^n \mid a_j^T x < b_j, j = 1, \dots, m\}$. For $t > 0$, the corresponding barrier problem is*

$$\min_{x \in \text{dom } \phi} \quad t c^T x + \phi(x). \tag{8.27}$$

Proof. This is a direct specialization of the general logarithmic barrier $\phi(x) = -\sum_j \ln(-g_j(x))$ with inequality constraints $g_j(x) = a_j^T x - b_j$. □

Gradient and Hessian of the barrier.

Specializing the barrier framework to linear programs, tracking the central path numerically—and analyzing its curvature—requires the first two derivatives of the logarithmic barrier. Both admit compact expressions in terms of the constraint slacks $b_j - a_j^T x$, which we record next; the Hessian in particular is the coefficient matrix of the Newton steps used by path-following methods.

Theorem 8.28 (Gradient and Hessian Structure). *Let $d \in \mathbb{R}^m$ be defined componentwise by $d_j := \frac{1}{b_j - a_j^T x}$. Then the gradient and Hessian of the barrier function are given by*

$$\begin{aligned}\nabla \phi(x) &= \sum_{j=1}^m \frac{1}{b_j - a_j^T x} a_j = A^T d, \\ \nabla^2 \phi(x) &= \sum_{j=1}^m \frac{1}{(b_j - a_j^T x)^2} a_j a_j^T = A^T \text{diag}(d)^2 A.\end{aligned}$$

Proof. Differentiating each term $-\ln(b_j - a_j^T x)$ yields

$$\nabla [-\ln(b_j - a_j^T x)] = \frac{1}{b_j - a_j^T x} a_j,$$

and differentiating once more gives

$$\nabla^2 [-\ln(b_j - a_j^T x)] = \frac{1}{(b_j - a_j^T x)^2} a_j a_j^T.$$

Summing over j produces the stated expressions. $\square$

Corollary 8.29 (Nonsingularity of the Hessian). *Since $d_j > 0$ for all $x \in \text{dom } \phi$, the Hessian $\nabla^2 \phi(x)$ is nonsingular if and only if $\text{rank}(A) = n$.*

Proof. Because $\text{diag}(d)^2$ is positive definite, the Hessian $A^T \text{diag}(d)^2 A$ is positive definite if and only if A has full column rank. $\square$

The central path condition.

With the gradient of the barrier now available, the central path of a linear program can be summarized by a single stationarity equation. The following result states this condition and gives it a geometric reading: at each central point the barrier gradient is aligned with the cost vector.

Theorem 8.30 (Central Path Equation for LPs). *A point $x^*(t)$ lies on the central path of (8.26) if and only if $tc + \sum_{j=1}^m \frac{1}{b_j - a_j^T x^*(t)} a_j = 0$. Equivalently, $\nabla \phi(x^*(t)) = -tc$.*

Proof. The first-order optimality condition for (8.27) is $t\nabla(c^T x) + \nabla \phi(x) = 0$. Since $\nabla(c^T x) = c$, substitution of the expression for $\nabla \phi(x)$ from Theorem 8.28 yields the stated condition. $\square$

Theorem 8.31 (Geometric Interpretation of the Central Path). *At a point $x^*(t)$ on the central path, the gradient of the barrier function $\nabla \phi(x^*(t))$ is parallel to $-c$. Consequently, the hyperplane $c^T x = c^T x^*(t)$ is tangent to the level set $\{x \mid \phi(x) = \phi(x^*(t))\}$ at $x^*(t)$.*

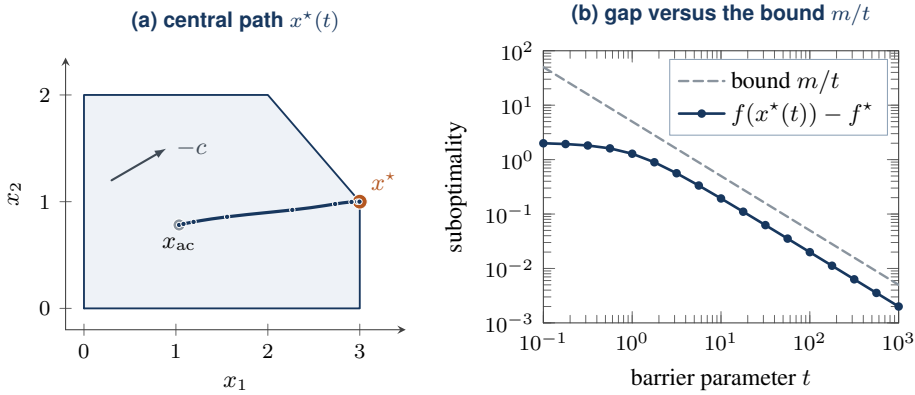


Figure 8.3: Logarithmic-barrier central path for the linear program of Example 8.32 ($m = 5$ inequality constraints). (a) As t increases from 0 the minimizer $x^*(t)$ leaves the analytic center x_{ac} and follows a smooth curve to the optimal vertex $x^* = (3, 1)$, staying strictly interior throughout; at every point the barrier level set is tangent to an objective hyperplane $c^\top x = \text{const}$. (b) The suboptimality $f(x^*(t)) - f^*$ computed along the path stays below the guaranteed bound m/t (dashed), which decays as $O(1/t)$.

Proof. By Theorem 8.30, $\nabla \phi(x^*(t)) = -tc$, so the two vectors are collinear. Since gradients are normal vectors to level sets, the objective hyperplane $c^\top x = \text{const}$ is tangent to the level set of ϕ at $x^*(t)$. $\square$

A worked example.

Example 8.32 Central Path of a Two-Dimensional Linear Program.

Consider the linear program $\min_x c^\top x$ with $c = (-1, -0.5)$ —equivalently, $\max x_1 + 0.5x_2$ —over the polytope

$$x_1 \leq 3, \quad x_2 \leq 2, \quad x_1 + x_2 \leq 4, \quad x_1 \geq 0, \quad x_2 \geq 0,$$

which has $m = 5$ inequality constraints. The unique optimal vertex is $x^* = (3, 1)$ with $f^* = c^\top x^* = -3.5$.

For $t > 0$ the central point $x^*(t)$ solves the barrier stationarity condition of Theorem 8.30,

$$tc + \sum_{j=1}^5 \frac{1}{b_j - a_j^\top x^*(t)} a_j = 0,$$

which we solve by Newton's method along a grid of increasing t . As $t \downarrow 0$ the objective term vanishes and $x^*(t)$ approaches the *analytic center* $x_{ac} \approx (1.034, 0.782)$, the point that maximizes the product of slacks. As t grows the objective dominates and $x^*(t)$ traces the smooth curve of Figure 8.3(a), always strictly interior, converging to the optimal vertex $(3, 1)$.

Theorem 8.26 guarantees $f(x^*(t)) - f^* \leq m/t$; for instance $t = 10$ certifies suboptimality at most 0.5, and $t = 100$ at most 0.05. Figure 8.3(b) shows this $O(1/t)$ decay, quantifying how far along the path one must travel for a prescribed accuracy. $\square$

Remark 8.33 (Connection to the Figure). The feasible polytope is defined by the intersection of the half-spaces $a_j^T x \leq b_j$. The level sets of the logarithmic barrier form smooth, strictly convex curves inside the polytope. The central path traces the locus of tangency points between these level sets and objective hyperplanes $c^T x = \text{const}$. As $t \rightarrow \infty$, these tangency points converge to the optimal vertex of the linear program.

8.3.5 ■ Barrier Methods as Homotopy Methods

Barrier methods admit a precise interpretation within the general framework of *homotopy* (or *continuation*) methods. Homotopy methods address a difficult problem by embedding it into a smooth, parameterized family of problems that continuously interpolates between an easily solvable problem and the target problem of interest. The solution of the original problem is obtained by tracking the solution trajectory of this family as the homotopy parameter varies.

General homotopy principle.

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a (possibly nonlinear) mapping, and consider the problem of solving

$$F(x) = 0. \quad (8.28)$$

A *homotopy* embeds (8.28) into a parameterized family of systems

$$H(x, \tau) = 0, \quad \tau \in [0, 1], \quad (8.29)$$

where $H : \mathbb{R}^n \times [0, 1] \rightarrow \mathbb{R}^n$ is continuously differentiable and satisfies $H(x, 0) = F(x)$, so that the endpoint $\tau = 0$ recovers the target system (8.28). The system $H(x, 1) = 0$ is chosen so that its solution is readily computable.

Theorem 8.34 (Local Homotopy Continuation). Suppose there exist $\tau_0 \in [0, 1]$ and $x_0 \in \mathbb{R}^n$ such that $H(x_0, \tau_0) = 0$, and that the Jacobian $\nabla_x H(x_0, \tau_0)$ is nonsingular. Then there exists a relatively open interval $I \subset [0, 1]$ containing τ_0 and a unique continuously differentiable function $x : I \rightarrow \mathbb{R}^n$ such that $x(\tau_0) = x_0$ and $H(x(\tau), \tau) = 0$ for all $\tau \in I$. Moreover, the solution path satisfies

$$\frac{dx(\tau)}{d\tau} = -[\nabla_x H(x(\tau), \tau)]^{-1} \nabla_\tau H(x(\tau), \tau). \quad (8.30)$$

Proof. Since H is continuously differentiable and $\nabla_x H(x_0, \tau_0)$ is nonsingular, the implicit function theorem applies to the equation $H(x, \tau) = 0$ in a neighborhood of (x_0, τ_0) . This yields a locally unique C^1 solution curve $\tau \mapsto x(\tau)$. Differentiating the identity $H(x(\tau), \tau) = 0$ with respect to τ gives (8.30). $\square$

Theorem 8.34 shows that, under regularity conditions, the solution set of (8.29) forms a smooth one-dimensional manifold. Homotopy methods exploit this structure by numerically tracking the solution curve from $\tau = 1$ toward $\tau = 0$ using predictor–corrector or Newton-type schemes. The original problem (8.28) is thus solved indirectly via *path following*.

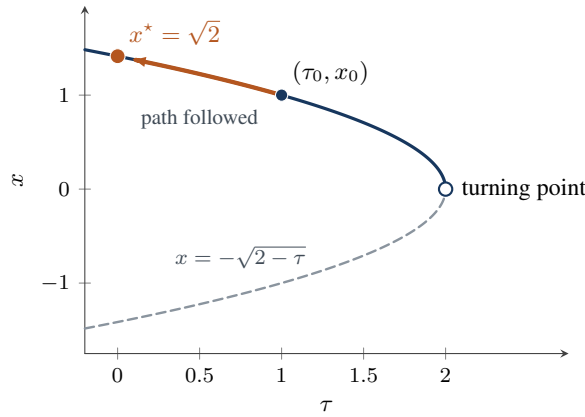


Figure 8.4: Homotopy continuation for the scalar system of Example 8.35, with $H(x, \tau) = x^2 - 2 + \tau$. The solution set is the parabola $x = \pm\sqrt{2 - \tau}$. Theorem 8.34 guarantees a unique smooth path (solid) through the start point $(\tau_0, x_0) = (1, 1)$; tracking it as $\tau \rightarrow 0$ (arrow) reaches the target root $x^* = \sqrt{2}$ of $F(x) = x^2 - 2$. The Jacobian $\nabla_x H = 2x$ vanishes only at the turning point $(\tau, x) = (2, 0)$, where the two branches meet and the implicit-function hypothesis fails, exactly the boundary of validity asserted by the theorem.

The next two examples make the theorem concrete: the first exhibits a continuation path in closed form and shows where the nonsingularity hypothesis fails, and the second identifies the central path of a barrier method as a special case.

Example 8.35 Continuation path for a scalar root.

To solve $F(x) = x^2 - 2 = 0$ by continuation, embed it in the family $H(x, \tau) = x^2 - 2 + \tau$, $\tau \in [0, 2)$, so that $H(x, 0) = F(x)$ and the start $\tau_0 = 1$ gives the trivially known root $x_0 = 1$ of $H(x, 1) = x^2 - 1$. Here $\nabla_x H = 2x$, which is nonsingular at $(x_0, \tau_0) = (1, 1)$, so Theorem 8.34 guarantees a locally unique C^1 path. Solving $H = 0$ explicitly gives $x(\tau) = \sqrt{2 - \tau}$, and one checks the path equation (8.30) directly:

$$\frac{dx}{d\tau} = -[\nabla_x H]^{-1} \nabla_\tau H = -\frac{1}{2x(\tau)} = -\frac{1}{2\sqrt{2 - \tau}},$$

which agrees with differentiating $\sqrt{2 - \tau}$. Tracking the path as $\tau \rightarrow 0$ yields the target root $x^* = \sqrt{2}$.

The example also exposes the *local* nature of the theorem. The full solution set $\{x = \pm\sqrt{2 - \tau}\}$ is the parabola of Figure 8.4; its two branches meet at the turning point $(\tau, x) = (2, 0)$, where $\nabla_x H = 2x$ vanishes. There the implicit-function hypothesis fails, the path cannot be written as a graph $x(\tau)$, and naive continuation in τ breaks down, which is precisely why practical path followers parameterize by arc length rather than by τ . $\square$

Example 8.36 The central path as a continuation trajectory.

Consider the one-dimensional barrier problem $\min_{x>0} tx - \ln x$, the logarithmic-barrier approximation of $\min_{x\geq 0} x$ with optimum $x = 0$. Its stationarity condition is

$$H(x, t) = t - \frac{1}{x} = 0,$$

a homotopy in the barrier parameter $t > 0$ playing the role of the continuation variable. The unique solution is the central path

$$x^*(t) = \frac{1}{t},$$

and since $\nabla_x H = 1/x^2 > 0$ is nonsingular for every $x > 0$, Theorem 8.34 applies at each t and the path is globally smooth. Its velocity matches (8.30):

$$\frac{dx^*}{dt} = -[\nabla_x H]^{-1} \nabla_t H = -x^2 \cdot 1 = -\frac{1}{t^2} = \frac{d}{dt} \left(\frac{1}{t} \right).$$

As $t \rightarrow \infty$ the central point $x^*(t) = 1/t$ approaches the constrained optimum $x = 0$ on the boundary, illustrating how the general continuation principle of Theorem 8.34 specializes to the barrier-induced homotopy developed next. Note that the direction is reversed relative to the generic continuation variable $\tau : 1 \rightarrow 0$: here large t plays the role of small τ (with $\tau = 1/t$), so path-following runs in the direction of increasing t .

□

Homotopy induced by the barrier parameter.

In constrained optimization, the Karush–Kuhn–Tucker (KKT) conditions define a system of equations and inequalities involving primal variables, dual variables, and the complementary slackness relations $\mu_j g_j(x) = 0$. These relations introduce nonsmoothness and degeneracy into the KKT system.

The logarithmic barrier replaces complementary slackness by the smooth, parameterized relation $\mu_j(t) g_j(x(t)) = -\frac{1}{t}$, thereby yielding a regularized KKT system indexed by $t > 0$.

Define the homotopy map

$$H(x, \mu, \lambda, t) = \begin{pmatrix} \nabla f(x) + \sum_{j=1}^m \mu_j \nabla g_j(x) + A^T \lambda \\ Ax - b \\ \text{diag}(\mu)g(x) + \frac{1}{t} \mathbf{1} \end{pmatrix}.$$

For each finite $t > 0$, the equation $H(x, \mu, \lambda, t) = 0$ characterizes the first-order optimality conditions of the barrier problem. As $t \rightarrow \infty$, the perturbation term $\frac{1}{t} \mathbf{1}$ vanishes, and the homotopy deforms continuously into the exact KKT system of the original constrained problem.

Central path as a continuation trajectory. Under convexity and regularity assumptions, the Jacobian of H with respect to (x, μ, λ) remains nonsingular along the solution set for all sufficiently large t . Consequently, the family $\{(x^*(t), \mu^*(t), \lambda^*(t))\}_{t>0}$ forms a smooth continuation path, known as the *central path*. This path connects strictly feasible, well-centered solutions of the barrier problem to a primal–dual KKT solution of the original constrained problem in the limit $t \rightarrow \infty$.

Geometrically, the central path is a smooth one-dimensional manifold embedded in the primal–dual space that regularizes the KKT conditions and avoids degeneracies associated with active constraints.

Variational inequality and dynamical viewpoints.

The Karush–Kuhn–Tucker (KKT) conditions of a convex optimization problem admit an equivalent formulation as a *variational inequality*. Specifically, let $X \subseteq \mathbb{R}^n$ denote the feasible set defined by the constraints in (8.24). Then a point $x^* \in X$ is optimal if and only if it satisfies

$$\langle \nabla f(x^*), x - x^* \rangle \geq 0, \quad \forall x \in X, \quad (8.31)$$

or, equivalently, $0 \in \nabla f(x^*) + N_X(x^*)$, where $N_X(x^*)$ denotes the normal cone to X at x^* . The normal cone operator is set-valued and generally nonsmooth, reflecting the nonsmooth nature of active constraints and complementary slackness.

The logarithmic barrier induces a smooth approximation of this variational inequality by replacing the normal cone $N_X(\cdot)$ with the gradient of a barrier function ϕ . For strictly feasible points $x \in \text{dom } \phi$, the barrier problem satisfies the smooth stationarity condition $\nabla f(x) + \frac{1}{t} \nabla \phi(x) = 0$, which can be interpreted as a single-valued, Lipschitz-continuous approximation of the inclusion $0 \in \nabla f(x) + N_X(x)$. In this sense, the barrier homotopy regularizes the variational inequality by replacing a nonsmooth, set-valued operator with a smooth vector field.

This smoothing perspective explains why Newton and quasi-Newton methods are effective along the central path: the barrier gradient provides curvature that prevents degeneracy near the boundary of the feasible set, while preserving the asymptotic structure of the original KKT conditions as $t \rightarrow \infty$.

A complementary viewpoint arises by interpreting the homotopy parameter t as a time-like variable. Differentiating the stationarity condition $\nabla f(x(t)) + \frac{1}{t} \nabla \phi(x(t)) = 0$ with respect to t yields an implicit ordinary differential equation governing the evolution of the central path:

$$\left[\nabla^2 f(x(t)) + \frac{1}{t} \nabla^2 \phi(x(t)) \right] \dot{x}(t) = \frac{1}{t^2} \nabla \phi(x(t)). \quad (8.32)$$

Under convexity and regularity assumptions, the matrix multiplying $\dot{x}(t)$ is positive definite, ensuring local well-posedness of the dynamics.

From this dynamical systems perspective, the central path is a trajectory of a smooth differential equation in primal–dual space. Interior-point methods may thus be interpreted as discrete-time predictor–corrector schemes that approximate the flow (8.32), combining Newton-type correction steps with controlled updates of the homotopy parameter. This interpretation provides a conceptual bridge between continuous-time optimization dynamics and practical path-following algorithms.

8.4 ■ Primal–Dual Algorithms

Primal–dual algorithms update *both* primal decision variables and dual multipliers in order to locate a saddle point of the Lagrangian. Their appeal lies in the fact that dual variables naturally encode constraint violations and often admit a distributed or economic interpretation (prices, congestion signals, shadow costs), while the primal update resembles unconstrained optimization perturbed by these signals.

From an algorithmic perspective, primal–dual methods are best understood not as constraint-handling tricks, but as explicit numerical schemes for solving a *saddle-point problem*.

8.4.1 ■ Problem setup and the saddle-point viewpoint

Consider the convex optimization problem

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) \\ \text{subject to} \quad & g(x) \preceq 0, \\ & Ax = b, \end{aligned} \tag{8.33}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and each component $g_j : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and continuously differentiable, $A \in \mathbb{R}^{p \times n}$, and Slater's condition holds. Under these assumptions, strong duality applies and Karush–Kuhn–Tucker (KKT) multipliers exist.

The Lagrangian associated with (8.33) is

$$\mathcal{L}(x, \mu, \lambda) = f(x) + \mu^\top g(x) + \lambda^\top (Ax - b), \quad \mu \succeq 0, \lambda \in \mathbb{R}^p. \tag{8.34}$$

A central fact, which we recall here for completeness, is that optimality of the primal problem is equivalent to the existence of a saddle point of L .

Theorem 8.37 (KKT conditions and saddle points). *Assume f and g_j are convex and differentiable, $Ax = b$ is affine, and Slater's condition holds for (8.33). Then x^* is an optimal solution of (8.33) if and only if there exist multipliers $\mu^* \succeq 0$ and λ^* such that*

$$\mathcal{L}(x^*, \mu, \lambda) \leq \mathcal{L}(x^*, \mu^*, \lambda^*) \leq \mathcal{L}(x, \mu^*, \lambda^*), \quad \forall x, \forall \mu \succeq 0, \forall \lambda. \tag{8.35}$$

Equivalently, (x^*, μ^*, λ^*) satisfies the KKT conditions

$$\begin{aligned} g(x^*) &\preceq 0, \quad Ax^* = b, \quad \mu^* \succeq 0, \\ \nabla f(x^*) + \sum_{j=1}^m \mu_j^* \nabla g_j(x^*) + A^\top \lambda^* &= 0, \\ (\mu^*)^\top g(x^*) &= 0. \end{aligned} \tag{8.36}$$

Proof. By Slater's condition and convexity, strong duality holds and KKT multipliers exist. The KKT conditions are necessary and sufficient for optimality. Stationarity implies that x^* minimizes $\mathcal{L}(\cdot, \mu^*, \lambda^*)$ over x , while feasibility, dual feasibility, and complementary slackness imply that (μ^*, λ^*) maximizes $\mathcal{L}(x^*, \cdot, \cdot)$ over $\mu \succeq 0$ and $\lambda \in \mathbb{R}^p$. These two statements together are equivalent to the saddle-point inequality (8.35). $\square$

This equivalence suggests a natural algorithmic principle: move *downhill* in the primal variables and *uphill* in the dual variables, while enforcing $\mu \succeq 0$ by projection.

To make the equivalence in Theorem 8.37 concrete, the following two examples exhibit the saddle point explicitly—first for an inequality constraint, then for an equality constraint—and verify the saddle inequality (8.35) in both directions.

Example 8.38 Saddle point of the Lagrangian: an inequality constraint.

Consider the one-dimensional convex program

$$\min_{x \in \mathbb{R}} \quad \frac{1}{2}x^2 \quad \text{subject to} \quad g(x) = 1 - x \leq 0,$$

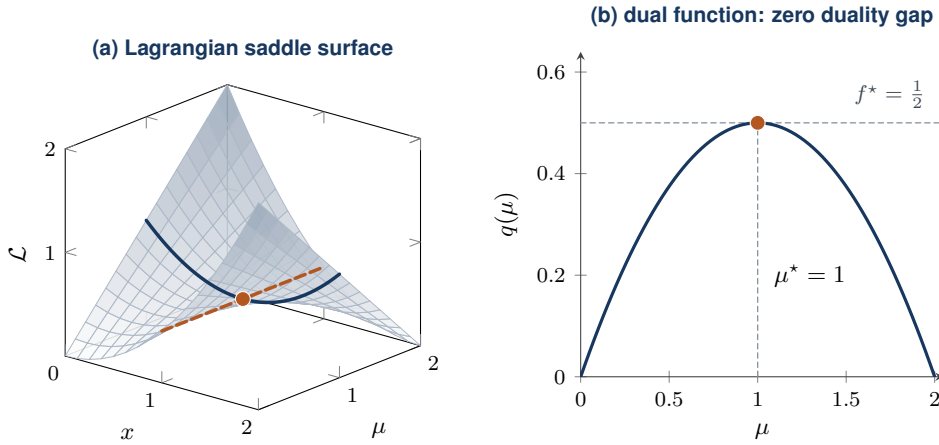


Figure 8.5: Saddle-point structure of the Lagrangian for the problem $\min \frac{1}{2}x^2$ subject to $x \geq 1$ (Example 8.38), illustrating Theorem 8.37. (a) $\mathcal{L}(x, \mu)$ is convex along x and affine, hence concave, along μ ; the KKT point $(x^*, \mu^*) = (1, 1)$ (dot) is a saddle: a minimum in the primal direction (solid cut $\mathcal{L}(x, \mu^*)$) and a maximum in the dual direction (dashed cut $\mathcal{L}(x^*, \mu)$, constant by complementary slackness). (b) The dual function $q(\mu) = \min_x \mathcal{L}(x, \mu) = \mu - \frac{1}{2}\mu^2$ is concave and attains its maximum $q(\mu^*) = \frac{1}{2}$ exactly at the primal optimal value $f^* = \frac{1}{2}$, so strong duality holds with zero gap.

that is, $x \geq 1$, with no equality constraints. The Lagrangian is $\mathcal{L}(x, \mu) = \frac{1}{2}x^2 + \mu(1 - x)$ with $\mu \geq 0$.

For fixed μ the inner minimization $\min_x \mathcal{L}(x, \mu)$ has stationary point $x = \mu$. The constraint is active at the optimum, so $x^* = 1$ and stationarity forces $\mu^* = 1$, giving $f^* = \frac{1}{2}$. The pair $(x^*, \mu^*) = (1, 1)$ satisfies the KKT conditions (8.36): $g(x^*) = 0$, $\mu^* \geq 0$, the stationarity $x^* - \mu^* = 0$ (i.e. $\nabla f + \mu^* \nabla g = 0$), and complementary slackness $\mu^* g(x^*) = 0$.

We verify the saddle inequality (8.35) directly. *Primal (right) inequality:* with $\mu^* = 1$ fixed, $\mathcal{L}(x, 1) = \frac{1}{2}x^2 + (1 - x) = \frac{1}{2}(x - 1)^2 + \frac{1}{2}$ is minimized at $x^* = 1$, so $\mathcal{L}(x^*, \mu^*) \leq \mathcal{L}(x, \mu^*)$ for all x . *Dual (left) inequality:* since $g(x^*) = 0$, we have $\mathcal{L}(x^*, \mu) = \frac{1}{2}$ for every $\mu \geq 0$; hence $\mathcal{L}(x^*, \mu) \leq \mathcal{L}(x^*, \mu^*)$ with equality: this is exactly complementary slackness. Thus $(1, 1)$ is a saddle point, as Theorem 8.37 predicts.

The min–max structure is displayed in Figure 8.5. The dual function $q(\mu) = \min_x \mathcal{L}(x, \mu) = \mathcal{L}(\mu, \mu) = \mu - \frac{1}{2}\mu^2$ is concave and attains its maximum $q(\mu^*) = \frac{1}{2} = f^*$ at $\mu^* = 1$: strong duality holds with zero gap, and $\max_{\mu \geq 0} \min_x \mathcal{L}(x, \mu) = \min_x \max_{\mu \geq 0} \mathcal{L}(x, \mu)$. $\square$

Example 8.39 Saddle point with an equality constraint.

Consider next an equality-constrained problem, which exercises the multiplier λ of Theorem 8.37:

$$\min_{x \in \mathbb{R}^2} \frac{1}{2} \|x\|^2 \quad \text{subject to} \quad x_1 + x_2 = 2.$$

Here $A = (1, 1)$, $b = 2$, and $\mathcal{L}(x, \lambda) = \frac{1}{2} \|x\|^2 + \lambda(x_1 + x_2 - 2)$ with $\lambda \in \mathbb{R}$ unrestricted in sign.

Stationarity $x + \lambda(1, 1)^\top = 0$ gives $x^* = (1, 1)$ and $\lambda^* = -1$, with $f^* = 1$; indeed $\nabla f(x^*) + A^\top \lambda^* = (1, 1) + (-1)(1, 1) = 0$ and $Ax^* = 2$. For the saddle inequality: with $\lambda^* = -1$ fixed, $\mathcal{L}(x, -1) = \frac{1}{2} \|x\|^2 - (x_1 + x_2) + 2$ is strictly convex with minimizer $x^* = (1, 1)$, which gives the right inequality; and since $Ax^* = b$, we have $\mathcal{L}(x^*, \lambda) = 1$ for all λ , so the left inequality holds with equality. Hence (x^*, λ^*) is a saddle point over $x \in \mathbb{R}^2, \lambda \in \mathbb{R}$.

The contrast with Example 8.38 is instructive: the inequality multiplier μ is confined to $\mu \geq 0$ and the dual inequality encodes complementary slackness, whereas the equality multiplier λ ranges over all of $\mathbb{R}$ and the dual inequality reduces to primal feasibility $Ax^* = b$. $\square$

8.4.2 ■ Projected primal–dual gradient method

The projected primal–dual gradient method performs gradient descent in the primal variable and gradient ascent in the dual variables:

$$\begin{aligned} x_{k+1} &= x_k - \gamma_k \nabla_x \mathcal{L}(x_k, \mu_k, \lambda_k), \\ \mu_{k+1} &= \Pi_{\mathbb{R}_+^m}(\mu_k + \beta_k g(x_k)), \\ \lambda_{k+1} &= \lambda_k + \beta_k (Ax_k - b), \end{aligned} \tag{8.37}$$

where $\Pi_{\mathbb{R}_+^m}$ denotes Euclidean projection onto the nonnegative orthant, and $\gamma_k > 0$ and $\beta_k > 0$ are the primal and dual step sizes.

Using (8.34), the primal gradient takes the form

$$\nabla_x \mathcal{L}(x, \mu, \lambda) = \nabla f(x) + \sum_{j=1}^m \mu_j \nabla g_j(x) + A^\top \lambda. \tag{8.38}$$

Each dual variable integrates a constraint violation:

$$g_j(x_k) > 0 \Rightarrow \mu_{j,k+1} \text{ increases}, \quad Ax_k \neq b \Rightarrow \lambda_{k+1} \text{ adjusts to penalize mismatch.}$$

Thus, dual variables act as adaptive penalties or prices, while the primal update balances objective descent against the current penalty signals.

Unlike projection-based feasible methods, the primal–dual gradient scheme does *not* maintain primal feasibility at every iteration. Feasibility is instead enforced asymptotically through dual ascent, a feature that is particularly attractive in large-scale or decentralized settings.

8.4.3 ■ Convergence in the strongly convex–concave case

We now present a convergence result that formalizes the stability and rate properties of the projected primal–dual gradient method under additional regularity assumptions.

Theorem 8.40 (Global linear convergence under strong convexity). *Consider the primal–dual iterations (8.37). Assume that the objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is σ -strongly convex and has an L_f -Lipschitz continuous gradient. Assume further that the inequality constraints are affine, $g(x) = Cx - d$, that the equality constraints are given by $Ax = b$, and that Slater’s condition holds. Assume finally the regularity condition that the stacked constraint matrix $\begin{pmatrix} C \\ A \end{pmatrix}$ has full row rank.*

Then the Lagrangian $\mathcal{L}(x, \mu, \lambda)$ is strongly convex in the primal variable x and concave in the dual variables (μ, λ) . The saddle point (x^, μ^*, λ^*) is unique. Moreover, there exist constants $\bar{\gamma} > 0$ and $\bar{\beta} > 0$ such that, for all constant stepsizes $0 < \gamma \leq \bar{\gamma}$ and $0 < \beta \leq \bar{\beta}$, the iterates generated by (8.37) converge linearly to (x^*, μ^*, λ^*) .*

Proof. Define the saddle-point operator

$$F(x, \mu, \lambda) = \begin{pmatrix} \nabla_x \mathcal{L}(x, \mu, \lambda) \\ -\nabla_\mu \mathcal{L}(x, \mu, \lambda) \\ -\nabla_\lambda \mathcal{L}(x, \mu, \lambda) \end{pmatrix} = \begin{pmatrix} \nabla f(x) + C^\top \mu + A^\top \lambda \\ -(Cx - d) \\ -(Ax - b) \end{pmatrix}. \quad (8.39)$$

Let $z = (x, \mu, \lambda)$ and define the feasible set

$$\mathcal{Z} = \mathbb{R}^n \times \mathbb{R}_+^m \times \mathbb{R}^p.$$

By Theorem 8.37, the saddle point (x^*, μ^*, λ^*) is characterized as the unique solution of the variational inequality

$$\langle F(z^*), z - z^* \rangle \geq 0, \quad \forall z \in \mathcal{Z}. \quad (8.40)$$

Since f is σ -strongly convex, its gradient is σ -strongly monotone, that is,

$$\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \sigma \|x - y\|^2, \quad \forall x, y \in \mathbb{R}^n.$$

It is tempting to conclude that F inherits strong monotonicity, but it does not. Writing $\Delta x = x - x'$, $\Delta \mu = \mu - \mu'$, $\Delta \lambda = \lambda - \lambda'$ and using affineness of the constraints,

$$\begin{aligned} \langle F(z) - F(z'), z - z' \rangle &= \langle \nabla f(x) - \nabla f(x'), \Delta x \rangle + \langle C^\top \Delta \mu, \Delta x \rangle - \langle C \Delta x, \Delta \mu \rangle \\ &\quad + \langle A^\top \Delta \lambda, \Delta x \rangle - \langle A \Delta x, \Delta \lambda \rangle = \langle \nabla f(x) - \nabla f(x'), \Delta x \rangle \geq \sigma \|\Delta x\|^2. \end{aligned}$$

The dual cross terms cancel *exactly*, so F is monotone, and strongly monotone only in the primal block: the bound carries no $\|\Delta \mu\|^2$ or $\|\Delta \lambda\|^2$. F is, however, Lipschitz continuous with a constant $L > 0$ depending on L_f , $\|C\|$ and $\|A\|$.

The primal–dual update (8.37) can be written compactly as

$$z_{k+1} = \Pi_{\mathcal{Z}}(z_k - \Gamma F(z_k)), \quad (8.41)$$

where $\Gamma = \text{diag}(\gamma I_n, \beta I_m, \beta I_p)$. The projection $\Pi_{\mathcal{Z}}$ is firmly nonexpansive. Monotonicity of F alone yields, for γ, β small enough, only the Fejér estimate $\|z_{k+1} - z^*\| \leq \|z_k - z^*\|$: the iterates cannot move away from the saddle point, but nothing so far forces the dual error to decrease, and indeed the plain primal–dual gradient scheme can cycle when the dual block is unconstrained by the data.

This is where the rank condition enters. Full row rank of $\begin{pmatrix} C \\ A \end{pmatrix}$ ensures that a nonzero dual error produces a nonzero primal residual, so that the primal contraction $\sigma \|\Delta x\|^2$ can be transferred to the dual variables. Concretely, there is a positive definite weighting P and $\rho \in (0, 1)$ such that the Lyapunov function $V(z) := \|z - z^*\|_P^2$ satisfies $V(z_{k+1}) \leq \rho V(z_k)$ for all sufficiently small γ, β ; the construction of P from σ, L and the smallest singular value of the constraint matrix is

standard and may be found in [15, 18]. Since $\|\cdot\|_P$ and $\|\cdot\|$ are equivalent norms, this establishes global linear convergence to the unique saddle point. $\square$

When f is convex but not strongly convex, the operator F is monotone but not strongly monotone, and constant stepsizes may lead to oscillatory behavior. Convergence can then be recovered through diminishing stepsizes, ergodic averaging, or stabilized primal–dual schemes such as the augmented Lagrangian, ADMM, or extra-gradient (mirror–prox) methods.

Remark 8.41 (Lyapunov-function perspective). An alternative proof of Theorem 8.40 can be obtained via a Lyapunov-function argument. Define the error vector $e_k := (x_k - x^*, \mu_k - \mu^*, \lambda_k - \lambda^*)$ and consider the quadratic Lyapunov function

$$V_k = \frac{1}{2}\|x_k - x^*\|^2 + \frac{1}{2}\|\mu_k - \mu^*\|^2 + \frac{1}{2}\|\lambda_k - \lambda^*\|^2.$$

Using the update equations (8.37) and expanding $\|x_{k+1} - x^*\|^2$, one obtains

$$\|x_{k+1} - x^*\|^2 = \|x_k - x^*\|^2 - 2\gamma\langle \nabla_x \mathcal{L}(z_k), x_k - x^* \rangle + \gamma^2 \|\nabla_x \mathcal{L}(z_k)\|^2.$$

Strong convexity of f implies

$$\langle \nabla_x \mathcal{L}(z_k) - \nabla_x \mathcal{L}(z^*), x_k - x^* \rangle \geq \sigma \|x_k - x^*\|^2.$$

For the dual variables, nonexpansiveness of the projection onto $\mathbb{R}_+^m$ yields

$$\|\mu_{k+1} - \mu^*\|^2 \leq \|\mu_k - \mu^*\|^2 + 2\beta\langle g(x_k) - g(x^*), \mu_k - \mu^* \rangle + \beta^2 \|g(x_k)\|^2,$$

and an analogous identity holds for λ_k .

Combining these inequalities and using Lipschitz continuity of the gradients, one can bound the quadratic terms and obtain, for sufficiently small constant stepsizes,

$$V_{k+1} \leq V_k - \eta \|e_k\|^2$$

for some $\eta > 0$. Since V_k is equivalent to $\|e_k\|^2$, this implies the existence of $\rho \in (0, 1)$ such that

$$V_{k+1} \leq (1 - \rho)V_k, \quad \forall k,$$

and therefore linear convergence of the primal–dual iterates.

This Lyapunov viewpoint highlights the primal–dual gradient method as a discrete-time stable feedback system driven by constraint violations, and it is particularly useful when extending the analysis to time-varying, stochastic, or control-theoretic settings.

Example 8.42 Projected Primal–Dual Gradient on a Constrained QP.

Consider the strongly convex quadratic program

$$\min_{x \in \mathbb{R}^2} \frac{1}{2}x^\top Qx + q^\top x \quad \text{s.t.} \quad \underbrace{x_1 + x_2 \leq 1}_{g(x) \leq 0}, \quad \underbrace{x_1 - x_2 = 0}_{Ax=b},$$

with $Q = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$ and $q = (-2, -5)$. Here f is σ -strongly convex with $\sigma = \frac{5-\sqrt{5}}{2} \approx 1.38$ and $L_f = \frac{5+\sqrt{5}}{2} \approx 3.62$, and the constraints are affine, so Theorem 8.40 applies.

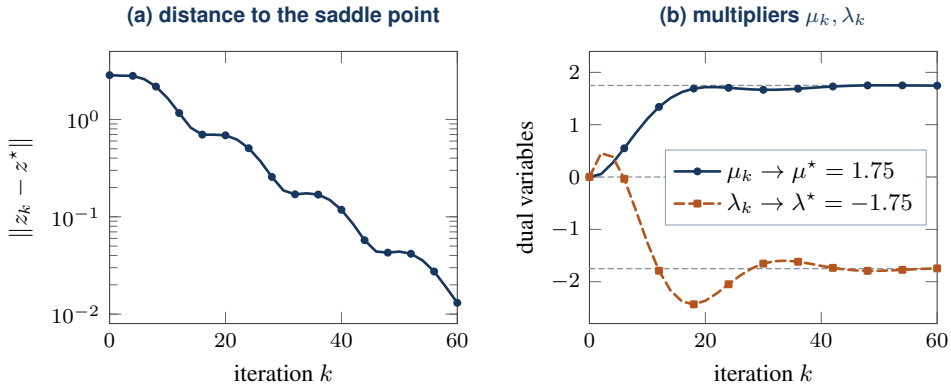


Figure 8.6: Projected primal–dual gradient method on the strongly convex QP of Example 8.42 with stepsizes $\gamma = \beta = 0.15$. (a) The distance $\|z_k - z^*\|$ of the joint variable $z_k = (x_k, \mu_k, \lambda_k)$ to the unique saddle point decays geometrically, i.e. linearly on the semilog scale, as guaranteed by Theorem 8.40. (b) The multipliers overshoot during the transient, dual ascent reacts to constraint violation, and then settle at the KKT values $\mu^* = \frac{7}{4}$ (inequality active) and $\lambda^* = -\frac{7}{4}$.

The inequality is active at the optimum. Enforcing $x_1 + x_2 = 1$ and $x_1 = x_2$ gives $x^* = (0.5, 0.5)$, and the stationarity condition $Qx^* + q + \mu^*a + \lambda^*e = 0$ with $a = (1, 1)$, $e = (1, -1)$ yields the KKT multipliers $\mu^* = \frac{7}{4}$ (nonnegative, confirming the inequality is binding) and $\lambda^* = -\frac{7}{4}$.

The updates (8.37),

$$\begin{aligned} x_{k+1} &= x_k - \gamma(Qx_k + q + \mu_k a + \lambda_k e), \\ \mu_{k+1} &= [\mu_k + \beta g(x_k)]_+, \quad \lambda_{k+1} = \lambda_k + \beta(x_{1,k} - x_{2,k}), \end{aligned}$$

were run with $\gamma = \beta = 0.15$ from the infeasible start $x_0 = (1.5, -0.5)$, $\mu_0 = \lambda_0 = 0$. Figure 8.6(a) confirms the linear convergence predicted by the theorem, while Figure 8.6(b) shows the multipliers overshooting during the transient, dual ascent responding to constraint violation, before settling at (μ^*, λ^*) . Note that feasibility holds only in the limit, in contrast to the projection methods of the previous section. $\square$

8.4.4 ■ Dual decomposition and separability

A central reason primal–dual algorithms play a foundational role in large-scale optimization is their ability to exploit *separability* through dual decomposition. When coupling among decision variables appears only through shared constraints, dual variables provide a mechanism for coordinating otherwise independent subsystems.

Consider optimization problems of the form

$$\begin{aligned}
 & \min_{x_1, \dots, x_N} \sum_{i=1}^N f_i(x_i) \\
 & \text{subject to} \quad \sum_{i=1}^N A_i x_i \preceq c, \\
 & \quad x_i \in X_i, \quad i = 1, \dots, N,
 \end{aligned} \tag{8.42}$$

where each $f_i : X_i \rightarrow \mathbb{R}$ is convex, each feasible set X_i is closed and convex, and the matrices A_i define a shared resource or coupling constraint. The objective function is additively separable across agents or subsystems, while the inequality constraint couples all decision variables through their aggregate effect.

From a centralized perspective, the constraint $\sum_{i=1}^N A_i x_i \preceq c$ prevents the problem from decomposing directly. Dual decomposition resolves this difficulty by transferring the coupling from the primal variables to the dual variables.

Introduce a Lagrange multiplier $\mu \succeq 0$ associated with the coupling constraint and define the (partial) Lagrangian

$$\begin{aligned}
 L(\{x_i\}, \mu) &= \sum_{i=1}^N f_i(x_i) + \mu^\top \left(\sum_{i=1}^N A_i x_i - c \right) \\
 &= \sum_{i=1}^N \left(f_i(x_i) + \mu^\top A_i x_i \right) - \mu^\top c.
 \end{aligned} \tag{8.43}$$

Crucially, for a fixed dual variable μ , the Lagrangian separates across the subsystems. The dependence on x_i appears only through the local term $f_i(x_i) + \mu^\top A_i x_i$.

As a result, the minimization of the Lagrangian over the primal variables decomposes into N independent subproblems:

$$x_i^*(\mu) \in \arg \min_{x_i \in X_i} \left(f_i(x_i) + \mu^\top A_i x_i \right), \quad i = 1, \dots, N. \tag{8.44}$$

Each subsystem solves a local optimization problem that depends only on its own decision variable and the current value of the dual multiplier μ . No direct information about other subsystems' decision variables is required.

The dual variable μ is then updated by ascending along the dual function, which amounts to correcting μ in the direction of the aggregate constraint violation:

$$\mu_{k+1} = \Pi_{\mathbb{R}_+^m} \left(\mu_k + \beta_k \left(\sum_{i=1}^N A_i x_{i,k} - c \right) \right). \tag{8.45}$$

The projection enforces the nonnegativity of the multiplier, while the update itself increases μ whenever the shared constraint is violated and decreases it when resources are underutilized.

This iterative interaction between local primal optimization and global dual updates constitutes the essence of dual decomposition. Each agent responds to a common signal μ_k by optimizing its own objective, while the dual variable aggregates the collective impact of all agents and feeds back a coordinating signal. In economic terms, μ_k can be interpreted as a vector of prices or congestion costs, and the update (8.45) implements a price-adjustment mechanism that equilibrates supply and demand.

From an algorithmic viewpoint, dual decomposition replaces a single large optimization problem by many smaller ones coupled only through low-dimensional dual variables. This structure underlies a wide range of distributed and networked optimization methods, including resource allocation in communication networks, power systems, and multi-agent control, and provides the conceptual bridge between primal–dual gradient methods and decentralized computation.

8.5 ■ Primal–Dual Algorithms for Network Utility Maximization

Consider a communication network consisting of a finite set of links $\mathcal{E}$ and a finite set of sources $\mathcal{S}$. Each source $r \in \mathcal{S}$ transmits data at rate $x_r \geq 0$ along a fixed route, which is represented as a subset of links $r \subseteq \mathcal{E}$. Each link $l \in \mathcal{E}$ has capacity $c_l > 0$. Each source r is associated with a utility function $U_r : \mathbb{R}_+ \rightarrow \mathbb{R}$ that quantifies the benefit of transmitting at rate x_r .

Assumption 8.43. *For each $r \in \mathcal{S}$, the utility function U_r is strictly increasing, continuously differentiable, and strictly concave.*

The *Network Utility Maximization* (NUM) problem is

$$\begin{aligned} & \max_{x \in \mathbb{R}_+^{|\mathcal{S}|}} \sum_{r \in \mathcal{S}} U_r(x_r) \\ & \text{subject to} \quad \sum_{r: l \in r} x_r \leq c_l, \quad \forall l \in \mathcal{E}. \end{aligned} \tag{8.46}$$

Since the objective is concave and the constraints are affine, problem (8.46) is a convex optimization problem (equivalently, a concave maximization problem). Under Slater's condition, it admits a unique optimal solution x^* .

Figure 8.7 illustrates this structure on a general network: the sources are flows routed across shared links, and two sources become coupled whenever their routes traverse a common link. The routing matrix R records which links each route uses, so that the capacity constraints in (8.46) simply state that the aggregate rate on every link stays within its capacity.

8.5.1 ■ Lagrangian and optimality conditions

Introduce Lagrange multipliers $p_l \geq 0$ associated with the link capacity constraints. The Lagrangian of problem (8.46) is given by

$$\mathcal{L}(x, p) = \sum_{r \in \mathcal{S}} U_r(x_r) - \sum_{l \in \mathcal{E}} p_l \left(\sum_{r: l \in r} x_r - c_l \right). \tag{8.47}$$

For each source r , define the *route price*

$$q_r := \sum_{l \in r} p_l. \tag{8.48}$$

The Karush–Kuhn–Tucker (KKT) conditions state that a pair (x^*, p^*) is optimal for (8.46) if

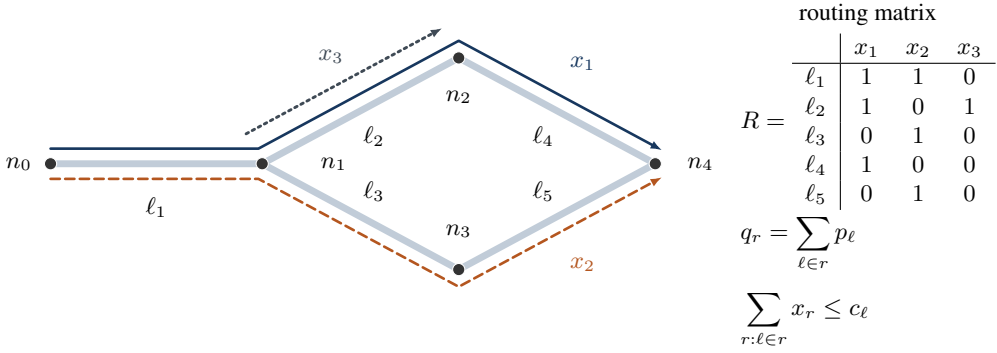


Figure 8.7: A general network utility maximization instance. The network is a set of links $\mathcal{E} = \{\ell_1, \dots, \ell_5\}$, each with capacity c_ℓ and a congestion price p_ℓ (the multiplier of its capacity constraint). Each source r sends at rate x_r along a fixed route $r \subseteq \mathcal{E}$ and earns utility $U_r(x_r)$; the routes are the columns of the routing matrix R , whose entry $R_{\ell r} = 1$ indicates that source r uses link ℓ . Sources sharing a link compete for its capacity, here source 1 meets source 2 on ℓ_1 and source 3 on ℓ_2 , which couples the constraints $\sum_{r: \ell \in r} x_r \leq c_\ell$. Each source reacts only to the aggregate route price $q_r = \sum_{\ell \in r} p_\ell$ fed back along its path.

and only if the following conditions hold:

$$U'_r(x_r^*) \leq q_r^*, \quad \forall r \in \mathcal{S}, \quad (8.49)$$

$$x_r^* (U'_r(x_r^*) - q_r^*) = 0, \quad \forall r \in \mathcal{S}, \quad (8.50)$$

$$\sum_{r: \ell \in r} x_r^* \leq c_\ell, \quad \forall \ell \in \mathcal{E}, \quad (8.51)$$

$$p_l^* \left(\sum_{r: \ell \in r} x_r^* - c_l \right) = 0, \quad \forall \ell \in \mathcal{E}, \quad (8.52)$$

$$p_l^* \geq 0, \quad \forall \ell \in \mathcal{E}.$$

In particular, whenever $x_r^* > 0$, the stationarity condition reduces to

$$U'_r(x_r^*) = q_r^*. \quad (8.53)$$

At optimality, each source increases its transmission rate until its marginal utility equals the total congestion price along its route. Links that are not fully utilized have zero price, while congested links carry strictly positive prices.

8.5.2 ■ Penalized primal formulation with link cost functions

Instead of imposing hard capacity constraints, one may incorporate congestion effects directly into the objective function via link cost penalties. Let $B_l : \mathbb{R}_+ \rightarrow \mathbb{R}$ be a continuously differentiable, convex function representing the congestion cost incurred on link l as a function of its aggregate load.

Define the penalized objective function

$$W(x) = \sum_{r \in \mathcal{S}} U_r(x_r) - \sum_{l \in \mathcal{E}} B_l \left(\sum_{r: \ell \in r} x_r \right). \quad (8.54)$$

Assume that each link cost function admits the integral representation

$$B_l(y) = \int_0^y f_l(z) dz, \quad (8.55)$$

where $f_l : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a nondecreasing congestion price function.

The first-order optimality condition for maximizing $W(x)$ is

$$U'_r(x_r) = \sum_{l \in r} f_l \left(\sum_{s: l \in s} x_s \right), \quad \forall r \in \mathcal{S}. \quad (8.56)$$

Connection to the dual formulation. Equation (8.56) is the primal analogue of the KKT stationarity condition (8.53). In the constrained formulation, congestion prices emerge endogenously as Lagrange multipliers enforcing link capacities. In the penalized formulation, congestion prices are embedded explicitly through the marginal cost functions $f_l(\cdot) = B'_l(\cdot)$.

Both formulations describe the same equilibrium principle: each source adapts its transmission rate until marginal utility balances aggregate congestion along its route. From an algorithmic viewpoint, this equivalence underlies the close relationship between dual ascent methods, primal–dual gradient dynamics, and distributed congestion control algorithms.

8.5.3 ■ Discrete-Time Primal–Dual Algorithms

We now develop a discrete-time primal–dual algorithm corresponding to the continuous-time saddle-point dynamics of the Network Utility Maximization (NUM) problem and establish its convergence under standard regularity assumptions.

Let $k \in \mathbb{N}$ denote the iteration index. Throughout this section the iteration index is carried as a *superscript*, since the subscript slot is reserved for the link label l and the source label r ; elsewhere in the chapter iterates are written x_k . For each link $l \in \mathcal{E}$ and source $r \in \mathcal{S}$, define the aggregate link load and route price as

$$y_l^k := \sum_{r: l \in r} x_r^k, \quad q_r^k := \sum_{l \in r} p_l^k.$$

The discrete-time primal–dual updates are given by

$$x_r^{k+1} = \Pi_{\mathbb{R}_+} \left(x_r^k + \gamma_r (U'_r(x_r^k) - q_r^k) \right), \quad r \in \mathcal{S}, \quad (8.57)$$

$$p_l^{k+1} = \Pi_{\mathbb{R}_+} \left(p_l^k + \gamma_l (y_l^k - c_l) \right), \quad l \in \mathcal{E}, \quad (8.58)$$

where $\Pi_{\mathbb{R}_+}$ denotes Euclidean projection onto the nonnegative orthant, and $\gamma_r > 0$, $\gamma_l > 0$ are stepsizes.

Equations (8.57)–(8.58) correspond to projected gradient ascent in the primal variables and projected gradient descent in the dual variables applied to the Lagrangian of the NUM problem.

Vector form. Let $x^k \in \mathbb{R}^{|\mathcal{S}|}$, $p^k \in \mathbb{R}^{|\mathcal{E}|}$, and let $R \in \{0, 1\}^{|\mathcal{E}| \times |\mathcal{S}|}$ denote the routing matrix with $R_{lr} = 1$ if link l is used by source r . Then the updates can be written as

$$x^{k+1} = \Pi_{\mathbb{R}_+^{|\mathcal{S}|}} \left(x^k + A(\nabla U(x^k) - R^\top p^k) \right), \quad (8.59)$$

$$p^{k+1} = \Pi_{\mathbb{R}_+^{|\mathcal{E}|}} \left(p^k + B(Rx^k - c) \right), \quad (8.60)$$

where $A = \text{diag}(\gamma_r)$ and $B = \text{diag}(\gamma_l)$.

Assumption 8.44. For each $r \in \mathcal{S}$, the utility function U_r is continuously differentiable with a Lipschitz continuous derivative, and is σ -strongly concave for some $\sigma > 0$ on the compact set of rate vectors reachable by the iteration. (Logarithmic utilities $U_r = \log x_r$ satisfy this on any set bounded away from $x_r = 0$, since $U_r'' = -1/x_r^2 \leq -1/\bar{x}^2$ there.) Slater's condition holds for the NUM problem (8.46), and the optimal primal–dual pair (x^*, p^*) is unique. The routing matrix R has full row rank, so that no link price is unidentifiable from the observed rates. The stepsizes γ_r and γ_l are positive and sufficiently small.

Saddle-point characterization

The Lagrangian of the NUM problem is

$$\mathcal{L}(x, p) = \sum_{r \in \mathcal{S}} U_r(x_r) - p^\top (Rx - c).$$

Lemma 8.45 (Saddle-point characterization). A pair (x^*, p^*) is optimal for the NUM problem if and only if

$$\mathcal{L}(x^*, p) \leq \mathcal{L}(x^*, p^*) \leq \mathcal{L}(x, p^*), \quad \forall x \geq 0, \forall p \geq 0.$$

Proof. The result follows from convexity of the primal problem, Slater's condition, and the Karush–Kuhn–Tucker conditions, which are necessary and sufficient for optimality. $\square$

Theorem 8.46 (Convergence of the discrete-time primal–dual algorithm). Under Assumption 8.44, there exist constants $\bar{\gamma}_s > 0$ and $\bar{\gamma}_l > 0$ such that for all stepsizes $\gamma_r \in (0, \bar{\gamma}_s]$ and $\gamma_l \in (0, \bar{\gamma}_l]$, the iterates generated by (8.57)–(8.58) converge to the unique saddle point (x^*, p^*) . In particular,

$$\lim_{k \rightarrow \infty} x^k = x^*, \quad \lim_{k \rightarrow \infty} p^k = p^*.$$

Proof. Define the Lyapunov function $V^k = \|x^k - x^*\|^2 + \|p^k - p^*\|^2$.

By nonexpansiveness of Euclidean projection onto a closed convex set, we have

$$\begin{aligned} \|x^{k+1} - x^*\|^2 &\leq \left\| x^k - x^* + A(\nabla_x \mathcal{L}(x^k, p^k) - \nabla_x \mathcal{L}(x^*, p^*)) \right\|^2, \\ \|p^{k+1} - p^*\|^2 &\leq \left\| p^k - p^* - B(\nabla_p \mathcal{L}(x^k, p^k) - \nabla_p \mathcal{L}(x^*, p^*)) \right\|^2, \end{aligned}$$

where $A = \text{diag}(\gamma_r)$ and $B = \text{diag}(\gamma_l)$.

Expanding the squares and summing yields

$$\begin{aligned} V^{k+1} &\leq V^k - 2 \begin{pmatrix} x^k - x^* \\ p^k - p^* \end{pmatrix}^\top \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} \nabla_x \mathcal{L}(x^k, p^k) - \nabla_x \mathcal{L}(x^*, p^*) \\ -\nabla_p \mathcal{L}(x^k, p^k) + \nabla_p \mathcal{L}(x^*, p^*) \end{pmatrix} \\ &\quad + \|A(\nabla_x \mathcal{L}(x^k, p^k) - \nabla_x \mathcal{L}(x^*, p^*))\|^2 + \|B(\nabla_p \mathcal{L}(x^k, p^k) - \nabla_p \mathcal{L}(x^*, p^*))\|^2. \end{aligned} \quad (8.61)$$

Define the saddle-point operator

$$F(x, p) = \begin{pmatrix} \nabla_x \mathcal{L}(x, p) \\ -\nabla_p \mathcal{L}(x, p) \end{pmatrix}.$$

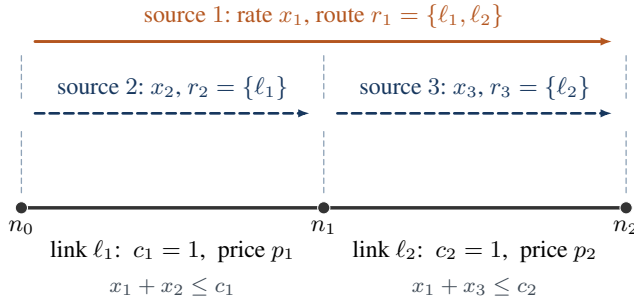


Figure 8.8: A network utility maximization instance: three sources sharing two unit-capacity links in tandem (the network of Example 8.47). Each source r sends at rate x_r along a fixed route $r \subseteq \mathcal{E}$ and earns utility $U_r(x_r)$; the spans above the link line are the rows of the routing matrix R . Source 1 traverses both bottlenecks, so it competes for capacity with source 2 on ℓ_1 and with source 3 on ℓ_2 , coupling the two capacity constraints $\sum_{r: \ell \in r} x_r \leq c_\ell$. Each link ℓ carries a congestion price p_ℓ (the constraint multiplier), and each source reacts only to the aggregate route price $q_r = \sum_{\ell \in r} p_\ell$.

Under Assumption 8.44, the utility functions are σ -strongly concave with Lipschitz continuous derivatives, and the constraints are affine. Because the coupling $-p^\top(Rx - c)$ is bilinear, its contributions to $\langle F(z) - F(z^*), z - z^* \rangle$ cancel exactly, and one obtains only

$$\langle F(x^k, p^k) - F(x^*, p^*), (x^k - x^*, p^k - p^*) \rangle \geq \mu \|x^k - x^*\|^2,$$

with no $\|p^k - p^*\|^2$ term: F is monotone, and strongly monotone in the *rate* block only. Strong monotonicity of the full operator would require the price block to be curvature-bounded as well, which a bilinear coupling never provides. The missing dual contraction is supplied instead by the routing matrix: under Assumption 8.44 R has full row rank, so $\|R^\top(p - p^*)\| \geq \varsigma \|p - p^*\|$ with $\varsigma := \sigma_{\min}(R) > 0$, and a price error therefore shows up as a rate residual of comparable size. Combining the two effects gives a modulus $\mu > 0$ for the *weighted* inner product used in V^k , which is what the estimate below requires.

Because $\nabla_x L$ and $\nabla_p L$ are Lipschitz continuous, there exists $L > 0$ such that

$$\|\nabla_x \mathcal{L}(x^k, p^k) - \nabla_x \mathcal{L}(x^*, p^*)\|^2 + \|\nabla_p \mathcal{L}(x^k, p^k) - \nabla_p \mathcal{L}(x^*, p^*)\|^2 \leq L^2 V^k.$$

Hence, the quadratic remainder terms in (8.61) are bounded by $(\|A\|^2 + \|B\|^2)L^2 V^k$.

Combining the above estimates, we obtain

$$V^{k+1} \leq (1 - 2\mu\lambda_{\min}(A, B) + L^2(\|A\|^2 + \|B\|^2))V^k,$$

where $\lambda_{\min}(A, B) = \min\{\min_r \gamma_r, \min_l \gamma_l\}$. By choosing γ_r and γ_l sufficiently small, the coefficient on the right-hand side is strictly less than 1, implying that $\{V^k\}$ is strictly decreasing and converges to zero.

Therefore,

$$\lim_{k \rightarrow \infty} x^k = x^*, \quad \lim_{k \rightarrow \infty} p^k = p^*,$$

which establishes convergence to the unique saddle point. $\square$

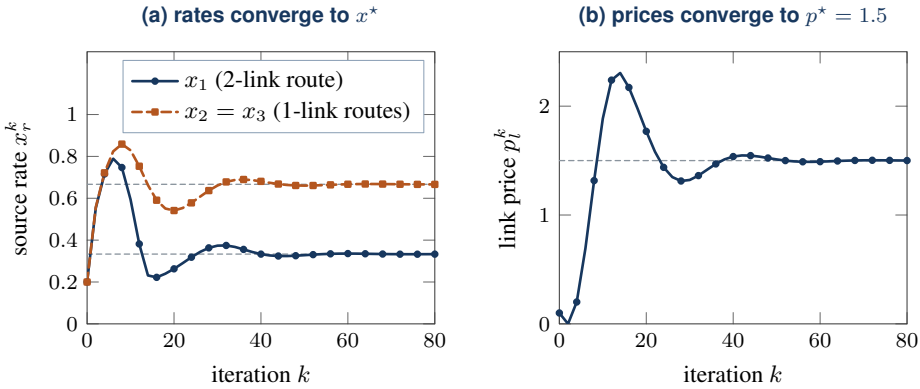


Figure 8.9: Distributed primal–dual (congestion-control) iteration (8.57)–(8.58) on the three-source, two-link network of Example 8.47 with log utilities, $\gamma_r = 0.05$, $\gamma_l = 0.5$. (a) The rates converge to the proportionally fair allocation $x^* = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$: the long route through both bottlenecks receives half the rate of each single-link route. (b) The common link price converges to $p^* = 1.5$, the shadow price of capacity; the initial overshoot is the price responding to transient overload $y_l^k > c_l$.

Example 8.47 Distributed Rate Control on a Three-Source Network.

Consider the canonical network with three sources and two links of unit capacity, $c_1 = c_2 = 1$. Source 1 uses the two-link route $\{1, 2\}$, while sources 2 and 3 use the single-link routes $\{1\}$ and $\{2\}$, respectively, giving the routing matrix

$$R = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad \text{constraints} \quad x_1 + x_2 \leq 1, \quad x_1 + x_3 \leq 1.$$

With proportionally fair (logarithmic) utilities $U_r(x_r) = \log x_r$, so $U'_r(x_r) = 1/x_r$, the KKT balance $U'_r(x_r^*) = q_r^*$ together with both capacity constraints tight yields the closed-form optimum $x^* = (\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$ and $p^* = (\frac{3}{2}, \frac{3}{2})$. The long-route source is penalized twice—its route price $q_1^* = p_1^* + p_2^* = 3$ is double that of the single-link sources ($q_2^* = q_3^* = \frac{3}{2}$)—and correspondingly receives half the rate.

The distributed iteration (8.57)–(8.58) was run with $\gamma_r = 0.05$, $\gamma_l = 0.5$ from $x_r^0 = 0.2$, $p_l^0 = 0.1$. Each source updates using only its locally observed route price $q_r^k = \sum_{l \in r} p_l^k$, and each link updates its price from its local excess load $y_l^k - c_l$; no global coordination is required. Figure 8.9 shows the rates and prices converging to (x^*, p^*) , with an initial price overshoot as the links react to transient overload $y_l^k > c_l$. This is precisely the feedback loop underlying TCP congestion control, discussed next. $\square$

Protocol-level implementation and connection to TCP/IP

A key strength of the primal–dual formulation is that it admits a fully distributed implementation that closely mirrors the structure of Internet congestion control protocols. In this interpretation, *sources* correspond to end hosts (TCP senders), while *links* correspond to routers with finite capacity and buffering. The algorithm operates without centralized coordination and relies only

on locally observable quantities and end-to-end feedback.

At a high level, the primal–dual dynamics implement a closed-loop feedback system: sources adjust their sending rates in response to congestion signals, while links generate congestion signals based on local overload. This is precisely the control architecture underlying TCP/IP.

Mapping of variables to networking primitives. The correspondence between optimization variables and protocol-level quantities is as follows: the primal variable x_r^k represents the sending rate (or congestion window, up to scaling by RTT) of source r ; the dual variable p_l^k represents the congestion price associated with link l , which can be interpreted as a queue length, packet loss probability, or explicit congestion signal (e.g., ECN marking probability). The aggregate load $y_l^k = \sum_{r:l \in r} x_r^k$ corresponds to the total incoming traffic at link l , while the capacity c_l represents the service rate of the link.

The route price $q_r^k = \sum_{l \in r} p_l^k$ is the end-to-end congestion signal observed by source r , accumulated across all bottleneck links along its path.

Distributed primal–dual protocol. The primal–dual algorithm can be implemented as the following distributed protocol.

ALGORITHM 8.2. Distributed Primal–Dual Congestion Control Protocol.

```

1: Initialization:  $x_r^0 \geq 0$  for all sources  $r \in \mathcal{S}$ ,  $p_l^0 \geq 0$  for all links  $l \in \mathcal{E}$ 
2: for each iteration  $k = 0, 1, 2, \dots$  do
3:   for each link  $l \in \mathcal{E}$  (in parallel) do
4:     Measure aggregate incoming rate  $y_l^k = \sum_{r:l \in r} x_r^k$ 
5:     Update congestion signal:  $p_l^{k+1} \leftarrow [p_l^k + \gamma_l(y_l^k - c_l)]^+$ 
6:   end for
7:   for each source  $r \in \mathcal{S}$  (in parallel) do
8:     Receive end-to-end congestion feedback  $q_r^k = \sum_{l \in r} p_l^k$ 
9:     Update sending rate:  $x_r^{k+1} \leftarrow [x_r^k + \gamma_r(U'_r(x_r^k) - q_r^k)]^+$ 
10:  end for
11: end for

```

Interpretation of the link update (dual step). The update $p_l^{k+1} = [p_l^k + \gamma_l(y_l^k - c_l)]^+$ is a discrete-time integrator driven by excess load. When incoming traffic exceeds capacity, the congestion signal increases; when the link is underutilized, the signal decreases. This is exactly the behavior of a router queue accumulating packets when arrivals exceed service and draining otherwise.

Remark 8.48 (Queue-length interpretation). If $\gamma_l = 1$ and p_l^k is interpreted as the queue length at link l , then the dual update coincides with the standard queue evolution equation $p_l^{k+1} = [p_l^k + y_l^k - c_l]^+$. In this case, congestion prices need not be explicitly computed: the physical queue itself implements the dual dynamics. Packet loss, delay, or ECN marking then serves as an implicit feedback signal to sources.

Interpretation of the source update (primal step). The source update $x_r^{k+1} = [x_r^k + \gamma_r(U_r'(x_r^k) - q_r^k)]^+$ adjusts the sending rate by comparing marginal utility to perceived congestion. When congestion is low (q_r^k small), the rate increases; when congestion is high, the rate decreases. For logarithmic utilities $U_r(x) = w_r \log x$, this update yields dynamics closely related to additive-increase multiplicative-decrease (AIMD) behavior after appropriate time-scale normalization.

In TCP, the route price q_r^k is not computed explicitly. Instead, it is inferred from packet loss events, RTT inflation, or ECN marks, which collectively approximate the sum of link congestion signals along the path.

Relation to TCP/IP congestion control. From this perspective, TCP/IP can be viewed as a distributed primal–dual algorithm operating under noisy, delayed, and quantized feedback: routers implement (approximate) dual updates through queues and packet marking, while end hosts implement (approximate) primal updates through window-based rate adaptation. The stability and convergence properties of TCP variants can thus be understood through the lens of saddle-point dynamics and Lyapunov stability.

Robustness and extensions. The convergence analysis of the primal–dual algorithm extends naturally to settings relevant for real networks, including diminishing or adaptive stepsizes, asynchronous updates across links and sources, delayed feedback due to RTTs, and stochastic perturbations arising from bursty traffic. These extensions can be handled using standard tools from stochastic approximation, projected dynamical systems, and input-to-state stability theory.

Remark 8.49. The primal–dual interpretation provides a unifying framework for understanding classical congestion control, modern ECN-based schemes, and newer optimization-driven protocols. It also offers a principled pathway for designing next-generation congestion control algorithms with provable stability and fairness guarantees.

8.6 ■ Further Reading

This chapter introduces only the foundational principles of constrained convex optimization. The following references provide more comprehensive treatments. For the theory and implementation of interior-point methods, see [89, 90, 26]. These works develop polynomial-time complexity theory, primal–dual frameworks, and practical algorithmic implementations. Barrier, continuation, and homotopy approaches to constrained optimization are treated in detail in [15, 90]. These texts connect classical penalty and barrier formulations to modern interior-point techniques.

Modern first-order methods for large-scale convex optimization, including projected gradient and proximal algorithms, are developed in [88, 11, 116]. These references emphasize complexity guarantees and scalability. For algorithmic and theoretical developments in nonconvex optimization, see [90, 15, 1]. These works address second-order methods, global convergence analysis, and optimization on manifolds. Optimization methods are central to modern machine learning. Representative references include [108, 53, 25]. These texts and surveys connect first-order and stochastic methods to large-scale learning problems.

8.7 ■ Exercises

Exercise 8.1 (Projection Operator). Let $S \subseteq \mathbb{R}^n$ be the set $S = \{x \in \mathbb{R}^n : a^T x \leq b\}$, where $b \in \mathbb{R}$ and $a \in \mathbb{R}^n$ is a nonzero vector.

Let $y \in \mathbb{R}^n$. Find an explicit expression for the projection of y onto the set S in terms of y , a , and b .

Hint: Formulate the projection as a convex optimization problem and derive a closed-form solution.

Exercise 8.2 (Backtracking Line Search). Consider the constrained optimization problem

$$\max_{x \in D} f(x) = \sum_{i=1}^3 w_i \log x_i - \frac{1}{(1 - x_1 - x_2 - x_3)^2}, \quad (8.62)$$

where $w_i \in \mathbb{R}_+$ for $i = 1, 2, 3$, and the feasible set is

$$D = \{x \in \mathbb{R}^3 \mid x_1 > 0, x_2 > 0, x_3 > 0, x_1 + x_2 + x_3 < 1\}.$$

This problem is smooth on its domain but becomes ill-conditioned near the boundary $x_1 + x_2 + x_3 = 1$ due to the barrier term. You are asked to solve it using a gradient-based method with *backtracking line search*.

Gradient ascent with backtracking line search. Starting from a feasible initial point $x^{(0)} \in D$, generate iterates $x^{(k+1)} = x^{(k)} + \alpha_k d^{(k)}$, where $d^{(k)} = \nabla f(x^{(k)})$ is the ascent direction and $\alpha_k > 0$ is the step size chosen by backtracking line search. Given parameters $\alpha_0 > 0, \beta \in (0, 1), c \in (0, 1)$, the backtracking line search proceeds as follows:

(i) Initialize $\alpha = \alpha_0$.

(ii) While either $x^{(k)} + \alpha d^{(k)} \notin D$ or the sufficient ascent condition

$$f(x^{(k)} + \alpha d^{(k)}) < f(x^{(k)}) + c\alpha \nabla f(x^{(k)})^\top d^{(k)} \quad (8.63)$$

is not satisfied, update

$$\alpha \leftarrow \beta\alpha.$$

(iii) Set $\alpha_k = \alpha$ and update $x^{(k+1)}$.

The algorithm is terminated when $\|\nabla f(x^{(k)})\|$ falls below a prescribed tolerance.

(1) Implement the above gradient ascent method with backtracking line search in MATLAB, C, or Java.

(2) Choose a feasible initial point and appropriate line-search parameters.

(3) Compute and report the optimal solution for the following cases:

(i) $w_1 = 1, w_2 = 1, w_3 = 1$.

(ii) $w_1 = 1, w_2 = 2, w_3 = 3$.

(4) Plot or tabulate the convergence behavior of the objective value.

Exercise 8.3 (Matching Pursuit). Matching pursuit is a greedy algorithm for solving the following ℓ^0 optimization problem:

$$\min_{x \in \mathbb{R}^n} \|x\|_0 \quad \text{subject to} \quad \Phi x = y, \quad (8.64)$$

where $\Phi \in \mathbb{R}^{m \times n}$ and $y \in \mathbb{R}^m$ are given. Let

$$\Phi = [\phi_1, \phi_2, \dots, \phi_n] \in \mathbb{R}^{m \times n}, \quad \phi_i \in \mathbb{R}^m.$$

The mutual coherence $\mu(\Phi)$ is defined as

$$\mu(\Phi) := \max_{\substack{i,j=1,\dots,n \\ i \neq j}} \frac{|\langle \phi_i, \phi_j \rangle|}{\|\phi_i\|_2 \|\phi_j\|_2}. \quad (8.65)$$

A classical result states that if there exists $x \in \mathbb{R}^n$ such that $\Phi x = y$ and

$$\|x\|_0 < \frac{1}{2} \left(1 + \frac{1}{\mu(\Phi)} \right), \quad (8.66)$$

then x is the sparsest solution.

Assume $\mu(\Phi) < 1$. The 1-sparse solution must be parallel to one of the columns ϕ_i . This leads to the problem of finding

$$e(i) := \min_{x \in \mathbb{R}} \|x\phi_i - y\|_2^2. \quad (8.67)$$

The minimum value of $e(i)$ is

$$e(i) = \|y\|_2^2 - \frac{\langle \phi_i, y \rangle^2}{\|\phi_i\|_2^2}. \quad (8.68)$$

Let i^* minimize $e(i)$. The corresponding 1-sparse vector is

$$x^* = [0, \dots, 0, x_{i^*}^*, 0, \dots, 0], \quad x_{i^*}^* = \frac{\langle \phi_{i^*}, y \rangle}{\|\phi_{i^*}\|_2^2}.$$

Matching pursuit iteratively refines the solution by updating the residual $r[k] = y - \Phi x[k]$.

At each iteration:

- (1) Find a 1-sparse vector minimizing $\|\Phi x - r[k]\|_2$.
- (2) Update $x[k+1] = x[k] + x^*$.

Write a program to implement the matching pursuit algorithm.

Exercise 8.4 (Projection onto the Probability Simplex). Many constrained problems require projecting onto the probability simplex

$$\Delta = \left\{ x \in \mathbb{R}^n : x_i \geq 0, \sum_{i=1}^n x_i = 1 \right\}.$$

- (1) Writing the projection of y as the solution of $\min_{x \in \Delta} \frac{1}{2} \|x - y\|_2^2$, form the Lagrangian for the equality constraint and use the KKT conditions to show that the projection has the form $x_i^* = \max\{y_i - \tau, 0\}$, where the scalar τ is chosen so that $\sum_{i=1}^n \max\{y_i - \tau, 0\} = 1$.
- (2) Show that the left-hand side above is continuous and nonincreasing in τ , so that τ is unique, and describe an $O(n \log n)$ algorithm that computes it by sorting the entries of y .
- (3) Implement the projection and verify it numerically against a general-purpose convex solver on several random vectors y .

Exercise 8.5 (Projected Gradient for Box-Constrained Least Squares). Consider the box-constrained least-squares problem

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - b\|_2^2 \quad \text{subject to} \quad \ell \leq x \leq u,$$

with $A \in \mathbb{R}^{m \times n}$, to be solved by the projected gradient method $x_{k+1} = P_X(x_k - \alpha A^\top (Ax_k - b))$.

- (1) Show that $\nabla f(x) = A^\top(Ax - b)$ is Lipschitz continuous with constant $L = \|A\|_2^2 = \lambda_{\max}(A^\top A)$, and give a closed form for the projection P_X onto the box $X = \{x : \ell \leq x \leq u\}$.
- (2) Implement the method in MATLAB, Python, or C for a random instance with $m > n$ and a box that makes the constraint active at the solution.
- (3) Run the iteration with $\alpha \in \{1/L, 1.9/L, 2.1/L\}$ and plot $f(x_k) - f^*$ versus k on a semilogarithmic scale. Relate your observations to the stability threshold $\alpha < 2/L$ of Theorem 8.20 and to Example 8.21.

Exercise 8.6 (Central Path and the Duality Gap). Consider the linear program $\min_x c^\top x$ subject to $a_j^\top x \leq b_j, j = 1, \dots, m$, with logarithmic barrier $\phi(x) = -\sum_j \ln(b_j - a_j^\top x)$.

- (1) Using $\mu_j^*(t) = 1/(t(b_j - a_j^\top x^*(t)))$ as the associated dual variables, show that the duality gap along the central path satisfies $c^\top x^*(t) - f^* \leq m/t$.
- (2) Take the two-dimensional polytope and objective of Example 8.32. Starting from the analytic center, use Newton's method on the barrier stationarity condition $tc + \nabla\phi(x) = 0$ to compute $x^*(t)$ for $t \in \{1, 2, 5, 10, 50, 200\}$.
- (3) Tabulate $x^*(t)$, the certified gap m/t , and the actual suboptimality $c^\top x^*(t) - f^*$, and confirm that the central path converges to the optimal vertex as $t \rightarrow \infty$.

Exercise 8.7 (Primal–Dual Gradient Method for an Equality-Constrained QP). Consider the strongly convex quadratic program

$$\min_{x \in \mathbb{R}^n} \frac{1}{2}x^\top Qx - c^\top x \quad \text{subject to} \quad Ax = b,$$

with $Q \succ 0$ and $A \in \mathbb{R}^{p \times n}$ of full row rank.

- (1) Write the Lagrangian and the KKT system, and give the closed-form solution (x^*, λ^*) .
- (2) The primal–dual gradient method applies gradient descent in x and gradient ascent in λ :

$$x_{k+1} = x_k - \alpha(Qx_k - c + A^\top \lambda_k), \quad \lambda_{k+1} = \lambda_k + \beta(Ax_k - b).$$

Writing $z_k = (x_k, \lambda_k)$, express the iteration in the affine form $z_{k+1} = z_k - S(z_k - z^*)$ and identify the matrix S .

- (3) Argue that for sufficiently small α, β the spectral radius of $I - S$ is strictly less than one, so the iterates converge linearly, in agreement with Theorem 8.40.
- (4) Implement the method and plot $\|z_k - z^*\|$ on a semilogarithmic scale, as in Example 8.42. Experiment with the stepsizes and report what happens when they are chosen too large.

Exercise 8.8 (Distributed Network Utility Maximization). Consider the three-source, two-link network of Figure 8.8, with unit link capacities $c_1 = c_2 = 1$ and weighted logarithmic utilities $U_r(x_r) = w_r \log x_r$.

- (1) Write the KKT conditions for the NUM problem and solve them in closed form for the optimal rates x^* and link prices p^* as functions of the weights w_1, w_2, w_3 .
- (2) Evaluate the optimal allocation for $w = (1, 1, 1)$ and for $w = (2, 1, 1)$, and explain how increasing the weight w_1 of the source that traverses both bottlenecks changes the allocation and the prices.

- (3) Implement the distributed primal–dual algorithm (8.57)–(8.58), in which each source updates using only its route price $q_r = \sum_{l \in r} p_l$ and each link updates from its local excess load $y_l - c_l$. Plot the rate and price trajectories and confirm convergence to (x^*, p^*) .

Part IV

Deterministic Optimal Control

Chapter 9

Calculus of Variations

This chapter presents a systematic introduction to the calculus of variations as a framework for infinite-dimensional optimization, with an emphasis on first-order necessary conditions. Beginning from classical geometric and physical problems, the exposition develops the functional-analytic setting underlying variational problems, including admissible sets, variations, and notions of weak and strong local extrema. The Euler–Lagrange equation is derived rigorously using the first variation and the fundamental lemma of the calculus of variations, highlighting its role as the infinite-dimensional analogue of stationarity in finite-dimensional optimization.

Several important structural consequences are explored, including cyclic variables, the Beltrami identity, and conservation laws arising from invariances of the Lagrangian. The framework is then extended to variational problems involving higher-order derivatives and vector-valued trajectories, where first-order necessary conditions already encode rich dynamical and geometric structure. A range of informative examples illustrates how smoothness penalties lead to spline solutions, how anisotropic metrics bend optimal paths, and how interaction energies induce coupled dynamics and conservation of collective quantities in multi-agent systems.

Throughout, the emphasis remains on extracting interior equations and boundary conditions directly from first-order optimality, without invoking second-order sufficiency or stability analysis. This perspective clarifies how variational principles unify geometry, mechanics, and optimization, while providing a conceptual bridge to optimal control and modern nonsmooth extensions.

9.1 ■ From Classical Examples to the Calculus of Variations

Many fundamental problems in mathematics, physics, and engineering can be formulated as optimization problems in which the decision variable is a *curve* rather than a finite-dimensional vector. Historically, such problems arose from geometry, mechanics, and optics, long before the formal development of modern calculus.

We begin with several classical examples and then show that they all share a common mathematical structure, which leads naturally to the calculus of variations.

9.1.1 ■ Dido’s Isoperimetric Problem

According to the legend about the foundation of Carthage (~ 850 B.C.), Dido purchased from a local king all the land that could be enclosed by the hide of an ox along the Northern African coastline. By slicing the hide into thin strips and tying them together, she was able to enclose a sizable area.

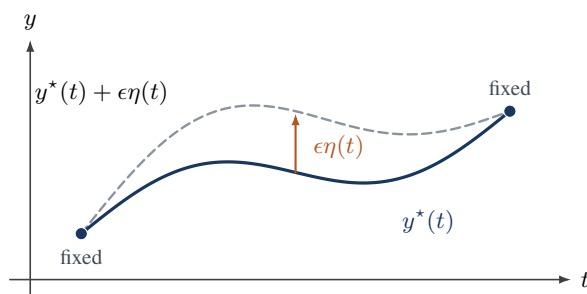


Figure 9.1: The calculus of variations studies how a functional changes under admissible perturbations of a curve, often with endpoint conditions fixed.

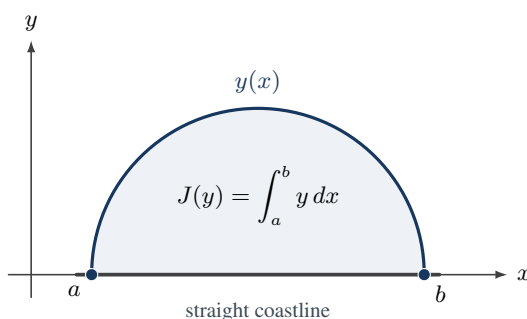


Figure 9.2: Dido's isoperimetric problem: among all curves $y(x)$ of a fixed length $\int_a^b \sqrt{1 + (y'(x))^2} dx = C$ that start and end on the straight coastline ($y(a) = y(b) = 0$), one seeks the curve enclosing the largest area $J(y) = \int_a^b y dx$ (shaded). The decision variable is the entire curve, subject to a global length constraint.

Assume the coastline is a straight segment. Let $y(x)$ describe the enclosing curve, with $y(a) = y(b) = 0$. The enclosed area is $J(y) = \int_a^b y(x) dx$, which we wish to maximize subject to the fixed-length constraint $\int_a^b \sqrt{1 + (y'(x))^2} dx = C$.

This is an optimization problem over curves with a global constraint.

9.1.2 ■ Light Reflection and Refraction

In free space, light travels along the path of shortest distance, which is a straight line. This is already a curve minimization principle.

Reflection can be derived by minimizing arclength, leading to equality of incident and reflected angles. Refraction is governed by Fermat's principle: light follows the path of *shortest time*. If the propagation speed depends on the medium, the relevant functional is $J = \int \frac{ds}{v}$, and Snell's law emerges as a necessary optimality condition.

9.1.3 ■ Catenary and Brachistochrone

The catenary problem asks for the shape of a uniform chain suspended between two points that minimizes gravitational potential energy. The corresponding functional is $J(y) = \int_a^b y(x) \sqrt{1 + (y'(x))^2} dx$, subject to a fixed-length constraint. The solution is

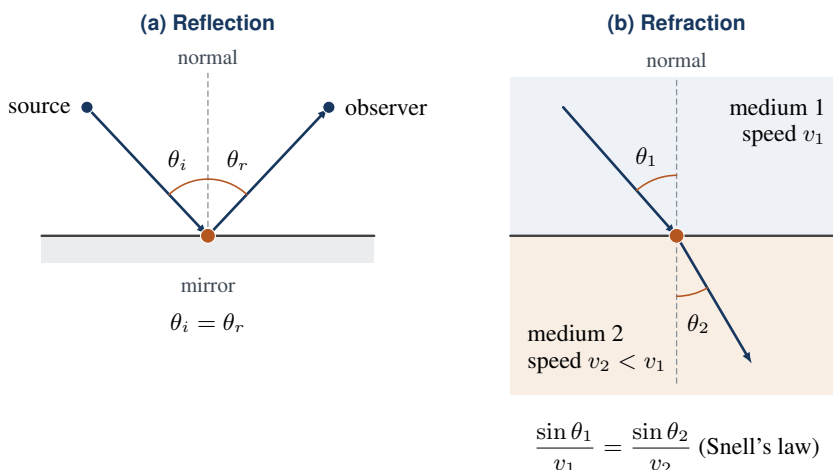


Figure 9.3: Fermat's principle as a variational statement. (a) Minimizing the optical path length of a reflected ray forces the angle of incidence to equal the angle of reflection. (b) Minimizing travel time $J = \int ds/v$ across an interface where the speed changes yields Snell's law of refraction; the ray bends toward the normal when entering the slower medium ($v_2 < v_1$).

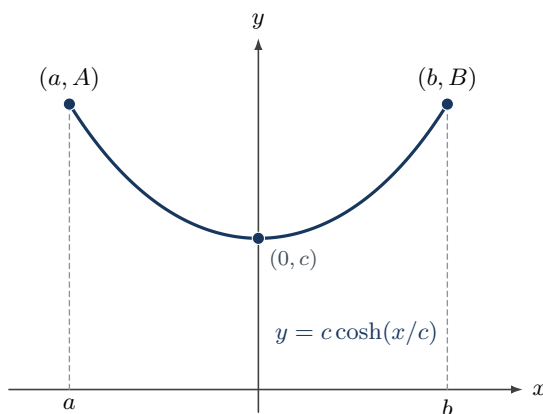


Figure 9.4: The catenary is the shape $y = c \cosh(x/c)$ of a uniform chain hung between two fixed points of equal height. Among all curves of a prescribed length joining the supports, it minimizes the gravitational potential energy $J(y) = \int_a^b y \sqrt{1 + (y')^2} dx$.

the catenary $y(x) = c \cosh(x/c)$.

The brachistochrone problem asks for the curve along which a particle slides under gravity between two points in the shortest time. The time functional is $J(y) = \int_a^b \frac{\sqrt{1 + (y'(x))^2}}{\sqrt{y(x)}} dx$, whose minimizer is a cycloid.

These examples illustrate that diverse physical problems lead to a common variational structure.

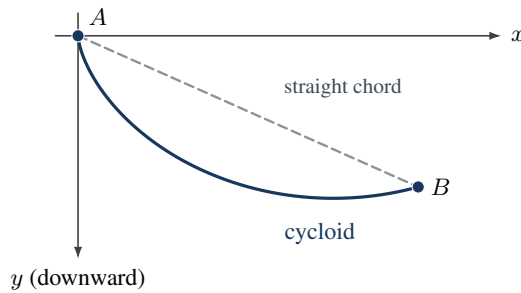


Figure 9.5: The brachistochrone is the curve of fastest descent between A and B under gravity, i.e. the minimizer of the travel-time functional $J(y) = \int_a^b \sqrt{1 + (y')^2} / \sqrt{y} \, dx$. Its minimizer is a cycloid (drawn exactly for these endpoints), whose steep initial drop builds speed early and beats the straight chord.

9.2 ■ The Calculus of Variations: Functional-Analytic Setting

Variational problems are formulated on infinite-dimensional spaces of functions. Unlike finite-dimensional optimization, where all norms are equivalent and compactness issues are relatively benign, the analysis of functionals defined on spaces of functions depends critically on the choice of both the underlying vector space and the topology imposed on it. In particular, notions such as local optimality, convergence of minimizing sequences, and differentiability of functionals are meaningful only after these choices are made explicit.

9.2.1 ■ Underlying Function Spaces

Throughout this chapter, we work on a compact interval $[a, b] \subset \mathbb{R}$ and consider real-valued functions defined on this interval. The natural ambient spaces for the calculus of variations are Banach spaces of functions. Recall that a Banach space is a normed vector space that is complete, meaning that every Cauchy sequence converges to a limit within the space. Completeness plays a central role in variational analysis: minimizing sequences, approximating sequences of variations, and infinitesimal perturbations are all constructed as sequences whose limits must remain admissible. Without completeness, such limits may fail to exist within the chosen space, rendering local optimality conditions, existence results, and convergence arguments ill defined.

The use of Banach spaces also provides a natural setting for treating variational functionals as mappings between normed spaces. In particular, first- and second-order optimality conditions can be formulated in terms of Gâteaux or Fréchet derivatives, notions that rely on the linear and topological structure of Banach spaces. For these reasons, specifying a Banach space framework is a foundational step in any rigorous treatment of the calculus of variations.

The normed and Banach spaces recalled here, together with completeness and its role in guaranteeing that limits of minimizing or approximating sequences remain admissible, were developed in Section 2.1.2. The calculus of variations is, in this sense, optimization carried out over these same spaces, now realized concretely as the function spaces $C([a, b])$, $C^1([a, b])$, $AC([a, b])$, and $H^1([a, b])$ on which variational functionals act.

We now describe the principal function spaces that will serve as ambient spaces for admissible curves and variations.

We build up the relevant spaces in order of increasing generality, beginning with the largest and least restrictive. Let $C([a, b])$ denote the space of continuous functions on the compact interval $[a, b]$. When equipped with the supremum norm $\|y\|_\infty := \max_{x \in [a, b]} |y(x)|$, this space

is a Banach space. Convergence with respect to this norm is uniform convergence, meaning that functions converge pointwise at every $x \in [a, b]$ with a uniform rate. The space $C([a, b])$ is therefore well suited for problems in which the functional depends only on the values of the function itself.

However, many variational problems involve derivatives of the unknown function. Uniform convergence alone does not control derivatives, nor does it guarantee their existence. As a result, $C([a, b])$ is generally insufficient as an ambient space for variational problems involving integrands that depend on y' .

A natural remedy is to work instead in the space of continuously differentiable functions. Let $C^1([a, b])$ denote the space of continuously differentiable functions on $[a, b]$. Endowed with the norm

$$\|y\|_{C^1} = \max_{x \in [a, b]} |y(x)| + \max_{x \in [a, b]} |y'(x)|,$$

this space is a Banach space. Convergence in the C^1 norm implies uniform convergence of both the function and its derivative.

The space $C^1([a, b])$ provides sufficient regularity to ensure that expressions involving y' are well defined pointwise and that classical analytical tools, such as integration by parts, may be applied without qualification. For this reason, $C^1([a, b])$ constitutes the classical setting for the calculus of variations and underlies the traditional derivation of the Euler–Lagrange equations.

At the same time, the C^1 framework is sometimes overly restrictive. In many problems, especially those arising in optimal control or nonsmooth mechanics, optimal trajectories may fail to be continuously differentiable, even though they possess derivatives almost everywhere. This motivates the introduction of weaker function spaces.

The weaker class that accommodates such behavior is the space of absolutely continuous functions. The space $AC([a, b])$ of absolutely continuous functions consists of all functions $y : [a, b] \rightarrow \mathbb{R}$ for which there exists an integrable function $v \in L^1([a, b])$ such that $y(x) = y(a) + \int_a^x v(t) dt$ for all $x \in [a, b]$. For such functions, the derivative $y'(x)$ exists almost everywhere and coincides with $v(x)$.

Absolute continuity represents the weakest regularity class for which the Fundamental Theorem of Calculus holds. In particular, absolutely continuous functions admit a meaningful notion of derivative almost everywhere and satisfy $\int_a^b y'(x) dx = y(b) - y(a)$. This property makes $AC([a, b])$ the natural ambient space for modern treatments of the calculus of variations and optimal control, where admissible trajectories may exhibit corners, switching behavior, or other nonsmooth features.

The space $AC([a, b])$ is not complete under the supremum norm alone. However, it becomes a Banach space when equipped with a suitable graph norm, such as

$$\|y\|_{AC} = |y(a)| + \int_a^b |y'(x)| dx.$$

Under this norm, convergence corresponds to convergence of the initial value and convergence of derivatives in $L^1([a, b])$, ensuring completeness and stability under limits of admissible perturbations.

9.2.2 ■ Admissible Set, Variations, and the Variational Functional

Let $A, B \in \mathbb{R}$ be prescribed boundary values. In a variational problem, admissible curves are required to satisfy these boundary conditions exactly. As a consequence, the set of admissible functions is not a vector space: adding two admissible curves or multiplying an admissible curve by a scalar generally violates the boundary constraints. Instead, the admissible set has the structure of an affine subspace of an appropriate Banach space of functions.

A canonical choice of admissible set is

$$\mathcal{A} = \{ y \in C^1([a, b]) \mid y(a) = A, y(b) = B \}.$$

This set consists of all continuously differentiable curves that interpolate the prescribed endpoint values. The choice of $C^1([a, b])$ ensures that both the curve and its derivative are well defined pointwise and continuous, which is sufficient for the classical formulation of the calculus of variations.

In many applications, however, such regularity is unnecessarily restrictive. In particular, in optimal control, problems with switching behavior, or problems with state constraints, optimal trajectories may fail to be continuously differentiable, even though they possess derivatives almost everywhere. For this reason, it is often natural to replace $C^1([a, b])$ by the space of absolutely continuous functions $AC([a, b])$ and to define instead $\mathcal{A} = \{ y \in AC([a, b]) \mid y(a) = A, y(b) = B \}$. This choice preserves the validity of the Fundamental Theorem of Calculus while allowing for nonsmooth behavior.

Regardless of the ambient function space, the admissible set $\mathcal{A}$ is an affine space. To make this structure explicit, fix an arbitrary admissible function $y^* \in \mathcal{A}$. Every other admissible function $y \in \mathcal{A}$ can then be written uniquely as $y = y^* + \eta$, where η belongs to the linear space $\mathcal{A}_0 = \{ \eta \in C^1([a, b]) \mid \eta(a) = \eta(b) = 0 \}$, or to its absolutely continuous analogue when working in $AC([a, b])$. The space $\mathcal{A}_0$ is a genuine vector space and represents the set of admissible perturbations of an admissible curve. From a geometric point of view, $\mathcal{A}_0$ plays the role of a tangent space to the affine manifold $\mathcal{A}$.

9.2.3 ■ The Variational Functional

Let $\mathcal{L} : [a, b] \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be a given function, called the *Lagrangian*. Unless stated otherwise, we assume that $\mathcal{L}$ is continuous in all its arguments and continuously differentiable with respect to its second and third arguments. When second-order optimality conditions are considered, $\mathcal{L}$ will be assumed to be twice continuously differentiable with respect to these arguments.

Given an admissible function $y \in \mathcal{A}$, we define the associated variational functional

$$J(y) = \int_a^b \mathcal{L}(x, y(x), y'(x)) dx,$$

provided that the integral is well defined. The regularity assumptions on $\mathcal{L}$ and the choice of admissible function space ensure that the integrand is measurable and integrable, so that $J(y)$ is finite for all $y \in \mathcal{A}$. Under these assumptions, the functional J defines a mapping $J : \mathcal{A} \rightarrow \mathbb{R}$.

9.2.4 ■ Local Extrema and Topology

The admissible set $\mathcal{A}$ is an infinite-dimensional subset of a Banach space of functions. In contrast to finite-dimensional spaces, where all norms induce the same topology, different norms on infinite-dimensional spaces may induce genuinely different notions of convergence and locality. As a result, the concept of a *local* extremum is not intrinsic, but depends essentially on the topology chosen on the ambient function space.

Let X be a Banach space of functions (for example $C([a, b])$, $C^1([a, b])$, or $AC([a, b])$ equipped with an appropriate norm), and let $\mathcal{A} \subset X$ be the admissible set. A norm $\| \cdot \|$ on X induces a metric and hence a topology on $\mathcal{A}$ by restriction. Local optimality is always understood with respect to this induced topology.

Definition 9.1 (Local Extremum). Let $\| \cdot \|$ be a norm on X and let $J : \mathcal{A} \rightarrow \mathbb{R}$ be a functional. A function $y^* \in \mathcal{A}$ is called a *local minimum* (respectively, a *local maximum*) of J with respect to

the norm $\|\cdot\|$ if there exists $\varepsilon > 0$ such that $J(y^*) \leq J(y)$ (respectively, $J(y^*) \geq J(y)$) for all $y \in \mathcal{A}$ satisfying $\|y - y^*\| < \varepsilon$.

This definition extends the classical notion of local optimality from finite-dimensional optimization to variational problems. The essential difference is that, in infinite-dimensional spaces, changing the norm changes the class of admissible perturbations that are considered “small.” Consequently, a function may be locally optimal with respect to one topology but fail to be locally optimal with respect to another.

9.2.5 ■ Strong and Weak Extrema

Two norms arise naturally when the admissible set is contained in $C^1([a, b])$. The first is the supremum norm $\|y\|_0 := \max_{x \in [a, b]} |y(x)|$, which measures uniform closeness of functions and induces the topology of uniform convergence. The second is the C^1 norm $\|y\|_1 := \max_{x \in [a, b]} |y(x)| + \max_{x \in [a, b]} |y'(x)|$, which measures uniform closeness of both the function and its derivative.

The $\|\cdot\|_1$ norm induces a strictly stronger topology than the $\|\cdot\|_0$ norm: convergence in $\|\cdot\|_1$ implies convergence in $\|\cdot\|_0$, but not conversely. Equivalently, for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|y - y^*\|_1 < \delta$ implies $\|y - y^*\|_0 < \varepsilon$.

Definition 9.2 (Strong and Weak Extrema). A local extremum of J with respect to the $\|\cdot\|_0$ norm is called a strong extremum. A local extremum of J with respect to the $\|\cdot\|_1$ norm is called a weak extremum.

Because the $\|\cdot\|_1$ topology is stronger, a strong minimum is a more restrictive notion than a weak minimum. Indeed, if y^* is a strong minimum and belongs to $C^1([a, b])$, then it is automatically a weak minimum. To see this, note that any sufficiently small $\|\cdot\|_1$ -neighborhood of y^* is contained in a $\|\cdot\|_0$ -neighborhood on which $J(y^*)$ is minimal.

The converse implication does not hold in general. A function may be a weak minimum without being a strong minimum, since weak minimality only excludes variations that are small in both value and derivative, while allowing variations that are uniformly small in value but highly oscillatory in derivative.

This distinction is not merely technical. In optimal control and nonsmooth variational problems, small perturbations of controls or switching times may produce trajectories that remain uniformly close in the $\|\cdot\|_0$ sense while exhibiting large variations in their derivatives. Strong extrema are designed to capture robustness with respect to such perturbations and therefore play a central role in control-theoretic formulations.

Remark 9.3 (Nonsmooth Optima and Bang–Bang Control). The preceding definitions of admissible sets and local extrema rely on an implicit assumption of regularity for admissible trajectories. In many important applications, most notably in optimal control, this assumption must be relaxed. A canonical example is *bang–bang control*, in which the control variable switches discontinuously between extreme values.

In such problems, it is essential to distinguish between the regularity of the control and that of the associated state trajectory. While a bang–bang control is generally discontinuous and therefore not absolutely continuous, the corresponding state trajectory satisfies an integral equation and remains continuous and, in fact, absolutely continuous under standard assumptions on the dynamics. Thus, nonsmoothness arises in the control variable rather than in the state trajectory.

This distinction explains why the classical admissible space $C^1([a, b])$ is too restrictive to

capture optimal solutions, whereas the space $AC([a, b])$ provides the minimal regularity required for the calculus of variations. Absolute continuity ensures the existence of derivatives almost everywhere and the validity of integral cost functionals, while allowing for trajectories with corners and other nonsmooth features.

From a variational perspective, this also clarifies the relationship between strong and weak extrema. Weak extrema defined with respect to the C^1 topology may fail to exist in such problems, even though strong extrema exist once the admissible set is enlarged to absolutely continuous functions and the topology is chosen accordingly.

Finally, this observation motivates the development of optimal control theory, where first-order necessary conditions are no longer expressed solely through the Euler–Lagrange equation but through Pontryagin’s Maximum Principle and related nonsmooth optimality conditions. In this sense, nonsmooth optima do not represent a breakdown of variational methods, but rather indicate the need to adopt a functional framework adapted to the regularity of the optimal solution.

9.3 ■ First-Order Necessary Conditions

The derivation of first-order necessary conditions in the calculus of variations rests on an appropriate notion of differentiation for functionals defined on infinite-dimensional spaces. Before stating the Euler–Lagrange equation, we make precise what is meant by “taking a derivative” of a functional and how this notion arises from one-dimensional variations.

9.3.1 ■ Variations and Directional Derivatives

Let X denote a Banach space of functions on $[a, b]$, such as $C^1([a, b])$ or $AC([a, b])$, and let $\mathcal{A} \subset X$ be the admissible set determined by boundary conditions. Recall that $\mathcal{A}$ is an affine subspace of X , modeled on the linear subspace $\mathcal{A}_0 = \{\eta \in X \mid \eta(a) = \eta(b) = 0\}$, which represents admissible perturbations that preserve the boundary values.

Fix an admissible curve $y^* \in \mathcal{A}$. Since $\mathcal{A}$ is affine, every admissible curve sufficiently close to y^* can be written uniquely as $y^* + \eta$ with $\eta \in \mathcal{A}_0$.

A *variation* of y^* in the direction $\eta \in \mathcal{A}_0$ is the one-parameter family of curves $y_\varepsilon(x) = y^*(x) + \varepsilon\eta(x)$, $\varepsilon \in \mathbb{R}$. Because $\eta(a) = \eta(b) = 0$, the boundary conditions are preserved for all ε , and for sufficiently small $|\varepsilon|$ the curve y_ε remains in $\mathcal{A}$. Thus, variations describe admissible infinitesimal displacements of y^* within the constraint set.

This construction reduces the original infinite-dimensional optimization problem for the functional J to a family of one-dimensional problems indexed by the scalar parameter ε .

9.3.2 ■ Differentiation of Functionals

Let $J : \mathcal{A} \rightarrow \mathbb{R}$ be a functional of the form $J(y) = \int_a^b \mathcal{L}(x, y(x), y'(x)) dx$, where the Lagrangian $\mathcal{L} : [a, b] \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is assumed to satisfy the following standing assumptions:

- (1) $\mathcal{L}(\cdot, y, p)$ is measurable in x for each (y, p) , and $\mathcal{L}(x, \cdot, \cdot)$ is continuously differentiable in (y, p) for each x ;
- (2) $\partial_y \mathcal{L}$ and $\partial_{y'} \mathcal{L}$ satisfy growth conditions ensuring integrability along admissible curves, so that differentiation under the integral sign is justified.

Under these assumptions, $J(y)$ is well defined for all $y \in \mathcal{A}$.

For a fixed admissible direction $\eta \in \mathcal{A}_0$, consider the scalar function $\phi_\eta(\varepsilon) := J(y^* + \varepsilon\eta)$. This is an ordinary real-valued function of the single variable ε . If y^* is a local extremum of J

with respect to the chosen topology on $\mathcal{A}$, then for every $\eta \in \mathcal{A}_0$ the function $\phi_\eta(\varepsilon)$ has a local extremum at $\varepsilon = 0$. Elementary calculus therefore yields the necessary condition

$$\left. \frac{d}{d\varepsilon} J(y^* + \varepsilon\eta) \right|_{\varepsilon=0} = 0 \quad \text{for all } \eta \in \mathcal{A}_0.$$

Definition 9.4 (First Variation). Let X be a Banach space of functions on $[a, b]$, let $\mathcal{A} \subset X$ be an affine admissible set, and let $J : \mathcal{A} \rightarrow \mathbb{R}$ be a functional. Let $\mathcal{A}_0$ denote the linear space of admissible directions associated with $\mathcal{A}$.

Fix $y^* \in \mathcal{A}$ and $\eta \in \mathcal{A}_0$. Assume that there exists $\varepsilon_0 > 0$ such that $y^* + \varepsilon\eta \in \mathcal{A}$ for all $|\varepsilon| < \varepsilon_0$, and that the limit $\lim_{\varepsilon \rightarrow 0} \frac{J(y^* + \varepsilon\eta) - J(y^*)}{\varepsilon}$ exists as a finite real number.

This limit is called the first variation of J at y^* in the direction η and is denoted by $\delta J(y^*; \eta)$.

Remark 9.5 (Derivative Interpretation of the First Variation). For a fixed direction $\eta \in \mathcal{A}_0$, define the scalar function $\phi_\eta(\varepsilon) := J(y^* + \varepsilon\eta)$, which is well defined for all sufficiently small $|\varepsilon|$.

If the function ϕ_η is differentiable at $\varepsilon = 0$ in the classical (one-dimensional) sense, then the first variation coincides with the ordinary derivative of ϕ_η at zero, namely,

$$\delta J(y^*; \eta) = \left. \frac{d}{d\varepsilon} J(y^* + \varepsilon\eta) \right|_{\varepsilon=0}.$$

In this case, the first variation may be computed using standard rules of differentiation.

From the viewpoint of functional analysis, the mapping $\eta \mapsto \delta J(y^*; \eta)$ is the *directional derivative* of J at y^* . Since it is defined direction by direction, without requiring uniform control over all directions, this derivative is naturally identified with the *Gâteaux derivative* of J at y^* .

In general, the existence of directional derivatives does not guarantee linearity or continuity with respect to η . However, under the smoothness assumptions on the Lagrangian stated above, the first variation admits an explicit integral representation that is linear in η and continuous with respect to the natural norms on $\mathcal{A}_0$. In this case, the Gâteaux derivative coincides with the *Fréchet derivative* of J at y^* .

These are not new objects. The Gâteaux and Fréchet differentials were introduced for general normed spaces in Section 2.3; the first variation is precisely their specialization to integral functionals $J(y) = \int_a^b \mathcal{L}(x, y, y') dx$ on a space of admissible curves. What distinguishes the variational setting is not the notion of derivative but the domain on which it acts: the perturbation directions η form an infinite-dimensional vector space rather than $\mathbb{R}^n$, so the derivative is a linear functional on that space instead of a gradient vector.

When such a Fréchet derivative exists, first-order optimality conditions take the familiar stationarity form: the derivative of J must vanish in all admissible directions. This condition is the infinite-dimensional analogue of the requirement $\nabla f(x^*) = 0$ in finite-dimensional optimization, the first-order necessary condition of the chapter on Classical Optimality Conditions. There, stationarity is the vanishing of the gradient (Section 2.3.3); here it is the vanishing of the first variation in every admissible direction, which the fundamental lemma will convert into the pointwise Euler–Lagrange equation.

Example 9.6 Integral depending only on the function value.

Let $J(y) = \int_a^b f(y(x)) dx$, where $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuously differentiable. Let the admissible set be $\mathcal{A} = \{y \in C([a, b]) \mid y(a) = A, y(b) = B\}$, with admissible

directions $\mathcal{A}_0 = \{\eta \in C([a, b]) \mid \eta(a) = \eta(b) = 0\}$.

Fix $y^* \in \mathcal{A}$ and $\eta \in \mathcal{A}_0$. For ε sufficiently small,

$$J(y^* + \varepsilon\eta) = \int_a^b f(y^*(x) + \varepsilon\eta(x)) dx.$$

Since $f \in C^1$, differentiation under the integral sign is justified, and the first variation exists and is given by

$$\delta J(y^*; \eta) = \int_a^b f'(y^*(x)) \eta(x) dx.$$

□

Example 9.7 Dirichlet energy.

Consider the functional $J(y) = \frac{1}{2} \int_a^b (y'(x))^2 dx$, with admissible set $\mathcal{A} = \{y \in C^1([a, b]) \mid y(a) = y(b) = 0\}$ and $\mathcal{A}_0 = \mathcal{A}$.

For $y^* \in \mathcal{A}$ and $\eta \in \mathcal{A}_0$,

$$J(y^* + \varepsilon\eta) = \frac{1}{2} \int_a^b (y^{*'}(x) + \varepsilon\eta'(x))^2 dx.$$

Differentiating with respect to ε at $\varepsilon = 0$ yields

$$\delta J(y^*; \eta) = \int_a^b y^{*'}(x) \eta'(x) dx.$$

If $y^* \in C^2([a, b])$, integration by parts gives

$$\delta J(y^*; \eta) = - \int_a^b y^{*''}(x) \eta(x) dx,$$

using the boundary conditions $\eta(a) = \eta(b) = 0$.

□

Example 9.8 General Lagrangian depending on y and y' .

Let $J(y) = \int_a^b \mathcal{L}(x, y(x), y'(x)) dx$, where $\mathcal{L} \in C^1([a, b] \times \mathbb{R} \times \mathbb{R})$. Let

$$\begin{aligned} \mathcal{A} &= \{y \in C^1([a, b]) \mid y(a) = A, y(b) = B\}, \\ \mathcal{A}_0 &= \{y \in C^1([a, b]) \mid \eta(a) = \eta(b) = 0\}. \end{aligned}$$

For $y^* \in \mathcal{A}$ and $\eta \in \mathcal{A}_0$,

$$J(y^* + \varepsilon\eta) = \int_a^b \mathcal{L}(x, y^* + \varepsilon\eta, y^{*\prime} + \varepsilon\eta') dx.$$

Differentiating under the integral sign yields the first variation

$$\delta J(y^*; \eta) = \int_a^b \left(\frac{\partial \mathcal{L}}{\partial y}(x, y^*, y^{*\prime}) \eta + \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*\prime}) \eta' \right) dx.$$

□

Example 9.9 Nonsmooth functional on $AC([a, b])$.

Consider the functional $J(y) = \int_a^b |y'(x)| dx$, with admissible set

$$\begin{aligned} \mathcal{A} &= \{y \in AC([a, b]) \mid y(a) = A, y(b) = B\}, \\ \mathcal{A}_0 &= \{\eta \in AC([a, b]) \mid \eta(a) = \eta(b) = 0\}. \end{aligned}$$

Assume that $y^* \in \mathcal{A}$ satisfies $y^{*\prime}(x) \neq 0$ almost everywhere. Then the first variation exists and is given by

$$\delta J(y^*; \eta) = \int_a^b \text{sign}(y^{*\prime}(x)) \eta'(x) dx.$$

This example illustrates that the first variation may exist even when the functional is not Fréchet differentiable, and it shows the role of directional derivatives in nonsmooth variational problems.

□

9.3.3 ■ Fundamental Lemma and the Euler–Lagrange Equation

A central mechanism in the calculus of variations is the passage from a global integral identity to a local, pointwise condition. This passage is justified by a fundamental result in analysis, commonly known as the *fundamental lemma of the calculus of variations*. Although it is traditionally introduced in variational contexts, the lemma is in fact a general statement about integrable functions and smooth test functions.

We begin with a formulation that is independent of any particular variational problem.

Lemma 9.10 (Fundamental Lemma of the Calculus of Variations). *Let $f \in L^1((a, b))$. If $\int_a^b f(x) \eta(x) dx = 0$ for all test functions $\eta \in C_c^\infty((a, b))$, then $f(x) = 0$ for almost every $x \in (a, b)$.*

This lemma asserts that smooth test functions with compact support are rich enough to detect any nonzero behavior of an integrable function in the interior of the interval. In other words, an L^1 function that annihilates all such test functions must vanish almost everywhere.

The assumption $f \in L^1((a, b))$ is minimal and natural. It ensures that the integral against

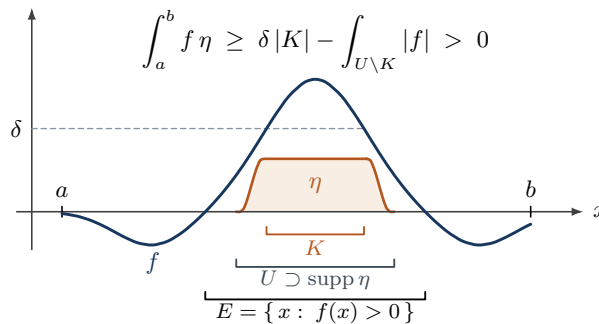


Figure 9.6: The idea behind the proof of the Fundamental Lemma (Lemma 9.10). If f does not vanish almost everywhere, then, after possibly replacing f by $-f$, it is positive on a set E of positive measure, and $f \geq \delta > 0$ on a compact set $K \subset E$ with $|K| > 0$. One selects a smooth test function $\eta \in C_c^\infty((a, b))$ with $0 \leq \eta \leq 1$, $\eta \equiv 1$ on K , and support in an open neighborhood U of K so thin that $\int_{U \setminus K} |f| < \frac{\delta}{2} |K|$. Then $\int_a^b f \eta \geq \delta |K| - \int_{U \setminus K} |f| > \frac{\delta}{2} |K| > 0$. This contradicts the hypothesis that f annihilates every test function; hence $f = 0$ almost everywhere. (In general E need not contain an interval, which is why η is built around K rather than inside E ; the drawing shows the simplest case.)

bounded test functions is well defined, without requiring any continuity or differentiability of f . The space $C_c^\infty((a, b))$ consists of infinitely differentiable functions whose support is compactly contained in the open interval (a, b) . Such functions vanish identically near the endpoints a and b , eliminating boundary effects and allowing one to localize arbitrarily within the interior of the domain.

Proof. We argue by contradiction. Assume that $f \in L^1((a, b))$ does not vanish almost everywhere on (a, b) . Then there exists a measurable set $E \subset (a, b)$ of positive Lebesgue measure such that either $f(x) > 0$ for almost every $x \in E$, or $f(x) < 0$ for almost every $x \in E$. Without loss of generality, we assume the first case.

Since E has positive measure, inner regularity of Lebesgue measure supplies a compact set $K \subset E$ with $|K| > 0$, and we may further shrink K so that $f \geq \delta$ on K for some $\delta > 0$ (replace E by $E \cap \{f \geq 1/n\}$ for n large, which still has positive measure for some n).

A measurable set of positive measure need not contain any interval, so no nonzero test function can be supported inside E itself; we therefore work with an open neighborhood of K instead. By outer regularity, choose an open set U with $K \subset U \subset (a, b)$ and

$$\int_{U \setminus K} |f(x)| dx < \frac{\delta}{2} |K|, \quad (9.1)$$

which is possible because $f \in L^1((a, b))$, so the map $A \mapsto \int_A |f|$ is absolutely continuous with respect to Lebesgue measure. By the standard smooth Urysohn construction there exists $\eta \in C_c^\infty((a, b))$ with

$$0 \leq \eta \leq 1, \quad \eta \equiv 1 \text{ on } K, \quad \text{supp}(\eta) \subset U.$$

This construction is illustrated in Figure 9.6: the bump η sits on a compact core K where f is bounded away from zero, and its spillover into $U \setminus K$ is made negligible by (9.1). Since $f \in L^1((a, b))$ and η is bounded, the product $f\eta$ is integrable, and splitting the integral over K

and $U \setminus K$ gives

$$\int_a^b f(x) \eta(x) dx = \int_K f(x) \eta(x) dx + \int_{U \setminus K} f(x) \eta(x) dx \geq \delta |K| - \int_{U \setminus K} |f(x)| dx > \frac{\delta}{2} |K| > 0.$$

This contradicts the hypothesis that $\int_a^b f(x) \eta(x) dx = 0$ for all $\eta \in C_c^\infty((a, b))$. Hence, no such set E can exist, and we conclude that $f(x) = 0$ for almost every $x \in (a, b)$. $\square$

9.3.4 ■ Derivation of the Euler–Lagrange Equation

We now apply the fundamental lemma of the calculus of variations to derive the classical first-order necessary condition for variational problems. The key analytical step consists in reducing the vanishing of the first variation to an integral identity of the form covered by the lemma and then verifying that all assumptions of the lemma are satisfied in the present setting.

Theorem 9.11 (Euler–Lagrange Equation). *Let $J(y) = \int_a^b \mathcal{L}(x, y(x), y'(x)) dx$, where $\mathcal{L} \in C^2([a, b] \times \mathbb{R} \times \mathbb{R})$. Let $\mathcal{A} = \{y \in C^1([a, b]) \mid y(a) = A, y(b) = B\}$, and suppose that $y^* \in \mathcal{A}$ is a weak local extremum of J which is moreover of class C^2 on $[a, b]$.*

Then y^ satisfies the Euler–Lagrange equation*

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(x, y^*(x), y^{*'}(x)) - \frac{\partial \mathcal{L}}{\partial y}(x, y^*(x), y^{*'}(x)) = 0 \quad \text{for all } x \in (a, b).$$

Remark 9.12 (Why C^2 is assumed, and how to avoid it). The differentiated form of the Euler–Lagrange equation involves $\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*'})$, whose existence already requires $y^{*''}$; for a merely C^1 extremal that derivative need not exist, and the integration by parts performed below is then unjustified. Two standard remedies are available. One may assume, as we do here, that the extremal is C^2 . Alternatively one integrates the *other* term by parts, which requires no extra smoothness and yields the *du Bois-Reymond* form

$$\frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*'}) = \int_a^x \frac{\partial \mathcal{L}}{\partial y}(s, y^*, y^{*'}) ds + C, \quad x \in [a, b],$$

valid for every C^1 weak extremum. Its right-hand side is C^1 , so $\partial \mathcal{L} / \partial y'$ is automatically C^1 along the extremal, and under the strengthened Legendre condition $\mathcal{L}_{y'y'}(x, y^*, y^{*'}) \neq 0$ Hilbert’s differentiability theorem then upgrades y^* to C^2 , after which differentiating recovers the form stated above. The same caveat applies verbatim to the free-endpoint, mixed-endpoint, transversality, corner, higher-order and vector versions proved below, each of which integrates $\partial \mathcal{L} / \partial y' \eta'$ by parts.

Proof. Since y^* is a weak local extremum of J , the first variation of J at y^* must vanish in every admissible direction. That is, $\delta J(y^*; \eta) = 0$ for all $\eta \in \mathcal{A}_0$, where $\mathcal{A}_0 = \{\eta \in C^1([a, b]) \mid \eta(a) = \eta(b) = 0\}$ is the vector space of admissible variations.

Under the stated regularity assumptions on $\mathcal{L}$ and y^* , the first variation can be computed by differentiating under the integral sign. This yields

$$\delta J(y^*; \eta) = \int_a^b \left(\frac{\partial \mathcal{L}}{\partial y}(x, y^*, y^{*'}) \eta + \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*'}) \eta' \right) dx.$$

We integrate the second term by parts. Since $\eta(a) = \eta(b) = 0$, the boundary terms vanish, and we obtain

$$\delta J(y^*; \eta) = \int_a^b \left[\frac{\partial \mathcal{L}}{\partial y}(x, y^*, y^{*\prime}) - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*\prime}) \right] \eta(x) dx.$$

At this point, we interpret the above identity as an integral pairing between the function

$$g(x) := \frac{\partial \mathcal{L}}{\partial y}(x, y^*, y^{*\prime}) - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*\prime})$$

and the admissible variations η . Because $\mathcal{L} \in C^2$ and $y^* \in C^2([a, b])$, the function g is continuous on (a, b) and, in particular, belongs to $L^1((a, b))$.

Although admissible variations are defined as functions $\eta \in C^1([a, b])$ satisfying the boundary conditions $\eta(a) = \eta(b) = 0$, the vanishing of the first variation holds for *all* such functions. In particular, it holds for any smaller class of functions contained in this admissible variation space.

Every test function $\eta \in C_c^\infty((a, b))$ is infinitely differentiable on (a, b) and has compact support strictly contained in the open interval (a, b) . Consequently, η vanishes identically in a neighborhood of the endpoints a and b , and hence satisfies $\eta(a) = \eta(b) = 0$. It follows that $C_c^\infty((a, b)) \subset \{\eta \in C^1([a, b]) \mid \eta(a) = \eta(b) = 0\}$.

Since the integral identity obtained from the first variation holds for all admissible variations, it must therefore hold in particular for all $\eta \in C_c^\infty((a, b))$. Consequently, $\int_a^b g(x) \eta(x) dx = 0$ for all $\eta \in C_c^\infty((a, b))$.

Since $g \in L^1((a, b))$, the fundamental lemma of the calculus of variations (Lemma 9.10) applies and implies that $g(x) = 0$ for almost every $x \in (a, b)$.

Finally, because g is continuous on (a, b) , the almost-everywhere equality upgrades to pointwise equality. Hence,

$$\frac{\partial \mathcal{L}}{\partial y}(x, y^*, y^{*\prime}) - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*\prime}) = 0 \quad \text{for all } x \in (a, b),$$

which is precisely the Euler–Lagrange equation. $\square$

The Euler–Lagrange equation is the infinite-dimensional analogue of the stationarity condition $\nabla f(x^*) = 0$ in finite-dimensional optimization. In both settings, optimality is characterized by the vanishing of the first-order derivative of the objective functional in all admissible directions. The essential difference lies in the nature of the derivative: in the calculus of variations, the first derivative of J is a linear functional acting on an infinite-dimensional space of variations rather than a finite-dimensional gradient vector.

Example 9.13 Why localization forces pointwise vanishing.

The purpose of this example is to illustrate why testing an integral identity against all compactly supported smooth functions forces the integrand to vanish pointwise.

Let $g \in C([a, b])$ and assume that $\int_a^b g(x) \eta(x) dx = 0$ for all $\eta \in C_c^\infty((a, b))$. We show that this condition implies $g(x) = 0$ for all $x \in (a, b)$.

Suppose, to the contrary, that g is not identically zero on (a, b) . Then there exists a point $x_0 \in (a, b)$ such that $g(x_0) \neq 0$. Without loss of generality, assume $g(x_0) > 0$. By continuity of g , there exists an open interval $I \subset (a, b)$ containing x_0 such that

$g(x) > 0$ for all $x \in I$.

The defining feature of the space $C_c^\infty((a, b))$ is its localization property: for any open interval $I \subset (a, b)$, there exists a nonzero function $\eta \in C_c^\infty((a, b))$ whose support is contained in I . Choose such a function with $\eta \geq 0$ and $\eta \not\equiv 0$.

For this choice of η , we have

$$\int_a^b g(x) \eta(x) dx = \int_I g(x) \eta(x) dx > 0,$$

since both g and η are strictly positive on a set of positive measure. This contradicts the assumption that the integral vanishes for all test functions.

Therefore, no such point x_0 can exist, and we conclude that $g(x) = 0$ for all $x \in (a, b)$.

This example shows that compactly supported smooth test functions can detect nonzero behavior of a function at arbitrarily small scales, eliminating the possibility of cancellation across the domain. $\square$

Example 9.14 Why boundary values cannot be detected.

Let $g(x) \equiv 1$ on (a, b) and consider the class of test functions $\eta \in C^1([a, b])$ satisfying $\eta(a) = \eta(b) = 0$. Then

$$\int_a^b g(x) \eta'(x) dx = \int_a^b \eta'(x) dx = \eta(b) - \eta(a) = 0 \quad \text{for all such } \eta.$$

Despite this identity holding for all admissible variations, the function g is clearly not zero. This shows that identities involving derivatives of variations do not directly yield pointwise information until integration by parts is performed and the resulting expression is tested against functions in $C_c^\infty((a, b))$.

The example highlights why the fundamental lemma is applied only after boundary terms have been eliminated. $\square$

Example 9.15 Failure when the test space is too small.

Let $g(x) = x - \frac{a+b}{2}$ on (a, b) . Then $\int_a^b g(x) dx = 0$. Thus, if one were to test only against constant functions $\eta \equiv 1$, the integral identity would hold even though g is not identically zero. This shows that a small test space cannot detect local variations of the integrand.

By contrast, the space $C_c^\infty((a, b))$ is rich enough to localize on arbitrary subintervals, ensuring that the vanishing of the integral against all test functions forces g to vanish almost everywhere. $\square$

Example 9.16 Why C^1 admissible variations are sufficient in practice.

Suppose that

$$\int_a^b g(x) \eta(x) dx = 0 \quad \text{for all } \eta \in C^1([a, b]) \text{ with } \eta(a) = \eta(b) = 0.$$

Since $C_c^\infty((a, b)) \subset C^1([a, b])$ and every function in $C_c^\infty((a, b))$ vanishes at the endpoints, the identity holds in particular for all $\eta \in C_c^\infty((a, b))$. The fundamental lemma therefore applies and yields $g = 0$ almost everywhere.

This example clarifies why admissible variations are defined in C^1 , while the analytical extraction of pointwise conditions relies on the smaller but more powerful test space $C_c^\infty((a, b))$. $\square$

The general Euler–Lagrange equation admits several important simplifications when the Lagrangian possesses additional structural properties. We collect these special cases as corollaries.

Corollary 9.17 (Cyclic variable). *Suppose that the Lagrangian is independent of y , i.e., $\mathcal{L}(x, y, y') = \mathcal{L}(x, y')$. Then any weak extremal y^* satisfies*

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(x, y^{*'}(x)) = 0,$$

and hence

$$\frac{\partial \mathcal{L}}{\partial y'}(x, y^{*'}(x)) = C,$$

for some constant $C \in \mathbb{R}$.

Proof. Since $\mathcal{L}$ does not depend explicitly on y , we have $\frac{\partial \mathcal{L}}{\partial y}(x, y, y') = 0$ for all (x, y, y') . Substituting this into the Euler–Lagrange equation,

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} - \frac{\partial \mathcal{L}}{\partial y} = 0,$$

we obtain

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(x, y^{*'}(x)) = 0 \quad \text{for all } x \in (a, b).$$

Therefore, the function

$$x \mapsto \frac{\partial \mathcal{L}}{\partial y'}(x, y^{*'}(x))$$

has zero derivative on (a, b) and is hence constant on that interval. $\square$

Corollary 9.18 (Beltrami identity). *Suppose that the Lagrangian is independent of x , i.e., $\mathcal{L}(x, y, y') = \mathcal{L}(y, y')$. Then any weak extremal y^* satisfies*

$$\mathcal{L}(y^*(x), y^{*'}(x)) - y^{*'}(x) \frac{\partial \mathcal{L}}{\partial y'}(y^*(x), y^{*'}(x)) = C,$$

for some constant $C \in \mathbb{R}$.

Proof. Define the function

$$H(x) := \mathcal{L}(y(x), y'(x)) - y'(x) \frac{\partial \mathcal{L}}{\partial y'}(y(x), y'(x)),$$

where y is a C^2 extremal (or C^1 with absolutely continuous derivative).

Since $\mathcal{L}$ does not depend explicitly on x , differentiation of $\mathcal{L}(y(x), y'(x))$ with respect to x yields

$$\frac{d}{dx} \mathcal{L}(y, y') = \frac{\partial \mathcal{L}}{\partial y}(y, y') y' + \frac{\partial \mathcal{L}}{\partial y'}(y, y') y''.$$

Similarly,

$$\frac{d}{dx} \left(y' \frac{\partial \mathcal{L}}{\partial y'} \right) = y'' \frac{\partial \mathcal{L}}{\partial y'} + y' \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}.$$

Subtracting these expressions, we obtain

$$\frac{dH}{dx} = \frac{\partial \mathcal{L}}{\partial y}(y, y') y' - y' \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(y, y').$$

Factoring out y' , this becomes

$$\frac{dH}{dx} = y' \left(\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right).$$

Along an extremal y^* , the Euler–Lagrange equation implies

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} = 0,$$

and hence $\frac{dH}{dx} = 0$ for all $x \in (a, b)$. Therefore, $H(x)$ is constant on (a, b) , which yields the stated identity. $\square$

Example 9.19 Straight line as the shortest path.

Consider the variational problem $J(y) = \int_a^b \sqrt{1 + (y'(x))^2} dx$, defined on the admissible set $\mathcal{A} = \{y \in C^1([a, b]) \mid y(a) = A, y(b) = B\}$. The functional $J(y)$ represents the arc length of the curve $y(x)$ connecting the points (a, A) and (b, B) .

The Lagrangian is $\mathcal{L}(y, y') = \sqrt{1 + (y')^2}$, which does not depend explicitly on y . The Euler–Lagrange equation therefore reduces to $\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} = 0$. Computing the derivative, $\frac{\partial \mathcal{L}}{\partial y'} = \frac{y'}{\sqrt{1 + (y')^2}}$, we conclude that $\frac{y'}{\sqrt{1 + (y')^2}} = C$, where C is a constant. Solving for y' , we find that y' is constant on $[a, b]$. Hence the minimizer is an affine function $y(x) = mx + c$, with constants m and c determined by the boundary conditions.

This recovers the classical geometric result that the shortest path between two points in the plane is a straight line. $\square$

Example 9.20 Classical mechanics and Newton's second law.

In this example, the independent variable represents time. That is, we identify $x = t$ and consider curves defined on a time interval $[t_0, t_1]$.

Let $y(t)$ denote the position of a particle of mass $m > 0$ moving along a line under the influence of a potential $U : \mathbb{R} \rightarrow \mathbb{R}$. Consider the action functional

$$J(y) = \int_{t_0}^{t_1} \left(\frac{1}{2} m (y'(t))^2 - U(y(t)) \right) dt,$$

defined on the admissible set $\mathcal{A} = \{y \in C^1([t_0, t_1]) \mid y(t_0) = y_0, y(t_1) = y_1\}$.

The associated Lagrangian is $\mathcal{L}(t, y, y') = \frac{1}{2} m (y')^2 - U(y)$, which is independent of the time variable t .

The Euler–Lagrange equation $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial y'} - \frac{\partial \mathcal{L}}{\partial y} = 0$ yields

$$\frac{\partial \mathcal{L}}{\partial y'} = my', \quad \frac{\partial \mathcal{L}}{\partial y} = -U'(y),$$

and therefore $my''(t) + U'(y(t)) = 0$, or equivalently, $my''(t) = -U'(y(t))$.

This is precisely Newton's second law, with force given by $F(y) = -U'(y)$. Thus, the classical equations of motion arise as Euler–Lagrange equations associated with the stationarity of the action functional. $\square$

9.4 ■ Extensions of the Euler–Lagrange Equation

The Euler–Lagrange equation derived in the previous section provides a necessary condition for optimality in the interior of the interval (a, b) . Its derivation depends essentially on the boundary conditions imposed on the admissible curves, since these conditions determine the class of admissible variations and, consequently, the boundary terms arising in the first variation.

We now present a systematic treatment of variational problems with fixed endpoints, free endpoints, and mixed endpoint conditions.

9.4.1 ■ Free Endpoints

We next consider variational problems in which no endpoint values are prescribed.

Definition 9.21 (Admissible set: free endpoints). The admissible set is $\mathcal{A} = C^1([a, b])$.

Definition 9.22 (Admissible variations: free endpoints). Admissible variations are all functions $\eta \in C^1([a, b])$, with no restrictions on $\eta(a)$ or $\eta(b)$.

Theorem 9.23 (Euler–Lagrange equation with free endpoints). Let J and $\mathcal{L}$ be as above, and let $y^* \in \mathcal{A}$ be a weak local extremum. Then y^* satisfies:

(1) the Euler–Lagrange equation

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} - \frac{\partial \mathcal{L}}{\partial y} = 0 \quad \text{for all } x \in (a, b);$$

(2) *the natural boundary conditions*

$$\frac{\partial \mathcal{L}}{\partial y'}(a, y^*(a), y^{*'}(a)) = 0, \quad \frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) = 0.$$

Proof. Let $y^* \in \mathcal{A} = C^1([a, b])$ be a weak local extremum of J . By definition of a weak extremum, the first variation of J at y^* must vanish in every admissible direction $\eta \in C^1([a, b])$, that is, $\delta J(y^*; \eta) = 0$ for all $\eta \in C^1([a, b])$.

To turn this stationarity requirement into pointwise conditions, we first express the first variation explicitly and then isolate the interior and boundary contributions. By differentiating under the integral sign, we obtain

$$\delta J(y^*; \eta) = \int_a^b \left(\frac{\partial \mathcal{L}}{\partial y}(x, y^*, y^{*'}) \eta(x) + \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*'}) \eta'(x) \right) dx.$$

Since $\eta \in C^1([a, b])$ is arbitrary and does *not* vanish at the endpoints, we integrate the second term by parts:

$$\int_a^b \frac{\partial \mathcal{L}}{\partial y'} \eta' dx = \left. \frac{\partial \mathcal{L}}{\partial y'} \eta \right|_a^b - \int_a^b \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \eta dx.$$

Substituting into the first variation yields

$$\delta J(y^*; \eta) = \int_a^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta(x) dx + \left. \frac{\partial \mathcal{L}}{\partial y'} \eta \right|_a^b.$$

We now exploit the freedom in choosing η to separate the interior equation from the boundary terms. The identity $\delta J(y^*; \eta) = 0$ must hold for *all* $\eta \in C^1([a, b])$. In particular, we may choose variations η with compact support contained in (a, b) . For such variations, $\eta(a) = \eta(b) = 0$, and the boundary term vanishes. Therefore,

$$\int_a^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta(x) dx = 0 \quad \text{for all } \eta \in C_c^\infty((a, b)).$$

By the fundamental lemma of the calculus of variations, it follows that

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} = 0 \quad \text{almost everywhere on } (a, b).$$

Since $\mathcal{L} \in C^2$, the expression above is continuous, and therefore the equality holds for all $x \in (a, b)$.

It remains to extract the conditions carried by the boundary term. Having established the Euler–Lagrange equation, we return to the full expression for the first variation. The interior integral now vanishes identically, leaving

$$\left. \frac{\partial \mathcal{L}}{\partial y'} \eta \right|_a^b = \frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) \eta(b) - \frac{\partial \mathcal{L}}{\partial y'}(a, y^*(a), y^{*'}(a)) \eta(a).$$

Since $\eta(a)$ and $\eta(b)$ are independent and arbitrary real numbers, the only way for this expression to vanish for all admissible variations is that

$$\frac{\partial \mathcal{L}}{\partial y'}(a, y^*(a), y^{*'}(a)) = 0, \quad \frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) = 0.$$

The weak local extremum y^* therefore satisfies both the Euler–Lagrange equation in the interior and the natural boundary conditions at the endpoints. $\square$

9.4.2 ■ Mixed Endpoint Conditions

Finally, we consider the case in which one endpoint is fixed and the other is free.

Definition 9.24 (Admissible set: mixed endpoints). Assume that $y(a) = A$ is prescribed while $y(b)$ is free. The admissible set is $\mathcal{A} = \{y \in C^1([a, b]) \mid y(a) = A\}$.

Definition 9.25 (Admissible variations: mixed endpoints). The corresponding admissible variations satisfy $\eta(a) = 0$, with $\eta(b)$ arbitrary.

Theorem 9.26 (Euler–Lagrange equation with mixed endpoints). Let J and $\mathcal{L}$ be as above, and let $y^* \in \mathcal{A}$ be a weak local extremum. Then y^* satisfies:

- (1) the Euler–Lagrange equation on (a, b) ;
- (2) the natural boundary condition at the free endpoint, $\frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) = 0$.

Proof. Let $y^* \in \mathcal{A}$ be a weak local extremum, so that the first variation vanishes in every admissible direction: $\delta J(y^*; \eta) = 0$ for all $\eta \in C^1([a, b])$ with $\eta(a) = 0$. Differentiating under the integral sign and integrating the η' term by parts gives

$$\delta J(y^*; \eta) = \int_a^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta(x) dx + \left. \frac{\partial \mathcal{L}}{\partial y'} \eta \right|_a^b.$$

Because every admissible variation satisfies $\eta(a) = 0$, the boundary contribution at $x = a$ vanishes, leaving

$$\delta J(y^*; \eta) = \int_a^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta(x) dx + \frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) \eta(b).$$

As in the free-endpoint case, we first isolate the interior equation. Restricting to variations $\eta \in C_c^\infty((a, b))$, which satisfy $\eta(a) = \eta(b) = 0$, the boundary term drops out, so that

$$\int_a^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta(x) dx = 0 \quad \text{for all } \eta \in C_c^\infty((a, b)).$$

Since $\mathcal{L} \in C^2$ and $y^* \in C^1$, the bracketed function is continuous, and the fundamental lemma of the calculus of variations yields the Euler–Lagrange equation

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} - \frac{\partial \mathcal{L}}{\partial y} = 0 \quad \text{on } (a, b).$$

With the interior integral now identically zero, the first variation reduces to its boundary contribution,

$$\delta J(y^*; \eta) = \frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) \eta(b).$$

This must vanish for every admissible variation. Since the terminal value is unconstrained, $\eta(b)$ may take any real value: for instance, the variation $\eta(x) = (x - a)/(b - a)$ is admissible and has $\eta(b) = 1$. Hence

$$\frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) = 0,$$

which is the natural boundary condition at the free endpoint. No boundary condition arises at $x = a$, where the value $y^*(a) = A$ is prescribed and all admissible variations vanish. $\square$

Example 9.27 Free-end elastic string and the Neumann condition.

Consider the functional

$$J(y) = \int_0^1 \frac{1}{2} ((y'(x))^2 + (y(x))^2) dx, \quad y(0) = 1, \quad y(1) \text{ free},$$

which models the equilibrium energy of a string with elastic restoring force, clamped at the left end and left free at the right. Here $\mathcal{L}(x, y, y') = \frac{1}{2}(y'^2 + y^2)$, so $\partial\mathcal{L}/\partial y' = y'$ and $\partial\mathcal{L}/\partial y = y$.

The Euler–Lagrange equation is

$$\frac{d}{dx}(y') - y = y'' - y = 0,$$

with general solution $y(x) = c_1 \cosh x + c_2 \sinh x$. The fixed endpoint gives $y(0) = c_1 = 1$. The mixed-endpoint theorem supplies the missing condition at the free end through the natural boundary condition $\partial\mathcal{L}/\partial y'(1) = y'(1) = 0$. Since $y'(x) = \sinh x + c_2 \cosh x$, this requires $\sinh 1 + c_2 \cosh 1 = 0$, i.e. $c_2 = -\tanh 1$. Hence

$$y^*(x) = \cosh x - \tanh 1 \sinh x = \frac{\cosh(1-x)}{\cosh 1},$$

whose graph leaves the right end horizontally, $y^{*'}(1) = 0$.

The natural boundary condition $y'(1) = 0$ is precisely a homogeneous *Neumann* condition: leaving the endpoint value free in the variational problem is equivalent to imposing vanishing slope there. Had the right end instead been *fixed* at some value $y(1) = B$, the constant c_2 would be determined by $c_1 \cosh 1 + c_2 \sinh 1 = B$ and the solution would generally meet the end with nonzero slope. The free-endpoint problem thus selects, among all extremals of $y'' = y$, the unique one satisfying the Neumann condition. $\square$

9.4.3 ■ Transversality Conditions

We now consider variational problems in which the endpoint is not fixed at a specific value, but is constrained to lie on a prescribed set. Such problems arise naturally when the terminal state is only partially specified, or when the terminal time or terminal position is free subject to a constraint.

To make this precise, consider the functional

$$J(y) = \int_a^b \mathcal{L}(x, y(x), y'(x)) dx, \quad \mathcal{L} \in C^2([a, b] \times \mathbb{R} \times \mathbb{R}),$$

in which the initial value is fixed, $y(a) = A$, while the terminal point $(b, y(b))$ is free to slide along a prescribed smooth curve $\Phi(b, y(b)) = 0$, where $\Phi \in C^1(\mathbb{R}^2)$ and $\nabla\Phi \neq 0$ on the constraint set. The admissible curves are therefore those meeting the fixed left endpoint and terminating anywhere on this target curve,

$$\mathcal{A} = \{y \in C^1([a, b]) \mid y(a) = A, \Phi(b, y(b)) = 0\}.$$

Because the terminal point may move, the admissible variations are no longer required to vanish at $x = b$; instead, the perturbed endpoint must stay on the target curve to first order. Because

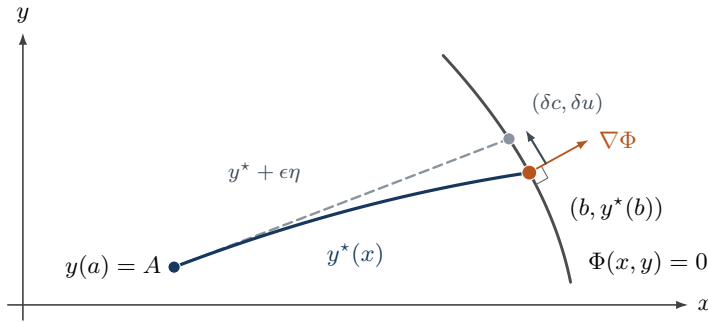


Figure 9.7: Geometry of the transversality condition. The initial point is pinned at $y(a) = A$, while the terminal point may slide along the target curve $\Phi(x, y) = 0$. A neighboring admissible curve (dashed) again terminates on Φ , so to first order the endpoint displacement $(\delta c, \delta u)$ is tangent to the target curve, $\Phi_x \delta c + \Phi_y \delta u = 0$, that is, orthogonal to $\nabla \Phi$. With the abscissa b held fixed ($\delta c = 0$, Theorem 9.28) stationarity gives $\partial \mathcal{L} / \partial y' = \lambda \partial \Phi / \partial y$; when b is free as well (Remark 9.29) it requires $(\mathcal{L} - y'^* \mathcal{L}_{y'}, \mathcal{L}_{y'})$ to be parallel to $\nabla \Phi$, which ties the slope of the extremal to the geometry of the target curve. When the endpoint is fully free, the condition reduces to the natural condition $\partial \mathcal{L} / \partial y' = 0$.

the terminal abscissa b is held fixed here, only the ordinate $y(b)$ moves, and differentiating the terminal constraint $\Phi(b, y(b)) = 0$ along a variation shows that an admissible η satisfies

$$\eta(a) = 0, \quad \frac{\partial \Phi}{\partial y}(b, y^*(b)) \eta(b) = 0,$$

which expresses the tangency of variations to the terminal constraint. (If the terminal *time* b is free as well, a second condition appears; see the example below and Remark 9.29.) Figure 9.7 illustrates this geometry: the optimal curve meets the target curve so that its slope is tied to the slope of the constraint through the transversality condition derived below.

Theorem 9.28 (Transversality condition). *Let $y^* \in \mathcal{A}$ be a weak local extremum of J . Then y^* satisfies:*

- (1) *the Euler–Lagrange equation on (a, b) ;*
- (2) *the transversality condition*

$$\frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) = \lambda \frac{\partial \Phi}{\partial y}(b, y^*(b)),$$

for some scalar multiplier $\lambda \in \mathbb{R}$.

Proof. Let $y^* \in \mathcal{A}$ be a weak local extremum of J , where the admissible set is $\mathcal{A} = \{y \in C^1([a, b]) \mid y(a) = A, \Phi(b, y(b)) = 0\}$, with $\Phi \in C^1(\mathbb{R}^2)$ and $\nabla \Phi \neq 0$ on the constraint set.

By definition of a weak local extremum, the first variation must vanish in every admissible direction η , that is, $\delta J(y^*; \eta) = 0$ for all admissible η .

We begin by identifying which variations are admissible. Admissible variations $\eta \in C^1([a, b])$ satisfy $\eta(a) = 0$, since the initial value is fixed. At the terminal point, admissibility requires that the perturbed curve $y_\varepsilon(b) = y^*(b) + \varepsilon \eta(b)$ remain on the constraint manifold $\Phi(b, y(b)) = 0$ for

all sufficiently small ε . Differentiating the constraint with respect to ε at $\varepsilon = 0$, and recalling that b is fixed so that only the second argument varies, yields the linearized endpoint condition

$$\frac{\partial \Phi}{\partial y}(b, y^*(b)) \eta(b) = 0.$$

Thus, admissible endpoint variations $\eta(b)$ are restricted to satisfy this linear constraint.

With the admissible directions characterized, we turn to the first variation itself. As in previous cases, the first variation is given by

$$\delta J(y^*; \eta) = \int_a^b \left(\frac{\partial \mathcal{L}}{\partial y}(x, y^*, y^{*'}) \eta(x) + \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*'}) \eta'(x) \right) dx.$$

Integrating by parts and using $\eta(a) = 0$, we obtain

$$\delta J(y^*; \eta) = \int_a^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta(x) dx + \frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) \eta(b).$$

As before, the interior equation is recovered first by testing against localized variations. By choosing variations η with compact support contained in (a, b) , which automatically satisfy the endpoint constraint, the boundary term vanishes. Thus,

$$\int_a^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta(x) dx = 0 \quad \text{for all } \eta \in C_c^\infty((a, b)).$$

By the fundamental lemma of the calculus of variations, it follows that

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} = 0 \quad \text{for all } x \in (a, b).$$

The remaining content of stationarity now lives entirely at the constrained endpoint. Having established the Euler–Lagrange equation, the first variation reduces to

$$\delta J(y^*; \eta) = \frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) \eta(b).$$

This expression must vanish for all admissible $\eta(b)$ satisfying the linearized constraint

$$\frac{\partial \Phi}{\partial y}(b, y^*(b)) \eta(b) = 0.$$

This is a finite-dimensional constrained optimality problem at the boundary. By the classical Lagrange multiplier theorem, there exists a scalar $\lambda \in \mathbb{R}$ such that

$$\frac{\partial \mathcal{L}}{\partial y'}(b, y^*(b), y^{*'}(b)) = \lambda \frac{\partial \Phi}{\partial y}(b, y^*(b)).$$

In summary, the weak local extremum y^* satisfies the Euler–Lagrange equation in the interior together with the stated transversality condition at the terminal point. $\square$

The transversality condition expresses a balance between the variational momentum $\partial \mathcal{L} / \partial y'$ and the geometry of the terminal constraint. When the endpoint is completely free, this reduces to the natural boundary condition $\partial \mathcal{L} / \partial y' = 0$. The multiplier λ is not incidental: the terminal condition is a finite-dimensional equality-constrained problem at the boundary, and λ is exactly

a Lagrange multiplier in the sense of Section 3.2. The transversality condition is thus the stationarity, with respect to endpoint displacements, of a Lagrangian pairing the variational momentum against the gradient $\nabla\Phi$ of the terminal constraint, the same multiplier rule that governs equality-constrained optimization in finite dimensions, transplanted to the free boundary of a variational problem.

Remark 9.29 (Free terminal abscissa). Theorem 9.28 keeps the terminal abscissa b fixed, so the only freedom at the right end is the ordinate $y(b)$ and a single multiplier suffices. It should be said plainly that in this fixed- b setting the transversality condition carries information only in the degenerate case $\Phi_y(b, y^*(b)) = 0$, where it forces the natural boundary condition $\mathcal{L}_{y'}(b) = 0$. If instead $\Phi_y \neq 0$, the linearized constraint $\Phi_y \eta(b) = 0$ forces $\eta(b) = 0$: the terminal ordinate is pinned by $\Phi(b, \cdot) = 0$, the problem is a fixed-endpoint problem in disguise, and the multiplier may simply be read off as $\lambda = \mathcal{L}_{y'}(b)/\Phi_y(b)$. The condition becomes substantive precisely when the abscissa is released. When b is also free—the situation in the example that follows, and the one that arises in optimal control with free terminal time—the endpoint displacement is a pair $(\delta c, \delta u)$ constrained to be tangent to the target curve, $\Phi_x \delta c + \Phi_y \delta u = 0$, and stationarity produces *two* conditions,

$$\frac{\partial \mathcal{L}}{\partial y'}(b) = \lambda \frac{\partial \Phi}{\partial y}(b), \quad \mathcal{L}(b) - y^{*'}(b) \frac{\partial \mathcal{L}}{\partial y'}(b) = \lambda \frac{\partial \Phi}{\partial x}(b),$$

the second being the companion generated by the free abscissa. Equivalently, the vector $(\mathcal{L} - y^{*'}\mathcal{L}_{y'}, \mathcal{L}_{y'})$ must be parallel to $\nabla\Phi$.

Example 9.30 Shortest path from a point to a line: orthogonal transversality.

Among all curves joining the fixed point $A = (0, 0)$ to a terminal point that is free to slide along the line $\ell : \Phi(x, y) = x + y - 2 = 0$, we seek the one of least Euclidean length,

$$J(y) = \int_0^b \sqrt{1 + (y'(x))^2} dx,$$

where both the terminal abscissa b and ordinate $y(b)$ are free, subject only to $(b, y(b)) \in \ell$.

Interior. The integrand $\mathcal{L} = \sqrt{1 + y'^2}$ depends on neither x nor y , so the Euler–Lagrange equation reduces to $\frac{d}{dx}(y'/\sqrt{1 + y'^2}) = 0$, giving $y' = \text{const}$. The extremals are straight lines; write $y^*(x) = mx$ to satisfy $y^*(0) = 0$.

Sliding endpoint. Because the terminal point moves both horizontally and vertically, we record the terminal contribution to the first variation under a displacement $(\delta c, \delta u)$ of the endpoint. Along an extremal, the vertical variation η satisfies $\eta(b) = \delta u - y^{*'}(b) \delta c$, and a short computation (changing the upper limit and integrating by parts) gives

$$\delta J = \frac{\partial \mathcal{L}}{\partial y'}(b) \delta u + \left(\mathcal{L}(b) - y^{*'}(b) \frac{\partial \mathcal{L}}{\partial y'}(b) \right) \delta c.$$

Stationarity requires $\delta J = 0$ for every displacement *tangent* to ℓ , that is, for all $(\delta c, \delta u)$ with $\Phi_x \delta c + \Phi_y \delta u = \delta c + \delta u = 0$. Equivalently, the vector $(\mathcal{L} - y^{*'}\mathcal{L}_{y'}, \mathcal{L}_{y'})$ must be parallel to $\nabla\Phi = (\Phi_x, \Phi_y)$; its second component reproduces the momentum relation $\mathcal{L}_{y'} = \lambda \Phi_y$ of Theorem 9.28, while its first

component $\mathcal{L} - y'^* \mathcal{L}_{y'} = \lambda \Phi_x$ is the companion condition generated by the free terminal abscissa.

Solution. For $\mathcal{L} = \sqrt{1 + y'^2}$ one has

$$\frac{\partial \mathcal{L}}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}}, \quad \mathcal{L} - y' \frac{\partial \mathcal{L}}{\partial y'} = \frac{1}{\sqrt{1 + y'^2}}.$$

Substituting $\delta u = -\delta c$ into the terminal variation gives

$$\delta J = \frac{1}{\sqrt{1 + m^2}} (\delta c + m \delta u) = \frac{1 - m}{\sqrt{1 + m^2}} \delta c,$$

which vanishes for all δc only if $m = 1$. Hence $y^*(x) = x$, meeting ℓ at $(1, 1)$. The tangent direction of the extremal, $(1, m) = (1, 1)$, is parallel to $\nabla \Phi = (1, 1)$: the least-length path crosses the target line *orthogonally*. Transversality is, in the Euclidean-length case, precisely the statement that the extremal meets the target set at a right angle. $\square$

9.4.4 ■ Corner Points and Weierstrass–Erdmann Conditions

We now consider admissible curves that are only piecewise continuously differentiable. Such curves arise naturally in optimal control, impact problems, and systems with switching dynamics.

To model such behavior, let $c \in (a, b)$ be an unknown interior point and consider curves $y \in C^1([a, c]) \cap C^1([c, b])$ that are continuous at $x = c$ but may have a discontinuous derivative there. The admissible curves are thus required to satisfy $y(c^-) = y(c^+)$ while allowing $y'(c^-) \neq y'(c^+)$, so that a corner may form at $x = c$ whose location is itself part of the unknown.

Theorem 9.31 (Weierstrass–Erdmann corner conditions). *Let y^* be a weak local extremum of J among piecewise C^1 curves with a corner at $x = c$. Then y^* satisfies:*

- (1) *the Euler–Lagrange equation on (a, c) and (c, b) ;*
- (2) *continuity of the conjugate momentum,*

$$\frac{\partial \mathcal{L}}{\partial y'}(c^-, y^*, y'^*) = \frac{\partial \mathcal{L}}{\partial y'}(c^+, y^*, y'^*);$$

- (3) *continuity of the Hamiltonian,*

$$\mathcal{L}(c^-, y^*, y'^*) - y'^*(c^-) \frac{\partial \mathcal{L}}{\partial y'}(c^-, y^*, y'^*) = \mathcal{L}(c^+, y^*, y'^*) - y'^*(c^+) \frac{\partial \mathcal{L}}{\partial y'}(c^+, y^*, y'^*).$$

Proof. Let y^* be a weak local extremum of J among curves that are piecewise C^1 with a single corner at $x = c$, that is, $y^* \in C^1([a, c]) \cap C^1([c, b])$ with $y^*(c^-) = y^*(c^+)$, while the derivatives $y'^*(c^-)$ and $y'^*(c^+)$ are not a priori equal.

By definition of a weak local extremum, the first variation of J must vanish for all admissible variations η . Admissible variations are functions $\eta \in C^1([a, c]) \cap C^1([c, b])$ satisfying $\eta(a) = \eta(b) = 0$ and $\eta(c^-) = \eta(c^+)$, so that the perturbed curve remains continuous at the corner.

Because the integrand is only piecewise smooth, we split the functional at the corner and treat each side separately. The functional decomposes as $J(y) = \int_a^c \mathcal{L}(x, y, y') dx + \int_c^b \mathcal{L}(x, y, y') dx$. Accordingly, the first variation at y^* in the direction η is

$$\delta J(y^*; \eta) = \delta J_1(y^*; \eta) + \delta J_2(y^*; \eta),$$

where

$$\delta J_1(y^*; \eta) = \int_a^c \left(\frac{\partial \mathcal{L}}{\partial y} \eta + \frac{\partial \mathcal{L}}{\partial y'} \eta' \right) dx, \quad \delta J_2(y^*; \eta) = \int_c^b \left(\frac{\partial \mathcal{L}}{\partial y} \eta + \frac{\partial \mathcal{L}}{\partial y'} \eta' \right) dx.$$

We integrate by parts on $[a, c]$ and $[c, b]$ separately. On $[a, c]$ we obtain

$$\delta J_1(y^*; \eta) = \int_a^c \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta dx + \left. \frac{\partial \mathcal{L}}{\partial y'} \eta \right|_a^{c^-}.$$

On $[c, b]$ we similarly obtain

$$\delta J_2(y^*; \eta) = \int_c^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta dx + \left. \frac{\partial \mathcal{L}}{\partial y'} \eta \right|_{c^+}^b.$$

Adding the two expressions and using $\eta(a) = \eta(b) = 0$ yields

$$\delta J(y^*; \eta) = \int_a^c \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta dx + \int_c^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta dx + \left[\frac{\partial \mathcal{L}}{\partial y'}(c^-) - \frac{\partial \mathcal{L}}{\partial y'}(c^+) \right] \eta(c).$$

As in the smooth case, localized variations first pin down the interior equations on each side of the corner. First, consider variations η whose support is contained entirely in (a, c) . For such variations, the integral over (c, b) and the boundary term at c vanish. By the fundamental lemma of the calculus of variations, we conclude that

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} = 0 \quad \text{on } (a, c).$$

An identical argument using variations supported in (c, b) yields

$$\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} = 0 \quad \text{on } (c, b).$$

The interior equations having removed the integral terms, what survives is the jump across the corner. Having established the Euler–Lagrange equation on both subintervals, the integral terms vanish. The first variation reduces to

$$\delta J(y^*; \eta) = \left[\frac{\partial \mathcal{L}}{\partial y'}(c^-) - \frac{\partial \mathcal{L}}{\partial y'}(c^+) \right] \eta(c).$$

Since $\eta(c)$ is an arbitrary real number (subject only to continuity), the vanishing of the first variation for all admissible η implies

$$\frac{\partial \mathcal{L}}{\partial y'}(c^-) = \frac{\partial \mathcal{L}}{\partial y'}(c^+),$$

which is the continuity of the conjugate momentum.

A second, independent condition arises because the location of the corner is itself free. To capture it, regard the broken extremal as determined by the position of its corner, namely the abscissa c and the ordinate $u = y^*(c)$; both are free parameters, since neither is prescribed by the admissible class. Write

$$S^-(c, u) = \int_a^c \mathcal{L}(x, y, y') dx, \quad S^+(c, u) = \int_c^b \mathcal{L}(x, y, y') dx,$$

where on each subinterval y is the extremal joining the fixed outer endpoint to the corner (c, u) , so that $J = S^- + S^+$.

We compute the differential of S^- under an infinitesimal displacement $(\delta c, \delta u)$ of the corner. Let η denote the induced vertical variation of the left segment at fixed x , so that the displacement of the endpoint ordinate is $\delta u = \eta(c^-) + y^{*\prime}(c^-) \delta c$. Expanding to first order,

$$\delta S^- = \mathcal{L}(c^-) \delta c + \int_a^c \left(\frac{\partial \mathcal{L}}{\partial y} \eta + \frac{\partial \mathcal{L}}{\partial y'} \eta' \right) dx = \mathcal{L}(c^-) \delta c + \frac{\partial \mathcal{L}}{\partial y'} \eta \Big|_a^{c^-} + \int_a^c \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \eta dx,$$

where the first term accounts for the change of the integration interval. The interior integral vanishes because the Euler–Lagrange equation holds on (a, c) , and $\eta(a) = 0$ since the left endpoint is fixed. Substituting $\eta(c^-) = \delta u - y^{*\prime}(c^-) \delta c$,

$$\delta S^- = \mathcal{L}(c^-) \delta c + \frac{\partial \mathcal{L}}{\partial y'}(c^-) (\delta u - y^{*\prime}(c^-) \delta c) = \frac{\partial \mathcal{L}}{\partial y'}(c^-) \delta u + \left(\mathcal{L}(c^-) - y^{*\prime}(c^-) \frac{\partial \mathcal{L}}{\partial y'}(c^-) \right) \delta c.$$

An identical computation for the right segment, whose *left* endpoint is the corner while its right endpoint $x = b$ is fixed, produces the opposite sign,

$$\delta S^+ = -\frac{\partial \mathcal{L}}{\partial y'}(c^+) \delta u - \left(\mathcal{L}(c^+) - y^{*\prime}(c^+) \frac{\partial \mathcal{L}}{\partial y'}(c^+) \right) \delta c.$$

Stationarity of $J = S^- + S^+$ requires $\delta S^- + \delta S^+ = 0$ for all independent choices of δu and δc . The coefficient of δu recovers the continuity of the conjugate momentum obtained above, while the coefficient of δc yields

$$\mathcal{L}(c^-) - y^{*\prime}(c^-) \frac{\partial \mathcal{L}}{\partial y'}(c^-) = \mathcal{L}(c^+) - y^{*\prime}(c^+) \frac{\partial \mathcal{L}}{\partial y'}(c^+).$$

This is the continuity of the Hamiltonian across the corner.

The weak local extremum y^* therefore satisfies the Euler–Lagrange equation on each smooth segment, together with continuity of both the conjugate momentum and the Hamiltonian at the corner point $x = c$. These are precisely the Weierstrass–Erdmann corner conditions. $\square$

The Weierstrass–Erdmann conditions ensure that optimal trajectories cannot have arbitrary kinks. The continuity of $\partial \mathcal{L} / \partial y'$ expresses conservation of generalized momentum across the corner, while the Hamiltonian continuity reflects energy balance. These conditions play a central role in nonsmooth variational problems and form a precursor to switching conditions in optimal control.

Example 9.32 Detailed illustration of the Weierstrass–Erdmann corner conditions.

We consider a simple but revealing variational problem in which corner points are *a priori* allowed. As the setup, let

$$J(y) = \int_0^1 \mathcal{L}(y'(x)) \, dx, \quad \mathcal{L}(y') = \frac{1}{2}(y')^2,$$

and define the admissible class as the set of continuous, piecewise C^1 functions $y : [0, 1] \rightarrow \mathbb{R}$ satisfying the fixed boundary conditions $y(0) = 0$ and $y(1) = 1$. No restriction is imposed on the derivative at interior points, so corner points are permitted.

The functional J represents the total squared slope of the curve and penalizes rapid changes in y .

We first extract the interior equation. Since the Lagrangian depends only on y' , the Euler–Lagrange equation reduces to $\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} = 0$, or equivalently, $y''(x) = 0$, on every interval where y is differentiable. Thus, any extremal must be affine on each smooth subinterval.

To probe whether a genuine corner can survive, we build a candidate with one. Fix an arbitrary point $c \in (0, 1)$ and consider a piecewise-linear function of the form

$$y(x) = \begin{cases} m_1 x, & x \in [0, c], \\ m_2(x - c) + m_1 c, & x \in [c, 1], \end{cases}$$

where the constants m_1 and m_2 are chosen so that y is continuous at $x = c$ and satisfies $y(1) = 1$. Continuity at $x = c$ is automatic by construction, while the endpoint condition imposes the relation $m_1 c + m_2(1 - c) = 1$.

This curve satisfies the Euler–Lagrange equation on $(0, c)$ and $(c, 1)$ for *any* choice of m_1 and m_2 . At this stage, nothing forbids a genuine corner at $x = c$.

The first corner condition, continuity of the momentum, now constrains the slopes. The conjugate momentum associated with the Lagrangian is $p = \frac{\partial \mathcal{L}}{\partial y'} = y'$. The Weierstrass–Erdmann momentum condition requires continuity of p at the corner: $p(c^-) = p(c^+)$, i.e., $y'(c^-) = y'(c^+)$. In terms of the parameters, this implies $m_1 = m_2$. Thus, the slope must be the same on both sides of the corner.

The second corner condition, continuity of the Hamiltonian, points to the same conclusion. The Hamiltonian associated with the Lagrangian is $H = y' \frac{\partial \mathcal{L}}{\partial y'} - \mathcal{L} = \frac{1}{2}(y')^2$. The second corner condition requires continuity of the Hamiltonian across c : $H(c^-) = H(c^+)$, which again yields $\frac{1}{2}m_1^2 = \frac{1}{2}m_2^2$, and hence $m_1 = m_2$.

Although piecewise-linear curves with different slopes satisfy the Euler–Lagrange equation on each subinterval, they fail to be extremals of the global variational problem unless the corner conditions are satisfied. The Weierstrass–Erdmann conditions therefore rule out spurious extremals that are locally stationary but globally inadmissible.

Imposing $m_1 = m_2$ and using the endpoint condition yields $m_1 = m_2 = 1$, and the unique admissible extremal is $y(x) = x$. This example shows that corner points are

not forbidden in variational problems, but they must satisfy strict matching conditions. The Weierstrass–Erdmann conditions enforce conservation of generalized momentum and energy across corners, ensuring global optimality. $\square$

Example 9.33 Stationary curves with and without corners.

Consider the variational problem

$$J(x) = \int_0^4 (\dot{x}(t)^2 - 1)^2 dt,$$

subject to the fixed endpoint conditions $x(0) = 0$ and $x(4) = 2$, with admissible curves $x \in AC([0, 4])$.

The Lagrangian is $\mathcal{L}(\dot{x}) = (\dot{x}^2 - 1)^2$, which depends only on the velocity. The Euler–Lagrange equation therefore reduces to $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = 0$. Computing

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = 4\dot{x}(\dot{x}^2 - 1),$$

we obtain the stationarity condition $4\dot{x}(\dot{x}^2 - 1) = C$, where C is constant on any interval where x is continuously differentiable.

Assume first that a stationary solution x is globally C^1 on $[0, 4]$. Then $\dot{x}$ must satisfy $\dot{x}(\dot{x}^2 - 1) = \text{constant}$.

One possibility is a constant velocity $\dot{x} \equiv v$, which yields $v(v^2 - 1) = \text{constant}$. Imposing the endpoint condition $\int_0^4 \dot{x}(t) dt = 2$ gives $v = \frac{1}{2}$. The corresponding smooth candidate is therefore $x_{\text{lin}}(t) = \frac{1}{2}t$.

Its cost is

$$J(x_{\text{lin}}) = \int_0^4 \left(\frac{1}{4} - 1\right)^2 dt = 4 \cdot \frac{9}{16} = \frac{9}{4}.$$

This curve *does* satisfy the Euler–Lagrange equation: since $\partial \mathcal{L} / \partial \dot{x} = 4\dot{x}(\dot{x}^2 - 1)$ is constant in t whenever $\dot{x}$ is constant, the value

$$4\dot{x}(\dot{x}^2 - 1) = 4 \cdot \frac{1}{2} \cdot \left(\frac{1}{4} - 1\right) = -\frac{3}{2}$$

is a (nonzero) constant, so $\frac{d}{dt} \partial \mathcal{L} / \partial \dot{x} = 0$ holds and x_{lin} is a genuine stationary curve. It is not, however, a minimizer: the integrand is nonconvex in $\dot{x}$, and the Legendre condition fails at $\dot{x} = \frac{1}{2}$, where $\partial^2 \mathcal{L} / \partial \dot{x}^2 = 12\dot{x}^2 - 4 = -1 < 0$. The smooth extremal is thus a saddle of the action, so to locate the minimizer we enlarge the admissible class to piecewise- C^1 curves.

We now enlarge the admissible class from globally C^1 curves to piecewise C^1 curves, allowing finitely many points at which the derivative may be discontinuous. Such points are called *corners*. This enlargement is essential in the present problem, since we have already seen that the globally smooth stationary curve $x_{\text{lin}} = t/2$, though it

satisfies the Euler–Lagrange equation and the endpoint conditions, fails the Legendre condition and therefore cannot be a minimizer.

On any open subinterval on which an admissible curve is continuously differentiable, the classical Euler–Lagrange equation applies. For the present problem, the Lagrangian is $\mathcal{L}(\dot{x}) = (\dot{x}^2 - 1)^2$, which depends only on the velocity. The Euler–Lagrange equation therefore reduces to $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = 0$, or equivalently,

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = 4\dot{x}(\dot{x}^2 - 1) = \text{constant} \quad \text{on each smooth subinterval.}$$

A particularly important case occurs when this constant is zero. Then $\dot{x}(\dot{x}^2 - 1) = 0$, which admits the constant-velocity solutions $\dot{x} = \pm 1$. For these velocities,

$$\dot{x}^2 - 1 = 0, \quad \frac{\partial \mathcal{L}}{\partial \dot{x}} = 0, \quad \mathcal{L} = 0.$$

Thus, any curve whose derivative takes only the values ± 1 almost everywhere is stationary on each smooth arc and incurs zero running cost.

We now show that a corner is *forced* by the endpoint constraints. Since $x(4) - x(0) = \int_0^4 \dot{x}(t) dt = 2$, any admissible curve with $\dot{x}(t) \in \{-1, 1\}$ almost everywhere must spend unequal amounts of time moving with velocity $+1$ and velocity -1 . Let $c \in (0, 4)$ denote the total time during which $\dot{x} = 1$. Then the remaining time $4 - c$ corresponds to $\dot{x} = -1$, and the net displacement condition becomes $(+1)c + (-1)(4 - c) = 2$. Solving this equation yields $2c - 4 = 2$, so that $c = 3$.

Thus, any stationary curve composed of constant-velocity arcs with $\dot{x} = \pm 1$ must spend a total time 3 at velocity $+1$. If, in addition, the curve is required to have a *single* corner, so that the $+1$ arc precedes the -1 arc, this pins the corner at $t = 3$. Without that restriction the corner location is not unique: any measurable partition of $[0, 4]$ into a set of measure 3 where $\dot{x} = +1$ and its complement produces an admissible curve of the same (zero) cost, possibly with several corners.

A representative stationary curve with a single corner is therefore

$$x^*(t) = \begin{cases} t, & 0 \leq t \leq 3, \\ 6 - t, & 3 \leq t \leq 4, \end{cases}$$

for which

$$\dot{x}^*(t) = \begin{cases} 1, & t < 3, \\ -1, & t > 3. \end{cases}$$

The curve is continuous, piecewise C^1 , satisfies the endpoint conditions $x^*(0) = 0$ and $x^*(4) = 2$, and has a genuine corner at $t = 3$. Figure 9.8 shows this broken extremal alongside the smooth linear competitor.

Having identified the corner point at $t = 3$, we now verify that the stationary curve x^* satisfies the Weierstrass–Erdmann corner conditions. These conditions ensure that the first variation vanishes despite the presence of a discontinuity in the derivative.

The conjugate momentum associated with the Lagrangian is

$$p(t) = \frac{\partial \mathcal{L}}{\partial \dot{x}} = 4\dot{x}(t)(\dot{x}(t)^2 - 1).$$

For the curve x^* , we have

$$p(3^-) = 4 \cdot 1 \cdot (1 - 1) = 0, \quad p(3^+) = 4 \cdot (-1) \cdot (1 - 1) = 0.$$

Thus, the conjugate momentum is continuous across the corner: $p(3^-) = p(3^+)$.

The Hamiltonian of the problem is given by

$$H(t) = \dot{x}(t) \frac{\partial \mathcal{L}}{\partial \dot{x}}(t) - \mathcal{L}(\dot{x}(t)).$$

For $\dot{x} = \pm 1$, both $\partial \mathcal{L} / \partial \dot{x}$ and $\mathcal{L}$ vanish identically, and therefore $H(3^-) = 0$ and $H(3^+) = 0$. Hence, the Hamiltonian is also continuous across the corner.

Since both the conjugate momentum and the Hamiltonian agree on the left and right limits at $t = 3$, the Weierstrass–Erdmann corner conditions are satisfied. The curve x^* is therefore a valid stationary curve in the class of piecewise C^1 admissible functions.

We now compare the value of the functional for the corner solution x^* and for a representative smooth admissible curve.

For the corner solution, the integrand vanishes identically: $(\dot{x}^*(t)^2 - 1)^2 = 0$ for almost all $t \in [0, 4]$. Consequently,

$$J(x^*) = \int_0^4 0 \, dt = 0.$$

By contrast, consider the smooth linear curve $x_{\text{lin}}(t) = \frac{1}{2}t$, which satisfies the endpoint conditions and the Euler–Lagrange equation but, as shown above, fails the Legendre condition and so is not a minimizer. Its cost is

$$J(x_{\text{lin}}) = \int_0^4 \left(\frac{1}{4} - 1\right)^2 dt = \frac{9}{4}.$$

Thus, allowing a corner reduces the cost from $\frac{9}{4}$ to 0. This shows that no globally smooth curve can be optimal, while the piecewise Euler–Lagrange solution with a corner achieves strictly lower cost.

□

9.5 ■ Second-Order Conditions

First-order necessary conditions characterize *stationary curves*, that is, curves at which the first variation vanishes. However, stationarity alone does not distinguish between local minima, local

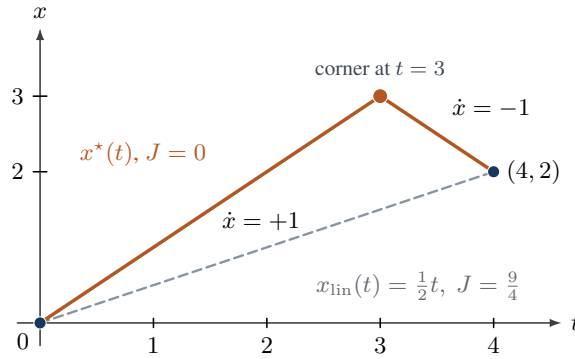


Figure 9.8: A forced corner. For $J(x) = \int_0^4 (\dot{x}^2 - 1)^2 dt$ with $x(0) = 0$, $x(4) = 2$, no globally smooth curve is a minimizer. The piecewise-linear extremal x^* (solid) travels with velocity $\dot{x} = +1$ until the corner at $t = 3$, then $\dot{x} = -1$, making the integrand vanish identically and achieving $J = 0$. The smooth competitor $x_{\text{lin}}(t) = t/2$ (dashed) satisfies the endpoints but costs $J = \frac{9}{4} > 0$. The Weierstrass–Erdmann conditions (continuity of the momentum p and Hamiltonian H , both zero here) certify the corner.

maxima, and saddle points. To obtain such distinctions, one must analyze the behavior of the functional under second-order perturbations. This leads to the notion of the *second variation*.

Throughout this section, we assume that:

- the Lagrangian $\mathcal{L} : [a, b] \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is of class C^2 in all arguments;
- the admissible set is $\mathcal{A} = \{y \in C^1([a, b]) \mid y(a) = A, y(b) = B\}$;
- $y^* \in \mathcal{A}$ is a stationary curve, i.e., it satisfies the Euler–Lagrange equation on (a, b) .

9.5.1 ■ Second Variation

Let $\mathcal{A}_0 = \{\eta \in C^1([a, b]) \mid \eta(a) = \eta(b) = 0\}$ denote the space of admissible variations. For $\eta \in \mathcal{A}_0$, consider the one-parameter family $y_\varepsilon = y^* + \varepsilon\eta$, $\varepsilon \in \mathbb{R}$. Since $\mathcal{L}$ is C^2 and $y^*, \eta \in C^1$, the mapping $\varepsilon \mapsto J(y_\varepsilon)$ is twice continuously differentiable in a neighborhood of $\varepsilon = 0$. Consequently, the functional admits the Taylor expansion

$$J(y_\varepsilon) = J(y^*) + \varepsilon \delta J(y^*; \eta) + \frac{\varepsilon^2}{2} \delta^2 J(y^*; \eta) + o(\varepsilon^2), \quad \varepsilon \rightarrow 0.$$

Since y^* is stationary, the first variation vanishes: $\delta J(y^*; \eta) = 0$ for all $\eta \in \mathcal{A}_0$. Thus, the leading-order behavior of $J(y_\varepsilon) - J(y^*)$ is governed by the second variation.

Definition 9.34 (Second variation). *The quantity*

$$\delta^2 J(y^*; \eta) := \left. \frac{d^2}{d\varepsilon^2} J(y^* + \varepsilon\eta) \right|_{\varepsilon=0}$$

is called the second variation of J at y^ in the direction η .*

A direct computation yields the explicit quadratic form

$$\delta^2 J(y^*; \eta) = \int_a^b (\mathcal{L}_{yy} \eta^2 + 2\mathcal{L}_{yy'} \eta \eta' + \mathcal{L}_{y'y'} (\eta')^2) dx,$$

where all partial derivatives of $\mathcal{L}$ are evaluated along the extremal $(x, y^*(x), y^{*'}(x))$.

The second variation is the infinite-dimensional counterpart of the Hessian quadratic form $\eta^\top \nabla^2 f(x^*) \eta$ from Section 2.3.3, and the analysis that follows parallels the finite-dimensional second-order theory of the chapter on Classical Optimality Conditions: nonnegativity of $\delta^2 J$ is the analogue of a positive semidefinite Hessian (a necessary condition), while its uniform positivity, coercivity, plays the role of a positive definite Hessian (a sufficient condition). The essential new feature, absent in $\mathbb{R}^n$, is that positive definiteness of the pointwise integrand no longer implies positivity of the quadratic form, which is what makes the Legendre and Jacobi conditions necessary.

9.5.2 ■ Second-Order Necessary Condition

The second variation must be nonnegative along any minimizing extremal.

Theorem 9.35 (Second-order necessary condition). *If y^* is a weak local minimizer of J , then $\delta^2 J(y^*; \eta) \geq 0$ for all $\eta \in \mathcal{A}_0$.*

Proof. Since y^* is a weak local minimum, there exists $\varepsilon_0 > 0$ such that $J(y^* + \varepsilon\eta) \geq J(y^*)$ for all $|\varepsilon| < \varepsilon_0$. Substituting the Taylor expansion and using $\delta J(y^*; \eta) = 0$, we obtain

$$\frac{\varepsilon^2}{2} \delta^2 J(y^*; \eta) + o(\varepsilon^2) \geq 0.$$

Dividing by $\varepsilon^2/2$ and letting $\varepsilon \rightarrow 0$ yields the claim. $\square$

9.5.3 ■ Legendre Necessary Condition

The second-order necessary condition implies a pointwise constraint on the Lagrangian.

Theorem 9.36 (Legendre necessary condition). *If y^* is a weak local minimizer of J , then $\mathcal{L}_{y'y'}(x, y^*(x), y^{*'}(x)) \geq 0$ for all $x \in [a, b]$.*

Proof. Abbreviate the coefficients of the second variation evaluated along the extremal by

$$R(x) := \mathcal{L}_{y'y'}(x, y^*, y^{*'}), \quad S(x) := \mathcal{L}_{yy'}(x, y^*, y^{*'}), \quad T(x) := \mathcal{L}_{yy}(x, y^*, y^{*'}),$$

which are continuous on $[a, b]$ because $\mathcal{L} \in C^2$ and $y^* \in C^1$, so that

$$\delta^2 J(y^*; \eta) = \int_a^b (R(\eta')^2 + 2S\eta\eta' + T\eta^2) dx.$$

Fix an interior point $x_0 \in (a, b)$ and a nonconstant profile $\varphi \in C_c^\infty((0, 1))$. For $h > 0$ small enough that $[x_0, x_0 + h] \subset (a, b)$, define the localized variation

$$\eta_h(x) = \varphi\left(\frac{x - x_0}{h}\right) \text{ on } [x_0, x_0 + h], \quad \eta_h(x) = 0 \text{ otherwise,}$$

which belongs to $\mathcal{A}_0$ (indeed to $C_c^\infty((a, b))$), as φ and all its derivatives vanish at the endpoints of $[0, 1]$. Substituting and changing variables through $s = (x - x_0)/h$, using $\eta_h'(x) = h^{-1}\varphi'(s)$, gives

$$\delta^2 J(y^*; \eta_h) = \frac{1}{h} \int_0^1 R(x_0 + hs) \varphi'(s)^2 ds + 2 \int_0^1 S(x_0 + hs) \varphi(s) \varphi'(s) ds + h \int_0^1 T(x_0 + hs) \varphi(s)^2 ds.$$

Multiplying by h and letting $h \rightarrow 0^+$, continuity of R, S, T collapses the right-hand side to its leading term,

$$\lim_{h \rightarrow 0^+} h \delta^2 J(y^*; \eta_h) = R(x_0) \int_0^1 \varphi'(s)^2 ds.$$

By the second-order necessary condition (Theorem 9.35), $\delta^2 J(y^*; \eta_h) \geq 0$ for every h , so the left-hand side is nonnegative. Since $\int_0^1 \varphi'(s)^2 ds > 0$ for the nonconstant φ , we conclude $R(x_0) \geq 0$. As $x_0 \in (a, b)$ was arbitrary and R is continuous, the inequality $\mathcal{L}_{y'y'}(x, y^*(x), y^{*\prime}(x)) \geq 0$ extends to all of $[a, b]$. $\square$

9.6 ■ Second-Order Sufficient Conditions

First-order necessary conditions characterize stationary curves but do not distinguish between minima, maxima, and saddle points. To obtain such distinctions, one must examine the behavior of the functional under second-order perturbations, leading to the notion of the *second variation*.

Throughout this section, we assume that the Lagrangian $\mathcal{L} : [a, b] \times \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is of class C^2 , and that $y^* \in \mathcal{A}$ is a stationary curve, i.e., a solution of the Euler–Lagrange equation with fixed endpoints.

9.6.1 ■ Abstract Second-Order Sufficient Condition

Nonnegativity of the second variation is not sufficient to guarantee minimality, as it does not preclude the possibility of flat directions or higher-order degeneracies. However, uniform positivity of the second variation with respect to an appropriate norm yields a sufficient condition for a strict weak local minimum. The term *coercivity* is used here in a sense related to, but distinct from, its use in the existence theory of Section 2.2: there, coercivity of the *functional* J forces minimizing sequences to stay bounded and thereby guarantees that a minimizer *exists*; here, coercivity of the *second variation* certifies that a given stationary curve, already known to exist, actually *is* a local minimizer. Both invoke the same underlying mechanism, a positive quadratic lower bound in the ambient norm.

Theorem 9.37 (Coercive second-order sufficient condition). *Let y^* be a stationary curve of J . Assume that there exists a constant $\alpha > 0$ such that*

$$\delta^2 J(y^*; \eta) \geq \alpha \|\eta\|_{H^1([a,b])}^2 \quad \text{for all } \eta \in \mathcal{A}_0.$$

Then y^ is a strict weak local minimum of J .*

Proof. It is not enough to expand J along each fixed ray $\varepsilon \mapsto y^* + \varepsilon\eta$: that would only show that J increases along every straight line through y^* , which in infinite dimensions does not imply a local minimum, because the rate at which the remainder becomes negligible could degrade as η varies. We therefore estimate the remainder uniformly over a whole C^1 -neighborhood.

Write an arbitrary competitor as $y = y^* + \eta$ with $\eta \in \mathcal{A}_0$, and fix $\delta_0 > 0$ so small that the compact tube

$$\mathcal{T} := \{(x, \xi, \zeta) : x \in [a, b], |\xi - y^*(x)| \leq \delta_0, |\zeta - y^{*\prime}(x)| \leq \delta_0\}$$

is contained in the domain of $\mathcal{L}$. Since $\mathcal{L} \in C^2$, its second-order partials $\mathcal{L}_{yy}, \mathcal{L}_{yy'}, \mathcal{L}_{y'y'}$ are uniformly continuous on $\mathcal{T}$; let ω denote a common modulus of continuity, so $\omega(t) \rightarrow 0$ as $t \rightarrow 0^+$.

Let $\|\eta\|_{C^1} \leq \delta \leq \delta_0$. Taylor's theorem with integral remainder, applied to $t \mapsto \mathcal{L}(x, y^* + t\eta, y^{*'} + t\eta')$ and integrated over $[a, b]$, gives

$$J(y^* + \eta) = J(y^*) + \delta J(y^*; \eta) + \frac{1}{2} \delta^2 J(y^*; \eta) + R(\eta), \quad (9.2)$$

where

$$R(\eta) = \int_a^b \int_0^1 (1-t) \left[\Delta_{yy}^t \eta^2 + 2\Delta_{yy'}^t \eta \eta' + \Delta_{y'y'}^t (\eta')^2 \right] dt dx,$$

and $\Delta_{yy}^t := \mathcal{L}_{yy}(x, y^* + t\eta, y^{*'} + t\eta') - \mathcal{L}_{yy}(x, y^*, y^{*'})$, with $\Delta_{yy'}^t$ and $\Delta_{y'y'}^t$ defined analogously. Every point $(x, y^* + t\eta, y^{*'} + t\eta')$ lies in $\mathcal{T}$ and is within distance δ of $(x, y^*, y^{*'})$, so each Δ^t is bounded by $\omega(\delta)$ in absolute value. Hence

$$|R(\eta)| \leq \frac{1}{2} \omega(\delta) \int_a^b (|\eta| + |\eta'|)^2 dx \leq \omega(\delta) \|\eta\|_{H^1([a,b])}^2. \quad (9.3)$$

The crucial point is that the bound (9.3) is quadratic in the *same* H^1 -norm that appears in the coercivity hypothesis, while its small factor $\omega(\delta)$ is controlled by the C^1 -size of η . This is exactly the pairing of norms that a weak local minimum calls for.

Since y^* is stationary, $\delta J(y^*; \eta) = 0$ for every $\eta \in \mathcal{A}_0$. Substituting the coercivity hypothesis and (9.3) into (9.2),

$$J(y^* + \eta) - J(y^*) \geq \frac{\alpha}{2} \|\eta\|_{H^1}^2 - \omega(\delta) \|\eta\|_{H^1}^2 = \left(\frac{\alpha}{2} - \omega(\delta) \right) \|\eta\|_{H^1}^2.$$

Choose $\delta \in (0, \delta_0]$ so small that $\omega(\delta) < \alpha/4$. Then

$$J(y^* + \eta) - J(y^*) \geq \frac{\alpha}{4} \|\eta\|_{H^1}^2 > 0 \quad \text{whenever } 0 < \|\eta\|_{C^1} \leq \delta,$$

the strict inequality because $\eta \neq 0$ in C^1 forces $\|\eta\|_{H^1} > 0$. Thus $J(y) > J(y^*)$ for every admissible $y \neq y^*$ in the C^1 -ball of radius δ about y^* , which is precisely the assertion that y^* is a strict weak local minimum of J . $\square$

Example 9.38 Directly verifying coercivity.

Consider

$$J(y) = \int_0^1 \frac{1}{2} ((y'(x))^2 + (y(x))^2) dx, \quad y(0) = y(1) = 0.$$

The Euler–Lagrange equation is $y'' - y = 0$, whose only solution vanishing at both endpoints is $y^* \equiv 0$; this is the unique stationary curve. Because J is already an exact quadratic functional, its second variation at y^* coincides with J acting on the variation,

$$\delta^2 J(y^*; \eta) = \int_0^1 ((\eta')^2 + \eta^2) dx = \|\eta\|_{H^1([0,1])}^2 \quad \text{for all } \eta \in \mathcal{A}_0.$$

Thus the coercivity hypothesis of Theorem 9.37 holds with $\alpha = 1$, and $y^* \equiv 0$ is a strict weak local minimum. Here coercivity is transparent because the quadratic form is *exactly* the squared H^1 -norm; the value of the theorem is that it delivers the same

conclusion whenever the second variation merely dominates a multiple of that norm, without requiring the functional to be quadratic. $\square$

9.6.2 ■ Legendre–Jacobi Sufficient Condition

The coercivity hypothesis in Theorem 9.37 is strong and not immediately verifiable from the Lagrangian. In classical calculus of variations, it is ensured by two structural conditions: pointwise convexity of the Lagrangian with respect to the velocity variable and the absence of conjugate points.

Theorem 9.39 (Legendre–Jacobi sufficient condition). *Let y^* be a stationary curve of $J(y) = \int_a^b \mathcal{L}(x, y, y') dx$ with fixed endpoints. Assume that:*

- (i) (Strict Legendre condition) $\mathcal{L}_{y'y'}(x, y^*(x), y^{*'}(x)) > 0$ for all $x \in [a, b]$;
- (ii) (Absence of conjugate points) *There exists no point $c \in (a, b]$ conjugate to a along y^* .*

Then there exists $\alpha > 0$ such that

$$\delta^2 J(y^*; \eta) \geq \alpha \|\eta\|_{H^1([a, b])}^2 \quad \text{for all } \eta \in \mathcal{A}_0,$$

and hence y^ is a strict weak local minimum of J .*

Proof. The proof links the second variation to a coercive quadratic form, in four steps: writing the second variation as a quadratic form, using the Legendre condition to control its leading term, ruling out degeneracy through the absence of conjugate points, and concluding coercivity. Since $\mathcal{L} \in C^2$ and y^* is a stationary curve with fixed endpoints, the second variation is given by

$$\delta^2 J(y^*; \eta) = \int_a^b (\mathcal{L}_{y'y'}(x, y^*, y^{*'}) (\eta')^2 + 2\mathcal{L}_{yy'}(x, y^*, y^{*'}) \eta \eta' + \mathcal{L}_{yy}(x, y^*, y^{*'}) \eta^2) dx.$$

Integrating the mixed term by parts and using $\eta(a) = \eta(b) = 0$, this expression can be rewritten in the standard quadratic form

$$\delta^2 J(y^*; \eta) = \int_a^b (P(x)(\eta')^2 + Q(x)\eta^2) dx,$$

where

$$P(x) := \mathcal{L}_{y'y'}(x, y^*, y^{*'}), \quad Q(x) := \mathcal{L}_{yy}(x, y^*, y^{*'}) - \frac{d}{dx} \mathcal{L}_{yy'}(x, y^*, y^{*'}).$$

The functions P and Q are continuous on $[a, b]$.

The strict Legendre condition now gives a uniform lower bound on the leading coefficient. By assumption, $P(x) > 0$ for all $x \in [a, b]$. Since P is continuous on a compact interval, there exists a constant $p_0 > 0$ such that $P(x) \geq p_0$ for all $x \in [a, b]$. Thus the second variation controls the derivative term:

$$\delta^2 J(y^*; \eta) \geq p_0 \int_a^b (\eta')^2 dx + \int_a^b Q(x)\eta^2 dx.$$

Positivity of the whole form, however, can still fail through the lower-order term; this is controlled by conjugate points via the Jacobi equation. Consider the associated Jacobi (or accessory) equation

$$-\frac{d}{dx} (P(x)v'(x)) + Q(x)v(x) = 0,$$

a second-order linear equation whose leading coefficient P is strictly positive. A point $c \in (a, b]$ is said to be *conjugate to a* if the nontrivial solution v_a determined by $v_a(a) = 0$, $v'_a(a) = 1$ satisfies $v_a(c) = 0$. (Every nontrivial solution vanishing at a is a scalar multiple of v_a , so this is independent of the normalization.)

Step 1: a nonvanishing Jacobi solution. By hypothesis a has no conjugate point in $(a, b]$, hence $v_a(x) \neq 0$ for every $x \in (a, b]$; fixing the sign so that $v'_a(a) = 1$, we have $v_a > 0$ on $(a, b]$. Consider now the solution v_ρ with data $v_\rho(a) = \rho > 0$, $v'_\rho(a) = 1$. Solutions of a linear second-order equation depend continuously on their initial data, uniformly on the compact interval $[a, b]$, so $v_\rho \rightarrow v_a$ uniformly as $\rho \rightarrow 0^+$. Since $v_a > 0$ on each compact set $[a + \delta, b]$ and $v'_\rho(a) = 1 > 0$ keeps v_ρ increasing on a fixed neighborhood of a where it starts at $v_\rho(a) = \rho > 0$, one may fix $\rho > 0$ small enough that

$$w := v_\rho \quad \text{satisfies} \quad w(x) > 0 \quad \text{for all } x \in [a, b].$$

Step 2: completing the square. Because $\eta(a) = \eta(b) = 0$, the integral of the total derivative of $x \mapsto W(x)\eta(x)^2$ vanishes for any $W \in C^1([a, b])$:

$$\int_a^b \frac{d}{dx} (W\eta^2) dx = \int_a^b (2W\eta\eta' + W'\eta^2) dx = 0.$$

Choose $W := -P w'/w$, which is well defined and C^1 on $[a, b]$ by Step 1. Using $(Pw')' = Qw$ from the Jacobi equation, a direct computation gives

$$W' = -\frac{(Pw')'}{w} + P\left(\frac{w'}{w}\right)^2 = -Q + \frac{W^2}{P}, \quad \text{so that} \quad Q + W' = \frac{W^2}{P}.$$

Adding the (vanishing) integral above to $\delta^2 J = \int_a^b (P(\eta')^2 + Q\eta^2) dx$ therefore yields a perfect square,

$$\delta^2 J(y^*; \eta) = \int_a^b \left(P(\eta')^2 + 2W\eta\eta' + \frac{W^2}{P} \eta^2 \right) dx = \int_a^b P(x) \left(\eta' + \frac{W}{P} \eta \right)^2 dx.$$

In particular $\delta^2 J(y^*; \eta) \geq 0$, and equality forces $\eta' + (W/P)\eta \equiv 0$ together with $\eta(a) = 0$, hence $\eta \equiv 0$. Thus the second variation is positive definite on $\mathcal{A}_0 \setminus \{0\}$.

Step 3: coercivity. Set $\xi := \eta' + (W/P)\eta$ and $\beta := W/P$, a continuous, hence bounded, function with $\|\beta\|_\infty \leq M$. From the square just obtained and $P \geq p_0 > 0$,

$$\delta^2 J(y^*; \eta) \geq p_0 \int_a^b \xi^2 dx = p_0 \|\xi\|_{L^2}^2.$$

Since $\eta(a) = 0$, solving the linear relation $\eta' = \xi - \beta\eta$ gives $\eta(x) = \int_a^x e^{-\int_s^x \beta} \xi(s) ds$, so by the Cauchy–Schwarz inequality

$$\|\eta\|_{L^2}^2 \leq C_0 \|\xi\|_{L^2}^2, \quad C_0 := (b-a)^2 e^{2M(b-a)},$$

and consequently

$$\|\eta'\|_{L^2}^2 = \|\xi - \beta\eta\|_{L^2}^2 \leq 2\|\xi\|_{L^2}^2 + 2M^2\|\eta\|_{L^2}^2 \leq (2 + 2M^2C_0) \|\xi\|_{L^2}^2.$$

Hence $\|\eta\|_{H^1}^2 \leq C_1 \|\xi\|_{L^2}^2$ with $C_1 := C_0 + 2 + 2M^2C_0$, and combining with the lower bound on $\delta^2 J$,

$$\delta^2 J(y^*; \eta) \geq \frac{p_0}{C_1} \|\eta\|_{H^1([a,b])}^2 =: \alpha \|\eta\|_{H^1([a,b])}^2, \quad \alpha > 0,$$

for all $\eta \in \mathcal{A}_0$. This is precisely the coercivity condition required in Theorem 9.37. Applying Theorem 9.37, we conclude that y^* is a strict weak local minimum of J . $\square$

The strict Legendre condition ensures local convexity of the Lagrangian, while the absence of conjugate points eliminates global degeneracies of the second variation. Together, these conditions yield coercivity, which is the essential ingredient for second-order sufficiency.

This analysis highlights a fundamental distinction between finite-dimensional optimization and the calculus of variations: in infinite-dimensional settings, pointwise convexity alone is insufficient, and global geometric properties of extremals play a decisive role in determining optimality. The strict Legendre condition $\mathcal{L}_{y'y'} > 0$ is precisely convexity of the Lagrangian in the velocity variable, in the sense of Section 4.26; were the objective globally convex on the admissible set, this local condition would already guarantee that every stationary curve is a global minimizer, exactly as convexity turns first-order stationarity into global optimality in finite dimensions. The role of the Jacobi (no-conjugate-point) condition is to supply, in the *absence* of global convexity, the missing control that convexity would otherwise provide.

Example 9.40 Conjugate points and the loss of minimality.

Consider the one-parameter family of problems

$$J(y) = \int_0^L \frac{1}{2} ((y'(x))^2 - (y(x))^2) dx, \quad y(0) = y(L) = 0,$$

where the interval length $L > 0$ is the parameter of interest. The Euler–Lagrange equation is $y'' + y = 0$; for $L \notin \{\pi, 2\pi, \dots\}$ its only solution meeting both endpoints is $y^* \equiv 0$, the unique stationary curve.

Legendre and Jacobi data. Along $y^* \equiv 0$,

$$P(x) = \mathcal{L}_{y'y'} = 1 > 0, \quad Q(x) = \mathcal{L}_{yy} - \frac{d}{dx} \mathcal{L}_{yy'} = -1,$$

so the strict Legendre condition holds everywhere, and the second variation is

$$\delta^2 J(y^*; \eta) = \int_0^L ((\eta')^2 - \eta^2) dx.$$

The Jacobi (accessory) equation $-(Pv')' + Qv = -v'' - v = 0$ with $v(0) = 0$, $v'(0) = 1$ has the solution $v(x) = \sin x$, whose first zero beyond the origin is at $x = \pi$. Hence the point conjugate to 0 along y^* is $c = \pi$.

Short intervals ($L < \pi$): a genuine minimum. There is no conjugate point in $(0, L]$, and the strict Legendre condition holds, so Theorem 9.39 guarantees that $y^* \equiv 0$ is a strict weak local minimum. This is confirmed directly by Wirtinger's inequality: for $\eta \in \mathcal{A}_0$ one has $\int_0^L \eta^2 \leq (L/\pi)^2 \int_0^L (\eta')^2$, whence

$$\delta^2 J(y^*; \eta) \geq \left(1 - \frac{L^2}{\pi^2}\right) \int_0^L (\eta')^2 dx > 0 \quad (\eta \neq 0).$$

Long intervals ($L > \pi$): minimality is lost. The interval now contains the conjugate point π , so the hypothesis of Theorem 9.39 fails. In fact $y^* \equiv 0$ is no longer a

minimizer: choosing the admissible variation $\eta(x) = \sin(\pi x/L)$ gives

$$\delta^2 J(y^*; \eta) = \frac{1}{2} \left(\frac{\pi^2}{L} - L \right) = \frac{\pi^2 - L^2}{2L} < 0,$$

so the second variation is indefinite and the extremal is a saddle.

Borderline ($L = \pi$). The conjugate point sits exactly at the right endpoint. The second variation is nonnegative but degenerate: $\eta(x) = \sin x$ yields $\delta^2 J(y^*; \eta) = 0$, so the form is not coercive and the sufficient condition just fails to apply. This example shows that the strict Legendre condition, pointwise convexity in y' , is by itself powerless to certify optimality; it is the *global* absence of conjugate points that decides the matter. $\square$

9.7 ■ Extensions: First-Order Necessary Conditions

In this section, we extend the Euler–Lagrange framework in two directions: (i) variational problems involving higher-order derivatives, and (ii) variational problems with vector-valued trajectories. Throughout, we focus exclusively on first-order necessary conditions for weak local extrema, with explicit attention to how different endpoint conditions affect the resulting optimality equations.

9.7.1 ■ Functionals Involving Higher-Order Derivatives

We first consider variational problems in which the Lagrangian depends on higher derivatives of the unknown function. Let $m \geq 2$ be an integer and consider the functional

$$J(y) = \int_a^b \mathcal{L}(x, y(x), y'(x), \dots, y^{(m)}(x)) dx,$$

where $\mathcal{L} \in C^m([a, b] \times \mathbb{R}^{m+1})$; this much smoothness is needed because the necessary condition below differentiates $\partial \mathcal{L} / \partial y^{(k)}$ up to k times.

To compute the first variation, one must specify which endpoint values are fixed. A standard choice is to prescribe derivatives up to order $m - 1$ at both endpoints:

$$y^{(k)}(a) = A_k, \quad y^{(k)}(b) = B_k, \quad k = 0, \dots, m - 1.$$

This corresponds to *fully fixed endpoints* and ensures that all boundary terms arising from repeated integration by parts vanish.

The admissible set is

$$\mathcal{A} = \left\{ y \in C^m([a, b]) \mid y^{(k)}(a) = A_k, y^{(k)}(b) = B_k, k = 0, \dots, m - 1 \right\},$$

with admissible variations

$$\mathcal{A}_0 = \left\{ \eta \in C^m([a, b]) \mid \eta^{(k)}(a) = \eta^{(k)}(b) = 0, k = 0, \dots, m - 1 \right\}.$$

Theorem 9.41 (Higher-order Euler–Lagrange equation). *If $y^* \in \mathcal{A}$ is a weak local extremum of J , then y^* satisfies*

$$\sum_{k=0}^m (-1)^k \frac{d^k}{dx^k} \frac{\partial \mathcal{L}}{\partial y^{(k)}}(x, y^*, y^{*'}, \dots, y^{*(m)}) = 0 \quad \text{for all } x \in (a, b).$$

Proof. Since y^* is a weak local extremum, the first variation must vanish: $\delta J(y^*; \eta) = 0$ for all $\eta \in \mathcal{A}_0$. Differentiation under the integral sign yields

$$\delta J(y^*; \eta) = \int_a^b \sum_{k=0}^m \frac{\partial \mathcal{L}}{\partial y^{(k)}} \eta^{(k)} dx.$$

Repeated integration by parts transfers derivatives from $\eta^{(k)}$ to the corresponding coefficients. All boundary terms vanish due to the endpoint conditions imposed on η . We obtain

$$\delta J(y^*; \eta) = \int_a^b \left[\sum_{k=0}^m (-1)^k \frac{d^k}{dx^k} \frac{\partial \mathcal{L}}{\partial y^{(k)}} \right] \eta(x) dx.$$

Since this identity holds for all $\eta \in C_c^\infty((a, b))$, the fundamental lemma of the calculus of variations implies the stated differential equation. $\square$

Remark 9.42 (Partially fixed or free endpoints). If some endpoint values $y^{(k)}(a)$ or $y^{(k)}(b)$ are not prescribed, then the corresponding boundary terms do not vanish. Stationarity of the first variation then yields additional first-order necessary conditions, known as *natural boundary conditions*, which arise from the coefficients of $\eta^{(k)}(a)$ or $\eta^{(k)}(b)$. These conditions complement the interior Euler–Lagrange equation but do not alter its form.

9.7.2 ■ Vector-Valued Variational Problems

We now extend the Euler–Lagrange equation to vector-valued trajectories. Let $y : [a, b] \rightarrow \mathbb{R}^n$ and consider the functional

$$J(y) = \int_a^b \mathcal{L}(x, y(x), y'(x)) dx,$$

where

$$\mathcal{L} : [a, b] \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathcal{L} \in C^1.$$

As before, the treatment of the boundary depends on which endpoints are prescribed. If both endpoints are fixed, $y(a) = A$ and $y(b) = B$, then admissible variations satisfy $\eta(a) = \eta(b) = 0$, and all boundary terms vanish after integration by parts.

If one or both endpoints are free, boundary terms involving $\partial \mathcal{L} / \partial y'$ remain and yield additional first-order transversality conditions. These conditions are again necessary for optimality but are separate from the interior Euler–Lagrange equation.

The admissible set is

$$\mathcal{A} = \{y \in C^1([a, b]; \mathbb{R}^n) \mid y(a) = A, y(b) = B\},$$

with admissible variations

$$\mathcal{A}_0 = \{\eta \in C^1([a, b]; \mathbb{R}^n) \mid \eta(a) = \eta(b) = 0\}.$$

Theorem 9.43 (Vector Euler–Lagrange equation). If $y^* \in \mathcal{A}$ is a weak local extremum of J , then it satisfies

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'}(x, y^*, y^{*'}) - \frac{\partial \mathcal{L}}{\partial y}(x, y^*, y^{*'}) = 0 \quad \text{on } (a, b),$$

where the derivatives are taken componentwise. Equivalently, for each $i = 1, \dots, n$,

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'_i} - \frac{\partial \mathcal{L}}{\partial y_i} = 0.$$

Proof. Let $\eta \in \mathcal{A}_0$ be arbitrary. The first variation is

$$\delta J(y^*; \eta) = \int_a^b \left(\frac{\partial \mathcal{L}}{\partial y} \cdot \eta + \frac{\partial \mathcal{L}}{\partial y'} \cdot \eta' \right) dx.$$

Integrating the second term by parts and using $\eta(a) = \eta(b) = 0$ yields

$$\delta J(y^*; \eta) = \int_a^b \left[\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'} \right] \cdot \eta(x) dx.$$

Since this holds for all $\eta \in C_c^\infty((a, b); \mathbb{R}^n)$, the fundamental lemma implies the vector Euler–Lagrange equation. $\square$

Example 9.44 Anisotropic shortest paths (Finsler geometry).

Let $y : [a, b] \rightarrow \mathbb{R}^2$ and consider the functional

$$J(y) = \int_a^b \sqrt{\alpha(x) (y'_1(x))^2 + \beta(x) (y'_2(x))^2} dx,$$

where $\alpha, \beta \in C^1([a, b])$ are positive and nonconstant.

We consider two cases.

(i) *Fixed endpoints.* If $y(a) = A, y(b) = B$, then admissible variations satisfy $\eta(a) = \eta(b) = 0$. All boundary terms vanish, and the first-order necessary conditions reduce to the interior Euler–Lagrange equations.

The Lagrangian $\mathcal{L}(x, y') = \sqrt{\alpha(x)(y'_1)^2 + \beta(x)(y'_2)^2}$ is independent of y , so the cyclic-variable condition applies componentwise:

$$\frac{d}{dx} \frac{\partial \mathcal{L}}{\partial y'_i} = 0, \quad i = 1, 2.$$

Computing the derivatives gives

$$\frac{\alpha(x)y'_1}{\mathcal{L}} = C_1, \quad \frac{\beta(x)y'_2}{\mathcal{L}} = C_2,$$

which implies

$$\frac{y'_2}{y'_1} = \frac{C_2}{C_1} \frac{\alpha(x)}{\beta(x)}.$$

Thus, even at first order, the Euler–Lagrange equations encode direction-dependent bending of optimal paths, generalizing Snell’s law.

(ii) *Terminal endpoint on a curve.* If $y(b)$ is constrained to a smooth curve Γ rather than fixed, the boundary term from the first variation yields the transversality condition

$$\frac{\partial \mathcal{L}}{\partial y'}(x, y'(x)) \Big|_{x=b} \cdot \tau = 0 \quad \text{for every } \tau \text{ tangent to } \Gamma \text{ at } y(b).$$

Because $\mathcal{L}$ is positively homogeneous of degree one in y' , the naive free-endpoint condition $\partial \mathcal{L} / \partial y' = 0$ has no nonzero solution (the covector $\partial \mathcal{L} / \partial y'$ is homogeneous of degree zero); it is this transversality form that is meaningful. It states that the terminal tangent is orthogonal, in the Finsler metric induced by $\mathcal{L}$, to Γ , giving a geometric characterization of the optimal exit direction. $\square$

Example 9.45 Coupled agents with interaction energy.

Let $y(x) = (y_1(x), \dots, y_n(x)) \in \mathbb{R}^n$ represent the positions of n interacting agents along a line. Consider the functional

$$J(y) = \int_0^T \left(\frac{1}{2} \sum_{i=1}^n (y'_i(x))^2 + \sum_{i < j} W(y_i(x) - y_j(x)) \right) dx,$$

where $W \in C^1(\mathbb{R})$ is a symmetric interaction potential.

Turning to the boundary, we again distinguish two representative cases.

(i) *Fixed initial and terminal configurations.* If $y(0) = y^0$ and $y(T) = y^T$, then admissible variations satisfy $\eta(0) = \eta(T) = 0$, and all boundary terms vanish. The vector Euler–Lagrange equation yields, for each i ,

$$y''_i(x) = \sum_{j \neq i} W'(y_i(x) - y_j(x)),$$

where the sign is $+$ because the cost integrates $T + V$ (not the mechanical action $T - V$), so the extremal equation is $y''_i = \partial V / \partial y_i$.

(ii) *Free terminal configuration.* If $y(T)$ is free while $y(0)$ is fixed, the first variation produces the natural boundary condition $y'_i(T) = 0$ for $i = 1, \dots, n$. Thus, each agent reaches the terminal time with zero velocity, reflecting the absence of terminal constraints or costs.

Let

$$W(s) = \frac{k}{2} s^2, \quad k > 0.$$

Then

$$y''_i(x) = k \sum_{j \neq i} (y_i(x) - y_j(x)),$$

or in vector form, $y''(x) = k L_{\text{graph}} y(x)$, where L_{graph} is the Laplacian of the complete interaction graph.

These endpoint conditions clarify the structure of the extremals. The interior Euler–Lagrange equations encode pairwise force balance through the interaction potential and determine the coupled dynamics of the agents. Fixed endpoint conditions enforce prescribed initial or terminal configurations and eliminate all boundary contributions to the first variation. By contrast, when terminal endpoints are free, stationarity yields natural boundary conditions that enforce zero terminal momentum for each agent, reflecting the absence of external constraints or terminal costs.

Even without invoking second-order optimality or stability analysis, the first-order necessary conditions already encode conservation laws, interaction structure, and boundary behavior in multi-agent systems.

□

9.7.3 ■ CoV and Optimal Control: Vector-Valued Formulation

We now present a vector-valued formulation showing how a restricted class of optimal control problems can be solved using the calculus of variations by eliminating the control variable and working directly with state trajectories. The elimination is exact only under structural assumptions that allow the control to be uniquely recovered from the state and its derivative.

Let $x : [0, T] \rightarrow \mathbb{R}^n$ denote the state and $u : [0, T] \rightarrow \mathbb{R}^n$ the control. Consider the control-affine system $\dot{x}(t) = f(x(t)) + G(x(t)) u(t)$, with fixed endpoint conditions $x(0) = x_0$ and $x(T) = x_T$, and cost functional

$$J(u) = \int_0^T \left(\frac{1}{2} \|u(t)\|^2 + U(x(t)) \right) dt,$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, $G : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$, and $U : \mathbb{R}^n \rightarrow \mathbb{R}$ are continuously differentiable.

We assume that $G(x)$ is invertible for all x along admissible trajectories. Under this assumption, the control is uniquely determined by the state and its derivative via $u(t) = G(x(t))^{-1} (\dot{x}(t) - f(x(t)))$.

Substituting this expression into the cost functional yields an equivalent variational problem over state trajectories:

$$J(x) = \int_0^T \left[\frac{1}{2} \|G(x(t))^{-1} (\dot{x}(t) - f(x(t)))\|^2 + U(x(t)) \right] dt.$$

The admissible set is

$$\mathcal{A} = \{x \in AC([0, T]; \mathbb{R}^n) \mid x(0) = x_0, x(T) = x_T\}.$$

The first-order necessary condition follows from the standard theory. The induced Lagrangian

$$\mathcal{L}(x, \dot{x}) = \frac{1}{2} \|G(x)^{-1} (\dot{x} - f(x))\|^2 + U(x)$$

defines a standard variational problem. If x^* is a weak local extremum, then it satisfies the Euler–Lagrange equation

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} - \frac{\partial \mathcal{L}}{\partial x} = 0 \quad \text{on } (0, T).$$

The quantity

$$p(t) := - \frac{\partial \mathcal{L}}{\partial \dot{x}}(x^*, \dot{x}^*)$$

acts as a costate variable and coincides with the adjoint variable of the Pontryagin Maximum Principle. The minus sign is dictated by the convention $H(x, p, u) = \ell(x, u) + p^\top f(x, u)$ adopted in Section 11.13: minimizing H over u for the control-affine system above gives $u = -G(x)^\top p$, whereas eliminating the control yields $\partial \mathcal{L} / \partial \dot{x} = G(x)^{-\top} u$, so the two differ by a sign. With this identification the Euler–Lagrange equation $\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{\partial \mathcal{L}}{\partial x}$ becomes exactly the adjoint equation $\dot{p} = -\nabla_x H$. Readers who prefer the opposite sign convention for H should drop the minus sign here; what matters is that the pairing be consistent throughout.

Example 9.46 Two-dimensional control system via CoV.

Consider the two-dimensional system

$$\dot{x}(t) = u(t), \quad x(t), u(t) \in \mathbb{R}^2,$$

with fixed endpoints $x(0) = x_0$ and $x(T) = x_T$, and quadratic cost

$$J(u) = \int_0^T \left(\frac{1}{2} \|u(t)\|^2 + U(x(t)) \right) dt,$$

where $U \in C^1(\mathbb{R}^2)$.

Since $u(t) = \dot{x}(t)$, the control can be eliminated exactly, yielding the variational problem

$$J(x) = \int_0^T \left(\frac{1}{2} \|\dot{x}(t)\|^2 + U(x(t)) \right) dt,$$

with admissible set

$$\mathcal{A} = \{x \in AC([0, T]; \mathbb{R}^2) \mid x(0) = x_0, x(T) = x_T\}.$$

The associated Lagrangian is $\mathcal{L}(x, \dot{x}) = \frac{1}{2} \|\dot{x}\|^2 + U(x)$. Applying the vector Euler–Lagrange equation yields $\ddot{x}(t) = \nabla U(x(t))$, which is Newton’s second law for a particle moving in the *inverted* potential $-U$. The sign differs from mechanics because the cost here integrates $\frac{1}{2} \|\dot{x}\|^2 + U$ rather than the mechanical action $\frac{1}{2} \|\dot{x}\|^2 - U$; minimizing the running cost drives the state up the gradient of U .

The optimal control is recovered as $u^*(t) = \dot{x}^*(t)$.

□

9.8 ■ Further Reading

The calculus of variations has a vast and well-established literature, with classical treatments emphasizing the derivation of the Euler–Lagrange equation, boundary conditions, and regularity of extremals. Standard introductory texts include the monographs of Gelfand and Fomin [49], Weinstock [123], and Sagan [107], which develop the foundational theory together with canonical examples arising from geometry, mechanics, and physics.

From a systems and optimization perspective, the calculus of variations provides the foundational framework for optimal control theory. The connection between variational first-order necessary conditions and state–costate dynamics is developed systematically in the classical text

of Athans and Falb [6], which emphasizes continuous-time systems and Hamiltonian methods. A modern and rigorous treatment of these ideas, with particular attention to nonsmooth systems and control-theoretic interpretations, is given by Liberzon [77]. Complementary insight from an optimization viewpoint can be found in Luenberger's text [79], which places the calculus of variations within a broader framework of constrained optimization and duality.

These perspectives culminate in the Pontryagin Maximum Principle and related Hamiltonian formulations of optimal control, developed in their definitive form in Pontryagin's seminal work [95] and presented from an engineering and algorithmic standpoint in the applied control text of Bryson [30].

Variational principles also play a central role in the theory of partial differential equations. Many elliptic, parabolic, and hyperbolic PDEs arise as Euler–Lagrange equations associated with energy functionals, and weak solutions of PDEs are naturally formulated in variational form. Standard references in this direction include Evans [44] and Dacorogna [37], which develop direct methods, lower semicontinuity, and existence theory for variational problems. A comprehensive treatment of regularity theory and variational integrals is provided by Giaquinta and Hildebrandt in their two-volume work [51, 50].

Modern developments extend classical variational methods to convex, nonsmooth, and infinite-dimensional settings, with applications in materials science, mechanics, and data-driven regularization. The optimization-oriented text of Ekeland and Temam [42] presents variational problems from the viewpoint of convex analysis, while Braides [27] provides an accessible introduction to Γ -convergence and asymptotic analysis of variational problems.

9.9 ■ Exercises

Exercise 9.1 (Stationary Curves). Find the stationary curves for the functional $J(y) = \int_{x_0}^{x_1} F(x, y, y') dx$, $y(x_0) = y_0$, $y(x_1) = y_1$, when $F(x, y, y')$ is given by:

$$(i) \quad F(x, y, y') = \frac{\sqrt{1 + (y')^2}}{y},$$

$$(ii) \quad F(x, y, y') = \sqrt{y(1 + (y')^2)},$$

$$(iii) \quad F(x, y, y') = y'(1 + x^2 y').$$

Exercise 9.2 (Vector-Valued Curves). Let $x(t) = [x_1(t), x_2(t)]^\top$. Find the stationary curve for the functional $J(x(t)) = \int_0^1 G(x(t), \dot{x}(t), t) dt$, with boundary conditions $x(0) = [1, 1]^\top$ and $x(1) = [0, 1]^\top$, where $G(x(t), \dot{x}(t), t) = \dot{x}_1 + x_2 + \dot{x}_1 x_2$.

Exercise 9.3 (Second-Order Functional). Let $x(t)$ be a scalar function. Find the stationary curve for the functional

$$J(x) = \int_0^{\pi/2} (\dot{x}^2 - x^2) dt, \quad x(0) = 0, \quad x\left(\frac{\pi}{2}\right) = 1.$$

Exercise 9.4 (Maximum Entropy Principle). Let X be an absolutely continuous random variable with density $\phi(x)$.

- (i) Let $\Omega = [a, b]$. Use the Euler–Lagrange equation to show that the uniform distribution $\phi^*(x) = \frac{1}{b-a}$, $a \leq x \leq b$, is a candidate density maximizing $H[X] = -\int_a^b \phi(x) \ln \phi(x) dx$, subject to the constraint $\int_a^b \phi(x) dx = 1$. Show that the associated entropy is $H[X^*] = \ln(b-a)$.

(ii) Let $\Omega = [0, \infty)$ and suppose $\mathbb{E}[X] = \beta$. Show that the exponential density $\phi^*(x) = \frac{1}{\beta} e^{-x/\beta}$, $x \geq 0$, maximizes $H[X] = -\int_0^\infty \phi(x) \ln \phi(x) dx$, subject to $\int_0^\infty \phi(x) dx = 1$, $\int_0^\infty x \phi(x) dx = \beta$. Compute the associated entropy $H[X^*] = 1 + \ln \beta$.

(iii) Let $\Omega = \mathbb{R}$ and suppose $\mathbb{E}[X] = \mu$, $\text{Var}(X) = \sigma^2$. Show that the Gaussian density

$$\phi^*(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

maximizes $H[X] = -\int_{-\infty}^\infty \phi(x) \ln \phi(x) dx$, subject to

$$\int_{-\infty}^\infty \phi(x) dx = 1, \quad \int_{-\infty}^\infty x \phi(x) dx = \mu, \quad \int_{-\infty}^\infty (x-\mu)^2 \phi(x) dx = \sigma^2.$$

Show that $H[X^*] = \frac{1}{2} + \ln(\sqrt{2\pi}\sigma)$.

Exercise 9.5 (Failure of Strong Local Minimality). Consider the minimization problem $\min J(x) = \int_0^\pi x(t)^2 (1 - \dot{x}(t)^2) dt$, $x(0) = x(\pi) = 0$.

(i) Derive the Euler–Lagrange equation.

(ii) Show that $x^0(t) = 0$ satisfies the Euler–Lagrange equation and compute $J(x^0)$.

(iii) Let $x^{(n)}(t) = \frac{1}{\sqrt{n}} \sin(nt)$. Show that for any $\varepsilon_0 > 0$, $x^{(n)} \in N_{\varepsilon_0}^s(x^0)$ (the *strong*, C^0 neighborhood, note $\sup_t |\dot{x}^{(n)}| = \sqrt{n} \rightarrow \infty$, so $x^{(n)}$ is *not* in any weak neighborhood) for sufficiently large n , and that $J(x^{(n)}) < 0$ for n large.

Conclude that x^0 is not a strong local minimizer.

Exercise 9.6 (Quadratic Integral Functional). Obtain the necessary condition for y to be a local minimizer of

$$J(y) = \int_a^b \int_a^b K(s, t) y(s) y(t) ds dt + \int_a^b y(t)^2 dt - 2 \int_a^b y(t) f(t) dt,$$

where $K(s, t)$ is continuous and symmetric, and $f(t)$ is continuous.

Exercise 9.7 (Higher-Order Euler–Lagrange Equation). Assume sufficient smoothness of f and derive the Euler–Lagrange equation associated with $\min \int_0^1 f(t, x(t), \dot{x}(t), \ddot{x}(t)) dt$, subject to $x(0) = a_0$, $\dot{x}(0) = b_0$, $x(1) = a_1$, $\dot{x}(1) = b_1$.

Exercise 9.8 (Fixed Endpoint Problem). Solve the Euler–Lagrange equation associated with

$$J(x) = \int_0^{\ln 2} x(t) \sqrt{1 + \dot{x}(t)^2} dt, \quad x(0) = 1, \quad x(\ln 2) = \frac{5}{4}.$$

Exercise 9.9 (Mixed Endpoint Conditions). Let $D^{(2)}[0, b]$ denote the space of twice continuously differentiable functions on $[0, b]$. Consider $J(x) = \int_0^b f(t, x(t), \dot{x}(t)) dt$, subject to the mixed boundary condition $x(0) + \beta x(b) = \alpha$, where $\alpha, \beta \neq 0$ are given scalars. Derive the complete set of first-order necessary conditions for a local extremum.

Exercise 9.10 (Variational Method and KL Divergence). KL divergence can be used as a measure of distance between two distributions. Let $x \in \mathcal{X}$, where $\mathcal{X} \subseteq \mathbb{R}^d$, be a continuous random variable defined over an appropriate probability space. Let $p(x)$ and $q(x)$ be two probability densities over $\mathcal{X}$. We seek q that minimizes its KL divergence from a given distribution p : $q^* = \arg \min_{q \in \mathcal{Q}} D_{\text{KL}}(q||p)$, where

$$D_{\text{KL}}(q||p) := \int_{\mathcal{X}} q(x) \log \left(\frac{q(x)}{p(x)} \right) dx.$$

(i) Let $\phi(x) = [\phi_1(x), \dots, \phi_d(x)]^\top : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a smooth vector function. Show that $\mathbb{E}_{x \sim p} [\mathcal{A}_p \phi(x)] = 0$ for any $\phi(x)$, where

$$\mathcal{A}_p \phi(x) = \nabla_x \log p(x) \phi(x)^\top + \nabla_x \phi(x).$$

Discuss the conditions on $\phi(x)$, $p(x)$, and $\mathcal{X}$ required for this identity to hold.

(ii) Let T be a small perturbation defined by $T(x) = x + \varepsilon \phi(x)$, where $\phi(x) \in \mathbb{R}^d$ is the direction of perturbation. Let $q_{[T]}(z)$ denote the density of the transformed random variable $z = T(x)$ when x has density $q(x)$.

Show that the first-order variation of the KL divergence is

$$\nabla_\varepsilon D_{\text{KL}}(q_{[T]}||p)|_{\varepsilon=0} = -\mathbb{E}_{x \sim q} [\text{trace}(\mathcal{A}_p \phi(x))].$$

Exercise 9.11 (Minimum-Distance Problem as a Constrained Functional). The minimum-distance problem seeks the shortest curve joining the origin to a point (b, c) in the positive orthant: minimize $\int_0^b \sqrt{1 + \dot{x}(t)^2} dt$ subject to $x(0) = 0$, $x(b) = c$, whose unique solution is the straight line. Formulate this as minimizing a functional $f(y)$ on $L^2[0, b]$ subject to a constraint $H(y) = \theta$, and, after identifying the appropriate H , show that the optimal solution is a *regular* point of H .

Exercise 9.12 (Regularity of Constraints in Optimal Control). For each of the following optimal control problems, discuss the regularity of the constraints at any candidate solution point.

(i) Minimize (over u) $\int_0^1 [x(t)^2 + u(t)^2] dt$ subject to $\dot{x}(t) = -x(t) + u(t)^2$, $x(0) = 1$, $x(1) = e^{-1}$.

(ii) The same, with the dynamic constraint replaced by $\dot{x}(t) = -x(t) + u(t)$.

Exercise 9.13 (Constrained Vector Calculus of Variations with Free Endpoints). Minimize $J(x, y) = \int_0^1 f[t, x(t), \dot{x}(t), y(t), \dot{y}(t)] dt$ over $(x, y) \in D[0, 1] \times D[0, 1]$ subject to $x(0) + y(0) = 1$ and $\int_0^1 g_i[t, x(t), y(t)] dt = 0$, $i = 1, 2$, where f and g_i are continuously differentiable and f is twice continuously differentiable in $\dot{x}, \dot{y}$, and $D[0, 1]$ carries the maximum norm (on the function and its derivative).

(i) For a local minimizer (x_o, y_o) , what conditions make it a *regular* point of the constraints?

(ii) Assuming regularity, obtain a complete set of first-order necessary conditions (note that $x(1)$ and $y(1)$ are free endpoints).

Chapter 10

Discrete-Time Optimal Control

This chapter develops the theory of optimal control for systems that evolve in discrete time. Two complementary methodologies organize the subject. The first is *variational*: by viewing the control trajectory as a finite-dimensional decision variable coupled to the state through the system equation, we derive first-order necessary conditions—forward state equations, backward adjoint equations, and stationarity of a Hamiltonian—that constitute the discrete-time maximum principle. The second is *recursive*: the principle of dynamic programming decomposes a multistage problem into a sequence of single-stage problems linked by the value function, yielding the Bellman recursion and feedback policies.

The variational viewpoint connects this chapter to the finite-dimensional optimization theory of Parts I and II: a discrete-time optimal control problem *is* a structured nonlinear program, and its adjoint system is a systematic bookkeeping device for Lagrange multipliers. The recursive viewpoint introduces genuinely new ideas—information, feedback, and the propagation of optimality backward in time—that carry over to the continuous-time and stochastic settings of the chapters that follow.

The chapter is organized as follows. We first derive the discrete-time maximum principle from first principles and record the associated necessary conditions. We then study two structured applications: the training of neural networks, formulated as a finite-dimensional constrained optimal control problem, and convex optimal control with state inequality constraints, including its infinite-horizon limit. The second half of the chapter develops the dynamic programming principle, the Bellman recursion, and optimal feedback policies, and illustrates the machinery on shortest-path problems, dynamic resource allocation, and optimal stopping. Exercises conclude the chapter.

10.1 ■ Discrete-Time Maximum Principle

This section studies optimization problems governed by discrete-time dynamic systems, that is, by vector difference equations. Such problems arise frequently in applications and are often challenging because the decision variables are coupled across time. Their continuous-time counterparts form the classical setting for the calculus of variations and the Pontryagin maximum principle. In the discrete-time setting, the same structural ideas lead to forward state equations, backward adjoint equations, and stationarity conditions that are central both to analysis and to computation.

Consider the problem of finding sequences $u = (u_0, u_1, \dots, u_{N-1})$ and $x = (x_1, x_2, \dots, x_N)$,

which minimize

$$g_N(x_N) + \sum_{i=0}^{N-1} \ell_i(x_i, u_i), \quad (10.1)$$

subject to the constraints

$$x_{i+1} = f_i(x_i, u_i), \quad i = 0, \dots, N-1, \quad (10.2)$$

x_0 : given,

$$x_i \in X_i \subset \mathbb{R}^n, \quad i = 1, \dots, N, \quad (10.3)$$

$$u_i \in U_i \subset \mathbb{R}^m, \quad i = 0, \dots, N-1. \quad (10.4)$$

The variables u_i are called control vectors, and $u = (u_0, u_1, \dots, u_{N-1})$ is called a control trajectory. Similarly, the variables x_i are state vectors, and the sequence $x = (x_0, x_1, \dots, x_N)$ is a state trajectory. The equation $x_{i+1} = f_i(x_i, u_i)$ is the system equation; for a given initial state x_0 , it uniquely determines the state trajectory associated with a given control trajectory. The functions $f_i : \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R}^n$ are given and they will be assumed once or twice differentiable for the most part of the following. The sets X_i and U_i specify constraints on the state and control vectors, respectively. They are usually represented by equations or inequalities. For the present discussion, their precise representation is left open. A more general constraint, used on occasion below, has the form $(x_i, u_i) \in \Omega_i \subset \mathbb{R}^n \times \mathbb{R}^m$, $i = 0, \dots, N-1$.

The functions $g_N : \mathbb{R}^n \mapsto \mathbb{R}$ and $\ell_i : \mathbb{R}^n \times \mathbb{R}^m \mapsto \mathbb{R}$, $i = 0, \dots, N-1$, are also given and, unless otherwise stated, are assumed to be once or twice differentiable.

For a fixed initial state x_0 , the system equation determines every state as a function of the control trajectory alone. We write this dependence as

$$x_i = \phi_i(u), \quad i = 1, \dots, N, \quad (10.5)$$

where each $\phi_i : \mathbb{R}^{Nm} \rightarrow \mathbb{R}^n$ is obtained by composing $f_0, \dots, f_{i-1}$. In particular, the cost becomes a function of u only, which is the point of view adopted throughout this section.

10.1.1 ■ First Derivation of the Gradient $\nabla J(u)$

First assume that the cost function has the form $J(u) = g_N(x_N) = g_N(\phi_N(u))$, so that the only cost is the terminal state cost. The general case can be reduced to this terminal-cost form by augmenting the state.

We have for every $i = 0, \dots, N-1$, using the chain rule,

$$\nabla_{u_i} J(u) = \nabla_{u_i} \phi_N \cdot \nabla g_N, \quad (10.6)$$

where the derivatives are calculated along the current control trajectory u and the corresponding state trajectory x , and where x and u are related by $x_i = \phi_i(u)$, $i = 1, \dots, N$, [cf. Eq. (10.5)]. From Eq. (10.6), we obtain, using again the chain rule,

$$\begin{aligned} \nabla_{u_i} J(u) &= \nabla_{u_i} x_{i+1} \cdot \nabla_{x_{i+1}} x_{i+1} \cdots \nabla_{x_N} x_N \cdot \nabla g_N \\ &= \nabla_{u_i} f_i \cdot \nabla_{x_{i+1}} f_{i+1} \cdots \nabla_{x_{N-1}} f_{N-1} \cdot \nabla g_N. \end{aligned}$$

Define the costate vectors $p_i \in \mathbb{R}^n$, $i = 1, \dots, N$, via the equations

$$p_i = \nabla_{x_i} f_i \cdots \nabla_{x_{N-1}} f_{N-1} \cdot \nabla g_N, \quad i = 1, \dots, N-1, \quad (10.7)$$

$$p_N = \nabla g_N, \quad (10.8)$$

we obtain

$$\nabla_{u_i} J(u) = \nabla_{u_i} f_i \cdot p_{i+1}, \quad i = 0, \dots, N-1. \quad (10.9)$$

The costates $p_1, \dots, p_N$ are generated backward in time by the adjoint equation

$$p_i = \nabla_{x_i} f_i \cdot p_{i+1}, \quad i = 1, \dots, N-1, \quad (10.10)$$

starting from

$$p_N = \nabla g_N. \quad (10.11)$$

Consider now the general case of the cost function

$$J(u) = g_N(x_N) + \sum_{i=0}^{N-1} \ell_i(x_i, u_i), \quad (10.12)$$

which involves the intermediate cost terms $\ell_i(x_i, u_i)$. We may reduce the corresponding problem to one which involves a terminal state cost only, by introducing an additional state variable y_i and a corresponding equation $y_{i+1} = y_i + \ell_i(x_i, u_i)$, $y_0 = 0$.

Defining a new state vector

$$\tilde{x}_i = \begin{pmatrix} x_i \\ y_i \end{pmatrix}, \quad (10.13)$$

and a new system equation

$$\tilde{x}_{i+1} = \begin{pmatrix} f_i(x_i, u_i) \\ y_i + \ell_i(x_i, u_i) \end{pmatrix} \triangleq \tilde{f}_i(\tilde{x}_i, u_i), \quad (10.14)$$

the cost function (10.12) takes the form

$$J(u) = g_N(x_N) + y_N \triangleq \tilde{g}_N(\tilde{x}_N). \quad (10.15)$$

This cost function involves only a terminal state cost and is of the type examined previously. We have from Eqs. (10.9)–(10.11),

$$\nabla_{u_i} J(u) = \nabla_{u_i} \tilde{f}_i \cdot \tilde{p}_{i+1}, \quad i = 0, \dots, N-1, \quad (10.16)$$

where $\tilde{p}_i = \nabla_{\tilde{x}_i} \tilde{f}_i \cdot \tilde{p}_{i+1}$, $i = 1, \dots, N-1$, $\tilde{p}_N = \nabla \tilde{g}_N$.

From Eqs. (10.13)–(10.15), we have by writing

$$\tilde{p}_i = \begin{pmatrix} p_i \\ z_i \end{pmatrix}, \quad p_i \in \mathbb{R}^n, \quad z_i \in \mathbb{R},$$

the following form of the adjoint equation

$$\begin{aligned} \tilde{p}_N &= \begin{pmatrix} p_N \\ z_N \end{pmatrix} = \begin{pmatrix} \nabla g_N \\ 1 \end{pmatrix}, \\ \tilde{p}_i &= \begin{pmatrix} p_i \\ z_i \end{pmatrix} = \begin{pmatrix} \nabla_{x_i} f_i & \nabla_{x_i} \ell_i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} p_{i+1} \\ z_{i+1} \end{pmatrix}. \end{aligned}$$

As a result, we have $z_i = 1$, $i = 1, \dots, N$, and we obtain the final form of the adjoint equation

$$p_i = \nabla_{x_i} f_i \cdot p_{i+1} + \nabla_{x_i} \ell_i, \quad i = 1, \dots, N-1, \quad (10.17)$$

$$p_N = \nabla g_N. \quad (10.18)$$

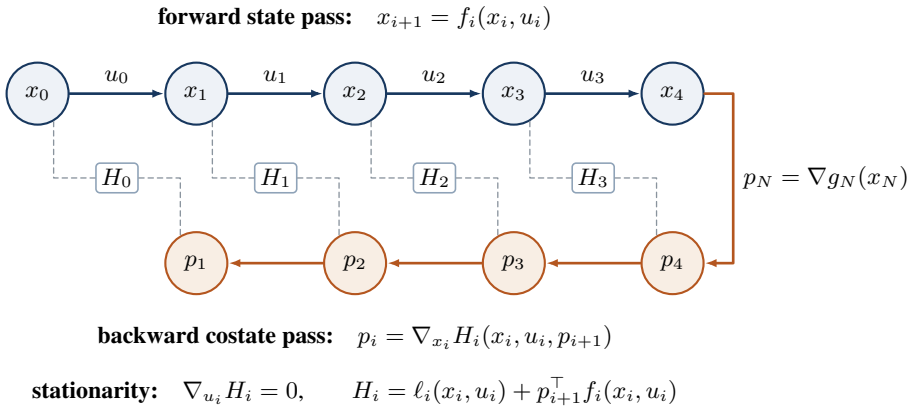


Figure 10.1: The discrete-time optimality conditions couple three ingredients. A *forward* pass propagates the state $x_{i+1} = f_i(x_i, u_i)$ (top row); a *backward* pass propagates the costate $p_i = \nabla_{x_i} H_i$ from the terminal condition $p_N = \nabla g_N$ (bottom row); and at each stage the current state x_i and next costate p_{i+1} meet in the Hamiltonian H_i (dashed links), whose stationarity $\nabla_{u_i} H_i = 0$ selects the control u_i . The same forward–then–backward pattern reappears as backpropagation in neural networks and as the Bellman recursion in dynamic programming.

Furthermore, from Eq. (10.16),

$$\nabla_{u_i} J(u) = (\nabla_{u_i} f_i \quad \nabla_{u_i} \ell_i) \begin{pmatrix} p_{i+1} \\ z_{i+1} \end{pmatrix}$$

and finally

$$\nabla_{u_i} J(u) = \nabla_{u_i} f_i \cdot p_{i+1} + \nabla_{u_i} \ell_i. \quad (10.19)$$

Equation (10.19) yields the cost gradient in terms of the costate vectors obtained from the adjoint equation (10.17) and (10.18), with all derivatives evaluated along the control trajectory u and the corresponding state trajectory under consideration. The computation of $\nabla J(u)$ therefore has a forward–backward structure, illustrated in Figure 10.1. One first propagates the system equation $x_{i+1} = f_i(x_i, u_i)$ forward to obtain the state trajectory associated with u . One then propagates the adjoint equation (10.17) backward, starting from the terminal condition $p_N = \nabla g_N$, with all partial derivatives evaluated along the current state and control trajectories. Finally, Eq. (10.19) gives each component $\nabla_{u_i} J(u)$ of the gradient. This two-sweep pattern, a forward state pass feeding a backward costate pass, recurs throughout the chapter, from backpropagation in neural networks to the Bellman recursion of dynamic programming.

10.1.2 ■ Lagrangian Derivation of $\nabla J(u)$ and $\nabla^2 J(u)$

The alternative method for calculating the derivatives of J is based on writing all the system equations $x_{i+1} = f_i(x_i, u_i)$ compactly as

$$h(\phi(u), u) = 0, \quad (10.20)$$

where $u = (u_0, u_1, \dots, u_{N-1})$ is the control trajectory and $\phi(u)$ is the corresponding state trajectory $x = \phi(u)$, where the function ϕ maps $\mathbb{R}^{Nm}$ into $\mathbb{R}^{Nn}$ and is defined in terms of the

functions ϕ_i of Eq. (10.5) as

$$\phi(u) = \begin{pmatrix} \phi_1(u) \\ \vdots \\ \phi_N(u) \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_N \end{pmatrix}. \quad (10.21)$$

The equation $h(\phi(u), u) = 0$ is a compact representation of the equations

$$\begin{aligned} f_0(x_0, u_0) - x_1 &= 0, \\ f_1(x_1, u_1) - x_2 &= 0, \\ &\vdots \\ f_{N-1}(x_{N-1}, u_{N-1}) - x_N &= 0, \end{aligned} \quad (10.22)$$

and is satisfied by every control trajectory u .

Similarly the cost function may be written abstractly as

$$F(\phi(u), u) = g_N(\phi_N(u)) + \sum_{i=0}^{N-1} \ell_i(\phi_i(u), u_i), \quad (10.23)$$

and the optimal control problem becomes $\min_{u \in \mathbb{R}^{Nm}} J(u) = F(\phi(u), u)$.

To obtain the derivatives of $J(u)$, introduce the identity, valid for any vector $p \in \mathbb{R}^{Nn}$, $J(u) = F(\phi(u), u) + h(\phi(u), u)^\top p$, [cf. Eq. (10.20)]. The multiplier p is then chosen so that the dependence of the derivative on the state sensitivity is eliminated. We have

$$\begin{aligned} \nabla J(u) &= \nabla \phi(u) (\nabla_x F(\phi(u), u) + \nabla_x h(\phi(u), u) \cdot p) \\ &\quad + \nabla_u F(\phi(u), u) + \nabla_u h(\phi(u), u) \cdot p. \end{aligned} \quad (10.24)$$

where ∇_x denotes gradient with respect to the first argument [$x = \phi(u)$], ∇_u denotes gradient with respect to the second argument u , and $\nabla \phi(u)$ is the $Nm \times Nn$ gradient matrix of ϕ evaluated at u , i.e.,

$$\nabla \phi(u) = [\nabla_u \phi_1(u) : \nabla_u \phi_2(u) : \cdots : \nabla_u \phi_N(u)].$$

Equation (10.24) holds for every $p \in \mathbb{R}^{Nn}$. Suppose that p is selected so that it satisfies the equation

$$\nabla_x F(\phi(u), u) + \nabla_x h(\phi(u), u) \cdot p(u) = 0, \quad (10.25)$$

where we denote by $p(u)$ the particular value of p satisfying the equation above. Equation (10.25) has a unique solution with respect to p , as follows from the backward adjoint recursion below, so the notation is justified. From Eq. (10.23),

$$\nabla J(u) = \nabla_u F(\phi(u), u) + \nabla_u h(\phi(u), u) \cdot p(u). \quad (10.26)$$

For notational convenience, we introduce the function $\mathcal{L} : \mathbb{R}^{Nn} \times \mathbb{R}^{Nm} \times \mathbb{R}^{Nn} \rightarrow \mathbb{R}$ given by

$$\mathcal{L}(x, u, p) = F(x, u) + h(x, u)^\top p. \quad (10.27)$$

We then have

$$\nabla J(u) = \nabla_u \mathcal{L}(\phi(u), u, p(u)). \quad (10.28)$$

To compute the Hessian matrix of the cost, we write Eq. (10.24) as

$$\nabla J(u) = \nabla \phi(u) \nabla_x \mathcal{L}(\phi(u), u, p) + \nabla_u \mathcal{L}(\phi(u), u, p)$$

and we differentiate this equation with respect to u . By using also the equation $\nabla_x \mathcal{L}(\phi(u), u, p(u)) = 0$ [cf. Eq. (10.25)] we obtain for $p = p(u)$

$$\begin{aligned} \nabla^2 J(u) &= \nabla \phi(u) \nabla_{xx}^2 \mathcal{L}(\phi(u), u, p(u)) \nabla \phi(u)^\top \\ &\quad + \nabla \phi(u) \nabla_{xu}^2 \mathcal{L}(\phi(u), u, p(u)) \\ &\quad + \nabla_{ux}^2 \mathcal{L}(\phi(u), u, p(u)) \nabla \phi(u)^\top \\ &\quad + \nabla_{uu}^2 \mathcal{L}(\phi(u), u, p(u)). \end{aligned} \quad (10.29)$$

In order to calculate the derivatives of J , we must calculate the solution $p(u)$ of Eq. (10.25) and then evaluate the derivatives of $\mathcal{L}$. We have:

$$\mathcal{L}(x, u, p) = g_N(x_N) + \sum_{i=0}^{N-1} \ell_i(x_i, u_i) + \sum_{i=1}^N p_i^\top (f_{i-1}(x_{i-1}, u_{i-1}) - x_i). \quad (10.30)$$

Equation (10.25) can be written as $\nabla_x \mathcal{L}(\phi(u), u, p(u)) = 0$, or by using Eq. (10.30),

$$\begin{pmatrix} \nabla g_N \\ \nabla_{x_{N-1}} \ell_{N-1} \\ \vdots \\ \nabla_{x_1} \ell_1 \end{pmatrix} + \begin{pmatrix} -I & 0 & \cdots & 0 \\ \nabla_{x_{N-1}} f_{N-1} & -I & \cdots & 0 \\ 0 & \nabla_{x_{N-2}} f_{N-2} & \ddots & \vdots \\ \vdots & \vdots & \ddots & -I \end{pmatrix} \begin{pmatrix} p_N \\ p_{N-1} \\ \vdots \\ p_1 \end{pmatrix} = 0.$$

By rewriting this equation, we have

$$p_i = \nabla_{x_i} f_i \cdot p_{i+1} + \nabla_{x_i} \ell_i, \quad i = 1, \dots, N-1, \quad (10.31)$$

$$p_N = \nabla g_N, \quad (10.32)$$

which is the same as the adjoint equation derived earlier [Eqs. (10.17) and (10.18)]. Thus, the adjoint equation specifies uniquely $p(u)$.

Also, from Eqs. (10.27), (10.28), and (10.30),

$$\nabla_{u_i} J(u) = \nabla_{u_i} f_i \cdot p_{i+1} + \nabla_{u_i} \ell_i, \quad i = 0, \dots, N-1, \quad (10.33)$$

which is the same equation for the gradient ∇J as the one obtained earlier.

The Hessian matrix $\nabla^2 J$ is given by Eqs. (10.29) and (10.30). For reference, the gradient matrix $\nabla \phi(u)$ has the form

$$\nabla \phi(u) = \begin{pmatrix} \nabla_{u_0} f_0 & \nabla_{x_1} f_1 \nabla_{u_0} f_0 & \cdots & \nabla_{x_{N-1}} f_{N-1} \cdots \nabla_{x_1} f_1 \nabla_{u_0} f_0 \\ 0 & \nabla_{u_1} f_1 & \cdots & \nabla_{x_{N-1}} f_{N-1} \cdots \nabla_{x_2} f_2 \nabla_{u_1} f_1 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \nabla_{u_{N-1}} f_{N-1} \end{pmatrix}.$$

Although the expression (10.29) for $\nabla^2 J$ is algebraically involved, it is useful for Newton-type methods because it displays explicitly how the state sensitivities enter the second derivative of the reduced cost.

10.1.3 ■ First-Order Necessary Conditions for Discrete-Time Optimal Control

Consider the finite-horizon discrete-time optimal control problem

$$\begin{aligned} \min_{\{u_i\}_{i=0}^{N-1}} \quad & J(u) := \sum_{i=0}^{N-1} \ell(x_i, u_i) + g_N(x_N) \\ \text{s.t.} \quad & x_{i+1} = f(x_i, u_i), \quad i = 0, \dots, N-1, \\ & x_0 \text{ given.} \end{aligned} \quad (10.34)$$

Here $x_i \in \mathbb{R}^n$ and $u_i \in \mathbb{R}^m$, and the functions

$$f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n, \quad \ell : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}, \quad g_N : \mathbb{R}^n \rightarrow \mathbb{R}$$

are assumed continuously differentiable.

Define the control vector $u := (u_0, u_1, \dots, u_{N-1})$.

Theorem 10.1 (First-Order Necessary Conditions). *Let $u^* = (u_0^*, \dots, u_{N-1}^*)$ be a local minimizer of problem (10.34), and let $x^* = (x_0^*, \dots, x_N^*)$ be the corresponding state trajectory. Define the stage Hamiltonian*

$$H_i(x_i, u_i, p_{i+1}) = \ell(x_i, u_i) + p_{i+1}^\top f(x_i, u_i). \quad (10.35)$$

Then there exist costate vectors $p_1^, \dots, p_N^* \in \mathbb{R}^n$ such that, for all i ,*

$$(\text{stationarity}) \quad \nabla_{u_i} H_i(x_i^*, u_i^*, p_{i+1}^*) = 0, \quad i = 0, \dots, N-1, \quad (10.36)$$

$$(\text{adjoint recursion}) \quad p_i^* = \nabla_{x_i} H_i(x_i^*, u_i^*, p_{i+1}^*), \quad i = 1, \dots, N-1, \quad (10.37)$$

$$(\text{terminal condition}) \quad p_N^* = \nabla g_N(x_N^*). \quad (10.38)$$

Proof. The three conditions are simply a repackaging of the vanishing-gradient condition $\nabla J(u^*) = 0$, expressed through the costates. We make the connection explicit.

Step 1: reduction to an unconstrained problem. The initial state x_0 is fixed, and the system equation $x_{i+1} = f(x_i, u_i)$ determines each state as a function of the controls alone, $x_i = \phi_i(u)$ [cf. Eq. (10.5)]. Substituting these relations into the cost turns problem (10.34) into the *unconstrained* minimization of the reduced cost $J(u) = \sum_{i=0}^{N-1} \ell(\phi_i(u), u_i) + g_N(\phi_N(u))$ over $u \in \mathbb{R}^{Nm}$. Because J is continuously differentiable and u^* is a local minimizer lying in the open set $\mathbb{R}^{Nm}$, Fermat's rule applies and every partial gradient vanishes,

$$\nabla_{u_i} J(u^*) = 0, \quad i = 0, \dots, N-1. \quad (10.39)$$

Step 2: the adjoint bookkeeping. Define the terminal costate by $p_N^* = \nabla g_N(x_N^*)$, which is exactly (10.38), and generate the remaining costates backward through

$$p_i^* = \nabla_{x_i} f(x_i^*, u_i^*) p_{i+1}^* + \nabla_{x_i} \ell(x_i^*, u_i^*), \quad i = N-1, \dots, 1. \quad (10.40)$$

This is the adjoint recursion (10.17)–(10.18) derived earlier (with $\ell_i \equiv \ell$), and it defines the p_i^* uniquely.

Step 3: identifying the Hamiltonian derivatives. By the forward–backward gradient formula (10.19), the gradient of the reduced cost is, along the optimal trajectory,

$$\nabla_{u_i} J(u^*) = \nabla_{u_i} f(x_i^*, u_i^*) p_{i+1}^* + \nabla_{u_i} \ell(x_i^*, u_i^*). \quad (10.41)$$

Now recall the Hamiltonian $H_i(x, u, p) = \ell(x, u) + p^\top f(x, u)$ of (10.35). Differentiating H_i in u and in x gives

$$\nabla_{u_i} H_i(x, u, p) = \nabla_{u_i} \ell(x, u) + \nabla_{u_i} f(x, u) p, \quad \nabla_{x_i} H_i(x, u, p) = \nabla_{x_i} \ell(x, u) + \nabla_{x_i} f(x, u) p. \quad (10.42)$$

Comparing (10.42) with (10.41) and (10.40) yields the two identities

$$\nabla_{u_i} J(u^*) = \nabla_{u_i} H_i(x_i^*, u_i^*, p_{i+1}^*), \quad p_i^* = \nabla_{x_i} H_i(x_i^*, u_i^*, p_{i+1}^*).$$

The second identity is precisely the adjoint condition (10.37). Substituting the first identity into the optimality condition (10.39) gives $\nabla_{u_i} H_i(x_i^*, u_i^*, p_{i+1}^*) = 0$, which is the stationarity condition (10.36). Together with the forward dynamics and the terminal condition of Step 2, this establishes all the stated conditions. $\square$

The theorem should be read as a change of variables rather than a new principle: the abstract statement $\nabla J(u^*) = 0$ is true for any unconstrained smooth minimization, and the costates p_i^* are the Lagrange multipliers that let us write that condition *stage by stage*, without ever forming the dense gradient ∇J explicitly. The Hamiltonian H_i collects, at each stage, the immediate cost ℓ and the future cost transmitted through the dynamics, weighted by the costate p_{i+1} ; the control u_i^* is chosen to make H_i stationary, balancing present against future. This is the discrete shadow of the Pontryagin maximum principle.

The following worked example carries the whole forward–backward machinery through on a small problem, so that the roles of the state pass, the adjoint pass, and the stationarity condition become concrete.

Example 10.2 Minimum-energy regulation with a terminal penalty.

Consider the scalar system

$$x_{i+1} = x_i + u_i, \quad x_0 = 2, \quad N = 2,$$

with cost $J = u_0^2 + u_1^2 + x_2^2$. Here $\ell(x, u) = u^2$ (there is no state running cost) and $g_2(x) = x^2$; the terminal state is penalized *softly* rather than constrained to the origin, so the optimum will leave a small residual $x_2^* \neq 0$. The stage Hamiltonian is

$$H_i(x_i, u_i, p_{i+1}) = u_i^2 + p_{i+1}(x_i + u_i).$$

Adjoint pass. Since $\nabla_{x_i} H_i = p_{i+1}$, the costate recursion is $p_i = p_{i+1}$, so the costate is constant, $p_1 = p_2$. The terminal condition gives $p_2 = g_2'(x_2) = 2x_2$. Hence $p_1 = p_2 = 2x_2$.

Stationarity. From $\partial H_i / \partial u_i = 2u_i + p_{i+1} = 0$ we obtain $u_0^* = -\frac{1}{2}p_1 = -x_2$ and $u_1^* = -\frac{1}{2}p_2 = -x_2$. Both controls are equal because the costate is constant, a discrete echo of the “spread the effort evenly” principle.

Closing the loop. Propagating the dynamics forward, $x_2 = x_0 + u_0^* + u_1^* = 2 - 2x_2$, so $3x_2 = 2$ and

$$x_2^* = \frac{2}{3}, \quad u_0^* = u_1^* = -\frac{2}{3}, \quad p_1^* = p_2^* = \frac{4}{3}.$$

The optimal cost is $J^* = (\frac{2}{3})^2 + (\frac{2}{3})^2 + (\frac{2}{3})^2 = \frac{4}{3}$. One checks directly that eliminating the state and minimizing $J(u_0, u_1) = u_0^2 + u_1^2 + (2 + u_0 + u_1)^2$ gives the

same answer, confirming that the maximum principle reproduces the unconstrained optimum. $\square$

10.2 ■ Neural Network Training as a Finite-Dimensional Constrained Optimization Problem

10.2.1 ■ Data and Architecture

Consider a supervised learning setting. Let $M \geq 1$ denote the number of training samples and consider the dataset $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^M$, where each input vector $x^{(i)}$ belongs to $\mathbb{R}^{n_0}$ and each target output $y^{(i)}$ belongs to $\mathbb{R}^{n_N}$. The integer n_0 represents the input dimension of the network, while n_N represents the output dimension. We fix the depth $N \geq 1$ and choose intermediate layer dimensions $n_1, \dots, n_{N-1}$. Thus the network has N layers (excluding the input layer) and widths $n_0, n_1, \dots, n_N$.

For each layer $k = 0, \dots, N-1$, the network contains a weight matrix $W_k \in \mathbb{R}^{n_{k+1} \times n_k}$ and a bias vector $b_k \in \mathbb{R}^{n_{k+1}}$. The matrix W_k maps activations from layer k to layer $k+1$, while b_k shifts the affine transformation. All parameters are collected into a single vector $\theta = (W_0, b_0, \dots, W_{N-1}, b_{N-1})$, which belongs to the parameter space

$$\Theta = \prod_{k=0}^{N-1} (\mathbb{R}^{n_{k+1} \times n_k} \times \mathbb{R}^{n_{k+1}}).$$

Since each W_k contains $n_{k+1}n_k$ entries and each b_k contains n_{k+1} entries, the total number of scalar parameters is

$$d = \sum_{k=0}^{N-1} (n_{k+1}n_k + n_{k+1}).$$

Hence neural network training is a finite-dimensional optimization problem in $\mathbb{R}^d$, and Figure 10.2 depicts the architecture as a sequence of state transformations.

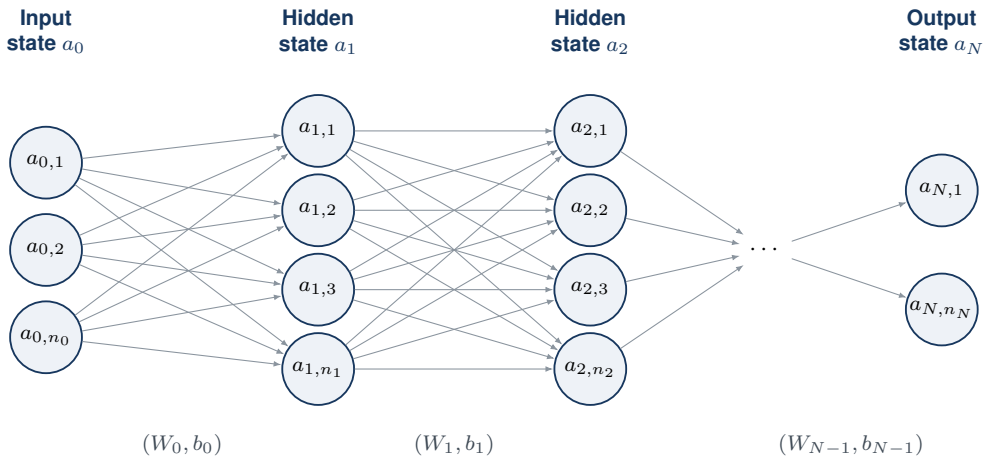


Figure 10.2: A feedforward neural network viewed as a finite-horizon discrete-time system. For each training sample, the activation vectors $a_0, a_1, \dots, a_N$ form a state trajectory, while the layer parameters (W_k, b_k) determine the transition from one layer to the next.

10.2.2 ■ Forward State Equations

Given a parameter vector θ and a sample index i , the network computes its output through a recursive forward pass. We initialize the process with the input $a_0^{(i)} = x^{(i)}$. For each layer $k = 0, \dots, N - 1$, we first compute the affine transformation $z_k^{(i)} = W_k a_k^{(i)} + b_k$, and then apply a nonlinear activation function σ_k : $a_{k+1}^{(i)} = \sigma_k(z_k^{(i)})$. Thus $a_k^{(i)}$ represents the activation (or state) of layer k for sample i . After N such steps, the final activation $a_N^{(i)}$ is the network output corresponding to the input $x^{(i)}$. From a dynamical systems viewpoint, the sequence $a_0^{(i)}, a_1^{(i)}, \dots, a_N^{(i)}$ is a discrete-time state trajectory driven by the parameters θ .

10.2.3 ■ Objective Function

To measure prediction quality, we introduce a loss function $\Phi : \mathbb{R}^{n_N} \times \mathbb{R}^{n_N} \rightarrow \mathbb{R}$, which compares the network output with the true label. Typical examples include squared error and cross-entropy loss. We assume Φ is continuously differentiable in its first argument.

The empirical risk is defined as the average loss over the dataset:

$$J(\theta) = \frac{1}{M} \sum_{i=1}^M \Phi(a_N^{(i)}, y^{(i)}).$$

The neural network training problem is therefore

$$\min_{\theta \in \Theta} J(\theta). \quad (10.43)$$

The function J depends on θ indirectly through the forward state equations. Under differentiability assumptions on σ_k and Φ , the function J is continuously differentiable on Θ .

10.2.4 ■ Common Activation Functions: Analytical and Structural Perspective

Activation functions σ_k are the source of nonlinearity in a neural network. If one were to remove the nonlinearities and keep only the affine transformations $z_k = W_k a_k + b_k$, then the composition of layers would remain affine. Indeed, a composition of affine maps is again affine, and therefore a deep linear network would be equivalent to a single linear transformation. Thus, the expressive power of deep architectures arises entirely from the nonlinear activation functions. Figure 10.3 collects the most common choices, each shown together with its derivative; as we detail below, the shape of the derivative, in particular whether it saturates, is what governs how gradients propagate backward through the network.

ReLU (Rectified Linear Unit). The ReLU activation is defined by $\sigma(x) = \max\{0, x\}$. It is piecewise linear: it coincides with the identity map for $x > 0$ and is identically zero for $x < 0$. Its derivative is equal to 1 for positive inputs and 0 for negative inputs, and it is not differentiable at $x = 0$. In practice one selects a subgradient at the origin.

ReLU preserves linear structure within each region $x > 0$ and $x < 0$, which leads to networks that are globally piecewise affine functions of the input. Moreover, because negative inputs are mapped to zero, many neurons become inactive, producing sparse representations. However, if a neuron consistently receives negative inputs, its gradient vanishes and its parameters may cease to update. This phenomenon is commonly referred to as the “dead neuron” problem.

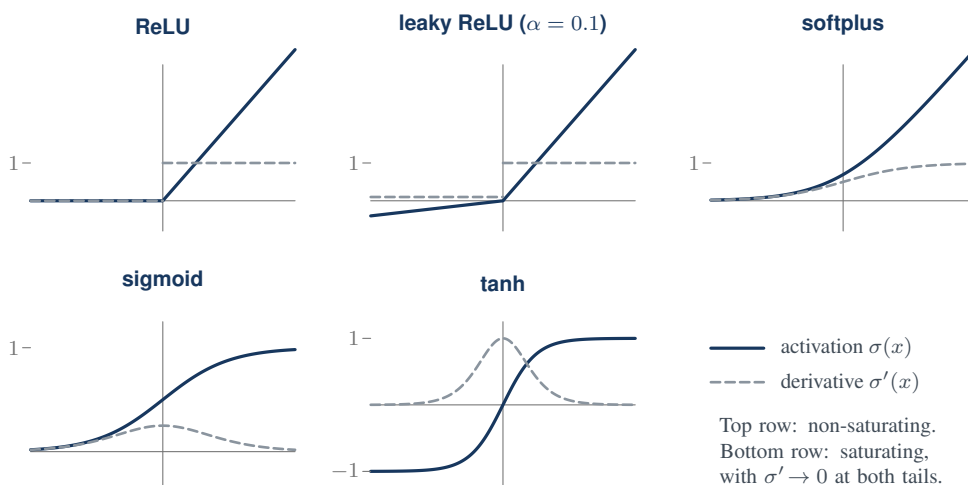


Figure 10.3: Common activation functions and their derivatives, one per panel (solid: σ ; dashed: σ' ; the tick marks level 1). The non-saturating family (top) keeps the derivative from vanishing for large positive inputs—it equals 1 there for ReLU and leaky ReLU, and tends to 1 for softplus, whose derivative is the sigmoid—so gradients pass backward undiminished through active units. The saturating family (bottom) has derivatives that vanish for large $|x|$, the mechanism behind vanishing gradients in deep networks; leaky ReLU’s nonzero negative slope additionally avoids the “dead neuron” failure of ReLU. The derivative is what multiplies the incoming costate in the adjoint recursion $\delta_k^{(i)} = D\sigma_k(z_k^{(i)})^\top p_{k+1}^{(i)}$.

Sigmoid. The sigmoid activation is given by $\sigma(x) = \frac{1}{1+e^{-x}}$. It maps the real line into the interval $(0, 1)$ and has an S-shaped profile. Its derivative satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x))$. For large positive or negative inputs, $\sigma'(x)$ becomes very small. When many such derivatives are multiplied across layers during backpropagation, the resulting product may approach zero, leading to the vanishing gradient phenomenon. For this reason, sigmoid activations are now used primarily in output layers for binary classification rather than in deep hidden layers.

Hyperbolic Tangent. The hyperbolic tangent activation is $\sigma(x) = \tanh(x)$, which maps $\mathbb{R}$ into $(-1, 1)$. It is related to the sigmoid by a simple rescaling and shift. Its derivative is $\sigma'(x) = 1 - \tanh^2(x)$. Because $\tanh$ is zero-centered, it often yields better optimization behavior than the sigmoid in hidden layers. Nevertheless, it also saturates for large $|x|$ and therefore suffers from vanishing gradients in deep networks.

Leaky ReLU. Leaky ReLU modifies the standard ReLU by introducing a small slope for negative inputs:

$$\sigma(x) = \begin{cases} x, & x \geq 0, \\ \alpha x, & x < 0, \end{cases} \quad \alpha > 0.$$

Its derivative equals 1 for positive inputs and α for negative inputs. Since α is nonzero, gradients continue to propagate even when the neuron operates in the negative region. This alleviates the dead neuron problem while preserving the computational simplicity and piecewise linear structure of ReLU.

Softplus. The softplus activation is defined by $\sigma(x) = \log(1 + e^x)$. It can be viewed as a smooth approximation of ReLU. Unlike ReLU, it is differentiable everywhere. Its derivative is the sigmoid function, $\sigma'(x) = \frac{1}{1+e^{-x}}$. Softplus maintains similar qualitative behavior to ReLU while providing smoother first and second derivatives, which can be advantageous in certain theoretical analyses.

Softmax (Final Layer). For multi-class classification, the softmax function is applied to a vector $z \in \mathbb{R}^m$:

$$(\sigma(z))_j = \frac{e^{z_j}}{\sum_{\ell=1}^m e^{z_\ell}}.$$

This transformation produces a probability distribution, since each component is positive and the components sum to one. Softmax is typically combined with cross-entropy loss, leading to a well-structured gradient expression that aligns naturally with maximum likelihood estimation.

Some activations, such as ReLU, are not differentiable at isolated points. However, they are differentiable almost everywhere and locally Lipschitz. Gradient-based optimization methods require differentiability only along trajectories encountered during training, and isolated nondifferentiable points do not prevent convergence in practice. When necessary, one may appeal to subgradient calculus to formalize first-order optimality conditions.

Smooth activations such as sigmoid and tanh produce smooth loss landscapes but may induce vanishing gradients. Piecewise-linear activations such as ReLU generate networks that are globally piecewise affine in both inputs and parameters. Consequently, the overall network can be interpreted as selecting, via activation patterns, one linear map from a large collection of region-dependent affine maps. The choice of activation function therefore directly influences the geometry of the optimization landscape, the stability of gradient propagation, and the expressive capacity of the model.

10.2.5 ■ Characterization of Optimal Solutions

Proposition 10.3 (First-Order Optimality System). *Assume that each activation function σ_k and the loss Φ are continuously differentiable. Let θ^* be a local minimizer of the training problem (10.43), and let $a_k^{(i)*}$ and $z_k^{(i)*}$ denote the corresponding forward trajectories generated by θ^* . Define the terminal adjoint variable for each sample i by*

$$p_N^{(i)} = \frac{1}{M} \nabla_1 \Phi(a_N^{(i)*}, y^{(i)}).$$

For $k = N - 1, \dots, 0$, define recursively

$$\delta_k^{(i)} = D\sigma_k(z_k^{(i)*})^\top p_{k+1}^{(i)}, \quad p_k^{(i)} = W_k^{*\top} \delta_k^{(i)}.$$

Then, for every $k = 0, \dots, N - 1$, the parameter gradients vanish:

$$(i) \text{ stationarity in the weights: } \nabla_{W_k} J(\theta^*) = \sum_{i=1}^M \delta_k^{(i)} (a_k^{(i)*})^\top = 0;$$

$$(ii) \text{ stationarity in the biases: } \nabla_{b_k} J(\theta^*) = \sum_{i=1}^M \delta_k^{(i)} = 0.$$

Conversely, any parameter vector θ^* satisfying these conditions is a first-order stationary point of J .

Proof. The forward equations $z_k^{(i)} = W_k a_k^{(i)} + b_k$ and $a_{k+1}^{(i)} = \sigma_k(z_k^{(i)})$ are the system dynamics, with the activations $a_k^{(i)}$ playing the role of states and the parameters (W_k, b_k) the role of controls; the empirical risk $J(\theta)$ is a pure terminal cost. We compute $\nabla_{W_k} J$ and $\nabla_{b_k} J$ by the chain rule and show that the backward recursion for $p_k^{(i)}$ is exactly the resulting adjoint pass. It suffices to treat a single sample i and sum over i at the end, since J is an average of per-sample terms.

Terminal sensitivity. The risk depends on $a_N^{(i)}$ through $\frac{1}{M} \Phi(a_N^{(i)}, y^{(i)})$, so the sensitivity of the cost to the output activation is $p_N^{(i)} = \frac{1}{M} \nabla_1 \Phi(a_N^{(i)*}, y^{(i)})$, as defined.

Propagation across a layer. Fix a layer k and view the cost as a function of the pre-activation $z_k^{(i)}$ and of $a_{k+1}^{(i)}$. Since $a_{k+1}^{(i)} = \sigma_k(z_k^{(i)})$, the chain rule gives the sensitivity with respect to $z_k^{(i)}$,

$$\delta_k^{(i)} := \nabla_{z_k^{(i)}} J = D\sigma_k(z_k^{(i)*})^\top \nabla_{a_{k+1}^{(i)}} J = D\sigma_k(z_k^{(i)*})^\top p_{k+1}^{(i)},$$

which is the stated definition of $\delta_k^{(i)}$, and $D\sigma_k$ is the (diagonal) Jacobian of the activation. Because $z_k^{(i)} = W_k a_k^{(i)} + b_k$ depends linearly on $a_k^{(i)}$ through W_k , propagating the sensitivity one more step back to the input activation of the layer yields

$$p_k^{(i)} := \nabla_{a_k^{(i)}} J = W_k^{\star\top} \nabla_{z_k^{(i)}} J = W_k^{\star\top} \delta_k^{(i)},$$

which is the stated backward recursion. Thus $\{p_k^{(i)}\}$ is precisely the adjoint (costate) sequence of the maximum principle, generated backward from $p_N^{(i)}$.

Parameter gradients. The parameters (W_k, b_k) enter the cost only through $z_k^{(i)} = W_k a_k^{(i)} + b_k$. Differentiating and using $\nabla_{z_k^{(i)}} J = \delta_k^{(i)}$ together with $\partial z_k^{(i)} / \partial W_k = (a_k^{(i)})^\top$ (a rank-one dependence) and $\partial z_k^{(i)} / \partial b_k = I$, we obtain, after summing over samples,

$$\nabla_{W_k} J(\theta^*) = \sum_{i=1}^M \delta_k^{(i)} (a_k^{(i)*})^\top, \quad \nabla_{b_k} J(\theta^*) = \sum_{i=1}^M \delta_k^{(i)}.$$

At a local minimizer θ^* lies in the open parameter space, so Fermat's rule forces both gradients to vanish, giving conditions (i) and (ii). Conversely, if these gradients vanish then $\nabla J(\theta^*) = 0$ by construction, so θ^* is a first-order stationary point. This is the announced equivalence. $\square$

The proposition shows that a local minimizer must satisfy a coupled forward-backward system. The forward recursion $a_{k+1}^{(i)} = \sigma_k(W_k^* a_k^{(i)*} + b_k^*)$ propagates the activations from input to output. The backward recursion defining $p_k^{(i)}$ propagates sensitivity information in reverse order, beginning from the terminal gradient $\nabla_1 \Phi$. The quantities $\delta_k^{(i)}$ represent layerwise error signals. Each gradient component is expressed as a sum of rank-one outer products $\delta_k^{(i)} (a_k^{(i)*})^\top$, which couple forward activations and backward sensitivities.

The forward dynamics, the adjoint recursion, and the stationarity conditions form a complete first-order optimality system. This system is structurally identical to the discrete-time maximum principle for finite-horizon optimal control, where the activations act as state variables, the parameters (W_k, b_k) act as control variables, and the loss Φ plays the role of a terminal cost. Backpropagation is therefore not an algorithmic heuristic; it is precisely the adjoint equation associated with this finite-dimensional nonlinear optimization problem.

The following example makes the forward and backward passes fully concrete on a minimal network, so that every quantity a_k , z_k , δ_k , and p_k carries a number.

Example 10.4 Forward and backward pass on a scalar network.

Take a single training sample with input $x = 1$ and target $y = 0$, and a two-layer scalar network ($M = 1, n_0 = n_1 = n_2 = 1$). Layer 0 uses the ReLU activation and layer 1 is affine (identity activation), with initial parameters $W_0 = 1, b_0 = 0, W_1 = 1, b_1 = 0$ and squared-error loss $\Phi(a, y) = \frac{1}{2}(a - y)^2$.

Forward pass. Starting from $a_0 = x = 1$,

$$z_0 = W_0 a_0 + b_0 = 1, \quad a_1 = \text{ReLU}(z_0) = 1, \quad z_1 = W_1 a_1 + b_1 = 1, \quad a_2 = z_1 = 1.$$

The loss is $\Phi(a_2, y) = \frac{1}{2}(1 - 0)^2 = \frac{1}{2}$.

Backward pass. The terminal adjoint is $p_2 = \nabla_1 \Phi(a_2, y) = a_2 - y = 1$. Propagating backward with $\sigma'_1 = 1$ and $\text{ReLU}'(z_0) = 1$ (since $z_0 = 1 > 0$),

$$\delta_1 = \sigma'_1(z_1) p_2 = 1, \quad p_1 = W_1 \delta_1 = 1, \quad \delta_0 = \text{ReLU}'(z_0) p_1 = 1.$$

Parameter gradients. The rank-one outer-product formulas give

$$\nabla_{W_1} J = \delta_1 a_1 = 1, \quad \nabla_{b_1} J = \delta_1 = 1, \quad \nabla_{W_0} J = \delta_0 a_0 = 1, \quad \nabla_{b_0} J = \delta_0 = 1.$$

None of the gradients vanish, so this parameter vector is *not* stationary, consistent with the fact that the network outputs $a_2 = 1$ while the target is 0. A gradient step $\theta \leftarrow \theta - \alpha \nabla J$ with, say, $\alpha = \frac{1}{2}$ sets W_1, b_1, W_0, b_0 to $\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}$; recomputing the forward pass gives $z_0 = 0, a_1 = 0, a_2 = -\frac{1}{2}$, and loss $\frac{1}{8} < \frac{1}{2}$, confirming descent. This is one iteration of backpropagation, read as one iteration of the adjoint-based gradient method of Section 10.2. $\square$

10.3 ■ Convex Optimal Control with State Inequality Constraints

Consider the finite-horizon convex optimal control problem

$$\begin{aligned} \min_{\{u_i\}_{i=0}^{N-1}} \quad & J(u) := \sum_{i=0}^{N-1} \ell(x_i, u_i) + g_N(x_N) \\ \text{s.t.} \quad & x_{i+1} = f(x_i, u_i), \quad i = 0, \dots, N-1, \\ & g_j(x_i) \leq 0, \quad j = 1, \dots, q, \quad i = 0, \dots, N, \\ & x_0 \text{ given.} \end{aligned} \tag{10.44}$$

Assume that f is affine in (x, u) , that $\ell(\cdot, \cdot)$ and $g_N(\cdot)$ are convex, that each state-constraint function $g_j(\cdot)$ is convex, and that all functions appearing in the problem are differentiable. A word on notation: the terminal cost g_N carries the horizon N as a *stage* index, whereas the state-constraint functions $g_1, \dots, g_q$ are indexed by a *constraint* counter j ; the two families are unrelated, and g_N never denotes a constraint even when $q \geq N$. Note also that the requirement $g_j(x_0) \leq 0$ involves only the given initial state, so it is a feasibility assumption on the data rather than a restriction on the decision variables; we assume throughout that x_0 satisfies it.

Under these assumptions, problem (10.44) is a convex optimization problem in the decision variables $(x_1, \dots, x_N, u_0, \dots, u_{N-1})$. Many applications impose limits on the state itself, a

temperature that may not exceed a safety bound, an inventory that must stay nonnegative, a vehicle that must remain within a corridor. Such requirements are captured by the inequalities $g_j(x_i) \leq 0$, and they interact with the dynamics: forcing a state to respect a bound at one stage constrains the controls at earlier stages. The role of this section is to show how the maximum principle absorbs these state constraints through additional multipliers in the costate recursion.

10.3.1 ■ Lagrangian and Hamiltonian

Two constraint families are present: the equality constraints imposed by the dynamics, and the inequality constraints on the state. We attach a costate multiplier $p_{i+1} \in \mathbb{R}^n$ to each dynamic constraint $x_{i+1} = f(x_i, u_i)$ and a *nonnegative* multiplier $\kappa_{i,j} \geq 0$ to each state inequality $g_j(x_i) \leq 0$, $i \geq 1$ —the stage-0 inequalities need no multiplier, since x_0 is not a decision variable. Collecting every term, the complete Lagrangian of problem (10.44) is

$$\mathcal{L} = \underbrace{\sum_{i=0}^{N-1} \ell(x_i, u_i)}_{\text{cost}} + \underbrace{\sum_{i=0}^{N-1} p_{i+1}^\top (f(x_i, u_i) - x_{i+1})}_{\text{dynamics}} + \underbrace{\sum_{i=1}^N \sum_{j=1}^q \kappa_{i,j} g_j(x_i)}_{\text{state constraints}}. \quad (10.45)$$

As in the unconstrained case it is convenient to group, at each stage, the immediate cost with the cost transmitted through the dynamics into the stage Hamiltonian

$$H_i(x_i, u_i, p_{i+1}) = \ell(x_i, u_i) + p_{i+1}^\top f(x_i, u_i), \quad (10.46)$$

so that $\mathcal{L} = \sum_i H_i(x_i, u_i, p_{i+1}) + g_N(x_N) - \sum_i p_{i+1}^\top x_{i+1} + \sum_{i,j} \kappa_{i,j} g_j(x_i)$: the Lagrangian is the sum of the Hamiltonian terms, the coupling terms $-p_{i+1}^\top x_{i+1}$, and the state-constraint penalties. Attaching the multiplier to $f - x_{i+1}$ (rather than $x_{i+1} - f$) is what makes the stationarity conditions read directly in terms of H_i . Setting the partial derivatives of $\mathcal{L}$ to zero will produce the stationarity and costate conditions below, with the multipliers $\kappa_{i,j}$ appearing wherever a state constraint is active.

Theorem 10.5 (KKT Conditions for Convex State-Constrained Optimal Control). *Let (x^*, u^*) be a feasible trajectory of (10.44), and assume Slater's condition holds for the state constraints. Then (x^*, u^*) is optimal if and only if there exist multipliers $p_1^*, \dots, p_N^*$ and $\kappa_{i,j}^* \geq 0$ such that the following conditions hold:*

- (i) primal feasibility: $x_{i+1}^* = f(x_i^*, u_i^*)$ and $g_j(x_i^*) \leq 0$;
- (ii) control stationarity: $\nabla_{u_i} H_i(x_i^*, u_i^*, p_{i+1}^*) = 0$, $i = 0, \dots, N-1$;
- (iii) costate recursion (incorporating the active state constraints), for $i = 1, \dots, N-1$,

$$p_i^* = \nabla_{x_i} H_i(x_i^*, u_i^*, p_{i+1}^*) + \sum_{j=1}^q \kappa_{i,j}^* \nabla g_j(x_i^*);$$

- (iv) terminal condition: $p_N^* = \nabla g_N(x_N^*) + \sum_{j=1}^q \kappa_{N,j}^* \nabla g_j(x_N^*)$;

- (v) complementary slackness: $\kappa_{i,j}^* \geq 0$ and $\kappa_{i,j}^* g_j(x_i^*) = 0$.

Proof. Under the stated assumptions the problem is convex and Slater's condition holds, so the KKT conditions are both necessary and sufficient for optimality; it therefore suffices to write out the stationarity of the Lagrangian (10.45) in the decision variables (x_i, u_i) and to append primal feasibility, dual feasibility, and complementary slackness.

Stationarity in u_i . The control u_i appears in $\mathcal{L}$ only through the stage term $\ell(x_i, u_i) + p_{i+1}^\top f(x_i, u_i) = H_i(x_i, u_i, p_{i+1})$. Hence

$$\nabla_{u_i} \mathcal{L} = \nabla_{u_i} \ell(x_i^*, u_i^*) + \nabla_{u_i} f(x_i^*, u_i^*) p_{i+1}^* = \nabla_{u_i} H_i(x_i^*, u_i^*, p_{i+1}^*) = 0,$$

which is the stated control stationarity.

Stationarity in x_i , $1 \leq i \leq N-1$. The interior state x_i appears in three places: the running cost $\ell(x_i, u_i)$, the coupling term $-p_i^\top x_i$ (from the dynamic constraint at stage $i-1$), and the term $p_{i+1}^\top f(x_i, u_i)$ (from the dynamic constraint at stage i), together with the active state penalties $\sum_j \kappa_{i,j} g_j(x_i)$. Collecting,

$$\nabla_{x_i} \mathcal{L} = \nabla_{x_i} \ell + \nabla_{x_i} f(x_i^*, u_i^*) p_{i+1}^* - p_i^* + \sum_{j=1}^q \kappa_{i,j}^* \nabla g_j(x_i^*) = 0.$$

Solving for p_i^* and recognizing $\nabla_{x_i} \ell + \nabla_{x_i} f p_{i+1}^* = \nabla_{x_i} H_i$ gives

$$p_i^* = \nabla_{x_i} H_i(x_i^*, u_i^*, p_{i+1}^*) + \sum_{j=1}^q \kappa_{i,j}^* \nabla g_j(x_i^*),$$

which is the stated costate recursion: the state constraints inject the extra term $\sum_j \kappa_{i,j}^* \nabla g_j$.

Stationarity in x_N . The terminal state appears in $g_N(x_N)$, in the coupling term $-p_N^\top x_N$, and in the terminal penalties $\sum_j \kappa_{N,j} g_j(x_N)$, whence

$$\nabla_{x_N} \mathcal{L} = \nabla g_N(x_N^*) - p_N^* + \sum_j \kappa_{N,j}^* \nabla g_j(x_N^*) = 0,$$

i.e. $p_N^* = \nabla g_N(x_N^*) + \sum_j \kappa_{N,j}^* \nabla g_j(x_N^*)$, the stated terminal condition.

Feasibility and slackness. Primal feasibility ($x_{i+1}^* = f(x_i^*, u_i^*)$, $g_j(x_i^*) \leq 0$) is the requirement that (x^*, u^*) be admissible; dual feasibility $\kappa_{i,j}^* \geq 0$ and complementary slackness $\kappa_{i,j}^* g_j(x_i^*) = 0$ are the standard KKT conditions for the inequality multipliers, expressing that a constraint carries a nonzero multiplier only when it is active. By convexity these conditions are sufficient as well as necessary, completing the proof. $\square$

The costate recursion makes the mechanism transparent: away from the constraint boundary all $\kappa_{i,j}^*$ vanish and the recursion collapses to the unconstrained adjoint equation, while on an active boundary the term $\sum_j \kappa_{i,j}^* \nabla g_j(x_i^*)$ acts as a *reaction force* that bends the costate—and hence, through stationarity, the control—so as to keep the trajectory feasible.

10.4 ■ Convex Optimal Control with State Constraints: Normal-Cone Formulation

The previous section encoded the state constraints through explicit multipliers $\kappa_{i,j}$ attached to each inequality $g_j(x_i) \leq 0$. There is a more geometric and often more compact way to say the same thing: instead of listing the active constraints, one records that the costate residual must point “outward” from the feasible set. This is the language of the normal cone, and it has the advantage of applying to any closed convex constraint set X , however it happens to be described.

10.4.1 ■ Problem, Assumptions, and the Normal Cone

Consider the finite-horizon optimal control problem

$$\begin{aligned} \min_{\{u_i\}_{i=0}^{N-1}} \quad & J(u) := \sum_{i=0}^{N-1} \ell(x_i, u_i) + g_N(x_N) \\ \text{s.t.} \quad & x_{i+1} = f(x_i, u_i), \quad i = 0, \dots, N-1, \\ & x_i \in X, \quad i = 0, \dots, N, \\ & x_0 \text{ given,} \end{aligned} \tag{10.47}$$

where the state-constraint set is $X := \{x \in \mathbb{R}^n \mid g_j(x) \leq 0, j = 1, \dots, q\}$. We assume, as before, that f is affine in (x, u) , that $\ell(\cdot, \cdot)$ and $g_N(\cdot)$ are convex, that each g_j is convex and differentiable, that X is closed and convex, and that Slater's condition holds for X (there exists $\bar{x}$ with $g_j(\bar{x}) < 0$ for all j). Under these assumptions problem (10.47) is convex in the stacked variables $(x_1, \dots, x_N, u_0, \dots, u_{N-1})$.

The geometric object that replaces the multipliers is the normal cone. For a closed convex set $C \subset \mathbb{R}^n$ and a point $x \in C$, it is

$$N_C(x) := \{v \in \mathbb{R}^n \mid \langle v, y - x \rangle \leq 0 \quad \forall y \in C\}, \tag{10.48}$$

equivalently $N_C(x) = (T_C(x))^\circ$, the polar of the tangent cone. Two special cases fix the intuition: if $x \in \text{int}(C)$ then $N_C(x) = \{0\}$ (no constraint is active, so nothing is added to the costate), whereas on the boundary $N_C(x)$ is the cone of outward directions, generated by the gradients of the active constraints, which is exactly the multiplier combination $\sum_j \kappa_{i,j} \nabla g_j$ of the previous section. With the same stage Hamiltonian

$$H_i(x_i, u_i, p_{i+1}) = \ell(x_i, u_i) + p_{i+1}^\top f(x_i, u_i),$$

we can now state the optimality conditions in normal-cone form.

Theorem 10.6 (First-Order Optimality: Normal-Cone Form). *Let (x^*, u^*) be a feasible trajectory of (10.47). Then (x^*, u^*) is optimal if and only if there exist costate vectors $p_1^*, \dots, p_N^*$ such that the following conditions hold:*

- (i) primal feasibility: $x_{i+1}^* = f(x_i^*, u_i^*)$ and $x_i^* \in X$;
- (ii) control stationarity: $\nabla_{u_i} H_i(x_i^*, u_i^*, p_{i+1}^*) = 0, \quad i = 0, \dots, N-1$;
- (iii) costate inclusion: for $i = 1, \dots, N-1$,

$$p_i^* - \nabla_{x_i} H_i(x_i^*, u_i^*, p_{i+1}^*) \in N_X(x_i^*);$$

- (iv) terminal inclusion: $p_N^* - \nabla g_N(x_N^*) \in N_X(x_N^*)$.

Proof. The problem is convex with a differentiable objective and a closed convex feasible set, so (x^*, u^*) is optimal if and only if it is stationary, i.e. the directional derivative of the objective is nonnegative along every feasible direction. We show that the three stated conditions are this stationarity written stage by stage.

Adjoin the dynamics with costates p_{i+1} to form the Lagrangian $\mathcal{L} = \sum_i \ell(x_i, u_i) + g_N(x_N) + \sum_i p_{i+1}^\top (f(x_i, u_i) - x_{i+1})$, keeping the constraints $x_i \in X$ explicit rather than dualized. Because the dynamics are equality constraints, the controls u_i are unconstrained directions, and stationarity

of $\mathcal{L}$ in u_i gives, exactly as in the unconstrained proof, $\nabla_{u_i} H_i(x_i^*, u_i^*, p_{i+1}^*) = 0$. The state x_i ($1 \leq i \leq N-1$), by contrast, is restricted to X . A convex function is minimized over a convex set X at x_i^* if and only if its gradient lies in the negative normal cone, i.e. the descent directions are blocked by X :

$$-\nabla_{x_i} \mathcal{L}|_{x_i^*} \in N_X(x_i^*) \iff -(\nabla_{x_i} \ell + \nabla_{x_i} f p_{i+1}^* - p_i^*) \in N_X(x_i^*).$$

Writing $\nabla_{x_i} H_i = \nabla_{x_i} \ell + \nabla_{x_i} f p_{i+1}^*$, the vector inside the parentheses is $\nabla_{x_i} H_i - p_i^*$, so its negative is $p_i^* - \nabla_{x_i} H_i$; the inclusion therefore reads

$$p_i^* - \nabla_{x_i} H_i(x_i^*, u_i^*, p_{i+1}^*) \in N_X(x_i^*),$$

the stated costate inclusion. The identical argument at $i = N$, where the only cost term is $g_N(x_N)$, gives $p_N^* - \nabla g_N(x_N^*) \in N_X(x_N^*)$. Feasibility is assumed. Since the problem is convex these stationarity conditions are also sufficient, proving the equivalence. $\square$

The normal-cone inclusion is intrinsic: it refers only to the set X , not to any particular inequality description of it. When X is presented explicitly through smooth constraints, however, one usually wants the multiplier form back. Slater's condition lets us expand the normal cone as a nonnegative combination of active-constraint gradients, and the next theorem records the resulting KKT system; it also accommodates equality constraints $h_k(x) = 0$ describing X , whose (free-sign) multipliers we denote $\nu_{i,k}$.

Theorem 10.7 (KKT Conditions with Standard Equality Form). *Let (x^*, u^*) be feasible for (10.47). Then (x^*, u^*) is optimal if and only if there exist $p_1^*, \dots, p_N^*$, $\nu_{i,k}^* \in \mathbb{R}$, and $\kappa_{i,j}^* \geq 0$ such that the following conditions hold:*

- (i) primal feasibility: $x_{i+1}^* = f(x_i^*, u_i^*)$, $h_k(x_i^*) = 0$, and $g_j(x_i^*) \leq 0$;
- (ii) control stationarity: $\nabla_{u_i} H_i(x_i^*, u_i^*, p_{i+1}^*) = 0$;
- (iii) costate recursion (collecting equality and inequality multipliers), for $i = 1, \dots, N-1$,

$$p_i^* = \nabla_{x_i} H_i(x_i^*, u_i^*, p_{i+1}^*) + \sum_{k=1}^r \nu_{i,k}^* \nabla h_k(x_i^*) + \sum_{j=1}^q \kappa_{i,j}^* \nabla g_j(x_i^*);$$

- (iv) terminal condition:

$$p_N^* = \nabla g_N(x_N^*) + \sum_{k=1}^r \nu_{N,k}^* \nabla h_k(x_N^*) + \sum_{j=1}^q \kappa_{N,j}^* \nabla g_j(x_N^*);$$

- (v) complementary slackness: $\kappa_{i,j}^* g_j(x_i^*) = 0$ and $\kappa_{i,j}^* \geq 0$.

Proof. This is the normal-cone theorem with $N_X(x_i^*)$ written explicitly. Under Slater's condition the normal cone of $X = \{x : h_k(x) = 0, g_j(x) \leq 0\}$ at a feasible point is

$$N_X(x_i^*) = \left\{ \sum_k \nu_{i,k} \nabla h_k(x_i^*) + \sum_j \kappa_{i,j} \nabla g_j(x_i^*) : \nu_{i,k} \in \mathbb{R}, \kappa_{i,j} \geq 0, \kappa_{i,j} g_j(x_i^*) = 0 \right\},$$

i.e. it is generated by the equality-constraint gradients (with free sign) and the *active* inequality gradients (with nonnegative sign). Substituting this representation into the inclusion $p_i^* - \nabla_{x_i} H_i \in N_X(x_i^*)$ of the previous theorem yields the stated equality-form costate recursion, and the requirement that inactive constraints contribute nothing is exactly complementary slackness $\kappa_{i,j}^* g_j(x_i^*) = 0$. The terminal condition is obtained the same way at $i = N$. $\square$

10.5 ■ Infinite-Horizon Limit and Stationary Case

10.5.1 ■ Infinite-Horizon Problem and Stationary Solutions

Consider the time-invariant infinite-horizon optimal control problem

$$\begin{aligned} \min_{\{u_i\}_{i=0}^{\infty}} \quad & \sum_{i=0}^{\infty} \ell(x_i, u_i) \\ \text{s.t.} \quad & x_{i+1} = f(x_i, u_i), \quad i \geq 0, \\ & x_i \in X, \quad i \geq 0, \\ & x_0 \text{ given,} \end{aligned} \tag{10.49}$$

where $X := \{x \in \mathbb{R}^n \mid g_j(x) \leq 0, j = 1, \dots, q\}$ is closed and convex. We keep the standing assumptions of the previous section: f is affine in (x, u) , ℓ is convex and differentiable, each g_j is convex and differentiable, and Slater's condition holds for X . In addition we assume that a stationary optimal solution exists, a hypothesis that must be posited separately, since an infinite sum of stage costs need not converge and a steady state need not be reachable.

As the horizon lengthens, one expects a well-posed time-invariant problem to settle into a steady state in which nothing changes from stage to stage. A *stationary solution* is a triple $(\bar{x}, \bar{u}, \bar{p})$ for which $x_i^* = \bar{x}$, $u_i^* = \bar{u}$, and $p_i^* = \bar{p}$ for all i . Constancy of the state forces the dynamics to be self-reproducing, $\bar{x} = f(\bar{x}, \bar{u})$: the control must exactly maintain the state. The next theorem shows what remains of the forward-backward optimality system once every quantity is frozen in time.

Theorem 10.8 (Stationary Reduction of the Normal-Cone Optimality System). *Assume the infinite-horizon problem (10.49) admits a stationary optimal solution $(\bar{x}, \bar{u}, \bar{p})$. Then the triple $(\bar{x}, \bar{u}, \bar{p})$ satisfies:*

- (i) steady-state feasibility: $\bar{x} = f(\bar{x}, \bar{u})$ and $\bar{x} \in X$;
- (ii) control stationarity: $\nabla_u \ell(\bar{x}, \bar{u}) + (D_u f(\bar{x}, \bar{u}))^\top \bar{p} = 0$;
- (iii) stationary costate inclusion:

$$\nabla_x \ell(\bar{x}, \bar{u}) + (D_x f(\bar{x}, \bar{u}))^\top \bar{p} - \bar{p} \in N_X(\bar{x}).$$

Proof. The claim follows by freezing the finite-horizon normal-cone optimality conditions (Theorem 10.6) at their common stationary value. For each stage that system gives control stationarity $\nabla_{u_i} H(x_i^*, u_i^*, p_{i+1}^*) = 0$ together with the costate inclusion $p_i^* - \nabla_{x_i} H(x_i^*, u_i^*, p_{i+1}^*) \in N_X(x_i^*)$. Substituting the stationary values $x_i^* = \bar{x}$, $u_i^* = \bar{u}$, and $p_i^* = \bar{p}$ and using $H(x, u, p) = \ell(x, u) + p^\top f(x, u)$, whose gradients are $\nabla_u H = \nabla_u \ell + (D_u f)^\top p$ and $\nabla_x H = \nabla_x \ell + (D_x f)^\top p$, the control condition becomes $\nabla_u \ell(\bar{x}, \bar{u}) + (D_u f(\bar{x}, \bar{u}))^\top \bar{p} = 0$, and the costate inclusion becomes

$$\bar{p} - \nabla_x H(\bar{x}, \bar{u}, \bar{p}) \in N_X(\bar{x}) \iff \nabla_x \ell(\bar{x}, \bar{u}) + (D_x f(\bar{x}, \bar{u}))^\top \bar{p} - \bar{p} \in N_X(\bar{x}).$$

These are conditions (ii) and (iii); feasibility (i) holds because a stationary optimal solution is by definition admissible, so $\bar{x} = f(\bar{x}, \bar{u})$ and $\bar{x} \in X$. $\square$

10.5.2 ■ Interpretation: Static KKT and the Role of the Normal Cone

The stationary conditions of Theorem 10.8 deserve to be read from three angles, which together explain why the infinite-horizon problem is, at its steady state, a purely static one.

Read first as an optimization problem in its own right, the stationary conditions are precisely the first-order optimality conditions for the static constrained optimization problem

$$\min_{x,u} \ell(x, u) \quad \text{s.t.} \quad x = f(x, u), \quad x \in X.$$

Indeed, control stationarity $\nabla_u \ell + (D_u f)^\top \bar{p} = 0$ and the costate inclusion are the stationarity of the Lagrangian $\ell(x, u) + \bar{p}^\top (x - f(x, u))$ of this static problem with respect to u and x . Thus, under stationarity, the infinite-horizon *dynamic* optimality conditions collapse into the KKT system of the steady-state problem: solving for a steady state and solving a one-shot optimization become the same task.

Read next as a concrete equation, the abstract inclusion becomes an explicit equation once we locate $\bar{x}$ relative to the boundary of X . If $\bar{x} \in \text{int}(X)$ then $N_X(\bar{x}) = \{0\}$ and the costate condition is the equality

$$\nabla_x \ell(\bar{x}, \bar{u}) + (D_x f(\bar{x}, \bar{u}))^\top \bar{p} - \bar{p} = 0.$$

If instead $\bar{x}$ lies on the boundary and Slater's condition holds, then $N_X(\bar{x}) = \text{cone}\{\nabla g_j(\bar{x}) \mid g_j(\bar{x}) = 0\}$, so there exist multipliers $\kappa_j \geq 0$ with

$$\nabla_x \ell(\bar{x}, \bar{u}) + (D_x f(\bar{x}, \bar{u}))^\top \bar{p} - \bar{p} = \sum_j \kappa_j \nabla g_j(\bar{x}),$$

which recovers the classical multiplier form.

Read finally as a balance of marginal effects, the stationary adjoint equation $\nabla_x \ell + (D_x f)^\top \bar{p} - \bar{p} \in N_X(\bar{x})$ is a *fixed-point balance condition*. The term $\nabla_x \ell$ is the marginal stage cost, $(D_x f)^\top \bar{p}$ is the marginal continuation effect transmitted through the dynamics, $-\bar{p}$ closes the loop because the costate reproduces itself, and the right-hand side supplies the reaction forces of the active state constraints. In this way infinite-horizon stationarity transforms the backward costate *recursion* into a steady-state *equilibrium*, the exact analogue of how an equilibrium replaces a trajectory in a dynamical system.

10.6 ■ Dynamic Programming Principle: Intuition, Structure, and Formalization

10.6.1 ■ Problem Setting

We consider the finite-horizon discrete-time optimal control problem

$$\min_{\{u_i\}_{i=0}^{N-1}} J(u) = g_N(x_N) + \sum_{i=0}^{N-1} \ell_i(x_i, u_i), \quad (10.50)$$

subject to the system dynamics

$$x_{i+1} = f_i(x_i, u_i), \quad i = 0, \dots, N-1, \quad (10.51)$$

with initial condition

$$x_0 \text{ given.} \quad (10.52)$$

The state sequence $(x_0, \dots, x_N)$ is uniquely determined by the control sequence $(u_0, \dots, u_{N-1})$ through the system equation.

10.6.2 ■ Time Indexing and Sequence of Events

Before analyzing the problem it is worth fixing the timing convention, since dynamic programming is sensitive to *when* the state is known and *when* the control is chosen. The time index runs over $i = 0, 1, \dots, N$, and at each time i the system occupies a state x_i . At each of the decision times $i = 0, \dots, N-1$ the current state x_i is observed, a control $u_i \in \mathcal{U}_i(x_i)$ is selected, a stage cost $\ell_i(x_i, u_i)$ is incurred, and the system moves to the next state $x_{i+1} = f_i(x_i, u_i)$. At the terminal time N no control is applied and only the terminal cost $g_N(x_N)$ is incurred. The realized history is thus the alternating sequence

$$x_0 \xrightarrow{u_0} x_1 \xrightarrow{u_1} x_2 \xrightarrow{u_2} \dots \xrightarrow{u_{N-1}} x_N, \quad \text{i.e.} \quad (x_0, u_0, x_1, u_1, \dots, x_{N-1}, u_{N-1}, x_N), \quad (10.53)$$

and the total cost accumulated along it is $J = \sum_{i=0}^{N-1} \ell_i(x_i, u_i) + g_N(x_N)$.

The essential timing convention is that the state x_i is *known* at the moment u_i is chosen. This makes it natural to let the control depend on the current state, $u_i = \mu_i(x_i)$, so that a decision rule is a sequence of feedback functions $\mu = (\mu_0, \dots, \mu_{N-1})$. This state-feedback information structure is precisely what allows the optimal cost from time i onward to be written recursively as

$$V_i(x_i) = \min_{u_i} [\ell_i(x_i, u_i) + V_{i+1}(x_{i+1})], \quad x_{i+1} = f_i(x_i, u_i), \quad (10.54)$$

a relation we now derive carefully.

10.6.3 ■ The Principle of Optimality

The key structural observation is that optimal trajectories are built from optimal sub-trajectories, a property known as *optimal substructure*, or Bellman's principle of optimality.

Proposition 10.9 (Principle of Optimality). *Let $(u_0^*, \dots, u_{N-1}^*)$ be an optimal control sequence for the problem started at x_0 , and let $(x_0^*, \dots, x_N^*)$ be the corresponding state trajectory. Then for every intermediate time k , the tail $(u_k^*, \dots, u_{N-1}^*)$ is optimal for the subproblem that starts at time k in state x_k^* , namely*

$$\min_{\{u_j\}_{j=k}^{N-1}} \left[g_N(x_N) + \sum_{j=k}^{N-1} \ell_j(x_j, u_j) \right], \quad x_k = x_k^*, \quad x_{j+1} = f_j(x_j, u_j).$$

Proof. The total cost splits at time k into a head and a tail that share only the state x_k^* :

$$J(u^*) = \underbrace{\sum_{j=0}^{k-1} \ell_j(x_j^*, u_j^*)}_{\text{head, fixed}} + \underbrace{\left[\sum_{j=k}^{N-1} \ell_j(x_j^*, u_j^*) + g_N(x_N^*) \right]}_{\text{tail} =: J_k(\text{tail from } x_k^*)}.$$

The head depends only on the controls $u_0^*, \dots, u_{k-1}^*$ and is unaffected by any change made from time k onward, because such a change leaves $x_0^*, \dots, x_k^*$ untouched (the dynamics up to time k have already been executed). Suppose, for contradiction, that the tail $(u_k^*, \dots, u_{N-1}^*)$ were *not* optimal for the subproblem started at (k, x_k^*) . Then there would exist an admissible tail $(\tilde{u}_k, \dots, \tilde{u}_{N-1})$, generating a trajectory $\tilde{x}_k = x_k^*, \tilde{x}_{k+1}, \dots, \tilde{x}_N$, with strictly smaller tail cost,

$$\sum_{j=k}^{N-1} \ell_j(\tilde{x}_j, \tilde{u}_j) + g_N(\tilde{x}_N) < \sum_{j=k}^{N-1} \ell_j(x_j^*, u_j^*) + g_N(x_N^*).$$

Splice this tail onto the original head to form the control sequence $(u_0^*, \dots, u_{k-1}^*, \tilde{u}_k, \dots, \tilde{u}_{N-1})$. Since the head cost is unchanged and the tail cost is strictly smaller, the spliced sequence has strictly smaller total cost than u^* , contradicting the optimality of u^* . Hence the tail is optimal for the subproblem. $\square$

This recursive property is the engine of dynamic programming: it licenses us to compute the optimal tail cost *once* for each state x_k and reuse it, rather than re-optimizing the future separately for every trajectory that passes through x_k .

10.6.4 ■ Value Function

To formalize the above intuition, we introduce the value function

$$V_i(x) = \min_{\{u_j\}_{j=i}^{N-1}} \left[g_N(x_N) + \sum_{j=i}^{N-1} \ell_j(x_j, u_j) \right], \quad (10.55)$$

where the state trajectory satisfies

$$x_{j+1} = f_j(x_j, u_j), \quad x_i = x. \quad (10.56)$$

Thus, $V_i(x)$ represents the minimal cost achievable from stage i onward, given that the system is currently in state x .

At the terminal stage, no further decisions are made, and therefore

$$V_N(x) = g_N(x). \quad (10.57)$$

10.6.5 ■ Dynamic Programming Principle

The recursive structure underlying the optimal control problem can be formalized through the dynamic programming principle. A two-stage problem provides the simplest useful illustration.

Consider first the two-stage problem

$$\min_{u_0, u_1} \left\{ \ell_0(x_0, u_0) + \ell_1(x_1, u_1) + g_2(x_2) \right\}, \quad (10.58)$$

subject to

$$x_1 = f_0(x_0, u_0), \quad x_2 = f_1(x_1, u_1). \quad (10.59)$$

Substituting the dynamics yields

$$\min_{u_0, u_1} \left\{ \ell_0(x_0, u_0) + g_1(f_0(x_0, u_0), u_1) + g_2(f_1(f_0(x_0, u_0), u_1)) \right\}. \quad (10.60)$$

We can separate the second-stage decision:

$$\min_{u_0} \left\{ \ell_0(x_0, u_0) + \underbrace{\min_{u_1} \left\{ g_1(f_0(x_0, u_0), u_1) + g_2(f_1(f_0(x_0, u_0), u_1)) \right\}}_{\text{optimal continuation from } x_1 = f_0(x_0, u_0)} \right\}. \quad (10.61)$$

Define the value function at stage 1:

$$V_1(x_1) = \min_{u_1} \left\{ \ell_1(x_1, u_1) + g_2(f_1(x_1, u_1)) \right\}. \quad (10.62)$$

Then the problem reduces to

$$V_0(x_0) = \min_{u_0} \left\{ \ell_0(x_0, u_0) + \underbrace{V_1(f_0(x_0, u_0))}_{\text{optimal cost-to-go}} \right\}. \quad (10.63)$$

This explicitly shows that the optimal decision at stage 0 depends on future decisions only through the value function V_1 , which summarizes the optimal continuation.

The same decomposition also explains why the Bellman condition is sufficient for optimality.

The sufficiency arises because the value function V_1 already encodes the *optimal continuation cost*. Thus, when optimizing u_0 , the future is not approximated or ignored; instead, each choice of u_0 is evaluated under the assumption that all subsequent decisions are taken optimally. In this sense, suboptimal continuations are excluded by the construction of the value function itself.

Let u_0^* satisfy

$$u_0^* \in \arg \min_{u_0} \left\{ \ell_0(x_0, u_0) + V_1(f_0(x_0, u_0)) \right\}, \quad (10.64)$$

and let $x_1^* = f_0(x_0, u_0^*)$. Suppose u_1^* satisfies

$$u_1^* \in \arg \min_{u_1} \left\{ \ell_1(x_1^*, u_1) + g_2(f_1(x_1^*, u_1)) \right\}. \quad (10.65)$$

Then, by definition of V_1 , we have

$$V_1(x_1^*) = \ell_1(x_1^*, u_1^*) + g_2(f_1(x_1^*, u_1^*)). \quad (10.66)$$

Hence,

$$V_0(x_0) = \ell_0(x_0, u_0^*) + \ell_1(x_1^*, u_1^*) + g_2(x_2^*), \quad (10.67)$$

where $x_2^* = f_1(x_1^*, u_1^*)$.

Now consider any admissible pair (u_0, u_1) . From the definition of V_1 ,

$$\ell_1(x_1, u_1) + g_2(x_2) \geq V_1(x_1), \quad (10.68)$$

which shows that no second-stage decision can improve upon the value function. Therefore,

$$\ell_0(x_0, u_0) + \ell_1(x_1, u_1) + g_2(x_2) \geq \ell_0(x_0, u_0) + V_1(f_0(x_0, u_0)). \quad (10.69)$$

By optimality of u_0^* ,

$$\ell_0(x_0, u_0) + V_1(f_0(x_0, u_0)) \geq \ell_0(x_0, u_0^*) + V_1(f_0(x_0, u_0^*)). \quad (10.70)$$

Combining the inequalities yields

$$\ell_0(x_0, u_0) + \ell_1(x_1, u_1) + g_2(x_2) \geq \ell_0(x_0, u_0^*) + \ell_1(x_1^*, u_1^*) + g_2(x_2^*). \quad (10.71)$$

Thus (u_0^*, u_1^*) is globally optimal.

The argument shows why the Bellman recursion is *sufficient* for optimality: it replaces the entire future optimization by its optimal value V_1 , so any alternative control sequence must incur a cost no smaller than this continuation value and therefore cannot outperform the policy

built by stagewise Bellman minimization. Consequently, if each stagewise decision minimizes $\ell_i(x, u) + V_{i+1}(f_i(x, u))$, the resulting policy is optimal for the overall problem.

Differentiating the same two-stage objective, rather than decomposing it, recovers the minimum principle of Section 10.1. Starting from

$$\min_{u_0, u_1} \left\{ \ell_0(x_0, u_0) + g_1(f_0(x_0, u_0), u_1) + g_2(f_1(f_0(x_0, u_0), u_1)) \right\}, \quad (10.72)$$

assume that all functions are differentiable and that the controls are unconstrained. Define

$$x_1 = f_0(x_0, u_0), \quad x_2 = f_1(x_1, u_1). \quad (10.73)$$

Then the objective can be written compactly as

$$J(u_0, u_1) = \ell_0(x_0, u_0) + \ell_1(x_1, u_1) + g_2(x_2). \quad (10.74)$$

Taking the gradient with respect to u_1 gives

$$\nabla_{u_1} J = \nabla_{u_1} \ell_1(x_1, u_1) + (\nabla_{u_1} f_1(x_1, u_1))^\top \nabla_{x_2} g_2(x_2). \quad (10.75)$$

Thus the first-order condition with respect to u_1 is

$$\nabla_{u_1} \ell_1(x_1, u_1) + (\nabla_{u_1} f_1(x_1, u_1))^\top \nabla_{x_2} g_2(x_2) = 0. \quad (10.76)$$

Next, taking the gradient with respect to u_0 gives

$$\nabla_{u_0} J = \nabla_{u_0} \ell_0(x_0, u_0) + (\nabla_{u_0} f_0(x_0, u_0))^\top \left[\nabla_{x_1} \ell_1(x_1, u_1) + (\nabla_{x_1} f_1(x_1, u_1))^\top \nabla_{x_2} g_2(x_2) \right]. \quad (10.77)$$

Hence the first-order condition with respect to u_0 is

$$\nabla_{u_0} \ell_0(x_0, u_0) + (\nabla_{u_0} f_0(x_0, u_0))^\top \left[\nabla_{x_1} \ell_1(x_1, u_1) + (\nabla_{x_1} f_1(x_1, u_1))^\top \nabla_{x_2} g_2(x_2) \right] = 0. \quad (10.78)$$

Now define the costates

$$\lambda_2 := \nabla_{x_2} g_2(x_2), \quad (10.79)$$

and

$$\lambda_1 := \nabla_{x_1} \ell_1(x_1, u_1) + (\nabla_{x_1} f_1(x_1, u_1))^\top \lambda_2. \quad (10.80)$$

Then the first-order conditions become

$$\nabla_{u_1} \ell_1(x_1, u_1) + (\nabla_{u_1} f_1(x_1, u_1))^\top \lambda_2 = 0, \quad (10.81)$$

and

$$\nabla_{u_0} \ell_0(x_0, u_0) + (\nabla_{u_0} f_0(x_0, u_0))^\top \lambda_1 = 0. \quad (10.82)$$

Equivalently, define the Hamiltonian

$$H_i(x_i, u_i, \lambda_{i+1}) := \ell_i(x_i, u_i) + \lambda_{i+1}^\top f_i(x_i, u_i). \quad (10.83)$$

Then the necessary conditions can be written as

$$x_{i+1} = f_i(x_i, u_i), \quad (10.84)$$

$$\lambda_i = \nabla_{x_i} H_i(x_i, u_i, \lambda_{i+1}), \quad (10.85)$$

$$0 = \nabla_{u_i} H_i(x_i, u_i, \lambda_{i+1}), \quad (10.86)$$

with terminal condition

$$\lambda_2 = \nabla_{x_2} g_2(x_2). \quad (10.87)$$

Thus, taking gradients of the two-stage objective yields the discrete-time minimum principle: the optimal control must minimize the Hamiltonian locally, while the state evolves forward and the costate evolves backward.

10.6.6 ■ Bellman Recursion

The dynamic programming principle provides a recursive characterization of the value function through a system of backward equations.

Definition 10.10 (Bellman Recursion). *The value functions $\{V_i\}_{i=0}^N$ satisfy the recursion*

$$V_N(x) = g_N(x), \quad x \in \mathcal{X}_N, \quad (10.88)$$

and for each stage $i = N - 1, \dots, 0$,

$$V_i(x) = \min_{u \in \mathcal{U}_i(x)} [\ell_i(x, u) + V_{i+1}(f_i(x, u))], \quad x \in \mathcal{X}_i. \quad (10.89)$$

The above system of equations is referred to as the *Bellman equation*. It expresses the optimal cost-to-go at stage i as the minimum, over all admissible controls, of the sum of the immediate cost and the optimal cost-to-go from the next state.

The Bellman recursion has a clear computational interpretation. Starting from the terminal condition V_N , one computes $V_{N-1}, V_{N-2}, \dots, V_0$ successively by solving a sequence of one-step optimization problems. In this way, a global optimization problem over an entire control sequence is reduced to a sequence of local optimization problems indexed by time.

The recursion also has a structural interpretation. For each state x at stage i , the value $V_i(x)$ aggregates all future consequences of decisions taken at that state. The dependence on future stages is captured entirely through the value function V_{i+1} , which serves as a summary of the remaining optimization problem.

10.6.7 ■ Optimal Policy

The Bellman equation not only characterizes the optimal value, but also yields a constructive description of optimal decisions.

For each stage i and state $x \in \mathcal{X}_i$, define the set of minimizers

$$\mathcal{U}_i^*(x) = \arg \min_{u \in \mathcal{U}_i(x)} [\ell_i(x, u) + V_{i+1}(f_i(x, u))]. \quad (10.90)$$

Any selection $u_i^*(x) \in \mathcal{U}_i^*(x)$ defines an optimal control action at stage i when the system is in state x .

An *optimal policy* is a sequence of state-feedback mappings

$$\mu^* = (\mu_0^*, \dots, \mu_{N-1}^*), \quad (10.91)$$

where each mapping satisfies

$$\mu_i^*(x) \in \mathcal{U}_i^*(x), \quad x \in \mathcal{X}_i. \quad (10.92)$$

Given an initial state x_0 , the optimal trajectory is generated recursively by the closed-loop system

$$x_{i+1}^* = f_i(x_i^*, \mu_i^*(x_i^*)), \quad i = 0, \dots, N - 1, \quad (10.93)$$

with $x_0^* = x_0$.

This construction shows that optimal control can be implemented in *feedback form*: at each stage, the control decision depends only on the current state, and not on the entire history of the system. The optimal policy therefore provides a decentralized and time-consistent decision rule.

Finally, the pair (V_i, μ_i^*) satisfies the fundamental relation

$$V_i(x) = g_i(x, \mu_i^*(x)) + V_{i+1}(f_i(x, \mu_i^*(x))), \quad (10.94)$$

which shows that the optimal value is achieved by applying the optimal policy at each stage.

10.7 ■ Shortest-Path Problems: A Constructive Illustration

The structure induced by the dynamic programming principle can be understood concretely through a shortest-path problem defined on a layered graph. This interpretation makes explicit the relationship between sequential decision-making and path optimization.

Consider a directed graph organized into layers indexed by $i = 0, 1, 2, 3$. Each layer corresponds to a stage in time, and each node in layer i represents a state $x_i \in \mathcal{X}_i$. For illustration, suppose

$$\mathcal{X}_0 = \{x_0, y_0\}, \quad \mathcal{X}_1 = \{x_1, y_1\}, \quad \mathcal{X}_2 = \{x_2, y_2\}, \quad \mathcal{X}_3 = \{A, B\}.$$

Transitions are allowed only between consecutive layers. A transition from x_i to x_{i+1} corresponds to the application of a control u_i satisfying the system equation $x_{i+1} = f_i(x_i, u_i)$. Each such transition incurs a cost $\ell_i(x_i, u_i) = c_i(x_i, x_{i+1})$. At the terminal stage $i = 3$, no further control is applied and each terminal node x_3 is assigned a terminal cost $g_3(x_3)$.

A control sequence (u_0, u_1, u_2) generates a state trajectory $x_0 \rightarrow x_1 \rightarrow x_2 \rightarrow x_3$, which corresponds to a path through the graph. The total cost of this trajectory is $\ell_0(x_0, u_0) + \ell_1(x_1, u_1) + \ell_2(x_2, u_2) + g_3(x_3)$, which is exactly the cost of the corresponding path.

To make this concrete, suppose the edge costs are given by

$$\begin{aligned} c_0(x_0, x_1) &= 1, & c_0(x_0, y_1) &= 4, \\ c_0(y_0, x_1) &= 2, & c_0(y_0, y_1) &= 1, \\ c_1(x_1, x_2) &= 3, & c_1(x_1, y_2) &= 2, \\ c_1(y_1, x_2) &= 4, & c_1(y_1, y_2) &= 1, \\ c_2(x_2, A) &= 2, & c_2(x_2, B) &= 5, \\ c_2(y_2, A) &= 3, & c_2(y_2, B) &= 1, \end{aligned}$$

and the terminal costs are $g_3(A) = 0, g_3(B) = 0$.

The goal is to determine the minimum-cost path from an initial node, say x_0 , to the terminal layer. Rather than enumerating all possible paths, dynamic programming computes the optimal cost recursively by working backward in time. Figure 10.4 shows the graph, the resulting cost-to-go values, and the optimal path.

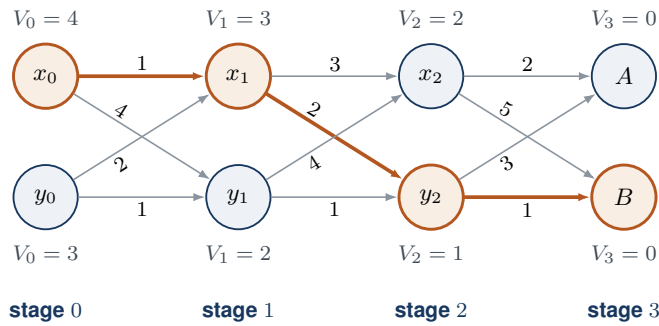


Figure 10.4: Shortest path on a layered graph. Each node carries its optimal cost-to-go V_i , computed backward by the Bellman recursion $V_i(x) = \min_{x_{i+1}} [c_i(x, x_{i+1}) + V_{i+1}(x_{i+1})]$; edge labels are the transition costs. The bold path $x_0 \rightarrow x_1 \rightarrow y_2 \rightarrow B$ is optimal, with total cost $V_0(x_0) = 4$.

At the terminal stage, the value function is given directly by $V_3(A) = 0, V_3(B) = 0$. At stage $i = 2$, the value function is obtained by minimizing over all possible successors: $V_2(x_2) = \min\{2 + V_3(A), 5 + V_3(B)\} = \min\{2, 5\} = 2$, $V_2(y_2) = \min\{3 + V_3(A), 1 + V_3(B)\} = \min\{3, 1\} = 1$. At stage $i = 1$, the same reasoning yields $V_1(x_1) = \min\{3 + V_2(x_2), 2 + V_2(y_2)\} = \min\{3 + 2, 2 + 1\} = 3$, $V_1(y_1) = \min\{4 + V_2(x_2), 1 + V_2(y_2)\} = \min\{4 + 2, 1 + 1\} = 2$. Finally, at stage $i = 0$, we obtain $V_0(x_0) = \min\{1 + V_1(x_1), 4 + V_1(y_1)\} = \min\{1 + 3, 4 + 2\} = 4$.

The value $V_0(x_0)$ represents the minimal cost of reaching the terminal layer starting from x_0 . The optimal path is recovered by selecting, at each stage, the successor that achieves the minimum in the recursion. In this example, the optimal trajectory is $x_0 \rightarrow x_1 \rightarrow y_2 \rightarrow B$, with total cost 4.

This computation illustrates that the value function $V_i(x)$ coincides with the shortest-path distance from node (i, x) to the terminal layer. The recursive computation performed above is an instance of the Bellman equation,

$$V_i(x) = \min_{x_{i+1}} [c_i(x, x_{i+1}) + V_{i+1}(x_{i+1})],$$

which characterizes shortest-path distances in a layered graph.

The example makes explicit the central idea of dynamic programming. The global optimization problem over an entire trajectory is decomposed into a sequence of local optimization problems, each involving only a single transition and the value of the remaining subproblem. The optimal path is constructed by combining locally optimal decisions, and the correctness of this construction follows from the principle of optimality.

10.8 ■ A Canonical Example: Resource Allocation over Time

This section illustrates the dynamic programming framework through a simple yet fundamental resource allocation problem. The example highlights the distinction between open-loop and feedback solutions, and demonstrates how the Bellman recursion reveals the structure of optimal decisions.

10.8.1 ■ Problem Formulation

Consider a finite horizon indexed by $i = 0, 1, \dots, N - 1$. A total resource $C > 0$ must be allocated over time.

At each stage i , a control variable $u_i \in \mathbb{R}_+$ represents the amount of resource used at that stage. The total resource constraint is

$$\sum_{i=0}^{N-1} u_i = C. \quad (10.95)$$

The objective is to minimize the quadratic cost

$$J(u) = \sum_{i=0}^{N-1} \ell_i(u_i), \quad \ell_i(u_i) = u_i^2. \quad (10.96)$$

The quadratic structure penalizes uneven allocations, as large values of u_i incur disproportionately higher cost.

10.8.2 ■ Open-Loop Solution: First Look

We first consider the problem as a static optimization over the control sequence $u = (u_0, \dots, u_{N-1})$:

$$\min_{u_i \geq 0} \sum_{i=0}^{N-1} u_i^2 \quad \text{s.t.} \quad \sum_{i=0}^{N-1} u_i = C. \quad (10.97)$$

This is a convex optimization problem. Introducing a Lagrange multiplier λ for the equality constraint, the optimality conditions are

$$2u_i - \lambda = 0, \quad i = 0, \dots, N-1. \quad (10.98)$$

Thus, all optimal controls are equal, and enforcing the constraint yields

$$u_i^* = \frac{C}{N}, \quad i = 0, \dots, N-1. \quad (10.99)$$

This solution is referred to as an *open-loop* solution, as the entire control sequence is determined in advance, based solely on the initial condition.

10.8.3 ■ Dynamic Programming Formulation

The problem can be reformulated as a sequential decision problem by introducing the state variable

$$x_i = \text{remaining resource at stage } i, \quad (10.100)$$

with initial condition $x_0 = C$.

The system evolves according to

$$x_{i+1} = f_i(x_i, u_i) = x_i - u_i, \quad (10.101)$$

subject to the constraint $0 \leq u_i \leq x_i$.

The cost function can be written as

$$J = \sum_{i=0}^{N-1} u_i^2, \quad (10.102)$$

with the terminal allocation condition

$$x_N = 0. \quad (10.103)$$

Equivalently, any unused resource at the terminal time is infeasible; all resource must be allocated by the end of the horizon.

10.8.4 ■ Value Function and Bellman Equation

Define the value function

$$V_i(x) = \min_{\{u_j\}_{j=i}^{N-1}} \left[\sum_{j=i}^{N-1} u_j^2 \right], \quad (10.104)$$

where the state trajectory satisfies $x_{j+1} = x_j - u_j$, $x_i = x$, and $x_N = 0$.

The Bellman recursion is given by

$$V_N(0) = 0, \quad V_N(x) = +\infty \quad \text{for } x \neq 0, \quad (10.105)$$

$$V_i(x) = \min_{0 \leq u \leq x} [u^2 + V_{i+1}(x - u)], \quad i = N - 1, \dots, 0. \quad (10.106)$$

10.8.5 ■ Quadratic Structure and Value Function

For $i = 0, \dots, N - 1$, seek a value function of the form

$$V_i(x) = m_i x^2, \quad (10.107)$$

for some coefficients $m_i > 0$.

At the last decision stage, the terminal condition $x_N = 0$ forces $u_{N-1} = x$, and hence $V_{N-1}(x) = x^2$ so that $m_{N-1} = 1$. For $i = 0, \dots, N - 2$, substituting the quadratic form into the Bellman equation yields

$$m_i x^2 = \min_{0 \leq u \leq x} [u^2 + m_{i+1}(x - u)^2]. \quad (10.108)$$

This is a quadratic minimization problem in u . Differentiating and setting the derivative to zero gives

$$2u - 2m_{i+1}(x - u) = 0, \quad (10.109)$$

which implies

$$u_i^*(x) = \frac{m_{i+1}}{1 + m_{i+1}} x. \quad (10.110)$$

Substituting the optimal control into the Bellman equation yields the backward recursion

$$m_i = \frac{m_{i+1}}{1 + m_{i+1}}, \quad m_{N-1} = 1. \quad (10.111)$$

Solving this recursion gives

$$m_i = \frac{1}{N - i}. \quad (10.112)$$

10.8.6 ■ Feedback Optimal Policy

The optimal control law is therefore

$$u_i^*(x_i) = \frac{1}{N - i} x_i, \quad i = 0, \dots, N - 1. \quad (10.113)$$

This policy has a natural interpretation. At stage i , the remaining resource is x_i , and the number of remaining stages is $N - i$. The optimal decision allocates the resource evenly across the remaining horizon.

The corresponding closed-loop trajectory satisfies

$$x_{i+1}^* = \left(1 - \frac{1}{N - i}\right) x_i^*, \quad x_0^* = C. \quad (10.114)$$

10.8.7 ■ Open-Loop Solution: Two Derivations

Returning to the open-loop problem of Section 10.8.2, we determine the optimal control sequence in open-loop form, that is, as a sequence $(u_0, \dots, u_{N-1})$ chosen at the initial time based solely on the initial resource C . The same solution can be derived using two complementary approaches: static optimization via KKT conditions, and dynamic optimization via the discrete-time minimum principle.

Static Optimization via KKT Conditions

Consider the problem

$$\min_{u_i \geq 0} \sum_{i=0}^{N-1} u_i^2 \quad \text{s.t.} \quad \sum_{i=0}^{N-1} u_i = C. \quad (10.115)$$

This is a convex optimization problem with a linear equality constraint. The Lagrangian is

$$\mathcal{L}(u, \lambda) = \sum_{i=0}^{N-1} u_i^2 + \lambda \left(\sum_{i=0}^{N-1} u_i - C \right). \quad (10.116)$$

The Karush–Kuhn–Tucker (KKT) stationarity conditions are

$$\frac{\partial \mathcal{L}}{\partial u_i} = 2u_i + \lambda = 0, \quad i = 0, \dots, N-1. \quad (10.117)$$

Thus, all optimal controls must be equal:

$$u_i = -\frac{\lambda}{2}, \quad i = 0, \dots, N-1. \quad (10.118)$$

Enforcing the constraint $\sum_{i=0}^{N-1} u_i = C$ yields

$$u_i^* = \frac{C}{N}, \quad i = 0, \dots, N-1. \quad (10.119)$$

Dynamic Optimization via the Minimum Principle

The same solution can be derived using a dynamic formulation. Define the state

$$x_{i+1} = x_i - u_i, \quad x_0 = C, \quad (10.120)$$

and consider the cost together with the terminal allocation condition

$$J = \sum_{i=0}^{N-1} u_i^2, \quad x_N = 0. \quad (10.121)$$

Define the Hamiltonian

$$H_i(x_i, u_i, p_{i+1}) = u_i^2 + p_{i+1}(x_i - u_i). \quad (10.122)$$

The discrete-time minimum principle requires:

Costate recursion

$$p_i = \frac{\partial H_i}{\partial x_i} = p_{i+1}, \quad (10.123)$$

which implies that the costate is constant:

$$p_i = p, \quad i = 0, \dots, N. \quad (10.124)$$

Stationarity condition

$$\frac{\partial H_i}{\partial u_i} = 2u_i - p = 0, \quad (10.125)$$

which yields

$$u_i^* = \frac{p}{2}, \quad i = 0, \dots, N-1. \quad (10.126)$$

Using the constraint $\sum_{i=0}^{N-1} u_i = C$, we obtain

$$u_i^* = \frac{C}{N}, \quad i = 0, \dots, N-1. \quad (10.127)$$

Equivalence and Interpretation

Both methods yield the same open-loop solution

$$u_i^* = \frac{C}{N}, \quad i = 0, \dots, N-1. \quad (10.128)$$

Moreover, comparing the stationarity conditions,

$$2u_i + \lambda = 0 \quad (\text{KKT}), \quad 2u_i - p = 0 \quad (\text{minimum principle}),$$

we obtain the correspondence

$$p = -\lambda. \quad (10.129)$$

Thus, the costate variable in the dynamic formulation coincides (up to sign) with the Lagrange multiplier in the static formulation.

This equivalence reveals that the minimum principle is a dynamic extension of the KKT conditions. The optimal allocation satisfies the condition that the marginal cost $\frac{d}{du_i}(u_i^2) = 2u_i$ is equal across all stages, leading to a uniform allocation of the resource.

10.8.8 ■ Equivalence of Open-Loop and Feedback Solutions

From the dynamic programming formulation, the optimal feedback policy is given by

$$u_i^*(x_i) = \frac{1}{N-i} x_i. \quad (10.130)$$

Substituting the closed-loop trajectory

$$x_i = C - \sum_{j=0}^{i-1} u_j \quad (10.131)$$

into the policy yields

$$u_i^* = \frac{1}{N-i} \left(C - \sum_{j=0}^{i-1} u_j^* \right). \quad (10.132)$$

By induction, one verifies that

$$u_i^* = \frac{C}{N}, \quad i = 0, \dots, N-1. \quad (10.133)$$

Thus, in the absence of uncertainty, the feedback policy reproduces the open-loop allocation.

10.8.9 ■ Performance Degradation under Process and Measurement Noise

The following analysis separates two distinct sources of uncertainty and their impact on performance.

The performance metric is

$$J = \sum_{i=0}^{N-1} u_i^2 + x_N^2, \quad J^* = \frac{C^2}{N}, \quad (10.134)$$

and the degradation is

$$\Delta J = \mathbb{E}[J - J^*]. \quad (10.135)$$

Case I: Process Noise Only

Consider

$$x_{i+1} = x_i - u_i + \varepsilon_i, \quad x_0 = C, \quad (10.136)$$

where $\{\varepsilon_i\}$ are independent with zero mean and variance σ^2 . We compare the two representations of the optimal solution: the open-loop sequence, which is committed at time zero, and the feedback law, which reacts to the realized state. Under the open-loop schedule the disturbances accumulate without correction.

Proposition 10.11 (Terminal spread under open-loop allocation). *Under $u_i = \frac{C}{N}$,*

$$x_N = \sum_{i=0}^{N-1} \varepsilon_i, \quad \text{Var}(x_N) = N\sigma^2. \quad (10.137)$$

Proof. Unrolling the recursion, $x_{i+1} = x_i - \frac{C}{N} + \varepsilon_i$. Summing from 0 to $N - 1$,

$$x_N = x_0 - \sum_{i=0}^{N-1} \frac{C}{N} + \sum_{i=0}^{N-1} \varepsilon_i = C - C + \sum_{i=0}^{N-1} \varepsilon_i.$$

The variance follows from independence. $\square$

The feedback law, by contrast, reallocates the remaining resource at every stage and so absorbs each disturbance as it occurs.

Proposition 10.12 (Cost of the feedback policy under process noise). *Under*

$$u_i^*(x_i) = \frac{1}{N-i} x_i, \quad (10.138)$$

the terminal state satisfies

$$x_N = \varepsilon_{N-1}, \quad (10.139)$$

and the expected cost is

$$\mathbb{E}[J_{\text{FB}}] = \frac{C^2}{N} + \sigma^2 \left(1 + \sum_{r=1}^{N-1} \frac{1}{r} \right). \quad (10.140)$$

Proof. Deviation dynamics. Decompose the state into its noise-free (nominal) value and a stochastic deviation, $x_i = \frac{N-i}{N}C + \tilde{x}_i$. Substituting into $x_{i+1} = x_i - u_i + \varepsilon_i$ with $u_i = \frac{1}{N-i}x_i$, the nominal parts cancel by the deterministic analysis, and the deviation obeys the linear recursion

$$\tilde{x}_{i+1} = \left(1 - \frac{1}{N-i}\right)\tilde{x}_i + \varepsilon_i = \frac{N-i-1}{N-i}\tilde{x}_i + \varepsilon_i, \quad \tilde{x}_0 = 0.$$

Solving the recursion (telescoping). Unrolling gives $\tilde{x}_i = \sum_{k=0}^{i-1} \left(\prod_{j=k+1}^{i-1} \frac{N-j-1}{N-j} \right) \varepsilon_k$. The product telescopes: numerator and denominator are consecutive integers, so

$$\prod_{j=k+1}^{i-1} \frac{N-j-1}{N-j} = \frac{(N-k-2)(N-k-3)\cdots(N-i)}{(N-k-1)(N-k-2)\cdots(N-i+1)} = \frac{N-i}{N-k-1}.$$

Hence the coefficient of ε_k in $\tilde{x}_i$ is $\frac{N-i}{N-k-1}$ for $k = 0, \dots, i-1$.

Control effort. The random part of the control $u_i = x_i/(N-i)$ is $\tilde{x}_i/(N-i)$, whose coefficient on ε_k is $\frac{1}{N-k-1}$. By independence of the ε_k ,

$$\text{Var}(u_i) = \sigma^2 \sum_{k=0}^{i-1} \frac{1}{(N-k-1)^2}.$$

Summing over all stages and *swapping the order of summation* (for a fixed k , the index i runs from $k+1$ to $N-1$, contributing $N-1-k$ equal terms),

$$\sum_{i=0}^{N-1} \text{Var}(u_i) = \sigma^2 \sum_{k=0}^{N-2} \frac{N-1-k}{(N-k-1)^2} = \sigma^2 \sum_{k=0}^{N-2} \frac{1}{N-k-1} = \sigma^2 \sum_{r=1}^{N-1} \frac{1}{r},$$

where the last step substitutes $r = N-k-1$. This is the additional expected control effort, growing only logarithmically in N (a harmonic sum), in sharp contrast to the open-loop growth $N\sigma^2$.

Terminal state. At the last stage the control $u_{N-1} = x_{N-1}$ removes the entire current state before the final disturbance is realized, so $x_N = x_{N-1} - u_{N-1} + \varepsilon_{N-1} = \varepsilon_{N-1}$ and $\mathbb{E}[x_N^2] = \sigma^2$.

Assembling the cost. Since $\mathbb{E}[u_i^2] = (C/N)^2 + \text{Var}(u_i)$, summing the nominal control cost $\sum_i (C/N)^2 = C^2/N$, the extra control effort $\sigma^2 \sum_{r=1}^{N-1} \frac{1}{r}$, and the terminal variance σ^2 yields $\mathbb{E}[J_{\text{FB}}] = \frac{C^2}{N} + \sigma^2 \left(1 + \sum_{r=1}^{N-1} \frac{1}{r}\right)$, as claimed. $\square$

Comparing the two policies,

$$\mathbb{E}[J_{\text{OL}}] = \frac{C^2}{N} + N\sigma^2, \quad \mathbb{E}[J_{\text{FB}}] = \frac{C^2}{N} + \sigma^2 \left(1 + \sum_{r=1}^{N-1} \frac{1}{r}\right), \quad (10.141)$$

we see that the open-loop excess cost $N\sigma^2$ grows linearly in the horizon, whereas the feedback excess cost grows only like $\sigma^2 \ln N$, since the harmonic sum satisfies $\sum_{r=1}^{N-1} \frac{1}{r} = \ln N + O(1)$. Figure 10.5 plots the two curves; the widening gap is the quantitative payoff of closing the loop.

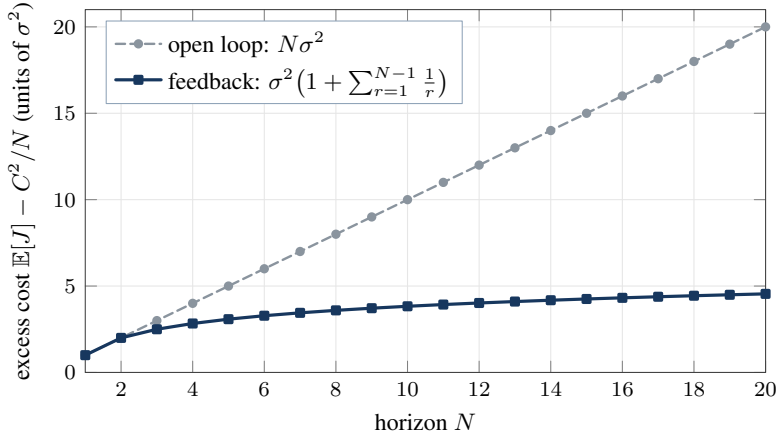


Figure 10.5: Cost of process noise under the two representations of the optimal resource-allocation solution. The pre-committed open-loop schedule lets disturbances accumulate in the terminal state, so its excess cost grows *linearly* in the horizon N . The feedback policy reallocates the remaining resource at every stage, and its excess cost grows only *logarithmically* (a harmonic sum). Along the noise-free trajectory the two policies coincide; the gap opens only once uncertainty is present.

Case II: Measurement Noise Only

Consider

$$x_{i+1} = x_i - u_i, \quad x_0 = C, \quad (10.142)$$

and

$$y_i = x_i + \eta_i, \quad (10.143)$$

where $\{\eta_i\}$ are independent with variance τ^2 . Here the state evolves without disturbance, but the controller can only act on a corrupted reading of it. Applying the feedback gain to the measurement gives the perturbed policy

$$u_i = \frac{1}{N-i} y_i. \quad (10.144)$$

Proposition 10.13 (Cost of the feedback policy under measurement noise). *The terminal state satisfies*

$$x_N = -\eta_{N-1}, \quad (10.145)$$

and the expected cost is

$$\mathbb{E}[J_{\text{FB}}] = \frac{C^2}{N} + \tau^2 \left(1 + \sum_{r=1}^N \frac{1}{r^2} + \sum_{r=2}^N \frac{1}{r^2(r-1)} \right). \quad (10.146)$$

Proof. *Perturbed closed loop.* The controller acts on the noisy measurement $y_i = x_i + \eta_i$, so $u_i = \frac{1}{N-i} x_i + \frac{1}{N-i} \eta_i$, and substituting into the (noise-free) dynamics gives

$$x_{i+1} = x_i - u_i = \left(1 - \frac{1}{N-i}\right) x_i - \frac{1}{N-i} \eta_i.$$

This has exactly the form of the process-noise recursion with the disturbance ε_i replaced by $-\frac{1}{N-i}\eta_i$; in particular the state deviation is again a telescoping sum of the past shocks with the same propagation coefficients $\frac{N-i}{N-k-1}$.

Terminal state. At the last stage $u_{N-1} = y_{N-1} = x_{N-1} + \eta_{N-1}$, so $x_N = x_{N-1} - u_{N-1} = -\eta_{N-1}$ and $\mathbb{E}[x_N^2] = \tau^2$.

Control effort. Each control carries two independent measurement-error contributions: the *current* error η_i entering through the term $\frac{1}{N-i}\eta_i$, and the *accumulated* past errors entering through the perturbed state x_i . Squaring u_i , taking expectations, and using independence separates these two channels. Summing the direct terms over the horizon and reindexing by $r = N - i$ produces $\tau^2 \sum_{r=1}^N 1/r^2$, while summing the accumulated terms, each weighted by the squared propagation coefficients and telescoped as in Case I, produces $\tau^2 \sum_{r=2}^N 1/(r^2(r-1))$. Both are *square-summable* series, so unlike the harmonic growth caused by process noise they remain bounded as $N \rightarrow \infty$. Adding the terminal variance τ^2 and the nominal cost C^2/N gives the stated expression. $\square$

Combined Noise

Proposition 10.14 (Cost under both noise sources). *When both noises are present,*

$$x_{i+1} = x_i - u_i + \varepsilon_i, \quad y_i = x_i + \eta_i, \quad (10.147)$$

the terminal state satisfies

$$x_N = \varepsilon_{N-1} - \eta_{N-1}, \quad (10.148)$$

and the expected cost is

$$\mathbb{E}[J_{\text{FB}}] = \frac{C^2}{N} + \sigma^2 \left(1 + \sum_{r=1}^{N-1} \frac{1}{r} \right) + \tau^2 \left(1 + \sum_{r=1}^N \frac{1}{r^2} + \sum_{r=2}^N \frac{1}{r^2(r-1)} \right). \quad (10.149)$$

Proof. The closed-loop system is linear in the disturbances and the feedback law is linear in the observation, so the deviation of the trajectory from its nominal value is the *superposition* of the deviation produced by the process noise $\{\varepsilon_i\}$ alone (Case I) and that produced by the measurement noise $\{\eta_i\}$ alone (Case II). Adding the two single-source terminal states $x_N = \varepsilon_{N-1}$ and $x_N = -\eta_{N-1}$ gives $x_N = \varepsilon_{N-1} - \eta_{N-1}$.

For the cost, write each control and the terminal state as (nominal) + (ε -part) + (η -part). Squaring and taking expectations, every cross term between an ε -part and an η -part carries a factor $\mathbb{E}[\varepsilon_k \eta_\ell] = 0$ (the two noise sequences are independent and zero-mean) and hence vanishes. The expected cost therefore splits additively into the nominal cost C^2/N , the process-noise contribution $\sigma^2(1 + \sum_{r=1}^{N-1} \frac{1}{r})$ computed in Case I, and the measurement-noise contribution $\tau^2(1 + \sum_{r=1}^N \frac{1}{r^2} + \sum_{r=2}^N \frac{1}{r^2(r-1)})$ computed in Case II. Summing these three pieces yields the stated expression. $\square$

Interpretation

Process noise and measurement noise affect the system through distinct channels:

$$\begin{aligned} \text{Process noise:} & \quad \text{state disturbance,} \\ \text{Measurement noise:} & \quad \text{control disturbance.} \end{aligned} \quad (10.150)$$

Feedback prevents disturbances from accumulating indefinitely in the terminal state because each subsequent decision reallocates the remaining resource. Process noise must still be corrected through future control actions, producing a harmonic growth in expected control effort, whereas measurement noise enters the control law directly and is governed by square-summable terms. Thus robust implementation requires both feedback and accurate state information.

10.9 ■ Time Consistency and Representation

This section presents the general theory of time consistency and solution representation in finite-horizon dynamic optimization, and then illustrates these concepts using the resource allocation example.

10.9.1 ■ Time Consistency

Consider a finite-horizon optimal control problem with dynamics

$$x_{i+1} = f_i(x_i, u_i), \quad i = 0, \dots, N-1, \quad (10.151)$$

and cost functional

$$J = \sum_{i=0}^{N-1} \ell_i(x_i, u_i) + g_N(x_N). \quad (10.152)$$

Let the value functions $\{V_i\}_{i=0}^N$ be defined recursively by the Bellman equations

$$V_N(x) = g_N(x), \quad (10.153)$$

$$V_i(x) = \min_{u \in \mathcal{U}_i(x)} [\ell_i(x, u) + V_{i+1}(f_i(x, u))], \quad i = N-1, \dots, 0. \quad (10.154)$$

Definition 10.15 (Time-Consistent Policy). A policy $\mu = (\mu_0, \dots, \mu_{N-1})$ is said to be time-consistent if, for every stage i and every state $x_i \in \mathcal{X}_i$, the truncated policy $(\mu_i, \dots, \mu_{N-1})$ is optimal for the problem starting at stage i with initial condition x_i .

Theorem 10.16 (Time Consistency of Bellman Policies). Let μ^* be a policy defined by

$$\mu_i^*(x) \in \arg \min_{u \in \mathcal{U}_i(x)} [\ell_i(x, u) + V_{i+1}(f_i(x, u))]. \quad (10.155)$$

Then μ^* is time-consistent.

Proof. Fix any stage i and state x_i . By the Bellman recursion, $V_i(x_i)$ is the optimal value of the subproblem starting at (i, x_i) . The control $\mu_i^*(x_i)$ attains the minimum in the Bellman equation, and the continuation policy is constructed recursively using the same optimality condition. Therefore, $(\mu_i^*, \dots, \mu_{N-1}^*)$ is optimal for the subproblem starting from x_i , establishing time consistency. $\square$

Remark 10.17. Time consistency is a direct consequence of the principle of optimality: an optimal policy remains optimal when the problem is reinitialized at any future stage.

10.9.2 ■ Representation of Solutions

Two representations of admissible controls are especially important.

Definition 10.18 (Open-Loop Control). *An open-loop control is a sequence $(u_0, \dots, u_{N-1})$ determined at the initial time as a function of the initial condition x_0 .*

Definition 10.19 (Feedback Policy). *A feedback policy is a sequence of mappings $\mu = (\mu_0, \dots, \mu_{N-1})$ such that*

$$u_i = \mu_i(x_i), \quad i = 0, \dots, N-1. \quad (10.156)$$

Proposition 10.20 (Representation Equivalence). *Let μ be a feedback policy and x_0 an initial state. Then the recursion*

$$x_{i+1} = f_i(x_i, \mu_i(x_i)) \quad (10.157)$$

generates a unique trajectory (x_i, u_i) , where $u_i = \mu_i(x_i)$, which defines an open-loop sequence.

Conversely, any open-loop sequence $(u_0, \dots, u_{N-1})$ can be represented as a feedback policy by defining $\mu_i(x) = u_i$ for all $x \in \mathcal{X}_i$.

Proof. The first statement follows from the deterministic evolution of the system under the feedback law. The second follows by constructing a constant mapping that reproduces the prescribed sequence. $\square$

Remark 10.21. Although the two representations coincide along a realized trajectory, they differ fundamentally in scope. A feedback policy prescribes actions for all states, whereas an open-loop control specifies only a single trajectory.

10.9.3 ■ Illustration: Resource Allocation Example

The preceding concepts can be illustrated using the resource allocation problem.

The optimal feedback policy is given by

$$u_i^*(x_i) = \frac{1}{N-i} x_i. \quad (10.158)$$

At any stage i , given the current state x_i , the policy allocates the remaining resource evenly across the remaining $N-i$ stages. Since this rule solves the subproblem starting from (i, x_i) , the policy is time-consistent.

Starting from $x_0 = C$, the closed-loop trajectory satisfies

$$x_i = \frac{N-i}{N} C, \quad (10.159)$$

which yields the control sequence

$$u_i^* = \frac{C}{N}, \quad i = 0, \dots, N-1. \quad (10.160)$$

Thus, in the deterministic setting, the feedback policy reproduces the open-loop solution along the nominal trajectory.

In the presence of disturbances, the state deviates from its nominal trajectory. The feedback policy adapts to the realized state, whereas the open-loop sequence remains fixed. Consequently, the two representations yield different outcomes under uncertainty.

Structural Insight

Feedback policies are state-dependent and time-consistent, whereas open-loop controls are trajectory-based and pre-committed. Consequently, time consistency is an intrinsic property of feedback solutions derived from dynamic programming, and it underpins their robustness in dynamic and uncertain environments.

10.9.4 ■ Dynamic Inconsistency and Information Structures**Dynamic Inconsistency**

Section 10.9.1 established that any policy generated by the Bellman recursion is time-consistent, so that it never requires revision as time evolves. We now ask what can break that conclusion.

Time consistency relies on the stability of preferences over time. When preferences evolve, the decision-maker may revise previously optimal plans.

Definition 10.22 (Dynamically Inconsistent Problem). *A decision problem is dynamically inconsistent if there exists an initial state x_0 such that an optimal policy at time 0 is not time-consistent.*

Information Structures and Partial Observation

The applicability of dynamic programming depends not only on the stability of preferences, but also on the information available to the decision-maker. When the system state is not fully observed, the formulation of the control problem must explicitly incorporate the information structure.

Let $\{\mathcal{I}_i\}_{i=0}^N$ be a filtration representing the information available to the decision-maker over time. A control policy must be adapted to this filtration, i.e.,

$$u_i = \mu_i(\mathcal{I}_i), \quad i = 0, \dots, N-1. \quad (10.161)$$

Different information structures arise depending on the relationship between $\mathcal{I}_i$ and the underlying state process $\{x_i\}$. Under perfect information, $\mathcal{I}_i = \sigma(x_i)$. Under delayed information, $\mathcal{I}_i = \sigma(x_{i-d})$ for some delay $d \geq 1$. Under imperfect or noisy information, $\mathcal{I}_i = \sigma(y_0, \dots, y_i)$, where y_i is a noisy observation of x_i . Under incomplete information, some system parameters are unknown and must be inferred from data.

We now focus on the case in which the state is not observed directly.

Definition 10.23 (Partially Observed Control Problem). *A control problem is partially observed if the state $\{x_i\}$ evolves according to*

$$x_{i+1} = f_i(x_i, u_i, \varepsilon_i), \quad (10.162)$$

while the decision-maker observes

$$y_i = h_i(x_i, \eta_i), \quad (10.163)$$

where $\{\varepsilon_i\}$ and $\{\eta_i\}$ are exogenous noise processes. The available information is

$$\mathcal{I}_i = \sigma(y_0, \dots, y_i, u_0, \dots, u_{i-1}). \quad (10.164)$$

In this setting, the control u_i must be measurable with respect to $\mathcal{I}_i$.

Define the conditional distribution

$$\pi_i(\cdot) := \mathbb{P}(x_i \in \cdot \mid \mathcal{I}_i), \quad (10.165)$$

which is referred to as the *information state* or *belief state*.

Proposition 10.24 (Sufficiency of the belief state). *The conditional distribution π_i is a sufficient statistic for decision-making. That is, there exists an optimal policy of the form*

$$u_i = \mu_i(\pi_i). \quad (10.166)$$

Proof. We prove by backward induction that the value function depends on the information $\mathcal{I}_i$ only through the belief π_i ; the minimizing control is then automatically a function of π_i .

Base case. At the terminal stage the cost-to-go is $V_N(\mathcal{I}_N) = \mathbb{E}[g_N(x_N) \mid \mathcal{I}_N] = \int g_N d\pi_N$, which depends on $\mathcal{I}_N$ only through π_N . Write $V_N(\mathcal{I}_N) = W_N(\pi_N)$.

Inductive step. Suppose $V_{i+1}(\mathcal{I}_{i+1}) = W_{i+1}(\pi_{i+1})$ for some functional W_{i+1} on beliefs. The dynamic programming equation reads

$$V_i(\mathcal{I}_i) = \inf_{u_i} \mathbb{E}[\ell_i(x_i, u_i) + W_{i+1}(\pi_{i+1}) \mid \mathcal{I}_i, u_i].$$

Two observations reduce the right-hand side to a function of (π_i, u_i) alone. First, the immediate cost $\mathbb{E}[\ell_i(x_i, u_i) \mid \mathcal{I}_i] = \int \ell_i(\cdot, u_i) d\pi_i$ depends on $\mathcal{I}_i$ only through π_i . Second, the updated belief $\pi_{i+1} = \Phi_i(\pi_i, u_i, y_{i+1})$ is, by Bayes' rule, a function of (π_i, u_i, y_{i+1}) ; moreover the conditional law of the next observation y_{i+1} given $(\mathcal{I}_i, u_i)$ is determined by π_i and u_i through the transition and observation kernels, say $\rho(\cdot \mid \pi_i, u_i)$. Hence

$$\mathbb{E}[W_{i+1}(\pi_{i+1}) \mid \mathcal{I}_i, u_i] = \int W_{i+1}(\Phi_i(\pi_i, u_i, y)) \rho(dy \mid \pi_i, u_i)$$

is a function of (π_i, u_i) only. Consequently $V_i(\mathcal{I}_i) = W_i(\pi_i)$ with

$$W_i(\pi_i) = \inf_{u_i} \left[\int \ell_i(\cdot, u_i) d\pi_i + \int W_{i+1}(\Phi_i(\pi_i, u_i, y)) \rho(dy \mid \pi_i, u_i) \right],$$

and any minimizer may be selected as a function of π_i . This closes the induction and exhibits an optimal policy of the form $u_i = \mu_i(\pi_i)$. $\square$

Using the belief state, the partially observed problem can be reformulated as a fully observed control problem in the space of probability measures. The value function satisfies

$$V_i(\pi_i) = \inf_{u_i} [\mathbb{E}[\ell_i(x_i, u_i) \mid \pi_i] + V_{i+1}(\pi_{i+1})], \quad (10.167)$$

where π_{i+1} is obtained from π_i via Bayesian updating:

$$\pi_{i+1} = \Phi_i(\pi_i, u_i, y_{i+1}), \quad (10.168)$$

for an appropriate update operator Φ_i .

Remark 10.25. The belief state π_i evolves as a controlled Markov process. Dynamic programming can therefore be applied, but in an infinite-dimensional state space.

As an illustration, consider the system

$$x_{i+1} = x_i + u_i + \varepsilon_i, \quad (10.169)$$

with observation model

$$y_i = x_i + \eta_i, \quad (10.170)$$

where $\varepsilon_i \sim \mathcal{N}(0, W)$ and $\eta_i \sim \mathcal{N}(0, V)$ are independent (we write W, V for the noise covariances, reserving Q, R for the cost weights).

In this case, the belief state is Gaussian and can be characterized by its mean $\hat{x}_i$ and covariance P_i . The mean evolves according to the Kalman filter.

Proposition 10.26 (Separation principle). *For linear-quadratic-Gaussian (LQG) problems, the optimal control law is given by*

$$u_i^* = \mu_i(\hat{x}_i), \quad (10.171)$$

where $\hat{x}_i = \mathbb{E}[x_i \mid \mathcal{I}_i]$ is computed via the Kalman filter.

Proof. Specialize the belief-state recursion to the linear-Gaussian dynamics with quadratic cost. Because all maps are linear and all noises Gaussian, the belief π_i is Gaussian and hence completely described by its mean $\hat{x}_i = \mathbb{E}[x_i \mid \mathcal{I}_i]$ and covariance P_i , propagated by the Kalman filter. Crucially, P_i obeys a Riccati recursion driven only by the system matrices and the noise covariances W, V , not by the controls or the realized observations. Thus P_i is a *deterministic* function of the stage index, and the only stochastic, control-relevant component of the belief is the mean $\hat{x}_i$. By the belief-state sufficiency, an optimal policy has the form $u_i = \mu_i(\pi_i) = \mu_i(\hat{x}_i, P_i)$, which collapses to $u_i = \mu_i(\hat{x}_i)$.

To identify this law, decompose the state as $x_i = \hat{x}_i + e_i$ with estimation error $e_i := x_i - \hat{x}_i$. The Kalman error is orthogonal to the information, so $\hat{x}_i$ and e_i are uncorrelated and e_i has the deterministic covariance P_i , independent of the control policy. Substituting into a quadratic cost and taking expectations, the cross terms vanish and the cost separates,

$$\mathbb{E} \left[\sum_i (\hat{x}_i^\top Q \hat{x}_i + u_i^\top R u_i) + \hat{x}_N^\top Q_N \hat{x}_N \right] + \mathbb{E} \left[\sum_i e_i^\top Q e_i + e_N^\top Q_N e_N \right].$$

The second bracket, the estimation-error cost, depends only on $\{P_i\}$ and is therefore unaffected by the choice of controls. Minimizing the total cost thus reduces to minimizing the first bracket, an LQ problem in the estimate $\hat{x}_i$, whose dynamics $\hat{x}_{i+1} = A\hat{x}_i + Bu_i + (\text{innovation})$ are driven by the control plus zero-mean innovations independent of the past. For such a problem the optimal control is the deterministic (certainty-equivalent) LQR feedback $u_i^* = -K_i \hat{x}_i$ obtained from the noise-free Riccati equations, independent of the noise statistics. This is the separation principle: estimation by the Kalman filter and control by the full-information optimal gain can be designed independently and composed. $\square$

Under partial observation, the decision-maker must solve an estimation problem and a control problem simultaneously: the state must be inferred from observations, and actions must then be chosen on the basis of the inferred state.

Dynamic programming remains applicable, but only after reformulating the problem in terms of the information state.

Limits of the Bellman Framework

Dynamic inconsistency and information constraints represent two fundamental challenges to the classical Bellman framework.

Dynamic inconsistency arises from evolving preferences and requires equilibrium-based extensions.

Information constraints arise from limited observability and require state augmentation via belief states.

In both cases, dynamic programming remains applicable only after an appropriate reformulation of the problem.

10.9.5 ■ Example: Inventory Control Problem

Consider a finite-horizon inventory control problem over periods $k = 0, 1, \dots, N - 1$. Let $x_k \in \{0, 1, 2\}$ denote the inventory level at the beginning of period k , let $u_k \in \{0, 1, 2\}$ be the quantity ordered at that time, and let $w_k \in \{0, 1, 2\}$ be the demand realized during the period. Storage is limited to two units, so the order quantity is constrained by the space that remains,

$$0 \leq u_k \leq 2 - x_k. \quad (10.172)$$

The demand is independent and identically distributed across periods, with

$$\mathbb{P}(w_k = 0) = 0.1, \quad \mathbb{P}(w_k = 1) = 0.7, \quad \mathbb{P}(w_k = 2) = 0.2, \quad (10.173)$$

so that a single unit is the typical requirement and the two extremes are comparatively rare. Orders arrive before the demand is served, and demand in excess of the stock on hand is lost rather than backlogged; the inventory carried into the next period is therefore

$$x_{k+1} = \max\{0, x_k + u_k - w_k\}. \quad (10.174)$$

Three effects contribute to the cost of a period: the purchase of the units ordered, a holding charge on whatever remains at the end of the period, and a shortage penalty for demand that could not be met. Charging shortages three times as heavily as holding gives the stage cost

$$\ell(x, u, w) = u + \max\{0, x + u - w\} + 3 \max\{0, w - x - u\}, \quad (10.175)$$

and, starting from an empty warehouse, the objective is to choose the order quantities so as to minimize the total expected cost over the horizon,

$$V_0(x_0) = \min_{\{u_k\}} \mathbb{E} \left[\sum_{k=0}^{N-1} \ell(x_k, u_k, w_k) \right], \quad x_0 = 0. \quad (10.176)$$

Because the demand is independent across periods, the successor state (10.174) depends on the past only through the current pair (x_k, u_k) , so the backward recursion of Section 10.6 applies verbatim once the expectation over w is inserted at each stage. Writing x' for the successor state, the value functions satisfy

$$V_k(x) = \min_{u \in \mathcal{U}(x)} \mathbb{E} [\ell(x, u, w) + V_{k+1}(x')], \quad x' = \max\{0, x + u - w\}, \quad (10.177)$$

with terminal condition and admissible control set

$$V_N(x) = 0, \quad \mathcal{U}(x) = \{0, 1, \dots, 2 - x\}. \quad (10.178)$$

Since both the state set and the control set are finite, the recursion (10.177) can be evaluated exactly, one stage at a time; we carry out the last two stages explicitly.

At the last decision stage there is no continuation value, so $V_N = 0$ and the minimization reduces to the one-period problem $V_{N-1}(x) = \min_u \mathbb{E}[\ell(x, u, w)]$. Evaluating the expectation for each admissible order quantity and taking the smallest gives

$$\begin{aligned} V_{N-1}(0) &= 1.7, & u_{N-1}^*(0) &= 1, \\ V_{N-1}(1) &= 0.7, & u_{N-1}^*(1) &= 0, \\ V_{N-1}(2) &= 0.9, & u_{N-1}^*(2) &= 0. \end{aligned} \quad (10.179)$$

One stage earlier the continuation value is no longer zero, and the ordering decision must weigh the cost of the current period against the inventory it leaves behind. It remains to compute

$$V_{N-2}(x) = \min_{u \in \mathcal{U}(x)} \mathbb{E}[\ell(x, u, w) + V_{N-1}(x')]. \quad (10.180)$$

Case $x = 0$. For each $u \in \{0, 1, 2\}$:

$u = 0$: $w = 0 : (0, x'=0)$, $w = 1 : (3, x'=0)$, $w = 2 : (6, x'=0)$, so $\mathbb{E}[\cdot] = 0.1(0 + 1.7) + 0.7(3 + 1.7) + 0.2(6 + 1.7) = 5.0$.

$u = 1$: $w = 0 : (2, x'=1)$, $w = 1 : (1, x'=0)$, $w = 2 : (4, x'=0)$, so $\mathbb{E}[\cdot] = 0.1(2 + 0.7) + 0.7(1 + 1.7) + 0.2(4 + 1.7) = 3.3$.

$u = 2$: $w = 0 : (4, x'=2)$, $w = 1 : (3, x'=1)$, $w = 2 : (2, x'=0)$, so $\mathbb{E}[\cdot] = 0.1(4 + 0.9) + 0.7(3 + 0.7) + 0.2(2 + 1.7) = 3.82$.

Thus,

$$V_{N-2}(0) = 3.3, \quad u_{N-2}^*(0) = 1. \quad (10.181)$$

Case $x = 1$. Controls $u \in \{0, 1\}$:

$u = 0$: $w = 0 : (1, x'=1)$, $w = 1 : (0, x'=0)$, $w = 2 : (3, x'=0)$, so $\mathbb{E}[\cdot] = 0.1(1 + 0.7) + 0.7(0 + 1.7) + 0.2(3 + 1.7) = 2.3$.

$u = 1$: $w = 0 : (3, x'=2)$, $w = 1 : (2, x'=1)$, $w = 2 : (1, x'=0)$, so $\mathbb{E}[\cdot] = 0.1(3 + 0.9) + 0.7(2 + 0.7) + 0.2(1 + 1.7) = 2.82$.

Thus,

$$V_{N-2}(1) = 2.3, \quad u_{N-2}^*(1) = 0. \quad (10.182)$$

Case $x = 2$. Only $u = 0$ is feasible: $w = 0 : (2, x'=2)$, $w = 1 : (1, x'=1)$, $w = 2 : (0, x'=0)$, so $\mathbb{E}[\cdot] = 0.1(2 + 0.9) + 0.7(1 + 0.7) + 0.2(0 + 1.7) = 1.82$ and

$$V_{N-2}(2) = 1.82, \quad u_{N-2}^*(2) = 0. \quad (10.183)$$

Collecting the minimizers found at the two stages, the backward recursion yields the policy

$$u^*(x) = \begin{cases} 1, & x = 0, \\ 0, & x = 1, \\ 0, & x = 2. \end{cases} \quad (10.184)$$

Thus, the optimal policy is a *threshold policy*:

$$\text{Order one unit if inventory is empty; otherwise, do not order.} \quad (10.185)$$

10.9.6 ■ Example: Optimal Exercise of a Call Option

A call option gives its holder the right, but not the obligation, to purchase an asset at a fixed *strike price* p . In an *American option*, this right can be exercised at any time prior to a fixed maturity date T . The holder faces a sequential decision problem: at each time, the choice is between exercising immediately and waiting. Exercising now yields an immediate payoff, whereas waiting preserves flexibility and may produce a higher payoff in the future. This trade-off naturally leads to an *optimal stopping problem*, which can be analyzed using dynamic programming.

Let $\{X_t\}_{t=0}^T$ denote the asset price process evolving in discrete time:

$$X_{t+1} = X_t + \varepsilon_t, \quad (10.186)$$

where $\{\varepsilon_t\}$ is an i.i.d. sequence with $\mathbb{E}[\varepsilon_t] = 0$ and $\mathbb{E}|\varepsilon_t| < \infty$. Let $p > 0$ denote the strike price. At each time $t = 0, 1, \dots, T$, the decision-maker may either exercise the option and receive the payoff $X_t - p$, or continue and postpone the decision to a later time.

The option must be exercised no later than time T .

Payoffs received later are worth less than payoffs received now: a payoff collected at time t is discounted by β^t , where $\beta \in (0, 1)$ is the per-period discount factor (for instance $\beta = 1/(1+r)$ for an interest rate $r > 0$). Let $\mathcal{T}$ denote the set of stopping times with values in $\{0, \dots, T\}$. The objective is

$$\sup_{\tau \in \mathcal{T}} \mathbb{E}[\beta^\tau (X_\tau - p)]. \quad (10.187)$$

This formulation highlights that the problem is one of choosing the optimal stopping time. Discounting is not a technicality here: we show at the end of this section that without it ($\beta = 1$) early exercise is never optimal, so the threshold structure derived below would be vacuous.

To analyze the problem we reformulate it recursively. It is convenient to index the value function by the *remaining* horizon rather than by calendar time, so we write $s = T - t$ for the number of periods still available and let

$$F_s(x) = \text{maximum expected payoff with } s \text{ periods remaining and current price } x. \quad (10.188)$$

At the final stage no further waiting is possible and the option must be exercised, which fixes the terminal condition

$$F_0(x) = \max(x - p, 0). \quad (10.189)$$

For $s \geq 1$ the holder compares the two available actions, and the value function accordingly satisfies the Bellman recursion

$$F_s(x) = \max \{x - p, \beta \mathbb{E}[F_{s-1}(x + \varepsilon)]\}. \quad (10.190)$$

The first term in the maximum is the payoff from immediate exercise, while the second is the expected value of continuing and making all future decisions optimally, discounted by one period.

Structural Properties of the Value Function

Several key properties of F_s follow from the Bellman recursion.

Proposition 10.27 (Monotonicity in the state). *For each s , the function $F_s(x)$ is nondecreasing in x .*

Proof. Both $x - p$ and $\beta \mathbb{E}[F_{s-1}(x + \varepsilon)]$ are nondecreasing in x (the latter by induction on s , since F_0 is). The maximum of nondecreasing functions is nondecreasing. $\square$

Proposition 10.28 (Monotonicity in time). *For each x , the function $F_s(x)$ is nondecreasing in s .*

Proof. We show $F_s \geq F_{s-1}$ pointwise by induction on s . For the base case, $F_1(x) = \max\{x - p, \beta \mathbb{E}[F_0(x + \varepsilon)]\} \geq x - p$, and since $F_0 = (x - p)^+ \geq 0$ we also have $F_1(x) \geq \beta \mathbb{E}[F_0(x + \varepsilon)] \geq 0$; together these give $F_1(x) \geq \max\{x - p, 0\} = F_0(x)$. For the inductive step, assume $F_s \geq F_{s-1}$. Monotonicity of the expectation gives $\beta \mathbb{E}[F_s(x + \varepsilon)] \geq \beta \mathbb{E}[F_{s-1}(x + \varepsilon)]$, and taking the maximum with the common exercise value $x - p$ preserves the inequality:

$$F_{s+1}(x) = \max\{x - p, \beta \mathbb{E}[F_s(x + \varepsilon)]\} \geq \max\{x - p, \beta \mathbb{E}[F_{s-1}(x + \varepsilon)]\} = F_s(x).$$

Hence $F_s(x)$ is nondecreasing in s . $\square$

To better understand the structure, define

$$G_s(x) := F_s(x) - x. \quad (10.191)$$

This transformation separates the linear growth in x from the option value. Note that $G_s(x) \geq -p$ for every s , because $F_s(x) \geq x - p$.

Proposition 10.29. *For each s , the function $G_s(x)$ is nonincreasing in x .*

Proof. Substituting $F_{s-1}(y) = y + G_{s-1}(y)$ into the Bellman equation and using $\mathbb{E}[\varepsilon] = 0$,

$$G_s(x) = \max\{-p, -(1 - \beta)x + \beta \mathbb{E}[G_{s-1}(x + \varepsilon)]\}. \quad (10.192)$$

The base case $G_0(x) = \max\{-p, -x\}$ is nonincreasing. If G_{s-1} is nonincreasing, so is $\beta \mathbb{E}[G_{s-1}(x + \varepsilon)]$, and $-(1 - \beta)x$ is nonincreasing because $\beta \leq 1$; the maximum of nonincreasing functions is nonincreasing, and the result follows by induction. $\square$

Optimal Exercise Policy

The optimal exercise policy is characterized by a threshold rule.

Theorem 10.30 (Threshold Structure). *Let $\beta \in (0, 1)$. For each $s \geq 1$, there exists a finite threshold $a_s \in \mathbb{R}$ such that*

$$\begin{cases} \text{exercise} & \text{if } x \geq a_s, \\ \text{continue} & \text{if } x < a_s. \end{cases} \quad (10.193)$$

Proof. Compare the two options through their difference,

$$D_s(x) = (x - p) - \beta \mathbb{E}[F_{s-1}(x + \varepsilon)], \quad (10.194)$$

so that exercising is optimal exactly when $D_s(x) \geq 0$. Using $F_{s-1}(y) = y + G_{s-1}(y)$ and $\mathbb{E}[\varepsilon] = 0$, we rewrite

$$D_s(x) = (x - p) - \beta(x + \mathbb{E}[G_{s-1}(x + \varepsilon)]) = (1 - \beta)x - p - \beta \mathbb{E}[G_{s-1}(x + \varepsilon)].$$

Because G_{s-1} is nonincreasing (previous proposition), the expectation $\mathbb{E}[G_{s-1}(x + \varepsilon)]$ is nonincreasing in x , so D_s is nondecreasing in x . Therefore the exercise set $\{x : D_s(x) \geq 0\}$ is an

upper interval with left endpoint $a_s = \inf\{x : D_s(x) \geq 0\}$; below a_s it is optimal to continue and at or above a_s it is optimal to exercise, which is the stated threshold rule.

It remains to show that a_s is finite, and this is where $\beta < 1$ enters. Since $G_{s-1} \geq -p$, we have $D_s(x) \leq (1 - \beta)x - (1 - \beta)p$, which is negative for $x < p$; hence $a_s \geq p$. For the other direction, an induction using $(y + \varepsilon)^+ \leq y^+ + |\varepsilon|$ gives $F_{s-1}(y) \leq y^+ + (s - 1)\mathbb{E}|\varepsilon|$, so for $x \geq 0$

$$D_s(x) \geq (x - p) - \beta(x + s\mathbb{E}|\varepsilon|) = (1 - \beta)x - p - \beta s\mathbb{E}|\varepsilon|,$$

which is positive for all sufficiently large x . The exercise set is therefore nonempty and bounded below, and $p \leq a_s < \infty$. $\square$

The threshold a_s is characterized by the indifference condition:

$$a_s - p = \beta \mathbb{E}[F_{s-1}(a_s + \varepsilon)]. \quad (10.195)$$

This condition states that at the threshold, the decision-maker is indifferent between exercising and continuing.

Proposition 10.31. *The threshold sequence $\{a_s\}$ is nondecreasing in s .*

Proof. Recall from (10.194) that $D_s(x) = (1 - \beta)x - p - \beta \mathbb{E}[G_{s-1}(x + \varepsilon)]$ is nondecreasing in x and that a_s is its left crossing point, $D_s(a_s) = 0$. By the monotonicity of F_s in s we have $F_s \geq F_{s-1}$, hence $G_s \geq G_{s-1}$ and $\mathbb{E}[G_s(x + \varepsilon)] \geq \mathbb{E}[G_{s-1}(x + \varepsilon)]$, so that

$$\begin{aligned} D_{s+1}(x) &= (1 - \beta)x - p - \beta \mathbb{E}[G_s(x + \varepsilon)] \\ &\leq (1 - \beta)x - p - \beta \mathbb{E}[G_{s-1}(x + \varepsilon)] = D_s(x) \quad \text{for every } x. \end{aligned}$$

Evaluating at $x = a_s$ gives $D_{s+1}(a_s) \leq D_s(a_s) = 0$. Since D_{s+1} is nondecreasing and vanishes at a_{s+1} , the inequality $D_{s+1}(a_s) \leq 0$ forces $a_s \leq a_{s+1}$. Thus the threshold rises with the number of stages to go: with more time remaining, the holder demands a higher price before exercising. (Equivalently, as maturity approaches and s decreases, the threshold falls, as depicted in Figure 10.6.) $\square$

The optimal stopping time is given by

$$\tau^* = \inf\{t \in \{0, \dots, T\} : X_t \geq a_{T-t}\}. \quad (10.196)$$

Thus, the option is exercised the first time the price crosses the threshold, as illustrated in Figure 10.6.

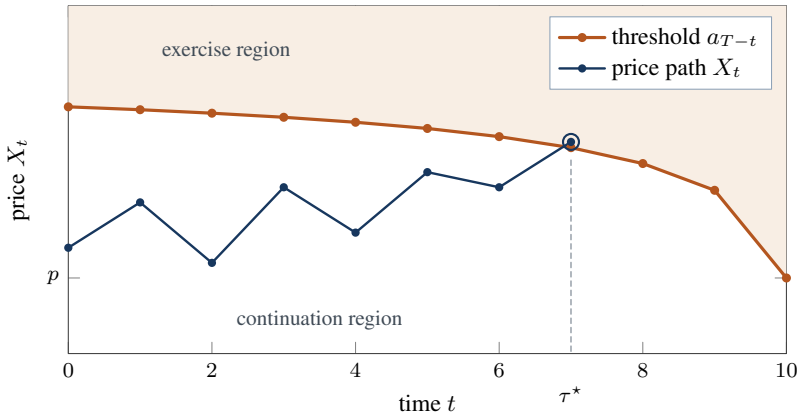


Figure 10.6: Optimal exercise of an American option as a threshold rule. The free boundary a_{T-t} separates the state space into a continuation region (below) and an exercise region (above), and it is nonincreasing as maturity approaches. The holder waits while $X_t < a_{T-t}$ and exercises at the first crossing time $\tau^* = \inf\{t : X_t \geq a_{T-t}\}$. (The boundary is computed from the discounted recursion $F_s(x) = \max\{x - p, \beta \mathbb{E}[F_{s-1}(x + \varepsilon)]\}$ with $p = 2$, $\beta = 0.95$, $\varepsilon \sim \mathcal{N}(0, 0.5^2)$, $T = 10$; the price path is illustrative. Without discounting the boundary recedes to $+\infty$ and exercise is deferred to maturity.)

Gaussian Model and Integral Representation

In the Gaussian case, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$, the Bellman recursion takes the form

$$F_s(x) = \max \left\{ x - p, \beta \int_{\mathbb{R}} F_{s-1}(x + z) \phi_{\sigma}(z) dz \right\}, \quad (10.197)$$

where $\phi_{\sigma}(z) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-z^2/(2\sigma^2)}$ is the Gaussian density. It is convenient to define the linear continuation operator $(\mathcal{K}F)(x) := \int_{\mathbb{R}} F(x + z) \phi_{\sigma}(z) dz$ (the letter $\mathcal{T}$ being reserved for the set of stopping times), so that the recursion can be written compactly as $F_s(x) = \max\{x - p, \beta(\mathcal{K}F_{s-1})(x)\}$.

The structural properties established earlier remain valid: F_s is nondecreasing in x , the function $F_s(x) - x$ is nonincreasing, and the optimal policy has a threshold form. In particular, there exists a finite threshold a_s such that it is optimal to exercise if $x \geq a_s$ and to continue otherwise. The threshold is characterized by the indifference condition

$$a_s - p = \beta \int_{\mathbb{R}} F_{s-1}(a_s + z) \phi_{\sigma}(z) dz, \quad (10.198)$$

which is a nonlinear integral equation defining the free boundary.

At the terminal stage, $F_0(x) = (x - p)^+$. For one step to go, the continuation value admits a closed-form expression:

$$\mathbb{E}[(x + \varepsilon - p)^+] = (x - p) \Phi\left(\frac{x - p}{\sigma}\right) + \sigma \phi\left(\frac{x - p}{\sigma}\right), \quad (10.199)$$

where Φ and ϕ denote the standard normal cumulative distribution and density functions. The first term represents the expected payoff when the option finishes in the money, while the second

term captures the additional value generated by uncertainty. Consequently a_1 is the root of the scalar equation

$$a_1 - p = \beta \left[(a_1 - p) \Phi \left(\frac{a_1 - p}{\sigma} \right) + \sigma \phi \left(\frac{a_1 - p}{\sigma} \right) \right];$$

for $p = 2$, $\beta = 0.95$, $\sigma = 0.5$ it gives $a_1 \approx 2.580$, and iterating the recursion numerically produces the thresholds plotted in Figure 10.6.

Why discounting is needed. Consider the undiscounted case $\beta = 1$. Since $\mathbb{E}[X_{t+1} | X_t = x] = x$, the price process is a martingale. By convexity of $(x - p)^+$ and Jensen's inequality, $\mathbb{E}[(X_{t+1} - p)^+ | X_t = x] \geq (x - p)^+$, with strict inequality for Gaussian noise, so waiting strictly dominates immediate exercise. Consequently, for $\beta = 1$,

$$F_s(x) = \mathbb{E}[(X_T - p)^+ | X_t = x], \quad a_s = +\infty, \quad (10.200)$$

and early exercise is never optimal: the lower bound $(1 - \beta)x - p - \beta s \mathbb{E}[\varepsilon]$ used in the proof of the threshold theorem no longer grows with x . With $\beta \in (0, 1)$, waiting is penalized, the threshold a_s is finite, and early exercise is optimal once the price is high enough. The threshold is still characterized only implicitly, through the integral equation above, and does not admit a closed-form solution for $s \geq 2$ in general.

10.10 ■ Further Reading

The discrete-time optimal control problems studied in this chapter are governed by the principle of optimality and the associated dynamic programming recursion, whose origins trace back to Bellman. The definitive modern treatment is Bertsekas [17, 16], which develops the finite- and infinite-horizon theory, the linear-quadratic regulator, and the structural properties of value functions in a unified and rigorous manner. A concise systems-oriented presentation that connects the discrete-time recursions to the continuous-time theory is given by Bryson and Ho [30], while Athans and Falb [6] offer a classical exposition of the underlying optimal control principles.

Readers interested in the stochastic and infinite-horizon extensions of the deterministic recursions developed here will find the theory of Markov decision processes in Puterman [99] and the measure-theoretic foundations of stochastic dynamic programming in Bertsekas and Shreve [19]. For a systems-theoretic account emphasizing estimation, identification, and adaptive control alongside dynamic programming, Kumar and Varaiya [74] is recommended.

When the state or action spaces are large, exact dynamic programming becomes intractable and approximation is essential. The foundational references for this setting are Bertsekas and Tsitsiklis [20] on neuro-dynamic programming and Powell [98] on approximate dynamic programming, both of which extend the recursions of this chapter to high-dimensional and data-driven problems and provide a natural bridge to reinforcement learning.

10.11 ■ Exercises

Exercise 10.1 (Adjoint Equations for a Linear-Quadratic Problem). Consider the scalar system $x_{i+1} = ax_i + bu_i$, $i = 0, \dots, N - 1$, with x_0 given, and the cost

$$J(u) = \frac{1}{2}q x_N^2 + \frac{1}{2} \sum_{i=0}^{N-1} r u_i^2, \quad q, r > 0.$$

- (i) Write the Hamiltonian $H_i(x_i, u_i, p_{i+1})$ and derive the costate equations and the stationarity condition from the discrete-time maximum principle.

- (ii) Solve the resulting two-point boundary-value problem explicitly and determine the optimal control sequence in terms of the optimal terminal state.
- (iii) Verify that the same solution is obtained from the gradient formula $\nabla_{u_i} J(u) = \nabla_{u_i} f_i \cdot p_{i+1} + \nabla_{u_i} g_i$ by setting the gradient to zero.

Exercise 10.2 (Gradient Method for Optimal Control). Consider the two-dimensional system

$$x_{i+1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} x_i + \begin{pmatrix} 0 \\ 1 \end{pmatrix} u_i, \quad x_0 = \begin{pmatrix} 10 \\ 0 \end{pmatrix},$$

with cost $J(u) = \|x_N\|^2 + \sum_{i=0}^{N-1} u_i^2$ and horizon $N = 3$.

- (i) Using the adjoint recursion, compute $\nabla J(u)$ at $u = (0, 0, 0)$.
- (ii) Perform one step of the gradient method with step size $\alpha = 1/4$ and evaluate the resulting cost.
- (iii) Explain why, for this problem, the gradient method converges to the global minimizer regardless of the initial control trajectory.

Exercise 10.3 (Backward Induction for a Finite Control Set). A machine can be in state $x_i \in \{\text{good}, \text{bad}\}$. In each period one chooses $u_i \in \{\text{maintain}, \text{skip}\}$. Maintenance costs $m > 0$ and keeps a good machine good; skipping is free but a good machine becomes bad with certainty at the next stage. A bad machine yields an operating loss $\ell > 0$ at every stage $i = 1, \dots, N$ it is bad (including the terminal stage N) and remains bad. Over a horizon N , with a good machine at stage 0,

- (i) set up the Bellman recursion for the minimal total cost;
- (ii) compute the value function for $N = 3$, $m = 1$, $\ell = 3$, and exhibit an optimal policy;
- (iii) determine, for general N , the threshold relation between m and ℓ under which “always maintain” is optimal.

Exercise 10.4 (Principle of Optimality on a Graph). Consider a shortest-path problem on a directed staged graph with node set partitioned into stages $0, 1, \dots, N$, all arcs pointing from stage i to stage $i + 1$, and nonnegative arc lengths.

- (i) Show that the tail of any shortest path, restricted to stages $k, \dots, N$, is itself a shortest path from its starting node, and explain precisely where the staged structure is used.
- (ii) Give an example of a graph with a negative-length arc for which the backward Bellman recursion still returns the correct shortest path, and explain why staged graphs are immune to negative-length difficulties.

Exercise 10.5 (Discrete-Time Riccati Recursion). For the linear-quadratic problem

$$\min_u x_N^\top Q_N x_N + \sum_{i=0}^{N-1} (x_i^\top Q x_i + u_i^\top R u_i), \quad x_{i+1} = A x_i + B u_i,$$

with $Q_N, Q \succeq 0$ and $R \succ 0$:

- (i) Using the quadratic value-function ansatz $V_i(x) = x^\top K_i x$, derive the backward Riccati recursion for K_i and the feedback law $u_i^* = -(R + B^\top K_{i+1} B)^{-1} B^\top K_{i+1} A x_i$.

- (ii) Show by induction that $K_i \succeq 0$ for all i .
- (iii) For the scalar case $A = 1$, $B = 1$, $Q = 0$, $Q_N = 1$, $R = 1$, compute K_i in closed form and identify its limit as the horizon grows.

Exercise 10.6 (Open-Loop versus Feedback Solutions). Consider the resource allocation problem of this chapter with dynamics $x_{i+1} = x_i - u_i$ and cost $\sum_{i=0}^{N-1} u_i^2 + \lambda x_N^2$, $\lambda > 0$.

- (i) Compute the open-loop optimal control sequence by eliminating the states and minimizing over $(u_0, \dots, u_{N-1})$ directly.
- (ii) Compute the optimal feedback policy by dynamic programming.
- (iii) Verify that along the nominal (noise-free) trajectory the two solutions generate identical controls, and explain why they differ once an additive disturbance enters the dynamics.

Exercise 10.7 (State Augmentation). Show that the general cost $g_N(x_N) + \sum_{i=0}^{N-1} g_i(x_i, u_i)$ can be reduced to a purely terminal cost by augmenting the state with a running cost accumulator $z_{i+1} = z_i + g_i(x_i, u_i)$, $z_0 = 0$. Write the augmented system and terminal cost explicitly, and verify that the adjoint equations of the augmented problem reproduce the costate recursion $p_i = \nabla_{x_i} f_i \cdot p_{i+1} + \nabla_{x_i} g_i$ of the original one.

Exercise 10.8 (Optimal Stopping with Discounting). Let $\{X_t\}$ be the random walk $X_{t+1} = X_t + \varepsilon_t$ with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ i.i.d., and consider the discounted optimal stopping problem with reward $(x - p)^+$ upon stopping and discount factor $\beta \in (0, 1)$.

- (i) Write the Bellman equation for the finite-horizon value functions F_s (s stages to go).
- (ii) Show that each F_s is convex and nondecreasing in x , and that $F_s \geq F_{s-1}$ pointwise.
- (iii) Prove that for $\beta < 1$ the continuation region at each stage is an interval of the form $(-\infty, a_s)$, so the optimal rule is a threshold rule, and show that without discounting ($\beta = 1$) early exercise is never optimal.

Chapter 11

Continuous-Time Optimal Control

This chapter develops the fundamental concepts of continuous-time optimal control. We formulate optimal control problems as optimization over functions subject to dynamical constraints and develop the associated analytical framework. We present Pontryagin's Minimum Principle as a set of first-order necessary conditions derived from a variational perspective, where the costate arises as a Lagrange multiplier for the dynamics. We also introduce the dynamic programming approach and the Hamilton–Jacobi–Bellman equation, which characterizes optimal controls in feedback form. These complementary viewpoints provide a unified understanding of optimal control and form the basis for analyzing canonical problems such as linear-quadratic regulation, constrained control, and minimum-time problems.

Figure 11.1 previews the structure of the chapter: the two analytical routes, the Hamiltonian that unifies them, and the canonical applications, including Kalman filtering, that follow from the same principles.

11.1 ■ Optimal Control Problems and the Minimum Principle

The structure of optimal control problems can be understood as a natural extension of classical optimization. In finite-dimensional calculus, an optimal point is characterized by first-order conditions, typically expressed through vanishing derivatives. When constraints are present, these conditions are refined through the introduction of Lagrange multipliers, which capture the sensitivity of the objective with respect to the constraints.

In optimal control, the decision variable is no longer a finite-dimensional vector but a function $u(\cdot)$ defined over a time interval. The constraint is given by a differential equation describing the evolution of the state, and admissible solutions must satisfy this constraint at every time. Consequently, the problem becomes one of optimization over a space of functions subject to dynamical constraints.

The Lagrange multiplier principle extends naturally to this setting. The multiplier associated with the dynamical constraint becomes a function of time, leading to the introduction of the costate variable $p(t)$, which can be interpreted as a time-dependent shadow price of the state trajectory. Pontryagin's Minimum Principle formalizes this idea and provides necessary conditions for optimality.

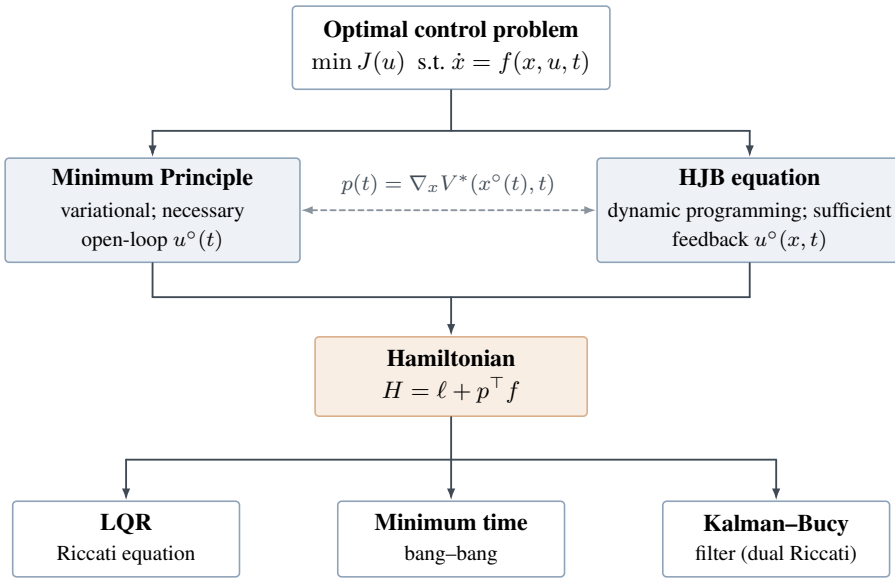


Figure 11.1: Where the pieces fit. Two complementary routes—the variational Minimum Principle and dynamic programming (the Hamilton–Jacobi–Bellman equation)—meet at the Hamiltonian and are joined by the identity $p = \nabla_x V^*$ along optimal trajectories. The canonical applications of this chapter, including the Kalman–Bucy filter, all descend from this common machinery.

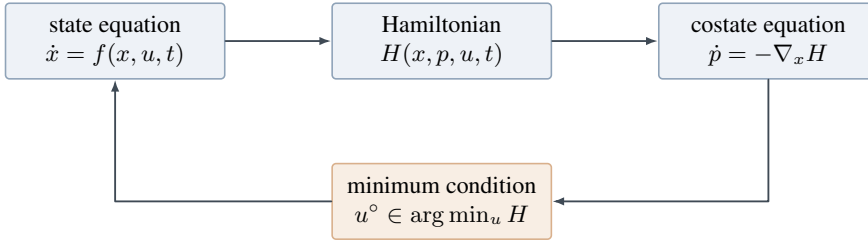


Figure 11.2: Pontryagin’s principle couples state dynamics, costate dynamics, and Hamiltonian minimization into a boundary-value optimality system.

11.1.1 ■ Equivalent Forms of Optimal Control Problems

To formulate optimal control problems in a mathematically rigorous way, it is necessary to specify the underlying function spaces and admissibility conditions. This provides the proper setting for variational analysis, multiplier theory, and the derivation of the Minimum Principle.

Let $[t_0, t_1] \subset \mathbb{R}$ be a compact interval and let $U \subset \mathbb{R}^m$ be a nonempty, closed set. We define the admissible control space

$$\mathcal{U} := \{u : [t_0, t_1] \rightarrow U \mid u \text{ is measurable and essentially bounded}\} \subset L^\infty([t_0, t_1]; \mathbb{R}^m),$$

and the admissible state space

$$\mathcal{X} := \{x : [t_0, t_1] \rightarrow \mathbb{R}^n \mid x \text{ is absolutely continuous, } x(t_0) = x_0\} = W^{1,1}([t_0, t_1]; \mathbb{R}^n).$$

We assume that the dynamics $f : \mathbb{R}^n \times U \times [t_0, t_1] \rightarrow \mathbb{R}^n$ satisfies the Carathéodory conditions: for each t , the map $(x, u) \mapsto f(x, u, t)$ is continuous; for each (x, u) , the map $t \mapsto f(x, u, t)$ is

measurable; and f is locally Lipschitz in x uniformly in (u, t) . Under these assumptions, for every $u \in \mathcal{U}$ there exists a unique $x \in \mathcal{X}$ satisfying

$$\dot{x}(t) = f(x(t), u(t), t), \quad x(t_0) = x_0, \quad (11.1)$$

for almost every $t \in [t_0, t_1]$.

We define the admissible set

$$\mathcal{A} := \{(x, u) \in \mathcal{X} \times \mathcal{U} \mid \dot{x}(t) = f(x(t), u(t), t) \text{ a.e.}\}. \quad (11.2)$$

The optimal control problem is therefore a constrained minimization problem over $\mathcal{A}$.

Bolza form. Let $\ell : \mathbb{R}^n \times U \times [t_0, t_1] \rightarrow \mathbb{R}$ and $m : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous functions such that $\ell(x(\cdot), u(\cdot), \cdot) \in L^1([t_0, t_1])$ for all $(x, u) \in \mathcal{A}$. The Bolza problem consists of minimizing

$$J(x, u) = \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt + m(x(t_1)) \quad (11.3)$$

over $(x, u) \in \mathcal{A}$. This formulation combines a running cost and a terminal cost and is the most general among the classical forms.

Lagrange form. If $m \equiv 0$, the problem reduces to

$$J(x, u) = \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt, \quad (11.4)$$

which depends only on the accumulation of cost along the trajectory. The functional is well-defined provided $\ell(x(\cdot), u(\cdot), \cdot) \in L^1$.

Mayer form. If $\ell \equiv 0$, the problem reduces to

$$J(x, u) = m(x(t_1)), \quad (11.5)$$

which depends only on the terminal state. This form is particularly convenient for theoretical developments since it eliminates the integral term.

Proposition 11.1 (Equivalence of Bolza, Lagrange, and Mayer forms). *The Bolza, Lagrange, and Mayer formulations are equivalent in the sense that any problem in one form can be transformed into another by augmenting the state space, while preserving admissible controls, trajectories, and optimal solutions.*

Proof. We show that any Bolza problem can be reduced to a Mayer problem. Let $(x, u) \in \mathcal{A}$. Define an auxiliary state $z \in W^{1,1}([t_0, t_1]; \mathbb{R})$ by

$$\dot{z}(t) = \ell(x(t), u(t), t), \quad z(t_0) = 0. \quad (11.6)$$

Since $\ell(x(\cdot), u(\cdot), \cdot) \in L^1$, the function z is well-defined and absolutely continuous.

Define the augmented state $\tilde{x}(t) = (x(t), z(t)) \in \mathbb{R}^{n+1}$. Then $\tilde{x}$ satisfies

$$\dot{\tilde{x}}(t) = \tilde{f}(\tilde{x}(t), u(t), t), \quad \tilde{f}(\tilde{x}, u, t) = \begin{pmatrix} f(x, u, t) \\ \ell(x, u, t) \end{pmatrix}. \quad (11.7)$$

The augmented dynamics satisfy the same regularity assumptions as f .

Integrating (11.6) gives $z(t_1) = \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt$, and therefore

$$J(x, u) = z(t_1) + m(x(t_1)) = \tilde{m}(\tilde{x}(t_1)), \quad (11.8)$$

where $\tilde{m}(x, z) = z + m(x)$. Thus the Bolza problem is transformed into a Mayer problem.

The correspondence between admissible pairs is one-to-one. For each $u \in \mathcal{U}$, the state x uniquely determines z , and conversely the original state x is recovered by projection from $\tilde{x}$. Therefore optimal controls and trajectories are preserved.

The Lagrange and Mayer forms are obtained as special cases by setting $m \equiv 0$ or $\ell \equiv 0$, respectively. $\square$

With the admissible pairs fixed, the optimal control problem can now be viewed as a constrained optimization problem in infinite-dimensional spaces:

$$\min_{(x,u) \in \mathcal{X} \times \mathcal{U}} J(x, u) \quad \text{subject to} \quad \mathcal{F}(x, u) = 0,$$

where the constraint operator $\mathcal{F}$ is defined by

$$\mathcal{F}(x, u)(t) = \dot{x}(t) - f(x(t), u(t), t), \quad \mathcal{F} : \mathcal{X} \times \mathcal{U} \rightarrow L^1([t_0, t_1]; \mathbb{R}^n). \quad (11.9)$$

The linearization of $\mathcal{F}$ at (x, u) is given by

$$D\mathcal{F}(x, u)[\eta, \psi] = \dot{\eta}(t) - \nabla_x f(x(t), u(t), t)\eta(t) - \nabla_u f(x(t), u(t), t)\psi(t), \quad (11.10)$$

which plays the role of the constraint Jacobian.

This formulation identifies the optimal control problem as an infinite-dimensional analogue of nonlinear programming. The costate variable arises as a Lagrange multiplier associated with the constraint $\mathcal{F}(x, u) = 0$, and the Hamiltonian framework emerges naturally from the corresponding saddle-point structure of the augmented functional.

In particular, under appropriate surjectivity conditions on the linearized operator $D\mathcal{F}(x^\circ, u^\circ)$, which serve as an infinite-dimensional analogue of the linear independence constraint qualification, one obtains the existence of nontrivial multipliers. This provides the rigorous foundation for the derivation of Pontryagin's Minimum Principle and its interpretation as a system of first-order optimality conditions.

The equivalence of the Bolza, Lagrange, and Mayer forms therefore ensures that all subsequent analytical developments using variational arguments, penalty methods, and saddle-point formulations can be carried out within a unified and well-posed functional framework.

11.1.2 ■ Minimum Principle: Variational Approach

Due to the equivalence established in Proposition 11.1, it suffices to consider the Bolza optimal control problem

$$\dot{x}(t) = f(x(t), u(t), t), \quad x(t_0) = x_0 \in \mathbb{R}^n, \quad (11.11)$$

$$J(u) = \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt + m(x(t_1)). \quad (11.12)$$

The objective is to determine an admissible control $u^\circ(\cdot)$ that minimizes the functional (11.12) subject to the dynamics (11.11).

A central role in the analysis is played by the Hamiltonian function

$$H(x, p, u, t) = \ell(x, u, t) + p^\top f(x, u, t). \quad (11.13)$$

This function arises by incorporating the dynamical constraint (11.11) into the objective functional through a time-dependent Lagrange multiplier $p(\cdot)$. The costate $p(t)$ represents the marginal value of the state and quantifies how infinitesimal perturbations of the state propagate through the dynamics to influence the total cost.

Theorem 11.2 (Pontryagin's Minimum Principle). *Let $U \subset \mathbb{R}^m$ be a convex set, and assume that f and ℓ are continuously differentiable in (x, u) and continuous in t . Suppose that $u^\circ(\cdot)$ is an optimal control for (11.11)–(11.12), with associated trajectory $x^\circ(\cdot)$, and assume that $x^\circ(\cdot)$ is continuously differentiable.*

Then there exists an absolutely continuous function $p : [t_0, t_1] \rightarrow \mathbb{R}^n$ such that

$$\dot{x}^\circ(t) = \nabla_p H(x^\circ(t), p(t), u^\circ(t), t), \quad (11.14)$$

$$\dot{p}(t) = -\nabla_x H(x^\circ(t), p(t), u^\circ(t), t), \quad (11.15)$$

for almost every $t \in [t_0, t_1]$, with boundary conditions

$$x^\circ(t_0) = x_0, \quad p(t_1) = \nabla_x m(x^\circ(t_1)), \quad (11.16)$$

and satisfying the minimum condition

$$H(x^\circ(t), p(t), u^\circ(t), t) = \min_{u \in U} H(x^\circ(t), p(t), u, t) \quad \text{for a.e. } t \in [t_0, t_1]. \quad (11.17)$$

Proof. We reformulate the constrained problem (11.11)–(11.12) as an unconstrained variational problem. Let $x(\cdot)$ and $u(\cdot)$ be admissible functions, and introduce a Lagrange multiplier $p(\cdot)$ associated with the constraint $\dot{x} - f(x, u, t) = 0$. Define the augmented functional

$$\widehat{J}(x, u, p) = \int_{t_0}^{t_1} [\ell(x, u, t) + p(t)^\top (f(x, u, t) - \dot{x}(t))] dt + m(x(t_1)). \quad (11.18)$$

Assume that (x°, u°) is optimal. Then there exists $p(\cdot)$ such that (x°, u°, p) is a stationary point of $\widehat{J}$ with respect to admissible variations.

We first eliminate the derivative on x by integration by parts. Using

$$\int_{t_0}^{t_1} -p^\top \dot{x} dt = -p^\top(t_1)x(t_1) + p^\top(t_0)x(t_0) + \int_{t_0}^{t_1} \dot{p}^\top x dt, \quad (11.19)$$

we rewrite (11.18) as

$$\widehat{J}(x, u, p) = \int_{t_0}^{t_1} [H(x, p, u, t) + \dot{p}(t)^\top x(t)] dt - p^\top(t_1)x(t_1) + p^\top(t_0)x(t_0) + m(x(t_1)). \quad (11.20)$$

Variation with respect to x . Let $\eta : [t_0, t_1] \rightarrow \mathbb{R}^n$ be an absolutely continuous function with $\eta(t_0) = 0$, and consider the variation $x_\varepsilon = x^\circ + \varepsilon\eta$. Differentiating (11.20) with respect to ε at $\varepsilon = 0$ yields

$$\begin{aligned} 0 &= \int_{t_0}^{t_1} [\nabla_x H(x^\circ, p, u^\circ, t) + \dot{p}(t)^\top] \eta(t) dt \\ &\quad + [\nabla_x m(x^\circ(t_1)) - p(t_1)^\top] \eta(t_1). \end{aligned} \quad (11.21)$$

Since $\eta(\cdot)$ is arbitrary except for $\eta(t_0) = 0$, the fundamental lemma of the calculus of variations implies

$$\dot{p}(t) = -\nabla_x H(x^\circ(t), p(t), u^\circ(t), t) \quad \text{a.e.}, \quad (11.22)$$

and

$$p(t_1) = \nabla_x m(x^\circ(t_1)). \quad (11.23)$$

Variation with respect to u . Let $\psi : [t_0, t_1] \rightarrow \mathbb{R}^m$ be measurable such that $u^\circ(t) + \delta\psi(t) \in U$ for sufficiently small δ . Consider the variation $u_\delta = u^\circ + \delta\psi$. Differentiating (11.18) gives

$$0 = \int_{t_0}^{t_1} \nabla_u H(x^\circ(t), p(t), u^\circ(t), t)^\top \psi(t) dt. \quad (11.24)$$

Since U is convex and ψ ranges over the admissible directions at $u^\circ(t)$, this implies the pointwise *variational inequality*

$$\nabla_u H(x^\circ(t), p(t), u^\circ(t), t)^\top (v - u^\circ(t)) \geq 0 \quad \text{for all } v \in U, \text{ a.e. } t, \quad (11.25)$$

which reduces to $\nabla_u H(x^\circ(t), p(t), u^\circ(t), t) = 0$ whenever $u^\circ(t)$ lies in the interior of U .

Variation with respect to p . Let $\xi : [t_0, t_1] \rightarrow \mathbb{R}^n$ be arbitrary and consider $p_\varepsilon = p + \varepsilon\xi$. Differentiating (11.18) yields

$$0 = \int_{t_0}^{t_1} \xi(t)^\top (f(x^\circ(t), u^\circ(t), t) - \dot{x}^\circ(t)) dt. \quad (11.26)$$

Since $\xi(\cdot)$ is arbitrary, it follows that

$$\dot{x}^\circ(t) = f(x^\circ(t), u^\circ(t), t) = \nabla_p H(x^\circ(t), p(t), u^\circ(t), t). \quad (11.27)$$

It remains to pass from (11.25) to the minimum condition (11.17). Under the standing hypothesis that $u \mapsto H(x^\circ(t), p(t), u, t)$ is convex on the convex set U —which holds, for instance, when $\ell(x, \cdot, t)$ is convex and $f(x, \cdot, t)$ is affine, as in every example of this chapter—the variational inequality (11.25) is precisely the first-order characterization of a global minimizer over U , and (11.17) follows. Combining this with the state and costate equations yields (11.14)–(11.17). $\square$

Remark 11.3 (Weak versus strong form of the principle). The derivation above uses *weak* (one-parameter, tangential) variations of the control, and consequently delivers the stationarity/variational-inequality form (11.25). Without convexity of H in u this is genuinely weaker than the global minimum condition (11.17): a stationary point of H need not minimize it. The unconditional form of Pontryagin's principle requires *needle* (spike) variations, which perturb the control by a large amount on a vanishingly short interval and therefore probe H globally in u ; that argument is what covers bang–bang problems, where the minimizer sits on the boundary of U and H is affine rather than convex-with-interior-minimum. The examples of Section 11.2 that use the minimum condition on a compact U , the double integrator and the minimum-time problem, should be read as invoking the strong form.

Remark 11.4 (Interpretation and structural meaning). The relations (11.14)–(11.15) define a coupled forward–backward dynamical system. The state equation (11.14) evolves forward in time from the prescribed initial condition, while the costate equation (11.15) evolves backward in time from the terminal condition (11.16). This forward–backward structure reflects the dependence of

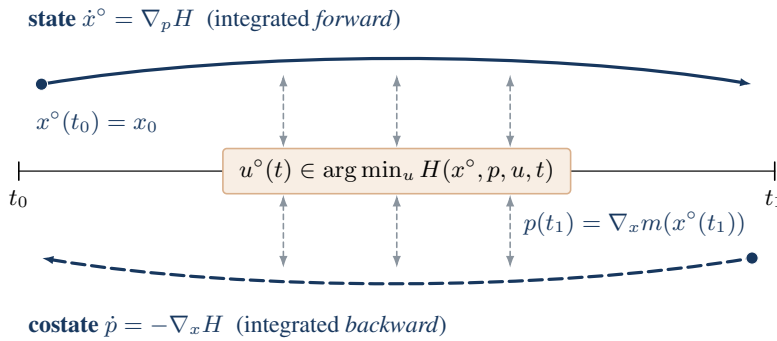


Figure 11.3: The optimality system is a two-point boundary-value problem. The state runs forward from the prescribed initial condition while the costate runs backward from the terminal (transversality) condition; the two are coupled at each instant through pointwise minimization of the Hamiltonian. This forward–backward coupling—not an initial-value problem—is what distinguishes optimal control from ordinary simulation.

the objective functional (11.12) on both the initial and terminal states. The minimum condition (11.17) determines the control at each time by minimizing the Hamiltonian, thereby balancing instantaneous cost with the impact of current decisions on future evolution.

The Hamiltonian (11.13) plays a central structural role. It can be interpreted as the integrand of the augmented functional obtained by incorporating the dynamical constraint (11.11) via a time-dependent Lagrange multiplier. The term $p(t)^\top f(x, u, t)$ evaluates how the control influences future evolution, weighted by the marginal value $p(t)$. Thus, the Hamiltonian encodes both immediate and future contributions to the objective.

The system (11.14)–(11.15) has the canonical structure

$$\dot{x} = \nabla_p H, \quad \dot{p} = -\nabla_x H,$$

which coincides with Hamilton’s equations in classical mechanics. In this analogy, x and p correspond to position and momentum, and the Hamiltonian governs their joint evolution.

The costate arises from the Lagrange multiplier principle applied to the dynamical constraint (11.11). In finite-dimensional optimization, the existence of multipliers relies on constraint qualifications such as LICQ. In the present setting, the constraint $\dot{x} - f(x, u, t) = 0$ defines an operator on function spaces, and its linearization

$$\delta \dot{x}(t) - \nabla_x f(x^\circ(t), u^\circ(t), t) \delta x(t) - \nabla_u f(x^\circ(t), u^\circ(t), t) \delta u(t)$$

must satisfy a suitable surjectivity condition. This condition plays the role of an infinite-dimensional analogue of LICQ and ensures the existence of a nontrivial costate. In the absence of such a condition, abnormal extremals may arise.

Consequently, Pontryagin’s Minimum Principle can be viewed as an infinite-dimensional analogue of the Karush–Kuhn–Tucker conditions. The costate corresponds to the Lagrange multiplier associated with the dynamical constraint, the adjoint equation arises from stationarity with respect to the state, and the minimum condition reflects stationarity with respect to the control.

11.1.3 ■ Minimum Principle via Penalty Approach

We present a derivation of Pontryagin's Minimum Principle based on a penalty formulation. This approach replaces the hard constraint (11.11) by a sequence of unconstrained problems whose minimizers asymptotically satisfy the dynamics.

For each $k > 0$, define the functional

$$\widehat{J}_k(x, u) = \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt + \frac{k}{2} \int_{t_0}^{t_1} \|\dot{x}(t) - f(x(t), u(t), t)\|^2 dt + m(x(t_1)). \quad (11.28)$$

We consider the unconstrained minimization problem $\min_{(x, u)} \widehat{J}_k(x, u)$, where $x(\cdot)$ is absolutely continuous and $u(\cdot)$ is measurable with values in U .

The penalty parameter controls the constraint violation, as the following estimate shows. Assume that ℓ and m are nonnegative. Let (x_k, u_k) be a minimizer of (11.28). Let (x°, u°) be an optimal pair for the original problem. Then

$$\widehat{J}_k(x_k, u_k) \leq \widehat{J}_k(x^\circ, u^\circ) = J(u^\circ). \quad (11.29)$$

Consequently,

$$\int_{t_0}^{t_1} \|\dot{x}_k(t) - f(x_k(t), u_k(t), t)\|^2 dt \leq \frac{2}{k} J(u^\circ). \quad (11.30)$$

Hence, $\dot{x}_k - f(x_k, u_k, t) \rightarrow 0$ in $L^2([t_0, t_1]; \mathbb{R}^n)$, as $k \rightarrow \infty$.

We now impose stationarity of $\widehat{J}_k$ in each of its arguments, beginning with the state. Fix u_k and consider variations $x_k^\varepsilon = x_k + \varepsilon \eta$, where η is absolutely continuous with $\eta(t_0) = 0$. Define $\Phi_k(\varepsilon) := \widehat{J}_k(x_k^\varepsilon, u_k)$. Optimality implies $\Phi'_k(0) = 0$. Differentiating yields

$$\begin{aligned} 0 &= \int_{t_0}^{t_1} \nabla_x \ell(x_k(t), u_k(t), t)^\top \eta(t) dt \\ &\quad + k \int_{t_0}^{t_1} (\dot{x}_k(t) - f(x_k(t), u_k(t), t))^\top (\dot{\eta}(t) - \nabla_x f(x_k(t), u_k(t), t) \eta(t)) dt \\ &\quad + \nabla_x m(x_k(t_1))^\top \eta(t_1). \end{aligned} \quad (11.31)$$

Define the adjoint (penalty costate)

$$p_k(t) := -k(\dot{x}_k(t) - f(x_k(t), u_k(t), t)). \quad (11.32)$$

Substituting into (11.31) gives

$$\begin{aligned} 0 &= \int_{t_0}^{t_1} \nabla_x \ell(x_k, u_k, t)^\top \eta(t) dt \\ &\quad - \int_{t_0}^{t_1} p_k(t)^\top (\dot{\eta}(t) - \nabla_x f(x_k, u_k, t) \eta(t)) dt + \nabla_x m(x_k(t_1))^\top \eta(t_1). \end{aligned} \quad (11.33)$$

Integrating by parts,

$$- \int_{t_0}^{t_1} p_k^\top \dot{\eta} dt = -p_k(t_1)^\top \eta(t_1) + \int_{t_0}^{t_1} \dot{p}_k^\top \eta dt,$$

we obtain

$$\begin{aligned} 0 &= \int_{t_0}^{t_1} [\nabla_x \ell(x_k, u_k, t) + \dot{p}_k(t) + \nabla_x f(x_k, u_k, t)^\top p_k(t)]^\top \eta(t) dt \\ &\quad + [\nabla_x m(x_k(t_1)) - p_k(t_1)]^\top \eta(t_1). \end{aligned} \quad (11.34)$$

Since η is arbitrary, it follows that

$$\dot{p}_k(t) = -\nabla_x \ell(x_k(t), u_k(t), t) - \nabla_x f(x_k(t), u_k(t), t)^\top p_k(t), \quad (11.35)$$

$$p_k(t_1) = \nabla_x m(x_k(t_1)). \quad (11.36)$$

Turning to the control, let ψ be an admissible variation of u_k . Differentiating (11.28) yields

$$0 = \int_{t_0}^{t_1} [\nabla_u \ell(x_k, u_k, t) + \nabla_u f(x_k, u_k, t)^\top p_k(t)]^\top \psi(t) dt, \quad (11.37)$$

Hence,

$$\nabla_u H(x_k(t), p_k(t), u_k(t), t) = 0 \quad \text{for a.e. } t. \quad (11.38)$$

It remains to let the penalty parameter grow. By (11.30), (x_k, u_k) is approximately feasible. Under standard compactness assumptions, there exists a subsequence such that $x_k \rightarrow x^\circ$, $u_k \rightarrow u^\circ$ and $p_k \rightarrow p$. Passing to the limit in (11.35)–(11.38), we obtain

$$\dot{x}^\circ(t) = f(x^\circ(t), u^\circ(t), t), \quad (11.39)$$

$$\dot{p}(t) = -\nabla_x H(x^\circ(t), p(t), u^\circ(t), t), \quad (11.40)$$

$$p(t_1) = \nabla_x m(x^\circ(t_1)), \quad (11.41)$$

$$\nabla_u H(x^\circ(t), p(t), u^\circ(t), t) = 0, \quad (11.42)$$

which coincide with the conditions of Theorem 11.2.

Remark 11.5 (Relation to multiplier theory). The adjoint variable $p_k(t) = -k(\dot{x}_k(t) - f(x_k(t), u_k(t), t))$ arises as a scaled constraint violation. As $k \rightarrow \infty$, this sequence converges to the Lagrange multiplier associated with the constraint (11.11). This is directly analogous to finite-dimensional penalty methods, where multipliers emerge as limits of scaled residuals.

The linearized constraint operator $\eta \mapsto \dot{\eta} - \nabla_x f(x^\circ, u^\circ, t)\eta$ plays the role of a Jacobian. A suitable surjectivity condition ensures the existence of a nontrivial limit $p(\cdot)$, serving as an infinite-dimensional analogue of LICQ. In this sense, Pontryagin's Minimum Principle can be viewed as the KKT system associated with the constrained optimal control problem.

11.1.4 ■ Variations of the Minimum Principle under Boundary Conditions

The form of Pontryagin's Minimum Principle depends on the specification of boundary conditions. We summarize the main variants in a unified framework. Consider the Bolza problem (11.11)–(11.12). Let (x°, u°) be an optimal pair, and let H be the Hamiltonian defined in (11.13). Then there exists a costate $p(\cdot)$ such that the state and costate satisfy

$$\dot{x}^\circ(t) = \nabla_p H(x^\circ(t), p(t), u^\circ(t), t), \quad \dot{p}(t) = -\nabla_x H(x^\circ(t), p(t), u^\circ(t), t),$$

and the control satisfies the minimum condition (11.17). The distinction between different formulations arises through the boundary conditions.

Proposition 11.6 (Minimum principle with general terminal conditions). Let $I \subset \{1, \dots, n\}$ denote the set of indices for which the terminal state is prescribed, i.e., $x_i(t_1) = x_{1i}$, $i \in I$, while the remaining components are free. Then the optimal pair (x°, u°) satisfies the following transversality conditions:

$$p_i(t_1) = \frac{\partial m}{\partial x_i}(x^\circ(t_1)), \quad i \notin I, \quad x_i(t_1) = x_{1i}, \quad i \in I.$$

Proof. The result follows from the variational argument applied to the augmented functional (11.18).

Let η be an admissible variation of the state. If $x_i(t_1)$ is fixed for $i \in I$, then $\eta_i(t_1) = 0$. If $x_i(t_1)$ is free for $i \notin I$, then $\eta_i(t_1)$ is arbitrary. From the variation with respect to x , we obtain

$$\int_{t_0}^{t_1} [\nabla_x H(x^\circ, p, u^\circ, t) + \dot{p}(t)]^\top \eta(t) dt + [\nabla_x m(x^\circ(t_1)) - p(t_1)]^\top \eta(t_1) = 0.$$

Since $\eta(\cdot)$ is arbitrary in the interior, we recover the adjoint equation. At t_1 , the arbitrariness of $\eta_i(t_1)$ for $i \notin I$ implies $p_i(t_1) = \frac{\partial m}{\partial x_i}(x^\circ(t_1))$, while for $i \in I$ no condition is imposed on $p_i(t_1)$, since $\eta_i(t_1) = 0$. This yields the stated result. $\square$

The general transversality conditions specialize to several classical formulations. If the terminal state is free, i.e., $I = \emptyset$, then the costate satisfies the boundary condition $p(t_1) = \nabla_x m(x^\circ(t_1))$, which coincides with the standard form of the Minimum Principle in Theorem 11.2.

If the terminal state is fully prescribed, i.e., $I = \{1, \dots, n\}$, then $x(t_1)$ is fixed and no terminal condition is imposed on $p(t_1)$. This corresponds to the fixed-endpoint formulation, where admissible variations vanish at t_1 .

If the terminal time t_1 is free, an additional scalar transversality condition is obtained. In this case, the optimal trajectory satisfies

$$\frac{\partial m}{\partial t}(x^\circ(t_1), t_1) + H(x^\circ(t_1), p(t_1), u^\circ(t_1), t_1) = 0. \quad (11.43)$$

This condition follows from stationarity with respect to variations of the terminal time and can be interpreted as a continuous-time envelope condition.

Remark 11.7 (Unified interpretation). All boundary conditions arise from the same variational principle applied to (11.18). Free components of the terminal state yield stationarity conditions involving the gradient of m , while fixed components impose constraints on admissible variations. The free terminal time introduces an additional degree of freedom, leading to the scalar condition (11.43). Thus, the various forms of the Minimum Principle differ only in the admissible class of variations, while the Hamiltonian structure and adjoint dynamics remain unchanged.

11.2 ■ Examples of Optimal Control Problems

We illustrate the application of Pontryagin's Minimum Principle through canonical examples. These examples demonstrate how the Hamiltonian system and optimality conditions lead to explicit solution structures.

11.2.1 ■ Linear-Quadratic Regulator (LQR)

The linear-quadratic regulator provides a fundamental test case for the Minimum Principle and yields an explicit characterization of optimal controls. Consider the linear system

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(t_0) = x_0, \quad (11.44)$$

with quadratic cost

$$J(u) = \int_{t_0}^{t_1} (x(t)^\top Qx(t) + u(t)^\top Ru(t)) dt + x(t_1)^\top Mx(t_1), \quad (11.45)$$

where $Q, M \succeq 0$ and $R \succ 0$.

The Hamiltonian (11.13) takes the form

$$H(x, p, u, t) = x^\top Qx + u^\top Ru + p^\top (Ax + Bu).$$

The minimum condition (11.17) gives

$$\nabla_u H = 2Ru + B^\top p = 0, \quad u^*(t) = -\frac{1}{2}R^{-1}B^\top p(t).$$

Substituting into the state equation yields

$$\dot{x}(t) = Ax(t) - \frac{1}{2}BR^{-1}B^\top p(t). \quad (11.46)$$

The adjoint equation (11.15) becomes

$$\dot{p}(t) = -2Qx(t) - A^\top p(t), \quad p(t_1) = 2Mx(t_1). \quad (11.47)$$

Thus (x, p) satisfy the linear Hamiltonian system

$$\frac{d}{dt} \begin{pmatrix} x \\ p \end{pmatrix} = \begin{pmatrix} A & -\frac{1}{2}BR^{-1}B^\top \\ -2Q & -A^\top \end{pmatrix} \begin{pmatrix} x \\ p \end{pmatrix}, \quad (11.48)$$

with two-point boundary conditions $x(t_0) = x_0$ and $p(t_1) = 2Mx(t_1)$.

The two-point boundary value problem can be collapsed into a single matrix differential equation. The terminal condition $p(t_1) = 2Mx(t_1)$ suggests seeking a relation of the form $p(t) = 2P(t)x(t)$ that holds along the whole trajectory; if such a $P(\cdot)$ exists, the costate is eliminated and the optimal control becomes a state feedback. Differentiating this ansatz gives $\dot{p}(t) = 2\dot{P}(t)x(t) + 2P(t)\dot{x}(t)$, and substituting $\dot{x}(t)$ from (11.46) yields

$$\dot{p}(t) = 2\dot{P}(t)x(t) + 2P(t)\left(Ax(t) - BR^{-1}B^\top P(t)x(t)\right).$$

On the other hand the adjoint equation (11.47) gives, after the same substitution $p = 2Px$, $\dot{p}(t) = -2Qx(t) - 2A^\top P(t)x(t)$. Equating the two expressions and dividing by 2 leaves

$$\dot{P}(t)x + P(t)Ax - P(t)BR^{-1}B^\top P(t)x = -Qx - A^\top P(t)x,$$

and since this must hold for every x the ansatz is consistent precisely when $P(\cdot)$ solves the Riccati differential equation

$$\dot{P}(t) = -Q - A^\top P(t) - P(t)A + P(t)BR^{-1}B^\top P(t), \quad P(t_1) = M, \quad (11.49)$$

the terminal value being read off from $p(t_1) = 2Mx(t_1)$.

With $P(\cdot)$ in hand the optimal control follows immediately from $u^* = -\frac{1}{2}R^{-1}B^\top p$ and $p = 2Px$, namely

$$u^*(t) = -R^{-1}B^\top P(t)x(t), \quad (11.50)$$

so that the optimal law is a linear, time-varying state feedback and the closed-loop system is

$$\dot{x}(t) = (A - BR^{-1}B^\top P(t))x(t).$$

11.2.2 ■ LQR with Fixed Terminal State

We now consider the same dynamics $\dot{x}(t) = Ax(t) + Bu(t)$, but with terminal constraint $x(t_1) = x_1$. The cost is taken as

$$J(u) = \frac{1}{2} \int_0^{t_1} (x(t)^\top Qx(t) + u(t)^\top Ru(t)) dt,$$

so that the terminal condition replaces the Mayer term. (The factor $\frac{1}{2}$ is a customary normalization; relative to the preceding subsection it rescales the costate by $\frac{1}{2}$ and leaves the optimal control unchanged.)

The Hamiltonian becomes

$$H(x, p, u, t) = p^\top (Ax + Bu) + \frac{1}{2} (x^\top Qx + u^\top Ru).$$

The minimum condition yields $u^\circ(t) = -R^{-1}B^\top p(t)$, and the adjoint equation is $\dot{p}(t) = -A^\top p(t) - Qx(t)$.

Thus the state–costate system is

$$\frac{d}{dt} \begin{pmatrix} x \\ p \end{pmatrix} = \begin{pmatrix} A & -BR^{-1}B^\top \\ -Q & -A^\top \end{pmatrix} \begin{pmatrix} x \\ p \end{pmatrix}, \quad x(0) = x_0, \quad x(t_1) = x_1. \quad (11.51)$$

In contrast to the previous case, the boundary condition at t_1 is imposed on x rather than p . As a result, (11.51) is a genuine two-point boundary value problem.

Let $\psi(t, \tau)$ denote the transition matrix of (11.51), partitioned as

$$\psi(t, \tau) = \begin{pmatrix} \psi_{11}(t, \tau) & \psi_{12}(t, \tau) \\ \psi_{21}(t, \tau) & \psi_{22}(t, \tau) \end{pmatrix}.$$

Then $x(t_1) = \psi_{11}(t_1, 0)x_0 + \psi_{12}(t_1, 0)p_0$.

Assuming $\psi_{12}(t_1, 0)$ is invertible, we obtain

$$p_0 = \psi_{12}(t_1, 0)^{-1} (x_1 - \psi_{11}(t_1, 0)x_0),$$

and hence

$$p(t) = \psi_{21}(t, 0)x_0 + \psi_{22}(t, 0)p_0, \quad u^\circ(t) = -R^{-1}B^\top p(t).$$

The optimal trajectory is $x^\circ(t) = \psi_{11}(t, 0)x_0 + \psi_{12}(t, 0)p_0$.

This formulation highlights the distinction between free and fixed terminal conditions: in the former, the costate is specified at t_1 via (11.47), leading to the Riccati equation (11.49); in the latter, the terminal state is fixed, and the problem is solved by propagating the Hamiltonian system and matching boundary conditions through the transition matrix.

11.2.3 ■ Double Integrator and Bang–Bang Control

We consider the second-order system $\ddot{y} = u$, written in state-space form as $\dot{x}_1(t) = x_2(t)$, $\dot{x}_2(t) = u(t)$, with control constraint $|u(t)| \leq 1$. This is a canonical example illustrating the emergence of bang–bang optimal controls under input constraints.

Let the running cost be identically zero and suppose that the objective is to steer the system between prescribed boundary conditions. The Hamiltonian (11.13) reduces to $H(x, p, u) = p_1x_2 + p_2u$.

The minimum condition (11.17) requires $u^\circ(t) \in \arg \min_{|u| \leq 1} p_2(t)u$, which yields the bang–bang control law $u^\circ(t) = -\operatorname{sgn}(p_2(t))$, with the convention that any $u \in [-1, 1]$ is admissible when $p_2(t) = 0$.

The adjoint equations (11.15) are

$$\dot{p}_1(t) = -\frac{\partial H}{\partial x_1} = 0, \quad \dot{p}_2(t) = -\frac{\partial H}{\partial x_2} = -p_1(t).$$

Integrating gives $p_1(t) = c_1$ and $p_2(t) = -c_1 t + c_2$, for constants c_1, c_2 determined by boundary conditions.

Since $p_2(t)$ is affine in t , it can change sign at most once. Hence the control $u^\circ(t)$ switches at most once, and the optimal control is of bang–bang type with a single switching time.

On intervals where $u^\circ(t) = \pm 1$, the state dynamics reduce to $\dot{x}_1(t) = x_2(t), \dot{x}_2(t) = \pm 1$. Integrating twice yields $x_2(t) = \pm t + c$ and $x_1(t) = \pm \frac{1}{2}t^2 + ct + d$, so the optimal trajectories are piecewise quadratic curves. The switching time is determined by matching boundary conditions and continuity of the state.

This example illustrates that when the Hamiltonian is affine in the control and the admissible set is compact, the Minimum Principle naturally leads to bang–bang controls, with switching determined by the sign of the switching function $p_2(t)$.

Example 11.8 Time-optimal synthesis for the double integrator.

We solve the minimum-time problem of steering the double integrator from rest at position $a > 0$, i.e., $(x_1(0), x_2(0)) = (a, 0)$, to rest at the origin, with $|u| \leq 1$.

Since the optimal control is bang–bang with at most one switch, and the system must first move left and then brake, try $u^\circ(t) = -1$ on $[0, t_s)$ and $u^\circ(t) = +1$ on $(t_s, T]$. On the first arc, $x_2(t) = -t$ and $x_1(t) = a - \frac{1}{2}t^2$. On the final arc, integrating backward from the origin under $u = +1$ shows that every trajectory that reaches the origin with $u = +1$ lies on the curve $x_1 = \frac{1}{2}x_2^2$ with $x_2 \leq 0$. Matching the first arc to this curve at $t = t_s$ requires $a - \frac{1}{2}t_s^2 = \frac{1}{2}t_s^2$, hence $t_s = \sqrt{a}$ and $T = t_s + |x_2(t_s)| = 2\sqrt{a}$, since the final arc must bring the velocity $x_2(t_s) = -\sqrt{a}$ back to zero at unit rate. The costate consistency check succeeds: p_2 is affine and vanishes exactly once, at t_s .

The same construction for arbitrary initial conditions yields a complete feedback synthesis. Define the *switching curve*

$$\Gamma := \{(x_1, x_2) : x_1 = -\frac{1}{2}x_2|x_2|\},$$

composed of the two parabolic arcs that flow into the origin. The time-optimal feedback law is

$$u^\circ(x) = \begin{cases} -1, & \text{above } \Gamma \text{ or on } \Gamma \text{ with } x_2 > 0, \\ +1, & \text{below } \Gamma \text{ or on } \Gamma \text{ with } x_2 < 0. \end{cases}$$

Trajectories drive toward Γ under one extreme control, then slide along Γ into the origin under the other. This is among the earliest examples of an optimal feedback law synthesized globally from the Minimum Principle. $\square$

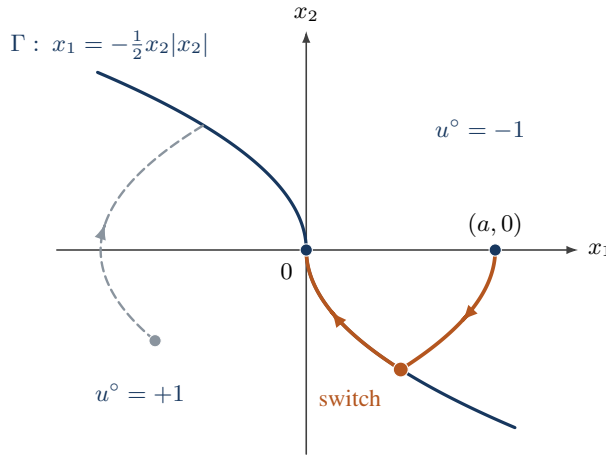


Figure 11.4: Time-optimal feedback synthesis for the double integrator $\ddot{y} = u$, $|u| \leq 1$. The switching curve Γ is made of the two parabolic arcs that flow into the origin; the optimal control is $u^\circ = -1$ above Γ and $u^\circ = +1$ below it. Off Γ the optimal control takes one extreme value, driving the state *toward* Γ (from $(a, 0)$, the arc with $u = -1$); upon reaching Γ the control switches once and the state slides into the origin along Γ with $u = +1$. A trajectory starting below Γ (dashed) uses $u = +1$ first and switches on the upper branch. This is a feedback law $u^\circ(x)$ synthesized globally from the Minimum Principle.

11.2.4 ■ Minimum-Time Control Problem

We next consider a minimum-time problem for the linear system $\dot{x}(t) = Ax(t) + bu(t)$, with constraint $|u(t)| \leq 1$, where the objective is to steer $x(0) = x_0$ to $x(t_1) = x_1$ in minimum time. The cost functional is $J(u) = \int_0^{t_1} 1 \, dt = t_1$.

The Hamiltonian is given by $H(x, p, u, t) = 1 + p^\top (Ax + bu)$. The minimum condition (11.17) yields $u^\circ(t) \in \arg \min_{|u| \leq 1} p(t)^\top b u$, which gives the bang–bang control law

$$u^\circ(t) = \begin{cases} 1, & p(t)^\top b < 0, \\ -1, & p(t)^\top b > 0. \end{cases}$$

The adjoint equation (11.15) is $\dot{p}(t) = -A^\top p(t)$, which is a linear homogeneous system. Hence $p(t) = \exp(-A^\top t)p(0)$.

Since the terminal time t_1 is free, the transversality condition (11.43) applies and gives $H(x^\circ(t_1), p(t_1), u^\circ(t_1), t_1) = 0$. Substituting the Hamiltonian yields $1 + p(t_1)^\top Ax^\circ(t_1) + p(t_1)^\top b u^\circ(t_1) = 0$. Using the optimal control law, this becomes $1 + p(t_1)^\top Ax^\circ(t_1) - |p(t_1)^\top b| = 0$.

The function $t \mapsto p(t)^\top b$ is called the *switching function*. Since $p(t)$ evolves linearly, the number of sign changes of $p(t)^\top b$ determines the number of switches in the optimal control.

If all eigenvalues of A are real, then $p(t)^\top b$ is a linear combination of functions of the form $t^j e^{-\lambda t}$ with λ real, and such a combination either vanishes identically or has at most $n - 1$ zeros. Consequently, the optimal control exhibits at most $n - 1$ switchings (Feldbaum's theorem). The hypothesis cannot be dropped: for the harmonic oscillator, whose eigenvalues are $\pm i\omega$, the switching function oscillates and time-optimal controls may switch arbitrarily many times on long horizons.

Example 11.9 Optimal consumption over a finite horizon.

An investor holds wealth $x(t)$ earning interest at rate $r > 0$ and consumes at rate $c(t) \geq 0$:

$$\dot{x}(t) = rx(t) - c(t), \quad x(0) = x_0 > 0, \quad x(T) = 0.$$

The objective is to maximize the discounted utility $\int_0^T e^{-\rho t} 2\sqrt{c(t)} dt$ with subjective discount rate $\rho > 0$; equivalently, we minimize $J(c) = -\int_0^T e^{-\rho t} 2\sqrt{c(t)} dt$. The Hamiltonian $H(x, p, c, t) = -2e^{-\rho t}\sqrt{c} + p(rx - c)$ is strictly convex in c , and the interior minimum condition $\partial H/\partial c = -e^{-\rho t}/\sqrt{c} - p = 0$ requires $p < 0$ and gives

$$c^\circ(t) = \frac{e^{-2\rho t}}{p(t)^2}.$$

The costate equation is $\dot{p} = -\partial H/\partial x = -rp$, so $p(t) = p_0 e^{-rt}$ with $p_0 < 0$, and therefore $c^\circ(t) = c_0 e^{2(r-\rho)t}$ with $c_0 = 1/p_0^2$. Optimal consumption grows exponentially when the return dominates impatience ($r > \rho$) and decays in the opposite case, a structural conclusion obtained *before* any boundary data are used. The constant c_0 is then fixed by the budget: solving the linear state equation and imposing $x(T) = 0$ yields

$$c_0 = \frac{x_0}{\int_0^T e^{(r-2\rho)s} ds},$$

and the wealth trajectory decreases to zero exactly at the horizon. Note how the Minimum Principle converts an infinite-dimensional decision, the entire consumption path, into a one-parameter family of extremals indexed by the initial costate p_0 , pinned down by the terminal condition $x(T) = 0$. $\square$

11.3 ■ Saddle-Point Maximum Principle

We consider a convex optimal control problem with path constraints and present a saddle-point characterization that generalizes the Pontryagin Minimum Principle in a convex setting.

11.3.1 ■ Saddle-Point Theorem**Theorem 11.10 (Convex Saddle-Point Maximum Principle with Path Constraint).**

Let $t_0 < t_1$ and consider the problem

$$\min_{(x,u)} J(x, u) = \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt + m(x(t_1)) \quad (11.52)$$

$$\text{s.t. } \dot{x}(t) = f(x(t), u(t), t) \quad \text{a.e. } t, \quad (11.53)$$

$$x(t_0) = x_0, \quad (11.54)$$

$$u(t) \in U \quad \text{a.e.}, \quad (11.55)$$

$$c(x(t), t) \leq 0 \quad \text{a.e. } t. \quad (11.56)$$

Let $X := W^{1,2}([t_0, t_1]; \mathbb{R}^n)$ and $\mathcal{U} := L^2([t_0, t_1]; \mathbb{R}^m)$.

Assume that $U \subset \mathbb{R}^m$ is nonempty, convex, and compact; $\ell(\cdot, \cdot, t)$ is convex in (x, u) and measurable in t ; m is convex; $f(\cdot, \cdot, t)$ is affine in (x, u) ; and $c(\cdot, t)$ is convex for each t . Assume further that a Slater point exists, i.e., there is $(\bar{x}, \bar{u})$ satisfying the constraints with $c(\bar{x}(t), t) < 0$ for a.e. t .

Then $(x^*, u^*) \in X \times \mathcal{U}$ is a global minimizer if and only if there exist

$$p^* \in W^{1,2}([t_0, t_1]; \mathbb{R}^n), \quad \mu^* \in L^2([t_0, t_1]), \quad \mu^*(t) \geq 0 \text{ a.e.},$$

such that the Lagrangian

$$\mathcal{L}(x, u, p, \mu) = \int_{t_0}^{t_1} \left[\ell(x, u, t) + p(t)^\top (\dot{x} - f(x, u, t)) + \mu(t)c(x, t) \right] dt + m(x(t_1)) \quad (11.57)$$

admits a saddle point at (x^*, u^*, p^*, μ^*) , i.e.,

$$\mathcal{L}(x^*, u^*, p, \mu) \leq \mathcal{L}(x^*, u^*, p^*, \mu^*) \leq \mathcal{L}(x, u, p^*, \mu^*). \quad (11.58)$$

Moreover, (x^*, u^*, p^*, μ^*) satisfies

$$\dot{x}^*(t) = f(x^*(t), u^*(t), t), \quad (11.59)$$

$$\dot{p}^*(t) = -\nabla_x H(x^*(t), p^*(t), u^*(t), \mu^*(t), t), \quad (11.60)$$

with terminal condition

$$p^*(t_1) = \nabla m(x^*(t_1)), \quad (11.61)$$

and

$$u^*(t) \in \arg \min_{u \in U} H(x^*(t), p^*(t), u, \mu^*(t), t), \quad (11.62)$$

where

$$H(x, p, u, \mu, t) = \ell(x, u, t) + p^\top f(x, u, t) + \mu c(x, t). \quad (11.63)$$

The complementarity conditions hold:

$$c(x^*(t), t) \leq 0, \quad \mu^*(t) \geq 0, \quad \mu^*(t)c(x^*(t), t) = 0 \quad \text{a.e.} \quad (11.64)$$

Proof. We formulate the problem as a convex optimization problem in the Hilbert space $X \times \mathcal{U}$, where $X = W^{1,2}([t_0, t_1]; \mathbb{R}^n)$ and $\mathcal{U} = L^2([t_0, t_1]; \mathbb{R}^m)$. Define the constraint operator $\mathcal{F}(x, u)(t) := \dot{x}(t) - f(x(t), u(t), t)$, which maps $X \times \mathcal{U}$ into $L^2([t_0, t_1]; \mathbb{R}^n)$. Under the assumption that f is affine in (x, u) , the operator $\mathcal{F}$ is affine and continuous.

Define the feasible set

$$\mathcal{K} = \{(x, u) \in X \times \mathcal{U} : \mathcal{F}(x, u) = 0, x(t_0) = x_0, u(t) \in U \text{ a.e.}, c(x(t), t) \leq 0 \text{ a.e.}\}.$$

Since f is affine, U is convex, and $c(\cdot, t)$ is convex, it follows that $\mathcal{K}$ is convex. The existence of a strictly feasible pair ensures that $\mathcal{K}$ is nonempty and satisfies a Slater-type condition. The cost functional

$$J(x, u) = \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt + m(x(t_1))$$

is convex in (x, u) by assumption. Moreover, under standard growth conditions on ℓ and m , the mapping J is lower semicontinuous on $X \times \mathcal{U}$ with respect to the weak topology. We consider the convex optimization problem

$$\min_{(x, u) \in X \times \mathcal{U}} J(x, u) \quad \text{subject to } (x, u) \in \mathcal{K}.$$

Let (x^*, u^*) be an optimal solution. Define the epigraph of the constrained problem

$$\mathcal{E} = \{(x, u, \alpha) \in X \times \mathcal{U} \times \mathbb{R} : (x, u) \in \mathcal{K}, J(x, u) \leq \alpha\}.$$

The set $\mathcal{E}$ is convex. Under the stated assumptions, it is also closed in the product topology induced by $X \times \mathcal{U} \times \mathbb{R}$. Define the strict sublevel set $\mathcal{S} = \{(x^*, u^*, \beta) : \beta < J(x^*, u^*)\}$. This set is convex, nonempty, and disjoint from $\mathcal{E}$.

By the Hahn–Banach separation theorem, there exists a nontrivial continuous linear functional on $X \times \mathcal{U} \times \mathbb{R}$ that separates $\mathcal{E}$ and $\mathcal{S}$. Hence, there exist $\lambda \in \mathbb{R}$, $p^* \in L^2([t_0, t_1]; \mathbb{R}^n)$, and $\mu^* \in L^2([t_0, t_1])$ such that $\mu^*(t) \geq 0$ almost everywhere and

$$\lambda\alpha + \int_{t_0}^{t_1} p^*(t)^\top (\dot{x}(t) - f(x(t), u(t), t)) dt + \int_{t_0}^{t_1} \mu^*(t) c(x(t), t) dt \geq \lambda\beta \quad (11.65)$$

for all $(x, u, \alpha) \in \mathcal{E}$ and $(x^*, u^*, \beta) \in \mathcal{S}$.

Letting $\alpha = J(x, u)$ and taking the supremum over $\beta < J(x^*, u^*)$ yields

$$\lambda J(x, u) + \int_{t_0}^{t_1} p^*(t)^\top (\dot{x} - f(x, u, t)) dt + \int_{t_0}^{t_1} \mu^*(t) c(x(t), t) dt \geq \lambda J(x^*, u^*) \quad (11.66)$$

for all $(x, u) \in X \times \mathcal{U}$.

We now show that $\lambda > 0$. If $\lambda = 0$, the inequality reduces to

$$\int_{t_0}^{t_1} p^*(t)^\top (\dot{x} - f(x, u, t)) dt + \int_{t_0}^{t_1} \mu^*(t) c(x(t), t) dt \geq 0$$

for all (x, u) , which contradicts the existence of a strictly feasible point (Slater condition). Hence $\lambda > 0$, and by normalization we may assume $\lambda = 1$.

This establishes the saddle inequality

$$\mathcal{L}(x^*, u^*, p, \mu) \leq \mathcal{L}(x^*, u^*, p^*, \mu^*) \leq \mathcal{L}(x, u, p^*, \mu^*),$$

where $\mu \geq 0$.

We now derive the first-order conditions. First, variation with respect to p yields $\int_{t_0}^{t_1} \xi(t)^\top (\dot{x}^*(t) - f(x^*(t), u^*(t), t)) dt = 0$ for all $\xi \in L^2$, hence $\dot{x}^*(t) = f(x^*(t), u^*(t), t)$ a.e.. Next, consider variation with respect to x . Let $\eta \in W^{1,2}$ with $\eta(t_0) = 0$. Differentiating the Lagrangian in direction η gives

$$\begin{aligned} 0 &= \int_{t_0}^{t_1} \left[\nabla_x \ell(x^*, u^*, t) + \mu^*(t) \nabla_x c(x^*, t) \right]^\top \eta(t) dt \\ &\quad + \int_{t_0}^{t_1} p^*(t)^\top \left(\dot{\eta}(t) - \nabla_x f(x^*, u^*, t) \eta(t) \right) dt + \nabla m(x^*(t_1))^\top \eta(t_1). \end{aligned} \quad (11.67)$$

Integrating by parts,

$$\int_{t_0}^{t_1} p^*(t)^\top \dot{\eta}(t) dt = p^*(t_1)^\top \eta(t_1) - \int_{t_0}^{t_1} \dot{p}^*(t)^\top \eta(t) dt,$$

we obtain

$$0 = \int_{t_0}^{t_1} \left[\nabla_x \ell + \mu^* \nabla_x c - \dot{p}^* - \nabla_x f^\top p^* \right]^\top \eta dt + (\nabla m(x^*(t_1)) - p^*(t_1))^\top \eta(t_1). \quad (11.68)$$

Since η is arbitrary, it follows that

$$\dot{p}^*(t) = -\nabla_x \ell(x^*, u^*, t) - \nabla_x f(x^*, u^*, t)^\top p^*(t) - \mu^*(t) \nabla_x c(x^*(t), t),$$

and $p^*(t_1) = \nabla m(x^*(t_1))$. Finally, variation with respect to u yields

$$\int_{t_0}^{t_1} \nabla_u H(x^*, p^*, u^*, \mu^*, t)^\top (u(t) - u^*(t)) dt \geq 0$$

for all admissible u . By convexity of U , this implies the pointwise minimization condition.

The complementarity conditions follow from convex duality: since μ^* is the multiplier associated with the inequality constraint, it belongs to the dual cone, yielding

$$\mu^*(t) \geq 0, \quad c(x^*(t), t) \leq 0, \quad \mu^*(t) c(x^*(t), t) = 0.$$

Conversely, suppose a saddle point exists. Then for any admissible (x, u) ,

$$J(x^*, u^*) = \mathcal{L}(x^*, u^*, p^*, \mu^*) \leq \mathcal{L}(x, u, p^*, \mu^*) \leq J(x, u),$$

where the last inequality follows from feasibility and $\mu^* \geq 0$. Hence (x^*, u^*) is a global minimizer. This completes the proof. $\square$

Remark 11.11 (Connection with finite-dimensional optimization). The above result is the infinite-dimensional analogue of the Karush–Kuhn–Tucker conditions. In finite dimensions, multipliers correspond to gradients of constraints, while in optimal control the costate $p(\cdot)$ is the multiplier associated with the dynamical constraint, and $\mu(\cdot)$ corresponds to path constraints. The adjoint equation plays the role of stationarity with respect to the state, and the minimum condition corresponds to stationarity with respect to the control. Thus, the optimality system can be interpreted as a continuous-time KKT system with a saddle-point structure.

11.3.2 ■ Scalar Linear-Quadratic Problem with a Linear State Constraint

We illustrate Theorem 11.10 through a scalar linear-quadratic optimal control problem with a linear state constraint. Consider the Bolza problem (11.52)–(11.56) given by

$$\min_{u(\cdot)} J(u) = \int_0^T \left(\frac{1}{2} q x(t)^2 + \frac{1}{2} r u(t)^2 \right) dt + \frac{1}{2} q_f x(T)^2, \quad (11.69)$$

$$\text{s.t. } \dot{x}(t) = ax(t) + bu(t), \quad x(0) = x_0, \quad (11.70)$$

$$x(t) \leq x_{\max} \quad \text{for a.e. } t \in [0, T], \quad (11.71)$$

where $r > 0$ and $q, q_f \geq 0$.

Proposition 11.12 (Free-boundary structure). Let (x^*, u^*) be an optimal solution with adjoint p^* and multiplier μ^* . Then $u^*(t) = -\frac{b}{r} p^*(t)$ and (x^*, p^*) satisfy

$$\dot{x}^* = ax^* - \frac{b^2}{r} p^*, \quad \dot{p}^* = -qx^* - ap^* - \mu^*, \quad p^*(T) = q_f x^*(T).$$

Moreover, there exists $t_s \in [0, T]$ such that the trajectory is free for $t < t_s$ and constrained for $t \geq t_s$, with $x^*(t) = x_{\max}$ on $[t_s, T]$ and

$$x_0 \exp \left(\int_0^{t_s} \left(a - \frac{b^2}{r} P(s) \right) ds \right) = x_{\max},$$

where P solves $\dot{P} = -q - 2aP + \frac{b^2}{r}P^2$, $P(T) = q_f$.

Proof. From the Hamiltonian, the minimum condition $\partial_u H = 0$ gives $ru^*(t) + bp^*(t) = 0$, hence $u^*(t) = -\frac{b}{r}p^*(t)$. Substituting into the state equation yields $\dot{x}^*(t) = ax^*(t) - \frac{b^2}{r}p^*(t)$, and the adjoint equation becomes $\dot{p}^*(t) = -qx^*(t) - ap^*(t) - \mu^*(t)$.

On any interval where $x^*(t) < x_{\max}$, the complementarity condition implies $\mu^*(t) = 0$, so the system reduces to

$$\dot{x}^*(t) = ax^*(t) - \frac{b^2}{r}p^*(t), \quad \dot{p}^*(t) = -qx^*(t) - ap^*(t).$$

Assume $x^*(t) \neq 0$ and set $p^*(t) = P(t)x^*(t)$. Differentiating gives $\dot{p}^*(t) = \dot{P}(t)x^*(t) + P(t)\dot{x}^*(t)$. Substituting $\dot{x}^*(t)$ yields

$$\dot{p}^*(t) = \dot{P}(t)x^*(t) + P(t) \left(ax^*(t) - \frac{b^2}{r}P(t)x^*(t) \right).$$

On the other hand, from the adjoint equation, $\dot{p}^*(t) = -(q + aP(t))x^*(t)$. Equating the two expressions and dividing by $x^*(t)$ gives $\dot{P}(t) + aP(t) - \frac{b^2}{r}P(t)^2 = -q - aP(t)$, which simplifies to the Riccati equation

$$\dot{P}(t) = -q - 2aP(t) + \frac{b^2}{r}P(t)^2, \quad P(T) = q_f.$$

Substituting back into the state equation yields $\dot{x}^*(t) = (a - \frac{b^2}{r}P(t))x^*(t)$, which integrates to

$$x^*(t) = x_0 \exp \left(\int_0^t \left(a - \frac{b^2}{r}P(s) \right) ds \right).$$

If the free trajectory satisfies $x^*(t) < x_{\max}$ for all t , then $\mu^* \equiv 0$ and the unconstrained solution is optimal. Otherwise, let t_s be the first time such that $x^*(t_s) = x_{\max}$. Continuity of x^* implies

$$x_0 \exp \left(\int_0^{t_s} \left(a - \frac{b^2}{r}P(s) \right) ds \right) = x_{\max}.$$

For $t \geq t_s$, feasibility requires $x^*(t) = x_{\max}$. Then $\dot{x}^*(t) = 0$, and substituting into the state equation gives $0 = ax_{\max} + bu^*(t)$, hence $u^*(t) = -\frac{a}{b}x_{\max}$. The adjoint equation becomes $\dot{p}^*(t) = -qx_{\max} - ap^*(t) - \mu^*(t)$, where $\mu^*(t) \geq 0$ ensures that the constraint remains active.

Since the problem is convex, once the trajectory reaches the boundary it cannot leave without violating optimality. This establishes the free-boundary structure and completes the proof. $\square$

11.4 ■ Kalman Filtering via the Minimum Principle

Optimal estimation is often taught as a subject apart from optimal control. It is not. As Figure 11.5 anticipates, the filtering problem can be recast as a deterministic optimal control problem and solved with the Minimum Principle developed above; the Kalman–Bucy filter then emerges as the optimal control law, dual to the linear-quadratic regulator.

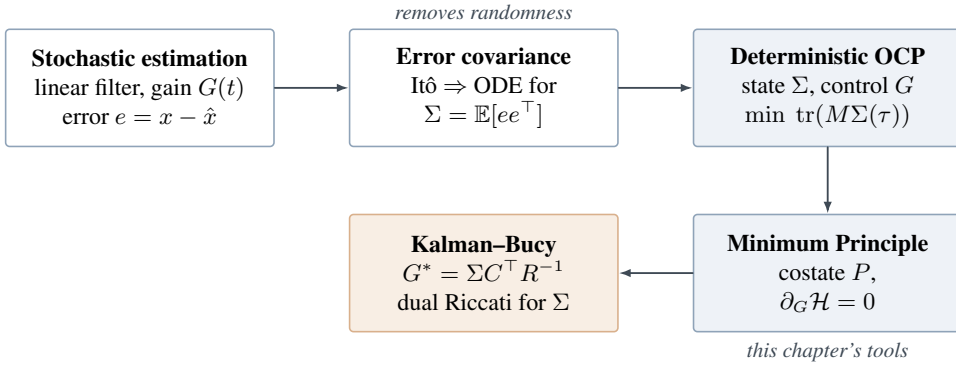


Figure 11.5: How Kalman filtering fits into optimal control. The stochastic estimation problem is converted, via the Itô dynamics of the error covariance, into a purely *deterministic* optimal control problem on the cone of symmetric matrices, with the covariance Σ as state and the gain G as control. Pontryagin’s Minimum Principle then delivers the optimal gain and the Kalman–Bucy Riccati equation, so estimation becomes an application of the very machinery developed for control.

11.4.1 ■ Stochastic Estimation Problem and Motivation

We begin with a continuous-time linear stochastic system observed through noisy measurements. Let $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}, \mathbb{P})$ be a filtered probability space, and consider

$$dx(t) = A(t)x(t)dt + B(t)dW(t), \quad (11.72)$$

$$dz(t) = C(t)x(t)dt + dV(t), \quad (11.73)$$

where $x(t) \in \mathbb{R}^n$ is the state and $z(t) \in \mathbb{R}^p$ is the observation process.

The processes $W(\cdot)$ and $V(\cdot)$ are independent Wiener processes with covariance structure $\mathbb{E}[dW(t)dW(t)^\top] = Q(t)dt$ and $\mathbb{E}[dV(t)dV(t)^\top] = R(t)dt$, where $Q(t) \succeq 0$ represents process noise and $R(t) \succ 0$ represents observation noise.

At each time t , the objective is to estimate the state $x(t)$ based on the observation history $\{z(s) : s \leq t\}$. Instead of addressing the general nonlinear filtering problem, we restrict attention to linear estimators of the form

$$d\hat{x}(t) = A(t)\hat{x}(t)dt + G(t)(dz(t) - C(t)\hat{x}(t)dt), \quad (11.74)$$

where $G(t) \in \mathbb{R}^{n \times p}$ is a deterministic gain to be designed.

The quantity $dz(t) - C(t)\hat{x}(t)dt$ is called the innovation. It represents the discrepancy between the observed signal and its prediction based on the current estimate. The gain $G(t)$ determines how this discrepancy is used to update the estimate.

Define the estimation error $e(t) = x(t) - \hat{x}(t)$. We derive its dynamics explicitly. Using (11.72) and (11.74),

$$\begin{aligned} de(t) &= dx(t) - d\hat{x}(t) \\ &= (A(t)x(t)dt + B(t)dW(t)) - (A(t)\hat{x}(t)dt + G(t)(dz(t) - C(t)\hat{x}(t)dt)). \end{aligned}$$

Substituting $dz(t) = C(t)x(t)dt + dV(t)$ yields

$$\begin{aligned} de(t) &= A(t)(x(t) - \hat{x}(t))dt - G(t)(C(t)x(t)dt + dV(t) - C(t)\hat{x}(t)dt) + B(t)dW(t) \\ &= (A(t) - G(t)C(t))e(t)dt + B(t)dW(t) - G(t)dV(t). \end{aligned}$$

Therefore,

$$de(t) = (A(t) - G(t)C(t))e(t) dt + B(t) dW(t) - G(t) dV(t). \quad (11.75)$$

The performance of the estimator is measured by the terminal mean-square error $J(G) = \mathbb{E}[e(\tau)^\top M e(\tau)]$ with $M \succ 0$. The objective is to determine a gain function $G(\cdot)$ that minimizes $J(G)$.

This defines a stochastic optimization problem over deterministic gain functions. The key step in the analysis is to reformulate this problem as a deterministic optimal control problem, which will be carried out in the next subsection.

11.4.2 ■ Deterministic Optimal Control Reformulation

A key observation is that, for linear systems with Gaussian noise, the distribution of the estimation error $e(t)$ is fully characterized by its second moment. In particular, the covariance matrix $\Sigma(t) = \mathbb{E}[e(t)e(t)^\top]$ contains all relevant information needed to evaluate the performance criterion.

We now derive the evolution equation for $\Sigma(t)$. Applying Itô's formula to the matrix-valued process $e(t)e(t)^\top$ yields

$$d(ee^\top) = (de)e^\top + e(de)^\top + (de)(de)^\top.$$

Substituting the error dynamics (11.75), we compute each term explicitly.

The drift contributions are given by

$$(de)e^\top + e(de)^\top = (A - GC)ee^\top dt + ee^\top (A - GC)^\top dt.$$

The quadratic variation term is obtained from the diffusion components:

$$(de)(de)^\top = (B dW - G dV)(B dW - G dV)^\top.$$

Using the independence of W and V , together with the covariance identities $\mathbb{E}[dW dW^\top] = Q dt$ and $\mathbb{E}[dV dV^\top] = R dt$, we obtain

$$\mathbb{E}[(de)(de)^\top] = BQB^\top dt + GRG^\top dt.$$

Taking expectations and using the fact that stochastic integrals have zero mean, we obtain the deterministic matrix differential equation

$$\dot{\Sigma}(t) = (A(t) - G(t)C(t))\Sigma(t) + \Sigma(t)(A(t) - G(t)C(t))^\top + B(t)Q(t)B(t)^\top + G(t)R(t)G(t)^\top, \quad (11.76)$$

with initial condition $\Sigma(t_0) = \Sigma_0$.

The cost functional can be expressed entirely in terms of $\Sigma(t)$. Using the identity $\mathbb{E}[x^\top M x] = \text{tr}(M \mathbb{E}[xx^\top])$, we obtain $J(G) = \text{tr}(M \Sigma(\tau))$.

Consequently, the stochastic estimation problem is equivalent to the deterministic optimal control problem

$$\begin{aligned} \min_{G(\cdot)} \quad & \text{tr}(M \Sigma(\tau)) \\ \text{s.t.} \quad & \dot{\Sigma} = (A - GC)\Sigma + \Sigma(A - GC)^\top + BQB^\top + GRG^\top, \\ & \Sigma(t_0) = \Sigma_0. \end{aligned} \quad (11.77)$$

In this formulation, the state variable is the covariance matrix $\Sigma(t) \in \mathbb{S}_+^n$, and the control variable is the gain $G(t) \in \mathbb{R}^{n \times p}$. The stochastic nature of the original problem has been eliminated; all uncertainty is captured by the deterministic evolution of $\Sigma(t)$.

This reformulation is fundamental. It converts the stochastic filtering problem into a deterministic optimal control problem on the space of symmetric matrices, thereby enabling the use of Pontryagin's Minimum Principle to characterize optimal estimators.

11.4.3 ■ Solution via the Minimum Principle

We now solve the deterministic optimal control problem (11.77) using Pontryagin's Minimum Principle. Let the state be $\Sigma(t) \in \mathbb{S}_+^n$ and the control be $G(t) \in \mathbb{R}^{n \times p}$. Introduce a costate matrix $P(t) \in \mathbb{S}^n$ and define the Hamiltonian

$$\mathcal{H}(\Sigma, G, P, t) = \text{tr} \left(P \left[(A - GC)\Sigma + \Sigma(A - GC)^\top + BQB^\top + GRG^\top \right] \right). \quad (11.78)$$

The Minimum Principle yields the necessary conditions

$$\dot{\Sigma}(t) = \frac{\partial \mathcal{H}}{\partial P}(\Sigma, G, P, t), \quad (11.79)$$

$$\dot{P}(t) = -\frac{\partial \mathcal{H}}{\partial \Sigma}(\Sigma, G, P, t), \quad (11.80)$$

$$0 = \frac{\partial \mathcal{H}}{\partial G}(\Sigma, G, P, t). \quad (11.81)$$

We compute each term explicitly.

State equation. From (11.78), differentiation with respect to P immediately gives

$$\dot{\Sigma} = (A - GC)\Sigma + \Sigma(A - GC)^\top + BQB^\top + GRG^\top,$$

which recovers (11.76).

Costate equation. We compute the derivative with respect to Σ . Using the identity $\text{tr}(P(A\Sigma)) = \text{tr}(A^\top P \Sigma)$, we rewrite

$$\mathcal{H} = \text{tr}(A^\top P \Sigma) + \text{tr}(P \Sigma A^\top) - \text{tr}(PGC\Sigma) - \text{tr}(P \Sigma C^\top G^\top) + \text{terms independent of } \Sigma.$$

Differentiating term-by-term yields

$$\frac{\partial \mathcal{H}}{\partial \Sigma} = A^\top P + PA - C^\top G^\top P - PGC.$$

Thus the costate equation becomes

$$\dot{P} = -A^\top P - PA + C^\top G^\top P + PGC = -(A - GC)^\top P - P(A - GC), \quad (11.82)$$

with the terminal condition supplied by the terminal cost $\text{tr}(M\Sigma(\tau))$,

$$P(\tau) = M. \quad (11.83)$$

Equation (11.82) is linear and homogeneous in P , so $P(t) = \Psi(\tau, t)^\top M \Psi(\tau, t)$, where Ψ is the transition matrix of $A - GC$; in particular $P(t) \succeq 0$ for all t , and $P(t)$ is nonsingular for all t precisely when $M \succ 0$.

Stationarity condition. We now compute the derivative with respect to G . Let δG be an arbitrary variation. Then

$$\delta \mathcal{H} = \text{tr}(P[-\delta G C \Sigma - \Sigma C^\top \delta G^\top + \delta G R G^\top + G R \delta G^\top]).$$

We rewrite each term using trace identities:

$$\text{tr}(P \delta G C \Sigma) = \text{tr}(C \Sigma P \delta G), \quad \text{tr}(P \Sigma C^\top \delta G^\top) = \text{tr}(P \Sigma C^\top \delta G^\top).$$

After rearranging and grouping terms, we obtain

$$\delta \mathcal{H} = \text{tr}\left(\left(-P \Sigma C^\top + P G R\right)^\top \delta G\right) + \text{tr}\left(\left(-P \Sigma C^\top + P G R\right) \delta G^\top\right).$$

Using the symmetry of P and Σ , this reduces to

$$\delta \mathcal{H} = 2 \text{tr}\left((P G R - P \Sigma C^\top)^\top \delta G\right).$$

Since δG is arbitrary, the stationarity condition (11.81) yields

$$P G R = P \Sigma C^\top. \quad (11.84)$$

Assuming $M \succ 0$ —so that $P(t)$ is nonsingular for every t by the representation just given—we may cancel P and obtain

$$G^*(t) = \Sigma^*(t) C(t)^\top R(t)^{-1}. \quad (11.85)$$

Riccati equation. Substituting (11.85) into the state equation (11.76), we obtain

$$\dot{\Sigma}^* = A \Sigma^* + \Sigma^* A^\top + B Q B^\top - \Sigma^* C^\top R^{-1} C \Sigma^*, \quad (11.86)$$

which is the Kalman–Bucy Riccati equation.

Optimal filter. Substituting the optimal gain into (11.74), we obtain

$$d\hat{x}(t) = A\hat{x}(t) dt + \Sigma^*(t) C(t)^\top R(t)^{-1} (dz(t) - C(t)\hat{x}(t) dt). \quad (11.87)$$

This establishes that the Kalman–Bucy filter arises as the optimal control law for the deterministic covariance control problem. The structure of (11.86) is dual to the Riccati equation in the linear-quadratic regulator, revealing a fundamental correspondence between optimal estimation and optimal control.

Theorem 11.13 (Kalman–Bucy filter via the Minimum Principle). *Consider the stochastic system (11.72)–(11.73) and the class of linear estimators (11.74). Let $J(G) = \mathbb{E}[e(\tau)^\top M e(\tau)]$ with $M \succ 0$, and let $\Sigma(\cdot)$ denote the covariance of the estimation error. (Positive definiteness of the terminal weight is what makes the costate $P(\cdot)$ nonsingular and hence determines the gain uniquely; for $M \succeq 0$ singular the stationarity condition (11.84) pins down G only on the range of M propagated backward.)*

Then the gain $G^(\cdot)$ that minimizes $J(G)$ is given by*

$$G^*(t) = \Sigma^*(t) C(t)^\top R(t)^{-1}, \quad (11.88)$$

where $\Sigma^(\cdot)$ is the unique solution of the Riccati equation*

$$\dot{\Sigma}^* = A \Sigma^* + \Sigma^* A^\top + B Q B^\top - \Sigma^* C^\top R^{-1} C \Sigma^*, \quad \Sigma^*(t_0) = \Sigma_0. \quad (11.89)$$

The corresponding optimal estimator is

$$d\hat{x}(t) = A(t)\hat{x}(t) dt + \Sigma^*(t) C(t)^\top R(t)^{-1} (dz(t) - C(t)\hat{x}(t) dt). \quad (11.90)$$

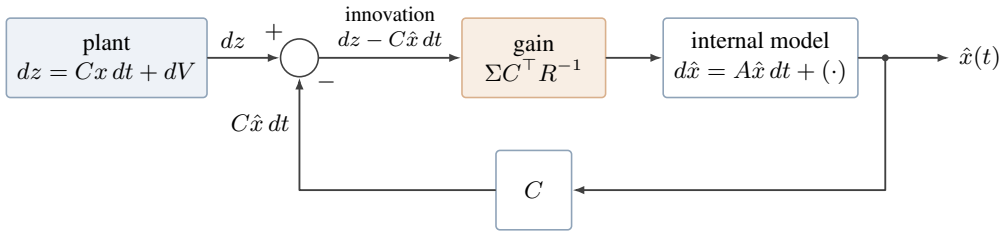


Figure 11.6: Structure of the Kalman–Bucy filter as an optimal feedback law. The filter carries an internal copy of the plant dynamics and drives it with the *innovation* $dz - C\hat{x} dt$ —the part of the measurement not predicted by the current estimate—weighted by the optimal gain $\Sigma C^\top R^{-1}$. The estimate $\hat{x}$ is fed back through C to form the innovation, closing a predictor–corrector loop that mirrors the state/costate feedback structure of the regulator.

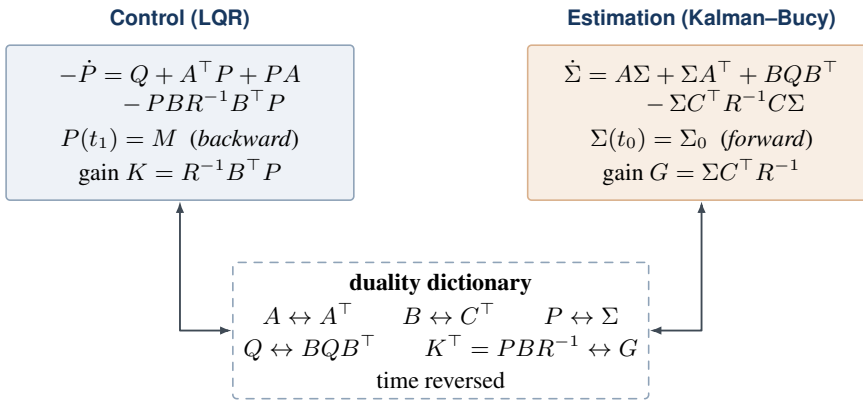


Figure 11.7: Control–estimation duality. The Kalman–Bucy Riccati equation is obtained from the regulator Riccati equation by transposing the state matrix, swapping the input matrix B for the output matrix $C^\top$ (and the state weight Q for the process-noise intensity $BQB^\top$), reversing time, and reading the cost matrix P as an error covariance Σ . The optimal feedback gain K and the optimal filter gain G are dual objects ($K^\top \leftrightarrow G$). This is why solving one problem solves the other.

Remark 11.14. The Riccati equation (11.89) is dual to the Riccati equation arising in the linear-quadratic regulator. In particular, the roles of the control and observation matrices are interchanged, revealing a structural duality between optimal control and optimal estimation. Figure 11.7 makes this dictionary explicit.

11.5 ■ Dynamic Programming and Hamilton–Jacobi Equations

We consider the optimal control of a dynamical system evolving in continuous time. Let the state $x(\cdot) \in \mathbb{R}^n$ evolve according to the nonlinear dynamics

$$\dot{x}(t) = f(x(t), u(t), t), \quad x(t_0) = x_0, \quad (11.91)$$

where $u(\cdot)$ is an admissible control taking values in a given set $\mathcal{U} \subset \mathbb{R}^m$, and f is sufficiently regular.

11.5.1 ■ Cost Functional and Value Function

We begin by formalizing the optimal control problem.

Let $\mathcal{U} \subset \mathbb{R}^m$ be a nonempty set. A control $u : [t_0, t_1] \rightarrow \mathcal{U}$ is said to be *admissible* if it is measurable and essentially bounded. We denote the set of admissible controls on $[t_0, t_1]$ by $\mathcal{A}[t_0, t_1]$.

Given $u \in \mathcal{A}[t_0, t_1]$, the corresponding state trajectory $x(\cdot)$ is defined as the (unique) absolutely continuous solution of

$$\dot{x}(t) = f(x(t), u(t), t), \quad x(t_0) = x_0. \quad (11.92)$$

Under standard assumptions (e.g., f Lipschitz in x uniformly in (u, t)), the solution exists and is unique.

For each admissible control $u \in \mathcal{A}[t_0, t_1]$, we define the cost functional

$$J_{t_0, x_0}(u) = \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt + m(x(t_1)), \quad (11.93)$$

where

$$\ell : \mathbb{R}^n \times \mathbb{R}^m \times [t_0, t_1] \rightarrow \mathbb{R} \quad \text{and} \quad m : \mathbb{R}^n \rightarrow \mathbb{R}$$

are continuous functions, referred to as the *running cost* and *terminal cost*, respectively.

The optimal control problem is to compute

$$\inf_{u \in \mathcal{A}[t_0, t_1]} J_{t_0, x_0}(u). \quad (11.94)$$

To capture the dependence of the optimal cost on the initial condition, we define the *value function*

$$V^*(x, t) = \inf_{u \in \mathcal{A}[t, t_1]} \left[\int_t^{t_1} \ell(x(\tau), u(\tau), \tau) d\tau + m(x(t_1)) \right], \quad (11.95)$$

where the trajectory $x(\cdot)$ satisfies (11.92) with initial condition $x(t) = x$.

Thus, $V^*(x, t)$ represents the minimal cost-to-go from state x at time t .

Remark 11.15. The infimum in (11.95) need not be attained in general. If an optimal control exists, it is denoted by $u^*(\cdot)$ and the corresponding trajectory by $x^*(\cdot)$.

11.5.2 ■ Principle of Optimality

A key structural property of the value function is the *principle of optimality*, which encodes the time-consistency of optimal decisions.

Proposition 11.16 (Principle of optimality). *Let $(x, t) \in \mathbb{R}^n \times [t_0, t_1]$ and let $t_m \in (t, t_1)$. Then*

$$V^*(x, t) = \inf_{u \in \mathcal{A}[t, t_m]} \left[\int_t^{t_m} \ell(x(\tau), u(\tau), \tau) d\tau + V^*(x(t_m), t_m) \right], \quad (11.96)$$

where $x(\cdot)$ satisfies (11.92) with initial condition $x(t) = x$.

Proof. Fix any admissible control $u \in \mathcal{A}[t, t_1]$. Let u_1 and u_2 denote its restrictions to $[t, t_m]$ and $[t_m, t_1]$, respectively. Then

$$J_{t, x}(u) = \int_t^{t_m} \ell(x(\tau), u_1(\tau), \tau) d\tau + J_{t_m, x(t_m)}(u_2).$$

Taking the infimum over u_2 yields

$$J_{t,x}(u) \geq \int_t^{t_m} \ell(x(\tau), u_1(\tau), \tau) d\tau + V^*(x(t_m), t_m).$$

Now taking the infimum over u_1 gives

$$V^*(x, t) \geq \inf_{u_1 \in \mathcal{A}[t, t_m]} \left[\int_t^{t_m} \ell(x(\tau), u_1(\tau), \tau) d\tau + V^*(x(t_m), t_m) \right].$$

The reverse inequality follows by concatenating an ε -optimal control on $[t, t_m]$ with an ε -optimal control on $[t_m, t_1]$, and letting $\varepsilon \rightarrow 0$. $\square$

Remark 11.17. The principle of optimality implies that an optimal policy is *time-consistent*: any restriction of an optimal control to a subinterval is itself optimal for the corresponding subproblem.

11.5.3 ■ Derivation of the Hamilton–Jacobi–Bellman Equation

We now derive a differential characterization of V^* .

Let $\Delta t > 0$ and define $t_m = t + \Delta t$ and $x_m = x(t + \Delta t)$.

From (11.96), we obtain

$$V^*(x, t) = \inf_{u \in \mathcal{A}[t, t+\Delta t]} \left[\int_t^{t+\Delta t} \ell(x(\tau), u(\tau), \tau) d\tau + V^*(x_m, t + \Delta t) \right]. \quad (11.97)$$

To turn (11.97) into a differential equation we expand each of its ingredients to first order in Δt . Assume that V^* is continuously differentiable. Then

$$V^*(x_m, t + \Delta t) = V^*(x, t) + \nabla_x V^*(x, t)^\top (x_m - x) + \partial_t V^*(x, t) \Delta t + o(\Delta t). \quad (11.98)$$

Using the dynamics, $x_m - x = f(x, u, t) \Delta t + o(\Delta t)$, and

$$\int_t^{t+\Delta t} \ell(x(\tau), u(\tau), \tau) d\tau = \ell(x, u, t) \Delta t + o(\Delta t).$$

Substituting these expansions into (11.97) yields

$$V^*(x, t) = \inf_u \left[V^*(x, t) + (\ell(x, u, t) + \nabla_x V^*(x, t)^\top f(x, u, t) + \partial_t V^*(x, t)) \Delta t + o(\Delta t) \right].$$

Canceling $V^*(x, t)$, dividing by Δt , and letting $\Delta t \rightarrow 0$, we obtain

$$0 = \inf_{u \in \mathcal{U}} \left[\ell(x, u, t) + \nabla_x V^*(x, t)^\top f(x, u, t) + \partial_t V^*(x, t) \right]. \quad (11.99)$$

Rearranging gives the Hamilton–Jacobi–Bellman equation:

$$-\partial_t V^*(x, t) = \inf_{u \in \mathcal{U}} \left[\ell(x, u, t) + \nabla_x V^*(x, t)^\top f(x, u, t) \right]. \quad (11.100)$$

Remark 11.18 (Formal vs rigorous derivation). The above derivation is formal and assumes differentiability of V^* . In general, V^* may fail to be smooth, and the correct notion of solution is that of a *viscosity solution*. The dynamic programming principle remains valid without smoothness and serves as the foundation for the viscosity framework.

11.5.4 ■ Hamiltonian Formulation

We introduce the Hamiltonian as the central object that connects the dynamic programming formulation with the calculus of variations and Pontryagin’s Minimum Principle.

Definition 11.19 (Hamiltonian). *The Hamiltonian associated with the control system is defined as the function*

$$H : \mathbb{R}^n \times \mathbb{R}^n \times \mathcal{U} \times [t_0, t_1] \rightarrow \mathbb{R}, \quad H(x, p, u, t) := \ell(x, u, t) + p^\top f(x, u, t), \quad (11.101)$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathcal{U}$ is the control, $t \in [t_0, t_1]$ is time, and $p \in \mathbb{R}^n$ is an auxiliary variable.

The vector p is interpreted as a costate variable or shadow price. It represents the sensitivity of the optimal cost with respect to perturbations of the state and will later be identified with the gradient of the value function along optimal trajectories.

In terms of the Hamiltonian, the Hamilton–Jacobi–Bellman equation can be expressed in the compact form

$$-\partial_t V^*(x, t) = \inf_{u \in \mathcal{U}} H(x, \nabla_x V^*(x, t), u, t), \quad (11.102)$$

for $(x, t) \in \mathbb{R}^n \times [t_0, t_1]$, together with the terminal condition

$$V^*(x, t_1) = m(x). \quad (11.103)$$

The structure of (11.102) shows that the value function evolves backward in time and that the control enters the problem only through a pointwise minimization of the Hamiltonian. This feature is fundamental to the synthesis of optimal feedback laws.

We now state a rigorous result that connects solutions of the HJB equation to optimal controls.

Theorem 11.20 (Verification theorem). *Suppose that $V \in C^1(\mathbb{R}^n \times [t_0, t_1])$ satisfies the Hamilton–Jacobi–Bellman equation*

$$-\partial_t V(x, t) = \inf_{u \in \mathcal{U}} H(x, \nabla_x V(x, t), u, t), \quad (11.104)$$

for all (x, t) , together with the terminal condition $V(x, t_1) = m(x)$. Let $u^*(\cdot)$ be an admissible control and let $x^*(\cdot)$ denote the corresponding state trajectory. Assume that for almost every $t \in [t_0, t_1]$ the control $u^*(t)$ satisfies

$$H(x^*(t), \nabla_x V(x^*(t), t), u^*(t), t) = \inf_{u \in \mathcal{U}} H(x^*(t), \nabla_x V(x^*(t), t), u, t). \quad (11.105)$$

Then u^* is an optimal control and the value function satisfies $V(x_0, t_0) = J_{t_0, x_0}(u^*)$. Conversely, if an optimal control exists and the associated value function is continuously differentiable, then it satisfies (11.104) and the optimal control satisfies (11.105).

Proof. Let $u(\cdot)$ be any admissible control and $x(\cdot)$ the corresponding trajectory. From (11.104) it follows that for all (x, t) and all $u \in \mathcal{U}$,

$$-\partial_t V(x, t) \leq H(x, \nabla_x V(x, t), u, t).$$

Evaluating along the trajectory $x(\cdot)$ and using the definition of the Hamiltonian yields

$$-\partial_t V(x, t) \leq \ell(x, u, t) + \nabla_x V(x, t)^\top f(x, u, t).$$

Applying the chain rule along the trajectory gives

$$\frac{d}{dt}V(x(t), t) = \partial_t V(x, t) + \nabla_x V(x, t)^\top \dot{x}(t).$$

Substituting $\dot{x}(t) = f(x, u, t)$ and combining with the previous inequality yields

$$\frac{d}{dt}V(x(t), t) \geq -\ell(x(t), u(t), t).$$

Integrating from t_0 to t_1 gives

$$V(x(t_1), t_1) - V(x_0, t_0) \geq - \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt.$$

Using the terminal condition $V(x(t_1), t_1) = m(x(t_1))$ yields

$$V(x_0, t_0) \leq \int_{t_0}^{t_1} \ell(x(t), u(t), t) dt + m(x(t_1)).$$

Thus $V(x_0, t_0)$ is a lower bound for the cost of any admissible control. If $u = u^*$ satisfies (11.105), then all inequalities above become equalities, and therefore u^* achieves the minimum.

Converse. Suppose now that an optimal control exists for each initial datum and that the value function V^* is continuously differentiable. By the principle of optimality (Proposition 11.16), V^* obeys the dynamic programming principle (11.97), and the local expansion carried out in the derivation of (11.100) applies verbatim with V^* in place of V , so V^* satisfies (11.104) together with $V^*(x, t_1) = m(x)$. To obtain (11.105), let $u^*(\cdot)$ be optimal from (x_0, t_0) with trajectory $x^*(\cdot)$. Applying the dynamic programming principle along this trajectory on $[t, t + \Delta t]$ and using optimality of u^* gives, without an infimum,

$$V^*(x^*(t), t) = \int_t^{t+\Delta t} \ell(x^*(s), u^*(s), s) ds + V^*(x^*(t + \Delta t), t + \Delta t).$$

Dividing by Δt and letting $\Delta t \rightarrow 0$, the C^1 regularity of V^* yields, for almost every t ,

$$0 = \ell(x^*, u^*, t) + \partial_t V^*(x^*, t) + \nabla_x V^*(x^*, t)^\top f(x^*, u^*, t) = \partial_t V^*(x^*, t) + H(x^*, \nabla_x V^*(x^*, t), u^*, t).$$

Comparing with (11.104) evaluated at $x^*(t)$ shows that $u^*(t)$ attains the infimum, which is (11.105). $\square$

The verification theorem shows that the HJB equation provides both necessary and sufficient conditions for optimality when a smooth solution exists. In particular, it implies that the optimal control can be synthesized in feedback form. Specifically, if V is known, then an optimal control law is given pointwise by $u^*(x, t) \in \arg \min_{u \in \mathcal{U}} H(x, \nabla_x V(x, t), u, t)$. This representation shows that the optimal control depends only on the current state and time, and not explicitly on the past trajectory.

11.5.5 ■ Connection between Dynamic Programming and the Minimum Principle

The two pillars of this chapter—the variational Minimum Principle (Theorem 11.2) and the dynamic programming principle above—are not independent. They are linked by a single identity:

the costate is the gradient of the value function along the optimal trajectory. This is the bridge $p = \nabla_x V^*$ displayed in Figure 11.1, and we now establish it.

Proposition 11.21 (Costate as the gradient of the value function). *Suppose f and ℓ are continuously differentiable, and suppose the value function V^* is twice continuously differentiable in a neighborhood of an optimal trajectory $x^*(\cdot)$. Assume the pointwise minimizer*

$$u^\circ(x, t) \in \arg \min_{u \in \mathcal{U}} H(x, \nabla_x V^*(x, t), u, t)$$

is well defined and continuous, and generates x^ from (x_0, t_0) . Define the costate along the optimal trajectory by*

$$p(t) := \nabla_x V^*(x^*(t), t). \quad (11.106)$$

Then (x^, p) satisfies the canonical Hamiltonian system*

$$\dot{x}^*(t) = \nabla_p H(x^*(t), p(t), u^\circ(t), t), \quad \dot{p}(t) = -\nabla_x H(x^*(t), p(t), u^\circ(t), t),$$

with $u^\circ(t) := u^\circ(x^(t), t)$, together with the minimum condition*

$$H(x^*(t), p(t), u^\circ(t), t) = \min_{u \in \mathcal{U}} H(x^*(t), p(t), u, t)$$

and the terminal condition $p(t_1) = \nabla_x m(x^(t_1))$. That is, the Minimum Principle of Theorem 11.2 is recovered from the HJB equation.*

Proof. The HJB equation (11.102) holds as an identity in (x, t) throughout the neighborhood:

$$\partial_t V^*(x, t) + H(x, \nabla_x V^*(x, t), u^\circ(x, t), t) = 0, \quad (11.107)$$

where $u^\circ(x, t)$ attains the minimum. Differentiate (11.107) with respect to x . Because $u^\circ(x, t)$ minimizes the Hamiltonian, the envelope theorem applies: the derivative of the minimized value through its explicit dependence on the minimizer vanishes, and only the dependence through the first argument and through $p = \nabla_x V^*(x, t)$ survives. Writing $\nabla_{xx}^2 V^*$ for the Hessian and evaluating the partial gradients of H at $(x, \nabla_x V^*(x, t), u^\circ(x, t), t)$,

$$\nabla_x (\partial_t V^*) + \nabla_x H + \nabla_{xx}^2 V^* \nabla_p H = 0. \quad (11.108)$$

Now differentiate the costate (11.106) in time along $x^*(\cdot)$. Using $\dot{x}^* = f = \nabla_p H$ and the equality of mixed partials ($\partial_t \nabla_x V^* = \nabla_x \partial_t V^*$, valid since $V^* \in C^2$),

$$\dot{p}(t) = \nabla_x (\partial_t V^*)(x^*(t), t) + \nabla_{xx}^2 V^*(x^*(t), t) \dot{x}^*(t) = \nabla_x (\partial_t V^*) + \nabla_{xx}^2 V^* \nabla_p H.$$

Comparing with (11.108) gives exactly

$$\dot{p}(t) = -\nabla_x H(x^*(t), p(t), u^\circ(t), t),$$

the adjoint equation. The state equation $\dot{x}^* = f = \nabla_p H$ holds by construction, and the minimum condition holds because u° was chosen as the pointwise minimizer. Finally, the terminal condition of the HJB equation, $V^*(x, t_1) = m(x)$, gives $p(t_1) = \nabla_x V^*(x^*(t_1), t_1) = \nabla_x m(x^*(t_1))$. $\square$

Remark 11.22 (Two viewpoints, one costate). Proposition 11.21 makes precise the interpretation introduced earlier: the costate $p(t)$ is the marginal value, the shadow price, of the state, since $\nabla_x V^*$ measures the sensitivity of the optimal cost-to-go to a perturbation of the current state. Dynamic programming delivers this sensitivity as a *field* $\nabla_x V^*(x, t)$ over all states, while the Minimum Principle propagates it as a *trajectory* $p(t)$ along the single optimal path. The two agree wherever V^* is smooth.

The implication is one-directional in its hypotheses but not in its value. The derivation above assumes $V^* \in C^2$, which fails in many problems, already in the minimum-time example of Example 11.24. The Minimum Principle, by contrast, requires no smoothness of V^* : it furnishes a finite-dimensional two-point boundary-value problem that remains valid at kinks of the value function. Thus the Minimum Principle is the more robust necessary condition, while the HJB equation, when a smooth solution exists, additionally certifies optimality and yields the control in feedback form. The costate identity (11.106) is precisely where the two theories touch.

Finally, we note that the HJB equation is a nonlinear first-order partial differential equation. In many problems, particularly those involving nonlinear dynamics or nonconvex costs, the value function fails to be differentiable. In such cases, the appropriate notion of solution is that of a viscosity solution, which preserves the validity of the dynamic programming principle without requiring classical differentiability.

11.5.6 ■ Example: A Nonlinear Scalar System

We consider a simple nonlinear optimal control problem that illustrates the structure of the Hamilton–Jacobi–Bellman equation.

Let the state $x(\cdot) \in \mathbb{R}$ evolve according to the dynamics

$$\dot{x}(t) = u(t), \quad t \in [0, t_1], \quad (11.109)$$

where the control $u(\cdot)$ is measurable and takes values in $\mathbb{R}$. The cost functional is defined by

$$J(u) = \int_0^{t_1} (u(t)^2 + x(t)^4) dt, \quad (11.110)$$

with no terminal cost.

The associated Hamiltonian is

$$H(x, p, u) = p u + u^2 + x^4. \quad (11.111)$$

For each fixed (x, p) , the Hamiltonian is strictly convex in u , and hence admits a unique minimizer. Differentiating with respect to u yields $\frac{\partial H}{\partial u}(x, p, u) = p + 2u$, and therefore the minimizing control is

$$u^*(x, p) = -\frac{1}{2}p. \quad (11.112)$$

With the minimizer in hand the HJB equation can be written explicitly. Let $V(x, t)$ denote the value function. Substituting the minimizing control into the HJB equation gives

$$-\partial_t V(x, t) = \inf_{u \in \mathbb{R}} \{p u + u^2 + x^4\} \Big|_{p=\partial_x V} \quad (11.113)$$

$$= -\frac{1}{4}(\partial_x V(x, t))^2 + x^4. \quad (11.114)$$

Thus the value function satisfies the nonlinear first-order PDE

$$-\partial_t V(x, t) = -\frac{1}{4}(\partial_x V(x, t))^2 + x^4, \quad V(x, t_1) = 0. \quad (11.115)$$

Remark 11.23. The equation (11.115) is a nonlinear Hamilton–Jacobi equation. In general, classical solutions may fail to exist globally due to the formation of singularities. A rigorous treatment therefore requires the framework of viscosity solutions, which ensures well-posedness under mild conditions.

Example 11.24 A nonsmooth value function.

Nonsmoothness is not a pathology: it already occurs in the simplest minimum-time problem. For $\dot{x} = u$ with $|u| \leq 1$, let $V(x)$ be the minimum time to reach the origin from $x \in \mathbb{R}$. Clearly $V(x) = |x|$: the optimal control is $u^\circ = -\text{sgn}(x)$. The stationary HJB equation for minimum-time problems,

$$\min_{|u| \leq 1} \{1 + V'(x)u\} = 0, \quad V(0) = 0,$$

reads $1 - |V'(x)| = 0$. The value function satisfies it classically for $x \neq 0$ but is not differentiable at the origin, where optimal trajectories from both sides merge; no globally C^1 solution exists. Worse, the equation $|V'| = 1$ admits infinitely many Lipschitz functions vanishing at the origin and satisfying it almost everywhere (e.g., sawtooth profiles), so “almost-everywhere solution” is too weak a notion to single out the value function. The viscosity solution framework resolves both difficulties: $V(x) = |x|$ is the unique viscosity solution, and the kink at the origin is precisely where the infimum in the definition of V switches between the two control regimes. $\square$

11.5.7 • Linear Quadratic Regulator

We now consider a fundamental class of problems for which the HJB equation admits an explicit solution.

Let the state $x(\cdot) \in \mathbb{R}^n$ evolve according to the linear system

$$\dot{x}(t) = A(t)x(t) + B(t)u(t), \quad x(t_0) = x_0, \quad (11.116)$$

where $A(\cdot)$ and $B(\cdot)$ are given matrix-valued functions of appropriate dimensions. The cost functional is

$$J(u) = \int_{t_0}^{t_1} (x(t)^\top Q(t)x(t) + u(t)^\top R(t)u(t)) dt + x(t_1)^\top Mx(t_1), \quad (11.117)$$

where $Q(t) = Q(t)^\top \succeq 0$, $R(t) = R(t)^\top \succ 0$, and $M = M^\top \succeq 0$.

The Hamiltonian associated with this data is

$$H(x, p, u, t) = x^\top Q(t)x + u^\top R(t)u + p^\top (A(t)x + B(t)u). \quad (11.118)$$

For each (x, p, t) , the Hamiltonian is strictly convex in u , and the minimizing control is obtained by solving $\nabla_u H(x, p, u, t) = 2R(t)u + B(t)^\top p = 0$. Thus,

$$u^*(x, p, t) = -\frac{1}{2}R(t)^{-1}B(t)^\top p. \quad (11.119)$$

Quadratic Value Function Ansatz

Motivated by the quadratic structure of the cost, we seek a value function of the form

$$V(x, t) = x^\top P(t)x, \quad (11.120)$$

where $P(t) \in \mathbb{R}^{n \times n}$ is symmetric.

Then $\nabla_x V(x, t) = 2P(t)x$ and $\partial_t V(x, t) = x^\top \dot{P}(t)x$.

It remains to determine $P(\cdot)$, and the HJB equation does so. Substituting $p = \nabla_x V = 2P(t)x$ into (11.119) yields $u^*(x, t) = -R(t)^{-1}B(t)^\top P(t)x$. Substituting the ansatz (11.120) into the HJB equation gives

$$-x^\top \dot{P}(t)x = x^\top Q(t)x + x^\top (P(t)A(t) + A(t)^\top P(t))x - x^\top P(t)B(t)R(t)^{-1}B(t)^\top P(t)x. \quad (11.121)$$

Since this identity holds for all $x \in \mathbb{R}^n$, we obtain the matrix differential equation

$$-\dot{P}(t) = Q(t) + P(t)A(t) + A(t)^\top P(t) - P(t)B(t)R(t)^{-1}B(t)^\top P(t), \quad (11.122)$$

with terminal condition

$$P(t_1) = M. \quad (11.123)$$

Closed-Loop Dynamics

The optimal control is given in feedback form by

$$u^*(t) = -K(t)x(t), \quad K(t) := R(t)^{-1}B(t)^\top P(t). \quad (11.124)$$

Substituting into (11.116) yields the closed-loop system

$$\dot{x}(t) = (A(t) - B(t)K(t))x(t). \quad (11.125)$$

Solution of the LQR Problem

Theorem 11.25 (Linear quadratic regulator). *Consider the problem (11.116)–(11.117). Assume that $A(\cdot)$, $B(\cdot)$, $Q(\cdot)$, $R(\cdot)$ are piecewise continuous and that, for all $t \in [t_0, t_1]$,*

$$Q(t) = Q(t)^\top \succeq 0, \quad R(t) = R(t)^\top \succ 0, \quad M = M^\top \succeq 0.$$

The semidefiniteness of Q and M is not cosmetic: it is what rules out a finite escape time for P in Step 3 below. Without it the conclusion fails—for $n = m = 1$ with $A = 0$, $B = 1$, $R = 1$, $M = 0$ and the indefinite choice $Q = -1$, the Riccati equation reads $\dot{P} = 1 + P^2$ with $P(t_1) = 0$, whose solution $P(t) = -\tan(t_1 - t)$ blows up once the horizon exceeds $\pi/2$. Then the following statements hold.

First, the Riccati equation (11.122) with terminal condition (11.123) admits a unique symmetric solution $P(\cdot)$ on $[t_0, t_1]$.

Second, the optimal control is given by the feedback law (11.124), and the corresponding trajectory is the unique solution of the closed-loop system (11.125).

Third, the value function is given by

$$V^*(x, t) = x^\top P(t)x. \quad (11.126)$$

Proof. The argument rests on a single algebraic identity, completion of squares along trajectories, from which existence, optimality, and the value function all follow.

Step 1: a trajectory identity. The Riccati equation (11.122) is a quadratic (hence locally Lipschitz) matrix differential equation, so it has a unique solution $P(\cdot)$ on some maximal subinterval $(t_-, t_1] \subseteq [t_0, t_1]$. Because the map $P \mapsto Q + PA + A^\top P - PBR^{-1}B^\top P$ sends symmetric

matrices to symmetric matrices and the terminal datum $P(t_1) = M$ is symmetric, $P(t) = P(t)^\top$ throughout. Fix $t \in (t_-, t_1]$ and $x \in \mathbb{R}^n$, let $u(\cdot)$ be any admissible control on $[t, t_1]$, and let $x(\cdot)$ be the corresponding trajectory of (11.116) with $x(t) = x$. Differentiating the quadratic form along the trajectory and using (11.116),

$$\frac{d}{ds}(x(s)^\top P(s)x(s)) = x^\top (\dot{P} + A^\top P + PA)x + 2x^\top PBu.$$

Substituting $\dot{P} = -Q - A^\top P - PA + PBR^{-1}B^\top P$ from (11.122) gives $\dot{P} + A^\top P + PA = -Q + PBR^{-1}B^\top P$, so, adding the running cost $x^\top Qx + u^\top Ru$ to both sides and completing the square in u ,

$$\begin{aligned} \frac{d}{ds}(x^\top Px) + (x^\top Qx + u^\top Ru) &= x^\top PBR^{-1}B^\top Px + 2x^\top PBu + u^\top Ru \\ &= (u + R^{-1}B^\top Px)^\top R(u + R^{-1}B^\top Px). \end{aligned} \quad (11.127)$$

Step 2: value function and optimality. Integrate (11.127) from t to t_1 and use $P(t_1) = M$ so that $x(t_1)^\top P(t_1)x(t_1)$ is exactly the terminal cost. Rearranging,

$$J(u) = x^\top P(t)x + \int_t^{t_1} (u(s) + R^{-1}B^\top P(s)x(s))^\top R(s)(u(s) + R^{-1}B^\top P(s)x(s)) ds. \quad (11.128)$$

Since $R(s) \succ 0$, the integral is nonnegative and vanishes if and only if $u(s) = -R^{-1}(s)B(s)^\top P(s)x(s)$ for almost every s , which is precisely the feedback law (11.124). Hence $J(u) \geq x^\top P(t)x$ for every admissible u , with equality attained by (11.124). This simultaneously proves the third claim, $V^*(x, t) = x^\top P(t)x$, and the optimality of the feedback control. Under this control the closed-loop dynamics (11.125) form a linear time-varying system with piecewise-continuous coefficients, so its solution from $x(t_0) = x_0$ exists and is unique, establishing the second claim.

Step 3: global existence of P . It remains to show $t_- \leq t_0$, i.e., that P does not escape to infinity on $[t_0, t_1]$. Evaluating (11.128) with the admissible choice $u \equiv 0$ shows that, on $(t_-, t_1]$,

$$0 \leq x^\top P(t)x = \min_u J(u) \leq J(0) = \int_t^{t_1} x_0(s)^\top Q(s)x_0(s) ds + x_0(t_1)^\top Mx_0(t_1),$$

where $x_0(\cdot)$ is the uncontrolled trajectory. The leftmost inequality uses $Q \succeq 0$ and $M \succeq 0$, which make $J(u) \geq 0$ for every admissible u . The right-hand side is a continuous function of t on the compact interval $[t_0, t_1]$ and, since $x_0(\cdot)$ depends linearly on the initial state x , is bounded by $c|x|^2$ for some constant c independent of t . As $P(t)$ is symmetric, $0 \preceq P(t) \preceq cI$ uniformly, so $P(\cdot)$ remains bounded and cannot have a finite escape time. Therefore the solution extends to all of $[t_0, t_1]$, and by local Lipschitz continuity it is unique, proving the first claim. $\square$

Remark 11.26. The Riccati equation (11.122) provides a finite-dimensional characterization of the infinite-dimensional HJB equation. This reduction is a consequence of the quadratic structure of both the dynamics and the cost, and it underlies the tractability of the LQR problem.

Remark 11.27. In the time-invariant case, when A, B, Q, R are constant and the horizon tends to infinity, the solution $P(t)$ converges to a constant matrix satisfying the algebraic Riccati equation. This yields a time-invariant optimal feedback law.

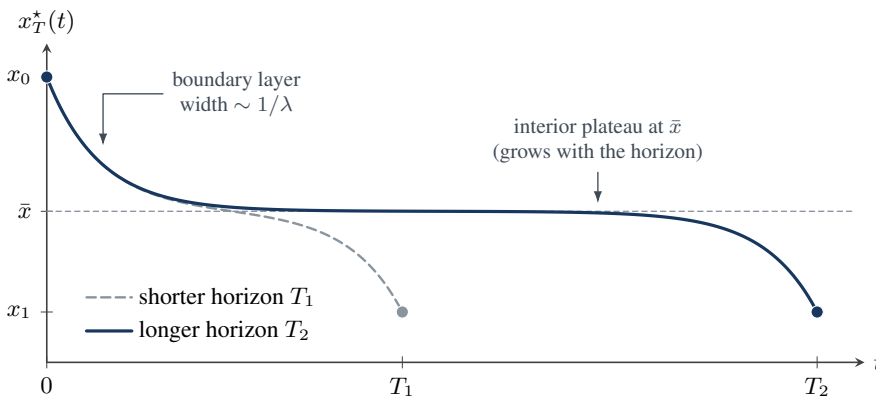


Figure 11.8: The turnpike phenomenon for the scalar problem of Example 11.38 ($\bar{x}$ the turnpike, rate $\lambda = \sqrt{a^2 + b^2}$). Two optimal trajectories with the *same* boundary data $x(0) = x_0, x(T) = x_1$ but different horizons are shown; each is the exact solution $x_T^*(t) = \bar{x} + [\xi_0 \sinh \lambda(T - t) + \xi_T \sinh \lambda t] / \sinh \lambda T$ of the two-point boundary-value problem. Both leave x_0 through an initial boundary layer of horizon-independent width $O(1/\lambda)$, rest near the turnpike $\bar{x}$ for the bulk of the interval, and peel off to x_1 through a terminal layer of the same width. Lengthening the horizon only stretches the central plateau, the two transients are unchanged, which is the geometric content of the measure and exponential turnpike estimates (11.135) and (11.154).

11.6 ■ Turnpike Properties of Optimal Control

The remark above hints at a phenomenon that pervades long-horizon optimal control: as the horizon T grows, the optimal solution ceases to depend on T in its interior and instead settles onto a fixed configuration determined by a static optimization problem. A finite-horizon trajectory then consists of an initial transient that reaches this configuration, a long central phase spent near it, and a final transient that peels off to satisfy the terminal data. The central configuration is called the *turnpike*, following the metaphor of Dorfman, Samuelson, and Solow: to travel efficiently between two towns one drives to a high-speed turnpike, stays on it for the bulk of the journey, and exits only near the destination, and the longer the trip, the larger the fraction spent on the turnpike.

This section makes the phenomenon rigorous, working first in full generality and then specializing. We begin with a general nonlinear problem: we identify the turnpike as the solution of a steady-state optimization problem and as the equilibrium of the Hamiltonian flow, we prove a *measure* turnpike theorem from strict dissipativity, and we obtain a *local exponential* turnpike theorem by linearizing the optimality system. We then specialize to the linear-quadratic problem, where the steady state, the dissipation inequality, the Hamiltonian spectrum, and the resulting *global exponential* estimate are all explicit and self-contained. We close with the interpretation and ramifications of the results. Figure 11.8 shows the phenomenon we are after: optimal trajectories that leave the initial state, settle onto a horizon-independent plateau, and depart only near the end.

11.6.1 ■ The Steady-State Problem and the Turnpike

We develop the theory for a general nonlinear problem and specialize to the linear-quadratic case in Section 11.6.4. Consider

$$\min_u J_T(u) = \int_0^T \ell(x(t), u(t)) dt, \quad \dot{x} = f(x, u), \quad x(0) = x_0, \quad u(t) \in U, \quad (11.129)$$

with $f, \ell \in C^2$ and $U \subseteq \mathbb{R}^m$ closed; in the sharpest results a terminal state $x(T) = x_1$ is also prescribed.

Freezing the dynamics at equilibrium defines the static counterpart of (11.129),

$$\min_{(x,u)} \ell(x,u) \quad \text{subject to} \quad f(x,u) = 0, \quad u \in U. \quad (11.130)$$

Its minimizer $(\bar{x}, \bar{u})$ is the *turnpike*.

Assumption 11.28. *The static problem (11.130) has a minimizer $(\bar{x}, \bar{u})$ with $\bar{u} \in \text{int } U$ at which $\nabla_x f(\bar{x}, \bar{u})$ is nonsingular, so that a unique multiplier $\bar{p} \in \mathbb{R}^n$ exists. Moreover the turnpike is uniformly reachable: there are $\tau, K > 0$ such that the boundary states can be joined to $\bar{x}$ by admissible controls, steering $x_0 \rightarrow \bar{x}$ and $\bar{x} \rightarrow x_1$ in time τ , each of cost at most K , with τ, K independent of the horizon T .*

Writing the Hamiltonian $H(x, p, u) = \ell(x, u) + p^\top f(x, u)$, the multiplier $\bar{p}$ is characterized by the steady-state optimality conditions

$$\nabla_x H(\bar{x}, \bar{p}, \bar{u}) = 0, \quad \bar{u} \in \arg \min_{u \in U} H(\bar{x}, \bar{p}, u), \quad f(\bar{x}, \bar{u}) = 0. \quad (11.131)$$

We call $\bar{p}$ the *turnpike costate*.

The turnpike is more than the solution of a static problem: it is the *equilibrium* of the optimality flow. Let $u^*(x, p) \in \arg \min_{u \in U} H(x, p, u)$ denote the pointwise minimizer, assumed well defined and C^1 near the turnpike, and form the canonical system of the Minimum Principle,

$$\dot{x} = \nabla_p H(x, p, u^*(x, p)) = f(x, u^*(x, p)), \quad \dot{p} = -\nabla_x H(x, p, u^*(x, p)). \quad (11.132)$$

Evaluating (11.132) at $(\bar{x}, \bar{p})$ and using (11.131) gives $\dot{x} = f(\bar{x}, \bar{u}) = 0$ and $\dot{p} = -\nabla_x H(\bar{x}, \bar{p}, \bar{u}) = 0$: the turnpike is exactly the *equilibrium* of the optimality flow. Thus the static optimizer, the Lagrange multiplier of the frozen problem, and the rest point of the canonical system all coincide. Every turnpike theorem is a statement about how finite-horizon optimal orbits, solutions of (11.132) constrained by boundary data, approach and depart from this equilibrium.

The coincidence just observed is not a coincidence at all: the steady-state optimality conditions (11.131) are *term by term* the necessary conditions of the dynamic problem, the Minimum Principle of Theorem 11.2, with the time derivatives set to zero. Writing $H = \ell + p^\top f$, so that $\nabla_p H = f$ and $\nabla_x H = \nabla_x \ell + \nabla_x f^\top p$, the three conditions carried along an optimal trajectory freeze, at a rest point of the flow, into the three Karush–Kuhn–Tucker conditions of the static problem (11.130):

<i>dynamic</i> (Minimum Principle)	$\xrightarrow{\quad \quad \quad}$	<i>static</i> (KKT at the turnpike)	
$\dot{x} = \nabla_p H$	$\xrightarrow{\dot{x}=0}$	$f(\bar{x}, \bar{u}) = 0,$	(11.133)
$\dot{p} = -\nabla_x H$	$\xrightarrow{\dot{p}=0}$	$\nabla_x \ell(\bar{x}, \bar{u}) + \nabla_x f(\bar{x}, \bar{u})^\top \bar{p} = 0,$	
$u \in \arg \min_{u \in U} H$	$\xrightarrow{\text{unchanged}}$	$\bar{u} \in \arg \min_{u \in U} H(\bar{x}, \bar{p}, u).$	

The state equation frozen at $\dot{x} = 0$ is feasibility; the costate equation frozen at $\dot{p} = 0$ is stationarity of the Lagrangian; and the minimum condition passes through unchanged. In other words, the static Karush–Kuhn–Tucker system (11.131) is the dynamic Minimum Principle evaluated at a rest point, and the static multiplier $\bar{p}$ is the value the costate takes there.

This gives the turnpike costate an operational meaning. Recall from Proposition 11.21 that along an optimal trajectory the costate is the gradient of the value function, $p(t) = \nabla_x V^*(x^*(t), t)$,

the marginal cost of the state, its *shadow price*. The turnpike is therefore the unique state at which this shadow price stops moving: $\dot{p} = 0$ says that the marginal value of the state is in balance, the running-cost gradient $\nabla_x \ell$ exactly offsetting the sensitivity $\nabla_x f^\top \bar{p}$ of the dynamics through which the state must be held in place. This is the control-theoretic form of the economist's *golden rule*, the configuration a planner would choose to hold forever, were there no initial or terminal obligations. A finite-horizon problem inherits those obligations only at its two ends; in between, having nothing better to do than sit at the best sittable state, the optimal trajectory does exactly that, and (11.131) is why the state it sits at is $\bar{x}$ and the price it pays is $\bar{p}$.

11.6.2 ■ Strict Dissipativity and the Measure Turnpike

The first turnpike theorem requires no spectral information about the optimality flow; it follows from a single structural hypothesis of convexity type.

Definition 11.29 (Strict dissipativity). *Problem (11.129) is strictly dissipative at the turnpike $(\bar{x}, \bar{u})$ if there exist a continuously differentiable storage function $S : \mathbb{R}^n \rightarrow \mathbb{R}$ and a class- $\mathcal{K}_\infty$ function α such that the dissipation inequality*

$$\nabla S(x)^\top f(x, u) \leq \ell(x, u) - \ell(\bar{x}, \bar{u}) - \alpha(\|x - \bar{x}\|) \quad (11.134)$$

holds for all $x \in \mathbb{R}^n$ and $u \in U$.

Inequality (11.134) says that, measured through the storage S , no admissible motion can sustain the turnpike cost rate $\ell(\bar{x}, \bar{u})$ while staying away from $\bar{x}$: an irrecoverable deficit $\alpha(\|x - \bar{x}\|)$ is charged for every excursion. Integrating this deficit against the horizon-independent cost of the reachability comparison yields the turnpike.

Theorem 11.30 (Measure turnpike). *Let Assumption 11.28 hold, suppose (11.129) is strictly dissipative at $(\bar{x}, \bar{u})$ in the sense of Definition 11.29, and assume either that the terminal state is fixed at $x(T) = x_1$ or that S is bounded below. Then there is a constant C , independent of the horizon T , such that every optimal trajectory $x_T^*(\cdot)$ satisfies*

$$\int_0^T \alpha(\|x_T^*(t) - \bar{x}\|) dt \leq C, \quad \text{hence} \quad |\{t \in [0, T] : \|x_T^*(t) - \bar{x}\| > \varepsilon\}| \leq \frac{C}{\alpha(\varepsilon)} \quad (11.135)$$

for every $\varepsilon > 0$.

Proof. *Upper bound on the optimal value.* By the reachability in Assumption 11.28, concatenating a control that steers $x_0 \rightarrow \bar{x}$ on $[0, \tau]$ (cost $\leq K$), the constant turnpike control $\bar{u}$ on $[\tau, T - \tau]$ (cost $(T - 2\tau)\ell(\bar{x}, \bar{u})$), and a control that steers $\bar{x} \rightarrow x_1$ on $[T - \tau, T]$ (cost $\leq K$) yields an admissible comparison control. Optimality of u_T^* therefore gives

$$J_T(u_T^*) \leq T\ell(\bar{x}, \bar{u}) + C_1, \quad C_1 := 2K - 2\tau\ell(\bar{x}, \bar{u}), \quad (11.136)$$

with C_1 independent of T .

Lower bound via dissipation. Integrate the dissipation inequality (11.134) along the optimal pair (x_T^*, u_T^*) , using $\int_0^T \nabla S(x_T^*)^\top f(x_T^*, u_T^*) dt = S(x_T^*(T)) - S(x_0)$:

$$\int_0^T \alpha(\|x_T^* - \bar{x}\|) dt \leq J_T(u_T^*) - T\ell(\bar{x}, \bar{u}) + S(x_0) - S(x_T^*(T)).$$

Combining with (11.136) and bounding the boundary term—by $S(x_0) - S(x_1)$ when $x(T) = x_1$ is fixed, or by $S(x_0) - \inf S$ when S is bounded below—gives the horizon-independent bound

$\int_0^T \alpha(\|x_T^* - \bar{x}\|) dt \leq C$. Since α is increasing, on the set A_ε in (11.135) the integrand is at least $\alpha(\varepsilon)$, so $C \geq \alpha(\varepsilon) |A_\varepsilon|$, which is the second bound. $\square$

The measure turnpike (11.135) already captures the essence of the phenomenon: an $O(1)$ time budget, independent of T , is all an optimal trajectory can afford away from $\bar{x}$. It uses only convexity-in-the-large, encoded as dissipativity, together with reachability, no spectral structure. It does not, however, locate the excursions or give a decay rate. For that we linearize.

11.6.3 ■ Linearization and the Exponential Turnpike

Near the turnpike the optimality flow is governed by its linearization. Assume the strengthened second-order data at $(\bar{x}, \bar{u}, \bar{p})$:

$$A := \nabla_x f, \quad B := \nabla_u f, \quad H_{xx} := \nabla_{xx}^2 H, \quad H_{uu} := \nabla_{uu}^2 H, \quad H_{xu} := \nabla_{xu}^2 H,$$

all evaluated at the turnpike, with $H_{uu} \succ 0$ (the Legendre–Clebsch condition). Eliminating the control from (11.132) by the implicit function theorem and linearizing yields, for the deviations $\xi = x - \bar{x}$ and $\pi = p - \bar{p}$, the linear canonical system

$$\frac{d}{dt} \begin{pmatrix} \xi \\ \pi \end{pmatrix} = \mathcal{H} \begin{pmatrix} \xi \\ \pi \end{pmatrix},$$

with the *reduced Hamiltonian matrix*

$$\mathcal{H} = \begin{pmatrix} \tilde{A} & -\tilde{S} \\ -\tilde{Q} & -\tilde{A}^\top \end{pmatrix}, \quad \begin{aligned} \tilde{A} &:= A - BH_{uu}^{-1}H_{xu}^\top, \\ \tilde{S} &:= BH_{uu}^{-1}B^\top \succeq 0, \\ \tilde{Q} &:= H_{xx} - H_{xu}H_{uu}^{-1}H_{xu}^\top. \end{aligned} \quad (11.137)$$

As in the linear-quadratic case treated below, $J\mathcal{H}$ is symmetric, so the spectrum of $\mathcal{H}$ is symmetric about the imaginary axis. The following second-order sufficient condition removes the axis and produces the sharp estimate.

Theorem 11.31 (Local exponential turnpike). *Suppose Assumption 11.28 holds, (11.129) is strictly dissipative at $(\bar{x}, \bar{u})$, $H_{uu} \succ 0$, and the reduced Hamiltonian matrix (11.137) is hyperbolic (no eigenvalue on the imaginary axis); when in addition $\tilde{Q} \succeq 0$, hyperbolicity is equivalent to the reduced pair $(\tilde{A}, B)$ being stabilizable and $(\tilde{A}, \tilde{Q}^{1/2})$ detectable. Then there exist constants $C, \lambda, \delta, T_0 > 0$, independent of T , such that whenever $\|x_0 - \bar{x}\| \leq \delta$ and $\|x_1 - \bar{x}\| \leq \delta$, the optimal state–costate pair obeys, for all $T \geq T_0$ and $t \in [0, T]$,*

$$\|x_T^*(t) - \bar{x}\| + \|p_T^*(t) - \bar{p}\| \leq C \left(e^{-\lambda t} \|x_0 - \bar{x}\| + e^{-\lambda(T-t)} \|x_1 - \bar{x}\| \right). \quad (11.138)$$

We do not prove Theorem 11.31 in full; the argument reduces to two ingredients. First, the linearized problem is exactly a linear-quadratic turnpike problem with data $(\tilde{A}, B, \tilde{Q}, H_{uu})$, for which (11.138) is established *globally* and constructively in Theorem 11.37 below. Second, the nonlinear remainder along the two-point boundary-value problem is controlled by a contraction (implicit function) argument that exploits the same stable/unstable dichotomy; the measure turnpike of Theorem 11.30 supplies the a priori proximity to $\bar{x}$ that localizes the fixed point. The details are carried out in the references cited in Section 11.7. We now turn to the linear-quadratic case, where every hypothesis of Theorem 11.31 is explicit and the estimate is global.

11.6.4 ■ The Linear-Quadratic Turnpike

Problem data and the turnpike

We now consider the time-invariant linear-quadratic problem

$$\min_u J_T(u) = \int_0^T \ell(x(t), u(t)) dt, \quad \dot{x}(t) = Ax(t) + Bu(t), \quad (11.139)$$

with running cost¹

$$\ell(x, u) = \frac{1}{2}x^\top Qx + q^\top x + \frac{1}{2}u^\top Ru, \quad Q = Q^\top \succeq 0, \quad R = R^\top \succ 0. \quad (11.140)$$

The boundary data are the initial condition $x(0) = x_0$ and, in the sharpest results, a prescribed terminal state $x(T) = x_1$. We write $S := BR^{-1}B^\top \succeq 0$.

Assumption 11.32. *The pair (A, B) is controllable and $Q \succ 0$.*

Controllability guarantees that the fixed-endpoint problem is feasible for every (x_0, x_1) and every $T > 0$, and, together with $Q \succ 0$, that the two algebraic Riccati solutions used below exist and are ordered. The condition $Q \succ 0$ is stronger than needed—detectability of $(A, Q^{1/2})$ suffices—but it streamlines the statements.

The turnpike is defined by the static counterpart of (11.139), in which the dynamics are frozen at equilibrium:

$$\min_{(x,u) \in \mathbb{R}^n \times \mathbb{R}^m} \ell(x, u) \quad \text{subject to} \quad Ax + Bu = 0. \quad (11.141)$$

Since ℓ is strictly convex in u and convex in x , and the constraint is affine, (11.141) is a convex program; under Assumption 11.32 it admits a unique minimizer $(\bar{x}, \bar{u})$. Introducing a multiplier $\bar{p} \in \mathbb{R}^n$ for the equality constraint, the Karush–Kuhn–Tucker conditions read

$$Q\bar{x} + q + A^\top \bar{p} = 0, \quad R\bar{u} + B^\top \bar{p} = 0, \quad A\bar{x} + B\bar{u} = 0. \quad (11.142)$$

We call $(\bar{x}, \bar{u})$ the *turnpike* and $\bar{p}$ the associated *turnpike costate*.

As in the nonlinear case, the turnpike is the rest point of the optimality flow. The Hamiltonian of (11.139) is $H(x, p, u) = \ell(x, u) + p^\top (Ax + Bu)$, and pointwise minimization over u gives $u^*(x, p) = -R^{-1}B^\top p$. The resulting canonical system is

$$\frac{d}{dt} \begin{pmatrix} x \\ p \end{pmatrix} = \underbrace{\begin{pmatrix} A & -S \\ -Q & -A^\top \end{pmatrix}}_{=: \mathcal{H}} \begin{pmatrix} x \\ p \end{pmatrix} + \begin{pmatrix} 0 \\ -q \end{pmatrix}, \quad (11.143)$$

where $\mathcal{H}$ is the *Hamiltonian matrix*. Comparing (11.142) with (11.143) shows that $(\bar{x}, \bar{p})$ is precisely the equilibrium of the canonical flow,

$$\mathcal{H} \begin{pmatrix} \bar{x} \\ \bar{p} \end{pmatrix} + \begin{pmatrix} 0 \\ -q \end{pmatrix} = 0. \quad (11.144)$$

Thus the static optimizer, the Lagrange multiplier of the frozen problem, and the rest point of the optimality system all coincide. This identity is the germ of every turnpike theorem: the finite-horizon optimal trajectory is an orbit of (11.143) constrained by boundary data, and its behavior is governed by how orbits approach or depart from the equilibrium $(\bar{x}, \bar{p})$.

¹Relative to Section 11.5.7 we insert the factors $\frac{1}{2}$ and admit a linear term $q^\top x$. The factors are a harmless normalization; the associated algebraic Riccati equation coincides with the stationary form of (11.122) under $S := BR^{-1}B^\top$, and the linear term is what renders the turnpike nontrivial.

Strict dissipativity and the integral turnpike

We verify that the linear-quadratic problem satisfies Definition 11.29 globally, with a linear storage function, and that the abstract measure turnpike then sharpens to an L^2 estimate. Everything rests on one algebraic identity that shifts the cost to the turnpike.

Lemma 11.33 (Rotated-cost identity). *For all $(x, u) \in \mathbb{R}^n \times \mathbb{R}^m$,*

$$\ell(x, u) - \ell(\bar{x}, \bar{u}) + \bar{p}^\top (Ax + Bu) = \frac{1}{2}(x - \bar{x})^\top Q(x - \bar{x}) + \frac{1}{2}(u - \bar{u})^\top R(u - \bar{u}). \quad (11.145)$$

Consequently, with the linear storage function $\lambda(x) := -\bar{p}^\top (x - \bar{x})$ and $\sigma := \lambda_{\min}(Q) > 0$, every admissible pair satisfies the strict dissipation inequality

$$\frac{d}{dt} \lambda(x(t)) \leq \ell(x(t), u(t)) - \ell(\bar{x}, \bar{u}) - \frac{\sigma}{2} \|x(t) - \bar{x}\|^2. \quad (11.146)$$

Proof. Write $\delta x = x - \bar{x}$, $\delta u = u - \bar{u}$. Expanding the quadratic ℓ about $(\bar{x}, \bar{u})$,

$$\ell(x, u) - \ell(\bar{x}, \bar{u}) = (Q\bar{x} + q)^\top \delta x + (R\bar{u})^\top \delta u + \frac{1}{2}\delta x^\top Q\delta x + \frac{1}{2}\delta u^\top R\delta u.$$

Because $A\bar{x} + B\bar{u} = 0$, we have $Ax + Bu = A\delta x + B\delta u$, so $\bar{p}^\top (Ax + Bu) = (A^\top \bar{p})^\top \delta x + (B^\top \bar{p})^\top \delta u$. Adding this to the previous display, the linear terms carry the coefficients $Q\bar{x} + q + A^\top \bar{p}$ and $R\bar{u} + B^\top \bar{p}$, both of which vanish by the stationarity conditions (11.142). Only the quadratic remainder survives, which is (11.145). For the second claim, note that along a trajectory $\frac{d}{dt} \lambda(x) = -\bar{p}^\top \dot{x} = -\bar{p}^\top (Ax + Bu)$; substituting (11.145) and discarding the nonnegative term $\frac{1}{2}\delta u^\top R\delta u$ yields (11.146), using $\delta x^\top Q\delta x \geq \sigma \|\delta x\|^2$. $\square$

Inequality (11.146) says that the problem is *strictly dissipative* at $(\bar{x}, \bar{u})$ with supply rate $\ell(x, u) - \ell(\bar{x}, \bar{u})$ and storage λ : no admissible trajectory can accumulate cost at the turnpike rate $\ell(\bar{x}, \bar{u})$ while remaining away from $\bar{x}$, because the deficit $\frac{\sigma}{2} \|x - \bar{x}\|^2$ is irrecoverable. Integrating this observation produces the first turnpike theorem.

Theorem 11.34 (Integral and measure turnpike). *Let Assumption 11.32 hold and consider the fixed-endpoint problem (11.139) with $x(0) = x_0$, $x(T) = x_1$. There is a constant $C = C(x_0, x_1)$, independent of T , such that the optimal trajectory $x_T^*(\cdot)$ satisfies, for every $T \geq 2$,*

$$\int_0^T \|x_T^*(t) - \bar{x}\|^2 dt \leq C. \quad (11.147)$$

In particular, for every $\varepsilon > 0$ the amount of time the optimal state spends outside the ball $B_\varepsilon(\bar{x})$ is bounded uniformly in the horizon:

$$|\{t \in [0, T] : \|x_T^*(t) - \bar{x}\| > \varepsilon\}| \leq \frac{C}{\varepsilon^2}. \quad (11.148)$$

Proof. *Upper bound on the optimal value.* By controllability there exist controls steering x_0 to $\bar{x}$ on $[0, 1]$ and $\bar{x}$ to x_1 on $[T - 1, T]$, with costs $c_0(x_0)$ and $c_1(x_1)$ that depend on the endpoints but not on T . Concatenating these with the constant turnpike control $\bar{u}$ on $[1, T - 1]$ yields an admissible comparison control of cost $(T - 2)\ell(\bar{x}, \bar{u}) + c_0(x_0) + c_1(x_1)$. Hence

$$J_T(u_T^*) \leq T\ell(\bar{x}, \bar{u}) + C_1, \quad C_1 := c_0(x_0) + c_1(x_1) - 2\ell(\bar{x}, \bar{u}). \quad (11.149)$$

Lower bound via the rotated cost. Integrating the identity (11.145) along the optimal pair and using $\int_0^T \bar{p}^\top (Ax_T^* + Bu_T^*) dt = \bar{p}^\top (x_T^*(T) - x_T^*(0)) = \bar{p}^\top (x_1 - x_0)$,

$$J_T(u_T^*) = T \ell(\bar{x}, \bar{u}) - \bar{p}^\top (x_1 - x_0) + \int_0^T \left(\frac{1}{2} \delta x^\top Q \delta x + \frac{1}{2} \delta u^\top R \delta u \right) dt,$$

where $\delta x = x_T^* - \bar{x}$, $\delta u = u_T^* - \bar{u}$. Combining with (11.149) and discarding the nonnegative control term,

$$\frac{\sigma}{2} \int_0^T \|x_T^*(t) - \bar{x}\|^2 dt \leq \int_0^T \frac{1}{2} \delta x^\top Q \delta x dt \leq C_1 + \bar{p}^\top (x_1 - x_0) =: \frac{\sigma}{2} C.$$

This is (11.147). Finally, if A_ε denotes the set in (11.148), then $C \geq \int_{A_\varepsilon} \|x_T^* - \bar{x}\|^2 dt \geq \varepsilon^2 |A_\varepsilon|$, which is Chebyshev's inequality in disguise and gives (11.148). $\square$

Remark 11.35. Theorem 11.34 is the linear-quadratic instance of the abstract measure turnpike (Theorem 11.30), with quadratic α and the linear storage $\lambda(x) = -\bar{p}^\top (x - \bar{x})$; the quadratic supply rate upgrades the measure bound to a full L^2 estimate. Like its abstract counterpart it uses only convexity and controllability, not the spectral structure of $\mathcal{H}$, and it says how *much* time, but not *where*, the optimal trajectory spends away from $\bar{x}$. To localize the excursions at the two ends of the interval and to sharpen $\|x - \bar{x}\| \in L^2$ to a pointwise exponential decay, we turn to the Hamiltonian matrix.

Hyperbolicity and the two Riccati solutions

The exponential turnpike rests on a spectral dichotomy for the Hamiltonian matrix $\mathcal{H}$ of (11.143). Its defining structural property is that $J\mathcal{H}$ is symmetric, where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$; a direct computation gives

$$J\mathcal{H} = \begin{pmatrix} -Q & -A^\top \\ -A & S \end{pmatrix} = (J\mathcal{H})^\top.$$

Such matrices are called *Hamiltonian* in the algebraic sense, and their spectrum is symmetric with respect to the imaginary axis: if μ is an eigenvalue, so is $-\mu$ (and, since $\mathcal{H}$ is real, so are $\bar{\mu}$, $-\bar{\mu}$). The next lemma shows that under Assumption 11.32 none of these eigenvalues lies on the axis itself.

Lemma 11.36 (Hyperbolicity). *Under Assumption 11.32 the Hamiltonian matrix $\mathcal{H}$ has no eigenvalue on the imaginary axis. Consequently it has exactly n eigenvalues with negative real part and n with positive real part.*

Proof. Suppose $\mathcal{H} \begin{pmatrix} x \\ p \end{pmatrix} = i\omega \begin{pmatrix} x \\ p \end{pmatrix}$ for some $\omega \in \mathbb{R}$ and $(x, p) \neq 0$, i.e.

$$Ax - Sp = i\omega x, \quad -Qx - A^\top p = i\omega p. \quad (11.150)$$

Left-multiply the first equation by p^* and the conjugate transpose of the second by x (using that A, Q, S are real and Q, S symmetric):

$$p^* Ax - p^* Sp = i\omega p^* x, \quad -x^* Qx - p^* Ax = -i\omega p^* x.$$

Adding the two eliminates the cross terms $p^* Ax$ and $\pm i\omega p^* x$, leaving $p^* Sp + x^* Qx = 0$. Both quadratic forms are nonnegative, so $p^* Sp = 0$ and $x^* Qx = 0$. Since $R \succ 0$, $p^* Sp =$

$\|R^{-1/2}B^\top p\|^2 = 0$ forces $B^\top p = 0$; since $Q \succ 0$, $x^*Qx = 0$ forces $x = 0$. With $x = 0$ the second equation of (11.150) reads $A^\top p = -i\omega p$ with $B^\top p = 0$; by the Popov–Belevitch–Hautus test, controllability of (A, B) admits no such left eigenvector unless $p = 0$. Hence $(x, p) = 0$, a contradiction. The count of eigenvalues then follows from the axis symmetry of the spectrum. $\square$

Lemma 11.36 endows $\mathcal{H}$ with complementary invariant subspaces: the *stable* subspace $\mathcal{V}_-$ spanned by the generalized eigenvectors of the n eigenvalues in the open left half-plane, and the *unstable* subspace $\mathcal{V}_+$ for those in the right half-plane. Each is n -dimensional and, being Lagrangian, is the graph of a symmetric matrix: there exist symmetric $\bar{P}$ and $\bar{P}^-$ with

$$\mathcal{V}_- = \{(x, \bar{P}x) : x \in \mathbb{R}^n\}, \quad \mathcal{V}_+ = \{(x, \bar{P}^-x) : x \in \mathbb{R}^n\}. \quad (11.151)$$

Here $\bar{P}$ is the *stabilizing* solution of the algebraic Riccati equation

$$A^\top P + PA - PSP + Q = 0, \quad (11.152)$$

characterized by $A_- := A - S\bar{P}$ Hurwitz, and $\bar{P}^-$ is the *anti-stabilizing* solution, characterized by $A_+ := A - S\bar{P}^-$ anti-Hurwitz. Under Assumption 11.32 both exist, and controllability together with $Q \succ 0$ yields the strict ordering

$$D := \bar{P} - \bar{P}^- \succ 0. \quad (11.153)$$

The matrix A_- is exactly the closed-loop generator of the infinite-horizon regulator of Section 11.5.7; the anti-stabilizing A_+ is its time-reversed counterpart. Geometrically, $\mathcal{V}_-$ and $\mathcal{V}_+$ are the stable and anti-stable manifolds of the turnpike, which is a *saddle point* of the Hamiltonian flow; an optimal orbit enters along $\mathcal{V}_-$ and leaves along $\mathcal{V}_+$, as illustrated in Figure 11.9. These two objects diagonalize the Hamiltonian flow, as we now exploit.

The exponential turnpike theorem

The stable/anti-stable splitting of the previous subsection converts the optimality boundary-value problem into two decoupled, exponentially contracting flows, yielding the sharp two-sided estimate, globally, and with explicit rate.

Theorem 11.37 (Exponential turnpike, linear-quadratic). *Let Assumption 11.32 hold and consider the fixed-endpoint problem (11.139) with $x(0) = x_0$, $x(T) = x_1$. There exist constants $C > 0$, $\lambda > 0$, and $T_0 \geq 2$, all independent of T , x_0 , and x_1 , such that for every $T \geq T_0$ the optimal state–costate pair $(x_T^*(t), p_T^*(t))$ obeys*

$$\|x_T^*(t) - \bar{x}\| + \|p_T^*(t) - \bar{p}\| \leq C \left(e^{-\lambda t} \|x_0 - \bar{x}\| + e^{-\lambda(T-t)} \|x_1 - \bar{x}\| \right), \quad t \in [0, T]. \quad (11.154)$$

The estimate (11.154) is the precise form of the turnpike metaphor: two boundary layers of width $O(1/\lambda)$ at the ends of the interval, joined by an exponentially small plateau at the turnpike $(\bar{x}, \bar{p})$ throughout the interior.

Proof. Write the deviations $\xi := x_T^* - \bar{x}$ and $\pi := p_T^* - \bar{p}$. Subtracting the equilibrium relation (11.144) from the canonical system (11.143) removes the affine term, so (ξ, π) solves the *homogeneous* linear system

$$\dot{\xi} = A\xi - S\pi, \quad \dot{\pi} = -Q\xi - A^\top\pi, \quad (11.155)$$

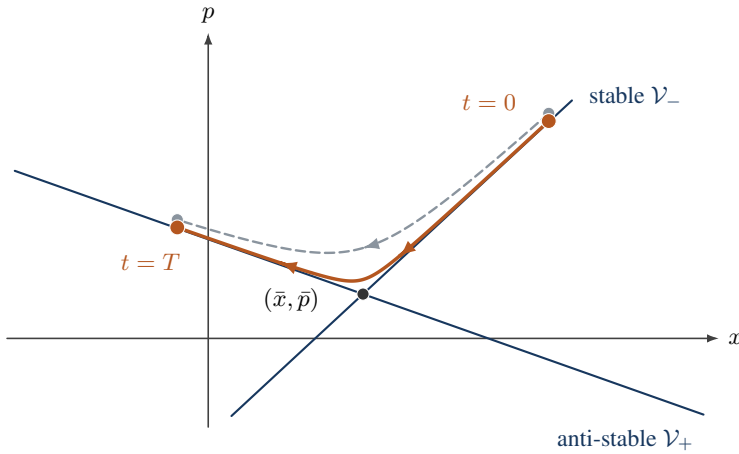


Figure 11.9: The turnpike as a saddle point of the Hamiltonian flow, drawn for the scalar problem of Example 11.38 in the (x, p) plane. Hyperbolicity (Lemma 11.36) splits the flow into a stable manifold $\mathcal{V}_-$ (slope $\bar{P}$, the stabilizing Riccati solution) and an anti-stable manifold $\mathcal{V}_+$ (slope $\bar{P}^-$), crossing at the turnpike $(\bar{x}, \bar{p})$. A finite-horizon optimal orbit (solid) with fixed endpoints starts at $t = 0$, is drawn in along $\mathcal{V}_-$ toward the saddle, dwells near $(\bar{x}, \bar{p})$, and is expelled along $\mathcal{V}_+$ to meet the terminal condition at $t = T$. The two exponential rates $\pm\lambda$ of the approach and departure are exactly the boundary-layer rates of Figure 11.8; the longer the horizon, the closer the orbit passes to the saddle, as the orbit for a shorter horizon with the same endpoint states (dashed) shows.

with boundary data $\xi(0) = x_0 - \bar{x} =: \xi_0$ and $\xi(T) = x_1 - \bar{x} =: \xi_T$.

Step 1: decoupling along the Riccati graphs. Introduce the coordinates $w := \pi - \bar{P}\xi$ and $v := \pi - \bar{P}^-\xi$, measuring the deviation of the orbit from the stable and unstable subspaces (11.151). Differentiating w along (11.155) and substituting $\pi = w + \bar{P}\xi$,

$$\dot{w} = (-Q - A^\top \bar{P} - \bar{P}A + \bar{P}S\bar{P})\xi + (-A^\top + \bar{P}S)w.$$

The coefficient of ξ vanishes identically by the Riccati equation (11.152), leaving the decoupled scalarizable equation $\dot{w} = -(A - S\bar{P})^\top w = -A_-^\top w$. The identical computation with $\bar{P}^-$ gives $\dot{v} = -A_+^\top v$. Because A_- is Hurwitz and A_+ is anti-Hurwitz, there are constants $\kappa \geq 1$ and $\lambda > 0$ (any λ below the spectral abscissa of $-A_-^\top$ and of $A_+^\top$) with

$$\|e^{A_-^\top s}\| \leq \kappa e^{-\lambda s}, \quad \|e^{-A_+^\top s}\| \leq \kappa e^{-\lambda s}, \quad s \geq 0. \quad (11.156)$$

Hence v decays forward and w decays backward in time:

$$\|v(t)\| \leq \kappa e^{-\lambda t} \|v(0)\|, \quad \|w(t)\| \leq \kappa e^{-\lambda(T-t)} \|w(T)\|. \quad (11.157)$$

Subtracting the definitions of v and w recovers the state deviation,

$$\xi(t) = D^{-1}(v(t) - w(t)), \quad D = \bar{P} - \bar{P}^- \succ 0, \quad (11.158)$$

and $\pi = w + \bar{P}\xi$ recovers the costate deviation; both are therefore controlled once $\|v(0)\|$ and $\|w(T)\|$ are bounded by the boundary data.

Step 2: solving for the boundary modes. Evaluate (11.158) at $t = 0$ and $t = T$, using $w(0) = e^{A^\top T} w(T) =: E_1 w(T)$ and $v(T) = e^{-A^\top T} v(0) =: E_2 v(0)$ from (11.157), where $\|E_1\|, \|E_2\| \leq \kappa e^{-\lambda T}$. This gives the linear system

$$\begin{pmatrix} I & -E_1 \\ E_2 & -I \end{pmatrix} \begin{pmatrix} v(0) \\ w(T) \end{pmatrix} = \begin{pmatrix} D\xi_0 \\ D\xi_T \end{pmatrix}.$$

Choose T_0 so that $\kappa^2 e^{-\lambda T_0} \leq \frac{1}{2}$; then for $T \geq T_0$ the Schur complement $I - E_1 E_2$ is invertible with $\|(I - E_1 E_2)^{-1}\| \leq 2$, and solving the system yields

$$v(0) = (I - E_1 E_2)^{-1} (D\xi_0 - E_1 D\xi_T), \quad w(T) = E_2 v(0) - D\xi_T,$$

whence, with $c_0 := 2\|D\|$,

$$\|v(0)\| \leq c_0 (\|\xi_0\| + e^{-\lambda T} \|\xi_T\|), \quad \|w(T)\| \leq c_0 (\|\xi_T\| + e^{-\lambda T} \|\xi_0\|). \quad (11.159)$$

Step 3: assembling the estimate. Insert (11.159) into (11.157) and then into (11.158):

$$\|\xi(t)\| \leq \|D^{-1}\| (\|v(t)\| + \|w(t)\|) \leq \|D^{-1}\| \kappa c_0 \left(e^{-\lambda t} \|\xi_0\| + e^{-\lambda(T-t)} \|\xi_T\| + e^{-\lambda t} (e^{-\lambda t} \|\xi_T\| + e^{-\lambda(T-t)} \|\xi_0\|) \right)$$

For $t \in [0, T]$ one has $e^{-\lambda T} e^{-\lambda t} \leq e^{-\lambda(T-t)}$ and $e^{-\lambda T} e^{-\lambda(T-t)} \leq e^{-\lambda t}$, so the cross terms are absorbed into the first two, giving the bound (11.154) for ξ with $C = 2\|D^{-1}\| \kappa c_0$. The costate deviation satisfies $\|\pi(t)\| \leq \|w(t)\| + \|\bar{P}\| \|\xi(t)\|$, which obeys the same bound after enlarging C . This proves (11.154). $\square$

Example 11.38 Scalar turnpike.

Consider $\dot{x} = ax + bu$ with $b \neq 0$ and cost $\ell(x, u) = \frac{1}{2}x^2 + qx + \frac{1}{2}u^2$, so that $Q = 1$, $R = 1$, $S = b^2$. The Hamiltonian matrix $\mathcal{H} = \begin{pmatrix} a & -b^2 \\ -1 & -a \end{pmatrix}$ has eigenvalues $\pm\lambda$ with $\lambda = \sqrt{a^2 + b^2}$, so the turnpike is approached and left at the exponential rate λ . The stabilizing Riccati solution is $\bar{P} = (a + \lambda)/b^2 > 0$ and the anti-stabilizing one is $\bar{P}^- = (a - \lambda)/b^2$, giving $D = \bar{P} - \bar{P}^- = 2\lambda/b^2 > 0$ as required. The turnpike itself solves the 2×2 system (11.144), namely

$$\bar{x} = -\frac{qb^2}{a^2 + b^2}, \quad \bar{p} = -\frac{qa}{a^2 + b^2}, \quad \bar{u} = -b\bar{p} = \frac{qab}{a^2 + b^2}.$$

For a long horizon with $x(0) = x_0$ and $x(T) = x_1$, the optimal state hugs $\bar{x}$ except within $O(1/\lambda)$ of each endpoint, and (11.154) quantifies the two boundary layers. $\square$

Remark 11.39 (Free terminal state). If instead of $x(T) = x_1$ the terminal state is free, the terminal condition becomes $p_T^*(T) = 0$, i.e. $\pi(T) = -\bar{p}$, and the proof goes through verbatim with $D\xi_T$ replaced by the datum $-\bar{p}$ in Step 2. The conclusion is a one-sided estimate: a single initial boundary layer of width $O(1/\lambda)$ followed by convergence to the turnpike, with the trajectory departing slightly near $t = T$ because, absent a terminal reward, there is no incentive to remain at $\bar{x}$. The finite-horizon Riccati matrix $P(t)$ of Section 11.5.7 correspondingly satisfies $\|P(t) - \bar{P}\| \leq Ce^{-2\lambda(T-t)}$, the Riccati turnpike underlying the convergence to the algebraic Riccati equation noted in Remark after Theorem 11.25.

Remark 11.40 (Consistency with the general theory). Theorem 11.37 is the global, constructive realization of the local statement in Theorem 11.31: here the reduced Hamiltonian matrix (11.137) is $\mathcal{H} = \begin{pmatrix} A & -S \\ -Q & -A^\top \end{pmatrix}$ (since $H_{xu} = 0$, $H_{uu} = R$, $H_{xx} = Q$), its hyperbolicity is Lemma 11.36, and the storage is linear, so no localization and no implicit-function step are needed, the estimate holds for all x_0, x_1 and the rate λ is the spectral gap of $\mathcal{H}$. The linear-quadratic problem is thus the exactly solvable prototype behind the general nonlinear turnpike.

11.6.5 ■ Interpretation and Ramifications

The turnpike theorem originated in mathematical economics, where x is a capital stock, ℓ a discounted-utility deficit, and $(\bar{x}, \bar{u})$ the optimal balanced-growth or golden-rule configuration. Dorfman, Samuelson, and Solow observed that optimal accumulation programs over a long planning horizon stay close to the golden-rule state regardless of the initial endowment or the terminal target, and McKenzie established this rigorously for a broad class of models. The costate $\bar{p}$ is then the vector of efficiency prices that support the golden-rule state, consistent with its reading as a shadow price throughout this chapter.

The turnpike property is the theoretical backbone of model predictive control. Because the optimal trajectory reaches a neighborhood of $\bar{x}$ after a horizon-independent transient, the first portion of a finite-horizon optimal control, the part actually implemented in a receding-horizon scheme, is nearly independent of the horizon length and of the terminal ingredients. This explains why short prediction horizons already deliver near-optimal closed-loop performance and underlies modern stability and performance guarantees for economic model predictive control.

Finally, (11.154) has direct computational value. The two-point boundary-value problem (11.155) is exponentially ill-conditioned as stated, it contains growing modes of size $e^{\lambda T}$, but the stable/unstable decoupling of Step 1 in the proof of Theorem 11.37 is precisely the change of variables that stabilizes it. Discretizations that respect this splitting, or that initialize the interior at the turnpike and resolve only the two boundary layers, compute long-horizon optimal trajectories at a cost essentially independent of T .

11.7 ■ Further Reading

The foundations of optimal control theory can be traced to the classical text by Athans and Falb, which provides a comprehensive introduction to both the theory and its applications [6]. The role of saddle point formulations and functional analytic methods in optimization is systematically developed in Luenberger's seminal work [79], which offers a rigorous treatment of optimization in vector spaces and underlies many variational formulations used in control theory.

A direct and elegant derivation of the Kalman filter via the maximum principle is given in the influential paper by Athans and Tse [7], which highlights the deep connections between estimation and optimal control. We further recommend standard modern references on optimal control, including [77] and [3], which provide complementary perspectives on deterministic control theory and linear quadratic methods.

The turnpike property has its origins in mathematical economics: the metaphor and the first analyses appear in Dorfman, Samuelson, and Solow [39], and the classical rigorous theory of optimal accumulation is due to McKenzie [84]. Its modern control-theoretic form, including the local exponential estimates for the state, control, and costate (Theorem 11.31) and the role of Hamiltonian hyperbolicity, together with the implicit-function argument controlling the nonlinear remainder, is developed by Trélat and Zuazua [115]. The equivalence between strict dissipativity (Definition 11.29) and the measure turnpike property (Theorem 11.30) is established by Grüne and Müller [54] and surveyed, together with its consequences for model predictive control, by

Faulwasser and coauthors [46]. The two algebraic Riccati solutions and the spectral geometry of the Hamiltonian matrix used in Section 11.6.4 are treated in depth by Lancaster and Rodman [76].

11.8 ■ Exercises

Exercise 11.1 (Finite-horizon optimal control for a scalar linear system). Consider the one-dimensional linear system $\dot{x}(t) = ax(t) + bu(t)$, $x(0) = x_0$, where $u(t), x(t) \in \mathbb{R}$, and a, b, x_0 are given scalars. Use Pontryagin's maximum principle to determine an optimal control over the interval $[0, T]$ that minimizes the quadratic cost $J = \frac{1}{2}q(x(T))^2 + \frac{1}{2}\int_0^T u(t)^2 dt$, where $q > 0$ is a given scalar.

Exercise 11.2 (Numerical solution of a multi-dimensional linear-quadratic regulator). Use Matlab to obtain a numerical solution to the following linear-quadratic regulator problem:

$$\dot{x} = \begin{pmatrix} -0.2 & 0.5 & 0 & 0 & 0 \\ 0 & -0.5 & 1.6 & 0 & 0 \\ 0 & 0 & -\frac{1}{7} & \frac{6}{7} & 0 \\ 0 & 0 & 0 & -0.25 & 7.5 \\ 0 & 0 & 0 & 0 & -0.1 \end{pmatrix} x + \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0.3 \end{pmatrix} u,$$

with cost $J = \int_0^\infty (x_1^2(t) + u^2(t)) dt$.

Determine the optimal (minimizing) control and the eigenvalues of the resulting closed-loop system. Then write down the Hamiltonian associated with this problem and compute its eigenvalues. Compare the Hamiltonian eigenvalues with those of the closed-loop system.

Exercise 11.3 (Constrained optimal control and switching structure). Consider the minimization of the functional $\int_0^2 (2x^2(t) + u^2(t)) dt$, subject to $\dot{x}(t) = x(t) + 2u(t)$, $x(0) = 1$, $u(t) \in [-1, 1]$.

Use Pontryagin's minimum principle to determine the optimal control and corresponding trajectory. Show that the only candidate optimal solution has a switching structure of the form $u^*(t) = -1$ on $[0, t_1]$ and $u^*(t) = -\lambda^*(t)$ on $[t_1, 2]$, for some costate $\lambda^*(t)$, where the switching time $t_1 \in (0, 2)$ satisfies $n(t_1) := 2e^{-12+4t_1} - 2e^{-2t_1} - 6e^{-12+3t_1} + 3e^{-3t_1} = 0$.

Exercise 11.4 (Hamilton–Jacobi–Bellman equation for a scalar integrator). Consider the scalar system $\dot{x}(t) = u(t)$, $x(0) = x_0$, with cost

$$J(u) = \frac{1}{2} \int_0^T u(t)^2 dt + \frac{1}{2} s x(T)^2, \quad s > 0.$$

- (i) Write the Hamilton–Jacobi–Bellman equation for the value function $V(x, t)$ and carry out the pointwise minimization over u to eliminate the control.
- (ii) Using the quadratic ansatz $V(x, t) = \frac{1}{2}P(t)x^2$, show that P satisfies the Riccati equation $\dot{P}(t) = P(t)^2$ with $P(T) = s$, and solve it to obtain

$$P(t) = \frac{1}{(T-t) + 1/s}.$$

- (iii) Deduce the optimal feedback law $u^*(t) = -P(t)x(t)$ and the closed-loop trajectory.
- (iv) Solve the same problem with Pontryagin's Minimum Principle and verify that the costate satisfies $p(t) = P(t)x^*(t) = \partial_x V(x^*(t), t)$, in agreement with Proposition 11.21.

Exercise 11.5 (Minimum-energy transfer for the double integrator). Consider $\ddot{y}(t) = u(t)$, written as $\dot{x}_1 = x_2$, $\dot{x}_2 = u$, and the fixed-endpoint problem of minimizing $\int_0^T u(t)^2 dt$ subject to $(x_1(0), x_2(0)) = (y_0, 0)$ and $(x_1(T), x_2(T)) = (0, 0)$.

- (i) Form the Hamiltonian and show, from the adjoint equations, that the costate components satisfy $p_1(t) = c_1$ and $p_2(t) = -c_1 t + c_2$, so that the optimal control $u^*(t) = -\frac{1}{2}p_2(t)$ is *affine* in time.
- (ii) Impose the four boundary conditions and show that

$$u^*(t) = \frac{6y_0}{T^2} \left(\frac{2t}{T} - 1 \right).$$

- (iii) Explain why this is a genuine two-point boundary-value problem rather than an initial-value problem, and relate it to the fixed-terminal-state LQR formulation of Section 11.2.

Exercise 11.6 (Time-optimal control of the double integrator from a moving state). For $\ddot{y} = u$ with $|u| \leq 1$, consider steering the state from $(x_1(0), x_2(0)) = (0, v_0)$, with $v_0 > 0$, to the origin in minimum time.

- (i) Using the switching curve $\Gamma = \{x_1 = -\frac{1}{2}x_2|x_2|\}$, argue that the optimal control is bang-bang with exactly one switch, applying $u = -1$ first.
- (ii) Show that the switch occurs at $t_s = v_0(1 + \frac{1}{\sqrt{2}})$, where the trajectory meets Γ with $x_2 = -v_0/\sqrt{2}$.
- (iii) Conclude that the minimum time is $T = (1 + \sqrt{2})v_0$, and sketch the trajectory in the phase plane relative to Γ .

Exercise 11.7 (Scalar Kalman–Bucy filter and its steady state). A scalar process and its noisy measurement evolve as

$$dx(t) = -a x(t) dt + dW(t), \quad dz(t) = x(t) dt + dV(t),$$

with $a > 0$, $\mathbb{E}[dW^2] = q dt$ and $\mathbb{E}[dV^2] = r dt$, $q, r > 0$.

- (i) Write the Kalman–Bucy variance (Riccati) equation for $\Sigma(t) = \mathbb{E}[e(t)^2]$ and the corresponding optimal gain $G(t)$.
- (ii) Find the steady-state error variance Σ_∞ by setting $\dot{\Sigma} = 0$, and show that

$$\Sigma_\infty = r \left(-a + \sqrt{a^2 + q/r} \right), \quad G_\infty = \frac{\Sigma_\infty}{r}.$$

- (iii) Interpret the limits $r \rightarrow 0$ (accurate sensor) and $q \rightarrow 0$ (accurate model), and comment on how each affects the gain.

Exercise 11.8 (Turnpike of a scalar long-horizon regulator). Consider $\dot{x} = ax + bu$ with $b \neq 0$ and running cost $\ell(x, u) = \frac{1}{2}x^2 + qx + \frac{1}{2}u^2$, on the horizon $[0, T]$ with fixed endpoints $x(0) = x_0$, $x(T) = x_1$.

- (i) Solve the steady-state problem $\min\{\ell(x, u) : ax + bu = 0\}$ and show that the turnpike is $\bar{x} = -qb^2/(a^2 + b^2)$, with turnpike costate $\bar{p} = -qa/(a^2 + b^2)$.

- (ii) Write the Hamiltonian matrix $\mathcal{H}$ and show its eigenvalues are $\pm\lambda$ with $\lambda = \sqrt{a^2 + b^2}$. Identify the stabilizing and anti-stabilizing solutions of the scalar algebraic Riccati equation $2aP - b^2P^2 + 1 = 0$ and verify $\bar{P} - \bar{P}^- = 2\lambda/b^2 > 0$.
- (iii) Using the rotated-cost identity (11.145), show that $\int_0^T (x^*(t) - \bar{x})^2 dt$ is bounded independently of T .
- (iv) Solve the two-point boundary-value problem (11.155) explicitly and confirm the two-boundary-layer estimate (11.154), identifying the decay rate with λ .

Exercise 11.9 (Control–estimation duality). Recall the regulator Riccati equation $-\dot{P} = Q + A^\top P + PA - PBR^{-1}B^\top P$ and the filter Riccati equation $\dot{\Sigma} = A\Sigma + \Sigma A^\top + BQB^\top - \Sigma C^\top R^{-1}C\Sigma$.

- (i) Using the substitutions $A \mapsto A^\top$, $B \mapsto C^\top$, and time reversal $t \mapsto t_1 - t$, show that one Riccati equation is transformed into the other, and identify the correspondence between the feedback gain $K = R^{-1}B^\top P$ and the filter gain $G = \Sigma C^\top R^{-1}$.
- (ii) Given the scalar filter of the previous exercise, write down the scalar LQR problem (its A, B, Q, R) whose Riccati equation is dual to it, and verify that the two steady-state solutions coincide.

Chapter 12

Optimal Control under Incomplete Information

When the state of a dynamical system cannot be observed directly, the controller must act on the basis of the information actually available to it: the record of past measurements and past controls. This chapter develops optimal control under such *incomplete information* in a deterministic, set-membership framework: the initial state—and, more generally, the disturbances and measurement errors—are unknown but bounded, and the object that summarizes the controller’s knowledge is the *compatible state set* (or *information set*), the set of all states consistent with the model and the data observed so far.

The central conceptual move parallels the stochastic theory of Chapters 14 and 15: the compatible state set evolves according to a prediction–correction recursion driven by the measurements, and the control problem is re-posed as a fully observed problem on the space of information states. On that space we develop two complementary tools. The first is a minimum principle adapted to the information structure, in which the classical Hamiltonian and adjoint recursion survive intact but the admissible control variations are constrained to be measurable with respect to the observed data. The second is a dynamic-programming synthesis built on two value functions—a *cost-to-go*, computed backward and offline, and a *cost-to-come*, computed forward and online from the measurements—whose sum is minimized over the compatible set to select a surrogate state; feeding that state to the perfect-information feedback law yields the deterministic *certainty-equivalence* controller, whose optimality we establish through a monotonicity-and-invariance argument on the selected value.

Because the problem is non-adversarial, the residual uncertainty is resolved *optimistically*, by scoring each situation with the least cost consistent with the data; this best-case convention is what makes the value decompose additively. The chapter then treats the time-invariant infinite-horizon case, where the construction collapses to two algebraic Riccati equations—one for control, one for estimation—coupled only through the selection step; the resulting stationary controller is an observer feeding a detuned linear gain, and its closed-loop spectrum separates into controller and observer modes. The final section carries the theory to continuous time, where the measurement record is a function rather than a list and the certainty-equivalence controller arises as the fixed point of an implicit relation; the optimality proof rests on a Danskin-type monotonicity argument, and the linear-quadratic specialization couples the LQR Riccati equation to the Kalman–Bucy filter through the same optimistic selection. A concluding section of notes places the construction in its historical context and details its relationship to the minimax theory of robust and H^∞ control, of which it is the best-case counterpart. Exercises on set-membership recursions, the forward cost-to-come recursion, the value of information under optimism, and the stationary linear-quadratic controller conclude the chapter.

12.1 ■ Discrete-Time Optimal Control under Imperfect State Measurements

In every optimal control problem considered so far the controller had access to the current state x_k and could therefore implement a state-feedback law $u_k = \mu_k(x_k)$. In many applications this assumption is untenable: the state is observed only through a sensor that reports a partial, and possibly ambiguous, function of it. The controller must then act on the basis of the *information actually available*, the record of past measurements and of its own past controls, rather than on the state itself. The optimization is consequently carried out over *causal output-feedback policies* instead of over state-feedback laws or open-loop control sequences.

This section and the next develop the deterministic output-feedback problem in discrete time. The controller's ignorance of the true state is modeled by set membership, the initial state is known only to lie in a given set, and, the problem being non-adversarial, this residual uncertainty is resolved *optimistically*: each situation is scored by the least cost consistent with the observed data. The present section formulates the problem, introduces the objects that summarize the controller's knowledge, the *information history* and the *compatible state set*, and establishes a minimum principle adapted to the information structure. Section 12.2 then gives a dynamic-programming synthesis that culminates in a *certainty-equivalence* controller, and extends it to the infinite horizon. The origins of this construction, and its relationship to the minimax theory of robust and H^∞ control, are taken up in the notes at the end of the chapter (Section 12.4).

12.1.1 ■ Problem Formulation

Consider the discrete-time nonlinear system

$$x_{k+1} = f_k(x_k, u_k), \quad k = 0, 1, \dots, N-1, \quad (12.1)$$

with state $x_k \in \mathcal{X} \subseteq \mathbb{R}^n$ and control $u_k \in \mathcal{U} \subseteq \mathbb{R}^m$. In contrast to the perfect-information setting, the initial state is *not* known to the controller; it is known only to belong to a prescribed set of admissible initial conditions,

$$x_0 \in X_0 \subseteq \mathcal{X}. \quad (12.2)$$

The set X_0 is the sole source of incomplete information: it is what the controller does not know, and every subsequent measurement serves to narrow it down. Throughout we assume that X_0 is nonempty and *compact*, that f_k and h_k are continuous, and that ℓ_k and Φ are lower semicontinuous; these hypotheses make every compatible state set (12.9) compact and every infimum over it attained, so that the minima written below are genuine minima and the certainty-equivalent selection (12.23) exists. (Section 12.2.7 indicates how process disturbances and measurement errors are incorporated; they enlarge the unknowns but do not change the structure.)

The controller does not observe the state directly. At each stage it receives a measurement

$$y_k = h_k(x_k), \quad y_k \in \mathcal{Y}, \quad (12.3)$$

where the output map h_k is in general *not* injective: distinct states may produce the same measurement, so a single reading y_k does not determine x_k .

The objective is the finite-horizon cost

$$J(\mu) = \Phi(x_N) + \sum_{k=0}^{N-1} \ell_k(x_k, u_k), \quad (12.4)$$

with terminal cost Φ and stage costs ℓ_k . Because the state is hidden, admissible controls may depend only on the data available when they are applied. Formally, the control at stage k is

generated by a causal decision rule

$$u_k = \mu_k(y^k), \quad y^k := (y_0, y_1, \dots, y_k), \quad (12.5)$$

so that the optimization variable is not the control sequence $\{u_k\}$ but the sequence of maps

$$\mu = (\mu_0, \mu_1, \dots, \mu_{N-1}), \quad \mu_k : \mathcal{Y}^{k+1} \rightarrow \mathcal{U}. \quad (12.6)$$

The output-feedback optimal control problem is therefore

$$\min_{\mu \in \mathcal{M}} J(\mu), \quad \mathcal{M} := \left\{ (\mu_0, \dots, \mu_{N-1}) : \mu_k : \mathcal{Y}^{k+1} \rightarrow \mathcal{U} \right\}, \quad (12.7)$$

where $\mathcal{M}$ is the set of admissible causal policies.

Two features distinguish (12.7) from the classical (perfect-information) problem. First, the cost (12.4) is evaluated along the trajectory generated by the true initial state $x_0 \in X_0$, which the controller does not know; the value of the problem must therefore be defined so as to reflect this ignorance, a point we return to in Section 12.2. Second, the decision variable is an infinite-dimensional object, a policy, rather than a finite control sequence. The remedy for both difficulties is to identify a state variable that is available to the controller and with respect to which the problem becomes fully observed. That variable is the information state introduced next.

12.1.2 ■ Information Structure: Histories and Compatible State Sets

At each stage the controller knows the measurements it has received and the controls it has issued. This record is the raw information on which any admissible policy must be based.

Definition 12.1 (Information history). *The information history available to the controller at stage k is the pair*

$$I_k = (u^{k-1}, y^k), \quad u^{k-1} = (u_0, \dots, u_{k-1}), \quad y^k = (y_0, \dots, y_k), \quad (12.8)$$

with $I_0 = (y_0)$. We write $\mathbb{I}_k$ for the set of information histories that can occur at stage k . (The script $\mathcal{I}_k$ is reserved, in the stochastic chapters of Part V, for the σ -algebra generated by the data; here the setting is deterministic and I_k is the data record itself.)

Since an admissible policy generates each u_i from y^i , the control record u^{k-1} is itself a function of y^{k-1} ; including it explicitly in I_k is nonetheless convenient, because reconstructing which states are consistent with the data requires knowing which controls were applied.

The measurements do not reveal the state, but they do exclude states that could not have produced the observed data. What the controller genuinely knows about x_k is thus a *set*.

Definition 12.2 (Compatible state set). *The compatible state set associated with the information history I_k is*

$$\mathcal{X}_k(I_k) = \left\{ \xi \in \mathcal{X} : \begin{array}{l} \exists x_0 \in X_0 \text{ and a trajectory } x_0, \dots, x_k \text{ with} \\ x_{i+1} = f_i(x_i, u_i), \quad y_i = h_i(x_i) \quad (0 \leq i \leq k), \quad x_k = \xi \end{array} \right\}. \quad (12.9)$$

Equivalently, $\mathcal{X}_k(I_k)$ is the set of all states in which the system could actually be, given the model, the applied controls u^{k-1} , and the observed outputs y^k .

The compatible state set is the deterministic, set-membership analogue of a Bayesian posterior: it is the exact “belief” the controller holds about the hidden state. It is computable from I_k alone

and evolves by a *prediction–correction* recursion: the current set is first pushed forward through the dynamics (prediction),

$$\text{Pred}_k(\mathcal{X}_k(I_k), u_k) := \{f_k(\xi, u_k) : \xi \in \mathcal{X}_k(I_k)\}, \quad (12.10)$$

and then intersected with the states consistent with the new measurement (correction),

$$\mathcal{X}_{k+1}(I_{k+1}) = \text{Pred}_k(\mathcal{X}_k(I_k), u_k) \cap h_{k+1}^{-1}(y_{k+1}), \quad h_{k+1}^{-1}(y) := \{\xi : h_{k+1}(\xi) = y\}. \quad (12.11)$$

The recursion (12.11) shows that $\mathcal{X}_{k+1}$ is determined by $\mathcal{X}_k$, the control u_k , and the new datum y_{k+1} alone; the compatible state set is therefore a bona fide state for the controller's knowledge. The following properties make this precise.

Proposition 12.3 (Properties of the compatible state sets). *Along any trajectory of (12.1)–(12.3) generated by an admissible policy, the sets $\{\mathcal{X}_k(I_k)\}_{k=0}^N$ satisfy:*

- (i) **Consistency.** *The true state always lies in the compatible set, $x_k \in \mathcal{X}_k(I_k)$; in particular $\mathcal{X}_k(I_k) \neq \emptyset$.*
- (ii) **Perfect recall.** *Information can only accumulate: if I_{k+1} extends I_k , then, under the forward map $f_k(\cdot, u_k)$, $\mathcal{X}_{k+1}(I_{k+1}) \subseteq \text{Pred}_k(\mathcal{X}_k(I_k), u_k)$, so each new measurement can only shrink the set of explanations.*
- (iii) **Nonanticipativity.** *$\mathcal{X}_k(I_k)$ depends only on the past data (u^{k-1}, y^k) and not on future controls or measurements.*

Proof. (i) The realized trajectory $x_0, \dots, x_k$ with x_0 the true initial state satisfies every constraint in (12.9), so x_k is compatible. (ii) Immediate from (12.11), since the intersection with $h_{k+1}^{-1}(y_{k+1})$ can only remove points from the predicted set. (iii) The defining constraints in (12.9) involve only indices $0 \leq i \leq k$. $\square$

Example 12.4 Persistent ambiguity and the dual effect of control.

Let $x_{k+1} = x_k + u_k$ with $X_0 = \{-1, +1\}$ and the magnitude sensor $h_k(x) = x^2$. The first reading is $y_0 = 1$ whichever state is true, so $\mathcal{X}_0(I_0) = \{-1, +1\}$: the measurement is uninformative, and perfect recall holds *with equality*, the corrected set equals the predicted one, and no information is gained.

Under the control $u \equiv 0$ this ambiguity persists forever: every prediction is $\{-1, +1\}$, every reading is 1, and $\mathcal{X}_k(I_k) = \{-1, +1\}$ for all k . All three properties of Proposition 12.3 are on display in their least favorable form: the true state is always in the set (consistency), the sets are nested but never shrink (perfect recall need not be strict), and each set is determined by the, here unchanging, past data alone (nonanticipativity). The compatible set need not converge to a singleton: distinguishability is a property of the data the system emits, not of the passage of time.

Now apply $u_0 = 1$ instead. The prediction step gives $\text{Pred}_0(\{-1, 1\}, 1) = \{0, 2\}$, and the next reading separates the two candidates: $y_1 = h_1(x_1) \in \{0, 4\}$, so the correction step returns the singleton $\{0\}$ or $\{2\}$, and the state is known exactly from then on. The control thus has a *dual effect*: here it cannot shrink the compatible

set directly—the prediction step (12.10) maps it bijectively, because $f_k(\cdot, u_k)$ is a translation—but it can steer the set into a region where the output map h_{k+1} tells its elements apart. (For non-injective dynamics the prediction step can itself merge states and so shrink the set; the two mechanisms are then both available.) In linear problems whose measurement map is affine, $h_k(x) = Cx + d_k$ with a known offset (the noisy variant $y_k = Cx_k + v_k$ with unknown-but-bounded v_k is treated in Section 12.2.7), this second channel is absent—translating the compatible set translates the measurement by a known amount and reveals nothing new, as the exercises make precise—and its absence is what ultimately permits the clean separation between estimation and control in the linear-quadratic theory of Section 12.2.6. $\square$

Property (iii) is what legitimizes the reformulation to come: because $\mathcal{X}_k(I_k)$ is nonanticipative and evolves by the Markov recursion (12.11), it can serve as the state of an equivalent, fully observed control problem.

Example 12.5 Prediction–correction with a binary sensor.

Let $x_{k+1} = x_k + u_k$ on $\mathcal{X} = \mathbb{R}$, with initial uncertainty $X_0 = [-2, 2]$ and the one-bit sensor

$$h_k(x) = \begin{cases} +1, & x \geq 0, \\ -1, & x < 0. \end{cases}$$

Suppose $y_0 = +1$. The correction step (12.11) at stage 0 (applied to X_0 itself) gives $\mathcal{X}_0(I_0) = [-2, 2] \cap [0, \infty) = [0, 2]$: half of the initial uncertainty is eliminated by a single bit. Now apply $u_0 = -1$. The prediction step translates the set, $\text{Pred}_0(\mathcal{X}_0, u_0) = [-1, 1]$, and if the next reading is $y_1 = -1$, the correction step intersects it with $(-\infty, 0)$, giving $\mathcal{X}_1(I_1) = [-1, 1] \cap (-\infty, 0) = [-1, 0)$. All three properties of Proposition 12.3 are visible: the true state lies in every set produced (consistency); each corrected set is contained in the corresponding prediction (perfect recall); and each set is computed from past data alone (nonanticipativity). Observe also that the control u_0 translated the compatible set but did not shrink it, in this problem, information is gained only through measurements, a point pursued in the exercises. $\square$

12.1.3 ■ Admissible Variations and the Information Constraint

The essential difference between output feedback and state feedback appears most sharply when one perturbs a candidate policy. Consider $\mu = (\mu_0, \dots, \mu_{N-1})$ and a perturbation $\delta\mu = (\delta\mu_0, \dots, \delta\mu_{N-1})$. The induced control variation at stage k is

$$\delta u_k = \delta\mu_k(I_k). \quad (12.12)$$

The variation depends on the state only through the information history I_k . Consequently, if two distinct trajectories give rise to the *same* I_k —that is, if two states $\xi, \xi' \in \mathcal{X}_k(I_k)$ are both compatible—then they necessarily receive the *same* control perturbation.

This is the *information constraint*. In the classical Pontryagin principle, where $u_k = \mu_k(x_k)$, the perturbation δu_k may be chosen independently at every state, and optimality reduces to a pointwise condition on the Hamiltonian at each x_k . Under imperfect measurements, admissible

perturbations must be constant on each compatible state set: they are *measurable with respect to* I_k . As we shall see, this single restriction is responsible for every structural change between the two theories.

12.1.4 ■ A Minimum Principle under Imperfect Measurements

We first record the first-order (minimum-principle) characterization of optimality. Define the discrete-time Hamiltonian

$$H_k(x, u, \lambda) = \ell_k(x, u) + \lambda^\top f_k(x, u), \quad \lambda \in \mathbb{R}^n, \quad (12.13)$$

where λ is the adjoint (costate) variable. Recall that in the perfect-information problem the optimal control minimizes H_k pointwise in u at the current state. The imperfect-information version differs only in that the minimization is subject to the information constraint of Section 12.1.3.

Theorem 12.6 (Minimum principle under imperfect measurements). *Assume that f_k, h_k, ℓ_k and Φ are continuously differentiable, that $\mathcal{U}$ is convex, and that the policy maps μ_k^* are continuously differentiable in their arguments. Let $\mu^* \in \mathcal{M}$ be a (locally) optimal policy for (12.7), let $x_0^* \in X_0$ be the realized initial state, and let $\{x_k^*\}_{k=0}^N$ be the resulting trajectory under $u_k^* = \mu_k^*(y^k)$. For an admissible perturbation $\delta\mu = (\delta\mu_0, \dots, \delta\mu_{N-1})$ with each $\delta\mu_k$ continuously differentiable, and such that $\mu^* + \varepsilon \delta\mu$ is admissible for all small $\varepsilon > 0$, let $(\delta x_k, \delta u_k)$ denote the induced first-order variation, defined by the forward linear recursion*

$$\begin{aligned} \delta x_0 &= 0, & \delta x_{k+1} &= \nabla_x f_k^* \delta x_k + \nabla_u f_k^* \delta u_k, \\ \delta u_k &= \delta\mu_k(I_k^*) + \sum_{j=0}^k D_{y_j} \mu_k^*(y^{*k}) \nabla h_j(x_j^*) \delta x_j, \end{aligned} \quad (12.14)$$

where $\nabla_x f_k^*, \nabla_u f_k^*$ denote Jacobians evaluated along the optimal trajectory. Then there exists an adjoint sequence $\{\lambda_k\}_{k=0}^N$ such that the following hold.

(i) **State equation.**

$$x_{k+1}^* = f_k(x_k^*, u_k^*), \quad x_0^* \in X_0.$$

(ii) **Adjoint equation.**

$$\lambda_N = \nabla \Phi(x_N^*), \quad \lambda_k = \nabla_x \ell_k(x_k^*, u_k^*) + [\nabla_x f_k(x_k^*, u_k^*)]^\top \lambda_{k+1}, \quad k = N-1, \dots, 0.$$

(iii) **Variational inequality (information-constrained stationarity).** *For every admissible perturbation $\delta\mu$ as above whose induced control variation points into $\mathcal{U}$ at each stage,*

$$\sum_{k=0}^{N-1} \nabla_u H_k(x_k^*, u_k^*, \lambda_{k+1})^\top \delta u_k \geq 0.$$

Proof. Define the adjoint sequence $\{\lambda_k\}$ by the terminal condition and backward recursion in (ii); this is well defined since the data are C^1 .

For $\varepsilon \geq 0$ let $\mu^\varepsilon := \mu^* + \varepsilon \delta\mu$, and let $\{x_k^\varepsilon\}, \{u_k^\varepsilon\}, \{y_k^\varepsilon\}$ be the trajectory, controls, and outputs generated by μ^ε from the same initial state x_0^* . Since f_k, h_k, μ_k^* , and $\delta\mu_k$ are C^1 , the maps $\varepsilon \mapsto x_k^\varepsilon$ and $\varepsilon \mapsto u_k^\varepsilon$ are differentiable at $\varepsilon = 0^+$, by induction on k : for $k = 0$ this is trivial, and if $\delta x_k := \frac{d}{d\varepsilon} x_k^\varepsilon|_{0^+}$ exists, then differentiating $u_k^\varepsilon = \mu_k^*(y^{\varepsilon,k}) + \varepsilon \delta\mu_k(y^{\varepsilon,k})$ and $x_{k+1}^\varepsilon = f_k(x_k^\varepsilon, u_k^\varepsilon)$ by the chain rule, using $\delta y_j = \nabla h_j(x_j^*) \delta x_j$, yields exactly (12.14). Note

that the second term in the expression for δu_k is the feedback of the state perturbation through the optimal policy; it is present because perturbing the policy perturbs the measurements, which the unperturbed portion μ_k^* of the policy continues to process.

The first variation of the cost is

$$\begin{aligned}\delta J &:= \left. \frac{d}{d\varepsilon} J(\mu^\varepsilon) \right|_{\varepsilon=0+} \\ &= \nabla \Phi(x_N^*)^\top \delta x_N + \sum_{k=0}^{N-1} \left[\nabla_x \ell_k^* \delta x_k + \nabla_u \ell_k^* \delta u_k \right].\end{aligned}$$

We eliminate the state variations by summation by parts. Using $\delta x_0 = 0$ and telescoping,

$$\nabla \Phi(x_N^*)^\top \delta x_N = \lambda_N^\top \delta x_N = \sum_{k=0}^{N-1} [\lambda_{k+1}^\top \delta x_{k+1} - \lambda_k^\top \delta x_k].$$

Substituting the linearized dynamics from (12.14) and the adjoint recursion (ii),

$$\begin{aligned}\lambda_{k+1}^\top \delta x_{k+1} - \lambda_k^\top \delta x_k &= \lambda_{k+1}^\top \nabla_x f_k^* \delta x_k + \lambda_{k+1}^\top \nabla_u f_k^* \delta u_k - [\nabla_x \ell_k^* + \nabla_x f_k^{*\top} \lambda_{k+1}]^\top \delta x_k \\ &= -\nabla_x \ell_k^* \delta x_k + \lambda_{k+1}^\top \nabla_u f_k^* \delta u_k.\end{aligned}$$

Summing over k and inserting into δJ , the terms in δx_k cancel, leaving

$$\delta J = \sum_{k=0}^{N-1} [\nabla_u \ell_k^* + \nabla_u f_k^{*\top} \lambda_{k+1}]^\top \delta u_k = \sum_{k=0}^{N-1} \nabla_u H_k(x_k^*, u_k^*, \lambda_{k+1})^\top \delta u_k.$$

Finally, since $\mathcal{U}$ is convex and the induced variation points into $\mathcal{U}$, the perturbed policy is admissible and feasible for all small $\varepsilon > 0$, and local optimality of μ^* gives $J(\mu^\varepsilon) \geq J(\mu^*)$; hence $\delta J \geq 0$, which is (iii). $\square$

Remark 12.7 (Information-measurability of the induced variation). Every quantity on the right-hand side of (12.14) is a composition of causal maps of the nominal data: δx_j is generated forward from I_j^* , and the policy Jacobians are evaluated at y^{*k} . Hence the induced variation δu_k is itself a function of the information history I_k^* , the perturbed policy never uses information that the original policy could not access. The cone of admissible variations in (iii) is thus strictly smaller than in the perfect-information minimum principle, where δu_k may be an arbitrary function of x_k^* ; this is the precise sense in which the information constraint restricts the first-order optimality condition to information-measurable directions.

The state and adjoint equations, and the Hamiltonian itself, are exactly as in the perfect-information problem: partial observation changes neither the dynamics nor the sensitivity of the cost to the state. What changes is condition (iii). The admissible directions $\delta u_k = \delta \mu_k(I_k^*)$ are constrained to be functions of the information history. When the compatible set $\mathcal{X}_k(I_k^*)$ is a singleton, i.e. when the past measurements have already pinned down the state, the constraint is vacuous and (iii) reduces to the classical pointwise condition $u_k^* \in \arg \min_{u \in \mathcal{U}} H_k(x_k^*, u, \lambda_{k+1})$. When several states remain compatible, a single variation δu_k perturbs the Hamiltonian along every consistent trajectory at once, and the shape of the optimality condition depends on how the objective scores those trajectories. Under the best-case convention adopted in Section 12.2, only the selected (cheapest) compatible trajectory governs optimality, and the pointwise condition holds along it, the remaining trajectories contributing residual Hamiltonian gradients that the

controller optimistically discounts. Under an objective that weights the whole compatible set—an average over it, or its worst case—no trajectory is privileged, and the sharp condition is that the Hamiltonian gradient *aggregated over the compatible set* be stationary in every admissible direction. Either way, the operative restriction is that δu_k be a function of I_k : this is the first-order shadow of the information constraint.

Example 12.8 The minimum principle with a blind sensor.

Take the single-stage problem ($N = 1$) with dynamics $x_1 = x_0 + u_0$, cost $J = \frac{1}{2}u_0^2 + x_1^2$, initial uncertainty $X_0 = \{0, 1\}$, and a blind sensor, $h_0 \equiv 0$, so that $\mathcal{X}_0(I_0) = X_0$ and an admissible policy is a single number $c = \mu_0(y_0)$.

Optimal policy. Under the best-case convention of Section 12.2, the cost of the policy c is

$$J(c) = \frac{1}{2}c^2 + \min\{c^2, (1+c)^2\} = \frac{3}{2}c^2 \quad \text{for } |c| < \frac{1}{2},$$

the inner minimum resolving the unknown initial state optimistically. Hence $c^* = 0$, with selected initial state $\hat{x}_0 = 0$: the optimistic controller does nothing, betting that the state already sits at the cost-free target.

Verification of Theorem 12.6. Along the realized trajectory from $x_0^* = \hat{x}_0 = 0$: the state equation gives $x_1^* = 0$; the adjoint is $\lambda_1 = \nabla\Phi(x_1^*) = 2x_1^* = 0$; and the Hamiltonian gradient is $\nabla_u H_0 = u_0^* + \lambda_1 = 0$, so the variational inequality (iii) holds with equality, as it must at an interior optimum, consistent with $J'(0) = 0$ above.

Where the information constraint bites. Along the *companion* compatible trajectory from $x_0 = 1$, the same control produces $x_1 = 1$, the companion adjoint is $\lambda'_1 = 2$, and $\nabla_u H_0 = u_0^* + \lambda'_1 = 2 \neq 0$: the pointwise Hamiltonian condition fails there. A perfect-information controller would apply $u = 0$ at $x_0 = 0$ but $u = -\frac{2}{3}$ at $x_0 = 1$; under the blind sensor these two states are informationally indistinguishable, so the single admissible control c cannot be right for both. The best-case objective resolves the conflict in favor of the cheaper state $\hat{x}_0 = 0$, along which the pointwise condition $\nabla_u H_0 = 0$ holds; the companion's residual $\nabla_u H_0 = 2$ is simply the price of a scenario the controller optimistically discounts. This is the elementary form of the general pattern developed in Example 12.9. $\square$

The single-stage example isolates the mechanism but hides the adjoint dynamics. The next example is fully worked over several stages: it exhibits the backward adjoint recursion, its agreement with the cost-to-go gradient, and the precise sense in which the shared control is optimal for the selected trajectory while failing the pointwise Hamiltonian test on every other compatible one.

Example 12.9 A multi-stage regulator under a coarse sensor.

Consider the scalar integrator $x_{k+1} = x_k + u_k$ over $N = 2$ stages, with running cost $\ell_k(x, u) = \frac{1}{2}(x^2 + u^2)$ for $k = 0, 1$ and terminal cost $\Phi(x) = \frac{1}{2}x^2$. The initial state is uncertain, $X_0 = [1, 2]$, and a coarse sensor reports only whether $x_0 \geq 1$; observing

y_0 that the threshold is met leaves the compatible set $\mathcal{X}_0(I_0) = [1, 2]$ but nothing finer. No cost precedes the first control, so $W_0 \equiv 0$ on $[1, 2]$.

Cost-to-go (offline). Writing $V_k(x) = \frac{1}{2}M_k x^2$, the backward recursion (12.17) gives

$$M_2 = 1, \quad \mu_1^{\text{fs}}(x) = -\frac{1}{2}x; \quad M_1 = \frac{3}{2}, \quad \mu_0^{\text{fs}}(x) = -\frac{3}{5}x; \quad M_0 = \frac{8}{5}.$$

Selection and closed loop. Since $W_0 \equiv 0$, the selection (12.23) minimizes $V_0(x) = \frac{4}{5}x^2$ over $[1, 2]$, giving the optimistic $\hat{x}_0 = 1$. The certainty-equivalence controller then generates

$$u_0 = \mu_0^{\text{fs}}(1) = -\frac{3}{5}, \quad \hat{x}_1 = \frac{2}{5}, \quad u_1 = \mu_1^{\text{fs}}\left(\frac{2}{5}\right) = -\frac{1}{5}, \quad \hat{x}_2 = \frac{1}{5}.$$

Adjoint (Theorem 12.6). Here $\nabla_x L_k = x$ and $\nabla_x f_k = 1$, so the adjoint recursion reads $\lambda_k = \hat{x}_k + \lambda_{k+1}$ with $\lambda_2 = \nabla \Phi(\hat{x}_2) = \hat{x}_2$. Propagating backward,

$$\lambda_2 = \frac{1}{5}, \quad \lambda_1 = \hat{x}_1 + \lambda_2 = \frac{3}{5}, \quad \lambda_0 = \hat{x}_0 + \lambda_1 = \frac{8}{5}.$$

These match the cost-to-go gradient exactly, $\lambda_k = \nabla_x V_k(\hat{x}_k) = M_k \hat{x}_k (1 \cdot \frac{1}{5}, \frac{3}{2} \cdot \frac{2}{5}, \frac{8}{5} \cdot 1)$, confirming the link between the minimum principle and the dynamic-programming synthesis anticipated in Section 12.2. The Hamiltonian gradient $\nabla_u H_k = u_k + \lambda_{k+1}$ vanishes at each stage, $-\frac{3}{5} + \frac{3}{5} = 0$ and $-\frac{1}{5} + \frac{1}{5} = 0$: along the selected trajectory the control is a genuine (pointwise) Hamiltonian minimizer.

Where the information constraint bites. The state $x_0 = 2$ is equally compatible with the data. The controller cannot tell it from $\hat{x}_0 = 1$, so it applies the *same* controls $(-\frac{3}{5}, -\frac{1}{5})$, producing the companion trajectory $x'_0 = 2, x'_1 = \frac{7}{5}, x'_2 = \frac{6}{5}$. Its adjoint $\lambda'_k = x'_k + \lambda'_{k+1}$ (with $\lambda'_2 = x'_2 = \frac{6}{5}$) gives $\lambda'_2 = \frac{6}{5}, \lambda'_1 = \frac{13}{5}$, and the Hamiltonian gradients

$$\nabla_u H'_1 = u_1 + \lambda'_2 = -\frac{1}{5} + \frac{6}{5} = 1, \quad \nabla_u H'_0 = u_0 + \lambda'_1 = -\frac{3}{5} + \frac{13}{5} = 2,$$

both far from zero. A full-information controller would apply $\mu_0^{\text{fs}}(2) = -\frac{6}{5}$ at $x_0 = 2$, not $-\frac{3}{5}$; the shared control is thus badly suboptimal for the companion. Under the best-case objective this costs nothing, because optimality is measured against the selected (cheapest) compatible trajectory alone, and along that one the principle holds pointwise. Were the objective instead an average over the compatible set, no single trajectory would be privileged, and the optimal shared control would have to *balance* the two gradients, setting $\nabla_u H_k + \nabla_u H'_k$ (suitably weighted) to zero rather than $\nabla_u H_k$ alone. This is the genuine aggregation over the compatible set, and it is the exact first-order counterpart of the information constraint of Section 12.1.3. $\square$

Theorem 12.6 is a genuine necessary condition, but it is not directly constructive: it is stated along the realized trajectory $\{x_k^*\}$, which the controller cannot observe. To synthesize an implementable controller we must work on a state the controller *does* possess, the information history, and this is precisely what dynamic programming accomplishes.

12.2 ■ Dynamic Programming and Certainty Equivalence

We now develop a constructive synthesis by dynamic programming on the information history. The plan is to introduce a *cost-to-come* function on the trajectories consistent with the observations, pair it with the ordinary *cost-to-go*, and recover the controller from an auxiliary optimization over the compatible set. Because the problem is non-adversarial, the cost-to-come is a *best-case* (least-cost) quantity and the auxiliary problem is a minimization.

12.2.1 ■ The Information Value Function

The perfect-information problem is solved by dynamic programming because the state x_k is a *sufficient statistic*: the optimal cost from stage k onward depends on the past only through x_k , and x_k evolves by a Markov recursion. Under imperfect measurements this is exactly what fails, the controller does not know x_k , so it cannot condition on it. What the controller *does* know, and what *is* a sufficient statistic, is the compatible state set $\mathcal{X}_k(I_k)$: by Proposition 12.3 it is computable from the data, it summarizes everything the observations reveal about the state, and it evolves by the Markov recursion (12.11). The remedy is therefore to run dynamic programming not on the unobservable physical state but on the information state, the deterministic, set-membership analogue of the belief-state reduction used for partially observed Markov decision processes.

One obstacle remains before an objective can be assigned to an information state. The cost (12.4) is evaluated along the true trajectory, which depends on the initial state the controller cannot identify; knowing only $\mathcal{X}_k(I_k)$, the controller faces a whole family of possible costs, one per compatible state. To obtain a scalar we resolve this ambiguity *optimistically*: we score an information state by the *least* cost attainable over the trajectories it admits. This optimistic convention is what makes the problem non-adversarial, and, being an infimum throughout, it is what will let the value split additively into a past and a future part. (Replacing this infimum by a supremum yields the robust, worst-case formulation; see the notes, Section 12.4.)

Definition 12.10 (Information value function). *The information value function is the least future cost achievable from stage k onward, given the information history I_k :*

$$\mathcal{V}_k(I_k) := \inf_{\mu_k, \dots, \mu_{N-1}} \inf_{\xi \in \mathcal{X}_k(I_k)} \left[\Phi(x_N) + \sum_{i=k}^{N-1} \ell_i(x_i, u_i) \right], \quad (12.15)$$

where the trajectory starts from a compatible state $x_k = \xi$, evolves by (12.1), and the controls obey $u_i = \mu_i(I_i)$. The terminal value is $\mathcal{V}_N(I_N) = \inf_{\xi \in \mathcal{X}_N(I_N)} \Phi(\xi)$.

The double infimum in (12.15) looks formidable, but it collapses to a single minimization against the ordinary optimal-control value function. Let V_k denote the *perfect-information cost-to-go*, the optimal cost from a *known* state x at stage k , i.e. the solution of the standard Bellman recursion (12.17) below.

Proposition 12.11 (Information value as a selection of the perfect-information value). *For every information history I_k ,*

$$\mathcal{V}_k(I_k) = \min_{\xi \in \mathcal{X}_k(I_k)} V_k(\xi). \quad (12.16)$$

In particular, if the measurements determine the state, $\mathcal{X}_k(I_k) = \{x_k\}$, then $\mathcal{V}_k(I_k) = V_k(x_k)$: the information value function reduces to the perfect-information value, and the construction of this section collapses to ordinary dynamic programming.

Proof. Interchange the two infima in (12.15): $\mathcal{V}_k(I_k) = \inf_{\xi \in \mathcal{X}_k(I_k)} [\inf_{\mu} (\text{cost of the trajectory from } \xi)]$. For a *fixed* starting state ξ the system is deterministic, so the cost is determined by the control sequence alone; minimizing it over admissible policies gives exactly the perfect-information optimal cost $V_k(\xi)$, the lower bound $V_k(\xi)$ is attained by the open-loop optimal sequence for ξ , which is an admissible (constant, hence causal) policy. Thus the inner infimum equals $V_k(\xi)$, and the outer infimum over the compact set $\mathcal{X}_k(I_k)$ is attained, giving (12.16). The singleton case is immediate. $\square$

Proposition 12.11 is the conceptual heart of the reduction: the hard-looking output-feedback value is nothing but the perfect-information value function V_k , minimized over the states the data leave open. It computes V_k once, offline, and evaluates it on the compatible set. Two things are still missing for a controller, however. First, the identity gives the *future* value but not *which* compatible state to steer from, and, as we will see, past cost must break the tie. Second, it presumes V_k in hand; we record its defining recursion next.

12.2.2 ■ The Cost-to-Go

The bridge (12.16) rests on the perfect-information cost-to-go, which we now record. It is the ordinary optimal-control value function, the cost of the fully observed problem, solved backward and offline, with no reference to any measurement:

$$V_N(x) = \Phi(x), \quad V_k(x) = \min_{u \in \mathcal{U}} [\ell_k(x, u) + V_{k+1}(f_k(x, u))]. \quad (12.17)$$

It answers the question: if the state were *known* to equal x at stage k , what is the least cost achievable from then on? With V_k in hand, Proposition 12.11 evaluates the output-feedback value in one line, $\mathcal{V}_k(I_k) = \min_{\xi \in \mathcal{X}_k(I_k)} V_k(\xi)$: solve the perfect-information problem once, then read the answer off the compatible set.

Remark 12.12 (An intrinsic recursion on histories). One may bypass V_k and let the information value satisfy its own backward recursion directly on histories. Appending a control u and the ensuing measurement y_{k+1} gives $I_{k+1} = (I_k, u, y_{k+1})$, with the compatible set propagating by (12.11); since both operations in (12.15) are infima they combine into

$$\mathcal{V}_k(I_k) = \min_{u \in \mathcal{U}} \inf_{\xi \in \mathcal{X}_k(I_k)} [\ell_k(\xi, u) + \mathcal{V}_{k+1}(I_k, u, h_{k+1}(f_k(\xi, u)))]. \quad (12.18)$$

This is exact but abstract—its argument is the (generally infinite-dimensional) information history—and it is the deterministic output-feedback Bellman equation. Equation (12.16) is the practical alternative: it confines all the infinite-dimensional bookkeeping to the compatible set $\mathcal{X}_k(I_k)$ and delegates the optimization to the finite-dimensional recursion (12.17).

12.2.3 ■ The Cost-to-Come and the Total Value

Proposition 12.11 values the *future* but leaves a gap for *action*. To apply a control the certainty-equivalent way, the controller must commit to one compatible state and steer from it, and the future value V_k alone is the wrong tie-breaker. Minimizing V_k over $\mathcal{X}_k(I_k)$ would pick the state with the cheapest continuation regardless of whether the observed data could have reached it cheaply; but the best-case cost the controller is entitled to claim is that of the cheapest *complete* compatible trajectory, past included. What is missing is a record of how expensively each compatible state was reached. That record is the cost-to-come, and adding it to the cost-to-go yields the object the controller should actually minimize; Figure 12.1 depicts the resulting past–future split.

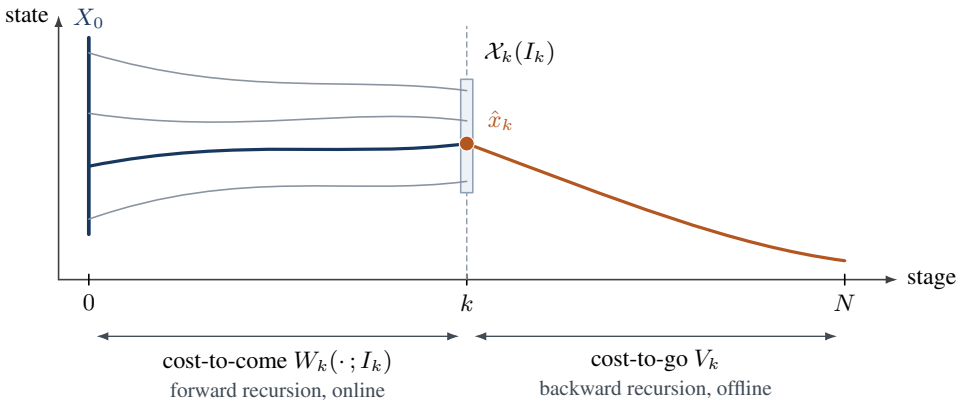


Figure 12.1: The past–future decomposition at the present stage k . Thin curves are state trajectories compatible with the observed data; they emanate from the initial set X_0 and terminate in the compatible set $\mathcal{X}_k(I_k)$. The cost-to-come $W_k(x; I_k)$ is the least past cost among compatible trajectories ending at x ; the cost-to-go $V_k(x)$ is the least future cost from x . The certainty-equivalent state $\hat{x}_k$ minimizes their sum over the compatible set, splicing the cheapest explanation of the past (thick, left) onto the cheapest continuation into the future (thick, right).

The *cost-to-come* $W_k(x; I_k)$ is the least cost accumulated up to stage k among all trajectories that are consistent with the information history I_k and end at the state x :

$$W_k(x; I_k) = \inf \left\{ \sum_{i=0}^{k-1} \ell_i(x_i, u_i) : \begin{array}{l} x_0 \in X_0, \ x_k = x, \ x_{i+1} = f_i(x_i, u_i), \\ y_i = h_i(x_i) \ (0 \leq i \leq k) \end{array} \right\}, \quad (12.19)$$

with the convention $W_k(x; I_k) = +\infty$ when no compatible trajectory ends at x , i.e. when $x \notin \mathcal{X}_k(I_k)$. It is initialized by

$$W_0(x; I_0) = \begin{cases} 0, & x \in X_0 \cap h_0^{-1}(y_0), \\ +\infty, & \text{otherwise,} \end{cases} \quad (12.20)$$

and propagated forward. In (12.19) the controls are the recorded ones, u_i being the i th entry of u^{k-1} ; only the trajectory explaining them is unknown.

The forward propagation is a dynamic-programming principle run in the direction of increasing time, and it mirrors the two-step evolution (12.11) of the compatible set. Given the cost-to-come at stage k , one step consists of a *prediction* and a *correction*, illustrated in Figure 12.2. In the prediction, each compatible state ξ is carried one step forward under the recorded control to $x^+ = f_k(\xi, u_k)$, and its accumulated cost is increased by the stage cost $\ell_k(\xi, u_k)$; when the dynamics are not injective, several predecessors reach the same x^+ , and the cost-to-come retains the cheapest of them. In the correction, the new measurement discards every predicted state inconsistent with y_{k+1} , keeping only those on $h_{k+1}^{-1}(y_{k+1})$. The next lemma states this precisely; its two inequalities are the algebraic form of the two readings of the figure, *appending* a step to a compatible past (left-to-right) and *truncating* a compatible future (right-to-left).

Lemma 12.13 (Forward dynamic programming for the cost-to-come). For $k = 0, \dots, N-1$, with u_k the recorded control and y_{k+1} the recorded measurement in $I_{k+1} = (I_k, u_k, y_{k+1})$,

$$W_{k+1}(x^+; I_{k+1}) = \inf_{\substack{\xi \in \mathcal{X}_k(I_k) \\ x^+ = f_k(\xi, u_k)}} [W_k(\xi; I_k) + \ell_k(\xi, u_k)] \quad \text{if } h_{k+1}(x^+) = y_{k+1}, \quad (12.21)$$

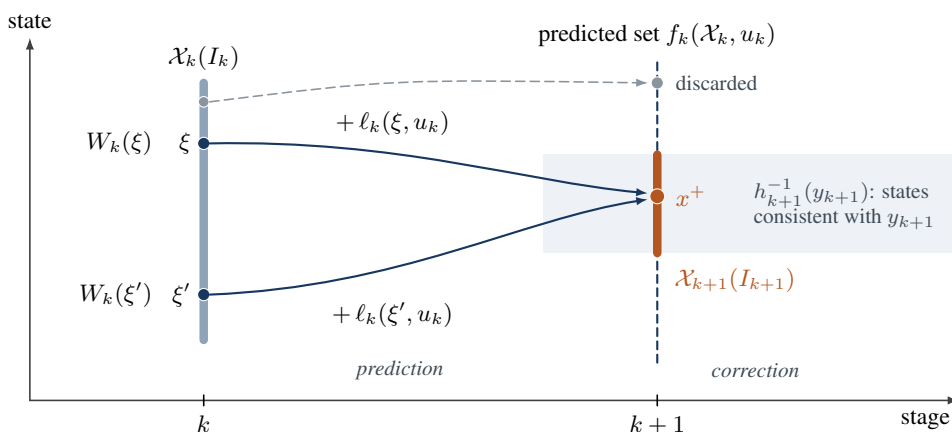


Figure 12.2: One step of the forward recursion for the cost-to-come (Lemma 12.13). Each state ξ in the compatible set $\mathcal{X}_k(I_k)$ carries a cost-to-come value $W_k(\xi)$. In the *prediction* step it is pushed one step forward under the recorded control to $x^+ = f_k(\xi, u_k)$, its cost increased by $\ell_k(\xi, u_k)$; when the dynamics are not injective, several predecessors ξ, ξ' land on the same successor x^+ , and the cost-to-come keeps the cheapest: $W_{k+1}(x^+)$ is the minimum of $W_k(\xi) + \ell_k(\xi, u_k)$ over those predecessors. In the *correction* step the new measurement admits only the states in the shaded band $h_{k+1}^{-1}(y_{k+1})$; its intersection with the predicted set (dashed) is the new compatible set $\mathcal{X}_{k+1}(I_{k+1})$ (solid), and predicted states outside the band are discarded. Reading the diagram right-to-left recovers the truncation argument of the proof.

and $W_{k+1}(x^+; I_{k+1}) = +\infty$ otherwise (with $\inf \emptyset = +\infty$).

Proof. If $h_{k+1}(x^+) \neq y_{k+1}$, no trajectory compatible with I_{k+1} ends at x^+ , and both sides are $+\infty$ by convention. Suppose then $h_{k+1}(x^+) = y_{k+1}$.

($\leq$) Let $\xi \in \mathcal{X}_k(I_k)$ with $f_k(\xi, u_k) = x^+$, and let $x_0, \dots, x_k = \xi$ be any trajectory feasible for the problem defining $W_k(\xi; I_k)$, with past cost c . Appending the transition $\xi \mapsto x^+$ under u_k produces a trajectory feasible for $W_{k+1}(x^+; I_{k+1})$ —the dynamics constraint holds by construction and the measurement constraint holds since $h_{k+1}(x^+) = y_{k+1}$ —with past cost $c + \ell_k(\xi, u_k)$. Taking infima over trajectories and then over ξ gives $W_{k+1}(x^+; I_{k+1}) \leq \text{RHS}$.

($\geq$) Conversely, let $x_0, \dots, x_k, x_{k+1} = x^+$ be any trajectory feasible for $W_{k+1}(x^+; I_{k+1})$, with cost $\sum_{i=0}^k \ell_i(x_i, u_i)$. Its truncation $x_0, \dots, x_k$ is feasible for $W_k(x_k; I_k)$, all constraints of (12.19) up to stage k are among those up to stage $k+1$, and $x_k \in \mathcal{X}_k(I_k)$ with $f_k(x_k, u_k) = x^+$. Hence $\sum_{i=0}^k \ell_i \geq W_k(x_k; I_k) + \ell_k(x_k, u_k) \geq \text{RHS}$, and taking the infimum over feasible trajectories gives $W_{k+1}(x^+; I_{k+1}) \geq \text{RHS}$. $\square$

Whereas V_k is computed backward and offline, W_k is computed forward and online, driven by the incoming measurements. It is the deterministic set-membership analogue of a minimum-energy estimator: the least accumulated cost among all trajectories that could have produced the data and now occupy state x .

The two value functions now assemble the object of interest. Summing the cheapest past ending at x and the cheapest future from x gives the cost of the cheapest complete compatible trajectory through x , and minimizing over the compatible set gives the *total best-case value*

$$m_k(I_k) := \min_{x \in \mathcal{X}_k(I_k)} [W_k(x; I_k) + V_k(x)]. \quad (12.22)$$

This is the object the controller minimizes, and it stands in a clean relation to the future value of Section 12.2.1: whereas $\mathcal{V}_k(I_k) = \min_x V_k(x)$ weights every compatible state equally, $m_k(I_k)$ weights it by the past cost $W_k(x)$ needed to reach it, so the two coincide precisely when W_k is constant on $\mathcal{X}_k(I_k)$, for instance when no cost is charged before the present stage, as at $k = 0$, where $m_0(I_0) = \mathcal{V}_0(I_0)$. The role of the cost-to-come is exactly to tilt the selection away from states that only *look* cheap in the future toward those that are cheap *overall*. When the measurements determine the state, $\mathcal{X}_k(I_k) = \{x_k\}$, both value functions are evaluated at a single point, $m_k = W_k(x_k) + V_k(x_k)$, the selection is forced, and, as we verify next, the controller reduces to the perfect-information law $\mu_k^{\text{fs}}(x_k)$.

12.2.4 ■ The Certainty-Equivalence Controller

The two value functions combine to select a single representative state. Because $W_k(x; I_k)$ is the best compatible explanation of the past ending at x and $V_k(x)$ is the best continuation from x , their sum is the best total cost of a compatible trajectory passing through x at stage k . Minimizing over the compatible set identifies the most favorable such state,

$$\hat{x}_k \in \arg \min_{x \in \mathcal{X}_k(I_k)} [W_k(x; I_k) + V_k(x)], \quad (12.23)$$

and the controller then acts *as if* the state were $\hat{x}_k$, applying the perfect-information feedback law evaluated there,

$$u_k = \mu_k^{\text{fs}}(\hat{x}_k), \quad \mu_k^{\text{fs}}(x) \in \arg \min_{u \in \mathcal{U}} [\ell_k(x, u) + V_{k+1}(f_k(x, u))]. \quad (12.24)$$

This is the *certainty-equivalence* principle: the output-feedback problem is solved by combining an information-processing step, which produces $\hat{x}_k$, with the certainty-equivalent state-feedback law μ_k^{fs} . The synthesis has the transparent pipeline

$$I_k \longmapsto \mathcal{X}_k(I_k) \longmapsto W_k(\cdot; I_k) \longmapsto \hat{x}_k \longmapsto u_k. \quad (12.25)$$

The information-processing block, compatible set and cost-to-come, is computed online from the data; the control block, cost-to-go and feedback law, is computed offline. Their interface is the single vector $\hat{x}_k$.

We now make the optimality of this construction precise. Recall the total best-case value $m_k(I_k) = \min_{x \in \mathcal{X}_k(I_k)} [W_k(x; I_k) + V_k(x)]$ of (12.22), the quantity minimized by the selection (12.23); we also call it the *selected value*. The first result identifies m_k with the least total cost compatible with the data.

Lemma 12.14 (Past–future decomposition). *Fix an information history I_k with recorded controls u^{k-1} , and consider the class $\mathcal{T}_k(I_k)$ of complete state trajectories $x_0, \dots, x_N$ such that (a) $x_0 \in X_0$, (b) $x_{i+1} = f_i(x_i, u_i)$ and $y_i = h_i(x_i)$ for $0 \leq i \leq k$ with the recorded u_i, y_i , and (c) $x_{i+1} = f_i(x_i, u_i)$ for $k \leq i \leq N-1$ for some controls $u_k, \dots, u_{N-1} \in \mathcal{U}$. Then*

$$\inf_{\mathcal{T}_k(I_k)} \left[\Phi(x_N) + \sum_{i=0}^{N-1} \ell_i(x_i, u_i) \right] = m_k(I_k).$$

Proof. Split the total cost of any trajectory in $\mathcal{T}_k(I_k)$ at stage k :

$$\Phi(x_N) + \sum_{i=0}^{N-1} \ell_i = \underbrace{\sum_{i=0}^{k-1} \ell_i(x_i, u_i)}_{\text{past}} + \underbrace{\Phi(x_N) + \sum_{i=k}^{N-1} \ell_i(x_i, u_i)}_{\text{future}}.$$

The past and future portions share only the state x_k : for fixed $x_k = x$, the past constraints (a)–(b) and the future constraint (c) involve disjoint sets of decision variables. Hence the joint infimum may be iterated,

$$\inf_{\mathcal{T}_k(I_k)} [\cdot] = \inf_{x \in \mathcal{X}_k(I_k)} \left(\inf_{\text{past ending at } x} \sum_{i < k} \ell_i + \inf_{\text{future from } x} [\Phi + \sum_{i \geq k} \ell_i] \right).$$

The inner past infimum is $W_k(x; I_k)$ by definition (12.19), and the inner future infimum is $V_k(x)$: by the classical dynamic-programming verification theorem for the perfect-information problem, the value of the open-loop minimization from a known state x at stage k is the solution V_k of the backward recursion (12.17). The claim follows. $\square$

Theorem 12.15 (Deterministic certainty-equivalence principle). *Assume that for every information history I_k the compatible set $\mathcal{X}_k(I_k)$ is nonempty, that $W_k(\cdot; I_k)$ and V_k are finite on $\mathcal{X}_k(I_k)$, and that the infima in (12.17), (12.19), and (12.22) are attained. Let $\hat{x}_k$ be a selection (12.23) and let $\hat{\mu} = (\hat{\mu}_0, \dots, \hat{\mu}_{N-1})$, $\hat{\mu}_k(I_k) := \mu_k^{\text{fs}}(\hat{x}_k)$, be the induced certainty-equivalence policy. Then:*

- (i) **(Value identity)** *The optimal value of the best-case output-feedback problem equals the least total cost compatible with the data, $\mathcal{V}_0(I_0) = m_0(I_0)$; in particular the information constraint entails no loss of best-case value relative to jointly choosing the trajectory and the controls with full knowledge.*
- (ii) **(Monotonicity)** *Along any realized play, any controls and any measurements, the selected value is nondecreasing: $m_{k+1}(I_{k+1}) \geq m_k(I_k)$.*
- (iii) **(Invariance and attainment)** *Under the policy $\hat{\mu}$, if the measurement at stage $k+1$ is the one generated by the selected trajectory, $y_{k+1} = h_{k+1}(f_k(\hat{x}_k, \hat{\mu}_k(I_k)))$, then $m_{k+1}(I_{k+1}) = m_k(I_k)$, and the stage- $(k+1)$ selection may be taken as $\hat{x}_{k+1} = f_k(\hat{x}_k, \hat{\mu}_k(I_k))$. Consequently, when the true initial state is the one identified by the stage-0 selection, the policy $\hat{\mu}$ realizes the total cost $m_0(I_0)$, which by (i) is optimal.*
- (iv) **(Uniqueness)** *If for each k the map $x \mapsto W_k(x; I_k) + V_k(x)$ is strictly convex on the convex set $\mathcal{X}_k(I_k)$, the selection $\hat{x}_k$ is unique and $\hat{\mu}$ is well defined.*

Proof. (i) We first show that policies cannot beat the joint minimization and can emulate it. For any policy $\mu \in \mathcal{M}$ and any compatible initial state, the pair (initial state, realized control sequence) generates a trajectory in the class $\mathcal{T}_0(I_0)$ of Lemma 12.14 with the same cost; hence $\mathcal{V}_0(I_0) \geq \inf_{\mathcal{T}_0(I_0)} [\cdot] = m_0(I_0)$, the last equality by Lemma 12.14 with $k = 0$ (where the past portion is empty and W_0 is the indicator (12.20)). Conversely, let $(x_0^\circ, u_0^\circ, \dots, u_{N-1}^\circ)$ be ϵ -optimal for the joint problem. The open-loop policy $\mu_k(\cdot) \equiv u_k^\circ$ is admissible (it is a constant, hence causal, function of y^k), and its best-case cost (12.15) is at most the cost of the trajectory from x_0° , i.e. at most $m_0(I_0) + \epsilon$. Letting $\epsilon \downarrow 0$ gives $\mathcal{V}_0(I_0) \leq m_0(I_0)$.

(ii) Let u_k be the control applied at stage k and y_{k+1} the measurement received. Let $x \in \mathcal{X}_{k+1}(I_{k+1})$ attain $m_{k+1}(I_{k+1}) = W_{k+1}(x; I_{k+1}) + V_{k+1}(x)$. By Lemma 12.13 there exists $\xi \in \mathcal{X}_k(I_k)$ with $f_k(\xi, u_k) = x$ and $W_{k+1}(x; I_{k+1}) = W_k(\xi; I_k) + \ell_k(\xi, u_k)$ (the infimum in (12.21) being attained by assumption). By the backward recursion (12.17), $V_k(\xi) \leq \ell_k(\xi, u_k) + V_{k+1}(f_k(\xi, u_k)) = \ell_k(\xi, u_k) + V_{k+1}(x)$. Combining,

$$m_{k+1}(I_{k+1}) = W_k(\xi; I_k) + \ell_k(\xi, u_k) + V_{k+1}(x) \geq W_k(\xi; I_k) + V_k(\xi) \geq m_k(I_k).$$

(iii) Under $\hat{\mu}$ the applied control is $u_k = \mu_k^{\text{fs}}(\hat{x}_k)$, which attains the minimum in (12.17) at $x = \hat{x}_k$: $V_k(\hat{x}_k) = \ell_k(\hat{x}_k, u_k) + V_{k+1}(x^+)$ with $x^+ := f_k(\hat{x}_k, u_k)$. If the received measurement is $y_{k+1} = h_{k+1}(x^+)$, then $x^+ \in \mathcal{X}_{k+1}(I_{k+1})$, and Lemma 12.13 (choosing $\xi = \hat{x}_k$) gives $W_{k+1}(x^+; I_{k+1}) \leq W_k(\hat{x}_k; I_k) + \ell_k(\hat{x}_k, u_k)$. Hence

$$\begin{aligned} m_{k+1}(I_{k+1}) &\leq W_{k+1}(x^+; I_{k+1}) + V_{k+1}(x^+) \\ &\leq W_k(\hat{x}_k; I_k) + \ell_k(\hat{x}_k, u_k) + V_{k+1}(x^+) = m_k(I_k), \end{aligned}$$

which together with (ii) forces equality throughout; in particular x^+ attains m_{k+1} , so $\hat{x}_{k+1} = x^+$ is a valid selection. Iterating from $k = 0$: when the true initial state is the one explaining $\hat{x}_0$ (so that the realized measurements coincide with the selected ones at every stage), the realized trajectory is the selected one, the realized controls are optimal for it stage by stage, and the realized total cost equals $m_N(I_N) = m_0(I_0)$, using $V_N = \Phi$ at the final stage. By (i) this is the optimal value.

(iv) A strictly convex function on a convex set has at most one minimizer; combined with attainment, exactly one. $\square$

Remark 12.16 (Reading the monotonicity). Parts (ii)–(iii) say that the optimistic estimate m_k of the total cost can only *rise* as data arrive: each measurement either confirms the selected trajectory (equality, by (iii)) or refutes it, forcing the controller to settle for a costlier explanation. Optimism is thus self-consistent when right and self-correcting when wrong, but it never has to be revised downward. The mechanism is transparent: m_k is a *minimum* over the compatible set, and that set only shrinks as observations accumulate (perfect recall, Proposition 12.3(ii)), so its minimum can only increase.

Example 12.17 Tracking under a coarse measurement.

Consider the scalar system $x_{k+1} = x_k + u_k$ over the horizon $N = 2$, with the tracking cost

$$J = \frac{1}{2}(x_2 - r)^2 + \frac{1}{2} \sum_{k=0}^1 u_k^2, \quad r = 2.$$

The initial state is known only to lie in $X_0 = [-1, 2]$, and the first measurement is coarse: h_0 is constant on $[0, 1]$, so after observing y_0 the controller learns $\mathcal{X}_0(I_0) = [0, 1]$ but nothing finer.

Cost-to-go (offline). Solving (12.17) backward gives $V_2(x) = \frac{1}{2}(x - r)^2$, then

$$V_1(x) = \frac{1}{4}(x - r)^2, \quad \mu_1^{\text{fs}}(x) = \frac{r-x}{2}, \quad V_0(x) = \frac{1}{6}(x - r)^2, \quad \mu_0^{\text{fs}}(x) = \frac{r-x}{3}.$$

Cost-to-come (online). No cost is charged before the first control, so $W_0(x; I_0) = 0$ on the compatible set $[0, 1]$ and $+\infty$ elsewhere.

Selection and control. The selection (12.23) minimizes $W_0(x) + V_0(x) = \frac{1}{6}(x - 2)^2$ over $x \in [0, 1]$. Since $(x - 2)^2$ decreases on $[0, 1]$, the optimistic best-case state is $\hat{x}_0 = 1$, and the certainty-equivalent control is $u_0 = \mu_0^{\text{fs}}(\hat{x}_0) = \frac{2-1}{3} = \frac{1}{3}$. The controller assumes the most favorable compatible state, the one nearest the target, and drives toward the goal accordingly.

The best-case convention is exactly what makes this an optimal-control, rather than a robust, design. Had we instead *maximized* $W_0 + V_0$ over the compatible set, committing to the least favorable explanation of the data, the selected state would be $\hat{x}_0 = 0$ (the farthest from the target, hence the most expensive), giving the more cautious control $u_0 = \mu_0^{\text{fs}}(0) = \frac{2}{3}$. The two designs share the identical machinery—compatible set, cost-to-come, cost-to-go, certainty-equivalent feedback—and differ only in whether the compatible set is resolved by an infimum or a supremum; the latter is the robust variant discussed in the notes (Section 12.4). $\square$

The next example continues the previous one and makes the monotonicity–invariance dichotomy of Theorem 12.15 concrete: a measurement either confirms the selected trajectory, leaving the selected value untouched, or refutes it, forcing the value upward.

Example 12.18 Confirmation versus refutation of the selection.

In the setting of Example 12.17, the stage-0 selection is $\hat{x}_0 = 1$ with selected value $m_0 = \frac{1}{6}(1-2)^2 = \frac{1}{6} = \frac{6}{36}$, and the control $u_0 = \frac{1}{3}$ predicts the next state $\hat{x}_1 = \hat{x}_0 + u_0 = \frac{4}{3}$. The compatible set propagates to $\mathcal{X}_1 = [0, 1] + \frac{1}{3} = [\frac{1}{3}, \frac{4}{3}]$ before any new measurement, and the cost-to-come at stage 1 is *constant* on it, $W_1 \equiv \frac{1}{2}u_0^2 = \frac{1}{18}$, because the stage cost $\frac{1}{2}u^2$ charges only the (known) control, not the state.

Suppose the stage-1 sensor reports one bit: whether $x_1 \leq 1$.

Confirmation. If the reading is “ $x_1 > 1$ ”, consistent with the selected trajectory, the compatible set shrinks to $(1, \frac{4}{3}]$ and

$$m_1 = \frac{1}{18} + \min_{x \in (1, 4/3]} \frac{1}{4}(x-2)^2 = \frac{1}{18} + \frac{1}{4}\left(\frac{4}{3} - 2\right)^2 = \frac{1}{18} + \frac{1}{9} = \frac{6}{36} = m_0 :$$

the selected value is invariant, the selection continues as $\hat{x}_1 = \frac{4}{3}$, and the controller proceeds with $u_1 = \mu_1^{\text{fs}}(\frac{4}{3}) = \frac{1}{3}$, exactly as Theorem 12.15 (iii) asserts.

Refutation. If instead the reading is “ $x_1 \leq 1$ ”, the selected trajectory is refuted; the compatible set becomes $[\frac{1}{3}, 1]$ and

$$m_1 = \frac{1}{18} + \min_{x \in [1/3, 1]} \frac{1}{4}(x-2)^2 = \frac{1}{18} + \frac{1}{4}(1-2)^2 = \frac{2}{36} + \frac{9}{36} = \frac{11}{36} > \frac{6}{36} = m_0 :$$

the selected value jumps upward, in accordance with Theorem 12.15 (ii). The controller re-selects $\hat{x}_1 = 1$, the most favorable state still compatible with the data, and applies $u_1 = \mu_1^{\text{fs}}(1) = \frac{1}{2}$, working harder to make up for the optimism that the measurement disallowed. $\square$

12.2.5 ■ Relation to the Minimum Principle

The minimum principle of Section 12.1.4 and the dynamic-programming synthesis of this section play complementary roles. Theorem 12.6 gives necessary first-order conditions, the Hamiltonian (12.13) and the adjoint recursion, that any optimal trajectory must satisfy, but it presupposes

knowledge of that trajectory and is therefore not implementable as stated. Theorem 12.15 gives a constructive controller that never requires the true state: it manufactures the surrogate state $\hat{x}_k$ from the information history and feeds it to the certainty-equivalent law. We now show that these two descriptions are not merely analogous but coincide, in a precise sense, along the trajectory the controller actually generates.

The link between the two theorems is the classical identification of the Pontryagin costate with the gradient of the Bellman value function, here attached to the *selected* trajectory rather than to the unobserved true one.

Proposition 12.19 (Costate–gradient identity). *Assume the cost-to-go functions V_k of (12.17) are continuously differentiable and that for each (k, x) the minimization defining $\mu_k^{\text{fs}}(x)$ in (12.24) is attained at an interior point. Let $\{\hat{x}_k\}_{k=0}^N$ be the trajectory generated by the certainty-equivalence controller of Theorem 12.15 under self-consistent play, with controls $u_k = \mu_k^{\text{fs}}(\hat{x}_k)$, and define*

$$\lambda_k := \nabla_x V_k(\hat{x}_k), \quad k = 0, \dots, N. \quad (12.26)$$

Then $\{\lambda_k\}$ satisfies the terminal and adjoint conditions of Theorem 12.6,

$$\lambda_N = \nabla \Phi(\hat{x}_N), \quad \lambda_k = \nabla_x L_k(\hat{x}_k, u_k) + [\nabla_x f_k(\hat{x}_k, u_k)]^\top \lambda_{k+1}, \quad (12.27)$$

and the control obeys the Hamiltonian stationarity condition

$$\nabla_u H_k(\hat{x}_k, u_k, \lambda_{k+1}) = 0. \quad (12.28)$$

Thus the certainty-equivalence trajectory satisfies the minimum principle, with the costate given by the cost-to-go gradient at the selected state.

Proof. Write $g_k(x, u) := \ell_k(x, u) + V_{k+1}(f_k(x, u))$, so that $V_k(x) = \min_u g_k(x, u)$ and $\mu_k^{\text{fs}}(x) \in \arg \min_u g_k(x, u)$. The terminal identity is immediate: $V_N = \Phi$ gives $\lambda_N = \nabla \Phi(\hat{x}_N)$.

Fix $k < N$. Self-consistent play means $\hat{x}_{k+1} = f_k(\hat{x}_k, u_k)$ (Theorem 12.15(iii)), hence $\nabla V_{k+1}(f_k(\hat{x}_k, u_k)) = \nabla V_{k+1}(\hat{x}_{k+1}) = \lambda_{k+1}$. Interior optimality of u_k gives the first-order condition $\nabla_u g_k(\hat{x}_k, u_k) = 0$, i.e.

$$\nabla_u L_k(\hat{x}_k, u_k) + [\nabla_u f_k(\hat{x}_k, u_k)]^\top \lambda_{k+1} = \nabla_u H_k(\hat{x}_k, u_k, \lambda_{k+1}) = 0,$$

which is (12.28). For the adjoint recursion, apply the envelope theorem to $V_k(x) = g_k(x, \mu_k^{\text{fs}}(x))$: because the inner variable is at a minimizer, its variation contributes nothing to first order, so

$$\nabla_x V_k(x) = \nabla_x g_k(x, u) \big|_{u=\mu_k^{\text{fs}}(x)} = \nabla_x L_k(x, u) + [\nabla_x f_k(x, u)]^\top \nabla V_{k+1}(f_k(x, u)).$$

Evaluating at $x = \hat{x}_k$ (so $u = u_k$ and $\nabla V_{k+1}(f_k(\hat{x}_k, u_k)) = \lambda_{k+1}$) and using (12.26) yields (12.27). $\square$

Proposition 12.19 closes the loop between the two theorems. The minimum principle is a *certificate*: it lists conditions that an optimal trajectory must meet. Dynamic programming is a *construction*: it produces, from the data alone, the trajectory $\{\hat{x}_k\}$ and proves it optimal (Theorem 12.15). The proposition verifies that the constructed trajectory carries a costate, the cost-to-go gradient, satisfying the certificate. The only trace of incomplete information is *where* the certificate is anchored: not at the true state x_k^* , which the controller cannot see, but at the surrogate $\hat{x}_k = \arg \min_x [W_k(x; I_k) + V_k(x)]$, computed from the information history. Example 12.9 exhibits exactly this: the adjoint sequence $(\frac{8}{5}, \frac{3}{5}, \frac{1}{5})$ obtained from the backward recursion equals $M_k \hat{x}_k = \nabla_x V_k(\hat{x}_k)$ stage by stage, and the certainty-equivalent control makes $\nabla_u H_k$ vanish.

Taken together, the two theorems express a single principle: an output-feedback optimal control problem is solved by lifting it to a fully observed problem on the space of information states, whose “state” is the compatible set and whose value decomposes into a cost-to-come and a cost-to-go. The minimum principle certifies optimality along one trajectory; dynamic programming constructs that trajectory, the selected one, from the data, and Proposition 12.19 shows the two meet, with the costate equal to the cost-to-go gradient at the selected state.

12.2.6 ■ The Infinite-Horizon Case

We now specialize to the time-invariant linear-quadratic setting and let the horizon tend to infinity, seeking a *stationary* output-feedback controller. The cost-to-go and the cost-to-come each become a fixed quadratic, governed by an algebraic Riccati equation—one for control, one for estimation—and the certainty-equivalence controller reduces to a stationary observer feeding a linear gain. As we shall see, the construction is solvable under the standard structural conditions alone, with no auxiliary threshold to verify.

Stationary Problem

Consider the time-invariant linear system, driven by an unknown-but-persistent disturbance and observed through a noisy channel,

$$x_{k+1} = Ax_k + Bu_k + Dw_k, \quad y_k = Cx_k + v_k, \quad (12.29)$$

with quadratic running cost

$$L(x, u) = x^\top Qx + u^\top Ru, \quad Q = H^\top H \succeq 0, \quad R \succ 0. \quad (12.30)$$

The system is imagined to have been evolving from the infinitely remote past and to continue into the infinite future, with $x_k \rightarrow 0$ as $|k| \rightarrow \infty$. Fix the present instant and split the cost at it: the *future* portion $\sum_{k \geq 0} \ell(x_k, u_k)$ is the cost still to be paid, and the *past* portion is the cost of reconciling the model with the data already observed. Exactly as in the finite-horizon theory, the value decomposes into a cost-to-go and a cost-to-come, each of which is now *stationary*, a fixed function of the current state rather than a time-indexed sequence.

Stationary Cost-to-Go: the Control ARE

The cost-to-go is the perfect-information infinite-horizon LQR value. As the horizon grows, the backward recursion (12.17) converges, under stabilizability of (A, B) and detectability of (A, H) , to the stationary quadratic

$$V(x) = x^\top Mx, \quad (12.31)$$

where $M \succeq 0$ is the minimal stabilizing solution of the *control* algebraic Riccati equation

$$\begin{aligned} M &= A^\top MA - A^\top MB(R + B^\top MB)^{-1}B^\top MA + Q \\ &= A^\top (M^{-1} + BR^{-1}B^\top)^{-1}A + Q, \end{aligned} \quad (12.32)$$

the second (information) form assuming $M \succ 0$. The certainty-equivalent feedback law is $\mu^{\text{fs}}(x) = -K_c x$ with

$$K_c = (R + B^\top MB)^{-1}B^\top MA, \quad (12.33)$$

and $A - BK_c$ is Schur (spectral radius below one). Equation (12.32) is the standard discrete-time LQR algebraic Riccati equation; the disturbance Dw plays no role in the cost-to-go, which depends only on the dynamics the control can steer.

Stationary Cost-to-Come: the Estimation ARE

The cost-to-come is the stationary limit of the forward recursion (12.21). In the non-adversarial problem it is a *best-case* (least-energy) reconstruction of the past: among all disturbance and measurement-error sequences consistent with the observed outputs and leaving the state at the value x now, it retains the one of least total energy,

$$W(x) = \inf \left\{ \sum_{k < 0} (w_k^\top w_k + v_k^\top N^{-1} v_k) : \begin{array}{l} x_{k+1} = Ax_k + Bu_k + Dw_k, \quad y_k = Cx_k + v_k, \\ x_0 = x, \quad x_{-\infty} = 0 \end{array} \right\}, \quad (12.34)$$

with $N \succ 0$ weighting the measurement residuals. This is the deterministic minimum-energy estimation problem, the exact dual of the LQR problem (12.31)–(12.33). Throughout this subsection we assume that

$$\text{rank} \begin{bmatrix} A & D \end{bmatrix} = n, \quad (12.35)$$

i.e. that $\begin{bmatrix} A & D \end{bmatrix}$ is surjective, and that (A, D) is reachable and (A, C) detectable. Under these hypotheses the value of (12.34) is the *convex* quadratic

$$W(x) = (x - \tilde{x})^\top K (x - \tilde{x}) + \text{const}, \quad K = \Sigma^{-1} \succ 0, \quad (12.36)$$

where $\Sigma \succ 0$ is the minimal stabilizing solution of the dual, or *estimation*, algebraic Riccati equation

$$\Sigma = A(\Sigma^{-1} + C^\top N^{-1} C)^{-1} A^\top + DD^\top, \quad (12.37)$$

whose existence follows from the standard theory of the filtering ARE (the dual of the control ARE under the substitution $(A, B, Q, R) \mapsto (A^\top, C^\top, DD^\top, N)$; reachability of (A, D) makes $\Sigma \succ 0$). The reconstructed state $\tilde{x}$ is generated online by the stationary observer

$$\tilde{x}_{k+1} = A\tilde{x}_k + Bu_k + L(y_k - C\tilde{x}_k), \quad L = A\Sigma C^\top (C\Sigma C^\top + N)^{-1}, \quad (12.38)$$

whose error dynamics $A - LC$ are Schur. The following lemma verifies (12.36)–(12.38) by exhibiting the quadratic form as the stationary point of the forward dynamic-programming recursion of Lemma 12.13. We use the least-norm identity: for $T \succ 0$ and F of full row rank,

$$\min_{z: Fz=g} z^\top Tz = g^\top (FT^{-1}F^\top)^{-1}g, \quad z^* = T^{-1}F^\top (FT^{-1}F^\top)^{-1}g, \quad (12.39)$$

which follows by eliminating the multiplier in the Lagrange conditions $Tz = F^\top \nu$, $Fz = g$.

Lemma 12.20 (Stationary cost-to-come). *Assume (12.35), $N \succ 0$, and let $\Sigma \succ 0$ solve (12.37). Suppose at the current stage the cost-to-come is $W(x) = (x - \tilde{x})^\top \Sigma^{-1}(x - \tilde{x}) + c$. Then, after applying the control u and receiving the measurement y , the updated cost-to-come delivered by the forward recursion,*

$$W^+(x^+) = \min_{x, w, v} \left\{ W(x) + w^\top w + v^\top N^{-1} v : x^+ = Ax + Bu + Dw, \quad y = Cx + v \right\}, \quad (12.40)$$

is again of the same form, $W^+(x^+) = (x^+ - \tilde{x}^+)^\top \Sigma^{-1}(x^+ - \tilde{x}^+) + c^+$, with the center propagated by the observer (12.38): $\tilde{x}^+ = A\tilde{x} + Bu + L(y - C\tilde{x})$.

Proof. Eliminate $v = y - Cx$ and carry out the minimization in two steps.

Step 1 (measurement update). Collect the terms in x :

$$W(x) + (y - Cx)^\top N^{-1}(y - Cx) = (x - \bar{x})^\top \bar{K}(x - \bar{x}) + \bar{c},$$

by completing the square, where

$$\bar{K} = \Sigma^{-1} + C^\top N^{-1} C \succ 0, \quad \bar{x} = \bar{K}^{-1}(\Sigma^{-1} \tilde{x} + C^\top N^{-1} y),$$

and $\bar{c}$ collects terms independent of x . The push-through identities

$$\bar{K}^{-1} C^\top N^{-1} = \Sigma C^\top (C \Sigma C^\top + N)^{-1}, \quad \bar{K}^{-1} \Sigma^{-1} = I - \Sigma C^\top (C \Sigma C^\top + N)^{-1} C,$$

both verified by multiplying through by $\bar{K}$ and simplifying, give the update in innovation form:

$$\bar{x} = \tilde{x} + \Sigma C^\top (C \Sigma C^\top + N)^{-1} (y - C \tilde{x}). \quad (12.41)$$

Step 2 (time update). It remains to compute

$$W^+(x^+) = \min_{x, w : Ax + Dw = x^+ - Bu} [(x - \bar{x})^\top \bar{K} (x - \bar{x}) + w^\top w] + \bar{c}.$$

Set $z = (x - \bar{x}, w)$, $T = \text{blkdiag}(\bar{K}, I)$, $F = [A \ D]$, and $g = x^+ - Bu - A\bar{x}$; by (12.35), F has full row rank, and (12.39) yields

$$W^+(x^+) = |x^+ - A\bar{x} - Bu|_{(A\bar{K}^{-1}A^\top + DD^\top)^{-1}}^2 + \bar{c},$$

using the weighted-norm notation $|z|_S^2 := z^\top S z$. Since Σ solves (12.37), $A\bar{K}^{-1}A^\top + DD^\top = A(\Sigma^{-1} + C^\top N^{-1}C)^{-1}A^\top + DD^\top = \Sigma$, so the quadratic form reproduces itself with the same matrix Σ^{-1} and new center $\tilde{x}^+ = A\bar{x} + Bu$; substituting (12.41) gives precisely the observer recursion (12.38) with the gain $L = A\Sigma C^\top (C\Sigma C^\top + N)^{-1}$. $\square$

The estimator ARE (12.37) is the standard (Kalman/minimum-energy) filter equation: it does not involve the cost weightings Q, R at all. Estimation and control are therefore computed independently; their only coupling appears in the selection step, to which we now turn.

The Stationary Certainty-Equivalence Controller

Applying the selection (12.23) in the stationary regime, the certainty-equivalent state minimizes $W(x) + V(x)$ over $\mathbb{R}^n$:

$$\begin{aligned} \hat{x} &= \arg \min_x [(x - \tilde{x})^\top K (x - \tilde{x}) + x^\top M x] \\ &= (K + M)^{-1} K \tilde{x} = (I + \Sigma M)^{-1} \tilde{x}, \end{aligned} \quad (12.42)$$

and the stationary output-feedback control is

$$u_k = -K_c \hat{x}_k = -K_c (I + \Sigma M)^{-1} \tilde{x}_k. \quad (12.43)$$

The controller is therefore a stationary observer (12.38) feeding a *detuned* state-feedback gain $\tilde{K} := K_c(I + \Sigma M)^{-1}$. As established in the proof of Theorem 12.21 below, the spectrum of ΣM is real and nonnegative, so the factor $(I + \Sigma M)^{-1}$ is a well-defined contraction of the estimate: the certainty-equivalent state is pulled toward the origin relative to $\tilde{x}$. This is the quantitative signature of *optimism*, the best-case reading assumes the true state to be closer to the (cost-free) origin than the raw reconstruction indicates. Figure 12.3 illustrates the mechanism: the selection balances the “data pull” of the cost-to-come, anchored at $\tilde{x}$, against the “control pull” of the cost-to-go, anchored at the origin, and settles strictly between the two anchors.

Theorem 12.21 (Stationary certainty-equivalence controller). *Suppose (A, B) is stabilizable and (A, H) is detectable, so that the control ARE (12.32) has a minimal stabilizing solution $M \succeq 0$; and suppose (12.35) holds, (A, D) is reachable, and (A, C) is detectable, so that the estimation ARE (12.37) has a minimal stabilizing solution $\Sigma \succ 0$. Then:*

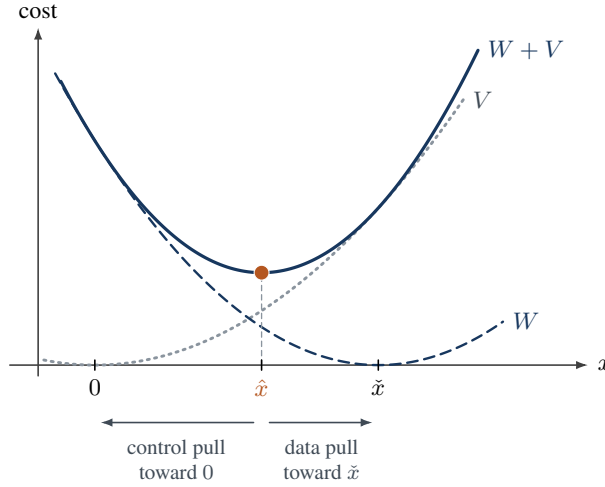


Figure 12.3: The selection step in the stationary linear-quadratic case. The cost-to-go $V(x) = x^\top M x$ (dotted) is anchored at the cost-free origin; the cost-to-come $W(x) = (x - \tilde{x})^\top \Sigma^{-1} (x - \tilde{x})$ (dashed) is anchored at the least-energy reconstruction $\tilde{x}$ of the data. The certainty-equivalent state $\hat{x} = (I + \Sigma M)^{-1} \tilde{x}$ minimizes their sum (solid) and lies strictly between the two anchors: the optimistic controller trusts the data, discounted toward the cheapest state.

(i) $\sigma(\Sigma M) \subset [0, \infty)$; in particular $I + \Sigma M$ is invertible and the selection (12.42) is well defined.

(ii) The stationary selected value is

$$\min_x [W(x) + V(x)] = \tilde{x}^\top M (I + \Sigma M)^{-1} \tilde{x} + \text{const},$$

where $M(I + \Sigma M)^{-1}$ is symmetric positive semidefinite, and the minimum is attained at $\hat{x} = (I + \Sigma M)^{-1} \tilde{x}$ alone.

(iii) The selected value is attained by the certainty-equivalence controller (12.38)–(12.43), in the sense of Theorem 12.15 (iii) applied stationarily.

(iv) The closed-loop dynamics of (12.29) under (12.38)–(12.43) have spectrum $\sigma(\text{closed loop}) = \sigma(A - B\tilde{K}) \cup \sigma(A - LC)$, where $A - LC$ is Schur; the controller is internally stabilizing if and only if the detuned matrix $A - B\tilde{K}$ is Schur as well.

Proof. (i) For any square matrices P, S the nonzero eigenvalues of PS and SP coincide. Applying this with $P = \Sigma^{1/2}$ and $S = \Sigma^{1/2} M$,

$$\sigma(\Sigma M) \setminus \{0\} = \sigma(\Sigma^{1/2} M \Sigma^{1/2}) \setminus \{0\} \subset [0, \infty),$$

since $\Sigma^{1/2} M \Sigma^{1/2} \succeq 0$. Hence $\sigma(\Sigma M) \subset [0, \infty)$ and $-1 \notin \sigma(\Sigma M)$, so $I + \Sigma M$ is invertible.

(ii) With $K = \Sigma^{-1} \succ 0$ and $M \succeq 0$, the function $x \mapsto (x - \tilde{x})^\top K (x - \tilde{x}) + x^\top M x$ is strictly convex (its Hessian $2(K + M) \succ 0$), so it has the unique minimizer determined by the stationarity condition $(K + M)x = K\tilde{x}$. Factoring $K + M = \Sigma^{-1}(I + \Sigma M)$ gives $(K + M)^{-1} = (I + \Sigma M)^{-1} \Sigma$ and hence $\hat{x} = (K + M)^{-1} K \tilde{x} = (I + \Sigma M)^{-1} \tilde{x}$, which is

(12.42). The minimum value is

$$\begin{aligned} (\hat{x} - \tilde{x})^\top K(\hat{x} - \tilde{x}) + \hat{x}^\top M \hat{x} &= \tilde{x}^\top [K - K(K + M)^{-1}K] \tilde{x} \\ &= \tilde{x}^\top M(K + M)^{-1}K \tilde{x} = \tilde{x}^\top M(I + \Sigma M)^{-1} \tilde{x}, \end{aligned}$$

where the middle identity follows from $M(K + M)^{-1}K = (K + M - K)(K + M)^{-1}K = K - K(K + M)^{-1}K$. The same identity shows $M(K + M)^{-1}K = [K(K + M)^{-1}M]^\top$ equals its own transpose, so the value matrix is symmetric; it is positive semidefinite because the minimum of a sum of nonnegative quadratics is nonnegative for every $\tilde{x}$.

(iii) By Lemma 12.20, the pair $(W, \tilde{x})$ reproduces itself under the forward recursion, and by the stationarity of V under the backward recursion (12.17), the hypotheses of Theorem 12.15 hold at every stage with the same pair of value functions; Theorem 12.15 (iii) then gives the invariance of the selected value and its attainment along self-consistent play.

(iv) Write the loop in the coordinates $(e, \tilde{x})$ with $e := x - \tilde{x}$. Using (12.29), (12.38), and $u_k = -\tilde{K}\tilde{x}_k$,

$$e_{k+1} = (A - LC)e_k + Dw_k - Lv_k, \quad \tilde{x}_{k+1} = (A - B\tilde{K})\tilde{x}_k + LCe_k + Lv_k.$$

The unforced dynamics matrix is block lower-triangular with diagonal blocks $A - LC$ and $A - B\tilde{K}$, whence the spectrum is the union claimed. The matrix $A - LC$ is Schur because Σ is the stabilizing ARE solution; the loop is then internally stable precisely when the remaining block $A - B\tilde{K}$ is Schur. $\square$

No auxiliary condition beyond the structural hypotheses is required: because the selection matrix is $I + \Sigma M$ with $\Sigma, M \succeq 0$, it is always invertible (part (i) of the theorem), so the two AREs and the selection (12.42) are always well-defined. The one genuine requirement is closed-loop stability of the detuned gain $\tilde{K}$; since $\tilde{K}$ is smaller than the LQR gain K_c , stability can fail only when the open loop is so unstable that the optimistic de-tuning drops the gain below the stabilizing threshold.

Optimism and the Detuned Gain

It is illuminating to place the stationary controller beside the risk-neutral LQG regulator of the stochastic theory (Chapters 14 and 15), which regulates the same linear plant from noisy measurements. The two share the identical estimator (12.38) and the identical LQR gain K_c , and differ only in how the reconstructed state is used:

$$\hat{x} = (I + \Sigma M)^{-1} \tilde{x} \quad (\text{here}), \quad \hat{x} = \tilde{x} \quad (\text{risk-neutral, LQG}).$$

The LQG controller feeds the estimate $\tilde{x}$ to the feedback gain verbatim, estimation and control separate exactly. The optimistic controller instead *contracts* the estimate toward the cost-free origin through the factor $(I + \Sigma M)^{-1}$: the best-case reading of the data presumes the true state to be closer to the origin, hence cheaper, than the raw reconstruction suggests. The coupling $(I + \Sigma M)^{-1}$ is precisely the residue of incomplete information: it vanishes ($\rightarrow I$) when either the estimation uncertainty Σ or the control stiffness M is negligible.

Example 12.22 The scalar plant over an infinite horizon.

Return to the scalar plant $x_{k+1} = x_k + u_k + w_k$, $y_k = x_k + v_k$, with $Q = R = N = 1$ (so $A = B = C = D = 1$), now over the infinite horizon. Both AREs reduce to $M^2 - M - 1 = 0$, whose stabilizing root is the golden ratio

$$M = \Sigma = \frac{1 + \sqrt{5}}{2} \approx 1.618.$$

The LQR gain and observer gain are $K_c = M/(1 + M) = L = \Sigma/(1 + \Sigma) \approx 0.618$, and the coupling is $1 + \Sigma M = 1 + M^2 \approx 3.618$. The certainty-equivalent state and control are

$$\hat{x}_k = (1 + \Sigma M)^{-1} \tilde{x}_k \approx 0.276 \tilde{x}_k, \quad u_k = -\tilde{K} \tilde{x}_k, \quad \tilde{K} = \frac{K_c}{1 + \Sigma M} \approx 0.171.$$

The closed loop separates into the observer mode $A - LC = 1 - 0.618 = 0.382$ and the controller mode $A - B\tilde{K} = 1 - 0.171 = 0.829$, both inside the unit circle, so the controller is stabilizing. The optimistic contraction pulls the certainty-equivalent state to $\hat{x} \approx 0.276 \tilde{x}$, well inside the raw estimate; the risk-neutral LQG controller for the same plant would instead use $\tilde{x}$ itself and apply the full gain $K_c \approx 0.618$. $\square$

12.2.7 ■ Extensions

The development above isolates the incomplete information in the unknown initial state $x_0 \in X_0$. The same framework accommodates richer models of uncertainty. If the dynamics are driven by an unknown-but-bounded disturbance, $x_{k+1} = f_k(x_k, u_k, w_k)$ with $w_k \in W_k$, and the measurements carry error, $y_k = h_k(x_k, v_k)$ with $v_k \in V_k$, then the compatible state set (12.9) is defined by quantifying over admissible disturbance and error sequences in addition to x_0 , and the prediction step (12.10) inflates the set by the disturbance range before the measurement intersection. The cost-to-come (12.19) then optimizes over these sequences as well. When f_k and h_k are linear and the uncertainty sets are ellipsoids, the compatible sets remain (outer-bounding) ellipsoids and the entire recursion collapses to a finite-dimensional update of a center and a shape matrix, the set-membership analogue of the Kalman-filter/Riccati structure. These reductions, and the value of information they quantify, are pursued in the exercises.

12.3 ■ Continuous-Time Optimal Control under Imperfect Measurements

We now develop the continuous-time counterpart of Sections 12.1 and 12.2. The best-case convention and the cost-to-come/cost-to-go decomposition carry over unchanged, but continuous time brings two genuinely new phenomena. First, the measurement record is a *function*, not a finite list, so the information structure must be handled through causal functionals rather than finite histories. Second, the certainty-equivalence controller is defined by an *implicit relation*, the control at time t depends on a selected state that itself depends on the control applied up to time t , whose well-posedness must be assumed, or verified, as a fixed-point property. We shall see that the monotonicity structure driving the optimality proof survives, now expressed through a Danskin-type argument on the selected value.

12.3.1 ■ Problem Formulation

Consider the deterministic system

$$\dot{x}(t) = f(t, x(t), u(t)), \quad x(0) = x_0 \in X_0, \quad t \in [0, T], \quad (12.44)$$

with state $x(t) \in \mathbb{R}^n$, control $u(t) \in \mathcal{U} \subseteq \mathbb{R}^m$, and the imperfect measurement

$$y(t) = h(t, x(t)) \in \mathbb{R}^p. \quad (12.45)$$

As in discrete time, the initial state is known only to lie in the set X_0 , and the output map h is in general not injective. The cost is

$$J = \Phi(x(T)) + \int_0^T \ell(t, x(t), u(t)) dt. \quad (12.46)$$

The information available to the controller at time t is the record

$$I_t := (y_{[0,t]}, u_{[0,t]}), \quad (12.47)$$

and an *admissible policy* is a causal functional μ generating

$$u(t) = \mu_t(y_{[0,t]}), \quad (12.48)$$

such that the closed loop (12.44)–(12.48) has a unique solution for every $x_0 \in X_0$. The set of admissible policies is denoted $\mathcal{M}$.

Assumption 12.23 (Regularity). *The maps f , h , ℓ , and Φ are of class C^1 , and f satisfies a growth condition guaranteeing, for each $(x_0, u(\cdot))$, a unique solution of (12.44) on $[0, T]$ depending continuously on $(x_0, u(\cdot))$. Moreover, for each admissible policy, the compatible initial conditions can be constrained to a compact subset of X_0 on which the map $x_0 \mapsto x(\cdot)$ is continuous.*

The compactness clause guarantees that the infima over compatible trajectories introduced below are attained and that the envelope (Danskin) argument in the main proof applies.

12.3.2 ■ Information Structure in Continuous Time

Definition 12.24 (Compatible state set). *For an information record I_t , the compatible state set is*

$$\mathcal{X}(t, I_t) := \left\{ \xi \in \mathbb{R}^n : \begin{array}{l} \exists x_0 \in X_0 \text{ such that the solution of (12.44) from } x_0 \\ \text{under } u_{[0,t]} \text{ satisfies } h(s, x(s)) = y(s) \forall s \in [0, t] \text{ and } x(t) = \xi \end{array} \right\}. \quad (12.49)$$

Proposition 12.25 (Properties of the information structure). *Along any closed-loop trajectory generated by an admissible policy:*

- (i) **Consistency.** $x(t) \in \mathcal{X}(t, I_t)$ for all t ; in particular $\mathcal{X}(t, I_t) \neq \emptyset$.
- (ii) **Perfect recall.** For $t' \geq t$, every state in $\mathcal{X}(t', I_{t'})$ is reached at time t' by a trajectory that passes through $\mathcal{X}(t, I_t)$ at time t : explanations can be refined but never created.

- (iii) **Nonanticipativity.** $\mathcal{X}(t, I_t)$ depends on (y, u) only through their restrictions to $[0, t]$ and $[0, t)$.

Proof. (i) The realized trajectory itself satisfies every requirement in (12.49) with $\xi = x(t)$. (ii) If $\xi' \in \mathcal{X}(t', I_{t'})$ is reached by a compatible trajectory $x(\cdot)$ from $x_0 \in X_0$, then the same trajectory satisfies the constraints of (12.49) on the subinterval $[0, t]$, so $x(t) \in \mathcal{X}(t, I_t)$. (iii) The constraints in (12.49) involve only $s \in [0, t]$. $\square$

Remark 12.26 (No prediction–correction split in continuous time). In discrete time the compatible set evolves by the two-stage recursion (12.11): propagate, then intersect. In continuous time the measurement constraint $h(t, x(t)) = y(t)$ holds at *every* instant, so propagation and intersection are fused: the compatible set is carried by the flow of (12.44) while being confined to the moving constraint set $\{x : h(t, x) = y(t)\}$. Differentiating the constraint along the flow shows that compatibility restricts the admissible *velocities* as well as the positions, a differential-algebraic, rather than sequential, structure. This is why the continuous-time theory works directly with the compatible trajectories (through the cost-to-come function below) rather than with a set-valued recursion.

12.3.3 ■ A Minimum Principle under Imperfect Measurements

Define the Hamiltonian

$$H(t, x, u, \lambda) := \ell(t, x, u) + \lambda^\top f(t, x, u). \quad (12.50)$$

Theorem 12.27 (Information-constrained minimum principle). *Let Assumption 12.23 hold, let $\mathcal{U}$ be convex, and let $\mu^* \in \mathcal{M}$ be a locally optimal policy whose maps μ_t^* are continuously (Fréchet) differentiable causal functionals of the output record. Let (x^*, u^*) be the trajectory and control realized from the initial state $x_0^* \in X_0$. For an admissible perturbation $\delta\mu$ (continuously differentiable, with $\mu^* + \varepsilon \delta\mu$ admissible for small $\varepsilon > 0$), define the induced variation $(\delta x, \delta u)$ as the unique solution of the causal linear system*

$$\begin{aligned} \delta \dot{x}(t) &= f_x(t) \delta x(t) + f_u(t) \delta u(t), & \delta x(0) &= 0, \\ \delta u(t) &= \delta \mu_t(y_{[0,t]}^*) + [D\mu_t^*(h_x \delta x)_{[0,t]}], \end{aligned} \quad (12.51)$$

where f_x, f_u, h_x are Jacobians along (x^*, u^*) and $D\mu_t^*$ is the Fréchet derivative of μ_t^* at $y_{[0,t]}^*$. Then there exists an adjoint arc $\lambda(\cdot)$ satisfying

$$-\dot{\lambda}(t) = \nabla_x H(t, x^*(t), u^*(t), \lambda(t)), \quad \lambda(T) = \nabla \Phi(x^*(T)), \quad (12.52)$$

and, for every admissible perturbation whose induced control variation points into $\mathcal{U}$,

$$\int_0^T \nabla_u H(t, x^*(t), u^*(t), \lambda(t))^\top \delta u(t) dt \geq 0. \quad (12.53)$$

Proof. *Step 1 (well-posedness of the induced variation).* Substituting the expression for δu into the linearized dynamics makes (12.51) a linear Volterra integral equation for δx with bounded causal kernel; by the standard Picard iteration it has a unique solution, and δu is then determined. As in discrete time (Remark 12.7), $\delta u(t)$ is a causal functional of the nominal record I_t^* : the perturbed policy uses no information the original one could not access.

Step 2 (differentiability). Let $\mu^\varepsilon = \mu^* + \varepsilon \delta\mu$ and let $(x^\varepsilon, u^\varepsilon, y^\varepsilon)$ be the closed-loop solution from x_0^* . By Assumption 12.23 and the differentiability of μ^* and $\delta\mu$, the map $\varepsilon \mapsto (x^\varepsilon, u^\varepsilon)$ is differentiable at $\varepsilon = 0^+$ in the uniform norm, and its derivative satisfies (12.51): differentiating $u^\varepsilon(t) = \mu_t^*(y_{[0,t]}^\varepsilon) + \varepsilon \delta\mu_t(y_{[0,t]}^\varepsilon)$ produces the direct term $\delta\mu_t(y_{[0,t]}^*)$ and the feedback term $D\mu_t^*(\delta y_{[0,t]})$ with $\delta y = h_x \delta x$.

Step 3 (first variation and integration by parts). Define λ by (12.52); the equation is linear with continuous coefficients, so λ exists on $[0, T]$. The first variation of the cost is

$$\delta J = \nabla \Phi(x^*(T))^\top \delta x(T) + \int_0^T [\ell_x(t)^\top \delta x(t) + \ell_u(t)^\top \delta u(t)] dt.$$

Since $\delta x(0) = 0$,

$$\lambda(T)^\top \delta x(T) = \int_0^T \frac{d}{dt} [\lambda^\top \delta x] dt = \int_0^T [\dot{\lambda}^\top \delta x + \lambda^\top (f_x \delta x + f_u \delta u)] dt.$$

Substituting $\dot{\lambda} = -\ell_x - f_x^\top \lambda$ from (12.52), the terms in δx cancel against those in δJ , leaving

$$\delta J = \int_0^T [\ell_u(t) + f_u(t)^\top \lambda(t)]^\top \delta u(t) dt = \int_0^T \nabla_u H(t, x^*, u^*, \lambda)^\top \delta u(t) dt.$$

Step 4 (conclusion). By convexity of $\mathcal{U}$ the perturbed policy is feasible for small $\varepsilon > 0$, and local optimality gives $J(\mu^\varepsilon) \geq J(\mu^*)$, hence $\delta J \geq 0$, which is (12.53). $\square$

As in discrete time, the adjoint equation and the Hamiltonian are those of the perfect-information Pontryagin principle; only the cone of admissible variations has changed. Pointwise minimization of the Hamiltonian cannot be required at every compatible state simultaneously, the blind-sensor mechanism of Example 12.8 operates verbatim here, and under the best-case objective the principle holds along the selected trajectory while failing pointwise on every other compatible one, exactly as in the discrete multi-stage Example 12.9. The next example makes this concrete in continuous time, with every quantity available in closed form.

Example 12.28 Minimum principle for a scalar regulator on $[0, T]$.

Consider $\dot{x} = u$ on $[0, T]$ with cost $J = \frac{1}{2} \int_0^T (x(t)^2 + u(t)^2) dt$ (so $a = 0, b = 1, q = 1, s_f = 0$). The perfect-information cost-to-go is the standard LQR quadratic $V(t, x) = \frac{1}{2} Z(t) x^2$, where Z solves $\dot{Z} = Z^2 - 1, Z(T) = 0$, i.e. $Z(t) = \tanh(T-t)$ and $\mu^{\text{fs}}(t, x) = -Z(t)x$. The initial state is uncertain, $x(0) \in [1, 2]$, and a coarse sensor reports only that $x(0) \geq 1$, so the compatible set is the interval $[1, 2]$ and the cost-to-come vanishes on it.

Selection and closed loop. Minimizing $V(0, x) = \frac{1}{2} \tanh(T) x^2$ over $[1, 2]$ gives the optimistic $\hat{x}(0) = 1$. The certainty-equivalence control $u = -Z\hat{x}$ drives the selected trajectory $\hat{\dot{x}} = -\tanh(T-t)\hat{x}$, whose closed form is

$$\hat{x}(t) = \frac{\cosh(T-t)}{\cosh T}, \quad \hat{x}(0) = 1, \quad \hat{x}(T) = \frac{1}{\cosh T}.$$

Adjoint (Theorem 12.27). With $H = \frac{1}{2}(x^2 + u^2) + \lambda u$, the adjoint equation is

$-\dot{\lambda} = \nabla_x H = x$ with $\lambda(T) = \nabla \Phi(\hat{x}(T)) = 0$. One verifies directly that

$$\lambda(t) = Z(t)\hat{x}(t) = \frac{\sinh(T-t)}{\cosh T}$$

satisfies both: $\dot{\lambda} = \dot{Z}\hat{x} + Z\dot{\hat{x}} = (Z^2 - 1)\hat{x} - Z^2\hat{x} = -\hat{x}$, and $\lambda(T) = 0$. Thus $\lambda = \nabla_x V(t, \hat{x})$, and the stationarity condition $\nabla_u H = u + \lambda = -Z\hat{x} + Z\hat{x} = 0$ holds along the selected trajectory, the continuous-time image of the discrete verification in Example 12.9.

Where the information constraint bites. The companion initial state $x(0) = 2$ is equally compatible. Because the applied control $u(t) = -Z(t)\hat{x}(t)$ depends on the data, not on the (unknown) true state, the companion obeys $\dot{x}' = u(t) = \dot{\hat{x}}(t)$, so it is a rigid translate $x'(t) = \hat{x}(t) + 1$. Its adjoint solves $-\dot{\lambda}' = x'$, $\lambda'(T) = 0$, and the Hamiltonian gradient along the companion is $\nabla_u H'(t) = u(t) + \lambda'(t) = \lambda'(t) - \lambda(t) = T - t$, since $\frac{d}{dt}(\lambda' - \lambda) = -(x' - \hat{x}) = -1$ with $(\lambda' - \lambda)(T) = 0$. The pointwise condition thus fails everywhere on $(0, T)$ along the companion, by the deficit $T - t$, largest at the initial time, where the optimistic commitment to $\hat{x}(0) = 1$ is most consequential, and vanishing at $t = T$, where the terminal cost pins both trajectories. As in discrete time, this deficit is harmless under the best-case objective and would become the aggregated stationarity condition under an averaging one. $\square$

12.3.4 ■ Certainty-Equivalent Synthesis

We now construct the controller, exactly as in discrete time. Three objects are needed: the perfect-information cost-to-go, the cost-to-come along compatible trajectories, and the selection that couples them.

Assumption 12.29 (Verification solution). *There exists $V \in C^1([0, T] \times \mathbb{R}^n)$ solving the Hamilton–Jacobi–Bellman equation*

$$-\partial_t V(t, x) = \min_{u \in \mathcal{U}} \{ \ell(t, x, u) + \nabla_x V(t, x)^\top f(t, x, u) \}, \quad V(T, x) = \Phi(x), \quad (12.54)$$

with the minimum attained at a unique, continuous feedback $\mu^{\text{fs}}(t, x)$.

This is nothing but the standard requirement that the perfect-information problem admit a classical (verification) solution. It supplies, for every $u \in \mathcal{U}$, the *HJB inequality*

$$\partial_t V(t, x) + \ell(t, x, u) + \nabla_x V(t, x)^\top f(t, x, u) \geq 0, \quad (12.55)$$

with equality if and only if $u = \mu^{\text{fs}}(t, x)$.

Definition 12.30 (Cost-to-come). *For an information record I_t with recorded control $u_{[0,t]}$,*

$$W(t, x; I_t) := \inf \left\{ \int_0^t L(s, x(s), u(s)) ds : \begin{array}{l} x(\cdot) \text{ solves (12.44) from some } x_0 \in X_0, \\ h(s, x(s)) = y(s) \ \forall s \in [0, t], \ x(t) = x \end{array} \right\}, \quad (12.56)$$

with $W(t, x; I_t) = +\infty$ if $x \notin \mathcal{X}(t, I_t)$.

Lemma 12.31 (Forward dynamic programming). For $0 \leq t \leq t' \leq T$ and $x' \in \mathcal{X}(t', I_{t'})$,

$$W(t', x'; I_{t'}) = \inf \left\{ W(t, \xi(t); I_t) + \int_t^{t'} \ell(s, \xi(s), u(s)) ds \right\},$$

the infimum being over arcs $\xi(\cdot)$ on $[t, t']$ solving (12.44) under the recorded control, satisfying $h(s, \xi(s)) = y(s)$ on $[t, t']$, $\xi(t) \in \mathcal{X}(t, I_t)$, and $\xi(t') = x'$.

Proof. Identical in structure to Lemma 12.13: concatenating an ϵ -optimal past for $W(t, \xi(t); I_t)$ with an admissible arc on $[t, t']$ produces a trajectory feasible for $W(t', x'; I_{t'})$, giving “ $\leq$ ”; truncating a feasible trajectory for $W(t', x'; I_{t'})$ at time t produces a feasible past and an admissible arc, giving “ $\geq$ ”. $\square$

The Selected Value and the Fixed-Point Controller

Define, for each t , the *selected value*

$$m(t) := \inf_{x \in \mathcal{X}(t, I_t)} [W(t, x; I_t) + V(t, x)], \quad (12.57)$$

and let $\hat{x}(t)$ denote a minimizer. The certainty-equivalence prescription is

$$u(t) = \mu^{\text{fs}}(t, \hat{x}(t)). \quad (12.58)$$

Unlike its discrete-time counterpart, (12.58) is an *implicit* relation: the compatible set and the cost-to-come at time t depend on the control applied on $[0, t]$, which (12.58) itself prescribes. Writing $\hat{u} = \hat{\mu}(u_{[0,t]}, y_{[0,t]})$ for the map that carries a control record into the right-hand side of (12.58), the certainty-equivalence controller is the fixed point of $\hat{\mu}$. Its existence and uniqueness must be posited (and, in the linear-quadratic case below, are verified by construction).

Assumption 12.32 (Well-posedness of the selection). For every $t \in [0, T]$ and every record generated as below, the infimum in (12.57) is attained at a unique $\hat{x}(t)$, the fixed point of $\hat{\mu}$ exists and is unique, and along the resulting closed loop the map $t \mapsto \hat{x}(t)$ is continuous.

Theorem 12.33 (Certainty-equivalence principle, continuous time). Let Assumptions 12.23, 12.29, and 12.32 hold. Then:

- (i) (**Monotonicity**) Along the closed loop generated by any admissible policy, $t \mapsto m(t)$ is nondecreasing on $[0, T]$.
- (ii) (**Invariance**) Along the closed loop generated by the certainty-equivalence controller (12.58), if the record is the one generated by the selected trajectory, then m is constant.
- (iii) (**Value**) The best-case value of the output-feedback problem is $m(0) = \inf_{x \in \mathcal{X}(0, I_0)} V(0, x)$, and under (ii) the realized total cost equals $m(T) = m(0)$; no admissible policy achieves a smaller best-case cost.

Proof. Fix a closed loop, and for each compatible initial state x_0 write $x_{x_0}(\cdot)$ for the trajectory it generates under the realized control (all compatible trajectories experience the same control, since the control is a functional of the common record). Define

$$\bar{G}(t, x_0) := \int_0^t L(s, x_{x_0}(s), u(s)) ds + V(t, x_{x_0}(t)),$$

so that, by Lemma 12.31 applied at time t and the definition (12.56), $m(t) = \min_{x_0 \in \Omega_t} \bar{G}(t, x_0)$, where $\Omega_t \subseteq X_0$ is the (compact, by Assumption 12.23) set of initial states compatible with the record up to time t .

(i) Two mechanisms act in the same direction. First, for fixed x_0 , $t \mapsto \bar{G}(t, x_0)$ is nondecreasing: it is differentiable with

$$\begin{aligned} \frac{\partial \bar{G}}{\partial t}(t, x_0) &= \ell(t, x_{x_0}(t), u(t)) + \partial_t V(t, x_{x_0}(t)) \\ &\quad + \nabla_x V(t, x_{x_0}(t))^\top f(t, x_{x_0}(t), u(t)) \geq 0 \end{aligned}$$

by the HJB inequality (12.55). Second, by perfect recall (Proposition 12.25(ii)) the constraint set shrinks, $\Omega_{t'} \subseteq \Omega_t$ for $t' \geq t$. Hence

$$m(t') = \min_{\Omega_{t'}} \bar{G}(t', \cdot) \geq \min_{\Omega_{t'}} \bar{G}(t, \cdot) \geq \min_{\Omega_t} \bar{G}(t, \cdot) = m(t).$$

(ii) By (i) it suffices to show $D^+ m(t) \leq 0$, where D^+ is the upper right Dini derivative. Since Ω_t is compact, $\bar{G}$ is C^1 in t and continuous in x_0 , and the record is self-consistent (so that $\Omega_{t'}$ contains the initial state $\hat{x}_0$ explaining the selected trajectory for all t'), Danskin's theorem gives

$$\begin{aligned} D^+ m(t) &\leq \frac{\partial \bar{G}}{\partial t}(t, \hat{x}_0) \\ &= \ell(t, \hat{x}(t), u(t)) + \partial_t V(t, \hat{x}(t)) + \nabla_x V(t, \hat{x}(t))^\top f(t, \hat{x}(t), u(t)), \end{aligned}$$

where $\hat{x}(t) = x_{\hat{x}_0}(t)$ is the selected state, a minimizer in (12.57) by Lemma 12.31. Under the certainty-equivalence control $u(t) = \mu^{\text{fs}}(t, \hat{x}(t))$, the HJB inequality holds with *equality* at $(t, \hat{x}(t))$, so the right-hand side vanishes and $D^+ m(t) \leq 0$. With (i), m is constant. As in discrete time, the best-case value can only rise as data arrive, and the certainty-equivalence controller is the one that never lets it: it holds m level along the trajectory it commits to.

(iii) At $t = 0$ the cost-to-come vanishes on the compatible set, so $m(0) = \inf_{x \in \mathcal{X}(0, I_0)} V(0, x)$; at $t = T$, $V(T, \cdot) = \Phi$, so $m(T)$ is the least total cost among trajectories compatible with the full record, the best-case cost realized by the policy. For any admissible policy, (i) gives (best-case cost) $= m(T) \geq m(0)$, and $m(0)$ does not depend on the policy; under (ii) the certainty-equivalence controller attains $m(T) = m(0)$, hence its optimality. $\square$

Remark 12.34 (Reading the fixed point). The implicit character of (12.58) is not a technical blemish; it expresses the fact that in continuous time the controller's optimism is *self-referential*: the state it selects as most favorable depends on the control history that its own optimism generated. Assumption 12.32 demands that this self-reference resolve uniquely. In the linear-quadratic specialization below the fixed point is resolved explicitly, the selected state is generated by a differential equation driven by the data, so the assumption is verified by construction rather than imposed.

12.3.5 ■ The Linear-Quadratic Case

We now specialize to the linear-quadratic problem and obtain an explicit, implementable controller. To give the past-reconciliation problem enough freedom, we enlarge the model exactly as in the discrete-time stationary theory (Section 12.2.6): the dynamics are driven by an unknown disturbance and the measurement carries an unknown error,

$$\dot{x} = A(t)x + B(t)u + D(t)w, \quad y = C(t)x + v, \quad (12.59)$$

and the unknowns $(x_0, w(\cdot), v(\cdot))$ are penalized by the reconciliation energy

$$E_{\text{past}} = |x_0 - \bar{x}_0|_{Q_0}^2 + \int_0^t (|w(s)|^2 + |v(s)|_{N^{-1}(s)}^2) ds, \quad Q_0 \succ 0, \quad N(t) \succ 0, \quad (12.60)$$

while the control cost to be minimized is

$$J = |x(T)|_{Q_f}^2 + \int_0^T (|x(t)|_{Q(t)}^2 + |u(t)|^2) dt, \quad Q_f \succeq 0, \quad Q(t) \succeq 0. \quad (12.61)$$

The general theory of the preceding subsection applies verbatim to this enlarged model, with the unknown $(x_0, w(\cdot), v(\cdot))$ playing the role of the unknown initial state, the compatibility constraint $y = Cx + v$ replacing $y = h(t, x)$, and the reconciliation energy (12.60) serving as the past cost in the cost-to-come; consistency, perfect recall, and nonanticipativity hold for the enlarged uncertainty exactly as in Proposition 12.25, and Theorem 12.33 carries over with the same proof.

Cost-to-go. The perfect-information value is the standard LQR quadratic $V(t, x) = x^\top Z(t)x$, where Z solves the Riccati differential equation

$$\dot{Z} + ZA + A^\top Z - ZBB^\top Z + Q = 0, \quad Z(T) = Q_f, \quad (12.62)$$

with feedback $\mu^{\text{fs}}(t, x) = -B^\top Z(t)x$; this is the classical LQR Riccati differential equation, and it exists on all of $[0, T]$ under the standard hypotheses, with no conjugate point in $[0, T]$.

Cost-to-come. The least reconciliation energy compatible with the data and ending at x is again a convex quadratic.

Lemma 12.35 (Stationary form of the cost-to-come). *Let $\Sigma(t) \succ 0$ solve the filter Riccati equation*

$$\dot{\Sigma} = A\Sigma + \Sigma A^\top - \Sigma C^\top N^{-1} C \Sigma + DD^\top, \quad \Sigma(0) = Q_0^{-1}, \quad (12.63)$$

and let $\tilde{x}$ be generated by the observer

$$\dot{\tilde{x}} = A\tilde{x} + Bu + \Sigma C^\top N^{-1}(y - C\tilde{x}), \quad \tilde{x}(0) = \bar{x}_0. \quad (12.64)$$

Then the cost-to-come associated with (12.60) is

$$W(t, x) = (x - \tilde{x}(t))^\top \Sigma^{-1}(t) (x - \tilde{x}(t)) + c(t), \quad \dot{c} = |y - C\tilde{x}|_{N^{-1}}^2, \quad c(0) = 0. \quad (12.65)$$

Proof. W satisfies the forward Hamilton–Jacobi equation obtained from Lemma 12.31 by shrinking the interval: eliminating $v = y - Cx$ and minimizing over the free disturbance w ,

$$\begin{aligned} \partial_t W &= \min_w \left[-\nabla_x W^\top (Ax + Bu + Dw) + |w|^2 \right] + |y - Cx|_{N^{-1}}^2 \\ &= -\nabla_x W^\top (Ax + Bu) - \frac{1}{4} |D^\top \nabla_x W|^2 + |y - Cx|_{N^{-1}}^2, \end{aligned} \quad (12.66)$$

the minimizing disturbance being $w^\circ = \frac{1}{2} D^\top \nabla_x W$. Substitute the ansatz $W = (x - \tilde{x})^\top P(x - \tilde{x}) + c$ with $P := \Sigma^{-1}$, so $\nabla_x W = 2P(x - \tilde{x})$, and expand both sides in powers of $(x - \tilde{x})$, writing $Ax = A(x - \tilde{x}) + A\tilde{x}$ and $y - Cx = (y - C\tilde{x}) - C(x - \tilde{x})$. Matching coefficients:

$$\text{quadratic: } \dot{P} = -PA - A^\top P - PDD^\top P + C^\top N^{-1} C,$$

$$\text{linear: } \dot{\tilde{x}} = A\tilde{x} + Bu + P^{-1} C^\top N^{-1} (y - C\tilde{x}),$$

$$\text{constant: } \dot{c} = |y - C\tilde{x}|_{N^{-1}}^2.$$

The first equation is (12.63) in information form ($\dot{P} = -P\dot{\Sigma}P$), the second is (12.64), and the initial data $W(0, x) = |x - \bar{x}_0|_{Q_0}^2$ give $P(0) = Q_0$, i.e. $\Sigma(0) = Q_0^{-1}$, and $c(0) = 0$. $\square$

Equation (12.63) is precisely the *Kalman–Bucy* Riccati equation of Chapter 14, and (12.64) the Kalman–Bucy filter: the best-case reading of the past reproduces the least-squares state reconstruction exactly, and it involves neither the cost weightings nor the control.

Theorem 12.36 (Certainty-equivalence controller, LQ case). *Let Z and Σ solve (12.62) and (12.63) on $[0, T]$. Then for every t the matrix $I + \Sigma(t)Z(t)$ is invertible, the selection (12.57) is the unique point*

$$\hat{x}(t) = (I + \Sigma(t)Z(t))^{-1} \check{x}(t), \quad (12.67)$$

the certainty-equivalence controller is

$$u(t) = -B^\top(t)Z(t)\hat{x}(t) = -B^\top(t)Z(t)(I + \Sigma(t)Z(t))^{-1}\check{x}(t), \quad (12.68)$$

with $\check{x}$ generated by (12.64) under this same u (a well-posed linear loop, verifying Assumption 12.32), and the selected value is

$$m(t) = \check{x}(t)^\top Z(t)(I + \Sigma(t)Z(t))^{-1}\check{x}(t) + c(t). \quad (12.69)$$

Proof. Invertibility of $I + \Sigma Z$ and the identities (12.67)–(12.69) follow verbatim from parts (i)–(ii) of Theorem 12.21, with $(\Sigma(t), Z(t))$ in place of (Σ, M) : the spectrum of ΣZ is real and nonnegative, the map $x \mapsto W(t, x) + V(t, x)$ is strictly convex with Hessian $2(\Sigma^{-1} + Z) \succ 0$, and completion of squares gives the minimizer and value. Substituting (12.68) into (12.64) yields a linear time-varying differential equation for $\check{x}$ driven by y , which has a unique solution; hence the fixed point of the implicit relation (12.58) exists, is unique, and is given by (12.68), verifying Assumption 12.32 by construction. $\square$

Remark 12.37 (Optimism and the risk-neutral filter). The selection factor $(I + \Sigma(t)Z(t))^{-1}$ requires no condition to be well defined: with $\Sigma, Z \succeq 0$ the matrix $I + \Sigma Z$ is always invertible. Its geometry is that of the discrete case: the optimistic controller *contracts* the estimate $\check{x}$ toward the cost-free origin, whereas the risk-neutral LQG controller of Chapters 14 and 15, built on the same Kalman–Bucy filter, uses $\check{x}$ verbatim, letting estimation and control separate exactly. The contraction is the whole-trajectory signature of the best-case reading of the data.

Remark 12.38 (The role of the process disturbance). The disturbance term Dw is not required by the general theory: everything in Sections 12.3.1–12.3.4, and in the discrete-time development of Sections 12.1 and 12.2, runs on the unknown initial state alone, and the disturbance is merely one more unknown to be reconciled with the data. It earns its place in the linear-quadratic specialization for two reasons. First, duality: with Dw present, the cost-to-come Riccati equation (12.63) is the exact dual of the control equation (12.62) under $(A, B, Q) \mapsto (A^\top, C^\top, DD^\top)$, and the estimator is exactly the Kalman–Bucy filter. Second, and decisively for the stationary theory of Section 12.2.6: without the source term $DD^\top$, the solution of (12.63) decays to zero under detectability, the data eventually pin the state down and the problem reverts to perfect information, so no nondegenerate stationary regime exists. Persistent disturbance is what keeps the imperfect-information problem imperfect. In the finite-horizon problem, by contrast, one may set $D \equiv 0$ throughout this subsection: all formulas remain valid with $DD^\top$ deleted, and Σ simply contracts monotonically as the measurements accumulate.

Remark 12.39 (Innovations measure the price of optimism). By (12.65), $\dot{c} = |y - C\hat{x}|_{N-1}^2$: the constant term of the selected value grows exactly at the rate of the squared *innovation*. Measurements that agree with the observer's prediction leave the selected value untouched; measurements that surprise it force the optimistic estimate of the total cost upward, in quantitative agreement with the monotonicity of Theorem 12.33 and with the discrete refutation mechanism of Example 12.18.

Example 12.40 A scalar plant with hyperbolic Riccati flows.

Let $A = 0$, $B = C = D = N = Q = 1$, $Q_f = 0$, $Q_0 = 2$, on the horizon $[0, T]$. The two Riccati equations become

$$\dot{Z} = Z^2 - 1, \quad Z(T) = 0 \quad \text{and} \quad \dot{\Sigma} = 1 - \Sigma^2, \quad \Sigma(0) = \frac{1}{2},$$

with the closed-form solutions

$$Z(t) = \tanh(T - t), \quad \Sigma(t) = \tanh(t + t_0), \quad t_0 = \operatorname{artanh} \frac{1}{2} \approx 0.549.$$

The cost-to-go stiffness Z decays to zero as the terminal time approaches (nothing left to pay), while the reconciliation stiffness Σ^{-1} relaxes from $Q_0 = 2$ toward 1 as data accumulate. The controller (12.68) is

$$u(t) = - \frac{\tanh(T - t)}{1 + \tanh(t + t_0) \tanh(T - t)} \tilde{x}(t), \quad \dot{\tilde{x}} = u + \tanh(t + t_0) (y - \tilde{x}),$$

an observer with saturating gain feeding a feedback gain that is *doubly* attenuated: by the vanishing of Z near the terminal time and by the optimism factor $(1 + \Sigma Z)^{-1}$, which is smallest, the contraction strongest, in the middle of the horizon, where both past uncertainty and future cost are substantial. $\square$

12.4 ■ Notes and Further Reading

The developments of this chapter are the deterministic, non-adversarial image of a body of work on *worst-case* (minimax) control under imperfect measurements, and it is worth making the connection explicit, both to credit the source of the method and to locate the present results within the wider theory of robust control.

The minimax origin. The certainty-equivalence construction—lift the output-feedback problem to the space of information states, split the value into a cost-to-come and a cost-to-go, and recover the controller from an auxiliary optimization over the compatible set—is due, in the discrete- and continuous-time minimax settings, to Başar and Bernhard [9]. There the incomplete information is an *adversarial disturbance* w , together with an unknown initial state, and the design goal is H^∞ disturbance attenuation: to keep the ratio of a weighted output energy to the disturbance energy below a level γ . The controller must then guard against the *worst* disturbance compatible with the data, so the compatible set is resolved by a *supremum*, the cost-to-come is a worst-case (largest) past cost, and the auxiliary problem is a maximization, $\Gamma_\tau = \max_x [V + W]$, whose per-stage kernel is an Isaacs saddle point.

From worst case to best case. The optimal-control problem of this chapter is obtained from that theory by a single formal substitution: replace every worst-case supremum over the compatible set by a best-case infimum. The disturbance, no longer an adversary, becomes just one more unknown to be reconciled with the data; the $-\gamma^{-2}$ coupling terms that the adversary induces in the game Riccati equations vanish; and the theory becomes self-contained, requiring no attenuation level and no threshold γ^* . The correspondence is term-for-term, and the following dictionary records it.

Minimax / H^∞ theory [9]	Optimal-control counterpart (this chapter)
adversarial disturbance $\omega = (x_1, w)$	unknown initial state $x_0 \in X_0$
worst-case resolution $\sup_{\omega \in \Omega_\tau}$	best-case resolution $\inf_{x \in \mathcal{X}_k(I_k)}$
worst past cost $W_\tau(x) = \sup(\cdots)$	best cost-to-come $W_k(x; I_k) = \inf(\cdots)$
auxiliary problem $\Gamma_\tau = \max_x [V + W]$	selection $m_k(I_k) = \min_x [V_k + W_k]$
guaranteed value Γ_τ <i>nonincreasing</i>	selected value m_k <i>nondecreasing</i>
per-stage Isaacs saddle point	per-stage Bellman minimum
concavity of the per-stage kernel in x	convexity of $W_k + V_k$ in x
attenuation threshold γ^* ; needs $\rho(\Sigma M) < \gamma^2$	no threshold; $I + \Sigma M$ always invertible
game Riccati (with $-\gamma^{-2}DD^\top$)	control Riccati M (LQR)
game filter Riccati (with $-\gamma^{-2}Q$)	estimation / Kalman Riccati Σ

The most instructive entry is the direction of monotonicity. In both theories the guaranteed value is an extremum of a stage-family over the compatible set, and in both the compatible set shrinks as data accumulate (perfect recall). But a *maximum* over a shrinking set can only fall, while a *minimum* over a shrinking set can only rise, so the minimax bound Γ_τ decreases toward the achieved cost as the controller learns, whereas the optimistic estimate m_k of this chapter climbs toward the truth, each measurement either confirming the selected trajectory or refuting it. Robustness drives the guaranteed cost down; optimism has its bet corrected upward.

Optimism, robustness, and risk-neutrality. The three canonical readings of the same linear-quadratic plant differ only in how the reconstructed state $\tilde{x}$ (the output of a common Kalman filter) is fed to a common LQR gain:

$$\hat{x} = (I + \Sigma M)^{-1} \tilde{x} \text{ (best case, this chapter), } \quad \hat{x} = (I - \gamma^{-2} \Sigma M)^{-1} \tilde{x} \text{ (worst case, } H^\infty), \quad \hat{x} = \tilde{x} \text{ (risk-neutral)}$$

The optimistic controller *contracts* the estimate toward the cost-free origin; the robust H^∞ controller *expands* it, hedging against the adversary; and the risk-neutral controller of the stochastic theory (Chapters 14 and 15) uses it verbatim, so that estimation and control separate exactly. The three sit on a single one-parameter family, with the coupling $(I - \gamma^{-2} \Sigma M)^{-1}$ interpolating: it reduces to the risk-neutral identity as $\gamma \rightarrow \infty$, and to the optimistic contraction when the sign of the γ^{-2} term is reversed. This unification of estimation and robust/risk-sensitive control is developed at length in [9]; the risk-sensitive (LEQG) limit connecting the stochastic and minimax pictures is treated there and in the references it gives.

Further pointers. The information-state (belief-state) reduction that underlies the whole construction is the deterministic, set-membership analogue of the reduction of a partially observed Markov decision process to a fully observed problem on the space of posterior distributions; the stochastic theory is developed in Chapters 14 and 15 and, at book length, in the standard references on dynamic programming and stochastic control. The set-membership (unknown-but-bounded) model of uncertainty, and the ellipsoidal and polytopic approximations of the compatible set that make the recursions finite-dimensional, originate in set-membership estimation and viability

theory; the exercises pursue the ellipsoidal case, whose linear-quadratic collapse parallels the separation structure of the Gaussian problem.

12.5 ■ Exercises

Exercise 12.1 (Compatible-Set Recursion for a Scalar System). Consider the scalar system $x_{k+1} = x_k + u_k + w_k$ with disturbance bound $|w_k| \leq \delta$ and measurements $y_k = x_k + v_k$ with $|v_k| \leq \epsilon$, so that the compatible state set evolves by the prediction–correction update of the chapter, with the prediction step inflated by the disturbance range. Suppose the initial compatible set is the interval $\mathcal{X}_0 = [\underline{x}_0, \bar{x}_0]$.

- (i) Show that if $\mathcal{X}_k$ is an interval, then the predicted set $\text{Pred}(\mathcal{X}_k, u_k)$ and the updated set $\mathcal{X}_{k+1} = \text{Pred}(\mathcal{X}_k, u_k) \cap [y_{k+1} - \epsilon, y_{k+1} + \epsilon]$ are again intervals, and write the recursion for their endpoints.
- (ii) Show that the width of $\mathcal{X}_{k+1}$ is at most $\min\{\text{width}(\mathcal{X}_k) + 2\delta, 2\epsilon\}$, and interpret the two regimes.
- (iii) Does the control u_k influence the width of the compatible set? Relate your answer to the pipeline $I_k \mapsto \mathcal{X}_k \mapsto W_k \mapsto \hat{x}_k \mapsto u_k$ of the certainty-equivalence synthesis: which blocks depend on the control and which do not?

Exercise 12.2 (Forward Recursion for the Cost-to-Come). Consider the scalar linear-quadratic problem $x_{k+1} = ax_k + u_k$ with known controls, unknown initial state $x_0 \in \mathbb{R}$ penalized by $W_0(x) = q_0 x^2$ with $q_0 > 0$, measurements $y_k = x_k + v_k$, and past cost charging v_k^2 for the measurement residuals, in the spirit of the stationary construction of the chapter.

- (i) Using the forward dynamic-programming recursion, show by induction that the cost-to-come remains a convex quadratic, $W_k(x) = \kappa_k (x - \tilde{x}_k)^2 + c_k$ with $\kappa_k > 0$.
- (ii) Derive the scalar recursions for κ_k and $\tilde{x}_k$, and verify that $\tilde{x}_k$ obeys an observer recursion driven by the innovation $y_k - \tilde{x}_k$.
- (iii) Compare your recursion for κ_k^{-1} with the estimation algebraic Riccati equation of the chapter, and identify the limit of κ_k as $k \rightarrow \infty$ when $|a| < 1$ and when $|a| > 1$.

Exercise 12.3 (Best-Case Value and the Certainty-Equivalent Selection). Consider the two-stage scalar problem $x_{k+1} = x_k + u_k$ with cost $J = (x_2 - r)^2 + u_0^2 + u_1^2$, target $r > 0$, and no measurements after time 0, so that the compatible set remains $\mathcal{X}_0 = [\alpha, \beta]$ throughout (translated by the applied controls).

- (i) Compute the perfect-information cost-to-go functions V_2, V_1, V_0 and the feedback laws $\mu_1^{\text{fs}}, \mu_0^{\text{fs}}$.
- (ii) Carry out the selection $\hat{x}_0 \in \arg \min_{x \in [\alpha, \beta]} [W_0(x) + V_0(x)]$ with $W_0 \equiv 0$ on the compatible set, distinguishing the cases $r < \alpha$, $r \in [\alpha, \beta]$, and $r > \beta$, and give the certainty-equivalence control u_0 in each case.
- (iii) Verify the monotonicity–invariance dichotomy of the certainty-equivalence theorem on this example: show that the selected value m_k is constant along play when the true initial state equals $\hat{x}_0$, and strictly increases at the first stage whose measurement refutes the selection when it does not.

Exercise 12.4 (The Value of Information under Optimism). Let $\mathcal{X} \subseteq \mathcal{X}'$ be two possible compatible sets at some stage k (the first arising from a finer measurement structure), and let $m(\mathcal{X}) = \inf_{x \in \mathcal{X}} [W(x) + V(x)]$ denote the selected value, with the same W and V in both cases.

- (i) Prove that $m(\mathcal{X}) \geq m(\mathcal{X}')$: under the best-case convention, *coarser* information yields a *lower* (more favorable) selected value.
- (ii) Show that if the infimum over the compatible set is replaced by a supremum (the robust reading of the data), the inequality reverses, so that *finer* information yields a *lower* (tighter) guaranteed worst-case value.
- (iii) Reconcile (i) with intuition: in what operational sense is finer information still valuable to the optimistic controller, even though it can only raise the selected value? Base your answer on the monotonicity and invariance properties of the certainty-equivalence theorem, in particular, on which of the two quantities (selected value versus realized cost) each convention controls.

Exercise 12.5 (The Stationary Controller for the Scalar Plant). Consider the scalar plant $x_{k+1} = x_k + u_k + w_k$, $y_k = x_k + v_k$, with cost weights $Q = R = 1$ and unit process- and measurement-noise intensities $W = V = 1$, over the infinite horizon, as in the concluding example of the chapter.

- (i) Verify that both the control and the estimation algebraic Riccati equations reduce to $M^2 - M - 1 = 0$, and hence that $M = \Sigma = (1 + \sqrt{5})/2$, the golden ratio.
- (ii) Compute the LQR gain K_c , the observer gain L , the coupling factor $(1 + \Sigma M)^{-1}$, and the detuned gain $\tilde{K}$, and verify that the closed-loop modes $1 - \tilde{K}$ and $1 - L$ both lie strictly inside the unit interval.
- (iii) Compare the optimistic controller with the risk-neutral LQG controller for the same plant, which uses $\hat{x} = \tilde{x}$ and the full gain K_c : order the two feedback gains and interpret the difference. (For the robust variant, which replaces the coupling by $(1 - \gamma^{-2}\Sigma M)^{-1}$, see the notes, Section 12.4.)

Exercise 12.6 (Detuning Can Destabilize). The stationary certainty-equivalence controller applies the detuned gain $\tilde{K} = K_c(I + \Sigma M)^{-1}$, which is smaller than the LQR gain.

- (i) For the scalar plant $x_{k+1} = ax_k + u_k + w_k$, $y_k = x_k + v_k$ with cost weights $Q = R = 1$ and unit noise intensities $W = V = 1$, express the stationary M , Σ , K_c , and $\tilde{K}$ as functions of a , and derive the closed-loop controller mode $a - \tilde{K}$.
- (ii) Show numerically or analytically that for sufficiently large $|a|$ the detuned mode satisfies $|a - \tilde{K}| > 1$ even though $|a - K_c| < 1$: optimism can destabilize a plant that the LQR gain would stabilize.
- (iii) Explain the mechanism in terms of the selection $\hat{x} = (I + \Sigma M)^{-1}\tilde{x}$: why does a strongly unstable plant make the optimistic contraction more severe, and why is this harmless in the finite-horizon problem but fatal in the stationary one?

Exercise 12.7 (Ellipsoidal Information States). Let $x_{k+1} = Ax_k + Bu_k + w_k$ with w_k constrained to the ellipsoid $w_k^\top Q^{-1}w_k \leq 1$, and suppose the compatible set at time k is the ellipsoid $E_k = \{x : (x - c_k)^\top P_k^{-1}(x - c_k) \leq 1\}$.

-
- (i) Show that the exact predicted set $AE_k + Bu_k + \{w : w^\top Q^{-1}w \leq 1\}$ is in general not an ellipsoid.
 - (ii) Verify that for any $\lambda \in (0, 1)$, the ellipsoid with center $Ac_k + Bu_k$ and shape matrix $\frac{1}{\lambda}AP_kA^\top + \frac{1}{1-\lambda}Q$ contains the exact predicted set, so an outer-ellipsoidal recursion is well defined.
 - (iii) Explain how such parametric recursions render the entire certainty-equivalence pipeline finite-dimensional, and compare the resulting center–shape recursion with the exact quadratic cost-to-come recursion obtained in the linear-quadratic case.

Part V

Stochastic Optimal Control

Chapter 13

Estimation Theory

Estimation is a best-approximation problem. A hidden quantity is not observed directly; instead, one observes related data and chooses an estimate that is closest to the hidden quantity in a prescribed error metric. For mean-square error, the natural geometry is Hilbert-space geometry: estimates are projections, errors are orthogonal residuals, and normal equations are the algebraic form of orthogonality.

This chapter develops that viewpoint from first principles. We begin with pre-Hilbert and Hilbert spaces, prove the projection theorem, and then specialize orthogonal projection to least squares, Gram–Schmidt innovations, scalar and vector MMSE estimation, and sequential linear least squares updates. The final formulas are the static counterparts of the update equations that later appear in Kalman filtering.

13.1 ■ Pre-Hilbert and Hilbert Spaces

Inner-product spaces provide the geometric backbone of modern analysis. They formalize notions such as length, angle, and orthogonality, and they allow one to generalize Euclidean geometry to abstract vector spaces arising in approximation theory, signal processing, quantum mechanics, and control.

This first section supplies only the tools needed later: a norm, an inequality that controls inner products, completeness, and orthogonality. The estimation sections use these tools repeatedly, but always with the same geometric picture: an estimate is a point in a subspace, and the error is the perpendicular leftover.

13.1.1 ■ Definition of a Pre-Hilbert Space

Definition 13.1 (Pre-Hilbert space). A pre-Hilbert space is a linear vector space X over $\mathbb{R}$ or $\mathbb{C}$ together with an inner product $\langle \cdot, \cdot \rangle : X \times X \rightarrow \mathbb{F}$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, satisfying the following axioms for all $x, y, z \in X$ and all scalars λ :

- (1) **Conjugate symmetry:** $\langle x, y \rangle = \overline{\langle y, x \rangle}$.
- (2) **Additivity in the first argument:** $\langle x + y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$.
- (3) **Homogeneity in the first argument:** $\langle \lambda x, y \rangle = \lambda \langle x, y \rangle$.
- (4) **Positive definiteness:**

$$\langle x, x \rangle \geq 0, \quad \langle x, x \rangle = 0 \iff x = 0.$$

The inner product generalizes the Euclidean dot product $\langle x, y \rangle = x^\top y$ on $\mathbb{R}^n$. Axioms (2) and (3) together imply linearity in the first argument, while axiom (1) ensures symmetry in the real case and conjugate symmetry in the complex case. Axiom (4) guarantees that the inner product induces a meaningful notion of length and excludes degenerate directions.

A pre-Hilbert space thus has a rich geometric structure, but it may lack certain topological completeness properties. This distinction becomes crucial when considering limits of sequences and approximation procedures.

Norm Induced by an Inner Product

Every inner product induces a natural notion of length.

Definition 13.2 (Induced norm). For $x \in X$, define $\|x\| := \sqrt{\langle x, x \rangle}$.

Basic properties. Axioms (1) and (3) imply absolute homogeneity, $\|\lambda x\| = |\lambda| \|x\|$, while axiom (4) implies $\|x\| > 0$ for $x \neq 0$. The remaining requirement for a norm is the triangle inequality, which follows from the Cauchy–Schwarz inequality.

The Cauchy–Schwarz inequality is the fundamental estimate linking the inner product to the induced norm. It is the analytic expression of the geometric fact that the absolute value of a projection cannot exceed the product of lengths.

Lemma 13.3 (Cauchy–Schwarz inequality). For all x, y in a pre-Hilbert space, $|\langle x, y \rangle| \leq \|x\| \|y\|$. Equality holds if and only if $x = \lambda y$ for some scalar λ , or $y = 0$.

Proof. If $y = 0$, the inequality is immediate. Assume $y \neq 0$. For any scalar λ , positive definiteness gives

$$0 \leq \langle x - \lambda y, x - \lambda y \rangle = \langle x, x \rangle - \lambda \langle y, x \rangle - \bar{\lambda} \langle x, y \rangle + |\lambda|^2 \langle y, y \rangle.$$

Choosing $\lambda = \langle x, y \rangle / \langle y, y \rangle$ minimizes the right-hand side and yields $0 \leq \langle x, x \rangle - \frac{|\langle x, y \rangle|^2}{\langle y, y \rangle}$, which implies $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$. Taking square roots gives the desired inequality. $\square$

Proposition 13.4 (The induced norm is a norm). Let X be a pre-Hilbert space with inner product $\langle \cdot, \cdot \rangle$. Then the function $\|x\| := \sqrt{\langle x, x \rangle}$ satisfies the three norm axioms: absolute homogeneity, positive definiteness, and the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$. Consequently every pre-Hilbert space is a normed space, and its geometry may be discussed in metric terms.

Proof. Only the triangle inequality requires verification. For $x, y \in X$,

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2.$$

By the Cauchy–Schwarz inequality,

$$\langle x, y \rangle + \langle y, x \rangle = 2 \operatorname{Re} \langle x, y \rangle \leq 2 |\langle x, y \rangle| \leq 2 \|x\| \|y\|.$$

Hence, $\|x + y\|^2 \leq (\|x\| + \|y\|)^2$, and taking square roots yields $\|x + y\| \leq \|x\| + \|y\|$. $\square$

The induced norm depends crucially on the inner product. Different inner products on the same vector space may generate different norms and hence different topologies. When a pre-Hilbert space is complete with respect to this norm, it becomes a *Hilbert space*, a setting in which orthogonal projections, Fourier expansions, and variational methods attain their full power.

Example 13.5 Euclidean space $\mathbb{R}^n$.

The space $\mathbb{R}^n$ with inner product $\langle x, y \rangle = \sum_{i=1}^n x_i y_i$ is a pre-Hilbert space. The induced norm is the Euclidean norm

$$\|x\| = \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2}.$$

□

Example 13.6 The sequence space ℓ^2 .

The real space ℓ^2 of square-summable sequences becomes a pre-Hilbert space with $\langle x, y \rangle = \sum_{i=1}^{\infty} x_i y_i$. The associated norm is the usual ℓ^2 norm. □

Example 13.7 The function space $L^2[a, b]$.

The real space $L^2[a, b]$ with inner product $\langle x, y \rangle = \int_a^b x(t)y(t) dt$ is a pre-Hilbert space (its elements being equivalence classes of functions equal almost everywhere). The Cauchy–Schwarz inequality (Hölder with $p = q = 2$) ensures that the inner product is finite. □

Example 13.8 Polynomial subspace of $L^2[a, b]$.

The space of polynomial functions on $[a, b]$, equipped with the same integral inner product, is a pre-Hilbert space and a subspace of $L^2[a, b]$. □

Basic Consequences

Lemma 13.9. *If $\langle x, y \rangle = 0$ for all $y \in X$, then $x = 0$.*

Proof. Taking $y = x$ yields $\langle x, x \rangle = 0$. By positive definiteness of the inner product, this implies $x = 0$. □

Lemma 13.10 (Parallelogram law). *For all x, y in a pre-Hilbert space, $\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$.*

Proof. By definition of the norm induced by the inner product, $\|x + y\|^2 = \langle x + y, x + y \rangle$ and $\|x - y\|^2 = \langle x - y, x - y \rangle$. Expanding each inner product using linearity in the first argument and conjugate symmetry, we obtain $\langle x + y, x + y \rangle = \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle$, and

$\langle x - y, x - y \rangle = \langle x, x \rangle - \langle x, y \rangle - \langle y, x \rangle + \langle y, y \rangle$. Adding these two expressions gives

$$\|x + y\|^2 + \|x - y\|^2 = 2\langle x, x \rangle + 2\langle y, y \rangle = 2\|x\|^2 + 2\|y\|^2,$$

which establishes the parallelogram law. $\square$

13.1.2 ■ Hilbert Spaces

Definition 13.11 (Hilbert space). A Hilbert space is a complete pre-Hilbert space.

Completeness means that every Cauchy sequence with respect to the norm converges to a limit within the space. Thus, a Hilbert space is a Banach space whose norm is induced by an inner product. In such spaces, both the geometric structure (lengths, angles, orthogonality) and the topological structure (limits, convergence) are fully compatible.

Classical examples of Hilbert spaces include $\mathbb{R}^n$ with the Euclidean inner product, the sequence space ℓ^2 , and the function space $L^2[a, b]$. By contrast, the space of polynomials on $[a, b]$, equipped with the L^2 inner product, is not complete and hence forms only a pre-Hilbert space.

Lemma 13.12 (Continuity of the inner product). If $x_n \rightarrow x$ and $y_n \rightarrow y$ in a pre-Hilbert space, then $\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$.

Proof. Since x_n converges, it is bounded: there exists $M > 0$ such that $\|x_n\| \leq M$ for all n . Then

$$\begin{aligned} |\langle x_n, y_n \rangle - \langle x, y \rangle| &\leq |\langle x_n, y_n - y \rangle| + |\langle x_n - x, y \rangle| \\ &\leq \|x_n\| \|y_n - y\| + \|x_n - x\| \|y\|. \end{aligned}$$

Both terms converge to zero as $n \rightarrow \infty$, hence $\langle x_n, y_n \rangle \rightarrow \langle x, y \rangle$. $\square$

13.2 ■ The Projection Theorem

The projection theorem is one of the central structural results of Hilbert space theory. It formalizes the geometric intuition that, given a point and a linear subspace, there exists a unique “closest point” in the subspace obtained by dropping a perpendicular. This principle underlies least-squares approximation, orthogonal decompositions, Fourier expansions, and variational methods.

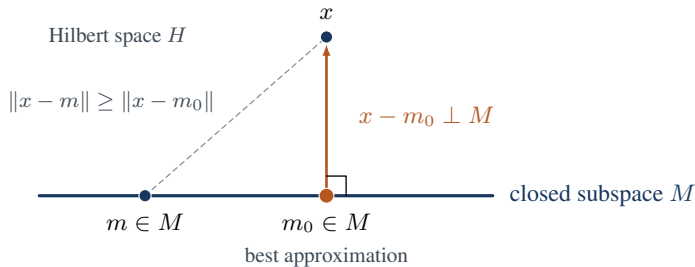


Figure 13.1: The projection theorem decomposes x into its best approximation m_0 in a closed subspace M and an orthogonal residual $x - m_0$; every other $m \in M$ lies farther from x .

13.2.1 ■ Orthogonality

Definition 13.13 (Orthogonality). Let X be a pre-Hilbert space. Two vectors $x, y \in X$ are said to be orthogonal if $\langle x, y \rangle = 0$. We write $x \perp y$. A vector x is said to be orthogonal to a set $S \subset X$ (written $x \perp S$) if $x \perp s$ for every $s \in S$.

Orthogonality generalizes the notion of perpendicular vectors in Euclidean geometry. Although pre-Hilbert spaces may be infinite-dimensional, the inner product allows one to retain a notion of angle and perpendicularity, enabling many familiar geometric arguments.

One of the most important consequences of orthogonality is the validity of the Pythagorean theorem.

Lemma 13.14 (Pythagorean identity). If $x \perp y$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$.

Proof. Using the definition of the norm induced by the inner product,

$$\|x + y\|^2 = \langle x + y, x + y \rangle = \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle.$$

Since $\langle x, y \rangle = 0$ and $\langle y, x \rangle = \overline{\langle x, y \rangle} = 0$, we obtain $\|x + y\|^2 = \|x\|^2 + \|y\|^2$. $\square$

13.2.2 ■ The Best Approximation Problem

Let X be a pre-Hilbert space, M a linear subspace of X , and $x \in X$. We consider the optimization problem $\min_{m \in M} \|x - m\|$. This is the problem of approximating x by an element of M as closely as possible in norm.

If $x \in M$, the solution is trivial. Otherwise, three fundamental questions arise:

- (1) Does a minimizer exist?
- (2) Is the minimizer unique?
- (3) How can the minimizer be characterized geometrically?

Pre-Hilbert versus Hilbert spaces. The inner product structure provides strong geometric constraints, but *completeness plays a decisive role*. In a pre-Hilbert space, the inner product allows one to characterize a minimizer if it exists, but it does not guarantee that such a minimizer actually lies in the space. In contrast, in a Hilbert space, completeness ensures that minimizing sequences converge within the space, guaranteeing existence.

Theorem 13.15 (Projection theorem in pre-Hilbert spaces). Let X be a pre-Hilbert space, M a subspace of X , and $x \in X$. If there exists $m_0 \in M$ such that $\|x - m_0\| \leq \|x - m\|$ for all $m \in M$, then m_0 is unique. Moreover, a necessary and sufficient condition for m_0 to be the unique minimizing vector is $x - m_0 \perp M$.

This result is purely geometric. It states that if a best approximation exists, then it is uniquely determined and obtained by orthogonality. However, the theorem makes no claim about existence; in a pre-Hilbert space, minimizing sequences may converge to limits lying outside the space.

Proof. Necessity. Assume that m_0 is a minimizer and suppose, for contradiction, that $x - m_0$ is not orthogonal to M . Then there exists $m \in M$ with $\|m\| = 1$ such that $\langle x - m_0, m \rangle = \delta \neq 0$.

Define $m_1 := m_0 + \delta m \in M$. Then

$$\begin{aligned}\|x - m_1\|^2 &= \|x - m_0 - \delta m\|^2 \\ &= \|x - m_0\|^2 - \langle x - m_0, \delta m \rangle - \langle \delta m, x - m_0 \rangle + |\delta|^2 \|m\|^2 \\ &= \|x - m_0\|^2 - |\delta|^2 < \|x - m_0\|^2,\end{aligned}$$

which contradicts the minimality of m_0 .

Sufficiency. Conversely, suppose $x - m_0 \perp M$. For any $m \in M$, we have $m - m_0 \in M$, and hence $(x - m_0) \perp (m - m_0)$. By the Pythagorean identity, $\|x - m\|^2 = \|x - m_0\|^2 + \|m - m_0\|^2$. This expression is minimized uniquely when $m = m_0$, proving the claim. $\square$

13.2.3 ■ The Classical Projection Theorem

The previous theorem establishes uniqueness and geometric characterization, *assuming existence*. In Hilbert spaces, completeness ensures that such a minimizer always exists for closed subspaces.

Theorem 13.16 (Projection theorem in Hilbert spaces). *Let H be a Hilbert space and M a closed subspace of H . For every $x \in H$ there exists a unique vector $m_0 \in M$ such that $\|x - m_0\| = \inf_{m \in M} \|x - m\|$. Moreover, m_0 is characterized by $x - m_0 \perp M$.*

Completeness guarantees that every minimizing sequence converges within the space. Thus, Hilbert spaces are precisely the settings in which best-approximation problems are always well posed for closed subspaces.

Proof. Uniqueness and orthogonality follow from the previous theorem. We prove existence.

If $x \in M$, then $m_0 = x$. Otherwise define $\delta := \inf_{m \in M} \|x - m\|$. Choose a sequence $(m_i) \subset M$ such that $\|x - m_i\| \rightarrow \delta$.

Using the parallelogram law,

$$\|m_j - m_i\|^2 = 2\|m_j - x\|^2 + 2\|x - m_i\|^2 - 4\left\|x - \frac{m_i + m_j}{2}\right\|^2.$$

Since $(m_i + m_j)/2 \in M$, we have $\|x - (m_i + m_j)/2\| \geq \delta$, and therefore $\|m_j - m_i\|^2 \leq 2\|m_j - x\|^2 + 2\|x - m_i\|^2 - 4\delta^2 \rightarrow 0$. Thus (m_i) is a Cauchy sequence in M . Because M is closed and H is complete, there exists $m_0 \in M$ such that $m_i \rightarrow m_0$. By continuity of the norm, $\|x - m_0\| = \delta$. $\square$

Pre-Hilbert spaces provide the geometry of projections, while Hilbert spaces guarantee their existence. This distinction explains why orthogonal projections, least-squares methods, and variational principles fundamentally belong to Hilbert space theory and do not extend to arbitrary Banach spaces.

Note 13.17 (What the projection theorem gives us). The theorem has two jobs. First, it tells us that a closest point in a closed subspace exists and is unique. Second, it gives the usable test for that point: the residual must be orthogonal to the whole subspace. In least squares and estimation, this orthogonality test becomes the normal equations.

13.2.4 ■ Example: MMSE Estimation as an Orthogonal Projection

We generalize the single-observation case to multiple observations and show that minimum mean-square error (MMSE) estimation arises naturally as an orthogonal projection in a Hilbert space. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $X, Y_1, Y_2, \dots, Y_n \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ be real-valued random variables satisfying $\mathbb{E}[X] = 0$ and $\mathbb{E}[Y_i] = 0$ for $i = 1, \dots, n$, with $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y_i^2] < \infty$ for $i = 1, \dots, n$.

Hilbert space formulation. Define the real Hilbert space

$$\mathcal{H} := \{Z \in L^2(\Omega, \mathcal{F}, \mathbb{P}) \mid \mathbb{E}[Z] = 0\},$$

which is a closed subspace of $L^2(\Omega)$. The inner product on $\mathcal{H}$ is defined by

$$\langle Z_1, Z_2 \rangle_{\mathcal{H}} = \mathbb{E}[Z_1 Z_2] = \int_{\Omega} Z_1(\omega) Z_2(\omega) d\mathbb{P}(\omega),$$

with induced norm

$$\|Z\|_{\mathcal{H}} = (\mathbb{E}[Z^2])^{1/2} = \left(\int_{\Omega} |Z(\omega)|^2 d\mathbb{P}(\omega) \right)^{1/2}.$$

Clearly, $X, Y_1, \dots, Y_n \in \mathcal{H}$.

Linear estimation subspace. Define the finite-dimensional linear subspace

$$M := \text{span}\{Y_1, \dots, Y_n\} = \left\{ \sum_{i=1}^n \alpha_i Y_i \mid \alpha_i \in \mathbb{R} \right\} \subset \mathcal{H}.$$

Since M is finite-dimensional, it is closed in $\mathcal{H}$.

Best approximation problem. We seek a linear estimator

$$\hat{X} = \sum_{i=1}^n \alpha_i Y_i \in M$$

that minimizes the mean-square error

$$\mathbb{E} \left[\left(X - \sum_{i=1}^n \alpha_i Y_i \right)^2 \right] = \int_{\Omega} \left| X(\omega) - \sum_{i=1}^n \alpha_i Y_i(\omega) \right|^2 d\mathbb{P}(\omega).$$

Equivalently, we consider the best approximation problem $\min_{m \in M} \|X - m\|_{\mathcal{H}}$.

Application of the projection theorem. Because $\mathcal{H}$ is a Hilbert space and M is a closed subspace, the projection theorem guarantees the existence of a unique minimizer $\hat{X} = \Pi_M X$, where Π_M denotes the orthogonal projection onto M . Moreover, the projection is characterized by the orthogonality condition $X - \hat{X} \perp M$.

Since $Y_1, \dots, Y_n$ span M , this condition is equivalent to $\langle X - \hat{X}, Y_i \rangle_{\mathcal{H}} = 0$ for $i = 1, \dots, n$, that is,

$$\int_{\Omega} \left(X(\omega) - \sum_{j=1}^n \alpha_j Y_j(\omega) \right) Y_i(\omega) d\mathbb{P}(\omega) = 0, \quad i = 1, \dots, n.$$

Normal equations. Let $\alpha = (\alpha_1, \dots, \alpha_n)^\top$ and $Y = (Y_1, \dots, Y_n)^\top$. Define the covariance matrix

$$R := \mathbb{E}[YY^\top] = \int_{\Omega} Y(\omega)Y(\omega)^\top d\mathbb{P}(\omega),$$

and the cross-correlation vector

$$r := \mathbb{E}[XY] = \int_{\Omega} X(\omega)Y(\omega) d\mathbb{P}(\omega).$$

The orthogonality conditions yield the normal equations $R\alpha = r$. If R is invertible, the solution is $\alpha = R^{-1}r$, and hence

$$\hat{X}(\omega) = \mathbb{E}[XY^\top] \mathbb{E}[YY^\top]^{-1}Y(\omega), \quad \omega \in \Omega.$$

Geometric interpretation. The linear least squares estimator $\hat{X}$ is the orthogonal projection of X onto the finite-dimensional subspace $M \subset \mathcal{H}$. The estimation error $X - \hat{X}$ is orthogonal to every linear combination of the observations, meaning that no further linear adjustment based on $(Y_1, \dots, Y_n)$ can reduce the mean-square error.

Connection to conditional expectation. If we allow estimators to be arbitrary square-integrable measurable functions of the observations, it is cleaner to work in the full Hilbert space $L^2(\Omega, \mathcal{F}, \mathbb{P})$. The relevant closed subspace is

$$\mathcal{M} := \{Z \in L^2(\Omega) \mid Z \text{ is } \sigma(Y_1, \dots, Y_n)\text{-measurable}\},$$

where $\sigma(Y_1, \dots, Y_n)$ is the information generated by the observations. In this case, the projection theorem yields $\hat{X} = \mathbb{E}[X \mid Y_1, \dots, Y_n]$, recovering the MMSE estimator as the orthogonal projection of X onto the space of all square-integrable functions of the observations. When $\mathbb{E}[X] = 0$, this projection also has mean zero.

Note 13.18 (Two estimator classes). Linear MMSE restricts the estimate to the span of the observations, so the subspace is finite-dimensional and the answer is found from normal equations. Full MMSE allows every square-integrable function of the observations, so the subspace is larger and the projection is the conditional expectation $\mathbb{E}[X \mid Y_1, \dots, Y_n]$.

13.3 ■ The Gram–Schmidt Procedure

The projection theorem tells us what the best approximation must satisfy. Gram–Schmidt tells us how to choose coordinates that make the projection easy to compute. Instead of working with correlated or non-orthogonal directions, we replace them by orthogonal directions that span the same subspaces.

Definition 13.19 (Orthogonal and orthonormal sets). Let $(X, \langle \cdot, \cdot \rangle)$ be a pre-Hilbert space.

A subset $S \subset X$ is said to be orthogonal if $\langle x, y \rangle = 0$ for all distinct $x, y \in S$.

The set S is said to be orthonormal if it is orthogonal and, in addition, $\|x\| = 1$ for all $x \in S$.

Orthogonal sets generalize the notion of mutually perpendicular vectors in Euclidean space. Their importance stems from the geometric simplifications induced by orthogonality: inner products between distinct directions vanish, norms decompose additively, and many optimization and approximation problems decouple across coordinates.

In particular, if $\{e_i\}_{i \in I}$ is an orthonormal set and $x = \sum_{i \in I} \alpha_i e_i$ is a finite linear combination, then

$$\|x\|^2 = \left\langle \sum_i \alpha_i e_i, \sum_j \alpha_j e_j \right\rangle = \sum_i |\alpha_i|^2,$$

which is the abstract form of the Pythagorean identity familiar from $\mathbb{R}^n$.

Proposition 13.20 (Orthogonal sets are linearly independent). *Let X be a pre-Hilbert space and let $S \subset X$ be an orthogonal set, $\langle x, y \rangle = 0$ for all distinct $x, y \in S$, none of whose elements is zero. Then S is linearly independent. Orthogonality is thus a strictly stronger requirement than independence, which is what makes orthogonal families usable as coordinate systems.*

Proof. Let $\{x_1, \dots, x_n\}$ be a finite orthogonal set with $x_i \neq 0$ for all i , and suppose that $\sum_{i=1}^n \alpha_i x_i = 0$ for some scalars $\alpha_1, \dots, \alpha_n$. Taking the inner product with x_k yields

$$\left\langle \sum_{i=1}^n \alpha_i x_i, x_k \right\rangle = \alpha_k \langle x_k, x_k \rangle,$$

since $\langle x_i, x_k \rangle = 0$ for $i \neq k$. Because $\langle x_k, x_k \rangle > 0$, it follows that $\alpha_k = 0$. As this holds for every $k = 1, \dots, n$, all coefficients vanish and the vectors are linearly independent. $\square$

13.3.1 ■ The Gram–Schmidt Theorem

The Gram–Schmidt procedure provides a systematic method for converting a linearly independent family of vectors into an orthonormal family, while preserving the subspaces generated at each finite stage.

Theorem 13.21 (Gram–Schmidt orthonormalization). *Let X be a pre-Hilbert space and let $\{x_i\}_{i \in I}$ be a finite or countable sequence of linearly independent vectors in X . Then there exists an orthonormal sequence $\{e_i\}_{i \in I}$ such that, for every positive integer n ,*

$$\text{span}\{e_1, \dots, e_n\} = \text{span}\{x_1, \dots, x_n\}.$$

Proof. We construct the sequence $\{e_i\}$ inductively.

First, define $e_1 := \frac{x_1}{\|x_1\|}$, which is well-defined since $x_1 \neq 0$.

Suppose that orthonormal vectors $e_1, \dots, e_{n-1}$ have been constructed and satisfy $\text{span}\{e_1, \dots, e_{n-1}\} = \text{span}\{x_1, \dots, x_{n-1}\}$. Define

$$z_n := x_n - \sum_{i=1}^{n-1} \langle x_n, e_i \rangle e_i, \quad e_n := \frac{z_n}{\|z_n\|}.$$

By construction, z_n is orthogonal to $e_1, \dots, e_{n-1}$. If $z_n = 0$, then

$$x_n = \sum_{i=1}^{n-1} \langle x_n, e_i \rangle e_i \in \text{span}\{x_1, \dots, x_{n-1}\},$$

which contradicts the linear independence of $\{x_i\}$. Hence $z_n \neq 0$, and e_n is well-defined.

It follows that $\{e_i\}$ is an orthonormal sequence. Moreover, since each x_n is a linear combination of $e_1, \dots, e_n$ and each e_n lies in $\text{span}\{x_1, \dots, x_n\}$, we obtain $\text{span}\{e_1, \dots, e_n\} = \text{span}\{x_1, \dots, x_n\}$ for all n , completing the proof. $\square$

At each step, the vector x_n is decomposed as

$$x_n = \underbrace{\sum_{i=1}^{n-1} \langle x_n, e_i \rangle e_i}_{\text{orthogonal projection onto the existing subspace}} + \underbrace{z_n}_{\text{orthogonal remainder}}.$$

The Gram–Schmidt procedure removes the component of x_n lying in the previously constructed subspace and retains only the new orthogonal direction, which is then normalized to unit length. In this way, redundancy is eliminated while preserving the geometry of the generated subspaces.

Note 13.22 (Information viewpoint). In estimation, the vector z_n is an *innovation*: it is the part of the new observation that could not already be predicted from the previous observations. If $z_n = 0$, the new observation is redundant for linear least-squares estimation.

13.3.2 ■ Approximation and Normal Equations

Let H be a Hilbert space and let $y_1, \dots, y_n \in H$ generate the finite-dimensional subspace $M := \text{span}\{y_1, \dots, y_n\} \subset H$. Given a vector $x \in H$, we consider the *best approximation problem* $\min_{\hat{x} \in M} \|x - \hat{x}\|$.

Since M is finite-dimensional, it is closed in H , and the Projection Theorem guarantees the existence of a unique minimizer $\hat{x} \in M$.

Writing $\hat{x}$ in coordinates relative to the spanning family, $\hat{x} = \sum_{j=1}^n \alpha_j y_j$ with $\alpha_j \in \mathbb{F}$, the orthogonality characterization of the projection implies $x - \hat{x} \perp M$. Equivalently, $\langle x - \hat{x}, y_i \rangle = 0$ for $i = 1, \dots, n$. Substituting the expansion of $\hat{x}$ yields the linear system

$$\sum_{j=1}^n \langle y_j, y_i \rangle \alpha_j = \langle x, y_i \rangle, \quad i = 1, \dots, n.$$

The above system of n linear equations in the unknown coefficients $\alpha_1, \dots, \alpha_n$ is called the *normal equations*. These equations encode the geometric condition that the residual $x - \hat{x}$ is orthogonal to the approximation subspace M .

Reading the normal equations. Each equation says that the error has zero inner product with one spanning direction y_i . If the spanning vectors are not orthogonal, the equations are coupled through the Gram matrix. If they are orthogonal, the equations decouple and each coefficient is just “correlation divided by energy.”

13.3.3 ■ Gram Matrices

Definition 13.23 (Gram matrix). Let $y_1, \dots, y_n$ be vectors in a pre-Hilbert space. The $n \times n$ matrix

$$G = G(y_1, \dots, y_n) := (\langle y_j, y_i \rangle)_{i,j=1}^n$$

is called the *Gram matrix* associated with $y_1, \dots, y_n$. Its determinant $\Delta(y_1, \dots, y_n) := \det G(y_1, \dots, y_n)$ is called the *Gram determinant*. The same notation is used for any finite family: $G(z_1, \dots, z_k) := (\langle z_j, z_i \rangle)_{i,j=1}^k$ and $\Delta(z_1, \dots, z_k) := \det G(z_1, \dots, z_k)$.

In a real inner product space, the Gram matrix is symmetric; in a complex inner product space, it is Hermitian. In either case, G is positive semidefinite.

Proposition 13.24 (Gram determinant test for independence). *Let $y_1, \dots, y_n$ be vectors in a pre-Hilbert space and let $\Delta(y_1, \dots, y_n) = \det G(y_1, \dots, y_n)$ be their Gram determinant. Then $\Delta(y_1, \dots, y_n) \neq 0$ if and only if $y_1, \dots, y_n$ are linearly independent; equivalently, the Gram matrix is positive definite exactly on independent families and singular otherwise. This is the criterion under which the normal equations of Section 13.3.2 have a unique solution.*

Proof. If the vectors are linearly dependent, then there exists a nonzero vector $\alpha = (\alpha_1, \dots, \alpha_n)^\top$ such that $\sum_{i=1}^n \alpha_i y_i = 0$. Taking inner products with y_j yields $G\alpha = 0$, hence $\det G = 0$.

Conversely, if $\det G = 0$, then there exists a nonzero α such that $G\alpha = 0$. Consequently,

$$\left\| \sum_{i=1}^n \alpha_i y_i \right\|^2 = \alpha^* G \alpha = 0,$$

which implies $\sum_{i=1}^n \alpha_i y_i = 0$ and hence linear dependence. $\square$

13.3.4 ■ Distance Formula via Gram Determinants

We now derive an explicit expression for the squared distance from a vector to a finite-dimensional subspace in terms of Gram determinants. This result provides a closed-form characterization of the minimum approximation error and clarifies its geometric structure.

Theorem 13.25 (Distance to a subspace via Gram determinants). *Let H be a real Hilbert space and let $y_1, \dots, y_n \in H$ be linearly independent vectors. Define the subspace $M := \text{span}\{y_1, \dots, y_n\}$. For any $x \in H$, define the distance*

$$\delta := \min_{\alpha_1, \dots, \alpha_n \in \mathbb{R}} \left\| x - \sum_{i=1}^n \alpha_i y_i \right\|.$$

Then

$$\delta^2 = \frac{\Delta(y_1, \dots, y_n, x)}{\Delta(y_1, \dots, y_n)}, \quad (13.1)$$

where $\Delta(y_1, \dots, y_n)$ denotes the Gram determinant of $\{y_1, \dots, y_n\}$ and $\Delta(y_1, \dots, y_n, x)$ denotes the Gram determinant of the $(n+1)$ -tuple $\{y_1, \dots, y_n, x\}$.

Proof. Let $\hat{x}$ denote the orthogonal projection of x onto M . By the Projection Theorem, $\delta^2 = \|x - \hat{x}\|^2$.

Since $\hat{x} \in M$, it admits a representation

$$\hat{x} = \sum_{i=1}^n \alpha_i y_i,$$

where the coefficient vector $\alpha = (\alpha_1, \dots, \alpha_n)^\top$ satisfies the normal equations

$$\sum_{j=1}^n \langle y_j, y_i \rangle \alpha_j = \langle x, y_i \rangle, \quad i = 1, \dots, n. \quad (13.2)$$

In matrix form,

$$G(y_1, \dots, y_n) \alpha = \begin{bmatrix} \langle x, y_1 \rangle \\ \vdots \\ \langle x, y_n \rangle \end{bmatrix}.$$

Because $y_1, \dots, y_n$ are linearly independent, the Gram matrix $G(y_1, \dots, y_n)$ is invertible, and the system has a unique solution.

Next, observe that orthogonality of the residual implies

$$x - \hat{x} \perp M, \quad \text{hence} \quad \delta^2 = \langle x - \hat{x}, x \rangle.$$

Substituting $\hat{x} = \sum_{i=1}^n \alpha_i y_i$ yields

$$\delta^2 = \langle x, x \rangle - \sum_{i=1}^n \alpha_i \langle x, y_i \rangle.$$

Write $G := G(y_1, \dots, y_n)$ and collect the right-hand sides of (13.2) into the vector $b := (\langle x, y_1 \rangle, \dots, \langle x, y_n \rangle)^\top$, so that the normal equations read $G\alpha = b$ and, since G is invertible, $\alpha = G^{-1}b$. Substituting into the displayed expression for δ^2 gives

$$\delta^2 = \langle x, x \rangle - \sum_{i=1}^n \alpha_i \langle x, y_i \rangle = \langle x, x \rangle - b^\top \alpha = \langle x, x \rangle - b^\top G^{-1}b. \quad (13.3)$$

It remains to recognize the right-hand side as a ratio of determinants. Ordering the $(n+1)$ -tuple as $(y_1, \dots, y_n, x)$ and using $\langle y_i, x \rangle = \langle x, y_i \rangle$ in the real space H , the enlarged Gram matrix has the block form

$$G(y_1, \dots, y_n, x) = \begin{bmatrix} G & b \\ b^\top & \langle x, x \rangle \end{bmatrix},$$

whose leading block G is invertible. The determinant of a block matrix with an invertible leading block factors through its Schur complement,

$$\det \begin{bmatrix} G & b \\ b^\top & \langle x, x \rangle \end{bmatrix} = \det G \cdot (\langle x, x \rangle - b^\top G^{-1}b),$$

so that, by (13.3), $\Delta(y_1, \dots, y_n, x) = \Delta(y_1, \dots, y_n) \delta^2$. Dividing by $\Delta(y_1, \dots, y_n) \neq 0$, which is nonzero because $y_1, \dots, y_n$ are linearly independent, yields (13.1). $\square$

The numerator $\Delta(y_1, \dots, y_n, x)$ measures the squared $(n+1)$ -dimensional volume of the parallelotope spanned by $\{y_1, \dots, y_n, x\}$, while the denominator $\Delta(y_1, \dots, y_n)$ measures the squared n -dimensional volume spanned by $\{y_1, \dots, y_n\}$. Their ratio therefore quantifies the energy of the component of x orthogonal to M .

Applying Gram–Schmidt orthogonalization to $\{y_1, \dots, y_n\}$ diagonalizes the Gram matrix, reducing the determinant ratio in (13.1) to a sum of squared orthogonal residuals. This explains why orthonormal bases yield numerically stable projection formulas.

Although determinant formulas provide valuable theoretical insight, they are numerically unstable for large n . In practice, one computes δ^2 using orthonormal bases obtained via Gram–Schmidt or QR factorization, which replace the determinant ratio by a simple Pythagorean decomposition.

13.3.5 ■ Two Observations: From Batch to Recursive Estimation

Before treating many observations, it is helpful to see the idea with two. The batch solution uses (Y_1, Y_2) together and solves one coupled system. The recursive solution first keeps the part explained by Y_1 , then adds only the new part of Y_2 that is orthogonal to Y_1 . Both methods produce the same projection; the recursive form is easier to update.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and define the real Hilbert space

$$\mathcal{H} := \{Z \in L^2(\Omega, \mathcal{F}, \mathbb{P}) \mid \mathbb{E}[Z] = 0\}, \quad \langle U, V \rangle := \mathbb{E}[UV].$$

Let $X, Y_1, Y_2 \in \mathcal{H}$, where X is the quantity to be estimated and Y_1, Y_2 are two available observations.

We consider linear MMSE estimation of X from (Y_1, Y_2) . The linear estimator is sought in the form

$$\hat{X} = \alpha_1 Y_1 + \alpha_2 Y_2 \in \text{span}\{Y_1, Y_2\}, \quad (13.4)$$

where $\text{span}\{Y_1, Y_2\} := \text{span}\{Y_1, Y_2\} \subset \mathcal{H}$.

By the Projection Theorem, $\hat{X}$ is characterized by the orthogonality condition $X - \hat{X} \perp \text{span}\{Y_1, Y_2\}$, which yields the normal equations

$$\begin{bmatrix} \langle X, Y_1 \rangle \\ \langle X, Y_2 \rangle \end{bmatrix} = \begin{bmatrix} \langle Y_1, Y_1 \rangle & \langle Y_2, Y_1 \rangle \\ \langle Y_1, Y_2 \rangle & \langle Y_2, Y_2 \rangle \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}. \quad (13.5)$$

This is the standard *batch* MMSE solution expressed in the (generally non-orthogonal) coordinate system $\{Y_1, Y_2\}$.

If $Y_1 \perp Y_2$, then the Gram matrix in (13.5) is diagonal and the estimator decomposes as

$$\hat{X} = \frac{\langle X, Y_1 \rangle}{\langle Y_1, Y_1 \rangle} Y_1 + \frac{\langle X, Y_2 \rangle}{\langle Y_2, Y_2 \rangle} Y_2. \quad (13.6)$$

This observation motivates an orthogonalization of the observation space.

Orthogonal Decomposition of Observations

Definition 13.26 (Innovation variable). Let $Y_1, Y_2 \in \mathcal{H}$ with $Y_1 \neq 0$. Define the innovation of Y_2 relative to Y_1 as

$$\tilde{Y}_2 := Y_2 - \Pi_{\text{span}\{Y_1\}} Y_2 = Y_2 - \frac{\langle Y_2, Y_1 \rangle}{\langle Y_1, Y_1 \rangle} Y_1. \quad (13.7)$$

Lemma 13.27 (Innovation orthogonality). The innovation $\tilde{Y}_2$ satisfies

$$\langle \tilde{Y}_2, Y_1 \rangle = 0, \quad \text{span}\{Y_1, Y_2\} = \text{span}\{Y_1, \tilde{Y}_2\}.$$

Proof. Orthogonality follows by direct computation:

$$\langle \tilde{Y}_2, Y_1 \rangle = \langle Y_2, Y_1 \rangle - \frac{\langle Y_2, Y_1 \rangle}{\langle Y_1, Y_1 \rangle} \langle Y_1, Y_1 \rangle = 0.$$

Since $\tilde{Y}_2$ differs from Y_2 by an element of $\text{span}\{Y_1\}$, both pairs generate the same linear subspace. $\square$

The transformation $(Y_1, Y_2) \mapsto (Y_1, \tilde{Y}_2)$ is exactly the two-dimensional Gram–Schmidt orthogonalization of the ordered family $\{Y_1, Y_2\}$.

Recursive MMSE Estimator Decomposition

Since $\hat{X} \in \text{span}\{Y_1, \tilde{Y}_2\}$, we may write

$$\hat{X} = \beta_1 Y_1 + \beta_2 \tilde{Y}_2. \quad (13.8)$$

Applying the orthogonality principle $X - \hat{X} \perp \text{span}\{Y_1, \tilde{Y}_2\}$, and using $Y_1 \perp \tilde{Y}_2$, we obtain the diagonal normal equations

$$\begin{bmatrix} \langle X, Y_1 \rangle \\ \langle X, \tilde{Y}_2 \rangle \end{bmatrix} = \begin{bmatrix} \langle Y_1, Y_1 \rangle & 0 \\ 0 & \langle \tilde{Y}_2, \tilde{Y}_2 \rangle \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix}. \quad (13.9)$$

If $\tilde{Y}_2 \neq 0$, then

$$\beta_1 = \frac{\langle X, Y_1 \rangle}{\langle Y_1, Y_1 \rangle}, \quad \beta_2 = \frac{\langle X, \tilde{Y}_2 \rangle}{\langle \tilde{Y}_2, \tilde{Y}_2 \rangle}. \quad (13.10)$$

Substitution into (13.8) yields the recursive MMSE estimator

$$\hat{X} = \underbrace{\frac{\langle X, Y_1 \rangle}{\langle Y_1, Y_1 \rangle} Y_1}_{\hat{X}_1} + \underbrace{\frac{\langle X, \tilde{Y}_2 \rangle}{\langle \tilde{Y}_2, \tilde{Y}_2 \rangle} \tilde{Y}_2}_{\hat{X}_2}. \quad (13.11)$$

If $\tilde{Y}_2 = 0$, then $Y_2 \in \text{span}\{Y_1\}$ and the second observation adds no new linear information; the correction term is omitted.

The estimator $\hat{X}_1$ is the MMSE estimate based solely on Y_1 . The second term $\hat{X}_2$ is the incremental correction obtained from the innovation $\tilde{Y}_2$, which contains exactly the component of Y_2 orthogonal to previously used information. This representation is algebraically equivalent to the batch solution (13.4)–(13.5), but expressed in the orthogonal Gram–Schmidt coordinate system.

13.3.6 ■ Multiple Observations: Iterative MMSE via Gram–Schmidt

We generalize the two-observation batch MMSE solution to an arbitrary number of observations. The key idea is to orthogonalize the observation space using the Gram–Schmidt procedure, which yields a recursive MMSE estimator and an additive decomposition of the minimum mean-square error.

Hilbert space setting. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and define the real Hilbert space

$$\mathcal{H} := \{Z \in L^2(\Omega, \mathcal{F}, \mathbb{P}) : \mathbb{E}[Z] = 0\}, \quad \langle U, V \rangle := \mathbb{E}[UV].$$

Let $X, Y_1, \dots, Y_n \in \mathcal{H}$ be square-integrable, zero-mean random variables. In this subsection we assume that the observations $Y_1, \dots, Y_n$ are linearly independent in $\mathcal{H}$; if not, redundant observations can be removed without changing the span. Define the linear estimation subspace $M_n := \text{span}\{Y_1, \dots, Y_n\} \subset \mathcal{H}$.

The batch linear MMSE estimator is the orthogonal projection

$$\hat{X}_n = \arg \min_{Z \in M_n} \|X - Z\|_{\mathcal{H}}^2 = \Pi_{M_n} X.$$

Orthogonalization of the observation space

Lemma 13.28 (Gram–Schmidt innovations). *Define recursively*

$$\tilde{Y}_1 := Y_1, \quad (13.12)$$

$$\tilde{Y}_k := Y_k - \sum_{i=1}^{k-1} \frac{\langle Y_k, \tilde{Y}_i \rangle}{\langle \tilde{Y}_i, \tilde{Y}_i \rangle} \tilde{Y}_i, \quad k = 2, \dots, n. \quad (13.13)$$

Then the following hold:

- (1) $\langle \tilde{Y}_i, \tilde{Y}_j \rangle = 0$ for $i \neq j$;
- (2) $\text{span}\{Y_1, \dots, Y_k\} = \text{span}\{\tilde{Y}_1, \dots, \tilde{Y}_k\}$ for all k ;
- (3) $\{\tilde{Y}_1, \dots, \tilde{Y}_n\}$ is an orthogonal basis of M_n .

Proof. This is precisely the Gram–Schmidt procedure applied to the ordered family $\{Y_1, \dots, Y_n\}$. Orthogonality follows by construction. Each $\tilde{Y}_k$ differs from Y_k by an element of $\text{span}\{Y_1, \dots, Y_{k-1}\}$, hence the spans are preserved at every step. $\square$

Recursive representation of the MMSE estimator

Theorem 13.29 (Recursive MMSE estimator). *Let $\{\tilde{Y}_1, \dots, \tilde{Y}_n\}$ be the orthogonal innovations from Lemma 13.28. Then the linear MMSE estimator admits the expansion*

$$\hat{X}_n = \sum_{k=1}^n \frac{\langle X, \tilde{Y}_k \rangle}{\langle \tilde{Y}_k, \tilde{Y}_k \rangle} \tilde{Y}_k. \quad (13.14)$$

Moreover, $\hat{X}_n$ satisfies the recursion

$$\hat{X}_k = \hat{X}_{k-1} + \frac{\langle X, \tilde{Y}_k \rangle}{\langle \tilde{Y}_k, \tilde{Y}_k \rangle} \tilde{Y}_k, \quad k = 1, \dots, n,$$

with $\hat{X}_0 := 0$.

Proof. Since $\hat{X}_n \in M_n$ and $\{\tilde{Y}_k\}$ is an orthogonal basis of M_n , we may write $\hat{X}_n = \sum_{k=1}^n \beta_k \tilde{Y}_k$. By the projection theorem, $X - \hat{X}_n \perp M_n$, which is equivalent to $\langle X - \hat{X}_n, \tilde{Y}_k \rangle = 0$ for $k = 1, \dots, n$. Substituting the expansion yields $\langle X, \tilde{Y}_k \rangle = \beta_k \langle \tilde{Y}_k, \tilde{Y}_k \rangle$, and hence

$$\beta_k = \frac{\langle X, \tilde{Y}_k \rangle}{\langle \tilde{Y}_k, \tilde{Y}_k \rangle}.$$

The recursive form follows immediately by grouping terms. $\square$

Additive decomposition of the MMSE error

Theorem 13.30 (MMSE error decomposition). *Let $\hat{X}_n$ be given by (13.14). Define the minimum mean-square error $\delta_n^2 := \mathbb{E}[(X - \hat{X}_n)^2]$. Then*

$$\delta_n^2 = \mathbb{E}[X^2] - \sum_{k=1}^n \frac{\langle X, \tilde{Y}_k \rangle^2}{\langle \tilde{Y}_k, \tilde{Y}_k \rangle}. \quad (13.15)$$

In particular, δ_n^2 is nonincreasing in n .

Proof. Since $X - \hat{X}_n \perp M_n$, the Pythagorean identity gives

$$\|X\|_{\mathcal{H}}^2 = \|\hat{X}_n\|_{\mathcal{H}}^2 + \|X - \hat{X}_n\|_{\mathcal{H}}^2.$$

Because the $\tilde{Y}_k$ are orthogonal,

$$\|\hat{X}_n\|_{\mathcal{H}}^2 = \sum_{k=1}^n \frac{\langle X, \tilde{Y}_k \rangle^2}{\langle \tilde{Y}_k, \tilde{Y}_k \rangle}.$$

Substituting into the identity yields (13.15). Each summand is nonnegative, hence the error decreases monotonically. $\square$

The Gram–Schmidt procedure converts correlated observations into orthogonal *innovations*. Each innovation explains an independent component of X , and the MMSE is exactly the energy of the component of X orthogonal to the observation subspace.

The recursion in Theorem 13.29 is the static analogue of innovation-based filters such as the Kalman filter, where new information updates the estimate through orthogonal residuals.

Note 13.31 (What to remember from Gram–Schmidt). The batch estimator and the recursive estimator are the same projection written in different coordinates. Batch coordinates use the original observations and lead to a Gram matrix. Recursive coordinates use innovations and lead to orthogonal correction terms.

13.4 ■ Estimation of Random Variables and Vectors

This section summarizes the estimation results developed earlier from a Hilbert-space perspective and specializes them to random variables. Rather than repeating proofs, we recall the main principles and illustrate them through representative examples.

The main distinction is the size of the estimator class. MMSE estimation uses all square-integrable functions of the observations. LLSE estimation uses only linear combinations of the observations. The LLSE is easier to compute; the MMSE is generally better unless the conditional expectation happens to be linear.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $X, Y_1, \dots, Y_n \in L^2(\Omega)$ be real-valued random variables with finite second moments. Unless stated otherwise, we assume $\mathbb{E}[X] = 0$ and $\mathbb{E}[Y_i] = 0$ for $i = 1, \dots, n$.

13.4.1 ■ Mean-Square Error and Optimal Estimation

An estimator of X based on $(Y_1, \dots, Y_n)$ is any random variable of the form $\hat{X} = g(Y_1, \dots, Y_n)$, where g is measurable. Its performance is measured by the mean-square error $\mathbb{E}[(X - \hat{X})^2]$.

Definition 13.32 (MMSE estimator). An estimator $\hat{X}^*$ is called a minimum mean-square error (MMSE) estimator if

$$\mathbb{E}[(X - \hat{X}^*)^2] \leq \mathbb{E}[(X - \hat{X})^2] \quad \text{for all estimators } \hat{X}.$$

13.4.2 ■ Hilbert-Space Characterization

For unrestricted MMSE estimation, use the real Hilbert space $L^2(\Omega)$ with inner product $\langle U, V \rangle := \mathbb{E}[UV]$. Let

$$\mathcal{M} := \{Z \in L^2(\Omega) : Z \text{ is } \sigma(Y_1, \dots, Y_n)\text{-measurable}\}$$

denote the closed subspace of square-integrable random variables determined by the observations.

Theorem 13.33 (MMSE as orthogonal projection). The MMSE estimator of X based on $(Y_1, \dots, Y_n)$ is given by

$$\hat{X}_{\text{MMSE}} = \Pi_{\mathcal{M}} X = \mathbb{E}[X \mid Y_1, \dots, Y_n],$$

that is, the orthogonal projection of X onto $\mathcal{M}$ (cf. the Projection Theorem and Section 13.2.4).

Thus, MMSE estimation is a best-approximation problem in $L^2(\Omega)$.

Note 13.34 (Why conditional expectation appears). The notation $\mathbb{E}[X \mid Y_1, \dots, Y_n]$ can look probabilistic rather than geometric. In this chapter it should be read as a projection: it is the closest square-integrable random variable to X among all random variables whose value is determined by the observations.

13.4.3 ■ Linear Least Squares Estimation

Restricting attention to linear estimators yields a finite-dimensional approximation problem. For the zero-mean variables assumed in this section, we work in the closed subspace

$$\mathcal{H} := \{Z \in L^2(\Omega) : \mathbb{E}[Z] = 0\}, \quad \langle U, V \rangle := \mathbb{E}[UV].$$

Definition 13.35 (Linear estimator). A linear estimator of X based on $Y_1, \dots, Y_n$ is an estimator of the form

$$\hat{X} = \sum_{i=1}^n \alpha_i Y_i, \quad \alpha_i \in \mathbb{R}.$$

Let

$$\text{span}\{Y_1, \dots, Y_n\} := \text{span}\{Y_1, \dots, Y_n\} \subset \mathcal{H}.$$

Theorem 13.36 (Linear least squares estimator). The linear least squares estimator (LLSE) of X is the orthogonal projection of X onto $\text{span}\{Y_1, \dots, Y_n\}$. It is uniquely characterized by the orthogonality conditions $\mathbb{E}[(X - \hat{X})Y_i] = 0$ for $i = 1, \dots, n$, equivalently by the normal equations

$$\sum_{j=1}^n \mathbb{E}[Y_j Y_i] \alpha_j = \mathbb{E}[X Y_i], \quad i = 1, \dots, n.$$

In vector form, letting $Y = (Y_1, \dots, Y_n)^\top$, the estimator is

$$\hat{X} = \mathbb{E}[XY^\top] \mathbb{E}[YY^\top]^{-1} Y,$$

whenever $\mathbb{E}[YY^\top]$ is invertible.

Note 13.37 (MMSE versus LLSE). The LLSE is the best estimator only within the linear class. The MMSE is the best estimator among all measurable functions of the observations. They agree when the conditional expectation is linear, as happens for jointly Gaussian random vectors.

Example 13.38 Truncated exponential model.

Let (X, Y) have joint density $f_{X,Y}(x, y) = \lambda^2 e^{-\lambda x}$ for $0 \leq y \leq x$. A direct calculation yields $\mathbb{E}[X | Y] = Y + \frac{1}{\lambda}$. Thus, the MMSE estimator is an affine function of the observation. $\square$

Example 13.39 Nonlinear observation with exponential noise.

Let

$$X \sim \text{Unif}[0, 1], \quad Y = \log \frac{1}{X} + V, \quad V \sim \text{Exp}(1),$$

with X and V independent. The MMSE estimator is

$$\mathbb{E}[X | Y] = \frac{1 - e^{-Y}}{Y},$$

illustrating that the optimal estimator is generally nonlinear in the observations. $\square$

13.4.4 ■ Vector Estimation

Let $X : \Omega \rightarrow \mathbb{R}^n$ and $Y : \Omega \rightarrow \mathbb{R}^m$ be random vectors defined on a common probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and satisfying the finite-energy conditions $\mathbb{E}[\|X\|^2] < \infty$ and $\mathbb{E}[\|Y\|^2] < \infty$. Unless stated otherwise we assume that both vectors are centered, $\mathbb{E}[X] = 0$ and $\mathbb{E}[Y] = 0$; the nonzero-mean case is recovered by centering and is treated in Theorem 13.50.

Our goal is to reconstruct the unobserved vector X from the observation Y using a linear rule, whose quality we measure by the resulting mean-square error. We therefore begin by fixing the class of estimators under consideration.

Definition 13.40 (Linear vector estimator). A linear estimator of X based on Y is a random vector of the form

$$\hat{X} = AY, \tag{13.16}$$

where $A \in \mathbb{R}^{n \times m}$.

The estimation performance is measured by the mean-square error

$$\mathbb{E}[\|X - \hat{X}\|^2]. \tag{13.17}$$

The vector linear least squares estimation problem is

$$\min_{A \in \mathbb{R}^{n \times m}} \mathbb{E}[\|X - AY\|^2]. \tag{13.18}$$

As a dimension check, note that X has n components and Y has m components, so the estimator matrix A must be $n \times m$ in order for AY to have the same dimension as X ; keeping this bookkeeping in mind is helpful when reading the covariance formulas below.

To solve problem (13.18) we recast it as a best-approximation problem in a suitable Hilbert space, so that the orthogonal projection machinery developed in the scalar case applies verbatim. Define the real Hilbert space

$$\mathcal{H}_X := \left\{ Z : \Omega \rightarrow \mathbb{R}^n \mid \mathbb{E}[Z] = 0, \mathbb{E}[\|Z\|^2] < \infty \right\},$$

equipped with the inner product

$$\langle Z_1, Z_2 \rangle_{\mathcal{H}_X} := \mathbb{E}[Z_1^\top Z_2]. \quad (13.19)$$

The induced norm is $\|Z\|_{\mathcal{H}_X} = (\mathbb{E}[\|Z\|^2])^{1/2}$.

Definition 13.41 (Linear estimation subspace). Define the linear subspace

$$\mathcal{M} := \{AY : A \in \mathbb{R}^{n \times m}\} \subset \mathcal{H}_X.$$

Since $\mathcal{M}$ is finite-dimensional, it is closed in $\mathcal{H}_X$.

Proposition 13.42 (Geometric formulation). Let $\mathcal{H}_X$ be the Hilbert space (13.19) of centered, square-integrable $\mathbb{R}^n$ -valued random vectors and let $\mathcal{M} = \{AY : A \in \mathbb{R}^{n \times m}\}$ be the linear estimation subspace. Then the matrix optimization problem (13.18) is equivalent to the best-approximation problem $\min_{\hat{X} \in \mathcal{M}} \|X - \hat{X}\|_{\mathcal{H}_X}$, in the sense that A minimizes $\mathbb{E}\|X - AY\|^2$ if and only if AY attains this minimum distance. The projection theorem of Section 13.2 therefore applies verbatim to the vector estimation problem.

Orthogonality principle

Theorem 13.43 (Vector projection theorem). There exists a unique linear estimator $\hat{X} \in \mathcal{M}$ minimizing (13.18). Moreover, $\hat{X} = AY$ is optimal if and only if

$$X - \hat{X} \perp \mathcal{M}, \quad (13.20)$$

that is,

$$\mathbb{E}[(X - AY)Y^\top] = 0. \quad (13.21)$$

Proof. The subspace $\mathcal{M}$ is finite-dimensional, hence closed in the Hilbert space $\mathcal{H}_X$, so the Projection Theorem yields a unique minimizer $\hat{X} \in \mathcal{M}$ of the map $\hat{X} \mapsto \|X - \hat{X}\|_{\mathcal{H}_X}$. It remains to establish the orthogonality characterization (13.20) directly. Fix a candidate $\hat{X} = AY \in \mathcal{M}$ and an arbitrary perturbation direction $BY \in \mathcal{M}$ with $B \in \mathbb{R}^{n \times m}$. Expanding the squared norm gives, for every $t \in \mathbb{R}$,

$$\|X - AY - tBY\|_{\mathcal{H}_X}^2 = \|X - AY\|_{\mathcal{H}_X}^2 - 2t \langle X - AY, BY \rangle_{\mathcal{H}_X} + t^2 \|BY\|_{\mathcal{H}_X}^2.$$

If $\hat{X} = AY$ is optimal, this quadratic in t is minimized at $t = 0$; setting its derivative at $t = 0$ to zero forces $\langle X - AY, BY \rangle_{\mathcal{H}_X} = 0$. Conversely, if this inner product vanishes for every B , then choosing $t = 1$ gives $\|X - AY - BY\|_{\mathcal{H}_X}^2 = \|X - AY\|_{\mathcal{H}_X}^2 + \|BY\|_{\mathcal{H}_X}^2 \geq \|X - AY\|_{\mathcal{H}_X}^2$ for

all $BY \in \mathcal{M}$, so AY is optimal. Optimality is therefore equivalent to $X - \hat{X} \perp \mathcal{M}$, which is (13.20).

To convert this into the moment condition, we expand the inner product. Since the scalar $(X - AY)^\top BY$ equals its own trace, cyclicity of the trace gives

$$\langle X - AY, BY \rangle_{\mathcal{H}_X} = \mathbb{E}[(X - AY)^\top BY] = \text{tr}(B^\top \mathbb{E}[(X - AY)Y^\top]) = \langle B, \mathbb{E}[(X - AY)Y^\top] \rangle_F,$$

where $\langle \cdot, \cdot \rangle_F$ denotes the Frobenius inner product on $\mathbb{R}^{n \times m}$. This quantity vanishes for all $B \in \mathbb{R}^{n \times m}$ if and only if the matrix $\mathbb{E}[(X - AY)Y^\top]$ is zero, which is precisely (13.21). $\square$

Normal Equations and Closed-Form Solution

Corollary 13.44 (Normal equations). *Let $\hat{X} = AY$ be the linear least squares estimator of X based on Y . Then the optimal matrix A satisfies the normal equation*

$$A \mathbb{E}[YY^\top] = \mathbb{E}[XY^\top]. \quad (13.22)$$

Proof. By Theorem 13.43, the estimator $\hat{X} = AY$ is optimal if and only if the orthogonality condition $\mathbb{E}[(X - AY)Y^\top] = 0$ holds. Expanding this expression yields $\mathbb{E}[XY^\top] - A \mathbb{E}[YY^\top] = 0$, which is equivalent to the stated normal equation. $\square$

Theorem 13.45 (Closed-form LLSE). *Assume that the covariance matrix $\mathbb{E}[YY^\top]$ is invertible. Then the normal equation (13.22) admits a unique solution, and the linear least squares estimator is given by*

$$\hat{X} = \mathbb{E}[XY^\top] \mathbb{E}[YY^\top]^{-1} Y. \quad (13.23)$$

Proof. Since $\mathbb{E}[YY^\top]$ is invertible, the linear matrix equation $A \mathbb{E}[YY^\top] = \mathbb{E}[XY^\top]$ has the unique solution $A = \mathbb{E}[XY^\top] \mathbb{E}[YY^\top]^{-1}$. Substituting this expression for A into $\hat{X} = AY$ yields the stated closed-form estimator.

Uniqueness follows directly from the invertibility of $\mathbb{E}[YY^\top]$ and from the uniqueness of the orthogonal projection onto the subspace $\mathcal{M} = \{AY\} \subset \mathcal{H}_X$. $\square$

Note 13.46 (Reading the closed form). The matrix $\mathbb{E}[XY^\top]$ measures how the target and observation move together, while $\mathbb{E}[YY^\top]^{-1}$ normalizes by the observation energy and correlation structure. Thus the estimator is “cross-covariance times observation covariance inverse times observation.”

Estimation Error and Covariance

Define the estimation error $\tilde{X} := X - \hat{X}$, where $\hat{X}$ denotes the linear least squares estimator of X based on Y .

Proposition 13.47 (Error orthogonality). *The estimation error satisfies $\mathbb{E}[\tilde{X} Y^\top] = 0$, that is, $\tilde{X}$ is uncorrelated with all linear functions of the observations.*

Proof. By construction, $\hat{X}$ is the orthogonal projection of X onto the linear subspace $\mathcal{M} = \{AY : A \in \mathbb{R}^{n \times m}\} \subset \mathcal{H}_X$. Hence, by the projection theorem, $X - \hat{X} \perp \mathcal{M}$. In particular, for every

$B \in \mathbb{R}^{n \times m}$, $\langle X - \hat{X}, BY \rangle_{\mathcal{H}_X} = 0$. Using the definition of the inner product on $\mathcal{H}_X$,

$$\langle X - \hat{X}, BY \rangle_{\mathcal{H}_X} = \mathbb{E}[(X - \hat{X})^\top BY] = \text{tr}(B^\top \mathbb{E}[\tilde{X} Y^\top]).$$

Since this holds for all matrices B , it follows that $\mathbb{E}[\tilde{X} Y^\top] = 0$, which proves the claim. $\square$

Proposition 13.48 (Error covariance). *Assume that $\mathbb{E}[YY^\top]$ is invertible. Then the covariance matrix of the estimation error $\tilde{X} := X - \hat{X}$ is*

$$\Sigma_{\tilde{X}} = \mathbb{E}[\tilde{X}\tilde{X}^\top] = \mathbb{E}[XX^\top] - \mathbb{E}[XY^\top] \mathbb{E}[YY^\top]^{-1} \mathbb{E}[YX^\top]. \quad (13.24)$$

Proof. Let

$$R_Y := \mathbb{E}[YY^\top], \quad R_{XY} := \mathbb{E}[XY^\top], \quad R_{YX} := \mathbb{E}[YX^\top] = R_{XY}^\top.$$

By Theorem 13.45, the linear least squares estimator admits the representation $\hat{X} = AY$ with $A = R_{XY} R_Y^{-1}$.

By definition of the estimation error covariance, $\Sigma_{\tilde{X}} = \mathbb{E}[(X - \hat{X})(X - \hat{X})^\top]$. Expanding the product and using linearity of expectation yields

$$\Sigma_{\tilde{X}} = \mathbb{E}[XX^\top] - \mathbb{E}[X\hat{X}^\top] - \mathbb{E}[\hat{X}X^\top] + \mathbb{E}[\hat{X}\hat{X}^\top].$$

Since $\hat{X}^\top = Y^\top A^\top$, we have $X\hat{X}^\top = (XY^\top)A^\top$, and therefore $\mathbb{E}[X\hat{X}^\top] = R_{XY} A^\top$. Using $A = R_{XY} R_Y^{-1}$ and the symmetry of R_Y , it follows that

$$A^\top = R_Y^{-1} R_{YX}, \quad \mathbb{E}[X\hat{X}^\top] = R_{XY} R_Y^{-1} R_{YX}.$$

An analogous calculation gives $\mathbb{E}[\hat{X}X^\top] = R_{XY} R_Y^{-1} R_{YX}$.

Next, observe that

$$\hat{X}\hat{X}^\top = (AY)(AY)^\top = A(YY^\top)A^\top,$$

and hence

$$\mathbb{E}[\hat{X}\hat{X}^\top] = AR_Y A^\top = R_{XY} R_Y^{-1} R_{YX}.$$

Substituting these expressions into the expansion of $\Sigma_{\tilde{X}}$, we obtain

$$\Sigma_{\tilde{X}} = \mathbb{E}[XX^\top] - R_{XY} R_Y^{-1} R_{YX},$$

which is precisely (13.24). $\square$

Proposition 13.49 (Pythagorean identity for estimation). *The following identity holds:*

$$\mathbb{E}[\|X\|^2] = \mathbb{E}[\|\hat{X}\|^2] + \mathbb{E}[\|\tilde{X}\|^2].$$

Consequently, the mean-square error $\mathbb{E}[\|\tilde{X}\|^2]$ equals the squared distance from X to the subspace $\mathcal{M}$.

Proof. Since $\hat{X} \in \mathcal{M}$ and $\tilde{X} = X - \hat{X} \perp \mathcal{M}$, the Pythagorean theorem in the Hilbert space $\mathcal{H}_X$ yields

$$\|X\|_{\mathcal{H}_X}^2 = \|\hat{X}\|_{\mathcal{H}_X}^2 + \|\tilde{X}\|_{\mathcal{H}_X}^2.$$

Recalling that $\|Z\|_{\mathcal{H}_X}^2 = \mathbb{E}[\|Z\|^2]$ for any $Z \in \mathcal{H}_X$, the stated identity follows immediately. $\square$

13.4.5 ■ Extensions

We collect several important extensions and structural properties of the vector linear least squares estimator.

These results explain when the same formula can be reused without redoing the whole projection argument: nonzero means are handled by centering, Gaussian models make LLSE equal to MMSE, and linear transformations commute with LLSE.

Nonzero mean random vectors

Theorem 13.50 (Affine linear least squares estimator). *Let $X \in \mathbb{R}^n$ and $Y \in \mathbb{R}^m$ be random vectors with finite second moments and arbitrary means $\mathbb{E}[X] = m_X$ and $\mathbb{E}[Y] = m_Y$. Define the centered random vectors $X_c := X - m_X$ and $Y_c := Y - m_Y$ (distinct from the estimation error $\tilde{X} = X - \hat{X}$ of Section 13.4.4). Then the linear least squares estimator of X based on Y is affine and given by*

$$\hat{X} = m_X + \text{cov}(X, Y) \text{cov}(Y)^{-1} (Y - m_Y), \quad (13.25)$$

provided that $\text{cov}(Y)$ is invertible.

Proof. An affine estimator of X from Y has the general form $\hat{X} = AY + b$ with $A \in \mathbb{R}^{n \times m}$ and $b \in \mathbb{R}^n$. Writing $AY + b = AY_c + c$ with $c := Am_Y + b$ and recalling $X = X_c + m_X$, the mean-square error decomposes as

$$\mathbb{E}[\|X - \hat{X}\|^2] = \mathbb{E}[\|X_c - AY_c - (c - m_X)\|^2] = \mathbb{E}[\|X_c - AY_c\|^2] + \|c - m_X\|^2,$$

where the cross term vanishes because the centered vector $X_c - AY_c$ has zero mean. This expression is minimized by taking $c = m_X$, which eliminates the second term, and by choosing A to minimize $\mathbb{E}[\|X_c - AY_c\|^2]$. The pair (X_c, Y_c) is centered with cross- and auto-covariances $\mathbb{E}[X_c Y_c^\top] = \text{cov}(X, Y)$ and $\mathbb{E}[Y_c Y_c^\top] = \text{cov}(Y)$, so the zero-mean closed-form solution of Theorem 13.45 gives $A = \text{cov}(X, Y) \text{cov}(Y)^{-1}$, valid because $\text{cov}(Y)$ is invertible. Substituting $c = m_X$ and this A into $\hat{X} = A(Y - m_Y) + c$ yields (13.25). $\square$

Gaussian random vectors

Theorem 13.51 (Equivalence of LLSE and MMSE for Gaussian vectors). *Let (X, Y) be jointly Gaussian random vectors with finite second moments. Then the conditional expectation $\mathbb{E}[X | Y]$ is an affine function of Y , and the linear least squares estimator coincides with the MMSE estimator:*

$$\hat{X}_{\text{LLSE}} = \hat{X}_{\text{MMSE}} = \mathbb{E}[X | Y].$$

Proof. Write $m_X = \mathbb{E}[X]$ and $m_Y = \mathbb{E}[Y]$, and set $\Sigma_Y = \text{cov}(Y)$ and $\Sigma_{XY} = \text{cov}(X, Y)$, with Σ_Y assumed invertible. For jointly Gaussian (X, Y) a direct computation with the joint Gaussian density shows that the conditional law of X given Y is again Gaussian, with mean

$$\mathbb{E}[X | Y] = m_X + \Sigma_{XY} \Sigma_Y^{-1} (Y - m_Y),$$

which is an affine function of Y . On the other hand, the MMSE estimator is the conditional expectation itself: for any estimator $g(Y)$ with $\mathbb{E}[\|g(Y)\|^2] < \infty$, conditioning and the tower property give the orthogonal decomposition

$$\mathbb{E}[\|X - g(Y)\|^2] = \mathbb{E}[\|X - \mathbb{E}[X | Y]\|^2] + \mathbb{E}[\|\mathbb{E}[X | Y] - g(Y)\|^2] \geq \mathbb{E}[\|X - \mathbb{E}[X | Y]\|^2],$$

with equality if and only if $g(Y) = \mathbb{E}[X | Y]$ almost surely; hence $\hat{X}_{\text{MMSE}} = \mathbb{E}[X | Y]$. Comparing the displayed conditional mean with the affine estimator (13.25) of Theorem 13.50, the two expressions are identical, so $\hat{X}_{\text{LLSE}} = \hat{X}_{\text{MMSE}} = \mathbb{E}[X | Y]$. $\square$

Equivariance under linear transformations

Theorem 13.52 (Equivariance of the LLSE). *Let $\hat{X}$ be the linear least squares estimator of $X \in \mathbb{R}^n$ based on observations Y . For any matrix $T \in \mathbb{R}^{k \times n}$, the estimator $\hat{Z} := T\hat{X}$ is the linear least squares estimator of $Z := TX$ based on the same observations Y .*

Proof. Write $\hat{X} = A^*Y$ with $A^* \in \mathbb{R}^{n \times m}$. By the orthogonality principle of Theorem 13.43, $\mathbb{E}[(X - A^*Y)Y^\top] = 0$. Left-multiplying this matrix identity by T and using linearity of expectation gives $\mathbb{E}[(TX - TA^*Y)Y^\top] = T\mathbb{E}[(X - A^*Y)Y^\top] = 0$. Now $T\hat{X} = (TA^*)Y$ belongs to the estimation subspace $\mathcal{M}_Z := \{CY : C \in \mathbb{R}^{k \times m}\}$ for $Z := TX$, and it satisfies the orthogonality condition $\mathbb{E}[(Z - TA^*Y)Y^\top] = 0$. By Theorem 13.43 applied to the pair (Z, Y) , this condition characterizes the unique linear least squares estimator of Z from Y ; hence $\hat{Z} = T\hat{X}$. $\square$

13.5 ■ Sequential Linear Least Squares Estimation

In Sections 13.2–13.3.5, linear least squares estimation was formulated as an orthogonal projection problem in the Hilbert space

$$\mathcal{H} = \{Z : \mathbb{E}[Z] = 0, \mathbb{E}[\|Z\|^2] < \infty\}, \quad \langle U, V \rangle := \mathbb{E}[U^\top V].$$

Given a collection of observations, the estimator is obtained by projecting the quantity of interest onto the closed linear subspace generated by those observations.

In many applications, observations arrive *sequentially*. Rather than recomputing the projection from scratch, one seeks a recursive update that incorporates each new observation while preserving optimality. As shown earlier, this is achieved by orthogonalizing new data against past information and updating the estimate via its *innovation* component.

In this section we summarize the sequential structure and present results that were not derived previously, namely the covariance updates and the innovation form underlying Kalman filtering.

Goal of the section. The batch estimator answers the question, “What is the best projection onto all observations at once?” The sequential estimator answers the update question, “Given the old projection, what correction is supplied by the part of the new measurement that was not predictable from the past?”

13.5.1 ■ Innovation-based decomposition

Let $X \in \mathbb{R}^n$ be the quantity to be estimated. All random vectors are assumed to be zero mean and square-integrable. Consider the Hilbert space

$$\mathcal{H}_X := \{Z \in L^2(\Omega; \mathbb{R}^n) \mid \mathbb{E}[Z] = 0\}, \quad \langle U, V \rangle := \mathbb{E}[U^\top V].$$

Suppose an estimate $\hat{X}_1 \in \mathcal{H}_X$ has been formed using the observation Y_1 , and define its error covariance $P_1 := \mathbb{E}[(X - \hat{X}_1)(X - \hat{X}_1)^\top]$. Let

$$\text{span}\{Y_1\} := \{AY_1 : A \in \mathbb{R}^{n \times m_1}\} \subset \mathcal{H}_X$$

denote the closed linear subspace generated by Y_1 . By construction,

$$\hat{X}_1 = \Pi_{\text{span}\{Y_1\}} X, \quad X - \hat{X}_1 \perp \text{span}\{Y_1\}.$$

A new observation Y_2 becomes available. As shown in Section 13.3.5, only the component of Y_2 that is orthogonal to $\text{span}\{Y_1\}$ can further reduce the estimation error. This motivates the definition of the *innovation* as the projection residual

$$\tilde{Y}_2 := Y_2 - \Pi_{\text{span}\{Y_1\}} Y_2 = Y_2 - \mathbb{E}[Y_2 Y_1^\top] \mathbb{E}[Y_1 Y_1^\top]^{-1} Y_1.$$

By construction,

$$\tilde{Y}_2 \perp \text{span}\{Y_1\}, \quad \text{span}\{Y_1, Y_2\} = \text{span}\{Y_1, \tilde{Y}_2\}.$$

Define the enlarged observation subspace

$$\text{span}\{Y_1, Y_2\} = \text{span}\{Y_1\} \oplus \text{span}\{\tilde{Y}_2\}.$$

The updated linear least squares estimate is $\hat{X} = \Pi_{\text{span}\{Y_1, Y_2\}} X$. Using the orthogonal direct-sum structure, the projection decomposes as

$$\hat{X} = \Pi_{\text{span}\{Y_1\}} X + \Pi_{\text{span}\{\tilde{Y}_2\}} X = \hat{X}_1 + \hat{X}_2,$$

where $\hat{X}_2 := \Pi_{\text{span}\{\tilde{Y}_2\}} X$. Orthogonality of the direct sum implies $\hat{X}_1 \perp \hat{X}_2$.

Thus, the updated estimate consists of the previous estimate plus a correction term that depends solely on the innovation $\tilde{Y}_2$, i.e., on the new information not already contained in Y_1 .

13.5.2 ■ Error covariance reduction

The orthogonal decomposition of the estimate leads directly to a precise characterization of how the estimation error covariance is reduced.

Proposition 13.53 (Error covariance update). *Let $P := \mathbb{E}[(X - \hat{X})(X - \hat{X})^\top]$ denote the error covariance after incorporating the new observation. Then*

$$P_1 - P = \mathbb{E}[X \tilde{Y}_2^\top] \mathbb{E}[\tilde{Y}_2 \tilde{Y}_2^\top]^{-1} \mathbb{E}[\tilde{Y}_2 X^\top] \succeq 0. \quad (13.26)$$

Proof. From the orthogonal decomposition $\hat{X} = \hat{X}_1 + \hat{X}_2$ with $\hat{X}_1 \perp \hat{X}_2$, let $e_1 := X - \hat{X}_1$ and $e := X - \hat{X}$. Then $e_1 = e + \hat{X}_2$. The updated residual e is orthogonal to the enlarged observation space and, in particular, to $\text{span}\{\tilde{Y}_2\}$. Since $\hat{X}_2 \in \text{span}\{\tilde{Y}_2\}$, the matrix cross terms vanish: $\mathbb{E}[e \hat{X}_2^\top] = 0$ and $\mathbb{E}[\hat{X}_2 e^\top] = 0$. Therefore

$$P_1 = \mathbb{E}[e_1 e_1^\top] = \mathbb{E}[(e + \hat{X}_2)(e + \hat{X}_2)^\top] = P + \mathbb{E}[\hat{X}_2 \hat{X}_2^\top],$$

so $P_1 - P = \mathbb{E}[\hat{X}_2 \hat{X}_2^\top]$.

Because $\hat{X}_2$ is the orthogonal projection of X onto the one-step innovation subspace $\text{span}\{\tilde{Y}_2\}$, the projection formula from Section 13.4.4 gives

$$\hat{X}_2 = \mathbb{E}[X \tilde{Y}_2^\top] \mathbb{E}[\tilde{Y}_2 \tilde{Y}_2^\top]^{-1} \tilde{Y}_2.$$

Taking second moments yields

$$\mathbb{E}[\hat{X}_2 \hat{X}_2^\top] = \mathbb{E}[X \tilde{Y}_2^\top] \mathbb{E}[\tilde{Y}_2 \tilde{Y}_2^\top]^{-1} \mathbb{E}[\tilde{Y}_2 X^\top],$$

which establishes (13.26). Positive semidefiniteness follows immediately. $\square$

Corollary 13.54 (Monotonic improvement). *Let P_1 and P be the error covariances of the linear least squares estimates of X before and after incorporating a new observation Y_2 , as in Proposition 13.53. Then $P \preceq P_1$, with equality if and only if $\mathbb{E}[XY_2^\top] = 0$, that is, if and only if the innovation $\tilde{Y}_2$ is uncorrelated with X . Sequential incorporation of observations can therefore never increase the estimation error covariance, and it strictly decreases it exactly when the new datum carries information about X that the past did not already contain.*

Proof. By (13.26), $P_1 - P = \mathbb{E}[X\tilde{Y}_2^\top] \mathbb{E}[\tilde{Y}_2\tilde{Y}_2^\top]^{-1} \mathbb{E}[\tilde{Y}_2X^\top]$, which is of the form $MR^{-1}M^\top$ with $R \succ 0$ and $M = \mathbb{E}[X\tilde{Y}_2^\top]$; such a matrix is positive semidefinite, and it vanishes precisely when $M = 0$. $\square$

13.5.3 ■ Measurement model and innovation form

Consider the linear measurement model

$$Y = CX + V, \quad (13.27)$$

where $C \in \mathbb{R}^{m \times n}$ and $V \in \mathbb{R}^m$ is a zero-mean measurement noise vector satisfying

$$\mathbb{E}[V] = 0, \quad \mathbb{E}[VV^\top] = R \succ 0, \quad V \perp \text{span}\{\text{past observations}, X\}.$$

Let $\hat{X}_1$ denote the linear least squares estimate of X based on all past observations, and define the associated error covariance $P_1 := \mathbb{E}[(X - \hat{X}_1)(X - \hat{X}_1)^\top]$.

Define the *innovation* as

$$\tilde{Y} := Y - C\hat{X}_1 = C(X - \hat{X}_1) + V. \quad (13.28)$$

Lemma 13.55 (Innovation covariances). *The innovation satisfies*

$$\mathbb{E}[X\tilde{Y}^\top] = P_1C^\top, \quad (13.29)$$

$$\mathbb{E}[\tilde{Y}\tilde{Y}^\top] = CP_1C^\top + R. \quad (13.30)$$

Proof. We compute each covariance explicitly.

Cross-covariance. Using (13.28),

$$\mathbb{E}[X\tilde{Y}^\top] = \mathbb{E}\left[X(X - \hat{X}_1)^\top C^\top\right] + \mathbb{E}[XV^\top].$$

Since V is orthogonal to $\text{span}\{\text{past observations}, X\}$ and has zero mean, we have $\mathbb{E}[XV^\top] = 0$. To evaluate the first term, write $X = (X - \hat{X}_1) + \hat{X}_1$. Substitution yields

$$\mathbb{E}[X(X - \hat{X}_1)^\top] = \mathbb{E}[(X - \hat{X}_1)(X - \hat{X}_1)^\top] + \mathbb{E}[\hat{X}_1(X - \hat{X}_1)^\top].$$

Because $\hat{X}_1$ is the orthogonal projection of X onto the closed subspace generated by past observations, the estimation error $X - \hat{X}_1$ is orthogonal to that subspace. Since $\hat{X}_1$ lies in the same subspace, it follows that $\mathbb{E}[\hat{X}_1(X - \hat{X}_1)^\top] = 0$. Therefore, $\mathbb{E}[X(X - \hat{X}_1)^\top] = P_1$, and hence $\mathbb{E}[X\tilde{Y}^\top] = P_1C^\top$.

Innovation covariance. Using again (13.28),

$$\begin{aligned}\mathbb{E}[\tilde{Y}\tilde{Y}^\top] &= \mathbb{E}\left[C(X - \hat{X}_1)(X - \hat{X}_1)^\top C^\top\right] + \mathbb{E}[VV^\top] \\ &\quad + \mathbb{E}[C(X - \hat{X}_1)V^\top] + \mathbb{E}[V(X - \hat{X}_1)^\top C^\top].\end{aligned}$$

The last two terms vanish because V is orthogonal to $\text{span}\{\text{past observations}, X\}$. Therefore $\mathbb{E}[\tilde{Y}\tilde{Y}^\top] = CP_1C^\top + R$, which proves (13.30). $\square$

13.5.4 ■ Recursive LLSE update

Theorem 13.56 (Sequential linear least squares update). *The linear least squares estimate of X after incorporating the new measurement Y is given by*

$$\hat{X} = \hat{X}_1 + P_1C^\top(CP_1C^\top + R)^{-1}\tilde{Y}, \quad (13.31)$$

$$P = P_1 - P_1C^\top(CP_1C^\top + R)^{-1}CP_1. \quad (13.32)$$

Proof. By construction, the innovation $\tilde{Y}$ is orthogonal to the closed linear subspace generated by past observations. Consequently, the enlarged observation subspace admits the orthogonal decomposition

$$\text{span}\{\text{past observations}, Y\} = \text{span}\{\text{past observations}\} \oplus \text{span}\{\tilde{Y}\}.$$

Since $\hat{X}_1$ is the orthogonal projection of X onto $\text{span}\{\text{past observations}\}$, the updated estimate must have the form

$$\hat{X} = \hat{X}_1 + \hat{X}_2, \quad \hat{X}_2 \in \text{span}\{\tilde{Y}\}.$$

The correction term $\hat{X}_2$ is therefore the linear least squares estimator of X based solely on $\tilde{Y}$. By the closed-form LLSE formula (Theorem 13.45),

$$\hat{X}_2 = \mathbb{E}[X\tilde{Y}^\top] \mathbb{E}[\tilde{Y}\tilde{Y}^\top]^{-1} \tilde{Y}.$$

Substituting the covariance identities from Lemma 13.55 yields

$$\hat{X}_2 = P_1C^\top(CP_1C^\top + R)^{-1}\tilde{Y},$$

which establishes (13.31).

Finally, the error covariance update follows from the general projection identity (Proposition 13.53), which gives

$$P = P_1 - \mathbb{E}[X\tilde{Y}^\top] \mathbb{E}[\tilde{Y}\tilde{Y}^\top]^{-1} \mathbb{E}[\tilde{Y}X^\top].$$

Substitution of the expressions in Lemma 13.55 yields (13.32). $\square$

Note 13.57 (Kalman-filter reading). The matrix $P_1C^\top(CP_1C^\top + R)^{-1}$ is the gain multiplying the innovation $\tilde{Y} = Y - C\hat{X}_1$. The covariance update subtracts a positive semidefinite matrix, so the new measurement can only reduce, not increase, the linear least-squares error covariance.

13.6 ■ Further Reading

For Hilbert-space foundations, a useful path is to review inner product spaces, orthogonal projection, and least-squares approximation in a functional analysis or vector-space optimization text. Luenberger's vector-space treatment [79] is especially close to the geometric style used here, while Rudin [104] and Brezis [28] provide broader analytical foundations. The projection theorem is the key bridge from abstract geometry to estimation: once an estimator class is identified as a closed subspace of L^2 , optimality is equivalent to orthogonality of the error.

For statistical estimation and stochastic systems, standard references on estimation, identification, and stochastic control develop the same ideas from the probabilistic side. Kay [68] and Poor [97] emphasize estimation theory and statistical signal processing, while Kumar and Varaiya [74] connect estimation with stochastic systems and control. In that literature the orthogonality principle appears as the normal-equation condition for linear least squares and as the defining property of conditional expectation for MMSE estimation.

Texts on Kalman filtering and adaptive filtering extend the innovation viewpoint in this chapter to dynamic systems, where each new measurement contributes only its component orthogonal to the past. Standard references include Anderson and Moore [2], Maybeck [83], Kailath, Sayed, and Hassibi [62], and Haykin [56].

Readers who want a computational perspective should compare the determinant formula for distance with QR factorization and numerically stable Gram–Schmidt variants. The mathematics is the same, but stable algorithms avoid forming ill-conditioned Gram determinants or normal equations directly; see Golub and Van Loan [52] for the standard matrix-computation treatment.

13.7 ■ Exercises

Exercise 13.1 (Cauchy–Schwarz via Minimization). Let X be a pre-Hilbert space. Prove the Cauchy–Schwarz inequality by minimizing $\|x - \lambda y\|^2$ over λ , and identify precisely when equality holds.

Exercise 13.2 (Incompleteness of the Polynomial Space). Let $P[a, b]$ be the space of polynomial functions on $[a, b]$ with the L^2 inner product. Explain why $P[a, b]$ is not complete by using the density of polynomials in $L^2[a, b]$ and a non-polynomial function in $L^2[a, b]$.

Exercise 13.3 (Normal Equations). Let $M = \text{span}\{y_1, \dots, y_n\}$ in a Hilbert space. Starting from the condition $x - \hat{x} \perp M$, derive the normal equations for the coefficients of $\hat{x} = \sum_{j=1}^n \alpha_j y_j$.

Exercise 13.4 (Distance to a One-Dimensional Subspace). For $M = \text{span}\{y\}$ with $y \neq 0$, show that

$$\text{dist}(x, M)^2 = \|x\|^2 - \frac{|(x | y)|^2}{\|y\|^2}.$$

Verify that this agrees with the Gram determinant formula in Theorem 13.25.

Exercise 13.5 (Scalar Linear Least Squares). Let X, Y be zero-mean square-integrable random variables with $\mathbb{E}[Y^2] > 0$. Find the scalar linear least squares estimator $\hat{X} = \alpha Y$ and compute the minimum mean-square error.

Exercise 13.6 (Conditional Expectation for a Truncated Exponential Density). For the density $f_{X,Y}(x, y) = \lambda^2 e^{-\lambda x}$ for $0 \leq y \leq x$, compute $f_Y(y)$, $f_{X|Y}(x | y)$, and verify that $\mathbb{E}[X | Y] = Y + 1/\lambda$.

Exercise 13.7 (A Nonlinear Estimator). Let $X \sim \text{Unif}[0, 1]$, let $V \sim \text{Exp}(1)$ be independent of X , and let $Y = \log(1/X) + V$. Show that $\mathbb{E}[X | Y = y] = \frac{1 - e^{-y}}{y}$ for $y > 0$, with value 1 at $y = 0$ by continuity.

Exercise 13.8 (Vector Least Squares and Error Covariance). Let $X \in \mathbb{R}^n$ and $Y \in \mathbb{R}^m$ be zero-mean random vectors with finite second moments and nonsingular $\mathbb{E}[YY^\top]$. Derive $\hat{X} = \mathbb{E}[XY^\top]\mathbb{E}[YY^\top]^{-1}Y$ and the error covariance formula in Proposition 13.48.

Exercise 13.9 (Innovations and Recursive Estimation). For two observations Y_1, Y_2 , form the innovation $\tilde{Y}_2 = Y_2 - \frac{\langle Y_2, Y_1 \rangle}{\langle Y_1, Y_1 \rangle} Y_1$. Show that the recursive estimator based on $(Y_1, \tilde{Y}_2)$ is algebraically equivalent to the batch estimator based on (Y_1, Y_2) .

Exercise 13.10 (Derivation of the Update Gain). Consider the measurement model $Y = CX + V$ with V zero mean, covariance R , and orthogonal to the previous estimation error. Starting from the innovation $\tilde{Y} = Y - C\hat{X}_1$, derive the gain $K = P_1 C^\top (CP_1 C^\top + R)^{-1}$ and the covariance update $P = P_1 - KCP_1$.

Exercise 13.11 (Minimum-Norm Random Variable with Moment Constraints). Let Y_1 and Y_2 be uncorrelated second-order random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $L^2(\Omega, \mathcal{F}; C[0, 1])$ be the space of parametrized (in t) random variables $X(t; \omega)$, where for fixed $t \in [0, 1]$, $X(t; \cdot)$ is a second-order random variable and for fixed $\omega \in \Omega$, $X(\cdot; \omega) \in C[0, 1]$. Endow it with the inner product

$$(X, Z) = \mathbb{E} \left[\int_0^1 X(t; \omega) Z(t; \omega) w(t) dt \right],$$

where $w(\cdot) > 0$ is in $C[0, 1]$. Determine a process $\hat{X}(t; \omega) \in L^2(\Omega, \mathcal{F}; C[0, 1])$ of *minimum norm* satisfying

$$\mathbb{E} \left[\int_0^1 \hat{X}(t; \omega) k_i(t) Y_i(\omega) dt \right] = c_i, \quad i = 1, 2,$$

where k_1, k_2 are linearly independent elements of $C[0, 1]$ and c_1, c_2 are given constants. Is the solution unique?

Exercise 13.12 (Linear Least-Squares Estimation over a Process). Let X be a second-order random variable on $(\Omega, \mathcal{F}, \mathbb{P})$, and let $Y(t; \omega)$ be a second-order stochastic process on the same space, $t \in [0, 2]$, correlated with X through the cross-correlation $R_{XY}(t) = \mathbb{E}[XY(t)]$. Let $R_{YY}(t, s)$ denote the autocorrelation of Y . We seek a linear least-squares (l.l.s.) estimate of X from the measurement $Y(t; \omega)$ on $[0, 2]$, of the form $m(\omega) = \int_0^2 K(t) Y(t; \omega) dt$.

- (i) Show that a unique such l.l.s. estimate exists, and obtain the equation satisfied by an optimal $K(\cdot)$ in terms of R_{XY} and R_{YY} . Under what conditions is the optimum $K(\cdot)$ unique?
- (ii) Redo (i) when $K(\cdot)$ is restricted to be a constant (independent of time).

Exercise 13.13 (Recursive Linear Least-Squares over Random Vectors). Let $L^2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R}^n)$ be the Hilbert space of second-order random n -vectors with inner product $(x, z) = \mathbb{E}[x^\top Qz]$, $Q \succ 0$. Let $\{y_0, \dots, y_i\}$ be zero-mean uncorrelated m -dimensional random vectors, and $\mathcal{M}_{nm}$ the class of $n \times m$ matrices with bounded entries. For given x , consider $\|x - \sum_{j=0}^i \hat{K}_j y_j\| = \inf_{K_j \in \mathcal{M}_{nm}} \|x - \sum_{j=0}^i K_j y_j\|$.

- (i) Solve for the optimal $\hat{K}_j$, $j = 0, \dots, i$. Is the solution unique?
- (ii) With $\epsilon_k = \|x - \sum_{j=0}^k \hat{K}_j y_j\|^2$, obtain a recursive (first-order difference) equation for ϵ_k .

Exercise 13.14 (Best Estimate over a Convex Cone). Let x, y_1, y_2 be zero-mean second-order random variables with y_1, y_2 uncorrelated, and let $\mathcal{Z} = \{z = a_1 y_1 + a_2 y_2 : a_1, a_2 \geq 0\}$. Seek a best minimum-mean-square estimate $\hat{x} \in \mathcal{Z}$, i.e. $\inf_{z \in \mathcal{Z}} \mathbb{E}[(x - z)^2] = \mathbb{E}[(x - \hat{x})^2]$.

- (i) Formulate this as minimum distance to a convex set in a Hilbert space.
- (ii) Does a unique solution exist? Justify.
- (iii) Compute $\hat{x}$ and $\mathbb{E}[(x - \hat{x})^2]$ when $\mathbb{E}[y_1^2] = \mathbb{E}[y_2^2] = \mathbb{E}[x^2] = 1$, $\mathbb{E}[xy_1] = 0.3$, $\mathbb{E}[xy_2] = -0.4$.

Exercise 13.15 (Minimum-Second-Moment Random Variable). Let y be a random variable with $\mathbb{E}[y] = 1$ and $\mathbb{E}[y^2] = 2$. Find a random variable x on the same space of minimum second moment (minimum norm) satisfying $\mathbb{E}[xy] = -3$ and $\mathbb{E}[x] = 2$.

- (i) Does the problem admit a (unique) solution? Justify.
- (ii) Obtain the solution if it exists.
- (iii) What is the solution if the constraint $\mathbb{E}[x] = 2$ is relaxed to $\mathbb{E}[x] \leq 2$?

Chapter 14

Kalman Filtering

The Kalman filter is the centerpiece of linear estimation theory: a recursive algorithm that maintains the best linear least-squares estimate of the state of a stochastic linear system, together with the covariance of the estimation error, as measurements arrive one at a time. This chapter develops the filter from first principles and then uses it to solve the linear-quadratic-Gaussian (LQG) control problem, in which a quadratic cost must be minimized on the basis of noisy, partial observations of the state.

The development rests on a single geometric idea, inherited from Chapter 13: least-squares estimation is orthogonal projection in the Hilbert space of square-integrable random variables. There we proved the projection theorem (Section 13.2), and from it the orthogonality principle (Theorem 13.43) and the closed-form linear least-squares estimator (Theorem 13.45); we also saw how an estimate can be refreshed one observation at a time by projecting onto the part of each new datum that is orthogonal to what came before (Section 13.5). The Kalman filter is what that recursive projection becomes when the data are generated by a dynamical system rather than by a fixed collection of random variables. Nothing new needs to be assumed: the filter's predict-update cycle is the same projection, applied over and over, organized around the *innovations*, the components of each new measurement that are orthogonal to all past data. Once this is understood, the celebrated Riccati recursion for the error covariance, its steady-state algebraic form, and the whiteness of the innovations sequence are not separate results to be memorized but consequences of one geometric fact, obtained by direct calculation.

The chapter is organized as follows. We first state the standing assumptions and derive the measurement update, the predictor, and the covariance recursions, including the case of correlated process and measurement noise; worked scalar examples make the formulas concrete. We then turn to the stochastic linear regulator, revisiting the deterministic dynamic programming of Chapters 10 and 11 in the presence of noise, and establish the perhaps surprising fact that additive white noise leaves the optimal feedback gain unchanged. Combining the two developments, an estimator that reconstructs the state and a regulator that would be optimal if the state were known, we prove the separation theorem for partially observed linear-quadratic problems, and conclude with the stability analysis of the resulting closed-loop system. Exercises are collected at the end of the chapter.

Table 14.1: Dimensions of the recurring symbols. Throughout, n is the state dimension, p the measurement dimension, r and s the process- and measurement-noise dimensions, and m the control dimension.

Symbol	Dimension	Meaning
<i>Filtering (Section 14.1)</i>		
x_k	$\mathbb{R}^n$	state
z_k	$\mathbb{R}^p$	measurement
w_k, v_k	$\mathbb{R}^r, \mathbb{R}^s$	process, measurement noise
A	$n \times n$	state transition
Γ	$n \times r$	process-noise input
C	$p \times n$	measurement map
H	$p \times s$	measurement-noise input
W, V	$r \times r, s \times s$	noise covariances
$T = \mathbb{E}[w_k v_k^\top]$	$r \times s$	noise cross-covariance
$\Sigma_{k k-1}, \Sigma_{k k}$	$n \times n$	prediction, filtered error covariances
ν_k	$\mathbb{R}^p$	innovation
$\Lambda_k = C\Sigma_{k k-1}C^\top + HVH^\top$	$p \times p$	innovation covariance
K_k^p (predictor), G_k (update)	$n \times p$	Kalman gains
<i>Control (Section 14.2)</i>		
u_k	$\mathbb{R}^m$	control
B	$n \times m$	control input
Q_k, R_k	$n \times n, m \times m$	state, control cost weights
N_k	$n \times m$	state–control cross weight
Q_N	$n \times n$	terminal cost weight
K_k	$n \times n$	value-function (Riccati) matrix
L_k	$m \times n$	LQR feedback gain

14.1 ■ Discrete-Time Kalman Filter

We work with the stochastic linear system

$$x_{k+1} = Ax_k + \Gamma w_k, \quad (14.1)$$

$$z_k = Cx_k + Hv_k, \quad (14.2)$$

whose dimensions we fix once and for all, since keeping them straight is half the battle with the formulas that follow. The state is $x_k \in \mathbb{R}^n$, the measurement is $z_k \in \mathbb{R}^p$, and the process and measurement noises are $w_k \in \mathbb{R}^r$ and $v_k \in \mathbb{R}^s$; the system matrices are therefore $A \in \mathbb{R}^{n \times n}$, $\Gamma \in \mathbb{R}^{n \times r}$, $C \in \mathbb{R}^{p \times n}$, and $H \in \mathbb{R}^{p \times s}$. The noise covariances $W = \mathbb{E}[w_k w_k^\top] \in \mathbb{R}^{r \times r}$ and $V = \mathbb{E}[v_k v_k^\top] \in \mathbb{R}^{s \times s}$ are symmetric and positive semidefinite, and the same-time cross-covariance is $T = \mathbb{E}[w_k v_k^\top] \in \mathbb{R}^{r \times s}$. Every error covariance $\Sigma_{k|k-1}$ and $\Sigma_{k|k}$ is an $n \times n$ symmetric matrix; the innovation $\nu_k \in \mathbb{R}^p$ carries the $p \times p$ covariance $\Lambda_k = C\Sigma_{k|k-1}C^\top + HVH^\top$; and both the predictor gain K_k^p and the filtered-update gain G_k are $n \times p$. A reliable way to keep the shapes straight is to remember that a gain multiplying an innovation must convert a p -vector of measurement residuals into a correction of the n -dimensional state, so it can only be $n \times p$. Table 14.1 collects these dimensions, together with the control-side symbols introduced in Section 14.2, as a running reference.

With these conventions we define the information sets

$$\mathcal{I}_k := \sigma(z_{k_0}, \dots, z_k), \quad \mathcal{I}_{k-1} := \sigma(z_{k_0}, \dots, z_{k-1}).$$

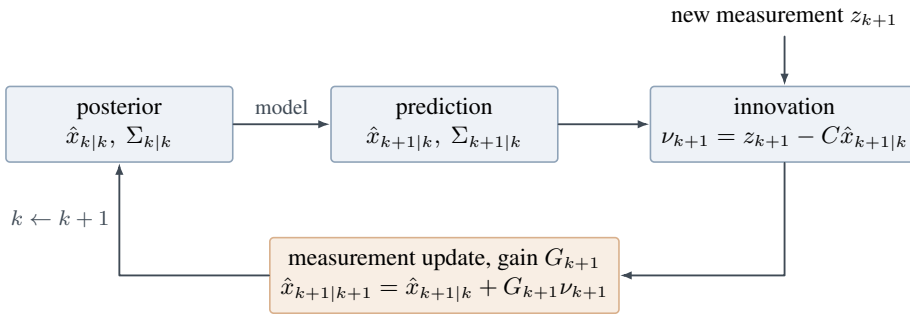


Figure 14.1: The Kalman filter alternates prediction through the model and correction through the innovation generated by new observations.

We seek the *linear least-squares* (LLS) estimate of x_k given $\mathcal{I}_{k-1}$, denoted $\hat{x}_{k|k-1} := \text{proj}(x_k \mid \mathcal{I}_{k-1})$, i.e., the orthogonal projection of x_k onto the closed linear subspace generated by $\{z_{k_0}, \dots, z_{k-1}\}$ in L^2 .

14.1.1 ■ Assumptions

Before writing down the assumptions it is worth asking what the estimator actually needs from the probabilistic model, because the answer explains the peculiar shape of the list that follows. The estimate we seek is a linear least-squares projection, and a projection in L^2 is determined entirely by inner products: that is, by means and covariances. Nothing in the construction ever calls on a probability density. This is why the standing assumptions below constrain only the first two moments of the noises and the initial state and say nothing about their distributions: for the linear filter, first and second moments are the whole story. The one place distributions will matter is the linear-quadratic-Gaussian problem of the later sections, where Gaussianity upgrades the linear estimate to the genuinely optimal one; we postpone that hypothesis until it is needed.

The two whiteness assumptions are the load-bearing ones. Requiring the process and measurement noises to be temporally uncorrelated is precisely what makes each innovation *new*: if the noise carried memory, part of z_k would be predictable from past data through the noise itself, the residual ν_k would no longer be orthogonal to $\mathcal{I}_{k-1}$, and the tidy recursion would collapse into a growing batch problem. Whiteness is therefore not a convenience but the hypothesis that buys recursiveness, and Chapter 13 already showed, in the Gram–Schmidt construction, how orthogonalizing the data is what turns a batch estimator into a sequential one. The remaining assumptions play supporting roles. Fixing the mean and covariance of x_0 supplies the recursion’s starting point, the prior the first measurement will correct. Mutual uncorrelatedness across the noises and the initial state, with the single same-time cross-term $\mathbb{E}[w_k v_k^\top] = T$ retained, keeps the bookkeeping honest while still covering the practically important case in which the process and sensor noises share a common disturbance; we carry T through the derivation precisely so that this case is not lost. Finally, invertibility of $HVH^\top$, and hence of the innovation covariance $\text{cov}(\nu_k)$, is what allows the normal equations to be solved for the gain: it says every measurement direction carries some genuine noise, so no direction of the innovation is a deterministic function of the state that would make the covariance singular and the gain ill-defined.

Assumption 14.1 (Standing assumptions).

- (1) $\{w_k\}$ is zero-mean and uncorrelated (white): $\mathbb{E}[w_k] = 0$, $\mathbb{E}[w_k w_j^\top] = 0$ for $k \neq j$, and $\mathbb{E}[w_k w_k^\top] = W$.

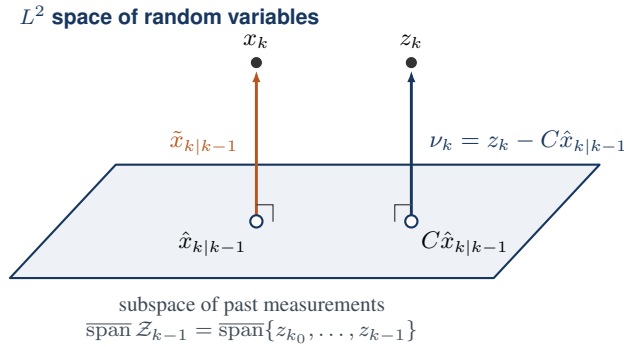


Figure 14.2: The estimator geometry of the Kalman filter, read directly from the projection theorem of Chapter 13. The subspace spanned by past measurements is drawn as a plane. The predictor $\hat{x}_{k|k-1}$ is the foot of the perpendicular dropped from the state x_k onto that plane, so the prediction error $\tilde{x}_{k|k-1}$ stands orthogonal to every past datum. The innovation ν_k is likewise the part of the new measurement z_k orthogonal to the plane, and it is the only direction along which the estimate is corrected.

- (2) $\{v_k\}$ is zero-mean and uncorrelated (white): $\mathbb{E}[v_k] = 0$, $\mathbb{E}[v_k v_j^\top] = 0$ for $k \neq j$, and $\mathbb{E}[v_k v_k^\top] = V$.
- (3) x_0 has mean m_0 and covariance Σ_0 .
- (4) $\{w_k\}, \{v_k\}, x_0$ are mutually uncorrelated for all indices except possibly same-time cross-correlation $\mathbb{E}[w_k v_k^\top] = T$.
- (5) $HVH^\top$ is nonsingular. Since $\text{cov}(\nu_k) = C\Sigma_{k|k-1}C^\top + HVH^\top$ with $\Sigma_{k|k-1} \succeq 0$, this makes every innovation covariance invertible, which is what the normal equations for the gain require.

14.1.2 ■ Preliminaries: projection lemma and LLS formula

Everything in this section is a specialization of Chapter 13, and it is worth pausing to see exactly which results we are reusing before the notation of the dynamical setting hides them. Two facts do all the work. The first is the *orthogonality principle*: the least-squares estimate is the unique point of the data subspace whose error is orthogonal to that subspace. The second is the *normal equations*, which turn that geometric statement into a formula, a matrix of covariances to be inverted. Figure 14.2 draws the picture we will keep returning to: the estimate is the foot of a perpendicular, and the innovation is the leftover direction along which a new measurement can still move it. We restate the two facts here in the form we need, so that the recursions below can cite them by name.

Lemma 14.2 (Orthogonal projection characterization). *Let $\mathcal{S} \subset L^2$ be a closed linear subspace and $x \in L^2$. Then $z^* \in \mathcal{S}$ minimizes $\mathbb{E}\|x - z\|^2$ over $z \in \mathcal{S}$ if and only if the error is orthogonal to the subspace,*

$$x - z^* \perp \mathcal{S}, \quad \text{i.e.} \quad \mathbb{E}[(x - z^*)s^\top] = 0 \text{ for every } s \in \mathcal{S},$$

and this $z^* = \text{proj}(x \mid \mathcal{S})$ is unique.

Proof. For any $z \in \mathcal{S}$ write $x - z = (x - z^*) + (z^* - z)$ with $z^* - z \in \mathcal{S}$, so

$$\mathbb{E}\|x - z\|^2 = \mathbb{E}\|x - z^*\|^2 + 2\mathbb{E}[(x - z^*)^\top (z^* - z)] + \mathbb{E}\|z^* - z\|^2.$$

If $x - z^* \perp \mathcal{S}$ the middle term vanishes for all z , giving $\mathbb{E}\|x - z\|^2 = \mathbb{E}\|x - z^*\|^2 + \mathbb{E}\|z^* - z\|^2 \geq \mathbb{E}\|x - z^*\|^2$, with equality iff $z = z^*$; hence z^* is the unique minimizer. Conversely, if z^* minimizes, then for every $s \in \mathcal{S}$ and $t \in \mathbb{R}$ the scalar $t \mapsto \mathbb{E}\|x - z^* - ts\|^2 = \mathbb{E}\|x - z^*\|^2 - 2t\mathbb{E}[(x - z^*)^\top s] + t^2\mathbb{E}\|s\|^2$ is minimized at $t = 0$, which forces $\mathbb{E}[(x - z^*)^\top s] = 0$; taking s to range over the coordinate directions gives $\mathbb{E}[(x - z^*)s^\top] = 0$. $\square$

This is precisely the orthogonality principle of Theorem 13.43; the inner product is the same $\langle u, v \rangle = \mathbb{E}[u^\top v]$ on L^2 used throughout Chapter 13.

Lemma 14.3 (Linear regression / normal equations). *Let x and y be square-integrable random vectors with $\mathbb{E}[y] = 0$ and $\Sigma_{yy} := \mathbb{E}[yy^\top]$ nonsingular. Among all affine estimators $\hat{x} = a + Gy$, the unique minimizer of $\mathbb{E}\|x - \hat{x}\|^2$ is*

$$\hat{x} = \mathbb{E}[x] + \Sigma_{xy}\Sigma_{yy}^{-1}(y - \mathbb{E}[y]), \quad \Sigma_{xy} := \mathbb{E}[(x - \mathbb{E}[x])(y - \mathbb{E}[y])^\top].$$

If $\mathbb{E}[x] = \mathbb{E}[y] = 0$, this reduces to $\hat{x} = \Sigma_{xy}\Sigma_{yy}^{-1}y$.

Proof. Write $J(a, G) := \mathbb{E}\|x - a - Ly\|^2$. Differentiate w.r.t. a :

$$\frac{\partial J}{\partial a} = -2\mathbb{E}[x - a - Ly] = 0 \quad \Rightarrow \quad a = \mathbb{E}[x] - G\mathbb{E}[y].$$

Substitute into J with centered variables $\bar{x} := x - \mathbb{E}[x]$, $\bar{y} := y - \mathbb{E}[y]$:

$$J(G) = \mathbb{E}\|\bar{x} - G\bar{y}\|^2 = \text{tr}(\Sigma_{xx} - 2G\Sigma_{yx} + G\Sigma_{yy}G^\top).$$

Differentiate w.r.t. G and set to zero: $-2\Sigma_{xy} + 2G\Sigma_{yy} = 0 \Rightarrow G = \Sigma_{xy}\Sigma_{yy}^{-1}$. Uniqueness follows from positive definiteness of Σ_{yy} . $\square$

This is the closed-form linear least-squares estimator of Chapter 13 (Theorem 13.45): estimate equals prior mean plus a gain times the centered data, with the gain built from a cross-covariance and the inverse of the data covariance. The entire Kalman filter is obtained by applying this one formula repeatedly, with a shrewd choice of “data”: at each step we regress not on the raw measurement z_k , whose informative part overlaps everything already seen, but on the *innovation*, the fresh component orthogonal to the past. Because successive innovations are mutually orthogonal, the gain never has to account for old data, and a batch problem that would grow without bound collapses into a fixed-size recursion.

14.1.3 ■ Filtered update $\hat{x}_{k|k}$ and covariance $\Sigma_{k|k}$

Definition 14.4 (Innovations). *The innovation (measurement residual) at time k is*

$$\nu_k := z_k - C\hat{x}_{k|k-1}. \quad (14.3)$$

It is the part of the new measurement z_k not predictable from past data.

We will show that the best LLS correction of $\hat{x}_{k|k-1}$ given the new datum z_k is linear in ν_k . Two elementary facts about the innovation make this possible, and it is worth isolating them,

because the correction along ν_k alone, rather than along all of z_k , is the single step that turns the batch estimator of Chapter 13 into a recursion. Both facts are consequences of the projection geometry of Chapter 13 and of the standing assumptions; crucially, neither uses the cross-time whiteness of Theorem 14.10, so the development that follows is free of circularity.

Lemma 14.5 (Innovation is orthogonal to the past). *Let $\tilde{x}_{k|k-1} := x_k - \hat{x}_{k|k-1}$ be the one-step prediction error. Then*

$$\nu_k = C\tilde{x}_{k|k-1} + Hv_k, \quad (14.4)$$

the innovation is orthogonal to every past datum,

$$\mathbb{E}[\nu_k s^\top] = 0 \quad \text{for all } s \in \overline{\text{span}} \mathcal{I}_{k-1}, \quad (14.5)$$

and the data subspace splits as the orthogonal direct sum

$$\overline{\text{span}} \mathcal{I}_k = \overline{\text{span}} \mathcal{I}_{k-1} \oplus \text{span}\{\nu_k\}. \quad (14.6)$$

Proof. Substituting $z_k = Cx_k + Hv_k$ into (14.3) and inserting $\pm C\hat{x}_{k|k-1}$ gives $\nu_k = C(x_k - \hat{x}_{k|k-1}) + Hv_k = C\tilde{x}_{k|k-1} + Hv_k$, which is (14.4). For (14.5), fix $s \in \overline{\text{span}} \mathcal{I}_{k-1}$. The prediction error is orthogonal to the past data subspace by the orthogonality principle (Lemma 14.2), so $\mathbb{E}[\tilde{x}_{k|k-1} s^\top] = 0$. The measurement noise v_k is uncorrelated with $z_{k_0}, \dots, z_{k-1}$: by Assumption 14.1 it is white and uncorrelated with x_0 and with $w_{k_0}, \dots, w_{k-1}$, and these variables generate $\overline{\text{span}} \mathcal{I}_{k-1}$ linearly, whence $\mathbb{E}[v_k s^\top] = 0$. Applying $\mathbb{E}[\cdot s^\top]$ to (14.4) therefore yields $\mathbb{E}[\nu_k s^\top] = C\mathbb{E}[\tilde{x}_{k|k-1} s^\top] + H\mathbb{E}[v_k s^\top] = 0$. Finally, from $z_k = \nu_k + C\hat{x}_{k|k-1}$ with $C\hat{x}_{k|k-1} \in \overline{\text{span}} \mathcal{I}_{k-1}$, adjoining z_k to $\mathcal{I}_{k-1}$ adjoins exactly the new direction ν_k , which by (14.5) is orthogonal to $\mathcal{I}_{k-1}$; this is (14.6). $\square$

Proposition 14.6 (Structure of the update). *The filtered estimate admits the representation*

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + G_k \nu_k \quad (14.7)$$

for some matrix G_k , and the G_k produced by this identity is exactly the one minimizing $\mathbb{E}\|x_k - \hat{x}_{k|k}\|^2$.

Proof. By definition $\hat{x}_{k|k} = \text{proj}(x_k \mid \mathcal{I}_k)$ is the orthogonal projection of x_k onto $\overline{\text{span}} \mathcal{I}_k$. By (14.6) this subspace is the orthogonal direct sum of $\overline{\text{span}} \mathcal{I}_{k-1}$ and $\text{span}\{\nu_k\}$, and the projection onto an orthogonal direct sum is the sum of the projections onto the summands: this is precisely the sequential projection update of Chapter 13 (Section 13.5). Hence

$$\hat{x}_{k|k} = \text{proj}(x_k \mid \mathcal{I}_{k-1}) + \text{proj}(x_k \mid \nu_k) = \hat{x}_{k|k-1} + G_k \nu_k,$$

where $\text{proj}(x_k \mid \nu_k) = G_k \nu_k$ is the regression of x_k onto the single vector ν_k (Lemma 14.3). Because orthogonal projection is by construction the minimizer of the mean-square error over the subspace (Lemma 14.2), the resulting G_k minimizes $\mathbb{E}\|x_k - \hat{x}_{k|k}\|^2$. $\square$

Now determine G_k by linear regression of x_k on ν_k (Lemma 14.3 with $y = \nu_k$). Since $\hat{x}_{k|k-1}$ is $\mathcal{I}_{k-1}$ -measurable, $x_k - \hat{x}_{k|k-1} =: \tilde{x}_{k|k-1}$ is the prediction error. Then (14.7) becomes $\tilde{x}_{k|k} := x_k - \hat{x}_{k|k} = \tilde{x}_{k|k-1} - G_k \nu_k$. Hence minimizing $\mathbb{E}\|\tilde{x}_{k|k}\|^2$ over G_k yields

$$G_k = \Sigma_{\tilde{x}\nu} \Sigma_{\nu\nu}^{-1}, \quad (14.8)$$

where

$$\Sigma_{\tilde{x}\nu} := \mathbb{E}[\tilde{x}_{k|k-1} \nu_k^\top], \quad \Sigma_{\nu\nu} := \mathbb{E}[\nu_k \nu_k^\top].$$

Compute $\Sigma_{\tilde{x}\nu}$. Using $z_k = Cx_k + Hv_k$ and $\nu_k = z_k - C\hat{x}_{k|k-1}$: $\nu_k = C(x_k - \hat{x}_{k|k-1}) + Hv_k = C\tilde{x}_{k|k-1} + Hv_k$. Therefore

$$\begin{aligned}\Sigma_{\tilde{x}\nu} &= \mathbb{E}[\tilde{x}_{k|k-1}(C\tilde{x}_{k|k-1} + Hv_k)^\top] \\ &= \mathbb{E}[\tilde{x}_{k|k-1}\tilde{x}_{k|k-1}^\top]C^\top + \mathbb{E}[\tilde{x}_{k|k-1}v_k^\top]H^\top.\end{aligned}$$

By assumptions, v_k is orthogonal to $\mathcal{I}_{k-1}$ and to x_k 's projection error (since the latter is a linear function of $x_0, w_0, \dots, w_{k-1}$, all uncorrelated with v_k), hence $\mathbb{E}[\tilde{x}_{k|k-1}v_k^\top] = 0$. Thus

$$\Sigma_{\tilde{x}\nu} = \Sigma_{k|k-1}C^\top. \quad (14.9)$$

Compute $\Sigma_{\nu\nu}$. Similarly,

$$\begin{aligned}\Sigma_{\nu\nu} &= \mathbb{E}[(C\tilde{x}_{k|k-1} + Hv_k)(C\tilde{x}_{k|k-1} + Hv_k)^\top] \\ &= C\Sigma_{k|k-1}C^\top + H\mathbb{E}[v_kv_k^\top]H^\top + C\mathbb{E}[\tilde{x}_{k|k-1}v_k^\top]H^\top + H\mathbb{E}[v_k\tilde{x}_{k|k-1}^\top]C^\top \\ &= C\Sigma_{k|k-1}C^\top + HVH^\top.\end{aligned}$$

Therefore

$$\Sigma_{\nu\nu} = C\Sigma_{k|k-1}C^\top + HVH^\top. \quad (14.10)$$

Theorem 14.7 (Measurement update). *Under Assumption 14.1, the filtered LLS estimate and its error covariance are given by*

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + \Sigma_{k|k-1}C^\top(C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1}\nu_k, \quad (14.11)$$

$$\Sigma_{k|k} = \Sigma_{k|k-1} - \Sigma_{k|k-1}C^\top(C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1}C\Sigma_{k|k-1}. \quad (14.12)$$

Proof. From (14.8)–(14.10), $G_k = \Sigma_{k|k-1}C^\top(C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1}$, and substituting into (14.7) yields (14.11).

For the covariance, use $\tilde{x}_{k|k} = \tilde{x}_{k|k-1} - G_k\nu_k$:

$$\begin{aligned}\Sigma_{k|k} &= \mathbb{E}[\tilde{x}_{k|k}\tilde{x}_{k|k}^\top] \\ &= \Sigma_{k|k-1} - G_k\mathbb{E}[\nu_k\tilde{x}_{k|k-1}^\top] - \mathbb{E}[\tilde{x}_{k|k-1}\nu_k^\top]G_k^\top + G_k\mathbb{E}[\nu_k\nu_k^\top]G_k^\top.\end{aligned} \quad (14.13)$$

But $\mathbb{E}[\tilde{x}_{k|k-1}\nu_k^\top] = \Sigma_{k|k-1}C^\top$ and $\mathbb{E}[\nu_k\nu_k^\top] = \Lambda_k := C\Sigma_{k|k-1}C^\top + HVH^\top$, hence $\Sigma_{k|k} = \Sigma_{k|k-1} - G_kC\Sigma_{k|k-1} - \Sigma_{k|k-1}C^\top G_k^\top + G_k\Lambda_k G_k^\top$. With $G_k = \Sigma_{k|k-1}C^\top\Lambda_k^{-1}$, one checks

$$G_k\Lambda_k G_k^\top = \Sigma_{k|k-1}C^\top\Lambda_k^{-1}C\Sigma_{k|k-1}, \quad G_kC\Sigma_{k|k-1} = \Sigma_{k|k-1}C^\top\Lambda_k^{-1}C\Sigma_{k|k-1},$$

so the last three terms collapse to a single subtraction, yielding (14.12). $\square$

It is worth recognizing these formulas. Equations (14.11)–(14.12) are, term for term, the single-observation update derived in Chapter 13 (equations (13.31)–(13.32)), with the prior covariance $\Sigma_{k|k-1}$ in place of Ψ_1 and the measurement noise $HVH^\top$ in place of R . The only new element is that the “prior” is now supplied by the dynamics: it is the prediction carried forward from the previous time step rather than a fixed initial guess. The covariance formula (14.12) says something reassuring and something to watch for. Reassuringly, $\Sigma_{k|k} \preceq \Sigma_{k|k-1}$: measuring never increases uncertainty, because we subtract a positive-semidefinite term. What to watch for is how large that subtraction is: it is governed by Λ_k , the innovation covariance, which weighs how much of the innovation is genuine signal ($C\Sigma_{k|k-1}C^\top$) against how much is sensor noise ($HVH^\top$). A quiet sensor lets the correction be aggressive; a noisy one forces it to be timid.

Remark 14.8 (Relation to the conditional-mean filter of the sequel). The update (14.11)–(14.12) was obtained purely as an orthogonal projection in L^2 , using only first and second moments. When the noises and the initial state are jointly Gaussian, this linear least-squares estimate coincides with the conditional mean $\mathbb{E}[x_k | \mathcal{I}_k]$, and the error covariance coincides with the conditional covariance, because for jointly Gaussian vectors least-squares and minimum-mean-square estimation agree (Chapter 13, Theorem 13.51). The fully probabilistic derivation of the same recursion, built directly from the Gaussian conditioning formula rather than from projection, is carried out in the stochastic-control chapter as the measurement update of Proposition 15.72, culminating in the Kalman filter equations of Theorem 15.73. With the notation aligned to that chapter—measurements z_k , prediction covariance $\Sigma_{k|k-1}$, innovation covariance Λ_k , and noise covariances W, V —the two derivations yield the identical recursion; the only surface difference is that the measurement-update gain written G_k here appears as Γ_k there, and, with $H = I$, the sensor term $HVH^\top$ reduces to V . The present chapter is thus the projection-side preview of that development; the two agree wherever both apply.

14.1.4 ■ Predictor $\hat{x}_{k+1|k}$ when $\mathbb{E}[w_k v_k^\top] = T$

We compute $\hat{x}_{k+1|k} = \text{proj}(x_{k+1} | \mathcal{I}_k)$, the linear least-squares estimate; it is *not* in general the conditional mean $\mathbb{E}[x_{k+1} | \mathcal{I}_k]$, and the two coincide only under the Gaussian hypothesis of Remark 14.8. From (14.1), $x_{k+1} = Ax_k + \Gamma w_k$, so

$$\hat{x}_{k+1|k} = A\hat{x}_{k|k} + \Gamma\hat{w}_{k|k}, \quad \hat{w}_{k|k} := \text{proj}(w_k | \mathcal{I}_k).$$

Because w_k is uncorrelated with $\mathcal{I}_{k-1}$ (it is “future” relative to $z_{k_0}, \dots, z_{k-1}$), the only dependence in $\mathcal{I}_k$ arises through z_k , or more precisely through the new component ν_k . Thus we can regress w_k on ν_k : $\hat{w}_{k|k} = \mathbb{E}[w_k \nu_k^\top] \mathbb{E}[\nu_k \nu_k^\top]^{-1} \nu_k$. Now $\nu_k = C\tilde{x}_{k|k-1} + Hv_k$, and w_k is uncorrelated with $\tilde{x}_{k|k-1}$, so $\mathbb{E}[w_k \nu_k^\top] = \mathbb{E}[w_k v_k^\top] H^\top = TH^\top$. Also $\mathbb{E}[\nu_k \nu_k^\top] = C\Sigma_{k|k-1}C^\top + HVH^\top$ as before. Hence

$$\hat{w}_{k|k} = TH^\top (C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1} \nu_k. \quad (14.14)$$

Therefore

$$\hat{x}_{k+1|k} = A\hat{x}_{k|k} + \Gamma TH^\top (C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1} \nu_k, \quad (14.15)$$

which matches the text.

To express the predictor in terms of $\hat{x}_{k|k-1}$, substitute $\hat{x}_{k|k} = \hat{x}_{k|k-1} + G_k \nu_k$ with G_k from (14.11):

$$\begin{aligned} \hat{x}_{k+1|k} &= A\hat{x}_{k|k-1} + \left(AG_k + \Gamma TH^\top \Lambda_k^{-1} \right) \nu_k \\ &= A\hat{x}_{k|k-1} + \left(A\Sigma_{k|k-1}C^\top + \Gamma TH^\top \right) \Lambda_k^{-1} \nu_k, \end{aligned}$$

where $\Lambda_k := C\Sigma_{k|k-1}C^\top + HVH^\top$. Define the Kalman gain

$$K_k^P := (A\Sigma_{k|k-1}C^\top + \Gamma TH^\top) (C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1}, \quad (14.16)$$

and obtain the compact predictor

$$\hat{x}_{k+1|k} = A\hat{x}_{k|k-1} + K_k^P (z_k - C\hat{x}_{k|k-1}). \quad (14.17)$$

14.1.5 ■ Error dynamics and Riccati recursion for $\Sigma_{k+1|k}$

Define the one-step prediction errors $\tilde{x}_{k|k-1} := x_k - \hat{x}_{k|k-1}$ and $\tilde{x}_{k+1|k} := x_{k+1} - \hat{x}_{k+1|k}$. Using $x_{k+1} = Ax_k + \Gamma w_k$ and (14.17),

$$\begin{aligned}\tilde{x}_{k+1|k} &= Ax_k + \Gamma w_k - \left(A\hat{x}_{k|k-1} + K_k^p(z_k - C\hat{x}_{k|k-1}) \right) \\ &= A(x_k - \hat{x}_{k|k-1}) + \Gamma w_k - K_k^p(Cx_k + Hv_k - C\hat{x}_{k|k-1}) \\ &= (A - K_k^p C)\tilde{x}_{k|k-1} + \Gamma w_k - K_k^p H v_k,\end{aligned}$$

which proves

$$\tilde{x}_{k+1|k} = (A - K_k^p C)\tilde{x}_{k|k-1} + \Gamma w_k - K_k^p H v_k. \quad (14.18)$$

Before computing the covariance we isolate the orthogonality that makes the cross-terms vanish.

Lemma 14.9 (Past error is orthogonal to current noise). *Under Assumption 14.1, $\mathbb{E}[\tilde{x}_{k|k-1} w_k^\top] = 0$ and $\mathbb{E}[\tilde{x}_{k|k-1} v_k^\top] = 0$.*

Proof. The state x_k is a fixed linear combination of x_0 and $w_{k_0}, \dots, w_{k-1}$, and $\hat{x}_{k|k-1} \in \overline{\text{span}} \mathcal{I}_{k-1}$ is a fixed linear combination of $x_0, w_{k_0}, \dots, w_{k-1}$, and $v_{k_0}, \dots, v_{k-1}$. Hence the prediction error $\tilde{x}_{k|k-1} = x_k - \hat{x}_{k|k-1}$ is a linear combination of those same variables. By Assumption 14.1 each of them is uncorrelated with w_k and with v_k : the sequences $\{w_k\}$ and $\{v_k\}$ are white, they are mutually uncorrelated with x_0 , and the only permitted same-time coupling $\mathbb{E}[w_k v_k^\top] = T$ pairs w_k with v_k at the *same* index k , never with a past index. Therefore both cross-covariances vanish. $\square$

Now compute the covariance $\Sigma_{k+1|k} = \mathbb{E}[\tilde{x}_{k+1|k} \tilde{x}_{k+1|k}^\top]$ directly from the error dynamics (14.18). Writing $\tilde{x}_{k+1|k} = (A - K_k^p C)\tilde{x}_{k|k-1} + \Gamma w_k - K_k^p H v_k$ and expanding the outer product into its nine terms, every term coupling the past error $\tilde{x}_{k|k-1}$ to a current noise w_k or v_k vanishes by Lemma 14.9, leaving

$$\begin{aligned}\Sigma_{k+1|k} &= (A - K_k^p C)\Sigma_{k|k-1}(A - K_k^p C)^\top + \Gamma \mathbb{E}[w_k w_k^\top] \Gamma^\top + K_k^p H \mathbb{E}[v_k v_k^\top] H^\top (K_k^p)^\top \\ &\quad - \Gamma \mathbb{E}[w_k v_k^\top] H^\top (K_k^p)^\top - K_k^p H \mathbb{E}[v_k w_k^\top] \Gamma^\top.\end{aligned} \quad (14.19)$$

Substituting $\mathbb{E}[w_k w_k^\top] = W$, $\mathbb{E}[v_k v_k^\top] = V$, and the same-time cross-covariance $\mathbb{E}[w_k v_k^\top] = T$ (so that $\mathbb{E}[v_k w_k^\top] = T^\top$) gives

$$\begin{aligned}\Sigma_{k+1|k} &= (A - K_k^p C)\Sigma_{k|k-1}(A - K_k^p C)^\top + \Gamma W \Gamma^\top \\ &\quad - \Gamma T H^\top (K_k^p)^\top - K_k^p H T^\top \Gamma^\top + K_k^p H V H^\top (K_k^p)^\top.\end{aligned} \quad (14.20)$$

Equivalent “Riccati” form. Let $\Lambda_k := C\Sigma_{k|k-1}C^\top + HVH^\top$ and $K_k^p = (A\Sigma_{k|k-1}C^\top + \Gamma TH^\top)\Lambda_k^{-1}$. The identity (14.20) can be recast as

$$\Sigma_{k+1|k} = A\Sigma_{k|k-1}A^\top + \Gamma W \Gamma^\top - (A\Sigma_{k|k-1}C^\top + \Gamma TH^\top)\Lambda_k^{-1}(C\Sigma_{k|k-1}A^\top + HT^\top \Gamma^\top), \quad (14.21)$$

as the following calculation shows.

Proof. [Derivation of (14.21) from (14.20)] Abbreviate the current prediction covariance by $\Sigma := \Sigma_{k|k-1}$ and set

$$U := A\Sigma C^\top + \Gamma TH^\top, \quad \text{so that} \quad U^\top = C\Sigma A^\top + HT^\top \Gamma^\top, \quad K_k^p = U\Lambda_k^{-1},$$

using the symmetry $\Sigma = \Sigma^\top$ and $\Lambda_k = \Lambda_k^\top$. Expand the first term of (14.20),

$$(A - K_k^p C) \Sigma (A - K_k^p C)^\top = A \Sigma A^\top - A \Sigma C^\top (K_k^p)^\top - K_k^p C \Sigma A^\top + K_k^p C \Sigma C^\top (K_k^p)^\top,$$

and substitute into (14.20). Collecting terms shows that the factors of $(K_k^p)^\top$ on the right combine as $-A \Sigma C^\top (K_k^p)^\top - \Gamma T H^\top (K_k^p)^\top = -U (K_k^p)^\top$; the factors of K_k^p on the left combine as $-K_k^p C \Sigma A^\top - K_k^p H T^\top \Gamma^\top = -K_k^p U^\top$; and the two quadratic terms combine as $K_k^p C \Sigma C^\top (K_k^p)^\top + K_k^p H V H^\top (K_k^p)^\top = K_k^p (C \Sigma C^\top + H V H^\top) (K_k^p)^\top = K_k^p \Lambda_k (K_k^p)^\top$. Hence

$$\Sigma_{k+1|k} = A \Sigma A^\top + \Gamma W \Gamma^\top - U (K_k^p)^\top - K_k^p U^\top + K_k^p \Lambda_k (K_k^p)^\top. \quad (14.22)$$

Now insert $K_k^p = U \Lambda_k^{-1}$, so that $(K_k^p)^\top = \Lambda_k^{-1} U^\top$ and $K_k^p \Lambda_k (K_k^p)^\top = U \Lambda_k^{-1} \Lambda_k \Lambda_k^{-1} U^\top = U \Lambda_k^{-1} U^\top$:

$$-U (K_k^p)^\top - K_k^p U^\top + K_k^p \Lambda_k (K_k^p)^\top = -U \Lambda_k^{-1} U^\top - U \Lambda_k^{-1} U^\top + U \Lambda_k^{-1} U^\top = -U \Lambda_k^{-1} U^\top.$$

Substituting this into (14.22) gives $\Sigma_{k+1|k} = A \Sigma A^\top + \Gamma W \Gamma^\top - U \Lambda_k^{-1} U^\top$, which is (14.21). $\square$ This is the discrete-time Riccati difference equation associated with the Kalman filter, and it

deserves to be read as a story about competing effects. The term $A \Sigma_{k|k-1} A^\top$ is how uncertainty grows when yesterday's error is propagated through the dynamics; $\Gamma W \Gamma^\top$ is fresh uncertainty injected by process noise; and the subtracted term is the reduction bought by the measurement at time k . The predict–update cycle of Figure 14.1 is exactly this equation in two moves: prediction inflates the covariance through the model, and the update deflates it through the innovation. A crucial feature, which sets this apart from the estimate itself, is that (14.21) involves no measurements, only the system matrices and noise statistics. The covariance sequence, and hence the gain, can be computed *offline*, before any data arrive. We will see the same equation again, transposed, when we solve the linear regulator: the algebra of optimal filtering and the algebra of optimal control are one and the same.

14.1.6 ■ Innovation whiteness and covariance formula

The chapter opened with the claim that the innovations are the “new information” in each measurement. We now make that precise: successive innovations are mutually orthogonal, so the sequence $\{\nu_k\}$ carries no redundancy, each ν_k is genuinely fresh, orthogonal to everything the filter has already absorbed. This is what justified, back in Proposition 14.6, correcting the estimate along ν_k alone, and it is the exact dynamical analogue of the Gram–Schmidt innovations of Chapter 13 (Lemma 13.27). Recall $\nu_k = z_k - C \hat{x}_{k|k-1} = C \tilde{x}_{k|k-1} + H v_k$.

Theorem 14.10 (Innovations orthogonality / whiteness). *For $j < k$, $\mathbb{E}[\nu_k \nu_j^\top] = 0$ (innovations are orthogonal across time), and $\text{cov}(\nu_k) = \mathbb{E}[\nu_k \nu_k^\top] = C \Sigma_{k|k-1} C^\top + H V H^\top$.*

Proof. Fix $j < k$. Since ν_j is a linear function of $(z_{k_0}, \dots, z_j)$, it is $\mathcal{I}_j$ -measurable, hence also $\mathcal{I}_{k-1}$ -measurable. By Lemma 14.2, the prediction error $\tilde{x}_{k|k-1} = x_k - \hat{x}_{k|k-1}$ is orthogonal to $\mathcal{I}_{k-1}$, therefore $\mathbb{E}[\tilde{x}_{k|k-1} \nu_j^\top] = 0$. Also v_k is uncorrelated with $\mathcal{I}_j$ (because the measurement noises are white and independent across time), so $\mathbb{E}[v_k \nu_j^\top] = 0$. Using $\nu_k = C \tilde{x}_{k|k-1} + H v_k$,

$$\mathbb{E}[\nu_k \nu_j^\top] = C \mathbb{E}[\tilde{x}_{k|k-1} \nu_j^\top] + H \mathbb{E}[v_k \nu_j^\top] = 0,$$

proving orthogonality for $k \neq j$.

For the covariance, expand:

$$\mathbb{E}[\nu_k \nu_k^\top] = \mathbb{E}[(C\tilde{x}_{k|k-1} + H v_k)(C\tilde{x}_{k|k-1} + H v_k)^\top] = C\Sigma_{k|k-1}C^\top + H V H^\top,$$

since $\mathbb{E}[\tilde{x}_{k|k-1} v_k^\top] = 0$. $\square$

Whiteness is not merely elegant; it is the filter's built-in self-check. If the model (A, C, W, V) is correct, the residuals ν_k that the filter produces in operation should look like white noise with the predicted covariance Λ_k . Feeding real residuals through a whiteness test therefore audits the model: temporal correlation in the residuals is a signature of a misspecified system—a wrong A , an underestimated W , an unmodeled input—because a correct filter would already have extracted any such predictable structure. This diagnostic is pursued in the exercises.

14.1.7 ■ Steady-state algebraic Riccati equation

In the time-invariant case one expects the offline covariance recursion to settle down, and it does. If $\Sigma_{k|k-1} \rightarrow \Sigma$ as $k \rightarrow \infty$, taking limits in (14.21) yields the algebraic Riccati equation (ARE)

$$\Sigma = A\Sigma A^\top + \Gamma W \Gamma^\top - (A\Sigma C^\top + \Gamma T H^\top)(C\Sigma C^\top + H V H^\top)^{-1}(C\Sigma A^\top + H T^\top \Gamma^\top). \quad (14.23)$$

At the fixed point the two opposing effects of the Riccati recursion—the inflation of uncertainty by the dynamics and process noise, and its deflation by each measurement—come exactly into balance, and the filter runs with a constant gain $K^p = (A\Sigma C^\top + \Gamma T H^\top)(C\Sigma C^\top + H V H^\top)^{-1}$. This is the version of the filter one usually implements: no on-line covariance propagation, just a fixed gain applied to the innovation. Whether the balance point exists and is stable is settled by detectability and stabilizability conditions dual to those of the regulator. To state them, introduce the noise-reduced filter data

$$\bar{A}_f := A - \Gamma T H^\top (H V H^\top)^{-1} C, \quad \bar{\Gamma}_f \bar{\Gamma}_f^\top := \Gamma(W - T H^\top (H V H^\top)^{-1} H T^\top) \Gamma^\top, \quad (14.24)$$

the exact duals, under (14.61), of the reduced regulator data $(\bar{A}, \bar{Q}^{1/2})$ of (14.62); when $T = 0$ they collapse to $\bar{A}_f = A$ and $\bar{\Gamma}_f = \Gamma W^{1/2}$.

Theorem 14.11 (Steady-state filter algebraic Riccati equation). *Suppose (C, A) is detectable and $(\bar{A}_f, \bar{\Gamma}_f)$ is stabilizable. Then the filter algebraic Riccati equation (14.23) has a unique symmetric positive-semidefinite solution Σ ; from every initialization $\Sigma_{k_0|k_0-1} \succeq 0$ the prediction covariances generated by (14.21) converge to Σ ; and the steady-state predictor error matrix*

$$A - K^p C, \quad K^p = (A\Sigma C^\top + \Gamma T H^\top)(C\Sigma C^\top + H V H^\top)^{-1}, \quad (14.25)$$

is Schur stable.

Proof. By Proposition 14.20, the filter Riccati recursion (14.21) is the image under the substitution (14.61) of the regulator Riccati recursion (14.52) for the auxiliary data $(A, B, Q, R, N) = (A^\top, C^\top, \Gamma W \Gamma^\top, H V H^\top, \Gamma T H^\top)$; concretely, substituting these into (14.63) reproduces (14.23) verbatim with Σ in the role of K . For this auxiliary regulator, stabilizability of $(A^\top, C^\top)$ is equivalent to detectability of (C, A) , and detectability of the reduced pair $(\bar{Q}^{1/2}, \bar{A})$ translates, transposing (14.62) through (14.61), into stabilizability of $(\bar{A}_f, \bar{\Gamma}_f)$ in (14.24). The hypotheses of Theorem 14.22 thus hold for the auxiliary regulator, yielding a unique stabilizing solution, convergence of the iterates, and Schur stability of the auxiliary closed-loop matrix $A^\top - C^\top (K^p)^\top$. Transposing returns the three claims for the filter, with $A - K^p C$ Schur. $\square$

The same conclusion is reached, without duality and directly for the conditional covariance, as the steady-state Kalman filter theorem of the stochastic-control chapter (Section 15.10); the two are the projection-side and conditional-mean-side statements of a single fact.

14.1.8 ■ Worked examples (scalar cases)

Example 14.12 Constant random variable observed in noise.

This is the simplest possible filtering problem, and it is instructive precisely because it is the recursive form of a problem already solved statically in Chapter 13: estimating one unknown scalar from a growing list of noisy measurements. Chapter 13 would form the estimate in a single batch once all the data are in hand; here we watch the same estimate being assembled one measurement at a time, and we get to see the uncertainty shrink as it happens. Consider the scalar system

$$x_{k+1} = x_k, \quad z_k = x_k + v_k, \quad (14.26)$$

where $\{v_k\}$ is a zero-mean white-noise sequence with $\mathbb{E}[v_k^2] = r > 0$, and the initial state x_0 is random with $\mathbb{E}[x_0] = m_0$ and $\text{var}(x_0) = p_0$. We denote the one-step prediction error variance by $p_k := \Sigma_{k|k-1} = \mathbb{E}[(x_k - \hat{x}_{k|k-1})^2]$. For (14.26) we have $A = 1$, $C = 1$, $\Gamma = 0$, $H = 1$, and no process noise ($W = 0$).

Innovations and the scalar Kalman gain. The innovation (residual) is $\nu_k := z_k - \hat{x}_{k|k-1}$. Since $z_k = x_k + v_k$, we can write $\nu_k = (x_k - \hat{x}_{k|k-1}) + v_k = \tilde{x}_{k|k-1} + v_k$, where $\tilde{x}_{k|k-1} := x_k - \hat{x}_{k|k-1}$ is the prediction error.

The innovation variance is therefore

$$\text{var}(\nu_k) = \text{var}(\tilde{x}_{k|k-1}) + \text{var}(v_k) = p_k + r,$$

using the orthogonality of $\tilde{x}_{k|k-1}$ and v_k .

The scalar Kalman gain for the *filtered* update is

$$g_k = \frac{\text{cov}(x_k, \nu_k)}{\text{var}(\nu_k)} = \frac{\text{cov}(\tilde{x}_{k|k-1} + \hat{x}_{k|k-1}, \tilde{x}_{k|k-1} + v_k)}{p_k + r} = \frac{p_k}{p_k + r}. \quad (14.27)$$

Filtered estimate and filtered error variance. The LLS filtered estimate is

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + g_k \nu_k = \hat{x}_{k|k-1} + \frac{p_k}{p_k + r} (z_k - \hat{x}_{k|k-1}). \quad (14.28)$$

Equivalently,

$$\hat{x}_{k|k} = \frac{r}{p_k + r} \hat{x}_{k|k-1} + \frac{p_k}{p_k + r} z_k,$$

so the filtered estimate is a convex combination of the prior prediction and the new measurement.

The filtered error variance is

$$\Sigma_{k|k} = p_k - g_k^2 \text{var}(\nu_k) = p_k - \left(\frac{p_k}{p_k + r} \right)^2 (p_k + r) = p_k - \frac{p_k^2}{p_k + r} = \frac{p_k r}{p_k + r}.$$

(As expected, $\Sigma_{k|k} \leq p_k$.)

Prediction and the Riccati recursion. Because $x_{k+1} = x_k$ and there is no process noise, the one-step predictor is simply

$$\hat{x}_{k+1|k} = \hat{x}_{k|k}. \quad (14.29)$$

Likewise, the one-step prediction error variance at time $k + 1$ equals the filtered variance at time k : $p_{k+1} = \Sigma_{k+1|k} = \Sigma_{k|k}$. Hence we obtain the scalar Riccati difference equation

$$p_{k+1} = p_k - \frac{p_k^2}{p_k + r} = \frac{p_k r}{p_k + r}. \quad (14.30)$$

Closed-form solution of the Riccati recursion. Invert (14.30):

$$\frac{1}{p_{k+1}} = \frac{p_k + r}{p_k r} = \frac{1}{p_k} + \frac{1}{r}.$$

Iterating gives

$$\frac{1}{p_k} = \frac{1}{p_0} + \frac{k}{r}, \quad \implies \quad p_k = \frac{p_0 r}{r + p_0 k}. \quad (14.31)$$

Predictor recursion in explicit time-varying form. Using (14.29) and (14.28),

$$\hat{x}_{k+1|k} = \hat{x}_{k|k-1} + \frac{p_k}{p_k + r} (z_k - \hat{x}_{k|k-1}). \quad (14.32)$$

Substitute (14.31):

$$\frac{p_k}{p_k + r} = \frac{\frac{p_0 r}{r + p_0 k}}{\frac{p_0 r}{r + p_0 k} + r} = \frac{p_0}{r + p_0(k + 1)}.$$

Hence

$$\begin{aligned} \hat{x}_{k+1|k} &= \left(1 - \frac{p_0}{r + p_0(k + 1)}\right) \hat{x}_{k|k-1} + \frac{p_0}{r + p_0(k + 1)} z_k \\ &= \frac{r + p_0 k}{r + p_0(k + 1)} \hat{x}_{k|k-1} + \frac{p_0}{r + p_0(k + 1)} z_k. \end{aligned} \quad (14.33)$$

Unrolling the recursion as weighted averaging. The recursion (14.33) implies $\hat{x}_{k|k-1}$ is a weighted average of the prior mean and past measurements. In particular,

$$\hat{x}_{k|k-1} = \frac{r}{r + p_0 k} m_0 + \frac{p_0}{r + p_0 k} \sum_{j=0}^{k-1} z_j, \quad k \geq 1. \quad (14.34)$$

Proof. [Proof of (14.34) by induction on k] For $k = 1$, the predictor recursion (14.33) with $k = 0$ and the initialization $\hat{x}_{0|-1} = m_0$ give

$$\hat{x}_{1|0} = \frac{r}{r + p_0} m_0 + \frac{p_0}{r + p_0} z_0,$$

which is (14.34) at $k = 1$ (the sum has the single term z_0). Assume (14.34) holds at some $k \geq 1$. Applying (14.33) and the inductive hypothesis,

$$\begin{aligned}\hat{x}_{k+1|k} &= \frac{r + p_0 k}{r + p_0(k+1)} \hat{x}_{k|k-1} + \frac{p_0}{r + p_0(k+1)} z_k \\ &= \frac{r + p_0 k}{r + p_0(k+1)} \left(\frac{r}{r + p_0 k} m_0 + \frac{p_0}{r + p_0 k} \sum_{j=0}^{k-1} z_j \right) + \frac{p_0}{r + p_0(k+1)} z_k \\ &= \frac{r}{r + p_0(k+1)} m_0 + \frac{p_0}{r + p_0(k+1)} \sum_{j=0}^{k-1} z_j + \frac{p_0}{r + p_0(k+1)} z_k \\ &= \frac{r}{r + p_0(k+1)} m_0 + \frac{p_0}{r + p_0(k+1)} \sum_{j=0}^k z_j,\end{aligned}$$

which is (14.34) at $k + 1$, where the factor $\frac{r+p_0k}{r+p_0(k+1)}$ cancels the common denominator $r + p_0 k$ in both weights. This closes the induction. $\square$

Interpretation: the estimator behaves like a running average of measurements, with the weight on the initial mean m_0 decaying as k grows. This is the sample mean of the observations, and the filter has rediscovered it without ever storing the data: the innovations form let the whole history be summarized by two running numbers, the current estimate and its variance. It is exactly the batch least-squares answer of Chapter 13 delivered recursively.

Asymptotics. From (14.31), $p_k \rightarrow 0$ as $k \rightarrow \infty$: the uncertainty collapses, and repeated noisy observations of a constant latent variable eventually determine it arbitrarily well. Figure 14.3 shows this decay for several sensor noise levels. The message is worth internalizing, because it recurs throughout the chapter: the noisier the sensor (the larger r), the more slowly the variance falls, but the limit is always zero. Sensor quality sets the *rate* at which certainty accumulates, not the *ceiling*, an unchanging quantity, watched long enough, is knowable exactly. $\square$

Example 14.13 Scalar process with perfect observation.

The previous example had a static state and a noisy sensor; this one is the opposite extreme — a moving state and a perfect sensor — and the two together bracket what the filter does. With no measurement noise the update is trivial (the measurement *is* the state), and all the action is in the prediction, where process noise re-injects uncertainty at every step. Watching where the uncertainty lives in each case is the quickest route to intuition for the general filter, in which the two mechanisms operate simultaneously. Consider the scalar process

$$x_{k+1} = ax_k + w_k, \quad z_k = x_k, \quad (14.35)$$

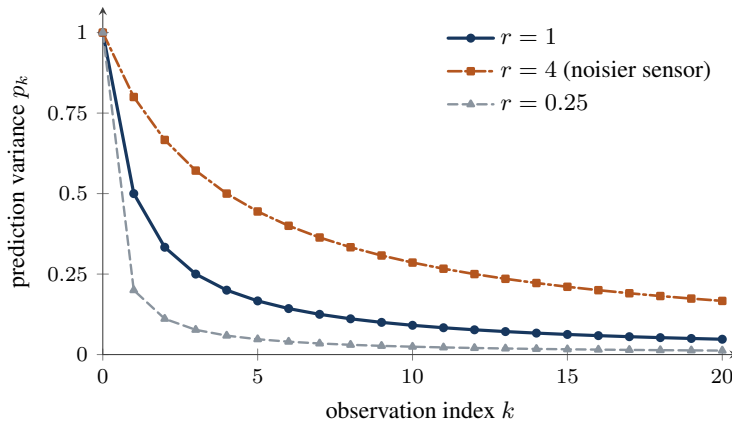


Figure 14.3: Collapse of the prediction variance for a constant scalar state observed in white noise (Example 14.12), from the closed-form solution $p_k = p_0 r / (r + p_0 k)$ with $p_0 = 1$. Every new measurement removes a further fraction of the remaining uncertainty, and $p_k \rightarrow 0$: repeated noisy looks at an unchanging quantity eventually pin it down exactly. A noisier sensor (larger r) only slows the decay; it does not change the limit.

where $\{w_k\}$ is zero-mean white noise with variance $\mathbb{E}[w_k^2] = q \geq 0$. This corresponds to (14.1)–(14.2) with $A = a$, $\Gamma = 1$, $C = 1$, and $H = 0$, so there is *no measurement noise*. This is a degenerate case of Assumption 14.1: with $H = 0$ the matrix $HVH^\top$ is singular, so the innovation covariance Λ_k is not invertible and the gain formula (14.16) does not apply. Nothing is lost — with a perfect sensor the measurement update is trivial, as we now verify directly — but the example is best read as the limit $V \downarrow 0$ of the standing model rather than an instance of it.

Exact filtering. Because $z_k = x_k$ is observed without noise, the best estimate of x_k given $\mathcal{I}_k$ is simply

$$\hat{x}_{k|k} = z_k = x_k. \quad (14.36)$$

Therefore the filtered error is identically zero:

$$\tilde{x}_{k|k} := x_k - \hat{x}_{k|k} = 0, \quad \implies \quad \Sigma_{k|k} = \mathbb{E}[\tilde{x}_{k|k}^2] = 0. \quad (14.37)$$

Prediction. Using the state equation in (14.35) and (14.36),

$$\hat{x}_{k+1|k} = \text{proj}(x_{k+1} \mid \mathcal{I}_k) = \text{proj}(ax_k + w_k \mid \mathcal{I}_k) = ax_k + \text{proj}(w_k \mid \mathcal{I}_k) = az_k, \quad (14.38)$$

since x_k is $\mathcal{I}_k$ -measurable and w_k is orthogonal to $\mathcal{I}_k$, so its projection vanishes. (Under the standing assumptions the projection is the linear least-squares estimate; it coincides with the conditional expectation $\mathbb{E}[x_{k+1} \mid \mathcal{I}_k]$ only when the primitives are jointly Gaussian, cf. Remark 14.8.)

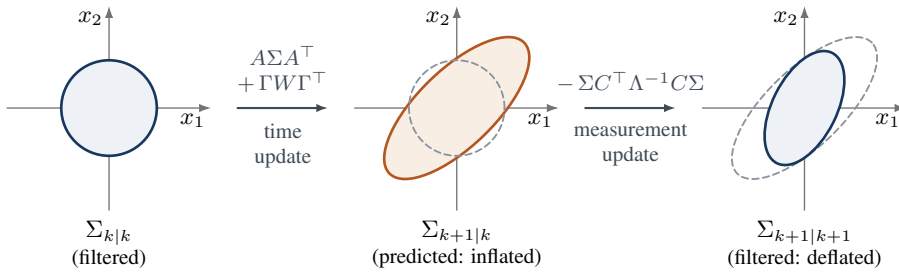


Figure 14.4: One predict–update cycle of the error covariance, the geometry behind the Riccati recursion (14.21), drawn as one-standard-deviation ellipses (dashed: the ellipse before each step) for $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, $\Gamma = \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$, $W = 0.12$, $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$, $V = 0.1$, and $\Sigma_{k|k} = 0.1 I$. The time update propagates the filtered ellipse through the dynamics and inflates it by the process noise $\Gamma W \Gamma^\top$; the measurement update deflates it along the measured direction x_1 by the positive-semidefinite term $\Sigma C^\top \Lambda^{-1} C \Sigma$, and the tilt (correlation) introduced by A lets a measurement of x_1 also shrink the uncertainty in the unmeasured x_2 . Uncertainty grows through prediction and is bought back by each measurement.

Prediction error variance. The prediction error is $\tilde{x}_{k+1|k} = x_{k+1} - \hat{x}_{k+1|k} = (ax_k + w_k) - ax_k = w_k$, hence

$$\Sigma_{k+1|k} = \mathbb{E}[w_k^2] = q. \quad (14.39)$$

So in the perfect-observation case, *all uncertainty is due to the new process noise injected between k and $k + 1$* . Immediately upon observing $z_{k+1} = x_{k+1}$, the error resets to zero again: $\Sigma_{k+1|k+1} = 0$.

Summary. Perfect observation yields

$$\hat{x}_{k|k} = z_k, \quad \Sigma_{k|k} = 0, \quad \hat{x}_{k+1|k} = az_k, \quad \Sigma_{k+1|k} = q,$$

which aligns with physical intuition: the filter “knows” the current state exactly, but the next-step prediction is uncertain only because of process noise. $\square$

Example 14.14 Constant-velocity target tracking: a vector case.

The two scalar examples isolated the two mechanisms of the filter — a static state observed in noise, and a moving state observed exactly. A genuinely two-dimensional problem lets them act together and, more strikingly, shows the filter reconstructing a quantity it never measures. Track a target moving along a line, sampled every h seconds. With state $x_k = [p_k, \dot{p}_k]^\top$ (position and velocity) the constant-velocity

model is

$$x_{k+1} = \underbrace{\begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix}}_A x_k + \Gamma w_k, \quad z_k = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_C x_k + v_k, \quad (14.40)$$

so *only position is measured*. A random acceleration drives the state: $\Gamma = [\frac{h^2}{2}, h]^\top$ with scalar intensity $\mathbb{E}[w_k^2] = W$, and the sensor has $\mathbb{E}[v_k^2] = V$.

The filter couples the two coordinates. Although $C = [1 \ 0]$ sees only p_k , the prediction step $\Sigma_{k+1|k} = A\Sigma_{k|k}A^\top + \Gamma W \Gamma^\top$ populates the off-diagonal of the covariance: because A feeds velocity into the next position, the predicted position and velocity errors become *correlated*, and a position measurement therefore sharpens the velocity estimate as well. Concretely the predictor gain

$$K_k^p = A\Sigma_{k|k-1}C^\top\Lambda_k^{-1}, \quad \Lambda_k = C\Sigma_{k|k-1}C^\top + V = [\Sigma_{k|k-1}]_{11} + V,$$

has a nonzero second (velocity) component precisely because $[\Sigma_{k|k-1}]_{21} \neq 0$, even though C has a zero there. This is the vector face of the covariance bookkeeping in the Riccati proof of Section 14.1.5: Figure 14.4 shows the error ellipse inflating through the dynamics and then deflating along the measured axis $x_1 = p$, with the tilt carrying the reduction over to the unmeasured $x_2 = \dot{p}$.

Observability and the steady state. The pair (C, A) is observable, since the observability matrix $\begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & h \end{bmatrix}$ is nonsingular for $h \neq 0$; hence (C, A) is detectable and Theorem 14.11 supplies a unique steady-state covariance and a constant gain. In steady state the gain takes the classical α - β form

$$G = \Sigma C^\top \Lambda^{-1} = \begin{bmatrix} \alpha \\ \beta/h \end{bmatrix}, \quad \hat{p}_{k|k} = \hat{p}_{k|k-1} + \alpha \nu_k, \quad \hat{\dot{p}}_{k|k} = \hat{\dot{p}}_{k|k-1} + \frac{\beta}{h} \nu_k, \quad (14.41)$$

where Σ and $\Lambda = C\Sigma C^\top + V$ are the steady-state prediction and innovation covariances, and G is the *filtered* (measurement-update) gain, related to the steady-state predictor gain by $K^p = AG$; the α - β form applies G in the correction step. Here α and β lie in $(0, 1)$ and are increasing functions of a dimensionless *tracking index* $\lambda \propto h^2\sqrt{W}/\sqrt{V}$ that weighs process noise against measurement noise. A quiet target (small λ) gives small α, β and heavy smoothing; an erratic one (large λ) gives gains near unity and a filter that defers to each new measurement. The steady-state α - β filter is what one implements in practice: a fixed two-line recursion that estimates velocity from position alone, the payoff of the coupling that the Riccati recursion creates and Figure 14.4 depicts. $\square$

14.2 ■ The Stochastic Linear Regulator Problem

We now change subjects, from estimation to control, but not methods. Chapters 10 and 11 solved the deterministic optimal control problem by dynamic programming: guess a value function, propagate it backward through Bellman's principle of optimality, and read off the optimal feedback law. For the linear-quadratic case that recipe produces a quadratic value function, a Riccati

equation, and a linear state-feedback gain, in discrete time in Chapter 10, and in the continuous-time treatment of Chapter 11 (Theorem 11.25, with Riccati equation (11.122) and gain (11.124)). The question of this section is what happens when the dynamics are pushed around by random noise. The answer is cleaner than one might fear, and it is the reason the linear-quadratic-Gaussian problem is tractable at all: the backward recursion and the feedback gain come out *identical* to the deterministic case, and the noise leaves its mark only on the achieved cost. Establishing this carefully is the goal here; it is also the half of the LQG puzzle that, joined to the filter, gives the separation theorem.

14.2.1 ■ Problem formulation

The system evolves according to

$$x_{k+1} = Ax_k + Bu_k + \Gamma w_k, \quad x_{k_0} = x_0, \quad (14.42)$$

where $\{w_k\}$ is an independent sequence of random vectors satisfying

$$\mathbb{E}[w_k] = 0, \quad \mathbb{E}[w_k w_k^\top] = W,$$

and x_0 is independent of $\{w_k\}$ with $\mathbb{E}[x_0] = m_0$ and $\text{cov}(x_0) = \Sigma_0$. The cost criterion is the quadratic form

$$J = \mathbb{E} \left\{ x_N^\top Q_N x_N + \sum_{k=k_0}^{N-1} \left(x_k^\top Q_k x_k + 2x_k^\top N_k u_k + u_k^\top R_k u_k \right) \right\}, \quad (14.43)$$

in which the stage cost is written compactly as the joint quadratic

$$x_k^\top Q_k x_k + 2x_k^\top N_k u_k + u_k^\top R_k u_k = \begin{bmatrix} x_k \\ u_k \end{bmatrix}^\top \mathcal{W}_k \begin{bmatrix} x_k \\ u_k \end{bmatrix}, \quad \mathcal{W}_k := \begin{bmatrix} Q_k & N_k \\ N_k^\top & R_k \end{bmatrix}. \quad (14.44)$$

We assume $R_k \succ 0$, $Q_N \succeq 0$, and $\mathcal{W}_k \succeq 0$ for every k (so each stage cost is nonnegative); the control set is all of $\mathbb{R}^m$ (unconstrained controls).

The control enters through $u_k \in \mathbb{R}^m$ and $B \in \mathbb{R}^{n \times m}$, while the state, process noise, and their matrices A, Γ, W keep the dimensions of the filtering problem. The cost weights are the symmetric state weight $Q_k \in \mathbb{R}^{n \times n}$, the control weight $R_k \in \mathbb{R}^{m \times m}$, the state-control cross weight $N_k \in \mathbb{R}^{n \times m}$, and the terminal weight $Q_N \in \mathbb{R}^{n \times n}$; positive definiteness of R_k makes the control genuinely costly and the minimizations below well posed, and joint positive semidefiniteness of $\mathcal{W}_k$ keeps the stage cost a bona fide penalty. The special case $\mathcal{W}_k = \begin{bmatrix} D \\ F \end{bmatrix}^\top \begin{bmatrix} D \\ F \end{bmatrix}$, i.e. $Q_k = D^\top D$, $N_k = D^\top F$, $R_k = F^\top F$, recovers the regulated-output cost $\|Dx_k + Fu_k\|^2$. Two symbols are deliberately recycled from the filtering half, and their dimensions are the cleanest way to tell the meanings apart: the Riccati (value-function) matrix K_k is now $n \times n$, not the $p \times p$ innovation covariance of Section 14.1, and the feedback gain L_k is $m \times n$, mapping an n -dimensional state to an m -dimensional control, rather than the $n \times p$ update gain of the filter. Context always disambiguates, but when in doubt the shapes settle the matter.

14.2.2 ■ Dynamic programming equation

The principle of optimality carries over from the deterministic theory of Chapters 10 and 11 with one modification: because the next state $Ax + Bu + \Gamma w_k$ is random, the cost-to-go from it must be *averaged* over the process noise before we minimize. Writing $V_k(x)$ for the optimal expected

cost-to-go from state x at time k , the deterministic discrete-time Bellman recursion of Chapter 10 (whose continuous-time form is Chapter 11's equation (11.96)) becomes its stochastic counterpart

$$V_k(x) = \min_{u \in \mathbb{R}^m} \left\{ x^\top Q_k x + 2x^\top N_k u + u^\top R_k u + \mathbb{E}[V_{k+1}(Ax + Bu + \Gamma w_k)] \right\}, \quad (14.45)$$

with terminal condition

$$V_N(x) = x^\top Q_N x. \quad (14.46)$$

The single expectation operator in (14.45) is where all the stochasticity enters, and tracking what it does to a quadratic value function is the whole content of the derivation that follows. We record two elementary facts we will use repeatedly.

14.2.3 ■ Two preliminary results

Lemma 14.15 (Quadratic expectation identity). *For any random vector x with mean m and covariance Π , and any symmetric matrix $\Xi \succeq 0$,*

$$\begin{aligned} \mathbb{E}[x^\top \Xi x] &= \mathbb{E}[(x - m)^\top \Xi (x - m)] + m^\top \Xi m \\ &= \text{tr}(\Xi \Pi) + m^\top \Xi m. \end{aligned} \quad (14.47)$$

Proof. Write $x = m + (x - m)$ and expand: $x^\top \Xi x = (x - m)^\top \Xi (x - m) + 2m^\top \Xi (x - m) + m^\top \Xi m$. Take expectations; $\mathbb{E}[x - m] = 0$ eliminates the cross term, and

$$\mathbb{E}[(x - m)^\top \Xi (x - m)] = \text{tr}(\Xi \mathbb{E}[(x - m)(x - m)^\top]) = \text{tr}(\Xi \Pi).$$

□

Lemma 14.16 (Completing the square). *Let $R_1 \succ 0$, and let R_2 be any matrix and R_3 symmetric. Consider*

$$g(u) = u^\top R_1 u + u^\top R_2 x + x^\top R_2^\top u + x^\top R_3 x. \quad (14.48)$$

Then

$$g(u) = (u + R_1^{-1} R_2 x)^\top R_1 (u + R_1^{-1} R_2 x) + x^\top (R_3 - R_2^\top R_1^{-1} R_2) x, \quad (14.49)$$

so the minimizer is

$$u^*(x) = -R_1^{-1} R_2 x, \quad (14.50)$$

and the minimum value is

$$\min_u g(u) = x^\top (R_3 - R_2^\top R_1^{-1} R_2) x. \quad (14.51)$$

Proof. Expand the right-hand side of (14.49):

$$(u + R_1^{-1} R_2 x)^\top R_1 (u + R_1^{-1} R_2 x) = u^\top R_1 u + u^\top R_2 x + x^\top R_2^\top u + x^\top R_2^\top R_1^{-1} R_2 x.$$

Subtract $x^\top R_2^\top R_1^{-1} R_2 x$ and add $x^\top R_3 x$ to match (14.48). Since $R_1 \succ 0$, the quadratic term is minimized at $u = -R_1^{-1} R_2 x$. □

14.2.4 ■ Optimal value function, feedback law, and Riccati recursion

We now solve the dynamic program in closed form. The quadratic terminal cost and stage cost make a quadratic value function the natural candidate, and the two lemmas verify that the candidate propagates.

Theorem 14.17 (Optimal linear-quadratic regulator). *Define K_k and q_k backward by $K_N = Q_N$, $q_N = 0$, and, for $k = N - 1, \dots, k_0$, the Riccati recursion*

$$K_k = Q_k + A^\top K_{k+1} A - (A^\top K_{k+1} B + N_k) (R_k + B^\top K_{k+1} B)^{-1} (B^\top K_{k+1} A + N_k^\top) \quad (14.52)$$

together with the scalar recursion

$$q_k = q_{k+1} + \text{tr}(K_{k+1} \Gamma W \Gamma^\top). \quad (14.53)$$

Then each K_k is symmetric, $R_k + B^\top K_{k+1} B \succ 0$, and the optimal expected cost-to-go is the quadratic

$$V_k(x) = x^\top K_k x + q_k, \quad (14.54)$$

attained by the Markov policy $\mu^ = \{\mu_k^*\}$ with*

$$u_k = \mu_k^*(x_k) = L_k x_k, \quad L_k := -(R_k + B^\top K_{k+1} B)^{-1} (B^\top K_{k+1} A + N_k^\top). \quad (14.55)$$

Proof. We argue by backward induction on k , proving that V_k has the stated form and that (14.55) attains the minimum in the dynamic programming equation (14.45).

Base case. At $k = N$ the terminal condition (14.46) gives $V_N(x) = x^\top Q_N x = x^\top K_N x + q_N$ with $K_N = Q_N \succeq 0$ and $q_N = 0$.

Inductive step. Suppose $V_{k+1}(x) = x^\top K_{k+1} x + q_{k+1}$ with K_{k+1} symmetric and positive semidefinite; the base case supplies both, since $K_N = Q_N \succeq 0$. The next state from (x, u) is $Ax + Bu + \Gamma w_k$, which has conditional mean $Ax + Bu$ and covariance $\Gamma W \Gamma^\top$. Lemma 14.15 with $\Xi = K_{k+1}$ therefore gives

$$\mathbb{E}[V_{k+1}(Ax + Bu + \Gamma w_k)] = (Ax + Bu)^\top K_{k+1} (Ax + Bu) + \text{tr}(K_{k+1} \Gamma W \Gamma^\top) + q_{k+1}. \quad (14.56)$$

Substituting (14.56) into (14.45),

$$\begin{aligned} V_k(x) = \min_u \bigg\{ & x^\top Q_k x + 2x^\top N_k u + u^\top R_k u \\ & + (Ax + Bu)^\top K_{k+1} (Ax + Bu) + \text{tr}(K_{k+1} \Gamma W \Gamma^\top) + q_{k+1} \bigg\}. \end{aligned} \quad (14.57)$$

Collecting the terms in u , using

$$\begin{aligned} & (Ax + Bu)^\top K_{k+1} (Ax + Bu) \\ & = x^\top A^\top K_{k+1} A x + u^\top B^\top K_{k+1} B u + 2u^\top B^\top K_{k+1} A x, \end{aligned}$$

the bracketed expression is exactly $g(u) + \text{tr}(K_{k+1} \Gamma W \Gamma^\top) + q_{k+1}$ with g of the form (14.48) and

$$R_1 = R_k + B^\top K_{k+1} B, \quad R_2 = B^\top K_{k+1} A + N_k^\top, \quad R_3 = Q_k + A^\top K_{k+1} A.$$

Since $R_k \succ 0$ and $K_{k+1} \succeq 0$ by the induction hypothesis, we have $R_1 = R_k + B^\top K_{k+1} B \succ 0$. Lemma 14.16 then gives the unique minimizer $u = -R_1^{-1} R_2 x = L_k x$, which is (14.55), and the minimum value $x^\top (R_3 - R_2^\top R_1^{-1} R_2) x$. Hence

$$V_k(x) = x^\top (R_3 - R_2^\top R_1^{-1} R_2) x + \text{tr}(K_{k+1} \Gamma W \Gamma^\top) + q_{k+1} = x^\top K_k x + q_k,$$

with $K_k = R_3 - R_2^\top R_1^{-1} R_2$ equal to (14.52) and q_k equal to (14.53). Symmetry of K_k is manifest from that expression. For positive semidefiniteness, read the completed square backwards: by Lemma 14.16, for every x ,

$$x^\top K_k x = \min_{u \in \mathbb{R}^m} \left\{ x^\top Q_k x + 2x^\top N_k u + u^\top R_k u + (Ax + Bu)^\top K_{k+1} (Ax + Bu) \right\},$$

and each term in the brace is nonnegative—the first three because $\mathcal{W}_k \succeq 0$, the last because $K_{k+1} \succeq 0$ —so the minimum is nonnegative and $K_k \succeq 0$. This restores the induction hypothesis and closes the induction. $\square$

Iterating the scalar recursion (14.53) from $q_N = 0$ gives the offset in closed form,

$$q_k = \sum_{j=k}^{N-1} \text{tr}(K_{j+1} \Gamma W \Gamma^\top). \quad (14.58)$$

14.2.5 ■ Optimal cost

Averaging the value function over the initial state turns Theorem 14.17 into a formula for the minimal expected cost.

Proposition 14.18 (Optimal cost of the stochastic regulator). *Under the policy μ^* of Theorem 14.17, the minimal value of the cost (14.43) is*

$$J^* = \mathbb{E}[V_{k_0}(x_0)] = \text{tr}(K_{k_0} \Sigma_0) + m_0^\top K_{k_0} m_0 + \sum_{j=k_0}^{N-1} \text{tr}(K_{j+1} \Gamma W \Gamma^\top). \quad (14.59)$$

Consequently $J^* = m_0^\top K_{k_0} m_0$ in the deterministic case ($\Sigma_0 = 0$, $W = 0$); the two remaining terms $\text{tr}(K_{k_0} \Sigma_0)$ and $\sum_{j=k_0}^{N-1} \text{tr}(K_{j+1} \Gamma W \Gamma^\top)$ are the price of the random initial condition and of the process noise, respectively.

Proof. By the dynamic programming principle, $\min_\mu J = \mathbb{E}[V_{k_0}(x_0)]$, and Theorem 14.17 gives $V_{k_0}(x) = x^\top K_{k_0} x + q_{k_0}$. Hence $J^* = \mathbb{E}[x_0^\top K_{k_0} x_0] + q_{k_0}$. The quadratic expectation identity (Lemma 14.15) with mean m_0 and covariance Σ_0 gives $\mathbb{E}[x_0^\top K_{k_0} x_0] = \text{tr}(K_{k_0} \Sigma_0) + m_0^\top K_{k_0} m_0$, and $q_{k_0} = \sum_{j=k_0}^{N-1} \text{tr}(K_{j+1} \Gamma W \Gamma^\top)$ by (14.58). Adding the two yields (14.59); the deterministic specialization and the interpretation of the two extra terms are immediate. $\square$

14.2.6 ■ Alternative derivation: completion of squares

The dynamic programming argument of Theorem 14.17 can be replaced by a single algebraic identity that exhibits the cost, for every admissible control, as a fixed part plus a nonnegative penalty for departing from the feedback law. This “completion of squares” re-proves optimality without invoking the principle of optimality, and it makes the role of the Riccati matrix K_k

transparent: it is exactly the matrix that makes the running cost telescope. It is the trajectory-wide version of the pointwise Lemma 14.16 used inside the dynamic program.

Proposition 14.19 (Completion-of-squares identity). *Let K_k solve the Riccati recursion (14.52) with $K_N = Q_N$, and set $S_k := R_k + B^\top K_{k+1} B \succ 0$ and $L_k := -S_k^{-1}(B^\top K_{k+1} A + N_k^\top)$ as in (14.55). Then for every control sequence $\{u_k\}$ adapted to the state (each u_k a function of $x_{k_0}, \dots, x_k$), the cost (14.43) satisfies*

$$J = m_0^\top K_{k_0} m_0 + \text{tr}(K_{k_0} \Sigma_0) + \sum_{k=k_0}^{N-1} \text{tr}(K_{k+1} \Gamma W \Gamma^\top) + \sum_{k=k_0}^{N-1} \mathbb{E}[(u_k - L_k x_k)^\top S_k (u_k - L_k x_k)]. \quad (14.60)$$

Proof. Because $K_N = Q_N$, the terminal cost telescopes pathwise,

$$x_N^\top Q_N x_N = x_{k_0}^\top K_{k_0} x_{k_0} + \sum_{k=k_0}^{N-1} (x_{k+1}^\top K_{k+1} x_{k+1} - x_k^\top K_k x_k).$$

Substituting into (14.43) gives $J = \mathbb{E}[x_{k_0}^\top K_{k_0} x_{k_0} + \sum_k \Delta_k]$ with

$$\Delta_k := x_{k+1}^\top K_{k+1} x_{k+1} - x_k^\top K_k x_k + x_k^\top Q_k x_k + 2x_k^\top N_k u_k + u_k^\top R_k u_k.$$

Condition on (x_k, u_k) . Since $x_{k+1} = Ax_k + Bu_k + \Gamma w_k$ with w_k zero-mean, covariance W , and independent of (x_k, u_k) ,

$$\mathbb{E}[x_{k+1}^\top K_{k+1} x_{k+1} \mid x_k, u_k] = (Ax_k + Bu_k)^\top K_{k+1} (Ax_k + Bu_k) + \text{tr}(K_{k+1} \Gamma W \Gamma^\top).$$

Expanding the quadratic and collecting terms, and writing $\Psi_k := A^\top K_{k+1} B + N_k$,

$$\mathbb{E}[\Delta_k \mid x_k, u_k] = x_k^\top (A^\top K_{k+1} A + Q_k - K_k) x_k + 2x_k^\top \Psi_k u_k + u_k^\top S_k u_k + \text{tr}(K_{k+1} \Gamma W \Gamma^\top).$$

The Riccati equation (14.52) is exactly $A^\top K_{k+1} A + Q_k - K_k = \Psi_k S_k^{-1} \Psi_k^\top$, so the first three terms are a perfect square,

$$\begin{aligned} & x_k^\top \Psi_k S_k^{-1} \Psi_k^\top x_k + 2x_k^\top \Psi_k u_k + u_k^\top S_k u_k \\ &= (u_k + S_k^{-1} \Psi_k^\top x_k)^\top S_k (u_k + S_k^{-1} \Psi_k^\top x_k) \\ &= (u_k - L_k x_k)^\top S_k (u_k - L_k x_k), \end{aligned}$$

using $L_k = -S_k^{-1} \Psi_k^\top$. Hence $\mathbb{E}[\Delta_k \mid x_k, u_k] = (u_k - L_k x_k)^\top S_k (u_k - L_k x_k) + \text{tr}(K_{k+1} \Gamma W \Gamma^\top)$. Taking total expectations, summing over k , and evaluating $\mathbb{E}[x_{k_0}^\top K_{k_0} x_{k_0}] = m_0^\top K_{k_0} m_0 + \text{tr}(K_{k_0} \Sigma_0)$ by Lemma 14.15 yields (14.60). $\square$

The identity (14.60) settles both halves of the problem at once. Every $S_k \succ 0$, so the last sum is nonnegative and vanishes if and only if $u_k = L_k x_k$ for all k ; therefore the feedback law $u_k = L_k x_k$ is optimal among *all* adapted controls, not merely Markov ones, which reproves Theorem 14.17, and the minimal value is the leading three terms, which reproves the optimal cost of Proposition 14.18. Beyond re-deriving the answer, the identity itemizes it: K_{k_0} carries the deterministic value, the traces $\text{tr}(K_{k+1} \Gamma W \Gamma^\top)$ are the price of process noise, and the final sum is precisely the penalty for using any gain other than L_k . The same accounting the dynamic program produced backward in time is here visible in a single forward identity.

Table 14.2: The regulator–filter duality dictionary. Replacing each regulator quantity on the left by its filter counterpart on the right turns the regulator Riccati recursion (14.52) into the filter Riccati recursion (14.21), term for term; Proposition 14.20 makes this precise. The last row is the induced correspondence of the cross-term-eliminated weights.

Regulator	Filter
A	$A^\top$
B	$C^\top$
Q_k	$\Gamma W \Gamma^\top$
R_k	$H V H^\top$
N_k	$\Gamma T H^\top$
K_{k+1}, K_k	$\Sigma_{k k-1}, \Sigma_{k+1 k}$
$Q_k - N_k R_k^{-1} N_k^\top$	$\Gamma (W - T H^\top (H V H^\top)^{-1} H T^\top) \Gamma^\top$

14.2.7 ■ Remarks and filter–control duality

Three lessons stand out, and they are best seen together, because each explains a different way in which this problem is “the same” as one we have already solved.

The first is that the backward recursion (14.52) is a discrete-time Riccati difference equation, and it is not merely reminiscent of the filter’s covariance recursion (14.21): it is that recursion in disguise. Replacing each control-side quantity by its filter-side counterpart turns one recursion into the other, according to the correspondence summarized in Table 14.2.

The regulator runs backward in time from the terminal cost, $k \leq N$; the filter runs forward from the initial covariance, $k \geq k_0$. Reading A against $A^\top$ and the input matrix B against the output matrix $C^\top$, one sees that controlling a system through its inputs and estimating it through its outputs are dual operations. This duality is more than a curiosity: it lets every theorem about one problem be imported, with the dictionary above, as a theorem about the other. The correspondence is not merely suggestive but exact, as the next proposition records.

Proposition 14.20 (Estimation–control duality). *Consider the substitution*

$$A \mapsto A^\top, \quad B \mapsto C^\top, \quad Q_k \mapsto \Gamma W \Gamma^\top, \quad R_k \mapsto H V H^\top, \quad N_k \mapsto \Gamma T H^\top, \quad (14.61)$$

together with $K_{k+1} \mapsto \Sigma_{k|k-1}$ and $K_k \mapsto \Sigma_{k+1|k}$. Under (14.61) the regulator Riccati recursion (14.52) becomes, term for term, the filter Riccati recursion (14.21). Moreover the regulator gain L_k of (14.55) and the filter gain $K_k^p := (A \Sigma_{k|k-1} C^\top + \Gamma T H^\top) (C \Sigma_{k|k-1} C^\top + H V H^\top)^{-1}$ correspond through $-L_k^\top \mapsto K_k^p$.

Proof. Write $\Sigma := \Sigma_{k|k-1}$ for the current prediction covariance and apply (14.61) to each block of (14.52), using $N_k^\top \mapsto (\Gamma T H^\top)^\top = H T^\top \Gamma^\top$. The quadratic-growth term maps as $A^\top K_{k+1} A \mapsto (A^\top)^\top \Sigma (A^\top) = A \Sigma A^\top$, and the additive weight as $Q_k \mapsto \Gamma W \Gamma^\top$. The two off-diagonal factors map as

$$A^\top K_{k+1} B + N_k \mapsto (A^\top)^\top \Sigma (C^\top) + \Gamma T H^\top = A \Sigma C^\top + \Gamma T H^\top,$$

$$B^\top K_{k+1} A + N_k^\top \mapsto (C^\top)^\top \Sigma (A^\top) + H T^\top \Gamma^\top = C \Sigma A^\top + H T^\top \Gamma^\top,$$

and the middle (inverse) factor as

$$R_k + B^\top K_{k+1} B \mapsto H V H^\top + (C^\top)^\top \Sigma (C^\top) = C \Sigma C^\top + H V H^\top = \Lambda_k.$$

Substituting these images into (14.52) yields exactly

$$\Sigma_{k+1|k} = A\Sigma A^\top + \Gamma W \Gamma^\top - (A\Sigma C^\top + \Gamma T H^\top) \Lambda_k^{-1} (C\Sigma A^\top + H T^\top \Gamma^\top),$$

which is (14.21). For the gains, transpose L_k from (14.55) and apply the same images:

$$-L_k^\top = (B^\top K_{k+1} A + N_k^\top)^\top (R_k + B^\top K_{k+1} B)^{-1} = (A^\top K_{k+1} B + N_k) (R_k + B^\top K_{k+1} B)^{-1},$$

whose image under (14.61) is $(A\Sigma C^\top + \Gamma T H^\top) (C\Sigma C^\top + H V H^\top)^{-1} = K_k^p$. $\square$

The last row of the dictionary is the induced correspondence of the cross-term-eliminated weights: completing the square in the control removes the N_k coupling and leaves the effective state weight $Q_k - N_k R_k^{-1} N_k^\top$, whose dual is the effective process-noise weight $\Gamma(W - T H^\top (H V H^\top)^{-1} H T^\top) \Gamma^\top$ obtained by the same elimination of the $\Gamma T H^\top$ coupling; these are exactly the reduced data $(\bar{A}, \bar{Q})$ of Section 14.3 and their filter analogues $(\bar{A}_f, \bar{\Gamma}_f)$ of (14.24). The upshot is that every convergence statement proved for one recursion transfers verbatim to the other under (14.61); we invoke this in Section 14.3 and again for the steady-state filter.

The second lesson is the invariance of the optimal gain under the noise, which we now state precisely.

Proposition 14.21 (Certainty equivalence of the gain). *The Riccati recursion (14.52) and hence the optimal feedback gains $\{L_k\}$ of (14.55) depend only on the system matrices A, B and the cost weights (Q_k, N_k, R_k, Q_N) ; they do not depend on the process-noise covariance W , the input matrix Γ , or the initial statistics (m_0, Σ_0) . Consequently the optimal policy μ^* coincides with the optimal policy of the deterministic problem obtained by setting $w_k \equiv 0$ and fixing x_0 ; the noise affects only the additive cost offset q_k .*

Proof. In the backward recursion of Theorem 14.17 the process noise enters a single place: through the term $\text{tr}(K_{k+1} \Gamma W \Gamma^\top)$, which is a scalar independent of u and is absorbed entirely into the offset q_k by (14.53). It does not appear in (14.52) or in the minimizer (14.55), both of which are built from (A, B, Q_k, N_k, R_k) and K_{k+1} alone. By induction K_k , and therefore L_k , are functions of the cost weights and system matrices only. Setting $W = 0$, $\Gamma = 0$, and $x_0 \equiv m_0$ leaves (14.52)–(14.55) unchanged, so the deterministic optimizer is the same $L_k x_k$; the noise moves only q_k , and hence only the additive terms of the optimal cost (14.59). $\square$

The third lesson locates precisely where the randomness did leave a mark, and it is exactly the decomposition of the optimal cost proved in Proposition 14.18: relative to the deterministic value $m^\top K_{k_0} m$, the random initial condition adds $\text{tr}(K_{k_0} \Sigma_0)$ and the process noise adds $\sum_{j=k_0}^{N-1} \text{tr}(K_{j+1} \Gamma W \Gamma^\top)$, one contribution for each step at which noise is injected. Noise cannot be steered away, only paid for; the controller's job is to keep the state where that payment is smallest.

14.3 ■ Asymptotic Properties of the Linear Regulator

In the time-invariant case the weights are constant, $Q_k \equiv Q$, $N_k \equiv N$, $R_k \equiv R$, and the asymptotic properties of the regulator are governed by the Riccati recursion. Eliminating the cross-weight N through the control, define the reduced data

$$\bar{A} := A - B R^{-1} N^\top, \quad \bar{Q} := Q - N R^{-1} N^\top \succeq 0, \quad (14.62)$$

where $\bar{Q} \succeq 0$ because $\mathcal{W} = \begin{bmatrix} Q & N \\ N^\top & R \end{bmatrix} \succeq 0$ and $R \succ 0$ (Schur complement). The matrix $\bar{A}$ is the closed-loop matrix after cross-weight elimination and $\bar{Q}^{1/2}$ the reduced state weight; they are the

objects to which the standard Riccati-convergence theory of Appendix A applies, and the exact duals, under (14.61), of the filter's $(\bar{A}_f, \bar{\Gamma}_f)$ in (14.24).

Theorem 14.22 (Regulator algebraic Riccati equation). *If (A, B) is stabilizable and $(\bar{Q}^{1/2}, \bar{A})$ is detectable, then the algebraic Riccati equation*

$$K = Q + A^\top K A - (A^\top K B + N)(R + B^\top K B)^{-1}(B^\top K A + N^\top) \quad (14.63)$$

has a unique symmetric positive-semidefinite solution K , the associated closed-loop matrix

$$A_{cl} := A - B(R + B^\top K B)^{-1}(B^\top K A + N^\top) \quad (14.64)$$

is Schur stable, and for every terminal weight $Q_N \succeq 0$ the finite-horizon Riccati iterates (14.52) converge to K as the horizon grows (equivalently $K_k \rightarrow K$ as $k \rightarrow \infty$).

Proof. Completing the square in (14.52) to eliminate the cross-weight N writes the recursion in the standard form $K_k = \bar{A}^\top K_{k+1} \bar{A} + \bar{Q} - \bar{A}^\top K_{k+1} B(R + B^\top K_{k+1} B)^{-1} B^\top K_{k+1} \bar{A}$ with $\bar{A}, \bar{Q}$ as in (14.62); this is precisely the discrete-time control Riccati recursion for the data $(\bar{A}, B, \bar{Q}^{1/2}, R)$ with no cross term. Under stabilizability of (A, B) —equivalently of $(\bar{A}, B)$, since the two differ by a state feedback—and detectability of $(\bar{Q}^{1/2}, \bar{A})$, the monotone-convergence theorem for the discrete-time Riccati recursion (Theorem 15.88, proved independently of the present chapter) supplies a unique stabilizing solution $K \succeq 0$, convergence of the iterates from any $Q_N \succeq 0$, and Schur stability of the closed-loop matrix (14.64). $\square$

The corresponding stationary optimal policy μ^* is the feedback law

$$\mu^*(x) = -(R + B^\top K B)^{-1}(B^\top K A + N^\top)x. \quad (14.65)$$

Under (14.65), the closed-loop system is

$$x_{k+1} = \left(A - B(R + B^\top K B)^{-1}(B^\top K A + N^\top) \right) x_k + \Gamma w_k. \quad (14.66)$$

If $\Pi_k := \text{cov}(x_k)$, then stability of (14.66) implies $\Pi_k \rightarrow \Pi$ for some steady-state covariance Π . In particular, if x_0 is Gaussian and w_k is white Gaussian, then x_k is Gaussian for all k and converges (in distribution) to a stationary Gaussian process. Note that because of the noise input, x_k does not converge to zero almost surely even if the closed-loop is stable.

14.4 ■ Stochastic Control of Linear Systems with Partial Observations

Everything so far has assumed the controller can see the state. The realistic situation, and the one that makes this chapter's two halves finally meet, is that it cannot: the state is hidden, and only noisy measurements are available. This is where the filter of the first half and the regulator of the second half must be used together. The remarkable conclusion, the *separation theorem*, is that they can be designed *separately*, each in ignorance of the other, and the naive combination is nonetheless optimal. Build the best estimator as if no control were being applied, build the best regulator as if the state were perfectly known, and then simply let the regulator act on the estimate. Figure 14.5 is the whole result in one picture; the rest of the section earns it. We now consider the case where the state is not directly observed:

$$x_{k+1} = A x_k + B u_k + \Gamma w_k, \quad (14.67)$$

$$z_k = C x_k + H v_k, \quad (14.68)$$

$$x_{k_0} = x_0,$$

where $\{w_k\}$ and $\{v_k\}$ are independent Gaussian white-noise sequences with

$$\mathbb{E}[w_k] = \mathbb{E}[v_k] = 0, \quad \mathbb{E}[w_k w_j^\top] = W \delta_{kj}, \quad \mathbb{E}[v_k v_j^\top] = V \delta_{kj}, \quad V \succ 0,$$

and $\mathbb{E}[w_k v_j^\top] = T \delta_{kj}$. Assume x_0 is Gaussian with mean m_0 and covariance Σ_0 , independent of $\{w_k, v_k\}$.

The cost is

$$J = \mathbb{E} \left\{ x_N^\top Q_N x_N + \sum_{k=k_0}^{N-1} \left(x_k^\top Q_k x_k + 2x_k^\top N_k u_k + u_k^\top R_k u_k \right) \right\}, \quad (14.69)$$

with the same weights (Q_k, N_k, R_k, Q_N) and joint positive semidefiniteness $\mathcal{W}_k \succeq 0$, $R_k \succ 0$ as in (14.43)–(14.44).

14.4.1 ■ Admissible policies and information structure

Let $\mathcal{I}_k := \sigma\{z(s) : k_0 \leq s \leq k\}$ be the sigma-field generated by observations up to time k . We impose a one-step delay: the control at time k may depend only on $\mathcal{I}_{k-1}$. Thus an admissible policy is a sequence $\mu = \{\mu_{k_0}, \dots, \mu_{N-1}\}$ such that $u_k = \mu_k(\mathcal{I}_{k-1})$ for $k = k_0, \dots, N-1$, with μ_k Borel measurable; the set of admissible policies is denoted $\mathcal{M}$.

14.4.2 ■ A useful decomposition and an information lemma

Introduce an auxiliary system

$$\bar{x}_{k+1} = A\bar{x}_k + \Gamma w_k, \quad \bar{x}_{k_0} = x_0, \quad (14.70)$$

$$\bar{z}_k = C\bar{x}_k + H v_k. \quad (14.71)$$

Lemma 14.23 (The control does not change the information). *For any admissible policy μ , the observation sigma-fields satisfy $\bar{\mathcal{I}}_k = \mathcal{I}_k$ for $k = k_0, \dots, N-1$. In other words, $\bar{\mathcal{I}}_k$ contains the same information as $\mathcal{I}_k$.*

Proof. Let $\tilde{x}_k := x_k - \bar{x}_k$. Then subtracting (14.70) from (14.67) yields

$$\tilde{x}_{k+1} = A\tilde{x}_k + B u_k, \quad \tilde{x}_{k_0} = 0. \quad (14.72)$$

We claim $\tilde{x}_k$ depends only on $\mathcal{I}_{k-2}$. This is true for $k = k_0 + 1$ since $\tilde{x}_{k_0+1} = B u_{k_0}$ and u_{k_0} is $\mathcal{I}_{k_0-1}$ -measurable (no observations). Assume the claim holds for $\tilde{x}_k$. Then (14.72) shows $\tilde{x}_{k+1}$ depends only on $\tilde{x}_k$ and u_k , hence only on $\mathcal{I}_{k-1}$. Now observe that

$$z_k = C x_k + H v_k = C \bar{x}_k + H v_k + C \tilde{x}_k = \bar{z}_k + C \tilde{x}_k. \quad (14.73)$$

Since $C \tilde{x}_k$ is $\mathcal{I}_{k-1}$ -measurable, (14.73) implies z_k is measurable w.r.t. $\bar{\mathcal{I}}_k$ and conversely $\bar{z}_k$ is measurable w.r.t. $\mathcal{I}_k$. Thus the generated sigma-fields coincide. $\square$

14.4.3 ■ Kalman predictor with control input

Let $\hat{x}_{k|k-1} := \mathbb{E}[x_k \mid \mathcal{I}_{k-1}]$ denote the one-step predictor. The controlled dynamics (14.67) differ from the uncontrolled filtering model only by the additive term $B u_k$, and because u_k is

$\mathcal{I}_{k-1}$ -measurable this term passes unchanged through every projection. This yields the controlled predictor recursion

$$\hat{x}_{k+1|k} = A\hat{x}_{k|k-1} + Bu_k + K_k^p(z_k - C\hat{x}_{k|k-1}), \quad (14.74)$$

where K_k^p is the Kalman gain (14.16) of the filtering problem and $\nu_k := z_k - C\hat{x}_{k|k-1}$ is the innovation.

Proof. [Derivation of (14.74)] Project the state equation (14.67) onto $\mathcal{I}_k$. By linearity of the projection, and since u_k is $\mathcal{I}_{k-1} \subseteq \mathcal{I}_k$ measurable so that $\text{proj}(Bu_k | \mathcal{I}_k) = Bu_k$,

$$\hat{x}_{k+1|k} = A\hat{x}_{k|k} + Bu_k + \Gamma\hat{w}_{k|k}, \quad \hat{w}_{k|k} := \text{proj}(w_k | \mathcal{I}_k).$$

The two facts about the innovation used in the uncontrolled case are unaffected by the known shift Bu_k : the innovation is still $\nu_k = z_k - C\hat{x}_{k|k-1} = C\tilde{x}_{k|k-1} + Hv_k$, since the Bu_k contribution to x_k and to $\hat{x}_{k|k-1}$ cancels in $\tilde{x}_{k|k-1}$, so $\mathbb{E}[w_k\nu_k^\top] = TH^\top$ and $\mathbb{E}[\nu_k\nu_k^\top] = \Lambda_k$ exactly as in (14.14). Substituting the filtered update $\hat{x}_{k|k} = \hat{x}_{k|k-1} + G_k\nu_k$ and $\hat{w}_{k|k} = TH^\top\Lambda_k^{-1}\nu_k$ and collecting the innovation terms into $K_k^p = (A\Sigma_{k|k-1}C^\top + \Gamma TH^\top)\Lambda_k^{-1}$, exactly as in the derivation of (14.17), gives (14.74). In particular Bu_k enters additively and leaves the gain and the innovation structure unchanged. $\square$

This is the projection-side counterpart of the time update Proposition 15.71 in the stochastic-control chapter, where the same recursion is obtained for the conditional mean.

14.4.4 ■ Simplifying the cost and separation

The linear-Gaussian assumptions do two things for us here, and it is worth naming both. First, they make the conditional expectation $\hat{x}_{k|k-1} = \mathbb{E}[x_k | \mathcal{I}_{k-1}]$ coincide with the *linear* predictor computed by the Kalman filter: this is the equivalence of least-squares and minimum-mean-square estimation for jointly Gaussian vectors established in Chapter 13 (Theorem 13.51). So the object the regulator will act on is exactly the one the filter produces. Second, the estimation error $x_k - \hat{x}_{k|k-1}$ is orthogonal to $\mathcal{I}_{k-1}$, which lets us split any quadratic in x_k into a piece that lives in the estimate and a piece that is pure estimation error and cannot be influenced by the control. This is made precise by the following reduction.

Proposition 14.24 (Reduction to the certainty-equivalent regulator). *Under the linear-Gaussian assumptions, for any admissible policy the cost (14.69) equals*

$$J = \underbrace{\mathbb{E}\left\{\hat{x}_{N|N-1}^\top Q_N \hat{x}_{N|N-1} + \sum_{k=k_0}^{N-1} \left(\hat{x}_{k|k-1}^\top Q_k \hat{x}_{k|k-1} + 2\hat{x}_{k|k-1}^\top N_k u_k + u_k^\top R_k u_k\right)\right\}}_{=: \hat{J}} + c, \quad (14.75)$$

where $c = \text{tr}(Q_N \Sigma_{N|N-1}) + \sum_{k=k_0}^{N-1} \text{tr}(Q_k \Sigma_{k|k-1})$ is a deterministic constant independent of the policy. Consequently minimizing J over admissible policies is equivalent to minimizing the fully observed regulator cost $\hat{J}$ with the state replaced by the predictor $\hat{x}_{k|k-1}$.

Proof. Fix k and condition on $\mathcal{I}_{k-1}$. The prediction error $e_k := x_k - \hat{x}_{k|k-1}$ satisfies $\mathbb{E}[e_k | \mathcal{I}_{k-1}] = 0$ and, because under the Gaussian hypothesis $\hat{x}_{k|k-1} = \mathbb{E}[x_k | \mathcal{I}_{k-1}]$ is the conditional mean (Theorem 13.51) and the conditional covariance is the deterministic matrix $\Sigma_{k|k-1}$, $\mathbb{E}[e_k e_k^\top |$

$\mathcal{I}_{k-1}] = \Sigma_{k|k-1}$. Writing $x_k = \hat{x}_{k|k-1} + e_k$ and using that $\hat{x}_{k|k-1}$ and u_k are $\mathcal{I}_{k-1}$ -measurable,

$$\mathbb{E}[x_k^\top Q_k x_k | \mathcal{I}_{k-1}] = \hat{x}_{k|k-1}^\top Q_k \hat{x}_{k|k-1} + \mathbb{E}[e_k^\top Q_k e_k | \mathcal{I}_{k-1}] \quad (14.76)$$

$$= \hat{x}_{k|k-1}^\top Q_k \hat{x}_{k|k-1} + \text{tr}(Q_k \Sigma_{k|k-1}), \quad (14.77)$$

while the cross and control terms are unchanged because they are linear in e_k or free of x_k :

$$\mathbb{E}[2x_k^\top N_k u_k | \mathcal{I}_{k-1}] = 2\hat{x}_{k|k-1}^\top N_k u_k, \quad \mathbb{E}[u_k^\top R_k u_k | \mathcal{I}_{k-1}] = u_k^\top R_k u_k. \quad (14.78)$$

Summing (14.77)–(14.78), taking total expectation, and treating the terminal term x_N the same way (with weight Q_N) gives $J = \hat{J} + c$ with c as stated. Since $\Sigma_{k|k-1}$ is generated by the covariance recursion (14.21), which involves neither the measurements nor the controls, c is deterministic and policy-independent; minimizing J therefore coincides with minimizing $\hat{J}$. $\square$

The reduced problem $\hat{J}$ is the fully observed regulator of Section 14.2 driven by the predictor dynamics (14.74), in which the innovation ν_k plays the role of a zero-mean white process noise. By the gain-invariance of Proposition 14.21 that noise does not change the optimal gain, so Theorem 14.17 applied to $\hat{J}$ yields the optimal law

$$u_k = L_k \hat{x}_{k|k-1} = \mu_k(\mathcal{I}_{k-1}), \quad L_k = -(R_k + B^\top K_{k+1} B)^{-1} (B^\top K_{k+1} A + N_k^\top), \quad (14.79)$$

which is admissible because $\hat{x}_{k|k-1}$ is $\mathcal{I}_{k-1}$ -measurable.

Theorem 14.25 (Separation theorem). *Under the linear-Gaussian assumptions, the optimal controller for the partially observed problem is obtained by:*

- (1) *designing the optimal state estimator (Kalman filter / predictor) to compute $\hat{x}_{k|k-1}$, and*
- (2) *applying the same LQR feedback gain as in the fully observed regulator problem, but evaluated at $\hat{x}_{k|k-1}$.*

Concretely, the optimal control is $u_k = L_k \hat{x}_{k|k-1}$ with L_k the regulator gain (14.55) and $\hat{x}_{k|k-1}$ the Kalman predictor (14.74). Thus estimation and control design separate.

Proof. By Proposition 14.24, minimizing the cost (14.69) over admissible policies is equivalent to minimizing the certainty-equivalent cost $\hat{J}$ of (14.75), which differs from J only by the policy-independent constant c . The cost $\hat{J}$ is the fully observed regulator cost (14.43) with the state replaced by the predictor $\hat{x}_{k|k-1}$, whose controlled dynamics (14.74), $\hat{x}_{k+1|k} = A\hat{x}_{k|k-1} + Bu_k + K_k^p \nu_k$, are those of the regulator plant driven by the innovation ν_k , a zero-mean white sequence (Theorem 14.10) independent of the past. Applying Theorem 14.17 to this regulator problem gives the optimal law $u_k = L_k \hat{x}_{k|k-1}$; by Proposition 14.21 the driving noise $K_k^p \nu_k$ leaves the gains $\{L_k\}$ equal to those of the noise-free regulator, so L_k is exactly the fully observed gain (14.55). This control is admissible, since $\hat{x}_{k|k-1}$ is $\mathcal{I}_{k-1}$ -measurable and generated by the Kalman recursion (14.74). Hence the optimal controller is the Kalman predictor followed by the fully observed LQR gain, which are designed by their own (decoupled) Riccati equations (14.21) and (14.52); this is the asserted separation. $\square$

The same separation is established, from the dynamic-programming side and with the conditional law carried explicitly through the recursion, in the finite-horizon LQG case study of the stochastic-control chapter (Section 15.9): there the estimate $\hat{x}_{k|k-1}$ is the conditional mean produced by the Kalman filter of Theorem 15.73, and the cost splits by the conditional orthogonal

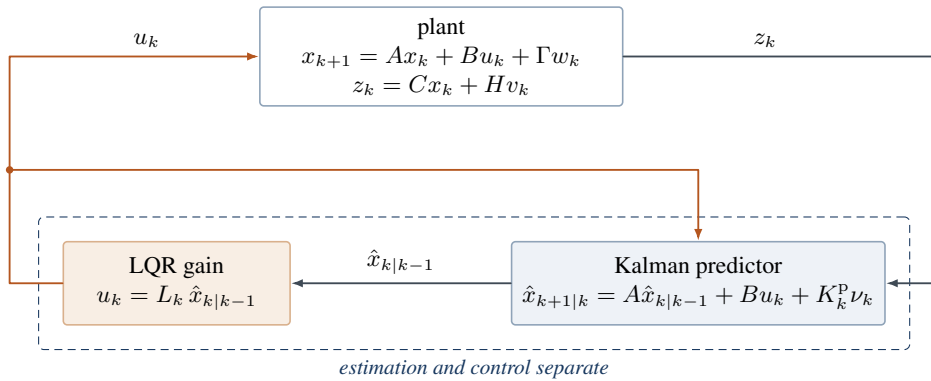


Figure 14.5: The separation structure of the LQG controller (Theorem 14.25). The Kalman predictor produces the state estimate $\hat{x}_{k|k-1}$ from the measurements and the known controls, with a gain and covariance computed as though no control were present; the LQR feedback then acts on that estimate exactly as it would act on the true state, as though the estimate were exact. The dashed box is the certainty-equivalent controller: two blocks designed in isolation, yet jointly optimal.

decomposition of Lemma 15.68, which is the measure-theoretic form of the projection splitting (14.77) used here.

It is worth being clear about what the theorem is and is not saying, because it is easy to overstate. It does not say that partial observation is free: the hidden state costs us through the estimation-error terms $\text{tr}(Q_k \Sigma_{k|k-1})$ in the cost (the constant c of Proposition 14.24), a contribution no controller can remove. What separates is the *design*, not the performance. The gain is computed from the regulator Riccati equation alone, with no reference to the noise covariances; the filter is computed from its own Riccati equation, with no reference to the cost weights; and the two are wired together only at run time, when the gain reads the estimate. This is the discrete-time, quadratic instance of the certainty-equivalence and separation principle that Chapter 10 states for dynamic programming under imperfect information, and it is special: as the next remark and the exercises stress, it is the joint linear-quadratic-Gaussian structure that makes it hold, and it fails the moment the dynamics are nonlinear or the noise non-Gaussian, because then the conditional expectation is no longer a linear function of the data and the cost no longer splits so cleanly.

Remark 14.26 (No information delay). If controls may depend on $\mathcal{I}_k$ (no information delay) and additionally $\mathbb{E}[w_k v_k^\top] = 0$ (i.e. $T = 0$), then the same development yields

$$u_k = L_k \hat{x}_{k|k} = -(R_k + B^\top K_{k+1} B)^{-1} (B^\top K_{k+1} A + N_k^\top) \hat{x}_{k|k}.$$

14.5 ■ Stability of the Closed-Loop System

Combining the plant (14.67) with the separated control law (14.79) gives the closed-loop system

$$x_{k+1} = A x_k + B L_k \hat{x}_{k|k-1} + \Gamma w_k, \quad L_k = -(R_k + B^\top K_{k+1} B)^{-1} (B^\top K_{k+1} A + N_k^\top), \quad (14.80)$$

where K_k solves the regulator Riccati recursion (14.52). Let the estimation error be $e_{k|k-1} := x_k - \hat{x}_{k|k-1}$. Subtracting the controlled predictor (14.74) from the state equation (14.67) and

using $z_k = Cx_k + Hv_k$,

$$\begin{aligned} e_{k+1|k} &= (Ax_k + Bu_k + \Gamma w_k) - (A\hat{x}_{k|k-1} + Bu_k + K_k^p(Cx_k + Hv_k - C\hat{x}_{k|k-1})) \\ &= A e_{k|k-1} + \Gamma w_k - K_k^p(C e_{k|k-1} + Hv_k), \end{aligned}$$

in which the control term Bu_k cancels identically, the estimation error is insensitive to the applied control, exactly as in the open-loop filter (14.18). Substituting $K_k^p = (A\Sigma_{k|k-1}C^\top + \Gamma TH^\top)(C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1}$ gives

$$\begin{aligned} e_{k+1|k} &= A e_{k|k-1} - (A\Sigma_{k|k-1}C^\top + \Gamma TH^\top)(C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1}C e_{k|k-1} \\ &\quad - (A\Sigma_{k|k-1}C^\top + \Gamma TH^\top)(C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1}Hv_k + \Gamma w_k. \end{aligned} \quad (14.81)$$

Define

$$\begin{aligned} L_k &:= -(R_k + B^\top K_{k+1}B)^{-1}(B^\top K_{k+1}A + N_k^\top), \\ K_k^p &:= (A\Sigma_{k|k-1}C^\top + \Gamma TH^\top)(C\Sigma_{k|k-1}C^\top + HVH^\top)^{-1}. \end{aligned} \quad (14.82)$$

Then (14.80) and (14.81) combine into the block-triangular system

$$\begin{bmatrix} x_{k+1} \\ e_{k+1|k} \end{bmatrix} = \begin{bmatrix} A + BL_k & -BL_k \\ 0 & A - K_k^p C \end{bmatrix} \begin{bmatrix} x_k \\ e_{k|k-1} \end{bmatrix} + \begin{bmatrix} \Gamma w_k \\ \Gamma w_k - K_k^p H v_k \end{bmatrix}. \quad (14.83)$$

The block-triangular shape of (14.83) is the entire stability story in one glance, and it repays a moment's attention. Because the lower-left block is zero, the state x_k influences the estimation error not at all; the error evolves on its own through $A - K_k^p C$, exactly the filter error dynamics (14.18), while the state evolves through the regulator matrix $A + BL_k$ driven by that error. A triangular transition matrix has, as its spectrum, the union of the spectra of its diagonal blocks, a standard fact from the linear system theory of Appendix A. The closed-loop poles are therefore precisely the regulator poles of $A + BL_k$ together with the filter poles of $A - K_k^p C$, with no cross-coupling. This is the dynamical face of the separation theorem: not only do we design the regulator and the filter separately, their closed-loop modes do not interfere either. Each subsystem is stable on its own under the standard conditions, so the whole is stable.

If the algebraic Riccati equations associated with K_k and $\Sigma_{k|k-1}$ admit unique stabilizing solutions, then the stationary separated controller and stationary filter yield

$$u_k = -(R + B^\top KB)^{-1}(B^\top KA + N^\top) \hat{x}_{k|k-1}, \quad (14.84)$$

where K solves (14.63) and $\hat{x}_{k|k-1}$ is generated by the stationary filter. Let

$$L := -(R + B^\top KB)^{-1}(B^\top KA + N^\top), \quad K^p := (A\Sigma C^\top + \Gamma TH^\top)(C\Sigma C^\top + HVH^\top)^{-1}. \quad (14.85)$$

Then the closed-loop takes the stationary form

$$\begin{bmatrix} x_{k+1} \\ e_{k+1|k}^s \end{bmatrix} = \begin{bmatrix} A + BL & -BL \\ 0 & A - K^p C \end{bmatrix} \begin{bmatrix} x_k \\ e_{k|k-1}^s \end{bmatrix} + \begin{bmatrix} \Gamma w_k \\ \Gamma w_k - K^p H v_k \end{bmatrix}. \quad (14.86)$$

Theorem 14.27 (Closed-loop stability under separation). Suppose (A, B) is stabilizable and $(\bar{Q}^{1/2}, \bar{A})$ is detectable (the regulator conditions of Theorem 14.22), and (C, A) is detectable and $(\bar{A}_f, \bar{\Gamma}_f)$ is stabilizable (the filter conditions of Theorem 14.11), with $\bar{A}, \bar{Q}$ as in (14.62) and $\bar{A}_f, \bar{\Gamma}_f$ as in (14.24). Then the stationary separated control law $u_k = -(R + B^\top KB)^{-1}(B^\top KA + N^\top) \hat{x}_{k|k-1}$, where K is the stabilizing solution of (14.63) and $\hat{x}_{k|k-1}$

is generated by the stationary Kalman filter, produces a stable closed-loop system: the transition matrix $\mathcal{A}$ of (14.86) is Schur, and the augmented covariance $\mathbb{E}[\xi_k \xi_k^\top]$ stays bounded and converges.

Proof. The closed loop is the block-triangular system (14.86) with transition matrix

$$\mathcal{A} = \begin{bmatrix} A + BL & -BL \\ 0 & A - K^p C \end{bmatrix}.$$

Because the $(2, 1)$ block vanishes, the characteristic polynomial factors, $\det(zI - \mathcal{A}) = \det(zI - (A + BL)) \det(zI - (A - K^p C))$, so the spectrum of $\mathcal{A}$ is the union of the spectra of its diagonal blocks (Appendix A). Under the regulator hypotheses, Theorem 14.22 makes $A + BL$ Schur; under the filter hypotheses, Theorem 14.11 makes $A - K^p C$ Schur. Hence every eigenvalue of $\mathcal{A}$ lies strictly inside the unit disc, i.e. $\mathcal{A}$ is Schur. The driving sequence in (14.86) is zero-mean white with finite covariance $\mathcal{Q}$, so $\Pi_k := \mathbb{E}[\xi_k \xi_k^\top]$ obeys the linear recursion $\Pi_{k+1} = \mathcal{A} \Pi_k \mathcal{A}^\top + \mathcal{Q}$; since $\rho(\mathcal{A}) < 1$ its linear part has spectral radius $\rho(\mathcal{A})^2 < 1$, so the recursion is a contraction and Π_k remains bounded and converges to the unique solution of the associated discrete Lyapunov equation. Thus the closed-loop state and estimation error have bounded, convergent second moments. (The limiting covariance and its consequences for the average cost are worked out in Proposition 14.28.) $\square$

14.5.1 ■ Average cost for the infinite-horizon problem

The finite-horizon value (14.59) carried a running sum of trace terms $\sum_{j=k_0}^{N-1} \text{tr}(K_{j+1} \Gamma W \Gamma^\top)$, one for each instant at which process noise is injected. Under persistent excitation this sum grows linearly in the horizon, so the total cost diverges as $N \rightarrow \infty$ and can no longer serve as a criterion: every admissible policy scores infinity. What remains finite, and what one actually cares about in steady operation, is the cost *per unit time*. The appropriate infinite-horizon objective is therefore the long-run average cost per stage,

$$J_{\text{avg}} := \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=0}^{N-1} \mathbb{E}[x_k^\top Q x_k + 2x_k^\top N u_k + u_k^\top R u_k], \quad (14.87)$$

with the constant steady-state weights (Q, N, R) , and the task is to show that this limit exists under the separated stationary controller and to identify its value.

The device that makes this precise is the closed-loop description (14.86). Stack the state and the estimation error into the augmented vector

$$\xi_k := \begin{bmatrix} x_k \\ e_{k|k-1}^s \end{bmatrix}, \quad \xi_{k+1} = \mathcal{A} \xi_k + \eta_k, \quad \mathcal{A} := \begin{bmatrix} A + BL & -BL \\ 0 & A - K^p C \end{bmatrix}, \quad (14.88)$$

where the driving noise factors through the primitive disturbances as

$$\eta_k = \underbrace{\begin{bmatrix} \Gamma & 0 \\ \Gamma & -K^p H \end{bmatrix}}_{=: \mathcal{G}} \begin{bmatrix} w_k \\ v_k \end{bmatrix}, \quad \mathcal{Q} := \mathbb{E}[\eta_k \eta_k^\top] = \mathcal{G} \begin{bmatrix} W & T \\ T^\top & V \end{bmatrix} \mathcal{G}^\top, \quad (14.89)$$

a zero-mean white sequence whose covariance is the primitive noise covariance $\begin{bmatrix} W & T \\ T^\top & V \end{bmatrix}$ shaped by the closed-loop input map $\mathcal{G}$. The dimensions of the augmented objects are inherited from

the originals: the stacked state $\xi_k \in \mathbb{R}^{2n}$ and the transition matrix $\mathcal{A} \in \mathbb{R}^{2n \times 2n}$, while the input map $\mathcal{G} \in \mathbb{R}^{2n \times (r+s)}$ acts on the stacked primitive noise $[w_k^\top, v_k^\top]^\top \in \mathbb{R}^{r+s}$, whose covariance $\begin{bmatrix} W & T \\ T^\top & V \end{bmatrix}$ is $(r+s) \times (r+s)$; hence $\mathcal{Q} \in \mathbb{R}^{2n \times 2n}$. The stage cost is itself a quadratic form in ξ_k . Substituting the stationary control law $u_k = \hat{L}x_{k|k-1} = L(x_k - e_{k|k-1}^s)$ expresses the cost pair $[x_k^\top, u_k^\top]^\top$ as a linear image of ξ_k ,

$$\begin{bmatrix} x_k \\ u_k \end{bmatrix} = M_L \xi_k, \quad M_L := \begin{bmatrix} I & 0 \\ L & -L \end{bmatrix}, \quad \text{so} \quad (14.90)$$

$$x_k^\top Q x_k + 2x_k^\top N u_k + u_k^\top R u_k = \xi_k^\top \Theta \xi_k, \quad \Theta := M_L^\top \mathcal{W} M_L,$$

where $\mathcal{W} = \begin{bmatrix} Q & N \\ N^\top & R \end{bmatrix}$ is the joint steady-state weight and $\Theta \in \mathbb{R}^{2n \times 2n}$ is symmetric positive semidefinite. (When $\mathcal{W} = \begin{bmatrix} D & F \\ F^\top & R \end{bmatrix}$ this is the regulated-output form $\Theta = \Upsilon^\top \Upsilon$ with $\Upsilon = [D + FL \quad -FL]$.) Hence $\mathbb{E}[x_k^\top Q x_k + 2x_k^\top N u_k + u_k^\top R u_k] = \text{tr}(\Theta \Pi_k)$ with the augmented covariance $\Pi_k := \mathbb{E}[\xi_k \xi_k^\top] \in \mathbb{R}^{2n \times 2n}$. Everything now hinges on the convergence of Π_k , which is a statement about the closed-loop dynamics (14.88) alone.

Proposition 14.28 (Stationary average cost). *Suppose the hypotheses of Theorem 14.27 hold, so that the block matrix $\mathcal{A}$ in (14.88) is stable (Schur). Then:*

- (1) *the augmented covariance Π_k converges, for every initial condition, to the unique positive-semidefinite solution Π of the discrete Lyapunov equation*

$$\Pi = \mathcal{A} \Pi \mathcal{A}^\top + \mathcal{Q}; \quad (14.91)$$

- (2) *the per-stage cost converges, $\mathbb{E}[x_k^\top Q x_k + 2x_k^\top N u_k + u_k^\top R u_k] \rightarrow \text{tr}(\Theta \Pi)$, and consequently the average cost (14.87) exists with value*

$$J_{\text{avg}} = \text{tr}(\Theta \Pi); \quad (14.92)$$

- (3) *since the error block of (14.88) is autonomous and driven by $A - K^p C$, the lower-right block of Π is exactly the steady-state prediction covariance Σ solving the filter algebraic Riccati equation (14.23).*

Proof. Because $\mathcal{A}$ is Schur, iterating (14.88) and using the whiteness of η_k gives $\Pi_{k+1} = \mathcal{A} \Pi_k \mathcal{A}^\top + \mathcal{Q}$; the map $X \mapsto \mathcal{A} X \mathcal{A}^\top + \mathcal{Q}$ is a contraction in the sense that its linear part has spectral radius $\rho(\mathcal{A})^2 < 1$, so it has a unique fixed point $\Pi \succeq 0$ and $\Pi_k \rightarrow \Pi$ from any start. This is (14.91). Continuity of the trace then yields $\mathbb{E}[x_k^\top Q x_k + 2x_k^\top N u_k + u_k^\top R u_k] = \text{tr}(\Theta \Pi_k) \rightarrow \text{tr}(\Theta \Pi)$, and a convergent sequence has the same Cesàro limit as its terms, giving (14.92). Finally, the lower-right block of (14.91) decouples, because the $(2, 1)$ block of $\mathcal{A}$ vanishes, into $\Sigma = (A - K^p C) \Sigma (A - K^p C)^\top + \Gamma W \Gamma^\top - \Gamma T H^\top (K^p)^\top - K^p H T^\top \Gamma^\top + K^p H V H^\top (K^p)^\top$, which is the closed-loop form (14.20) of the filter Riccati recursion at its fixed point. $\square$

Proposition 14.28 completes the arc of the chapter. The average cost (14.92) is the steady-state stage cost of the closed loop, and its two ingredients are exactly the two Riccati equations we have solved: the control-side solution K enters through the gain L inside Θ , and the filter-side solution Σ enters through the gain K^p and, via part (iii), through the error block of Π . The estimation error that partial observation forces on us is thus not merely an aesthetic blemish but a quantifiable, irreducible contribution to the running cost, the price of not seeing the state, paid at every stage forever. Under the hypotheses of Theorem 14.27 the stationary separated controller minimizes (14.87), and the duality of Section 14.2.7 guarantees that both Riccati equations supplying Π have the stabilizing solutions the proposition requires.

14.6 ■ Further Reading

The Kalman filter and the linear estimation theory developed in this chapter admit several complementary treatments. The projection-theoretic and innovations-based viewpoint adopted here is developed in depth by Kailath, Sayed, and Hassibi [62], which remains the authoritative reference on linear estimation and its connections to the geometry of Hilbert spaces. The classical state-space account, including the duality between estimation and linear-quadratic control and the steady-state Riccati theory, is given by Anderson and Moore [3] and by Bryson and Ho [30].

The maximum-principle derivation of the optimal linear filter, which parallels the variational treatment of control, is presented by Athans and Tse [7]. A systems-theoretic development that situates the filter within estimation, identification, and adaptive control is given by Kumar and Varaiya [74], while the underlying linear-systems machinery on which the observer and Riccati recursions rest is treated by Kailath [61].

The separation of estimation and control that underlies the output-feedback results of this chapter, together with its measure-theoretic foundations, is developed by Bertsekas [16] and, in full generality, by Bertsekas and Shreve [19]; these references also point toward the nonlinear and partially observed extensions of the linear-Gaussian theory.

14.7 ■ Exercises

Exercise 14.1 (Projection Lemma Warm-Up). Let x, y_1, y_2 be zero-mean, square-integrable scalar random variables with $\mathbb{E}[y_1 y_2] = 0$, $\mathbb{E}[y_i^2] = 1$, and $\mathbb{E}[x y_i] = c_i$ for $i = 1, 2$.

- (i) Compute $\text{proj}(x \mid \text{span}\{y_1, y_2\})$ using the normal equations.
- (ii) Verify the orthogonality condition $\mathbb{E}[(x - \hat{x}) y_i] = 0$ directly.
- (iii) Show that the projection can be computed sequentially: first project onto y_1 , then add the projection of the residual onto y_2 . Explain how this prefigures the innovations structure of the Kalman filter.

Exercise 14.2 (Scalar Kalman Filter by Hand). Consider the scalar system $x_{k+1} = a x_k + w_k$, $z_k = x_k + v_k$, with $\mathbb{E}[w_k^2] = q$, $\mathbb{E}[v_k^2] = r$, $\mathbb{E}[w_k v_k] = 0$, and initial variance Σ_0 .

- (i) Write the measurement update and prediction equations for $\hat{x}_{k|k}$, $\hat{x}_{k+1|k}$, $\Sigma_{k|k}$, and $\Sigma_{k+1|k}$.
- (ii) For $a = 1$, $q = 0$, and $\Sigma_0 = \sigma^2$, show by induction that $\Sigma_{k|k} = \frac{\sigma^2 r}{r + (k+1)\sigma^2}$, and interpret the limit $k \rightarrow \infty$.
- (iii) For $a = 1$ and $q > 0$, find the steady-state prediction variance $\bar{\Sigma}$ by solving the scalar algebraic Riccati equation, and verify $\bar{\Sigma} > 0$.

Exercise 14.3 (Innovations are White). Let $\{\nu_k\}$ be the innovations sequence of the Kalman filter under the standing assumptions of this chapter.

- (i) Prove that $\mathbb{E}[\nu_k] = 0$ and that $\mathbb{E}[\nu_k \nu_j^\top] = 0$ for $j < k$.
- (ii) Explain why whiteness of the innovations is a practical diagnostic: describe a test on filter residuals that detects an incorrectly specified model, and what correlated residuals would indicate.

Exercise 14.4 (Correlated Process and Measurement Noise). Suppose $\mathbb{E}[w_k v_k^\top] = T \neq 0$.

- (i) Rederive the predictor gain and show that it becomes

$$K_k = (A\Sigma_{k|k-1}C^\top + GTH^\top)(C\Sigma_{k|k-1}C^\top + HVBH^\top)^{-1}.$$

- (ii) Specialize to the scalar case $A = G = C = H = 1$ and examine how the sign of T shifts the gain relative to the uncorrelated case. Give an intuitive explanation.

Exercise 14.5 (Steady-State Filter and Detectability). Consider

$$x_{k+1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} x_k + w_k, \quad z_k = \begin{pmatrix} 1 & 0 \end{pmatrix} x_k + v_k,$$

with $\text{cov}(w_k) = \text{diag}(0, q)$, $q > 0$, and $\text{cov}(v_k) = r > 0$.

- (i) Verify that the pair (A, C) is detectable (in fact observable).
- (ii) Write the algebraic Riccati equation satisfied by the steady-state prediction covariance $\bar{\Sigma}$ and reduce it to scalar equations in the entries of $\bar{\Sigma}$.
- (iii) Argue that the steady-state Kalman filter is asymptotically stable, i.e., that $A - \bar{K}C$ has spectral radius less than one, where $\bar{K}$ is the steady-state gain.

Exercise 14.6 (Certainty Equivalence and Separation). Consider the partially observed LQG problem of this chapter with control law restricted to $u_k = \mu_k(\mathcal{Z}_{k-1})$.

- (i) Show that the optimal cost decomposes into a deterministic LQR part and an estimation part, and identify precisely which term the controller can influence.
- (ii) Explain why the LQR gain is computed as if the state were perfectly observed (certainty equivalence), and why the estimator is designed as if there were no control (separation). Where does the linear-Gaussian structure enter?
- (iii) Give an example of a system (nonlinear or non-Gaussian) where certainty equivalence fails, and explain the mechanism of failure.

Exercise 14.7 (Filter–Regulator Duality). Compare the Riccati recursion of the Kalman predictor with the Riccati recursion of the discrete-time linear regulator from this chapter.

- (i) Exhibit an explicit correspondence $(A, C, GWG^\top, HVBH^\top) \leftrightarrow (A^\top, B^\top, Q, R)$ under which one recursion maps to the other.
- (ii) Use the correspondence to translate the stabilizability/detectability conditions guaranteeing convergence of the regulator Riccati recursion into the corresponding conditions for the filter.

Chapter 15

Stochastic Control

Every optimization problem studied so far in this book shares a silent assumption: when a decision is made, its consequences are known. This chapter removes that assumption. A control chosen today influences a state that also depends on random disturbances, and future decisions may react to what has actually happened. Uncertainty thus changes the very nature of a solution: the object of choice is no longer a sequence of numbers, as in the deterministic optimal control of Chapter 10, but a *policy*, a rule that maps observed information to actions. A thermostat, an inventory manager, an investor rebalancing a portfolio: each acts repeatedly on partial, noisy knowledge, and each can outperform any fixed plan precisely because it reacts.

This chapter develops the resulting theory of discrete-time stochastic control. We formulate stochastic dynamical systems on a probability space, define admissible (nonanticipative) control policies, and contrast open-loop and feedback control, quantifying their divergence under uncertainty. Using the controlled Markov property, we develop the dynamic programming framework (value functions, the principle of optimality, and the Bellman recursion), which extends the deterministic theory of Chapter 10 stage by stage. Linear-Gaussian systems are treated as the central tractable class; here the machinery of estimation theory (Chapter 13) and the Kalman filter (Chapter 14) combines with dynamic programming to produce the celebrated linear-quadratic-Gaussian (LQG) theory, in which estimation and control decouple.

Case studies (stochastic resource allocation, optimal stopping, and dual control) illustrate the key structural phenomena: certainty equivalence, feedback adaptation, threshold rules, and the value of information. The chapter closes with partially observed systems, where the conditional distribution of the state (the *belief*) serves as an information state and reduces the problem to a fully observed Markov decision process; the separation principle emerges as the linear-Gaussian instance of this reduction. The same information-state viewpoint, pushed to set-membership uncertainty, is the subject of Chapter 12.

15.1 ■ Elements of Stochastic Control Systems

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. All random variables and stochastic processes introduced below are defined on this space.

We consider a discrete-time stochastic dynamical system described by

$$x_{k+1} = f_k(x_k, u_k, w_k), \quad k \geq 0, \quad (15.1)$$

where $x_k : \Omega \rightarrow \mathbb{R}^n$ is the state random vector, u_k is a control input taking values in a given control set $U \subseteq \mathbb{R}^m$, and $w_k : \Omega \rightarrow \mathbb{W}$ is a disturbance random variable defined on a measurable

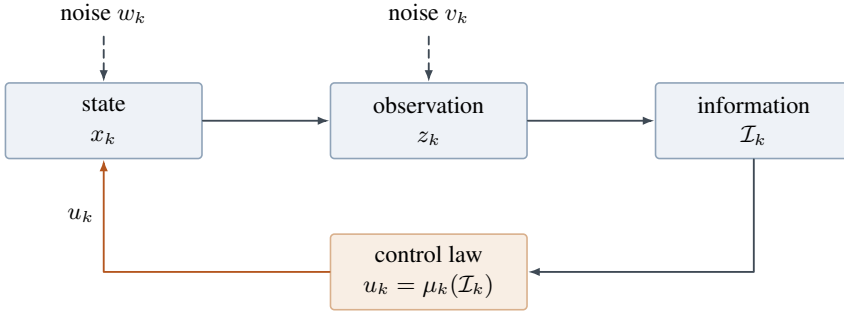


Figure 15.1: Stochastic control is organized around the information available to the controller; observations update information, and information determines admissible controls.

space $(\mathbb{W}, \mathcal{B}(\mathbb{W}))$. For each $k \geq 0$, the mapping $f_k : \mathbb{R}^n \times U \times \mathbb{W} \rightarrow \mathbb{R}^n$ is assumed to be Borel measurable.

The sequence $\{w_k\}_{k \geq 0}$ represents exogenous uncertainty and is modeled as a stochastic process with a prescribed joint probability law. The fundamental sources of randomness in the system are the random variables $x_0, w_0, w_1, \dots, v_0, v_1, \dots$, where x_0 denotes the initial state, $\{w_k\}$ the process noise, and $\{v_k\}$ a sequence of measurement noise variables. We assume that these random variables are defined on $(\Omega, \mathcal{F}, \mathbb{P})$ with a given joint distribution. All other random variables in the model are constructed as measurable functions of these basic variables through the system and observation equations.

The controller does not, in general, observe the state directly. Instead, it has access to an observation process $\{z_k\}_{k \geq 0}$ defined by

$$z_k = h_k(x_k, v_k), \quad k \geq 0, \quad (15.2)$$

where $z_k : \Omega \rightarrow \mathbb{R}^p$ is the observation, $v_k : \Omega \rightarrow \mathbb{V}$ is the measurement noise taking values in a measurable space $(\mathbb{V}, \mathcal{B}(\mathbb{V}))$, and $h_k : \mathbb{R}^n \times \mathbb{V} \rightarrow \mathbb{R}^p$ is a Borel-measurable mapping.

A frequently encountered special case is the linear observation model

$$z_k = Cx_k + v_k, \quad (15.3)$$

where $C \in \mathbb{R}^{p \times n}$ is a given matrix. If $z_k = x_k$ for all k , the system is said to have full state observation. Otherwise, it is partially observed. In the latter case, the initial state x_0 is typically modeled as a nondegenerate random variable to capture initial uncertainty. Let $\mathcal{F}_k^0 := \sigma(x_0, w_0, \dots, w_{k-1}, v_0, \dots, v_k)$ denote the σ -algebra generated by the basic random variables up to time k . All processes introduced below will be adapted to appropriate sub- σ -algebras of $\{\mathcal{F}_k^0\}$.

Definition 15.1. A discrete-time stochastic control system is specified by the tuple $(\{f_k\}_{k \geq 0}, \{h_k\}_{k \geq 0}, \mathbb{P})$, where $\{f_k\}$ and $\{h_k\}$ are measurable mappings defining the state and observation equations (15.1) and (15.2), and $\mathbb{P}$ is the probability law of the basic random variables $x_0, w_0, w_1, \dots, v_0, v_1, \dots$.

15.1.1 ■ State and Observation Processes

Let $u = \{u_k\}_{k \geq 0}$ be a fixed control sequence. The state process $\{x_k\}_{k \geq 0}$ is defined recursively by (15.1) with initial condition x_0 . By induction, for each $k \geq 0$, the random variable x_k is $\sigma(x_0, w_0, \dots, w_{k-1})$ -measurable, and hence $\mathcal{F}_k^0$ -measurable.

The observation process $\{z_k\}_{k \geq 0}$ is defined by (15.2). For each $k \geq 0$, the random variable z_k is $\sigma(x_0, w_0, \dots, w_{k-1}, v_0, \dots, v_k)$ -measurable. Thus, the processes $\{x_k\}$ and $\{z_k\}$ are stochastic processes induced by the control sequence u and the basic random variables. Their joint distribution is uniquely determined by the probability law of the basic random variables together with the measurable mappings f_k and h_k . In particular, all finite-dimensional distributions of $\{(x_k, z_k)\}$ can be obtained by successive application of transformation rules for random variables, without explicit reference to the underlying sample space.

15.1.2 ■ Information Structure and Admissible Controls

In stochastic control systems, the controller does not have access to the full realization of the underlying randomness. Instead, decisions must be based on the information available up to the current time. This motivates the introduction of an information structure.

For each $k \geq 0$, define the *information available to the controller at time k* by

$$\mathcal{I}_k := \sigma(z_0, \dots, z_k, u_0, \dots, u_{k-1}), \quad (15.4)$$

with the convention that $\mathcal{I}_0 = \sigma(z_0)$. The sequence $\{\mathcal{I}_k\}_{k \geq 0}$ is an increasing family of σ -algebras, i.e., $\mathcal{I}_0 \subseteq \mathcal{I}_1 \subseteq \dots \subseteq \mathcal{F}$. Thus, $\{\mathcal{I}_k\}$ defines a filtration representing the flow of information available to the controller.

A control process $\{u_k\}_{k \geq 0}$ is said to be *adapted* to the information structure if, for each $k \geq 0$, the control input u_k is $\mathcal{I}_k$ -measurable. This expresses the nonanticipativity constraint that the control at time k may depend only on information available up to time k .

Definition 15.2 (Admissible control law). A control law (or policy) $\mu = \{\mu_k\}_{k \geq 0}$ is said to be *admissible* if, for each $k \geq 0$, there exists a Borel-measurable mapping $\mu_k : (\mathbb{R}^p)^{k+1} \times (\mathbb{R}^m)^k \rightarrow U$ such that the control process satisfies

$$u_k = \mu_k(z_0, \dots, z_k, u_0, \dots, u_{k-1}), \quad (15.5)$$

and is adapted to the filtration $\{\mathcal{I}_k\}$.

In many cases, it is sufficient to consider policies that depend only on the observation history, namely

$$u_k = \mu_k(z_0, \dots, z_k), \quad (15.6)$$

in which case $\mathcal{I}_k = \sigma(z_0, \dots, z_k)$.

Under an admissible control law μ , the control process $\{u_k\}$ becomes a sequence of random variables adapted to $\{\mathcal{I}_k\}$, and the closed-loop system defines a stochastic process $\{(x_k, z_k, u_k)\}_{k \geq 0}$ whose law depends on μ .

The nonanticipativity condition encoded by $\mathcal{I}_k$ is fundamental in stochastic control, as it ensures that decisions are made causally and precludes the use of future information. This structure will play a central role in the formulation of optimal control problems and in the derivation of dynamic programming equations.

It is worth pausing on the picture behind the definition. As time advances, the controller accumulates data: at stage k it has seen the observations $z_0, \dots, z_k$ and has itself chosen the controls $u_0, \dots, u_{k-1}$. The σ -algebra $\mathcal{I}_k$ is precisely the mathematical record of this accumulated knowledge, and because nothing is ever forgotten, the family is *nested*: $\mathcal{I}_0 \subseteq \mathcal{I}_1 \subseteq \dots$, a *filtration*. Figure 15.2 depicts this growth. The requirement that u_k be $\mathcal{I}_k$ -measurable is the formal expression of a physical constraint, the controller cannot act on information it has not yet received, and it is exactly this constraint that separates the feedback policies of this chapter from the open-loop plans of deterministic optimal control.

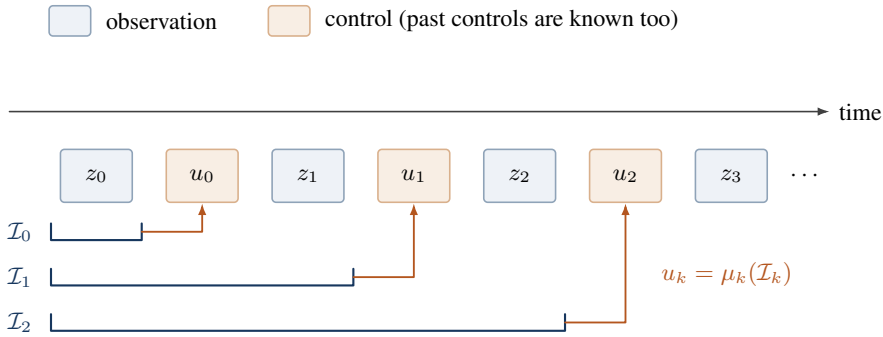


Figure 15.2: The information filtration. At each stage the controller first sees the observation z_k and then chooses the control u_k , so the data arrive in the interleaved order $z_0, u_0, z_1, u_1, \dots$. The information available when u_k is selected is $\mathcal{I}_k = \sigma(z_0, \dots, z_k, u_0, \dots, u_{k-1})$, every bracket “cups” exactly the quantities it contains. Since each bracket extends the one above it, the family is nested, $\mathcal{I}_0 \subseteq \mathcal{I}_1 \subseteq \mathcal{I}_2 \subseteq \dots$ (a filtration), and the admissibility constraint $u_k = \mu_k(\mathcal{I}_k)$ forbids the controller from using anything to the right of the $\mathcal{I}_k$ bracket, that is, any information it has not yet received.

Example 15.3 Portfolio growth as a stochastic control system.

An investor’s wealth evolves as

$$x_{k+1} = x_k(1 + u_k r_k),$$

where $x_k > 0$ is wealth, $u_k \in U = [0, 1]$ is the fraction invested in a risky asset, and r_k is the random return, i.i.d. with known distribution on $\mathbb{W} = (-1, \infty)$. Here $f_k(x, u, w) = x(1 + uw)$ is continuous, hence Borel measurable, and the disturbance enters multiplicatively. A nonanticipative policy may let u_k depend on the observed wealth history $(x_0, \dots, x_k)$ but not on the future returns $r_k, r_{k+1}, \dots$, the formal content of the admissibility requirement introduced above. $\square$

15.2 ■ Open-Loop and Feedback Control

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the underlying probability space, and let the stochastic system be defined by (15.1) and (15.2). An *open-loop control* is a sequence $u = \{u_k\}_{k \geq 0}$ such that, for each k , $u_k \in \mathbb{R}^m$ is deterministic (or, more generally, independent of the observation process). In this case, the control sequence is fixed in advance and does not depend on the realization of the system.

In contrast, a *feedback control law* (or *policy*) is a sequence of measurable mappings $\mu = \{\mu_k\}_{k \geq 0}$, where, for each $k \geq 0$, $\mu_k : (\mathbb{R}^p)^{k+1} \rightarrow \mathbb{R}^m$ is a Borel-measurable function. The control action at time k is defined by

$$u_k = \mu_k(z^k), \quad z^k := (z_0, \dots, z_k), \quad (15.7)$$

so that u_k is a random variable measurable with respect to the σ -algebra generated by the observation history.

A special case arises when the control depends only on the current observation, namely

$$u_k = \mu_k(z_k), \quad (15.8)$$

which corresponds to a memoryless feedback law.

Such a law is admissible in the sense of Definition 15.2: the nonanticipativity condition there requires u_k to be $\mathcal{I}_k$ -measurable, and when the policy is written as a function of the observation history alone this reduces to u_k being $\sigma(z^k)$ -measurable for every $k \geq 0$, so that the control at time k depends only on information available up to time k .

15.2.1 ■ Closed-Loop System Dynamics

Let $\mu = \{\mu_k\}_{k \geq 0}$ be a given admissible feedback law. The closed-loop system is defined recursively by

$$x_{k+1} = f_k(x_k, u_k, w_k), \quad (15.9)$$

$$z_k = h_k(x_k, v_k), \quad (15.10)$$

$$u_k = \mu_k(z^k), \quad (15.11)$$

for $k \geq 0$, with initial condition x_0 .

Under the feedback law μ , the control sequence $\{u_k\}$ becomes a stochastic process. Indeed, since z^k is a random variable, it follows from (15.11) that u_k is also a random variable measurable with respect to the σ -algebra generated by $x_0, w_0, \dots, w_{k-1}, v_0, \dots, v_k$. The state, observation, and control processes induced by the feedback law μ are denoted by $\{x_k^\mu\}_{k \geq 0}$, $\{z_k^\mu\}_{k \geq 0}$, $\{u_k^\mu\}_{k \geq 0}$, and are defined recursively by (15.9) and (15.11).

These processes are measurable functions of the basic random variables $x_0, w_0, w_1, \dots, v_0, v_1, \dots$, and therefore admit a well-defined joint probability distribution. We denote by $\mathbb{P}^\mu$ the probability law on the path space induced by the feedback law μ . This law is obtained as the pushforward of $\mathbb{P}$ under the mapping that associates to each $\omega \in \Omega$ the trajectory

$$(x_0^\mu(\omega), x_1^\mu(\omega), \dots; z_0^\mu(\omega), z_1^\mu(\omega), \dots; u_0^\mu(\omega), u_1^\mu(\omega), \dots).$$

While the induced processes depend on the choice of μ , the distribution of the basic random variables remains unchanged.

15.2.2 ■ Equivalence of Open-Loop and Feedback Control in Deterministic Systems

We consider a deterministic discrete-time system

$$x_{k+1} = f_k(x_k, u_k), \quad k \geq 0, \quad (15.12)$$

with given initial condition $x_0 \in \mathbb{R}^n$, where each mapping $f_k : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ is deterministic.

Proposition 15.4 (Equivalence of open-loop and feedback control in deterministic systems).

Consider the deterministic system (15.12), $x_{k+1} = f_k(x_k, u_k)$, with a fixed initial condition $x_0 \in \mathbb{R}^n$ and full state observation $z_k = x_k$ for all k . Let $\mu = \{\mu_k\}_{k \geq 0}$ be any feedback law, where each $\mu_k : (\mathbb{R}^n)^{k+1} \rightarrow \mathbb{R}^m$ maps the observed history $(z_0, \dots, z_k)$ to a control. Then there exists a deterministic open-loop control sequence $u = \{u_k\}_{k \geq 0}$, fixed in advance and independent of the observations, such that the state trajectory generated by u coincides at every time k with the closed-loop trajectory induced by μ . Consequently, under deterministic dynamics with full observation, feedback produces no state trajectory that open-loop control cannot already realize.

Proof. Since the system (15.12) is deterministic and x_0 is fixed, any sequence of controls uniquely determines a state trajectory. Let μ be a feedback law. Define recursively the closed-loop trajectory

$\{x_k^\mu\}$ and control sequence $\{u_k^\mu\}$ by

$$u_k^\mu = \mu_k(x_0^\mu, \dots, x_k^\mu), \quad x_{k+1}^\mu = f_k(x_k^\mu, u_k^\mu), \quad k \geq 0, \quad (15.13)$$

with $x_0^\mu = x_0$. By induction, the sequences $\{x_k^\mu\}$ and $\{u_k^\mu\}$ are uniquely determined and deterministic.

Now define an open-loop control sequence $u = \{u_k\}_{k \geq 0}$ by $u_k := u_k^\mu, k \geq 0$. Let $\{x_k\}$ denote the state trajectory generated by u through (15.12). We show by induction that $x_k = x_k^\mu$ for all k . For $k = 0$, we have $x_0 = x_0^\mu$. Suppose that $x_k = x_k^\mu$ for some k . Then, $x_{k+1} = f_k(x_k, u_k) = f_k(x_k^\mu, u_k^\mu) = x_{k+1}^\mu$. Thus, by induction, $x_k = x_k^\mu$ for all $k \geq 0$. Hence, the open-loop control sequence u reproduces exactly the same state trajectory as the feedback law μ , which completes the proof. $\square$

Remark 15.5. The proposition shows that, in deterministic systems with full state observation, feedback control does not enlarge the set of achievable state trajectories relative to open-loop control. In contrast, as will be seen in stochastic systems, this equivalence fails due to the presence of uncertainty, and feedback control becomes strictly more expressive.

15.2.3 ■ Non-Equivalence of Open-Loop and Feedback Control in Stochastic Systems

We consider the one-dimensional stochastic system

$$x_{k+1} = x_k + u_k + w_k, \quad k \geq 0, \quad (15.14)$$

with full state observation $z_k = x_k$ and initial condition x_0 . The basic random variables $\{x_0, w_0, w_1, \dots\}$ are assumed to be mutually independent, with

$$\mathbb{E}[x_0] = 0, \quad \mathbb{E}[w_k] = 0, \quad \text{Var}(x_0) = \text{Var}(w_k) = \sigma^2 > 0.$$

Proposition 15.6 (Feedback cannot be replicated by open-loop control). *Consider the scalar stochastic system (15.14), $x_{k+1} = x_k + u_k + w_k$, with full state observation $z_k = x_k$ and mutually independent, zero-mean primitives $x_0, w_0, w_1, \dots$ of common variance $\sigma^2 > 0$. Then there exists an admissible feedback law μ whose induced state process $\{x_k^\mu\}$ has a probability distribution that no deterministic open-loop control sequence can reproduce: for every open-loop control $u = \{u_k\}_{k \geq 0}$, the law of the resulting process $\{x_k^u\}$ differs from that of $\{x_k^\mu\}$. In contrast to the deterministic case (Proposition 15.4), feedback is therefore strictly more expressive once process noise is present.*

Proof. Define the feedback law $\mu = \{\mu_k\}_{k \geq 0}$ by

$$u_k^\mu = \mu_k(z^k) := -x_k, \quad k \geq 0. \quad (15.15)$$

Since $z_k = x_k$, this is an admissible feedback law. The closed-loop dynamics are given by $x_{k+1}^\mu = x_k^\mu + u_k^\mu + w_k = x_k^\mu - x_k^\mu + w_k = w_k$. By recursion, we obtain $x_0^\mu = x_0$ and $x_k^\mu = w_{k-1}$ for $k \geq 1$. Hence, for $k \geq 1$, the random variables $\{x_k^\mu\}$ are independent, with $\mathbb{E}[x_k^\mu] = 0$ and $\text{Var}(x_k^\mu) = \sigma^2$.

Now consider any deterministic open-loop control sequence $u = \{u_k\}_{k \geq 0}$. The corresponding state process $\{x_k^u\}$ satisfies $x_k^u = x_0 + \sum_{m=0}^{k-1} u_m + \sum_{m=0}^{k-1} w_m$ for $k \geq 1$. Taking expectations, we

obtain $\mathbb{E}[x_k^u] = \sum_{m=0}^{k-1} u_m$, and, using independence and the fact that $\text{Var}(x_0) = \sigma^2$, $\text{Var}(w_m) = \sigma^2$, we have

$$\text{Var}(x_k^u) = \text{Var}(x_0) + \sum_{m=0}^{k-1} \text{Var}(w_m) = (k+1)\sigma^2.$$

Suppose, by contradiction, that there exists a deterministic sequence u such that the processes $\{x_k^u\}$ and $\{x_k^\mu\}$ have the same distribution. Then, in particular, their second moments must coincide. However, for $k \geq 1$ we have $\text{Var}(x_k^\mu) = \sigma^2$ while $\text{Var}(x_k^u) = (k+1)\sigma^2$, which is impossible since $(k+1)\sigma^2 > \sigma^2$ for all $k \geq 1$.

This contradiction shows that no deterministic open-loop control sequence can reproduce the state process induced by the feedback law μ . $\square$

Remark 15.7. The above example demonstrates that, in stochastic systems, feedback control can fundamentally alter the distributional properties of the state process in ways that cannot be replicated by open-loop control. In particular, feedback enables the cancellation of accumulated uncertainty, whereas open-loop control cannot prevent the growth of variance over time. This establishes that feedback control is strictly more expressive than open-loop control in stochastic settings.

15.2.4 ■ Performance Comparison Between Feedback and Open-Loop Control

We consider the stochastic system (15.14) under the assumptions stated therein. Fix a finite horizon $K \geq 1$ and define the performance criterion

$$J(\mu) := \mathbb{E} \left[\sum_{k=0}^K (x_k^\mu)^2 \right], \quad J(u) := \mathbb{E} \left[\sum_{k=0}^K (x_k^u)^2 \right], \quad (15.16)$$

for a feedback law μ and an open-loop control sequence u , respectively.

Proposition 15.8 (Performance gap between feedback and open-loop control). *Consider the scalar system (15.14) with the primitives of Proposition 15.6 and the horizon $K \geq 1$ fixed above. Let μ be the feedback law defined by $u_k^\mu = -x_k$. Then*

$$J(\mu) = (K+1)\sigma^2. \quad (15.17)$$

Moreover, for any deterministic open-loop control sequence $u = \{u_k\}_{k \geq 0}$,

$$J(u) \geq \frac{1}{2}(K+1)(K+2)\sigma^2, \quad (15.18)$$

and therefore

$$J(u) > J(\mu) \quad \text{for every } K \geq 1. \quad (15.19)$$

The restriction $K \geq 1$ is not cosmetic: at $K = 0$ the cost is $\mathbb{E}[x_0^2] = \sigma^2$ whatever the control, since no decision influences x_0 , and the two classes tie. Feedback pays only once there is a disturbance left to cancel.

Proof. Under the feedback law μ , the state process satisfies $x_0^\mu = x_0$ and $x_k^\mu = w_{k-1}$ for $k \geq 1$. Using independence and the moment assumptions, we obtain

$$\mathbb{E}[(x_0^\mu)^2] = \sigma^2, \quad \mathbb{E}[(x_k^\mu)^2] = \sigma^2 \quad \text{for all } k \geq 1.$$

Substituting into (15.16) yields $J(\mu) = \sigma^2 + \sum_{k=1}^K \sigma^2 = (K+1)\sigma^2$.

Now consider any deterministic open-loop control sequence $u = \{u_k\}$. The corresponding state satisfies $x_k^u = x_0 + \sum_{m=0}^{k-1} u_m + \sum_{m=0}^{k-1} w_m$ for $k \geq 1$. Define $\bar{x}_k := \sum_{m=0}^{k-1} u_m$. Then $\mathbb{E}[x_k^u] = \bar{x}_k$ and, by independence,

$$\text{Var}(x_k^u) = \text{Var}(x_0) + \sum_{m=0}^{k-1} \text{Var}(w_m) = (k+1)\sigma^2.$$

Hence,

$$\mathbb{E}[(x_k^u)^2] = \text{Var}(x_k^u) + (\mathbb{E}[x_k^u])^2 = (k+1)\sigma^2 + \bar{x}_k^2.$$

Therefore,

$$J(u) = \sum_{k=0}^K \mathbb{E}[(x_k^u)^2] = \sum_{k=0}^K ((k+1)\sigma^2 + \bar{x}_k^2) \geq \sum_{k=0}^K (k+1)\sigma^2.$$

Since $\bar{x}_k^2 \geq 0$, we obtain the lower bound $J(u) \geq \sigma^2 \sum_{k=0}^K (k+1) = \sigma^2 \frac{(K+1)(K+2)}{2}$. Finally, for all $K \geq 1$, $\frac{(K+1)(K+2)}{2} > (K+1)$, which implies $J(u) > J(\mu)$. $\square$

Remark 15.9. The above result shows that feedback control achieves a linear growth of the expected quadratic cost, whereas any open-loop control incurs a quadratic growth in the horizon length. This performance gap reflects the ability of feedback to counteract uncertainty in real time, preventing the accumulation of variance that is unavoidable under open-loop control.

The mechanism is easiest to see in pictures. Under the open-loop plan the state executes an uncorrected random walk, so its variance accumulates disturbance-by-disturbance and its trajectories fan out; under the feedback law $u_k = -x_k$ the accumulated state is cancelled at every step, leaving only the most recent shock, $x_{k+1} = w_k$, so the spread stays bounded. Figure 15.3 contrasts the two: representative trajectories on the left, and the resulting linear-versus-quadratic growth of the expected cost on the right. The gap is not a matter of choosing better open-loop numbers: it is structural, and it grows without bound with the horizon.

Example 15.10 Feedback strictly outperforms open loop.

Let $x_{k+1} = x_k + u_k + w_k$ with $x_0 = 0$, where w_0, w_1 are uncorrelated, zero-mean, with variance σ^2 , and consider minimizing $\mathbb{E}[x_2^2]$ over two stages with unconstrained controls. An open-loop plan fixes numbers (u_0, u_1) in advance, so $x_2 = u_0 + u_1 + w_0 + w_1$ and the best achievable cost is

$$\min_{u_0, u_1 \in \mathbb{R}} \mathbb{E}[x_2^2] = \mathbb{E}[(w_0 + w_1)^2] = 2\sigma^2.$$

A feedback policy may react to x_1 : choosing $u_1 = -x_1$ gives $x_2 = w_1$ and cost σ^2 , half the open-loop value. The advantage is entirely due to the policy's ability to cancel the realized disturbance w_0 after observing it, information that no open-loop plan can use. $\square$

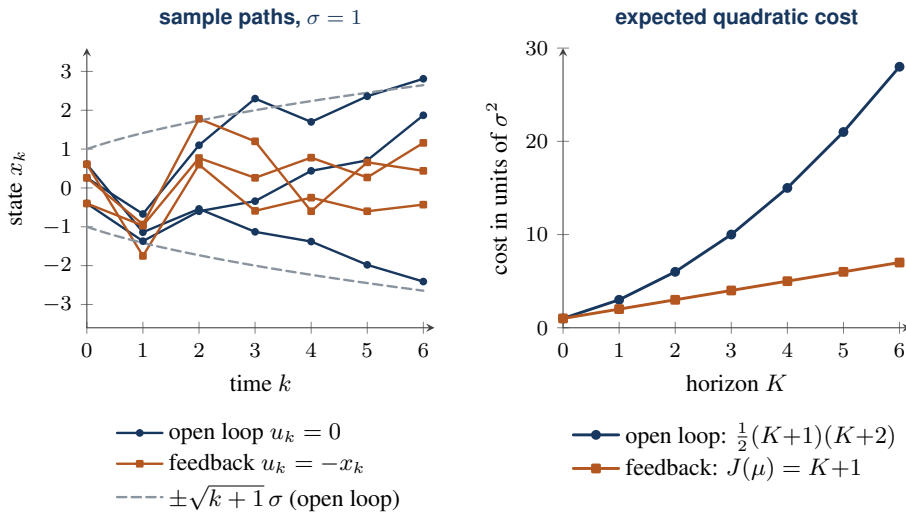


Figure 15.3: Open-loop versus feedback control for $x_{k+1} = x_k + u_k + w_k$ with $\text{Var}(w_k) = \sigma^2$. *Left:* sample trajectories starting from an initial state x_0 of standard deviation σ ($\sigma = 1$), with the three open-loop and the three feedback paths driven by the same three draws of x_0 and of the disturbances. Without feedback ($u_k = 0$) the state performs a random walk with $\text{Var}(x_k) = (k+1)\sigma^2$, so its standard deviation grows like $\sqrt{k+1}\sigma$ (dashed $\pm 1\sigma$ envelope); the feedback law $u_k = -x_k$ cancels the accumulated state each step, so $x_{k+1} = w_k$ stays contained. *Right:* the expected quadratic cost grows *quadratically* in the horizon under the best open-loop plan, $J(u) \geq \frac{1}{2}(K+1)(K+2)\sigma^2$, but only *linearly* under feedback, $J(\mu) = (K+1)\sigma^2$, the performance gap quantified in Proposition 15.8.

15.3 ■ Transition Probabilities and the Markov Property

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\xi_1 : \Omega \rightarrow \mathbb{R}^{n_1}$ and $\xi_2 : \Omega \rightarrow \mathbb{R}^{n_2}$ be random vectors. Their joint distribution is the probability measure P_{ξ_1, ξ_2} on $\mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ defined by

$$P_{\xi_1, \xi_2}(B) := \mathbb{P}((\xi_1, \xi_2) \in B), \quad B \in \mathcal{B}(\mathbb{R}^{n_1+n_2}), \quad (15.20)$$

where $\mathcal{B}(\cdot)$ denotes the Borel σ -algebra. In particular, for measurable sets $Z_1 \subseteq \mathbb{R}^{n_1}$ and $Z_2 \subseteq \mathbb{R}^{n_2}$,

$$P_{\xi_1, \xi_2}(Z_1 \times Z_2) = \mathbb{P}\{\xi_1 \in Z_1, \xi_2 \in Z_2\}. \quad (15.21)$$

For any measurable function $q : \mathbb{R}^{n_1} \times \mathbb{R}^{n_2} \rightarrow \mathbb{R}$ such that $q(\xi_1, \xi_2)$ is integrable, the expectation is given by

$$\mathbb{E}[q(\xi_1, \xi_2)] = \int_{\mathbb{R}^{n_1+n_2}} q(\xi_1, \xi_2) P_{\xi_1, \xi_2}(d(\xi_1, \xi_2)). \quad (15.22)$$

The mean and covariance of a random vector ξ_i are defined by

$$\bar{\xi}_i := \mathbb{E}[\xi_i], \quad \text{cov}(\xi_i) := \mathbb{E}[(\xi_i - \bar{\xi}_i)(\xi_i - \bar{\xi}_i)^\top], \quad (15.23)$$

and the cross-covariance between ξ_1 and ξ_2 is

$$\text{cov}(\xi_1, \xi_2) := \mathbb{E}[(\xi_1 - \bar{\xi}_1)(\xi_2 - \bar{\xi}_2)^\top]. \quad (15.24)$$

Conditional distributions are defined via regular conditional probabilities. In particular, there exists a stochastic kernel $P_{\xi_1|\xi_2}(\cdot | \xi_2)$ such that for all measurable $A \subseteq \mathbb{R}^{n_1}$,

$$\mathbb{P}\{\xi_1 \in A | \xi_2\} = P_{\xi_1|\xi_2}(A | \xi_2) \quad \text{a.s.} \quad (15.25)$$

This induces the conditional expectation

$$\mathbb{E}[q(\xi_1, \xi_2) | \xi_2] = \int q(\xi_1, \xi_2) P_{\xi_1|\xi_2}(d\xi_1 | \xi_2), \quad (15.26)$$

whenever the integral is well defined.

Controlled Transition Probabilities

Consider the stochastic control system

$$x_{k+1} = f_k(x_k, u_k, w_k), \quad z_k = h_k(x_k, v_k), \quad (15.27)$$

with basic random variables $x_0, w_0, w_1, \dots, v_0, v_1, \dots$ defined on $(\Omega, \mathcal{F}, \mathbb{P})$.

Assume that the sequence $\{w_k\}$ is independent of $(x_0, w_0, \dots, w_{k-1}, v_0, \dots, v_k)$ for each $k \geq 0$. Then, for each k , there exists a stochastic kernel

$$P_k(\cdot | x, u) : \mathcal{B}(\mathbb{R}^n) \rightarrow [0, 1], \quad (15.28)$$

such that

$$P_k(X | x, u) = \mathbb{P}\{f_k(x, u, w_k) \in X\}, \quad X \in \mathcal{B}(\mathbb{R}^n). \quad (15.29)$$

This kernel characterizes the conditional distribution of x_{k+1} given (x_k, u_k) .

The rigorous statement rests on the following elementary substitution rule, which we isolate for later reuse.

Lemma 15.11 (Freezing lemma). *Let $\mathcal{G} \subseteq \mathcal{F}$ be a σ -algebra, let Y be a $\mathcal{G}$ -measurable random vector, and let W be a random vector independent of $\mathcal{G}$. Then for every bounded (or nonnegative) Borel function ϕ ,*

$$\mathbb{E}[\phi(Y, W) | \mathcal{G}] = \Phi(Y) \quad \text{a.s.}, \quad \Phi(z) := \mathbb{E}[\phi(z, W)].$$

Proof. First suppose $\phi(z, w) = a(z)b(w)$ with a, b bounded Borel. Then $a(Y)$ is $\mathcal{G}$ -measurable and $b(W)$ is independent of $\mathcal{G}$, so

$$\mathbb{E}[a(Y)b(W) | \mathcal{G}] = a(Y) \mathbb{E}[b(W)] = \Phi(Y), \quad \Phi(z) = a(z) \mathbb{E}[b(W)] = \mathbb{E}[\phi(z, W)].$$

Finite linear combinations of such products form an algebra that generates the product Borel σ -algebra and is closed under multiplication; by the functional monotone class theorem the identity extends to all bounded Borel ϕ , and then to nonnegative ϕ by monotone convergence. $\square$

Lemma 15.12 (Controlled Markov Property). *Suppose that the disturbance sequence $\{w_k\}_{k \geq 0}$ is independent across time and independent of x_0 and $\{v_k\}$. Let μ be any admissible control law, and write $\mathcal{H}_k := \sigma(x_0, \dots, x_k, u_0, \dots, u_k)$ for the state-control history. Then, for all $k \geq 0$ and all measurable sets $X \subseteq \mathbb{R}^n$,*

$$\mathbb{P}(x_{k+1} \in X | \mathcal{H}_k) = \mathbb{P}(x_{k+1} \in X | x_k, u_k) = P_k(X | x_k, u_k) \quad \text{a.s.}, \quad (15.30)$$

and the kernel $P_k(\cdot \mid x_k, u_k)$ does not depend on the control law μ . In particular, under complete state information ($z_k = x_k$, so that $\mathcal{I}_k = \mathcal{H}_k$) the identity holds with $\mathcal{H}_k$ replaced by $\mathcal{I}_k$.

Proof. Both x_k and $u_k = \mu_k(z_0, \dots, z_k, u_0, \dots, u_{k-1})$ are measurable functions of the primitive variables $(x_0, w_0, \dots, w_{k-1}, v_0, \dots, v_k)$; hence $\mathcal{H}_k \subseteq \sigma(x_0, w_0, \dots, w_{k-1}, v_0, \dots, v_k)$. By hypothesis w_k is independent of this larger σ -algebra, and therefore independent of $\mathcal{H}_k$. The pair $Y := (x_k, u_k)$ is $\mathcal{H}_k$ -measurable.

Apply the freezing lemma (Lemma 15.11) with $W := w_k$, $\mathcal{G} := \mathcal{H}_k$, and $\phi((x, u), w) := \mathbf{1}_X(f_k(x, u, w))$. Since $x_{k+1} = f_k(x_k, u_k, w_k)$,

$$\mathbb{P}(x_{k+1} \in X \mid \mathcal{H}_k) = \mathbb{E}[\mathbf{1}_X(f_k(x_k, u_k, w_k)) \mid \mathcal{H}_k] = \Phi(x_k, u_k),$$

where

$$\Phi(x, u) = \mathbb{E}[\mathbf{1}_X(f_k(x, u, w_k))] = \mathbb{P}(f_k(x, u, w_k) \in X) = P_k(X \mid x, u).$$

The random variable $\Phi(x_k, u_k)$ is $\sigma(x_k, u_k)$ -measurable; because $\sigma(x_k, u_k) \subseteq \mathcal{H}_k$, conditioning the displayed identity on $\sigma(x_k, u_k)$ and using the tower property yields $\mathbb{P}(x_{k+1} \in X \mid x_k, u_k) = \Phi(x_k, u_k)$ as well. This establishes both equalities. Finally, $\Phi = P_k(\cdot \mid \cdot, \cdot)$ is determined by the map f_k and the law of w_k alone, hence is independent of the policy μ . $\square$

Remark 15.13. Lemma 15.12 shows that the state process $\{x_k\}$ is a controlled Markov process. The future evolution of the state depends on the past only through the current state and control input. This property is fundamental for the applicability of dynamic programming and underlies the structure of optimal control policies.

Example 15.14 Transition kernel of a controlled inventory.

Let $x_k \in \{0, 1, 2, \dots\}$ be the stock at the start of period k , let $u_k \in \{0, 1\}$ be the quantity ordered (delivered immediately), and let demand D_k be i.i.d. Bernoulli with $\mathbb{P}(D_k = 1) = p$. Unmet demand is lost, so

$$x_{k+1} = \max\{x_k + u_k - D_k, 0\}.$$

The transition kernel is, for $s := x + u \geq 1$,

$$\mathbb{P}(x_{k+1} = s \mid x_k = x, u_k = u) = 1 - p, \quad \mathbb{P}(x_{k+1} = s - 1 \mid x_k = x, u_k = u) = p,$$

while for $s = 0$ the state remains 0 with probability one. The next state depends on the past only through the current pair (x_k, u_k) , the controlled Markov property in its simplest form. $\square$

15.4 ■ Linear Gaussian Stochastic Systems

We develop fundamental properties of discrete-time linear stochastic systems driven by Gaussian disturbances and measurement noise. In this setting, conditional and marginal distributions admit closed-form expressions. These computations are the probabilistic backbone of everything that follows: they supply the transition kernels on which dynamic programming operates, they underlie the projection formulas of estimation theory (Chapter 13), and they are exactly the calculations iterated by the Kalman filter of Chapter 14. The reader should view this section as building the “Gaussian calculus” that makes linear-quadratic-Gaussian (LQG) control solvable in closed form.

15.4.1 ■ Gaussian Random Vectors

Let $\xi : \Omega \rightarrow \mathbb{R}^n$ be a random vector. We say that ξ is Gaussian with mean $\bar{\xi} \in \mathbb{R}^n$ and covariance $\Sigma \in \mathbb{S}_+^n$ if its distribution is the probability measure $\mathcal{N}(\bar{\xi}, \Sigma)$, i.e., $\xi \sim \mathcal{N}(\bar{\xi}, \Sigma)$. If Σ is positive definite, then ξ admits the density

$$p(\xi) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\xi - \bar{\xi})^\top \Sigma^{-1}(\xi - \bar{\xi})\right), \quad (15.31)$$

with respect to Lebesgue measure on $\mathbb{R}^n$.

Gaussian random vectors are closed under affine transformations: if $\xi \sim \mathcal{N}(\bar{\xi}, \Sigma)$ and A, B are matrices of appropriate dimensions, then $A\xi + b \sim \mathcal{N}(A\bar{\xi} + b, A\Sigma A^\top)$.

15.4.2 ■ Linear Stochastic System Model

Consider the linear system

$$x_{k+1} = Ax_k + Bu_k + Gw_k, \quad (15.32)$$

$$z_k = Cx_k + Hv_k, \quad (15.33)$$

where $x_k \in \mathbb{R}^n$, $u_k \in U \subseteq \mathbb{R}^m$, and $z_k \in \mathbb{R}^p$. The noise processes $\{w_k\}$ and $\{v_k\}$ take values in $\mathbb{R}^r$ and $\mathbb{R}^s$, respectively.

The basic random variables are assumed mutually independent and Gaussian:

$$x_0 \sim \mathcal{N}(\bar{x}_0, \Sigma_0), \quad w_k \sim \mathcal{N}(0, W), \quad v_k \sim \mathcal{N}(0, V), \quad (15.34)$$

with $W \succeq 0$, $V \succeq 0$.

15.4.3 ■ One-Step Transition and Observation Kernels

Proposition 15.15 (One-step conditional distributions). *For each $k \geq 0$, the conditional distribution of x_{k+1} given (x_k, u_k) is Gaussian:*

$$x_{k+1} \mid (x_k, u_k) \sim \mathcal{N}(Ax_k + Bu_k, GWG^\top). \quad (15.35)$$

Similarly, the conditional distribution of z_k given x_k is

$$z_k \mid x_k \sim \mathcal{N}(Cx_k, HVH^\top). \quad (15.36)$$

Proof. The pair (x_k, u_k) is a measurable function of $(x_0, w_0, \dots, w_{k-1}, v_0, \dots, v_k)$, whereas w_k is independent of this collection by assumption. Hence w_k is independent of (x_k, u_k) , and conditionally on (x_k, u_k) the right-hand side of (15.32) is the sum of the constant $Ax_k + Bu_k$ and the Gaussian vector Gw_k . An affine image of a Gaussian vector is Gaussian: $Gw_k \sim \mathcal{N}(0, GWG^\top)$. Therefore $x_{k+1} \mid (x_k, u_k) \sim \mathcal{N}(Ax_k + Bu_k, GWG^\top)$. The same argument applied to the observation equation, using the independence of v_k from x_k , gives $z_k \mid x_k \sim \mathcal{N}(Cx_k, HVH^\top)$. $\square$

Equivalently, these define stochastic kernels

$$P_k(\cdot \mid x, u) = \mathcal{N}(Ax + Bu, GWG^\top), \quad \Psi_k(\cdot \mid x) = \mathcal{N}(Cx, HVH^\top).$$

15.4.4 ■ m -Step Transition Distribution

Iterating (15.32), we obtain

$$x_{k+m} = A^m x_k + \sum_{j=0}^{m-1} A^{m-1-j} B u_{k+j} + \sum_{j=0}^{m-1} A^{m-1-j} G w_{k+j}. \quad (15.37)$$

Proposition 15.16 (m -step transition). *The conditional distribution of x_{k+m} given $(x_k, u_k, \dots, u_{k+m-1})$ is Gaussian:*

$$x_{k+m} \mid (x_k, u_k, \dots, u_{k+m-1}) \sim \mathcal{N}(\bar{x}_{k+m|k}, \Sigma^{(m)}), \quad (15.38)$$

where

$$\bar{x}_{k+m|k} = A^m x_k + \sum_{j=0}^{m-1} A^{m-1-j} B u_{k+j}, \quad (15.39)$$

$$\Sigma^{(m)} = \sum_{j=0}^{m-1} A^{m-1-j} G W G^\top (A^{m-1-j})^\top. \quad (15.40)$$

Proof. Conditionally on $(x_k, u_k, \dots, u_{k+m-1})$, the representation (15.37) expresses x_{k+m} as the constant $\bar{x}_{k+m|k}$ plus the noise term $\zeta_m := \sum_{j=0}^{m-1} A^{m-1-j} G w_{k+j}$. The vectors $w_k, \dots, w_{k+m-1}$ are i.i.d. $\mathcal{N}(0, W)$ and independent of the conditioning variables, so ζ_m is a sum of independent Gaussian vectors and is therefore Gaussian with mean zero. Its covariance is the sum of the covariances of the summands,

$$\text{cov}(\zeta_m) = \sum_{j=0}^{m-1} A^{m-1-j} G \text{cov}(w_{k+j}) G^\top (A^{m-1-j})^\top = \Sigma^{(m)},$$

since cross terms vanish by independence. This is the assertion. $\square$

The covariance $\Sigma^{(m)}$ satisfies the recursion

$$\Sigma^{(m+1)} = A \Sigma^{(m)} A^\top + G W G^\top, \quad \Sigma^{(0)} = 0. \quad (15.41)$$

15.4.5 ■ Unconditional State Distribution under Open-Loop Control

Assume the control sequence $\{u_k\}$ is deterministic. Then x_k is Gaussian for all k .

Proposition 15.17 (Unconditional state distribution). *Consider the linear system (15.32) driven by the Gaussian primitives (15.34) under a deterministic (open-loop) control sequence $\{u_k\}$. Then for each $k \geq 0$,*

$$x_k \sim \mathcal{N}(\bar{x}_k, \Sigma_k), \quad (15.42)$$

where

$$\bar{x}_k = A^k \bar{x}_0 + \sum_{j=0}^{k-1} A^{k-1-j} B u_j, \quad (15.43)$$

$$\Sigma_k = A \Sigma_{k-1} A^\top + G W G^\top, \quad k \geq 1, \quad (15.44)$$

where $x_0 \sim \mathcal{N}(\bar{x}_0, \Sigma_0)$ is the given Gaussian initial condition.

Proof. We argue by induction on k . For $k = 0$ the claim is the assumption on the initial condition. Suppose $x_k \sim \mathcal{N}(\bar{x}_k, \Sigma_k)$. Since the control u_k is deterministic, (15.32) expresses $x_{k+1} = Ax_k + Bu_k + Gw_k$ as an affine function of the independent Gaussian vectors x_k and w_k ; hence x_{k+1} is Gaussian. Its mean is $\bar{x}_{k+1} = A\bar{x}_k + Bu_k$, which unrolls to the stated formula, and by independence its covariance is

$$\Sigma_{k+1} = A \text{cov}(x_k) A^\top + G \text{cov}(w_k) G^\top = A\Sigma_k A^\top + GWG^\top.$$

□

In particular, Σ_k does not depend on the control sequence.

15.4.6 ■ State Covariance Function

Define $\Sigma_{k+m,k} = \mathbb{E}[(x_{k+m} - \bar{x}_{k+m})(x_k - \bar{x}_k)^\top]$.

Proposition 15.18 (State covariance function). *For the linear Gaussian system (15.32) under a deterministic control sequence, let $\Sigma_k = \text{cov}(x_k)$ be the state covariance of Proposition 15.17 and let $\Sigma_{k+m,k} = \mathbb{E}[(x_{k+m} - \bar{x}_{k+m})(x_k - \bar{x}_k)^\top]$ be the lagged cross-covariance. Then*

$$\Sigma_{k+m,k} = A^m \Sigma_k, \quad m \geq 1. \quad (15.45)$$

The correlation between two instants therefore decays exactly as the deterministic dynamics propagate: memory in the state process is carried entirely by the powers of A .

Proof. Subtracting means in (15.37) (the control terms cancel) gives

$$x_{k+m} - \bar{x}_{k+m} = A^m(x_k - \bar{x}_k) + \sum_{j=0}^{m-1} A^{m-1-j} Gw_{k+j}.$$

Each w_{k+j} with $j \geq 0$ is independent of x_k and has zero mean, so the noise sum is uncorrelated with $x_k - \bar{x}_k$. Multiplying by $(x_k - \bar{x}_k)^\top$ and taking expectations,

$$\Sigma_{k+m,k} = A^m \mathbb{E}[(x_k - \bar{x}_k)(x_k - \bar{x}_k)^\top] = A^m \Sigma_k.$$

□

15.4.7 ■ Observation Statistics

Proposition 15.19 (Observation statistics). *For the linear Gaussian system (15.32)–(15.33) under a deterministic control sequence, with $\bar{x}_k, \Sigma_k$ as in Proposition 15.17, the observation is Gaussian for each $k \geq 0$:*

$$z_k \sim \mathcal{N}(\bar{z}_k, \Sigma_k^z), \quad (15.46)$$

where

$$\bar{z}_k = C\bar{x}_k, \quad (15.47)$$

$$\Sigma_k^z = C\Sigma_k C^\top + HVH^\top. \quad (15.48)$$

Moreover, for $m \geq 1$,

$$\Sigma_{k+m,k}^z = \mathbb{E}[(z_{k+m} - \bar{z}_{k+m})(z_k - \bar{z}_k)^\top] = CA^m \Sigma_k C^\top. \quad (15.49)$$

Proof. From $z_k = Cx_k + Hv_k$ with $v_k \sim \mathcal{N}(0, V)$ independent of x_k , the observation is an affine function of independent Gaussian vectors, hence Gaussian, with mean $\bar{z}_k = C\bar{x}_k$ and covariance

$$\Sigma_k^z = C \operatorname{cov}(x_k) C^\top + H \operatorname{cov}(v_k) H^\top = C \Sigma_k C^\top + HVH^\top.$$

For the cross covariance with $m \geq 1$, write $z_{k+m} - \bar{z}_{k+m} = C(x_{k+m} - \bar{x}_{k+m}) + Hv_{k+m}$ and $z_k - \bar{z}_k = C(x_k - \bar{x}_k) + Hv_k$. The terms involving v_{k+m} and v_k drop out in expectation: v_{k+m} is independent of (x_k, v_k) , v_k is independent of x_{k+m} , and v_{k+m} is independent of v_k since $m \geq 1$. Using (15.45),

$$\Sigma_{k+m,k}^z = C \mathbb{E}[(x_{k+m} - \bar{x}_{k+m})(x_k - \bar{x}_k)^\top] C^\top = CA^m \Sigma_k C^\top.$$

□

Remark 15.20. The linear–Gaussian structure implies that all relevant distributions are characterized by first and second moments. This property underlies the tractability of Kalman filtering and the separation principle in LQG control.

Example 15.21 Mean and covariance propagation.

For the scalar linear Gaussian system $x_{k+1} = ax_k + w_k$ with $w_k \sim \mathcal{N}(0, q)$ i.i.d. and $x_0 \sim \mathcal{N}(m_0, P_0)$, the state remains Gaussian for all k , with $m_{k+1} = am_k$ and $P_{k+1} = a^2 P_k + q$. If $|a| < 1$, the variance recursion is a contraction and $P_k \rightarrow \bar{P} = q/(1 - a^2)$ regardless of P_0 : the state distribution converges to the stationary law $\mathcal{N}(0, \bar{P})$. If $|a| \geq 1$, the variance diverges: uncertainty accumulates faster than the stable dynamics can dissipate it. This dichotomy is the scalar seed of the stability theory developed later in the chapter. □

15.5 ■ Optimal Control for Stochastic Systems Under Complete Information

We now turn to optimization. The dynamic programming principle developed here is the stochastic counterpart of the deterministic Bellman recursion of Chapter 10: the new ingredient is that the minimization at each stage is interleaved with an expectation over the disturbance, so the recursion propagates *expected* cost-to-go rather than cost-to-go. We consider a discrete-time stochastic control system defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with state and observation equations

$$x_{k+1} = f_k(x_k, u_k, w_k), \quad (15.50)$$

$$z_k = h_k(x_k, v_k), \quad k = 0, 1, \dots, \quad (15.51)$$

where $x_k \in \mathbb{R}^n$ is the state, $u_k \in U \subset \mathbb{R}^m$ is the control input, and $z_k \in \mathbb{R}^p$ is the observation. The random variables w_k and v_k represent process disturbances and measurement noise, respectively, and are assumed to be mutually independent across time. The mappings f_k and h_k are assumed to be measurable.

At each time k , the control input must be selected based on the information available to the controller. Let $z^k := (z_0, \dots, z_k)$ denote the observation history. A *control law* (or policy) is a sequence of measurable mappings $\mu = \{\mu_k\}_{k \geq 0}$ such that $u_k = \mu_k(z^k)$, and $\mu_k(z^k) \in U$ for all admissible histories z^k . The set of all such policies is denoted by $\mathcal{M}$.

Fix a finite horizon $N < \infty$. To each policy $\mu \in \mathcal{M}$ we associate the cost functional

$$C^\mu := \sum_{k=0}^{N-1} c_k(x_k^\mu, u_k^\mu) + c_N(x_N^\mu), \quad (15.52)$$

where $c_k : \mathbb{R}^n \times U \rightarrow \mathbb{R}$ is the stage cost and $c_N : \mathbb{R}^n \rightarrow \mathbb{R}$ is the terminal cost. The processes $\{x_k^\mu, u_k^\mu\}$ are those induced by the policy μ .

Since C^μ is a random variable, policies are compared via their expected cost $J(\mu) := \mathbb{E}[C^\mu]$.

Definition 15.22 (Optimal policy). A policy $\mu^* \in \mathcal{M}$ is optimal if $J(\mu^*) = \inf_{\mu \in \mathcal{M}} J(\mu)$. The quantity $J^* := \inf_{\mu \in \mathcal{M}} J(\mu)$ is called the optimal value.

Two standing conventions keep the statements below free of measure-theoretic qualifications.

Assumption 15.23 (Regularity and measurable selection). Throughout this section the stage costs c_k and the terminal cost c_N are bounded below and Borel measurable, so that every expectation appearing in the dynamic programming recursion is well defined; and for each k the infimum in the Bellman recursion (15.54) is attained by a Borel-measurable selector $\mu_k : \mathbb{R}^n \rightarrow U$. The latter holds, for instance, when U is finite, and more generally when U is compact, $c_k(x, \cdot)$ is lower semicontinuous and $f_k(x, \cdot, w)$ continuous, by a measurable selection theorem. Without it, “arg min” should be read as “ ε -argmin” and every optimality statement holds up to an arbitrarily small ε .

Remark 15.24 (Disturbance-dependent stage costs). Some models, the resource-allocation case study of Section 15.6 among them, charge a stage cost $c_k(x_k, u_k, w_k)$ that also depends on the realized disturbance. Nothing changes: since w_k is independent of (x_k, u_k) , one may replace c_k throughout by its averaged version $\bar{c}_k(x, u) := \mathbb{E}[c_k(x, u, w_k)]$, which is again a function of (x, u) alone, and all the results of this section apply verbatim to $\bar{c}_k$.

15.5.1 ■ Complete Information and Markov Policies

We consider the case of complete state information, i.e., $z_k = x_k$ for all $k \geq 0$. In this setting, the information available at time k is $\mathcal{I}_k = \sigma(x_0, \dots, x_k)$. Although admissible controls may depend on the entire state history, the controlled Markov property implies that the conditional distribution of x_{k+1} given the past depends only on (x_k, u_k) . Consequently, it is sufficient to restrict attention to policies depending only on the current state.

Definition 15.25 (Markov policy). A policy $\mu = \{\mu_k\}_{k=0}^{N-1}$ is called Markov if there exist measurable mappings $\mu_k : \mathbb{R}^n \rightarrow U$ such that $u_k = \mu_k(x_k)$ for $k = 0, \dots, N-1$. The set of Markov policies is denoted by $\mathcal{M}_M \subset \mathcal{M}$.

For a policy μ and time k , define the cost-to-go

$$J_k^\mu := \mathbb{E} \left[\sum_{l=k}^{N-1} c_l(x_l^\mu, u_l^\mu) + c_N(x_N^\mu) \middle| \mathcal{I}_k \right].$$

For a Markov policy μ , define recursively $V_N^\mu(x) := c_N(x)$ and

$$V_k^\mu(x) = c_k(x, \mu_k(x)) + \mathbb{E}[V_{k+1}^\mu(f_k(x, \mu_k(x), w_k))].$$

Lemma 15.26 (Cost-to-go equals the policy value recursion). *Consider the complete-information problem ($z_k = x_k$), and let $\mu \in \mathcal{M}_M$ be a Markov policy with cost-to-go J_k^μ . Let $\{V_k^\mu\}$ be the functions generated by the backward recursion $V_N^\mu(x) = c_N(x)$ and $V_k^\mu(x) = c_k(x, \mu_k(x)) + \mathbb{E}[V_{k+1}^\mu(f_k(x, \mu_k(x), w_k))]$. Then the random cost-to-go coincides almost surely with this deterministic function evaluated along the closed-loop trajectory: $V_k^\mu(x_k^\mu) = J_k^\mu$ almost surely, for $k = 0, \dots, N$. In particular $J(\mu) = \mathbb{E}[V_0^\mu(x_0)]$, so the performance of a Markov policy is obtained by a single backward pass rather than by averaging the cost over trajectories.*

Proof. We argue by backward induction. At $k = N$, we have $V_N^\mu(x_N^\mu) = c_N(x_N^\mu) = J_N^\mu$. Assume the claim holds at time $k + 1$. Then,

$$\begin{aligned} J_k^\mu &= \mathbb{E}[c_k(x_k^\mu, u_k^\mu) + J_{k+1}^\mu \mid \mathcal{I}_k] \\ &= c_k(x_k^\mu, \mu_k(x_k^\mu)) + \mathbb{E}[J_{k+1}^\mu \mid \mathcal{I}_k]. \end{aligned}$$

By the induction hypothesis, $J_{k+1}^\mu = V_{k+1}^\mu(x_{k+1}^\mu)$ almost surely, hence

$$J_k^\mu = c_k(x_k^\mu, \mu_k(x_k^\mu)) + \mathbb{E}[V_{k+1}^\mu(x_{k+1}^\mu) \mid \mathcal{I}_k].$$

By the controlled Markov property, the conditional expectation depends only on (x_k^μ, u_k^μ) , yielding

$$\mathbb{E}[V_{k+1}^\mu(x_{k+1}^\mu) \mid \mathcal{I}_k] = \mathbb{E}[V_{k+1}^\mu(f_k(x_k^\mu, \mu_k(x_k^\mu), w_k))].$$

Thus $J_k^\mu = V_k^\mu(x_k^\mu)$, completing the induction. $\square$

15.5.2 ■ Comparison Principle

The value functions produced by dynamic programming are characterized by a one-sided variational inequality. The following lemma isolates the underlying principle: any family of functions that underestimates the Bellman recursion underestimates the cost of every policy, and hence bounds the optimal value from below.

Lemma 15.27 (Comparison principle). *Let $\{V_k\}_{k=0}^N$ be measurable functions that are pointwise sub-solutions of the dynamic programming recursion, that is,*

$$V_N(x) \leq c_N(x), \quad V_k(x) \leq c_k(x, u) + \mathbb{E}[V_{k+1}(f_k(x, u, w_k))] \quad \text{for all } u \in U.$$

Then $\{V_k\}$ underestimates the cost-to-go of every admissible policy: for any $\mu \in \mathcal{M}$, $V_k(x_k^\mu) \leq J_k^\mu$ almost surely, $k = 0, \dots, N$. In particular, taking $k = 0$ and expectations, $\mathbb{E}[V_0(x_0)] \leq J(\mu)$ for every μ , so any sub-solution furnishes a lower bound on the optimal value J^ .*

Proof. We proceed by backward induction. At $k = N$, $V_N(x_N^\mu) \leq c_N(x_N^\mu) = J_N^\mu$. Assume the statement holds at $k + 1$. Then,

$$V_k(x_k^\mu) \leq c_k(x_k^\mu, u_k^\mu) + \mathbb{E}[V_{k+1}(x_{k+1}^\mu) \mid \mathcal{I}_k].$$

By the induction hypothesis and monotonicity of conditional expectation,

$$\mathbb{E}[V_{k+1}(x_{k+1}^\mu) \mid \mathcal{I}_k] \leq \mathbb{E}[J_{k+1}^\mu \mid \mathcal{I}_k].$$

Hence $V_k(x_k^\mu) \leq J_k^\mu$. $\square$

Corollary 15.28 (Sufficient condition for optimality). *Let $\{V_k\}_{k=0}^N$ be any family of sub-solutions of the dynamic programming recursion in the sense of Lemma 15.27. If an admissible policy μ attains the resulting lower bound at the initial time, i.e. $J_0^\mu = V_0(x_0)$ almost surely, then μ is optimal and the optimal value is $J^* = \mathbb{E}[V_0(x_0)]$. This is a verification principle: exhibiting a sub-solution together with a policy that meets it certifies optimality without comparing μ to every competitor directly.*

15.5.3 ■ Dynamic Programming Principle

Define the optimal value functions recursively by

$$V_N(x) := c_N(x), \quad (15.53)$$

$$V_k(x) := \inf_{u \in U} \{c_k(x, u) + \mathbb{E}[V_{k+1}(f_k(x, u, w_k))]\}. \quad (15.54)$$

Theorem 15.29 (Dynamic programming). *The following statements hold:*

(1) *For any admissible policy μ , $V_k(x_k^\mu) \leq J_k^\mu$ almost surely.*

(2) *If a Markov policy μ satisfies*

$$\mu_k(x) \in \arg \min_{u \in U} \{c_k(x, u) + \mathbb{E}[V_{k+1}(f_k(x, u, w_k))]\},$$

then μ is optimal.

(3) *Conversely, any optimal Markov policy satisfies the Bellman equality*

$$V_k(x_k^\mu) = c_k(x_k^\mu, \mu_k(x_k^\mu)) + \mathbb{E}[V_{k+1}(x_{k+1}^\mu) \mid \mathcal{I}_k] \quad \text{almost surely.}$$

Proof. Throughout, $\{V_k\}$ are the functions defined by (15.53) and (15.54). We first record a telescoping identity. For any admissible Markov policy μ , since $V_N = c_N$ the sum telescopes path-wise:

$$C^\mu - V_0(x_0^\mu) = \sum_{k=0}^{N-1} [c_k(x_k^\mu, u_k^\mu) + V_{k+1}(x_{k+1}^\mu) - V_k(x_k^\mu)]. \quad (15.55)$$

Under complete information $\mathcal{I}_k = \sigma(x_0^\mu, \dots, x_k^\mu)$, so $x_k^\mu, u_k^\mu = \mu_k(x_k^\mu)$ and hence $c_k(x_k^\mu, u_k^\mu)$ and $V_k(x_k^\mu)$ are $\mathcal{I}_k$ -measurable. Taking $\mathbb{E}[\cdot \mid \mathcal{I}_k]$ of the k -th summand and using the controlled Markov property (Lemma 15.12) to evaluate $\mathbb{E}[V_{k+1}(x_{k+1}^\mu) \mid \mathcal{I}_k] = \mathbb{E}[V_{k+1}(f_k(x_k^\mu, u_k^\mu, w_k))]$, define

$$\Delta_k := c_k(x_k^\mu, u_k^\mu) + \mathbb{E}[V_{k+1}(x_{k+1}^\mu) \mid \mathcal{I}_k] - V_k(x_k^\mu). \quad (15.56)$$

By the recursion (15.54), $V_k(x) = \inf_{u \in U} \{c_k(x, u) + \mathbb{E}[V_{k+1}(f_k(x, u, w_k))]\}$, so evaluating the bracket at the particular control $u = u_k^\mu$ gives $\Delta_k \geq 0$ almost surely. Taking expectations in (15.55) and using $x_0^\mu = x_0$,

$$J(\mu) - \mathbb{E}[V_0(x_0)] = \sum_{k=0}^{N-1} \mathbb{E}[\Delta_k] \geq 0. \quad (15.57)$$

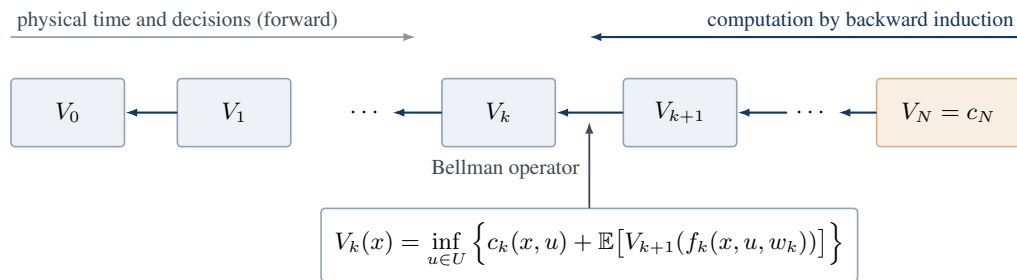


Figure 15.4: Backward dynamic programming. The value functions are computed *backward* in time, starting from the terminal cost $V_N = c_N$ and applying the Bellman operator at each stage. Physical decisions are then made *forward*: at state x and time k the controller selects any minimizer u of the one-stage problem. The expectation inside the operator averages over the disturbance w_k , so the recursion propagates *expected* cost-to-go.

Part (1). The functions $\{V_k\}$ satisfy the hypotheses of Lemma 15.27 (with equality replaced by “ $\leq$ ”): indeed $V_N = c_N$, and $V_k(x) \leq c_k(x, u) + \mathbb{E}[V_{k+1}(f_k(x, u, w_k))]$ for every $u \in U$ because V_k is the infimum over u . Hence $V_k(x_k^\mu) \leq J_k^\mu$ almost surely for every admissible μ . In particular, taking $k = 0$ and expectations, $\mathbb{E}[V_0(x_0)] \leq J(\mu)$ for all μ , so $\mathbb{E}[V_0(x_0)] \leq J^*$.

Part (2). Let μ be a Markov policy with $\mu_k(x) \in \arg \min_{u \in U} \{c_k(x, u) + \mathbb{E}[V_{k+1}(f_k(x, u, w_k))]\}$. Then the bracket in (15.56) attains the infimum, so $\Delta_k = 0$ almost surely for every k . By (15.57), $J(\mu) = \mathbb{E}[V_0(x_0)]$. Combined with Part (1), $J(\mu) = \mathbb{E}[V_0(x_0)] \leq J(\mu')$ for every admissible μ' ; hence μ is optimal and $J^* = \mathbb{E}[V_0(x_0)]$.

Part (3). Let μ be an optimal Markov policy, so $J(\mu) = J^* = \mathbb{E}[V_0(x_0)]$ by Part (2). Then (15.57) gives $\sum_{k=0}^{N-1} \mathbb{E}[\Delta_k] = 0$; since each $\Delta_k \geq 0$ almost surely, we conclude $\mathbb{E}[\Delta_k] = 0$ and therefore $\Delta_k = 0$ almost surely for every k . By the definition (15.56), this is precisely the Bellman equality

$$V_k(x_k^\mu) = c_k(x_k^\mu, \mu_k(x_k^\mu)) + \mathbb{E}[V_{k+1}(x_{k+1}^\mu) \mid \mathcal{I}_k] \quad \text{almost surely.}$$

□

15.5.4 ■ Backward Dynamic Programming Algorithm

The dynamic programming principle turns a single optimization over policies, an object living in a huge function space, into a sequence of small optimizations, one per stage, each over the control set U alone. The price is that the stages must be solved in reverse: to know the value of being at (k, x) we must already know the value of every state reachable at time $k + 1$. Computation therefore flows *backward* in time even though the system, and the eventual decisions, run *forward*. Figure 15.4 summarizes this two-directional structure.

The value functions and an optimal policy can be computed by backward induction. Initialize $V_N(x) = c_N(x)$, and for $k = N - 1, \dots, 0$ define

$$V_k(x) = \inf_{u \in U} \{c_k(x, u) + \mathbb{E}[V_{k+1}(f_k(x, u, w_k))]\}.$$

An optimal policy is obtained by selecting measurable minimizers

$$\mu_k(x) \in \arg \min_{u \in U} \{c_k(x, u) + \mathbb{E}[V_{k+1}(f_k(x, u, w_k))]\}.$$

Definition 15.30 (Value functions). The functions $\{V_k\}_{k=0}^N$ generated by the backward recursion (15.53)–(15.54) are called the value functions of the problem: $V_k(x)$ is the least expected cost incurred from stage k onward when the state at that stage is x .

Example 15.31 Two-stage stochastic regulator by backward induction.

Let $x_{k+1} = x_k + u_k + w_k$ with $\mathbb{E}[w_k] = 0$, $\mathbb{E}[w_k^2] = \sigma^2$, and cost $\mathbb{E}[x_2^2 + u_0^2 + u_1^2]$. Backward induction gives $V_2(x) = x^2$ and

$$V_1(x) = \min_u \{u^2 + \mathbb{E}[(x + u + w_1)^2]\} = \min_u \{u^2 + (x + u)^2\} + \sigma^2 = \frac{1}{2}x^2 + \sigma^2,$$

attained at $u_1^* = -x/2$. One more step yields

$$V_0(x) = \min_u \{u^2 + \frac{1}{2}(x + u)^2\} + \frac{3}{2}\sigma^2 = \frac{1}{3}x^2 + \frac{3}{2}\sigma^2, \quad u_0^* = -x/3.$$

The feedback gains $-1/2$ and $-1/3$ are exactly those of the deterministic problem ($\sigma = 0$); the noise contributes only the additive constant $\frac{3}{2}\sigma^2$. This is certainty equivalence, established in general for linear-quadratic problems later in the chapter.

□

15.6 ■ Case Studies of Optimal Control Under Complete Information

The dynamic programming principle of the previous section is a general recipe: define the value function as the optimal cost-to-go, propagate it backward through the Bellman recursion, and read off an optimal policy as the stagewise minimizer. Its power, however, is best appreciated on concrete problems, where the abstract recursion turns into explicit value functions and closed-form feedback laws. This section collects several such case studies, all under *complete state information* ($z_k = x_k$), so that attention rests squarely on the optimization structure rather than on estimation.

The two examples are chosen to illustrate complementary themes that will reappear throughout the chapter. The stochastic resource-allocation problem is a *how much* decision: it shows how a terminal constraint enters the recursion, how backward induction yields a quadratic value function and an “equal-split” feedback law, and how feedback and open-loop control coincide in expectation yet diverge sharply once the state is perturbed, foreshadowing the certainty-equivalence phenomenon made precise later for linear-quadratic problems. The option-pricing problem is instead a *when to act* decision, an optimal stopping problem in which at each stage one chooses between exercising now and continuing; here dynamic programming produces not a control value but a *threshold policy*, splitting the state space into a stopping region and a continuation region. Together these studies show the same backward recursion specializing to two qualitatively different kinds of decision, and they set the stage for the partially observed problems taken up in the remainder of the chapter.

15.6.1 ■ Case Study: Stochastic Resource Allocation via Dynamic Programming

We consider a finite-horizon stochastic control problem in which a fixed amount of resource must be allocated over time in the presence of uncertainty. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. The

state process $\{x_k\}_{k=t_0}^{t_1+1}$ evolves according to

$$x_{k+1} = x_k - u_k, \quad k = t_0, \dots, t_1, \quad (15.58)$$

with initial condition $x_{t_0} = c > 0$ and terminal constraint $x_{t_1+1} = 0$. The control satisfies $u_k \geq 0$.

The stage cost is given by

$$c_k(x_k, u_k, w_k) = u_k^2 - \theta_k w_k u_k, \quad (15.59)$$

where $\{w_k\}$ is a sequence of independent random variables with $\mathbb{E}[w_k] = 0$ and $\text{Var}(w_k) = \sigma_k^2$.

The objective is to minimize

$$J(\mu) = \mathbb{E}^\mu \left[\sum_{k=t_0}^{t_1} c_k(x_k, u_k, w_k) \right] \quad (15.60)$$

over admissible policies.

Lemma 15.32 (Feasibility of resource schedules). *Consider the resource-allocation dynamics (15.58), $x_{k+1} = x_k - u_k$, with initial stock $x_{t_0} = c > 0$, nonnegativity $u_k \geq 0$, and terminal constraint $x_{t_1+1} = 0$. Then a control sequence $\{u_k\}_{k=t_0}^{t_1}$ is feasible if and only if it is nonnegative and exhausts the stock,*

$$\sum_{k=t_0}^{t_1} u_k = c.$$

In other words, the terminal constraint is exactly a budget constraint: the withdrawals must sum to the initial resource c , and any nonnegative schedule meeting this budget produces a feasible trajectory.

Proof. Summing (15.58) from t_0 to t_1 yields $x_{t_1+1} = x_{t_0} - \sum_{k=t_0}^{t_1} u_k$. Using $x_{t_1+1} = 0$ and $x_{t_0} = c$ gives the result. $\square$

Admissible policies. A policy $\mu = \{\mu_k\}$ is admissible if $u_k = \mu_k(\mathcal{I}_k)$, $\mathcal{I}_k = \sigma(x_{t_0}, w_{t_0}, \dots, w_{k-1})$. Since the state is fully observed and deterministic given controls, it suffices to consider Markov policies $u_k = \mu_k(x_k)$.

Value function. Define

$$V_k(x) := \inf_{\mu} \mathbb{E} \left[\sum_{\ell=k}^{t_1} c_\ell(x_\ell, u_\ell, w_\ell) \mid x_k = x \right]. \quad (15.61)$$

Proposition 15.33 (Dynamic programming equation). *The value function satisfies*

$$V_k(x) = \inf_{0 \leq u \leq x} \mathbb{E} [u^2 - \theta_k w_k u + V_{k+1}(x - u)], \quad (15.62)$$

with terminal condition

$$V_{t_1+1}(x) = \begin{cases} 0, & x = 0, \\ +\infty, & x \neq 0, \end{cases}$$

which encodes the terminal constraint $x_{t_1+1} = 0$. In particular, $V_{t_1}(x) = x^2$, since the constraint forces $u_{t_1} = x$ at the last stage.

Proof. Fix k and x . For any admissible policy, conditioning on the first action $u \in [0, x]$ and on w_k splits the cost as

$$\mathbb{E} \left[\sum_{\ell=k}^{t_1} c_\ell \middle| x_k = x \right] = \mathbb{E} [u^2 - \theta_k w_k u] + \mathbb{E} \left[\sum_{\ell=k+1}^{t_1} c_\ell \middle| x_{k+1} = x - u \right],$$

where the second term uses that the state transition $x_{k+1} = x - u$ is deterministic and that the disturbances $(w_\ell)_{\ell > k}$ are independent of w_k . Taking the infimum over the continuation policy first, and then over u , yields $V_k(x) \leq$ the right-hand side of (15.62); conversely, any ε -optimal policy for V_k induces a feasible first action and an admissible continuation whose value is bounded below by the right-hand side, giving the reverse inequality. The terminal condition restates the constraint $x_{t_1+1} = 0$, and evaluating the recursion at $k = t_1$ forces $u = x$, so $V_{t_1}(x) = x^2$. $\square$

Lemma 15.34 (Certainty-equivalence reduction). *Under $\mathbb{E}[w_k] = 0$, the Bellman equation reduces to*

$$V_k(x) = \inf_{0 \leq u \leq x} [u^2 + V_{k+1}(x - u)]. \quad (15.63)$$

Proof. Inside the infimum in (15.62), the decision variable u is a deterministic quantity chosen before w_k is realized, and $V_{k+1}(x - u)$ contains no randomness because the transition is deterministic. Hence the only random term is $-\theta_k w_k u$, whose expectation vanishes: $\mathbb{E}[-\theta_k w_k u] = -\theta_k u \mathbb{E}[w_k] = 0$. The disturbance therefore disappears from the recursion, which reduces to (15.63). $\square$

Proposition 15.35 (Structure of the value function). *For the resource-allocation problem (15.58)–(15.59), the value function is quadratic and strictly convex on $[0, \infty)$: for each stage k there is a constant $a_k > 0$ with $V_k(x) = a_k x^2$ for $x \geq 0$, where $a_{t_1} = 1$ and the coefficients obey the backward recursion $a_k = a_{k+1}/(1 + a_{k+1})$. Strict convexity in x follows from $a_k > 0$ and guarantees that each Bellman minimization has a unique minimizer.*

Proof. We argue by backward induction that $V_k(x) = a_k x^2$ with $a_k > 0$ for $x \geq 0$; strict convexity is then immediate since $a_k > 0$. At $k = t_1$ the terminal constraint forces $u_{t_1} = x$, so $V_{t_1}(x) = x^2$ and $a_{t_1} = 1 > 0$. Suppose $V_{k+1}(x) = a_{k+1} x^2$ with $a_{k+1} > 0$. By the reduced Bellman equation (15.63),

$$V_k(x) = \inf_{0 \leq u \leq x} [u^2 + a_{k+1}(x - u)^2].$$

The objective $u \mapsto u^2 + a_{k+1}(x - u)^2$ is strictly convex, and its unconstrained stationary point $u^* = \frac{a_{k+1}}{1 + a_{k+1}} x$ lies in $[0, x]$, so the constraint is inactive. Substituting u^* yields

$$V_k(x) = \frac{a_{k+1}}{1 + a_{k+1}} x^2 =: a_k x^2, \quad a_k = \frac{a_{k+1}}{1 + a_{k+1}} > 0,$$

which completes the induction. The explicit closed form of a_k and the associated policy are computed in Proposition 15.36. $\square$

Proposition 15.36 (Optimal policy). *For the resource-allocation problem (15.58)–(15.59), the unique optimal policy is the state feedback that splits the remaining stock evenly over the stages that remain,*

$$u_k^*(x_k) = \frac{x_k}{t_1 - k + 1}, \quad (15.64)$$

and the resulting optimal cost-to-go is $V_k(x) = x^2/(t_1 - k + 1)$. Uniqueness follows from the strict convexity of the Bellman objective established in Proposition 15.35.

Proof. Write $V_{k+1}(x) = a_{k+1}x^2$ with $a_{t_1} = 1$ (from $V_{t_1}(x) = x^2$). The reduced Bellman equation requires minimizing the strictly convex function $u \mapsto u^2 + a_{k+1}(x - u)^2$ over $[0, x]$. The unconstrained stationarity condition $2u - 2a_{k+1}(x - u) = 0$ gives

$$u^*(x) = \frac{a_{k+1}}{1 + a_{k+1}} x \in [0, x],$$

so the constraint is inactive. Substituting back,

$$V_k(x) = \left(\frac{a_{k+1}^2}{(1 + a_{k+1})^2} + \frac{a_{k+1}}{(1 + a_{k+1})^2} \right) x^2 = \frac{a_{k+1}}{1 + a_{k+1}} x^2, \quad \text{i.e.} \quad a_k = \frac{a_{k+1}}{1 + a_{k+1}}.$$

The map $a \mapsto a/(1 + a)$ applied to $a_{t_1} = 1$ gives, by induction, $a_k = 1/(t_1 - k + 1)$. Therefore

$$u_k^*(x) = \frac{a_{k+1}}{1 + a_{k+1}} x = \frac{1/(t_1 - k)}{1 + 1/(t_1 - k)} x = \frac{x}{t_1 - k + 1},$$

which is the stated policy; strict convexity gives uniqueness. $\square$

Example 15.37 Resource allocation with concrete numbers.

Let $c = 12$, $t_0 = 0$, $t_1 = 2$ (three stages), and $\theta_k \equiv 0$. The coefficients of the value function $V_k(x) = a_k x^2$ evolve backward by $a_k = a_{k+1}/(1 + a_{k+1})$ from $a_2 = 1$, giving $a_2 = 1$, $a_1 = \frac{1}{2}$ and $a_0 = \frac{1}{3}$, so the optimal total cost from $x_0 = 12$ is $V_0(12) = \frac{1}{3} \cdot 144 = 48$. The feedback policy $u_k^* = x_k/(t_1 - k + 1)$ produces

$$u_0^* = \frac{12}{3} = 4, \quad x_1 = 8, \quad u_1^* = \frac{8}{2} = 4, \quad x_2 = 4, \quad u_2^* = \frac{4}{1} = 4,$$

the equal split predicted by the open-loop analysis, with cost $3 \times 16 = 48$. If, however, a disturbance perturbs the state, say $x_1 = 10$ instead of 8, the feedback policy re-optimizes automatically ($u_1^* = 5, u_2^* = 5$; continuation cost 50), whereas the open-loop schedule (4, 4, 4) would violate the terminal constraint entirely. Feedback provides not only optimality but also feasibility under perturbation. $\square$

Feedback versus Open-Loop Policies

Proposition 15.38 (Optimal open-loop control). *For the same resource-allocation problem, restrict attention to deterministic (open-loop) schedules $\{u_k\}$ chosen in advance and subject to the feasibility constraint $\sum_{k=t_0}^{t_1} u_k = c$ of Lemma 15.32. Then the optimal open-loop schedule is the constant equal split*

$$u_k^{\text{OL}} = \frac{c}{t_1 - t_0 + 1}, \quad k = t_0, \dots, t_1,$$

and it is unique. Along the undisturbed trajectory this agrees stage by stage with the feedback policy of Proposition 15.36; the two part ways only once the state is perturbed, when the feedback law re-optimizes on the realized stock while the open-loop schedule, fixed in advance, cannot.

Proof. For a deterministic sequence, $\mathbb{E}[-\theta_k w_k u_k] = 0$ for every k , so $J(u) = \sum_{k=t_0}^{t_1} u_k^2$. Let $n = t_1 - t_0 + 1$. By the Cauchy-Schwarz inequality,

$$c^2 = \left(\sum_{k=t_0}^{t_1} u_k \right)^2 \leq n \sum_{k=t_0}^{t_1} u_k^2,$$

with equality if and only if all u_k are equal. Hence $\sum u_k^2 \geq c^2/n$, attained uniquely at $u_k = c/n$. $\square$

Proposition 15.39 (Equivalence in expectation). *For the resource-allocation problem with deterministic dynamics $x_{k+1} = x_k - u_k$ and zero-mean disturbances entering only the cost (15.59), the optimal feedback policy μ^* of Proposition 15.36 and the optimal open-loop schedule u^{OL} of Proposition 15.38 incur the same expected cost,*

$$J(\mu^*) = J(u^{\text{OL}}) = \frac{c^2}{t_1 - t_0 + 1}.$$

In fact, along the undisturbed dynamics the feedback law reproduces the constant open-loop allocation $u_k = c/(t_1 - t_0 + 1)$ exactly, so the two policies are indistinguishable in performance. This is the certainty-equivalence situation: disturbances that are zero-mean and affect only the cost change neither the optimal actions nor confer any advantage on feedback.

Proof. Under the deterministic dynamics $x_{k+1} = x_k - u_k$, the feedback policy $u_k^* = x_k/(t_1 - k + 1)$ generates a deterministic trajectory. We claim it produces the constant allocation $u_k^* = c/n$ with $n = t_1 - t_0 + 1$. Indeed, if $x_k = c(t_1 - k + 1)/n$ (true at $k = t_0$), then $u_k^* = x_k/(t_1 - k + 1) = c/n$ and $x_{k+1} = x_k - c/n = c(t_1 - k)/n$, closing the induction. Thus both policies produce the identical allocation $(c/n, \dots, c/n)$; since the disturbance enters the cost linearly with zero mean, both incur the same expected cost c^2/n . $\square$

Proposition 15.40 (Role of feedback under stochastic dynamics). *Suppose the resource dynamics are perturbed by process noise that enters the state, $x_{k+1} = x_k - u_k + w_k$, with $\{w_k\}$ zero-mean and of positive variance. Then the equivalence of Proposition 15.39 breaks down: since open-loop schedules form a strict subclass of feedback policies (those that ignore the observed state), the optimal feedback policy incurs strictly lower expected cost than the best open-loop schedule in general. The mechanism is structural: once noise enters the state, a feedback law sees the realized x_k and re-optimizes, cancelling accumulated disturbances, whereas an open-loop plan is committed in advance and can only hedge against the mean.*

Proof. Open-loop sequences are exactly the feedback policies whose maps μ_k are constant, so the optimal feedback cost never exceeds the optimal open-loop cost; only strictness needs an argument, and for that it is enough to exhibit one instance of the perturbed resource problem in which the gap is positive.

Take two stages, $t_0 = 0$ and $t_1 = 1$, initial stock $x_0 = c > 0$, dynamics $x_{k+1} = x_k - u_k + w_k$ with w_0 zero-mean of variance $\sigma^2 > 0$, the running cost $u_0^2 + u_1^2$ of (15.59) with $\theta_k \equiv 0$, and the terminal requirement $x_2 = 0$, which forces $u_1 = x_1$. Because the terminal constraint must hold

for every realization of w_0 , an open-loop pair (u_0, u_1) fixed in advance is feasible only on a null set of outcomes unless w_0 is degenerate; the honest comparison therefore charges the terminal miss, so replace the hard constraint by the penalty ρx_2^2 with $\rho > 0$.

Under feedback, u_1 may see the realized $x_1 = c - u_0 + w_0$; minimizing $u_1^2 + \rho(x_1 - u_1)^2$ gives $u_1^* = \rho x_1 / (1 + \rho)$ and continuation cost $\rho x_1^2 / (1 + \rho)$, so the feedback value is $\min_{u_0} \{u_0^2 + \frac{\rho}{1+\rho} \mathbb{E}[(c - u_0 + w_0)^2]\} = \min_{u_0} \{u_0^2 + \frac{\rho}{1+\rho} (c - u_0)^2\} + \frac{\rho}{1+\rho} \sigma^2$. An open-loop pair must commit to u_1 before seeing w_0 , and incurs $u_0^2 + u_1^2 + \rho \mathbb{E}[(c - u_0 + w_0 - u_1)^2] = u_0^2 + u_1^2 + \rho(c - u_0 - u_1)^2 + \rho\sigma^2$. Minimizing the deterministic parts gives the same first-stage split in both cases, so the two values differ by exactly

$$\rho\sigma^2 - \frac{\rho}{1+\rho} \sigma^2 = \frac{\rho^2}{1+\rho} \sigma^2 > 0.$$

The feedback policy is therefore strictly better, and the gap is precisely the variance it can absorb by waiting to see w_0 . The mechanism is general: once the realized disturbance enters the state, a feedback policy can cancel it, whereas an open-loop plan can only optimize against its mean. $\square$

This example illustrates that when uncertainty enters only through a zero-mean additive term in the cost, the optimal policy exhibits a certainty-equivalence structure and coincides with the deterministic solution. However, when uncertainty affects the state dynamics, feedback becomes essential, as it enables adaptation to realized disturbances and improves performance.

15.6.2 ■ Option Pricing as a Stochastic Control Problem (Optimal Stopping)

We formulate the option pricing problem as a finite-horizon stochastic control problem with a stopping decision. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\{x_k\}_{k=0}^N$ be a Markov process satisfying

$$x_{k+1} = f_k(x_k, w_k), \quad (15.65)$$

where $\{w_k\}$ is a sequence of independent random variables. The state at time k is (k, x_k) , where k represents the time index (equivalently, time to maturity $N - k$).

Let $g_k : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ denote the payoff function. The objective is to choose a stopping time $\tau \in \{0, \dots, N\}$ to maximize

$$\mathbb{E}[g_\tau(x_\tau)]. \quad (15.66)$$

For each k and x , define

$$V_k(x) := \sup_{\tau \in \mathcal{T}_k} \mathbb{E}[g_\tau(x_\tau) \mid x_k = x], \quad (15.67)$$

where $\mathcal{T}_k$ is the set of stopping times taking values in $\{k, \dots, N\}$.

Proposition 15.41 (Dynamic programming equation). *The value function satisfies*

$$V_k(x) = \max \{g_k(x), \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x]\}, \quad (15.68)$$

with terminal condition $V_N(x) = g_N(x)$.

Proof. Fix (k, x) . Any admissible stopping time $\tau \in \mathcal{T}_k$ either satisfies $\tau = k$ or $\tau > k$. In the first case, the payoff is $g_k(x)$. In the second case, conditioning on $x_k = x$ and using the Markov property,

$$\mathbb{E}[g_\tau(x_\tau) \mid x_k = x] = \mathbb{E}[\mathbb{E}[g_\tau(x_\tau) \mid x_{k+1}] \mid x_k = x] \leq \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x].$$

Taking the supremum over τ shows that $V_k(x)$ is at most the right-hand side of (15.68). For the reverse inequality, the choice $\tau = k$ achieves $g_k(x)$; and for any $\varepsilon > 0$, choosing for each state x' a stopping time $\tau_{x'} \in \mathcal{T}_{k+1}$ that is ε -optimal for $V_{k+1}(x')$ and applying it after one transition produces an admissible stopping time in $\mathcal{T}_k$ with value at least $\mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x] - \varepsilon$. Since ε is arbitrary, $V_k(x)$ dominates both terms of the maximum, and equality holds. $\square$

Proposition 15.42 (Optimal stopping rule). *For the optimal stopping problem (15.67) with value functions V_k solving the Bellman equation (15.68), an optimal stopping time is the first stage at which the immediate payoff is at least the expected continuation value,*

$$\tau^* = \inf \{k \geq 0 : g_k(x_k) \geq \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k]\}, \quad V_{N+1} \equiv -\infty, \quad (15.69)$$

the convention $V_{N+1} \equiv -\infty$ making the test automatically true at $k = N$, so that $\tau^ \leq N$ and the rule is well defined. Equivalently, one stops as soon as exercising the payoff $g_k(x_k)$ weakly dominates holding for one more step, and continues otherwise. The rule is a pure feedback policy on the current state: it requires no look-ahead beyond the value function V_{k+1} .*

Proof. We verify by backward induction that the rule $\tau_k^* := \inf \{\ell \geq k : g_\ell(x_\ell) \geq \mathbb{E}[V_{\ell+1}(x_{\ell+1}) \mid x_\ell]\}$ attains V_k for every starting stage k . At $k = N$ the rule stops immediately and attains $V_N = g_N$. Assume τ_{k+1}^* attains $V_{k+1}(x_{k+1})$. Consider stage k at state x . If $g_k(x) \geq \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x]$, the Bellman equation gives $V_k(x) = g_k(x)$, and $\tau_k^* = k$ attains it. Otherwise $V_k(x) = \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x]$, the rule continues, and by the induction hypothesis the continuation attains $V_{k+1}(x_{k+1})$ almost surely, so the tower property yields $\mathbb{E}[g_{\tau_k^*}(x_{\tau_k^*}) \mid x_k = x] = V_k(x)$. Applying the claim at $k = 0$ proves the optimality of τ^* . $\square$

Structural Properties of the Value Function

We now establish several fundamental properties of V_k .

Lemma 15.43 (Dominance). *For the optimal stopping problem (15.67), the value function dominates the immediate payoff at every stage and state,*

$$V_k(x) \geq g_k(x) \quad \text{for all } k \text{ and } x. \quad (15.70)$$

Intuitively, the option of continuing can never be worth less than stopping now, because stopping immediately ($\tau = k$) is always one of the admissible choices in the definition of V_k .

Proof. Choosing $\tau = k$ in (15.67) yields $V_k(x) \geq g_k(x)$. $\square$

Lemma 15.44 (Time monotonicity). *For the optimal stopping problem (15.67), the value function dominates its own expected continuation,*

$$V_k(x) \geq \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x] \quad \text{for all } k \text{ and } x, \quad (15.71)$$

so waiting one step never raises the conditional value above the current value. If in addition the dynamics are time homogeneous, $f_k \equiv f$ with $\{w_k\}$ identically distributed, and the payoff is time invariant, $g_k \equiv g$, then the value also decreases as the deadline approaches,

$$V_k(x) \geq V_{k+1}(x) \quad \text{for all } k \text{ and } x : \quad (15.72)$$

fewer remaining opportunities can only lower the option value.

Proof. The displayed inequality is immediate from (15.68), since a maximum dominates each of its arguments.

For (15.72) we argue by backward induction on k , the hypotheses $f_k \equiv f$ and $g_k \equiv g$ being what make the two sides comparable at all. At $k = N - 1$ we have, using $V_N = g$ and the previous inequality applied at stage $N - 1$, $V_{N-1}(x) = \max\{g(x), \mathbb{E}[V_N(x_N) \mid x_{N-1} = x]\} \geq g(x) = V_N(x)$. Assume $V_{k+1} \geq V_{k+2}$ pointwise. Monotonicity of the conditional expectation gives $\mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x] \geq \mathbb{E}[V_{k+2}(x_{k+1}) \mid x_k = x]$, and because the transition law does not depend on the stage, the right-hand side is exactly the continuation value appearing in the Bellman equation at stage $k + 1$. Taking the maximum with the common payoff $g(x)$ on both sides yields $V_k(x) \geq V_{k+1}(x)$, which closes the induction. $\square$

Proposition 15.45 (Monotonicity in the state). *Consider the optimal stopping problem (15.67) and suppose the payoff $g_k(\cdot)$ is nondecreasing for every k and the transition (15.65) preserves stochastic order (a larger current state induces a stochastically larger next state). Then each value function $V_k(\cdot)$ is nondecreasing in the state. Thus a more favourable current state is worth at least as much, a monotonicity that underlies the threshold structure of the optimal policy.*

Proof. We proceed by backward induction. The property holds at $k = N$ since $V_N = g_N$. Assume it holds at $k + 1$. Then for $x \leq x'$,

$$\mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x] \leq \mathbb{E}[V_{k+1}(x'_{k+1}) \mid x_k = x'],$$

by the monotonicity of V_{k+1} and the stochastic order preservation. Taking maxima in (15.68) preserves the inequality. $\square$

Proposition 15.46 (Convexity). *Consider the optimal stopping problem (15.67) and suppose the payoff $g_k(\cdot)$ is convex for every k and the dynamics (15.65) are affine in the state. Then each value function $V_k(\cdot)$ is convex in the state. Convexity is inherited backward through the Bellman recursion: affine dynamics and expectation preserve convexity, and the pointwise maximum of convex functions is again convex.*

Proof. The result follows by backward induction. Convexity holds at $k = N$. If V_{k+1} is convex, then the mapping $x \mapsto \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x]$ is convex due to affine dynamics and linearity of expectation. Since the maximum of convex functions is convex, V_k is convex. $\square$

Stopping and Continuation Regions

Define the stopping region $\mathcal{S}_k := \{x : g_k(x) = V_k(x)\}$ and the continuation region $\mathcal{C}_k := \{x : g_k(x) < V_k(x)\}$.

Proposition 15.47 (Characterization of the optimal policy). *With the stopping region $\mathcal{S}_k = \{x : g_k(x) = V_k(x)\}$ and continuation region $\mathcal{C}_k = \{x : g_k(x) < V_k(x)\}$ defined above, which partition the state space at each stage, the optimal stopping time is the first time the state enters a stopping region,*

$$\tau^* = \inf\{k \geq 0 : x_k \in \mathcal{S}_k\}.$$

The optimal policy is therefore geometric rather than numerical: at each stage one stops exactly on $\mathcal{S}_k$ and continues on $\mathcal{C}_k$, so solving the problem reduces to locating these regions. Under the monotonicity and convexity of Propositions 15.45 and 15.46, the continuation region is an interval and the policy reduces to a time-dependent threshold.

Proof. That $\mathcal{S}_k$ and $\mathcal{C}_k$ partition the state space is immediate from Lemma 15.43, which gives $V_k \geq g_k$ pointwise, so that exactly one of $g_k(x) = V_k(x)$ and $g_k(x) < V_k(x)$ holds at each x . By the Bellman equation (15.68), $x \in \mathcal{S}_k$ precisely when $g_k(x) \geq \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x]$, which is the test defining τ^* in Proposition 15.42; the two descriptions of the optimal rule therefore agree, and optimality is Proposition 15.42.

For the threshold claim, work on $\mathcal{X} \subseteq \mathbb{R}$ and set $D_k(x) := g_k(x) - \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x]$, so that $\mathcal{S}_k = \{x : D_k(x) \geq 0\}$. Under the hypotheses of Proposition 15.46 the continuation value $x \mapsto \mathbb{E}[V_{k+1}(x_{k+1}) \mid x_k = x]$ is convex, and it is nondecreasing by Proposition 15.45. If the payoff g_k is in addition concave and nondecreasing, as it is for the standard call and put payoffs, then D_k is concave, so its superlevel set $\{D_k \geq 0\}$ is an interval and the continuation region $\mathcal{C}_k = \{D_k < 0\}$ is the complement of an interval. When moreover g_k grows faster than the continuation value at $+\infty$, that interval is a half-line $[b_k, \infty)$, and the optimal rule reduces to the time-dependent threshold “stop as soon as $x_k \geq b_k$ ”. The asset-selling example below exhibits exactly this structure, with the thresholds computed numerically. $\square$

Example 15.48 Selling an asset: numerical thresholds.

Offers x_k arrive i.i.d. and uniform on $\{0, 1, \dots, 9\}$; at each of $N = 3$ stages ($k = 0, 1, 2$) the seller may accept the current offer ($g_k(x) = x$) or wait; an offer refused is lost, and $V_3 \equiv$ value of the final forced sale, taken here as the last offer. Backward induction gives the continuation values

$$\begin{aligned}\mathbb{E}[V_3] &= \mathbb{E}[x] = 4.5, & \mathbb{E}[V_2] &= \mathbb{E}[\max(x, 4.5)] = \frac{4.5 \cdot 5 + 5 + 6 + 7 + 8 + 9}{10} = 5.75, \\ \mathbb{E}[V_1] &= \mathbb{E}[\max(x, 5.75)] = \frac{5.75 \cdot 6 + 6 + 7 + 8 + 9}{10} = 6.45.\end{aligned}$$

The optimal rule is therefore a time-dependent threshold policy:

at $k = 2$ accept iff $x \geq 5$, at $k = 1$ accept iff $x \geq 6$, at $k = 0$ accept iff $x \geq 7$,

and the value of the opportunity before any offer is seen is $\mathbb{E}[V_0] = \mathbb{E}[\max(x, 6.45)] = 6.915$ (the number 6.45 is the stage-0 reservation value, i.e. the worth of holding out past the first offer). The thresholds *decrease* as the deadline approaches: with fewer chances remaining, the seller becomes less demanding. This is the discrete, computable prototype of the threshold structure established abstractly in the preceding propositions. $\square$

Figure 15.5 draws this policy. The state space splits into a *stopping region* $\mathcal{S}_k = \{x : g_k(x) = V_k(x)\}$, where accepting is optimal, and a *continuation region* $\mathcal{C}_k$, where waiting is worth more than the offer in hand. For monotone payoffs these regions are separated by a threshold, and the optimal rule is simply “stop the first time the state enters $\mathcal{S}_k$.” The receding thresholds visualize the value of the remaining optionality: early on, the option to wait is valuable and the bar is high; near the deadline it is nearly worthless and the bar drops to meet the current offer.

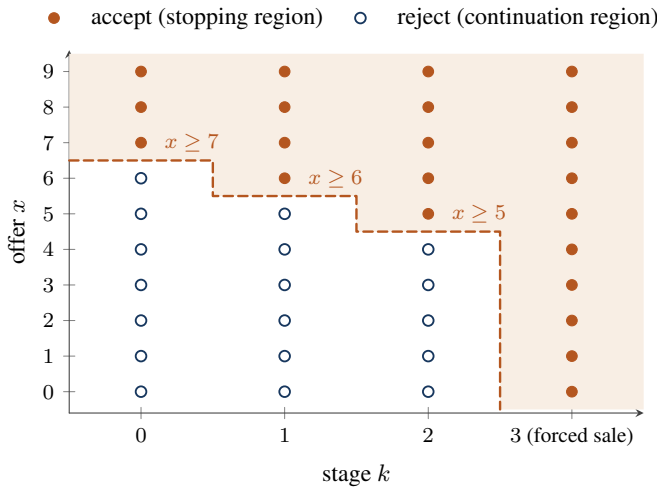


Figure 15.5: Optimal stopping as a threshold policy, for the asset-selling example (offers i.i.d. uniform on $\{0, \dots, 9\}$, $N = 3$ stages). At each stage the seller accepts the current offer if it lies in the shaded *stopping region* (filled dots, on or above the threshold) and otherwise waits (open dots); at $k = 3$ the last offer is taken whatever it is. The optimal thresholds 7, 6, 5 *decrease* as the deadline approaches: with fewer chances left, the seller grows less demanding. The boundary of the stopping region $\mathcal{S}_k = \{x : g_k(x) = V_k(x)\}$ is exactly the threshold computed by backward induction.

Feedback versus Open-Loop Stopping

Proposition 15.49 (Suboptimality of open-loop stopping). *For the optimal stopping problem (15.67), contrast the adaptive optimal rule τ^* with an open-loop stopping time $\tau \equiv t$ that fixes the exercise date t in advance, ignoring the realized trajectory. Then no such deterministic time can exceed the optimal value, $\mathbb{E}[g_t(x_t)] \leq V_0(x_0)$, and the inequality is strict unless the constant time t is itself optimal, i.e. unless stopping at t regardless of the path attains the supremum. Generically the gap is strict at every fixed t : the value of the option arises precisely from the freedom to adapt the stopping decision to the observed state, which an open-loop schedule forgoes.*

Proof. A constant time is an admissible stopping time, so the inequality is immediate from the definition of V_0 as a supremum over $\mathcal{T}_0$. If the inequality holds with equality, then $\tau \equiv t$ attains the supremum and is by definition optimal. Conversely, whenever no constant time is optimal, for instance, when the stopping region $\mathcal{S}_k$ is a proper subset of the state space visited with positive probability at every stage, the inequality is strict, because the adaptive rule τ^* strictly improves on every constant time by stopping early exactly on the event that the reward is already high. $\square$

The optimal stopping problem fundamentally relies on feedback. The decision to exercise depends on the observed state, and the value of the option arises precisely from the flexibility to adapt to the realized evolution of the process. This contrasts sharply with problems in which open-loop policies suffice, and highlights the essential role of information in stochastic control.

15.7 ■ Information States and Dynamic Programming under Partial Information

When the state is not observed directly, the controller must act on a summary of the information actually available. This section makes that idea precise through the notion of an *information state*: a statistic that is (i) computable recursively from the data and (ii) sufficient for both prediction and optimization. For finite-state systems the natural information state is the conditional distribution of the state, the *belief*, and dynamic programming lifts to the space of beliefs. The same principle reappears twice more in this book: for linear-Gaussian systems the belief is Gaussian and collapses to its mean (the Kalman filter of Chapter 14), and for set-membership uncertainty the belief becomes an information *set* (Chapter 12).

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Consider a discrete-time stochastic system with state process $\{x_k\}_{k \geq 0}$ taking values in a finite set $\mathcal{X} := \{1, \dots, I\}$, and control process $\{u_k\}_{k \geq 0}$ taking values in a constraint set U . The controlled state evolution is specified by a stochastic kernel

$$P(u) = (P_{ij}(u)), \quad P_{ij}(u) := \mathbb{P}(x_{k+1} = j \mid x_k = i, u_k = u).$$

The controller observes an observation process $\{z_k\}_{k \geq 0}$ with values in a measurable space $\mathcal{Z}$, such that $\mathbb{P}(z_k \in dy \mid x_k = i) = p(z \mid i) dy$, where $p(\cdot \mid i)$ is the observation density in state i . The data available at time k are collected in

$$\mathcal{D}_k := (z_0, \dots, z_k, u_0, \dots, u_{k-1}), \quad \mathcal{I}_k := \sigma(\mathcal{D}_k),$$

and a control is admissible when u_k is $\mathcal{I}_k$ -measurable for every k .

15.7.1 ■ Belief State as a Sufficient Statistic

Definition 15.50 (Belief state). Let $\Pi := \{\pi \in \mathbb{R}_+^I : \sum_i \pi(i) = 1\}$ denote the probability simplex on $\mathcal{X} = \{1, \dots, I\}$. The belief (posterior distribution) at time k is the random element $\pi_{k|k} \in \Pi$ with components

$$\pi_{k|k}(i) = p_{k|k}(i \mid \mathcal{D}_k) := \mathbb{P}(x_k = i \mid \mathcal{I}_k), \quad i = 1, \dots, I.$$

Thus $\pi_{k|k}$ is the whole conditional law of the hidden state given everything the controller has seen, compressed into a single point of the simplex; the two notations $\pi_{k|k}(i)$ and $p_{k|k}(i \mid \mathcal{D}_k)$ are used interchangeably below.

Theorem 15.51 (Sufficiency of the belief state). For any admissible policy μ , the conditional distribution of the future given $\mathcal{I}_k$ depends on the past only through the belief:

$$\mathbb{P}(x_{k+1}, z_{k+1} \mid \mathcal{I}_k, u_k) = \mathbb{P}(x_{k+1}, z_{k+1} \mid \pi_{k|k}, u_k).$$

Proof. Fix an admissible policy; the control u_k is $\mathcal{I}_k$ -measurable. Let $\Psi(\cdot \mid j)$ denote the observation kernel, i.e. the conditional law of z_{k+1} given $x_{k+1} = j$, which by construction does not depend on the past given x_{k+1} (the observation noise v_{k+1} is independent of the history). For any next state j and any measurable observation set D , condition successively on x_{k+1} and on x_k , using the controlled Markov property of $\{x_k\}$ (Lemma 15.12):

$$\begin{aligned} \mathbb{P}(x_{k+1} = j, z_{k+1} \in D \mid \mathcal{I}_k, u_k) &= \Psi(D \mid j) \mathbb{P}(x_{k+1} = j \mid \mathcal{I}_k, u_k) \\ &= \Psi(D \mid j) \sum_i \mathbb{P}(x_k = i \mid \mathcal{I}_k) P_{ij}(u_k) = \Psi(D \mid j) \sum_i \pi_{k|k}(i) P_{ij}(u_k), \end{aligned}$$

where $\pi_{k|k}(i) = \mathbb{P}(x_k = i \mid \mathcal{I}_k)$. The right-hand side depends on the conditioning information $\mathcal{I}_k$ only through the belief vector $\pi_{k|k}$ and on the applied control u_k . Since j and D were arbitrary, the entire joint conditional law of (x_{k+1}, z_{k+1}) given $(\mathcal{I}_k, u_k)$ is a fixed function of $(\pi_{k|k}, u_k)$, which is the assertion. $\square$

Remark 15.52. This theorem shows that the belief state is a *sufficient statistic* for control. It plays the same role under partial information as the state does under full information.

Proposition 15.53 (Belief update). *Writing $\pi_{k|k} = (p_{k|k}(i))_i$ for the belief vector with components $p_{k|k}(i) := \mathbb{P}(x_k = i \mid \mathcal{I}_k)$, the belief evolves according to*

$$p_{k+1|k}(i) = \sum_j p_{k|k}(j) P_{ji}(u_k), \quad (15.73)$$

$$p_{k+1|k+1}(i) = \frac{p(z_{k+1} \mid i) p_{k+1|k}(i)}{\sum_{i'} p(z_{k+1} \mid i') p_{k+1|k}(i')}. \quad (15.74)$$

Theorem 15.54 (Belief-state Markov property). *The belief process $\{\pi_{k|k}\}$ is a controlled Markov process, i.e.,*

$$\mathbb{P}(\pi_{k+1|k+1} \in B \mid \mathcal{I}_k, u_k) = \mathbb{P}(\pi_{k+1|k+1} \in B \mid \pi_{k|k}, u_k).$$

Proof. The belief-update proposition expresses the updated belief as a fixed measurable map $\pi_{k+1|k+1} = T(\pi_{k|k}, z_{k+1}, u_k)$ of the current belief, the next observation, and the applied control. It therefore suffices to show that the conditional law of z_{k+1} given $(\mathcal{I}_k, u_k)$ depends on the past only through $(\pi_{k|k}, u_k)$. Conditioning first on the hidden state,

$$\mathbb{P}(z_{k+1} \in \cdot \mid \mathcal{I}_k, u_k) = \sum_j \sum_i \pi_{k|k}(j) P_{ji}(u_k) \Psi(\cdot \mid i),$$

where $\Psi(\cdot \mid i)$ is the observation kernel: the inner sums use the controlled Markov property of $\{x_k\}$ and the fact that z_{k+1} depends on the past only through x_{k+1} . The right-hand side is a function of $(\pi_{k|k}, u_k)$ alone. Hence for any measurable B ,

$$\mathbb{P}(\pi_{k+1|k+1} \in B \mid \mathcal{I}_k, u_k) = \int \mathbf{1}_B(T(\pi_{k|k}, z, u_k)) \mathbb{P}(dy \mid \pi_{k|k}, u_k),$$

which depends only on $(\pi_{k|k}, u_k)$, as claimed. $\square$

15.7.2 ■ Dynamic Programming with Partial Information

We consider a finite-horizon stochastic control problem over horizon $N \in \mathbb{N}$. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the underlying probability space, and let $\{\mathcal{I}_k\}_{k \geq 0}$ denote the information filtration generated by $\mathcal{I}_k := \sigma(z_0, \dots, z_k, u_0, \dots, u_{k-1})$.

Admissible policies. A control policy $\mu = \{\mu_k\}_{k=0}^{N-1}$ is said to be *admissible* if, for each k , $u_k = \mu_k(z_0, \dots, z_k, u_0, \dots, u_{k-1})$ is $\mathcal{I}_k$ -measurable. The set of admissible policies is denoted by $\mathcal{M}$.

Performance criterion. Given a policy $\mu \in \mathcal{M}$, define the cost functional

$$J(\mu) = \mathbb{E}^\mu \left[\sum_{k=0}^{N-1} c_k(x_k, u_k) + c_N(x_N) \right]. \quad (15.75)$$

For each k , define the cost-to-go

$$J_k^\mu := \mathbb{E}^\mu \left[\sum_{\ell=k}^{N-1} c_\ell(x_\ell, u_\ell) + c_N(x_N) \mid \mathcal{I}_k \right]. \quad (15.76)$$

Separated policies. Since the state is not directly observed, policies depend on the full information history. However, using the belief state $\pi_{k|k}$, we define:

Definition 15.55 (Separated policy). A policy μ is called *separated* if there exist measurable mappings $\mu_k : \Pi \rightarrow U$ such that $u_k = \mu_k(\pi_{k|k})$.

Remark 15.56. Separated policies restrict attention to controls that depend on the past only through the belief state. The key question is whether this restriction is without loss of optimality.

15.7.3 ■ Reduction to a Fully Observed Markov Decision Process

We now formalize the reduction of the partially observed control problem to a fully observed Markov decision process (MDP) defined on the belief space.

The idea is the partial-information analogue of the controlled Markov property. Under full observation the current state x_k summarizes the past for the purpose of prediction and control; under partial observation no single state does, but the *belief* $\pi_{k|k}$, the posterior distribution of the hidden state given everything seen so far, does. Two facts make this work (Theorems 15.51 and 15.54): the belief is a sufficient statistic, so the optimal action never needs more than $\pi_{k|k}$; and the belief is itself a controlled Markov process, propagated by the Bayesian filter. Together they replace the ever-growing information history by a single, recursively updated state. Figure 15.6 shows the resulting architecture: a Bayesian filter runs alongside the plant, and the controller acts on its output exactly as if the belief were an observed state.

The belief-space value functions to which the reduction leads are defined first, so that the theorem can refer to them.

Dynamic programming on the belief space. Define the value functions $V_k : \Pi \rightarrow \mathbb{R}$ recursively by

$$V_N(\pi) = \tilde{c}_N(\pi) = \sum_{i=1}^I c_N(i) \pi(i), \quad (15.77)$$

$$V_k(\pi) = \inf_{u \in U} \left\{ \tilde{c}_k(\pi, u) + \int V_{k+1}(T(\pi, z, u)) p(z \mid \pi, u) dy \right\}. \quad (15.78)$$

Remark 15.57. The expectation in (15.78) is taken with respect to the predictive distribution

$$p(z \mid \pi, u) = \sum_{i,j} \pi(i) P_{ij}(u) p(z \mid j),$$

which captures uncertainty arising from both the state transition and the observation mechanism.

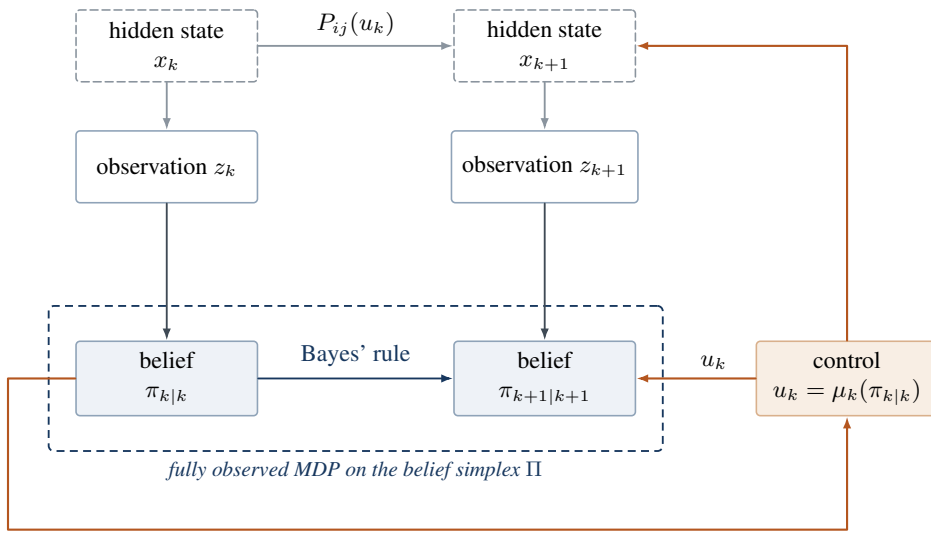


Figure 15.6: Reduction of a partially observed problem to a belief-state MDP. The true state x_k is hidden; only the observations z_k are seen. The controller maintains the *belief* $\pi_{k|k} = \mathbb{P}(x_k \in \cdot \mid \mathcal{I}_k)$, updated recursively by Bayes' rule as new observations arrive. Because the belief is a sufficient statistic, an optimal policy may depend on the past only through $\pi_{k|k}$: the intractable problem over information histories collapses to a *fully observed* Markov decision process whose state is the belief.

Theorem 15.58 (Reduction to a belief-state MDP). Consider the partially observed control problem described above. Then the following statements hold:

- (1) The belief process $\{\pi_{k|k}\}$ is a controlled Markov process with state space Π .
- (2) For any admissible policy μ , the cost-to-go admits the representation

$$J_k^\mu = \mathbb{E}^\mu \left[\sum_{\ell=k}^{N-1} \tilde{c}_\ell(\pi_{\ell|k}, u_\ell) + \tilde{c}_N(\pi_{N|N}) \mid \mathcal{I}_k \right],$$

where the belief-dependent cost functions are defined by

$$\tilde{c}_k(\pi, u) := \sum_{i=1}^I c_k(i, u) \pi(i), \quad \tilde{c}_N(\pi) := \sum_{i=1}^I c_N(i) \pi(i).$$

- (3) Consequently, the partially observed control problem is equivalent to a fully observed MDP with state $\pi_{k|k}$, control u_k , transition kernel induced by the belief update, and stage cost $\tilde{c}_k$.
- (4) There exists an optimal policy that is separated, i.e., depends on the information history only through the belief state.

Proof. (1) The Markov property follows from the belief update equation $\pi_{k+1|k+1} = T(\pi_{k|k}, z_{k+1}, u_k)$, together with the fact that the conditional distribution of z_{k+1} depends only on $(\pi_{k|k}, u_k)$.

(2) By the definition of conditional expectation,

$$\mathbb{E}[c_k(x_k, u_k) \mid \mathcal{I}_k] = \sum_{i=1}^I c_k(i, u_k) \mathbb{P}(x_k = i \mid \mathcal{I}_k) = \tilde{c}_k(\pi_{k|k}, u_k).$$

Applying the tower property recursively yields the stated representation of J_k^μ .

(3) Combining (1) and (2), the system dynamics and cost depend on the history only through the belief state, establishing the equivalence with a fully observed MDP.

(4) By (1) to (3) the problem is a finite-horizon Markov decision process on the belief space Π , with stage costs $\tilde{c}_k$ and the transition kernel induced by the belief update T . Let $\{V_k\}$ be the value functions defined by the belief-space recursion (15.78). Whenever the infimum in (15.78) is attained by a measurable selector $\mu_k : \Pi \rightarrow U$ for each k , which holds, for instance, when U is finite, or, more generally, when U is compact, $\tilde{c}_k(\pi, \cdot)$ is lower semicontinuous, and the belief kernel depends continuously on u , by a measurable selection theorem, the resulting separated policy $\mu = \{\mu_k\}_{k=0}^{N-1}$ satisfies the optimality selector (15.82) and is therefore optimal by Theorem 15.60 below. (That theorem is proved from the comparison principle, Lemma 15.59, and does not use the present theorem, so the argument is not circular.) This exhibits an optimal separated policy. $\square$

15.7.4 ■ Comparison Principle

Lemma 15.59 (Comparison principle). *Let $\{V_k\}_{k=0}^N$ be measurable functions on Π satisfying*

$$V_N(\pi) \leq \tilde{c}_N(\pi), \quad (15.79)$$

$$V_k(\pi) \leq \tilde{c}_k(\pi, u) + \mathbb{E}[V_{k+1}(\pi_{k+1|k+1}) \mid \pi_{k|k} = \pi, u_k = u], \quad \forall u \in U. \quad (15.80)$$

Then for any admissible policy μ , $V_k(\pi_{k|k}) \leq J_k^\mu$ a.s.

Proof. We argue by backward induction on k , proving $V_k(\pi_{k|k}) \leq J_k^\mu$ almost surely. *Terminal step:* at $k = N$, the terminal inequality gives $V_N(\pi_{N|N}) \leq \tilde{c}_N(\pi_{N|N}) = \mathbb{E}[c_N(x_N) \mid \mathcal{I}_N] = J_N^\mu$. *Inductive step:* suppose $V_{k+1}(\pi_{k+1|k+1}) \leq J_{k+1}^\mu$ almost surely. Since $\mathbb{E}[\cdot \mid \mathcal{I}_k]$ is monotone, this yields $\mathbb{E}[V_{k+1}(\pi_{k+1|k+1}) \mid \mathcal{I}_k] \leq \mathbb{E}[J_{k+1}^\mu \mid \mathcal{I}_k]$. Substituting into the stage- k sub-solution inequality and using $\tilde{c}_k(\pi_{k|k}, u_k) = \mathbb{E}[c_k(x_k, u_k) \mid \mathcal{I}_k]$ together with the tower property,

$$V_k(\pi_{k|k}) \leq \tilde{c}_k(\pi_{k|k}, u_k) + \mathbb{E}[V_{k+1}(\pi_{k+1|k+1}) \mid \mathcal{I}_k] \leq \mathbb{E}[c_k(x_k, u_k) \mid \mathcal{I}_k] + \mathbb{E}[J_{k+1}^\mu \mid \mathcal{I}_k] = J_k^\mu,$$

which completes the induction. $\square$

15.7.5 ■ Separation Principle

Theorem 15.60 (Separation principle). *Let $\{V_k\}_{k=0}^N$ be the value functions defined by the dynamic programming recursion on the belief space. Then, for any admissible policy μ and any $k = 0, \dots, N$,*

$$V_k(\pi_{k|k}) \leq \mathbb{E}^\mu \left[\sum_{\ell=k}^{N-1} c_\ell(x_\ell, u_\ell) + c_N(x_N) \mid \mathcal{I}_k \right] \quad a.s. \quad (15.81)$$

Moreover, suppose there exists a separated policy $\mu = \{\mu_k\}_{k=0}^{N-1}$ such that, for every $\pi \in \Pi$ and every k ,

$$\mu_k(\pi) \in \arg \min_{u \in U} \left\{ \tilde{c}_k(\pi, u) + \int V_{k+1}(T(\pi, z, u)) p(z \mid \pi, u) dy \right\}. \quad (15.82)$$

Then μ is optimal and satisfies

$$J(\mu) = \mathbb{E}[V_0(\pi_{0|0})]. \quad (15.83)$$

Proof. The inequality (15.81) follows from the comparison principle applied to the sequence $\{V_k\}$. Assume now that μ is a separated policy satisfying (15.82). By the definition of the value functions, one has almost surely

$$V_k(\pi_{k|k}) = \tilde{c}_k(\pi_{k|k}, u_k) + \mathbb{E}^\mu [V_{k+1}(\pi_{k+1|k+1}) \mid \mathcal{I}_k]. \quad (15.84)$$

We claim that $V_k(\pi_{k|k}) = J_k^\mu$ almost surely for all k . This is shown by backward induction.

At the terminal time,

$$V_N(\pi_{N|N}) = \tilde{c}_N(\pi_{N|N}) = \mathbb{E}[c_N(x_N) \mid \mathcal{I}_N] = J_N^\mu.$$

Assume that $V_{k+1}(\pi_{k+1|k+1}) = J_{k+1}^\mu$ almost surely. Substituting this into (15.84) yields

$$\begin{aligned} V_k(\pi_{k|k}) &= \tilde{c}_k(\pi_{k|k}, u_k) + \mathbb{E}^\mu [J_{k+1}^\mu \mid \mathcal{I}_k] \\ &= \mathbb{E}[c_k(x_k, u_k) \mid \mathcal{I}_k] + \mathbb{E}^\mu [J_{k+1}^\mu \mid \mathcal{I}_k] \\ &= J_k^\mu. \end{aligned}$$

Thus $V_k(\pi_{k|k}) = J_k^\mu$ holds for all k , and in particular $J(\mu) = \mathbb{E}[J_0^\mu] = \mathbb{E}[V_0(\pi_{0|0})]$. Combining this equality with (15.81) shows that μ is optimal. $\square$

Remark 15.61. The separation principle asserts that optimal control under partial observation can be implemented by first computing the belief state $\pi_{k|k}$ through Bayesian filtering, and then applying a control law that depends only on this belief. In this sense, the partially observed control problem is reduced to a fully observed control problem on the belief space. While estimation and control remain coupled through the belief dynamics, the optimal policy depends on the past only through the sufficient statistic $\pi_{k|k}$.

Example 15.62 Belief update for a hidden two-state chain.

Let $x_k \in \{1, 2\}$ evolve as an uncontrolled Markov chain with $\mathbb{P}(x_{k+1} = 2 \mid x_k = 1) = \alpha$ and $\mathbb{P}(x_{k+1} = 1 \mid x_k = 2) = \beta$, and suppose the observation $z_k \in \{1, 2\}$ reports the true state with probability $1 - \epsilon$. Write $\pi_k := \mathbb{P}(x_k = 1 \mid z_1, \dots, z_k)$. The belief updates in two steps. Prediction propagates the chain,

$$\pi_{k+1}^- = (1 - \alpha) \pi_k + \beta (1 - \pi_k),$$

and Bayes' correction reweights by the likelihood of z_{k+1} :

$$\pi_{k+1} = \frac{L \pi_{k+1}^-}{L \pi_{k+1}^- + (1 - L)(1 - \pi_{k+1}^-)}, \quad L := \begin{cases} 1 - \epsilon, & z_{k+1} = 1, \\ \epsilon, & z_{k+1} = 2. \end{cases}$$

The pair (predict, correct) is the finite-state counterpart of the Kalman filter recursion, and π_k is precisely the information state on which the dynamic programming of this section operates. $\square$

15.8 ■ Case Study: Dual Control with Unknown Parameter and Observed State

This case study examines a special class of stochastic control problems in which the system exhibits an *unknown parameter, observed state* structure. This setting lies at the intersection of stochastic control and statistical learning and provides a canonical framework for understanding dual control.

Unlike classical stochastic control problems, where uncertainty enters through exogenous disturbances, here the uncertainty resides in the *model itself*. The controller must therefore simultaneously control the system and learn the underlying dynamics.

15.8.1 ■ Special Structure: Unknown Parameter, Observed State

Consider a discrete-time controlled system defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The state process $\{x_k\}_{k \geq 0}$ takes values in a finite set $\mathcal{X} := \{1, \dots, I\}$, and the control process $\{u_k\}_{k \geq 0}$ takes values in a constraint set U . The system dynamics are governed by a transition kernel that depends on an unknown parameter $\theta \in \Theta = \{\theta^1, \dots, \theta^J\}$:

$$P(u, \theta) = (P_{ij}(u, \theta)), \quad P_{ij}(u, \theta) = \mathbb{P}\{x_{k+1} = j \mid x_k = i, u_k = u, \theta\}.$$

The parameter θ is fixed but unknown. We assume a prior distribution $\pi_0(\alpha) := \mathbb{P}\{\theta = \alpha\}$ on Θ . A defining feature of this model is that the state x_k is observed perfectly at each time. Thus, the information available at time k is

$$\mathcal{D}_k = (x_0, \dots, x_k, u_0, \dots, u_{k-1}), \quad \mathcal{I}_k := \sigma(\mathcal{D}_k),$$

and admissible controls satisfy $u_k = \mu_k(\mathcal{D}_k)$.

The objective is to minimize the expected finite-horizon cost

$$J(\mu) = \mathbb{E}^\mu \left[\sum_{k=0}^{N-1} c_k(x_k, u_k) + c_N(x_N) \right]. \quad (15.85)$$

Although the state is fully observed, the problem is not one of complete information, since the transition law depends on the unknown parameter θ .

Remark 15.63 (Structural distinction). In contrast to a classical POMDP, where the state is hidden, here the state is observed exactly and only the parameter is uncertain. Consequently, the relevant information state is a belief over Θ , rather than over $\mathcal{X}$. This structure is characteristic of Bayesian adaptive control problems.

15.8.2 ■ Augmented State Representation

To embed the problem into a Markovian framework, we introduce the augmented state $(x_k, \theta_k) \in \mathcal{X} \times \Theta$ with $\theta_{k+1} = \theta_k$.

The transition probabilities for the augmented state are

$$\mathbb{P}\{x_{k+1} = j, \theta_{k+1} = \beta \mid x_k = i, \theta_k = \alpha, u_k = u\} = P_{ij}(u, \alpha) \delta(\beta, \alpha), \quad (15.86)$$

where δ is the Kronecker delta. The initial distribution is assumed to factor as $p(x_0, \theta_0) = p(x_0) \pi_0(\theta_0)$.

15.8.3 ■ Information State and Bayesian Update

The information state is defined as the conditional distribution of the augmented state:

$$p_{k|k}(i, \alpha \mid \mathcal{D}_k) = \mathbb{P}\{x_k = i, \theta_k = \alpha \mid \mathcal{D}_k\}. \quad (15.87)$$

Since x_k is observed, this simplifies to

$$p_{k|k}(i, \alpha \mid \mathcal{D}_k) = \delta(i, x_k) \pi_{k|k}(\alpha \mid \mathcal{D}_k), \quad (15.88)$$

where $\pi_{k|k}$ is the posterior distribution of θ .

Using Bayes' rule, the posterior evolves according to

$$\pi_{k+1|k+1}(\theta \mid \mathcal{D}_{k+1}) = \frac{P_{x_k x_{k+1}}(u_k, \theta) \pi_{k|k}(\theta \mid \mathcal{D}_k)}{\sum_{\theta'} P_{x_k x_{k+1}}(u_k, \theta') \pi_{k|k}(\theta' \mid \mathcal{D}_k)}. \quad (15.89)$$

Thus, the information state is completely characterized by $(x_k, \pi_{k|k}) \in \mathcal{X} \times \Pi$, where Π is the probability simplex over Θ .

The dynamic programming of the next subsection rests on the fact that this pair is not merely a convenient summary but a genuine controlled Markov state, which we record next. It is the exact analogue, for the augmented state, of Theorems 15.51 and 15.54.

Lemma 15.64 (Sufficiency and Markov property of the dual state). *Consider the unknown-parameter model of this section, with observed state x_k , unknown parameter $\theta \in \Theta$, prior π_0 , and admissible controls $u_k = \mu_k(\mathcal{D}_k)$. Then for every admissible policy, every $j \in \mathcal{X}$ and every measurable $B \subseteq \Pi$,*

$$\mathbb{P}(x_{k+1} = j, \pi_{k+1|k+1} \in B \mid \mathcal{I}_k, u_k) = \mathbb{P}(x_{k+1} = j, \pi_{k+1|k+1} \in B \mid x_k, \pi_{k|k}, u_k) \quad a.s.$$

That is, $(x_k, \pi_{k|k})$ is a controlled Markov process on $\mathcal{X} \times \Pi$, and the conditional law of the next information state depends on the past only through the current one and the applied control.

Proof. Fix an admissible policy, so that u_k is $\mathcal{I}_k$ -measurable, and note that x_k and $\pi_{k|k}$ are $\mathcal{I}_k$ -measurable by construction. Condition first on the parameter: by (15.86) the transition law of x_{k+1} given the whole past and $\theta = \alpha$ is $P_{x_k j}(u_k, \alpha)$, which depends on the past only through (x_k, u_k) . Averaging over the posterior of θ , which is precisely $\pi_{k|k}$,

$$\mathbb{P}(x_{k+1} = j \mid \mathcal{I}_k, u_k) = \sum_{\alpha \in \Theta} \pi_{k|k}(\alpha) P_{x_k j}(u_k, \alpha) =: q(j \mid x_k, \pi_{k|k}, u_k), \quad (15.90)$$

a fixed function of $(x_k, \pi_{k|k}, u_k)$ alone. Next, the Bayesian update (15.89) exhibits the next posterior as a fixed measurable map $\pi_{k+1|k+1} = S_k(\pi_{k|k}, x_k, x_{k+1}, u_k)$ of the current posterior, the observed transition, and the control: it uses no other feature of the history. Combining the two, for any j and B ,

$$\mathbb{P}(x_{k+1} = j, \pi_{k+1|k+1} \in B \mid \mathcal{I}_k, u_k) = \mathbf{1}_B(S_k(\pi_{k|k}, x_k, j, u_k)) q(j \mid x_k, \pi_{k|k}, u_k),$$

whose right-hand side is a function of $(x_k, \pi_{k|k}, u_k)$ only. Since $\sigma(x_k, \pi_{k|k}) \subseteq \mathcal{I}_k$, conditioning this identity further on $\sigma(x_k, \pi_{k|k}, u_k)$ and using the tower property gives the same expression for the right-hand conditional probability, which proves the claim. $\square$

15.8.4 ■ Dynamic Programming with Dual State

The problem can now be reformulated as a fully observed Markov decision process on $\mathcal{X} \times \Pi$. Define the value functions $V_k(i, \pi)$ for $0 \leq k \leq N$, with terminal condition $V_N(i, \pi) = c_N(i)$. The dynamic programming recursion is

$$V_k(i, \pi) = \inf_{u \in U} \left[c_k(i, u) + \mathbb{E} \left(V_{k+1}(x_{k+1}, S_k(\pi, i, x_{k+1}, u)) \mid x_k = i, \pi_{k|k} = \pi, u_k = u \right) \right], \quad (15.91)$$

where S_k denotes the Bayesian update (15.89).

A fundamental feature of this formulation is that the control input u_k influences the system in two distinct ways. It affects the evolution of the state through the transition probabilities and simultaneously affects the evolution of the belief through the Bayesian update. Since the belief determines future decisions, control actions influence not only the physical trajectory but also the future information available to the controller.

This dual role creates a tradeoff between immediate cost minimization and information acquisition. Actions that are suboptimal in the short term may be optimal in the long term because they improve the estimate of θ .

15.8.5 ■ Optimal Policies and Dual Control

A policy is called *separated* if it depends on the information history only through the information state, i.e., $u_k = \mu_k(x_k, \pi_{k|k})$.

Theorem 15.65 (Optimal dual control policy). *Let $\{V_k\}_{k=0}^N$ be defined by the dynamic programming recursion (15.91). Then, for any admissible policy μ and any $k = 0, \dots, N$,*

$$V_k(x_k, \pi_{k|k}) \leq \mathbb{E}^\mu \left[\sum_{\ell=k}^{N-1} c_\ell(x_\ell, u_\ell) + c_N(x_N) \mid \mathcal{I}_k \right] \quad a.s. \quad (15.92)$$

Moreover, if there exists a separated policy $\mu = \{\mu_k\}$ such that for all (i, π) ,

$$\mu_k(i, \pi) \in \arg \min_{u \in U} \left\{ c_k(i, u) + \mathbb{E} [V_{k+1}(x_{k+1}, S_k(\pi, i, x_{k+1}, u)) \mid x_k = i, \pi_{k|k} = \pi, u_k = u] \right\},$$

then μ is optimal and satisfies

$$J(\mu) = \mathbb{E}[V_0(x_0, \pi_{0|0})]. \quad (15.93)$$

Proof. We first establish the inequality (15.92). Define the cost-to-go under policy μ as

$$J_k^\mu := \mathbb{E}^\mu \left[\sum_{\ell=k}^{N-1} c_\ell(x_\ell, u_\ell) + c_N(x_N) \mid \mathcal{I}_k \right].$$

We show by backward induction that $V_k(x_k, \pi_{k|k}) \leq J_k^\mu$ almost surely. At the terminal time $k = N$, one has $V_N(x_N, \pi_{N|N}) = c_N(x_N) = J_N^\mu$, so the claim holds. Assume now that $V_{k+1}(x_{k+1}, \pi_{k+1|k+1}) \leq J_{k+1}^\mu$ almost surely. By Lemma 15.64 the conditional law of $(x_{k+1}, \pi_{k+1|k+1})$ given $(\mathcal{I}_k, u_k)$ coincides with its law given $(x_k, \pi_{k|k}, u_k)$, so the expectation appearing in (15.91) may equally be conditioned on $\mathcal{I}_k$. Hence, by the definition of V_k , for any admissible control u_k one has

$$V_k(x_k, \pi_{k|k}) \leq c_k(x_k, u_k) + \mathbb{E} [V_{k+1}(x_{k+1}, \pi_{k+1|k+1}) \mid \mathcal{I}_k].$$

Applying the induction hypothesis inside the conditional expectation yields

$$\mathbb{E}[V_{k+1}(x_{k+1}, \pi_{k+1|k+1}) \mid \mathcal{I}_k] \leq \mathbb{E}[J_{k+1}^\mu \mid \mathcal{I}_k].$$

Combining the above inequalities gives

$$V_k(x_k, \pi_{k|k}) \leq c_k(x_k, u_k) + \mathbb{E}[J_{k+1}^\mu \mid \mathcal{I}_k] = J_k^\mu,$$

which completes the induction.

We now prove optimality. Suppose μ is a separated policy attaining the minimum in the dynamic programming recursion. Then, for each k , the Bellman equality holds:

$$V_k(x_k, \pi_{k|k}) = c_k(x_k, u_k) + \mathbb{E}^\mu[V_{k+1}(x_{k+1}, \pi_{k+1|k+1}) \mid \mathcal{I}_k].$$

We again argue by backward induction that $V_k(x_k, \pi_{k|k}) = J_k^\mu$ almost surely.

At $k = N$, equality holds. Assuming equality at time $k + 1$, substitution yields

$$V_k(x_k, \pi_{k|k}) = c_k(x_k, u_k) + \mathbb{E}^\mu[J_{k+1}^\mu \mid \mathcal{I}_k] = J_k^\mu.$$

Thus $V_k(x_k, \pi_{k|k}) = J_k^\mu$ for all k , and in particular $J(\mu) = \mathbb{E}[J_0^\mu] = \mathbb{E}[V_0(x_0, \pi_{0|0})]$. Combining this equality with the inequality (15.92) shows that μ is optimal. $\square$

Remark 15.66. The theorem shows that the control problem admits an optimal policy that depends on the past only through the information state $(x_k, \pi_{k|k})$. However, unlike classical separation results, the control law must explicitly account for how actions influence the evolution of the belief. Thus estimation and control remain intrinsically coupled, which is the hallmark of dual control.

15.9 ■ Case Study: Finite-Horizon Linear-Quadratic Control with Imperfect State Information

We consider a discrete-time stochastic control problem with imperfect state information and quadratic performance criteria, and derive the dynamic programming recursion together with the structure of the optimal policy. This case study assembles the entire toolkit of the last three chapters: the conditional-mean estimates are those of Chapter 13, their recursive computation is the Kalman filter of Chapter 14, and the backward induction is the stochastic dynamic programming developed in this chapter. The punchline, the separation of estimation from control, is among the most consequential structural results of linear systems theory.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space carrying all random variables below. Fix a horizon $N \in \mathbb{N}$ and consider the linear stochastic system

$$x_{k+1} = A_k x_k + B_k u_k + w_k, \quad k = 0, \dots, N-1, \quad (15.94)$$

with state $x_k \in \mathbb{R}^n$, control $u_k \in \mathbb{R}^m$, and known matrices $A_k \in \mathbb{R}^{n \times n}$, $B_k \in \mathbb{R}^{n \times m}$. The controller does not observe the state; it receives the measurements

$$z_k = C_k x_k + v_k, \quad k = 0, \dots, N-1, \quad (15.95)$$

with $z_k \in \mathbb{R}^p$ and known $C_k \in \mathbb{R}^{p \times n}$.

The performance criterion is

$$J(\mu) = \mathbb{E} \left[x_N^\top Q_N x_N + \sum_{k=0}^{N-1} (x_k^\top Q_k x_k + u_k^\top R_k u_k) \right], \quad (15.96)$$

the expectation being taken under the control policy μ specified below.

Assumption 15.67 (Standing assumptions). *Throughout this section we assume the following.*

(i) *The cost weights satisfy $Q_k = Q_k^\top \succeq 0$ for $k = 0, \dots, N$ and $R_k = R_k^\top \succ 0$ for $k = 0, \dots, N-1$.*

(ii) *The initial state and the noises are jointly Gaussian and mutually independent,*

$$x_0 \sim \mathcal{N}(\bar{x}_0, \Sigma_0), \quad w_k \sim \mathcal{N}(0, W_k), \quad v_k \sim \mathcal{N}(0, V_k),$$

with the families $\{w_k\}$ and $\{v_k\}$ independent across time and of x_0 , and with $W_k \succeq 0$, $V_k \succ 0$.

Define the observation filtration

$$\mathcal{I}_k := \sigma(z_0, \dots, z_k, u_0, \dots, u_{k-1}), \quad \mathcal{I}_{-1} := \{\emptyset, \Omega\}, \quad (15.97)$$

which records everything available to the controller at the instant u_k is chosen. An *admissible policy* is a sequence $\mu = (\mu_0, \dots, \mu_{N-1})$ of Borel functions such that the control $u_k = \mu_k(\mathcal{I}_k)$ is $\mathcal{I}_k$ -measurable and $\mathbb{E}\|u_k\|^2 < \infty$ for every k ; the set of admissible policies is denoted $\mathcal{M}$. Under Assumption 15.67 induction on (15.94) shows that every admissible policy yields $\mathbb{E}\|x_k\|^2 < \infty$ for all k , so the criterion (15.96) is finite and every conditional expectation below is well defined.

Throughout we write

$$\hat{x}_k := \mathbb{E}[x_k \mid \mathcal{I}_k], \quad e_k := x_k - \hat{x}_k, \quad \Sigma_k := \mathbb{E}[e_k e_k^\top \mid \mathcal{I}_k]$$

for the conditional mean, the estimation error, and the conditional error covariance. We use two facts, both established constructively by the Kalman filter derived in Section 15.9.1 below. First, under Assumption 15.67 the conditional law of x_k given $\mathcal{I}_k$ is Gaussian, $x_k \mid \mathcal{I}_k \sim \mathcal{N}(\hat{x}_k, \Sigma_k)$, and $\hat{x}_k$ is generated recursively by the Kalman filter (Theorem 15.73). Second, the conditional covariance Σ_k is a *deterministic* matrix: it is a function of the model data $(A_\ell, C_\ell, W_\ell, V_\ell)_{\ell \leq k}$ and of Σ_0 only, and in particular is unaffected by the realized measurements and by the applied controls.

The following elementary identity, a direct consequence of the orthogonality of the estimation error to the observed information, is used repeatedly.

Lemma 15.68 (Conditional orthogonal decomposition). *Let x be a square-integrable random vector, $\mathcal{G} \subseteq \mathcal{F}$ a σ -algebra, $\hat{x} := \mathbb{E}[x \mid \mathcal{G}]$, and $e := x - \hat{x}$. Then for every symmetric matrix M ,*

$$\mathbb{E}[x^\top M x \mid \mathcal{G}] = \hat{x}^\top M \hat{x} + \mathbb{E}[e^\top M e \mid \mathcal{G}].$$

Proof. Write $x = \hat{x} + e$ and expand $x^\top M x = \hat{x}^\top M \hat{x} + 2\hat{x}^\top M e + e^\top M e$. Because $\hat{x}$ is $\mathcal{G}$ -measurable, $\mathbb{E}[\hat{x}^\top M e \mid \mathcal{G}] = \hat{x}^\top M \mathbb{E}[e \mid \mathcal{G}]$, and $\mathbb{E}[e \mid \mathcal{G}] = \mathbb{E}[x \mid \mathcal{G}] - \hat{x} = 0$. Taking $\mathbb{E}[\cdot \mid \mathcal{G}]$ of the expansion yields the identity. $\square$

15.9.1 ■ Recursive State Estimation: The Kalman Filter

The dynamic programming analysis below feeds back on the conditional mean $\hat{x}_k = \mathbb{E}[x_k \mid \mathcal{I}_k]$ and charges the cost with the conditional error covariance $\Sigma_k = \mathbb{E}[e_k e_k^\top \mid \mathcal{I}_k]$. We now derive the

recursion that produces these two quantities directly from the measurements, using only the model (15.94) and (15.95) and Assumption 15.67, in particular, without reference to the cost (15.96). The recursion alternates a *time update*, which propagates the estimate through the dynamics, with a *measurement update*, which corrects it using the new observation.

It is convenient to separate the two half-steps notationally. Write

$$\hat{x}_{k|k} := \hat{x}_k = \mathbb{E}[x_k | \mathcal{I}_k], \quad \Sigma_{k|k} := \Sigma_k = \mathbb{E}[e_k e_k^\top | \mathcal{I}_k]$$

for the *filtered* mean and covariance, and introduce the *one-step predictions*

$$\hat{x}_{k|k-1} := \mathbb{E}[x_k | \mathcal{I}_{k-1}], \quad \Sigma_{k|k-1} := \mathbb{E}[(x_k - \hat{x}_{k|k-1})(x_k - \hat{x}_{k|k-1})^\top | \mathcal{I}_{k-1}].$$

With $\mathcal{I}_{-1} = \{\emptyset, \Omega\}$ the recursion is initialized at the prior of Assumption 15.67: $\hat{x}_{0|-1} = \bar{x}_0$ and $\Sigma_{0|-1} = \Sigma_0$. Throughout we use that, because $u_k = \mu_k(\mathcal{I}_k)$ is $\mathcal{I}_k$ -measurable, adjoining u_k to the data enlarges nothing: $\mathcal{I}_k = \sigma(\mathcal{I}_{k-1}, z_k)$ for every $k \geq 0$.

The whole derivation rests on a single fact about Gaussian vectors, recalled from the estimation theory of Chapter 13: for jointly Gaussian variables, conditioning is an *affine* operation on the observation, and the posterior covariance does not depend on the value observed. We state it in the form used below.

Lemma 15.69 (Gaussian conditioning). *Let $\xi \in \mathbb{R}^n$ and $\eta \in \mathbb{R}^p$ be jointly Gaussian with means $\bar{\xi}, \bar{\eta}$ and covariance blocks*

$$\text{Cov}(\xi) = \Xi_{\xi\xi}, \quad \text{Cov}(\eta) = \Xi_{\eta\eta} \succ 0, \quad \text{Cov}(\xi, \eta) = \Xi_{\xi\eta} = \Xi_{\eta\xi}^\top.$$

Then $\xi | \eta$ is Gaussian, with

$$\mathbb{E}[\xi | \eta] = \bar{\xi} + G(\eta - \bar{\eta}), \quad \text{Cov}(\xi | \eta) = \Xi_{\xi\xi} - \Xi_{\xi\eta} \Xi_{\eta\eta}^{-1} \Xi_{\eta\xi}, \quad G := \Xi_{\xi\eta} \Xi_{\eta\eta}^{-1}.$$

Proof. Seek the affine estimator $\hat{\xi} = \bar{\xi} + G(\eta - \bar{\eta})$ whose error $\xi - \hat{\xi}$ is uncorrelated with the data:

$$\text{Cov}(\xi - \hat{\xi}, \eta) = \text{Cov}(\xi, \eta) - G \text{Cov}(\eta) = \Xi_{\xi\eta} - G \Xi_{\eta\eta} \stackrel{!}{=} 0 \implies G = \Xi_{\xi\eta} \Xi_{\eta\eta}^{-1}.$$

For jointly Gaussian vectors, uncorrelated is equivalent to independent, so $\xi - \hat{\xi}$ is independent of η ; hence $\mathbb{E}[\xi - \hat{\xi} | \eta] = 0$, i.e. $\mathbb{E}[\xi | \eta] = \hat{\xi}$, and $\text{Cov}(\xi | \eta) = \text{Cov}(\xi - \hat{\xi})$. Expanding $\xi - \hat{\xi} = (\xi - \bar{\xi}) - G(\eta - \bar{\eta})$ and using $G \Xi_{\eta\eta} = \Xi_{\xi\eta}$ gives $\text{Cov}(\xi - \hat{\xi}) = \Xi_{\xi\xi} - \Xi_{\xi\eta} \Xi_{\eta\eta}^{-1} \Xi_{\eta\xi}$. $\square$

The gain G is exactly the matrix that makes the estimation error orthogonal to the data, the orthogonality principle of least squares, and the posterior covariance is the corresponding Schur complement, a constant independent of the observed η . The Kalman filter is nothing but this lemma applied afresh at each new measurement. We first record that its hypothesis holds at every step.

Lemma 15.70 (Gaussian conditional law). *Under Assumption 15.67, for every $k \geq 0$ the conditional laws of x_k given $\mathcal{I}_{k-1}$ and given $\mathcal{I}_k$ are Gaussian,*

$$x_k | \mathcal{I}_{k-1} \sim \mathcal{N}(\hat{x}_{k|k-1}, \Sigma_{k|k-1}), \quad x_k | \mathcal{I}_k \sim \mathcal{N}(\hat{x}_{k|k}, \Sigma_{k|k}).$$

Proof. Induction on k . For $k = 0$, $\mathcal{I}_{-1}$ is trivial and $x_0 \sim \mathcal{N}(\bar{x}_0, \Sigma_0)$ by Assumption 15.67, which is the prediction claim; the update claim is the $k = 0$ instance of the argument below.

Assume $x_k \mid \mathcal{I}_k$ is Gaussian. Conditionally on $\mathcal{I}_k$, the term $B_k u_k$ in (15.94) is a known constant, so $x_{k+1} = A_k x_k + B_k u_k + w_k$ is an affine image of the Gaussian $x_k \mid \mathcal{I}_k$ plus the independent Gaussian w_k ; hence $x_{k+1} \mid \mathcal{I}_k$ is Gaussian. Adjoining $z_{k+1} = C_{k+1} x_{k+1} + v_{k+1}$ with v_{k+1} independent Gaussian, the pair (x_{k+1}, z_{k+1}) is jointly Gaussian given $\mathcal{I}_k$. Conditioning a jointly Gaussian pair on one block leaves the other Gaussian; since $\mathcal{I}_{k+1} = \sigma(\mathcal{I}_k, z_{k+1})$, both $x_{k+1} \mid \mathcal{I}_k$ and $x_{k+1} \mid \mathcal{I}_{k+1}$ are Gaussian, closing the induction. $\square$

Proposition 15.71 (Time update). *The time update propagates the filtered estimate and its error covariance through the dynamics (15.94) to form the one-step-ahead prediction: for $k = 0, \dots, N-1$,*

$$\hat{x}_{k+1|k} = A_k \hat{x}_{k|k} + B_k u_k, \quad \Sigma_{k+1|k} = A_k \Sigma_{k|k} A_k^\top + W_k.$$

The predicted mean is the deterministic image of the current estimate under the dynamics (the known input $B_k u_k$ shifts it), while the prediction covariance inflates the propagated estimation error $A_k \Sigma_{k|k} A_k^\top$ by the process-noise covariance W_k .

Proof. The control u_k is $\mathcal{I}_k$ -measurable and w_k is independent of $\mathcal{I}_k$ with $\mathbb{E}[w_k] = 0$, so taking $\mathbb{E}[\cdot \mid \mathcal{I}_k]$ in (15.94) gives $\hat{x}_{k+1|k} = A_k \hat{x}_{k|k} + B_k u_k$. Subtracting this from (15.94), the one-step prediction error is

$$x_{k+1} - \hat{x}_{k+1|k} = A_k (x_k - \hat{x}_{k|k}) + w_k = A_k e_k + w_k.$$

Here $\mathbb{E}[e_k \mid \mathcal{I}_k] = 0$, while w_k is independent of $\sigma(\mathcal{I}_k, x_k) \ni e_k$ and has mean zero; the two terms are therefore conditionally uncorrelated given $\mathcal{I}_k$, and

$$\Sigma_{k+1|k} = A_k \mathbb{E}[e_k e_k^\top \mid \mathcal{I}_k] A_k^\top + \mathbb{E}[w_k w_k^\top] = A_k \Sigma_{k|k} A_k^\top + W_k.$$

$\square$

Proposition 15.72 (Measurement update). *For $k = 0, \dots, N-1$, define the innovation, its covariance, and the Kalman gain*

$$\nu_k := z_k - C_k \hat{x}_{k|k-1}, \quad \Lambda_k := C_k \Sigma_{k|k-1} C_k^\top + V_k \succ 0, \quad G_k := \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1}.$$

Then

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + G_k \nu_k, \quad \Sigma_{k|k} = (I - G_k C_k) \Sigma_{k|k-1} = \Sigma_{k|k-1} - \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1} C_k \Sigma_{k|k-1}.$$

Proof. We apply Lemma 15.69 conditionally on $\mathcal{I}_{k-1}$, with the state $\xi = x_k$ in the role of the unknown and the new measurement $\eta = z_k$ in the role of the data. By Lemma 15.70, $x_k \mid \mathcal{I}_{k-1}$ is Gaussian; since $z_k = C_k x_k + v_k$ with v_k independent Gaussian, the pair (x_k, z_k) is jointly Gaussian given $\mathcal{I}_{k-1}$, so the lemma applies. It remains to identify the three covariance blocks.

Write $e_{k|k-1} := x_k - \hat{x}_{k|k-1}$ for the prediction error, so that $\mathbb{E}[e_{k|k-1} \mid \mathcal{I}_{k-1}] = 0$ and, by definition, $\text{Cov}(x_k \mid \mathcal{I}_{k-1}) = \mathbb{E}[e_{k|k-1} e_{k|k-1}^\top \mid \mathcal{I}_{k-1}] = \Sigma_{k|k-1}$. The measurement has conditional mean $\mathbb{E}[z_k \mid \mathcal{I}_{k-1}] = C_k \hat{x}_{k|k-1}$, so its centered form is exactly the innovation,

$$z_k - \mathbb{E}[z_k \mid \mathcal{I}_{k-1}] = C_k e_{k|k-1} + v_k = \nu_k.$$

Because v_k is independent of $(\mathcal{I}_{k-1}, x_k)$ with mean zero and covariance V_k , all cross terms between $e_{k|k-1}$ and v_k vanish in conditional expectation, giving

$$\text{Cov}(z_k | \mathcal{I}_{k-1}) = \mathbb{E}[(C_k e_{k|k-1} + v_k)(C_k e_{k|k-1} + v_k)^\top | \mathcal{I}_{k-1}] = C_k \Sigma_{k|k-1} C_k^\top + V_k = \Lambda_k,$$

$$\text{Cov}(x_k, z_k | \mathcal{I}_{k-1}) = \mathbb{E}[e_{k|k-1}(C_k e_{k|k-1} + v_k)^\top | \mathcal{I}_{k-1}] = \Sigma_{k|k-1} C_k^\top.$$

As $V_k \succ 0$ and $\Sigma_{k|k-1} \succeq 0$, the innovation covariance $\Lambda_k \succ 0$ is invertible. The gain of Lemma 15.69 is thus $G = \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1} = G_k$, and, since $\mathcal{I}_k = \sigma(\mathcal{I}_{k-1}, z_k)$ so that conditioning on z_k supplies precisely the new information in $\mathcal{I}_k$, the lemma yields

$$\hat{x}_{k|k} = \mathbb{E}[x_k | \mathcal{I}_k] = \hat{x}_{k|k-1} + G_k(z_k - C_k \hat{x}_{k|k-1}) = \hat{x}_{k|k-1} + G_k \nu_k,$$

$$\Sigma_{k|k} = \Sigma_{k|k-1} - \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1} C_k \Sigma_{k|k-1} = \Sigma_{k|k-1} - G_k C_k \Sigma_{k|k-1} = (I - G_k C_k) \Sigma_{k|k-1},$$

where the middle equality substitutes $G_k = \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1}$. $\square$

Combining the two half-steps yields the *Kalman filter*: the explicit, closed-form recursion for the conditional mean and covariance stated next. It is the central result of this subsection.

Theorem 15.73 (Kalman filter equations). *Under Assumption 15.67, the filtered conditional mean $\hat{x}_k = \hat{x}_{k|k}$ and covariance $\Sigma_k = \Sigma_{k|k}$ are computed by the following recursion. Initialize the prediction with the prior, $\hat{x}_{0|-1} = \bar{x}_0$ and $\Sigma_{0|-1} = \Sigma_0$. For $k = 0, 1, 2, \dots$, first perform the measurement update (correction), which folds the new observation z_k into the prediction,*

$$\nu_k = z_k - C_k \hat{x}_{k|k-1}, \quad (\text{innovation}) \quad (15.98a)$$

$$\Lambda_k = C_k \Sigma_{k|k-1} C_k^\top + V_k, \quad (\text{innovation covariance}) \quad (15.98b)$$

$$G_k = \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1}, \quad (\text{Kalman gain}) \quad (15.98c)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + G_k \nu_k, \quad (\text{state update}) \quad (15.98d)$$

$$\Sigma_{k|k} = (I - G_k C_k) \Sigma_{k|k-1}, \quad (\text{covariance update}) \quad (15.98e)$$

and then the time update (prediction), which propagates the estimate through the dynamics,

$$\hat{x}_{k+1|k} = A_k \hat{x}_{k|k} + B_k u_k, \quad (\text{state prediction}) \quad (15.98f)$$

$$\Sigma_{k+1|k} = A_k \Sigma_{k|k} A_k^\top + W_k. \quad (\text{covariance prediction}) \quad (15.98g)$$

Equations (15.98) are the Kalman filter. The covariance relations (15.98b), (15.98c), (15.98e), (15.98g) form a closed recursion that involves neither the measurements z_k nor the controls u_k ; hence $\{\Sigma_{k|k-1}, \Sigma_{k|k}\}$ are deterministic, and eliminating $\Sigma_{k|k}$ collapses them into the single filtering Riccati recursion

$$\Sigma_{k+1|k} = A_k \left[\Sigma_{k|k-1} - \Sigma_{k|k-1} C_k^\top (C_k \Sigma_{k|k-1} C_k^\top + V_k)^{-1} C_k \Sigma_{k|k-1} \right] A_k^\top + W_k. \quad (15.99)$$

Proof. The measurement update (15.98a) and (15.98e) is Proposition 15.72 and the time update (15.98f) and (15.98g) is Proposition 15.71; the initialization is the $k = 0$ prediction, valid because $\mathcal{I}_{-1}$ is trivial and $x_0 \sim \mathcal{N}(\bar{x}_0, \Sigma_0)$. The four covariance relations depend only on the model matrices (A_k, C_k, W_k, V_k) and the previous covariance, not on z_k or u_k ; substituting $\Sigma_{k|k} = (I - G_k C_k) \Sigma_{k|k-1}$ with $G_k = \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1}$ into $\Sigma_{k+1|k} = A_k \Sigma_{k|k} A_k^\top + W_k$ gives (15.99). $\square$

For reference, the operational cycle (initialize once, then alternate correct and predict at every time step) is collected in Algorithm 15.1.

ALGORITHM 15.1. Kalman filter.

Input: model matrices (A_k, B_k, C_k) and noise covariances (W_k, V_k) ; prior mean and covariance $(\bar{x}_0, \Sigma_0)$; online, the applied controls u_k and the measurements z_k .

Initialize the prediction: $\hat{x}_{0|-1} = \bar{x}_0$, $\Sigma_{0|-1} = \Sigma_0$.

For $k = 0, 1, 2, \dots$ **repeat**:

Measurement update (correct the prediction with z_k):

$$\Lambda_k = C_k \Sigma_{k|k-1} C_k^\top + V_k, \quad G_k = \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1},$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + G_k (z_k - C_k \hat{x}_{k|k-1}), \quad \Sigma_{k|k} = (I - G_k C_k) \Sigma_{k|k-1}.$$

Output the current estimate $\hat{x}_k = \hat{x}_{k|k}$ with covariance $\Sigma_k = \Sigma_{k|k}$, and apply the control $u_k = \mu_k(\mathcal{I}_k)$.

Time update (predict the next state through the dynamics):

$$\hat{x}_{k+1|k} = A_k \hat{x}_{k|k} + B_k u_k, \quad \Sigma_{k+1|k} = A_k \Sigma_{k|k} A_k^\top + W_k.$$

Remark 15.74 (The innovation and the meaning of the gain). The correction term is driven by the *innovation* $\nu_k = z_k - C_k \hat{x}_{k|k-1}$, the part of the measurement z_k that could not be anticipated from the past data. It is a zero-mean *white* sequence: from $\mathbb{E}[\nu_k | \mathcal{I}_{k-1}] = 0$ we get $\mathbb{E}[\nu_k] = 0$ and orthogonality to the past, and for $j < k$ the vector ν_j is $\mathcal{I}_{k-1}$ -measurable, whence $\mathbb{E}[\nu_k \nu_j^\top] = \mathbb{E}[\mathbb{E}[\nu_k | \mathcal{I}_{k-1}] \nu_j^\top] = 0$; its covariance is $\mathbb{E}[\nu_k \nu_k^\top] = \Lambda_k$. The update $\hat{x}_{k|k} = \hat{x}_{k|k-1} + G_k \nu_k$ thus revises the prediction in proportion to this surprise, and the gain $G_k = \Sigma_{k|k-1} C_k^\top \Lambda_k^{-1}$ arbitrates between model and sensor: when the sensor is noisy (V_k , hence Λ_k , large) the gain is small and the filter trusts its prediction, whereas when the prediction is uncertain ($\Sigma_{k|k-1}$ large) the gain is large and the filter defers to the data. Every measurement can only sharpen the estimate, $\Sigma_{k|k} = (I - G_k C_k) \Sigma_{k|k-1} \preceq \Sigma_{k|k-1}$, while the intervening time update reinflates the covariance through the dynamics and process noise.

Remark 15.75 (Control-invariance, no dual effect, and the Joseph form). Theorem 15.73 re-derives constructively the two properties that the control analysis relies on: the conditional covariance $\Sigma_k = \Sigma_{k|k}$ is a deterministic matrix (compare Lemma 15.80), and the estimator uses the control only through the known additive term $B_k u_k$ in the time update. The applied control thus shifts the conditional mean but leaves every covariance, and hence the Kalman gain G_k , unchanged: the problem exhibits no *dual effect*, so the controller cannot probe to improve future state information. For computation the covariance update is often written in the numerically robust *Joseph form*

$$\Sigma_{k|k} = (I - G_k C_k) \Sigma_{k|k-1} (I - G_k C_k)^\top + G_k V_k G_k^\top,$$

equivalent to Proposition 15.72 but manifestly symmetric and positive semidefinite.

Example 15.76 Scalar filter.

Take $n = p = 1$ with constant coefficients $A_k \equiv a$, $C_k \equiv c$, $W_k \equiv w > 0$, $V_k \equiv v > 0$, and no control ($u_k \equiv 0$). Writing $\sigma_{k|k-1}$ and $\sigma_{k|k}$ for the scalar

prediction and filtered variances, Theorem 15.73 becomes

$$g_k = \frac{c \sigma_{k|k-1}}{c^2 \sigma_{k|k-1} + v}, \quad \sigma_{k|k} = (1 - g_k c) \sigma_{k|k-1} = \frac{v \sigma_{k|k-1}}{c^2 \sigma_{k|k-1} + v}, \quad \sigma_{k+1|k} = a^2 \sigma_{k|k} + w.$$

The filtered variance is transparent in *precision* (inverse-variance) form,

$$\frac{1}{\sigma_{k|k}} = \frac{1}{\sigma_{k|k-1}} + \frac{c^2}{v},$$

so each measurement adds a fixed amount of precision c^2/v , after which the time update reinflates the variance by a^2 and adds the process noise w . If $|a| < 1$ the prediction variance converges from any nonnegative start to the positive root $\bar{\sigma}$ of $\bar{\sigma} = a^2 v \bar{\sigma} / (c^2 \bar{\sigma} + v) + w$, the steady-state filter analyzed in the next case study (Section 15.10), whereas if $|a| \geq 1$ the observations must be informative enough ($c \neq 0$) for the variance to stay bounded. $\square$

15.9.2 ■ Dynamic Programming at the Terminal Stage

We first solve the last stage in full. The computation displays, in a transparent special case, every mechanism used in the general induction below. Define the terminal cost-to-go

$$J_{N-1}(\mathcal{I}_{N-1}) = \inf_{u_{N-1} \in \mathcal{U}_{N-1}} \mathbb{E}[x_N^\top Q_N x_N + x_{N-1}^\top Q_{N-1} x_{N-1} + u_{N-1}^\top R_{N-1} u_{N-1} \mid \mathcal{I}_{N-1}],$$

where $\mathcal{U}_{N-1}$ is the set of $\mathbb{R}^m$ -valued, $\mathcal{I}_{N-1}$ -measurable, square-integrable controls.

Proposition 15.77 (Quadratic reduction). *Under Assumption 15.67,*

$$J_{N-1}(\mathcal{I}_{N-1}) = \mathbb{E}[x_{N-1}^\top \bar{Q}_{N-1} x_{N-1} \mid \mathcal{I}_{N-1}] + \text{tr}(Q_N W_{N-1}) + \min_{u \in \mathbb{R}^m} \phi(u),$$

where $\bar{Q}_{N-1} := A_{N-1}^\top Q_N A_{N-1} + Q_{N-1}$ and $\phi(u) := u^\top H u + 2b^\top u$ with

$$H := B_{N-1}^\top Q_N B_{N-1} + R_{N-1} \succ 0, \quad b := B_{N-1}^\top Q_N A_{N-1} \hat{x}_{N-1}.$$

Thus the stage reduces to a deterministic, strictly convex quadratic program in u_{N-1} .

Proof. To lighten notation write $A := A_{N-1}$, $B := B_{N-1}$, $Q := Q_N$, $x := x_{N-1}$, $u := u_{N-1}$, $w := w_{N-1}$, and $I := \mathcal{I}_{N-1}$. By the dynamics (15.94), $x_N = Ax + Bu + w$, so

$$x_N^\top Q x_N = (Ax + Bu)^\top Q (Ax + Bu) + 2(Ax + Bu)^\top Q w + w^\top Q w,$$

and, expanding the first summand,

$$(Ax + Bu)^\top Q (Ax + Bu) = x^\top A^\top Q A x + u^\top B^\top Q B u + 2x^\top A^\top Q B u.$$

We now take $\mathbb{E}[\cdot \mid I]$ of each term. The control u is I -measurable and therefore factors out of the conditional expectation. For the noise terms, note that $w = w_{N-1}$ is independent of the σ -algebra $\sigma(I, x) = \sigma(I, x_{N-1})$: indeed I and x_{N-1} are, by (15.94) and (15.95), functions of $(x_0, w_0, \dots, w_{N-2}, v_0, \dots, v_{N-1})$, all independent of w_{N-1} under Assumption 15.67. Consequently

$$\mathbb{E}[(Ax + Bu)^\top Q w \mid I] = \mathbb{E}[(Ax + Bu)^\top Q \mathbb{E}[w \mid I, x] \mid I] = 0,$$

because $\mathbb{E}[w \mid I, x] = \mathbb{E}[w] = 0$, and

$$\mathbb{E}[w^\top Q w \mid I] = \mathbb{E}[w^\top Q w] = \text{tr}(Q W_{N-1}).$$

Adding the two remaining stage terms $\mathbb{E}[x^\top Q_{N-1} x \mid I]$ and $u^\top R_{N-1} u$ and collecting,

$$\mathbb{E}[\cdots \mid I] = \mathbb{E}\left[x^\top \underbrace{(A^\top Q A + Q_{N-1})}_{\bar{Q}_{N-1}} x \mid I\right] + \text{tr}(Q W_{N-1}) + u^\top H u + 2 u^\top B^\top Q A \hat{x}_{N-1},$$

where the cross term used $\mathbb{E}[x \mid I] = \hat{x}_{N-1}$ and we set $H = B^\top Q B + R_{N-1}$. The first two summands do not depend on u ; the last two equal $\phi(u)$. Taking the infimum over u gives the stated identity. Finally $H \succ 0$ because $R_{N-1} \succ 0$ and $B^\top Q B \succeq 0$ (as $Q = Q_N \succeq 0$), so ϕ is strictly convex. $\square$

Proposition 15.78 (Optimal terminal control). *The minimizer of ϕ is unique, and the optimal terminal control is*

$$u_{N-1}^* = -(B_{N-1}^\top Q_N B_{N-1} + R_{N-1})^{-1} B_{N-1}^\top Q_N A_{N-1} \hat{x}_{N-1} =: L_{N-1} \hat{x}_{N-1}.$$

It is admissible.

Proof. By Proposition 15.77, $\phi(u) = u^\top H u + 2b^\top u$ with $H \succ 0$, so ϕ is strictly convex and its unique stationary point is the global minimizer. Solving $\nabla \phi(u) = 2H u + 2b = 0$ gives $u = -H^{-1}b = -H^{-1}B_{N-1}^\top Q_N A_{N-1} \hat{x}_{N-1}$, which is the stated expression with $L_{N-1} := -H^{-1}B_{N-1}^\top Q_N A_{N-1}$. Since $\hat{x}_{N-1} = \mathbb{E}[x_{N-1} \mid \mathcal{I}_{N-1}]$ is $\mathcal{I}_{N-1}$ -measurable and square-integrable, so is $u_{N-1}^* = L_{N-1} \hat{x}_{N-1}$; hence it is admissible. $\square$

Substituting the minimizer places J_{N-1} in the canonical quadratic form and, crucially, exposes the estimation-error penalty explicitly.

Proposition 15.79 (Terminal value function). *Let*

$$K_{N-1} := A_{N-1}^\top \left(Q_N - Q_N B_{N-1} (R_{N-1} + B_{N-1}^\top Q_N B_{N-1})^{-1} B_{N-1}^\top Q_N \right) A_{N-1} + Q_{N-1}$$

and $P_{N-1} := A_{N-1}^\top Q_N B_{N-1} H^{-1} B_{N-1}^\top Q_N A_{N-1} \succeq 0$. Then

$$J_{N-1}(\mathcal{I}_{N-1}) = \mathbb{E}[x_{N-1}^\top K_{N-1} x_{N-1} \mid \mathcal{I}_{N-1}] + \text{tr}(P_{N-1} \Sigma_{N-1}) + \text{tr}(Q_N W_{N-1}).$$

Equivalently, $J_{N-1}(\mathcal{I}_{N-1}) = \mathbb{E}[x_{N-1}^\top K_{N-1} x_{N-1} \mid \mathcal{I}_{N-1}] + c_{N-1}$ with the constant $c_{N-1} := \text{tr}(P_{N-1} \Sigma_{N-1}) + \text{tr}(Q_N W_{N-1})$.

Proof. The minimum value of a strictly convex quadratic $\phi(u) = u^\top H u + 2b^\top u$ is $\phi(-H^{-1}b) = -b^\top H^{-1}b$. With $b = B_{N-1}^\top Q_N A_{N-1} \hat{x}_{N-1}$ this gives

$$\min_u \phi(u) = -\hat{x}_{N-1}^\top A_{N-1}^\top Q_N B_{N-1} H^{-1} B_{N-1}^\top Q_N A_{N-1} \hat{x}_{N-1} = -\hat{x}_{N-1}^\top P_{N-1} \hat{x}_{N-1}.$$

Substituting into Proposition 15.77,

$$J_{N-1}(\mathcal{I}_{N-1}) = \mathbb{E}[x_{N-1}^\top \bar{Q}_{N-1} x_{N-1} \mid \mathcal{I}_{N-1}] - \hat{x}_{N-1}^\top P_{N-1} \hat{x}_{N-1} + \text{tr}(Q_N W_{N-1}).$$

Apply Lemma 15.68 with $M = P_{N-1}$ and $\mathcal{G} = \mathcal{I}_{N-1}$:

$$\hat{x}_{N-1}^\top P_{N-1} \hat{x}_{N-1} = \mathbb{E}[x_{N-1}^\top P_{N-1} x_{N-1} \mid \mathcal{I}_{N-1}] - \mathbb{E}[e_{N-1}^\top P_{N-1} e_{N-1} \mid \mathcal{I}_{N-1}].$$

Since $\bar{Q}_{N-1} - P_{N-1} = K_{N-1}$ by construction, the two conditional-expectation quadratics combine into $\mathbb{E}[x_{N-1}^\top K_{N-1} x_{N-1} \mid \mathcal{I}_{N-1}]$, leaving

$$J_{N-1}(\mathcal{I}_{N-1}) = \mathbb{E}[x_{N-1}^\top K_{N-1} x_{N-1} \mid \mathcal{I}_{N-1}] + \mathbb{E}[e_{N-1}^\top P_{N-1} e_{N-1} \mid \mathcal{I}_{N-1}] + \text{tr}(Q_N W_{N-1}).$$

By the definition of the conditional error covariance, $\mathbb{E}[e_{N-1}^\top P_{N-1} e_{N-1} \mid \mathcal{I}_{N-1}] = \text{tr}(P_{N-1} \Sigma_{N-1})$, and Σ_{N-1} is deterministic; moreover $P_{N-1} \succeq 0$ because $H \succ 0$. This is precisely the asserted identity. $\square$

The decomposition isolates three contributions. The term $\mathbb{E}[x_{N-1}^\top K_{N-1} x_{N-1} \mid \mathcal{I}_{N-1}]$ propagates the optimal cost structure to the previous stage; $\text{tr}(P_{N-1} \Sigma_{N-1})$ is the unavoidable penalty incurred because x_{N-1} must be inferred from noisy data; and $\text{tr}(Q_N W_{N-1})$ is the cost of the terminal process noise. Only the first term depends on the state, and *none* of the three depends on the control: the optimal action feeds back on the estimate $\hat{x}_{N-1}$ alone. The results that follow show that this structure persists at every stage.

Lemma 15.80 (Control-invariance of the estimation error). *Under Assumption 15.67, for every admissible policy $\mu \in \mathcal{M}$ and every $k = 0, \dots, N-1$, the conditional error covariance $\Sigma_k = \mathbb{E}[e_k e_k^\top \mid \mathcal{I}_k]$ is a deterministic matrix that is the same for all policies. Consequently, for any symmetric matrix M ,*

$$\mathbb{E}[e_k^\top M e_k \mid \mathcal{I}_k] = \text{tr}(M \Sigma_k)$$

is a deterministic constant, and its one-step predictor $\mathbb{E}[\text{tr}(M \Sigma_k) \mid \mathcal{I}_{k-1}, u_{k-1}]$ does not depend on the control u_{k-1} .

Proof. Fix an admissible policy μ . Because μ is adapted, each applied control $u_\ell = \mu_\ell(\mathcal{I}_\ell)$ is $\mathcal{I}_\ell$ -measurable. Iterating (15.94),

$$x_k = \Phi_{k,0} x_0 + \sum_{\ell=0}^{k-1} \Phi_{k,\ell+1} (B_\ell u_\ell + w_\ell), \quad \Phi_{k,\ell} := A_{k-1} A_{k-2} \cdots A_\ell, \quad \Phi_{k,k} := I,$$

so that

$$x_k = c_k^\mu + \xi_k, \quad c_k^\mu := \sum_{\ell=0}^{k-1} \Phi_{k,\ell+1} B_\ell u_\ell, \quad \xi_k := \Phi_{k,0} x_0 + \sum_{\ell=0}^{k-1} \Phi_{k,\ell+1} w_\ell,$$

where c_k^μ is $\mathcal{I}_{k-1} \subseteq \mathcal{I}_k$ -measurable (a known function of the past controls) and ξ_k is a function of the primitives $(x_0, w_0, \dots, w_{k-1})$ alone. Likewise, for each $\ell \leq k$,

$$z_\ell = C_\ell x_\ell + v_\ell = C_\ell c_\ell^\mu + \tilde{z}_\ell, \quad \tilde{z}_\ell := C_\ell \xi_\ell + v_\ell,$$

and $\tilde{z}_\ell$ is a function of $(x_0, w_0, \dots, w_{\ell-1}, v_\ell)$ alone. Since the correction $C_\ell c_\ell^\mu$ is $\mathcal{I}_{\ell-1}$ -measurable and hence known, the control-free observations $\tilde{z}_0, \dots, \tilde{z}_k$ together with the controls generate the same information as $\mathcal{I}_k$. Because c_k^μ is $\mathcal{I}_k$ -measurable, it cancels in the estimation error:

$$e_k = x_k - \mathbb{E}[x_k \mid \mathcal{I}_k] = \xi_k - \mathbb{E}[\xi_k \mid \mathcal{I}_k] = \xi_k - \mathbb{E}[\xi_k \mid \tilde{z}_0, \dots, \tilde{z}_k].$$

Now ξ_k and $(\tilde{z}_0, \dots, \tilde{z}_k)$ are fixed linear functions of the primitive Gaussian vector $(x_0, w_0, \dots, w_{k-1}, v_0, \dots, v_k)$, whose joint law does not involve the controls. Conditioning on

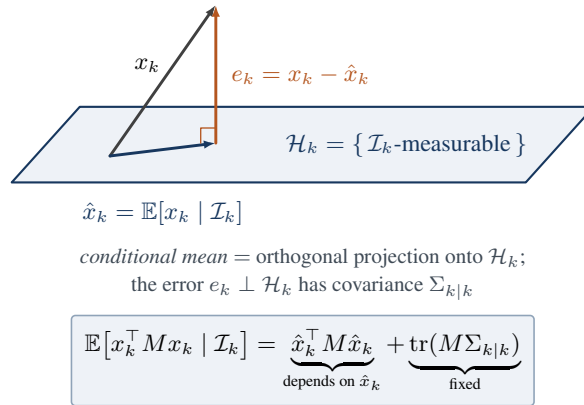


Figure 15.7: The orthogonal decomposition behind certainty equivalence. Conditioning on the information $\mathcal{I}_k$ is an orthogonal projection: the conditional mean $\hat{x}_k = \mathbb{E}[x_k \mid \mathcal{I}_k]$ is the projection of the state x_k onto the subspace $\mathcal{H}_k$ of $\mathcal{I}_k$ -measurable functions, and the estimation error $e_k = x_k - \hat{x}_k$ is orthogonal to it (covariance $\Sigma_{k|k}$). By the Pythagorean identity, every conditional quadratic cost splits into a term carried entirely by the estimate $\hat{x}_k$ and a term $\text{tr}(M \Sigma_{k|k})$ fixed by the error covariance alone. The first term is what the control can shape; the second, by Lemma 15.80, it cannot. This split is what propagates through the dynamic program in Figure 15.8.

jointly Gaussian block on another produces an error whose covariance is the (deterministic) Schur complement of the joint covariance matrix; it depends only on $(A_\ell, C_\ell, W_\ell, V_\ell)_{\ell \leq k}$ and Σ_0 . Hence $\Sigma_k = \text{Cov}(e_k \mid \tilde{z}_0, \dots, \tilde{z}_k)$ is a fixed matrix, identical for every admissible policy, and $\mathbb{E}[e_k^\top M e_k \mid \mathcal{I}_k] = \text{tr}(M \Sigma_k)$ is a deterministic constant. A deterministic quantity is its own conditional expectation, so $\mathbb{E}[\text{tr}(M \Sigma_k) \mid \mathcal{I}_{k-1}, u_{k-1}] = \text{tr}(M \Sigma_k)$ irrespective of u_{k-1} . $\square$

Theorem 15.81 (Reduction to certainty-equivalent dynamic programming). *Under Assumption 15.67, at each stage k the Bellman minimization over u_k depends on the information $\mathcal{I}_k$ only through the conditional mean $\hat{x}_k$, and its minimizer coincides with the solution of the full-information linear-quadratic problem with the state x_k replaced by $\hat{x}_k$. The estimation-error contributions enter the optimal cost only as additive deterministic constants.*

Proof. This is precisely the content of the backward induction carried out in Theorem 15.82 below; we record the two structural reasons here. First, by Lemma 15.68, every conditional quadratic splits as $\mathbb{E}[x_k^\top M x_k \mid \mathcal{I}_k] = \hat{x}_k^\top M \hat{x}_k + \text{tr}(M \Sigma_k)$, so the state dependence is carried entirely by $\hat{x}_k$. Second, by Lemma 15.80, each trace term $\text{tr}(M \Sigma_k)$ is a deterministic constant, unaffected by the control. Hence the trace terms are irrelevant to the argmin, and what remains is a deterministic strictly convex quadratic in u_k whose only data is $\hat{x}_k$, exactly the full-information Bellman step. $\square$

Figure 15.7 captures the mechanism. The orthogonal decomposition $x_k = \hat{x}_k + e_k$ splits every conditional quadratic into a term in the estimate $\hat{x}_k$ and a term fixed by the error covariance $\Sigma_{k|k}$; the latter is control-invariant (Lemma 15.80), so what remains to be optimized is a full-information quadratic in $\hat{x}_k$, and the optimizer feeds back on the estimate alone.

15.9.3 ■ Optimal Policy and Separation Principle

We now derive the structure of the optimal control law via dynamic programming.

Theorem 15.82 (Quadratic value function and optimal policy). *Under Assumption 15.67, define the symmetric matrices $\{K_k\}_{k=0}^N$ by the backward Riccati recursion*

$$K_N = Q_N, \quad K_k = A_k^\top (K_{k+1} - K_{k+1} B_k (R_k + B_k^\top K_{k+1} B_k)^{-1} B_k^\top K_{k+1}) A_k + Q_k. \quad (15.100)$$

Then $K_k \succeq 0$ for every k , the matrices $S_k := R_k + B_k^\top K_{k+1} B_k$ are positive definite, and the optimal cost-to-go

$$J_k^*(\mathcal{I}_k) := \inf_{\mu_k, \dots, \mu_{N-1}} \mathbb{E} \left[x_N^\top Q_N x_N + \sum_{j=k}^{N-1} (x_j^\top Q_j x_j + u_j^\top R_j u_j) \mid \mathcal{I}_k \right]$$

satisfies

$$J_k^*(\mathcal{I}_k) = \mathbb{E}[x_k^\top K_k x_k \mid \mathcal{I}_k] + c_k,$$

with constants defined backward by $c_N = 0$ and

$$c_k = c_{k+1} + \text{tr}(K_{k+1} W_k) + \text{tr}(P_k \Sigma_k), \quad P_k := A_k^\top K_{k+1} B_k S_k^{-1} B_k^\top K_{k+1} A_k.$$

Moreover, the unique optimal control law is

$$u_k^* = L_k \hat{x}_k, \quad L_k = -S_k^{-1} B_k^\top K_{k+1} A_k = -(R_k + B_k^\top K_{k+1} B_k)^{-1} B_k^\top K_{k+1} A_k.$$

Proof. We argue by backward induction on k , establishing simultaneously that $K_k \succeq 0$, that $S_k \succ 0$, and that J_k^* has the stated form.

Base case $k = N$. By definition $J_N^*(\mathcal{I}_N) = \mathbb{E}[x_N^\top Q_N x_N \mid \mathcal{I}_N]$, so the claim holds with $K_N = Q_N \succeq 0$ and $c_N = 0$.

Positivity of S_k and K_k . Assume $K_{k+1} \succeq 0$. Then $S_k = R_k + B_k^\top K_{k+1} B_k \succ 0$, being the sum of $R_k \succ 0$ and the positive semidefinite matrix $B_k^\top K_{k+1} B_k$; in particular S_k is invertible and L_k is well defined. For any $\xi \in \mathbb{R}^n$, completing the square in $v \in \mathbb{R}^m$ gives the identity

$$\min_{v \in \mathbb{R}^m} \left\{ (\xi + B_k v)^\top K_{k+1} (\xi + B_k v) + v^\top R_k v \right\} = \xi^\top \left(K_{k+1} - K_{k+1} B_k S_k^{-1} B_k^\top K_{k+1} \right) \xi,$$

with minimizer $v^* = -S_k^{-1} B_k^\top K_{k+1} \xi$. The left-hand side is nonnegative because $K_{k+1} \succeq 0$ and $R_k \succ 0$; hence the bracketed matrix is positive semidefinite, and $K_k = A_k^\top (\dots) A_k + Q_k \succeq 0$.

Inductive step for the value function. Suppose

$$J_{k+1}^*(\mathcal{I}_{k+1}) = \mathbb{E}[x_{k+1}^\top K_{k+1} x_{k+1} \mid \mathcal{I}_{k+1}] + c_{k+1}.$$

The information state here is the pair $(\hat{x}_k, \Sigma_k)$: by Lemma 15.70 the conditional law of x_k given $\mathcal{I}_k$ is $\mathcal{N}(\hat{x}_k, \Sigma_k)$, by Lemma 15.80 the covariance Σ_k is a deterministic function of k alone, and by Proposition 15.71 and Proposition 15.72 the estimate $\hat{x}_k$ evolves as a controlled Markov process driven by the innovations. The reduction of Theorem 15.58 therefore applies with the belief parameterized by $\hat{x}_k$ —the argument given there for a finite state space used only the sufficiency and Markov property of the information state, both of which hold here—and the resulting Bellman equation for the optimal cost-to-go reads

$$J_k^*(\mathcal{I}_k) = \inf_{u_k \in \mathcal{U}_k} \mathbb{E}[x_k^\top Q_k x_k + u_k^\top R_k u_k + J_{k+1}^*(\mathcal{I}_{k+1}) \mid \mathcal{I}_k].$$

Since $\mathcal{I}_k \subseteq \mathcal{I}_{k+1}$, the tower property yields $\mathbb{E}[\mathbb{E}[x_{k+1}^\top K_{k+1} x_{k+1} \mid \mathcal{I}_{k+1}] \mid \mathcal{I}_k] = \mathbb{E}[x_{k+1}^\top K_{k+1} x_{k+1} \mid \mathcal{I}_k]$, so

$$J_k^*(\mathcal{I}_k) = \inf_{u_k} \mathbb{E}[x_k^\top Q_k x_k + u_k^\top R_k u_k + x_{k+1}^\top K_{k+1} x_{k+1} \mid \mathcal{I}_k] + c_{k+1}.$$

Substituting $x_{k+1} = A_k x_k + B_k u_k + w_k$,

$$x_{k+1}^\top K_{k+1} x_{k+1} = (A_k x_k + B_k u_k)^\top K_{k+1} (A_k x_k + B_k u_k) + 2(A_k x_k + B_k u_k)^\top K_{k+1} w_k + w_k^\top K_{k+1} w_k.$$

Exactly as in the proof of Proposition 15.77, w_k is independent of $\sigma(\mathcal{I}_k, x_k)$ while u_k is $\mathcal{I}_k$ -measurable, so the middle term has zero conditional mean and $\mathbb{E}[w_k^\top K_{k+1} w_k \mid \mathcal{I}_k] = \text{tr}(K_{k+1} W_k)$. Using $\mathbb{E}[x_k \mid \mathcal{I}_k] = \hat{x}_k$ in the resulting cross term and collecting the part that depends on u_k ,

$$\begin{aligned} \mathbb{E}[\cdots \mid \mathcal{I}_k] &= \mathbb{E}[x_k^\top (Q_k + A_k^\top K_{k+1} A_k) x_k \mid \mathcal{I}_k] + \text{tr}(K_{k+1} W_k) \\ &\quad + u_k^\top S_k u_k + 2 u_k^\top B_k^\top K_{k+1} A_k \hat{x}_k. \end{aligned}$$

The first line is independent of u_k . The second line is a strictly convex quadratic in u_k (since $S_k \succ 0$), minimized uniquely at

$$u_k^* = -S_k^{-1} B_k^\top K_{k+1} A_k \hat{x}_k = L_k \hat{x}_k,$$

which is admissible because $\hat{x}_k$ is $\mathcal{I}_k$ -measurable and square-integrable; its minimal value is $-\hat{x}_k^\top P_k \hat{x}_k$ with $P_k = A_k^\top K_{k+1} B_k S_k^{-1} B_k^\top K_{k+1} A_k \succeq 0$. Therefore

$$J_k^*(\mathcal{I}_k) = \mathbb{E}[x_k^\top (Q_k + A_k^\top K_{k+1} A_k) x_k \mid \mathcal{I}_k] - \hat{x}_k^\top P_k \hat{x}_k + \text{tr}(K_{k+1} W_k) + c_{k+1}.$$

Apply Lemma 15.68 with $M = P_k$ to write $\hat{x}_k^\top P_k \hat{x}_k = \mathbb{E}[x_k^\top P_k x_k \mid \mathcal{I}_k] - \mathbb{E}[e_k^\top P_k e_k \mid \mathcal{I}_k]$. Because $Q_k + A_k^\top K_{k+1} A_k - P_k = K_k$ by (15.100), the two conditional-expectation quadratics merge into $\mathbb{E}[x_k^\top K_k x_k \mid \mathcal{I}_k]$, giving

$$J_k^*(\mathcal{I}_k) = \mathbb{E}[x_k^\top K_k x_k \mid \mathcal{I}_k] + \underbrace{\mathbb{E}[e_k^\top P_k e_k \mid \mathcal{I}_k] + \text{tr}(K_{k+1} W_k) + c_{k+1}}_{=: c_k}.$$

By Lemma 15.80, $\mathbb{E}[e_k^\top P_k e_k \mid \mathcal{I}_k] = \text{tr}(P_k \Sigma_k)$ is a deterministic constant, so $c_k = c_{k+1} + \text{tr}(K_{k+1} W_k) + \text{tr}(P_k \Sigma_k)$ is indeed constant and J_k^* has the asserted quadratic form. This closes the induction and, along the way, establishes the optimality and uniqueness of $u_k^* = L_k \hat{x}_k$ at every stage. $\square$

The value function $\hat{x}_k^\top K_k \hat{x}_k$ carried by this recursion is exactly the cost-to-go of the full-information regulator of Chapter 10, now evaluated at the estimate rather than the true state. This exposes the separation principle as a statement about the two directions of dynamic programming, drawn in Figure 15.8. The backward Riccati K_k is the *cost-to-go*: it plans the future and supplies the gain L_k , exactly as in the deterministic problem, and never sees the noise statistics. The estimate $\hat{x}_k$ is produced by the forward filter Riccati Σ_k , the estimation *cost-to-come*, which summarizes the past into a sufficient statistic and never sees the cost weights. Certainty equivalence joins the two at the present step, $u_k^* = L_k \hat{x}_k$, while the leftover $\text{tr}(P_k \Sigma_k)$ terms are the control-free price of estimation.

Theorem 15.83 (Separation theorem). *Under Assumption 15.67, the optimal LQG controller decomposes into an estimator followed by a static feedback gain:*

$$u_k^* = L_k \hat{x}_k, \quad \hat{x}_k = \mathbb{E}[x_k \mid \mathcal{I}_k],$$

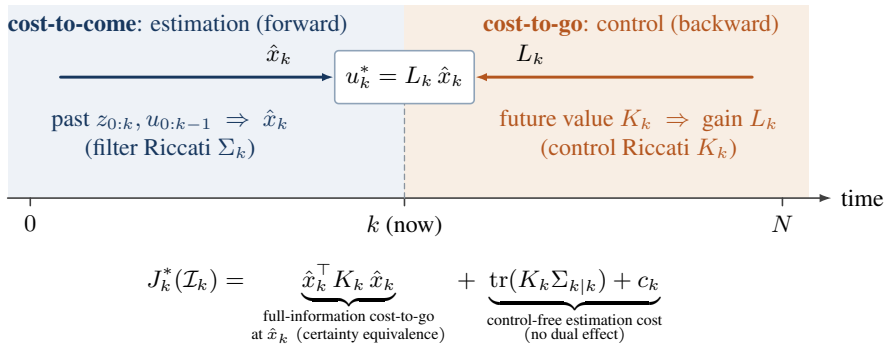


Figure 15.8: Cost-to-go and cost-to-come meet at the present. The optimal cost-to-go $J_k^*(\mathcal{I}_k)$ is built from the *same* backward value function K_k as the full-information linear-quadratic regulator of Chapter 10: the control can shape only the term $\hat{x}_k^\top K_k \hat{x}_k$, the deterministic cost-to-go evaluated at the current estimate, and it minimizes this by the certainty-equivalent law $u_k^* = L_k \hat{x}_k$. The estimate $\hat{x}_k$ is supplied by the dual, forward computation, the estimation *cost-to-come*, which compresses all past data into the sufficient statistic $\hat{x}_k$ through the filter Riccati Σ_k . The two Riccati recursions run in opposite directions in time and share no data: the future is planned by the cost-to-go (control), the past is summarized by the cost-to-come (estimation), and certainty equivalence joins them at the present step. The remaining term $\text{tr}(K_k \Sigma_{k|k}) + c_k$ is set by the estimation accuracy alone and is untouched by the control, the absence of a dual effect.

where $\hat{x}_k$ is produced by the Kalman filter of Section 15.9.1 (Theorem 15.73) and $L_k = -(R_k + B_k^\top K_{k+1} B_k)^{-1} B_k^\top K_{k+1} A_k$ is obtained from the Riccati recursion (15.100). The control gains $\{L_k\}$ depend only on the pair (A_k, B_k) and the cost weights (Q_k, R_k) ; the estimator depends only on (A_k, B_k, C_k) and the statistics (W_k, V_k, Σ_0) . Consequently the two designs may be carried out separately, and the control gain is identical to the one obtained under perfect state information.

Proof. By Theorem 15.82, the optimal control is $u_k^* = L_k \hat{x}_k$, and L_k is a function of K_{k+1} and (A_k, B_k, R_k) only. The recursion (15.100) determines $\{K_k\}$ from (A_k, B_k, Q_k, R_k) and the terminal weight Q_N alone; the observation matrices C_k and the noise covariances W_k, V_k never enter it. Conversely, under Assumption 15.67 the conditional mean $\hat{x}_k$ and its error covariance Σ_k are generated by the Kalman filter of Section 15.9.1, whose gain $G_k = \Sigma_{k|k-1} C_k^\top (C_k \Sigma_{k|k-1} C_k^\top + V_k)^{-1}$ is determined by $(A_k, B_k, C_k, W_k, V_k, \Sigma_0)$ and is independent of (Q_k, R_k) . Thus the regulator design and the estimator design are decoupled. Finally, when the state is directly observed, so that x_k is $\mathcal{I}_k$ -measurable and $\hat{x}_k = x_k$, the same law reads $u_k^{\text{full}} = L_k x_k$, which is the optimal full-information LQR feedback; since the gains $\{L_k\}$ are unchanged, the control gain coincides with and without perfect information. $\square$

Figure 15.9 renders the theorem as a block diagram. The optimal controller is literally two boxes wired in series: a Kalman filter that turns measurements into the estimate $\hat{x}_k$, and the deterministic LQR gain L_k applied to that estimate. The remarkable content of the theorem is not that such a controller works, but that each box can be designed in complete ignorance of the other, the Riccati recursion for L_k never sees the noise covariances, and the filter never sees the cost weights. This is what makes LQG design modular in practice.

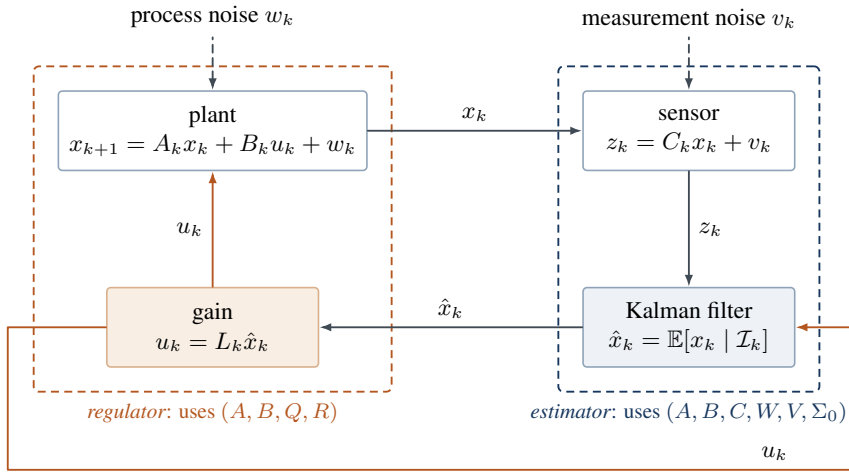


Figure 15.9: The separation principle for LQG control. The optimal controller factorizes into two independently designed blocks: a Kalman *estimator* that computes the conditional mean $\hat{x}_k = \mathbb{E}[x_k | \mathcal{I}_k]$ from the noisy measurements and the applied controls, and a static *regulator* that applies the full-information feedback gain L_k to the estimate, $u_k = L_k \hat{x}_k$. The gain L_k depends only on the dynamics and cost weights (A, B, Q, R) ; the filter depends only on the dynamics, observation model, and noise statistics $(A, B, C, W, V, \Sigma_0)$. Neither design needs the other's data: this is *certainty equivalence* in block-diagram form.

15.9.4 ■ Certainty Equivalence

The optimal policy $u_k^* = L_k \hat{x}_k$ coincides with the control law obtained under perfect state information after replacing the unobserved state x_k by its conditional mean $\hat{x}_k$. We state this precisely.

Proposition 15.84 (Certainty equivalence). *Let $u_k^{\text{full}} = L_k x_k$ be the optimal control under perfect state information, with the same gains L_k as in Theorem 15.83. Then the optimal partially observed control satisfies*

$$u_k^* = L_k \hat{x}_k = \mathbb{E}[u_k^{\text{full}} | \mathcal{I}_k].$$

That is, the optimal controller applies the perfect-information feedback law to the estimate, treating $\hat{x}_k$ as if it were the true state.

Proof. By Theorem 15.82, $u_k^* = L_k \hat{x}_k$. Since L_k is a constant matrix and $\hat{x}_k = \mathbb{E}[x_k | \mathcal{I}_k]$, linearity of conditional expectation gives $L_k \hat{x}_k = \mathbb{E}[L_k x_k | \mathcal{I}_k] = \mathbb{E}[u_k^{\text{full}} | \mathcal{I}_k]$, which is the claimed identity. $\square$

The result rests on exactly two structural features. The linearity of the dynamics (15.94) makes the conditional mean $\hat{x}_k$ propagate linearly and remain the natural state estimate; the quadratic form of the cost (15.96), through Lemma 15.68, splits every conditional expectation into a term in $\hat{x}_k$ and a control-independent estimation-error term. Together they render the conditional mean a *sufficient statistic* for control: no finer feature of the posterior law of x_k influences the optimal action. This is the partially observed counterpart of the deterministic LQR feedback, and it closes the loop with the complete-information case study developed earlier in this chapter.

15.10 ■ Case Study: Infinite-Horizon Steady-State LQG Control and Stability

We consider the infinite-horizon linear-quadratic Gaussian (LQG) control problem with partial state observations. Consider the linear stochastic system

$$x_{k+1} = Ax_k + Bu_k + w_k, \quad (15.101)$$

$$z_k = Cx_k + v_k, \quad (15.102)$$

where $x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}^m$, and $z_k \in \mathbb{R}^p$. The objective is to minimize

$$J = \lim_{N \rightarrow \infty} \frac{1}{N} \mathbb{E} \left[\sum_{k=0}^{N-1} (x_k^\top Q x_k + u_k^\top R u_k) \right], \quad (15.103)$$

the *average cost per stage*, where $Q \succeq 0$ and $R \succ 0$; with persistent process noise the total cost grows linearly in the horizon, so the undiscounted infinite sum diverges and the time average is the appropriate criterion.

Assumption 15.85 (Gaussian uncertainties). *The random variables x_0 , $\{w_k\}$, and $\{v_k\}$ are mutually independent and independent across time, Gaussian with zero means and covariances $x_0 \sim \mathcal{N}(0, \Sigma_0)$, $w_k \sim \mathcal{N}(0, W)$, and $v_k \sim \mathcal{N}(0, V)$, with $W \succeq 0$ and $V \succ 0$. (These are the time-invariant counterparts of the finite-horizon data W_k, V_k of Section 15.9.)*

Lemma 15.86 (Sufficient statistic). *Consider the infinite-horizon LQG system (15.101)–(15.102) under the Gaussian assumptions of this section, and let $\mathcal{I}_k := \sigma(z_0, \dots, z_k, u_0, \dots, u_{k-1})$ be the information available at time k . Then the conditional law of the state given the data is Gaussian,*

$$x_k \mid \mathcal{I}_k \sim \mathcal{N}(\hat{x}_k, \Sigma_k), \quad \hat{x}_k := \mathbb{E}[x_k \mid \mathcal{I}_k],$$

with an error covariance Σ_k that evolves deterministically, independently of the observations and of the applied controls. Consequently the conditional mean $\hat{x}_k$ is a sufficient statistic for control: the whole posterior is pinned down by $\hat{x}_k$ alone, so an optimal policy may depend on $\mathcal{I}_k$ only through $\hat{x}_k$.

Proof. Fix an admissible policy and condition on $\mathcal{I}_k$. The past controls $u_0, \dots, u_{k-1}$ are $\mathcal{I}_k$ -measurable, so given $\mathcal{I}_k$ they act as known constants. Iterating the dynamics, x_k equals a known (control-dependent) vector plus a fixed linear combination of $(x_0, w_0, \dots, w_{k-1})$, and each observation z_ℓ equals a known vector plus a fixed linear combination of $(x_0, w_0, \dots, w_{\ell-1}, v_\ell)$. The underlying primitive variables are jointly Gaussian, and conditioning a Gaussian vector on a component-wise affine function of the same primitives yields a Gaussian conditional law whose mean is affine in the data and whose covariance Σ_k is a deterministic matrix, independent of both the data and the applied controls. Hence $x_k \mid \mathcal{I}_k \sim \mathcal{N}(\hat{x}_k, \Sigma_k)$.

Sufficiency now follows because the conditional distribution, the natural information state, is parameterized by $\hat{x}_k$ alone: the covariance Σ_k evolves deterministically and carries no decision-relevant information. For the quadratic stage costs at hand this is explicit: $\mathbb{E}[x_k^\top Q x_k \mid \mathcal{I}_k] = \hat{x}_k^\top Q \hat{x}_k + \text{tr}(Q \Sigma_k)$, where the trace term is a constant unaffected by the control. $\square$

15.10.1 ■ Kalman Filter

The estimator is the time-invariant specialization of the finite-horizon Kalman filter derived in Section 15.9.1. With constant data (A, C, W, V) the predict/update recursion of Theorem 15.73

reads, writing $\Sigma_{k|k-1}$ and $\Sigma_{k|k}$ for the prediction and filtered error covariances and $\nu_k = z_k - C\hat{x}_{k|k-1}$ for the innovation,

$$\begin{aligned} G_k &= \Sigma_{k|k-1} C^\top (C \Sigma_{k|k-1} C^\top + V)^{-1}, & \hat{x}_{k+1|k} &= A \hat{x}_{k|k} + B u_k, \\ \hat{x}_{k|k} &= \hat{x}_{k|k-1} + G_k \nu_k, & \Sigma_{k+1|k} &= A \Sigma_{k|k} A^\top + W. \\ \underbrace{\Sigma_{k|k} &= (I - G_k C) \Sigma_{k|k-1}}_{\text{measurement update}}, & \underbrace{\Sigma_{k+1|k} &= A \Sigma_{k|k} A^\top + W}_{\text{time update}}. \end{aligned}$$

The next theorem shows that, unlike the finite-horizon case, these recursions settle to a *stationary* filter with a constant gain.

Theorem 15.87 (Steady-state Kalman filter). *Suppose (A, C) is detectable and $(A, W^{1/2})$ is stabilizable. Then, from any initial $\Sigma_{0|-1} \succeq 0$, the prediction covariance converges, $\Sigma_{k|k-1} \rightarrow \Sigma$, to the unique positive-semidefinite solution of the filtering algebraic Riccati equation*

$$\Sigma = A \left[\Sigma - \Sigma C^\top (C \Sigma C^\top + V)^{-1} C \Sigma \right] A^\top + W,$$

and the gain converges to $G = \Sigma C^\top (C \Sigma C^\top + V)^{-1}$. The stationary filter is

$$\nu_k = z_k - C \hat{x}_{k|k-1}, \quad \hat{x}_{k|k} = \hat{x}_{k|k-1} + G \nu_k, \quad \hat{x}_{k+1|k} = A \hat{x}_{k|k} + B u_k,$$

and its error $e_k := x_k - \hat{x}_{k|k}$ obeys

$$e_{k+1} = (I - GC) A e_k + (I - GC) w_k - G v_{k+1},$$

whose transition matrix $(I - GC)A$ is stable.

Proof. The transient recursion is Theorem 15.73 with time-invariant data; the control terms Bu_k are $\mathcal{I}_k$ -measurable and enter only the prediction of the mean. Eliminating $\Sigma_{k|k} = (I - G_k C) \Sigma_{k|k-1}$ leaves the prediction-covariance recursion

$$\Sigma_{k+1|k} = A \left[\Sigma_{k|k-1} - \Sigma_{k|k-1} C^\top (C \Sigma_{k|k-1} C^\top + V)^{-1} C \Sigma_{k|k-1} \right] A^\top + W,$$

which is exactly the control Riccati recursion (15.100) under the duality $(A, B, Q, R) \leftrightarrow (A^\top, C^\top, W, V)$. Convergence to the unique stabilizing solution Σ therefore follows from the control-ARE theorem below applied to the dual data, using detectability of (A, C) and stabilizability of $(A, W^{1/2})$; the gain converges to $G = \Sigma C^\top (C \Sigma C^\top + V)^{-1}$. For the error dynamics, the prediction error is $e_{k|k-1} := x_k - \hat{x}_{k|k-1} = A e_{k-1} + w_{k-1}$, and subtracting the update $\hat{x}_{k|k} = \hat{x}_{k|k-1} + G(C e_{k|k-1} + v_k)$ from x_k gives $e_k = (I - GC) e_{k|k-1} - G v_k$; substituting $e_{k|k-1} = A e_{k-1} + w_{k-1}$ and shifting the index yields the stated recursion. Its transition matrix $(I - GC)A$ has the same characteristic polynomial as $A(I - GC) = A - (AG)C$ —for square matrices $\det(\lambda I - XY) = \det(\lambda I - YX)$ —and the latter is stable by the stabilizing property of the filter ARE. $\square$

Theorem 15.88 (Control algebraic Riccati equation). *Consider the infinite-horizon average-cost regulator for the deterministic linear system $x_{k+1} = A x_k + B u_k$ with stage cost $x_k^\top Q x_k + u_k^\top R u_k$, where $Q \succeq 0$ and $R \succ 0$. Assume the pair (A, B) is stabilizable and the pair $(A, Q^{1/2})$ is detectable. Then the control algebraic Riccati equation*

$$K = A^\top [K - KB(R + B^\top KB)^{-1} B^\top K] A + Q$$

admits a unique positive-semidefinite solution K that is stabilizing, meaning that the induced feedback gain $L = -(R + B^\top KB)^{-1}B^\top KA$ renders the closed-loop matrix $A + BL$ Schur stable (all eigenvalues strictly inside the unit disc). This K is the limit of the finite-horizon Riccati recursion (15.100) started from any $K^{(0)} \succeq 0$, and the stationary feedback $u_k = Lx_k$ is optimal for the average-cost regulator.

Proof. Consider the finite-horizon Riccati recursion $K^{(j+1)} = A^\top [K^{(j)} - K^{(j)}B(R + B^\top K^{(j)}B)^{-1}B^\top K^{(j)}]A + Q$ initialized at $K^{(0)} = 0$.

Monotonicity. The Riccati map is monotone with respect to the positive-semidefinite order (it is the value-iteration operator of the LQR problem, and larger terminal cost yields larger value), and $K^{(1)} = Q \succeq 0 = K^{(0)}$; by induction the sequence $\{K^{(j)}\}$ is nondecreasing.

Boundedness. Since (A, B) is stabilizable, there exists a gain F with $A + BF$ stable, and the stationary policy $u_k = Fx_k$ incurs a finite quadratic cost $x_0^\top \bar{K}x_0$ with $\bar{K}$ the solution of a Lyapunov equation. Since $K^{(j)}$ is the optimal j -stage cost matrix, $K^{(j)} \preceq \bar{K}$ for all j .

A nondecreasing, bounded sequence of symmetric matrices converges; the limit K satisfies the algebraic Riccati equation by continuity of the Riccati map.

Stabilizing property and uniqueness. Let $L = -(R + B^\top KB)^{-1}B^\top KA$ be the induced gain. Rearranging the ARE gives the Lyapunov identity $K = (A + BL)^\top K(A + BL) + Q + L^\top RL$. If $(A + BL)\xi = \lambda\xi$ with $|\lambda| \geq 1$, then evaluating the identity at ξ forces $Q^{1/2}\xi = 0$ and $L\xi = 0$, hence $A\xi = \lambda\xi$, contradicting detectability of $(A, Q^{1/2})$. Thus $A + BL$ is stable. Uniqueness among stabilizing solutions follows by the standard comparison argument: if K_1, K_2 are two stabilizing solutions, subtracting their AREs yields a Stein equation for $K_1 - K_2$ with stable coefficient matrices on both sides, whose only solution is 0. $\square$

Theorem 15.89 (Separation principle for steady-state LQG). Consider the infinite-horizon LQG problem (15.101)–(15.103) under the Gaussian assumptions of this section, and suppose (A, B) is stabilizable, $(A, Q^{1/2})$ is detectable, (A, C) is detectable, and $(A, W^{1/2})$ is stabilizable. Let $K \succeq 0$ be the stabilizing solution of the control algebraic Riccati equation (Theorem 15.88), with induced gain $L = -(R + B^\top KB)^{-1}B^\top KA$, and let $\hat{x}_k = \mathbb{E}[x_k | \mathcal{I}_k]$ be the state estimate produced by the steady-state Kalman filter (Theorem 15.87). Then the admissible policy that minimizes the average cost (15.103) is $u_k^* = L\hat{x}_k$, the deterministic optimal regulator gain applied to the filtered estimate. Consequently the optimal controller separates into two pieces that may be designed independently: a Kalman filter that computes $\hat{x}_k$ from the measurements, whose gain G depends only on the dynamics and noise data (A, C, W, V) , and a state-feedback regulator whose gain L depends only on the dynamics and cost data (A, B, Q, R) . Neither design requires knowledge of the other's data.

Proof. By the sufficient statistic lemma, the conditional law of x_k given $\mathcal{I}_k$ is $\mathcal{N}(\hat{x}_k, \Sigma_k)$ with Σ_k deterministic, so the problem is equivalent to a fully observed problem with state $\hat{x}_k$. The filter recursion gives the dynamics of this new state,

$$\hat{x}_{k+1} = A\hat{x}_k + Bu_k + G\nu_{k+1},$$

in which the innovation sequence $\{\nu_k\}$ is zero-mean, white, and independent of the applied controls (Remark following Theorem 15.73); thus $\hat{x}_k$ follows a linear system driven by white noise. Expanding the stage cost with $x_k = \hat{x}_k + e_k$ and using $\mathbb{E}[e_k | \mathcal{I}_k] = 0$,

$$\mathbb{E}[x_k^\top Qx_k + u_k^\top Ru_k] = \mathbb{E}[\hat{x}_k^\top Q\hat{x}_k + u_k^\top Ru_k] + \mathbb{E}[e_k^\top Qe_k],$$

where the estimation-error term is a policy-independent constant determined by the stationary error covariance. The remaining problem is an average-cost LQR in $\hat{x}_k$, driven by the additive white

innovation $G\nu_{k+1}$. That noise is independent of the control and enters neither the Riccati recursion (15.100) nor its minimizer—it contributes only the additive trace term $\text{tr}(KG\Lambda G^\top)$ to the average cost, exactly as the process noise did in the fully observed problem—so Theorem 15.88, although stated for the deterministic plant, applies verbatim and its optimal policy is the stationary linear feedback $u_k^* = L\hat{x}_k$ with the stated gain. The gain L depends only on (A, B, Q, R) through the control ARE, while G depends only on (A, C, W, V) through the filter ARE, which is the asserted decoupling. $\square$

Remark 15.90. The controller gain L is independent of the noise statistics, while the estimator gain G is independent of (Q, R) .

15.10.2 ■ Closed-Loop Stability

Let $e_k := x_k - \hat{x}_k$ be the estimation error. Applying $u_k = L\hat{x}_k = L(x_k - e_k)$ to the dynamics and using the stationary filter error recursion,

$$x_{k+1} = (A + BL)x_k - BL e_k + w_k, \quad e_{k+1} = (I - GC)A e_k + (I - GC)w_k - G v_{k+1}.$$

Theorem 15.91 (Mean-square stability of the closed loop). *Apply the optimal separated policy $u_k = L\hat{x}_k$ of Theorem 15.89 to the system (15.101)–(15.102), and let $e_k := x_k - \hat{x}_k$ denote the estimation error. Under the stabilizability and detectability assumptions of that theorem, the augmented closed-loop state (x_k, e_k) evolves as a stable linear system driven by white noise: the regulator matrix $A + BL$ and the filter matrix $(I - GC)A$ are both Schur stable, and the system matrix of (x_k, e_k) is block-triangular with these two blocks on its diagonal. In particular the closed loop is mean-square stable: the second moment $\mathbb{E}[\|(x_k, e_k)\|^2]$ remains bounded for all k and, from any initial covariance, converges to the unique positive-semidefinite solution of the associated discrete Lyapunov equation, which is the invariant covariance of the closed loop.*

Proof. Stack the closed-loop state as (x_k, e_k) . The displayed equations give the block-triangular linear system

$$\begin{pmatrix} x_{k+1} \\ e_{k+1} \end{pmatrix} = \begin{pmatrix} A + BL & -BL \\ 0 & (I - GC)A \end{pmatrix} \begin{pmatrix} x_k \\ e_k \end{pmatrix} + \begin{pmatrix} w_k \\ (I - GC)w_k - G v_{k+1} \end{pmatrix}.$$

The spectrum of a block-triangular matrix is the union of the spectra of its diagonal blocks: $A + BL$ is stable by the ARE theorem (stabilizing solution), and $(I - GC)A$ is stable by the filter ARE under detectability of (A, C) . Hence the composite matrix is stable. A stable linear system driven by white noise with bounded covariance is mean-square stable, and its state covariance converges to the unique positive-semidefinite solution of the associated discrete Lyapunov equation, which is the claimed invariant covariance. $\square$

Corollary 15.92 (Certainty equivalence). *In the setting of Theorem 15.89, let $u_k^{\text{full}} = Lx_k$ be the optimal feedback for the corresponding full-information regulator, using the same gain L . Then the optimal partially observed control applies this law to the estimate:*

$$u_k^* = L\hat{x}_k = \mathbb{E}[Lx_k \mid \mathcal{I}_k] = \mathbb{E}[u_k^{\text{full}} \mid \mathcal{I}_k].$$

That is, the controller acts as if the conditional mean $\hat{x}_k$ were the true state; the unobserved state in the perfect-information law is simply replaced by its best estimate, with no correction for the estimation uncertainty Σ_k .

Proof. Linearity of expectation yields $\mathbb{E}[Lx_k | \mathcal{I}_k] = L\mathbb{E}[x_k | \mathcal{I}_k]$. $\square$

Example 15.93 Steady-state LQG with golden-ratio gains.

Let all quantities be scalar: $A = B = C = 1$, cost weights $Q = R = 1$, and noise covariances $W = V = 1$. The control ARE $K = K - K^2/(1 + K) + 1$ simplifies to $K^2 - K - 1 = 0$, whose positive root is the golden ratio

$$K = \frac{1 + \sqrt{5}}{2} \approx 1.618, \quad L = -\frac{K}{1 + K} \approx -0.618.$$

By the duality of the two Riccati equations, the filter ARE with $W = V = 1$ has the same solution $\Sigma = K \approx 1.618$ and gain $G = \Sigma/(1 + \Sigma) \approx 0.618$. The optimal controller is therefore

$$\hat{x}_{k+1} = \hat{x}_k + u_k + 0.618(z_{k+1} - \hat{x}_k - u_k), \quad u_k = -0.618 \hat{x}_k,$$

and both closed-loop matrices equal $1 - 0.618 = 0.382 = 1/K^2$, comfortably inside the unit circle. The average cost decomposes as the full-information LQR cost plus the price of estimation, each computable from the two AREs; the numerical coincidence of controller and filter quantities is the scalar manifestation of LQR-filter duality. $\square$

Proposition 15.94 (Optimality among linear controllers without Gaussianity). *Suppose the initial state x_0 and the process and measurement noises $\{w_k\}$, $\{v_k\}$ are mutually uncorrelated with the same zero means and covariances (Σ_0, W, V) as before, but are not assumed Gaussian. Restrict attention to controllers that are linear functions of the measurement history. Then the average cost (15.103) is minimized within this class by the same law $u_k = L \hat{x}_k^{\text{LLS}}$, where L is the regulator gain of Theorem 15.89 and $\hat{x}_k^{\text{LLS}}$ is the linear least-squares estimate generated by the Kalman recursions. Gaussianity is thus needed only to conclude optimality among all controllers; among linear ones the same controller is optimal for every noise distribution with these second moments.*

Proof. Restrict attention to controllers that are linear functions of the observation history. Under any such controller the closed-loop system is linear, so all signals have well-defined first and second moments determined by (A, B, C, W, V) and the controller coefficients, and the quadratic cost depends on the controller only through these moments. Since second moments do not distinguish the actual noise distribution from a Gaussian one with the same covariances, the optimal linear controller coincides with the optimal linear controller of the Gaussian problem, namely $u_k = L \hat{x}_k^{\text{LLS}}$, where $\hat{x}_k^{\text{LLS}}$ is the linear least-squares estimate produced by the Kalman recursions (which are moment computations, valid without Gaussianity, cf. Chapter 14). $\square$

Proposition 15.95 (Sufficiency of the conditional mean under a general cost). *Consider the linear-Gaussian system (15.101)–(15.102) with a stage cost that is a general (possibly non-quadratic) function of state and control. Because the conditional law of x_k given $\mathcal{I}_k$ is Gaussian with a deterministic covariance Σ_k that evolves independently of the observations and controls, the belief is parameterized by its mean alone. Hence there is an optimal control that depends on*

the information $\mathcal{I}_k$ only through the conditional mean $\hat{x}_k = \mathbb{E}[x_k | \mathcal{I}_k]$: the conditional mean is a sufficient statistic even when the cost is not quadratic. Unlike the quadratic case, however, the optimal dependence of u_k on $\hat{x}_k$ is in general nonlinear.

Proof. The natural information state for a partially observed problem is the conditional distribution of the state given the data. In the linear-Gaussian case this distribution is $\mathcal{N}(\hat{x}_k, \Sigma_k)$, and the covariance Σ_k evolves by the deterministic Riccati recursion, independently of the observations and controls. The belief is therefore parameterized by the single data-dependent quantity $\hat{x}_k$, and the dynamic program on beliefs collapses to a dynamic program on $\hat{x}_k$. Consequently the optimal control (for *any* stage cost for which the problem is well posed, not only quadratic) is a function of $\hat{x}_k$ (and the deterministic quantities k, Σ_k). What fails for general costs is only certainty equivalence: the optimal map from $\hat{x}_k$ to the control need not coincide with the full-information law evaluated at $\hat{x}_k$. $\square$

15.11 ■ Further Reading

The materials in this chapter are primarily based on classical references in stochastic control and dynamic programming. In particular, we draw on the foundational treatments in [74] and [16], which provide a comprehensive introduction to stochastic systems, estimation, and optimal control.

We further encourage the reader to consult [19] for a rigorous measure-theoretic treatment of discrete-time stochastic optimal control, including dynamic programming, Markov decision processes, and optimal stopping theory. Additional perspectives from economic theory and optimization can be found in [58], which connects optimization methods with economic decision-making.

15.12 ■ Exercises

Exercise 15.1 (Steady-State Kalman Filtering). Consider the stochastic system

$$x_{k+1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} x_k + w_k, \quad (15.104)$$

$$z_k = \begin{pmatrix} 1 & 0 \end{pmatrix} x_k + v_k. \quad (15.105)$$

Here, $\{w_k\}$ and $\{v_k\}$ are independent Gaussian white noise processes with covariance matrices $\text{cov}(w_k) = \begin{pmatrix} q_1 & 0 \\ 0 & q_2 \end{pmatrix}$ and $\text{cov}(v_k) = 2$.

- (i) Derive the equations satisfied by the components of the steady-state error covariance matrix Σ in terms of the parameters q_1 and q_2 .
- (ii) Solve these equations explicitly for the case $q_1 = 0$ and $q_2 = 1$.
- (iii) Verify your solution numerically using the `dlqe` routine in MATLAB.

Exercise 15.2 (LQG Control with One-Step Information Delay). Consider the finite-horizon LQG problem with quadratic running cost $\sum_k \|Dx_k + Fu_k\|^2$ (so that $D^\top D$, $F^\top F$, and $D^\top F$ are the state, control, and cross-weight matrices) and let K_{k+1} denote the associated control Riccati matrix. Suppose the control at time k is based on *delayed* information, so that admissible control laws take the form $u_k = \mu_k(\mathcal{D}_{k-1})$, where $\mathcal{D}_{k-1} := (z_0, \dots, z_{k-1})$. Assume that the noise processes satisfy $\mathbb{E}(w_k v_k^\top) = 0$.

- (i) Following the derivation of the standard LQG solution, show that the optimal control law for this delayed-information case is

$$u_k = - (F^\top F + B^\top K_{k+1} B)^{-1} (B^\top K_{k+1} A + F^\top D) \hat{x}_{k|k-1}.$$

- (ii) Explain how the presence of information delay modifies the structure of the optimal control law compared to the standard LQG setting.

Exercise 15.3 (Controlled Markov Chain with Discounted Reward). The Markov chain $\{x_k\}$ under control has two states $\{1, 2\}$. In state 1, there are three possible actions $n = 1, 2, 3$, where action n yields a return $(5 - n)/4$ and the system remains in state 1 with probability $(n - 1)/2$. In state 2, there is only one action, yielding a return 0, and the system remains in state 2 with probability 1.

Suppose the performance criterion is to maximize the discounted return $J = \mathbb{E} \sum_{k=0}^{\infty} \alpha^k L(x_k, u_k)$, where $0 < \alpha < 1$.

- (i) Show that the value functions under the three stationary policies μ_1, μ_2, μ_3 , defined by $\mu_1(1) = 1, \mu_2(1) = 2, \mu_3(1) = 3$, and $\mu_i(2) = 1$ for $i = 1, 2, 3$, are given by

$$V_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 3/4 \\ 1 - \alpha/2 \\ 0 \end{pmatrix}, \quad V_3 = \begin{pmatrix} 1 \\ 2(1 - \alpha) \\ 0 \end{pmatrix}.$$

- (ii) Determine the optimal control law as a function of $\alpha \in (0, 1)$.

Exercise 15.4 (Stochastic Control with Multiplicative Noise). Suppose the process $\{x_k\}$ satisfies $x_{k+1} = w_k u_k$, where the control u_k is constrained to lie in $[0, x_k]$, and $\{w_k\}$ is a nonnegative i.i.d. sequence. The objective is to maximize the discounted reward $\mathbb{E} \sum_{k=0}^{\infty} \alpha^k (x_k - u_k)^{1/2}$, where $0 < \alpha < 1$ and $\alpha \mathbb{E}[w_k^{1/2}] < 1$.

- (i) Derive the dynamic programming equation for the optimal value function $V(x)$.
- (ii) Show that the Bellman equation can be written in the fixed-point form $V = TV$, and compute the first two iterates $v_1(x)$ and $v_2(x)$.
- (iii) Based on the form of $v_1(x)$ and $v_2(x)$, conjecture the functional form of $V(x)$ and solve the Bellman equation explicitly.
- (iv) Verify that the function obtained in part (iii) is the optimal value function and determine the corresponding optimal control law.

Exercise 15.5 (Gaussian Conditioning and the Measurement Update). Let $x \in \mathbb{R}^n$ be Gaussian, $x \sim \mathcal{N}(\bar{x}, \Sigma)$, and let the measurement be $z = Cx + v$ with $v \sim \mathcal{N}(0, V)$ independent of x and $V \succ 0$. This exercise builds the Kalman measurement update (Proposition 15.72) from scratch.

- (i) Show that (x, z) is jointly Gaussian and compute the covariance blocks $\text{Cov}(x)$, $\text{Cov}(z)$, and $\text{Cov}(x, z)$.
- (ii) Apply the Gaussian conditioning lemma (Lemma 15.69) to show $x \mid z \sim \mathcal{N}(\hat{x}, \Sigma^+)$ with

$$\hat{x} = \bar{x} + G(z - C\bar{x}), \quad G = \Sigma C^\top (C \Sigma C^\top + V)^{-1}, \quad \Sigma^+ = (I - GC) \Sigma.$$

- (iii) Prove that the *Joseph form* $(I - GC)\Sigma(I - GC)^\top + GVG^\top$ equals $(I - GC)\Sigma$, and explain why the Joseph form is preferred in floating-point computation.
- (iv) Show $\Sigma^+ \preceq \Sigma$, with equality iff $C = 0$: a measurement never increases uncertainty.
- (v) (*Information form.*) Assuming $\Sigma \succ 0$, use the matrix inversion lemma to prove $(\Sigma^+)^{-1} = \Sigma^{-1} + C^\top V^{-1}C$ and $\hat{x} = \Sigma^+(\Sigma^{-1}\bar{x} + C^\top V^{-1}z)$. Interpret the update as the addition of *precisions*.

Exercise 15.6 (Scalar Kalman Filter and Its Steady State). Consider the scalar system $x_{k+1} = ax_k + w_k$, $z_k = x_k + v_k$, with $w_k \sim \mathcal{N}(0, q)$ and $v_k \sim \mathcal{N}(0, r)$ independent, $q, r > 0$, and $x_0 \sim \mathcal{N}(\bar{x}_0, p_0)$.

- (i) Specialize the Kalman filter of Theorem 15.73 to this scalar case, writing the recursions for the prediction variance $p_{k|k-1}$, the gain g_k , the filtered variance $p_{k|k}$, and the estimate $\hat{x}_{k|k}$.
- (ii) Show that the prediction variance satisfies $p_{k+1|k} = a^2 \frac{r p_{k|k-1}}{p_{k|k-1} + r} + q$.
- (iii) Show the steady-state prediction variance $\bar{p}$ is the positive root of $\bar{p}^2 + (r - q - a^2 r)\bar{p} - qr = 0$, and give the steady-state gain $\bar{g} = \bar{p}/(\bar{p} + r)$.
- (iv) For $a = 1$ and $q = r = 1$, show $\bar{p}$ is the golden ratio $(1 + \sqrt{5})/2 \approx 1.618$ and $\bar{g} \approx 0.618$, recovering the numbers of the steady-state example in Section 15.10.
- (v) Show the stationary filter is stable, i.e. $|(1 - \bar{g})a| < 1$, and verify it for the numbers in (iv).

Exercise 15.7 (The Innovations Are White). For the Kalman filter of Theorem 15.73, let $\nu_k = z_k - C_k \hat{x}_{k|k-1}$ be the innovation and $\Lambda_k = C_k \Sigma_{k|k-1} C_k^\top + V_k$.

- (i) Show $\mathbb{E}[\nu_k | \mathcal{I}_{k-1}] = 0$, and deduce $\mathbb{E}[\nu_k] = 0$ and $\mathbb{E}[\nu_k h(\mathcal{I}_{k-1})^\top] = 0$ for every bounded measurable h .
- (ii) Prove whiteness: $\mathbb{E}[\nu_k \nu_j^\top] = 0$ for $j \neq k$, and $\mathbb{E}[\nu_k \nu_k^\top] = \Lambda_k$. [Hint: for $j < k$, ν_j is $\mathcal{I}_{k-1}$ -measurable; use the tower property.]
- (iii) Conclude that $\{\Lambda_k^{-1/2} \nu_k\}$ is a standardized white sequence, and that the map $\{z_k\} \mapsto \{\nu_k\}$ is causal and causally invertible (a *whitening filter*).
- (iv) Explain why, in practice, testing the measured innovations for zero mean and whiteness is used to detect filter or model mis-specification.

Exercise 15.8 (Certainty Equivalence and the Price of Information). Consider the scalar finite-horizon LQG problem $x_{k+1} = ax_k + bu_k + w_k$, $z_k = cx_k + v_k$ (with $w_k \sim \mathcal{N}(0, W)$ and measurement noise $v_k \sim \mathcal{N}(0, \rho)$), with cost $\mathbb{E}[q_N x_N^2 + \sum_{k=0}^{N-1} (q x_k^2 + r u_k^2)]$ under the standing Gaussian assumptions.

- (i) Write the control Riccati recursion for K_k and the gain L_k , and the filtering Riccati for $\Sigma_{k|k}$ from Theorem 15.73.
- (ii) State the optimal policy $u_k^* = L_k \hat{x}_{k|k}$ and explain, citing certainty equivalence (Proposition 15.84), why $\hat{x}_{k|k}$ replaces the unavailable x_k .
- (iii) Using the value-function form of Theorem 15.82, show that the optimal cost is $J^* = \mathbb{E}[x_0^\top K_0 x_0] + \sum_{k=0}^{N-1} (K_{k+1} W_k + P_k \Sigma_{k|k})$ with $P_k = A_k^\top K_{k+1} B_k S_k^{-1} B_k^\top K_{k+1} A_k$ (scalars here), and identify $\sum_k P_k \Sigma_{k|k}$ as the *price of imperfect information*.

- (iv) Show this price vanishes as $c \rightarrow \infty$ (perfect observation) and increases with the measurement-noise variance ρ .
- (v) For $a = b = c = 1$, $q = r = q_N = 1$, $N = 3$, and w, v, p_0 of your choice, compute K_k , L_k , $\Sigma_{k|k}$, and the information penalty numerically.

Exercise 15.9 (No Dual Effect, and a System That Has One). (i) For the LQG model (15.94) and (15.95), prove directly that the conditional error covariance $\Sigma_{k|k}$ is unaffected by the past controls $u_0, \dots, u_{k-1}$ (cf. Lemma 15.80). [Hint: subtract the known, I_k -measurable control contribution from both x_k and each z_ℓ , and observe that the residuals do not involve the controls.]

- (ii) Now let the observation gain be control-dependent, $z_k = (c + d u_{k-1})x_k + v_k$ (scalar). Show that the posterior variance now depends on the applied controls, so the controller can act to *improve* future state estimation, the *dual effect*.
- (iii) Explain why certainty equivalence fails for the system in (ii), even though the dynamics and cost are linear-quadratic.

Exercise 15.10 (Open-Loop versus Feedback Control). Consider $x_{k+1} = x_k + u_k + w_k$ for $k = 0, \dots, N-1$, with $x_0 = 0$, i.i.d. $w_k \sim \mathcal{N}(0, \sigma^2)$, full-state observation $z_k = x_k$, and cost $J = \mathbb{E}[\sum_{k=1}^N x_k^2]$.

- (i) Compute the optimal *open-loop* control (a deterministic sequence fixed at $k = 0$) and show its cost is $\sigma^2 N(N+1)/2$.
- (ii) Compute the optimal *feedback* law $u_k = \mu_k(x_k)$ and show its cost is $N\sigma^2$.
- (iii) Conclude the feedback controller is strictly better for $N \geq 2$, and interpret the gap $\sigma^2 N(N-1)/2$ as the value of acting on new information, a concrete instance of the non-equivalence of open-loop and feedback control in stochastic systems.

Exercise 15.11 (Belief-State Filter for a Two-State POMDP). A machine is *good* ($x = 1$) or *failed* ($x = 2$). It fails irreversibly with probability β per step: $P_{11} = 1 - \beta$, $P_{12} = \beta$, $P_{22} = 1$. Each step emits a binary signal $z_k \in \{0, 1\}$ with $\mathbb{P}(z = 1 \mid x = 1) = 1 - \varepsilon$ and $\mathbb{P}(z = 1 \mid x = 2) = \eta$. Let $\pi_k = \Pr(x_k = 1 \mid z_1, \dots, z_k)$.

- (i) Derive the belief recursion in closed form: a prediction step $\pi_k^- = (1 - \beta)\pi_{k-1}$ followed by the Bayes update conditioned on z_k .
- (ii) Argue that π_k is a sufficient statistic for any decision problem on this chain (cf. the information-state reduction of the partial-information theory).
- (iii) Suppose running the failed machine costs $c > 0$ per step and a one-time inspection (stop) costs $s > 0$. Formulate the optimal-stopping problem on the belief π_k , and argue that the optimal rule is a threshold: stop once π_k falls below some π^* .

Exercise 15.12 (Duality of the Control and Filter Riccati Equations). (i) Write the control algebraic Riccati equation $K = A^\top [K - KB(R + B^\top KB)^{-1}B^\top K]A + Q$ and the filtering algebraic Riccati equation $\Sigma = A[\Sigma - \Sigma C^\top (C\Sigma C^\top + V)^{-1}C\Sigma]A^\top + W$.

- (ii) Show that the filtering equation is exactly the control equation under the substitution $(A, B, Q, R) \mapsto (A^\top, C^\top, W, V)$.

- (iii) Deduce that every existence/uniqueness and convergence result for the LQR Riccati equation transfers verbatim to the Kalman filter, and that one solver computes both. Verify on the scalar case $a = b = c = 1$, $q = r = w = v = 1$ that the control and filter equations share the golden-ratio solution.

Appendices

Appendix A

Linear System Theory

This appendix collects the results from linear system theory that support the developments in the main text, in particular the treatment of optimal control, the linear-quadratic regulator, Kalman filtering, and linear-quadratic–Gaussian design. Our aim is not encyclopedic coverage but a self-contained account of the concepts that recur throughout the book: the state-space model, the solution of linear differential and difference equations, internal stability, and the structural notions of controllability, observability, stabilizability, and detectability. Wherever possible we develop the continuous-time and discrete-time theories in parallel, since the optimal control and estimation chapters use both. Standard references for this material are [61, 29, 105, 5, 129].

Throughout, $\mathbb{R}^n$ denotes the n -dimensional real Euclidean space, $\mathbb{R}^{n \times m}$ the space of real $n \times m$ matrices, and I_n the $n \times n$ identity (the subscript is dropped when the dimension is clear). For a matrix A we write $A^\top$ for its transpose, $\text{rank } A$ for its rank, $\det A$ for its determinant, $\text{tr } A$ for its trace, and $\sigma(A) \subset \mathbb{C}$ for its spectrum (the set of eigenvalues). A symmetric matrix $P = P^\top$ is positive semidefinite, written $P \succeq 0$, if $x^\top P x \geq 0$ for all x , and positive definite, written $P \succ 0$, if the inequality is strict for all $x \neq 0$. The open left half of the complex plane is $\mathbb{C}^- = \{s \in \mathbb{C} : \text{Re } s < 0\}$, and the open unit disk is $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$.

A.1 ■ State-Space Models

A dynamical system is described by a set of internal variables, the *state*, whose knowledge at an initial instant, together with the future inputs, determines the future evolution of the system. The state-space model formalizes this idea and is the representation used throughout the book.

A.1.1 ■ Continuous-Time Linear Systems

A finite-dimensional, continuous-time, linear time-varying (LTV) system is described by

$$\begin{aligned}\dot{x}(t) &= A(t)x(t) + B(t)u(t), & x(t_0) &= x_0, \\ y(t) &= C(t)x(t) + D(t)u(t),\end{aligned}\tag{A.1}$$

where $x(t) \in \mathbb{R}^n$ is the state, $u(t) \in \mathbb{R}^m$ is the control (or input), and $y(t) \in \mathbb{R}^p$ is the output. The matrices $A(t) \in \mathbb{R}^{n \times n}$, $B(t) \in \mathbb{R}^{n \times m}$, $C(t) \in \mathbb{R}^{p \times n}$, and $D(t) \in \mathbb{R}^{p \times m}$ are assumed piecewise continuous in t . When the matrices do not depend on time the system is linear time-invariant (LTI),

$$\dot{x}(t) = Ax(t) + Bu(t), \quad y(t) = Cx(t) + Du(t),\tag{A.2}$$

and we refer to the quadruple (A, B, C, D) , or simply (A, B, C) when $D = 0$, as a *realization*. The integer n is the *order* of the system.

A.1.2 ■ Discrete-Time Linear Systems

The discrete-time counterpart replaces the differential equation by a difference equation,

$$\begin{aligned} x(k+1) &= A(k)x(k) + B(k)u(k), & x(k_0) &= x_0, \\ y(k) &= C(k)x(k) + D(k)u(k), \end{aligned} \quad (\text{A.3})$$

with $k \in \mathbb{Z}$, and the time-invariant case

$$x(k+1) = Ax(k) + Bu(k), \quad y(k) = Cx(k) + Du(k). \quad (\text{A.4})$$

Discrete-time models arise both as descriptions of inherently sampled systems and as the result of discretizing a continuous-time model, as in the discrete-time optimal control and Kalman filtering chapters.

Remark A.1. Nonlinear models $\dot{x} = f(x, u, t)$ are reduced to the form (A.1) by linearization about a nominal trajectory $(\bar{x}(\cdot), \bar{u}(\cdot))$: writing $x = \bar{x} + \delta x$, $u = \bar{u} + \delta u$ and retaining first-order terms yields $\delta \dot{x} = A(t)\delta x + B(t)\delta u$ with $A(t) = \partial_x f(\bar{x}, \bar{u}, t)$ and $B(t) = \partial_u f(\bar{x}, \bar{u}, t)$. This is why the linear theory developed here governs the local behavior of the nonlinear systems studied elsewhere in the text.

A.2 ■ Solution of the State Equation

We now describe the trajectories generated by the state equation. The central object is the transition operator that propagates the state from one time to another.

A.2.1 ■ The Matrix Exponential and the LTI Case

For the homogeneous LTI equation $\dot{x} = Ax$, the solution is expressed through the matrix exponential

$$e^{At} := \sum_{k=0}^{\infty} \frac{(At)^k}{k!}, \quad (\text{A.5})$$

a series that converges for every A and t . Its basic properties follow directly from the definition.

Proposition A.2 (Properties of the matrix exponential). For $A \in \mathbb{R}^{n \times n}$ and $s, t \in \mathbb{R}$:

- (1) $e^{A \cdot 0} = I$ and $\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A$;
- (2) $e^{A(s+t)} = e^{As}e^{At}$, and e^{At} is invertible with $(e^{At})^{-1} = e^{-At}$;
- (3) if $AB = BA$ then $e^{(A+B)t} = e^{At}e^{Bt}$;
- (4) $e^{TAT^{-1}t} = Te^{At}T^{-1}$ for any invertible T ;
- (5) $\sigma(e^{At}) = \{e^{\lambda t} : \lambda \in \sigma(A)\}$.

Proof. Properties (1)–(4) follow by differentiating or manipulating the series (A.5); for (2), $\frac{d}{dt}(e^{A(s+t)}) = A e^{A(s+t)}$ and $e^{A(s+t)}|_{t=0} = e^{As}$, and the same initial-value problem is solved by $e^{As}e^{At}$, so the two agree by uniqueness. Statement (5) follows from (4) by reducing A to Jordan form, since the exponential of a Jordan block $J = \lambda I + N$ is $e^{\lambda t}e^{Nt}$ with N nilpotent. $\square$

Theorem A.3 (Variation of constants, LTI). *The unique solution of the LTI state equation (A.2) is*

$$x(t) = e^{A(t-t_0)}x_0 + \int_{t_0}^t e^{A(t-\tau)}B u(\tau) d\tau, \quad (\text{A.6})$$

and the corresponding output is $y(t) = C e^{A(t-t_0)}x_0 + \int_{t_0}^t C e^{A(t-\tau)}B u(\tau) d\tau + Du(t)$.

Proof. Differentiating (A.6) using Proposition A.2(1) and the Leibniz rule gives $\dot{x} = A e^{A(t-t_0)}x_0 + A \int_{t_0}^t e^{A(t-\tau)}B u(\tau) d\tau + B u(t) = Ax + Bu$, and $x(t_0) = x_0$. Uniqueness follows from the Lipschitz continuity of the right-hand side in x . $\square$

The first term of (A.6) is the *zero-input response* and the second the *zero-state response*. The map $t \mapsto C e^{At}B$ (for $t \geq 0$), together with $D\delta(t)$, is the impulse response of the system.

A.2.2 ■ The State-Transition Matrix, LTV Case

For the time-varying homogeneous equation $\dot{x} = A(t)x$ the exponential is replaced by the state-transition matrix.

Definition A.4 (State-transition matrix). *The state-transition matrix $\Phi(t, t_0) \in \mathbb{R}^{n \times n}$ associated with (A.1) is the unique solution of the matrix differential equation*

$$\frac{\partial}{\partial t}\Phi(t, t_0) = A(t)\Phi(t, t_0), \quad \Phi(t_0, t_0) = I. \quad (\text{A.7})$$

Proposition A.5 (Properties of Φ). *The state-transition matrix satisfies:*

- (1) (semigroup) $\Phi(t_2, t_0) = \Phi(t_2, t_1)\Phi(t_1, t_0)$ for all t_0, t_1, t_2 ;
- (2) (invertibility) $\Phi(t, t_0)$ is invertible and $\Phi(t, t_0)^{-1} = \Phi(t_0, t)$;
- (3) $\frac{\partial}{\partial t_0}\Phi(t, t_0) = -\Phi(t, t_0)A(t_0)$;
- (4) (Liouville) $\det \Phi(t, t_0) = \exp\left(\int_{t_0}^t \text{tr } A(\tau) d\tau\right)$;
- (5) in the LTI case $\Phi(t, t_0) = e^{A(t-t_0)}$.

Proof. For (1), both sides solve (A.7) in t_2 with value $\Phi(t_1, t_0)$ at $t_2 = t_1$, so they coincide by uniqueness. Setting $t_2 = t_0$ in (1) gives (2). Differentiating the identity $\Phi(t, t_0)\Phi(t_0, t) = I$ yields (3). For (4), let $w(t) = \det \Phi(t, t_0)$; Jacobi's formula gives $\dot{w} = \text{tr}(A(t))w$, and $w(t_0) = 1$. Statement (5) is immediate from Proposition A.2. $\square$

Theorem A.6 (Variation of constants, LTV). *The unique solution of (A.1) is*

$$x(t) = \Phi(t, t_0)x_0 + \int_{t_0}^t \Phi(t, \tau)B(\tau)u(\tau) d\tau. \quad (\text{A.8})$$

Proof. Identical to the proof of Theorem A.3, using Proposition A.5 in place of Proposition A.2. $\square$

A.2.3 ■ The Discrete-Time Transition Matrix

For the discrete-time system (A.3), define the transition matrix by the product

$$\Phi(k, k_0) = A(k-1)A(k-2) \cdots A(k_0), \quad \Phi(k_0, k_0) = I, \quad (\text{A.9})$$

so that $\Phi(k, k_0) = A^{k-k_0}$ in the time-invariant case. The solution of (A.3) is

$$x(k) = \Phi(k, k_0) x_0 + \sum_{j=k_0}^{k-1} \Phi(k, j+1) B(j) u(j). \quad (\text{A.10})$$

Unlike the continuous-time case, $\Phi(k, k_0)$ need not be invertible, since the $A(k)$ may be singular; the transition therefore runs only forward in time.

Example A.7 Computing e^{At} for a 2×2 system.

Consider $A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & 0 \end{pmatrix}$, the model of an undamped oscillator. Since $A^2 = -\omega^2 I$, the series (A.5) separates into the Taylor series of cos and sin:

$$e^{At} = \cos(\omega t) I + \frac{\sin(\omega t)}{\omega} A = \begin{pmatrix} \cos \omega t & \frac{1}{\omega} \sin \omega t \\ -\omega \sin \omega t & \cos \omega t \end{pmatrix}.$$

The eigenvalues of A are $\pm i\omega$, so by Proposition A.2(5) those of e^{At} are $e^{\pm i\omega t}$, consistent with the purely oscillatory (marginally stable) behavior. More generally e^{At} can be computed from the Laplace transform, $e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$, or by diagonalization when A is diagonalizable. $\square$

A.3 ■ Stability

Stability concerns the qualitative behavior of trajectories as $t \rightarrow \infty$. We focus on the internal (Lyapunov) stability of the equilibrium $x = 0$ of the unforced system; input–output stability is discussed briefly at the end.

A.3.1 ■ Stability of LTI Systems

Definition A.8 (Stability notions). The equilibrium $x = 0$ of $\dot{x} = Ax$ is

- stable (in the sense of Lyapunov) if for every $\varepsilon > 0$ there is $\delta > 0$ such that $\|x_0\| < \delta$ implies $\|x(t)\| < \varepsilon$ for all $t \geq 0$;
- asymptotically stable if it is stable and $x(t) \rightarrow 0$ as $t \rightarrow \infty$ for all x_0 in a neighborhood of the origin;
- (globally) exponentially stable if there exist $M \geq 1$, $\alpha > 0$ with $\|x(t)\| \leq Me^{-\alpha t} \|x_0\|$ for all x_0 and $t \geq 0$.

Because the system is linear, asymptotic stability is automatically global and exponential. The following spectral characterization is fundamental. We call a matrix *Hurwitz* if $\sigma(A) \subset \mathbb{C}^-$.

Theorem A.9 (Spectral criterion, continuous time). For $\dot{x} = Ax$:

- (1) the origin is asymptotically stable if and only if A is Hurwitz, i.e. $\operatorname{Re} \lambda < 0$ for every $\lambda \in \sigma(A)$;
- (2) the origin is stable if and only if $\operatorname{Re} \lambda \leq 0$ for every eigenvalue and every eigenvalue with $\operatorname{Re} \lambda = 0$ is semisimple (its algebraic and geometric multiplicities coincide).

Proof. Transform A to Jordan form $A = TJT^{-1}$; then $e^{At} = Te^{Jt}T^{-1}$ and the entries of e^{Jt} are of the form $t^j e^{\lambda t}$ with $0 \leq j < (\text{size of the Jordan block for } \lambda)$. Such terms tend to zero for all j iff $\operatorname{Re} \lambda < 0$, giving (1). They remain bounded iff $\operatorname{Re} \lambda < 0$, or $\operatorname{Re} \lambda = 0$ with $j = 0$, i.e. the block has size one; this is (2). $\square$

For discrete time the role of $\mathbb{C}^-$ is played by the unit disk; a matrix with $\sigma(A) \subset \mathbb{D}$ is called *Schur* (or *stable*).

Theorem A.10 (Spectral criterion, discrete time). For $x(k+1) = Ax(k)$, the origin is asymptotically stable if and only if A is Schur, i.e. $|\lambda| < 1$ for every $\lambda \in \sigma(A)$; it is stable iff $|\lambda| \leq 1$ for all eigenvalues and every eigenvalue with $|\lambda| = 1$ is semisimple.

A.3.2 ■ Lyapunov Equations

Spectral tests require eigenvalue computation. The Lyapunov approach provides an equivalent algebraic characterization that is central to the LQR, Kalman filtering, and stochastic control chapters, where the Lyapunov and Riccati equations govern cost-to-go and covariance propagation.

Theorem A.11 (Lyapunov theorem, continuous time). The matrix A is Hurwitz if and only if for some (equivalently, every) $Q \succ 0$ the Lyapunov equation

$$A^\top P + PA = -Q \quad (\text{A.11})$$

has a unique solution P , and that solution satisfies $P \succ 0$. If A is Hurwitz, the solution is given explicitly by

$$P = \int_0^\infty e^{A^\top t} Q e^{At} dt. \quad (\text{A.12})$$

Proof. Suppose A is Hurwitz. The integral (A.12) converges because $\|e^{At}\| \leq Me^{-\alpha t}$. Substituting into $A^\top P + PA = \int_0^\infty \frac{d}{dt} (e^{A^\top t} Q e^{At}) dt = [e^{A^\top t} Q e^{At}]_0^\infty = -Q$ verifies (A.11). Positivity: $x^\top Px = \int_0^\infty \|Q^{1/2} e^{At} x\|^2 dt > 0$ for $x \neq 0$. Uniqueness holds because the linear map $P \mapsto A^\top P + PA$ has spectrum $\{\lambda_i + \lambda_j\}$, which excludes 0 when A is Hurwitz.

Conversely, suppose (A.11) holds with $P \succ 0$ and $Q \succ 0$. Let $\lambda \in \sigma(A)$ with eigenvector $v \neq 0$, $Av = \lambda v$. Then $v^*(A^\top P + PA)v = (\bar{\lambda} + \lambda)v^* P v = -v^* Q v < 0$, so $2 \operatorname{Re} \lambda (v^* P v) < 0$; since $v^* P v > 0$ we conclude $\operatorname{Re} \lambda < 0$. $\square$

The function $V(x) = x^\top Px$ is a quadratic Lyapunov function: along trajectories $\dot{V} = x^\top (A^\top P + PA)x = -x^\top Q x < 0$, certifying asymptotic stability. The discrete-time analogue replaces (A.11) by the *Stein equation*.

Theorem A.12 (Lyapunov theorem, discrete time). *The matrix A is Schur if and only if for some (equivalently, every) $Q \succ 0$ the Stein equation*

$$A^\top P A - P = -Q \quad (\text{A.13})$$

has a unique solution $P \succ 0$, given by $P = \sum_{k=0}^{\infty} (A^\top)^k Q A^k$.

Proof. Parallel to Theorem A.11. Convergence of the series uses $\|A^k\| \leq M\rho^k$ with $\rho < 1$; telescoping $A^\top P A - P = \sum_{k \geq 0} ((A^\top)^{k+1} Q A^{k+1} - (A^\top)^k Q A^k) = -Q$. The Lyapunov function is $V(x) = x^\top P x$, with increment $V(Ax) - V(x) = -x^\top Q x < 0$. $\square$

Remark A.13. In the LTV setting stability is characterized through the transition matrix: $\dot{x} = A(t)x$ is uniformly asymptotically stable iff there exist $M \geq 1, \alpha > 0$ with $\|\Phi(t, t_0)\| \leq M e^{-\alpha(t-t_0)}$ for all $t \geq t_0$. Eigenvalue conditions on $A(t)$ are neither necessary nor sufficient for LTV stability, a classical subtlety; time-varying Lyapunov functions $V(t, x) = x^\top P(t)x$ remain the appropriate tool.

A.4 ■ Controllability and Reachability

Controllability quantifies the extent to which the input can steer the state. It is one of the two structural properties (with observability) that determine whether the control and estimation problems of the main text are well posed.

Definition A.14 (Controllability). *The system (A.2), or the pair (A, B) , is controllable if for every pair of states $x_0, x_1 \in \mathbb{R}^n$ and every $T > 0$ there exists an input $u : [0, T] \rightarrow \mathbb{R}^m$ steering the state from $x(0) = x_0$ to $x(T) = x_1$. The reachable set from the origin is $\mathcal{R} = \{x_1 : \exists T > 0, u(\cdot) \text{ with } x(0) = 0, x(T) = x_1\}$.*

For LTI systems the reachable subspace is characterized algebraically through the *controllability matrix*

$$\mathcal{C} = [B \quad AB \quad A^2B \quad \cdots \quad A^{n-1}B] \in \mathbb{R}^{n \times nm}, \quad (\text{A.14})$$

and the *controllability Gramian*

$$W_c(T) = \int_0^T e^{A\tau} B B^\top e^{A^\top \tau} d\tau. \quad (\text{A.15})$$

Theorem A.15 (Controllability tests). *For the LTI pair (A, B) the following are equivalent:*

- (1) (A, B) is controllable;
- (2) $\text{rank } \mathcal{C} = n$ (Kalman rank condition);
- (3) $W_c(T) \succ 0$ for some (equivalently every) $T > 0$;
- (4) $\text{rank } [sI - A \quad B] = n$ for every $s \in \mathbb{C}$ (Popov–Belevitch–Hautus, or PBH, test);
- (5) no left eigenvector of A is orthogonal to the columns of B : there is no $w \neq 0$ with $w^\top A = \lambda w^\top$ and $w^\top B = 0$.

Moreover $\mathcal{R} = \text{Im } \mathcal{C} = \text{Im } W_c(T)$, and this reachable subspace is A -invariant.

Proof. (3) $\Rightarrow$ (1): if $W_c(T) \succ 0$, the input $u(\tau) = B^\top e^{A^\top(T-\tau)} W_c(T)^{-1} (x_1 - e^{AT} x_0)$ substituted into (A.6) gives $x(T) = x_1$; this is in fact the minimum-energy control. (1) $\Rightarrow$ (3): if $W_c(T)$ is singular there is $v \neq 0$ with $v^\top W_c(T) v = \int_0^T \|B^\top e^{A^\top \tau} v\|^2 d\tau = 0$, hence $v^\top e^{A\tau} B = 0$ for all $\tau \in [0, T]$; by (A.6) the zero-state response satisfies $v^\top x(T) = 0$, so $v \perp \mathcal{R}$ and the system is not controllable. (2) $\Leftrightarrow$ (3): since $v^\top e^{A\tau} B \equiv 0$ on an interval iff (by repeated differentiation at $\tau = 0$) $v^\top A^k B = 0$ for $k = 0, \dots, n-1$, we have $\ker W_c(T) = \ker \mathcal{C}^\top$; the Cayley–Hamilton theorem shows higher powers add nothing, so $\text{Im } \mathcal{C} = \text{Im } W_c(T)$. (4) $\Leftrightarrow$ (5): $\text{rank}[sI - A \ B] < n$ iff there is $w \neq 0$ with $w^\top [sI - A \ B] = 0$, i.e. $w^\top A = sw^\top$ and $w^\top B = 0$, which forces $s \in \sigma(A)$. (5) $\Rightarrow$ (2): if $\text{rank } \mathcal{C} < n$ there is $w \neq 0$ with $w^\top \mathcal{C} = 0$, so $w^\top A^k B = 0$ for all k ; the A -invariant space $(\text{Im } \mathcal{C})^\perp$ then contains an eigenvector of $A^\top$, contradicting (5). The converse is analogous. $\square$

Remark A.16. The tests are independent of the horizon T : an LTI system controllable on some interval is controllable on every interval. This is a distinctly linear, finite-dimensional phenomenon. For the LTV system (A.1) the Gramian generalizes to $W_c(t_0, t_1) = \int_{t_0}^{t_1} \Phi(t_1, \tau) B(\tau) B(\tau)^\top \Phi(t_1, \tau)^\top d\tau$, and controllability on $[t_0, t_1]$ is equivalent to $W_c(t_0, t_1) \succ 0$; there is no purely algebraic rank test in general.

Definition A.17 (Stabilizability). *The pair (A, B) is stabilizable if there exists $K \in \mathbb{R}^{m \times n}$ such that $A + BK$ is Hurwitz (in discrete time, Schur).*

Stabilizability is the weakest property needed for the infinite-horizon LQR problem to admit a stabilizing solution. It relaxes controllability to require control authority only over the unstable modes.

Proposition A.18 (PBH test for stabilizability). *(A, B) is stabilizable if and only if $\text{rank} [sI - A \ B] = n$ for every $s \in \mathbb{C}$ with $\text{Re } s \geq 0$ (in discrete time, $|s| \geq 1$). Equivalently, every uncontrollable mode of A is stable.*

A.4.1 ■ State Feedback and Pole Placement

Controllability is exactly the condition under which the closed-loop dynamics can be shaped arbitrarily by state feedback.

Theorem A.19 (Pole placement). *The pair (A, B) is controllable if and only if for every self-conjugate set $\Lambda = \{\mu_1, \dots, \mu_n\} \subset \mathbb{C}$ there exists $K \in \mathbb{R}^{m \times n}$ such that $\sigma(A + BK) = \Lambda$. In the single-input case the gain K is unique.*

We omit the constructive proof (via the controllable canonical form and Ackermann's formula in the single-input case); see [61, 5]. The result explains why controllability is the standing assumption in state-feedback design and, in weakened form (stabilizability), in the LQR problem.

A.5 ■ Observability and Duality

Observability is the dual notion: it asks whether the state can be reconstructed from measured outputs. It is the structural property underlying state estimation and, in particular, the Kalman filter.

Definition A.20 (Observability). The system (A.2), or the pair (C, A) , is observable if the initial state $x(0)$ can be uniquely determined from the knowledge of $u(\cdot)$ and $y(\cdot)$ on any interval $[0, T]$, $T > 0$.

Because the effect of the known input can be subtracted, observability concerns the map $x_0 \mapsto y(t) = Ce^{At}x_0$. Define the *observability matrix*

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix} \in \mathbb{R}^{np \times n}, \quad (\text{A.16})$$

and the *observability Gramian* $W_o(T) = \int_0^T e^{A^\top \tau} C^\top C e^{A\tau} d\tau$.

Theorem A.21 (Observability tests and duality). For the LTI pair (C, A) the following are equivalent:

- (1) (C, A) is observable;
- (2) $\text{rank } \mathcal{O} = n$;
- (3) $W_o(T) \succ 0$ for some (every) $T > 0$;
- (4) $\text{rank} \begin{bmatrix} sI - A \\ C \end{bmatrix} = n$ for every $s \in \mathbb{C}$ (PBH test).

Moreover, (C, A) is observable if and only if $(A^\top, C^\top)$ is controllable, and the unobservable subspace is $\ker \mathcal{O} = \ker W_o(T)$.

Proof. The duality statement is verified by comparing (A.14) with (A.16)[⊤]: $\mathcal{O}(C, A)^\top = \mathcal{C}(A^\top, C^\top)$, so $\text{rank } \mathcal{O} = n$ iff $(A^\top, C^\top)$ is controllable. Every equivalence in the list is then the transpose of the corresponding statement in Theorem A.15. Directly: if $x_0 \in \ker \mathcal{O}$ then $Ce^{At}x_0 \equiv 0$, so x_0 produces the zero output and cannot be distinguished from the origin; conversely if $\mathcal{O}$ has full rank, the least-squares inversion $x_0 = W_o(T)^{-1} \int_0^T e^{A^\top \tau} C^\top \bar{y}(\tau) d\tau$ (with $\bar{y}$ the input-corrected output) recovers the state. $\square$

Definition A.22 (Detectability). The pair (C, A) is detectable if there exists $L \in \mathbb{R}^{n \times p}$ such that $A + LC$ is Hurwitz (discrete time: Schur); equivalently, every unobservable mode is stable, i.e. $\text{rank} \begin{bmatrix} sI - A \\ C \end{bmatrix} = n$ for all s with $\text{Re } s \geq 0$.

Detectability is the dual of stabilizability and is the minimal condition under which a stable state observer (and the steady-state Kalman filter) exists. The observer $\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x})$ produces an error $e = x - \hat{x}$ obeying $\dot{e} = (A + LC)e$, which decays iff $A + LC$ is Hurwitz; by duality with pole placement, L can assign $\sigma(A + LC)$ arbitrarily iff (C, A) is observable.

Example A.23 A structurally deficient system.

Consider

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad C = (1 \quad 0).$$

The controllability matrix is $\mathcal{C} = \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$ with rank 1, so the system is not controllable; the mode at $s = 2$ is uncontrollable. Since that mode is unstable ($\operatorname{Re} 2 > 0$), (A, B) is *not* stabilizable, and no state feedback can stabilize the system. Likewise $\mathcal{O}$ has rank 1 and the unstable mode is unobservable, so the pair is not detectable. This system exhibits precisely the pathology ruled out by the standing stabilizability/detectability assumptions of the LQR and Kalman chapters. $\square$

A.6 ■ Canonical Decomposition and Realizations

The structural properties above induce a canonical decomposition of the state space that clarifies which parts of a system are “visible” from the input–output behavior.

Theorem A.24 (Kalman decomposition). *For any LTI system (A, B, C, D) there exists an invertible T such that, in the coordinates $\tilde{x} = T^{-1}x$, the data take the block form*

$$\tilde{A} = \begin{pmatrix} A_{c\bar{o}} & A_{12} & A_{13} & A_{14} \\ 0 & A_{co} & 0 & A_{24} \\ 0 & 0 & A_{\bar{c}\bar{o}} & A_{34} \\ 0 & 0 & 0 & A_{\bar{c}o} \end{pmatrix}, \quad \tilde{B} = \begin{pmatrix} B_{c\bar{o}} \\ B_{co} \\ 0 \\ 0 \end{pmatrix}, \quad \tilde{C} = (0 \quad C_{co} \quad 0 \quad C_{\bar{c}o}),$$

where the four blocks correspond to the controllable/uncontrollable and observable/unobservable subspaces. The transfer function depends only on the controllable-and-observable part $(A_{co}, B_{co}, C_{co}, D)$: $G(s) = C_{co}(sI - A_{co})^{-1}B_{co} + D$.

Definition A.25 (Minimal realization). *A realization (A, B, C, D) of a transfer function $G(s)$ is minimal if its order n is the smallest among all realizations of G .*

Theorem A.26 (Minimality). *A realization (A, B, C, D) is minimal if and only if (A, B) is controllable and (C, A) is observable. All minimal realizations of a given $G(s)$ have the same order and are related by a unique similarity transformation.*

Proof. [Sketch] By Theorem A.24, any realization can be reduced to its controllable-observable part without changing G , so a minimal realization must be controllable and observable. Conversely, if two controllable and observable realizations have the same transfer function, equating Markov parameters $CA^k B$ shows $\mathcal{O}_1 T = \mathcal{O}_2$ and $T C_1 = C_2$ for a unique invertible T , establishing uniqueness up to similarity and equality of orders. $\square$

A.6.1 ■ Transfer Functions

Applying the Laplace transform (continuous time) or the z -transform (discrete time) to (A.2)–(A.4) with zero initial condition yields the input–output description

$$Y(s) = G(s)U(s), \quad G(s) = C(sI - A)^{-1}B + D, \quad (\text{A.17})$$

with s replaced by z in discrete time. The entries of G are proper rational functions; the poles of G are a subset of $\sigma(A)$ (equal to it when the realization is minimal). The correspondence between the internal (state-space) and external (transfer-function) descriptions, mediated by controllability and observability, is the organizing principle of linear system theory, and it is the bridge to the frequency-domain methods of robust and H_∞ design referenced in the notes below.

A.6.2 ■ Connection to the Riccati Equation

The infinite-horizon LQR problem of the optimal control chapters minimizes $\int_0^\infty (x^\top Qx + u^\top Ru) dt$ subject to (A.2), with $Q \succeq 0$, $R \succ 0$. Its solution rests on the *algebraic Riccati equation* (ARE)

$$A^\top P + PA - PBR^{-1}B^\top P + Q = 0. \quad (\text{A.18})$$

The linear-system structure of this appendix governs its solvability, which we record here for reference by the main text.

Theorem A.27 (Stabilizing solution of the ARE). *Suppose (A, B) is stabilizable and (H, A) is detectable, where $Q = H^\top H$. Then (A.18) has a unique symmetric positive semidefinite solution P , and the closed-loop matrix $A - BR^{-1}B^\top P$ is Hurwitz. The optimal control is the state feedback $u = -R^{-1}B^\top P x =: -Kx$.*

The hypotheses are exactly the structural conditions of this appendix: stabilizability guarantees a stabilizing control exists, and detectability of (H, A) ensures that any mode left unpenalized by the cost is nonetheless stable, so that a finite optimal cost forces closed-loop stability. The dual statement, with (A, B) and (H, A) replaced by their observer counterparts, yields the existence of the steady-state Kalman filter; this duality between control and estimation is developed in the Kalman filtering and LQG chapters.

A.7 ■ Notes and Further Reading

The state-space viewpoint adopted here originates with Kalman's work of the 1960s, which introduced controllability, observability, and their duality, and tied them to minimal realizations and to the Riccati equations of optimal control and filtering. Comprehensive treatments are given by Kailath [61], Rugh [105], and Antsaklis and Michel [5]; the geometric theory of (A, B) -invariant subspaces is developed in Wonham [127], and the connection to H_∞ and robust control in Zhou, Doyle, and Glover [129]. The Lyapunov and Riccati equations recur throughout the chapters on optimal control, Kalman filtering, and stochastic control, where the structural results of this appendix are used without further comment.

Exercises

Exercise A.1 (Matrix exponential). Let $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Compute e^{At} directly from the series (A.5) and verify the result using $e^{At} = \mathcal{L}^{-1}\{(sI - A)^{-1}\}$. Is the origin of $\dot{x} = Ax$ stable? Asymptotically stable? Justify using Theorem A.9.

Exercise A.2 (Lyapunov equation). For $A = \begin{pmatrix} 0 & 1 \\ -2 & -3 \end{pmatrix}$ and $Q = I$, solve the Lyapunov equation (A.11) for P and verify $P \succ 0$. Confirm that A is Hurwitz both by its eigenvalues and by the sign-definiteness of P .

Exercise A.3 (Controllability and stabilizability). Consider $A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Determine whether (A, B) is controllable using both the rank condition (Theorem A.15) and the PBH test. If

it is controllable, find a gain K placing the closed-loop eigenvalues at $\{-1, -2\}$.

Exercise A.4 (Duality). Let (C, A) be observable. Prove directly from Definition A.4 and Theorem A.21 that $(A^\top, C^\top)$ is controllable, and construct an observer gain L such that $A + LC$ has a prescribed spectrum when (C, A) is observable.

Exercise A.5 (Minimal realization). Find a minimal state-space realization of the scalar transfer function $G(s) = \frac{s+2}{s^2+3s+2}$. What is its order? Explain, using Theorem A.26, why any lower-order realization is impossible.

Exercise A.6 (Discrete-time stability). For $x(k+1) = Ax(k)$ with $A = \begin{pmatrix} 0 & 1 \\ -a & -b \end{pmatrix}$, use Theorem A.10 to determine the range of (a, b) for which the origin is asymptotically stable. Verify one case by solving the Stein equation (A.13).

Appendix B

Probability Theory

This appendix collects the probabilistic background used in the stochastic half of the book, estimation (Chapter 13), Kalman filtering (Chapter 14), stochastic control (Chapter 15), and control under incomplete information (Chapter 12). The treatment is self-contained but selective: it develops in full only what those chapters actually call upon, and it does so in the language they use. Three ideas carry most of the weight and are therefore given the most care: the *Gaussian random vector* and its behaviour under affine maps and conditioning; *conditional expectation* and its interpretation as a least-squares projection; and the *Hilbert space of square-integrable random variables*, in which the orthogonality principle of linear estimation is simply the projection theorem. Measure theory is used where it clarifies (a probability is a measure, a random variable is measurable, an event of probability zero is negligible) but the measure-theoretic subtleties—existence of conditional expectations, regular conditional distributions, and the like—are stated rather than constructed, with pointers to the standard references in the notes at the end.

B.1 ■ Probability Spaces and Events

The primitive object is a *probability space*, a triple $(\Omega, \mathcal{F}, \mathbb{P})$ that formalizes a random experiment.

Definition B.1 (Probability space). A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ in which

- (1) Ω is a nonempty set, the sample space, whose points $\omega \in \Omega$ are the elementary outcomes;
- (2) $\mathcal{F}$ is a σ -algebra on Ω : a collection of subsets of Ω (the events) that contains Ω and is closed under complementation and countable unions,

$$\Omega \in \mathcal{F}, \quad A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}, \quad A_1, A_2, \dots \in \mathcal{F} \Rightarrow \bigcup_{i \geq 1} A_i \in \mathcal{F};$$

- (3) $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure: it satisfies $\mathbb{P}(\Omega) = 1$ and is countably additive, meaning that for pairwise disjoint events $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \mathbb{P}(A_i).$$

The σ -algebra is the mathematical carrier of *information*: it is exactly the collection of statements about the outcome to which a probability can be assigned. This reading is central to the

later chapters, where a σ -algebra generated by past measurements represents “what the controller knows,” and where conditioning on that σ -algebra (Section B.5) is the operation of best-guessing the unknown state from the available data.

From the axioms one derives the familiar elementary rules: $\mathbb{P}(\emptyset) = 0$; $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$; monotonicity, $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$; finite and countable subadditivity (the *union bound*) $\mathbb{P}(\bigcup_i A_i) \leq \sum_i \mathbb{P}(A_i)$; and *continuity* along monotone sequences, $A_i \uparrow A \Rightarrow \mathbb{P}(A_i) \uparrow \mathbb{P}(A)$ and $A_i \downarrow A \Rightarrow \mathbb{P}(A_i) \downarrow \mathbb{P}(A)$. An event N with $\mathbb{P}(N) = 0$ is called *null*; a statement that holds for all ω outside some null event is said to hold *almost surely* (abbreviated a.s.), or with probability one.

B.1.1 ■ Conditional Probability, Total Probability, and Bayes’ Rule

Definition B.2 (Conditional probability). For events $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the conditional probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (\text{B.1})$$

Fixing B , the map $A \mapsto \mathbb{P}(A \mid B)$ is itself a probability measure on $(\Omega, \mathcal{F})$; it is the measure that describes the experiment once the information “ B occurred” has been incorporated. Rearranging (B.1) gives the *chain rule* $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \mathbb{P}(B)$, and its n -fold extension

$$\mathbb{P}(A_1 \cap \cdots \cap A_n) = \mathbb{P}(A_1) \mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}),$$

which underlies the recursive factorizations of joint densities used in filtering.

Proposition B.3 (Total probability and Bayes’ rule). Let $\{B_i\}_{i \geq 1}$ be a countable partition of Ω into events with $\mathbb{P}(B_i) > 0$. Then for every $A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{i \geq 1} \mathbb{P}(A \mid B_i) \mathbb{P}(B_i), \quad (\text{B.2})$$

and, whenever $\mathbb{P}(A) > 0$,

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j) \mathbb{P}(B_j)}{\sum_{i \geq 1} \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)}. \quad (\text{B.3})$$

Proof. The events $A \cap B_i$ are disjoint with union A , so countable additivity gives $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)$, which is (B.2). For (B.3), write $\mathbb{P}(B_j \mid A) = \mathbb{P}(A \cap B_j) / \mathbb{P}(A) = \mathbb{P}(A \mid B_j) \mathbb{P}(B_j) / \mathbb{P}(A)$ and substitute (B.2) for the denominator. $\square$

Bayes’ rule (B.3) is the engine of recursive estimation: read B_j as a hypothesis about the state and A as a measurement, and it converts a *prior* $\mathbb{P}(B_j)$ and a *likelihood* $\mathbb{P}(A \mid B_j)$ into a *posterior* $\mathbb{P}(B_j \mid A)$. The measurement-update step of the Kalman filter (Chapter 14) and the belief-state recursion of stochastic control (Chapter 15) are density-valued versions of exactly this formula.

B.1.2 ■ Independence

Definition B.4 (Independence of events). Events $A_1, \dots, A_n$ are (mutually) independent if for every index subset $S \subseteq \{1, \dots, n\}$,

$$\mathbb{P}\left(\bigcap_{i \in S} A_i\right) = \prod_{i \in S} \mathbb{P}(A_i).$$

When $\mathbb{P}(B) > 0$, independence of A and B is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$: knowing that B occurred conveys nothing about A . Pairwise independence (the condition for two-element S) is strictly weaker than mutual independence. Independence extends from events to random variables and to σ -algebras in Section B.2.3; it is the assumption that makes disturbance and measurement noise sequences “white,” and that decouples the prediction and correction steps of the filter.

B.2 ■ Random Variables and Distributions

A random variable attaches a numerical readout to each outcome. So that probabilities of readouts are well defined, the readout map must be *measurable*.

Definition B.5 (Random variable and random vector). A random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ is a function $X : \Omega \rightarrow \mathbb{R}$ that is $\mathcal{F}$ -measurable: for every Borel set $B \subseteq \mathbb{R}$, the preimage $X^{-1}(B) = \{\omega : X(\omega) \in B\}$ belongs to $\mathcal{F}$. An n -dimensional random vector $X = (X_1, \dots, X_n)^\top : \Omega \rightarrow \mathbb{R}^n$ is a map each of whose components is a random variable, equivalently a map that is measurable with respect to the Borel σ -algebra of $\mathbb{R}^n$.

Every random vector X generates a σ -algebra

$$\sigma(X) = \{X^{-1}(B) : B \subseteq \mathbb{R}^n \text{ Borel}\} \subseteq \mathcal{F}, \quad (\text{B.4})$$

the smallest σ -algebra with respect to which X is measurable. It encodes precisely the information carried by observing X . The following standard fact—that a function of the data is exactly a $\sigma(X)$ -measurable random variable—is the formal statement of “estimators are functions of the observations,” and it recurs throughout Chapters 12–15.

Lemma B.6 (Doob–Dynkin). A random variable Y is $\sigma(X)$ -measurable if and only if there exists a Borel function g with $Y = g(X)$.

B.2.1 ■ Distributions, Distribution Functions, and Densities

Definition B.7 (Distribution and distribution function). The distribution (or law) of a random vector X is the probability measure P_X on $\mathbb{R}^n$ defined by push-forward, $P_X(B) = \mathbb{P}(X \in B)$. For a scalar X the cumulative distribution function (cdf) is $F_X(x) = \mathbb{P}(X \leq x)$, $x \in \mathbb{R}$, a nondecreasing, right-continuous function with $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$.

The distribution, not the underlying space $(\Omega, \mathcal{F}, \mathbb{P})$, is what one computes with. Two mutually exclusive special cases dominate the applications.

Discrete random variables take values in a countable set $\{x_1, x_2, \dots\}$ and are described by a *probability mass function* (pmf) $p_X(x_i) = \mathbb{P}(X = x_i)$, with $p_X(x_i) \geq 0$ and $\sum_i p_X(x_i) = 1$. The mode/state variables of the Markov-chain models in Chapter 15 are of this kind.

Continuous random vectors possess a probability density function (pdf) $f_X : \mathbb{R}^n \rightarrow [0, \infty)$ for which

$$\mathbb{P}(X \in B) = \int_B f_X(x) dx, \quad \int_{\mathbb{R}^n} f_X(x) dx = 1. \quad (\text{B.5})$$

The Gaussian density of Section B.4 is the recurring example.

B.2.2 ■ Joint, Marginal, and Conditional Distributions

For a pair (X, Y) of random vectors the *joint* distribution is the law of the stacked vector (X, Y) ; in the continuous case it has a joint density $f_{X,Y}(x, y)$. The *marginal* density of X is recovered by integrating out Y ,

$$f_X(x) = \int_{\mathbb{R}^m} f_{X,Y}(x, y) dy, \quad (\text{B.6})$$

and the *conditional* density of X given $Y = y$ is, wherever $f_Y(y) > 0$,

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}. \quad (\text{B.7})$$

Equations (B.6)–(B.7) are the density-level analogues of total probability and conditional probability; Bayes' rule takes the form $f_{X|Y}(x | y) = f_{Y|X}(y | x) f_X(x) / f_Y(y)$, the exact statement used to propagate the conditional density of the state in nonlinear filtering.

B.2.3 ■ Independence of Random Variables

Definition B.8 (Independence). Random vectors $X_1, \dots, X_n$ are independent if their joint distribution factors into the product of the marginals: for all Borel sets B_i ,

$$\mathbb{P}(X_1 \in B_1, \dots, X_n \in B_n) = \prod_{i=1}^n \mathbb{P}(X_i \in B_i).$$

Equivalently, the generated σ -algebras $\sigma(X_1), \dots, \sigma(X_n)$ are independent. In the continuous case independence is the factorization $f_{X_1, \dots, X_n}(x_1, \dots, x_n) = \prod_i f_{X_i}(x_i)$.

A sequence is *independent and identically distributed* (i.i.d.) if its members are mutually independent and share a common law. Independence of X and Y implies $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$ for all integrable g, h ; in particular independent random variables are uncorrelated (Section B.3). The converse fails in general but holds for jointly Gaussian vectors, a fact of central importance (Corollary B.15).

B.3 ■ Expectation, Moments, and Covariance

Definition B.9 (Expectation). The expectation (mean) of a random variable X is the Lebesgue integral

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) d\mathbb{P}(\omega),$$

defined whenever $\mathbb{E}|X| < \infty$, in which case X is called integrable. For a random vector the expectation is taken componentwise, $\mathbb{E}[X] = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_n])^\top$.

Expectation is *linear*, $\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y]$; *monotone*, $X \leq Y \Rightarrow \mathbb{E}[X] \leq \mathbb{E}[Y]$; and reduces to a weighted sum or an integral against the distribution, the *law of the unconscious statistician*:

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}^n} g(x) f_X(x) dx \quad (\text{continuous case}), \quad (\text{B.8})$$

with the integral replaced by $\sum_i g(x_i) p_X(x_i)$ in the discrete case. Note that linearity requires no independence, a fact used constantly when manipulating quadratic costs.

B.3.1 ■ Variance and the Covariance Matrix

Definition B.10 (Variance, covariance, covariance matrix). For square-integrable random variables (finite second moment) the variance and covariance are

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2], \quad \text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)].$$

For a random vector $X \in \mathbb{R}^n$ with mean $\mu = \mathbb{E}[X]$, the covariance matrix is

$$\Sigma = \text{cov}(X) = \mathbb{E}[(X - \mu)(X - \mu)^\top] \in \mathbb{R}^{n \times n}, \quad (\text{B.9})$$

and for two vectors $X \in \mathbb{R}^n$, $Y \in \mathbb{R}^m$ the cross-covariance is $\Sigma_{XY} = \text{cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)^\top] \in \mathbb{R}^{n \times m}$.

Expanding the product gives the frequently used identities $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$ and $\Sigma = \mathbb{E}[XX^\top] - \mu\mu^\top$. Two structural facts are used throughout the estimation and control chapters.

Proposition B.11 (Properties of the covariance matrix). Let $X \in \mathbb{R}^n$ be square-integrable with covariance Σ , and let $A \in \mathbb{R}^{p \times n}$, $b \in \mathbb{R}^p$. Then:

(1) Σ is symmetric and positive semidefinite, $\Sigma = \Sigma^\top \succeq 0$;

(2) affine maps transform the mean and covariance by

$$\mathbb{E}[AX + b] = A \mathbb{E}[X] + b, \quad \text{cov}(AX + b) = A \Sigma A^\top; \quad (\text{B.10})$$

(3) Σ is positive definite if and only if X is not supported on any proper affine subspace of $\mathbb{R}^n$ (no nontrivial linear combination $a^\top X$ is a.s. constant).

Proof. Symmetry is immediate from (B.9). For any $a \in \mathbb{R}^n$, $a^\top \Sigma a = \mathbb{E}[(a^\top (X - \mu))^2] = \text{Var}(a^\top X) \geq 0$, giving $\Sigma \succeq 0$ and, since this variance is zero exactly when $a^\top X$ is a.s. constant, also (3). Part (2) follows from linearity of expectation: $\text{cov}(AX + b) = \mathbb{E}[A(X - \mu)(X - \mu)^\top A^\top] = A \Sigma A^\top$. $\square$

The transformation law (B.10) is the single most-used computation in the linear-quadratic theory: it is how noise covariances propagate through the system dynamics $x_{k+1} = Ax_k + Bu_k + Gw_k$ and how the estimation-error covariance is pushed through the Kalman recursion.

B.3.2 ■ Correlation and Useful Inequalities

The *correlation coefficient* $\text{corr}(X, Y) = \text{cov}(X, Y) / \sqrt{\text{Var}(X) \text{Var}(Y)} \in [-1, 1]$ measures linear association; X and Y are *uncorrelated* when $\text{cov}(X, Y) = 0$. The following inequalities are used to bound moments and to justify convergence arguments.

Proposition B.12 (Moment inequalities). *Let X, Y be random variables with the indicated moments finite.*

- (1) Cauchy–Schwarz: $|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2] \mathbb{E}[Y^2]}$; consequently $|\text{corr}(X, Y)| \leq 1$.
- (2) Markov: if $X \geq 0$ then $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$ for $a > 0$.
- (3) Chebyshev: $\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \text{Var}(X)/a^2$ for $a > 0$.
- (4) Jensen: if φ is convex then $\varphi(\mathbb{E}[X]) \leq \mathbb{E}[\varphi(X)]$.

Chebyshev’s inequality (a corollary of Markov’s applied to $(X - \mathbb{E}X)^2$) is the quantitative link between a small variance and concentration about the mean, and it yields the weak law of large numbers in Section B.7. Jensen’s inequality justifies, among other things, that minimizing an expected convex cost is a well-posed convex problem.

B.3.3 ■ Characteristic and Moment Generating Functions

The characteristic function of $X \in \mathbb{R}^n$,

$$\phi_X(t) = \mathbb{E}[e^{it^\top X}], \quad t \in \mathbb{R}^n, \quad (\text{B.11})$$

always exists and determines the distribution uniquely; independent summands have multiplying characteristic functions, $\phi_{X+Y} = \phi_X \phi_Y$. The *moment generating function* $M_X(t) = \mathbb{E}[e^{t^\top X}]$, when finite near the origin, plays the same role and generates the moments by differentiation. These transforms give the cleanest route to the defining property of Gaussian vectors and to the central limit theorem, both taken up below.

B.4 ■ Gaussian Random Vectors

The Gaussian (normal) law is the backbone of the linear estimation and control theory: it is preserved under affine maps and under conditioning, and for Gaussian vectors the optimal estimator is linear. This section develops the properties the main text uses.

Definition B.13 (Gaussian random vector). *A random vector $X \in \mathbb{R}^n$ is Gaussian (normal) if every scalar linear combination $a^\top X$, $a \in \mathbb{R}^n$, is a (possibly degenerate) scalar Gaussian random variable. Equivalently, X has characteristic function*

$$\phi_X(t) = \exp\left(it^\top \mu - \frac{1}{2} t^\top \Sigma t\right), \quad t \in \mathbb{R}^n, \quad (\text{B.12})$$

for some $\mu \in \mathbb{R}^n$ and some symmetric $\Sigma \succeq 0$. We write $X \sim \mathcal{N}(\mu, \Sigma)$.

The parameters are exactly the mean and covariance, $\mu = \mathbb{E}[X]$ and $\Sigma = \text{cov}(X)$, so a Gaussian law is completely specified by its first two moments. When $\Sigma \succ 0$ the law is *nondegenerate* and possesses the density

$$f_X(x) = \frac{1}{(2\pi)^{n/2} (\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu)\right). \quad (\text{B.13})$$

When Σ is singular the vector is concentrated on the affine subspace $\mu + \text{range}(\Sigma)$ and has no density on $\mathbb{R}^n$, but (B.12) remains valid; carrying the degenerate case along via the characteristic function avoids the need to invert Σ .

Proposition B.14 (Affine maps and marginals). *Let $X \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^n$, $A \in \mathbb{R}^{p \times n}$, $b \in \mathbb{R}^p$. Then*

$$AX + b \sim \mathcal{N}(A\mu + b, A\Sigma A^\top). \quad (\text{B.14})$$

In particular every subvector of a Gaussian vector is Gaussian, with mean and covariance read off from the corresponding blocks of μ and Σ .

Proof. The characteristic function of $AX + b$ is $\mathbb{E}[e^{it^\top (AX+b)}] = e^{it^\top b} \phi_X(A^\top t) = \exp(it^\top (A\mu + b) - \frac{1}{2}t^\top A\Sigma A^\top t)$, which is (B.12) with parameters $A\mu + b$ and $A\Sigma A^\top$. Marginals are the special case in which A selects coordinates. $\square$

That the Gaussian family is *closed* under affine transformation (B.14) is why a linear system driven by Gaussian noise keeps a Gaussian state, and why the Kalman filter needs to propagate only means and covariances. Two further closure facts complete the picture.

Corollary B.15 (Sums and uncorrelatedness).

- (1) *If $X \sim \mathcal{N}(\mu_X, \Sigma_X)$ and $Y \sim \mathcal{N}(\mu_Y, \Sigma_Y)$ are independent, then $X + Y \sim \mathcal{N}(\mu_X + \mu_Y, \Sigma_X + \Sigma_Y)$.*
- (2) *If (X, Y) is jointly Gaussian, then X and Y are independent if and only if they are uncorrelated, $\Sigma_{XY} = 0$.*

Proof. (1) Multiply characteristic functions and use independence. (2) Independence always implies uncorrelatedness. Conversely, if (X, Y) is jointly Gaussian with $\Sigma_{XY} = 0$ its joint covariance is block diagonal, so (B.12) factors as $\phi_X(s)\phi_Y(t)$, which is equivalent to independence. $\square$

The equivalence in part (2) is special to the Gaussian world and is what makes the *innovations* of the Kalman filter, being jointly Gaussian and uncorrelated across time, an independent (“white”) sequence. The qualifier *jointly* is essential: uncorrelated Gaussian marginals need not be independent if the pair is not jointly Gaussian.

B.5 - Conditional Expectation and Conditional Distributions

Conditional expectation is the mathematical form of an optimal estimate: it is the best guess of a random variable given partial information, where “information” is a sub- σ -algebra and “best” is in mean square. It is the object that Chapters 13–15 estimate and that Chapter 12 replaces, in the non-stochastic setting, by an optimistic selection.

Definition B.16 (Conditional expectation). *Let X be an integrable random variable and $\mathcal{G} \subseteq \mathcal{F}$ a sub- σ -algebra. The conditional expectation of X given $\mathcal{G}$, written $\mathbb{E}[X \mid \mathcal{G}]$, is the a.s. unique random variable satisfying*

- (1) *measurability: $\mathbb{E}[X \mid \mathcal{G}]$ is $\mathcal{G}$ -measurable;*
- (2) *partial averaging: for every $G \in \mathcal{G}$,*

$$\int_G \mathbb{E}[X \mid \mathcal{G}] d\mathbb{P} = \int_G X d\mathbb{P}. \quad (\text{B.15})$$

Existence and uniqueness (up to a null event) are guaranteed by the Radon–Nikodym theorem. We abbreviate $\mathbb{E}[X | Y] := \mathbb{E}[X | \sigma(Y)]$; by Lemma B.6 this equals $g(Y)$ for some Borel g , and $\mathbb{E}[X | Y = y] := g(y)$.

Condition (1) says the estimate uses only the available information; condition (2) says it gets the averages right on every information event. When $\mathcal{G} = \sigma(Y)$ and a joint density exists, the estimate is the mean of the conditional density (B.7),

$$\mathbb{E}[X | Y = y] = \int x f_{X|Y}(x | y) dx, \quad (\text{B.16})$$

which reconciles the abstract definition with the elementary one.

Proposition B.17 (Properties of conditional expectation). *Let X, Y be integrable and $\mathcal{G}, \mathcal{H}$ sub- σ -algebras. The following hold almost surely.*

- (1) **Linearity:** $\mathbb{E}[aX + bY | \mathcal{G}] = a \mathbb{E}[X | \mathcal{G}] + b \mathbb{E}[Y | \mathcal{G}]$.
- (2) **Tower / smoothing:** if $\mathcal{H} \subseteq \mathcal{G}$ then $\mathbb{E}[\mathbb{E}[X | \mathcal{G}] | \mathcal{H}] = \mathbb{E}[X | \mathcal{H}]$; in particular $\mathbb{E}[\mathbb{E}[X | \mathcal{G}]] = \mathbb{E}[X]$.
- (3) **Taking out what is known:** if Z is $\mathcal{G}$ -measurable and ZX is integrable, then $\mathbb{E}[ZX | \mathcal{G}] = Z \mathbb{E}[X | \mathcal{G}]$.
- (4) **Independence:** if X is independent of $\mathcal{G}$ then $\mathbb{E}[X | \mathcal{G}] = \mathbb{E}[X]$.
- (5) **Positivity and monotonicity:** $X \geq 0 \Rightarrow \mathbb{E}[X | \mathcal{G}] \geq 0$; Jensen's inequality holds conditionally, $\varphi(\mathbb{E}[X | \mathcal{G}]) \leq \mathbb{E}[\varphi(X) | \mathcal{G}]$ for convex φ .

The tower property (2) is the *law of total expectation*; it is the identity that telescopes a multi-stage estimation or dynamic-programming recursion, and in the form $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]]$ it justifies computing an expectation by conditioning first on whatever is convenient. Property (3) is the rule that lets known (measured) quantities pass through the estimation operator, and (4) is the formal statement that independent noise cannot be predicted from the data.

B.5.1 ■ Conditional Expectation as Least-Squares Estimation

The estimation-theoretic meaning of $\mathbb{E}[X | \mathcal{G}]$ is that it is the *minimum mean-square-error* (MMSE) estimate of X among all $\mathcal{G}$ -measurable functions.

Theorem B.18 (Conditional expectation is the MMSE estimate). *Let X be square-integrable and $\mathcal{G} \subseteq \mathcal{F}$. Then $\hat{X} = \mathbb{E}[X | \mathcal{G}]$ minimizes the mean-square error over all square-integrable $\mathcal{G}$ -measurable random variables Z :*

$$\mathbb{E}[(X - \hat{X})^2] \leq \mathbb{E}[(X - Z)^2] \quad \text{for all such } Z, \quad (\text{B.17})$$

with equality only if $Z = \hat{X}$ a.s. Equivalently, the error $X - \hat{X}$ is orthogonal to every $\mathcal{G}$ -measurable Z : $\mathbb{E}[(X - \hat{X})Z] = 0$.

Proof. For any $\mathcal{G}$ -measurable Z write $X - Z = (X - \hat{X}) + (\hat{X} - Z)$. The cross term vanishes: $\hat{X} - Z$ is $\mathcal{G}$ -measurable, and by “taking out what is known” and (B.15), $\mathbb{E}[(X - \hat{X})(\hat{X} - Z)] =$

$\mathbb{E}[(\hat{X} - Z)\mathbb{E}[X - \hat{X} \mid \mathcal{G}]] = 0$ since $\mathbb{E}[X - \hat{X} \mid \mathcal{G}] = 0$. Hence $\mathbb{E}[(X - Z)^2] = \mathbb{E}[(X - \hat{X})^2] + \mathbb{E}[(\hat{X} - Z)^2] \geq \mathbb{E}[(X - \hat{X})^2]$, with equality iff $\hat{X} = Z$ a.s. $\square$

Theorem B.18 is the abstract *orthogonality principle*. It identifies the best estimate geometrically, as an orthogonal projection, and the geometry is made explicit in Section B.6. The practical trouble is that the minimizing Z ranges over *all* functions of the data, so $\mathbb{E}[X \mid \mathcal{G}]$ is generally nonlinear and hard to compute. Two escapes are used in the book: restrict the estimator to be *affine* in the data (linear least-squares estimation, Chapter 13), or assume the variables are *jointly Gaussian*, in which case the two coincide.

B.5.2 ■ Gaussian Conditioning

The following theorem is the analytical heart of the Kalman filter and of the LQG results of Chapter 15: for jointly Gaussian vectors the conditional distribution is again Gaussian, its mean is an affine function of the conditioning variable, and its covariance does not depend on the observed value.

Theorem B.19 (Gaussian conditioning). *Let (X, Y) be jointly Gaussian,*

$$\begin{bmatrix} X \\ Y \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_X \\ \mu_Y \end{bmatrix}, \begin{bmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{bmatrix}\right),$$

with $\Sigma_{YY} \succ 0$. Then the conditional distribution of X given $Y = y$ is Gaussian,

$$X \mid Y = y \sim \mathcal{N}(\mu_{X|Y}, \Sigma_{X|Y}), \quad (\text{B.18})$$

with conditional mean and covariance

$$\mu_{X|Y} = \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(y - \mu_Y), \quad (\text{B.19})$$

$$\Sigma_{X|Y} = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}. \quad (\text{B.20})$$

In particular the MMSE estimate $\mathbb{E}[X \mid Y]$ is the affine function (B.19) of Y , and the error covariance $\Sigma_{X|Y}$ is deterministic, independent of the realized y .

Proof. Define $\tilde{X} = X - \mu_X - \Sigma_{XY}\Sigma_{YY}^{-1}(Y - \mu_Y)$. As an affine image of the jointly Gaussian pair, $(\tilde{X}, Y)$ is jointly Gaussian. A direct computation gives $\mathbb{E}[\tilde{X}] = 0$ and $\text{cov}(\tilde{X}, Y) = \Sigma_{XY} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX} = 0$, so by Corollary B.15(2) $\tilde{X}$ is independent of Y . Consequently, conditioning on $Y = y$ leaves the law of $\tilde{X}$ unchanged: $\tilde{X} \mid Y = y \sim \mathcal{N}(0, \Sigma_{X|Y})$ with $\Sigma_{X|Y} = \text{cov}(\tilde{X}) = \Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}$. Since $X = \tilde{X} + \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(Y - \mu_Y)$ and the second term is a constant once $Y = y$, (B.18)–(B.20) follow. $\square$

Two features of (B.19)–(B.20) drive the recursive filter. The gain $K = \Sigma_{XY}\Sigma_{YY}^{-1}$ that multiplies the innovation $y - \mu_Y$ is precisely the Kalman gain; and because $\Sigma_{X|Y}$ does not depend on y , the error covariance can be computed *offline*, in advance of any data, the structural fact that makes the Riccati recursion of Chapter 14 possible. The Schur-complement form of (B.20) also shows $\Sigma_{X|Y} \preceq \Sigma_{XX}$: conditioning never increases uncertainty.

B.6 ■ The Hilbert Space of Random Variables and the Projection Theorem

The orthogonality principle acquires its full force when the square-integrable random variables are viewed as vectors in a Hilbert space. This is the setting of the linear least-squares estimation

theory of Chapter 13.

Definition B.20 (The space L^2). Let $L^2(\Omega, \mathcal{F}, \mathbb{P})$ denote the set of (equivalence classes of) random variables X with $\mathbb{E}[X^2] < \infty$, identified when equal almost surely. Equipped with the inner product and norm

$$\langle X, Y \rangle = \mathbb{E}[XY], \quad \|X\| = (\mathbb{E}[X^2])^{1/2}, \quad (\text{B.21})$$

it is a Hilbert space. Two random variables are orthogonal, $X \perp Y$, when $\mathbb{E}[XY] = 0$.

In this language the covariance is an inner product of centered variables, $\text{cov}(X, Y) = \langle X - \mathbb{E}X, Y - \mathbb{E}Y \rangle$; uncorrelated centered variables are orthogonal; and the variance is a squared length. The Cauchy–Schwarz inequality of Proposition B.12 is the Hilbert-space inequality $|\langle X, Y \rangle| \leq \|X\| \|Y\|$. The general principle behind every least-squares estimate is the following.

Theorem B.21 (Projection theorem). Let $\mathcal{M} \subseteq L^2$ be a closed subspace and $X \in L^2$. There is a unique $\hat{X} \in \mathcal{M}$ nearest to X , $\|X - \hat{X}\| \leq \|X - Z\|$ for all $Z \in \mathcal{M}$, and $\hat{X}$ is characterized by the orthogonality condition

$$X - \hat{X} \perp \mathcal{M}, \quad \text{i.e.} \quad \mathbb{E}[(X - \hat{X})Z] = 0 \quad \text{for all } Z \in \mathcal{M}. \quad (\text{B.22})$$

The map $X \mapsto \hat{X}$ is the orthogonal projection onto $\mathcal{M}$.

Theorem B.18 is the instance in which $\mathcal{M}$ is the subspace of all $\mathcal{G}$ -measurable square-integrable variables; the conditional expectation is the orthogonal projection onto that subspace. Restricting instead to the (smaller, closed) subspace of *affine* functions of the data yields the linear least-squares estimator.

Proposition B.22 (Linear least-squares estimation). Let $X \in \mathbb{R}^n$ and $Y \in \mathbb{R}^m$ be square-integrable with means μ_X, μ_Y , covariances Σ_{XX}, Σ_{YY} , and cross-covariance Σ_{XY} , with $\Sigma_{YY} \succ 0$. Among all affine estimators $\hat{X} = AY + b$, the mean-square error $\mathbb{E}\|X - \hat{X}\|^2$ is minimized by

$$\hat{X} = \mu_X + \Sigma_{XY}\Sigma_{YY}^{-1}(Y - \mu_Y), \quad (\text{B.23})$$

with resulting error covariance $\Sigma_{XX} - \Sigma_{XY}\Sigma_{YY}^{-1}\Sigma_{YX}$.

Proof. Unbiasedness $\mathbb{E}[\hat{X}] = \mu_X$ forces $b = \mu_X - A\mu_Y$, so $X - \hat{X} = (X - \mu_X) - A(Y - \mu_Y)$. The orthogonality condition (B.22) against the components of $Y - \mu_Y$ reads $\mathbb{E}[(X - \hat{X})(Y - \mu_Y)^\top] = 0$, i.e. $\Sigma_{XY} - A\Sigma_{YY} = 0$, giving $A = \Sigma_{XY}\Sigma_{YY}^{-1}$ and (B.23). The error covariance follows by substitution. $\square$

Comparing (B.23) with (B.19)–(B.20) makes the key point precise: the best *affine* estimator and the best *overall* (conditional-expectation) estimator have identical formulas. In general the affine estimator is only a suboptimal, computable surrogate; for jointly Gaussian (X, Y) the two coincide, so the linear estimator is globally optimal. This coincidence is precisely why the Kalman filter, a recursive implementation of (B.23), is the optimal estimator in the Gaussian setting and merely the optimal *linear* estimator otherwise.

B.7 ■ Sequences of Random Variables and Limit Theorems

Asymptotic statements—steady-state covariances, ergodic averages, consistency of estimators—rest on notions of convergence for random variables. Four are standard.

Definition B.23 (Modes of convergence). Let $X_1, X_2, \dots$ and X be random variables. Then $X_n \rightarrow X$

- (1) almost surely if $\mathbb{P}(\{\omega : X_n(\omega) \rightarrow X(\omega)\}) = 1$;
- (2) in mean square (L^2) if $\mathbb{E}[(X_n - X)^2] \rightarrow 0$;
- (3) in probability if $\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$;
- (4) in distribution if $F_{X_n}(x) \rightarrow F_X(x)$ at every continuity point x of F_X .

Almost-sure convergence and L^2 convergence each imply convergence in probability, which in turn implies convergence in distribution; no other implications hold in general. The two classical limit theorems govern the average of an i.i.d. sequence $\{X_i\}$ with mean μ and variance σ^2 , written $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

Theorem B.24 (Laws of large numbers and central limit theorem). Let $X_1, X_2, \dots$ be i.i.d. with mean μ .

- (1) (Weak law) If $\sigma^2 < \infty$, then $\bar{X}_n \rightarrow \mu$ in probability; more generally integrability alone suffices, and in fact $\bar{X}_n \rightarrow \mu$ almost surely (strong law).
- (2) (Central limit theorem) If $0 < \sigma^2 < \infty$, then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2). \quad (\text{B.24})$$

The weak law is immediate from Chebyshev's inequality: $\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$. The central limit theorem (B.24), provable through the characteristic function (B.11), explains the ubiquity of the Gaussian model: sums of many small independent effects are approximately Gaussian regardless of the individual laws, which is the standard justification for modeling process and measurement noise as Gaussian in Chapters 14 and 15.

B.8 - Stochastic Processes

The dynamic models of the book evolve in time and are therefore *stochastic processes*: indexed families of random variables on a common probability space. This final section fixes the vocabulary—filtrations, white noise, the Markov property, and martingales—used to describe them.

Definition B.25 (Stochastic process and filtration). A (discrete-time) stochastic process is a sequence $\{X_k\}_{k \geq 0}$ of random vectors on $(\Omega, \mathcal{F}, \mathbb{P})$. A filtration is an increasing family of sub- σ -algebras $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \dots \subseteq \mathcal{F}$, interpreted as the information available through time k . The process is adapted to $\{\mathcal{F}_k\}$ if each X_k is $\mathcal{F}_k$ -measurable. The natural filtration generated by a process $\{Y_k\}$ is $\mathcal{F}_k^Y = \sigma(Y_0, \dots, Y_k)$.

The filtration is the running record of what is known, and adaptedness is the requirement that a quantity not depend on the future. In the estimation and control chapters $\mathcal{F}_k^Y$ is the *measurement* filtration: admissible estimators and controls at time k are $\mathcal{F}_k^Y$ -measurable, functions of the data seen so far, which is the probabilistic form of the causality/information constraint that Chapter 12 imposes set-theoretically.

Definition B.26 (White noise and Gaussian process). A zero-mean process $\{w_k\}$ is white if it is uncorrelated across time with constant second moment, $\mathbb{E}[w_k] = 0$ and $\mathbb{E}[w_k w_\ell^T] = Q \delta_{k\ell}$, where δ is the Kronecker delta. A process is Gaussian if all its finite-dimensional distributions are Gaussian. A Gaussian white-noise sequence, white and Gaussian, is, by Corollary B.15(2), an i.i.d. $\mathcal{N}(0, Q)$ sequence.

The last observation is the reason “white” and “independent” are used almost interchangeably in the linear–Gaussian theory: for Gaussian sequences the distinction disappears. It also explains the diagnostic role of the Kalman filter’s *innovations*, which are constructed to be a Gaussian white sequence precisely when the filter is correctly tuned.

Definition B.27 (Markov process). An adapted process $\{X_k\}$ has the Markov property if the future is conditionally independent of the past given the present: for every bounded Borel g ,

$$\mathbb{E}[g(X_{k+1}) \mid \mathcal{F}_k^X] = \mathbb{E}[g(X_{k+1}) \mid X_k] \quad a.s. \quad (\text{B.25})$$

Its law is then determined by the initial distribution of X_0 together with the one-step transition kernels $P_k(x, B) = \mathbb{P}(X_{k+1} \in B \mid X_k = x)$.

Equation (B.25) is the property that makes the current state a sufficient statistic for prediction, and it is what allows dynamic programming to condition only on the present state rather than the entire history. Any system driven by independent noise, $x_{k+1} = f_k(x_k, u_k, w_k)$ with $\{w_k\}$ independent, is Markov, the structural fact underlying the controlled-Markov formulation of Chapter 15. When the state is finite-valued the kernel is a *transition matrix* and the process is a *Markov chain*.

Definition B.28 (Martingale). An adapted, integrable process $\{M_k\}$ is a martingale with respect to $\{\mathcal{F}_k\}$ if

$$\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] = M_k \quad a.s. \quad (\text{B.26})$$

for all k ; it is a submartingale if “=” is replaced by “ $\geq$ ” and a supermartingale if by “ $\leq$.”

A martingale models a “fair” accumulation whose best forecast of the next value is its current value; by the tower property its mean is constant, $\mathbb{E}[M_k] = \mathbb{E}[M_0]$. Martingales formalize the sense in which the innovations sequence carries no predictable component and in which certain value-function and error processes are conserved in expectation, and they are the discrete-time shadow of the continuous-time noise models, Brownian motion and the Wiener process, that appear in the continuous-time filtering and control of Chapters 12 and 14.

Notes and Further Reading

The measure-theoretic foundations summarized in Sections B.1–B.5 are developed in full in the standard graduate texts of Billingsley [21], Durrett [40], and Williams [125], the last especially readable on conditional expectation and martingales; the engineering-oriented account of Papoulis and Pillai [91] covers the same ground with an emphasis on random variables and stochastic processes. The Hilbert-space view of estimation in Section B.6, the projection theorem, and the orthogonality principle follow Luenberger [79]; the linear-estimation development in the signals-and-systems idiom used here is that of Kailath, Sayed, and Hassibi [62]. Gaussian conditioning (Theorem B.19) and its role in the linear-quadratic–Gaussian theory are central to Anderson and Moore [3], and the measure-theoretic treatment of stochastic control that this material serves is

that of Bertsekas and Shreve [19]. The appendix is arranged to support Chapters 12–15 directly, and the cross-references there point back to the numbered results above.

Exercises

Exercise B.1 (Bayes' rule and total probability). A binary source transmits $X \in \{0, 1\}$ with $\Pr(X = 1) = p$ over a channel that flips the bit with probability $\varepsilon \in (0, \frac{1}{2})$, producing the output Y . Using the law of total probability and Bayes' rule (Section B.1), compute $\Pr(Y = 1)$ and the posterior $\Pr(X = 1 \mid Y = 1)$. Show that the posterior is increasing in p and decreasing in ε , and interpret the two limiting cases $\varepsilon \rightarrow 0$ and $\varepsilon \rightarrow \frac{1}{2}$.

Exercise B.2 (Expectation, variance, and inequalities). Let X be a nonnegative random variable with $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$. Using the inequalities of Section B.3, (a) bound $\Pr(X \geq a)$ for $a > \mu$ by Markov's inequality, (b) bound $\Pr(|X - \mu| \geq k\sigma)$ by Chebyshev's inequality, and (c) exhibit a distribution for which the Chebyshev bound holds with equality, confirming that it cannot be improved without further assumptions.

Exercise B.3 (Covariance matrices and positive semidefiniteness). Let $X \in \mathbb{R}^n$ be a random vector with covariance matrix Σ . Prove directly from the definition that Σ is symmetric positive semidefinite, and show that Σ is singular if and only if some linear combination $a^\top X$ is almost surely constant. Relate this degeneracy to the geometry of the Hilbert space of random variables in Section B.6.

Exercise B.4 (Gaussian conditioning). Let $\begin{pmatrix} X \\ Y \end{pmatrix}$ be jointly Gaussian with mean $\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}$ and covariance $\begin{pmatrix} \Sigma_{XX} & \Sigma_{XY} \\ \Sigma_{YX} & \Sigma_{YY} \end{pmatrix}$ with $\Sigma_{YY} \succ 0$. Using Theorem B.19, derive the conditional mean and covariance of X given $Y = y$, and verify that the conditional covariance does not depend on y . Explain how this result underlies the measurement-update step of the Kalman filter.

Exercise B.5 (Conditional expectation as least-squares estimation). Let X and Y be square-integrable random variables. Show that the conditional expectation $\mathbb{E}[X \mid Y]$ is the minimizer of $\mathbb{E}[(X - g(Y))^2]$ over all measurable functions g , and prove the orthogonality principle $\mathbb{E}[(X - \mathbb{E}[X \mid Y])h(Y)] = 0$ for every bounded measurable h . Identify precisely when the optimal estimator is linear in Y , connecting the result to Section B.5.

Exercise B.6 (Convergence and limit theorems). Let $X_1, X_2, \dots$ be i.i.d. with mean μ and variance σ^2 , and let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Using the results of Section B.7, (a) show that $\bar{X}_n \rightarrow \mu$ in mean square and in probability, and (b) state the central limit theorem for $\sqrt{n}(\bar{X}_n - \mu)$ and use it to give an approximate confidence interval for μ . Explain the distinction between convergence in probability and convergence in distribution illustrated by these two statements.

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Index

- H^∞ control, 581, 613
- absolutely continuous function, 439
- action functional, 452
- activation function, 492
- active set, 122, 392
- adaptive control, 720
- additive-increase
 - multiplicative-decrease (AIMD), 428
- adjoint equation, 483, 537
 - backpropagation, 494
- adjoint operator, 317
- adjoint variable, *see* costate variable, 533
- admissible policy, 676, 687
- admissible set, 439
- ADMM, *see* alternating direction method of multipliers
- AIMD, *see* additive-increase multiplicative-decrease
- ALM, *see* augmented Lagrangian method
- α - β filter, 667
- alternating direction method of multipliers (ADMM), 418
- analytic center, 409
- Armijo condition, 339
- augmented Lagrangian method, 135, 418
- average cost, 681, 737
- backpropagation, 494
- backtracking line search, 339
- backward induction, 507
- ball
 - Euclidean, 168
- Banach space, 56, 298, 438
- bang–bang control, 441
- bang–bang control, 544
 - switching function, 545
- barrier function
 - logarithmic, 405
- barrier method, 405–413
- Bayes’ rule, 715, 721, 762
- belief state, 521, 583, 714
 - MDP reduction, 716
 - update, 715
- Bellman
 - equation, 507
 - principle of optimality, 503
- Bellman equation, 25, 507, 668, 702
 - continuous-time, 558
- Bellman recursion, 507
- Beltrami identity, 450
- best approximation problem, 625
- best-case cost, 590
- BFGS update, 337
- Bolza problem, 535
- Borel measurability, 676
- boundary conditions
 - transversality, 541
- bounded linear functional, 77
- bounded variation, 45
- brachistochrone problem, 22, 437
- broken extremal, 464
- $C[0, 1]$ space, 57
- calculus of variations, 15, 435, 483
- canonical form, 757
- Carathéodory’s theorem, 171
- catenary, 436
- Cauchy sequence, 54, 624
- Cauchy–Schwarz inequality, 622, 766
- central limit theorem, 771
- central path, 406–413
- certainty equivalence, 522, 581, 590–604, 674, 679, 704, 732
 - continuous-time, 608
 - controller, 594
 - deterministic, 595
 - fixed-point relation, 609
 - linear-quadratic, 612
 - LQG, 736
 - principle, 595, 609
 - stationary controller, 601
- change of variables, 220
- characteristic function, 766
- Charnes–Cooper transformation, 223
- Chebyshev center, 267
- Chebyshev inequality, 766
- closed set, 46
- closed-loop control, 507
- closed-loop spectrum
 - separation, 601
- closed-loop stability, 679, 740
- closure point, 46
- coercive function, *see* coercivity
- coercivity, 69, 468
 - verification, 469
- compactness, 64
- comparison principle, 701
- compatible state set, 581, 583
 - continuous-time, 605
- complementary slackness, 140, 193, 310, 406, 497
- completeness, 56, 624
- condition number, 349
- conditional expectation, 628, 677, 694, 767
- conditional probability, 762
- cone, 84, 300
 - convex, 169, 300

- critical, 98, 155
- descent, 169
- dual, 301
- interior points, 301
- negative, 300
- normal, 89, 124, 251, 398, 499
- of directions of constancy, 259
- optimization over, 316
- polar, 145, 259
- positive, 300
- positive semidefinite, 303
- tangent, 85, 122, 250, 499
- cone-convex mapping, 303
- congestion control, 426
 - TCP/IP, 426
- congestion price, 421
- conjugate exponent, 300
- conjugate function, *see* Fenchel
 - conjugate, *see* Fenchel
 - conjugate
- conjugate momentum, 459
- conjugate point, 471
 - example, 472
- conservation law, 450
- constrained variational problem, 436
- constraint, 13
 - active, 122, 189
 - effective, 252
- constraint qualification, 29, 120, 189, 318
 - linear independence, 261
 - linear independence (LICQ), 124
- Mangasarian–Fromovitz (MFCQ), 29, 149
- Slater’s, 29, 140
- constraint set, *see* feasible set
- continuation method, *see* homotopy method
- continuity, 51
 - of linear maps, 52
- control constraint, 544
- control trajectory, 484
- controllability, 754
 - Gramian, 754
 - tests, 754
- convergence, 48
 - linear, 347, 348
 - modes of, 771
 - quadratic, 382
 - sublinear, 348
 - superlinear, 337
- convex envelope, 228
- convex function, 174
 - continuity of, 177
 - first-order condition, 179
 - second-order condition, 180
 - strictly, 177
 - strongly, 182
- convex hull, 171
- convex mapping
 - with respect to a cone, 303
- convex optimal control, 496
- convex optimization, 167–238, 250
 - in vector spaces, 297
- convex program
 - general, 306
- convex set, 168
- convexification, 218
- corner point, 459
- correlation, 765
- cost-to-come, 581, 592
 - continuous-time, 608
 - forward recursion, 592
- cost-to-go, 504, 581, 591, 700
 - continuous-time, 557
- costate, 483, 533
 - as value-function gradient, 561, 598
 - continuous-time, 537
 - discrete-time, 489
- costate variable, 478
- covariance, 765
- covariance matrix, 765
- curvature condition, 338
- cyclic variable, 450
- cycloid, 22, 437
- Danskin’s theorem, 609
- decision variable, 13
- decomposition
 - dual, 226
 - primal, 225
- deep learning
 - as optimal control, 491
- descent direction, 336
- descent lemma, 341
- descent method
 - iterative, 336
- detectability, 661, 756
- detuned gain, 601, 612
- differentiability, 121
- Dinkelbach’s method, 224
- directional derivative, 72, 443
- Dirichlet energy, 444
- discrete-time maximum
 - principle, 483
- dissipativity
 - strict, 568, 571
- distribution, 763
- distribution function, 763
- disturbance attenuation, 613
- double integrator, 544
- dual ascent, 416
- dual control, 720
 - optimal policy, 722
- dual decomposition, 419–421
- dual function, 202, 312
 - complete, 271
 - concavity, 313
 - concavity of, 203
 - restricted, 270
- dual problem, 202, 312
 - complete, 272
 - restricted, 270
- dual space, 77, 298
- duality, 103–110
 - control and estimation, 551
 - controllability–observability, 756
 - estimation and control, 673
 - Fenchel dual problem, 107
 - game-theoretic interpretation, 212
 - gap, 108
 - strong, 109, 213, 270
 - weak, 108, 203, 270
- dynamic game, 613
- dynamic inconsistency, 520
- dynamic programming, 18, 502–508, 667, 699
 - Bellman equation, 702
 - Bellman recursion, 507
 - continuous-time, 557
 - cost-to-go, 591
 - discrete-time, 502
 - forward, 592, 608
 - incomplete information, 590
 - inventory control, 523
 - optimal stopping, 525
 - principle of optimality, 503
 - shortest path, 508
- ellipsoid, 168
- entropy optimization, 264
- epidemic control, 23
- epigraph, 174
- equality constraints, 314

- error covariance, 641
- estimation
 - batch, 633
 - recursive, 633
- estimation theory, 621
- Euler–Lagrange condition, 123
- Euler–Lagrange equation, 447
 - free endpoint, 452
 - higher-order, 473
 - mixed endpoints, 454
 - vector, 474
- Euler–Lagrange equations, 15
- exhaustible resource extraction, 321
- expectation, 764
- extragradient method, 418
- extreme point, 171
- extremum
 - strong, 441
 - weak, 441
- Farkas’ lemma, 143
- feasible descent direction, 392
- feasible direction, 83, 122
- feasible direction method, 391
- feasible point
 - strictly, 189
- feasible set, 13, 42, 120, 188
 - base, 250
 - decomposition of, 249
 - locally polyhedral, 98
- feedback
 - value of, 514
- feedback control, 688
- feedback policy, 503
- Fejer monotonicity, 402
- Fenchel conjugate, 104, 282
- Fenchel duality, 106
- Fenchel–Young inequality, 282
- Fermat’s principle, 436
- filtration, 367, 687, 771
- finite-horizon problem, 489
- finite-length property, 357
- Finsler geometry, 475
- first variation, 443
- fixed-point controller, 609
- fixed-point iteration, 399
- Fréchet derivative, 443
- fractional programming, 223
- Fréchet derivative, 299
- Fréchet differential, 75
- Fréchet–Taylor expansion, 81
- free-boundary problem, 550
- Fritz–John conditions
 - abnormal case, 126
 - functional, 15, 50, 440
 - fundamental lemma of the
 - calculus of variations, 445
- Gâteaux derivative, 443
- Gâteaux differential, 72
- Gaussian process, 772
- Gaussian random vector, 642, 696, 766
 - conditioning, 677, 769
 - uncorrelatedness and independence, 767
- generalized inequality
 - constraint, 306
- geodesic, 451
 - anisotropic, 475
- geometric programming, 222
- Gordan’s theorem, 141, 190
- gradient, 80, 121
 - monotone, 180
- gradient descent, 337
 - linear convergence, 347
 - nonconvex, 352
- gradient inequality, 179
- Gram determinant, 630
 - distance to a subspace, 631
- Gram matrix, 630
- Gram–Schmidt procedure, 628–636
- Gronwall inequality, 368
 - differential form, 368
 - discrete-time, 369
- half-space, 168
- Hamilton–Jacobi–Bellman
 - equation, 17, 558–565
 - and maximum principle, 561
 - incomplete information, 608
 - verification theorem, 559
- Hamiltonian, 459, 537
 - continuous-time, 537, 606
 - discrete-time, 489
 - imperfect measurements, 586
- Hamiltonian matrix, 569, 570
 - hyperbolicity, 572
 - spectral symmetry, 572
- Hamiltonian system, 543
- Hessian, 81, 131, 180
 - reduced, 131
- Hessian matrix, 337
- Hilbert space, 299, 624
 - of random variables, 770
- Hölder’s inequality, 300
- homotopy method, 410
- hyperplane
 - separating, 173
 - supporting, 172
- imperfect state information, 723
- imperfect state measurement, 582
- incomplete information, 581
 - continuous-time, 604
- independence, 763
 - of random variables, 764
- indicator function, 285
- infimum, 14
- infinite-horizon problem, 501
 - incomplete information, 599
- information constraint, 585
 - continuous-time, 606
- information history, 583
- information set, 581, 583
- information state, 521, 583, 714
 - sufficient statistic, 590
- information structure, 520, 676, 687
 - output feedback, 583
 - state feedback, 503
- information value function, 590
- inner product, 621
- innovation process, 726
- innovations, 630, 655
 - covariance, 645
 - variable, 633
 - whiteness, 660
- input–output behavior, 16
- interaction potential, 476
- interior point, 45
- interior-point method, 405, 407
- inventory control, 523
- Isaacs equation, 613
- isoperimetric problem, 160, 436
- Jacobi condition, 470
 - example, 472
- Jacobi equation, 470
- Jensen’s inequality, 177
- Joseph form, 728
- Kalman decomposition, 757
- Kalman filter, 522, 636, 651
 - finite horizon, 725
 - gain, 646
 - measurement update, 657
 - predictor, 658
 - steady-state, 738
- Kalman gain, 658, 726

- Kalman–Bucy filter, 551, 581, 610
 - via maximum principle, 555
- Kantorovich potentials, 268
- Karush–Kuhn–Tucker
 - conditions, 29, 150, 194, 261, 317, 393, 405
 - continuous-time, 547
 - discrete-time optimal control, 497
 - nonsmooth, 290
 - system, 412
- KKT, *see* Karush–Kuhn–Tucker
 - conditions, *see*
 - Karush–Kuhn–Tucker conditions, *see*
 - Karush–Kuhn–Tucker conditions, *see*
 - Karush–Kuhn–Tucker conditions, *see*
 - Karush–Kuhn–Tucker conditions, *see*
 - Karush–Kuhn–Tucker conditions
- L-BFGS, 337
- L^2 space, 623, 770
- Lagrange candidate, 127
- Lagrange form, 535
- Lagrange multiplier, 126, 191, 283
 - adjoint, 486
 - generalized, 306
 - state constraint, 497
 - uniqueness of, 273
 - unrestricted, 314
- Lagrange multiplier theorem, 311
- Lagrange multipliers, 28
- Lagrange, method of, 123
- Lagrangian, 126, 191, 306, 413, 440
 - complete, 271
 - discrete-time optimal control, 486
 - restricted, 269
- law of large numbers, 771
- learning rate, *see* step size
- least-squares estimation, 621
 - affine, 642
 - equivariance, 643
 - linear, 770
 - linear (LLS), 653
 - linear (LLSE), 637
 - sequential, 643
 - vector, 638
- Legendre condition, 467
 - strict, 470
- Legendre transform, *see* Fenchel
 - conjugate
- LICQ, *see* constraint
 - qualification, linear independence (LICQ), *see* constraint qualification
- line search, 338
 - Armijo backtracking, 339
 - exact, 338
 - Wolfe conditions, 338
- linear estimator, 637
- linear functional, 297
 - bounded, 298
 - continuous, 298
- linear matrix inequality, 220, 303
- linear programming, 28
 - duality, 209
- linear regulator
 - stochastic, 667
- linear time-invariant (LTI)
 - system, 749
- linear time-varying (LTV)
 - system, 749
- linear transformation, 50
- linear-quadratic control
 - incomplete information, 599, 610
- linear-quadratic regulator, 277, 542–544
 - continuous-time, 542
 - discrete-time, 522
 - fixed terminal state, 544
 - infinite-horizon, 599
 - via HJB, 563
- linear-quadratic regulator (LQR)
 - feedback gain, 670
- linear-quadratic-Gaussian control, 522, 614
- linear-quadratic-Gaussian control (LQG), 651, 695
 - finite horizon, 723
 - infinite horizon, 737
- Lipschitz constant, 341
- Lipschitz continuity
 - of the gradient, 341
- LLSE, *see* least-squares estimation
- local extremum, 440
- log-determinant function, 184
- log-sum-exp function, 223
- Łojasiewicz exponent, 354
- Łojasiewicz gradient inequality, 353
- Löwner order, 308
- ℓ^p space, 57
- L^p space, 58, 300
- ℓ^p space, 300
- LQG, *see* linear-quadratic-Gaussian control
 - (LQG)
- Lyapunov equation, 682, 740, 753
 - discrete (Stein), 754
- Lyapunov function, 371, 418
- Lyapunov stability, 371
- Markov decision process, 716
- Markov inequality, 766
- Markov policy, 700
- Markov process, 709, 772
- Markov property, 772
 - controlled, 694
- martingale, 367, 772
 - convergence theorem, 367
- martingale difference sequence, 364
- matrix exponential, 750
- maximum principle, *see* Pontryagin maximum principle
- Mayer form, 535
- measurability constraint, 585
- measurable selection, 700
- measurement noise, 514
- method of multipliers, *see* augmented Lagrangian method
- MFCQ, *see* constraint qualification, Mangasarian–Fromovitz (MFCQ), *see* constraint qualification, Mangasarian–Fromovitz (MFCQ)
- minimax control, 581, 613
- minimax theorem, 214, 311
 - Sion, 214
 - von Neumann, 214
- minimizer, 14
 - global, 83, 120
 - isolated local, 83, 120
 - local, 83, 120
 - local versus global, 188
 - strict local, 120

- uniqueness of, 188
- minimum mean-square error
 - estimation, 627–628, 768
 - estimator, 637
 - Gaussian case, 642
- minimum-time control, 546
- Minkowski sum, 171
- mirror-prox method, *see*
 - extragradient method
- MMSE, *see* minimum
 - mean-square error
 - estimation
- monomial, 222
- monotone operator, 399
- monotonicity argument, 609
- Motzkin's transposition
 - theorem, 144
- multi-agent system, 476
- Nash equilibrium, 311
- natural boundary condition, 452
 - Neumann, 455
- necessary conditions
 - first-order, 489
 - Pontryagin maximum
 - principle, 537
- negative entropy, 286
- network utility maximization,
 - 421–428
- neural network training, 491
- Newton decrement, 380
- Newton direction, 379
- Newton's method, 337, 409
 - damped, 380
 - global convergence, 381
 - pure, 379
 - quadratic convergence, 382
- Newton's second law, 452
- nonconvex optimization, 352
- nonsmooth optimization, 283
- norm
 - Euclidean, 44
 - induced by inner product, 622
 - supremum, 44
- normal equations, 628, 654
- normed space, 43
- NUM, *see* network utility
 - maximization
- objective function, 13
- observability, 756
 - Gramian, 756
- observer, 601
- ODE method, 365
- open set, 45
- open-loop control, 510, 688
 - versus feedback, 519
- operator norm, 298
- optimal control, 13, 477
 - Bolza form, 535
 - continuous-time, 533
 - discrete-time, 274, 483
 - imperfect measurements, 582,
 - 604
 - incomplete information, 581
 - Mayer form, 535
 - partially observed, 520
 - path-constrained, 547
 - state-constrained, 496
 - terminal inequality
 - constraints, 319
- optimal estimation, 551
- optimal policy, 507
- optimal solution, 14
- optimal stopping, 25, 525, 709
 - stopping region, 711
 - threshold policy, 526
- optimal substructure, 503
- optimal transport, 268
- optimal value, 14
- optimal value function, *see*
 - value function
- optimality conditions
 - first-order, 91, 123, 316
 - KKT system, 317
 - nonsmooth, 287
 - second-order, 131
 - second-order necessary, 95
 - second-order sufficient, 96
 - tangent-cone form, 250
 - without Slater's condition,
 - 256
- optimistic valuation, 590
 - detuning, 601
- optimization, 13
- option pricing, 709
 - American option, 525
- orthogonal projection, 627
- orthogonal set, 628
- orthogonality, 625
- orthogonality principle, 634,
 - 654, 768
- orthonormal set, 628
- output feedback, 582
- parallelogram law, 623
- partial order
 - induced by a cone, 300
- partially observed control
 - problem, 520
- partially observed system, 675
- path constraint, 547
- path following, 410
- PBH test, *see* controllability
- penalty method, 422
 - link cost, 422
- penalty method, quadratic, 135
 - penalty parameter, 137
- perspective function, 224
- perturbed gradient method, 361
- pole placement, 755
- policy, 687
 - optimal, 507, 700
 - separated, 716
 - time-consistent, 518
- polyhedron
 - inscribed ball, 267
- Pontryagin maximum principle,
 - 15, 478, 483, 533,
 - 537–542
 - and dynamic programming,
 - 598
 - imperfect measurements, 586,
 - 606
- portfolio optimization, 20
- posynomial, 222
- pre-Hilbert space, 621
- prediction–correction recursion,
 - 583
- predictor–corrector method, 410
- predictor–corrector, 725
- primal–dual algorithm, 413–428
 - projected gradient, 416
- principle of optimality, 503, 668
 - continuous-time, 557
- probability measure, 761
- probability space, 685, 761
- process noise, 514
- projected gradient method,
 - 401–405
- projection
 - Euclidean, 397
 - onto affine subspace, 398
 - onto nonnegative orthant, 398
- projection inequality, 401
- projection method, 396–405
- projection theorem, 624–626,
 - 770
- proper function, 104
- proportional fairness, 20
- Pythagorean identity, 625
- Pythagorean inequality
 - generalized, 401
- quadratic program

- equality-constrained, 274
- quadratic upper bound, 341
- quadratically constrained
 - quadratic program, 227
- quasi-Newton method, 337
- random variable, 763
- random vector, 763
- reachability, 754
- realization, 750
 - minimal, 757
- reformulation
 - equivalent, 219
- regularity condition, 121
- relative interior, 47
- relaxation
 - convex, 219
 - Lagrangian, 230
 - semidefinite, 227
- resource allocation, 20, 509
- Riccati equation, 279
 - algebraic (ARE), 565, 599, 661, 738, 758
 - continuous-time, 543, 612
 - control algebraic, 599
 - differential, 543
 - discrete-time, 522
 - estimation algebraic, 600
 - filter, 555, 660
 - filtering, 727
 - regulator, 670, 733
- Riesz representation theorem, 299
- Riesz–Markov representation theorem, 78
- risk-sensitive control, 614
- Robbins–Monro algorithm, 364
- robust control, 581, 613
- route price, 421
- routing matrix, 423
- saddle point, 121, 191, 252, 308, 413, 613
 - without Slater’s condition, 253
- saddle-point maximum
 - principle, 547
- Schur stability, 682
- search direction, 336
- secant condition, 337
- second variation, 466
- second-order necessary
 - condition, 467
- second-order sufficient
 - condition, 132, 468
- selected value, 609
- semicontinuity
 - lower, 64
 - upper, 64
- semidefinite program, 220
- semidefinite programming, 308
- sensitivity analysis, 215, 283
- separability, 419
- separated policy, 716
- separating hyperplane theorem, 310
- separation principle, 522, 581, 594, 614, 678, 718, 739
- LQG, 734
- sequential quadratic programming, 119
- set-membership estimation, 581, 583
 - continuous-time, 605
 - ellipsoidal, 604
 - least-energy, 600
- shadow price, 215, 321, 533
- Shor relaxation, 228
- shortest path, 451, 458
- shortest-path problem, 508
- σ -algebra, 628, 676, 686, 761
- Sion’s minimax theorem, 311
- SIR model, 23
- Slater’s condition, 140, 189, 306, 393, 497
 - absence of, 252
- smoothness
 - L-smooth function, 346
- Snell’s law, 436
- soft thresholding, 288
- sparsity, 288
- stability, 752
 - Hurwitz, 753
 - Lyapunov, 752
 - mean-square, 740
 - Schur, 753
- stabilizability, 661, 755
- state constraint, 496
 - continuous-time, 550
- state feedback, 755
- state trajectory, 484
- state-space model, 749
 - stochastic, 652
- state-transition matrix, 751
- stationarity
 - variational, 447
- stationarity condition, 317
- stationary point, 336
- stationary solution, 501
- steady-state filter, 661
- steepest descent, 337
- step size, 338
 - constant, 338
 - diminishing, 338
- stochastic approximation, 363
 - ODE method, 365
- stochastic control, 667, 685
 - complete information, 699
 - discrete-time, 685
 - infinite-horizon, 681
 - partial information, 714
 - partial observations, 675
- stochastic gradient descent, 361
 - strongly convex objective, 375
- stochastic kernel, 694
- stochastic process, 685, 771
- stopping time, 25
- storage function, 568, 571
- strong convexity, 346, 348
- strong duality, 308, 414
- subdifferential, 284
 - calculus, 287
- subgradient, 283
- sublevel set, 69
- subspace, linear, 43
- sufficient conditions
 - optimal control, 559
- sufficient statistic, 521, 590, 715
- sunk-cost fallacy, 26
- support function, 177
- support vector, 212
- support vector machine, 210
- surrogate state, 594
- SVM, *see* support vector machine
- switching curve, 545
- system, 16
- system equation, 484
- terminal condition
 - fixed, 544
 - free, 541
- test function, 445
- theorems of the alternative, 141–146
 - Farkas’, 143
 - Gordan’s, 141
 - Motzkin’s, 144
- threshold policy, 526
- time consistency, 518
 - dynamic inconsistency, 520
- time-optimal control, 546
- total probability, 762

- transfer function, 757
- transition probability, 694
- transversality condition, 456, 541
 - orthogonality, 458
- turnpike property, 566–576
 - exponential, 569, 573
 - integral, 571
 - measure, 568, 571
 - steady-state problem, 567, 570
 - turnpike costate, 567, 570
- two-point boundary value problem, 543
- unconstrained optimization, 335
- unknown-but-bounded disturbance, 604
- value function, 25, 280, 504, 670, 700, 704
 - continuous-time, 557
 - information state, 590
- value of information, 604
- variance, 765
- variation
 - admissible, 442
- variation of constants, 751
- variational functional, 440
- variational inequality, 101, 413
- vector space, 42
- vector-space optimization, 297
- verification theorem, 559
- VI, *see* variational inequality
- Wasserstein barycenter, 268
- water-filling, 201
- weak duality, 313
- Weierstrass extreme value theorem, 65
- Weierstrass–Erdmann corner conditions, 459
- white noise, 653, 772
- Wolfe conditions, 338
- worst-case control, 613
- zero-sum game, 214, 311